Axisymmetric (-1)-homogeneous solutions of the stationary Navier–Stokes equations with swirl: a complete picture of the solution set

Preface

The problem. In 1944 Landau found an explicit family of solutions of the stationary Navier–Stokes equations in $\mathbb{R}^3 \setminus \{0\}$ that are homogeneous of degree -1, smooth away from the origin, and singular only at the origin: a jet of viscous fluid emerging from a point source of momentum. The reduced ordinary differential equations had already been integrated by Slezkin, and the family was found again by Squire [42, 45]; these are among the first exact solutions of the Navier–Stokes equations that are genuinely nonlinear [21]. Every (-1)-homogeneous solution that is smooth on the whole unit sphere is a member of this family: this is a theorem of Tian and Xin for axisymmetric solutions [47] and of Šverák in general [49]. The question this book studies begins where those theorems stop. What are the (-1)-homogeneous solutions that are allowed to be singular on the two poles of the sphere, that is, on a whole axis through the origin? Li, Li and Yan answered it for solutions without swirl, that is, without azimuthal velocity: in a series of papers they studied solutions with one and with two singularities on the sphere, classified the axisymmetric no-swirl solutions, and constructed swirling solutions near the no-swirl ones [24–26]. Recently Yan has classified the local solutions near a singular ray with logarithmic behaviour, including solutions with swirl [56]. Physicists studied conical flows with swirl long before, in the form of swirling jets and vortices, and found folds and hysteresis in them [10, 36, 39–41, 44]; but a description of all global axisymmetric solutions with swirl of arbitrary size was missing. This book gives such a description, with one structural point left open (Chapter 13).

What the book does. Writing $x = \cos\theta$ for the cosine of the polar angle, the axisymmetric (-1)-homogeneous solutions form a six-dimensional family of solutions of a system of ordinary differential equations on $(-1, 1)$ for the meridional circulation U and the swirl f, with the poles $x = \pm 1$ as singular endpoints. We determine which of them are global (defined on all of $(-1, 1)$ and of class C^2 on the sphere minus the poles), with the following results.

- The set Ω of global solutions is a closed subset of $\mathbb{R}^6$. Every global solution is a no-swirl solution of Li–Li–Yan, or lies in class Z (the swirl changes sign once) or in class N (it does not).
- Class Z is parametrised by a cone angle, the Taylor constants of a potential, the slope κ of the swirl and the value γ of the meridional flow, in a region (the *lens*) bounded by continuous thresholds $\gamma^- \le \gamma \le \gamma^+$ and $\kappa \le \kappa_{\max}$; the thresholds are characterised and estimated, not given in closed form.

Class N is covered by two overlapping charts, each a product of an edge with a half-line, given by one-parameter shift flows (which commute in the overlap of the charts).

- For every strength of the swirl, in suitable effective coordinates, there is a solution of type G at both poles, and the number of such solutions is odd whenever the data are regular.
- For solutions of type G at both poles, uniqueness holds for moderate swirl, for small swirl, for large swirl (when $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$) and outside a window of folds; it fails in general. We prove, with a computer-assisted certificate that is part of the book, that there are data for which at least three distinct solutions exist.
- The two regimes of small and large swirl are described in detail: a $\sqrt{\kappa}$ law with a thin logistic layer for large swirl, a critical bubble with logistic layers at the poles for small swirl, and laws with sharp exponents (and, where proved, constants) for the thresholds when the constants of the problem approach the boundary of the Li–Li–Yan set.

The precise statements are in Chapter 1. What we could not prove is collected in Chapter 13, each question once, together with what is known about it and a suggestion of how to attack it.

How it is done. The classification is built on a reduction to scalar problems on the line $\xi = \operatorname{artanh} x$: the swirl enters through a single nonlocal potential Q, the meridional flow is the logarithmic derivative of a positive solution z of a Schrödinger equation $z'' = (1 + \frac{1}{2}Q)z$, and the swirl is recovered from $f' = \kappa z^{-2}$. Positivity of Green kernels and Wronskian identities then replace the *a priori* estimates of the no-swirl theory. Near the poles every solution belongs to one of four strata, with explicit normal forms; away from them one compares solutions by monotone-systems arguments. Existence for every swirl strength comes from a Leray–Schauder degree that equals $+1$ for all data, and the parity of the number of solutions follows from it. We have written the proofs in full, in a style that a strong upper-level undergraduate with real analysis, linear ordinary differential equations and basic functional analysis can follow; the standard theorems that are used are collected in Appendix A.

For whom. The book is addressed to readers of two kinds. Analysts working on the Navier–Stokes equations or on nonlinear ordinary differential equations with singular endpoints will find a description of a non-trivial class of solutions, with quantitative laws. Readers coming from fluid mechanics will find that the folds and hysteresis of conical swirling flows, known from experiments and numerical computation, appear here in the class of conical flows as a theorem (the three-solution certificate, which proves multiplicity at one datum) and as numerical observations (the ridge, located only between two values of the parameter). Hysteresis and symmetry breaking are not treated. Nothing in the book is a claim about the time-dependent equations; stability is among the open problems.

How the book is organised. *Part I* (Chapters 1–3) puts the problem on the line. Chapter 1 states the problem and the answer; Chapter 2 derives the reduced system and its reformulations (the Q-system, the Riccati and Schrödinger forms, the first integral, the seven normalisations of the swirl constant); Chapter 3 treats positive solutions on the line, Jost solutions, and the no-swirl solutions in closed form. *Part II* (Chapters 4–5) is about the poles and the global solutions. Chapter 4 gives the limits, exponents, the four strata and the pole charts; Chapter 5 defines the set Ω, proves that it is closed, develops the comparison principles and describes class Z (the lens). *Part III* (Chapters 6–7) completes the classification. Chapter 6 constructs the shifts and the half-cylinder structure of class N; Chapter 7 studies the edges of Ω: their regularity, the walls where the pole type changes, and the global picture of Ω as one stratified set. *Part IV* (Chapters 8–9) uses effective coordinates. Chapter 8 proves existence for every swirl strength and the index criterion; Chapter 9 treats uniqueness and non-uniqueness: the fold set, the ridge, and the computer-assisted proof of three solutions. *Part V* (Chapters 10–12) is about the asymptotic regimes: large swirl (the $\sqrt{\kappa}$ law and the layer), small swirl (the critical bubble, and the laws near the no-swirl window) and thresholds (how large the swirl can be as the constants approach the boundary of the admissible set). *Part VI* (Chapter 13) lists the conjectures and open problems. The appendices contain background theorems (Appendix A), special functions (Appendix B), numerical methods (Appendix C), the certificate of the computer-assisted proof (Appendix D) and a description of the code and data (Appendix E).

What is new, and what is not. The no-swirl classification, the Landau family and the regularity theory for (-1)-homogeneous solutions are due to the authors cited above, and we use them as the starting point. The results listed above for solutions with swirl of arbitrary size are proved in this book; the reductions of Chapter 2 are the tools by which this is done. The local classification of solutions near a singular ray, which Chapter 4 carries out in the form needed here, overlaps with [56], which treats the logarithmic types including swirl; we claim no priority there. We do not claim priority for the physical phenomena (folds, hysteresis, symmetry breaking), which are due to the physics literature cited above; in the axisymmetric class, multiple solutions and the fold structure are proved to occur, and the folds are located within a bounded window that is not sharp. Hysteresis (a statement about stability) and symmetry breaking (non-axisymmetric solutions) are not treated. Numerical computations are used in two ways only: as illustrations and evidence, always in a clearly marked environment, and as part of the one computer-assisted proof, whose method is explained in full. All the code is in the repository described in Appendix E.

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How to read this book

Prerequisites. Real analysis (uniform convergence, dominated convergence, the Arzelà–Ascoli theorem), linear ordinary differential equations (Wronskians, Sturm comparison, first-order systems) and the first notions of functional analysis (Banach spaces, compact operators, the contraction principle) suffice. The more specialised theorems that the proofs use (Leray–Schauder degree, Krawczyk's test, Kamke's comparison theorem for cooperative systems, the Whitney fold and cusp normal forms, Fuchsian equations) are stated precisely and proved or referenced in Appendix A and Appendix B.

The structure of a chapter. Every chapter opens with an *In words* paragraph, in plain language, followed by a *Roadmap* listing its sections; every section has its own *In words*. A new tool is introduced on a *toy example* before it is used. A chapter with numerical content collects its numerical observations in an atlas at its end.

Reading paths.

The fast path to the main theorems.: Read Chapter 1, which states the problem and the answer; then the definitions of the reduced system in Chapter 2 (the Q-system, the line variable $\xi = \operatorname{artanh} x$, the table of κ-normalisations); then the statements, not the proofs, of the main results in the order of the parts: the structure of Ω and the lens (Chapters 5 and 7), class N (Chapter 6), existence (Chapter 8), non-uniqueness (Theorem 9.68 in Chapter 9), the asymptotic laws (Theorem 10.11 in Chapter 10, Theorem 11.58 in Chapter 11, Theorem 12.11 in Chapter 12). Finish with Chapter 13.

The classification.: Chapters 2 to 7 in order. Chapter 4 (the pole) is technical; on a first reading take its statements and the toy examples and return to the proofs when a later chapter uses them.

Existence and uniqueness.: Chapters 2 and 3 (the Jost solutions), then Chapters 8 and 9 (with the statements of Chapters 4 and 5, whose pole types and set Ω they use). Appendix D contains the certificate of the computer-assisted proof and can be skipped by a reader who accepts Theorem 9.68.

Asymptotic regimes.: Chapter 3, then Chapters 10, 11 and 12; they build on the effective coordinates of Chapter 8, the lens of Chapter 5 and the folds of Chapter 9.

Numerics and reproducibility.: The atlas at the end of each chapter, Appendix C and Appendix E.

The main dependencies are as follows: the table lists, for each chapter, the earlier chapters and appendices whose results its proofs cite (a chapter may also quote a definition or a remark from elsewhere; every result is proved from earlier chapters and appendices only).

chapter	proofs use
2	–
3	2, Appendix B
4	2, 3; Appendix A
5	2–4; Appendix A
6	2–5; Appendix A
7	2–6; Appendix A
8	2–5; Appendix A
9	2–5, 8; Appendices A, D
10	2, 3, 5, 8, 9; Appendix A
11	3, 8–10; Appendix B
12	2, 3, 5–7, 11; Appendix B
13	all chapters

Status of a statement. Only statements called *Theorem*, *Proposition*, *Lemma* and *Corollary* are proved in this book, and each of them has a complete proof (in Appendix A the classical results are proved or referenced). There is one exception in method, not in rigour: Theorem 9.68 is proved by a computer certificate, whose method is explained in full in Chapter 9 and Appendix D. Everything else carries its own name and its own numbering. A *Numerical observation* is a computed fact, with its method, tolerance and code, and proves nothing. A *Heuristic* is a non-rigorous argument that explains a phenomenon. A *Formal computation* is a calculation whose steps are not all justified. A *Toy example* is a simple model on which a tool is first shown, and an *Atlas* collects the numerical observations of a chapter. A *Conjecture* is a precise statement that is believed but not proved, and an *Open problem* is a question that the book cannot answer; both occur only in Chapter 13, each once, and the earlier chapters point to them by their labels. A *Remark* may carry a claim; if it does, the claim is proved or points to a proof.

Conventions.

- *Labels.* Labels carry a prefix that tells the kind of statement: `thm:`, `prop:`, `lem:`, `cor:` (proved), `def:`, `ex:`, `toy:`, `rem:`, `no:` and `numobs:` (numerical observation), `heur:`, `form:`, `atlas:`, `conj:` and `op:` (Chapter 13 only), `sec:`, `eq:`, `fig:`, `tab:`, `key:` (a Key result of Chapter 1), `stmt:` (an earlier result recalled in a later chapter), `ch:` and `app:` (chapters and appendices). Cross-references print the name and the number, for example "Theorem 5.3"; sections print as 5.3, appendices as "Appendix A.4".

- *Variables.* $x = \cos\theta \in (-1, 1)$, the north pole is $x = 1$ and the south pole $x = -1$; $\xi = \operatorname{artanh} x$, and $'$ always means $\mathrm{d}/\mathrm{d}\xi$ from Chapter 2 on, with ∂_x written explicitly. $U = u_\theta \sin\theta$ is the meridional circulation, $f = u_\varphi \sin\theta$ the swirl, Q the potential of the Q-system and $\boldsymbol{c} = (c_1, c_2, c_3)$ its Taylor constants at a base point. The Wronskian is $\mathrm{Wr}(a, b) = ab' - a'b$, and $w_0 = \zeta_- - \zeta_+ = \mathrm{Wr}(u_+, u_-) > 0$ for the Jost solutions normalised by $u_\pm(0) = 1$. The notation table (*Notation*, after this chapter) lists every symbol once, with the macro that produces it.
- *The swirl constant κ.* The constant in $f' = \kappa z^{-2}$ depends on how the positive solution z is normalised, and different parts of the book (and of the literature) normalise differently. There are seven normalisations, collected with their conversion formulas in Table 2.1 of Chapter 2; a bare κ is used only after the chapter or section has declared which one it means. The symbols κ_x, $\varkappa$, κ_0 and κ_{D} always denote normalisations 1 to 4 of the table.
- *Constants.* A generic constant that depends on the data is written C and may change from line to line; named constants (such as κ_{lo}) are used consistently within a chapter.

The code repository. Every figure and every number in a numerical observation can be reproduced from the scripts in the repository `code/` that accompanies the book, which has one directory per chapter and per appendix; Appendix E lists which script produces which figure and which number. From the top directory of the repository, `make -C code figures` regenerates the figures, `make -C code checks` reruns the verification scripts and `make -C code all` does both and rebuilds the tables. The computer-assisted proof of Theorem 9.68 is a separate computation of a few minutes, described in Appendix D together with the hashes of its output. Appendix E describes the layout of the repository, the dependencies, the running times, and which script produces each figure and number; Appendix C describes the numerical methods and their tolerances.

Notation

Macros are in `macros.tex`; a bare κ is used only after its normalisation is declared (Section "The seven κ-normalisations").

Variables, data, functions

Symbol (macro)	Meaning	Remarks
θ, φ; $x = \cos\theta$	spherical angles; $x \in (-1, 1)$	
$\xi = \operatorname{artanh} x$	line variable; poles at $\xi = \pm\infty$; $' = d/d\xi$ always, ∂_x explicit	
$U = U_\theta = u_\theta \sin\theta$ (`\Uth`)	meridional circulation	
$f = U_\varphi = u_\varphi \sin\theta$	swirl	
Q, $P_{\boldsymbol{c}}$, $\boldsymbol{c} = (c_1, c_2, c_3)$ (`\cc`, `\Pc`)	potential and its polynomial part, $P_c = c_1(1-x)+c_2(1+x)+c_3(1-x^2)$; $L[P_c] = -2c_1 - 2c_2 - 4c_3$	as in Li–Li–Yan; at a base point these are the Taylor constants of Q
$\boldsymbol{c}'$ (`\Peff`)	**effective constants** $c'_i = \frac{1}{2}Q(\mp 1)$, $c'_3 = -\frac{1}{4}(\lambda + 2c'_1 + 2c'_2)$; $Q = P_{c'} + Q_{\mathrm{sw}}$	c' is used only where the distinction from c matters (Chapters 8–11)
λ	first-integral constant, $L[Q] + f^2 = \lambda$, $\lambda = L[P_c] + A^2$ at a base point with $f = A$	**not** λ_L
λ_L	Landau parameter, $U = 2(1-x^2)/(\lambda_L + x)$	
$W = 1 + \frac{1}{2}Q$, $W_{\mathrm{e}} = 1 + \frac{1}{2}P_{c'}$, $W_c = 1 + \frac{1}{2}P_c$, $W_{\mathrm{crit}} = W_{(c_1, c_2, \bar{c}_3)}$ (`\Weff`, `\Wcrit`)	line potentials	

Symbol (macro)	Meaning	Remarks
$z,\ u_\pm,\ \zeta_\pm = u'_\pm/u_\pm,\ w_0 = \zeta_- - \zeta_+ = \mathrm{Wr}(u_+, u_-) > 0$ (`\zm, \zp, \wzero`)	Jost data, $u_\pm(0) = 1$; $h = z'/z$ the Riccati log of a solution	
ϑ (`\mix`)	mixing ratio $z = u_- + \vartheta u_+$	
$k_i = \sqrt{1 + c_i}$, $k'_i = \sqrt{1 + c'_i}$	explicit exponents at the LLY level	μ is the exponent of an actual solution
$\bar{c}_3(c_1, c_2) = -\frac{1}{2}(k_1 + k_2)(k_1+k_2+2)$ (`\cbar`)	critical value (checked: $= -4$ at $c = 0$)	as in Li–Li–Yan
$\varepsilon := c_3 - \bar{c}_3$ (`\eps`)	distance to the critical surface ($W_c = W_{\mathrm{crit}} + \frac{\varepsilon}{2}\mathrm{sech}^2$); $\varepsilon' = c'_3 - \bar{c}_3(c'_1, c'_2)$	neighbourhood sizes are called δ
$\mathcal{J}$ (`\Jadm`)	Li–Li–Yan admissible set of constants	
$J = f(+\infty) - f(-\infty)$ (`\Jump`)	total swirl jump, $J = \kappa_D/(\vartheta w_0)$	**not** the admissible set
M (`\Mswirl`)	$M = 2\sqrt{\bar{c}_3 - c'_3}$ (critical swirl height)	other bounds are called C_{bd}
$A = f(x_b)$	swirl at the base point; $A = 0$ $\Leftrightarrow$ class-Z base	
γ	$U(x_b)$ at the base point; $\gamma^\pm(\kappa)$ lens walls (`\gm`,`\gp`)	the critical-function constant is γ_{c}
$\gamma_*(\mu)$	limit constant in the class-N law	
μ	pole exponent, $\mu = \sqrt{1 + Q(\pm 1)/2}$; $\sigma = \mu^2$	
η, τ	dummy variables; τ also the layer coordinate	

The seven κ-normalisations

Every positive solution z of $z'' = Wz$ gives a constant $\kappa_W[z] := f'z^2$ (so $f' = \kappa_W[z]z^{-2}$, and $\kappa_W[sz] = s^2\kappa_W[z]$). With $f_x = f'\cosh^2\xi$ and $y = z/\cosh\xi$:

#	Symbol (macro)	Definition	Used in	Conversion
1	κ_x (\kx; written κ in the regularity literature)	$f_x(x_b)$ with $y(x_b)=1$, i.e. $\kappa_W[z]$ for $z(\xi_b)=\cosh\xi_b$; $=\mathrm{Wr}(y,w)$ for $f=w/y$	Chapters 5, 12	$\kappa_x=\varkappa\cosh^2\xi_b$
2	$\varkappa[\xi_b]$ (\vk)	$f'(\xi_b)$ for $z(\xi_b)=1$; $\kappa_W[z]$ for $z(\xi_b)=1$	Chapters 7, 11, 12	$\varkappa=(1-x_b^2)\kappa_x$; $\kappa_{\max}=\cosh^2\xi_0\,\varkappa_{\max}$; at $x_b=0$ equal to κ_x
3	$\tilde\kappa=\kappa_0$ (\kz)	equator slope $f'(0)=f_x(0)$ with $z(0)=1$	Chapters 3, 9 ($x_b=0$)	$\tilde\kappa=\kappa_x(0)=\varkappa[0]$
4	κ_D (\kD)	$\kappa_W[u_-+\vartheta u_+]$, $u_\pm(0)=1$ (effective data)	Chapters 8–10	$\kappa_D=(1+\vartheta)^2\tilde\kappa$; $J=\kappa_D/(\vartheta w_0)$
5	$\kappa_W[z]$ (\kW)	for an arbitrary positive solution z	all chapters	$\kappa_W[sz]=s^2\kappa_W[z]$
6	κ_{ch} (\kch)	the coordinate κ of the pole charts of the pole charts: $\kappa_W[z]$ for the chart normalisation of z ($ze^{\mu\xi}\to a$ or the frame of the stratum)	Chapter 4	see Table 2.1
7	κ_{ξ_1} (\kcut)	$f'(\xi_1)$ for $z(\xi_1)=1$ (fixed-cut frame of a germ)	Chapter 11	$=\varkappa[\xi_1]$; $\kappa_{\xi_1}=\kappa\,z(\xi_1)^{-2}$ in any other normalisation

Convention of the book: a bare κ is allowed only inside a statement that begins "with the normalisation $\kappa=\kappa_x$ (base point x_b)", or in Chapters 8–10 where $\kappa:=\kappa_D$ is declared at the start of each chapter. The lens quantities $\kappa_{\max}(x_0,c)$, $\gamma^{\pm}(x_0,c,\kappa)$ are in normalisation 1; the threshold results are stated in $\varkappa$ and converted once (Chapter 12).

The two centrifugal functions

Symbol (macro)	Definition	Sign / role	Relation
$Q_{\mathrm{sw}} := Q - P_{c'}$ (\Qsw)	centrifugal part, kernel $\frac{1}{4}\cosh^{-2}\xi\, e^{-2\lvert\xi-\eta\rvert}$ applied to f^2	≥ 0, vanishes at both poles; *raises* W: $W = W_{\mathrm{e}} + \frac{1}{2}Q_{\mathrm{sw}}$	used in Chapters 8–10 (effective coordinates)
$\psi_c := P_c - Q$ (\psic)	swirl integral measured from the base point; $L[\psi_c] = f^2 - A^2$, $\psi_c = O((x-x_b)^3)$	≥ 0 and nondecreasing in $\lvert\xi-\xi_b\rvert$ at a zero of f; *lowers* W: $W = W_c - \frac{1}{2}\psi_c$	used in Chapters 5, 11, 12 (lens, thresholds); this is the function ψ of the threshold literature
relation	$c_i' = c_i - \frac{1}{2}\psi_c(\mp\infty)$, $c_3' = c_3 - \frac{1}{4}A^2 + \frac{1}{4}(\psi_c(-\infty)+\psi_c(+\infty))$, and $Q_{\mathrm{sw}} = (P_c - P_{c'}) - \psi_c$ with $P_c - P_{c'} = \frac{1}{2}\psi_-(1-x) + \frac{1}{2}\psi_+(1+x) + \frac{1}{4}(A^2 - \psi_- - \psi_+)(1-x^2)$, $\psi_\mp = \psi_c(\mp\infty)$		at a class-Z base point ($A=0$) this gives $\varepsilon' = \varepsilon - \frac{1}{2}[(k_1-k_1')(k_2+k_2'+2) + (k_2-k_2')(k_1+k_1'+2)]$

Note that ψ_c and Q_{sw} differ by the polynomial $P_c - P_{c'}$ and by sign.

Operators, spaces and the many G, L, Γ, E, A

Symbol (macro)	Meaning	Remarks
$\mathsf{L}[g] = (1-x^2)g_{xx} + 2xg_x - 2g$ (\Lop)	third-order operator's factor ($\partial_x \mathsf{L}[g] = (1-x^2)g_{xxx}$, $\ker = \mathrm{span}\{x, (1+x)^2\}$; checked)	stratum names are sans-serif $\mathsf{L}_1, \mathsf{L}_2$
$\mathsf{G}, \mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$ (\tG, \tR, \tLone, \tLtwo)	pole types	Green operator is G_{L} (\Gsw) — never bare G
$\mathcal{G}$	space of germs	subsets $\mathcal{G}_{\mathsf{G}}$ etc.
K_{L} (\Kcau)	Cauchy kernel of L	
$E(\xi) = e^{-2\lvert\xi\rvert}$ (\Epole)	pole weight (Chapters 4, 6, 7, 11, 12)	the identity is (E)

Symbol (macro)	Meaning	Remarks
$E : \mathrm{Sol} \to \mathbb{D}_+$	effective-data map $\omega \mapsto D$ (Chapters 8–9, 10; Chapter 9: proper, folds, non-injective)	always written with its argument or the domain Sol, never with a variable ξ; no macro
$\omega = (\boldsymbol{c}, \gamma, A, \kappa) \in \mathbb{R}^6$	Cauchy datum of a solution (Chapters 5, 6, 7, 9); $\mathrm{Sol} \subset \Omega \subset \mathbb{R}^6$	the Jost-plane coordinate ω (with $\sigma = \mu^2$, region $\widehat{\Gamma}$) is used in Chapter 4 and in the corner-chart sections of Chapter 7; the Cauchy datum in Chapters 5, 6, 9 and in the smooth-dependence section of Chapter 7; no section uses both. Auxiliary functions called ω (the factor in $u_P = e^{\mu\xi}\omega$, Chapter 6; ω_C; the growth function ω_k, Chapter 11) are local and carry subscripts or are declared at the point of use
Sol (`\Sol`)	set of global solutions with genuine swirl, $\mathrm{Sol} = \mathsf{Z} \sqcup \mathsf{N}$ ($\kappa \neq 0$)	Chapter 1, 5–9
Z, N, N_0 (`\clZ`, `\clN`, `\clNz`)	classes: f has a zero in $(-1, 1)$; f has no zero; fundamental domain of $\mathbb{Z}_2^2$ on N ($0 \le f(-1) < f$)	defined in Chapters 1, 5
Φ_{fold}, $\Phi_{\mathrm{fold}}^{\det}$ (`\Fold`)	spectral fold set $\{\omega \in \mathrm{Sol} : 1 \in \operatorname{spec} A_\omega\}$ and Jacobian fold set $\{\det \mathrm{D}E = 0\}$	Chapter 9
$\operatorname{int}\mathcal{J} = \operatorname{int}\mathcal{J}$ (`\intJ`)	interior of the admissible set of constants	
$\kappa_{\max} = \kappa_{\max}(x_0, \boldsymbol{c})$ (`\kmax`)	supremum of the swirl constants $\kappa = \kappa_x$ of the lens (normalisation 1); $\varkappa_{\max} = \kappa_{\max}/\cosh^2 \xi_0$	
$f_- = f_-$, $f_+ = f_+$ (`\fm`, `\fp`)	swirl at the south, north pole, $f(\mp\infty)$; $f_- \ge 0$ in class N_0	
Ω, Ω_{G} (`\Om`,`\OmG`)	region of Cauchy data	open sets in degree theory are called $\mathcal{O}$
$\mathbb{D} = \mathbb{R}^3 \times (0,\infty) \times \mathbb{R} \times (0,\infty)$ (`\Dsp`)	effective-data space	
T_D (`\TD`), Fix, $\deg_{\mathrm{LS}}$	fixed-point map, fixed points, degree	

Symbol (macro)	Meaning	Remarks
FS, BS, GS; Lens, Dia	safe sets; lens and diamond	
T_α, $\mathcal{R}_C$ (\Tsh, \Rshift)	t^2-shift and recessive shift	
$\alpha_- = c^-$, Θ^- (\athr, \Thr); dep	thresholds; depth	
$\mathrm{Wr}(a,b) = ab' - a'b$	Wronskian	
$\mathbb{Z}_2^2 = \langle \sigma_1, \sigma_2 \rangle$ (\symm)	symmetry group, $\sigma_1 : f \mapsto -f$, $\sigma_2 : x \mapsto -x$	group elements always carry subscripts

Overload register (resolutions for $\Gamma, \sigma, \omega, \nu, D, T, M$ and friends)

Symbol	Competing meanings	Resolution in the book (macro)
Γ	(i) class-N edge; (ii) Gamma function; (iii) Jost region $\widehat{\Gamma}$; (iv) rate constant; (v) kernel integrals; (vi) base Γ_T of the t^2-chart	(i) Γ_{N} (\Edge); (ii) \Gam(s) always with argument; (iii) \Jplane; (iv) Γ_{rate} (\Grate); (v) Γ_{ker}, Γ^U_{ker} (\Gker); (vi) Γ_T keeps its subscript (\BaseT)
σ	(i) Jost-plane coordinate $\sigma = \mu^2$; (ii) symmetries $\sigma_1, \sigma_2, \sigma_{12}$; (iii) logistic function $\sigma(s) = (1+e^{-s})^{-1}$; (iv) spectrum $\sigma(A)$	(i) σ (Jost chapters only); (ii) σ_j with subscript (\symm); (iii) $\lg(s)$ (\logistic); (iv) $\operatorname{spec} A$
ω	Jost-plane coordinate; $\omega(\xi) = \sqrt{W(\xi)}$; ω relaxation parameter (damped iteration)	(i) Jost-plane coordinate ω (Chapter 4); (ii) Cauchy datum $\omega \in \mathbb{R}^6$ (Chapters 5, 6, 9); (iii) $\varpi(\xi) = \sqrt{W}$ and its bounds $\varpi_1, \varpi_{\max}$ (\wfreq); (iv) damping ϱ
ν	leading eigenvalue of DT_D ($\nu_{\max}, \nu_b$); Bessel order; weight exponent ($E^{1-2\nu}$); an exponent $\nu \in (0,1]$	eigenvalues $\nu_{\max}, \nu_{\mathrm{b}}$ (\numax); Bessel order ν_{B} (\nuB); weight exponent ς; the last exponent is ν_0
D	effective data $D = (c', \vartheta, f_-, \kappa_D)$; the space $\mathbb{D}$; $D = 1 + \frac{q}{2} = \mu^2$; D derivative (as in DT_D); D^* the certified datum; a discriminant	data D (\Dat); space \Dsp; $1 + \frac{q}{2} = \mu^2$ is written σ; derivative D (upright, \Dif); D^* (\Dstar)

Symbol	Competing meanings	Resolution in the book (macro)
T	fixed-point map T_D; shifts T_c, T_α; truncation/cut point T (cut lemma, layers); tail operators $\mathbf{T}$	T_D; T_α; cut time t_{cut}; tails $\mathbf{T}$ (bold)
M	critical swirl height; sup-bounds M', M_B; matrices	M only for the critical height (`\Mswirl`); bounds C_{bd} (`\Cbd`)
E	pole weight; identity (E); effective-data map; $e^{2\xi}$	pole weight $E(\xi) = e^{-2\lvert\xi\rvert}$ (`\Epole`) and effective-data map $E : \mathrm{Sol} \to \mathbb{D}_+$, see the table of operators above
A	swirl at base point; Wronskian-type constants $A_\pm$	A = base swirl; the limits of the constants $A_\pm$ are $A_+^{\lim}$ (`\Alim`)
G	stratum G; Green operator; critical function g; $G(\xi) = \int_0^\xi g^{-2}$; G_i pole constants	stratum G; $\mathsf{G_L}$; g; $\mathcal{G}_{\mathrm{int}}(\xi)$ (`\Gint`); pole constants $\mathfrak{g}_i$
L	operator L; $L(\kappa)$ moderate-swirl number; $L_s(\kappa)$	L; $\Lambda_{\mathrm{mod}}(\kappa)$ (`\Lmod`)
Λ	lens; depth; $\Lambda_k(t)$ growth function	Lens; dep; Λ_k keeps its index

Part 1

The problem on the line

CHAPTER 1

The problem and the answer

In words. *A fluid that streams out of (or into) a point, possibly while rotating about an axis, is described at leading order by a flow that is homogeneous of degree* -1*: scaling the space by* ℓ *scales the velocity by* $1/\ell$*, and the flow is determined by its values on the unit sphere. If the flow is also symmetric about the* X_3*-axis, the unknowns are two functions of the polar angle: the meridional circulation* U *and the swirl* f *(the azimuthal velocity, times* $\sin\theta$*). The Navier–Stokes equations become one sixth-order ordinary differential equation, whose local solutions form a six-parameter family.* Which of them are defined on the whole sphere minus the two poles? *This book describes the solution set as one stratified closed set* Ω *of* $\mathbb{R}^6$*, in bijection with all solutions; the one structural point left open is the interval property of its fibres when the swirl does not vanish at the base point (Chapter 13). It gives a second, pole-adapted set of coordinates in which solutions are counted; a complete coordinate description of each of the two classes of solutions with genuine swirl (a* lens *when the swirl changes sign, a* half-cylinder *when it does not), whose thresholds are characterised and estimated, not given in closed form; an existence theorem for* every *prescribed strength of the swirl; asymptotics with sharp exponents and, where proved, constants in the three regimes of weak, strong and critical swirl; and a computer-assisted proof that more than one solution can occur at the same data. This chapter states these results precisely and points to the chapters that prove them; it proves nothing.*

Roadmap. Section 1.1 states the problem, its history and what is known from the work of Li, Li and Yan. Section 1.2 explains the reduction to one Schrödinger equation on the line, which is the main tool of the book. Section 1.3 states the global answer: the set Ω, its strata, the lens (class Z) and the half-cylinder (class N), and how the swirl solutions attach to the no-swirl solutions of Li–Li–Yan. Section 1.4 is about existence and multiplicity, Section 1.5 about asymptotics and thresholds. Section 1.6 is a short tour of what is not known, and Section 1.7 explains how the book is organised. The statements are called *Key results*; each one is followed by the labels of the theorems of later chapters that contain it and its proof, and it is never stronger than the theorem it points to.

1.1. The problem

In words. *We look for solutions of the stationary Navier–Stokes equations in* $\mathbb{R}^3 \setminus \{0\}$ *that are invariant under the natural scaling, symmetric about the* X_3*-axis, and smooth except possibly on the axis. The sphere carries the unknowns; the two poles are the places where the axis meets it.*

The stationary incompressible Navier–Stokes equations (unit viscosity, no force) are

$$-\Delta u + (u\cdot\nabla)u + \nabla p = 0, \qquad \operatorname{div} u = 0 \quad \text{in } \mathbb{R}^3\setminus\{0\}. \tag{1.1}$$

If (u,p) solves (1.1), so does $(\ell\, u(\ell\,\cdot),\ \ell^2 p(\ell\,\cdot))$ for every $\ell>0$. The solutions that are invariant under this scaling, i.e. $u(X)=r^{-1}u(X/r)$ and $p(X)=r^{-2}p(X/r)$ with $r=|X|$, are called (-1)-*homogeneous*. They are the model for the leading behaviour of a flow near an isolated singularity: a point source of momentum produces a velocity that decays like r^{-1}, which is the decay dictated by the scaling. The Landau jet [21], found also by Squire [45], is the classical example: for each number λ_L with $|\lambda_L|>1$ the formula

$$u_r,\ u_\theta \quad\text{with}\quad U = u_\theta\sin\theta = \frac{2(1-x^2)}{\lambda_L + x}, \qquad x=\cos\theta,$$

(Example 2.29) gives a flow that is smooth on all of the unit sphere. Šverák [49] proved that the nonzero (-1)-homogeneous solutions that are smooth on the whole sphere are exactly Landau's (Tian and Xin [47] had proved this for axisymmetric solutions), and that there are no such solutions in dimension $n\ge 4$.

Allowing singular rays. To go beyond Landau's family one allows u to be singular on some rays, i.e. at some points of the sphere. In this book we allow exactly the two rays of the symmetry axis, the north and south poles N and S of the sphere, and we consider solutions of the form

$$\begin{aligned} &u = r^{-1}\big(u_r(\theta)\,\mathbf{e}_r + u_\theta(\theta)\,\mathbf{e}_\theta + u_\varphi(\theta)\,\mathbf{e}_\varphi\big),\\ &p = r^{-2}p(\theta), \qquad u_r,u_\theta,u_\varphi,p\in C^2(0,\pi). \end{aligned} \tag{1.2}$$

The component u_φ is the *swirl velocity*; the flow *has no swirl* if $u_\varphi\equiv 0$. A *genuine* swirl is one that is not just the trivial one $u_\varphi = A/\sin\theta$, which is a harmless potential vortex along the axis that does not influence the other components (Example 2.30). Swirling conical flows with a singular axis have been studied in fluid mechanics since Serrin's paper on the swirling vortex [39]; see also [10, 40] for the physical picture (tornado-like and bathtub-like flows, hysteresis in swirling jets). The mathematical question is:

> *Which sixth-order data determine a flow* (1.2) *that is defined for all polar angles* $\theta\in(0,\pi)$*, and how many solutions are there for a given set of physically meaningful data?*

The six parameters. After the reduction of Section 1.2 the equations form an ordinary differential system for (U,f) on $x\in(-1,1)$ whose solutions are determined by their state at one base point x_b (we use $x_b=0$, the equator). The state is a point of $\mathbb{R}^6$: three constants $\boldsymbol{c}=(c_1,c_2,c_3)$ that come from integrating the pressure equation (they are the Taylor coefficients of a potential Q, see below), the circulation $\gamma=U(0)$, the swirl $A=f(0)$ and the swirl rate κ ($=f_x(0)$ in the normalisation of the book). Every local solution is a point $\omega=(\boldsymbol{c},\gamma,A,\kappa)\in\mathbb{R}^6$; the solution may or may not extend to all of $(-1,1)$.

What was known: Li–Li–Yan. The programme of classifying the solutions with singularities on the sphere is due to Li, Li and Yan. The three papers of the series relevant for us are the following.

- Part I [24] (one singularity, $S^2\setminus\{S\}$): the no-swirl solutions that are smooth except at one point are classified; a first set of asymptotic results at the singular point is proved, among them the existence of the limit of U at the pole and

the restriction $U_\theta(1) > -3$ for solutions with genuine swirl (see Theorem 4.9, which reproves it); and curves of axisymmetric swirling solutions are shown to emanate from every interior point of the no-swirl family and from the part of its boundary where the value of U_θ at the singular point lies below the threshold 3 (in our normalisation), and from no other boundary point.

- Part II [25] (two singularities): the axisymmetric *no-swirl* solutions on $S^2\backslash\{N, S\}$ are classified: they form a four-dimensional surface with boundary $\{(\boldsymbol{c}, \gamma)\}$ in the space of the parameters (Theorem 3.44 below).
- Part III [26] (two singularities, swirl): existence, non-existence and local uniqueness of axisymmetric solutions with swirl *near* the no-swirl surface. The threshold 3 enters again, as ± 3 for the pole values $U_\theta(\pm 1)$ of the no-swirl solution: the proofs construct smooth families of swirl solutions branching off the no-swirl surface, and, at the remaining parameter values, a rigidity statement says that any sequence of solutions converging to a no-swirl solution is eventually a no-swirl solution plus a multiple of the trivial swirl.

What these papers do not give is a description of *all* solutions: the swirl solutions are constructed in a neighbourhood of the no-swirl surface, by perturbation. For the local behaviour of a solution near a singular ray, with logarithmic growth and with swirl, see also [56], which overlaps with the pole analysis of Chapter 4. The present book describes the global structure of the solution set: Section 1.3 shows that the no-swirl surface is a five-dimensional face of the single closed set $\Omega \subset \mathbb{R}^6$ that contains *all* solutions, and the threshold 3 becomes the statement that an exponent at the pole is $\frac{1}{2}$ (Key result 1.7).

1.2. Reduction to one Schrödinger equation on the line

In words. *The key idea is a sequence of four rewritings. Each one separates one difficulty: first the Riccati structure of the meridional equation, then the nonlocal term, then the position of the poles, then the sign of the swirl feedback. At the end, "the solution is global" means "a one-dimensional Schrödinger equation has a positive solution on the whole line".*

Write $x = \cos\theta$ and

$$U = u_\theta \sin\theta, \qquad f = u_\varphi \sin\theta. \tag{1.3}$$

The four viewpoints that organise the book are the following (Chapter 2).

(1) *Riccati linearisation.* Put $\frac{1}{2}U/(1-x^2) = y_x/y$. The equation for U becomes the linear equation $2(1-x^2)^2 y_{xx} = Qy$ and the equation for f integrates once: $y^2 f_x = \kappa$ is constant. Thus f is a ratio of two solutions of one linear equation with Wronskian κ.

(2) *The Q-system.* The pressure and the nonlocal term are eliminated. The equations for (U, f) are equivalent to the following system for (U, f, Q), in which Q is defined by the first equation:

$$(1-x^2)U_x + 2xU + \tfrac{1}{2}U^2 = Q, \qquad (1-x^2)f_{xx} + Uf_x = 0,$$
$$(1-x^2)Q_{xxx} = -2ff_x.$$

The third equation is *local*. Its three integrations give the three constants $\boldsymbol{c}$: Q is a quadratic polynomial $P_{\boldsymbol{c}}(x) = c_1(1-x) + c_2(1+x) + c_3(1-x^2)$ plus a correction produced by the swirl.

(3) *The line.* With $x = \tanh\xi$ the poles go to $\xi = \pm\infty$, and the linear equation becomes the Schrödinger equation

$$z'' = Wz, \qquad W = 1 + \tfrac{1}{2}Q, \qquad U = 2\Big(\frac{z'}{z} - \tanh\xi\Big), \qquad f' = \kappa z^{-2}. \tag{1.4}$$

The solution is *global* (defined on all of $(-1,1)$) iff $z > 0$ on all of $\mathbb{R}$. The potential obeys the first integral $\mathsf{L}[Q] + f^2 = \lambda$, with L an explicit second-order operator.

(4) *Centrifugal decomposition.* Measured from the poles, $Q = P_{\boldsymbol{c}'} + Q_{\mathrm{sw}}$ with a *nonnegative* correction Q_{sw} given by an explicit positive kernel applied to f^2: swirl only raises the potential, $W = W_{\mathrm{e}} + \frac{1}{2}Q_{\mathrm{sw}} \geq W_{\mathrm{e}}$. Measured from a zero of the swirl, the correction has the opposite sign: swirl lowers the potential below $W_{\boldsymbol{c}} = 1 + \frac{1}{2}P_{\boldsymbol{c}}$. The first picture drives existence (Chapter 8), the second the lens of class Z (Chapter 5).

Figure 1.1 summarises the chain.

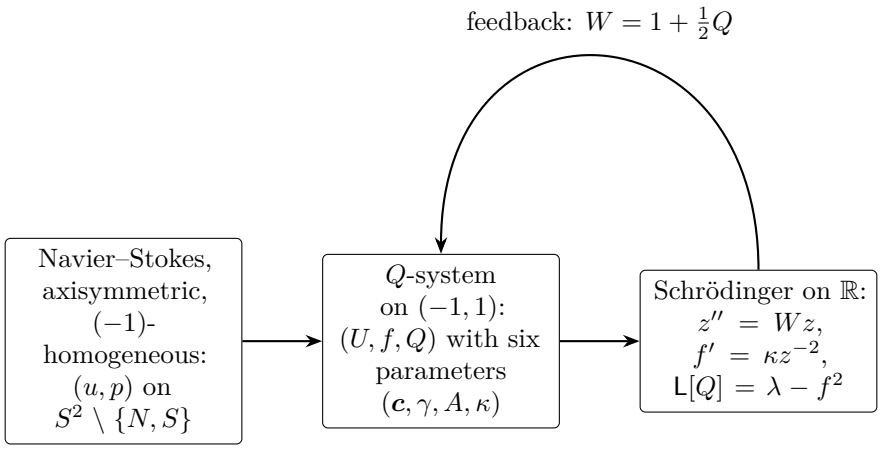


FIGURE 1.1. The chain of reformulations (Chapter 2). The feedback arrow is the whole difficulty: the swirl f, which is built from the solution z of the Schrödinger equation, changes the potential W of the same equation.

Key result 1.1 (the reduction)**.** (a) *A pair (u,p) of the form (1.2) solves (1.1) in $\mathbb{R}^3 \setminus \{X_1 = X_2 = 0\}$ if and only if $(U,f) \in C^1(-1,1) \times C^2(-1,1)$ of (1.3) solves, for some $\boldsymbol{c} \in \mathbb{R}^3$, the system $(1-x^2)U_x + 2xU + \frac{1}{2}U^2 + \psi = P_{\boldsymbol{c}}$, $(1-x^2)f_{xx} + Uf_x = 0$, where ψ is the explicit triple integral of $2ff_x/(1-x^2)$ (so that $Q = P_{\boldsymbol{c}} - \psi$ solves the Q-system).*

(b) *In the variable $\xi = \operatorname{artanh} x$ a solution of the Q-system is given by (1.4), and it is global (extends to $(-1,1)$) iff $z > 0$ on $\mathbb{R}$; the swirl is either constant or strictly monotone and has at most one zero.*

(c) *$\mathsf{L}[Q] + f^2$ is constant, and for a solution with bounded f and finite limits $Q(\pm 1)$ (for instance a global solution of type G or L_1 at both poles, Definition 4.16) one has $W = W_{\mathrm{e}} + \frac{1}{2}Q_{\mathrm{sw}} \geq W_{\mathrm{e}}$.*

Proofs: *Proposition 2.1, Lemma 2.14, Corollary 2.25, Corollary 2.10, Proposition 2.20, Proposition 2.27.*

Example 1.2 (the rest state)**.** For $\boldsymbol{c} = 0$ and $f \equiv A$ the potential is $Q = 0$, $W \equiv 1$, and $z = ae^{\xi} + be^{-\xi} = (a+b)\cosh\xi + (a-b)\sinh\xi$. This is positive on $\mathbb{R}$ iff $a, b \geq 0$ and $(a,b) \neq (0,0)$, and then $U = 2(z'/z - \tanh\xi) = 2(1-x^2)/(\lambda_L + x)$ with $\lambda_L = (a+b)/(a-b)$ for $a \neq b$: this is the Landau family, $|\lambda_L| \geq 1$, together with $U \equiv 0$ ($a = b$). The two "one-ray" solutions $z = e^{\pm\xi}$ ($\lambda_L = \pm 1$) are singular at one pole. At the pole $\xi \to +\infty$ a dominant solution ($a > 0$) gives $U(1) = 0$, the recessive one ($a = 0$) gives $U(1) = -4$. The dichotomy dominant/recessive at $\pm\infty$ is what the whole classification is about (Example 2.32).

1.3. The global answer: the set $\Omega \subset \mathbb{R}^6$

In words. *Take all solutions of the sixth-order system that exist on the whole sphere minus the poles. Record each of them by its state $(\boldsymbol{c}, \gamma, A, \kappa) \in \mathbb{R}^6$ at the equator. The collection is a closed subset Ω of $\mathbb{R}^6$ that contains everything: the no-swirl solutions of Li–Li–Yan as a five-dimensional face, the swirl solutions in two classes, and the boundary where the pole behaviour degenerates. This section describes Ω.*

1.3.1. The poles: four types of behaviour. Near a pole a local solution with genuine swirl behaves in one of four ways. Let $Q(1) = \lim_{x\to 1} Q$ and $\mu = \sqrt{1 + Q(1)/2}$.

Key result 1.3 (the pole)**.** *Let (U, f, Q) be a solution near $x = 1$ with $\kappa \neq 0$.*

(a) *$Q(1)$ exists and is finite, $Q(1) \geq -2$, so $\mu \geq 0$ is defined; $U(1) = \lim_{x\to 1} U$ exists and $U(1) \in \{-2 + 2\mu,\ -2 - 2\mu\}$.*

(b) *Accordingly the pole is of exactly one of four* types*: G (generic: $U(1) = -2 + 2\mu > -2$), R (recessive: $U(1) = -2 - 2\mu$ with $0 < \mu < \frac{1}{2}$, so $U(1) \in (-3, -2)$), L_1 and L_2 (logarithmic: $\mu = 0$, $U(1) = -2$; $f(1)$ finite for L_1, infinite for L_2).*

(c) *Every solution with genuine swirl satisfies $U(1) > -3$ at the north pole and, symmetrically, $U(-1) < 3$ at the south pole.*

(d) *The local solutions of the four types are parametrised, near the pole, by open sets of dimension 6 (G), 5 (R), 5 (L_1) and 4 (L_2) through injective C^∞ immersions into the scale-free state space $\mathbb{R}^6$ (C^∞ charts).*

Proofs: *Theorem 4.5, Definition 4.16, Theorem 4.9, Theorem 4.38, Theorem 4.40.*

So the generic pole is the one where the solution is of the dominant type; the recessive types are exceptional and are exactly what defines the *boundary* of Ω. The number -3 is the threshold of Li–Li–Yan; for us it is just $U(1) = -2 - 2\mu$ with $\mu = \frac{1}{2}$ (Corollary 7.73).

1.3.2. The set Ω. Fix the base point $x_b = 0$. Let

$$\Omega = \Big\{ \omega = (\boldsymbol{c}, \gamma, A, \kappa) \in \mathbb{R}^6 : \begin{array}{c} \text{the solution with state } \omega \text{ at } x = 0 \\ \text{exists on all of } (-1, 1) \end{array} \Big\}.$$

Let $\Omega_0 = \Omega \cap \{\kappa = 0\}$ be the face of solutions without (genuine) swirl, let Sol be the set of global solutions with $\kappa \neq 0$, and let $\Omega_{\mathsf{G}} \subset \Omega$ be the set of solutions that are of type G at both poles. Three symmetries act: σ_1 changes the sign of f (it maps

$(A,\kappa)\mapsto(-A,-\kappa)$), σ_2 reflects the sphere $x\mapsto -x$, and together they generate the group $\mathbb{Z}_2^2$.

Key result 1.4 (Ω is one stratified closed set)**.** (a) *$\Omega\subset\mathbb{R}^6$ is closed, and $\omega\mapsto$ (the solution with state ω) is a bijection from Ω onto the set of* all *global solutions, with and without swirl.*

(b) *As a set, $\Omega=\Omega_0\sqcup\mathsf{Z}\sqcup\mathsf{N}$. Here* $\mathrm{Sol}=\mathsf{Z}\sqcup\mathsf{N}$ *is the dichotomy by the zero of the swirl: class Z if f has a zero in $(-1,1)$, class N if not. Moreover $\mathsf{Z}=\mathsf{Z}_+\sqcup\sigma_1\mathsf{Z}_+$, $\mathsf{N}=\mathsf{N}_0\sqcup\sigma_1\mathsf{N}_0\sqcup\sigma_2\mathsf{N}_0\sqcup\sigma_1\sigma_2\mathsf{N}_0$, with $\mathsf{Z}_+=\mathsf{Z}\cap\{\kappa>0\}$ and N_0 the fundamental domain of $\mathbb{Z}_2^2$ on N, on which $0\le f(-1)<f(x)$.*

(c) (The face.) *$\Omega_0=\{(\boldsymbol{c},\gamma,A,0):\ \boldsymbol{c}\in\mathcal{J},\ U_{\boldsymbol{c}}^-(0)\le\gamma\le U_{\boldsymbol{c}}^+(0),\ A\in\mathbb{R}\}$, where $\mathcal{J}=\{c_1\ge-1,\ c_2\ge-1,\ c_3\ge\bar{c}_3(c_1,c_2)\}$ and $U_{\boldsymbol{c}}^{\pm}$ are the Li–Li–Yan solutions of Theorem 3.44. Its points with $\boldsymbol{c}\in\operatorname{int}\mathcal{J}$ and $U_{\boldsymbol{c}}^-(0)<\gamma<U_{\boldsymbol{c}}^+(0)$ are interior points of Ω.*

(d) (Interior and boundary.) *The set Ω_{G} of solutions of type G at both poles is open and contained in the interior of Ω; the boundary $\partial\Omega\cap\{\kappa\neq0\}$ consists of solutions that are not of type G at both poles.*

(e) (The interface between the classes.) *Inside $\Omega_{\mathsf{G}}\cap\{\kappa>0\}$: $\mathsf{Z}=\{f(-1)<0<f(1)\}$, and the interface between Z and the two outer pieces of N is $\{f(-1)=0\}\cup\{f(1)=0\}$, a union of C^∞ hypersurfaces; it belongs to N.*

(f) (Fibres.) *For $A\kappa=0$ the fibre $\{\gamma:(\boldsymbol{c},\gamma,A,\kappa)\in\Omega\}$ is a closed interval (possibly empty or unbounded). For $A\kappa>0$ the corresponding statement is proved only locally, near edge points of type R and L_1, and under additional hypotheses; the general statement is not proved (Section 1.6).*

Proofs: *Theorem 7.90, which assembles* (a)*: Theorem 5.12;* (b)*: Proposition 5.4, Theorem 5.62, Theorem 6.61;* (c), (d)*: Theorem 3.44, Proposition 5.15;* (e)*: Proposition 5.16. Part* (f)*: Theorem 7.65.*

Figure 1.2 shows Ω at $\boldsymbol{c}=0$: on the left as the region between two sheets over the (A,κ)-plane with the face $\kappa=0$ as the black rectangle (the Landau window $-2\le\gamma\le2$), on the right a section showing the classes: a double wedge of class Z around the κ-axis (the blue wedge), flanked by the four other sectors that make up class N.

1.3.3. The strata. Table 1.1 lists the pieces of Ω by dimension. The generic part of Ω is six-dimensional and open; everything else is a boundary phenomenon in which at least one pole is of a recessive or logarithmic type.

1.3.4. Class Z: the lens. Class Z consists of the solutions whose swirl changes sign; at the zero x_0 of f the swirl vanishes and the data reduce to $(x_0,\boldsymbol{c},\kappa,\gamma)$. The description is complete, but the thresholds γ^-, γ^+ and $\kappa_{\max}$ are characterised by global existence and estimated, not given in closed form.

Key result 1.5 (class Z is the lens)**.** (a) *The map $S\mapsto(x_0,\boldsymbol{c},\kappa,\gamma)$ (x_0 of f, Taylor constants $\boldsymbol{c}$ of Q and circulation $\gamma=U$ at x_0, swirl rate $\kappa>0$ there) is a bijection from $\mathsf{Z}_+=\mathsf{Z}\cap\{\kappa>0\}$ onto*

$$\mathcal{L}=\Big\{(x_0,\boldsymbol{c},\kappa,\gamma):\ \begin{array}{c}x_0\in(-1,1),\ \boldsymbol{c}\in\operatorname{int}\mathcal{J},\ 0<\kappa\le\kappa_{\max}(x_0,\boldsymbol{c}),\\ \gamma^-(x_0,\boldsymbol{c},\kappa)\le\gamma\le\gamma^+(x_0,\boldsymbol{c},\kappa)\end{array}\Big\},$$

and $\mathsf{Z}=\mathsf{Z}_+\sqcup\sigma_1\mathsf{Z}_+$. For fixed $(x_0,\boldsymbol{c})$ the lens $\mathrm{Lens}(x_0,\boldsymbol{c})=\{(\kappa,\gamma)\}$ *is the region between the graph of γ^- (strictly increasing in κ) and that of γ^+ (strictly*

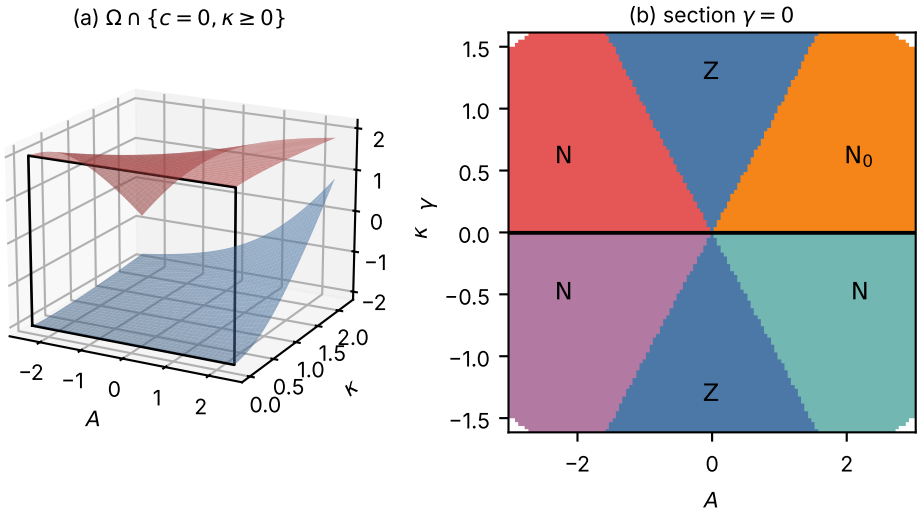


FIGURE 1.2. The set Ω at $\boldsymbol{c} = 0$ (reproduced from Figure 7.7). Left: the part $\kappa \geq 0$ between the sheets $\gamma = \Gamma^{\mp}(A, \kappa)$; the face $\kappa = 0$ is the black rectangle. Right: the section $\gamma = 0$ with both signs of κ: blue is class Z, the other colours the four sectors of class N (the orange one is N_0), black the face.

stratum	dim.	poles	where it is described
Ω_{G} (generic)	6	G / G	open set; Key result 1.4(d)
face Ω_0 ($\kappa = 0$)	5	no swirl	Li–Li–Yan; Key result 1.4(c)
edge $\mathfrak{E}$ of class Z (lens boundary)	5	R, L_1, L_2 at one pole	C^∞ at R, L_1 points; Key result 1.5
wall E_{L_2}	4	L_2 at one pole	C^3 hypersurface in the edge, across which the edge is $C^{1,1}$ but not C^2; Key result 1.5(d)
tips	4	both poles non-generic	the apex of each lens; Key result 1.5
edge Γ_{N} of class N	5	non-G at north pole	floor of the half-cylinder; Key result 1.6

TABLE 1.1. Strata of Ω (dimensions of the strata in $\mathbb{R}^6$; for class N and the face see the cited key results).

decreasing) over $(0, \kappa_{\max}]$*;* $\gamma^-, \gamma^+ \to U_{\boldsymbol{c}}^{\mp}(x_0)$ *as* $\kappa \downarrow 0$*, and the two graphs meet at one tip* $\kappa = \kappa_{\max}$*. The function* $\kappa_{\max}$ *is continuous and strictly increasing in each* c_i*, and* $\to 0$ *at* $\partial\mathcal{J}$*.*

(b) *The pole types are read off from the position in the lens: interior points are of type* G *at both poles; on the lower edge* $\gamma = \gamma^-$*,* $\kappa < \kappa_{\max}$*, the north pole is of*

type R, L_1 *or* L_2 *and the south pole of type* G*; on the upper edge the roles are exchanged by the reflection* $\sigma_1\sigma_2$*; at the tip both poles are non-generic.*

(c) *The interior of the lens,* $\gamma^- < \gamma < \gamma^+$, $\kappa < \kappa_{\max}$, *corresponds to solutions of* $\Omega_{\mathsf{G}} \subset \operatorname{int}\Omega$, *and the boundary of* Z_+ *in* Ω *is the edge hypersurface* $\mathfrak{E}$*, which is* C^∞ *at points of types* R *and* L_1 (*Theorem 7.12*), *where* $\partial_\kappa\gamma^- > 0$ *and* $\partial_{c_i}\gamma^- < 0$.

(d) (The wall.) *The set of edge points of type* L_2 *is closed and nowhere dense in the edge and is locally a* C^3 *hypersurface of it;* γ^- *is* $C^{1,1}$ *near the wall,* C^∞ *off it, and not* C^2 *at any point of it, the jump of the Hessian of* γ^- *across the wall being an explicit nonzero multiple of a rank-one matrix.*

Proofs: (a), (b)*: Theorem 5.62 (with Proposition 5.59);* (c)*: Theorem 7.90*(c)*, Proposition 7.42, Theorem 7.12;* (d)*: Theorem 7.32 and Theorem 7.90*(g)*.*

The lens is drawn in Figure 1.3.

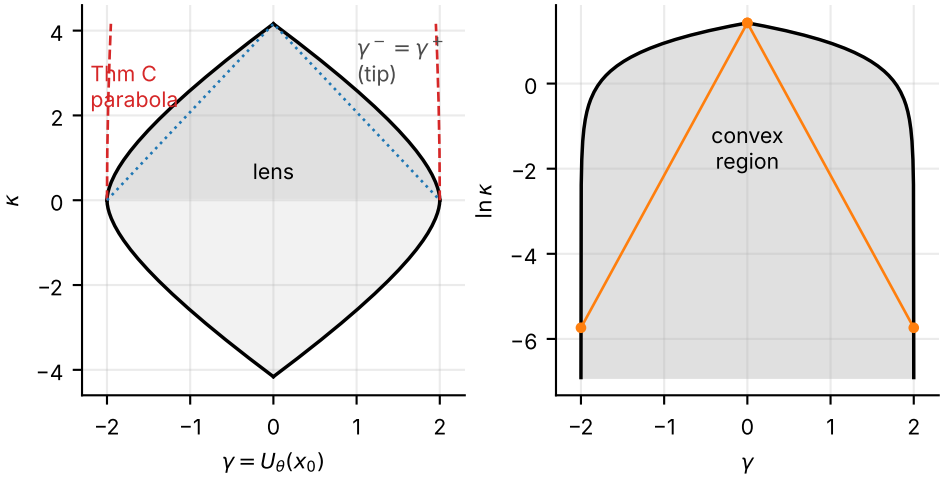


FIGURE 1.3. The lens at $x_0 = 0$, $\boldsymbol{c} = 0$ and its mirror image $\kappa \to -\kappa$ (reproduced from Figure 12.4). The apex is at $\kappa_{\max} = 4.1608$ (numerical value, Example 5.71) at $\gamma = 0$; the lens closes down to the Landau window $[-2, 2]$ as $\kappa \downarrow 0$. The right panel shows the same lens in the coordinates $(\gamma, \ln\kappa)$, where it is convex (Key result 1.15).

1.3.5. Class N: the half-cylinder. In class N the swirl does not change sign. Up to the symmetry group it is positive, with $0 \le f(-1) < f$. The solutions are no longer described by a bounded region at a base point, but by a *shift flow*. The pole at $x = +1$ is the "free" end: solutions are produced from solutions whose north pole is non-generic (the edge) by an explicit continuous flow in the data that moves them into the generic part.

Key result 1.6 (class N is a half-cylinder)**.** *Let* μ *be the exponent at the south pole.*

(a) *Every* $S \in \mathsf{N}_0$ *is of type* G *or* L_1 *at the south pole and of type* G, R (*with exponent* $\mu_+ < \frac{1}{2}$), L_1 *or* L_2 *at the north pole.*

(b) (Small μ.) $\mathsf{N}_0 \cap \{0 \le \mu < 2\} \cong \Gamma_T \times [0,\infty)$ *via the* t^2*-shift, where* Γ_T*, the* floor*, is the set of global solutions of this range that are not of type* G *at the north pole. The threshold function* α_- *(the position of the floor along each shift orbit) is finite and continuous, also across* $\mu = 0$*, i.e. across the passage from the generic to the logarithmic type at the south pole.*

(c) (Large μ.) $\mathsf{N}_0 \cap \{\mu > 1\} \cong B \times [0,\infty)$ *via the recessive shift, where* B *is an explicit open set of parameters, the threshold* Θ^- *tends to* $+\infty$ *at* ∂B*, and this includes* $\mu \ge 2$*.*

(d) *The two shifts commute and generate a free continuous action of* $\mathbb{R}^2$ *on the germs of type* G *with* $\frac{1}{2} < \mu < 2$*.*

(e) *In both charts the floor is the edge* Γ_{N} *(the solutions that are not of type* G *at the north pole); all other solutions of* N_0 *are of type* G *there.*

Proofs: *Theorem 6.61, with Theorem 6.32, Theorem 6.34, Theorem 6.50, Theorem 6.54, Corollary 6.55, Corollary 6.63.*

The two charts are sketched in Figure 1.4.

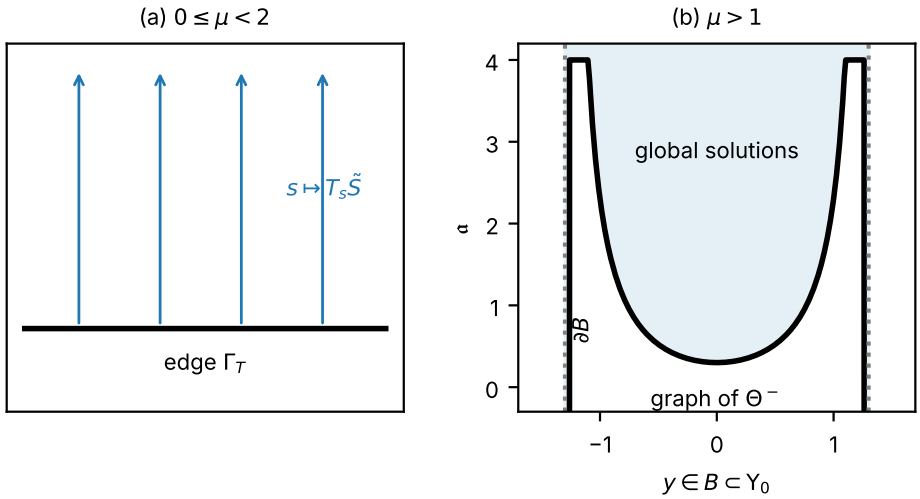


FIGURE 1.4. The two charts of class N_0 (reproduced from Figure 6.5, schematic). Left: $0 \le \mu < 2$, a half-cylinder over the floor Γ_T, the shift orbits being the vertical half-lines. Right: $\mu > 1$, the region above the graph of Θ^- over B.

1.3.6. How the picture relates to Li–Li–Yan. The face Ω_0 is the Li–Li–Yan surface times the free line of the constant swirl. Theorem 3.44 is their classification, reproved here by the linear picture of Chapter 3: for $\boldsymbol{c} \in \mathcal{J}$ the no-swirl solutions are exactly $U = 2(z'/z - \tanh\xi)$ with $z > 0$ a solution of $z'' = W_{\boldsymbol{c}} z$, i.e. they are parametrised by $\gamma \in [U^-_{\boldsymbol{c}}(0), U^+_{\boldsymbol{c}}(0)]$, the two endpoints being the recessive solutions at one of the poles. The swirl solutions attach to this face as follows.

Key result 1.7 (attachment to the no-swirl solutions)**.** (a) *If* $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ *and* $U^-_{\boldsymbol{c}}(0) < \gamma < U^+_{\boldsymbol{c}}(0)$*, then* $(\boldsymbol{c}, \gamma, A, 0)$ *is an interior point of* Ω *for every*

$A \in \mathbb{R}$. *Near such a point with* $A = 0$ *the class-*Z *solutions form the double wedge* $\{f(1)f(-1) < 0\}$, *where* $f(1) = A + \kappa F_+$, $f(-1) = A - \kappa F_-$ *with positive functions* $F_\pm$. *The two bounding hypersurfaces are smooth and tangent to* $A = -F_0(\pm 1)\kappa$, *with* $F_0(\pm 1) = \int_0^{\pm 1} y_0^{-2}\,\mathrm{d}x$ *explicit integrals of the no-swirl solution.*

(b) *Let* $\boldsymbol{c}^0 \in \operatorname{int}\mathcal{J}$ *with* $c_2^0 < -\frac{3}{4}$ (*the recessive endpoint solution has exponent* $k_2 = \sqrt{1 + c_2^0} < \frac{1}{2}$) *and let* $A_0 \geq 0$. *In the quadrant* $A, \kappa \geq 0$ *near* $(\boldsymbol{c}^0, A_0, 0)$, *and locally in* γ (*for* $|\gamma - U^-_{\boldsymbol{c}^0}(0)| < \varepsilon$), *the set* Ω *is the half-space above a* C^∞ *hypersurface* $\gamma = g(\boldsymbol{c}, A, \kappa)$ *with* $g(\boldsymbol{c}, A, 0) = U^-_{\boldsymbol{c}}(0)$, *and for* $\kappa > 0$ *the solutions on it are of type* R *at the north pole: the swirl solutions attach smoothly to the endpoint of the Li–Li–Yan interval.*

(c) *If the endpoint has* $k_2 > \frac{1}{2}$ (*i.e.* $U^-_{\boldsymbol{c}}(1) < -3$), *no swirl solution can approach it at the pole: every swirl solution has* $U(1) > -3$, *and* $U(1) - U^-_{\boldsymbol{c}}(1) > 2k_2 - 1 > 0$ *uniformly over all global swirl solutions. For* $A = 0$, $x_0 = 0$ *and* $c_2 > -\frac{3}{4}$ *the edge solutions of the lens converge to the endpoint solution at every fixed* $x \in [0, 1)$ *as* $\kappa \downarrow 0$, *but not at the pole: their pole value stays in* $(-3, -2]$, *because the swirl part* ψ *of* Q *concentrates in a boundary layer that is pushed into the pole.*

Proofs: (a)*: Proposition 5.15*(b)*, Proposition 5.67;* (b)*: Theorem 7.74, with Proposition 7.71;* (c)*: Corollary 5.17, Corollary 7.73, Corollary 7.81.*

So the Li–Li–Yan threshold 3 is exactly the value $\mu = \frac{1}{2}$ of the exponent of the swirl solution at the pole (Remark 7.72): the solutions with recessive behaviour at a pole are possible with swirl only for $\mu < \frac{1}{2}$. This is the dictionary between the two descriptions. What Li–Li–Yan prove by perturbation of a given no-swirl solution is seen here as the local structure of Ω at its face; the global description of the edge, the tips and class N is not in Li–Li–Yan (for a local classification near the singular ray see also [56]). For the second coordinate system see Section 1.4.

1.4. Existence and multiplicity

In words. *Given the strength of the swirl, is there a solution? For every strength, yes. Is it unique? For weak and for strong swirl, yes; in between, not always: there are solutions curves that fold back. A computer-assisted proof exhibits data with three solutions.*

The coordinates $(\boldsymbol{c}, \gamma, A, \kappa)$ at the equator are good for describing the shape of Ω but not for asking how many solutions there are for given data. For this the book uses a second coordinate system on the generic part $\Omega_{\mathsf{G}} \cap \{\kappa > 0\}$: the *effective data*

$$D = (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}}) \in \mathbb{R}^3 \times (0, \infty) \times \mathbb{R} \times (0, \infty) \qquad \text{(Definition 8.1)}.$$

Here $c_1' = \frac{1}{2}Q(-1)$, $c_2' = \frac{1}{2}Q(+1)$ and c_3' (determined by the first integral) are the constants of the potential *seen from the poles*; $f_- = f(-\infty)$ is the swirl at the south pole; $\vartheta > 0$ is the mixing ratio in $z = u_- + \vartheta u_+$, where u_- and u_+ are the positive Jost solutions of $z'' = Wz$ that are recessive at $-\infty$ and $+\infty$; and κ_{D} is the swirl constant. A solution of type G at both poles is determined by its swirl f, which is a fixed point of a map T_D, $f = f_- + \kappa_{\mathrm{D}} \int_{-\infty}^{\xi} z^{-2}$, where z is built from the Jost solutions of the potential $W_{\mathrm{e}} + \frac{1}{2}\mathcal{Q}[f^2]$ (Chapter 8). The map E that sends such a solution to its effective data D is a proper map between spaces of the same dimension 6, and a local diffeomorphism away from its fold set (Theorem 9.42); it is

not injective (Key result 1.10). Counting the solutions with given D is the question of uniqueness.

Key result 1.8 (existence for every swirl strength)**.** (a) *Let $c_1', c_2' > -1$, $c_3' \in \mathbb{R}$, $\vartheta > 0$, $f_- \in \mathbb{R}$. Then for* every *$\kappa_{\mathrm{D}} > 0$ there is a global solution of type G at both poles with effective data $(\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$.*

(b) *More precisely there are an open bounded set $\mathcal{O} \subset \mathrm{BC}(\mathbb{R})$ and a compact operator H such that all fixed points of T_D lie in $\mathcal{O}$, none on $\partial\mathcal{O}$, and $\deg_{\mathrm{LS}}(I - H, \mathcal{O}) = +1$; if all fixed points are nondegenerate there are finitely many and their number is odd; if no fixed point has a real eigenvalue > 1 or the eigenvalue 1 for $\mathrm{D}T_D$, the fixed point is unique, and conversely two or more fixed points force a fixed point with a real eigenvalue > 1 or $1 \in \operatorname{spec} \mathrm{D}T_D$.*

(c) *For $\boldsymbol{c}'$ below the no-swirl set ($c_3' < \bar{c}_3$) and $0 \le f_- < M = 2\sqrt{\bar{c}_3 - c_3'}$ every solution has $J = f(+\infty) - f_- \ge M - f_-$ (the swirl cannot be small), and for every $\kappa_{\mathrm{D}} > 0$ there is one.*

Proofs: *Theorem 8.20, Theorem 8.42, Corollary 8.45, Theorem 8.53.*

Key result 1.8(a) is proved by computing a Leray–Schauder degree; the point is that all fixed points admit *a priori* bounds, so that the degree is the same for all data (Chapter 8). Notice that no smallness of κ_{D}, of $\boldsymbol{c}'$ or of f_- is assumed: this answers for the class of solutions of type G at both poles the question of the existence of solutions with swirl of every strength.

Key result 1.9 (uniqueness regimes and the window)**.** *Let $D = (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$ be effective data.*

(a) (Moderate swirl.) *If $\Lambda_{\mathrm{mod}}(D) < 1$, where Λ_{mod} is an explicit number, then there is exactly one global solution of type G at both poles with data D, it is nondegenerate, and the Picard iteration converges to it.*

(b) (Small swirl.) *Let $c_1', c_2' > -1$, let $K \subset (0, \infty) \times \mathbb{R}$ be compact and assume $c_3' + \frac{1}{4} m_K^2 > \bar{c}_3$, $m_K = \min_K f_-^+$ (automatic if $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$). There is $\kappa_{\mathrm{lo}}(K) > 0$ such that for $\kappa_{\mathrm{D}} < \kappa_{\mathrm{lo}}(K)$ the solution is unique and nondegenerate and depends analytically on κ_{D}.*

(c) (Large swirl.) *If $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$ and $K \subset (0, \infty) \times \mathbb{R}$ is compact, there is $\kappa_{**}(K) < \infty$ such that for $\kappa_{\mathrm{D}} \ge \kappa_{**}$ the solution is unique, it is nondegenerate in the strong sense that $\operatorname{spec} \mathrm{D}T_D(f) \cap [\frac{1}{2}, \infty) = \emptyset$, and it depends real-analytically on $(\vartheta, f_-, \kappa_{\mathrm{D}})$. This includes class Z ($f_- < 0$).*

(d) (The window.) *Let $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$, K as in* (c) *and $(\vartheta, f_-) \in K$. The set $\mathcal{K}_{\mathrm{deg}}$ of values κ_{D} at which some fixed point is degenerate is closed and contained in $[\kappa_{\mathrm{lo}}(K), \kappa_{**}(K))$; with $\kappa_{\mathrm{first}} = \inf \mathcal{K}_{\mathrm{deg}}$ and $\kappa_{\mathrm{last}} = \sup \mathcal{K}_{\mathrm{deg}}$ the solution is unique for $\kappa_{\mathrm{D}} < \kappa_{\mathrm{first}}$ and for $\kappa_{\mathrm{D}} > \kappa_{\mathrm{last}}$ (for all κ_{D} if $\mathcal{K}_{\mathrm{deg}} = \emptyset$), and the number of solutions is odd for $\kappa_{\mathrm{D}} \notin \mathcal{K}_{\mathrm{deg}}$. If there is more than one solution at some κ_{D}, then κ_{first} and κ_{last} are degenerate values and all such κ_{D} lie in $[\kappa_{\mathrm{first}}, \kappa_{\mathrm{last}}]$. For the map E: at regular values the number of solutions is odd, and across a regular fold it changes by two.*

Proofs: (a)*: Theorem 8.34;* (b), (c), (d)*: Theorem 11.58* (*with Theorem 9.51 and Theorem 10.39*)*; the parity and the fold count: Theorem 9.42.*

Uniqueness thus holds at both ends and can only fail in a bounded window, which is where the fold structure of Theorem 9.42 lives. The second half of the story is that the window can be *nonempty*.

Key result 1.10 (three solutions: a computer-assisted theorem)**.** *Let* $D^* = (\boldsymbol{c}' = (\frac{5}{4}, -\frac{15}{16}, -\frac{517}{160}),\ \vartheta = 40,\ f_- = 0,\ \kappa_{\mathrm{D}} = 1400)$*, so that* $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$ *with* $c_3' = \bar{c}_3 + \frac{1}{20}$*. Then* T_{D^*} *has* at least three *distinct fixed points* f_1, f_2, f_3*, each nondecreasing with* $f_k(-\infty) = 0$*; their jumps* $J_k = f_k(+\infty)$ *lie in the intervals* $19.040738660117701 \pm 6 \cdot 10^{-16}$*,* $33.58919484747842 \pm 2 \cdot 10^{-15}$*,* $46.31852825252179 \pm 5 \cdot 10^{-15}$*. Equivalently there are at least three distinct global solutions of class* N_0*, of type* G *at both poles, with effective data* D^**, and the effective-data map* E *is not injective. On the line* $\kappa_{\mathrm{D}} \mapsto (\boldsymbol{c}', 40, 0, \kappa_{\mathrm{D}})$ *there are degenerate fixed points at some* $\kappa_{\mathrm{D}} \le 1400$ *and at some* $\kappa_{\mathrm{D}} \ge 1400$*, and every* κ_{D} *with more than one solution lies between them.*

The theorem does not *say that there are exactly three solutions at* D^**, that the three are nondegenerate, or anything about the spectrum of* $\mathrm{D}T_{D^*}$*.*

Proofs: *Theorem 9.68, Corollary 9.69, Corollary 11.60; the certificate and the method are in Appendix D.*

Where the theorems stop, numerical evidence goes on. In class Z the folds already occur at *moderate* data: for $\boldsymbol{c}' = (-0.15,\ 0.09,\ \bar{c}_3 + 0.04)$, $\vartheta = 4.49$, $f_- = -2.64$ there are three solutions at $\kappa_{\mathrm{D}} = 10$ and folds at $\kappa_{\mathrm{D}} \approx 6.30477$ and 11.35302 (Numerical observation 9.58, Numerical observation 9.59); in class N_0 a *ridge* of folds appears at large ϑ even at the rest state $\boldsymbol{c}' = 0$, $f_- = 0$ (between $\vartheta = 2300$ and 2400; Numerical observation 9.63). The number of solutions at regular data is always odd (Key result 1.9(d)) and the S-shaped branches of Figure 1.5 and Figure 9.3 are the generic way of having three; compare the hysteresis reported for swirling jets in [40]. A precise statement of what remains open is in Section 1.6.

1.5. Asymptotics and thresholds

In words. *Three limits control the picture: swirl large, swirl small, and the swirl near its largest admissible value (the tip of the lens). In each there is a sharp law, with an explicit power and, where the proof reaches, an explicit constant.*

1.5.1. Large swirl: the $\sqrt{\kappa_{\mathrm{D}}}$ law and the logistic layer. When the swirl rate κ_{D} is large, the total jump $J = f(+\infty) - f(-\infty)$ is of order $\sqrt{\kappa_{\mathrm{D}}}$, the swirl rises from f_- to $f_- + J$ in a thin *logistic layer* around the equator, and the circulation U_θ develops a thin jet-like layer there.

Key result 1.11 (the $\sqrt{\kappa}$ law and the layer)**.** *Let* $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$*,* $\vartheta > 0$*,* $f_- \in \mathbb{R}$*.*

(a) *There are* κ_* *and* C*, bounded on compact sets of* (ϑ, f_-)*, such that every global solution of type* G *at both poles with data* $(\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$*,* $\kappa_{\mathrm{D}} > \kappa_*$*, has*

$$\left| J - \sqrt{2\kappa_{\mathrm{D}}/\vartheta}\,\right| \le C, \qquad \text{so} \qquad J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta}\,\left(1 + O(\kappa_{\mathrm{D}}^{-1/2})\right)$$

uniformly in the solution.

(b) (Next order.) $J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta} + 1 + \ln\vartheta - f_-/2 + O(\kappa_{\mathrm{D}}^{-1/2})$ *for* κ_{D} *large.*

(c) (The layer.) *The layer centre* ξ_c *(defined by* $\int_0^{\xi_c}(\zeta_- - \zeta_+) = \ln\vartheta$*) is* $2\ln\vartheta/J + O(J^{-2})$*; f differs from the step* $f_- + J\mathbf{1}_{\xi > \xi_c}$ *by at most* $Je^{-\rho_{\min}\min(|\xi - \xi_c|, 3/8)}$ *for a constant* $\rho_{\min} > 0$*; and in the inner variable* $t = J\xi/2 - \ln\vartheta$ *it is the logistic function* $(1 + e^{-t})^{-1}$ *near the equator* $(|\xi| \le \frac{1}{4})$*, up to the error bounds of Theorem 10.28*(c)*;* $U_\theta = \frac{J}{2}\tanh\frac{t}{2} + O(1)$ *for* $|t| \le T$*, with constants depending on* T*.*

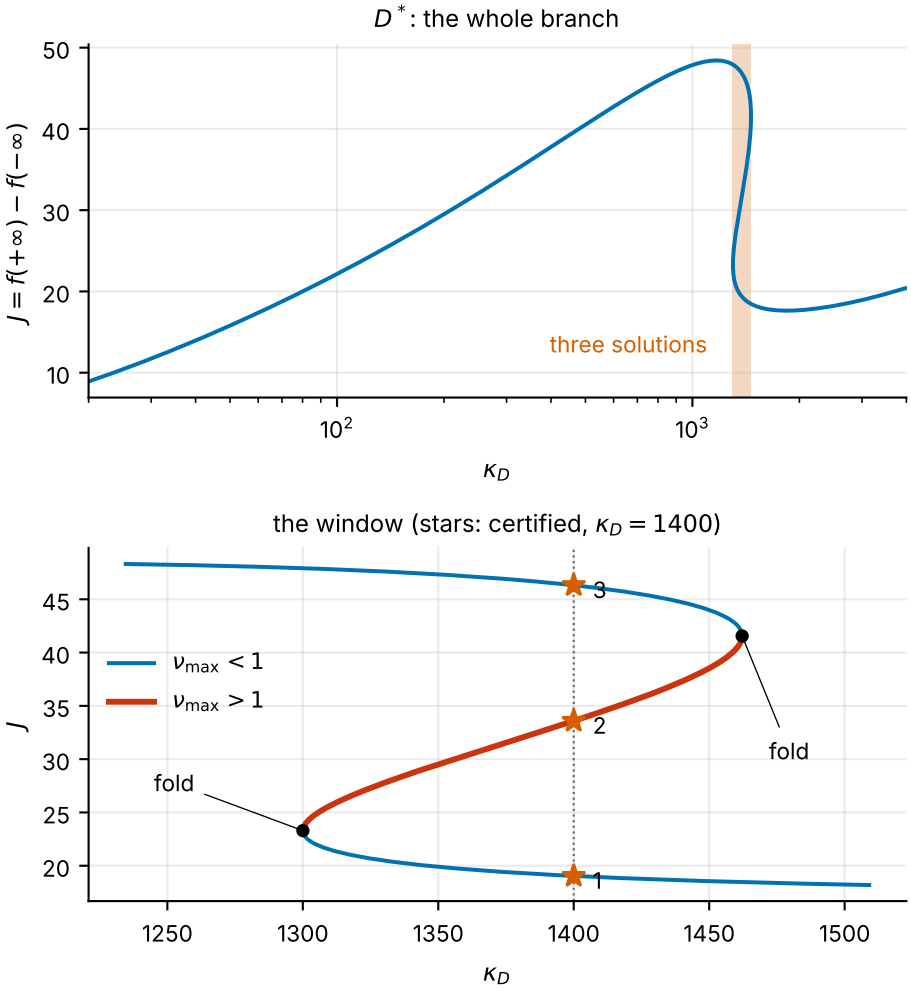


FIGURE 1.5. The solution branch of T_D in κ_{D} at the certified datum, as computed numerically (Numerical observation 9.75; reproduced from Figure 9.6). The jump J is a single-valued function of κ_{D} for small and for large κ_{D}; between the two folds, computed as $\kappa_{\mathrm{D}} = 1300.03$ and 1462.16, there are three solutions (stars at $\kappa_{\mathrm{D}} = 1400$), and the middle one is unstable under the iteration.

(d) (Uniqueness.) *The solution is unique for large* κ_{D} (*Key result 1.9*(c)), *and in class* Z ($f_- < 0$) *the zero of the swirl tends to the equator and the swirl rate at the zero is at least* $\frac{1}{2}|f_-|J(1+o(1))$, *so the lens at the zero grows at least like* $\sqrt{\kappa_{\mathrm{D}}}$.

Proofs: *Theorem 10.11*, *Theorem 10.22*, *Theorem 10.28*, *Theorem 10.39* (*with Proposition 10.36*), *Corollary 10.43*.

1.5.2. Small swirl: the critical bubble. If $c_3' < \bar{c}_3$ the data are *below* the no-swirl set, and by Key result 1.8(c) the swirl cannot be small: as $\kappa_D \to 0$ it settles at the value $M = 2\sqrt{\bar{c}_3 - c_3'}$ in the bulk. It rises from f_- to M in a thin layer near the south pole, stays near M between the two layers, which move out like $\ln(1/\kappa_D)$, and rises again to its final value in a layer near the north pole: a *bubble* at the critical potential.

Key result 1.12 (the critical bubble and its layers)**.** *Assume* (H)*:* $c_1', c_2' > -1$, $c_3' < \bar{c}_3(c_1', c_2')$, $\vartheta > 0$ *and* $0 \le f_- < M$. *Let* f_n *be solutions with data* $(\boldsymbol{c}', \vartheta, f_-, \kappa_n)$ *and* $\kappa_n \downarrow 0$.

(a) $J_n \to J_* = (1+\vartheta)(M - f_-)$ *and* $f_n \to M$ *locally uniformly on* $\mathbb{R}$*; the potentials converge to the critical potential* W_{crit} *and* $(U_\theta, U_\varphi) \to (U^*, M)$ *locally uniformly on* $(-1, 1)$*, where* U^* *is the critical no-swirl solution.*

(b) *The swirl rises through two logistic layers at explicit positions* $\xi_-(\kappa_D)$, $\xi_+(\kappa_D)$ *(of size* $\frac{1}{2k_i}\ln(1/\kappa_D)$ *at the south and north pole), and* $|J - J_*| \le C\varepsilon_k(\kappa_D)$ *with* $\varepsilon_k = \kappa_D$ *for* $k < 1$, $\kappa_D \ln^2(1/\kappa_D)$ *for* $k = 1$, $\kappa_D^{1/k}\ln(1/\kappa_D)$ *for* $k > 1$, $k = \max(\sqrt{1+c_1'}, \sqrt{1+c_2'})$*; for* $k < 1$ *the limit* $(J - J_*)/\kappa_D$ *exists and is given by an explicit formula.*

Proofs: (a)*: Theorem 11.6, and uniformly over compact sets of data Theorem 11.58*(b)*;* (b)*: Theorem 11.10.*

Figure 1.6 shows the bubble and its logistic layers for one choice of data.

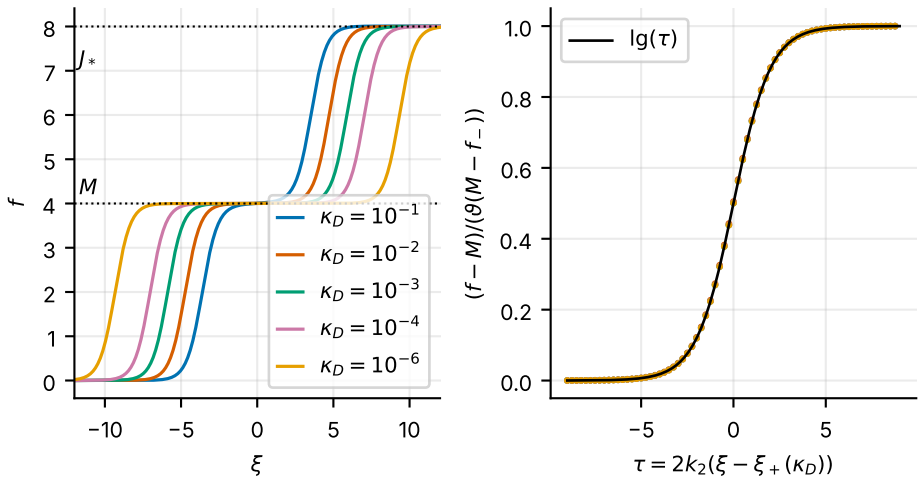


FIGURE 1.6. The bubble at $\boldsymbol{c}' = (0, 0, -8)$, $\vartheta = 1$, $f_- = 0$ ($M = 4$, $J_* = 8$; reproduced from Figure 11.2). Left: the swirl profiles for $\kappa_D = 10^{-1}, \dots, 10^{-6}$ with the plateau at $M = 4$. Right: the north layer in its inner variable, against the logistic function.

1.5.3. The thresholds: how large can the swirl be? For a lens, how large is the tip $\kappa_{\max}(x_0, \boldsymbol{c})$, and how does the lower edge leave the floor? As $c_3 \downarrow \bar{c}_3(c_1, c_2)$ the window of the no-swirl solutions closes, and so does the lens. First, the shape of the edge near the floor.

Key result 1.13 (three laws at the lower edge)**.** *Let $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$, $x_0 \in (-1, 1)$, $k = \sqrt{1 + c_2}$, $\varkappa = (1 - x_0^2)\kappa$ and $\Delta(\kappa) = \gamma^-(x_0, \boldsymbol{c}, \kappa) - U_{\boldsymbol{c}}^-(x_0)$. As $\kappa \downarrow 0$:*

$$\Delta = a_* \varkappa^2 + o(\varkappa^2) \qquad (k < 1),$$

$$\Delta = \frac{e^{-2\xi_0}}{12\, G_+^2}\, \varkappa^2 \ln \frac{1}{\varkappa}\, (1 + o(1)) \qquad (k = 1),$$

$$c_-\, \varkappa^{\frac{2k}{2k-1}} \le \Delta \le c_+\, \varkappa^{\frac{2k}{2k-1}} \qquad (k > 1),$$

with $a_* > 0$ an explicit integral (a digamma expression at the symmetric point), G_+ the explicit constant $\lim_{t\to\infty} e^t u_+(\xi_0 + t)$ and $0 < c_- \le c_+ < \infty$.

Proof: *Theorem 11.44.*

Next, the size of the tip. Write

$$\varepsilon = c_3 - \bar{c}_3(c_1, c_2), \qquad k = \max\left(\sqrt{1 + c_1}, \sqrt{1 + c_2}\right).$$

Key result 1.14 (the thresholds $\kappa_{\max}$)**.** *Fix $x_0 \in (-1, 1)$ and $c_1, c_2 > -1$.*

(a) (Two-sided law.) *As $\varepsilon \downarrow 0$,*

$$\kappa_{\max} \asymp \varepsilon^\beta \quad (\beta = \max(\tfrac{1}{2}, 1 - \tfrac{1}{2k}),\ k \ne 1),$$

$$\kappa_{\max} \asymp \left(\frac{\varepsilon}{\ln(1/\varepsilon)}\right)^{1/2} \quad (k = 1).$$

(b) (Constants.) *For $k < 1$ the limit of $\kappa_{\max}^2/\varepsilon$ exists and is an explicit ratio of two integrals built from the critical function g (the positive solution of $g'' = W_{\mathrm{crit}} g$). For $k = 1$, $\varepsilon/\kappa_{\max}^2 \sim \frac{1}{64} \ln \frac{1}{\varepsilon}$ at $x_0 = 0$, $c_1 = c_2 = 0$.*

(c) (Cones.) *For fixed $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$ and $x_0 \to 1$, $\kappa_{\max} \asymp (1 - x_0^2)^{-1/2}$, and for $\varepsilon \ge \varepsilon_* = k_1(1 + k_2)$ the limit of $\sqrt{1 - x_0^2}\, \kappa_{\max}$ exists and is explicit; the case $x_0 \to -1$ is the mirror image, with $\boldsymbol{c}$ replaced by its mirror $\boldsymbol{c}_m$ (so $\varepsilon_*(\boldsymbol{c}_m)$ and Λ_2 appear).*

(d) (Large constants.) *For $\boldsymbol{c} \in [0, \infty)^3$ with large constants, $\varkappa_{\max} = (1 - x_0^2)\kappa_{\max}$ is bounded above and below by explicit multiples of the square root of the size of the potential, and its ratio to an explicit WKB value $\varkappa_{\mathrm{WKB}}$ is at most $1 + O(\ln a / a)$, where $a \to \infty$ measures the constants (Theorem 12.31 gives the exact hypotheses and the lower bound).*

(e) (A parabola below the wall.) *For all $\kappa \in (0, \kappa_{\max}]$: $\gamma^-(\kappa) - U_{\boldsymbol{c}}^-(x_0) \ge A_{\mathrm{low}}\, \varkappa^2$ with an explicit $A_{\mathrm{low}} > 0$.*

Proofs: (a)*: Theorem 12.11 (upper bound Theorem 12.8);* (b)*: Theorem 12.15, Corollary 12.16;* (c)*: Theorem 12.37;* (d)*: Theorem 12.31;* (e)*: Theorem 12.53.*

Key result 1.15 (log-convexity of the lens)**.** *Fix $x_0 \in (-1, 1)$. The set $\{(\boldsymbol{c}, \ln\kappa, \gamma) : (\kappa, \gamma) \in \mathrm{Lens}(x_0, \boldsymbol{c})\} \subset \mathbb{R}^5$ is closed and convex. Consequently, for fixed $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$ the lens is convex in the coordinates $(\gamma, \ln\kappa)$, $\ln \kappa_{\max}$ is concave in $\boldsymbol{c}$, and $\bar{c}_3$ is a convex function of (c_1, c_2).*

Proofs: *Theorem 12.48, Corollary 12.49.*

The proof of log-convexity is a cooperative-comparison argument for the Q-system in the variables $(U, \ln g, \ln f, p)$ (Chapter 12); it does not reach convexity in the plain coordinates (κ, γ) (Section 1.6).

1.5.4. Class N: the law of the threshold. For class N the thresholds are those of the shift flows: how far one has to push a germ with a given recessive component before the solution becomes global. The limit law is sharp, with an explicit constant.

Key result 1.16 (the class-N threshold law)**.** *Let S be a germ of class N_0 of type G at the south pole with exponent $\mu \in (\frac{1}{2}, 2)$ and let $\mathcal{R}_C$ be the recessive shift. Then*

$$\frac{\alpha_-(\mathcal{R}_C S)}{|C|^{2/\mu}} \;\longrightarrow\; \gamma_*(\mu) = \ell_*(\mu)^{2/\mu} \qquad (C \to -\infty),$$

$$\ell_*(\mu) = \frac{8^{\mu/2}\,\Gamma(\mu/2)}{4\,\Gamma(1-\mu/2)},$$

and $\gamma_*(1) = \frac{1}{2}$, $\gamma_*(\mu)/(2-\mu) \to 1$ as $\mu \to 2$. The limit depends on S only through its exponent μ.

Proofs: *Theorem 12.61, with Proposition 12.66 (the Bessel model); see also Key result 12.60.*

1.6. What is open

In words. *The picture has gaps, listed in Chapter 13. They fall into four groups.*

The shape of Ω. Is every fibre $\{\gamma : (\boldsymbol{c}, \gamma, A, \kappa) \in \Omega\}$ an interval, so that Ω is the region between two graphs over the whole parameter space? This is proved for $A\kappa = 0$ and locally near the edges of types R and L_1 (Key result 1.4(f)); the general statement is Conjecture 13.1. The regularity of the edge, the location of the edge types and the topology of the closure of Ω_{G} are in Open problem 13.5. The shape of the lens is convex in $(\gamma, \ln\kappa)$ (Key result 1.15); whether it is convex in (γ, κ) is Conjecture 13.8.

Uniqueness and folds. Where exactly is $\mathrm{D}T_D$ degenerate, how large is the window of Key result 1.9(d), and do the fold curves end in Whitney cusps (Conjecture 13.18, Open problem 13.17)? Are there exactly three solutions at the certified datum D^* (Open problem 13.16)? What is the plateau mechanism that pins J at large ϑ in class Z (Open problem 13.21) and what replaces the $\sqrt{\kappa}_{\mathrm{D}}$ law when $\ln\vartheta \sim J$ (Open problem 13.20)?

Asymptotic constants. The sharp rates in Key result 1.12(b) for $\max k_i \geq 1$, the sharp constants for $k > 1$ in Key result 1.14, the saturation law below ε_* and the constants at the poles are in Chapter 13.

Beyond. Solutions with more singular rays, without axisymmetry, and their stability are collected in Open problem 13.24.

1.7. How the book is organised

In words. *The chapters follow the order of the arguments; the dependency diagram shows which chapters are needed for which.*

Figure 1.7 shows how the chapters depend on each other. Chapters 2–3 are the tools: the reduction and the Schrödinger equation with a prescribed potential (Jost solutions, Wronskians). Chapters 4–5 describe the poles and the set Ω, class Z

included; Chapter 6 is class N, Chapter 7 the edges, the wall and the global picture (Key result 1.4, Key result 1.7). Chapters 8 and 9 (effective coordinates, existence and uniqueness; Appendix D) form the second line; Chapters 10 to 12 are asymptotics; Chapter 13 is the list of open questions. The appendices are background results (Appendix A), special functions (Appendix B), numerical methods (Appendix C), the computer-assisted proof (Appendix D) and the repository (Appendix E).

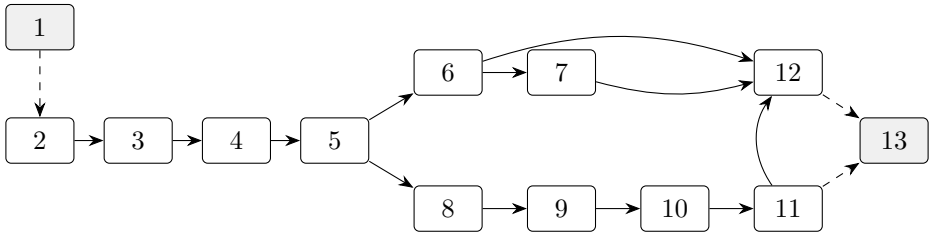


FIGURE 1.7. Dependencies between the chapters. Solid arrows: later chapters use results of earlier ones in their proofs (only the principal lines are drawn). Chapter 1 states the results; Chapter 13 collects the open questions that the proved results raise.

Reading paths. A reader interested in the *answer* reads this chapter and then Chapter 13. For the geometry of Ω: Chapters 2 to 5, then Chapters 6 and 7. For existence and multiplicity: Chapters 2 and 3, Chapters 8 and 9 and Appendix D, with the statements of Chapters 4 and 5. For the asymptotic laws: Chapter 8 and then Chapters 10 to 12 in any order. Each chapter opens with a paragraph *In words* and a *Roadmap*.

CHAPTER 2

The reduced system and its reformulations

In words. *Plugging the axisymmetric (-1)-homogeneous ansatz into the stationary Navier–Stokes equations leaves a sixth-order ordinary differential equation for the swirl f alone. It splits into three pieces: a Riccati equation for the meridional circulation U, which becomes* linear *after the substitution $U = 2(1-x^2)y_x/y$; a first-order equation for f, which integrates once to $y^2 f_x = \kappa$; and a* local *third-order equation for a potential Q. After the change of variable $x = \tanh\xi$ the poles $x = \pm 1$ move to $\xi = \pm\infty$, the linear equation becomes a Schrödinger equation $z'' = Wz$ on the whole line with $W = 1 + \frac{1}{2}Q$, and "the solution is global" means exactly "$z > 0$ on $\mathbb{R}$". The third-order equation has a first integral $\mathsf{L}[Q] + f^2 = \lambda$ with an explicit second-order operator L whose kernel consists of polynomials; consequently Q is a polynomial plus a nonnegative correction Q_{sw} produced by the swirl:* swirl only raises the potential. *The chapter ends with the dictionary between the many normalisations of the swirl constant κ that appear in the literature.*

Roadmap. Section 2.1 derives the reduced system (Proposition 2.1); every identity of the derivation is re-checked by computer algebra in `code/ch02_system/sympy_derivation.py`, which inserts the ansatz into the Navier–Stokes equations in Cartesian coordinates and never uses the spherical-coordinate formulas. Section 2.2 replaces the nonlocal term by a local one (the Q-system), discusses base points and symmetries; Section 2.3 linearises the Riccati equation and shows that f is a ratio of two solutions of one linear equation; Section 2.4 passes to the line variable and fixes, once and for all, the seven normalisations of κ (Lemma 2.15). Section 2.5 studies the operator L and the first integral; Section 2.6 proves the continuation lemma; Section 2.7 proves the centrifugal decomposition (Proposition 2.27), and Section 2.8 works out the examples that are used throughout the book (rest state, Landau family, trivial swirl, constant potential, one-ray solutions). Three tools are introduced with toy examples: the Liouville transformation (Toy example 2.13), the Wronskian (Toy example 2.11) and the Green kernel (Toy example 2.21). *Sources.* The reduced system (Proposition 2.1) and the classification without swirl are due to Li, Li and Yan [24–26]. *Conventions.* In this chapter a prime means $\mathrm{d}/\mathrm{d}\xi$ *from Section 2.4 on*; before that, and whenever it is more convenient, x-derivatives are written ∂_x or with a subscript x. In Sections 2.3 and 2.4 the letter κ denotes the constant belonging to the representative y (resp. z) in use; Section 2.4 makes the possible normalisations precise. Only inside Section 2.1 the symbol $m := \sin\theta = \sqrt{1-x^2}$ is used as a local abbreviation.

2.1. From Navier–Stokes to the reduced system

In words. *We write the flow in spherical coordinates, set $x = \cos\theta$, $U = u_\theta \sin\theta$, $f = u_\varphi \sin\theta$, and check, equation by equation, what the radial, polar and azimuthal*

components of the Navier–Stokes equations say. The azimuthal one is the swirl equation $(1-x^2)f_{xx}+Uf_x=0$*; the radial one determines the pressure; the polar one, after the pressure is eliminated, says that a certain function* Q *has* $(1-x^2)Q_{xxx}=-2ff_x$*. Integrating three times produces the three constants* $\boldsymbol{c}=(c_1,c_2,c_3)$*.*

We write points of $\mathbb{R}^3$ as X (the letter x is reserved for $\cos\theta$), $r=|X|$, and use spherical coordinates (r,θ,φ) with polar angle $\theta\in[0,\pi]$, north pole N at $\theta=0$ and south pole S at $\theta=\pi$; $\mathbf{e}_r,\mathbf{e}_\theta,\mathbf{e}_\varphi$ is the orthonormal frame. A pair (u,p) is *axisymmetric and* (-1)*-homogeneous* if

$$u(X)=r^{-1}\big(u_r(\theta)\,\mathbf{e}_r+u_\theta(\theta)\,\mathbf{e}_\theta+u_\varphi(\theta)\,\mathbf{e}_\varphi\big),\qquad p(X)=r^{-2}p(\theta). \tag{2.1}$$

The exponent -2 of p is forced by the scaling $u\mapsto \ell u(\ell\,\cdot)$, $p\mapsto\ell^2p(\ell\,\cdot)$ of the stationary equations $-\Delta u+(u\cdot\nabla)u+\nabla p=0$, $\operatorname{div}u=0$ (unit viscosity). The profiles live on the sphere; "$(u,p)\in C^2(S^2\setminus\{N,S\})$" means $u_r,u_\theta,u_\varphi,p\in C^2(0,\pi)$. Following Li–Li–Yan [24, 25] put

$$x=\cos\theta\in(-1,1),\qquad U(x)=u_\theta\sin\theta,\qquad f(x)=u_\varphi\sin\theta. \tag{2.2}$$

Since $\theta\mapsto\cos\theta$ is a C^∞ diffeomorphism $(0,\pi)\to(-1,1)$ and $\sin\theta=\sqrt{1-x^2}$ is smooth and positive there, regularity in θ and in x is the same thing. f is called the *swirl*; *genuine swirl* means that f is not constant (a constant swirl $f\equiv A$, i.e. $u_\varphi=A/\sin\theta$, is called *trivial*; it will be seen not to influence U).

Proposition 2.1 (Reduction; Li–Li–Yan)**.** *A pair* $(u,p)\in C^2(S^2\setminus\{N,S\})$ *of the form* (2.1) *is a solution of the stationary Navier–Stokes equations in* $\mathbb{R}^3\setminus\{X_1=X_2=0\}$ *if and only if* $(U,f)\in C^1(-1,1)\times C^2(-1,1)$ *solves, for some* $\boldsymbol{c}=(c_1,c_2,c_3)\in\mathbb{R}^3$*,*

$$(1-x^2)U_x+2xU+\tfrac12U^2+\psi=P_{\boldsymbol{c}}(x),\qquad \psi(x)=\int_0^x\!\!\int_0^l\!\!\int_0^t\frac{2f(s)f_x(s)}{1-s^2}\,\mathrm{d}s\,\mathrm{d}t\,\mathrm{d}l, \tag{2.3}$$

$$(1-x^2)f_{xx}+Uf_x=0, \tag{2.4}$$

where $P_{\boldsymbol{c}}(x)=c_1(1-x)+c_2(1+x)+c_3(1-x^2)$*. In that case* $u_r=U_x$*,* $u_\theta=U/\sin\theta$*,* $u_\varphi=f/\sin\theta$*, and the pressure is given by*

$$p=c_3+\tfrac12\psi_{xx}+U_x-\frac{U^2+f^2}{2(1-x^2)}\;=\;c_3+\tfrac12\psi_{xx}(\cos\theta)+u_r-\tfrac12(u_\theta^2+u_\varphi^2). \tag{2.5}$$

Remark 2.2. If $f\equiv A$ then $\psi\equiv0$ and (2.3) does not contain A. Thus the trivial swirl is invisible in U: it is the classical no-swirl problem of Landau [21], Šverák [49] and Tian–Xin [47] plus an added rotation $A/\sin\theta$.

Step 0: the equations in spherical coordinates. The stationary Navier–Stokes equations for axisymmetric fields $v=v_r\mathbf{e}_r+v_\theta\mathbf{e}_\theta+v_\varphi\mathbf{e}_\varphi$ (no φ-dependence) read, in spherical coordinates (see [2, Appendix 2]), with $D:=v_r\partial_r+\frac{v_\theta}{r}\partial_\theta$ and the scalar Laplacian $\Delta g=r^{-2}\partial_r(r^2\partial_r g)+(r^2\sin\theta)^{-1}\partial_\theta(\sin\theta\,\partial_\theta g)$:

$$Dv_r-\frac{v_\theta^2+v_\varphi^2}{r}+\partial_rp=\Delta v_r-\frac{2v_r}{r^2}-\frac{2}{r^2\sin\theta}\partial_\theta(\sin\theta\,v_\theta), \tag{2.6}$$

$$Dv_\theta+\frac{v_rv_\theta}{r}-\frac{v_\varphi^2\cot\theta}{r}+\frac1r\partial_\theta p=\Delta v_\theta+\frac{2}{r^2}\partial_\theta v_r-\frac{v_\theta}{r^2\sin^2\theta}, \tag{2.7}$$

$$Dv_\varphi+\frac{v_rv_\varphi}{r}+\frac{v_\theta v_\varphi\cot\theta}{r}=\Delta v_\varphi-\frac{v_\varphi}{r^2\sin^2\theta}, \tag{2.8}$$

$$\frac{1}{r^2}\partial_r(r^2 v_r) + \frac{1}{r\sin\theta}\partial_\theta(\sin\theta\, v_\theta) = 0. \tag{2.9}$$

These are the standard component forms of $\Delta\mathbf{v} = \nabla\operatorname{div}\mathbf{v} - \operatorname{curl}\operatorname{curl}\mathbf{v}$ and $(\mathbf{v}\cdot\nabla)\mathbf{v}$ in spherical coordinates; to guard against a misquoted formula, Part A of `sympy_derivation.py` recomputes everything independently in Cartesian coordinates and confirms the brackets (2.12)–(2.16) that are derived from them. In our case $v_r = u_r/r$, $v_\theta = u_\theta/r$, $v_\varphi = u_\varphi/r$ and the pressure is $p(\theta)/r^2$. (The The computer-algebra script does *not* start from these formulas: it computes $-\Delta v + (v\cdot\nabla)v + \nabla p$ in Cartesian coordinates for the ansatz, projects on $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\varphi$ and finds the same brackets as below.)

Step 1: calculus for (-1)-homogeneous fields. Put $m = \sin\theta = \sqrt{1-x^2}$. Then

$$\partial_\theta = -m\,\partial_x, \qquad \cot\theta = \frac{x}{m}, \qquad m_x = -\frac{x}{m}, \qquad \Big(\frac{1}{m}\Big)_x = \frac{x}{m^3}. \tag{2.10}$$

Lemma 2.3. *Let $g \in C^2(0,\pi)$ be regarded as a function of x, and let $u_r = U_x$. Then (a) $\Delta(g/r) = r^{-3}\big((1-x^2)g_x\big)_x$; (b) $D(g/r) = -r^{-3}(Ug)_x$; (c) $\partial_r(p/r^2) = -2p/r^3$ and $\frac{1}{r}\partial_\theta(p/r^2) = -r^{-3}m\,p_x$; (d) $(1-x^2)\big(g/m\big)_x = m\,g_x + xg/m$ and $\big((1-x^2)(g/m)_x\big)_x = m\,g_{xx} + g/m^3$.*

Proof. (a) The function g/r has $r^2\partial_r(g/r) = -g$, independent of r, so the radial part of Δ gives $r^{-2}\partial_r(-g) = 0$. In the angular part, $\sin\theta\,\partial_\theta(g/r) = -(1-x^2)g_x/r$ and $(\sin\theta)^{-1}\partial_\theta = -\partial_x$, which gives $r^{-3}((1-x^2)g_x)_x$. (b) $D(g/r) = \frac{u_r}{r}\cdot\big(-\frac{g}{r^2}\big) + \frac{u_\theta}{r^2}\partial_\theta g = r^{-3}(-U_x g - U m\,g_x/m) = -r^{-3}(Ug)_x$, because $u_\theta = U/m$ and $\partial_\theta g = -m g_x$. (c) is immediate from (2.10). (d) $(g/m)_x = g_x/m + xg/m^3$, so $(1-x^2)(g/m)_x = m\,g_x + xg/m$; differentiating with (2.10), $m\,g_{xx} - \frac{x}{m}g_x + \frac{x}{m}g_x + \frac{g}{m} + \frac{x^2 g}{m^3} = m\,g_{xx} + \frac{g(m^2+x^2)}{m^3} = m\,g_{xx} + \frac{g}{m^3}$. □

Step 2: incompressibility. In (2.9),

$$r^{-2}\partial_r(r^2\cdot u_r/r) = \frac{u_r}{r^2} \quad\text{and}\quad \frac{1}{r\sin\theta}\partial_\theta(\sin\theta\,u_\theta/r) = \frac{1}{r^2 m}\partial_\theta U = -\frac{U_x}{r^2}.$$

Hence

$$\operatorname{div} v = r^{-2}(u_r - U_x), \qquad \text{so} \quad \operatorname{div} v = 0 \iff u_r = U_x. \tag{2.11}$$

From now on $u_r = U_x$; Lemma 2.3(b) applies.

Step 3: the φ-equation. Multiply (2.8) by r^3 and move everything to the left. With $g = u_\varphi = f/m$, using Lemma 2.3 and $u_\theta^2 + u_\varphi^2 = (U^2+f^2)/m^2$, the terms are

$$r^3 Dv_\varphi = -(Uf/m)_x = -\frac{U_x f + U f_x}{m} - \frac{xUf}{m^3}, \qquad r^3\frac{v_r v_\varphi}{r} = \frac{U_x f}{m},$$
$$r^3\frac{v_\theta v_\varphi\cot\theta}{r} = \frac{xUf}{m^3}, \qquad -r^3\Delta v_\varphi = -m f_{xx} - \frac{f}{m^3},$$
$$r^3\frac{v_\varphi}{r^2\sin^2\theta} = \frac{f}{m^3}.$$

All terms but two cancel, and

$$r^3\cdot(\text{LHS} - \text{RHS of (2.8)}) = -\frac{Uf_x}{m} - m f_{xx} = -\frac{1}{m}\Big[(1-x^2)f_{xx} + U f_x\Big]. \tag{2.12}$$

So the φ-equation is exactly (2.4).

Step 4: the r-equation determines the pressure. The same procedure for (2.6) with $g = u_r = U_x$ gives

$$r^3 Dv_r = -(UU_x)_x, \qquad -r^3\frac{v_\theta^2+v_\varphi^2}{r} = -\frac{U^2+f^2}{1-x^2},$$
$$r^3\partial_r\frac{p}{r^2} = -2p, \qquad -r^3\Delta v_r = -\big((1-x^2)U_{xx}\big)_x,$$
$$r^3\frac{2v_r}{r^2} = 2U_x, \qquad r^3\frac{2\,\partial_\theta(\sin\theta\, v_\theta)}{r^2\sin\theta} = \frac{2\,\partial_\theta U}{m} = -2U_x,$$

and the last two cancel. Therefore

$$r^3\cdot(\text{LHS}-\text{RHS of (2.6)}) = -\big[(1-x^2)U_{xx}+UU_x\big]_x - \frac{U^2+f^2}{1-x^2} - 2p. \tag{2.13}$$

This is algebraic in p. Introduce

$$Q := (1-x^2)U_x + 2xU + \tfrac{1}{2}U^2, \qquad \text{so that} \qquad Q_x = (1-x^2)U_{xx} + 2U + UU_x \tag{2.14}$$

(the terms $\mp 2xU_x$ cancel). Then $(1-x^2)U_{xx}+UU_x = Q_x - 2U$, the r-equation is equivalent to

$$p = -\tfrac{1}{2}Q_{xx} + U_x - \frac{U^2+f^2}{2(1-x^2)}, \tag{2.15}$$

because $-(Q_x-2U)_x = -Q_{xx}+2U_x$.

Step 5: the θ-equation. For (2.7), with $g = u_\theta = U/m$ and $\partial_\theta v_r = -mU_{xx}/r$:

$$r^3 Dv_\theta = -\Big(\frac{U^2}{m}\Big)_x = -\frac{2UU_x}{m} - \frac{xU^2}{m^3}, \qquad r^3\frac{v_r v_\theta}{r} = \frac{UU_x}{m},$$
$$-r^3\frac{v_\varphi^2\cot\theta}{r} = -\frac{xf^2}{m^3}, \qquad r^3\frac{1}{r}\partial_\theta\frac{p}{r^2} = -m\,p_x,$$
$$-r^3\Delta v_\theta = -mU_{xx} - \frac{U}{m^3}, \qquad -r^3\frac{2\,\partial_\theta v_r}{r^2} = 2mU_{xx},$$
$$r^3\frac{v_\theta}{r^2\sin^2\theta} = \frac{U}{m^3}.$$

Adding up,

$$r^3\cdot(\text{LHS}-\text{RHS of (2.7)}) = \frac{1}{m}\Big[(1-x^2)U_{xx} - UU_x - \frac{x(U^2+f^2)}{1-x^2} - (1-x^2)p_x\Big]. \tag{2.16}$$

Now insert (2.15). Differentiating it and multiplying by $1-x^2$, using $\partial_x\frac{U^2+f^2}{2(1-x^2)} = \frac{UU_x+ff_x}{1-x^2} + \frac{x(U^2+f^2)}{(1-x^2)^2}$,

$$(1-x^2)p_x = -\tfrac{1}{2}(1-x^2)Q_{xxx} + (1-x^2)U_{xx} - (UU_x+ff_x) - \frac{x(U^2+f^2)}{1-x^2}.$$

Substituting into the bracket of (2.16), the terms $(1-x^2)U_{xx}$, $-UU_x$ and $-x(U^2+f^2)/(1-x^2)$ cancel and

$$r^3\cdot(\text{LHS}-\text{RHS of (2.7)}) = \frac{1}{2m}\Big[(1-x^2)Q_{xxx} + 2ff_x\Big]. \tag{2.17}$$

So, once p is given by (2.15), the θ-equation is equivalent to

$$(1-x^2)Q_{xxx} = -2ff_x. \tag{2.18}$$

Step 6: integrating three times. Let ψ be the triple integral in (2.3). Then $\psi_{xxx} = 2ff_x/(1-x^2)$ and $\psi(0) = \psi_x(0) = \psi_{xx}(0) = 0$. If (2.18) holds then $(Q+\psi)_{xxx} = 0$, so $Q+\psi$ is a polynomial of degree at most two. Every such polynomial $a+bx+dx^2$ is $P_{\boldsymbol{c}}$ for exactly one $\boldsymbol{c}$: the equation $a+bx+dx^2 = c_1(1-x)+c_2(1+x)+c_3(1-x^2)$ means $c_3 = -d$, $c_2-c_1 = b$, $c_1+c_2+c_3 = a$, which has exactly one solution. Hence $Q = P_{\boldsymbol{c}} - \psi$, which by (2.14) is (2.3). Conversely, (2.3) says $Q = P_{\boldsymbol{c}} - \psi$, hence $Q_{xxx} = -\psi_{xxx} = -2ff_x/(1-x^2)$, and $Q_{xx} = -2c_3 - \psi_{xx}$, so (2.15) becomes (2.5).

Step 7: regularity, and the proof of the proposition.

Lemma 2.4 (smoothness)**.** *If $(U,f) \in C^1(-1,1)\times C^2(-1,1)$ solves (2.3)–(2.4), then $U, f \in C^\infty(-1,1)$.*

Proof. We show by induction on $k \geq 1$ that $U \in C^k$ and $f \in C^{k+1}$; $k=1$ is the hypothesis. Assume it for k. Then $\psi_{xxx} = 2ff_x/(1-x^2) \in C^k$, so $\psi \in C^{k+3}$. By (2.3), $U_x = \big(P_{\boldsymbol{c}} - \psi - 2xU - \frac{1}{2}U^2\big)/(1-x^2) \in C^k$, so $U \in C^{k+1}$. By (2.4), $f_{xx} = -Uf_x/(1-x^2) \in C^k$, so $f \in C^{k+2}$. □

Proof of Proposition 2.1. Every term in (2.6)–(2.8) is r^{-3} times a function of x and (2.9) is r^{-2} times one (Lemma 2.3). So the equations hold in $\mathbb{R}^3 \setminus \{X_1 = X_2 = 0\}$ iff the bracketed functions of x in (2.11), (2.12), (2.13), (2.16) vanish on $(-1,1)$.

($\Rightarrow$) Let $(u,p) \in C^2$ solve the equations. Then $U = mu_\theta$ and $f = mu_\varphi$ are C^2, and $U_x = u_r \in C^2$ by Step 2, so $U \in C^3$. Step 3 gives (2.4); Step 4 gives (2.15), i.e. $Q_{xx} = 2U_x - 2p - (U^2+f^2)/(1-x^2) \in C^2$ (with Q from (2.14)), hence $Q \in C^3$ and the differentiation of Step 5 is legitimate. Step 5 gives (2.18), and Step 6 gives (2.3) with the unique $\boldsymbol{c}$ for which $P_{\boldsymbol{c}}$ is the Taylor polynomial of degree two of Q at 0; (2.5) is Step 6.

($\Leftarrow$) Let $(U,f) \in C^1 \times C^2$ solve (2.3)–(2.4). By Lemma 2.4, $U, f \in C^\infty$. Define $u_r = U_x$, $u_\theta = U/m$, $u_\varphi = f/m$ and p by (2.5); all are C^∞ on $(0,\pi)$. Then (2.11) and (2.12) vanish; (2.13) vanishes because (2.5) is (2.15) (Step 6); (2.16) vanishes by (2.17) and Step 6. □

The pressure is uniquely determined: a (-2)-homogeneous pressure has no free additive constant, and (2.15) fixes $p(\theta)$. *Verification by computer algebra:* `sympy_derivation.py`, Part A, inserts the ansatz (with profiles given as Taylor jets with symbolic coefficients) into $-\Delta v + (v\cdot\nabla)v + \nabla p$ in Cartesian coordinates and confirms (2.11), (2.12), (2.13), (2.16) as identities in the jets, and then (2.17); it checks $\operatorname{div}\mathbf{v}$ with u_r an independent jet (so that $u_r = U_x$ is derived, not assumed), Step 6 (unique $\boldsymbol{c}$, vanishing 2-jet of the triple integral) and the pressure formula (2.5).

2.2. The Q-system, base points and symmetries

In words. *Equation (2.3) contains the triple integral ψ, which remembers the whole history of f since $x = 0$. We trade it for a local equation: the third-order equation (2.18) for Q. The result is a* regular *ODE system for six unknowns on $(-1,1)$, which can be started at any base point x_b.*

Lemma 2.5 (the Q-system at any base point)**.** *Let $x_b \in (-1,1)$. A pair (U,f) solves (2.3)–(2.4) with the triple integral based at x_b (that is, $\int_0^x\int_0^l\int_0^t$ replaced by $\int_{x_b}^x\int_{x_b}^l\int_{x_b}^t$) and constants $\boldsymbol{c}$ if and only if there is $Q \in C^3(-1,1)$ with*

$$(1-x^2)U_x + 2xU + \tfrac{1}{2}U^2 = Q, \tag{2.19}$$

$$(1-x^2)f_{xx} + Uf_x = 0, \tag{2.20}$$

$$(1-x^2)Q_{xxx} = -2ff_x, \tag{2.21}$$

and $P_{\boldsymbol{c}}$ is the Taylor polynomial of degree two of Q at x_b. For $x_b = 0$ this is the original system of Proposition 2.1; for $x_b \neq 0$ the constants $\boldsymbol{c}$ are in general different from those at 0, and we always say at which base point they are taken (default: $x_b = 0$).

Proof. ($\Rightarrow$) Put $Q := P_{\boldsymbol{c}} - \psi$ with ψ based at x_b. Then (2.19) is the first equation, (2.20) is (2.4), and $Q_{xxx} = -\psi_{xxx} = -2ff_x/(1-x^2)$; $Q \in C^3$ because ψ_{xxx} is continuous. As ψ and its first two derivatives vanish at x_b, $P_{\boldsymbol{c}}$ and Q have the same 2-jet at x_b. ($\Leftarrow$) Let P be the degree-two Taylor polynomial of Q at x_b and $\psi := P-Q$. Then $\psi(x_b) = \psi_x(x_b) = \psi_{xx}(x_b) = 0$ and $\psi_{xxx} = -Q_{xxx} = 2ff_x/(1-x^2)$. The triple integral Ψ based at x_b has the same third derivative and the same vanishing 2-jet, so $\psi - \Psi$ is a polynomial of degree ≤ 2 with vanishing 2-jet at x_b, i.e. $\psi = \Psi$. Then (2.19) reads $(1-x^2)U_x + 2xU + \frac{1}{2}U^2 + \Psi = P$, and $P = P_{\boldsymbol{c}}$ for a unique $\boldsymbol{c}$ (Step 6 of Section 2.1). □

Remark 2.6 (the map $\boldsymbol{c} \leftrightarrow$ 2-jet)**.** For fixed x_b the map $\boldsymbol{c} \mapsto (P_{\boldsymbol{c}}, \partial_x P_{\boldsymbol{c}}, \partial_x^2 P_{\boldsymbol{c}})(x_b)$ is linear with matrix $\begin{pmatrix} 1-x_b & 1+x_b & 1-x_b^2 \\ -1 & 1 & -2x_b \\ 0 & 0 & -2 \end{pmatrix}$ of determinant $-2\cdot\big((1-x_b)+(1+x_b)\big) = -4 \neq 0$. So the constants $\boldsymbol{c}$ at x_b and the 2-jet $(Q, Q_x, Q_{xx})(x_b)$ determine each other. At $x_b = 0$: $Q(0) = c_1 + c_2 + c_3$, $Q_x(0) = c_2 - c_1$, $Q_{xx}(0) = -2c_3$, and conversely $c_3 = -\frac{1}{2}Q_{xx}(0)$, $c_{1,2} = \frac{1}{2}Q(0) \mp \frac{1}{2}Q_x(0) + \frac{1}{4}Q_{xx}(0)$.

The Q-system as a regular ODE. Solved for the highest derivatives, (2.19)–(2.21) read, for the state $Y = (U, f, f_x, Q, Q_x, Q_{xx}) \in \mathbb{R}^6$,

$$\begin{gathered} U_x = \frac{Q - 2xU - \frac{1}{2}U^2}{1-x^2}, \quad (f)_x = f_x, \quad (f_x)_x = -\frac{Uf_x}{1-x^2}, \\ (Q)_x = Q_x, \quad (Q_x)_x = Q_{xx}, \quad (Q_{xx})_x = -\frac{2ff_x}{1-x^2}. \end{gathered} \tag{2.22}$$

The right-hand side is polynomial in Y with coefficients C^∞ in $x \in (-1,1)$; it is singular only at $x = \pm 1$, which are not in the domain.

Remark 2.7 (ODE facts used throughout)**.** Let $\mathcal{D} \subset \mathbb{R}\times\mathbb{R}^n$ be open and $F : \mathcal{D} \to \mathbb{R}^n$ continuous and locally Lipschitz in Y. (i) (Picard–Lindelöf, [14, Ch. II, §1]) For each $(x_a, Y_a) \in \mathcal{D}$ the problem $Y' = F(x,Y)$, $Y(x_a) = Y_a$, has a unique solution on a maximal open interval (ω_-, ω_+). (ii) (Extension, [14, Ch. II, §3]) As $x \uparrow \omega_+$, $(x, Y(x))$ eventually leaves every compact subset of $\mathcal{D}$; equivalently, if $\omega_+ < \infty$ and Y stays in a compact set C with $[x_a, \omega_+] \times C \subset \mathcal{D}$, the solution extends beyond ω_+. Same at ω_-. (iii) (Continuous dependence, [14, Ch. V, §2]) The solution depends continuously on (x_a, Y_a), uniformly on compact subintervals of the maximal interval. For (2.22): $\mathcal{D} = (-1,1)\times\mathbb{R}^6$.

A solution of (2.22) has a maximal interval of existence; it is called *global* if this interval is $(-1,1)$. By Remark 2.6 and the fact that y (introduced in Section 2.3) is fixed up to a positive factor, a solution is determined by its *Cauchy data at* x_b: the constants $\boldsymbol{c}$ at x_b, the value $\gamma := U(x_b)$, the value $A := f(x_b)$ and the slope $\kappa_x = f_x(x_b)$ (six numbers; the set of those data giving global solutions is the region $\Omega \subset \mathbb{R}^6$ of Chapter 5).

Symmetries.

Lemma 2.8 (symmetries)**.** *The Q-system is invariant under*

$$\sigma_1 : (U,f,Q) \mapsto (U,-f,Q), \qquad \sigma_2 :\ (U,f,Q)(x) \mapsto \big(-U(-x),\ f(-x),\ Q(-x)\big). \tag{2.23}$$

σ_2 is the reflection $x \mapsto -x$ (north $\leftrightarrow$ south); at base point 0 it exchanges c_1 and c_2. If f is constant, $f \mapsto f + A$ is also a symmetry.

Proof. σ_1: (2.19) does not contain f, (2.20) is linear in f and (2.21) quadratic. σ_2: put $\tilde U(x) = -U(-x)$, $\tilde f(x) = f(-x)$, $\tilde Q(x) = Q(-x)$ and $\bar x = -x$. Then $\tilde U_x(x) = U_x(\bar x)$, $\tilde f_x(x) = -f_x(\bar x)$, $\tilde f_{xx}(x) = f_{xx}(\bar x)$, $\tilde Q_{xxx}(x) = -Q_{xxx}(\bar x)$, and

$$\begin{aligned}(1-x^2)\tilde U_x + 2x\tilde U + \tfrac12 \tilde U^2 &= (1-\bar x^2)U_x(\bar x) + 2\bar x U(\bar x) + \tfrac12 U(\bar x)^2 = Q(\bar x) = \tilde Q(x),\\ (1-x^2)\tilde f_{xx} + \tilde U \tilde f_x &= \big[(1-\bar x^2)f_{xx} + U f_x\big](\bar x) = 0,\\ (1-x^2)\tilde Q_{xxx} + 2\tilde f \tilde f_x &= -\big[(1-\bar x^2)Q_{xxx} + 2 f f_x\big](\bar x) = 0.\end{aligned}$$

The degree-two Taylor polynomial of $\tilde Q$ at 0 is $P_{\boldsymbol{c}}(-x) = c_1(1+x) + c_2(1-x) + c_3(1-x^2) = P_{(c_2,c_1,c_3)}(x)$. If $f \equiv A$ then $f_x \equiv 0$, and f enters (2.19)–(2.21) only through f_x and $f f_x = 0$. □

(In terms of Cauchy data at $x_b = 0$: $\sigma_1 : (A,\kappa_x) \mapsto (-A,-\kappa_x)$ and $\sigma_2 : (c_1,c_2,c_3,\gamma,A,\kappa_x) \mapsto (c_2,c_1,c_3,-\gamma,A,-\kappa_x)$; both are checked by computer algebra. The group they generate is $\sigma = \langle \sigma_1,\sigma_2\rangle \cong \mathbb{Z}_2^2$.)

2.3. Linearisation of the Riccati equation; f is a ratio of two solutions

In words. *Equation (2.19) is quadratic in U with U_x appearing linearly: a Riccati equation. Every Riccati equation is the logarithmic derivative of a linear one. Here the linear equation is $2(1-x^2)^2 y_{xx} = Qy$. Then "U blows up" means "y has a zero", and the swirl equation integrates once: $y^2 f_x$ is constant.*

Lemma 2.9 (linearisation)**.** *Let (U,f,Q) solve the Q-system on an interval $I \ni x_a$ and put $y(x) := \exp\int_{x_a}^{x} \frac{U(s)}{2(1-s^2)}\,\mathrm{d}s > 0$. Then*

$$U = \frac{2(1-x^2)y_x}{y}, \qquad 2(1-x^2)^2 y_{xx} = Q\,y, \qquad y^2 f_x \equiv \kappa \text{ (a constant).} \tag{2.24}$$

Conversely, if $y > 0$ solves $2(1-x^2)^2 y_{xx} = Qy$ on I, then $U := 2(1-x^2)y_x/y$ solves (2.19); and if $f_x = \kappa y^{-2}$ then (2.20) holds. Moreover, $f = w/y$ with $w := fy$ a solution of the same linear equation, and $\mathrm{Wr}(y,w) = y\,w_x - y_x w = y^2 f_x = \kappa$.

Proof. Taking the logarithm, $y_x/y = U/(2(1-x^2))$, the first identity. For any C^2 function $y > 0$ and $U := 2(1-x^2)y_x/y$ the product and quotient rules give

$U_x = -\frac{4xy_x}{y} + 2(1-x^2)\big(\frac{y_{xx}}{y} - \frac{y_x^2}{y^2}\big)$, hence

$$(1-x^2)U_x = -\frac{4x(1-x^2)y_x}{y} + \frac{2(1-x^2)^2 y_{xx}}{y} - \frac{2(1-x^2)^2 y_x^2}{y^2},$$
$$2xU = \frac{4x(1-x^2)y_x}{y}, \quad \tfrac{1}{2}U^2 = \frac{2(1-x^2)^2 y_x^2}{y^2},$$

and adding the three lines,

$$(1-x^2)U_x + 2xU + \tfrac{1}{2}U^2 = \frac{2(1-x^2)^2 y_{xx}}{y}. \tag{2.25}$$

So (2.19) holds iff $2(1-x^2)^2 y_{xx} = Qy$; this proves the second identity and the first converse. For the swirl equation, again with $U = 2(1-x^2)y_x/y$,

$$(y^2 f_x)_x = y^2 f_{xx} + 2y y_x f_x = y^2 f_{xx} + \frac{y^2}{1-x^2} U f_x = \frac{y^2}{1-x^2}\Big[(1-x^2)f_{xx} + U f_x\Big].$$

Since $y^2/(1-x^2) > 0$, (2.20) is equivalent to $(y^2 f_x)_x = 0$, i.e. $y^2 f_x \equiv \kappa$. Finally, with $w = fy$ and $f_{xx} = -2y_x f_x/y$ (from $y^2 f_x = \kappa$): $w_{xx} = f_{xx}y + 2f_x y_x + f y_{xx} = f y_{xx}$, so $2(1-x^2)^2 w_{xx} = Qw$, and $yw_x - y_x w = y(f_x y + f y_x) - y_x f y = y^2 f_x = \kappa$. □

The normalisation of y is free: $(y,\kappa) \mapsto (sy, s^2\kappa)$, $s > 0$, leaves (U, f, Q) unchanged, since U involves only y_x/y and $f_x = \kappa y^{-2}$. Changing the base point x_a to x_a' replaces y by $y/y(x_a')$, i.e. multiplies it by $s = y(x_a')^{-1}$, and κ by s^2. In particular the sign of κ and the question whether $\kappa = 0$ are intrinsic. (The normalisations used in the book are fixed in Section 2.4.)

Corollary 2.10 (monotone swirl)**.** *For every solution, f is either constant ($\kappa = 0$) or strictly monotone ($\kappa \neq 0$). In the latter case f has at most one zero, and $f f_x$ has a fixed sign near each endpoint of the interval of existence (it vanishes identically if $\kappa = 0$).*

Proof. By (2.24), $f_x = \kappa y^{-2}$ with $y > 0$. If $\kappa = 0$ then $f_x \equiv 0$. If $\kappa \neq 0$ then f_x has the sign of κ everywhere, so f is strictly monotone, hence injective, and has at most one zero x_0. Near an endpoint of the interval of existence (on the side away from x_0, if x_0 exists) f has a fixed sign and f_x has the sign of κ. □

Toy example 2.11 (the Wronskian, with a prescribed potential)**.** Here only (2.19)–(2.20) are used, with the potential Q treated as a *given* function (this is the point of view of Chapter 3); the third equation (2.21) is *not* imposed and, for $Q \equiv 0$, would force $f f_x \equiv 0$. Take $Q \equiv 0$. Then $y_{xx} = 0$, so $y = a + bx$ and every other solution of the linear equation is $w = \alpha + \beta x$. The swirl $f = w/y = (\alpha+\beta x)/(a+bx)$ is a Möbius function of x, and $f_x = (\beta a - \alpha b)/(a+bx)^2 = \kappa/y^2$ with $\kappa = \mathrm{Wr}(y,w) = a\beta - b\alpha$: the constant of the first integral *is* the Wronskian. Positivity of y on $(-1,1)$ means $a > 0$, $|b| \le a$. The toy illustrates $f = w/y$ and $\kappa = \mathrm{Wr}(y,w)$; it is not a solution of the full system unless $\kappa = 0$ (see Example 2.29 for the no-swirl case and Remark 2.33).

Remark 2.12 (the Schwarzian form)**.** If $\kappa \neq 0$ then $U = -(1-x^2)f_{xx}/f_x$, and (2.19) becomes $(1-x^2)^2\,\mathcal{S}(f) = -Q$ with $\mathcal{S}(f) = f_{xxx}/f_x - \frac{3}{2}(f_{xx}/f_x)^2$. Indeed with $\varphi = f_{xx}/f_x = -2y_x/y$ one has $\mathcal{S}(f) = \varphi_x - \frac{1}{2}\varphi^2 = -2y_{xx}/y = -Q/(1-x^2)^2$ by (2.24). With (2.21) the whole system is the single sixth-order equation $\big[(1-x^2)^2\mathcal{S}(f)\big]_{xxx} = 2f f_x/(1-x^2)$ for the swirl alone, and a global solution is a function f without critical points on $(-1,1)$ solving it. (Conversely, if $f_x \neq 0$ solves

it, $y := |f_x|^{-1/2}$ reproduces everything.) This explains why the problem has six parameters; we do not use the form.

2.4. The line variable, the Schrödinger equation, and the seven normalisations of κ

In words. *The substitution $x = \tanh\xi$ places the poles at infinity; with $y = z/\cosh\xi$ (Liouville) the linear equation loses its first-order term and becomes the Schrödinger equation $z'' = Wz$, $W = 1 + \frac{1}{2}Q$. A global solution is a positive z on all of $\mathbb{R}$. The price is that z and the swirl constant κ are only defined up to $(z,\kappa) \mapsto (sz, s^2\kappa)$; the second half of the section fixes what "κ" means in each chapter.*

Let $\xi = \operatorname{artanh} x \in \mathbb{R}$, i.e. $x = \tanh\xi$; this is an increasing bijection $(-1,1) \to \mathbb{R}$ with $x \to \pm 1 \iff \xi \to \pm\infty$. We have $\mathrm{d}x/\mathrm{d}\xi = \operatorname{sech}^2\xi = 1 - x^2$, $\cosh\xi = (1-x^2)^{-1/2}$, $\sinh\xi = x(1-x^2)^{-1/2}$, and $\partial_x = \cosh^2\xi\,\mathrm{d}/\mathrm{d}\xi$. *From now on a prime denotes* $\mathrm{d}/\mathrm{d}\xi$, and functions of x are tacitly regarded as functions of ξ.

Toy example 2.13 (the Liouville transformation)**.** Write the linear equation of (2.24) in ξ: from $y_x = \cosh^2\xi\, y'$ and $y_{xx} = \cosh^2\xi(\cosh^2\xi\, y')' = \cosh^4\xi\, y'' + 2\cosh^3\xi\sinh\xi\, y'$ it becomes

$$2y'' + 4\tanh\xi\, y' = Q\, y. \tag{2.26}$$

The first-order term is removed by $y = z\, e^{-\int \tanh\xi} = z/\cosh\xi$. Toy case $Q \equiv q$ constant: then $z'' = (1+\frac{q}{2})z$, a constant-coefficient equation with explicit exponential solutions, while (2.26) has the non-obvious solutions $y = e^{\pm\mu\xi}/\cosh\xi$, $\mu = \sqrt{1+q/2}$. (Checked by computer algebra.)

Lemma 2.14 (Schrödinger form)**.** *With $y = z/\cosh\xi$, the relations (2.24) become*

$$z'' = Wz, \quad W := 1 + \tfrac{1}{2}Q, \qquad U = 2\Big(\frac{z'}{z} - \tanh\xi\Big), \qquad f' = \kappa z^{-2}. \tag{2.27}$$

The logarithmic derivative $h := z'/z = \frac{1}{2}U + x$ solves the Riccati equation $h' + h^2 = W$. In particular U exists on an interval iff $z > 0$ there. The normalisation of z is free: $(z,\kappa) \mapsto (sz, s^2\kappa)$, $s > 0$, leaves (U, f, Q) unchanged.

Proof. Use $(\operatorname{sech}\xi)' = -\operatorname{sech}\xi\tanh\xi$, $(\tanh\xi)' = \operatorname{sech}^2\xi$. First, $y_x = \cosh^2\xi\,(z\operatorname{sech}\xi)' = \cosh\xi\,(z' - z\tanh\xi)$. Next,

$$\begin{aligned} y_{xx} &= \cosh^2\xi\Big[\cosh\xi\,(z' - z\tanh\xi)\Big]' \\ &= \cosh^2\xi\Big[\sinh\xi\,(z' - z\tanh\xi) + \cosh\xi\,(z'' - z'\tanh\xi - z\operatorname{sech}^2\xi)\Big] \\ &= \cosh^2\xi\Big[\cosh\xi\, z'' - z\,\frac{\sinh^2\xi + 1}{\cosh\xi}\Big] = \cosh^3\xi\,(z'' - z). \end{aligned}$$

Hence, as $(1-x^2)^2 = \operatorname{sech}^4\xi$ and $y = z\operatorname{sech}\xi$, $2(1-x^2)^2 y_{xx} - Qy = \frac{2(z''-z)}{\cosh\xi} - \frac{Qz}{\cosh\xi} = \frac{2}{\cosh\xi}\big(z'' - (1+\frac{1}{2}Q)z\big)$, so the linear equation of (2.24) is $z'' = Wz$. Further $U = 2(1-x^2)y_x/y = 2\operatorname{sech}^2\xi\cosh\xi\,(z' - z\tanh\xi)/(z\operatorname{sech}\xi) = 2(z'/z - \tanh\xi)$, and $f' = \operatorname{sech}^2\xi\, f_x = \operatorname{sech}^2\xi\,\kappa\cosh^2\xi/z^2 = \kappa/z^2$. Finally $h = \frac{1}{2}U + \tanh\xi$ and $h' = z''/z - h^2 = W - h^2$. The statement on s is clear: z'/z and κz^{-2} do not change under $(z,\kappa) \mapsto (sz, s^2\kappa)$. □

The system on the line. Together with the first integral of Section 2.5 below, $\mathsf{L}[Q]=\lambda-f^2$, where in the line variable $\mathsf{L}[g]=\cosh^2\xi\,g''+4\sinh\xi\cosh\xi\,g'-2g$, the reduced system reads

$$z''=\big(1+\tfrac12 Q\big)z,\qquad f'=\kappa z^{-2},\qquad \mathsf{L}[Q]=\lambda-f^2, \tag{2.28}$$

a first-order system for the state (z,z',f,Q,Q'):

$$\begin{gathered}(z)'=z',\ (z')'=\big(1+\tfrac12 Q\big)z,\ f'=\frac{\kappa}{z^2},\ (Q)'=Q',\\ (Q')'=\frac{\lambda-f^2+2Q-4\sinh\xi\cosh\xi\,Q'}{\cosh^2\xi},\end{gathered} \tag{2.29}$$

whose right-hand side is C^∞ on $\mathbb{R}\times\{z>0\}\times\mathbb{R}^4$. Its dimension count recovers six: five state variables plus (κ,λ), minus one for the normalisation of z. Near the poles we use

$$\begin{gathered}E=E(\xi):=e^{-2|\xi|},\qquad t:=1-x\ (\xi\ge0),\quad t:=1+x\ (\xi\le0),\\ E\le t=\frac{2E}{1+E}\le 2E.\end{gathered} \tag{2.30}$$

Indeed for $\xi\ge0$, $1-\tanh\xi=2e^{-2\xi}/(1+e^{-2\xi})$, and $0<E\le1$.

2.4.1. The seven normalisations of κ. The constant κ of $f'=\kappa z^{-2}$ depends on how z is normalised, and different parts of the literature normalise differently. Lemma 2.14 shows that the *intrinsic* object is the quantity

$$\begin{gathered}\kappa_W[z]:=f'z^2\quad\text{(a constant for each positive solution } z \text{ of } z''=Wz),\\ \kappa_W[sz]=s^2\kappa_W[z].\end{gathered} \tag{2.31}$$

Everything below is a statement about which positive solution z is used. The next lemma is stated exactly as in the conversion block; its proof (short, but the book relies on it constantly) follows.

For a positive solution z of $z''=Wz$ and $f'=\kappa z^{-2}$ put $\kappa_W[z]:=f'z^2$ (constant), so $\kappa_W[sz]=s^2\kappa_W[z]$. With $f_x=f'\cosh^2\xi$: (1) $\kappa_x=f_x(x_b)=\kappa_W[z]$ for $z(\xi_b)=\cosh\xi_b$; (2) $\varkappa=f'(\xi_b)=\kappa_W[z]$ for $z(\xi_b)=1$, $\varkappa=(1-x_b^2)\kappa_x$, $\kappa_{\max}=\cosh^2\xi_0\,\varkappa_{\max}$; (3) $\tilde\kappa=f'(0)=\kappa_x(0)$; (4) $\kappa_D=\kappa_W[u_-+\vartheta u_+]=(1+\vartheta)^2\tilde\kappa$, and, when u_- is recessive at $-\infty$, u_+ recessive at $+\infty$ and $w_0\neq0$, $J=\kappa_D/(\vartheta w_0)$; (5) $\kappa_W[z]$ for arbitrary z; (6) κ_{ch}, one normalisation per pole stratum: G: $ze^{-\mu\xi}\to1$; R: $ze^{\mu\xi}\to1$; L_1: $z'\to1$; L_2: $z(\xi_*)=1$ (in a cut frame $\kappa_{\xi_1}=\kappa_{\mathrm{ch}}z(\xi_1)^{-2}$); (7) $\kappa_{\xi_1}=f'(\xi_1)$ for $z(\xi_1)=1$.

The seven normalisations are collected in Table 2.1; the lemma below makes the words of the statement precise (the hypotheses on $u_\pm$ in (4) and the meaning of "stratum" in (6) are the only points where later chapters enter).

Lemma 2.15 (conventions: the seven normalisations of κ)**.** *Let (U,f,Q) be a solution of the Q-system on an interval $I\ni\xi_b$, $W=1+\frac12 Q$, and let $z>0$ solve $z''=Wz$ with $U=2(z'/z-\tanh\xi)$ (so $f'=\kappa_W[z]z^{-2}$). For $\xi_b\in I$ put $x_b=\tanh\xi_b$.*

(1) *$\kappa_x:=f_x(x_b)$ for the representative with $y(x_b)=1$, i.e. $z(\xi_b)=\cosh\xi_b$; then $\kappa_x=\kappa_W[z]$ and, for arbitrary z, $\kappa_x=\kappa_W[z]\cosh^2\xi_b/z(\xi_b)^2=\kappa_W[z]/y(\xi_b)^2$.*

(2) *$\varkappa[\xi_b]:=f'(\xi_b)$ for $z(\xi_b)=1$; then $\varkappa[\xi_b]=\kappa_W[z]$, and $\varkappa[\xi_b]=(1-x_b^2)\kappa_x$, $\kappa_x=\cosh^2\xi_b\,\varkappa[\xi_b]$. In particular, if the same set of solutions with base point x_0*

#	symbol (macro)	definition	used in	conversion
1	κ_x	$f_x(x_b)$, $y(x_b)=1$	Chapter 5, Chapter 7, Chapter 10, Chapter 11 (§§11.5–11.7)	$\kappa_x = \varkappa \cosh^2 \xi_b$
2	$\varkappa = \varkappa[\xi_b]$	$f'(\xi_b)$, $z(\xi_b)=1$	Chapter 10, Chapter 11 (§§11.5–11.7), Chapter 12	$\varkappa = (1-x_b^2)\kappa_x$, $\kappa_{\max} = \cosh^2 \xi_0 \, \varkappa_{\max}$
3	$\kappa_0 = \tilde\kappa$	$f'(0)=f_x(0)$, $z(0)=1$	Chapter 5, Chapter 9, Appendix D	$\kappa_0 = \kappa_x\vert_{x_b=0} = \varkappa[0]$
4	κ_{D}	$\kappa_W[u_- + \vartheta u_+]$	Ch. 8–11	$\kappa_{\mathrm{D}} = (1+\vartheta)^2\kappa_0$, $J = \kappa_{\mathrm{D}}/(\vartheta w_0)$
5	$\kappa_W[z]$	$f' z^2$, z arbitrary	everywhere	$\kappa_W[sz] = s^2 \kappa_W[z]$
6	κ_{ch}	per stratum $\mathsf{G}, \mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$	Chapter 4, Chapter 6, Chapter 7, Chapter 12 (Part C)	$\kappa_{\xi_1} = \kappa_{\mathrm{ch}}\, z(\xi_1)^{-2}$
7	κ_{ξ_1}	$f'(\xi_1)$, $z(\xi_1)=1$	Chapter 6	$\kappa_{\xi_1} = \varkappa[\xi_1] = \kappa\, z(\xi_1)^{-2}$

TABLE 2.1. The seven normalisations of the swirl constant. In the literature κ_x is written κ (in the regularity and global-picture literature), $\varkappa$ is $\varkappa$ (critical-surface and threshold papers), $\kappa_0 = \tilde\kappa$ (equator slope), κ_{D} is κ in the effective-coordinate papers. A bare κ is allowed in the book only inside a statement that begins "with the normalisation $\kappa = \kappa_x$ (base point x_b)", or in Chapters 8–10 where $\kappa := \kappa_{\mathrm{D}}$ is declared at the start of the chapter.

is described by $\kappa_x \in (0, \kappa_{\max}]$ and by $\varkappa \in (0, \varkappa_{\max}]$, then $\kappa_{\max} = \cosh^2 \xi_0 \, \varkappa_{\max}$ (the factor $\cosh^2 \xi_0$ is constant since the base point is fixed).

(3) *$\kappa_0 := f'(0) = f_x(0)$ for $z(0)=1$; it equals κ_x at $x_b = 0$ and $\varkappa[0]$.*

(4) *Suppose $u_\pm > 0$ are solutions of $z'' = Wz$ with $u_\pm(0) = 1$ such that u_- is recessive at $-\infty$ and u_+ recessive at $+\infty$ (i.e. $u_-/v \to 0$ at $-\infty$ and $u_+/v \to 0$ at $+\infty$ for every solution v independent of u_-, resp. u_+), and $w_0 := \mathrm{Wr}(u_+, u_-) = u_-'(0) - u_+'(0) \neq 0$. If $z = u_- + \vartheta u_+$ with $\vartheta > 0$ and $\kappa_{\mathrm{D}} := \kappa_W[z]$, then $\kappa_{\mathrm{D}} = (1+\vartheta)^2 \kappa_0$ (with κ_0 computed from the normalised solution $z/(1+\vartheta)$), and $f(\pm\infty)$ exist with*

$$
\begin{aligned}
f = f_- + J\,\frac{u_-}{z}, \qquad J := f(+\infty) - f(-\infty) = \frac{\kappa_{\mathrm{D}}}{\vartheta w_0}, \\
f(0) - f_- = \frac{J}{1+\vartheta}, \quad \textit{i.e. } \vartheta = \frac{J}{f(0)-f_-} - 1.
\end{aligned}
\tag{2.32}
$$

(5) $\kappa_W[z]$ *for an arbitrary positive solution* z*; any two normalisations differ by* $\kappa_W[sz] = s^2\kappa_W[z]$ *and* $\varkappa[\xi_b'] = \varkappa[\xi_b]\,\big(z(\xi_b)/z(\xi_b')\big)^2$ *for* any *positive representative* z *(change of base point). The constant* κ_x *at the base point* x_b' *changes accordingly:* $\kappa_x[x_b'] = \kappa_x[x_b]\,\big(y(\xi_b)/y(\xi_b')\big)^2$.

(6) *Let* ξ_1 *be a cut point and* z *the representative normalised by the stratum condition of the pole chart (Chapter 4):* G*:* $ze^{-\mu\xi} \to 1$*;* R*:* $ze^{\mu\xi} \to 1$*;* L_1*:* $z' \to 1$*;* L_2*:* $z(\xi_*) = 1$ *at the stratum's reference point* ξ_**; put* $\kappa_{\mathrm{ch}} := \kappa_W[z]$*. Then the constant in the cut frame* $z(\xi_1) = 1$ *is* $\kappa_{\xi_1} = \kappa_{\mathrm{ch}}\, z(\xi_1)^{-2}$.

(7) $\kappa_{\xi_1} := f'(\xi_1)$ *for* $z(\xi_1) = 1$*, i.e.* $\kappa_{\xi_1} = \varkappa[\xi_1] = \kappa_W[z]\, z(\xi_1)^{-2}$ *for any* z.

Proof. We only use (2.27): $f' = \kappa_W[z]z^{-2}$ and $\kappa_W[sz] = s^2\kappa_W[z]$. (5) is the definition and $\kappa_W[sz] = f'(sz)^2 = s^2 f' z^2$. For the change of base point, take the representative $\hat z := z/z(\xi_b')$ with $\hat z(\xi_b') = 1$: $\varkappa[\xi_b'] = \kappa_W[\hat z] = \kappa_W[z]z(\xi_b')^{-2}$, and likewise $\varkappa[\xi_b] = \kappa_W[z]z(\xi_b)^{-2}$; divide. (2) and (7): for the representative $\hat z = z/z(\xi_b)$ we have $f'(\xi_b) = \kappa_W[\hat z]\hat z(\xi_b)^{-2} = \kappa_W[\hat z] = \kappa_W[z]z(\xi_b)^{-2}$. (1): $f_x = f'\cosh^2\xi = \kappa_W[z]\cosh^2\xi/z^2$, and at ξ_b with $z(\xi_b) = \cosh\xi_b$ this is $\kappa_W[z]$. Comparing with (2), the representative $\cosh\xi_b\,\hat z$ gives $\kappa_x = \cosh^2\xi_b\,\kappa_W[\hat z] = \cosh^2\xi_b\,\varkappa[\xi_b]$, and $\cosh^2\xi_b = 1/(1-x_b^2)$. The statement on $\kappa_{\max}$ follows by applying this to each solution of the set. (3): at $\xi_b = 0$, $\cosh 0 = 1$ and $f_x(0) = f'(0)$ since $\mathrm{d}x/\mathrm{d}\xi = 1$ at 0. (6) is (7) with the factor $z(\xi_1)^{-2}$ from (5). (4): $z(0) = 1+\vartheta$, so the normalised solution $z/(1+\vartheta)$ has $\kappa_W[z] = \kappa_{\mathrm{D}}/(1+\vartheta)^2 = \kappa_0$, i.e. $\kappa_{\mathrm{D}} = (1+\vartheta)^2\kappa_0$. Next, $(u_-/z)' = (u_-'z - u_-z')/z^2 = \mathrm{Wr}(z,u_-)/z^2$ and, since $\mathrm{Wr}(u_-,u_-) = 0$ and Wronskians of solutions of $z'' = Wz$ are constant, $\mathrm{Wr}(z,u_-) = \vartheta\,\mathrm{Wr}(u_+,u_-) = \vartheta w_0$. Because u_+ is independent of u_- ($w_0 \neq 0$) and u_- is recessive at $-\infty$, $u_-/u_+ \to 0$ at $-\infty$, hence $u_-/z = 1/(1+\vartheta u_+/u_-) \to 0$; because u_+ is recessive at $+\infty$, $u_+/u_- \to 0$ and $u_-/z \to 1$ there. Therefore, for $\xi_0 < \xi$, $f(\xi) - f(\xi_0) = \kappa_{\mathrm{D}}\int_{\xi_0}^{\xi} z^{-2} = \frac{\kappa_{\mathrm{D}}}{\vartheta w_0}\Big[\frac{u_-}{z}\Big]_{\xi_0}^{\xi}$; letting $\xi_0 \to -\infty$ shows that $f_- := f(-\infty)$ exists and $f(\xi) = f_- + \frac{\kappa_{\mathrm{D}}}{\vartheta w_0}\frac{u_-(\xi)}{z(\xi)}$, and letting $\xi \to \infty$ gives $J = \kappa_{\mathrm{D}}/(\vartheta w_0)$. At $\xi = 0$, $u_-(0)/z(0) = 1/(1+\vartheta)$. □

Remark 2.16 ($w_0 > 0$ and the Jost solutions)**.** In Chapter 3 it is shown that for an admissible potential the Jost solutions $u_\pm$ exist, $u_\pm > 0$, and $w_0 = \zeta_-(0) - \zeta_+(0) > 0$ with $\zeta_\pm = u_\pm'/u_\pm$, so f is increasing for $\kappa > 0$ and $J > 0$. Note the convention: with $\mathrm{Wr}(a,b) = ab' - a'b$ one has $w_0 = \mathrm{Wr}(u_+,u_-)$, *not* $\mathrm{Wr}(u_-,u_+)$ (which is $-w_0$).

Toy example 2.17 (conversions at the rest state)**.** Let $Q \equiv 0$ ($W \equiv 1$) and take $z = \cosh\xi$, i.e. $y \equiv 1$, $U \equiv 0$, and $f' = \kappa/\cosh^2\xi$, $f = A + \kappa\tanh\xi = A + \kappa x$. Then for every base point $\kappa_x = f_x = \kappa$, while the representative with $z(\xi_b) = 1$ is $\hat z = \cosh\xi/\cosh\xi_b$ with $\varkappa[\xi_b] = \kappa_W[\hat z] = \kappa/\cosh^2\xi_b = (1-x_b^2)\kappa$: the slope in the line variable is smaller than the slope in x away from the equator, exactly by the factor $\mathrm{d}x/\mathrm{d}\xi = 1 - x^2$. At $x_b = 0$ all three coincide, $\kappa_0 = \kappa_x = \varkappa = \kappa$. (This is the source of the factor $\cosh^2\xi_0$ in $\kappa_{\max} = \cosh^2\xi_0\,\varkappa_{\max}$.)

Conversion of the remaining data. The data $(\boldsymbol{c}, \gamma, A, \kappa_x)$ at a base point and the *effective data* $D = (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$ of Chapter 8 are related by the following elementary formulas (the extraction of $\boldsymbol{c}'$ is the content of Proposition 2.27 below,

that of $(\vartheta, \kappa_{\mathrm{D}})$ is (2.32)); at base point 0 with $z(0) = 1$, $A = f(0)$:

$$c_1' = \tfrac{1}{2}Q(-1), \quad c_2' = \tfrac{1}{2}Q(1), \quad c_3' = -\tfrac{1}{4}(\lambda + 2c_1' + 2c_2'), \qquad \vartheta = \frac{J}{A - f_-} - 1, \quad \kappa_{\mathrm{D}} = (1+\vartheta)^2 f_x(0). \tag{2.33}$$

Here $\lambda = A^2 - 2c_1 - 2c_2 - 4c_3$ (Proposition 2.20). They are implemented in `code/common/swirl_conv.py` (together with all conversions of Lemma 2.15) and cross-checked in `sympy_derivation.py`, Part C, on explicit solutions (Jost solutions for $W = 1$ and for the Pöschl–Teller potential, the base-point rules, the factor $\cosh^2 \xi_0$, and the map to effective data for trivial and non-trivial swirl).

2.5. The operator L, the first integral and the Green operator at a pole

In words. *The equation $(1-x^2)Q_{xxx} = -2ff_x$ is the x-derivative of a second-order equation, $\mathsf{L}[Q] + f^2 = \lambda$, where $\mathsf{L}[g] = (1-x^2)g_{xx} + 2xg_x - 2g$ has polynomial kernel $\operatorname{span}\{x, (1+x)^2\}$. Hence the Green kernels of L are rational functions with visible signs, and, because the zeroth-order coefficient -2 is negative, L obeys a maximum principle. This is the source of all positivity statements about Q.*

Lemma 2.18 (the operator L)**.** *Let $\mathsf{L}[g] := (1-x^2)g_{xx} + 2xg_x - 2g$.*

(a) *$\partial_x \mathsf{L}[g] = (1-x^2)g_{xxx}$; in the line variable $\mathsf{L}[g] = \cosh^2\xi\, g'' + 4\sinh\xi\cosh\xi\, g' - 2g$.*

(b) *$\ker \mathsf{L} = \operatorname{span}\{x, (1+x)^2\} = \operatorname{span}\{x, (1-x)^2\}$ is two-dimensional; $\mathsf{L}[1] = -2$, $\mathsf{L}[1 \pm x] = -2$, $\mathsf{L}[1-x^2] = -4$, and $\mathsf{L}[P_{\boldsymbol{c}}] = -2c_1 - 2c_2 - 4c_3$.*

(c) *(Cauchy kernel) $K_{\mathsf{L}}(x,s) := \dfrac{(x-s)(1-xs)}{(1-s^2)^2}$ satisfies $\mathsf{L}_x K_{\mathsf{L}} = 0$, $K_{\mathsf{L}}(s,s) = 0$, $\partial_x K_{\mathsf{L}}(s,s) = (1-s^2)^{-1}$; moreover $\partial_x K_{\mathsf{L}}(x,s) = \dfrac{(x-s)^2 + 1 - x^2}{(1-s^2)^2} > 0$ for $x, s \in (-1,1)$, $K_{\mathsf{L}} > 0$ for $-1 < s < x < 1$, $K_{\mathsf{L}} < 0$ for $-1 < x < s < 1$, and $K_{\mathsf{L}}(1,s) = (1+s)^{-2}$, $K_{\mathsf{L}}(-1,s) = -(1-s)^{-2}$.*

(d) *(Cauchy formula) If $\mathsf{L}[g] = F$ on an interval containing x_a, with F continuous, then*

$$g(x) = g(x_a)A_1(x) + g_x(x_a)A_2(x) + \int_{x_a}^{x} K_{\mathsf{L}}(x,s)F(s)\,\mathrm{d}s, \tag{2.34}$$

with $A_1 = 1 + \frac{(x-x_a)^2}{1-x_a^2}$, $A_2 = \frac{(x-x_a)(1-x_a x)}{1-x_a^2} \in \ker \mathsf{L}$, and $\int_{x_a}^{x} K_{\mathsf{L}}(x,s)\,\mathrm{d}s = \frac{(x-x_a)^2}{2(1-x_a^2)}$. For $-1 < x_a \le x < 1$: $A_1 \ge 1$, $A_2 \ge 0$, $0 \le \int_{x_a}^{x} K_{\mathsf{L}}\,\mathrm{d}s \le \frac{1-x_a}{2(1+x_a)}$; if moreover $x_a \ge 0$ then $A_1 \le 2$ and $A_2 \le 1 - x_a$.

Proof. (a) $\partial_x\big[(1-x^2)g_{xx} + 2xg_x - 2g\big] = -2xg_{xx} + (1-x^2)g_{xxx} + 2g_x + 2xg_{xx} - 2g_x = (1-x^2)g_{xxx}$. For the line form, $g_x = \cosh^2\xi\, g'$ and $g_{xx} = \cosh^4\xi\, g'' + 2\cosh^3\xi\sinh\xi\, g'$ give $(1-x^2)g_{xx} = \cosh^2\xi\, g'' + 2\sinh\xi\cosh\xi\, g'$ and $2xg_x = 2\sinh\xi\cosh\xi\, g'$. (b) Direct substitution: $\mathsf{L}[1] = -2$, $\mathsf{L}[x] = 0 + 2x - 2x = 0$, $\mathsf{L}[x^2] = 2(1-x^2) + 4x^2 - 2x^2 = 2$. Hence $\mathsf{L}[(1+x)^2] = \mathsf{L}[1] + 2\mathsf{L}[x] + \mathsf{L}[x^2] = 0$, $\mathsf{L}[1 \pm x] = -2$, $\mathsf{L}[1-x^2] = -4$, and so $\mathsf{L}[P_{\boldsymbol{c}}] = -2c_1 - 2c_2 - 4c_3$; also $(1-x)^2 = (1+x)^2 - 4x$. The equation $\mathsf{L}[g] = 0$ is a linear second-order ODE with leading coefficient $1 - x^2 \neq 0$ on $(-1,1)$, so its solution space is two-dimensional [14, Ch. IV, §1]; x and $(1+x)^2$ are independent, since their Wronskian is $x \cdot 2(1+x) - (1+x)^2 = -(1-x^2) \neq 0$.

(c) $K_{\mathsf{L}} = \big(-sx^2+(1+s^2)x-s\big)/(1-s^2)^2$ is a quadratic polynomial in x, and by (b) $\mathsf{L}_x[-sx^2+(1+s^2)x-s] = -2s+0+2s = 0$. Clearly $K_{\mathsf{L}}(s,s) = 0$; and $\partial_x K_{\mathsf{L}} = \frac{(1-xs)-s(x-s)}{(1-s^2)^2} = \frac{1-2xs+s^2}{(1-s^2)^2} = \frac{(x-s)^2+1-x^2}{(1-s^2)^2} > 0$, equal to $\frac{1-s^2}{(1-s^2)^2} = \frac{1}{1-s^2}$ at $x=s$. For $s<x$ both $x-s$ and $1-xs$ are positive ($|xs|<1$), so $K_{\mathsf{L}} > 0$; for $x<s$, $x-s<0$. Finally $K_{\mathsf{L}}(1,s) = \frac{(1-s)^2}{(1-s^2)^2} = (1+s)^{-2}$ and $K_{\mathsf{L}}(-1,s) = -\frac{(1+s)^2}{(1-s^2)^2} = -(1-s)^{-2}$. (d) *Particular solution.* Put $g_p(x) = \int_{x_a}^x K_{\mathsf{L}}(x,s)F(s)\,\mathrm{d}s$. As K_{L} is a polynomial in x with coefficients continuous in s, we may differentiate under the integral: $\partial_x g_p = K_{\mathsf{L}}(x,x)F + \int_{x_a}^x \partial_x K_{\mathsf{L}}\,F\,\mathrm{d}s = \int_{x_a}^x \partial_x K_{\mathsf{L}}\,F\,\mathrm{d}s$ and $\partial_x^2 g_p = \frac{F(x)}{1-x^2} + \int_{x_a}^x \partial_x^2 K_{\mathsf{L}}\,F\,\mathrm{d}s$. Hence $\mathsf{L}[g_p] = F + \int_{x_a}^x \mathsf{L}_x K_{\mathsf{L}}\,F\,\mathrm{d}s = F$, and $g_p(x_a) = \partial_x g_p(x_a) = 0$. *Homogeneous part.* $A_1 \in \ker\mathsf{L}$ since $\mathsf{L}[1+x^2-2x_a x] = -2+2-0 = 0$, with $A_1(x_a) = 1$, $\partial_x A_1(x_a) = 0$; $A_2 = (1-x_a^2)K_{\mathsf{L}}(x,x_a) \in \ker\mathsf{L}$ with $A_2(x_a) = 0$, $\partial_x A_2(x_a) = 1$. So $g - g_p - g(x_a)A_1 - g_x(x_a)A_2 \in \ker\mathsf{L}$ has vanishing value and derivative at x_a, hence vanishes by uniqueness for the linear ODE. This is (2.34). *The integral of* K_{L} is g_p for $F\equiv 1$; since $\mathsf{L}[-\frac12] = 1$, (2.34) gives $\int_{x_a}^x K_{\mathsf{L}} = -\frac12 + \frac12 A_1 = \frac{(x-x_a)^2}{2(1-x_a^2)}$. *Bounds.* Let $-1 < x_a \le x < 1$. $A_1 \ge 1$ is clear; $A_2 \ge 0$ since $x - x_a \ge 0$ and $1 - x_a x > 0$; as $0 \le x - x_a \le 1 - x_a$, $\int_{x_a}^x K_{\mathsf{L}} \le \frac{(1-x_a)^2}{2(1-x_a^2)} = \frac{1-x_a}{2(1+x_a)}$. If $x_a \ge 0$: $A_1 \le 1 + \frac{(1-x_a)^2}{1-x_a^2} = 1 + \frac{1-x_a}{1+x_a} \le 2$, and $(1-x_a) - (1-x_a x) = x_a(x-x_a) \ge 0$ gives $1 - x_a x \le 1 - x_a^2$, so $A_2 \le x - x_a \le 1 - x_a$. □

Lemma 2.19 (maximum principle for L)**.** *Let $-1 \le a < b \le 1$, $g \in C^2(a,b)$ with $\mathsf{L}[g] \le 0$ on (a,b), $\liminf_{x\downarrow a} g \ge 0$, $\liminf_{x\uparrow b} g \ge 0$. Then $g \ge 0$ on (a,b).*

Proof. Suppose $g(x_1) < 0$ for some $x_1 \in (a,b)$. By the hypotheses at the ends there are $a < \alpha < x_1 < \beta < b$ with $g > \frac12 g(x_1)$ on $(a,\alpha] \cup [\beta,b)$. The continuous function g attains its minimum on $[\alpha,\beta]$ at some x_0; since $g(x_0) \le g(x_1) < \frac12 g(x_1) < \min(g(\alpha), g(\beta))$, x_0 is an interior point. So $g_x(x_0) = 0$, $g_{xx}(x_0) \ge 0$ and $\mathsf{L}[g](x_0) = (1-x_0^2)g_{xx}(x_0) - 2g(x_0) \ge -2g(x_0) > 0$, contradicting $\mathsf{L}[g] \le 0$. □

Proposition 2.20 (first integral)**.** *Let (U,f,Q) solve (2.19)–(2.21) on an interval. Then (2.21) holds iff $\mathsf{L}[Q] + f^2 \equiv \lambda$ for a constant λ. If $P_{\boldsymbol{c}}$ is the degree-two Taylor polynomial of Q at x_b and $A = f(x_b)$ then*

$$\lambda = \mathsf{L}[P_{\boldsymbol{c}}](x_b) + A^2 = -2c_1 - 2c_2 - 4c_3 + A^2, \tag{2.35}$$

(For the Li–Li–Yan normalisation $x_b = 0$: $\lambda = f(0)^2 - 2c_1 - 2c_2 - 4c_3$.)

Proof. By Lemma 2.18(a), $\partial_x(\mathsf{L}[Q]+f^2) = (1-x^2)Q_{xxx} + 2ff_x$. A C^1 function on an interval is constant iff its derivative vanishes, so (2.21) holds iff $\mathsf{L}[Q]+f^2$ is constant. Since $\mathsf{L}[g](x_b)$ depends only on the 2-jet of g at x_b and Q, $P_{\boldsymbol{c}}$ have the same 2-jet there, $\lambda = \mathsf{L}[Q](x_b) + f(x_b)^2 = \mathsf{L}[P_{\boldsymbol{c}}](x_b) + A^2$, and $\mathsf{L}[P_{\boldsymbol{c}}] \equiv -2c_1 - 2c_2 - 4c_3$ by Lemma 2.18(b). □

Toy example 2.21 (a Green kernel by hand)**.** Solve $\mathsf{L}[g] = -1$. The Cauchy formula (2.34) with $x_a = 0$ and zero data gives the particular solution $g_0 = -\int_0^x K_{\mathsf{L}}(x,s)\,\mathrm{d}s = -x^2/2$ (check: $\mathsf{L}[-x^2/2] = -1$). Adding the kernel element $\frac14(1+x^2) = \frac18[(1+x)^2 + (1-x)^2]$ yields $g = \frac14(1-x^2)$, the solution that vanishes at *both* poles. By Lemma 2.19 it is the only such solution. This is exactly the mechanism behind $Q_{\mathrm{sw}}[\text{const } A] = \frac14 A^2(1-x^2)$ in Proposition 2.27.

Remark 2.22 (why the first integral is useful)**.** The first integral lowers the order of the Q-equation by one and turns it into the linear equation $\mathsf{L}[Q] = \lambda - f^2$ with a source of known sign ($-f^2 \le 0$). Together with the positivity of K_{L} this gives upper bounds for Q and W that are independent of the size of f (used in Section 2.6).

2.5.1. Green's operator at a pole. At the south pole we need the solution of $\mathsf{L}[R] = F$ with the base point moved to the singular endpoint $x = -1$. Recall $E = e^{-2|\xi|} = e^{2\xi}$ for $\xi \le 0$, and $t = 1 + x$.

Lemma 2.23 (Green operator at a pole)**.** *Let $x_1 \le 0$, $\xi_1 = \operatorname{artanh} x_1$, and let F be continuous on $(-1, x_1]$ with $\int_{-\infty}^{\xi_1} |F|\, E^{-1}\, \mathrm{d}\eta < \infty$. Then*

$$\mathsf{G}_{\mathsf{L}}[F](x) := \int_{-1}^{x} K_{\mathsf{L}}(x,s) F(s)\, \mathrm{d}s$$

is well defined, $\mathsf{L}[\mathsf{G}_{\mathsf{L}}[F]] = F$, $\partial_x \mathsf{G}_{\mathsf{L}}[F] = \int_{-1}^{x} \partial_x K_{\mathsf{L}}\, F\, \mathrm{d}s$, and for $\xi \le \xi_1$

$$|\mathsf{G}_{\mathsf{L}}[F](\xi)| \le 4\, E(\xi)^2 \int_{-\infty}^{\xi} |F(\eta)|\, E(\eta)^{-1}\, \mathrm{d}\eta. \tag{2.36}$$

In particular, if $|F| \le \Phi\, Z^{-2} E^{\,q}$ on $(-\infty, \xi_1]$ with $q > 1$ and $Z \ge 1$ non-increasing, then $|\mathsf{G}_{\mathsf{L}}[F]| \le \frac{2\Phi}{q-1} Z^{-2} E^{\,q+1}$. If R solves $\mathsf{L}[R] = F$ on $(-1, x_1]$ for such an F and $R \to 0$ as $x \to -1$, then $R = \mathsf{G}_{\mathsf{L}}[F] + \beta(1+x)^2$ for a unique $\beta \in \mathbb{R}$; hence $R = \mathsf{G}_{\mathsf{L}}[F]$ iff $R = o(E^{\,2})$. (By the symmetry σ_2 the same holds at the north pole with $(1-x)^2$.)

Proof. Throughout $-1 < s < x \le x_1 \le 0$, $t_x = 1 + x$, $t_s = 1 + s \in (0, 1]$, $E_x = E(\operatorname{artanh} x)$, $E_s = E(\eta)$ with $\eta = \operatorname{artanh} s$, so $\mathrm{d}s = (1 - s^2)\mathrm{d}\eta$.

Step 1: kernel bound. With $x = t_x - 1$, $s = t_s - 1$ we get $x - s = t_x - t_s$ and $1 - xs = t_x + t_s - t_x t_s$. Then

$$t_x^2 - (t_x - t_s)(t_x + t_s - t_x t_s) = t_s\big[t_x^2 + t_s(1 - t_x)\big] \ge 0$$

since $t_x \le 1$ (expand both sides). So $(x - s)(1 - xs) \le t_x^2$. Also $(1 - s^2)^2 = t_s^2(2 - t_s)^2 \ge t_s^2$. With Lemma 2.18(c),

$$0 \le K_{\mathsf{L}}(x,s) \le \left(\frac{t_x}{t_s}\right)^2 \le \left(\frac{E_x}{E_s}\right)^2,$$

the last step because $t = 2E/(1+E)$ gives $t_x/t_s = (E_x/E_s)(1+E_s)/(1+E_x)$ and $E_s < E_x$. Moreover $1 - s^2 = t_s(2 - t_s) \le 2t_s \le 4E_s$ by (2.30), so $\mathrm{d}s \le 4E_s\, \mathrm{d}\eta$.

Step 2: (2.36). By Step 1,

$$\int_{-1}^{x} |K_{\mathsf{L}} F|\, \mathrm{d}s \le \int_{-\infty}^{\xi} \frac{E_x^2}{E_\eta^2} |F|\, 4E_\eta\, \mathrm{d}\eta = 4E_x^2 \int_{-\infty}^{\xi} |F| E^{-1} \mathrm{d}\eta < \infty.$$

Step 3: $\mathsf{L}[\mathsf{G}_{\mathsf{L}}[F]] = F$. Write $K_{\mathsf{L}}(x,s) = \sum_{k=0}^{2} a_k(s) x^k$ with $a_0 = a_2 = -s/(1-s^2)^2$ and $a_1 = (1+s^2)/(1-s^2)^2$, so $|a_k| \le 2(1-s^2)^{-2}$. Since $1 - s^2 = t_s(2 - t_s) \ge t_s \ge E_s$, $\int_{-1}^{x_1} |F|(1-s^2)^{-2}\mathrm{d}s = \int_{-\infty}^{\xi_1} |F|(1-s^2)^{-1}\mathrm{d}\eta \le \int_{-\infty}^{\xi_1} |F| E^{-1}\mathrm{d}\eta < \infty$. Hence $m_k(x) := \int_{-1}^{x} a_k F\, \mathrm{d}s$ is C^1 with $m_k' = a_k F$, and $\mathsf{G}_{\mathsf{L}}[F] = \sum_k x^k m_k$. Then $\partial_x \mathsf{G}_{\mathsf{L}}[F] = \sum_k k x^{k-1} m_k + K_{\mathsf{L}}(x,x)F = \sum_k k x^{k-1} m_k = \int_{-1}^{x} \partial_x K_{\mathsf{L}} F\, \mathrm{d}s$ and $\partial_x^2 \mathsf{G}_{\mathsf{L}}[F] = \sum_k k(k-1) x^{k-2} m_k + \partial_x K_{\mathsf{L}}(x,x)F = \int_{-1}^{x} \partial_x^2 K_{\mathsf{L}} F\, \mathrm{d}s + \frac{F(x)}{1-x^2}$. So, exactly as in the proof of Lemma 2.18(d), $\mathsf{L}[\mathsf{G}_{\mathsf{L}}[F]] = F + \int_{-1}^{x} \mathsf{L}_x K_{\mathsf{L}}\, F\, \mathrm{d}s = F$.

Step 4: power-law sources. If $|F| \le \Phi Z^{-2} E^q$ with $Z \ge 1$ non-increasing, then $Z(\eta)^{-2} \le Z(\xi)^{-2}$ for $\eta \le \xi$, so by (2.36)

$$|\mathsf{G}_\mathsf{L}[F](\xi)| \le 4E(\xi)^2 \Phi Z(\xi)^{-2} \int_{-\infty}^{\xi} e^{2(q-1)\eta} \mathrm{d}\eta = \frac{2\Phi}{q-1} Z(\xi)^{-2} E(\xi)^{q+1}.$$

Step 5: uniqueness. The tail $\int_{-\infty}^{\xi} |F| E^{-1}$ of a convergent integral tends to 0 as $\xi \to -\infty$, so (2.36) gives $\mathsf{G}_\mathsf{L}[F] = o(E^2)$; in particular $\mathsf{G}_\mathsf{L}[F] \to 0$ at $x = -1$. If $\mathsf{L}[R] = F$ and $R \to 0$ then $R - \mathsf{G}_\mathsf{L}[F] \in \ker \mathsf{L}$, i.e. $= \alpha x + \beta(1+x)^2$; letting $x \to -1$ gives $0 = -\alpha$. So $R = \mathsf{G}_\mathsf{L}[F] + \beta(1+x)^2$ with unique β, and since $(1+x)^2 = t^2 \ge E^2$ is not $o(E^2)$, $R = o(E^2)$ iff $\beta = 0$. □

2.6. Continuation and global solutions

In words. *A solution of the Q-system can only stop existing at a finite point in one way: $z \to 0$ there (and then $U \to \mp\infty$). Therefore a solution is global iff z does not tend to zero at any finite point; from Chapter 3 on, where W is given a priori, this is the statement "$z > 0$ on the whole line". The proof uses the Cauchy formula to show that Q and Q' stay bounded as long as z stays away from 0, and that Q stays bounded* above *even if f blows up.*

Lemma 2.24 (continuation)**.** *Let a solution of (2.29) be defined on $[\xi_a, \xi^*)$ with $\xi^* < \infty$ and assume $\liminf_{\xi \to \xi^*} z > 0$. Then it extends beyond ξ^*; the same holds on $(\xi^*, \xi_a]$. Consequently an endpoint of the maximal interval of existence is finite only if $\liminf z = 0$ there, and then $U \to -\infty$ at a finite right endpoint ξ^* and $U \to +\infty$ at a finite left endpoint. Hence a maximal solution is global (maximal interval $\mathbb{R}$) iff z does not tend to 0 at a finite point.*

Proof. Put $x_a = \tanh \xi_a$, $x^* = \tanh \xi^* < 1$.

Extension. There is $\delta > 0$ with $z \ge \delta$ on $[\xi_a, \xi^*)$. Then $|f'| \le |\kappa| \delta^{-2}$, so f is bounded (the interval is bounded). By the Cauchy formula (2.34) for $g = Q$, $F = \lambda - f^2$ on $[x_a, x^*)$, $Q(x) = Q(x_a)A_1(x) + Q_x(x_a)A_2(x) + \int_{x_a}^{x} K_\mathsf{L}(x,s)(\lambda - f(s)^2)\,\mathrm{d}s$, and the same formula for Q_x with A_i, K_L replaced by $\partial_x A_i, \partial_x K_\mathsf{L}$. These functions are continuous, hence bounded, on the compact sets $[x_a, x^*]$, resp. $[x_a, x^*]^2$, and $\lambda - f^2$ is bounded; so Q and Q_x, hence $Q' = \cosh^2 \xi\, Q_x$, are bounded on $[\xi_a, \xi^*)$. Thus $|W| \le M_W$. For $v := z^2 + z'^2$, $v' = 2zz' + 2z'Wz \le (1 + M_W)v$, so $v(\xi) \le v(\xi_a) e^{(1+M_W)(\xi - \xi_a)}$ (Gronwall [13]). The state (z, z', f, Q, Q') therefore stays in a compact $C \subset \{z \ge \delta\} \times \mathbb{R}^4$, and $[\xi_a, \xi^*] \times C$ lies in the open domain $\mathbb{R} \times \{z > \delta/2\} \times \mathbb{R}^4$ of the locally Lipschitz field (2.29). By Remark 2.7(ii) the solution extends. The case $(\xi^*, \xi_a]$ is identical.

Consequences. If the maximal interval had a finite right endpoint ξ^* with $\liminf z > 0$, the solution would extend, a contradiction. So $\liminf z = 0$ at every finite endpoint, and a maximal solution is global iff there is no finite endpoint.

$U \to -\infty$ at a finite right endpoint. First, W is bounded above on $[\xi_a, \xi^*)$ even though f may be unbounded: in the Cauchy formula $K_\mathsf{L}(x,s) \ge 0$ for $x_a \le s \le x$ and $\lambda - f^2 \le \lambda$, so $\int_{x_a}^{x} K_\mathsf{L}(\lambda - f^2) \le \max(\lambda, 0) \int_{x_a}^{x} K_\mathsf{L}$, and all other terms are bounded. So $Q \le M_Q$ and $W \le M := 1 + \frac{1}{2}\max(M_Q, 0)$. Now $h = z'/z$ satisfies $h' = W - h^2 \le M - h^2$. Since $\ln z(\xi) = \ln z(\xi_a) + \int_{\xi_a}^{\xi} h$ and $\liminf \ln z = -\infty$ on a bounded interval, h is not bounded below. Let $n \ge \sqrt{M} + 1$ and pick ξ_n with $h(\xi_n) < -n$. At any point with $h = -n$ we would have $h' \le M - n^2 < 0$, so h

cannot cross the level $-n$ from below: $h < -n$ on $[\xi_n, \xi^*)$. Hence $h \to -\infty$ and $U = 2(h - \tanh\xi) \to -\infty$.

Left endpoint. Put $\tilde{h}(\eta) = -h(-\eta)$; then $\tilde{h}' = W(-\eta) - \tilde{h}^2$, the same Riccati inequality. For $x < x_a$ the integral term is $\int_{x_a}^{x} K_{\mathsf{L}}(\lambda - f^2) = \int_{x}^{x_a}(-K_{\mathsf{L}}(x,s))(\lambda - f^2)\,\mathrm{d}s$ with $-K_{\mathsf{L}}(x,s) \geq 0$ for $s \geq x$, so it is at most $\max(\lambda, 0)\int_{x}^{x_a}(-K_{\mathsf{L}})$, which is bounded; thus W is bounded above near the left endpoint, and the argument gives $\tilde{h} \to -\infty$, i.e. $h \to +\infty$ and $U \to +\infty$. □

Corollary 2.25 (global $\iff$ $z > 0$)**.** *A solution of the Q-system is global (maximal interval $(-1,1)$, i.e. $\xi \in \mathbb{R}$) iff z does not tend to 0 at a finite point. Equivalently: for a* given *potential W on $\mathbb{R}$, $U = 2(z'/z - \tanh\xi)$ exists on all of $\mathbb{R}$ iff the solution z of $z'' = Wz$ is positive on $\mathbb{R}$ (this is the formulation used from Chapter 3 on). The property is unchanged under $(z, \kappa) \mapsto (sz, s^2\kappa)$.*

Proof. This is the last sentence of Lemma 2.24, whose proof ("Consequences" and the endpoint paragraphs) shows the following. If the maximal interval had a finite right (left) endpoint, then $\liminf z = 0$ there and $h = z'/z \to -\infty$ $(+\infty)$ monotonically after some point, so z is monotone near the endpoint and $z \to 0$; conversely if $z \to 0$ at a finite point the solution cannot be continued across it, because $U = 2(z'/z - \tanh\xi)$ is unbounded. Hence the maximal solution is global iff z does not tend to 0 at a finite point. For a given potential W on $\mathbb{R}$ the solution z of $z'' = Wz$ exists on $\mathbb{R}$; if it vanishes at some ξ^* then $z'(\xi^*) \neq 0$ (uniqueness for the linear ODE, $z \not\equiv 0$), so $z'/z \to \mp\infty$ and U blows up at ξ^*; if $z > 0$ on $\mathbb{R}$, U is smooth on $\mathbb{R}$. Finally z'/z is unchanged by $z \mapsto sz$ $(s > 0)$, so global existence of U is unchanged, and κ transforms as $f'z^2$ requires. □

Remark 2.26. In the line formulation the poles are *not* points of the domain: "regularity at the poles" is a statement about the behaviour of z as $\xi \to \pm\infty$ and is the subject of Chapter 4. If W has limits $\mu_\pm^2$ at $\pm\infty$, positive solutions behave like $e^{\pm\mu\xi}$ (Example 2.31(i)), and $U(\pm 1)$ is read off from the limit of z'/z.

2.7. The centrifugal decomposition: swirl raises the potential

In words. *The swirl enters the potential only through $\mathsf{L}[Q] = \lambda - f^2$. Solving this with the right boundary conditions shows two things. Measured from the poles (the* effective *polynomial $P_{\mathbf{c}'}$ whose values at ∓ 1 are those of Q), the remainder Q_{sw} is a* nonnegative *integral of f^2 against a positive kernel: swirl raises W. Measured from a base point (the polynomial $P_{\mathbf{c}}$ of the Cauchy data), the remainder $\psi_{\mathbf{c}} = P_{\mathbf{c}} - Q$* lowers *$W$; it is nonnegative and increases away from the base point when the base point is a zero of f. The two are related by an explicit polynomial. Both pictures are used later; the first for existence theory, the second for the lens of class Z.*

Let (U, f, Q) be a solution of the Q-system on $(-1,1)$ (not necessarily global) with $\mathsf{L}[Q] + f^2 = \lambda$. Assume

$$f \text{ is bounded on } (-1,1) \quad \text{and} \quad Q(\pm 1) := \lim_{x \to \pm 1} Q \text{ exist and are finite.} \tag{2.37}$$

(For global solutions of type G or L_1 at both poles these hypotheses hold, as is shown in Chapter 4; nothing in this chapter depends on that verification, and part (ii)

below does not need (2.37) at all.) Define the *effective constants*

$$c_1' = \tfrac{1}{2}Q(-1), \qquad c_2' = \tfrac{1}{2}Q(1), \qquad c_3' = -\tfrac{1}{4}\big(\lambda + 2c_1' + 2c_2'\big), \\ P_{\boldsymbol{c}'} := P_{\boldsymbol{c}'}, \quad W_{\mathrm{e}} := 1 + \tfrac{1}{2}P_{\boldsymbol{c}'}, \tag{2.38}$$

and the *centrifugal part* $Q_{\mathrm{sw}} := Q - P_{\boldsymbol{c}'}$.

Proposition 2.27 (centrifugal decomposition)**.** *Under* (2.37)*:*

(i) *(Swirl raises the potential, measured at the poles.)*

$$Q_{\mathrm{sw}}(x) = \frac{1}{4}\int_{-1}^{1} \frac{(1+x_<)^2(1-x_>)^2}{(1-s^2)^2}\, f(s)^2\,\mathrm{d}s = \frac{1}{4\cosh^2\xi}\int_{\mathbb{R}} e^{-2|\xi-\eta|} f(\eta)^2\,\mathrm{d}\eta, \tag{2.39}$$

where $x_< = \min(x,s)$, $x_> = \max(x,s)$. In particular $Q_{\mathrm{sw}} \geq 0$; $Q_{\mathrm{sw}} > 0$ on $(-1,1)$ unless $f \equiv 0$; $Q_{\mathrm{sw}} \leq \frac{1}{4}(\sup f^2)(1-x^2)$; $Q_{\mathrm{sw}}[A] = \frac{1}{4}A^2(1-x^2)$ for $f \equiv A$; the map $f^2 \mapsto Q_{\mathrm{sw}}$ is linear, monotone ($f_1^2 \leq f_2^2 \Rightarrow Q_{\mathrm{sw}}[f_1] \leq Q_{\mathrm{sw}}[f_2]$) and Lipschitz: $|Q_{\mathrm{sw}}[f_1] - Q_{\mathrm{sw}}[f_2]| \leq \frac{1}{4}\|f_1^2 - f_2^2\|_\infty \operatorname{sech}^2\xi$. Consequently

$$W = W_{\mathrm{e}} + \tfrac{1}{2}Q_{\mathrm{sw}} \;\geq\; W_{\mathrm{e}}. \tag{2.40}$$

(ii) *(Swirl lowers the potential, measured from a base point.) Let $x_b \in (-1,1)$ be a base point, $\boldsymbol{c}$ the constants of Q at x_b, $A = f(x_b)$, $\psi_{\boldsymbol{c}} := P_{\boldsymbol{c}} - Q$. Then $\psi_{\boldsymbol{c}} = O((x-x_b)^3)$ at x_b, $W = W_{\boldsymbol{c}} - \frac{1}{2}\psi_{\boldsymbol{c}}$ with $W_{\boldsymbol{c}} := 1 + \frac{1}{2}P_{\boldsymbol{c}}$, and*

$$\mathsf{L}[\psi_{\boldsymbol{c}}] = f^2 - A^2. \tag{2.41}$$

If $f(x_b) = 0$ (class-Z base point, Chapter 5), then $\psi_{\boldsymbol{c}} \geq 0$, and $\psi_{\boldsymbol{c}}$ is nondecreasing in $|\xi - \xi_b|$ (nonincreasing in x on $(-1,x_b]$, nondecreasing on $[x_b,1)$); in particular $W \leq W_{\boldsymbol{c}}$.

(iii) *(Relation.) The limits $\psi_\mp := \psi_{\boldsymbol{c}}(\mp 1)$ are finite and*

$$c_i' = c_i - \tfrac{1}{2}\psi_\mp \;(i=1,2), \quad c_3' = c_3 - \tfrac{1}{4}A^2 + \tfrac{1}{4}(\psi_- + \psi_+), \quad Q_{\mathrm{sw}} = (P_{\boldsymbol{c}} - P_{\boldsymbol{c}'}) - \psi_{\boldsymbol{c}}, \tag{2.42}$$

$$P_{\boldsymbol{c}} - P_{\boldsymbol{c}'} = \tfrac{1}{2}\psi_-(1-x) + \tfrac{1}{2}\psi_+(1+x) + \tfrac{1}{4}\big(A^2 - \psi_- - \psi_+\big)(1-x^2). \tag{2.43}$$

At a class-Z base point ($A = 0$), with $k_i = \sqrt{1+c_i}$, $k_i' = \sqrt{1+c_i'}$ and $\varepsilon = c_3 - \bar{c}_3$, $\varepsilon' = c_3' - \bar{c}_3(c_1', c_2')$ (notation of Chapter 3), $\varepsilon' = \varepsilon - \frac{1}{2}\big[(k_1 - k_1')(k_2 + k_2' + 2) + (k_2 - k_2')(k_1 + k_1' + 2)\big]$.

Remark 2.28. Q_{sw} and $\psi_{\boldsymbol{c}}$ are *different* objects: $Q_{\mathrm{sw}} = (P_{\boldsymbol{c}} - P_{\boldsymbol{c}'}) - \psi_{\boldsymbol{c}}$ differs from $-\psi_{\boldsymbol{c}}$ by a polynomial and by sign convention ($Q_{\mathrm{sw}} \geq 0$ raises W, $\psi_{\boldsymbol{c}} \geq 0$ at a zero of f lowers it). The function called ψ in the threshold literature is $\psi_{\boldsymbol{c}}$.

Proof. *Step 1: the solution operator* S. For a bounded continuous ϕ on $(-1,1)$ put $y_1 = (1+x)^2$, $y_2 = (1-x)^2$ (both in $\ker\mathsf{L}$, vanishing at -1, resp. 1; $y_1y_2' - y_1'y_2 = -4(1-x^2)$), $\varphi(s) = \phi(s)/(4(1-s^2)^2)$ and

$$\mathsf{S}[\phi] := y_2 A + y_1 B,$$

$$A(x) = \int_{-1}^{x} y_1\varphi\,\mathrm{d}s = \int_{-1}^{x} \frac{\phi}{4(1-s)^2}\mathrm{d}s, \quad B(x) = \int_{x}^{1} y_2\varphi\,\mathrm{d}s = \int_{x}^{1} \frac{\phi}{4(1+s)^2}\mathrm{d}s.$$

Both integrals converge for $x \in (-1,1)$. With $A' = y_1\varphi$, $B' = -y_2\varphi$: $(\mathsf{S}[\phi])' = y_2'A + y_1'B$ (the terms $y_2y_1\varphi - y_1y_2\varphi$ cancel) and $(\mathsf{S}[\phi])'' = y_2''A + y_1''B + (y_2'y_1 - y_1'y_2)\varphi = y_2''A + y_1''B - 4(1-x^2)\varphi$ (here $' = \partial_x$). Hence $\mathsf{L}[\mathsf{S}[\phi]] = A\mathsf{L}[y_2] + B\mathsf{L}[y_1] - 4(1-x^2)^2\varphi =$

$-\phi$. For $|\phi| \le M$: $|A| \le \frac{M}{4}(\frac{1}{1-x} - \frac{1}{2})$, so $y_2|A| \le \frac{M}{4}(1-x)\frac{1+x}{2} = \frac{M}{8}(1-x^2)$, and likewise $y_1|B| \le \frac{M}{8}(1-x^2)$. So

$$|\mathsf{S}[\phi]| \le \tfrac{1}{4}\sup|\phi|\,(1-x^2) \to 0 \quad (x \to \pm 1). \tag{2.44}$$

Uniqueness: if $g_1, g_2 \in C^2$ both solve $\mathsf{L}[g] = -\phi$ and tend to 0 at ± 1, then $h = g_1 - g_2$ and $-h$ satisfy the hypotheses of Lemma 2.19 with $a = -1, b = 1$, so $h \equiv 0$. *Positivity:* if $\phi \ge 0$, then $\mathsf{L}[\mathsf{S}[\phi]] \le 0$ with zero boundary values, so $\mathsf{S}[\phi] \ge 0$. By linearity, $\phi_1 \le \phi_2 \Rightarrow \mathsf{S}[\phi_1] \le \mathsf{S}[\phi_2]$, and $|\mathsf{S}[\phi_1] - \mathsf{S}[\phi_2]| \le \mathsf{S}[|\phi_1 - \phi_2|] \le \frac{1}{4}\|\phi_1 - \phi_2\|_\infty(1-x^2)$ by (2.44) and monotonicity. (Alternatively, the bound follows from the maximum principle by comparing with $\frac{1}{4}\sup|\phi|\,(1-x^2)$, using $\mathsf{L}[1-x^2] = -4$ and Lemma 2.19.)

Step 2: $Q_{\mathrm{sw}} = \mathsf{S}[f^2]$. By Lemma 2.18(b) and (2.38), $\mathsf{L}[P_{\boldsymbol{c}'}] = -2c_1' - 2c_2' - 4c_3' = -2c_1' - 2c_2' + \lambda + 2c_1' + 2c_2' = \lambda$. Thus $g := Q - P_{\boldsymbol{c}'}$ satisfies $\mathsf{L}[g] = (\lambda - f^2) - \lambda = -f^2$; since $P_{\boldsymbol{c}'}(-1) = 2c_1' = Q(-1)$ and $P_{\boldsymbol{c}'}(1) = 2c_2' = Q(1)$, $g \to 0$ at ± 1. By uniqueness in Step 1, $g = \mathsf{S}[f^2]$. This gives $Q_{\mathrm{sw}} \ge 0$, the bound $Q_{\mathrm{sw}} \le \frac{1}{4}(\sup f^2)(1-x^2)$, monotonicity, linearity and the Lipschitz estimate (in the form with $(1-x^2) = \operatorname{sech}^2\xi$). For $f \equiv A$: $\mathsf{L}[\frac{1}{4}A^2(1-x^2)] = -A^2$ and the function vanishes at ± 1, so $Q_{\mathrm{sw}}[A] = \frac{1}{4}A^2(1-x^2)$ by uniqueness (cf. Toy example 2.21).

Step 3: the kernel form. By the definition of A, B,

$$\mathsf{S}[\phi](x) = \frac{1}{4}\int_{-1}^{1} \frac{y_1(x_<)y_2(x_>)}{(1-s^2)^2}\phi(s)\,\mathrm{d}s,$$

which is the first expression in (2.39). With $x = \tanh\xi$, $s = \tanh\eta$: $1 \pm s = e^{\pm\eta}/\cosh\eta$, $1 - s^2 = \operatorname{sech}^2\eta$, $\mathrm{d}s = \operatorname{sech}^2\eta\,\mathrm{d}\eta$. For $s < x$ ($\eta < \xi$),

$$\frac{(1+s)^2(1-x)^2}{(1-s^2)^2}\mathrm{d}s = \frac{e^{2\eta}}{\cosh^2\eta}\cdot\frac{e^{-2\xi}}{\cosh^2\xi}\cdot\cosh^4\eta\cdot\frac{\mathrm{d}\eta}{\cosh^2\eta} = \frac{e^{-2(\xi-\eta)}}{\cosh^2\xi}\mathrm{d}\eta,$$

and symmetrically $e^{-2(\eta-\xi)}\cosh^{-2}\xi\,\mathrm{d}\eta$ for $s > x$; this is the second expression. Since $\int_{\mathbb{R}} e^{-2|\sigma|}\mathrm{d}\sigma = 1$, $Q_{\mathrm{sw}} \le \frac{\sup f^2}{4\cosh^2\xi}$, and the kernel being positive and f continuous, $Q_{\mathrm{sw}} > 0$ unless $f \equiv 0$. Finally $W = 1 + \frac{1}{2}Q = 1 + \frac{1}{2}P_{\boldsymbol{c}'} + \frac{1}{2}Q_{\mathrm{sw}}$.

Step 4: (ii). Q and $P_{\boldsymbol{c}}$ have the same 2-jet at x_b (Lemma 2.5), so $\psi_{\boldsymbol{c}} = P_{\boldsymbol{c}} - Q = O((x-x_b)^3)$; $W = 1 + \frac{1}{2}Q = W_{\boldsymbol{c}} - \frac{1}{2}\psi_{\boldsymbol{c}}$ is the definition. By Proposition 2.20, $\lambda = \mathsf{L}[P_{\boldsymbol{c}}] + A^2$, so $\mathsf{L}[\psi_{\boldsymbol{c}}] = \mathsf{L}[P_{\boldsymbol{c}}] - \mathsf{L}[Q] = \mathsf{L}[P_{\boldsymbol{c}}] - (\lambda - f^2) = f^2 - A^2$. Let $A = 0$. Then $\mathsf{L}[\psi_{\boldsymbol{c}}] = f^2 \ge 0$ with vanishing 2-jet at x_b, so (2.34) with $x_a = x_b$ gives $\psi_{\boldsymbol{c}}(x) = \int_{x_b}^{x} K_{\mathsf{L}}(x,s)f(s)^2\,\mathrm{d}s$. For $x > x_b$ the integrand is ≥ 0 by Lemma 2.18(c); for $x < x_b$, $\psi_{\boldsymbol{c}}(x) = \int_{x}^{x_b}(-K_{\mathsf{L}}(x,s))f(s)^2\,\mathrm{d}s \ge 0$ since $K_{\mathsf{L}}(x,s) < 0$ for $s > x$. Differentiating, $\partial_x\psi_{\boldsymbol{c}} = \int_{x_b}^{x}\partial_x K_{\mathsf{L}}\,f^2\,\mathrm{d}s$ (as $K_{\mathsf{L}}(x,x) = 0$), and $\partial_x K_{\mathsf{L}} > 0$, so $\partial_x\psi_{\boldsymbol{c}} \ge 0$ for $x \ge x_b$ and ≤ 0 for $x \le x_b$.

Step 5: (iii). Since $Q(\pm 1)$ are finite, so are $\psi_\mp = P_{\boldsymbol{c}}(\mp 1) - Q(\mp 1)$. As $P_{\boldsymbol{c}}(-1) = 2c_1$ and $P_{\boldsymbol{c}}(1) = 2c_2$: $Q(-1) = 2c_1 - \psi_-$, $Q(1) = 2c_2 - \psi_+$, i.e. $c_1' = c_1 - \frac{1}{2}\psi_-$, $c_2' = c_2 - \frac{1}{2}\psi_+$. By (2.35), $\lambda = -2c_1 - 2c_2 - 4c_3 + A^2$, so $c_3' = -\frac{1}{4}\big(-2c_1 - 2c_2 - 4c_3 + A^2 + 2c_1 - \psi_- + 2c_2 - \psi_+\big) = c_3 - \frac{1}{4}A^2 + \frac{1}{4}(\psi_- + \psi_+)$. Then $P_{\boldsymbol{c}} - P_{\boldsymbol{c}'}$ has coefficients $(c_1 - c_1', c_2 - c_2', c_3 - c_3') = (\frac{1}{2}\psi_-, \frac{1}{2}\psi_+, \frac{1}{4}(A^2 - \psi_- - \psi_+))$, which is (2.43); and $Q_{\mathrm{sw}} = Q - P_{\boldsymbol{c}'} = (P_{\boldsymbol{c}} - \psi_{\boldsymbol{c}}) - P_{\boldsymbol{c}'}$. The identity for ε' at $A = 0$ is algebra: with $\psi_i = 2(c_i - c_i') = 2(k_i^2 - k_i'^2)$, $\varepsilon' - \varepsilon = \frac{1}{4}(\psi_- + \psi_+) - (\bar{c}_3(c') - \bar{c}_3(c))$ and $\bar{c}_3 = -\frac{1}{2}(k_1 + k_2)(k_1 + k_2 + 2)$; expanding gives the stated formula (verified by computer algebra, `sympy_derivation.py` Part B). □

Figure 2.1 illustrates the centrifugal part Q_{sw} for an increasing swirl profile.

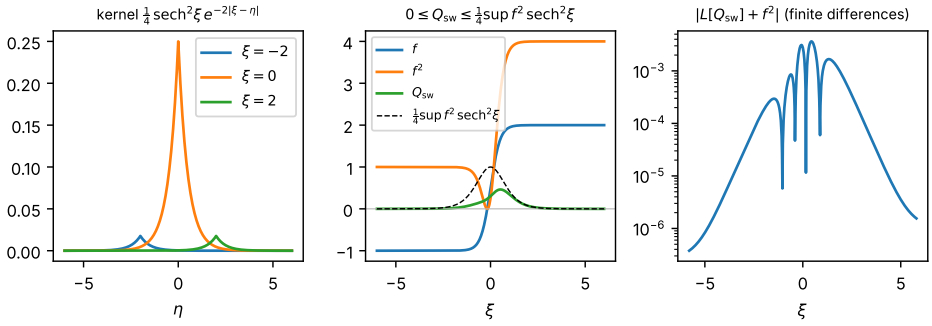


FIGURE 2.1. The centrifugal part. Left: the kernel $\frac{1}{4}\operatorname{sech}^2\xi\, e^{-2|\xi-\eta|}$ of (2.39) in η for $\xi=-2,0,2$ (its height decays like $\operatorname{sech}^2\xi$). Middle: an increasing swirl profile $f(\xi)=\frac{3}{2}\tanh 2\xi+\frac{1}{2}$ (illustration, not a solution), f^2, the resulting Q_{sw} and the bound $\frac{1}{4}\sup f^2\operatorname{sech}^2\xi$. Right: the residual $|\mathsf{L}[Q_{\mathrm{sw}}]+f^2|$ of the quadrature Q_{sw}, evaluated with second-order finite differences (step 0.02, coefficient $\cosh^2\xi$); it is discretisation error, and is not an independent proof of $\mathsf{L}[Q_{\mathrm{sw}}]=-f^2$ (which is proved in the text) (`code/ch02_system/make_qsw.py`).

2.8. Examples: the rest state, the Landau family, trivial swirl, constant potentials

In words. *Everything is explicit at $\boldsymbol{c}=0$. We record the examples that are used again and again: the Landau solutions $y=\lambda_L+x$, the one-ray solutions at the edge of the Landau family, the trivial swirl, and constant potentials, where positive solutions are exponentials and the value of U at a pole is read off from the limit of the potential.*

Example 2.29 (the rest state and the Landau family)**.** *(i) The rest state $\boldsymbol{c}=0$,* $f\equiv A$. Then $P_{\boldsymbol{c}}\equiv 0$, $\psi\equiv 0$, $Q\equiv 0$, $W\equiv 1$ and $z''=z$. The linear equation $y_{xx}=0$ gives $y=a+bx$, and

$$U=\frac{2b(1-x^2)}{a+bx}=\frac{2(1-x^2)}{\lambda_L+x},\qquad \lambda_L:=\frac{a}{b}.$$

By Corollary 2.25 U is global iff $y>0$ on $(-1,1)$, i.e. $a>0$, $a\ge|b|$; this gives $U\equiv 0$ ($b=0$) and the *Landau solutions* [21, 47, 49] $U=2(1-x^2)/(\lambda_L+x)$, $|\lambda_L|\ge 1$. For $|\lambda_L|>1$ these are smooth on the whole sphere; for $\lambda_L=\pm1$ one gets $U=2(1-x)$ and $U=-2(1+x)$, which are singular at one pole (the one-ray solutions of (iii)). In the line variable $z=\sinh\xi+\lambda_L\cosh\xi$ (so that $y=z/\cosh\xi=x+\lambda_L$). Parametrising instead by the value at the equator, $y=1+\frac{\gamma}{2}x$, $\gamma=U(0)$: $U=\gamma(1-x^2)/(1+\gamma x/2)$

and, the solution being global iff $|\gamma| \le 2$, the *Landau window* is $[-2, 2]$. The swirl is $f = A + \kappa F$ with

$$F(x) = \int_0^x \frac{\mathrm{d}s}{y(s)^2} = \frac{2x}{2+\gamma x}, \qquad F(1) = \frac{2}{\gamma+2}, \quad F(-1) = \frac{2}{\gamma-2},$$

by Lemma 2.9 with $y(0) = 1$ and $\kappa = \kappa_x = f_x(0)$ (the $x_b = 0$ normalisation). Thus $F(1) = +\infty$ exactly at $\gamma = -2$ and $F(-1) = -\infty$ exactly at $\gamma = +2$. These closed forms are the starting point of the wedge of Chapter 5.

The Landau profiles are plotted in Figure 2.2.

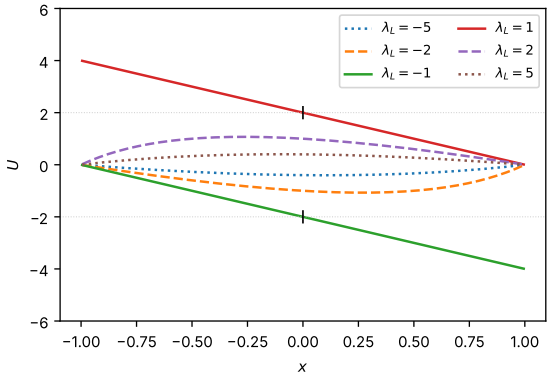


FIGURE 2.2. Landau profiles $U_{\lambda_L}(x) = 2(1-x^2)/(x+\lambda_L)$ for $\lambda_L = -5, -2, -1, 1, 2, 5$ (the reflection σ_2 maps $\lambda_L \mapsto -\lambda_L$, $U(x) \mapsto -U(-x)$). The one-ray edge solutions $\lambda_L = \pm 1$ are the straight lines $U = \pm 2(1 \mp x)$, and the values at the equator, $U(0) = 2/\lambda_L$, fill the Landau window $[-2, 2]$ (dotted lines). Produced by `code/ch02_system/make_landau.py`.

Example 2.30 (trivial swirl)**.** If $\kappa = 0$ then $f \equiv A$, $f_x \equiv 0$, $\psi_{xxx} = 0$, so $\psi \equiv 0$ and $Q = P_{\boldsymbol{c}}$ exactly: A enters only through the constant $\lambda = A^2 - 2c_1 - 2c_2 - 4c_3$ in $\mathsf{L}[Q] + f^2 = \lambda$, which is compensated by the term A^2 in λ itself. Therefore the U-equation is the *no-swirl* equation with potential $W_{\boldsymbol{c}} = 1 + \frac{1}{2}P_{\boldsymbol{c}}$, and the set of global trivial-swirl solutions is $\{\boldsymbol{c} : \ z'' = W_{\boldsymbol{c}} z \text{ has a positive solution}\} \times \mathbb{R}_A$, a product with the free line $A \in \mathbb{R}$. In effective constants, $\boldsymbol{c}'$ differs from $\boldsymbol{c}$ by the constant swirl: $c_i' = c_i$ $(i = 1, 2)$ and $c_3' = c_3 - \frac{1}{4}A^2$ (from (2.42) with $\psi_\mp = 0$) – consistent with $Q_{\mathrm{sw}}[A] = \frac{1}{4}A^2(1 - x^2)$: $P_{\boldsymbol{c}} = P_{\boldsymbol{c}'} + Q_{\mathrm{sw}}$.

Example 2.31 (constant potential)**.** Let f be constant and $c_1 = c_2 = q/2$, $c_3 = 0$, so $Q \equiv q$ and $W \equiv 1 + q/2$. (i) If $q > -2$, put $\mu = \sqrt{1 + q/2} > 0$. Then $z = \alpha e^{\mu\xi} + \beta e^{-\mu\xi}$ is positive on $\mathbb{R}$ iff $\alpha, \beta \ge 0$, not both zero. As $\xi \to +\infty$, $h = z'/z \to \mu$ if $\alpha > 0$, and $h \equiv -\mu$ if $\alpha = 0$; so $U(1) = 2(h(\infty) - 1) \in \{-2 + 2\mu, -2 - 2\mu\}$: the value of U at the pole is read off from the limit of the potential. This is the mechanism of Chapter 4. For $q = 0$, $z = a\cosh\xi + b\sinh\xi$ recovers the rest state $((a + bx)\cosh\xi = a\cosh\xi + b\sinh\xi)$. (ii) If $q = -2$: $z = \alpha + \beta\xi$ is positive on $\mathbb{R}$ only for $\beta = 0$; then $h \equiv 0$ and $U = -2x$. (iii) If $q < -2$: $z = \alpha\cos k\xi + \beta\sin k\xi$

with $k = \sqrt{-1 - q/2}$ vanishes on every interval of length π/k: there is no global solution (compare $q \geq -2$ in the pole theorem of Chapter 4).

Example 2.32 (one-ray solutions)**.** At the boundary $\lambda_L = \pm 1$ of the Landau family, $y = 1 \pm x$ (up to sign): $\lambda_L = 1$ gives $z = y \cosh \xi = e^{\xi}$ and $U = 2(1 - x)$; $\lambda_L = -1$ (take $y = 1 - x$) gives $z = e^{-\xi}$ and $U = -2(1 + x)$. Both are positive on all of $\mathbb{R}$, hence global in the sense of Corollary 2.25. For $W \equiv 1$ ($\mu = 1$) the solution $z = e^{-\xi}$ is *recessive* at $+\infty$ and dominant at $-\infty$: $U(1) = -4 = -2 - 2\mu$ and $U(-1) = 0 = -2 + 2\mu$, the recessive branch at the north pole and the dominant one at the south pole (Example 2.31(i)). As a flow on the sphere the solution is singular at one pole ($u_\theta = -2(1 + x)/\sin\theta \sim -4/\theta$ at $\theta = 0$, whereas for $|\lambda_L| > 1$, $u_\theta = O(\sin\theta)$): these *one-ray* (Slezkin–Landau [21, 42]) solutions are the two edge points of the Landau window $[-2, 2]$ of (i), and the distinction recessive/dominant at a pole is the starting point of the pole theory of Chapter 4.

Remark 2.33 (swirl feeds back into the potential)**.** If $Q \equiv 0$ then $\mathsf{L}[Q] = \lambda - f^2$ gives $f^2 \equiv \lambda$, so f is constant. Hence a solution with genuine swirl never has $Q \equiv 0$: the swirl feeds back into Q. This is the content of Proposition 2.27; for instance if $\boldsymbol{c}' = 0$ then $W = 1 + \frac{1}{2} Q_{\mathrm{sw}}[f] > 1$.

What we have, what comes next

We have the reduced system on the line, $z'' = Wz$, $f' = \kappa z^{-2}$, $\mathsf{L}[Q] = \lambda - f^2$, in which a solution is global iff z does not tend to 0 at a finite point, i.e. iff $z > 0$ on $\mathbb{R}$ for the given potential (Lemma 2.14, Lemma 2.24); the dictionary between the normalisations of κ (Lemma 2.15); and the two descriptions of the potential, $W = W_{\mathrm{e}} + \frac{1}{2} Q_{\mathrm{sw}} \geq W_{\mathrm{e}}$ and $W = W_{\boldsymbol{c}} - \frac{1}{2}\psi_{\boldsymbol{c}}$ (Proposition 2.27). Chapter 3 studies positive solutions of $z'' = Wz$ for a *given* potential: Sturm theory, the Riccati toolbox, Jost solutions and criticality, and applies them to the no-swirl case. The effective description of Proposition 2.27(i) is the basis of the existence theory of Chapter 8; the base-point description (ii) is the basis of the lens of class Z in Chapter 5.

CHAPTER 3

Positive solutions on the line; Jost theory; the no-swirl solutions

In words. *Every statement of this book that is not about the full nonlinear system is a statement about one* linear *equation on the line,*

$$z'' = W(\xi)\, z, \qquad ' = \frac{\mathrm{d}}{\mathrm{d}\xi},$$

because a global solution of the swirl problem is *a positive solution z of such an equation (Lemma 2.14). The potential $W = 1 + \frac{1}{2}Q$ is not known in advance, but it has two properties that make the linear theory powerful: it tends exponentially fast to constants k_1^2, k_2^2 at the two ends, and adding swirl can only* raise *it. This chapter builds the linear toolbox. At each end the solutions split into one* recessive *line (the decaying solution) and the* dominant *rest. The recessive solutions $u_\mp$ at $\mp\infty$ (the* Jost solutions*) decide everything: if u_- stays positive and grows at $+\infty$ the potential is* subcritical *and the positive solutions form the cone $\{au_- + bu_+ : a, b \geq 0\}$; if u_- is recessive at both ends the potential is* critical *and the cone is a single ray; if u_- has a zero the potential is* supercritical *and there is no positive solution at all. Raising the potential moves a supercritical W toward subcriticality (comparison of Jost solutions), and the sets of subcritical and critical potentials behave well under the natural topology (continuity of the Jost solutions in W). The last three sections apply the toolbox to the explicit potentials of the flows without swirl: the Li–Li–Yan classification, closed forms, and the identity (E) that measures how far a swirling solution sits from the edge of the no-swirl window.*

Roadmap. Section 3.1 recalls Sturm's disconjugacy argument ($z'' = Wz$ with $z > 0$ on a half-line excludes $W \leq -k^2 < 0$) and proves the two-line tools used in every later section: the Wronskian identity for two potentials (Lemma 3.4), reduction of order, and reflection. Section 3.2 is the *Riccati toolbox* (Lemma 3.10): comparison, comparison at a pole, tracking of $h = z'/z$ by $\pm\sqrt{W}$, the barrier bound and the cut-off integral, each proved from first principles and each with a toy; it is used in Chapters 4, 6, 10, 11, 12. Section 3.3 proves that recessive and dominant solutions exist when W is integrably close to a constant (a Levinson-type theorem). Section 3.4 defines Jost solutions and admissible potentials, proves the comparison lemma Lemma 3.23, and proves the *weighted-topology* lemmas: continuity of $u_\pm$ and of w_0 in W (also when the limits k_i move), and bounds that are uniform over a class of potentials. Section 3.5 treats the explicit potentials $W_{\boldsymbol{c}} = 1 + \frac{1}{2}P_{\boldsymbol{c}}$ (potential bounds, the critical function g), proves the trichotomy subcritical/critical/supercritical with its openness and closedness properties and the Pöschl–Teller toy (Toy example 3.34), and gives the subcriticality criterion. Section 3.6 constructs a Jost basis that survives $\mu \to 0$. Section 3.7 classifies the

no-swirl solutions for *all* $\boldsymbol{c} \in \mathbb{R}^3$ (Theorem 3.44). This is the second, independent proof of the trichotomy: Section 3.5 proves it for a general potential with $k_i > 0$ by comparison of Jost solutions (what Chapter 8 needs for potentials that are not explicit), whereas Section 3.7 proves it for $W_{\boldsymbol{c}}$ by reduction of order, which also covers the faces $c_i = -1$ ($k_i = 0$) and yields the slopes of the cone. Section 3.8 gives the closed forms by hypergeometric functions (Proposition 3.47), and Section 3.9 proves identity (E) (Proposition 3.51).

Conventions of the chapter. $W : \mathbb{R} \to \mathbb{R}$ is continuous; "solution" means a real C^2 solution of $z'' = Wz$ on an interval; $\operatorname{Wr}(a,b) = ab' - a'b$. For two solutions of the *same* equation $\operatorname{Wr}(a,b)' = ab'' - a''b = 0$, so the Wronskian is constant, and two solutions are linearly independent iff it is nonzero. A nontrivial solution has only simple zeros (if $z(\xi_0) = z'(\xi_0) = 0$ then $z \equiv 0$), hence isolated zeros. For positive a, b: $\operatorname{Wr}(a,b) = ab\,(b'/b - a'/a)$ and $(b/a)' = \operatorname{Wr}(a,b)/a^2$; the sign of the Wronskian is the direction in which the ratio b/a moves.

Jost solutions: one normalisation. For $W \in \mathcal{W}(k_1, k_2)$ (Definition 3.20) the Jost solutions $\varphi_\mp$ are fixed by their asymptotics $e^{\mp k_{1,2}\xi}$, and, when they are positive at 0,

$$u_\pm := \varphi_\pm/\varphi_\pm(0), \qquad \zeta_\pm := u_\pm'/u_\pm, \qquad w_0 := \operatorname{Wr}(u_+, u_-) = \zeta_-(0) - \zeta_+(0).$$

So $u_\pm(0) = 1$, and $w_0 > 0$ for admissible potentials (Lemma 3.21). The log-derivatives $\zeta_\pm$ do not depend on the normalisation. There are exactly two other normalisations in the chapter, and each is announced where it is used: the *principal solutions* $p_\pm$ with $p_\pm/g \to 1$ of Section 3.7 (multiples $2^{\bar{k}}\varphi_\pm$ of the Jost solutions), and the base-point normalisation $u_\pm(\xi_0) = 1$ of Section 3.9.

The explicit potentials. For $\boldsymbol{c} \in \mathbb{R}^3$ put $P_{\boldsymbol{c}}(x) = c_1(1-x) + c_2(1+x) + c_3(1-x^2)$ and $W_{\boldsymbol{c}}(\xi) = 1 + \frac{1}{2}P_{\boldsymbol{c}}(\tanh\xi)$, $x = \tanh\xi$. When $c_1, c_2 \geq -1$ put $k_i = \sqrt{1+c_i} \geq 0$, $\bar{c}_3 = \bar{c}_3(c_1, c_2) = -\frac{1}{2}(k_1 + k_2)(k_1 + k_2 + 2)$, $\varepsilon = c_3 - \bar{c}_3$ and $W_{\mathrm{crit}} = W_{(c_1, c_2, \bar{c}_3)}$. Symbols with two uses are kept apart by the context stated at each place: in Section 3.6 (and in Chapter 4, which uses the same hypotheses) C, S are the cosine-like and sine-like solutions and M is the amplitude constant of the perturbation w (not the critical swirl height M); elsewhere C is a constant, and the square root of the potential is $\varpi = \sqrt{W}$.

3.1. Disconjugacy, Wronskians and reflection

In words. *If $z'' = Wz$ has one positive solution on an interval, no other solution can wiggle there: it has at most one zero. And if W is very negative on a long interval* every *solution wiggles. So a positive solution on a half-line forces W not to be negative near $+\infty$. The same Wronskian computation, applied to solutions of two* different *equations, gives the identity $\operatorname{Wr}(a,b)' = (V_b - V_a)ab$: the difference of potentials, weighted by the product of the solutions. It is the engine of every comparison in this chapter.*

Toy example 3.1 (constant potentials)**.** Let $W \equiv q$. For $q = k^2 > 0$ the solutions are $ae^{k\xi} + be^{-k\xi}$: positive on $\mathbb{R}$ iff $a, b \geq 0$, $(a,b) \neq 0$; $e^{-k\xi}$ is the *recessive* one at $+\infty$ (smaller than every other solution), $e^{k\xi}$ at $-\infty$. For $q = 0$ the solutions are $a + b\xi$, positive on $\mathbb{R}$ iff $b = 0 < a$. For $q = -k^2 < 0$ they are $A\cos(k(\xi - \xi_1))$: every solution has a zero in every interval of length π/k. The next lemma says that the last phenomenon is stable under perturbation of W.

Lemma 3.2 (Sturm; disconjugacy)**.** (a) *If $z'' = Wz$ has a solution $z > 0$ on an interval I, then every nontrivial solution has at most one zero in I.*

(b) *If $W \le -k^2 < 0$ on $[a, a+\pi/k]$, then every nontrivial solution of $z'' = Wz$ has a zero in $[a, a+\pi/k]$.*

(c) *Consequently, if $z'' = Wz$ has a solution $z > 0$ on $[\xi_1, \infty)$, then $W \le -k^2$ on $[\xi_2, \infty)$ is impossible for every $k > 0$ and $\xi_2 \ge \xi_1$; in particular $\limsup_{\xi\to\infty} W \ge 0$. The same holds at $-\infty$ (Lemma 3.7).*

PROOF. (a) Let u be a nontrivial solution with two consecutive zeros $\xi_1 < \xi_2$ in I. Then $u'(\xi_1), u'(\xi_2) \neq 0$ and they have opposite signs (since u has a fixed sign on (ξ_1, ξ_2)). Hence $\mathrm{Wr}(z, u) = zu' - z'u$ takes the values $z(\xi_1)u'(\xi_1)$ and $z(\xi_2)u'(\xi_2)$, of opposite signs, at ξ_1, ξ_2, contradicting the constancy of the Wronskian.

(b) Suppose a nontrivial solution has no zero in $[a, a+\pi/k]$; replacing it by its negative we may assume $z > 0$ there. Put $s = \sin(k(\xi-a)) \ge 0$, $c = \cos(k(\xi-a))$ and $\beta = z's - kzc$. Then $\beta' = z''s + kz'c - kz'c + k^2zs = (W+k^2)zs \le 0$ on $[a, a+\pi/k]$, so β is nonincreasing there. But $\beta(a) = -kz(a) < 0$ and $\beta(a+\pi/k) = kz(a+\pi/k) > 0$, so β has increased: contradiction.

(c) If $z > 0$ on $[\xi_1, \infty)$ and $W \le -k^2$ on $[\xi_2, \infty)$, then z would have no zero in $[\xi_2, \xi_2 + \pi/k]$, contradicting (b). The statement at $-\infty$ follows by $\xi \mapsto -\xi$. □

Toy example 3.3 (the Wronskian of two exponentials)**.** Let $V_a \equiv \mu^2$, $V_b \equiv \nu^2$ be constants and $a = e^{-\mu s}$, $b = e^{-\nu s}$ on $[0, \infty)$. Then $\mathrm{Wr}(a, b) = (\mu - \nu)e^{-(\mu+\nu)s}$ and indeed $\mathrm{Wr}(a, b)' = -(\mu - \nu)(\mu + \nu)e^{-(\mu+\nu)s} = (\nu^2 - \mu^2)ab$. Since $\mathrm{Wr}(a, b) \to 0$ at $+\infty$, integration gives $\mathrm{Wr}(a, b)(0) = \mu - \nu = -\int_0^\infty (\nu^2 - \mu^2)ab\,ds$: *the difference $\nu - \mu$ of the two decay rates is the integral of the difference of the potentials against the product of the two decaying solutions.* The sign rule is visible: if $V_b > V_a$ (here $\nu > \mu$) the solution b decays faster, $b'/b < a'/a$, and $\mathrm{Wr}(a, b) < 0$.

Lemma 3.4 (Wronskian of two potentials)**.** *Let a, b be C^2 functions on an interval I with $a'' = V_a a$ and $b'' = V_b b$, where V_a, V_b are continuous on I. Then $\mathrm{Wr}(a, b)' = (V_b - V_a)\,ab$ on I, and consequently, for $\alpha < \beta$ in I,*

$$\mathrm{Wr}(a, b)(\beta) - \mathrm{Wr}(a, b)(\alpha) = \int_\alpha^\beta (V_b - V_a)\,ab\,d\xi.$$

PROOF. $\mathrm{Wr}(a, b)' = a'b' + ab'' - a''b - a'b' = ab'' - a''b = (V_b - V_a)ab$; integrate. □

Corollary 3.5 (half-line forms)**.** *In the situation of Lemma 3.4, if $I \supset [\alpha, \infty)$, $\mathrm{Wr}(a, b)(\beta) \to 0$ as $\beta \to \infty$ and $(V_b - V_a)ab$ has one sign on $[\alpha, \infty)$, then $\int_\alpha^\infty (V_b - V_a)ab$ converges and $\mathrm{Wr}(a, b)(\alpha) = -\int_\alpha^\infty (V_b - V_a)ab\,d\xi$. Mirror statement: if $I \supset (-\infty, \beta]$ and $\mathrm{Wr}(a, b)(\alpha) \to 0$ as $\alpha \to -\infty$, then $\mathrm{Wr}(a, b)(\beta) = \int_{-\infty}^\beta (V_b - V_a)ab\,d\xi$ (convergent under the same sign condition).*

PROOF. Let $\beta \to \infty$ in the integrated identity; the integrals $\int_\alpha^\beta$ are monotone in β and have the finite limit $-\mathrm{Wr}(a, b)(\alpha)$. The mirror case is the same. □

Corollary 3.6 (reduction of order)**.** *Let $g > 0$ solve $g'' = V_g g$ on I and write a C^2 function as $z = g\varphi$. Then $\mathrm{Wr}(g, z) = g^2\varphi'$, and $z'' = Vz$ holds if and only if*

$$(g^2\varphi')' = (V - V_g)\,g^2\varphi. \tag{3.1}$$

PROOF. $\mathrm{Wr}(g,z) = g(g'\varphi + g\varphi') - g'g\varphi = g^2\varphi'$. As in the proof of Lemma 3.4, $(g^2\varphi')' = \mathrm{Wr}(g,z)' = gz'' - g''z = g\,(z'' - V_g z)$. This equals $(V - V_g)g^2\varphi = (V - V_g)\,gz$ if and only if $z'' - V_g z = (V - V_g)z$, i.e. $z'' = Vz$ (as $g > 0$). □

Lemma 3.7 (reflection)**.** *Let $(Rz)(\xi) = z(-\xi)$. If $z'' = Wz$ then $(Rz)'' = (RW)(Rz)$, and $\mathrm{Wr}(Ra, Rb)(\xi) = -\mathrm{Wr}(a,b)(-\xi)$; for positive z the log-derivative of Rz is $-(z'/z)(-\xi)$. Hence every statement proved for $+\infty$ for all W holds at $-\infty$ with the signs of derivatives reversed.*

PROOF. Chain rule:

$$(Rz)'(\xi) = -z'(-\xi) \quad \text{and} \quad (Rz)''(\xi) = z''(-\xi) = W(-\xi)z(-\xi).$$

The Wronskian and log-derivative statements follow by inserting these formulas. □

Remark 3.8 (where the identity is used)**.** (i) With a, b the Jost solutions of two potentials it gives the comparison formulas of Lemma 3.23 and the order lemma Lemma 3.32(e). (ii) With $a = g$ the critical function (Lemma 3.30) and b a solution of $W_{\boldsymbol{c}}$, (3.1) has right side $\frac{\varepsilon}{2}\operatorname{sech}^2\xi\, g^2\varphi$ of one sign: this is the proof of the classification (Proposition 3.43) and of the subcriticality criterion. (iii) With $a = u_\pm$ Jost solutions of the no-swirl potential and $b = z$ a swirling solution, $V_b - V_a = -\frac{1}{2}\psi \le 0$: this is (E) (Section 3.9). (iv) With $a = g$ and $b = z$ over the whole line it is the balance identity (T0) (Theorem 12.5).

3.2. The Riccati toolbox

In words. *A positive solution z of $z'' = Wz$ is the same as a solution $h = z'/z$ of the first-order* Riccati equation *$h' = W - h^2$, and first-order equations can be compared with explicit ones. Where $W \approx \varpi^2$ is large and slowly varying, the Riccati flow is attracted at rate 2ϖ to $h \approx \varpi$ in forward time (and to $-\varpi$ in backward time), so h "tracks" $\pm\sqrt{W}$ up to an* absolute *error of order one, however large W is. This section proves the six facts of that kind that the later chapters use, once and for all.*

If $z > 0$ solves $z'' = Wz$ on an interval I then $h = z'/z$ is C^1 and $h' = (z''z - z'^2)/z^2 = W - h^2$. Conversely, if $h \in C^1(I)$ solves

$$h' = W - h^2, \tag{3.2}$$

then $z(\xi) = \exp\int_{\xi_0}^{\xi} h$ is positive, $z' = hz$ and $z'' = (h' + h^2)z = Wz$. So *positive solutions of $z'' = Wz$ on I, up to a positive factor, correspond one-to-one to C^1 solutions of (3.2) on I*; a solution h of (3.2) that ceases to exist at a finite point ξ^* (it tends to $\pm\infty$) corresponds to a z with a zero at ξ^*.

Toy example 3.9 (the phase line of $h' = m^2 - h^2$)**.** Let $W \equiv m^2$, $m > 0$. The equilibria are $h = \pm m$; $h' > 0$ for $|h| < m$ and $h' < 0$ for $|h| > m$. Thus m attracts and $-m$ repels in forward time: explicitly $h = m\tanh(m(\xi - \xi_0))$ for $|h(0)| < m$, $h = m\coth(m(\xi - \xi_0))$ for $h(0) > m$ (both tend to m), $h \equiv -m$, and $h = m\coth(m(\xi - \xi_*))$ with $\xi_* > 0$ for $h(0) < -m$, which tends to $-\infty$ at the finite point ξ_* (the corresponding $z = ae^{m\xi} + be^{-m\xi}$ has a zero). In terms of z: $h \to m$ iff the dominant part $ae^{m\xi}$ is present, and $h \equiv -m$ iff $z = be^{-m\xi}$ is recessive. Running the equation *backward* swaps the roles: $-m$ attracts. This asymmetry is why the Jost solution u_+ (recessive at $+\infty$) is computed by integrating ζ_+ from $+\infty$ *backward*. The solution $h = m\coth(m(\xi - a))$, which blows up to $+\infty$ at a (here

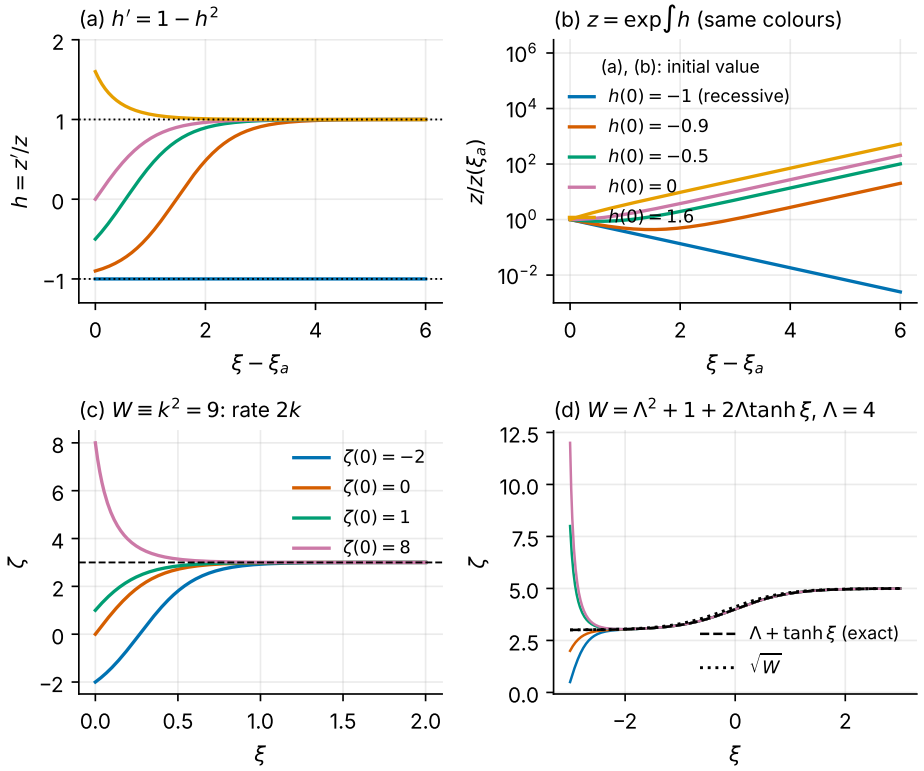


FIGURE 3.1. (a) The phase line of $h' = 1 - h^2$ ($m = 1$): solutions with $h(0) = -1, -0.9, -0.5, 0, 1.6$; only $h(0) = -1$ stays at $-m$ (recessive). (b) The corresponding $z = \exp\int h$ (same colours; $z(\xi_a) = 1$). (c) Tracking for the constant potential $W \equiv k^2 = 9$: all solutions of $\zeta' = W - \zeta^2$ with $\zeta(0) > -k$ converge to k at the rate $2k$. (d) Tracking for $W = \Lambda^2 + 1 + 2\Lambda\tanh\xi$, $\Lambda = 4$: all solutions collapse onto the exact slow solution $\Lambda + \tanh\xi$ (dashed); $\sqrt{W}$ (dotted) is only the first approximation (Lemma 3.10(c),(d)). Code: `code/ch03_line/make_riccati.py`.

$z = \sinh(m(\xi - a))$ has a zero at a), is the extreme case in the *pole-limit* comparison Lemma 3.10(b)(iv): every solution with $W \le m^2$ and bounded above near a lies below it. See Figure 3.1(a),(b).

Lemma 3.10 (Riccati toolbox)**.** (a) (Scalar comparison.) *Let $G(\xi, y)$ be continuous on $[a, b] \times \mathbb{R}$ and Lipschitz in y on bounded sets, uniformly in ξ. Let $y, Y \in C^1[a, b]$. (i) If $y' \ge G(\xi, y)$, $Y' \le G(\xi, Y)$ and $y(a) \ge Y(a)$, then $y \ge Y$ on $[a, b]$. (ii) If $y' \le G(\xi, y)$, $Y' \ge G(\xi, Y)$ and $y(b) \ge Y(b)$, then $y \ge Y$ on $[a, b]$.*

(b) (Comparison for Riccati sub- and supersolutions.) *Let $V \le \widetilde{V}$ be continuous on $[a,b]$ and $h, \tilde{h} \in C^1[a,b]$ with $h' \le V - h^2$ and $\tilde{h}' \ge \widetilde{V} - \tilde{h}^2$. (i) If $h(a) \le \tilde{h}(a)$ then $h \le \tilde{h}$ on $[a,b]$. (ii) If $h(b) \ge \tilde{h}(b)$ then $h \ge \tilde{h}$ on $[a,b]$. (iii)* (tanh barrier.) *Let $W \ge m^2 > 0$ on $[a,b]$ and $h' = W - h^2$ there. If $h(a) \ge -\lambda_0$ with $0 \le \lambda_0 < m$, then $h(\xi) \ge m \tanh\big(m(\xi - a) - \operatorname{artanh}(\lambda_0/m)\big)$ on $[a,b]$; if $h(b) \le \lambda_0$ with $0 \le \lambda_0 < m$, then $h(\xi) \le -m \tanh\big(m(b - \xi) - \operatorname{artanh}(\lambda_0/m)\big)$ on $[a,b]$. (iv)* (Pole limits.) *Let $V \le \widetilde{V}$ be continuous on $(a,b]$ and $h, \tilde{h} \in C^1(a,b]$ with $h' \le V - h^2$, $\tilde{h}' \ge \widetilde{V} - \tilde{h}^2$. If $\tilde{h}(\xi) \to +\infty$ as $\xi \downarrow a$ and h is bounded above near a, then $h \le \tilde{h}$ on $(a,b]$. Mirror: if $V \le \widetilde{V}$ are continuous on $[a,b)$, $h, \tilde{h} \in C^1[a,b)$ satisfy the same inequalities, $\tilde{h}(\xi) \to -\infty$ as $\xi \uparrow b$ and h is bounded below near b, then $h \ge \tilde{h}$ on $[a,b)$.*

(c) (First-order tracking.) *Let $\varpi \in C^1[a_0,b]$, $\varpi \ge m > 0$, $|\varpi'| \le C_{\mathrm{bd}}$, and let ζ solve $\zeta' = \varpi^2 - \zeta^2$ with $\zeta(a_0) > 0$. Then ζ exists on all of $[a_0,b]$, $\zeta > 0$ and*

$$|\zeta(\xi) - \varpi(\xi)| \le \frac{C_{\mathrm{bd}}}{m} + |\zeta(a_0) - \varpi(a_0)|\, e^{-m(\xi - a_0)} \qquad (\xi \in [a_0,b]).$$

*Moreover (*decaying solution*): let $I = [\xi_1, \infty)$, $W \in C^1(I)$ bounded with $W > 0$, $\varpi = \sqrt{W}$, $\varpi_* = \inf_I \varpi > 0$, $\mu = \sup_I |\varpi'|/\varpi < \varpi_*$. Then $u'' = Wu$ has a positive solution u on I, decaying exponentially at $+\infty$, whose logarithmic derivative $\zeta = u'/u$ satisfies $|\zeta + \varpi| \le \mu$ on I. Mirror statement on $(-\infty, \xi_1]$: there is a positive solution, decaying at $-\infty$, with $|\zeta - \varpi| \le \mu$.*

(d) (Second-order tracking.) *In addition to the hypotheses of the first part of (c) (with $\varpi \in C^2$) assume that the* lag *$a := -\varpi'/(2\varpi)$ and $F := a' + a^2$ satisfy $|a| \le m/2$, $|F| \le F_0$ on $[a_0,b]$. Then $r := \zeta - \varpi - a$ satisfies*

$$|r(\xi)| \le \frac{2F_0}{m} + |r(a_0)|\, e^{-m(\xi - a_0)/2}.$$

Symmetrically, if $\tilde{\zeta}$ solves the same Riccati equation on $[a_0,b]$ with $\tilde{\zeta}(b) < 0$ and $\tilde{\zeta} < 0$ on $[a_0,b]$ [automatic, see the proof], then $\tilde{r} := \tilde{\zeta} + \varpi - a$ satisfies $|\tilde{r}(\xi)| \le 2F_0/m + |\tilde{r}(b)| e^{-m(b-\xi)/2}$.

(e) (Barrier bound.) *Let W be continuous, $\xi \in \mathbb{R}$, $\ell, m > 0$, $W \ge m^2$ on $[\xi - \ell, \xi + \ell]$. If $u > 0$ solves $u'' = Wu$ on $[\xi, \infty)$, is bounded on $[\xi + \ell, \infty)$, and $W \ge 0$ on $[\xi + \ell, \infty)$, then $u'/u \le -m \tanh(m\ell)$ at ξ. If $u > 0$ solves $u'' = Wu$ on $(-\infty, \xi]$, is bounded on $(-\infty, \xi - \ell]$ and $W \ge 0$ there, then $u'/u \ge m \tanh(m\ell)$ at ξ. Consequently, for a pair of such solutions $\rho := \zeta_- - \zeta_+ \ge 2m \tanh(m\ell)$.*

(f) (Constants ≥ 0; the cut-off integral.) *(f1) Let $\boldsymbol{c} \in [0,\infty)^3$, $W_0 = W_{\boldsymbol{c}} = 1 + \Pi_1 + \Pi_2 + \Pi_3$ with $\Pi_1 = \frac{1}{2}c_1(1-x)$, $\Pi_2 = \frac{1}{2}c_2(1+x)$, $\Pi_3 = \frac{1}{2}c_3(1-x^2)$, $x = \tanh\xi$. Then (i) $W_0 \ge 1 + \min(c_1, c_2) \ge 1$; (ii) $|W_0'| \le 2(W_0 - 1)$, hence $\mu := \sup |(\sqrt{W_0})'|/\sqrt{W_0} \le \sup(W_0 - 1)/W_0 < 1$; (iii) the solutions u_+ decaying at $+\infty$ and u_- decaying at $-\infty$ exist, are positive on $\mathbb{R}$ and unique up to factors, and $|\zeta_+(\xi) + \sqrt{W_0(\xi)}| < 1$, $|\zeta_-(\xi) - \sqrt{W_0(\xi)}| < 1$ for all ξ, so $2\sqrt{W_0(\xi)} - 2 < \zeta_- - \zeta_+ < 2\sqrt{W_0(\xi)} + 2$; (iv) $W_0(\xi') \ge e^{-2|\xi' - \xi|} W_0(\xi)$. (f2) For $\alpha, \beta > 0$ and $0 < \varkappa < 1$ put $\mathcal{C}(\varkappa) = \int_0^\infty \min(\varkappa^2 e^{\beta t}, 1)\, e^{-\alpha t}\, dt$. As*

$\varkappa \to 0$: $\mathcal{C} = \frac{\varkappa^2}{\alpha-\beta} + O(\varkappa^{2\alpha/\beta})$ *if* $\beta < \alpha$; $\mathcal{C} = \frac{2}{\beta}\varkappa^2 \ln\frac{1}{\varkappa} + \frac{\varkappa^2}{\alpha}$ *if* $\beta = \alpha$; $\mathcal{C} = \frac{\beta}{\alpha(\beta-\alpha)}\varkappa^{2\alpha/\beta} - \frac{\varkappa^2}{\beta-\alpha}$ *if* $\beta > \alpha$.

PROOF. *(a)(i).* Put $d = y - Y$, so $d(a) \geq 0$. Let L be a Lipschitz constant of G in y on the bounded set $[a,b] \times [-R,R]$, $R = \max(\|y\|_\infty, \|Y\|_\infty)$. Then $d' \geq G(\xi,y) - G(\xi,Y) \geq -L|d|$. Suppose $d(\xi_1) < 0$ and let $\xi_0 = \sup\{\xi < \xi_1 : d(\xi) \geq 0\}$; then $d(\xi_0) = 0$, $d < 0$ on $(\xi_0,\xi_1]$, hence $d' \geq Ld$ there, so $(e^{-L\xi}d)' \geq 0$ and $e^{-L\xi_1}d(\xi_1) \geq e^{-L\xi_0}d(\xi_0) = 0$: contradiction. (ii) Put $\hat{y}(s) = Y(a+b-s)$, $\hat{Y}(s) = y(a+b-s)$, $\hat{G}(s,v) = -G(a+b-s,v)$. Then $\hat{y}'(s) = -Y'(a+b-s) \leq \hat{G}(s,\hat{y})$, $\hat{Y}'(s) = -y'(a+b-s) \geq \hat{G}(s,\hat{Y})$ and $\hat{Y}(a) = y(b) \geq Y(b) = \hat{y}(a)$; by (i) (roles of the two functions exchanged) $\hat{Y} \geq \hat{y}$, i.e. $y \geq Y$.

(b)(i),(ii). Put $G(\xi,y) = \widetilde{V}(\xi) - y^2$; it is locally Lipschitz in y. Then $\tilde{h}' \geq G(\xi,\tilde{h})$ and $h' \leq V - h^2 \leq G(\xi,h)$. Part (a)(i) with $(y,Y) = (\tilde{h},h)$ gives (i). For (ii) note that $h' \leq G(\xi,h)$ and $\tilde{h}' \geq G(\xi,\tilde{h})$, and use (a)(ii) with $(y,Y) = (h,\tilde{h})$. *(iii).* Let $c = \operatorname{artanh}(\lambda_0/m)$ and $Y(\xi) = m\tanh(m(\xi-a)-c)$, so $Y' = m^2 - Y^2$, $Y(a) = -\lambda_0 \leq h(a)$. Apply (i) with $V = m^2 \leq \widetilde{V} = W$, $(h,\tilde{h}) \to (Y,h)$. For the second bound let $\hat{h}(s) = -h(-s)$ on $[-b,-a]$: it solves $\hat{h}' = \hat{W} - \hat{h}^2$ with $\hat{W}(s) = W(-s) \geq m^2$ and $\hat{h}(-b) = -h(b) \geq -\lambda_0$ (Lemma 3.7); the first bound for $\hat{h}$ at $s = -\xi$ is the claim. *(iv).* Choose $\delta_0 > 0$ so small that h is bounded above on $(a, a+\delta_0]$. For $\delta \in (0,\delta_0)$ small enough $\tilde{h}(a+\delta) \geq h(a+\delta)$ because $\tilde{h}(a+\delta) \to +\infty$; apply (i) on $[a+\delta,b]$ (all functions are C^1 and $V, \widetilde{V}$ continuous there): $h \leq \tilde{h}$ on $[a+\delta,b]$. Let $\delta \downarrow 0$. The mirror statement: $h(b-\delta) \geq \tilde{h}(b-\delta)$ for small δ because $\tilde{h}(b-\delta) \to -\infty$ while h is bounded below; apply (ii) on $[a,b-\delta]$. (Only the *upper* bound of h at the pole enters; no information about h at the pole other than boundedness is used, and $\tilde{h}$ may be any supersolution that blows up, e.g. $m\coth(m(\xi-a))$ for $\widetilde{V} \equiv m^2$.)

(c), first part. If $0 < \zeta < m$ then $\zeta' \geq m^2 - \zeta^2 > 0$, and $\zeta' = \varpi^2 > 0$ at $\zeta = 0$; hence ζ stays positive (it cannot return to 0) and $\zeta \geq \min(\zeta(a_0), m)$. Since $\zeta' \leq \varpi^2$ is bounded on $[a_0,b]$ while $\zeta > 0$, the solution cannot blow up and exists on all of $[a_0,b]$: $\zeta' \leq \varpi^2 - \zeta^2$ is negative once $\zeta > \max\varpi$, so $\zeta \leq \max(\zeta(a_0), \max\varpi)$. Put $e = \zeta - \varpi$. Then $e' = \varpi^2 - (\varpi+e)^2 - \varpi' = -e(2\varpi+e) - \varpi'$. We may assume $C_{\mathrm{bd}} > 0$ (otherwise replace C_{bd} by $C'_{\mathrm{bd}} > 0$ and let $C'_{\mathrm{bd}} \to 0$). Let $E(\xi) = \frac{C_{\mathrm{bd}}}{m} + (|e(a_0)| - \frac{C_{\mathrm{bd}}}{m})_+ e^{-m(\xi-a_0)}$; then $E > 0$, $E' = C_{\mathrm{bd}} - mE$ and $E(a_0) \geq |e(a_0)|$. *Upper bound* $e \leq E$. If not, there is ξ_1 with $e(\xi_1) > E(\xi_1)$; let $\xi_0 = \sup\{\xi < \xi_1 : e \leq E\}$. On $(\xi_0,\xi_1]$, $e > E > 0$, so $2\varpi + e \geq 2m$ and $e' \leq -2me + C_{\mathrm{bd}} \leq C_{\mathrm{bd}} - me$. Hence $(e-E)' \leq -m(e-E) < 0$ there, and $(e-E)(\xi_0) = 0$ gives $(e-E)(\xi_1) < 0$: contradiction. *Lower bound* $e \geq -E$. If not, take ξ_1 with $-e(\xi_1) > E(\xi_1)$ and $\xi_0 = \sup\{\xi < \xi_1 : -e \leq E\}$. On $(\xi_0,\xi_1]$, $e < -E < 0$ and $\zeta > 0$ gives $|e| < \varpi$, so $(-e)' = e(2\varpi+e) + \varpi' = -|e|(2\varpi - |e|) + \varpi' \leq -m|e| + C_{\mathrm{bd}}$ (because $2\varpi - |e| \geq \varpi \geq m$). Thus $(-e-E)' \leq -m(-e-E)$, and the same contradiction results.

(c), decaying solution. Put $\delta = h + \varpi$ for a solution h of (3.2). Then $\delta' = h' + \varpi' = \varpi^2 - (\delta-\varpi)^2 + \varpi' = 2\varpi\delta - \delta^2 + \varpi'$ (checked by computer algebra). Fix $\mu_1 \in (\mu, \varpi_*)$. On the line $\delta = \mu_1$ we get $\delta' \geq \mu_1(2\varpi - \mu_1) - \mu\varpi > \mu_1\varpi - \mu\varpi > 0$ (as $\mu_1 < \varpi$); on $\delta = -\mu_1$, $\delta' \leq -\mu_1(2\varpi + \mu_1) + \mu\varpi < 0$. So the strip $|\delta| \leq \mu_1$ is positively invariant in *backward* time: if $|\delta(\xi^{(n)})| \leq \mu_1$, then $|\delta| \leq \mu_1$ on $[\xi_1, \xi^{(n)}]$ (at a first exit point, going backward, δ' would have the wrong sign). Let h_n be the

solution with $h_n(\xi^{(n)}) = -\varpi(\xi^{(n)})$ (that is, $\delta(\xi^{(n)}) = 0$), for a sequence $\xi^{(n)} \to \infty$. Then $|h_n + \varpi| \le \mu_1$ on $[\xi_1, \xi^{(n)}]$. The h_n are uniformly bounded, and $h_n' = W - h_n^2$ is uniformly bounded because W is bounded; by Arzelà–Ascoli (and a diagonal argument) a subsequence converges locally uniformly to a solution h_∞ of (3.2) on I with $|h_\infty + \varpi| \le \mu_1$. Then $u = \exp\int h_\infty$ is positive with $h_\infty \le -(\varpi_* - \mu_1) < 0$, so it decays exponentially. Since h_n does not depend on μ_1 and $\mu_1 \in (\mu, \varpi_*)$ was arbitrary, in fact $|h_n + \varpi| \le \mu$ on $[\xi_1, \xi^{(n)}]$, and so $|h_\infty + \varpi| \le \mu$ and $h_\infty \le -(\varpi_* - \mu)$. Thus $\zeta = h_\infty$ has the stated properties. The mirror statement follows from $\xi \mapsto -\xi$ (Lemma 3.7).

(d). Since $\varpi' = -2\varpi a$, with $\zeta = \varpi + a + r$,

$$r' = \zeta' - \varpi' - a' = \varpi^2 - (\varpi + a + r)^2 + 2\varpi a - a' \\ = -(a^2 + a') - r(2\varpi + 2a + r) = -F - r(\zeta + \varpi + a)$$

(checked by computer algebra). By (c) $\zeta > 0$, and $\varpi + a \ge m - m/2$, so $\kappa := \zeta + \varpi + a \ge m/2$. Let $E_2 = \frac{2F_0}{m} + (|r(a_0)| - \frac{2F_0}{m})_+ e^{-m(\xi - a_0)/2}$ (take $F_0 > 0$ w.l.o.g.), so $E_2' = F_0 - \frac{m}{2}E_2$. If $r > E_2$ at some point, take the last point ξ_0 before it with $r \le E_2$; on the interval between, $r' \le F_0 - \kappa r \le F_0 - \frac{m}{2}r$, so $(r - E_2)' \le -\frac{m}{2}(r - E_2) < 0$: contradiction as in (c). The lower bound $r \ge -E_2$ is identical: where $s := -r > 0$ we have $s' = F - \kappa s \le F_0 - \frac{m}{2}s$. For the symmetric statement set $\hat\zeta(\eta) = -\tilde\zeta(-\eta)$ on $[-b, -a_0]$. It solves $\hat\zeta' = \hat\varpi^2 - \hat\zeta^2$ with $\hat\varpi(\eta) = \varpi(-\eta)$, $\hat\zeta(-b) = -\tilde\zeta(b) > 0$, and $\hat\zeta > 0$ (by (c)); its lag is $\hat a(\eta) = -a(-\eta)$, so $|\hat a| \le m/2$ and $\hat F(\eta) = F(-\eta)$; moreover $\hat r(\eta) = \hat\zeta - \hat\varpi - \hat a = -\tilde r(-\eta)$. The first statement at the left end $-b$ gives the claim.

(e). Since $u'' = Wu \ge 0$ on $[\xi + \ell, \infty)$, u is convex there; if $u'(s) > 0$ at some $s \ge \xi + \ell$ then $u' \ge u'(s) > 0$ afterwards and u is unbounded. So $u' \le 0$ there, in particular $\zeta(\xi + \ell) \le 0$ for $\zeta = u'/u$. On $[\xi, \xi + \ell]$, $\zeta' = W - \zeta^2 \ge m^2 - \zeta^2 =: G(\cdot, \zeta)$, while $Y_m(s) = m\tanh(m(s - \xi - \ell))$ satisfies $Y_m' = G(s, Y_m)$ and $Y_m(\xi + \ell) = 0 \ge \zeta(\xi + \ell)$. By (a)(ii) $\zeta \le Y_m$ on $[\xi, \xi + \ell]$, and $Y_m(\xi) = -m\tanh(m\ell)$. The left statement is the mirror image.

(f1). (i) $\Pi_1 + \Pi_2 \ge \frac{1}{2}\min(c_1, c_2)[(1 - x) + (1 + x)]$ and $\Pi_3 \ge 0$. (ii) With $' = (1 - x^2)\partial_x$ we get $W_0' = \frac{1}{2}(c_2 - c_1)(1 - x^2) - c_3 x(1 - x^2)$. Now $(1 + x)\Pi_1 = \frac{1}{2}c_1(1 - x^2)$, $(1 - x)\Pi_2 = \frac{1}{2}c_2(1 - x^2)$ and $|c_3 x(1 - x^2)| = 2|x|\Pi_3$, so $|W_0'| \le (1+x)\Pi_1 + (1-x)\Pi_2 + 2|x|\Pi_3 \le 2(\Pi_1 + \Pi_2 + \Pi_3) = 2(W_0 - 1)$. Since $|(\sqrt{W_0})'|/\sqrt{W_0} = |W_0'|/(2W_0) \le (W_0 - 1)/W_0 = 1 - 1/W_0$ and W_0 is bounded, $\mu < 1$. (iii) Fix ξ and apply (c) on $[\xi, \infty)$ (here $\varpi_* \ge 1 > \mu$, W_0 bounded and C^1): there is a positive exponentially decaying solution with $|\zeta + \sqrt{W_0}| \le \mu < 1$ on $[\xi, \infty)$. Any two solutions decaying at $+\infty$ are proportional (Lemma 3.16(a) of Section 3.3, which does not use the toolbox; applicable since $W_0 \to 1 + c_2 > 0$ exponentially), so this solution is u_+ and $u_+ > 0$ on $[\xi, \infty)$; as ξ was arbitrary, $u_+ > 0$ on $\mathbb{R}$ and the bound holds everywhere. The same at $-\infty$. (iv) $|(\ln W_0)'| = |W_0'|/W_0 \le 2$.

(f2). Put $t_c = 2\ln(1/\varkappa)/\beta$, so that $\varkappa^2 e^{\beta t} \le 1$ iff $t \le t_c$, and note $\varkappa^2 e^{\beta t_c} = 1$, $e^{-\alpha t_c} = \varkappa^{2\alpha/\beta}$. Then $\mathcal{C} = \varkappa^2 \int_0^{t_c} e^{(\beta-\alpha)t}dt + \int_{t_c}^\infty e^{-\alpha t}dt$. For $\beta \ne \alpha$ the first integral is $(e^{(\beta-\alpha)t_c} - 1)/(\beta - \alpha)$ and $\varkappa^2 e^{(\beta-\alpha)t_c} = e^{-\alpha t_c}$, so $\mathcal{C} = \frac{\varkappa^{2\alpha/\beta} - \varkappa^2}{\beta - \alpha} + \frac{\varkappa^{2\alpha/\beta}}{\alpha}$; for $\beta < \alpha$ this equals $\frac{\varkappa^2}{\alpha - \beta} + \varkappa^{2\alpha/\beta}\big(\frac{1}{\alpha} - \frac{1}{\alpha - \beta}\big)$, for $\beta > \alpha$ it equals $\frac{\beta}{\alpha(\beta - \alpha)}\varkappa^{2\alpha/\beta} - \frac{\varkappa^2}{\beta - \alpha}$. For $\beta = \alpha$ it is $\varkappa^2 t_c + \varkappa^2/\alpha$. (Numerical check: `code/ch03_line/verify_ch03a.py`.) □

Toy example 3.11 (the cut-off integral)**.** The integrand $\min(\varkappa^2 e^{\beta t}, 1)e^{-\alpha t}$ grows like $\varkappa^2 e^{(\beta-\alpha)t}$ until the cap is reached at $t_c = 2\ln(1/\varkappa)/\beta$ and decays like $e^{-\alpha t}$ afterwards; which of the two pieces dominates depends on the sign of $\beta - \alpha$. For $\varkappa = \frac{1}{10}$: $(\alpha, \beta) = (2, 1)$ $(\beta < \alpha)$: $\mathcal{C} = \varkappa^2 - \frac{1}{2}\varkappa^4 = 0.00995$, essentially the uncapped value $\varkappa^2/(\alpha - \beta) = 0.01$; $(1, 1)$ $(\beta = \alpha)$: $\mathcal{C} = 2\varkappa^2 \ln 10 + \varkappa^2 = 0.05605$, larger than $\varkappa^2$ by the logarithmic factor; $(1, 2)$ $(\beta > \alpha)$: $\mathcal{C} = 2\varkappa - \varkappa^2 = 0.19$, of the order $\varkappa^{2\alpha/\beta} = \varkappa$, i.e. determined by the tail beyond the cap. (Values from direct quadrature agree to 10^{-12}: `verify_ch03a.py`.)

Remark 3.12 (lags cancel in $\zeta_- - \zeta_+$)**.** For the two Jost solutions of a large slowly varying potential, (c) and (d) give $\zeta_- = \varpi + a + r_-$ and $\zeta_+ = -\varpi + a + r_+$, so $\rho = \zeta_- - \zeta_+ = 2\varpi + r_- - r_+$: the lag a cancels, which is why $w_0 = \rho(0) = 2\sqrt{W(0)} + O(1/J)$ rather than $+O(1)$ in the large-swirl regime (Chapter 10).

Toy example 3.13 (a slowly varying potential with an exact slow solution)**.** Choose $\zeta_{\mathrm{ex}} = \Lambda + \tanh\xi$ and $W := \zeta_{\mathrm{ex}}' + \zeta_{\mathrm{ex}}^2 = \Lambda^2 + 1 + 2\Lambda\tanh\xi$ (since $\operatorname{sech}^2 + \tanh^2 = 1$). Every other solution is $\zeta_{\mathrm{ex}} + e$ with $e' = -e(2\zeta_{\mathrm{ex}} + e)$, so $e \sim e^{-2\Lambda\xi}$. Here $\varpi = \sqrt{W} = \Lambda + \tanh\xi + \frac{\operatorname{sech}^2\xi}{2\Lambda} + \dots$ and the lag is $a = -\frac{\operatorname{sech}^2\xi}{2\Lambda} + \dots$, so $\varpi + a = \Lambda + \tanh\xi + O(\Lambda^{-2})$: the lag is exactly the correction that turns $\sqrt{W}$ into the true slow solution (Figure 3.1(d)).

Lemma 3.14 (Riccati dichotomy)**.** *Let $m \geq 0$, $w \in C[\xi_0, \infty)$ with $w(\xi) \to 0$, and let $h \in C^1[\xi_0, \infty)$ be real-valued (finite on the whole half-line) with $h' = m^2 + w - h^2$. Then $h \to m$ or $h \to -m$ as $\xi \to \infty$.*

(The phase-line toy Toy example 3.9 is the model: for $w \equiv 0$ the only solution that does not tend to $+m$ is $h \equiv -m$.)

Proof. Fix $\varepsilon \in (0, m^2/4)$ (any $\varepsilon > 0$ if $m = 0$) and $\xi_\varepsilon \geq \xi_0$ with $|w| \leq \varepsilon$ on $[\xi_\varepsilon, \infty)$; put $\overline{m} = \sqrt{m^2 + \varepsilon}$ and, if $m > 0$, $\underline{m} = \sqrt{m^2 - \varepsilon}$. On $[\xi_\varepsilon, \infty)$,

$$\underline{m}^2 - h^2 \leq h' \leq \overline{m}^2 - h^2.$$

(i) $h \geq -\overline{m}$ on $[\xi_\varepsilon, \infty)$. Suppose $h(\xi_1) < -\overline{m}$, $\xi_1 \geq \xi_\varepsilon$. The solution $Y(\xi) = \overline{m}\coth(\overline{m}(\xi - \xi_*))$ of $Y' = \overline{m}^2 - Y^2$ with $\xi_* = \xi_1 + \frac{1}{\overline{m}}\operatorname{arcoth}(-h(\xi_1)/\overline{m}) > \xi_1$ satisfies $Y(\xi_1) = h(\xi_1)$ and $Y \to -\infty$ as $\xi \uparrow \xi_*$. Now $h' \leq \overline{m}^2 - h^2 = G(\xi, h)$ with $G(\xi, y) = \overline{m}^2 - y^2$, while $Y' = G(\xi, Y)$ and $Y(\xi_1) = h(\xi_1)$; by Lemma 3.10(a)(i) (with (y, Y) there replaced by (Y, h)) $h \leq Y$ on $[\xi_1, \xi_* - \delta]$ for each small $\delta > 0$. Hence $h(\xi) \to -\infty$ as $\xi \uparrow \xi_*$, contradicting the continuity of h at ξ_*. *(ii)* $\limsup h \leq \overline{m}$. By (i), $h(\xi_\varepsilon) \geq -\overline{m}$. If $h(\xi_\varepsilon) > -\overline{m}$ let Y solve $Y' = \overline{m}^2 - Y^2$, $Y(\xi_\varepsilon) = h(\xi_\varepsilon)$; then $Y \geq h$ by (a)(i), and $Y \to \overline{m}$ (Toy example 3.9). If $h(\xi_\varepsilon) = -\overline{m}$ take $Y \equiv -\overline{m}$; then (a)(i) gives $h \leq -\overline{m}$ on $[\xi_\varepsilon, \infty)$, which together with (i) says $h \equiv -\overline{m}$ there. (The repelling equilibrium $-\overline{m}$ enters only as an upper barrier: comparison needs Lipschitz uniqueness, not stability, and in this case it shows that h *stays* at $-\overline{m}$; the bound $\limsup h \leq \overline{m}$ is then trivial.) Hence $\limsup h \leq \overline{m}$ in all cases. Since $\overline{m} \to m$ as $\varepsilon \to 0$: $-m \leq \liminf h \leq \limsup h \leq m$. If $m = 0$ we are done. *(iii)* $m > 0$. Alternative (A_ε): $h(\xi_2) > -\underline{m}$ for some $\xi_2 \geq \xi_\varepsilon$. Then Lemma 3.10(b)(iii) on $[\xi_2, \xi_3]$ for every $\xi_3 > \xi_2$ (with $W = m^2 + w \geq \underline{m}^2$, $\lambda_0 = \max(0, -h(\xi_2)) < \underline{m}$) gives $h(\xi) \geq \underline{m}\tanh(\underline{m}(\xi - \xi_2) - c) \to \underline{m}$, so $\liminf h \geq \underline{m}$. Alternative (B_ε): $h \leq -\underline{m}$ on $[\xi_\varepsilon, \infty)$, so $\limsup h \leq -\underline{m}$. (A_ε) and $(\mathrm{B}_{\varepsilon'})$ cannot both hold (one gives $\liminf h > 0$, the other $\limsup h < 0$). If $(\mathrm{A}_{\varepsilon_0})$ holds, then (B_ε) fails and hence

(A_ε) holds for all $\varepsilon < \varepsilon_0$, so $\liminf h \ge \sqrt{m^2-\varepsilon} \to m$ and $h \to m$. Otherwise (B_ε) holds for all small ε and $\limsup h \le -\sqrt{m^2-\varepsilon} \to -m$, so $h \to -m$. □

3.3. Recessive and dominant solutions

In words. *If $W = \mu^2 + w$ with w small at infinity, the solutions of $z'' = Wz$ look like those of $z'' = \mu^2 z$: a dominant one $\sim e^{\mu\xi}$ and a recessive one $\sim e^{-\mu\xi}$, and the recessive one is unique up to a factor. The proof is a contraction argument in the form of a Volterra integral equation. At $\mu = 0$ the two solutions are ~ 1 and $\sim \xi$, which needs a little more decay of w.*

Toy example 3.15 (a Levinson example with Bessel functions)**.** For $z'' = (1+e^{-2\xi})z$ (so $\mu = 1$, $w = e^{-2\xi}$) put $t = e^{-\xi}$: the equation becomes the modified Bessel equation $t^2Z'' + tZ' - (1+t^2)Z = 0$ of order 1, so $z_- = 2I_1(e^{-\xi}) \sim e^{-\xi}$ and $z_+ = K_1(e^{-\xi}) \sim e^{\xi}$ as $\xi \to \infty$ (checked numerically, `verify_ch03a.py`). The next lemma says that the same picture holds for every w with $\int |w| < \infty$.

Lemma 3.16 (Asymptotic integration (Levinson type))**.** *Let w be continuous on $[\xi_0, \infty)$.*

(a) *Let $\mu > 0$ and $\int_{\xi_0}^\infty |w| < \infty$. Then $z'' = (\mu^2 + w)z$ has a solution $z_- = e^{-\mu\xi}\zeta$, where ζ is the unique bounded solution of the Volterra equation*

$$\zeta(\xi) = 1 + \int_\xi^\infty k(\xi,\eta)\, w(\eta)\, \zeta(\eta)\, d\eta, \qquad k(\xi,\eta) = \frac{1 - e^{-2\mu(\eta-\xi)}}{2\mu} \in \Big[0, \frac{1}{2\mu}\Big], \quad (3.3)$$

and $|\zeta| \le e^{\beta(\xi)}$, $|\zeta - 1| \le e^{\beta(\xi)} - 1$, where $\beta(\xi) := \frac{1}{2\mu}\int_\xi^\infty |w|$, and

$$\zeta'(\xi) = -\int_\xi^\infty e^{-2\mu(\eta-\xi)} w\zeta\, d\eta \to 0.$$

There is a second solution z_+ with $\mathrm{Wr}(z_-, z_+) = 2\mu$. They satisfy

$$e^{\mu\xi} z_- \to 1, \quad e^{\mu\xi} z_-' \to -\mu, \quad e^{-\mu\xi} z_+ \to 1, \quad e^{-\mu\xi} z_+' \to \mu \qquad (\xi \to \infty).$$

Every solution is $z = Az_+ + Bz_-$ with unique A, B. If $A \ne 0$ then $e^{-\mu\xi} z \to A$, $z \ne 0$ for large ξ and $z'/z \to \mu$; if $A = 0 \ne B$ then $z'/z \to -\mu$. In particular the solutions that tend to 0 at $+\infty$ (equivalently, with $z'/z \to -\mu$) are exactly the nonzero multiples of z_-.

(b) *Let $\mu = 0$ and $\int_{\xi_0}^\infty \xi^2 |w|\, d\xi < \infty$. Then $z'' = wz$ has solutions z_1, z_2 with $z_1 \to 1$, $z_1' \to 0$, $\int^\infty |z_1 - 1| < \infty$ and $z_2 - \xi \to c_2 \in \mathbb{R}$, $z_2' \to 1$. Consequently every solution satisfies $z = a + b\xi + o(1)$, $z' \to b$, and $z = az_1$ if $b = 0$.*

The mirror statements at $-\infty$ follow by applying the lemma to $\xi \mapsto W(-\xi)$ (Lemma 3.7).

Proof. (a) *The recessive solution.* Put $z = e^{-\mu\xi}\zeta$; then $z'' = (\mu^2 + w)z$ iff $\zeta'' - 2\mu\zeta' = w\zeta$. For continuous g with $\int_{\xi_0}^\infty |g| < \infty$ put $(\mathcal{I}g)(\xi) = \int_\xi^\infty k(\xi,\eta) g(\eta) d\eta$. Since $k(\xi,\xi) = 0$ and $\partial_\xi k = -e^{-2\mu(\eta-\xi)}$, $(\mathcal{I}g)'(\xi) = -\int_\xi^\infty e^{-2\mu(\eta-\xi)} g\, d\eta$ and $(\mathcal{I}g)'' = g + 2\mu(\mathcal{I}g)'$, i.e. $(\mathcal{I}g)'' - 2\mu(\mathcal{I}g)' = g$. So every bounded solution of (3.3) solves $\zeta'' - 2\mu\zeta' = w\zeta$. (3.3) is a Volterra equation $\zeta = 1 + V\zeta$; Picard iteration $\zeta_0 = 1$, $\zeta_{n+1} = 1 + V\zeta_n$ gives, by induction and $\beta' = -|w|/(2\mu)$, $|\zeta_{n+1} - \zeta_n|(\xi) \le$

$\beta(\xi)^{n+1}/(n+1)!$ (indeed $\int_\xi^\infty \frac{|w|}{2\mu}\frac{\beta^n}{n!}d\eta = \frac{\beta(\xi)^{n+1}}{(n+1)!}$). Hence $\zeta_n \to \zeta$ uniformly on $[\xi_0,\infty)$, ζ solves (3.3), $|\zeta| \le e^\beta$ and $|\zeta - 1| \le e^\beta - 1$. Uniqueness among bounded solutions: the difference Δ of two satisfies $|\Delta(\xi)| \le \int_\xi^\infty \frac{|w|}{2\mu}|\Delta|$, so $\Delta = 0$ by the Gronwall inequality (Remark 3.17). From the formula for $(\mathcal{I}g)'$ (with $g = w\zeta$): $|\zeta'| \le e^{\beta(\xi_0)}\int_\xi^\infty |w| \to 0$. Thus $e^{\mu\xi}z_- = \zeta \to 1$ and $e^{\mu\xi}z_-' = \zeta' - \mu\zeta \to -\mu$. *The dominant solution.* Choose ξ_2 with $\zeta \ge \frac12$ on $[\xi_2,\infty)$ and put $z_+ := 2\mu z_-\int_{\xi_2}^\xi z_-^{-2}$ (reduction of order), extended as a solution to $[\xi_0,\infty)$. With $v = \int^\xi z_-^{-2}$ one has $(z_-v)'' = z_-''v + 2z_-'v' + z_-v'' = (\mu^2+w)z_-v + 2z_-'z_-^{-2} - 2z_-'z_-^{-2}$, so it is a solution, and $\mathrm{Wr}(z_-,z_+) = 2\mu\,[z_-(z_-'v + z_-^{-1}) - z_-'z_-v] = 2\mu$. Since $z_-^{-2} = e^{2\mu\eta}\zeta^{-2}$ with $\zeta^{-2}\to 1$, $\int_{\xi_2}^\xi z_-^{-2} = \frac{e^{2\mu\xi}}{2\mu}(1+o(1))$ (split the integral where $|\zeta^{-2}-1| < \delta$), so $e^{-\mu\xi}z_+ = 2\mu\zeta e^{-2\mu\xi}\int_{\xi_2}^\xi z_-^{-2} \to 1$, and $e^{-\mu\xi}z_+' = \frac{z_-'}{z_-}e^{-\mu\xi}z_+ + \frac{2\mu}{\zeta} \to -\mu + 2\mu = \mu$. *Behaviour of z'/z.* Since $\mathrm{Wr}(z_-,z_+)\neq 0$, $z_\pm$ form a basis. If $z = Az_+ + Bz_-$ with $A\neq 0$: $e^{-\mu\xi}z \to A$, $e^{-\mu\xi}z' \to \mu A$, so $z'/z \to \mu$. If $A = 0 \neq B$, $z'/z = z_-'/z_- \to -\mu$.

(b) Choose $\xi_1 \ge \max(\xi_0,1)$ with $\int_{\xi_1}^\infty \eta|w| \le \frac12$ and put $(V_0\zeta)(\xi) = \int_\xi^\infty(\eta-\xi)w\zeta\,d\eta$. For $(\mathcal{I}_0 g)(\xi) = \int_\xi^\infty(\eta-\xi)g\,d\eta$ we have $(\mathcal{I}_0 g)' = -\int_\xi^\infty g$, $(\mathcal{I}_0 g)'' = g$, and $0 \le \eta-\xi \le \eta$ gives $\|V_0\| \le \frac12$ on bounded functions; so $z_1 = 1 + V_0 z_1$ has a unique bounded solution, $|z_1| \le 2$, it solves $z_1'' = wz_1$, and $|z_1(\xi)-1| \le 2\int_\xi^\infty \eta|w| \le \frac{2}{\xi}\int_\xi^\infty \eta^2|w|$, $|z_1'(\xi)| \le 2\int_\xi^\infty |w| \le \frac{2}{\xi}\int_\xi^\infty \eta|w|$; thus $z_1\to 1$, $z_1'\to 0$, and $\int_{\xi_1}^\infty |z_1-1| \le 2\int \eta(\eta-\xi_1)|w| < \infty$ by Tonelli. Take ξ_2 with $z_1 \in [\frac12, 2]$ on $[\xi_2,\infty)$ and $z_2 = z_1\int_{\xi_2}^\xi z_1^{-2}$ (a solution with $\mathrm{Wr}(z_1,z_2) = 1$). As $|z_1^{-2}-1| = |1-z_1||1+z_1|z_1^{-2} \le 12|z_1-1|$ is integrable, $\int_{\xi_2}^\xi z_1^{-2} = \xi + c + o(1)$, so $z_2 = z_1(\xi+c+o(1)) = \xi + c + o(1) + (z_1-1)(\xi+c+o(1)) = \xi + c + o(1)$ (since $\xi(z_1-1)\to 0$) and $z_2' = z_1'\int z_1^{-2} + z_1^{-1} \to 0+1$. Every solution is $\alpha z_1 + \beta z_2 = (\alpha+\beta c) + \beta\xi + o(1)$ with derivative $\to \beta$. □

Remark 3.17 (Gronwall on a half-line)**.** Let $b \ge 0$ be integrable on $[\xi_0,\infty)$ and $\varphi \ge 0$ bounded and continuous with $\varphi(\xi) \le a + \int_\xi^\infty b\varphi$. Then $\varphi(\xi) \le a\exp\int_\xi^\infty b$. Indeed, put $\Psi(\xi) = a + \int_\xi^\infty b\varphi$ and $b_\infty(\xi) = \int_\xi^\infty b$. Then $\Psi' = -b\varphi \ge -b\Psi$, so $(\Psi e^{-b_\infty})' = (\Psi' + b\Psi)e^{-b_\infty} \ge 0$; and $\Psi e^{-b_\infty} \to a$ at $+\infty$ because φ is bounded and b integrable. A nondecreasing function with limit a is $\le a$, so $\varphi \le \Psi \le ae^{b_\infty}$. The mirror statement: if $\varphi(\xi) \le a + \int_{-\infty}^\xi b\varphi$ on $(-\infty, L]$ then $\varphi(\xi) \le a\exp\int_{-\infty}^\xi b$.

Lemma 3.18 (Ordering of solutions at a fixed potential)**.** *Let W be continuous on $[\xi_1,\infty)$ and let z_1, z_2 solve $z'' = Wz$ with $z_1(\xi_1) = z_2(\xi_1) > 0$ and $z_1'(\xi_1) < z_2'(\xi_1)$.*

(a) *If $z_1 > 0$ on an interval $I \ni \xi_1$, $I \subset [\xi_1,\infty)$, then $z_2 > z_1$ and $h_2 > h_1$ on $I\setminus\{\xi_1\}$ resp. on I, where $h_i = z_i'/z_i$; hence $U_2 > U_1$ for $U_i = 2(h_i - \tanh\xi)$.*

(b) *Let $W = \mu^2 + w$ with $\mu > 0$, $\int_{\xi_1}^\infty |w| < \infty$ and let $z_r > 0$ on $[\xi_1,\infty)$ be recessive ($z_r'/z_r \to -\mu$). Then every solution with $z(\xi_1) = z_r(\xi_1)$ and $z'(\xi_1) > z_r'(\xi_1)$ is positive on $[\xi_1,\infty)$ and dominant ($z'/z \to \mu$), and every solution with $z(\xi_1) = z_r(\xi_1)$ and $z'(\xi_1) < z_r'(\xi_1)$ has a zero in (ξ_1,∞).*

Proof. (a) $d_{12} := \mathrm{Wr}(z_1,z_2) = z_1(\xi_1)(z_2'(\xi_1) - z_1'(\xi_1)) > 0$ is constant. On I, $(z_2/z_1)' = d_{12}/z_1^2 > 0$, so z_2/z_1 increases strictly from 1; thus $z_2 > z_1 > 0$ on $I\setminus\{\xi_1\}$, and $h_2 - h_1 = d_{12}/(z_1z_2) > 0$. (b) By Lemma 3.16(a), $z_r = Cz_-$ with $C > 0$ and $z = Az_+ + Bz_-$. Then $\mathrm{Wr}(z_r,z) = CA\ \mathrm{Wr}(z_-,z_+) = 2\mu CA$, and also

$\mathrm{Wr}(z_r, z) = z_r(\xi_1)(z'(\xi_1) - z_r'(\xi_1))$. If $z'(\xi_1) > z_r'(\xi_1)$, then $A > 0$, so $e^{-\mu\xi}z \to A > 0$: z is positive and dominant for large ξ, and positive everywhere on $[\xi_1, \infty)$ by (a) (with $z_1 = z_r$, $z_2 = z$). If $z'(\xi_1) < z_r'(\xi_1)$ then $A < 0$ and $e^{-\mu\xi}z \to A < 0$, so z becomes negative, hence vanishes. □

Toy example 3.19 (the lemma for $W \equiv \mu^2$)**.** With $z(\xi_1) = 1$ the solutions are $z_\beta = \cosh(\mu(\xi - \xi_1)) + \frac{\beta}{\mu}\sinh(\mu(\xi - \xi_1))$, $\beta = z'(\xi_1)$: positive and dominant for $\beta > -\mu$, equal to the recessive $e^{-\mu(\xi-\xi_1)}$ for $\beta = -\mu$, with a zero for $\beta < -\mu$. For $\mu = 0$: $z_\beta = 1 + \beta(\xi - \xi_1)$, positive for $\beta \geq 0$.

In the coupled problem $W = 1 + \frac{1}{2}Q$ depends on the solution itself, so Lemma 3.18 does not by itself order solutions of the nonlinear system; order properties of the coupled system are treated in Chapter 5.

3.4. Jost solutions, comparison, and the weighted topology

In words. *The potentials of the swirl problem tend to constants k_1^2, k_2^2 at the two ends* exponentially fast in ξ, *i.e. like $e^{-2|\xi|}$ (that is the size of the pole weight $E = e^{-2|\xi|}$). For such W there is a solution φ_- that decays like $e^{k_1\xi}$ at $-\infty$ and a solution φ_+ that decays like $e^{-k_2\xi}$ at $+\infty$: the* Jost solutions. *When φ_- stays positive and grows at $+\infty$ the potential is called* admissible, *and then both Jost solutions are positive, grow at the opposite end, and their Wronskian $w_0 > 0$ measures how far the potential is from the critical case. Three questions are answered in this section: how do the Jost solutions react to raising the potential (monotonically: Lemma 3.23); how do they depend on W (continuously, in a topology that only sees W near the middle, with uniform control of the tails, and also when the limits k_i move slightly: Lemma 3.25); and what bounds hold uniformly over a class of potentials (Lemma 3.28).*

Definition 3.20 (the exponential class; admissible potentials)**.** Let $k_1, k_2 > 0$ and $C \geq 0$. $\mathcal{W}(k_1, k_2; C)$ is the set of continuous $W : \mathbb{R} \to \mathbb{R}$ with

$$|W(\xi) - k_1^2| \leq Ce^{2\xi} \quad (\xi \leq 0), \qquad |W(\xi) - k_2^2| \leq Ce^{-2\xi} \quad (\xi \geq 0),$$

and $\mathcal{W}(k_1, k_2) = \bigcup_C \mathcal{W}(k_1, k_2; C)$. (By continuity this is the same as "$W - k_1^2 = O(e^{2\xi})$ at $-\infty$, $W - k_2^2 = O(e^{-2\xi})$ at $+\infty$".) For $W \in \mathcal{W}(k_1, k_2)$ let $\varphi_- = \varphi_-^W$ be the solution with $e^{-k_1\xi}\varphi_- \to 1$ at $-\infty$ and $\varphi_+ = \varphi_+^W$ the solution with $e^{k_2\xi}\varphi_+ \to 1$ at $+\infty$ (they exist and are unique by Lemma 3.16(a) and its mirror, because $\int_{-\infty}^0 |W - k_1^2| \leq C/2$ and $\int_0^\infty |W - k_2^2| \leq C/2$). W is *admissible* if $\varphi_-^W > 0$ on $\mathbb{R}$ and φ_-^W is unbounded at $+\infty$.

Lemma 3.21 (Jost solutions)**.** *Let $W \in \mathcal{W}(k_1, k_2; C)$.*

(a) *$\varphi_- = e^{k_1\xi}v$, where v is the unique bounded solution on $(-\infty, L]$ (every L) of $v(\xi) = 1 + \int_{-\infty}^{\xi} K_1(\xi, \eta)\,(W - k_1^2)(\eta)\,v(\eta)\,d\eta$, $K_1(\xi, \eta) = \frac{1 - e^{-2k_1(\xi-\eta)}}{2k_1} \in [0, \frac{1}{2k_1}]$; $|v| \leq e^\beta$, $|v - 1| \leq e^\beta - 1$ with $\beta(\xi) = \frac{1}{2k_1}\int_{-\infty}^{\xi} |W - k_1^2|$, and $v'(\xi) = \int_{-\infty}^{\xi} e^{-2k_1(\xi-\eta)}(W - k_1^2)v\,d\eta \to 0$; so $\varphi_-'/\varphi_- \to k_1$ at $-\infty$. Every solution y that is not a multiple of φ_- satisfies $e^{k_1\xi}y(\xi) \to \ell \neq 0$ as $\xi \to -\infty$; in particular φ_- is, up to a factor, the only solution tending to 0 at $-\infty$. Mirror statements hold for φ_+ at $+\infty$, with $\varphi_+'/\varphi_+ \to -k_2$ and $e^{-k_2\xi}y \to \ell \neq 0$.*

(b) *Let W be admissible. Then (i) $e^{-k_2\xi}\varphi_- \to a > 0$ at $+\infty$; (ii) $\varphi_+ > 0$ on $\mathbb{R}$, $e^{k_1\xi}\varphi_+ \to b > 0$ at $-\infty$, and φ_-, φ_+ are linearly independent; so $u_\pm := \varphi_\pm/\varphi_\pm(0) > 0$ are well defined and each Jost solution is dominant at the opposite end; (iii) $w_0 := \mathrm{Wr}(u_+, u_-) = \zeta_-(0) - \zeta_+(0) > 0$ and*

$$u_+(\xi) = w_0\, u_-(\xi) \int_\xi^\infty u_-^{-2}, \qquad u_-(\xi) = w_0\, u_+(\xi) \int_{-\infty}^\xi u_+^{-2};$$

(iv) $\rho := \zeta_- - \zeta_+ = \dfrac{w_0}{u_- u_+} = \left(u_-^2 \int_\xi^\infty u_-^{-2}\right)^{-1} = \left(u_+^2 \int_{-\infty}^\xi u_+^{-2}\right)^{-1}$ is continuous and positive, $\rho \to 2k_1$ at $-\infty$, $\rho \to 2k_2$ at $+\infty$; moreover $\zeta_- \to k_1$, $\zeta_+ \to -k_1$ at $-\infty$, $\zeta_- \to k_2$, $\zeta_+ \to -k_2$ at $+\infty$, and $\sup_{\mathbb{R}} 1/\rho < \infty$.

Proof. (a) This is Lemma 3.16(a) applied to $\xi \mapsto W(-\xi)$ with $\mu = k_1$ (the Volterra equation, the bounds and the formula for v' are the mirror images of (3.3) and of the formulas that follow it); a solution not proportional to φ_- has a nonzero dominant component A, and $e^{k_1\xi}y \to A$. (b) (i) φ_- is not a multiple of φ_+ (else it tends to 0 at $+\infty$, hence is bounded), so by (a) $e^{-k_2\xi}\varphi_- \to a \neq 0$; $a > 0$ since $\varphi_- > 0$. (ii) Since $\varphi_-^{-2} \sim a^{-2}e^{-2k_2\xi}$ is integrable at $+\infty$, $J_+(\xi) := \varphi_-(\xi)\int_\xi^\infty \varphi_-^{-2}$ is defined and positive on $\mathbb{R}$. Put $I = \int_\xi^\infty \varphi_-^{-2}$, so $I' = -\varphi_-^{-2}$ and $J_+' = \varphi_-' I - \varphi_-^{-1}$, $J_+'' = \varphi_-'' I - \varphi_-'\varphi_-^{-2} + \varphi_-'\varphi_-^{-2} = WJ_+$. Thus J_+ is a positive solution, and $J_+ \sim \varphi_- I \sim \frac{e^{-k_2\xi}}{2k_2 a} \to 0$. By (a) $J_+ = \lambda\varphi_+$ with $\lambda > 0$ (positive since both are positive near $+\infty$); hence $\varphi_+ > 0$ on $\mathbb{R}$. Further $\mathrm{Wr}(\varphi_-, J_+) = \varphi_-(\varphi_-' I - \varphi_-^{-1}) - \varphi_-'\varphi_- I = -1 \neq 0$, so φ_-, φ_+ are independent. By (a) at $-\infty$, $e^{k_1\xi}\varphi_+ \to b \neq 0$, and $b > 0$. (iii) Put $J_-(\xi) = u_+(\xi)\int_{-\infty}^\xi u_+^{-2}$ and $J_+^u(\xi) = u_-(\xi)\int_\xi^\infty u_-^{-2}$; the integrals converge because $u_+^{-2} = O(e^{2k_1\xi})$ at $-\infty$ and $u_-^{-2} = O(e^{-2k_2\xi})$ at $+\infty$ by (a), (i), (ii). The computation of (ii) shows that J_- and J_+^u are positive solutions, $J_- \to 0$ at $-\infty$ and $J_+^u \to 0$ at $+\infty$, with $\mathrm{Wr}(u_+, J_-) = 1$ and $\mathrm{Wr}(u_-, J_+^u) = -1$. By (a), $J_- = \lambda_- u_-$ and $J_+^u = \lambda_+ u_+$ with $\lambda_\pm > 0$. Now $\mathrm{Wr}(u_+, u_-) =: w_0$ (this is $u_+u_-' - u_+'u_-$, equal to $\zeta_-(0) - \zeta_+(0)$ at $\xi = 0$ because $u_\pm(0) = 1$), and $1 = \mathrm{Wr}(u_+, \lambda_- u_-) = \lambda_- w_0$, $-1 = \mathrm{Wr}(u_-, \lambda_+ u_+) = -\lambda_+ w_0$. Since $\lambda_\pm > 0$ we get $w_0 > 0$ and $\lambda_\pm = 1/w_0$, i.e. both formulas. (iv) $\zeta_- - \zeta_+ = \frac{u_-'u_+ - u_+'u_-}{u_-u_+} = \frac{w_0}{u_-u_+}$ and inserting (iii) gives the other two expressions. At $+\infty$ $u_- \sim a'e^{k_2\xi}$, so $u_-^2\int_\xi^\infty u_-^{-2} \to \frac{1}{2k_2}$; at $-\infty$ $u_+ \sim b'e^{-k_1\xi}$, so $u_+^2\int_{-\infty}^\xi u_+^{-2} \to \frac{1}{2k_1}$. The limits of ζ_- at $-\infty$ and of ζ_+ at $+\infty$ are in (a); the remaining two follow from $\zeta_+ = \zeta_- - \rho$ (and, at $+\infty$, from u_- being dominant: $\zeta_- \to k_2$). A continuous positive function with positive limits at $\pm\infty$ has a positive infimum. □

Example 3.22 (constant potential)**.** For $W \equiv k^2$ ($k_1 = k_2 = k$): the Volterra integrals vanish, $\varphi_\pm = e^{\mp k\xi}$, W is admissible, $u_\pm = e^{\mp k\xi}$, $\zeta_\pm = \mp k$, $w_0 = \zeta_- - \zeta_+ = 2k$, $\rho \equiv 2k$, and indeed $w_0 u_- \int_\xi^\infty u_-^{-2} = 2ke^{k\xi}\frac{e^{-2k\xi}}{2k} = e^{-k\xi} = u_+$. For $W \equiv -1$ nothing is admissible: every solution oscillates (Lemma 3.2). An example with a nontrivial potential, $W = k^2 - 2\,\mathrm{sech}^2\,\xi$, is worked out in Toy 3.34 (and, for the topology, in Toy 3.27).

Lemma 3.23 (Comparison of Jost solutions)**.** *Let $W \geq W_{\mathrm{ref}}$ be continuous with the same limits $k_1^2, k_2^2 > 0$ at $\mp\infty$ (at the rates of Definition 3.20) and let W_{ref} be admissible. Then W is admissible; before normalisation $u_-^W/u_-^{W_{\mathrm{ref}}}$ is nondecreasing*

and $u_+^W/u_+^{W_{\mathrm{ref}}}$ *is nonincreasing;* $\zeta_-^W \ge \zeta_-^{W_{\mathrm{ref}}}$, $\zeta_+^W \le \zeta_+^{W_{\mathrm{ref}}}$, $w_0(W) \ge w_0(W_{\mathrm{ref}})$*; and, with* $u_\pm(0) = 1$, $u_-^W \ge u_-^{W_{\mathrm{ref}}}$ *on* $[0,\infty)$ *and* $u_+^W \ge u_+^{W_{\mathrm{ref}}}$ *on* $(-\infty, 0]$*. (Second part.) For admissible* W_1, W_2 *with the same limits at these rates,* $u_i = u_-^{W_i}$, $v_i = u_+^{W_i}$*:* $\zeta_-^{W_1}(\eta) - \zeta_-^{W_2}(\eta) = \int_{-\infty}^{\eta} (W_1 - W_2)(r)\, \frac{u_1(r)u_2(r)}{u_1(\eta)u_2(\eta)}\, dr$ *and* $\zeta_+^{W_1}(\eta) - \zeta_+^{W_2}(\eta) = -\int_{\eta}^{\infty} (W_1 - W_2)(r)\, \frac{v_1(r)v_2(r)}{v_1(\eta)v_2(\eta)}\, dr$*.*

("Before normalisation" means for $\varphi_\pm$ instead of $u_\pm$.)

PROOF. We use the two-potential Wronskian identity Lemma 3.4, $\mathrm{Wr}(a,b)' = (V_b - V_a)ab$, and its half-line form Corollary 3.5.

Left end. Let $u = \varphi_-^W$, $v = \varphi_-^{W_{\mathrm{ref}}}$. Then $\mathrm{Wr}(v,u)' = (W - W_{\mathrm{ref}})uv$. As $\xi \to -\infty$, $u, v = O(e^{k_1\xi})$ and $u'/u, v'/v \to k_1$, so $\mathrm{Wr}(v,u) = uv\,(u'/u - v'/v) \to 0$; $uv = O(e^{2k_1\xi})$ and $W - W_{\mathrm{ref}}$ is bounded, so the integral below converges. Now $v > 0$ on $\mathbb{R}$ because W_{ref} is admissible. Let $\xi^* = \sup\{\xi : u > 0 \text{ on } (-\infty, \xi)\} \in (-\infty, \infty]$. On $(-\infty, \xi^*)$, where $u, v > 0$ and $W \ge W_{\mathrm{ref}}$,

$$\mathrm{Wr}(v,u)(\xi) = \int_{-\infty}^{\xi} (W - W_{\mathrm{ref}})uv \ \ge 0, \qquad \Big(\frac{u}{v}\Big)' = \frac{\mathrm{Wr}(v,u)}{v^2} \ge 0.$$

So u/v is nondecreasing on $(-\infty, \xi^*)$ with limit 1 at $-\infty$; hence $u \ge v$ there. If $\xi^* < \infty$, continuity would give $u(\xi^*) \ge v(\xi^*) > 0$, contradicting $u(\xi^*) = 0$. Thus $\xi^* = \infty$, $u > 0$ on $\mathbb{R}$, $u \ge v$ is unbounded at $+\infty$ (as v is), and since $W \in \mathcal{W}(k_1, k_2)$, W is admissible. Also u/v is nondecreasing on $\mathbb{R}$ and $\zeta_-^W - \zeta_-^{\mathrm{ref}} = u'/u - v'/v = \mathrm{Wr}(v,u)/(uv) \ge 0$. *Right end.* Now $u = \varphi_+^W$, $v = \varphi_+^{\mathrm{ref}}$ are positive (Lemma 3.21(b)(ii), both potentials being admissible), $\mathrm{Wr}(v,u)' = (W - W_{\mathrm{ref}})uv \ge 0$ and $\mathrm{Wr}(v,u) \to 0$ at $+\infty$ (as above, $uv = O(e^{-2k_2\xi})$, $u'/u, v'/v \to -k_2$); hence $\mathrm{Wr}(v,u)(\xi) = -\int_{\xi}^{\infty} (W - W_{\mathrm{ref}})uv \le 0$, $(u/v)' \le 0$ and $\zeta_+^W - \zeta_+^{\mathrm{ref}} = \mathrm{Wr}(v,u)/(uv) \le 0$. *Consequences.* $w_0(W) = \zeta_-^W(0) - \zeta_+^W(0) \ge \zeta_-^{\mathrm{ref}}(0) - \zeta_+^{\mathrm{ref}}(0) = w_0(W_{\mathrm{ref}})$. Normalising divides φ's by their values at 0: for $\xi \ge 0$ the monotonicity of $\varphi_-^W/\varphi_-^{\mathrm{ref}}$ gives $\varphi_-^W(\xi)/\varphi_-^{\mathrm{ref}}(\xi) \ge \varphi_-^W(0)/\varphi_-^{\mathrm{ref}}(0)$, i.e. $u_-^W(\xi) \ge u_-^{\mathrm{ref}}(\xi)$; similarly $u_+^W \ge u_+^{\mathrm{ref}}$ on $(-\infty, 0]$ from the nonincrease of the quotient.

Second part. With $u_i = u_-^{W_i}$: $\mathrm{Wr}(u_2, u_1)' = (W_1 - W_2)u_1u_2$, $\mathrm{Wr}(u_2, u_1) \to 0$ at $-\infty$ and the right side is integrable, so $\mathrm{Wr}(u_2, u_1)(\eta) = \int_{-\infty}^{\eta} (W_1 - W_2)u_1u_2$, while $\zeta_-^{W_1} - \zeta_-^{W_2} = \mathrm{Wr}(u_2, u_1)/(u_1u_2)$; the quotient does not depend on the normalisation of u_i. The formula for ζ_+ is the mirror image, with $\mathrm{Wr}(v_2, v_1)(\eta) = -\int_{\eta}^{\infty} (W_1 - W_2)v_1v_2$. □

Remark 3.24 (Gronwall in the stability lemma)**.** Both parts of the next lemma use Remark 3.17 (and its mirror image on $(-\infty, L]$).

Lemma 3.25 (Continuity of Jost solutions; the weighted topology)**.** *Fix* $0 < \underline{k} \le \overline{k}$ *and* $C \ge 0$*, and let* $L > 0$*. Write* $\mathcal{W}_{\mathrm{loc}} = \bigcup\{\mathcal{W}(k_1, k_2; C) : k_1, k_2 \in [\underline{k}, \overline{k}]\}$ *(the limits are allowed to vary in a compact subset of* $(0,\infty)$*) and, for* $W \in \mathcal{W}(k_1, k_2; C)$*,* $\widetilde{W} \in \mathcal{W}(\tilde{k}_1, \tilde{k}_2; C)$*,*

$$d_w(W, \widetilde{W}) = \sup_{\xi} \cosh^2 \xi\, |W - \widetilde{W}| \qquad (\textit{the} \text{ weighted distance}).$$

(a) *(*Lipschitz estimate.*) There is $K = K(\underline{k}, \overline{k}, C, L)$ such that for $W, \widetilde{W} \in \mathcal{W}_{\mathrm{loc}}$, with $v = e^{-k_1\xi}\varphi_-^W$, $\tilde{v} = e^{-\tilde{k}_1\xi}\varphi_-^{\widetilde{W}}$,*

$$\sup_{\xi\le L}\big(|v-\tilde{v}| + |v'-\tilde{v}'|\big) \le K\Big(\int_{-\infty}^{L} \big|(W-k_1^2) - (\widetilde{W}-\tilde{k}_1^2)\big|\,d\eta + |k_1-\tilde{k}_1|\Big),$$

and the integral on the right is at most $\int_{-T}^{L}|W-\widetilde{W}| + Ce^{-2T} + (L+T)\,|k_1^2-\tilde{k}_1^2|$ for every $T \ge 0$. The same holds for $v_+ = e^{k_2\xi}\varphi_+^W$ and $\tilde{v}_+$ on $[-L,\infty)$, with $\int_{-L}^{\infty}|(W-k_2^2)-(\widetilde{W}-\tilde{k}_2^2)|$ and $|k_2-\tilde{k}_2|$.

(b) *(*Weighted topology.*) If $W \in \mathcal{W}(k_1,k_2;C)$ and $d_w(W,\widetilde{W}) \le \delta$, then $\widetilde{W} \in \mathcal{W}(k_1,k_2;C+4\delta)$ and $\int_{\mathbb{R}}|W-\widetilde{W}| \le 2\delta$. Let $W_n \in \mathcal{W}(k_{1,n},k_{2,n};C)$ with $k_{i,n} \to k_i > 0$, and let W be continuous with $W_n \to W$ in L^1_{loc} (in particular if $W_n \to W$ locally uniformly, or in d_w). Then $W \in \mathcal{W}(k_1,k_2;C)$ and $\varphi_\pm^{W_n} \to \varphi_\pm^W$ in $C^1_{\mathrm{loc}}(\mathbb{R})$. If moreover all W_n, W are admissible, then $u_\pm^{W_n} \to u_\pm^W$ and $\zeta_\pm^{W_n} \to \zeta_\pm^W$ locally uniformly, and $w_0(W_n) \to w_0(W)$.*

(c) *(*Positive pairs.*) Let $\mathcal{P} = \{W \in \mathcal{W}(k_1,k_2),\ k_1,k_2>0 :\ \varphi_-^W > 0 \text{ on } \mathbb{R}\}$. For $W \in \mathcal{P}$ exactly one of the following holds: W is admissible, or $\varphi_-^W = c\,\varphi_+^W$ with $c>0$ (then W is called* critical*, $\zeta_- \equiv \zeta_+$); in both cases $\varphi_\pm^W(0) > 0$, and $w_0(W) := \zeta_-(0) - \zeta_+(0) \ge 0$ with equality iff W is critical. If $W_n \in \mathcal{P}$ are as in (b) and $W_n \to W$ in L^1_{loc}, then $W \in \mathcal{P}$ and $w_0(W_n) \to w_0(W)$.*

Proof. (a) Let $p = W-k_1^2$, $\tilde{p} = \widetilde{W}-\tilde{k}_1^2$, $k=k_1$, $\tilde{k}=\tilde{k}_1$, and $K_k(r) = (1-e^{-2kr})/(2k) \in [0, \frac{1}{2\underline{k}}]$. On $[0,L]$, $|p| \le |W-k_2^2| + |k_2^2-k_1^2| \le C + \overline{k}^2$, and $|p| \le Ce^{2\xi}$ on $(-\infty,0]$; so $P := \sup_{\xi\le L}\int_{-\infty}^{\xi}|p| \le \frac{C}{2} + L(C+\overline{k}^2)$, and the same for $\tilde{p}$. By Lemma 3.21(a), $|v|, |\tilde{v}| \le e^{P/(2\underline{k})} =: e^{\bar{\beta}}$ on $(-\infty, L]$. Both functions solve $v(\xi) = 1 + \int_{-\infty}^{\xi} K_k(\xi-\eta)p(\eta)v(\eta)\,d\eta$ (with k,p resp. $\tilde{k},\tilde{p}$). Subtracting, $\Delta = v - \tilde{v}$ satisfies

$$\Delta(\xi) = \int_{-\infty}^{\xi} K_k p\,\Delta\,d\eta + \int_{-\infty}^{\xi} K_k\,(p-\tilde{p})\,\tilde{v}\,d\eta + \int_{-\infty}^{\xi}(K_k - K_{\tilde{k}})\,\tilde{p}\,\tilde{v}\,d\eta.$$

Now $\partial_k K_k(r) = re^{-2kr}/k - K_k(r)/k$ gives $|K_k - K_{\tilde{k}}| \le |k-\tilde{k}|/\underline{k}^2$ (use $re^{-2kr} \le 1/(2ek)$). Hence $|\Delta(\xi)| \le A_0 + \int_{-\infty}^{\xi}\frac{|p|}{2\underline{k}}|\Delta|$ with $A_0 = \frac{e^{\bar{\beta}}}{2\underline{k}}\int_{-\infty}^{L}|p-\tilde{p}| + \frac{e^{\bar{\beta}}P}{\underline{k}^2}|k-\tilde{k}|$. Δ is bounded and $|p|/2\underline{k}$ is integrable, so Gronwall (Remark 3.17, mirror) gives $\sup_{\xi\le L}|\Delta| \le A_0 e^{\bar{\beta}}$. Differentiating (the kernel vanishes on the diagonal), $v'(\xi) = \int_{-\infty}^{\xi} e^{-2k(\xi-\eta)}p\,v\,d\eta$, and in the same way

$$|\Delta'| \le \sup|\Delta|\,P + e^{\bar{\beta}}\int_{-\infty}^{L}|p-\tilde{p}| + \frac{e^{\bar{\beta}}P}{e\,\underline{k}}|k-\tilde{k}|$$

(use $|e^{-2kr}-e^{-2\tilde{k}r}| \le 2re^{-2\underline{k}r}|k-\tilde{k}| \le |k-\tilde{k}|/(e\underline{k})$). Together this is the first claim, with K depending only on $(\underline{k},\overline{k},C,L)$. For the second claim: for $\eta \le -T \le 0$, $|p|, |\tilde{p}| \le Ce^{2\eta}$, so $\int_{-\infty}^{-T}|p-\tilde{p}| \le Ce^{-2T}$; on $[-T,L]$, $|p-\tilde{p}| \le |W-\widetilde{W}| + |k_1^2-\tilde{k}_1^2|$. The statement for φ_+ follows from $\xi \mapsto -\xi$ (Lemma 3.7). (b) Since $\cosh\xi \ge \frac{1}{2}e^{|\xi|}$, $d_w \le \delta$ gives $|W-\widetilde{W}| \le 4\delta e^{-2|\xi|}$, so $\widetilde{W}-k_1^2$ is bounded by $(C+4\delta)e^{2\xi}$ for $\xi \le 0$

(similarly at $+\infty$), and $\int|W-\widetilde{W}|\le\delta\int\operatorname{sech}^2=2\delta$. For the convergence statement: along a subsequence $W_n\to W$ a.e.; passing to the limit in $|W_n-k_{1,n}^2|\le Ce^{2\xi}$ ($\xi\le0$) and $|W_n-k_{2,n}^2|\le Ce^{-2\xi}$ ($\xi\ge0$) and using continuity of W gives $W\in\mathcal{W}(k_1,k_2;C)$. For large n the limits $k_{i,n},k_i$ lie in a compact subset $[\underline{k},\overline{k}]$ of $(0,\infty)$. Given $\epsilon>0$ choose T with $Ce^{-2T}<\epsilon$; by (a), $\sup_{\xi\le L}(|v_n-v|+|v_n'-v'|)\le K(\int_{-T}^{L}|W_n-W|+\epsilon+(L+T+1)\,2\overline{k}|k_{1,n}-k_1|)\le K(\epsilon+o(1))$. So $v_n\to v$, $v_n'\to v'$ uniformly on $(-\infty,L]$; since $\varphi_-^{W_n}=e^{k_{1,n}\xi}v_n$ and $k_{1,n}\to k_1$, $\varphi_-^{W_n}\to\varphi_-^{W}$ in C^1 on every $(-L',L]$. The same at $+\infty$. For admissible potentials $\varphi_\pm(0)>0$ (Lemma 3.21(b)), so the normalised $u_\pm$, the quotients $\zeta_\pm=u_\pm'/u_\pm$ and $w_0=\zeta_-(0)-\zeta_+(0)$ converge. (c) Let $W\in\mathcal{P}$. If φ_- is unbounded at $+\infty$, W is admissible. Otherwise, by Lemma 3.21(a) (mirror) $\varphi_-=c\varphi_+$, $c\neq0$, and $c>0$ because $\varphi_->0$ and $\varphi_+>0$ near $+\infty$; so $\varphi_+=\varphi_-/c>0$, $\zeta_-\equiv\zeta_+$, $w_0=0$. In the admissible case $w_0=\rho(0)>0$ and $\varphi_+>0$ by Lemma 3.21(b). Now let $W_n\in\mathcal{P}$, $W_n\to W$ in L^1_{loc}. By (b) $\varphi_-^{W}=\lim\varphi_-^{W_n}\ge0$. If $\varphi_-^{W}(\xi_0)=0$ then $(\varphi_-^{W})'(\xi_0)=0$ (an interior minimum of a C^1 function), so $\varphi_-^{W}\equiv0$, contradicting $e^{-k_1\xi}\varphi_-^{W}\to1$. Hence $W\in\mathcal{P}$, $\varphi_\pm^{W}(0)>0$, and since $\varphi_\pm^{W_n}\to\varphi_\pm^{W}$ in C^1_{loc} we get $\zeta_\pm^{W_n}(0)\to\zeta_\pm^{W}(0)$ and $w_0(W_n)\to w_0(W)$. □

Remark 3.26 (what is uniform)**.** The constant K of (a) depends on $(\underline{k},\overline{k},C,L)$ only, so everything holds uniformly for k_i in a compact subset of $(0,\infty)$ and for W in one class $\mathcal{W}(\cdot,\cdot;C)$. It degenerates as $\underline{k}\to0$ (the kernel K_k is not bounded uniformly), so the limits cannot be taken to $k_i=0$ in this lemma; the faces $c_i=-1$ are handled separately in Section 3.7.

Toy example 3.27 (the weighted topology on Pöschl–Teller potentials)**.** Take $k=2$ and $W=4-2\operatorname{sech}^2\xi$, the Pöschl–Teller potential with $\ell=1$ of Toy example 3.34; there $w_0(W)=2(k^2-1)/k=3$. Perturb it *inside* the class by $W_n=W+\frac{1}{n}\operatorname{sech}^2\xi\in\mathcal{W}(2,2;8)$. Then $d_w(W_n,W)=\frac{1}{n}$, $\int|W_n-W|=\frac{2}{n}$ and, by Lemma 3.25(a), w_0 moves by $O(\frac{1}{n})$; numerically (Riccati integration, `verify_ch03a.py`)

$$w_0(W_n)=3.5244,\ 3.0550,\ 3.00553,\ 3.000553\qquad(n=1,10,100,1000),$$

i.e. $n\,(w_0(W_n)-3)\to0.553$. To *move the limits* take $W^{(k)}=k^2-2\operatorname{sech}^2\xi\in\mathcal{W}(k,k;8)$ with $k\to2$: the explicit value $w_0=2(k^2-1)/k$ equals 3.2476 and 3.02498 for $k=2.1$, 2.01 and tends to 3, as (b) asserts. Finally, $\zeta_-(0)=(k^2-1)/k=1.5\le\overline{\varpi}=2$ is within the uniform bound Lemma 3.28(a). At $k=1$ the limit potential is critical, $w_0=0$ and $\mathcal{P}\ni W$; for $k<1$ the potential has left $\mathcal{P}$ (u_- has a zero): this is the boundary where admissibility ends (openness, Lemma 3.32(c), is about the interior).

3.4.1. Uniform bounds for admissible potentials. The remaining part of the weighted-topology package is a priori control that is *uniform* over a class of potentials; it is what makes sets of Jost data compact in the existence and degree arguments of Chapter 8 and in the small-swirl analysis of Chapter 11. Fix $0<\underline{k}\le\overline{\varpi}$, $C\ge0$, $m>0$ and $x_-,x_+\ge0$, and let $\mathcal{K}=\mathcal{K}(\underline{k},\overline{\varpi},C,m,x_\pm)$ be the set of $W\in\mathcal{W}(k_1,k_2;C)$ with $k_1,k_2\in[\underline{k},\overline{\varpi}]$, $W\le\overline{\varpi}^{\,2}$ on $\mathbb{R}$ and $W\ge m^2$ on $\mathbb{R}\setminus[-x_-,x_+]$. (For the potentials $W_{\mathrm{e}}+\frac12Q_{\mathrm{sw}}[f]$, $|f|\le S$, of Chapters 8–10 these constants depend only on $\boldsymbol{c}'$ and S; see Proposition 2.27.)

Lemma 3.28 (Uniform estimates)**.** *Let $W\in\mathcal{K}$ be admissible, $\vartheta>0$, $R=u_-/u_+$, $\Sigma=R/(R+\vartheta)$, $\rho=\zeta_--\zeta_+$ and $\rho_*=\min\big(2m,\ w_0\,e^{-2\overline{\varpi}\max(x_-,x_+)}\big)$.*

(a) $-\overline{\omega} \le \zeta_+ < \zeta_- \le \overline{\omega}$; *hence* $e^{-\overline{\omega}|\xi|} \le u_\pm(\xi) \le e^{\overline{\omega}|\xi|}$, $\rho \le 2\overline{\omega}$, $w_0 \le 2\overline{\omega}$.
(b) $\rho \ge \rho_*$ *on* $[x_+, \infty)$ *and on* $(-\infty, -x_-]$.
(c) $R(\xi) \ge e^{\rho_*(\xi - x_+)}$ *for* $\xi \ge x_+$ *and* $R(\xi) \le e^{-\rho_*(-x_- - \xi)}$ *for* $\xi \le -x_-$.
(d) Σ *is increasing,* $0 < \Sigma' \le \overline{\omega}/2$, $1 - \Sigma(\xi) \le \vartheta e^{-\rho_*(\xi - x_+)}$ *for* $\xi \ge x_+$ *and* $\Sigma(\xi) \le \vartheta^{-1} e^{-\rho_*(-x_- - \xi)}$ *for* $\xi \le -x_-$.
(e) *There are* $0 < c_{\mathrm{lo}} \le c_{\mathrm{hi}}$ *depending only on* $(\underline{k}, \overline{\omega}, C)$ *with* $c_{\mathrm{lo}} e^{-k_2\xi} \le u_+(\xi) \le c_{\mathrm{hi}} e^{-k_2\xi}$ *for* $\xi \ge 0$ *and* $c_{\mathrm{lo}} e^{k_1\xi} \le u_-(\xi) \le c_{\mathrm{hi}} e^{k_1\xi}$ *for* $\xi \le 0$.
(f) *For* $w_{\min} > 0$ *the set* $\{\Sigma^W : W \in \mathcal{K}$ *admissible,* $w_0(W) \ge w_{\min}\}$ *is relatively compact in* $\mathrm{BC}(\mathbb{R})$.

Proof. (a) ζ_- solves $\zeta' = W - \zeta^2$ and has limits $k_1, k_2 \le \overline{\omega}$ at $\mp\infty$. If $\sup \zeta_- > \overline{\omega}$, the supremum is attained at some ξ_0, where $\zeta_-' = 0$, i.e. $\zeta_-^2 = W \le \overline{\omega}^2$: contradiction. Likewise $\inf \zeta_+ \ge -\overline{\omega}$ (the limits are $-k_1, -k_2 \ge -\overline{\omega}$). $\rho > 0$ gives $\zeta_+ < \zeta_-$; integrating $\zeta_\pm$ from 0 gives the bounds on $u_\pm$; $\rho(0) = w_0$. (b) On $[x_+, \infty)$ we have $W \ge m^2$. First $\zeta_+ \le -m$ there: if $\zeta_+(\xi_0) > -m$ for some $\xi_0 \ge x_+$, Lemma 3.10(b)(iii) on $[\xi_0, \xi_3]$, every $\xi_3 > \xi_0$ (with $\lambda_0 := \max(0, -\zeta_+(\xi_0)) < m$), gives $\zeta_+(\xi) \ge m \tanh(m(\xi - \xi_0) - c) \to m > 0$, contradicting $\zeta_+ \to -k_2 < 0$. Now $\rho' = \zeta_-' - \zeta_+' = -\rho(\zeta_- + \zeta_+) = -\rho(\rho + 2\zeta_+) \ge \rho(2m - \rho)$ on $[x_+, \infty)$. Let $\rho_0 = \min(\rho(x_+), 2m)$ and Y the solution of $Y' = Y(2m - Y)$, $Y(x_+) = \rho_0$ (nondecreasing, $Y \ge \rho_0$). By Lemma 3.10(a)(i) with $G(\xi, y) = y(2m - y)$, $\rho \ge Y \ge \rho_0$. Finally $\rho(x_+) = w_0/(u_- u_+)(x_+) \ge w_0 e^{-2\overline{\omega}x_+}$ by (a). For the left half-line apply this to $W(-\xi)$ (which exchanges $\zeta_- \leftrightarrow -\zeta_+$, $\rho(\xi) \leftrightarrow \rho(-\xi)$, $x_- \leftrightarrow x_+$; Lemma 3.7). (c) $(\ln R)' = \zeta_- - \zeta_+ = \rho > 0$ and $R(0) = 1$; integrate and use (b). (d) $R' = R\rho$, so $\Sigma' = \vartheta R\rho/(R + \vartheta)^2 \le \rho/4 \le \overline{\omega}/2$ (since $(R + \vartheta)^2 \ge 4R\vartheta$); $1 - \Sigma = \vartheta/(R + \vartheta) \le \vartheta/R$ and $\Sigma \le R/\vartheta$; use (c). (e) By Lemma 3.21(a), $\varphi_+ = e^{-k_2\xi} v_+$ with $|v_+ - 1| \le e^{Ce^{-2\xi}/(4k_2)} - 1$ for $\xi \ge 0$. Choose $\xi_1 \ge 0$ (depending on $C, \underline{k}$) with $e^{Ce^{-2\xi_1}/(4\underline{k})} \le \frac{3}{2}$; then $v_+ \in [\frac{1}{2}, \frac{3}{2}]$ on $[\xi_1, \infty)$ and for $\xi \ge \xi_1$, $u_+(\xi) = u_+(\xi_1) e^{k_2\xi_1} e^{-k_2\xi} v_+(\xi)/v_+(\xi_1) \in [\frac{1}{3}, 3]\, u_+(\xi_1) e^{k_2\xi_1} e^{-k_2\xi}$ with $u_+(\xi_1) \in [e^{-\overline{\omega}\xi_1}, e^{\overline{\omega}\xi_1}]$. On $[0, \xi_1]$ use $e^{-\overline{\omega}\xi_1} \le u_+ \le e^{\overline{\omega}\xi_1}$ and $e^{-k_2\xi_1} \le e^{-k_2\xi} \le 1$. The bounds for u_- are the mirror image. (f) Let $\rho_{\min} = \min(2m, w_{\min} e^{-2\overline{\omega}\max(x_\pm)}) \le \rho_*$. By (d) the functions Σ^W take values in $(0, 1)$, are Lipschitz with constant $\overline{\omega}/2$, and satisfy the tail bounds $1 - \Sigma \le \vartheta e^{-\rho_{\min}(\xi - x_+)}$, $\Sigma \le \vartheta^{-1} e^{-\rho_{\min}(-x_- - \xi)}$, uniformly. Given a sequence, Arzelà–Ascoli on compact intervals and a diagonal argument give a subsequence converging locally uniformly; the uniform tails make it uniformly Cauchy on $\mathbb{R}$ (for $\epsilon > 0$ choose T with both tails $< \epsilon$ for $|\xi| \ge T$ and use uniform convergence on $[-T, T]$). As $\mathrm{BC}(\mathbb{R})$ is complete it converges there. □

3.5. Criticality: subcritical, critical, supercritical

In words. *Given a potential* $W \in \mathcal{W}(k_1, k_2)$*, look at the Jost solution* φ_- *that is recessive at* $-\infty$ *and follow it to* $+\infty$*. Three things can happen: it grows (*subcritical *= admissible: then* φ_- *and* φ_+ *are two independent positive solutions and every positive solution is a nonnegative combination of them), it decays again (*critical*: it coincides with* φ_+*, and positive solutions form one ray), or it changes sign (*supercritical*: there are no positive solutions at all). Raising the potential moves it toward subcriticality, and the three sets are arranged accordingly: the supercritical set is open, the subcritical set is open, the critical potentials form the wall between them. We begin with the family that matters for the no-swirl flows,* $W_{\mathbf{c}} = 1 + \frac{1}{2}P_{\mathbf{c}}$*:*

its bounds, and the explicit function g that is recessive at both ends exactly on the wall $c_3 = \bar{c}_3(c_1, c_2)$, so that $W_{\boldsymbol{c}} - W_{\mathrm{crit}} = \frac{\varepsilon}{2}\operatorname{sech}^2$ measures the distance from the wall.

Lemma 3.29 (bounds for the potential)**.** *Let $C_{\boldsymbol{c}} = |c_1 - c_2| + 2|c_3|$. For all $\boldsymbol{c} \in \mathbb{R}^3$ and all $\xi \in \mathbb{R}$*

$$|W_{\boldsymbol{c}}(\xi) - (1 + c_1)| \le C_{\boldsymbol{c}}\, e^{2\xi}, \qquad |W_{\boldsymbol{c}}(\xi) - (1 + c_2)| \le C_{\boldsymbol{c}}\, e^{-2\xi}.$$

In particular $W_{\boldsymbol{c}} \to 1 + c_1$ at $-\infty$ and $W_{\boldsymbol{c}} \to 1 + c_2$ at $+\infty$, exponentially fast; if $c_1, c_2 > -1$ then $W_{\boldsymbol{c}} \in \mathcal{W}(k_1, k_2; C_{\boldsymbol{c}})$.

Proof. $P_{\boldsymbol{c}}(x) - 2c_1 = (1 + x)\big(c_2 - c_1 + c_3(1 - x)\big)$ and $P_{\boldsymbol{c}}(x) - 2c_2 = (1 - x)\big(c_1 - c_2 + c_3(1 + x)\big)$. Since $|1 \mp x| \le 2$ the brackets are bounded by $C_{\boldsymbol{c}}$; and $1 + x = \frac{2e^{2\xi}}{1+e^{2\xi}} \le 2e^{2\xi}$, $1 - x = \frac{2e^{-2\xi}}{1+e^{-2\xi}} \le 2e^{-2\xi}$. Hence $|W_{\boldsymbol{c}} - (1 + c_1)| = \frac{1}{2}|P_{\boldsymbol{c}} - 2c_1| \le C_{\boldsymbol{c}} e^{2\xi}$, and likewise at $+\infty$. □

Lemma 3.30 (the critical function)**.** *Let $c_1, c_2 \ge -1$, $\bar{k} = \frac{1}{2}(k_1 + k_2)$, $\hat{k} = \frac{1}{2}(k_1 - k_2)$ and*

$$g(\xi) = (1 + x)^{k_1/2}(1 - x)^{k_2/2} = e^{\hat{k}\xi}(\cosh \xi)^{-\bar{k}} > 0 \qquad (x = \tanh \xi).$$

(a) *For every c_3: $W_{\boldsymbol{c}} - W_{\mathrm{crit}} = \frac{\varepsilon}{2}\operatorname{sech}^2 \xi$ with $\varepsilon = c_3 - \bar{c}_3$.*

(b) *$g'' = W_{\mathrm{crit}}\, g$; $g'/g = \hat{k} - \bar{k}\tanh \xi$ tends to k_1 at $-\infty$ and to $-k_2$ at $+\infty$; and*

$$e^{-k_2\xi} \le g(\xi) \le 2^{\bar{k}} e^{-k_2\xi}\ (\xi \ge 0), \qquad e^{k_1\xi} \le g(\xi) \le 2^{\bar{k}} e^{k_1\xi}\ (\xi \le 0). \tag{3.4}$$

In particular $g \le 2^{\bar{k}}$, $g^{-2} \ge 4^{-\bar{k}}$ on $\mathbb{R}$ and, with $\mathcal{G}_{\mathrm{int}}(\xi) = \int_0^\xi g^{-2}$, $\mathcal{G}_{\mathrm{int}}(\xi) \to \pm\infty$ as $\xi \to \pm\infty$.

Proof. (a) $P_{\boldsymbol{c}} - P_{(c_1, c_2, \bar{c}_3)} = (c_3 - \bar{c}_3)(1 - x^2)$ and $1 - \tanh^2 \xi = \operatorname{sech}^2 \xi$, so $W_{\boldsymbol{c}} - W_{\mathrm{crit}} = \frac{1}{2}(c_3 - \bar{c}_3)\operatorname{sech}^2 \xi$. (b) Put $p = g'/g = \hat{k} - \bar{k}x$; since $x' = 1 - x^2$, $g''/g = p^2 + p' = (\hat{k} - \bar{k}x)^2 - \bar{k}(1 - x^2) = (\hat{k}^2 - \bar{k}) - 2\hat{k}\bar{k}x + (\bar{k}^2 + \bar{k})x^2$. On the other hand, with $c_i = k_i^2 - 1$,

$$W_{\mathrm{crit}} = 1 + \tfrac{1}{2}\big[(k_1^2 + k_2^2 - 2) + (k_2^2 - k_1^2)x + \bar{c}_3(1 - x^2)\big], \qquad \bar{c}_3 = -2\bar{k}(\bar{k} + 1).$$

Coefficient of x: $\frac{1}{2}(k_2^2 - k_1^2) = -2\hat{k}\bar{k}$. Coefficient of x^2: $-\frac{1}{2}\bar{c}_3 = \bar{k}(\bar{k} + 1) = \bar{k}^2 + \bar{k}$. Constant term: $\frac{1}{2}(k_1^2 + k_2^2) + \frac{1}{2}\bar{c}_3 = \hat{k}^2 + \bar{k}^2 - \bar{k}^2 - \bar{k} = \hat{k}^2 - \bar{k}$ (as $\frac{1}{2}(k_1^2 + k_2^2) = \hat{k}^2 + \bar{k}^2$). So $g'' = W_{\mathrm{crit}}\, g$ (computer-algebra check: `verify_ch03a.py`). For (3.4): $\hat{k} - \bar{k} = -k_2$, $\hat{k} + \bar{k} = k_1$ and $\frac{1}{2}e^{|\xi|} \le \cosh \xi \le e^{|\xi|}$ give for $\xi \ge 0$: $g \in e^{(\hat{k} - \bar{k})\xi}[1, 2^{\bar{k}}]$, and for $\xi \le 0$: $g \in e^{(\hat{k} + \bar{k})\xi}[1, 2^{\bar{k}}]$. Finally $g^{-2} \ge 4^{-\bar{k}} e^{2k_2\xi} \ge 4^{-\bar{k}}$ for $\xi \ge 0$ and $g^{-2} \ge 4^{-\bar{k}} e^{-2k_1\xi} \ge 4^{-\bar{k}}$ for $\xi \le 0$, so $|\mathcal{G}_{\mathrm{int}}(\xi)| \ge 4^{-\bar{k}}|\xi|$. □

Definition 3.31 (subcritical, critical, supercritical)**.** Let $W \in \mathcal{W}(k_1, k_2)$. W is *subcritical* if it is admissible (Definition 3.20), *critical* if $\varphi_-^W > 0$ and φ_-^W is a multiple of φ_+^W (Lemma 3.25(c)), and *supercritical* if φ_-^W has a zero.

Lemma 3.32 (Criticality dichotomy and its stability)**.** *Let $k_1, k_2 > 0$ and $C \ge 0$.*

(a) *Every $W \in \mathcal{W}(k_1, k_2)$ is exactly one of subcritical, critical, supercritical.*

(b) *The positive solutions of $z'' = Wz$ on $\mathbb{R}$ are: $\{au_- + bu_+ : a, b \ge 0,\ (a, b) \ne (0, 0)\}$ if W is subcritical; the positive multiples of φ_- if W is critical; none if W is supercritical.*

(c) *(*Openness of admissibility.*) Let $W \in \mathcal{W}(k_1,k_2)$ be admissible and $\widetilde{C} \geq 0$. There are $L > 0$ and $\epsilon > 0$ such that every $\widetilde{W} \in \mathcal{W}(\tilde{k}_1,\tilde{k}_2;\widetilde{C})$ with*

$$|\tilde{k}_1 - k_1| + |\tilde{k}_2 - k_2| + \int_{-\infty}^{L} \left|(W - k_1^2) - (\widetilde{W} - \tilde{k}_1^2)\right| < \epsilon$$

is admissible. In particular (take $\tilde{k}_i = k_i$) $\int_{-\infty}^{L} |W - \widetilde{W}| < \epsilon$ suffices, and: if $W_n \to W$ in L^1_{loc} with $W_n \in \mathcal{W}(k_{1,n},k_{2,n};\widetilde{C})$, $k_{i,n} \to k_i$, then W_n is admissible for large n. Hence the admissible set is open for the L^1_{loc} topology and for d_w (a d_w-ball around $W \in \mathcal{W}(k_1,k_2;C)$ lies in $\mathcal{W}(k_1,k_2;C+4\delta)$, Lemma 3.25(b); the statement is for every $\widetilde{C}$).

(d) *(*Closedness.*) The set $\mathcal{P}$ of subcritical and critical potentials is closed under the limits of Lemma 3.25(c) (in particular closed in $\mathcal{W}(k_1,k_2;C)$ for L^1_{loc} convergence, hence for d_w), w_0 is continuous on $\mathcal{P}$, and the critical set $\{W \in \mathcal{P} : w_0(W) = 0\}$ is closed; the supercritical set is open.*

(e) *(*Order.*) Let $W_{\mathrm{ref}} \in \mathcal{W}(k_1,k_2)$ be critical and $W \in \mathcal{W}(k_1,k_2)$ with $W - W_{\mathrm{ref}}$ continuous, not identically zero and of one sign. If $W \geq W_{\mathrm{ref}}$ then W is subcritical; if $W \leq W_{\mathrm{ref}}$ then W is supercritical.*

Proof. (a) By Lemma 3.25(c), if $\varphi_- > 0$ then W is admissible or critical, and these are exclusive (bounded vs. unbounded); if φ_- has a zero, it is supercritical, and then it is neither. (b) *Subcritical.* $u_\pm > 0$ and $\mathrm{Wr}(u_-,u_+) = -w_0 < 0$, so every solution is $z = au_- + bu_+$; if $a,b \geq 0$ not both 0, $z > 0$. Conversely let $z > 0$. If $b < 0$, then $d_z := \mathrm{Wr}(u_-,z) = -bw_0 > 0$ and $(z/u_-)' = d_z/u_-^2$, so $\frac{z}{u_-}(\xi) = \frac{z}{u_-}(0) - d_z \int_\xi^0 u_-^{-2} \to -\infty$ as $\xi \to -\infty$ (because $u_-^{-2} \sim e^{-2k_1\xi}$ is not integrable at $-\infty$), contradicting $z > 0$. If $a < 0$, then $\mathrm{Wr}(u_+,z) = aw_0 < 0$ and $\frac{z}{u_+}(\xi) = \frac{z}{u_+}(0) + aw_0 \int_0^\xi u_+^{-2} \to -\infty$ as $\xi \to +\infty$: contradiction. *Critical.* Here $u_- = \varphi_-/\varphi_-(0) > 0$ is recessive at both ends, so $\int_{-\infty}^0 u_-^{-2} = \int_0^\infty u_-^{-2} = \infty$. If $z > 0$ and $d_z = \mathrm{Wr}(u_-,z) > 0$ then $z/u_- \to -\infty$ at $-\infty$; if $d_z < 0$ then $z/u_- \to -\infty$ at $+\infty$; so $d_z = 0$ and z is a multiple of u_-. *Supercritical.* Suppose $z > 0$ solves the equation. z is not a multiple of φ_- (which has a zero), so $d := \mathrm{Wr}(z,\varphi_-) \neq 0$ and, by Lemma 3.21(a), $e^{k_1\xi}z \to \ell \neq 0$, so $\varphi_-/z \to 0$ at $-\infty$. Moreover $\varphi_-/z > 0$ near $-\infty$ and $(\varphi_-/z)' = d/z^2$ has a constant nonzero sign. A function that is positive near $-\infty$, tends to 0 there and is strictly monotone is increasing, hence positive everywhere: $\varphi_- > 0$, a contradiction. (c) Let $C' = \max(C_W,\widetilde{C})$ where $W \in \mathcal{W}(k_1,k_2;C_W)$. Since $\zeta_- \to k_2 > 0$ at $+\infty$, choose $L > 0$ so large that $\zeta_-^W(L) > 0$ and $\widetilde{C}e^{-2L} \leq \frac{1}{4}k_2^2$. If $|\tilde{k}_2^2 - k_2^2| \leq \frac{1}{4}k_2^2$ then $\widetilde{W} \geq \tilde{k}_2^2 - \widetilde{C}e^{-2L} \geq \frac{1}{2}k_2^2$ on $[L,\infty)$. The function $v = e^{-k_1\xi}\varphi_-^W$ is continuous, positive on $(-\infty,L]$ and tends to 1 at $-\infty$, so $\inf_{(-\infty,L]} v =: v_0 > 0$. By Lemma 3.25(a) (applied in the class constant C', with $\underline{k} = \frac{1}{2}\min k_i$, $\overline{k} = 2\max k_i$), for ϵ small $|\tilde{v} - v| \leq v_0/2$ and $|\tilde{v}' - v'|$ is small on $(-\infty,L]$, where $\tilde{v} = e^{-\tilde{k}_1\xi}\varphi_-^{\widetilde{W}}$; so $\tilde{\varphi} := \varphi_-^{\widetilde{W}} = e^{\tilde{k}_1\xi}\tilde{v} > 0$ on $(-\infty,L]$ and $\tilde{\varphi}'(L) = e^{\tilde{k}_1 L}(\tilde{k}_1\tilde{v} + \tilde{v}')(L)$ is close to $\varphi'(L) = e^{k_1 L}(k_1 v + v')(L) = \zeta_-^W(L)\varphi_-^W(L) > 0$, hence $\tilde{\varphi}'(L) > 0$. For $\xi \geq L$, as long as $\tilde{\varphi}$ is positive, $\tilde{\varphi}'' = \widetilde{W}\tilde{\varphi} \geq \frac{1}{2}k_2^2\tilde{\varphi} > 0$, so $\tilde{\varphi}'$ stays positive and $\tilde{\varphi}$ increases; thus $\tilde{\varphi} > 0$ on $\mathbb{R}$, and $\tilde{\varphi}'' \geq \frac{1}{2}k_2^2\tilde{\varphi}$ forces $\tilde{\varphi} \to \infty$. Hence $\widetilde{W}$ is admissible. For the consequences: by Lemma 3.25(a) the quantity in the display is at most $\int_{-T}^{L} |W - W_n| + C'e^{-2T} + (L+T)|k_1^2 - k_{1,n}^2| + |k_1 - k_{1,n}| + |k_2 - k_{2,n}|$ for every T, which tends to 0 (first $n \to \infty$, then $T \to \infty$). (d) is Lemma 3.25(c) together

with (a): $\mathcal{P}$ is closed, w_0 is continuous on it and vanishes exactly on the critical set, and the supercritical set is the complement of $\mathcal{P}$. (e) Put $y=\varphi_-^{W_{\mathrm{ref}}}>0$ (recessive at both ends: $y\sim e^{k_1\xi}$ at $-\infty$, $y\sim ce^{-k_2\xi}$ at $+\infty$ with $c>0$) and $u=\varphi_-^{W}$. By Lemma 3.4, $\mathrm{Wr}(y,u)'=yu''-y''u=(W-W_{\mathrm{ref}})yu$, and $\mathrm{Wr}(y,u)\to 0$ at $-\infty$ (as in the proof of Lemma 3.23). Fix ξ_1 with $W-W_{\mathrm{ref}}\neq 0$ at ξ_1, hence on a neighbourhood. *Case* $W\geq W_{\mathrm{ref}}$. Let $\xi^*=\sup\{\xi : u>0 \text{ on } (-\infty,\xi)\}$. On $(-\infty,\xi^*)$, $\mathrm{Wr}(y,u)=\int_{-\infty}^{\xi}(W-W_{\mathrm{ref}})yu\geq 0$, so u/y is nondecreasing and $u\geq y$ (the limit of u/y at $-\infty$ is 1); as in Lemma 3.23, $\xi^*=\infty$ and $u>0$ on $\mathbb{R}$. Then $d_1:=\mathrm{Wr}(y,u)(\xi_1+1)>0$, $\mathrm{Wr}(y,u)$ is nondecreasing, and for $\xi\geq\xi_1+1$: $(u/y)'\geq d_1y^{-2}\geq d_1c_1e^{2k_2\xi}$ for large ξ ($y\leq c_2e^{-k_2\xi}$), so $u/y\geq c_3e^{2k_2\xi}$ and $u\geq c_3ye^{2k_2\xi}\geq c_4e^{k_2\xi}$ (using $y\geq c_5e^{-k_2\xi}$ for large ξ): u is unbounded and W is admissible. *Case* $W\leq W_{\mathrm{ref}}$. Suppose $u>0$ on $\mathbb{R}$. Then $\mathrm{Wr}(y,u)=\int_{-\infty}^{\xi}(W-W_{\mathrm{ref}})yu\leq 0$ is nonincreasing and <0 for $\xi>\xi_1$; with $d_1:=-\mathrm{Wr}(y,u)(\xi_1+1)>0$ we get $(u/y)'\leq -d_1y^{-2}\leq -d_1c_1e^{2k_2\xi}$ for large ξ, so $u/y\to-\infty$, contradicting $u>0$. Hence u has a zero. □

Remark 3.33 (two proofs of the trichotomy)**.** Part (b) describes the positive solutions of an arbitrary $W\in\mathcal{W}(k_1,k_2)$, $k_i>0$, through the Jost solutions; it is what Chapters 8 and 10 apply to potentials that are not explicit. For the explicit potentials $W_{\boldsymbol{c}}$ the same description (with the slopes $\zeta_+\leq z'/z\leq\zeta_-$) is proved again in Section 3.7 (Proposition 3.43) by reduction of order with the critical function g; that proof is independent of the present one and also covers the faces $c_i=-1$ ($k_i=0$), where the Jost solutions of this section are not available.

The potentials $k^2-\ell(\ell+1)\operatorname{sech}^2\xi$ are the simplest explicit family in which all three behaviours of Lemma 3.32 occur; they are also the model for the no-swirl potentials $W_{\boldsymbol{c}}$ near the critical surface (Example 3.46). Here u_- denotes (a multiple of) the Jost solution recessive at $-\infty$ and u_+ its mirror (Lemma 3.21).

Toy example 3.34 (Pöschl–Teller)**.** For $z''=(k^2-\ell(\ell+1)\operatorname{sech}^2\xi)z$ with $k,\ell>0$, $g=\operatorname{sech}^\ell\xi$ solves $g''=(\ell^2-\ell(\ell+1)\operatorname{sech}^2\xi)g$. Hence the potential is critical iff $k=\ell$, subcritical iff $k>\ell$ and supercritical iff $k<\ell$. For $\ell=1$ the Jost solutions are $u_\mp\propto e^{\pm k\xi}(k\mp\tanh\xi)$ and w_0 vanishes exactly at $k=1$.

Verification. Write $x=\tanh\xi$, so $x'=1-x^2=\operatorname{sech}^2\xi$.

The function g. $g'=-\ell xg$ and $g''=(\ell^2x^2-\ell\operatorname{sech}^2\xi)g=(\ell^2-\ell(\ell+1)\operatorname{sech}^2\xi)g$, using $x^2=1-\operatorname{sech}^2\xi$. The potential $W_k=k^2-\ell(\ell+1)\operatorname{sech}^2$ tends to k^2 at both ends with $|W_k-k^2|\leq 4\ell(\ell+1)e^{-2|\xi|}$, so the Jost solutions exist. Unlike the order lemma Lemma 3.32(e), the two potentials compared here have *different* limits (ℓ^2 for g, k^2 for W_k); the Wronskian argument still works because g is recessive at both ends. By Lemma 3.4 with $a=g$ (potential W_ℓ) and $b=u_-$ (potential W_k):

$$\mathrm{Wr}(g,u_-)'=(k^2-\ell^2)\,g\,u_-.$$

The boundary term. $g\leq 2^\ell e^{\ell\xi}$ for $\xi\leq 0$, $u_-=O(e^{k\xi})$ and $u_-'/u_-\to k$, $g'/g\to\ell$ as $\xi\to-\infty$; hence $\mathrm{Wr}(g,u_-)=gu_-(u_-'/u_--g'/g)=O(e^{(k+\ell)\xi})\to 0$. Therefore, on any interval $(-\infty,\xi]$ on which $u_->0$,

$$\mathrm{Wr}(g,u_-)(\xi)=(k^2-\ell^2)\int_{-\infty}^{\xi}g\,u_-\,d\eta,\qquad (u_-/g)'=\mathrm{Wr}(g,u_-)/g^2. \tag{3.5}$$

Case $k>\ell$. Then $\mathrm{Wr}(g,u_-)>0$ as long as $u_->0$. If u_- had a first zero ξ_*, then u_-/g would be increasing on $(-\infty,\xi_*)$, positive near $-\infty$, hence positive at

ξ_*: contradiction. So $u_- > 0$ on $\mathbb{R}$, $\mathrm{Wr}(g,u_-)$ is increasing and $\mathrm{Wr}(g,u_-)(\xi) \geq \mathrm{Wr}(g,u_-)(0) =: d_0 > 0$ for $\xi \geq 0$. Then $(u_-/g)' \geq d_0 g^{-2} = d_0 \cosh^{2\ell}\xi \geq d_0 4^{-\ell} e^{2\ell\xi}$, so $u_-/g \geq c\,e^{2\ell\xi}$ for $\xi \geq \xi_1$ (some $c, \xi_1 > 0$) and $u_- \geq g\,ce^{2\ell\xi} \geq c\,e^{\ell\xi} \to \infty$ ($g \geq e^{-\ell\xi}$ for $\xi \geq 0$): u_- is positive and unbounded, the potential is subcritical.

Case $k = \ell$. Then W_k is the potential of g, so g is a positive solution decaying like $e^{-\ell|\xi|}$ at both ends. A solution decaying at $-\infty$ is a multiple of u_- and one decaying at $+\infty$ a multiple of u_+ (uniqueness of the recessive solution in Lemma 3.21); so $u_- \propto g \propto u_+$: critical.

Case $k < \ell$. Now $\mathrm{Wr}(g,u_-) < 0$ while $u_- > 0$. Suppose $u_- > 0$ on $\mathbb{R}$; then $\mathrm{Wr}(g,u_-)$ is decreasing, $\mathrm{Wr}(g,u_-)(\xi) \leq \mathrm{Wr}(g,u_-)(0) =: -d_0 < 0$ for $\xi \geq 0$, and $(u_-/g)' \leq -d_0 g^{-2} \leq -d_0 4^{-\ell} e^{2\ell\xi}$ gives $u_-/g \to -\infty$: contradiction. So u_- has a zero: supercritical. The three cases are exhaustive and their conclusions mutually exclusive, so each is an equivalence.

The case $\ell = 1$. Put $v_-(\xi) = e^{k\xi}(k - x)$. Then $v_-' = e^{k\xi}\big(k(k-x) - (1-x^2)\big)$ and $v_-'' = e^{k\xi}\big(k^3 - k^2 x - 2k(1-x^2) + 2x(1-x^2)\big) = e^{k\xi}(k^2 - 2(1-x^2))(k-x) = (k^2 - 2\,\mathrm{sech}^2\xi)v_-$. Also $v_- \sim (k+1)e^{k\xi}$ at $-\infty$, so $u_- \propto v_-$. The potential is even, so $v_+(\xi) := v_-(-\xi) = e^{-k\xi}(k+x)$ is the solution recessive at $+\infty$. Normalising, $u_\mp = v_\mp/k$, and $v_-'(0) = k^2 - 1$, $v_+'(0) = 1 - k^2$, so $\zeta_-(0) = (k^2-1)/k$, $\zeta_+(0) = (1-k^2)/k$ and $w_0 = 2(k^2-1)/k$, which vanishes exactly at $k = 1$ (positive for $k > 1$; for $k < 1$ the number is negative and merely records $\mathrm{Wr}(u_+,u_-)$ of the two explicit solutions, since then u_- vanishes, at $\tanh\xi = k$, and the potential is supercritical, not admissible). All identities were also checked by computer algebra (`code/ch03_line/verify_ch03b.py`). □

The three behaviours of u_-, for $k < 1$, $k = 1$ and $k > 1$, are plotted in Figure 3.2.

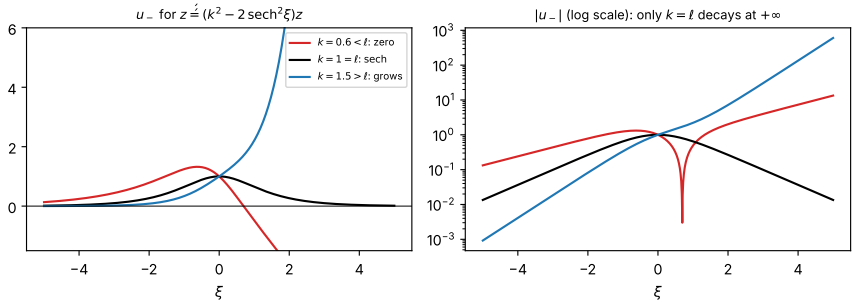


FIGURE 3.2. The solution u_- recessive at $-\infty$ of $z'' = (k^2 - 2\,\mathrm{sech}^2\xi)z$, $u_-(0) = 1$ (Toy example 3.34, $\ell = 1$). For $k < 1$ it crosses zero (supercritical), for $k = 1$ it is $\mathrm{sech}\,\xi$ (critical, decays at both ends) and for $k > 1$ it is positive and grows (subcritical). Code: `code/ch03_line/make_pt_toy.py`.

Order at $k = 1$: $\ell = 0, 1, 2$. Take $W_\ell = 1 - \ell(\ell+1)\,\mathrm{sech}^2\xi$ ($k_1 = k_2 = 1$). For $\ell = 0$, $W \equiv 1$ is subcritical ($u_\pm = e^{\pm\xi}$, $w_0 = 2$). For $\ell = 1$, $g = \mathrm{sech}\,\xi = 2\varphi_- = 2\varphi_+$ is positive and recessive at both ends: critical. For $\ell = 2$, $\varphi_- = -\frac{1}{2}\tanh\xi\,\mathrm{sech}\,\xi$ solves $\varphi'' = W_2\varphi$ (checked by computer algebra), $e^{-\xi}\varphi_- \to 1$ at $-\infty$, and vanishes at $\xi = 0$: supercritical. This is exactly the ordering of Lemma 3.32(e): $W_0 \geq W_1 \geq W_2$

with W_1 critical. The positive solutions are the cone $\{ae^{\xi}+be^{-\xi}\}$, the ray of $\operatorname{sech}\xi$, and none.

Lemma 3.35 (Subcriticality criterion for the explicit potentials)**.** *Let $\boldsymbol{c}\in\mathbb{R}^3$.*

(a) *If $c_1,c_2>-1$, then $W_{\boldsymbol{c}}\in\mathcal{W}(k_1,k_2;C_{\boldsymbol{c}})$ with $k_i=\sqrt{1+c_i}$ (Lemma 3.29).*

(b) *Let $c_1,c_2>-1$. Then $W_{\boldsymbol{c}}$ is subcritical iff $c_3>\bar c_3$ (then $u_-'/u_-\to k_2$), critical iff $c_3=\bar c_3$ (then $\varphi_\pm$ are positive multiples of g, and W_{crit} is the critical potential), supercritical iff $c_3<\bar c_3$ (φ_- has a zero).*

(c) *If $c_2<-1$ every nontrivial solution of $z''=W_{\boldsymbol{c}}z$ has zeros accumulating at $+\infty$. If $c_2=-1$ no solution has $z'/z\to L>0$ as $\xi\to+\infty$. (Mirror statements for c_1 at $-\infty$.)*

Proof. (a) is Lemma 3.29, as $1+c_i=k_i^2$. (b) By Lemma 3.30, $g''=W_{\mathrm{crit}}g$ and $g\sim 2^{\bar k}e^{k_1\xi}$ at $-\infty$, $g\sim 2^{\bar k}e^{-k_2\xi}$ at $+\infty$; by the uniqueness in Lemma 3.21(a), $g=2^{\bar k}\varphi_-^{W_{\mathrm{crit}}}=2^{\bar k}\varphi_+^{W_{\mathrm{crit}}}$. So W_{crit} is critical (it lies in $\mathcal{W}(k_1,k_2)$ by (a)). By Lemma 3.30(a), $W_{\boldsymbol{c}}-W_{\mathrm{crit}}=\frac{\varepsilon}{2}\operatorname{sech}^2\xi$ has the sign of $\varepsilon=c_3-\bar c_3$ and is not identically zero for $\varepsilon\neq0$; Lemma 3.32(e) (with $W_{\mathrm{ref}}=W_{\mathrm{crit}}$) gives: $\varepsilon>0$ subcritical, $\varepsilon<0$ supercritical; $\varepsilon=0$ is W_{crit} itself, critical. Finally if $W_{\boldsymbol{c}}$ is subcritical, $\zeta_-\to k_2$ at $+\infty$ by Lemma 3.21(b)(iv). (c) If $c_2<-1$, then $W_{\boldsymbol{c}}\to 1+c_2=:-\mu_2^2<0$ by Lemma 3.29, so $W_{\boldsymbol{c}}\le-\mu_2^2/2$ on some $[\xi_1,\infty)$; by Lemma 3.2(b) every nontrivial solution has a zero in each interval $[a,a+\pi\sqrt{2}/\mu_2]\subset[\xi_1,\infty)$. If $c_2=-1$ and $z'/z\to L>0$, then $z\neq0$ near $+\infty$ and $h=z'/z$ solves $h'=W_{\boldsymbol{c}}-h^2$ with $W_{\boldsymbol{c}}\to0$; Lemma 3.14 with $m=0$ gives $h\to0$: contradiction. The statements at $-\infty$ follow from $\xi\mapsto-\xi$, which maps $\boldsymbol{c}$ to (c_2,c_1,c_3) (Remark 3.36). □

Remark 3.36 (reflection of the explicit family)**.** For $\boldsymbol{c}=(c_1,c_2,c_3)$ put $\boldsymbol{c}^*=(c_2,c_1,c_3)$ and $(Rz)(\xi)=z(-\xi)$. Since $\tanh(-\xi)=-\tanh\xi$, $P_{\boldsymbol{c}}(-x)=c_1(1+x)+c_2(1-x)+c_3(1-x^2)=P_{\boldsymbol{c}^*}(x)$, hence $RW_{\boldsymbol{c}}=W_{\boldsymbol{c}^*}$; also $R\,W_{\mathrm{crit}}[\boldsymbol{c}]=W_{\mathrm{crit}}[\boldsymbol{c}^*]$, $Rg_{\boldsymbol{c}}=g_{\boldsymbol{c}^*}$, $k_1\leftrightarrow k_2$, $\bar c_3(c_1,c_2)=\bar c_3(c_2,c_1)$, so ε is unchanged, and $\mathcal{J}$ is invariant under $\boldsymbol{c}\mapsto\boldsymbol{c}^*$. (With Lemma 3.7 this reduces every statement about $+\infty$ for all $\boldsymbol{c}$ to one about $-\infty$.)

Toy example 3.37 (the $\varepsilon\operatorname{sech}^2$ perturbation of the critical g)**.** For $\boldsymbol{c}=(0,0,c_3)$: $k_1=k_2=1$, $\bar c_3=-4$, $g=\operatorname{sech}\xi$ and $W_{\mathrm{crit}}=1-2\operatorname{sech}^2\xi$ (Toy 3.34, $\ell=k=1$). Numerically (solution of $z''=W_{\boldsymbol{c}}z$ from $-\infty$ for $c_3=-3.9,-4,-4.1,-12$, `verify_ch03a.py`) the Jost solution φ_- is positive and grows like e^{ξ} for $c_3=-3.9$, equals $\frac12\operatorname{sech}\xi$ at $c_3=-4$, and has a zero for $c_3=-4.1$ and for $c_3=-12$ (where $W=1-6\operatorname{sech}^2\xi$, $\ell=2$): the sign of $\varepsilon=c_3+4$ decides, as $\operatorname{Wr}(g,\varphi_-)(\xi)=\frac{\varepsilon}{2}\int_{-\infty}^{\xi}\operatorname{sech}^3\varphi_-$ shows.

3.6. A Jost basis through $\sigma=0$

In words. *The Jost solutions $e^{\pm\mu\xi}(1+o(1))$ of $z''=(\mu^2+w)z$ become degenerate when $\mu\to0$ (both tend to 1). What does* not *degenerate is the pair "cosine-like" C, "sine-like" S with $C\to1$, $S\sim\xi$ at $\mu=0$, which depends analytically on $\sigma=\mu^2$ including $\sigma=0$: no square root of σ is ever needed. This is the basis in which the pole analysis of Chapter 4 lets the exponent pass through $\mu=0$.*

Standing assumptions (J). $\mu_* \in (0, \frac{1}{160}]$; $\xi_* \geq 1$ and $M > 0$ with $(M+1)e^{-\xi_*} \leq \frac{1}{16}$; $w \in C[\xi_*, \infty)$ with $|w(\xi)| \leq Me^{-\xi}$. Let B_{μ_*} be the Banach space of continuous φ on $[\xi_*, \infty)$ with $\|\varphi\|_B = \sup e^{-3\mu_*\xi}|\varphi| < \infty$. The margin $+1$ in $(M+1)e^{-\xi_*} \leq \frac{1}{16}$ is not needed for the lemma itself; it makes the lemma applicable to $w+h$ for every continuous h with $\sup e^{\xi}|h| \leq 1$ (the amplitude M is replaced by $M+1$), which is how Chapter 4 differentiates with respect to w. Here M is an amplitude, not the critical swirl height. For a continuous w put $N(w) = \sup_{\xi \geq \xi_*} e^{\xi}|w(\xi)|$. For complex σ, $|\sigma| \leq 4\mu_*^2$, and $\mu = \sqrt{\sigma}$ put

$$C_0(\xi;\sigma) = \sum_{n\geq 0} \frac{\sigma^n \xi^{2n}}{(2n)!} = \cosh \mu\xi, \qquad S_0(\xi;\sigma) = \sum_{n\geq 0} \frac{\sigma^n \xi^{2n+1}}{(2n+1)!} = \frac{\sinh \mu\xi}{\mu},$$

$$(\mathcal{V}_\sigma \varphi)(\xi) = \int_\xi^\infty S_0(\eta - \xi;\sigma)\, w(\eta)\varphi(\eta)\, d\eta.$$

Lemma 3.38 (Jost basis through $\sigma = 0$)**.** *Assume (J).*

(i) $\|\mathcal{V}_\sigma[w]\|_{B\to B} \leq (1-5\mu_*)^{-2} N(w) e^{-\xi_*}$ *for* $|\sigma| \leq 4\mu_*^2$, *in particular* $\leq \frac{1}{4}$ *under (J). Hence* $C := (1-\mathcal{V}_\sigma)^{-1} C_0$ *and* $S := (1-\mathcal{V}_\sigma)^{-1} S_0$ *are well defined, analytic in* σ *(for each* ξ*, in the open disc* $|\sigma| < 4\mu_*^2$, including $\sigma = 0$*; this disc, not the whole plane, is the domain of the lemma, and all later uses stay in* $|\sigma| \leq \mu_*^2$*), solve* $u'' = (\sigma + w)u$*, and* $\mathrm{Wr}(C, S) = CS' - C'S = 1$.

(ii) *For* $\sigma = \mu^2$, $0 < \mu \leq \mu_*$*:* $F_\pm := C \pm \mu S = e^{\pm\mu\xi}\zeta_\pm$ *with* $\zeta_\pm = 1 + \int_\xi^\infty k_\pm(\eta - \xi) w \zeta_\pm \, d\eta$, $k_-(r) = \frac{1-e^{-2\mu r}}{2\mu}$, $k_+(r) = \frac{e^{2\mu r}-1}{2\mu}$, $0 \leq k_\pm(r) \leq re^{2\mu_* r}$. *Consequently* $\zeta_\pm \in [\frac{2}{3}, \frac{4}{3}]$, $\zeta_\pm \to 1$, $\zeta_\pm' \to 0$, $F_\pm > 0$, $\mathrm{Wr}(F_+, F_-) = -2\mu$, *and* F_- *is the recessive solution* z_- *of Lemma 3.16(a).*

(iii) *For* $0 \leq \sigma \leq \mu_*^2$ *and* $\xi \geq \xi_*$*:* $\frac{2}{3}C_0 \leq C \leq \frac{4}{3}C_0$ *and* $\frac{2}{3}S_0 \leq S \leq \frac{4}{3}S_0$.

(iv) *At* $\sigma = 0$*:* $C \to 1$, $C' \to 0$, $S' \to 1$ *and* $S/\xi \to 1$ *as* $\xi \to \infty$.

(v) *(*Derivatives, joint continuity, dependence on w*.)* C', S' *are analytic in* σ *for each* ξ*, namely* $C' = \sigma S_0 + (\mathcal{V}_\sigma C)'$, $S' = C_0 + (\mathcal{V}_\sigma S)'$ *with* $(\mathcal{V}_\sigma\varphi)'(\xi) = -\int_\xi^\infty C_0(\eta - \xi;\sigma)\, w\varphi \, d\eta$, *and* $(\xi, \sigma) \mapsto (C, C', S, S')$ *is continuous on* $[\xi_*, \infty) \times \{|\sigma| < 4\mu_*^2\}$*. If* $w, \tilde{w}$ *both satisfy (J), then* $\|C_w - C_{\tilde{w}}\|_B + \|C_w' - C_{\tilde{w}}'\|_B \leq 4e^{-\xi_*} N(w - \tilde{w})$ *and* $\|S_w - S_{\tilde{w}}\|_B + \|S_w' - S_{\tilde{w}}'\|_B \leq \frac{4}{\mu_*} e^{-\xi_*} N(w - \tilde{w})$*, uniformly in* $|\sigma| \leq 4\mu_*^2$.

Proof. *(i), the operator.* For $|\sigma| \leq 4\mu_*^2$, $|S_0(r;\sigma)| \leq \sum |\sigma|^n r^{2n+1}/(2n+1)! \leq re^{2\mu_* r}$. For $\varphi \in B$ and $\eta = \xi + r$, $|w\varphi|(\eta) \leq M\|\varphi\|_B e^{-(1-3\mu_*)\xi} e^{-(1-3\mu_*)r}$, so $|\mathcal{V}_\sigma\varphi(\xi)| \leq M\|\varphi\|_B e^{-(1-3\mu_*)\xi} \int_0^\infty re^{-(1-5\mu_*)r} dr = M\|\varphi\|_B e^{-(1-3\mu_*)\xi}(1-5\mu_*)^{-2}$, i.e. $\|\mathcal{V}_\sigma\varphi\|_B \leq \frac{Me^{-\xi_*}}{(1-5\mu_*)^2}\|\varphi\|_B \leq \frac{1}{4}\|\varphi\|_B$ (as $Me^{-\xi_*} \leq \frac{1}{16}$, $5\mu_* \leq \frac{1}{32}$). Since $|C_0| \leq e^{2\mu_*\xi}$ and $|S_0| \leq \xi e^{2\mu_*\xi}$, both lie in B, and the Neumann series gives $C, S \in B$, $\|(1-\mathcal{V}_\sigma)^{-1}\| \leq \frac{4}{3}$. The kernel is entire in σ and the bound is uniform, so $\sigma \mapsto \mathcal{V}_\sigma \in \mathcal{L}(B)$ and hence C, S are analytic in σ (also analytic in w, since $\mathcal{V}$ is linear in w). *The equation.* Since $S_0(0) = 0$, $S_0'(0) = 1$, $S_0'' = \sigma S_0$: $(\mathcal{V}\varphi)' = -\int_\xi^\infty S_0'(\eta - \xi) w\varphi$ and $(\mathcal{V}\varphi)'' = w\varphi + \sigma\mathcal{V}\varphi$. With $C_0'' = \sigma C_0$, $C = C_0 + \mathcal{V}C$ gives $C'' = \sigma C + wC$; the same for S. *(ii).* $C_0 \pm \mu S_0 = e^{\pm\mu\xi}$, so $F_\pm = (1-\mathcal{V})^{-1} e^{\pm\mu\xi}$ is the fixed point of $F = e^{\pm\mu\xi} + \mathcal{V}F$ in B. Writing $F = e^{a\xi}\zeta$, $a = \pm\mu$, this reads $\zeta = 1 + \int_\xi^\infty e^{ar} S_0(r) w\zeta \, d\eta$ ($r = \eta - \xi$) and $e^{\pm\mu r}\sinh(\mu r)/\mu = k_\pm(r)$; $k_\pm(r) = \int_0^r e^{\pm 2\mu s} ds \leq re^{2\mu_* r}$. On bounded functions the operator has norm $\leq Me^{-\xi} \int_0^\infty re^{-(1-2\mu_*)r} dr \leq \frac{Me^{-\xi_*}}{(1-2\mu_*)^2} < \frac{1}{4}$, so the bounded

solution has $|\zeta_\pm - 1| \le \frac{1}{3}$ (and it is the one in B by uniqueness), $\zeta_\pm \to 1$, and $\zeta'_\pm = -\int_\xi^\infty k'_\pm(\eta-\xi)w\zeta_\pm \to 0$. Then $\mathrm{Wr}(F_+,F_-) = \zeta_+(-\mu\zeta_- + \zeta'_-) - (\mu\zeta_+ + \zeta'_+)\zeta_- \to -2\mu$. F_- is the bounded solution of the Volterra equation of Lemma 3.16(a) (same kernel k_-), i.e. $F_- = z_-$. *Wronskian of* (C,S). $C = \frac{1}{2}(F_+ + F_-)$, $S = \frac{1}{2\mu}(F_+ - F_-)$, so $\mathrm{Wr}(C,S) = -\frac{1}{2\mu}\mathrm{Wr}(F_+,F_-) = 1$ for $\sigma \in (0,\mu_*^2]$; since $\sigma \mapsto \mathrm{Wr}(C,S)(\xi_*)$ is analytic, it equals 1 for all $|\sigma| < 4\mu_*^2$. *(iii).* For $\sigma > 0$: $C = \frac{1}{2}(e^{\mu\xi}\zeta_+ + e^{-\mu\xi}\zeta_-) \in [\frac{2}{3}, \frac{4}{3}]\cosh\mu\xi$; $\sigma = 0$ follows by continuity in σ at fixed ξ. For S put $\hat S = S/S_0$ ($S_0 > 0$ for $\xi > 0$). Dividing $S = S_0 + \mathcal{V}S$ by $S_0(\xi)$: $\hat S = 1 + \int_\xi^\infty \hat K\, w\hat S\, d\eta$, $\hat K = S_0(\eta-\xi)S_0(\eta)/S_0(\xi) \ge 0$. Now $S_0(r) \le re^{\mu r}$ and, because $x \mapsto (1-e^{-2\mu x})/x$ is decreasing, $\frac{\sinh\mu\eta}{\sinh\mu\xi} \le \frac{\eta}{\xi}e^{\mu(\eta-\xi)} \le (1+r)e^{\mu r}$ ($\xi \ge 1$); at $\sigma = 0$, $S_0(\eta)/S_0(\xi) = \eta/\xi \le 1 + r$. So $\hat K \le r(1+r)e^{2\mu_* r}$ and the operator has norm $\le Me^{-\xi_*}\int_0^\infty r(1+r)e^{-(1-2\mu_*)r}dr \le 4Me^{-\xi_*} \le \frac{1}{4}$ on bounded functions: the bounded solution satisfies $|\hat S - 1| \le \frac{1}{3}$, and $S_0\hat S \in B$ solves $S = S_0 + \mathcal{V}S$, so it is S. *(iv).* At $\sigma = 0$ ($C_0 = 1$, $S_0 = \xi$): $C - 1 = \int_\xi^\infty(\eta-\xi)wC \to 0$, $C' = -\int_\xi^\infty wC \to 0$, $S' - 1 = -\int_\xi^\infty wS \to 0$ (the integrands are $O(\eta e^{-\eta})$ by (iii)), and $\hat S - 1 = \int \hat K w\hat S \to 0$ since $\int_\xi^\infty \hat K|w| \le 4Me^{-\xi}$, i.e. $S/\xi \to 1$. *(v).* For $|\sigma| \le 4\mu_*^2$, $|C_0(r;\sigma)| \le \cosh(2\mu_* r) \le e^{2\mu_* r}$, so $|(\mathcal{V}_\sigma\varphi)'(\xi)| \le N(w)\|\varphi\|_B\, e^{-(1-3\mu_*)\xi}\int_0^\infty e^{2\mu_* r}e^{-(1-3\mu_*)r}dr$, i.e. $\|(\mathcal{V}_\sigma\varphi)'\|_B \le (1-5\mu_*)^{-1}N(w)e^{-\xi_*}\|\varphi\|_B \le 1.1\,N(w)e^{-\xi_*}\|\varphi\|_B$ (the same computation as in (i)). The formulas for C', S' follow by differentiating $C = C_0 + \mathcal{V}C$, $S = S_0 + \mathcal{V}S$ with $C_0' = \sigma S_0$, $S_0' = C_0$. The map $\sigma \mapsto C(\cdot;\sigma) \in B$ is analytic, hence continuous, and convergence in B implies locally uniform convergence of continuous functions; so C (and S) are jointly continuous. For $(\mathcal{V}_\sigma C)'(\xi)$ the integrand is continuous in (ξ,η,σ) and dominated by $N\|C\|_B e^{2\mu_*(\eta-\xi)}e^{-(1-3\mu_*)\eta}$, so it is continuous in (ξ,σ) by dominated convergence and analytic in σ by Morera's theorem and Fubini. Dependence on w: $C_w - C_{\tilde w} = (1-\mathcal{V}_w)^{-1}\mathcal{V}_{w-\tilde w}C_{\tilde w}$, so $\|C_w - C_{\tilde w}\|_B \le \frac{4}{3}\cdot 1.07\, e^{-\xi_*}N(w-\tilde w)\|C_{\tilde w}\|_B \le 2e^{-\xi_*}N(w-\tilde w)$ (as $\|C_{\tilde w}\|_B \le \frac{4}{3}\|C_0\|_B \le \frac{4}{3}$); and, by the derivative bound, $\|C'_w - C'_{\tilde w}\|_B \le 1.1e^{-\xi_*}\big(N(w)\|C_w - C_{\tilde w}\|_B + N(w-\tilde w)\|C_{\tilde w}\|_B\big) \le 2e^{-\xi_*}N(w-\tilde w)$. For S the same computation with $\|S_0\|_B = \sup \xi e^{-\mu_*\xi} \le 1/\mu_*$ gives the stated bounds. □

Toy example 3.39 ($w \equiv 0$)**.** Here $\mathcal{V} = 0$, $F_\pm = e^{\pm\mu\xi}$, $C = \cosh\mu\xi$, $S = \sinh(\mu\xi)/\mu$ and $z = C + \omega S = \alpha e^{\mu\xi} + \beta e^{-\mu\xi}$ with $\alpha = \frac{1}{2}(1 + \omega/\mu)$, $\beta = \frac{1}{2}(1 - \omega/\mu)$. So z is recessive iff $\omega = -\mu$ and positive for large ξ iff $\omega \ge -\mu$; at $\sigma = 0$, $z = 1 + \omega\xi$ is positive iff $\omega \ge 0$. The coefficients α, β are *not* smooth at $\sigma = 0$, while $z = C + \omega S$ is: (ω,σ), not (α,μ), are the right coordinates (Chapter 4, Proposition 4.32).

3.7. The no-swirl solutions: the classification of Li–Li–Yan

In words. *Without swirl the problem is a single Riccati equation for U, which is a Schrödinger equation $z'' = W_{\mathbf{c}}z$ with an explicit potential in disguise. Positive solutions of $z'' = W_{\mathbf{c}}z$ on the line form a convex cone. It is empty when c_3 is below the critical value $\bar c_3$, it is a single ray (spanned by the explicit function $g = (1+x)^{k_1/2}(1-x)^{k_2/2}$) when $c_3 = \bar c_3$, and it is the wedge spanned by the two Jost solutions u_-, u_+ when $c_3 > \bar c_3$. At a point x_0 the slopes z'/z of the solutions in the wedge fill the interval $[\zeta_+(\xi_0), \zeta_-(\xi_0)]$, and in terms of U this is the interval $[U^-_{\mathbf{c}}(x_0), U^+_{\mathbf{c}}(x_0)]$ of values of no-swirl solutions: the "window" that the swirl will later squeeze into a lens. The proof below is a single computation with the reduction of order (3.1); it treats $k_i > 0$ and $k_i = 0$ (that is $c_i = -1$) at the same time.*

3.7.1. From the Riccati equation to a linear equation. A solution without swirl ($f \equiv$ const, so $\psi \equiv 0$) has $Q = P_{\boldsymbol{c}}$ for some $\boldsymbol{c} \in \mathbb{R}^3$ (Chapter 2), and the reduced system (Chapter 2) collapses to the single Riccati equation

$$(1-x^2)U_x + 2xU + \tfrac{1}{2}U^2 = P_{\boldsymbol{c}}(x), \qquad x \in (-1,1). \tag{3.6}$$

A *no-swirl solution* is a C^1 solution U of (3.6) on the whole interval $(-1,1)$; no condition is imposed at $x = \pm 1$. We write $\mathcal{J} = \{c \in \mathbb{R}^3 :\ c_1 \geq -1,\ c_2 \geq -1,\ c_3 \geq \bar{c}_3(c_1,c_2)\}$, so $\operatorname{int}\mathcal{J} = \{c_1, c_2 > -1,\ c_3 > \bar{c}_3\}$.

Lemma 3.40 (dictionary)**.** *Let $\boldsymbol{c} \in \mathbb{R}^3$. If $z > 0$ solves $z'' = W_{\boldsymbol{c}} z$ on $\mathbb{R}$, then $U = 2(z'/z - \tanh\xi)$, $\xi = \operatorname{artanh} x$, is a no-swirl solution, and sz ($s > 0$) gives the same U. Conversely every no-swirl solution arises this way, from a $z > 0$ unique up to a positive factor.*

Proof. Since $dx/d\xi = 1 - x^2$, we have $(1-x^2)U_x = U'$ ($' = d/d\xi$), and (3.6) reads $U' + 2TU + \frac{1}{2}U^2 = P_{\boldsymbol{c}}$ with $T = \tanh\xi = x$. Put $h = U/2 + T$ (so $U = 2(h - T)$). Using $T' = 1 - T^2$:

$$U' + 2TU + \tfrac{1}{2}U^2 = 2h' - 2(1-T^2) + 4Th - 4T^2 + 2h^2 - 4hT + 2T^2 = 2(h' + h^2) - 2.$$

Hence (3.6) is equivalent to $h' + h^2 = 1 + \frac{1}{2}P_{\boldsymbol{c}} = W_{\boldsymbol{c}}$. If $z > 0$ solves $z'' = W_{\boldsymbol{c}} z$ then $h = z'/z$ satisfies $h' = z''/z - h^2 = W_{\boldsymbol{c}} - h^2$. Conversely, if h is C^1 on $\mathbb{R}$ with $h' + h^2 = W_{\boldsymbol{c}}$, then $z = \exp\int_0^\xi h$ is positive, $z' = hz$, $z'' = (h' + h^2)z = W_{\boldsymbol{c}} z$; and $h = z'/z$ determines z up to a positive factor. □

(The same computation underlies Lemma 2.14, where it is done for the full system.)

If $c_i < -1$ then $W_{\boldsymbol{c}} \to 1 + c_i < 0$ at the corresponding end (Lemma 3.29) and by Lemma 3.2 no positive solution can exist on a half-line there. Thus $c_i \geq -1$ is necessary, and from now on in this section $c_1, c_2 \geq -1$ (so $k_i = \sqrt{1 + c_i} \geq 0$, $\bar{c}_3$, ε, W_{crit} and the critical function g of Lemma 3.30 are defined).

3.7.2. Reduction of order and principal solutions.

Toy example 3.41 (reduction of order when $g \equiv 1$)**.** The simplest instance is $\boldsymbol{c} = (-1,-1,c_3)$: then $k_1 = k_2 = 0$, $\bar{c}_3 = 0$, $g \equiv 1$ and $z'' = W_{\boldsymbol{c}} z$ with $W_{\boldsymbol{c}} = \frac{c_3}{2}\operatorname{sech}^2\xi$ is already of the form $\varphi'' = q\varphi$ with $q = \frac{c_3}{2}\operatorname{sech}^2\xi$ (here $\varepsilon = c_3$).

$c_3 < 0$ ($q < 0$). A positive solution would be a positive concave function on $\mathbb{R}$, hence constant, hence (as $\varphi'' = q\varphi \neq 0$) not a solution: there is none. For $c_3 = -4$ this is visible in closed form: $\tanh\xi$ and $\xi\tanh\xi - 1$ solve $\varphi'' = -2\operatorname{sech}^2\xi\,\varphi$, and $a\tanh\xi + b(\xi\tanh\xi - 1)$ has the value $-b$ at 0 and grows like $b|\xi|$ at both ends, so it vanishes somewhere for every $(a,b) \neq 0$.

$c_3 = 0$ ($q = 0$). $\varphi = a + b\xi$ is positive on $\mathbb{R}$ iff $b = 0$: a single ray.

$c_3 > 0$ ($q > 0$). Solutions are convex where positive. The solution bounded at $-\infty$ ($\chi_- \to 1$) is positive, increasing and grows at least linearly at $+\infty$; its mirror image has the same properties at the other end; and every positive solution is a nonnegative combination of the two. The general proof below follows exactly this pattern, with "concave positive functions on $\mathbb{R}$ are constant" replaced by "$g^2\varphi'$ is monotone and $\int g^{-2}$ diverges at both ends".

By Lemma 3.30(a), $W_{\boldsymbol{c}} = W_{\mathrm{crit}} + \frac{\varepsilon}{2}\operatorname{sech}^2\xi$, so by Corollary 3.6 the substitution $z = g\varphi$ turns $z'' = W_{\boldsymbol{c}} z$ into

$$(g^2\varphi')' = q\,\varphi, \qquad q := \tfrac{\varepsilon}{2}\,g^2\operatorname{sech}^2\xi, \qquad |q| \leq 2|\varepsilon|\,4^{\bar{k}}e^{-2|\xi|}. \tag{3.7}$$

The coefficient q is integrable and has the sign of ε; $1/g^2$ is not integrable at either end (Lemma 3.30). These two facts are all that the classification uses.

Lemma 3.42 (principal solutions)**.** *Let $c_1, c_2 \geq -1$ and $c_3 \in \mathbb{R}$ be arbitrary. (a) There is exactly one solution p_- of $z'' = W_{\boldsymbol{c}} z$ with $p_-/g \to 1$ as $\xi \to -\infty$. Writing $p_- = g\chi_-$ one has $\chi_-(\xi) = 1 + O(e^{2\xi})$, $\chi_-' = O(e^{2\xi})$ as $\xi \to -\infty$ and $g^2\chi_-'(\xi) = \int_{-\infty}^{\xi} q\chi_-$. (b) Every solution z for which z/g is bounded as $\xi \to -\infty$ is a multiple of p_-. (c) Mirror statements hold at $+\infty$: there is exactly one solution p_+ with $p_+/g \to 1$ at $+\infty$, $p_+ = g\chi_+$, $\chi_+ = 1 + O(e^{-2\xi})$, $g^2\chi_+'(\xi) = -\int_{\xi}^{\infty} q\chi_+$, and every solution with z/g bounded at $+\infty$ is a multiple of p_+. (d) If $k_1 > 0$, p_- is $2^{\bar{k}}$ times the Jost solution recessive at $-\infty$ of Lemma 3.21 (normalised by $e^{-k_1\xi}u \to 1$, since $g \sim 2^{\bar{k}} e^{k_1\xi}$); similarly $p_+ = 2^{\bar{k}}\times$ the Jost solution recessive at $+\infty$ if $k_2 > 0$. If $k_1 = 0$ then p_- is the (unique up to a factor) bounded solution at $-\infty$.*

Proof. (a) *Existence.* By (3.4), for $\eta \leq s \leq \xi \leq 0$ we have $g(s)^{-2} \leq e^{-2k_1 s} \leq e^{-2k_1\eta}$ ($k_1 \geq 0$) and $|q(\eta)| \leq 2|\varepsilon|4^{\bar{k}} e^{2\eta} e^{2k_1\eta}$ (using $g(\eta) \leq 2^{\bar{k}} e^{k_1\eta}$ and $\operatorname{sech}^2\eta \leq 4e^{2\eta}$). Hence the kernel $\kappa(\xi,\eta) = (\mathcal{G}_{\mathrm{int}}(\xi) - \mathcal{G}_{\mathrm{int}}(\eta))q(\eta)$, $\eta \leq \xi \leq 0$, satisfies $|\kappa| \leq C_0(\xi - \eta)e^{2\eta}$ with $C_0 = 2|\varepsilon|4^{\bar{k}}$. On the space X of bounded continuous functions on $(-\infty, \xi_2]$ define $(\mathcal{V}\varphi)(\xi) = \int_{-\infty}^{\xi} \kappa(\xi,\eta)\varphi(\eta)\,d\eta$. As $\int_{-\infty}^{\xi}(\xi-\eta)e^{2\eta}d\eta = \frac{1}{4}e^{2\xi}$, $\|\mathcal{V}\| \leq \frac{1}{4}C_0 e^{2\xi_2} \leq \frac{1}{2}$ once $\xi_2 \leq 0$ is chosen suitably. By Banach's fixed point theorem, $\varphi = 1 + \mathcal{V}\varphi$ has a unique solution $\chi_- \in X$, with $\|\chi_-\| \leq 2$ and $|\chi_- - 1| \leq \frac{1}{2}C_0 e^{2\xi}$. Differentiating the integral equation $\chi_-(\xi) = 1 + \int_{-\infty}^{\xi}\big(\mathcal{G}_{\mathrm{int}}(\xi) - \mathcal{G}_{\mathrm{int}}(\eta)\big)q(\eta)\chi_-(\eta)\,d\eta$ by Leibniz' rule (the boundary term has the factor $\mathcal{G}_{\mathrm{int}}(\xi) - \mathcal{G}_{\mathrm{int}}(\xi) = 0$, and $\mathcal{G}_{\mathrm{int}}' = g^{-2}$) gives $\chi_-' = g^{-2}m$ with $m(\xi) = \int_{-\infty}^{\xi} q\chi_-$; as $\|\chi_-\| \leq 2$, $|m(\xi)| \leq 4|\varepsilon|4^{\bar{k}}\int_{-\infty}^{\xi} e^{2(1+k_1)\eta}d\eta \leq Ce^{2(1+k_1)\xi}$ and $|\chi_-'| \leq e^{-2k_1\xi}|m| \leq Ce^{2\xi}$ (so $\chi_- \in C^2$ since $\chi_-' = g^{-2}m$ is C^1). Moreover $(g^2\chi_-')' = m' = q\chi_-$, so $p_- = g\chi_-$ solves $z'' = W_{\boldsymbol{c}}z$ on $(-\infty,\xi_2]$ by Corollary 3.6; extend p_- to $\mathbb{R}$ as the solution of the linear equation. The extension still satisfies (3.7), and $g^2\chi_-'$ and $\int_{-\infty}^{\xi} q\chi_-$ have the same derivative and agree for $\xi \leq \xi_2$, so they agree on $\mathbb{R}$.

Characterisation by boundedness (this gives (b) and the uniqueness in (a)). Let $z = g\varphi$ be a solution with φ bounded on $(-\infty,\xi_2]$, and $m = g^2\varphi'$. Then $m' = q\varphi$ is integrable, so $m \to m_0$ at $-\infty$. If $m_0 \neq 0$, then $|\varphi'| \geq \frac{|m_0|}{2}g^{-2}$ with a fixed sign near $-\infty$ and φ would be unbounded because $\mathcal{G}_{\mathrm{int}}(\xi) \to -\infty$. Hence $m_0 = 0$, $m(\xi) = \int_{-\infty}^{\xi} q\varphi$, so $|\varphi'| = g^{-2}|m| \leq Ce^{2\xi}$ and $\lambda := \lim_{-\infty}\varphi$ exists; then $\varphi - \lambda = \int_{-\infty}^{\xi}\varphi' = \int_{-\infty}^{\xi} g^{-2}(s)\int_{-\infty}^{s} q\varphi\,d\eta\,ds = \mathcal{V}\varphi$ (exchange the order of integration; $\int_{\eta}^{\xi} g^{-2} = \mathcal{G}_{\mathrm{int}}(\xi) - \mathcal{G}_{\mathrm{int}}(\eta)$). So $\varphi = \lambda + \mathcal{V}\varphi$, i.e. $\varphi = \lambda\chi_-$ by uniqueness of the fixed point. This proves (b); and if $z/g \to 1$ then $\lambda = 1$. (c) follows by the reflection (Lemma 3.7, Remark 3.36): $p_+^{\boldsymbol{c}}(\xi) = p_-^{\boldsymbol{c}^*}(-\xi)$, i.e. $\chi_+(\xi) = \chi_-^{\boldsymbol{c}^*}(-\xi)$. (d) If $k_1 > 0$, then $p_- = O(e^{k_1\xi})$ is recessive, hence a multiple of the Jost solution (uniqueness of the recessive solution, Lemma 3.21); the factor is $2^{\bar{k}}$ because $g \sim 2^{\bar{k}}e^{k_1\xi}$ at $-\infty$. For $k_1 = 0$ the claim is (b). □

3.7.3. The cone of positive solutions.

Proposition 3.43 (the linear picture of the no-swirl problem)**.** *Let $\boldsymbol{c} \in \mathbb{R}^3$.*

(a) If $c_1 < -1$ or $c_2 < -1$, or if $c_1, c_2 \geq -1$ and $\varepsilon < 0$, then $z'' = W_{\boldsymbol{c}}z$ has no positive solution on $\mathbb{R}$.

(b) If $c_1, c_2 \geq -1$ and $\varepsilon = 0$, the positive solutions are exactly the positive multiples of g; $p_- = p_+ = g$.

(c) If $c_1, c_2 \geq -1$ and $\varepsilon > 0$ (that is, $\mathbf{c} \in \mathcal{J}$ and $c_3 > \bar{c}_3$, which includes the faces $c_i = -1$): the solutions $p_\mp$ of Lemma 3.42 are positive on $\mathbb{R}$, p_-/g increases from 1 to ∞ and p_+/g decreases from ∞ to 1; p_-/p_+ increases from 0 to ∞, $\mathrm{Wr}(p_+, p_-) > 0$; and the positive solutions of $z'' = W_{\mathbf{c}} z$ are exactly

$$z = a\,p_- + b\,p_+, \qquad a, b \geq 0,\ (a,b) \neq (0,0).$$

For every $\xi_0 \in \mathbb{R}$ the slope $z'(\xi_0)/z(\xi_0)$ of these solutions fills the closed interval $[\zeta_+(\xi_0), \zeta_-(\xi_0)]$, $\zeta_\pm = p'_\pm/p_\pm$, $\zeta_+(\xi_0) < \zeta_-(\xi_0)$, each value being taken by exactly one ray $\{sz : s > 0\}$.

If $k_1, k_2 > 0$, the potential $W_{\mathbf{c}}$ is subcritical in case (c), critical in case (b) and supercritical for $\varepsilon < 0$, in the sense of Lemma 3.32 (recall that $p_\mp$ are positive multiples of the Jost solutions: Lemma 3.42(d)).

PROOF. For $k_1, k_2 > 0$ the trichotomy and the description of the positive solutions below are those of Lemma 3.32(b), Lemma 3.35(b) and Lemma 3.21, proved there by comparison of Jost solutions; the argument here is independent of them, uses only g and the reduction of order (3.7), and covers $k_i = 0$ as well. (a) If $c_2 < -1$ then $W_{\mathbf{c}} \to 1 + c_2 < 0$ at $+\infty$ (Lemma 3.29), so $W_{\mathbf{c}} \leq -\mu_2^2 < 0$ on some $[\xi_1, \infty)$ and no positive solution exists there by Lemma 3.2; likewise for $c_1 < -1$ at $-\infty$ (Lemma 3.7, Remark 3.36). Let now $c_1, c_2 \geq -1$, $\varepsilon < 0$, and suppose $z = g\varphi$ is positive. Put $m = g^2\varphi'$, so $m' = q\varphi < 0$ by (3.7): m is strictly decreasing. If $m(\xi_1) < 0$ for some ξ_1, then $m \leq m(\xi_1) =: -\delta < 0$ on $[\xi_1, \infty)$ and $\varphi(\xi) \leq \varphi(\xi_1) - \delta(\mathcal{G}_{\mathrm{int}}(\xi) - \mathcal{G}_{\mathrm{int}}(\xi_1)) \to -\infty$ as $\xi \to \infty$, contradicting $\varphi > 0$. So $m \geq 0$ everywhere, hence $m > 0$ (m strictly decreasing). Then for $\xi \leq 0$: $m(\xi) \geq m(0) > 0$ and $\varphi(0) - \varphi(\xi) \geq m(0)(\mathcal{G}_{\mathrm{int}}(0) - \mathcal{G}_{\mathrm{int}}(\xi)) \to \infty$ as $\xi \to -\infty$, so $\varphi(\xi) \to -\infty$: contradiction. In particular the solution p_- of Lemma 3.42, which is positive near $-\infty$ (as $p_-/g \to 1$), is not positive on $\mathbb{R}$: it has a zero. For $k_1, k_2 > 0$ this is the supercritical case of Lemma 3.32.

(b) Now $q \equiv 0$, so $g^2\varphi' = \mathrm{const} =: C$ and $\varphi = A + C\mathcal{G}_{\mathrm{int}}(\xi)$ with $\mathcal{G}_{\mathrm{int}}(\pm\infty) = \pm\infty$. If $C > 0$, $\varphi \to -\infty$ at $-\infty$; if $C < 0$, $\varphi \to -\infty$ at $+\infty$; so $C = 0$ and $z = Ag$, $A > 0$. Also $p_\pm/g \to 1$ forces $p_\pm = g$.

(c) Let $q > 0$. *Positivity and monotonicity of p_-.* Suppose χ_- has a first zero ξ_*. On $(-\infty, \xi_*)$, $\chi_- > 0$ and so $g^2\chi'_-(\xi) = \int_{-\infty}^{\xi} q\chi_- > 0$ (Lemma 3.42(a)): χ_- is increasing there, starting from the limit 1, so $\chi_-(\xi_*) > 1$: contradiction. Hence $\chi_- > 0$ on $\mathbb{R}$, $\chi'_- > 0$, $\chi_- > 1$. With $m_- := g^2\chi'_- = \int_{-\infty}^{\xi} q\chi_-$ increasing and positive, for $\xi \geq 0$: $\chi_-(\xi) \geq 1 + m_-(0)\mathcal{G}_{\mathrm{int}}(\xi) \to \infty$.

Mirror. $\chi_+ > 1$, $\chi'_+ < 0$, $\chi_+ \to \infty$ at $-\infty$ (reflect).

Wronskian and ratio. $(p_-/p_+) = \chi_-/\chi_+$ has derivative $(\chi'_-\chi_+ - \chi_-\chi'_+)/\chi_+^2 > 0$, tends to $1/\infty = 0$ at $-\infty$ and to $\infty/1 = \infty$ at $+\infty$; and $\mathrm{Wr}(p_+, p_-) = g^2(\chi_+\chi'_- - \chi'_+\chi_-) > 0$. In particular p_-, p_+ are linearly independent and form a basis of the two-dimensional solution space.

Positive solutions. If $a, b \geq 0$, not both 0, then $z = g(a\chi_- + b\chi_+) \geq (a+b)g > 0$. Conversely let $z = ap_- + bp_+ > 0$. At $+\infty$: $\chi_- \to \infty$, $\chi_+ \to 1$; if $a < 0$ then $a\chi_- + b\chi_+ \to -\infty$: contradiction. At $-\infty$ the roles are exchanged and $b < 0$ is impossible.

Slopes. For $a, b \geq 0$ and ξ_0 fixed, $z'/z = \tau\zeta_- + (1-\tau)\zeta_+$ with

$$\tau = \frac{ap_-(\xi_0)}{ap_-(\xi_0) + bp_+(\xi_0)} \in [0,1],$$

and τ runs through $[0,1]$ monotonically and bijectively with the ray $(a:b) \in [0,\infty]$. Finally $\zeta_- - \zeta_+ = \mathrm{Wr}(p_+, p_-)/(p_+p_-) > 0$.

Growth of p_-. In case (c), $p_- = g\chi_- \geq g\,(1 + m_-(0)\mathcal{G}_{\mathrm{int}})$ for $\xi \geq 0$, with $g \geq e^{-k_2\xi}$ and $\mathcal{G}_{\mathrm{int}}(\xi) \geq 4^{-\bar{k}} \int_0^\xi e^{2k_2 s} ds$; hence $p_- \geq c\, e^{k_2\xi}$ if $k_2 > 0$ and $p_- \geq c\,\xi$ if $k_2 = 0$: p_- is positive and unbounded, whereas $p_+ = O(e^{-k_2\xi})$ is recessive. □

THEOREM 3.44 (classification of the no-swirl solutions [25]). *Let $\boldsymbol{c} \in \mathbb{R}^3$ and consider (3.6).*

(i) If $\boldsymbol{c} \notin \mathcal{J}$ there is no solution on $(-1,1)$.

(ii) If $\boldsymbol{c} \in \mathcal{J}$ there are solutions $U_{\boldsymbol{c}}^- \leq U_{\boldsymbol{c}}^+$ on $(-1,1)$ such that, for any $x_0 \in (-1,1)$, the solutions on $(-1,1)$ are exactly those with $U_{\boldsymbol{c}}^-(x_0) \leq U(x_0) \leq U_{\boldsymbol{c}}^+(x_0)$, and for each γ in this interval there is exactly one solution $U_{\boldsymbol{c},\gamma}$ with $U(x_0) = \gamma$. The solutions are strictly ordered: if $U(x_0) < \tilde{U}(x_0)$ for one x_0 then $U < \tilde{U}$ on $(-1,1)$. Moreover $U_{\boldsymbol{c}}^- = U_{\boldsymbol{c}}^+$ if and only if $c_3 = \bar{c}_3$. In terms of z: $U_{\boldsymbol{c}}^\pm = 2(\zeta_\mp - \tanh\xi)$, where $\zeta_\mp = p_\mp'/p_\mp$ are the log-derivatives of the solutions of Lemma 3.42 (and $U_{\boldsymbol{c}}^+ = U_{\boldsymbol{c}}^- = 2(g'/g - \tanh\xi)$ if $c_3 = \bar{c}_3$).

(iii) If $c_3 > \bar{c}_3$: $U(-1) = 2 + 2k_1$ if $U = U_{\boldsymbol{c}}^+$ and $U(-1) = 2 - 2k_1$ otherwise; $U(1) = -2 - 2k_2$ if $U = U_{\boldsymbol{c}}^-$ and $U(1) = -2 + 2k_2$ otherwise.

(iv) If $c_3 = \bar{c}_3$, the unique solution is $U^* = (1+k_1)(1-x) - (1+k_2)(1+x)$, with $U^*(-1) = 2 + 2k_1$, $U^*(1) = -2 - 2k_2$.

Here $k_i = \sqrt{1+c_i}$, and the cases $c_1 = -1$ or $c_2 = -1$ ($k_i = 0$) are included.

PROOF. (i) If $\boldsymbol{c} \notin \mathcal{J}$, either some $c_i < -1$ or $c_1, c_2 \geq -1$ and $\varepsilon < 0$; by Proposition 3.43(a) there is no positive z, hence no no-swirl solution by Lemma 3.40.

(ii) Let $\boldsymbol{c} \in \mathcal{J}$. By Lemma 3.40 the solutions correspond to the rays of positive solutions of $z'' = W_{\boldsymbol{c}} z$, and by Proposition 3.43(b),(c) the slope $h(\xi_0) = z'/z(\xi_0)$ fills $[\zeta_+(\xi_0), \zeta_-(\xi_0)]$ with exactly one ray for each value (a single ray and a single value $g'/g(\xi_0)$ if $\varepsilon = 0$). Since $U(x_0) = 2(h(\xi_0) - x_0)$, $\xi_0 = \operatorname{artanh} x_0$, the first claims follow with $U_{\boldsymbol{c}}^\pm = 2(\zeta_\mp - \tanh\xi)$ (the case $\varepsilon = 0$: $\zeta_\pm = g'/g$). Strict ordering: (3.6) is $U_x = (P_{\boldsymbol{c}} - 2xU - \frac{1}{2}U^2)/(1-x^2)$ with smooth right side on $(-1,1)$, so two different solutions never meet; by continuity one stays above the other. $U_{\boldsymbol{c}}^- = U_{\boldsymbol{c}}^+$ iff $\zeta_+ = \zeta_-$, which holds for $\varepsilon = 0$ and fails for $\varepsilon > 0$ by Proposition 3.43(c).

(iv) For $\varepsilon = 0$: $U^* = 2(g'/g - \tanh\xi) = 2\big(\frac{k_1 - k_2}{2} - (\bar{k}+1)x\big) = (k_1 - k_2) - (k_1 + k_2 + 2)x = (1+k_1)(1-x) - (1+k_2)(1+x)$, and $U^*(\mp 1)$ are as stated ($g'/g \to k_1$, $-k_2$).

(iii) Let $\varepsilon > 0$. For $z = ap_- + bp_+$ write $z = g\varphi$, $\varphi = a\chi_- + b\chi_+$ and $h = z'/z = g'/g + \varphi'/\varphi$, $U = 2(h - \tanh\xi)$, so $U(-1) = 2h(-\infty) + 2$ and $U(1) = 2h(+\infty) - 2$. We determine $h(\pm\infty)$.

Claim 1: $h \to -k_2$ at $+\infty$ for $z = p_+$. $h = g'/g + \chi_+'/\chi_+$, $g'/g \to -k_2$, $\chi_+ \to 1$ and, by Lemma 3.42(c) and its mirror estimate, $\chi_+' = g^{-2} \cdot O(e^{-2(1+k_2)\xi}) = O(e^{-2\xi}) \to 0$ ($g^{-2} \leq e^{2k_2\xi}$ for $\xi \geq 0$).

Claim 2: $h \to k_2$ at $+\infty$ if $a > 0$. First, $m_-(\xi) = \int_{-\infty}^\xi q\chi_-$ is increasing and bounded: for $\xi \geq 0$, since $\chi_-(s) \leq \chi_-(0) + m_-(s)\mathcal{G}_{\mathrm{int}}(s)$ (m_- is increasing),

$m_-' = q\chi_- \le \chi_-(0)q + qG\,m_-$, and Gronwall's inequality gives $m_-(\xi) \le \big(m_-(0) + \chi_-(0)\int_0^\infty q\big)\exp\int_0^\infty qG$, finite because $q \le 2\varepsilon 4^{\bar k}e^{-2(1+k_2)s}$ and $\mathcal{G}_{\rm int}(s) \le se^{2k_2 s}$, so $qG \le 2\varepsilon 4^b s e^{-2s}$. So $m_- \uparrow \bar m_- > 0$. Also $m_+ = g^2\chi_+' = -\int_\xi^\infty q\chi_+ \to 0$. Hence $m := g^2\varphi' = am_- + bm_+ \to \bar m := a\bar m_- > 0$, and $\varphi \to \infty$. Given $\delta \in (0,1)$ choose ξ_δ with $(1-\delta)\bar m \le m \le (1+\delta)\bar m$ on $[\xi_\delta,\infty)$; integrating $\varphi' = g^{-2}m$, $\varphi(\xi) \in \varphi(\xi_\delta) + [(1-\delta),(1+\delta)]\,\bar m\,(\mathcal{G}_{\rm int}(\xi) - \mathcal{G}_{\rm int}(\xi_\delta))$ and since $\mathcal{G}_{\rm int} \to \infty$ this yields $\varphi/(\bar m\mathcal{G}_{\rm int}) \to 1$. Hence $\varphi'/\varphi = \dfrac{g^{-2}m}{\varphi} = \dfrac{G'}{G}\cdot\dfrac{m}{\bar m}\cdot\dfrac{\bar m\mathcal{G}_{\rm int}}{\varphi}$ has the same limit as $\mathcal{G}_{\rm int}'/\mathcal{G}_{\rm int}$. Now $g^{-2} = e^{-(k_1-k_2)\xi}(\cosh\xi)^{k_1+k_2} = 4^{-\bar k}e^{2k_2\xi}(1+e^{-2\xi})^{2\bar k}$. For $k_2 > 0$ this gives $\mathcal{G}_{\rm int}(\xi) \sim \frac{4^{-\bar k}}{2k_2}e^{2k_2\xi}$ and $\mathcal{G}_{\rm int}'/\mathcal{G}_{\rm int} \to 2k_2$; for $k_2 = 0$, $g^{-2} \to 4^{-\bar k}$, $\mathcal{G}_{\rm int} \sim 4^{-\bar k}\xi$ and $\mathcal{G}_{\rm int}'/\mathcal{G}_{\rm int} \sim 1/\xi \to 0 = 2k_2$. Thus $h \to -k_2 + 2k_2 = k_2$.

At $-\infty$. The reflection $\xi \mapsto -\xi$ exchanges $p_- \leftrightarrow p_+$, $k_1 \leftrightarrow k_2$ and changes the sign of h: so $h \to k_1$ for $z = p_-$ and $h \to -k_1$ if $b > 0$. Now $U_{\boldsymbol{c}}^+ \leftrightarrow z = p_-$ $(a > 0, b = 0)$: $U(-1) = 2 + 2k_1$, $U(1) = -2 + 2k_2$; $U_{\boldsymbol{c}}^- \leftrightarrow z = p_+$: $U(-1) = 2 - 2k_1$, $U(1) = -2 - 2k_2$; every other solution $(a, b > 0)$: $U(-1) = 2 - 2k_1$, $U(1) = -2 + 2k_2$. This is (iii). □

Remark 3.45 (the faces $c_i = -1$; two independent proofs)**.** (1) An alternative, more elementary but longer, proof for $c_1, c_2 > -1$ uses the Riccati dichotomy and Levinson-type asymptotic integration at both ends, with a separate treatment of the three signs of ε; the faces $c_1 = -1$ or $c_2 = -1$ are also treated in Li–Li–Yan [25]. The proof above treats all $k_i \ge 0$ at once because it uses only the two properties of g that survive at $k_i = 0$: $q \in L^1$ and $\mathcal{G}_{\rm int} \to \pm\infty$ (no exponential rate is used). In particular the faces $c_i = -1$ are proved here. (2) For $k_i > 0$ the trichotomy and the cone of Proposition 3.43 are also those of Lemma 3.32(b) and Lemma 3.35(b) (comparison of Jost solutions with g). The two proofs are independent: that one applies to every potential in $\mathcal{W}(k_1, k_2)$ with $k_i > 0$, this one to $W_{\boldsymbol{c}}$ with all $k_i \ge 0$, and it also gives the slopes. (3) The pole values in (iii) are the values $2 \pm 2k_1$, $-2 \pm 2k_2$ that the pole theory of Chapter 4 assigns to the solutions of type G and to the edge solutions; no pole theory is used here.

Example 3.46 (the no-swirl solutions in six cases)**.** Recall the critical function $g = (1+x)^{k_1/2}(1-x)^{k_2/2}$ with $g'' = W_{\rm crit}g$ (Lemma 3.30; equivalently $g = y_c\cosh\xi$ with $y_c = (1+x)^{r_1}(1-x)^{r_2}$, $r_i = \frac12(1+k_i)$), and the extremal solution $U^* = (1+k_1)(1-x) - (1+k_2)(1+x)$ at $c_3 = \bar c_3$.

(a) *The rest state $\boldsymbol{c} = 0$ and Landau's solutions.* $k_1 = k_2 = 1$, $\bar c_3 = -4 < 0 = c_3$, $W_0 \equiv 1$, $u_- = e^\xi$, $u_+ = e^{-\xi}$: positive solutions are $z = ae^\xi + be^{-\xi}$, $a, b \ge 0$. With $e^{\pm\xi}\sqrt{1-x^2} = 1 \pm x$ one finds

$$\frac{z'}{z} = \frac{a(1+x) - b(1-x)}{a(1+x) + b(1-x)},$$

$$U = 2\Big(\frac{z'}{z} - x\Big) = \frac{2(a-b)(1-x^2)}{(a+b) + (a-b)x} = \frac{2(1-x^2)}{\lambda_L + x} \quad (a \ne b),$$

with $\lambda_L = (a+b)/(a-b)$, $|\lambda_L| \ge 1$: Landau's solutions [21, 45] (the Landau parameter λ_L is not the constant λ of the first integral). $a = b$ gives $U \equiv 0$; $b = 0$ $(\lambda_L = 1)$ gives the edge $U_0^+ = 2(1-x)$, $a = 0$ $(\lambda_L = -1)$ gives $U_0^- = -2(1+x)$. At $x_0 = 0$, $U(0) = 2(a-b)/(a+b)$ fills $[-2, 2]$.

Pole values (Theorem 3.44(iii)): $U_0^+(-1) = 4 = 2 + 2k_1$, $U_0^-(1) = -4$, and every other solution has $U(\mp 1) = 0$.

(b) *The critical case* $\boldsymbol{c} = (0, 0, -4)$. $W = 1 - 2\operatorname{sech}^2 \xi$ is the Pöschl–Teller potential of Toy example 3.34 with $\ell = k = 1$; $g = \operatorname{sech} \xi$, the cone is the ray of g and $U^* = -4x$. (Check: $(1 - x^2)(-4) + 2x(-4x) + 8x^2 = -4(1 - x^2) = P_{\boldsymbol{c}}$.)

(c) *A potential below the critical value,* $\boldsymbol{c} = (0, 0, -12)$. $W = 1 - 6\operatorname{sech}^2 \xi$ is the same toy with $\ell = 2 > k = 1$: u_- has a zero at $\xi = 0$ (closed form in the toy), so no no-swirl solution exists. More generally $\boldsymbol{c} = (0, 0, c_3)$ with $c_3 < 0$ is the toy with $k = 1$ and $\ell(\ell + 1) = -c_3/2$, so (for $c_3 < 0$) $c_3 > -4 \Leftrightarrow \ell < 1$ is subcritical, in agreement with $\bar{c}_3(0, 0) = -4$.

(d) *An end exponent below the threshold.* For $\boldsymbol{c} = (0, -2, 0)$, $W_{\boldsymbol{c}} = -\tanh \xi \to -1$ at $+\infty$: all solutions oscillate, there is no no-swirl solution ($c_2 < -1$).

(e) *A face* $c_1 = -1$ *(the case* $k_1 = 0$*).* Take $\boldsymbol{c} = (-1, -1, c_3)$, $k_1 = k_2 = 0$, $\bar{c}_3 = 0$, $g \equiv 1$, $W_{\boldsymbol{c}} = \frac{c_3}{2} \operatorname{sech}^2 \xi$. For $c_3 < 0$ no solution exists; for $c_3 = 0$, $W \equiv 0$, $z = a + b\xi$ is positive on $\mathbb{R}$ iff $b = 0$, and $U^* = -2x$ is the only solution (check: $(1 - x^2)(-2) - 4x^2 + 2x^2 = -2 = P_{\boldsymbol{c}}$); for $c_3 > 0$ the solution bounded at $-\infty$ is positive, grows linearly at $+\infty$, and the window is open with $U(\mp 1) = \pm 2$ for all solutions by Theorem 3.44(iii) ($k_i = 0$).

(f) *An asymmetric critical case.* At $\boldsymbol{c} = (c_1, c_2, \bar{c}_3)$ with $k_1 \neq k_2$, $g'/g = \frac{k_1 - k_2}{2} - b \tanh \xi$, so $U^*(0) = k_1 - k_2$ is not zero: the symmetric center $U^* = 0$ of (a) is an artifact of $c_1 = c_2$.

$\boldsymbol{c}$	(k_1, k_2)	$\bar{c}_3$	ε	no-swirl solutions	$U_{\boldsymbol{c}}^-(0)$	$U_{\boldsymbol{c}}^+(0)$
$(0, 0, 0)$	$(1, 1)$	-4	4	window	-2	2
$(0, 0, -2)$	$(1, 1)$	-4	2	window	-1.1350	1.1350
$(0, 0, -4)$	$(1, 1)$	-4	0	one solution $U^* = -4x$	0	0
$(0, 0, -12)$	$(1, 1)$	-4	-8	none	–	–
$(-\frac{3}{4}, -\frac{3}{4}, \frac{3}{10})$	$(\frac{1}{2}, \frac{1}{2})$	-1.5	1.8	window	-1.1668	1.1668
$(2, -\frac{1}{2}, \frac{1}{5})$	$(\sqrt{3}, \frac{1}{\sqrt{2}})$	-5.4139	5.6139	window	-2.3276	2.9586
$(-1, -1, 1)$	$(0, 0)$	0	1	window (face $c_i = -1$)	-0.8445	0.8445
$(-1, -1, 3)$	$(0, 0)$	0	3	window (face $c_i = -1$)	-1.9898	1.9898
$(-1, -1, 0)$	$(0, 0)$	0	0	one solution $U^* = -2x$	0	0
$(0, -2, 0)$	–	–	–	none ($c_2 < -1$)	–	–

TABLE 3.1. The classification in ten cases: the number of no-swirl solutions and the window at $x_0 = 0$ (values computed from Proposition 3.47; the rows with $k_i = 0$ were also checked by direct shooting of the solution bounded at $-\infty$, see Numerical observation 3.49). Code: `code/ch03_line/verify_table_cases.py`.

Figure 3.3 shows the cone of positive solutions and the window of the no-swirl problem.

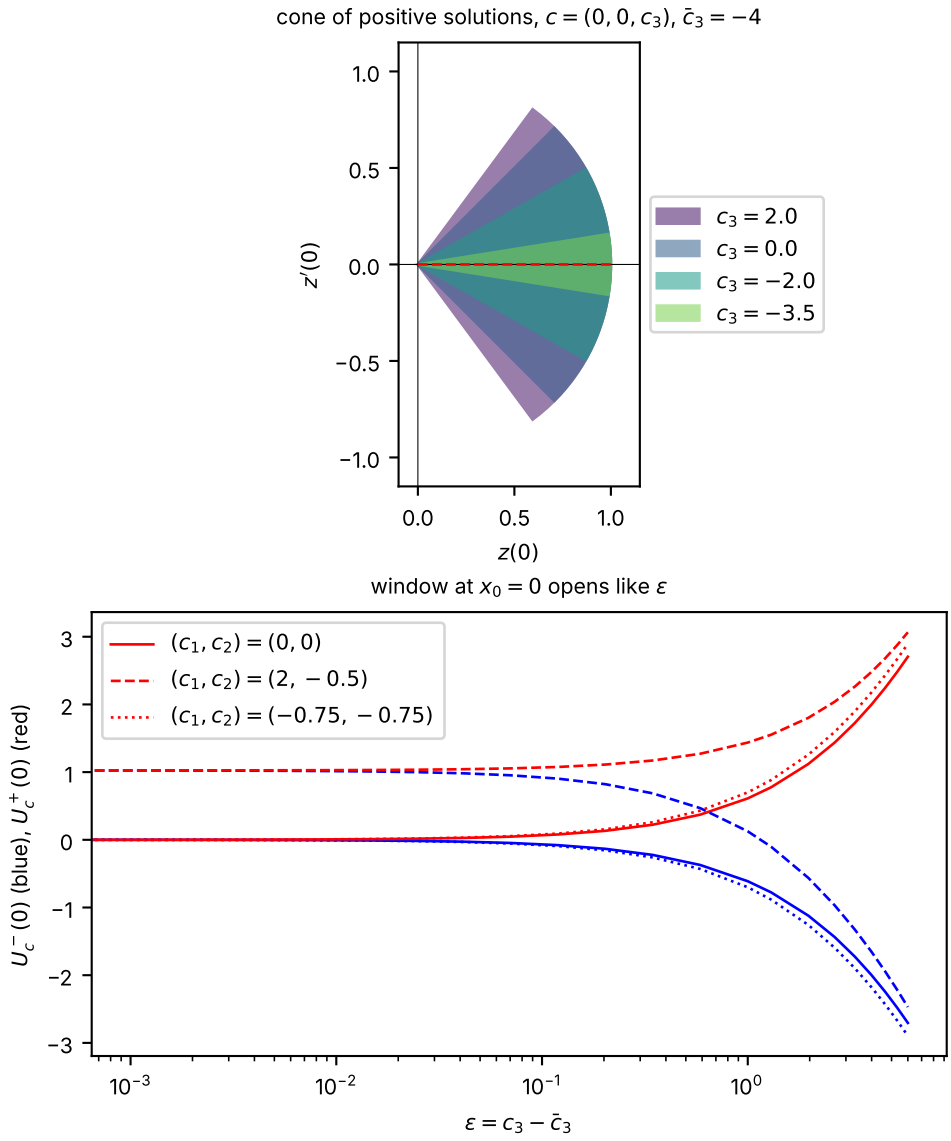


FIGURE 3.3. Left: the cone of positive solutions of $z'' = W_{\boldsymbol{c}} z$ in the plane $(z(0), z'(0))$ for $\boldsymbol{c} = (0, 0, c_3)$ ($\bar{c}_3 = -4$); it is the wedge between the slopes $\zeta_+(0)$ and $\zeta_-(0)$ and closes to the ray of $g = \operatorname{sech}$ (dashed, slope 0) as $c_3 \downarrow \bar{c}_3$. Right: the window $[U_{\boldsymbol{c}}^-(0), U_{\boldsymbol{c}}^+(0)]$ against $\varepsilon = c_3 - \bar{c}_3$ for three choices of (c_1, c_2); it opens linearly in ε (proved in Chapter 12, Proposition 12.3). Code: `code/ch03_line/make_noswirl_window.py`.

3.8. Closed forms: the Jost solutions by hypergeometric functions

In words. *The potential $W_{\boldsymbol{c}}$ is rational in $t=\frac{1+x}{2}$, and $z''=W_{\boldsymbol{c}}z$ has exactly three regular singular points $t=0,1,\infty$. It is therefore a Gauss hypergeometric equation in disguise, and the principal solution u_- and its mirror u_+ are explicit: the window $[U^-_{\boldsymbol{c}}(x_0),U^+_{\boldsymbol{c}}(x_0)]$ is a closed-form function of (x_0,c). This is an example, not a tool: later chapters use it for exact values (the Landau-type integrals of Chapter 11) and for numerical checks.*

Proposition 3.47 (Jost solutions of $W_{\boldsymbol{c}}$ by ${}_2F_1$)**.** *Let $c_1,c_2\ge -1$, $c_3\in\mathbb{R}$, $k_i=\sqrt{1+c_i}$, $s=\sqrt{1-2c_3}$ ($s\in i\mathbb{R}_{\ge 0}$ if $c_3>\frac12$), $t=\frac{1+x}{2}=\frac{e^\xi}{2\cosh\xi}\in(0,1)$, and*

$$a_\pm=\tfrac{k_1-k_2}{2}+\tfrac{1\pm s}{2},\qquad u^{\mathrm{hyp}}(\xi)=t^{k_1/2}(1-t)^{-k_2/2}\,{}_2F_1(a_+,a_-;1+k_1;t).$$

Then u^{hyp} is real, solves $z''=W_{\boldsymbol{c}}z$, and $u^{\mathrm{hyp}}=2^{-\bar k}p_-$ with p_- the principal solution of Lemma 3.42 ($\bar k=\frac12(k_1+k_2)$); for $k_1>0$ it is a multiple of the Jost solution recessive at $-\infty$. Its logarithmic derivative is

$$\zeta_-(\xi)=\frac{1-x^2}{2}\Big[\frac{k_1}{2t}+\frac{k_2}{2(1-t)}+\frac{a_+a_-}{1+k_1}\,\frac{{}_2F_1(a_++1,a_-+1;2+k_1;t)}{{}_2F_1(a_+,a_-;1+k_1;t)}\Big]. \tag{3.8}$$

The mirror solution is $p_+(\xi)=\tilde p_-(-\xi)$, where $\tilde p_-$ is the principal solution for $\boldsymbol{c}^*=(c_2,c_1,c_3)$ (computed by the same formulas with $k_1\leftrightarrow k_2$), so $\zeta_+(\xi)=-\tilde\zeta_-(-\xi)$. For $\boldsymbol{c}\in\mathcal{J}$ with $c_3>\bar c_3$:

$$U^+_{\boldsymbol{c}}(x_0)=2(\zeta_-(\xi_0)-x_0),\qquad U^-_{\boldsymbol{c}}(x_0)=2(\zeta_+(\xi_0)-x_0),$$
$$U^+_{\boldsymbol{c}}(x_0)-U^-_{\boldsymbol{c}}(x_0)=2\big(\zeta_-(\xi_0)-\zeta_+(\xi_0)\big).$$

Proof. *Reduction to Gauss' equation.* With $dt/d\xi=\frac12(1-x^2)=2t(1-t)$ the operator $d/d\xi$ is $D=2t(1-t)\,d/dt$, $1-x=2(1-t)$, $1+x=2t$, $1-x^2=4t(1-t)$, and with $1=(1-t)^2+t^2+2t(1-t)$ the potential becomes

$$W_{\boldsymbol{c}}=1+c_1(1-t)+c_2t+2c_3t(1-t)=k_1^2(1-t)^2+k_2^2t^2+(k_1^2+k_2^2+2c_3)\,t(1-t).$$

Write $z=e^\phi w(t)$ with $\phi=\alpha\ln t+\beta\ln(1-t)$, $\alpha=\frac{k_1}{2}$, $\beta=-\frac{k_2}{2}$, and put $A=t(1-t)\phi_t=\alpha(1-t)-\beta t$. Since $D^2z=4t^2(1-t)^2z_{tt}+4t(1-t)(1-2t)z_t$, a direct computation gives

$$e^{-\phi}\big(D^2z-W_{\boldsymbol{c}}z\big)=4t^2(1-t)^2w_{tt}+4t(1-t)\big[2A+1-2t\big]w_t$$
$$+\Big[4\big(t^2(1-t)^2(\phi_{tt}+\phi_t^2)+(1-2t)A\big)-W_{\boldsymbol{c}}\Big]w.$$

Here $2A+1-2t=(1+k_1)-(k_1-k_2+2)t$. For the coefficient of w: $t^2(1-t)^2\phi_{tt}=-\alpha(1-t)^2-\beta t^2$, $t^2(1-t)^2\phi_t^2=A^2$ and $1-2t=(1-t)-t$, so it equals

$$4\big[\alpha^2(1-t)^2+\beta^2t^2-(2\alpha\beta+\alpha+\beta)t(1-t)\big]-W_{\boldsymbol{c}}$$
$$=-t(1-t)\big[4(2\alpha\beta+\alpha+\beta)+k_1^2+k_2^2+2c_3\big],$$

because $4\alpha^2=k_1^2$, $4\beta^2=k_2^2$ cancel against the first two terms of $W_{\boldsymbol{c}}$. The bracket is $-2k_1k_2+2(k_1-k_2)+k_1^2+k_2^2+2c_3=(k_1-k_2+1)^2-s^2=4a_+a_-$, since $a_++a_-=k_1-k_2+1$ and $a_+a_-=\frac14[(k_1-k_2+1)^2-s^2]$. Hence $z''=W_{\boldsymbol{c}}z$ is equivalent to

$$t(1-t)w_{tt}+\big(1+k_1-(a_++a_-+1)t\big)w_t-a_+a_-w=0,$$

Gauss' equation with parameters $(a_+, a_-; 1+k_1)$ (the same reduction was verified by computer algebra).

The solution. The parameter $1+k_1 \geq 1$ is not a non-positive integer, so ${}_2F_1(a_+, a_-; 1+k_1; t)$, the solution analytic at $t=0$ with value 1 (convergent for $|t|<1$, which contains $t \in (0,1)$), solves it; it is real because $a_\pm$ are real or complex conjugate. Hence u^{hyp} solves $z'' = W_{\boldsymbol{c}} z$ for $\xi \in \mathbb{R}$.

Identification. As $\xi \to -\infty$, $t \to 0$ and ${}_2F_1 \to 1$, $(1-t)^{-k_2/2} \to 1$, so $u^{\mathrm{hyp}}/g = 2^{-\bar{k}}(1-t)^{-k_2}\,{}_2F_1 \to 2^{-\bar{k}}$ ($g = (2t)^{k_1/2}(2(1-t))^{k_2/2}$); by Lemma 3.42(b), $u^{\mathrm{hyp}} = 2^{-\bar{k}} p_-$.

Log-derivative. Since $\frac{d}{d\xi} = 2t(1-t)\frac{d}{dt}$ and $\partial_t {}_2F_1(a,b;c;t) = \frac{ab}{c}\,{}_2F_1(a+1, b+1; c+1; t)$,

$$\frac{d}{d\xi}\ln u^{\mathrm{hyp}} = 2t(1-t)\Big[\frac{k_1}{2t} + \frac{k_2}{2(1-t)} + \partial_t \ln {}_2F_1(a_+, a_-; 1+k_1; t)\Big],$$

and $2t(1-t) = \frac{1}{2}(1-x^2)$ turns this into (3.8). Finally the mirror statement is the reflection of Lemma 3.7 and Remark 3.36, and the formulas for $U_{\boldsymbol{c}}^{\pm}$ are Theorem 3.44(ii). □

Example 3.48 (rest state and symmetric potentials)**.** $\boldsymbol{c}=0$: $s=1$, $a_+=1$, $a_-=0$, so ${}_2F_1 = 1$ and $u^{\mathrm{hyp}} = t^{1/2}(1-t)^{-1/2} = e^{\xi}$, $\zeta_- = 1$ (Example 3.46(a)). At $x_0 = 0$ (so $t=\frac{1}{2}$) and $c_1 = c_2$, $k_1 = k_2 = k$, the symmetry $\zeta_+(0) = -\zeta_-(0)$ gives the window width in closed form,

$$U_{\boldsymbol{c}}^{+}(0) - U_{\boldsymbol{c}}^{-}(0) = 4\zeta_-(0) = 4k + \frac{2a_+a_-}{1+k}\,\frac{{}_2F_1(a_+ + 1, a_- + 1; 2+k; \frac{1}{2})}{{}_2F_1(a_+, a_-; 1+k; \frac{1}{2})},$$
$$a_\pm = \tfrac{1\pm s}{2}.$$

Numerical observation 3.49 (the closed form against numerical integration)**.** *We evaluated (3.8) and its mirror with 30-digit arithmetic (mpmath) and compared with the stable Riccati integration of $\zeta' = W_{\boldsymbol{c}} - \zeta^2$ (from $\mp\infty$, tolerance 10^{-12}). For six parameter sets and two base points ($x_0 = 0, 0.4$) the values $\zeta_\mp(\xi_0)$ agree to at most $2.3\cdot 10^{-13}$; four of the six sets are shown (the others are in the script):*

$\boldsymbol{c}$	x_0	$U_{\boldsymbol{c}}^{+} - U_{\boldsymbol{c}}^{-}$ *(hypergeom.)*	*max. error in* $\zeta_\mp$
$(0, 0, -2)$	0	2.2699124248	$2.3\cdot 10^{-13}$
$(-\frac{3}{4}, -\frac{3}{4}, \frac{3}{10})$	0	2.3335114312	$8.1\cdot 10^{-14}$
$(2, -\frac{1}{2}, \frac{1}{5})$	0	5.2861802780	$1.5\cdot 10^{-13}$
$(2, -\frac{1}{2}, \frac{3}{2})$ *($c_3 > \frac{1}{2}$, s imaginary)*	0	6.0993945514	$1.3\cdot 10^{-13}$

For the face $k_1 = k_2 = 0$ (not covered by the Riccati integration, which needs $k_i > 0$ to start) the principal solution was obtained independently by shooting $u'' = \frac{c_3}{2}\operatorname{sech}^2\xi\, u$ from $u(-18)=1$, $u'(-18)=0$: $U_{\boldsymbol{c}}^{+}(0) = 0.84454934$ for $c_3 = 1$ and 1.98981993 for $c_3 = 3$, in agreement with Table 3.1 (to 10^{-6}). This is a check of the formula and of the code, not part of any proof. Code: `code/ch03_line/verify_hyp_window.py`.

3.9. Identity (E): the distance of a swirling solution from the edge of the window

In words. *A global solution with swirl, based at the zero of the swirl, solves $z'' = (W_0 - \frac{1}{2}\psi)z$, where W_0 is the no-swirl potential and $\psi \geq 0$ the swirl integral. Swirl therefore* lowers *the potential, and the log-derivative $h = z'/z$ of such a solution*

lies above the log-derivative ζ_+ of the Jost solution of W_0 that is recessive at $+\infty$, and below ζ_-: the lens sits inside the window $[U_{\boldsymbol{c}}^-, U_{\boldsymbol{c}}^+]$. Identity (E) says by how much*: $h(\xi_0) - \zeta_+(\xi_0)$ is exactly one half of the integral of ψ against the product of the two solutions u_+ and z over $[\xi_0, \infty)$, and $\zeta_-(\xi_0) - h(\xi_0)$ is the same over $(-\infty, \xi_0]$. This is the Wronskian lemma integrated over a half-line; the only work is to show that the Wronskian dies out at the far end.*

3.9.1. Setting. We use the class-Z formulation of Chapter 2. Fix $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$ ($\varepsilon = c_3 - \bar{c}_3 > 0$), a base point $x_0 \in (-1, 1)$, $\xi_0 = \operatorname{artanh} x_0$, and put $W_0 = W_{\boldsymbol{c}} = W_{\mathrm{crit}} + \frac{\varepsilon}{2} \operatorname{sech}^2 \xi$, the no-swirl potential. By Proposition 3.43(c), W_0 has the positive solutions $p_\pm$; since $k_i > 0$ they are multiples of the Jost solutions (Lemma 3.42(d)). *In this section only*, $u_\pm$ denotes the Jost solutions normalised at the base point, $u_\pm(\xi_0) = 1$ (not at 0 as elsewhere in the chapter); thus $u_\pm = p_\pm / p_\pm(\xi_0)$, $\zeta_\pm = u_\pm'/u_\pm$ is unaffected, $\zeta_+ < \zeta_-$, $u_+ \le Ce^{-k_2 \xi}$ and $\zeta_+ \to -k_2$ at $+\infty$, $u_- \le Ce^{k_1 \xi}$ and $\zeta_- \to k_1$ at $-\infty$. The Wronskian w_0 of Lemma 3.21 is not used in this section. A *class-Z solution* with data $(\gamma, \varkappa)$, $\varkappa > 0$, is a solution of the system of Chapter 2 with $f(\xi_0) = 0$, $U(x_0) = \gamma$ and $f'(\xi_0) = \varkappa$ for $z(\xi_0) = 1$ (the constant $\varkappa = \varkappa[\xi_0]$, normalisation 2 of Lemma 2.15); it is *admissible* if it is global, which by Lemma 2.24 means $z > 0$ on $\mathbb{R}$. In these variables (Lemma 2.14, Proposition 2.27 with $A = f(\xi_0) = 0$):

- (Z1) $z'' = Wz$ with $W = W_0 - \frac{1}{2}\psi$, $z > 0$, $z(\xi_0) = 1$, $h(\xi_0) = \frac{1}{2}\gamma + x_0$, $U = 2(h - \tanh \xi)$;
- (Z2) $f' = \varkappa z^{-2}$, so $f > 0$ on (ξ_0, ∞) and $f < 0$ on $(-\infty, \xi_0)$;
- (Z3) $\psi = \psi_{\boldsymbol{c}}$ satisfies $\mathsf{L}[\psi] = f^2$ (in the variable x), $\psi(\xi_0) = \psi'(\xi_0) = 0$, $\psi \ge 0$, and ψ is nondecreasing on $[\xi_0, \infty)$ and nonincreasing on $(-\infty, \xi_0]$.

(For example, (Z3) in explicit form is $\psi(\xi) = \int_{\xi_0}^{\xi} \frac{\sinh^2(\xi - s)}{\cosh^2 \xi} f f'(s)\, ds$ for $\xi \ge \xi_0$ and the mirror formula for $\xi \le \xi_0$; it is not used beyond (Z3).)

Toy example 3.50 (a delta shell)**.** Take $W_0 \equiv k^2$ on $t = \xi - \xi_0 \ge 0$ (so $u_+ = e^{-kt}$) and replace the swirl integral by a thin shell at distance σ, $\psi = m\,\delta(t - \sigma)$, i.e. $W = k^2 - \frac{m}{2}\delta(t - \sigma)$ (a toy: the actual ψ is smooth and nondecreasing). The solution that decays at $+\infty$ is $z = z(\sigma) e^{-k(t-\sigma)}$ for $t > \sigma$, so $h(\sigma^+) = -k$; across the shell z' jumps by $-\frac{m}{2} z(\sigma)$, so $h(\sigma^-) = -k + \frac{m}{2}$. For $t < \sigma$ the potential is constant, and with $\Delta_0 := h(0) - \zeta_+(0)$ (the distance of h from the edge ζ_+ at $t = 0$) one has $z = e^{-kt}\tau(t)$, $\tau(t) = 1 + \Delta_0 \frac{e^{2kt} - 1}{2k}$, hence $h(\sigma^-) = -k + \Delta_0 e^{2k\sigma}/\tau(\sigma)$. Matching gives

$$\Delta_0 = \frac{m}{2}\, e^{-2k\sigma} \tau(\sigma), \qquad \tau(\sigma) = \frac{1}{1 - \frac{m}{4k}(1 - e^{-2k\sigma})}.$$

This is (E) in miniature: $\frac{1}{2}\int \psi u_+ z = \frac{m}{2} u_+(\sigma) z(\sigma) = \frac{m}{2} e^{-2k\sigma}\tau(\sigma)$. Two lessons. (i) The edge distance Δ_0 is the mass of the shell weighted by u_+^2 at the shell: a shell farther out moves the edge by exponentially less. (ii) The amplification $\tau = z/u_+$ is close to 1 when the shell is weak and blows up at $m = 4k/(1 - e^{-2k\sigma})$, where the swirl has closed the window (z ceases to be positive). Symbolic check: `code/ch03_line/verify_ch03b.py`.

3.9.2. The identity.

Proposition 3.51 (identity (E))**.** *Let $c \in \operatorname{int} \mathcal{J}$ ($\varepsilon = c_3 - \bar{c}_3 > 0$), $x_0 \in (-1, 1)$, $\xi_0 = \operatorname{artanh} x_0$, and let $(\gamma, \varkappa)$ with $\varkappa > 0$ ($\varkappa = \varkappa[\xi_0]$ in the normalisation $z(\xi_0) = 1$*

*of Lemma 2.15) be admissible at the class-*Z *base point ($f(\xi_0)=0$). Let $z>0$, $z(\xi_0)=1$, $h=z'/z$ be the corresponding solution of $z''=(W_0-\frac12\psi)z$, $W_0=W_{\rm crit}+\frac{\varepsilon}{2}\operatorname{sech}^2$, $\psi=\psi_{\mathbf c}$ the swirl integral ($L[\psi]=f^2$, $\psi(\xi_0)=\psi'(\xi_0)=0$), and let $u_\pm$ be the Jost solutions of $u''=W_0u$ with $u_\pm(\xi_0)=1$, $\zeta_\pm=u_\pm'/u_\pm$. Then $h(\xi_0)-\zeta_+(\xi_0)=\frac12\int_{\xi_0}^{\infty}\psi u_+z$ and $\zeta_-(\xi_0)-h(\xi_0)=\frac12\int_{-\infty}^{\xi_0}\psi u_-z$; in U-units $\gamma-U_c^-=\int_{\xi_0}^{\infty}\psi u_+z$ and $U_c^+-\gamma=\int_{-\infty}^{\xi_0}\psi u_-z$. (The proof uses $\operatorname{Wr}(u_+,z)\to0$ at $+\infty$, which needs $\psi_+>0$, i.e. $\varkappa>0$.) In displayed form, with all integrals finite:*

$$\begin{aligned} h(\xi_0)-\zeta_+(\xi_0)&=\tfrac12\int_{\xi_0}^{\infty}\psi\,u_+z\,d\xi,\\ \zeta_-(\xi_0)-h(\xi_0)&=\tfrac12\int_{-\infty}^{\xi_0}\psi\,u_-z\,d\xi. \end{aligned}\tag{E}$$

Proof. We write the proof for the right half-line; the left half-line is its mirror image (Step 4). By (Z1) and Lemma 3.4 with $a=u_+$ ($V_a=W_0$), $b=z$ ($V_b=W_0-\frac12\psi$),

$$\begin{aligned} \operatorname{Wr}(u_+,z)'&=-\tfrac12\psi\,u_+z\le0\quad\text{on }\mathbb{R},\\ \operatorname{Wr}(u_+,z)&=u_+z\,(h-\zeta_+),\qquad \Big(\frac{z}{u_+}\Big)'=\frac{\operatorname{Wr}(u_+,z)}{u_+^2}. \end{aligned}\tag{3.9}$$

So $\operatorname{Wr}(u_+,z)$ is nonincreasing, and has a limit $L\in[-\infty,\infty)$ at $+\infty$.

Step 1: $L\ge0$. Suppose $L<0$. Then there are ξ_1 and $\delta>0$ with $\operatorname{Wr}(u_+,z)\le-\delta$ on $[\xi_1,\infty)$, and by (3.9)

$$\frac{z}{u_+}(\xi)\le\frac{z}{u_+}(\xi_1)-\delta\int_{\xi_1}^{\xi}u_+^{-2}\,d\eta.$$

Since $u_+\le Ce^{-k_2\eta}$ for $\eta\ge\xi_1$ (Lemma 3.42(c), (3.4) and $u_+\propto p_+$), $u_+^{-2}\ge ce^{2k_2\eta}\ge c$ is not integrable at $+\infty$, so $z/u_+\to-\infty$, contradicting $z>0$. Consequently $\operatorname{Wr}(u_+,z)\ge L\ge0$ on $\mathbb{R}$, that is $h\ge\zeta_+$ everywhere.

Step 2: $L=0$, i.e. $\operatorname{Wr}(u_+,z)\to0$ at $+\infty$.

(a) ψ is positive to the right of ξ_0. $f>0$ on (ξ_0,∞), so $\mathsf L[\psi]=f^2>0$ there; if $\psi\equiv0$ on some interval $(\xi_0,\xi_0+\delta)$ then $\mathsf L[\psi]=0$ there, a contradiction. So $\psi(\xi_1)>0$ for some $\xi_1>\xi_0$, and by monotonicity (Z3) $\psi\ge\psi_1:=\min\{\psi(\xi_1),2k_2^2\}>0$ on $[\xi_1,\infty)$ (the cap $2k_2^2$ only guarantees $\lambda_1^2=k_2^2-\psi_1/4\ge k_2^2/2>0$ below).

(b) h stays away from k_2. Put $\lambda_1=\sqrt{k_2^2-\psi_1/4}\in[k_2/\sqrt2,k_2)$. By Lemma 3.29, $W_0\le k_2^2+C_{\mathbf c}e^{-2\xi}$, so there is $\xi_2\ge\xi_1$ with $W=W_0-\frac12\psi\le k_2^2+\frac{\psi_1}{4}-\frac{\psi_1}{2}=\lambda_1^2$ on $[\xi_2,\infty)$. Hence $h'=W-h^2\le\lambda_1^2-h^2$ there. Let $\lambda_2=\frac12(\lambda_1+k_2)\in(\lambda_1,k_2)$ and $\beta=\lambda_2^2-\lambda_1^2>0$. Wherever $h\ge\lambda_2$ we have $h'\le\lambda_1^2-h^2\le-\beta$. So if $h(\xi_2)>\lambda_2$, h decreases with speed at least β until it reaches λ_2 (this happens after a finite time since h is finite at ξ_2), and once $h\le\lambda_2$ it can never exceed λ_2 again (at a point where $h=\lambda_2$, $h'\le-\beta<0$). Therefore $h\le\lambda_2$ on $[\xi_3,\infty)$ for some ξ_3, and $z(\xi)\le z(\xi_3)e^{\lambda_2(\xi-\xi_3)}$ there.

(c) Conclusion. ζ_+ is bounded near $+\infty$ ($\zeta_+\to-k_2$), say $|\zeta_+|\le K$, and $u_+\le Ce^{-k_2\xi}$. By Step 1 and (b), for $\xi\ge\xi_3$,

$$0\le\operatorname{Wr}(u_+,z)=u_+z\,(h-\zeta_+)\le Ce^{-k_2\xi}\,z(\xi_3)e^{\lambda_2(\xi-\xi_3)}\,(\lambda_2+K)\longrightarrow0,$$

because $\lambda_2<k_2$. So $L=0$.

Step 3: integration. By (3.9) and $L = 0$, Corollary 3.5 gives (the integrand is ≥ 0)

$$\mathrm{Wr}(u_+, z)(\xi_0) = \tfrac{1}{2}\int_{\xi_0}^{\infty} \psi\, u_+ z\, d\xi,$$

and, since $u_+(\xi_0) = z(\xi_0) = 1$, $\mathrm{Wr}(u_+, z)(\xi_0) = h(\xi_0) - \zeta_+(\xi_0)$. This is the first identity of (E), with the integral finite.

Step 4: the left half-line. Let $\tilde\xi = -\xi$. The reflection of Lemma 3.7 maps the situation to one of the same kind: $\tilde{\boldsymbol{c}} = (c_2, c_1, c_3)$ (so $\tilde W_0(\tilde\xi) = W_0(-\tilde\xi)$), $\tilde z(\tilde\xi) = z(-\tilde\xi)$ solving $\tilde z'' = (\tilde W_0 - \frac{1}{2}\tilde\psi)\tilde z$ with $\tilde\psi(\tilde\xi) = \psi(-\tilde\xi)$ and $\tilde f(\tilde\xi) = -f(-\tilde\xi)$ (the operator L is invariant under $x \mapsto -x$, $\tilde f' = \varkappa \tilde z^{-2}$, $\tilde f^2 = f^2$), base point $-\xi_0$, the same $\varkappa$, and $\tilde u_+(\tilde\xi) = u_-(-\tilde\xi)$. Hence $\tilde h(-\xi_0) - \tilde\zeta_+(-\xi_0) = \frac{1}{2}\int_{-\xi_0}^{\infty} \tilde\psi \tilde u_+ \tilde z\, d\tilde\xi$ and, because $\tilde h = -h(-\cdot)$, $\tilde\zeta_+ = -\zeta_-(-\cdot)$, this reads $\zeta_-(\xi_0) - h(\xi_0) = \frac{1}{2}\int_{-\infty}^{\xi_0} \psi u_- z\, d\xi$.

Step 5: U-units. $U = 2(h - \tanh\xi)$ and $U^{\mp}_{\boldsymbol{c}}(x_0) = 2(\zeta_\pm(\xi_0) - x_0)$ (Theorem 3.44(ii)); multiplying (E) by 2 gives $\gamma - U^-_{\boldsymbol{c}} = \int_{\xi_0}^{\infty} \psi u_+ z$ and $U^+_{\boldsymbol{c}} - \gamma = \int_{-\infty}^{\xi_0} \psi u_- z$. □

Corollary 3.52 (the lens is inside the window)**.** *In the situation of Proposition 3.51, $U^-_{\boldsymbol{c}}(x_0) < \gamma < U^+_{\boldsymbol{c}}(x_0)$, and*

$$\zeta_-(\xi_0) - \zeta_+(\xi_0) = \tfrac{1}{2}\Big[\int_{\xi_0}^{\infty} \psi\, u_+ z + \int_{-\infty}^{\xi_0} \psi\, u_- z\Big].$$

Moreover $\zeta_+ \leq h \leq \zeta_-$ on all of $\mathbb{R}$ (Step 1 and its mirror).

Proof. Add the two identities of (E). Both integrals are strictly positive: $\psi > 0$ on $\mathbb{R} \setminus \{\xi_0\}$ (Step 2(a) and its mirror) and $u_\pm, z > 0$. □

Remark 3.53 (how (E) is used)**.** (i) At the lower edge of the lens, $\gamma = \gamma^-(\kappa)$, the solution is recessive at $+\infty$ (of type R or L), the integral $\int \psi u_+ z$ converges fast, and (E) turns the distance $\gamma^- - U^-_{\boldsymbol{c}}$ of the edge from the no-swirl endpoint into an explicit integral; this is how the small-swirl laws of Chapter 11 are derived. (ii) Adding the two halves, or integrating against the critical function g in place of $u_\pm$, gives the balance identity (T0) of Chapter 12 (Theorem 12.5); the three are the same Wronskian computation, with u_+, u_- or g as the reference solution. (iii) The hypothesis $\varkappa > 0$ enters only through Step 2: for $\varkappa = 0$ one has $f \equiv 0$, $\psi \equiv 0$, $z \in \mathrm{span}\{u_-, u_+\}$ and there is nothing to prove. (iv) The identity involves no smallness: ψ and z are those of the full nonlinear problem.

Numerical observation 3.54 ((E) at the lower edge)**.** *For four admissible data on the lower edge of the lens ($\gamma = \gamma^-(\kappa)$, $\kappa = \kappa_x$ the base-point normalisation, so $\varkappa = (1 - x_0^2)\kappa$), computed by shooting with the shooting code of Appendix E, the two sides of $\gamma - U^-_{\boldsymbol{c}} = \int_{\xi_0}^{\infty} \psi u_+ z$ were evaluated by quadrature on $[\xi_0, \xi_0 + 30]$:*

$\boldsymbol{c}$	(x_0, κ)	$\gamma^- - U^-_{\boldsymbol{c}}$	$\int \psi u_+ z$	*difference*
$(-\frac{1}{2}, -\frac{1}{2}, 0)$	$(0, 0.5)$	0.0614321180	0.0614321177	$3 \cdot 10^{-10}$
$(-\frac{1}{2}, -\frac{1}{2}, 0)$	$(0.4, 0.7)$	0.0364594374	0.0364594373	$4 \cdot 10^{-11}$
$(0, 0, 0)$	$(0.3, 1)$	0.1117877098	0.1117877098	$1 \cdot 10^{-12}$
$(1, -\frac{1}{2}, \frac{1}{2})$	$(-0.2, 1.5)$	0.9320982144	0.9320980934	$1 \cdot 10^{-7}$

(the larger difference in the last row is quadrature and shooting error at the larger swirl). Code: see Appendix E.

What we have, what comes next

This chapter hands over the following. (1) The Sturm lemma, the two-potential Wronskian identity with its half-line forms, reduction of order and reflection (Section 3.1). (2) The Riccati toolbox Lemma 3.10 (comparison, also at a pole; tracking; barrier bound; cut-off integral) and the dichotomy Lemma 3.14, used in Chapters 4, 6, 10, 11, 12; recessive and dominant solutions and ordering at a fixed potential (Lemma 3.16, Lemma 3.18). (3) Jost solutions, the comparison lemma Lemma 3.23, the weighted topology Lemma 3.25 (also with moving limits k_i) and the uniform estimates Lemma 3.28, used by Chapters 8 and 10; the trichotomy and its openness and closedness (Lemma 3.32), the potential bounds and the critical function (Lemmas 3.29 and 3.30), the Pöschl–Teller toy Toy example 3.34 and the criterion Lemma 3.35. (4) The Jost basis through $\sigma = 0$ (Lemma 3.38), the tool for the pole analysis of Chapter 4. (5) The complete no-swirl classification for *all* $\boldsymbol{c} \in \mathbb{R}^3$, including the faces $c_i = -1$ (Proposition 3.43, Theorem 3.44), with the principal solutions $p_\pm$ and their normalisations (Lemma 3.42); worked examples (Example 3.46, Example 3.48) and the closed forms of Proposition 3.47. (6) Identity (E) (Proposition 3.51, (E)). Chapter 4 studies the behaviour at the poles of the solutions with swirl, whose pole values are compared with the no-swirl values $2 \pm 2k_i$ of Theorem 3.44(iii); Chapter 5 uses the window $[U_{\boldsymbol{c}}^-, U_{\boldsymbol{c}}^+]$ as the base of the lens (class Z); Chapters 11 and 12 use (E) and its companion (T0).

Part 2

Poles and global solutions

CHAPTER 4

The pole: limits, exponents, strata and charts

In words. *A global solution lives on the whole line $\xi \in \mathbb{R}$, and the two poles of the sphere sit at $\xi = \pm\infty$. Near a pole the potential $W = 1 + \frac{1}{2}Q$ settles down to a constant μ^2, and the equation $z'' = Wz$ then looks like $z'' = \mu^2 z$: its solutions are, up to small corrections, $e^{\mu\xi}$ (*dominant*) and $e^{-\mu\xi}$ (*recessive*); when $\mu = 0$ they are ξ and 1. A positive solution is therefore either dominant, or recessive (a codimension-one coincidence), or logarithmic ($\mu = 0$). The swirl f is the integral of κz^{-2}: it stays bounded at the pole if z is dominant, and blows up if z is recessive. But the swirl feeds back into the potential, and the finiteness of the swirl energy $\int f^2 e^{-2\xi}\,\mathrm{d}\xi$, which comes from the first integral, forbids too violent a blow-up: the recessive branch is possible only for $\mu < \frac{1}{2}$, i.e. $U_\theta(1) > -3$. This chapter makes the picture precise (four* strata $\mathsf{G}, \mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$ *of pole behaviours), proves the a priori asymptotics of each stratum, and then parametrizes each stratum by a fixed point of a contraction (the* pole charts*). For a local classification near a singular ray with logarithmic behaviour, including swirl, see also [56].*

Roadmap. The chapter has two halves. This half (Section 4.1–Section 4.5) is qualitative and uses only the equations of Chapter 2 and the linear toolbox of Chapter 3. Section 4.1 fixes what a *local solution at a pole* is and works out the constant-potential toy. Section 4.2 proves that $q = Q(1)$ exists, $q \geq -2$, and that $h = z'/z \to \pm\mu$ (Theorem 4.5). Section 4.3 proves that on the recessive branch $\mu < \frac{1}{2}$ (Theorem 4.9), by a growth lemma (Lemma 4.7) and a sign-propagation argument at the borderline. Section 4.4 proves the weighted bounds for the Green operator of L at a pole (Lemma 4.13), which are the only analytic input for Section 4.5: the definition of the four types (Definition 4.16) and their a priori asymptotics (Proposition 4.19). The second half of the chapter places the types in the Jost plane (Proposition 4.32), proves the weighted Volterra lemma (Lemma 4.35), and builds the charts (Theorem 4.38, Theorem 4.40), the openness of type G (Corollary 4.42) and the local corner chart (Theorem 4.45); its toy is Toy example 4.36, and the numerical atlas of the chapter is Atlas 4.50. New tools of this half, each with a toy: the Euler (constant-potential) model (Toy example 4.3); the growth-versus-energy count behind the -3 threshold (same toy and Figure 4.1); the Dirichlet Green operator with pole weights (Toy example 4.12). *Conventions.* A prime is $\mathrm{d}/\mathrm{d}\xi$; $x = \tanh\xi$; $E(\xi) = e^{-2|\xi|}$. Everything is stated at the *north* pole $x = 1$ ($\xi = +\infty$); the south pole is obtained from the symmetry σ_2 (Lemma 2.8).

4.1. Local solutions at a pole

In words. *A* local solution at the north pole *is a solution of the Q-system on an interval $(x_0, 1)$ that need not extend to the other side. Near the pole it is the same thing as a positive solution z of $z'' = Wz$ on a half-line, together with the*

swirl $f' = \kappa z^{-2}$ and the first integral. The constant-potential model $W \equiv \mu^2$, whose solutions are explicit, shows all the phenomena of the chapter.

Definition 4.1 (local solution at the north pole)**.** A *local solution at the north pole* is a triple (U, f, Q) with $Q \in C^3[x_0, 1)$ solving (2.19)–(2.21) on $[x_0, 1)$ for some $x_0 \in (-1, 1)$ (a solution on an open interval $(x_0, 1)$ is restricted to a smaller closed one). Put $\xi_0 = \operatorname{artanh} x_0$ and $y = \exp \int^x U/(2(1-s^2))\,\mathrm{d}s$, $z = y\cosh\xi$. By Lemma 2.9 and Lemma 2.14, on $[\xi_0, \infty)$

$$\begin{gathered} z > 0, \qquad z'' = Wz, \qquad W = 1 + \tfrac{1}{2}Q, \qquad U = 2\Big(\frac{z'}{z} - \tanh\xi\Big), \\ f' = \kappa z^{-2}, \qquad \mathsf{L}[Q] + f^2 = \lambda, \end{gathered} \tag{4.1}$$

the last by Proposition 2.20; $h := z'/z$ solves $h' = W - h^2$. The solution has *genuine swirl* if $\kappa \neq 0$. Conversely a positive z on $[\xi_0, \infty)$, a function Q and f with (4.1) give a local solution.

Only the *vanishing* of κ matters in this half of the chapter, not its value, and the vanishing is independent of the normalisation of z (Lemma 2.15(5)); the normalisation conventions of the pole charts (normalisation 6 of that lemma) enter in the second half. If $\kappa = 0$ then $f \equiv A$ is constant, $Q = P_{\mathbf{c}}$ is a polynomial (Example 2.30), and we are in the no-swirl theory of Chapter 3; so the interesting case is $\kappa \neq 0$. By Corollary 2.10, f is then strictly monotone with at most one zero, and f and ff' have a fixed sign near $x = 1$.

Remark 4.2 (global solutions are local at both poles)**.** A global solution (Corollary 2.25: $z > 0$ on all of $\mathbb{R}$) restricts to a local solution at the north pole for every x_0, and, applying the symmetry σ_2 (Lemma 2.8), to a local solution at the south pole. The map σ_2 sends $U(x) \mapsto -U(-x)$, so a statement "$U(1) = v$" at the north pole reads "$U(-1) = -v$" at the south pole. The converse is false: that the two local half-solutions glue to a global one is a condition at the base point, and is the subject of Chapters 5 and 8.

Toy example 4.3 (the Euler model: constant potential on a half-line)**.** Let $Q \equiv q$ be constant and f the swirl with $f' = \kappa z^{-2}$ (we ignore for the moment that Q must satisfy the first integral with this f: the model *freezes* the potential at its limit). Then $z'' = (1 + q/2)z$ and we look for solutions that are positive for all large ξ (only the behaviour near the pole matters).

- $q < -2$: $z = C\cos(\nu\xi - \delta)$ with $\nu = \sqrt{-1 - q/2}$ vanishes on every interval of length π/ν; there is no positive solution.
- $q = -2$ ($\mu = 0$): $z = a + b\xi$ is positive for large ξ iff $b > 0$, or $b = 0$ and $a > 0$. If $b > 0$, $f' = \kappa(a + b\xi)^{-2}$ is integrable and $f(\infty)$ is finite (*logarithmic of the first kind*); if $b = 0$, $f = \kappa\xi/a^2 + f(0)$ is unbounded (*second kind*).
- $q > -2$, $\mu = \sqrt{1 + q/2} > 0$: $z = \alpha e^{\mu\xi} + \beta e^{-\mu\xi}$. If $\alpha > 0$ the solution is *dominant*: $z'/z \to \mu$, $\int^\infty z^{-2} < \infty$ and $f(\infty)$ is finite. If $\alpha = 0 < \beta$ it is *recessive*: $z'/z \equiv -\mu$, $f' = \kappa\beta^{-2}e^{2\mu\xi}$, and $f = \frac{\kappa}{2\mu\beta^2}(e^{2\mu\xi} - 1) + f(0)$ blows up.

Solutions with $\alpha < 0$ become negative. The recessive branch is a single ray in the two-parameter family (α, β): it costs one condition, and the other positive solutions are dominant, in agreement with the phase portrait of $h' = \mu^2 - h^2$ (Chapter 3, Toy example 3.9): $-\mu$ is repelling, $+\mu$ attracting. Now compute the *swirl energy* on the

recessive branch,

$$f^2 e^{-2\xi} \sim \left(\frac{\kappa}{2\mu\beta^2}\right)^2 e^{(4\mu-2)\xi}, \qquad \text{integrable on } [0,\infty) \iff \mu < \tfrac{1}{2}. \tag{4.2}$$

Since $U(1) = -2 - 2\mu$ on the recessive branch, $\mu < \frac{1}{2}$ is the same as $U(1) > -3$. Figure 4.1 shows z, f and $f^2 e^{-2\xi}$. This count, applied to the *true* potential, is what Section 4.3 makes rigorous.

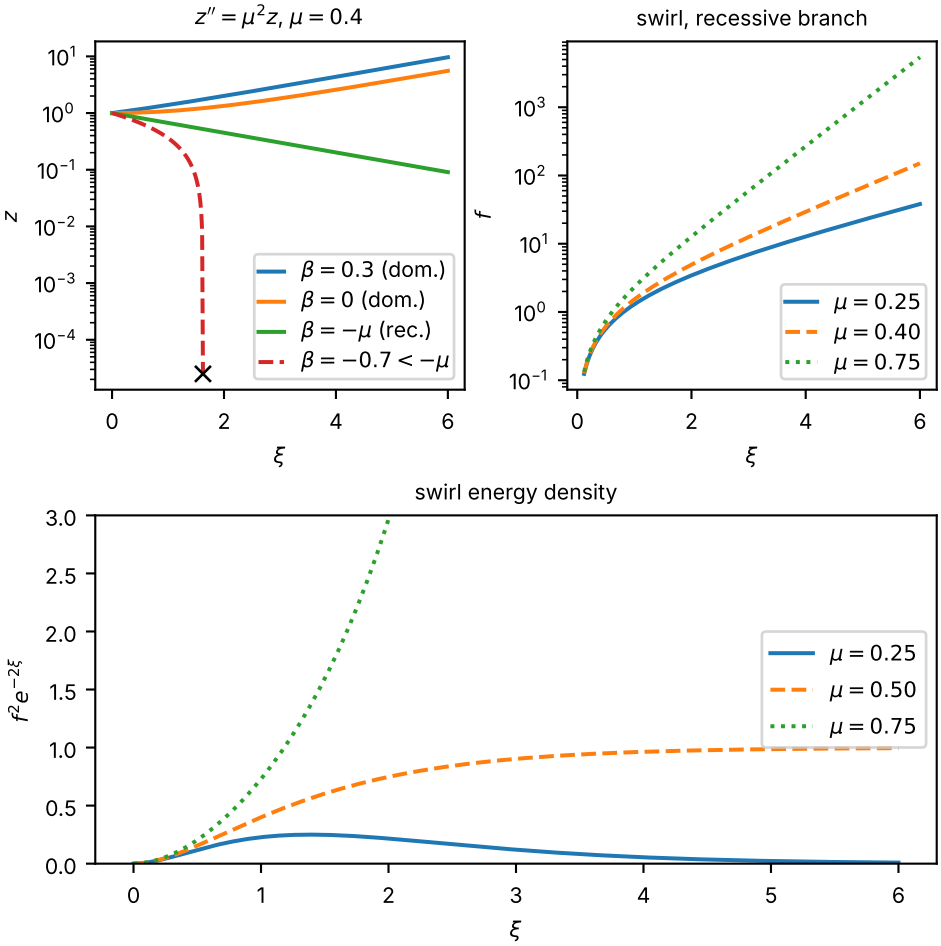


FIGURE 4.1. Toy example 4.3 (generated by `code/ch04_pole/make_a_toy_euler.py`, closed forms, no solver). Left: positive solutions of $z'' = \mu^2 z$, $\mu = 0.4$, with $z(0) = 1$, $z'(0) = \beta$; for $\beta < -\mu$ the solution has a zero (cross). Middle: swirl $f = \int_0^\xi \kappa z^{-2}$ on the recessive branch, $\kappa = 1$, $f = (e^{2\mu\xi} - 1)/(2\mu)$. Right: the swirl energy density $f^2 e^{-2\xi}$ on the recessive branch for $\mu = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$: it decays for $\mu < \frac{1}{2}$ and does not for $\mu \geq \frac{1}{2}$.

Example 4.4 (the one-ray solution: recessive branch without swirl)**.** The Landau solution $z = e^{-\xi}$ of the rest state ($Q \equiv 0$, $\mu = 1$; Example 2.32) is recessive at $+\infty$ with $U(1) = -4 < -3$. It has $\kappa = 0$. The theorem of Section 4.3 says that this value is possible *only* without swirl: a recessive branch with $\mu \geq \frac{1}{2}$ is a feature of the no-swirl theory (the edge of the Landau window) which disappears as soon as $\kappa \neq 0$.

4.2. Existence of the endpoint limits

In words. *Three things happen at a pole. The potential has a limit q (the swirl enters Q only through $(1-x^2)Q_{xxx} = -2ff_x$, whose sign is fixed near the pole, which makes Q monotone in a generalized sense); the limit cannot be below -2, because then $z'' = Wz$ oscillates and z cannot stay positive; and then the Riccati equation for $h = z'/z$ has only the two equilibria $\pm\mu$ as possible limits. The limit of f exists in $[-\infty, \infty]$ and is finite exactly when $\int z^{-2}$ converges.*

THEOREM 4.5 (endpoint limits). *Let (U, f, Q) be a local solution at the north pole (Definition 4.1).*

(a) *$q := \lim_{x\to 1} Q(x)$ exists, is finite, and $q \geq -2$. Equivalently, $\psi_{\mathbf{c}}(1) = P_{\mathbf{c}}(1) - q$ is finite for every polynomial $P_{\mathbf{c}}$.*

(b) *Put $\mu := \sqrt{1 + q/2} \geq 0$. Then $h = z'/z \to \mu$ or $h \to -\mu$ as $\xi \to \infty$; accordingly $U(1) := \lim_{x\to 1} U$ exists and equals $v_+ := -2 + 2\mu$ or $v_- := -2 - 2\mu$.*

(c) *$f(1) := \lim_{x\to 1} f \in [-\infty, \infty]$ exists; if $\kappa \neq 0$ it is finite if and only if $\int^{\infty} z^{-2}\,\mathrm{d}\xi < \infty$.*

PROOF. *(a), existence of the limit in $[-\infty, \infty)$.* If $\kappa = 0$, f is constant, so $Q_{xxx} = 0$ by (2.21): Q is a quadratic polynomial and has a finite limit. Let $\kappa \neq 0$. By Corollary 2.10 f is strictly monotone, so ff_x has a fixed strict sign on some $(x_1, 1)$; take $x_1 \geq \max(x_0, 0)$. Hence $Q_{xxx} = -2ff_x/(1-x^2)$ has a fixed strict sign on $(x_1, 1)$.

Case $Q_{xxx} > 0$ on $(x_1, 1)$. Then $(f^2)_x = 2ff_x < 0$: f^2 is decreasing and bounded below by 0, so $\int_{x_1}^1 |ff_x|\,\mathrm{d}s = \frac{1}{2}\big(f(x_1)^2 - \lim_{s\to 1} f(s)^2\big) \leq \frac{1}{2}f(x_1)^2 < \infty$. Taylor's formula with integral remainder gives, for $x_1 < x < 1$,

$$Q(x) = T(x) + \int_{x_1}^{x} \frac{(x-s)^2}{2} Q_{xxx}(s)\,\mathrm{d}s,$$
$$T(x) = Q(x_1) + Q_x(x_1)(x - x_1) + \tfrac{1}{2}Q_{xx}(x_1)(x - x_1)^2.$$

For $x_1 \leq s \leq x < 1$ we have $0 \leq x - s \leq 1 - s$ and $1 - s^2 = (1-s)(1+s)$, hence, as $s \geq 0$, $(x-s)^2|Q_{xxx}(s)| \leq (1-s)^2 \frac{2|ff_x|}{(1-s)(1+s)} = \frac{2(1-s)}{1+s}|ff_x| \leq 2|ff_x(s)|$. The integrand $\mathbf{1}_{\{s<x\}}\frac{(x-s)^2}{2}Q_{xxx}(s)$ converges pointwise as $x \to 1$ and is dominated by the integrable function $|ff_x|$; by dominated convergence the integral converges, and $T(x)$ converges. So Q has a finite limit.

Case $Q_{xxx} < 0$ on $(x_1, 1)$. Then Q_{xx} is strictly decreasing. If Q_{xx} is bounded below, it is bounded on $(x_1, 1)$, so $Q_x(x) = Q_x(x_1) + \int_{x_1}^x Q_{xx}$ is bounded, and Q has a finite limit (a bounded derivative makes Q Lipschitz, hence Cauchy at 1). If $Q_{xx} \to -\infty$, then $Q_{xx} < 0$ on some $(x_2, 1)$, so Q_x is strictly decreasing there. Either $Q_x > 0$ on all of $(x_2, 1)$: then Q is increasing and $Q(x) \leq Q(x_2) + Q_x(x_2)(x - x_2) \leq Q(x_2) + Q_x(x_2)$ (as $0 < x - x_2 < 1$), so Q converges; or $Q_x(x_3) \leq 0$ for some x_3,

then $Q_x < 0$ on $(x_3, 1)$, Q is decreasing, and its limit lies in $[-\infty, \infty)$. In all cases $q = \lim Q$ exists in $[-\infty, \infty)$.

(a), $q \geq -2$. Suppose $q < -2$ (this includes $q = -\infty$). Choose $k > 0$ with $1 + q/2 < -k^2$ if q is finite, and $k = 1$ if $q = -\infty$. Since $Q \to q$ there is ξ_1 with $W = 1 + \frac{1}{2}Q \leq -k^2$ on $[\xi_1, \infty)$. But $z > 0$ on $[\xi_0, \infty)$, contradicting Lemma 3.2(c). Hence $q \in [-2, \infty)$. As $P_{\boldsymbol{c}}$ is a polynomial, $\psi_{\boldsymbol{c}}(1) = P_{\boldsymbol{c}}(1) - q$ is finite; conversely finiteness of $\psi_{\boldsymbol{c}}(1)$ gives finiteness of q.

(b). By (a), $W = \mu^2 + w$ with $\mu^2 = 1 + q/2 \geq 0$ and $w = \frac{1}{2}(Q - q) \to 0$ $(\xi \to \infty)$. Since $z > 0$ on $[\xi_0, \infty)$, $h = z'/z \in C^1[\xi_0, \infty)$ is finite and $h' = \mu^2 + w - h^2$. Lemma 3.14 gives $h \to \mu$ or $h \to -\mu$, and $U = 2(h - \tanh\xi) \to 2(\pm\mu - 1) = -2 \pm 2\mu$.

(c). If $\kappa = 0$, f is constant. If $\kappa \neq 0$, $f' = \kappa z^{-2}$ has the fixed sign of κ, so f is monotone and has a limit in $[-\infty, \infty]$; and $f(\xi) - f(\xi_1) = \kappa \int_{\xi_1}^{\xi} z^{-2}$ with positive integrand, so the limit is finite iff $\int^{\infty} z^{-2} < \infty$. □

Remark 4.6 (what was used)**.** Only two properties of the solution enter: the monotonicity of f (hence the fixed sign of Q_{xxx} near the pole) and the positivity of z on a half-line. In particular the theorem holds for local solutions that do not extend to a global one. Parts (a), (b) are contained in Theorem 1.3 of Li–Li–Yan [24] (stated there at the south pole), whose proof uses a sign property of the triple integral; the proof above, through Q and the linear equation, is shorter. Since $\mu^2 = 1 + q/2$, the endpoint value determines the effective constant: $U(1) \in \{-2 \pm 2\sqrt{1 + c_2'}\}$ with $c_2' = q/2$, as in Proposition 2.27.

4.3. The recessive branch and the threshold −3

In words. *On the recessive branch $z \approx e^{-\mu\xi}$, the swirl $f' = \kappa z^{-2}$ grows like $e^{2\mu\xi}$, and $f^2 e^{-2\xi} \approx e^{(4\mu-2)\xi}$. On the other hand the first integral forces the swirl energy $\int f^2 e^{-2\xi}$ to be finite when f blows up. So the recessive branch needs $\mu < \frac{1}{2}$ (Toy example 4.3). At $\mu = \frac{1}{2}$ the exponents balance and one needs a sign argument on the correction $Q - q$. The result is that $U_\theta(1) = -2 - 2\mu$ cannot be ≤ -3 when there is swirl.*

The equation for the correction. Let (U, f, Q) be a local solution at the north pole with $\kappa \neq 0$ and q, μ as in Theorem 4.5. Put

$$R := Q - q, \qquad \Lambda_{\mathrm{p}} := \lambda + 2q, \qquad F := \Lambda_{\mathrm{p}} - f^2. \tag{4.3}$$

Since $\mathsf{L}[q] = q\mathsf{L}[1] = -2q$ (Lemma 2.18(b)) and $\mathsf{L}[Q] = \lambda - f^2$, subtracting gives

$$\mathsf{L}[R] = F, \qquad R \to 0 \ (\xi \to \infty), \qquad W = \mu^2 + \tfrac{1}{2}R. \tag{4.4}$$

For every C^2 function g, by the line form of L in Lemma 2.18(a),

$$(\cosh^4\xi\, g')' = \cosh^4\xi\, g'' + 4\cosh^3\xi \sinh\xi\, g' = \cosh^2\xi \left(\mathsf{L}[g] + 2g\right), \tag{4.5}$$

so (4.4) is equivalent to $(\cosh^4\xi\, R')' = \cosh^2\xi\,(F + 2R)$. Since $\operatorname{sech}^2\xi = 4E/(1 + E)^2 \in [E, 4E]$ for $\xi \geq 0$ (with $E = e^{-2\xi}$),

$$\int^{\infty} f^2 e^{-2\xi}\,\mathrm{d}\xi \;\leq\; \int^{1} f^2\,\mathrm{d}x \;\leq\; 4\int^{\infty} f^2 e^{-2\xi}\,\mathrm{d}\xi, \tag{4.6}$$

so finiteness of $\int f^2 e^{-2\xi}$ means finiteness of $\int^1 U_\varphi^2\,\mathrm{d}x$: the *swirl energy.*

Lemma 4.7 (growth bounds on the recessive branch; finite swirl energy)**.** *Let (U, f, Q) be a local solution at the north pole with $\kappa \neq 0$ and $h \to -\mu$, $\mu > 0$.*

(a) (Growth.) *For every* $\varepsilon \in (0,\mu)$ *there are* $c>0$ *and* ξ_ε *with* $|f| \ge c\, e^{2(\mu-\varepsilon)\xi}$ *for* $\xi \ge \xi_\varepsilon$*; and* $ff' > 0$ *eventually. In particular* $|f| \to \infty$.

(b) (Energy.) *If* $|f| \to \infty$ (*so in particular under* (a)), *then there is* ξ_2 *with* $R' < 0$ *and* $R > 0$ *on* $[\xi_2, \infty)$, *and* $\int^\infty f^2 e^{-2\xi}\,\mathrm{d}\xi < \infty$.

Proof. (a) There is ξ_1 with $h \le -(\mu-\varepsilon)$ on $[\xi_1,\infty)$. Integrating $(\ln z)' = h$ gives $z(\xi) \le z(\xi_1)e^{-(\mu-\varepsilon)(\xi-\xi_1)} = Ce^{-(\mu-\varepsilon)\xi}$, $C = z(\xi_1)e^{(\mu-\varepsilon)\xi_1}$. Hence $|f'| = |\kappa| z^{-2} \ge c' e^{2(\mu-\varepsilon)\xi}$ with $c' = |\kappa| C^{-2}$. Since f' has the constant sign of κ,

$$|f(\xi) - f(\xi_1)| = \int_{\xi_1}^{\xi} |f'| \ge \frac{c'}{2(\mu-\varepsilon)}\big(e^{2(\mu-\varepsilon)\xi} - e^{2(\mu-\varepsilon)\xi_1}\big),$$

so $|f(\xi)| \ge \frac{c'}{4(\mu-\varepsilon)} e^{2(\mu-\varepsilon)\xi} =: c\, e^{2(\mu-\varepsilon)\xi}$ for all ξ beyond some ξ_ε. Finally $f(\xi)-f(\xi_1)$ has the sign of κ and modulus $\to\infty$, so f eventually has the sign of κ, which is the sign of f': $ff' > 0$ eventually.

(b) Since $R \to 0$, Λ_{p} is constant and $|f| \to \infty$, there is ξ_1 with $\Lambda_{\mathrm{p}} + 2R \le \frac12 f^2$, i.e. $F + 2R \le -\frac12 f^2$, on $[\xi_1,\infty)$. Then $\Phi := \cosh^4\!\xi\, R'$ satisfies, by (4.5), $\Phi' = \cosh^2\!\xi\,(F+2R) \le -\frac12 \cosh^2\!\xi\, f^2$, so

$$\Phi(\xi) \le \Phi(\xi_1) - \tfrac12 I(\xi), \qquad I(\xi) := \int_{\xi_1}^{\xi} \cosh^2 s\, f(s)^2\,\mathrm{d}s \qquad (\xi \ge \xi_1). \tag{4.7}$$

$R' < 0$, $R > 0$. As $f^2 \to \infty$ the integrand of I is ≥ 1 for large s, so $I \to \infty$ and $\Phi \to -\infty$; thus $R' = \Phi/\cosh^4\!\xi < 0$ on some $[\xi_2,\infty)$. A strictly decreasing function with limit 0 is positive, so $R > 0$ there.

Finite energy. By (4.7), $-R'(\xi) = -\Phi(\xi)\cosh^{-4}\!\xi \ge \cosh^{-4}\!\xi\,\big(\frac12 I(\xi) - \Phi(\xi_1)\big)$. Choose $\xi_3 \ge \xi_1$ with $\frac12 I(\xi_3) \ge |\Phi(\xi_1)|$; as I is increasing the bracket is ≥ 0 on $[\xi_3,\infty)$. Integrating over $[\xi_3, X]$ and letting $X \to \infty$ (monotone convergence; $R(X) \to 0$),

$$R(\xi_3) = \int_{\xi_3}^{\infty} (-R')\,\mathrm{d}\xi \;\ge\; \int_{\xi_3}^{\infty} \cosh^{-4}\!\xi\,\big(\tfrac12 I(\xi) - \Phi(\xi_1)\big)\mathrm{d}\xi,$$

so $\int_{\xi_3}^{\infty} \cosh^{-4}\!\xi\, I(\xi)\,\mathrm{d}\xi \le 2R(\xi_3) + 2|\Phi(\xi_1)| \int_{\xi_3}^{\infty} \cosh^{-4} < \infty$. Using $I(\xi) \ge \int_{\xi_3}^{\xi} \cosh^2 s\, f^2\,\mathrm{d}s$ and Tonelli's theorem (nonnegative integrands),

$$\begin{aligned}\int_{\xi_3}^{\infty} \cosh^{-4}\!\xi\, I(\xi)\,\mathrm{d}\xi \;&\ge\; \int_{\xi_3}^{\infty} \cosh^2 s\, f(s)^2 \Big(\int_s^{\infty} \cosh^{-4}\!\xi\,\mathrm{d}\xi\Big)\mathrm{d}s \\ &\ge\; \frac{1}{16}\int_{\xi_3}^{\infty} f(s)^2 e^{-2s}\,\mathrm{d}s,\end{aligned}$$

because $\cosh^2 s \ge \frac14 e^{2s}$ and, as $\cosh\xi \le e^{\xi}$ for $\xi \ge 0$, $\int_s^\infty \cosh^{-4} \ge \int_s^\infty e^{-4\xi}\mathrm{d}\xi = \frac14 e^{-4s}$. □

Toy example 4.8 (a correction tilts the recessive solution)**.** Let $\mu = \frac12$ and consider $\eta' = \eta(1-\eta) + \frac12 R$ (the deviation equation (4.8) below) with a prescribed small correction $R(\xi) = \delta e^{-a\xi}$, $a > 0$. Near $\eta = 0$ the equation is $\eta' = \eta + \frac12 R$ up to $O(\eta^2)$, and $\eta = 0$ is *repelling* in forward time: the only solution that stays near 0 for all large ξ is $\eta_{\mathrm{lin}}(\xi) = -\frac12 \int_\xi^\infty e^{\xi - s} R(s)\,\mathrm{d}s = -\frac{\delta\, e^{-a\xi}}{2(1+a)}$ (check: $\eta_{\mathrm{lin}}' = \eta_{\mathrm{lin}} + \frac12 R$). Hence the recessive solution $z = \varphi e^{-\xi/2}$, $\varphi'/\varphi = \eta$, satisfies

$$\operatorname{sign}\eta = -\operatorname{sign} R:$$
$$R > 0 \implies \eta < 0,\ \varphi \text{ decreasing}; \qquad R < 0 \implies \varphi \text{ increasing}.$$

In the first case $z^{-4}e^{-2\xi} = \varphi^{-4}$ is bounded below and the swirl energy $\int f^2 e^{-2\xi} \gtrsim \int \varphi^{-4}$ diverges; in the second $\varphi \uparrow$ could make φ^{-4} integrable. The proof of Theorem 4.9 at $\mu = \frac{1}{2}$ shows that the swirl itself *forces* $R > 0$ (Lemma 4.7(b)), the bad sign for the recessive branch.

THEOREM 4.9 (no genuine swirl with $U_\theta(1) \le -3$). *Let (U, f, Q) be a local solution at the north pole with $\kappa \neq 0$. If $h \to -\mu$ (the recessive branch, $U(1) = -2 - 2\mu$), then $\mu < \frac{1}{2}$. Consequently every local solution with genuine swirl has $U_\theta(1) > -3$, and, at the south pole, $U_\theta(-1) < 3$.*

PROOF. *The consequences.* By Theorem 4.5(b), $U(1) = -2 + 2\mu \ge -2$ on the dominant branch and $U(1) = -2 - 2\mu$ on the recessive branch; hence $U(1) \le -3$ can happen only on the recessive branch with $-2 - 2\mu \le -3$, i.e. $\mu \ge \frac{1}{2}$, which the first statement excludes. At the south pole apply σ_2, which sends $U(1)$ to $-U(-1)$ (Remark 4.2).

The first statement. Suppose $\kappa \neq 0$, $h \to -\mu$ and $\mu \ge \frac{1}{2}$. By Lemma 4.7(a), $|f| \to \infty$, so Lemma 4.7(b) applies: $R > 0$ on $[\xi_2, \infty)$ and $\int f^2 e^{-2\xi} < \infty$.

Case $\mu > \frac{1}{2}$. Take $\varepsilon \in (0, \mu - \frac{1}{2})$. By Lemma 4.7(a), $f^2 e^{-2\xi} \ge c^2 e^{(4\mu - 4\varepsilon - 2)\xi} \ge c^2$ for large ξ, since $4\mu - 4\varepsilon - 2 > 0$: not integrable. Contradiction.

Case $\mu = \frac{1}{2}$. The exponent count is borderline and we argue with signs, in four steps.

Step 1 (the equation for the deviation). Write $h = -\frac{1}{2} + \eta$ with $\eta \to 0$. Since $W = \mu^2 + \frac{1}{2}R = \frac{1}{4} + \frac{1}{2}R$ and $h^2 = \frac{1}{4} - \eta + \eta^2$, the Riccati equation $h' = W - h^2$ becomes

$$\eta' = \eta(1 - \eta) + \tfrac{1}{2}R. \tag{4.8}$$

(The linearization $\eta' = \eta$ shows that η is the repelling mode; only its sign is needed.)

Step 2 ($\eta < 0$ eventually: sign propagation). Take $\xi_4 \ge \xi_2$ with $|\eta| < \frac{1}{2}$ on $[\xi_4, \infty)$. At any $\xi \ge \xi_4$ with $\eta(\xi) \ge 0$ we have $\eta(1 - \eta) \ge 0$ and $\frac{1}{2}R > 0$, hence $\eta'(\xi) > 0$. Suppose $\eta(\xi_5) \ge 0$ for some $\xi_5 \ge \xi_4$. We claim $\eta \ge 0$ on $[\xi_5, \infty)$. Otherwise there is $\xi_7 > \xi_5$ with $\eta(\xi_7) < 0$; let $\xi_8 = \sup\{\xi \in [\xi_5, \xi_7] : \eta(\xi) \ge 0\}$. By continuity $\eta(\xi_8) = 0$ and $\eta < 0$ on $(\xi_8, \xi_7]$; but $\eta'(\xi_8) > 0$ forces $\eta > 0$ just to the right of ξ_8, a contradiction. Hence $\eta \ge 0$, and therefore $\eta' > 0$, on $[\xi_5, \infty)$: η is strictly increasing there, so $\eta(\xi) \ge \eta(\xi_5 + 1) > \eta(\xi_5) \ge 0$ for $\xi \ge \xi_5 + 1$, contradicting $\eta \to 0$. Consequently

$$\eta < 0 \quad \text{on } [\xi_4, \infty).$$

Put $\varphi := ze^{\xi/2} > 0$. Then $\varphi'/\varphi = h + \frac{1}{2} = \eta < 0$, so φ is decreasing on $[\xi_4, \infty)$, and φ^{-4} is increasing and positive, hence bounded below by $\varphi(\xi_4)^{-4} > 0$.

Step 3 ($|f|$ is comparable to $|f'|$). Let $\rho := f/f'$ (local notation; $f' \neq 0$ everywhere). From $f' = \kappa z^{-2}$ we get $f'' = -2\kappa z^{-3} z'$, so $f''/f' = -2h = 1 - 2\eta$ and

$$\rho' = \frac{f'^2 - ff''}{f'^2} = 1 - \rho\,\frac{f''}{f'} = 1 - (1 - 2\eta)\rho.$$

By Lemma 4.7(a), $\rho > 0$ on some $[\xi_6, \infty)$; take $\xi_6 \ge \xi_4$. There $a := 1 - 2\eta \in (1, 2)$ (since $-\frac{1}{2} < \eta < 0$). With $\mathcal{A}(\xi) := \int_{\xi_6}^{\xi} a$ we have $(\rho e^{\mathcal{A}})' = (\rho' + a\rho)e^{\mathcal{A}} = e^{\mathcal{A}}$, hence for $\xi \ge \xi_6 + 1$

$$\rho(\xi)e^{\mathcal{A}(\xi)} = \rho(\xi_6) + \int_{\xi_6}^{\xi} e^{\mathcal{A}(s)}\mathrm{d}s \ge \int_{\xi-1}^{\xi} e^{\mathcal{A}(s)}\mathrm{d}s.$$

Dividing by $e^{\mathcal{A}(\xi)}$ and using $\mathcal{A}(\xi)-\mathcal{A}(s)=\int_s^\xi a\le 2(\xi-s)$, $\rho(\xi)\ge\int_{\xi-1}^{\xi}e^{-2(\xi-s)}\mathrm{d}s=\int_0^1 e^{-2u}\mathrm{d}u=:c_0>0$. Therefore $|f|=\rho|f'|\ge c_0|f'|=c_0|\kappa|z^{-2}$ on $[\xi_6+1,\infty)$.

Step 4 (the energy is infinite). By Step 3 and $z^{-4}e^{-2\xi}=(ze^{\xi/2})^{-4}=\varphi^{-4}$, $\int^\infty f^2e^{-2\xi}\,\mathrm{d}\xi\ge c_0^2\kappa^2\int^\infty\varphi^{-4}\,\mathrm{d}\xi=\infty$, because φ^{-4} is bounded below by a positive constant (Step 2). This contradicts Lemma 4.7(b). □

Remark 4.10 (what the proof shows; the mechanism at the borderline)**.** Theorem 4.9 is Theorem 1.4(iii) of [24] (including the borderline value 3); the proof given here is self-contained and uses the variables of the line. It shows more: the recessive solution of $z''=(1+\frac12 Q)z$ is excluded as soon as the feedback $f^2e^{-2\xi}$ fails to be integrable, and at $\mu=\frac12$ the sign $R>0$, forced by the swirl through Lemma 4.7(b), drives $\varphi=ze^{\xi/2}$ *down*, which makes $z^{-4}e^{-2\xi}=\varphi^{-4}$ non-integrable. (In the toy with $R\equiv0$ one has $\varphi\equiv$ const and $f^2e^{-2\xi}\to$ const $\neq0$, Figure 4.1, right: not integrable. The sign argument is needed because a correction $R\neq0$ of the wrong sign could in principle make φ increase and the energy converge.) Numerically, families of solutions whose no-swirl edge has exponent $k\to\frac12$ approach $U_\theta(1)=-3$ only as $\kappa\to0$; see Atlas 4.50. The relation of Theorem 4.9 to the threshold 3 of Li–Li–Yan [26] (the no-swirl endpoint solution U_c^- has $U_c^-(1)=-2-2k_2\le-3$ iff $k_2\ge\frac12$, so no swirl solution can approach it when $k_2>\frac12$) is worked out in Chapter 5.

Example 4.11 (the threshold in the no-swirl family $\boldsymbol{c}=(0,c_2,0)$)**.** For $\kappa=0$ the potential is the polynomial $W_{\boldsymbol{c}}$ and $q=Q(1)=2c_2$, $\mu=\sqrt{1+c_2}=k_2$. The no-swirl solutions of Theorem 3.44 have $U(1)=-2+2k_2$ (dominant) and, for the extremal solution U_c^-, $U_c^-(1)=-2-2k_2$ (recessive). The recessive value is ≤-3 exactly when $k_2\ge\frac12$, i.e. $c_2\ge-\frac34$. So for $c_2\in[-\frac34,\infty)$ the recessive pole value $-2-2k_2$ *occurs without swirl and is excluded as soon as* $\kappa\neq0$ (Theorem 4.9): in the Landau case $c=0$ ($k_2=1$) the edge $U_0^-=-2(1+x)$ with $U(1)=-4$ (Example 4.4) is replaced, for $\kappa>0$, by an edge of type L_1 (Numerical observation 4.28, second row). For $c_2\in(-1,-\frac34)$ ($k_2<\frac12$) the recessive value lies in $(-3,-2)$ and type R becomes possible for the swirl solutions: this is the window of exponents in which the lower edge of the class-Z lens is of type R (Chapter 7); the third row of Numerical observation 4.28 is $c_2=-0.9$, $k_2=0.316<\frac12$. The point is that $\mu=\frac12$ is a *wall between two behaviours*, not a technicality of the proof.

4.4. The Green operator at a pole, with weights

In words. *The only place where the swirl acts on the potential is the equation* $\mathsf{L}[R]=\Lambda_{\mathrm{p}}-f^2$. *To control* $R=Q-q$ *near the pole from bounds on* f *we need an explicit solution operator* $R=\mathfrak{r}_{R_*}F$ *with* $R\to0$ *at the pole and a prescribed value* R_* *at a cut point* ξ_*, *and weighted estimates for it. The kernel of* L *is spanned by* x *(which does not tend to* 0*) and* $t^2=(1-x)^2$ *(which does); variation of constants gives the formula and the estimates (Q1)–(Q3), which are the analytic input for the a priori asymptotics (Section 4.5) and for the contraction maps of the charts.*

Toy example 4.12 (Dirichlet data at a cut)**.** Take $F\equiv-2$ on $[\xi_*,\infty)$. The constant $R\equiv1$ solves $\mathsf{L}[R]=-2$ (Lemma 2.18(b)) but does not tend to 0. Since $\mathsf{L}[1-x]=-2$ as well, $R=1-x=:t$ solves the equation and $t\to0$ at the pole; indeed $1-x=t\le2E$. The general solution tending to 0 is $t+\beta t^2$, because $\ker\mathsf{L}=\operatorname{span}\{x,t^2\}$ and the coefficient of x must vanish; the value at the cut fixes β. This is the role of the Dirichlet datum R_* in Lemma 4.13: for $R_*=t_*$ one finds

$\beta = 0$ and $\mathfrak{r}_{t_*}(-2) = t$, and the bound (Q1) with $\Phi_0 = 2$ reads $t \le (6 + 4t_*/E_*)E$, true (since $t \le 2E$) though not sharp.

Lemma 4.13 (Green operator with pole weights)**.** *Let $\xi_* \ge 0$, $t := 1 - \tanh\xi$, $t_* = t(\xi_*)$, $E = e^{-2\xi}$, $E_* = e^{-2\xi_*}$. Let F be continuous on $[\xi_*, \infty)$ with $|F| \le \Phi$ for a positive function Φ with $\int_{\xi_*}^{\infty} \Phi E \,\mathrm{d}\eta < \infty$, and let $R_* \in \mathbb{R}$. Put*

$$\begin{gathered} \mathcal{A}(\xi) = -\int_{\xi}^{\infty} FE \,\mathrm{d}\eta, \qquad \mathcal{B}(\xi) = \mathcal{B}_* - \frac{1}{2}\int_{\xi_*}^{\xi} F \sinh(2\eta)\,\mathrm{d}\eta, \\ \mathcal{B}_* = \frac{R_* - \tanh(\xi_*)\mathcal{A}(\xi_*)}{t_*^2}, \end{gathered} \tag{4.9}$$

$$\mathfrak{r}_{R_*}F := \tanh(\xi)\,\mathcal{A} + t^2\mathcal{B}. \tag{4.10}$$

Then $\mathsf{L}[\mathfrak{r}_{R_*}F] = F$, $\mathfrak{r}_{R_*}F(\xi_*) = R_*$, $\mathfrak{r}_{R_*}F \to 0$ as $\xi \to \infty$, and for $\xi \ge \xi_*$

$$\begin{aligned} |\mathfrak{r}_{R_*}F(\xi)| \le \int_{\xi}^{\infty} \Phi E \,\mathrm{d}\eta + E(\xi)\int_{\xi_*}^{\xi} e^{-2(\xi-\eta)}\Phi(\eta)\,\mathrm{d}\eta \\ + 4\frac{E(\xi)^2}{E_*^2}\Big(|R_*| + \int_{\xi_*}^{\infty}\Phi E\,\mathrm{d}\eta\Big). \end{aligned} \tag{4.11}$$

If $R \in C^2[\xi_*, \infty)$ solves $\mathsf{L}[R] = F$ with $R \to 0$ and $R(\xi_*) = R_*$, then $R = \mathfrak{r}_{R_*}F$. In particular, with $m := 1 + \xi - \xi_*$:

(Q1) *if $|F| \le \Phi_0$ (a constant), then $|\mathfrak{r}_{R_*}F| \le (3\Phi_0 + 4|R_*|/E_*)\,E$;*

(Q2) *if $|F| \le F_0 E^{-2\nu}$ with $0 < \nu < \frac{1}{2}$, then $|\mathfrak{r}_{R_*}F| \le (F_0 K_\nu + 4|R_*|E_*^{2\nu-1})\,E^{1-2\nu}$, $K_\nu = \frac{5}{2-4\nu} + \frac{1}{2+4\nu}$;*

(Q3) *if $|F| \le F_0 m^2$, then $|\mathfrak{r}_{R_*}F| \le (7F_0 + 4|R_*|/E_*)\,Em^2$.*

Moreover $\mathfrak{r}_{R_*}F$ is jointly linear in (R_*, F); in particular $\mathfrak{r}_{R_*}F_1 - \mathfrak{r}_{R_*}F_2 = \mathfrak{r}_0(F_1 - F_2)$, so (Q1)–(Q3) *apply to differences with R_*-datum 0.*

Proof. We use the elementary identities

$$\begin{gathered} t = \frac{e^{-\xi}}{\cosh\xi} = \frac{2E}{1+E}, \quad t^2\cosh^2\xi = E, \\ t' = -\operatorname{sech}^2\xi, \quad E + 2t\sinh\xi\cosh\xi = 1, \end{gathered} \tag{4.12}$$

the last because $2t\sinh\xi\cosh\xi = 2e^{-\xi}\sinh\xi = 1 - e^{-2\xi}$. Note $E \le t \le 2E$ for $\xi \ge 0$.

The equation. Write $g = \tanh\xi\,\mathcal{A} + t^2\mathcal{B}$ with $\mathcal{A}' = FE$ and $\mathcal{B}' = -\frac{1}{2}F\sinh 2\xi = -F\sinh\xi\cosh\xi$. First, $\tanh\xi\,\mathcal{A}' + t^2\mathcal{B}' = F\big(\tanh\xi\, t^2\cosh^2\xi - t^2\sinh\xi\cosh\xi\big) = 0$ by (4.12). Hence $g' = (\tanh)'\mathcal{A} + (t^2)'\mathcal{B}$ and $g'' = (\tanh)''\mathcal{A} + (t^2)''\mathcal{B} + (\tanh)'\mathcal{A}' + (t^2)'\mathcal{B}'$. Since $\tanh\xi$ and $t^2 = (1-x)^2$ lie in $\ker\mathsf{L}$ (Lemma 2.18(b)), the line form $\mathsf{L}[g] = \cosh^2\xi\, g'' + 4\sinh\xi\cosh\xi\, g' - 2g$ gives

$$\begin{aligned} \mathsf{L}[g] &= \mathcal{A}\,\mathsf{L}[\tanh] + \mathcal{B}\,\mathsf{L}[t^2] + \cosh^2\xi\big((\tanh)'\mathcal{A}' + (t^2)'\mathcal{B}'\big) \\ &= \cosh^2\xi\big(\operatorname{sech}^2\xi\, FE + (-2t\operatorname{sech}^2\xi)(-F\sinh\xi\cosh\xi)\big) \\ &= F\,(E + 2t\sinh\xi\cosh\xi) = F. \end{aligned}$$

At ξ_*, $g(\xi_*) = \tanh\xi_*\mathcal{A}(\xi_*) + t_*^2\mathcal{B}_* = R_*$ by the choice of $\mathcal{B}_*$. (The integrals converge: $|FE| \le \Phi E$ is integrable, and $\mathcal{B}$ involves an integral over a bounded interval.) The identities above were also checked by computer algebra (`verify_a_ch04.py`).

The bound (4.11). Since $0 \le \tanh\xi \le 1$, $|\tanh\xi\,\mathcal{A}| \le \int_\xi^\infty \Phi E$. Next $|\sinh 2\eta| \le \frac{1}{2}e^{2\eta}$ and $t \le 2E$ give

$$t^2|\mathcal{B}-\mathcal{B}_*| \le 4E^2\cdot\frac{1}{2}\int_{\xi_*}^{\xi}\Phi\,\tfrac{1}{2}e^{2\eta}\mathrm{d}\eta = E^2\int_{\xi_*}^{\xi}\Phi e^{2\eta}\mathrm{d}\eta$$
$$= E(\xi)\int_{\xi_*}^{\xi} e^{-2(\xi-\eta)}\Phi(\eta)\,\mathrm{d}\eta.$$

Finally $t^2|\mathcal{B}_*| = (t/t_*)^2|R_* - \tanh\xi_*\mathcal{A}(\xi_*)| \le (t/t_*)^2\big(|R_*| + \int_{\xi_*}^\infty \Phi E\big)$ and $t/t_* \le 2E/E_*$ because $t \le 2E$ and $t_* \ge E_*$. Adding the three bounds gives (4.11).

Decay. Each term of (4.11) tends to 0: the first is the tail of a convergent integral; the third is $O(E^2)$; the second equals $\int_{\xi_*}^{\xi} e^{-4(\xi-\eta)}\Phi(\eta)E(\eta)\,\mathrm{d}\eta$, which tends to 0 by dominated convergence (ΦE is integrable and $e^{-4(\xi-\eta)} \to 0$ for each η, with $e^{-4(\xi-\eta)} \le 1$). So $\mathfrak{r}_{R_*}F \to 0$.

Uniqueness. If $\mathsf{L}[R] = F$, $R \to 0$, $R(\xi_*) = R_*$, then $D := R - \mathfrak{r}_{R_*}F$ solves $\mathsf{L}[D] = 0$, so $D = \alpha\tanh\xi + \beta t^2$ (Lemma 2.18(b)). Since $D \to 0$, $\tanh\xi \to 1$ and $t^2 \to 0$, $\alpha = 0$; then $D(\xi_*) = \beta t_*^2 = 0$ gives $\beta = 0$.

(Q1). With $\Phi = \Phi_0$: $\int_\xi^\infty \Phi_0 E = \frac{1}{2}\Phi_0 E$; $E\int_{\xi_*}^{\xi} e^{-2(\xi-\eta)}\Phi_0 \le \frac{1}{2}\Phi_0 E$; and, using $E \le E_*$, $4\frac{E^2}{E_*^2}\big(|R_*| + \frac{1}{2}\Phi_0 E_*\big) \le 4|R_*|\frac{E}{E_*} + 2\Phi_0 E$. The sum is $(\frac{1}{2}+\frac{1}{2}+2)\Phi_0 E + 4|R_*|E/E_*$.

(Q2). With $\Phi = F_0E^{-2\nu} = F_0e^{4\nu\eta}$, $0 < \nu < \frac{1}{2}$:

$$\int_\xi^\infty F_0e^{4\nu\eta}e^{-2\eta}\mathrm{d}\eta = \frac{F_0E^{1-2\nu}}{2-4\nu},$$
$$E\int_{\xi_*}^{\xi} e^{-2(\xi-\eta)}F_0e^{4\nu\eta}\mathrm{d}\eta \le F_0e^{-4\xi}\frac{e^{(2+4\nu)\xi}}{2+4\nu} = \frac{F_0E^{1-2\nu}}{2+4\nu}.$$

For the third term $(E/E_*)^2 \le (E/E_*)^{1-2\nu}$ (as $E/E_* \le 1$ and $2 \ge 1-2\nu$), so it is at most $4(E/E_*)^{1-2\nu}\big(|R_*| + \frac{F_0E_*^{1-2\nu}}{2-4\nu}\big) = 4|R_*|E_*^{2\nu-1}E^{1-2\nu} + \frac{4F_0}{2-4\nu}E^{1-2\nu}$. The coefficient of $F_0E^{1-2\nu}$ is $\frac{1}{2-4\nu} + \frac{1}{2+4\nu} + \frac{4}{2-4\nu} = K_\nu$.

(Q3). With $\Phi = F_0m^2$: for $\eta = \xi + s$, $m(\eta) = m + s \le m(1+s)$ (as $m \ge 1$), so $\int_\xi^\infty F_0m(\eta)^2e^{-2\eta}\mathrm{d}\eta \le F_0m^2E\int_0^\infty(1+s)^2e^{-2s}\mathrm{d}s = \frac{5}{4}F_0Em^2$; since $m(\eta) \le m(\xi)$ for $\eta \le \xi$, the second term is $\le F_0m^2E\int_0^\infty e^{-2u}\mathrm{d}u = \frac{1}{2}F_0Em^2$; and $\int_{\xi_*}^\infty F_0m^2E\,\mathrm{d}\eta = \frac{5}{4}F_0E_*$, so the third term is $\le 4\frac{E^2}{E_*^2}\big(|R_*| + \frac{5}{4}F_0E_*\big) \le 4|R_*|\frac{E}{E_*} + 5F_0E$ (using $E \le E_*$). Since $m \ge 1$ the total is $\le \big(\frac{5}{4}+\frac{1}{2}+5\big)F_0Em^2 + 4|R_*|Em^2/E_*$, and $\frac{27}{4} \le 7$. □

Example 4.14 (how sharp are the weights? exponential sources)**.** Take $F = e^{4\nu\xi} = E^{-2\nu}$ with $0 < \nu < \frac{1}{2}$. The function $R_0 = -\dfrac{E^{1-2\nu}}{1-4\nu^2} = -\dfrac{e^{-(2-4\nu)\xi}}{1-4\nu^2}$ satisfies, by the line form of L and a direct computation, $\mathsf{L}[R_0] = F + O(e^{(4\nu-2)\xi})$ (the remainder is explicit, `verify_a_ch04.py`), and $R_0 \to 0$. The difference $D = \mathfrak{r}_{R_*}F - R_0$ tends to 0 and satisfies $\mathsf{L}[D] = O(E^{1-2\nu})$, a bounded source, so $|D| \le CE = o(E^{1-2\nu})$ by (Q1) and uniqueness (Lemma 4.13); hence $\mathfrak{r}_{R_*}F = -\dfrac{E^{1-2\nu}}{1-4\nu^2}\,(1+o(1))$ for every R_*: the power $E^{1-2\nu}$ of (Q2) is *exactly* the true decay rate, and the best constant is $1/(1-4\nu^2)$ (negative sign: $\mathsf{L}[R] = F > 0$ and $R \to 0$ force $R < 0$, cf. Lemma 2.19), whereas the bound (Q2) gives $K_\nu = \frac{5}{2-4\nu} + \frac{1}{2+4\nu}$. Both constants blow up as $\nu \uparrow \frac{1}{2}$

(Figure 4.2, right): at $\nu = \frac{1}{2}$ the Green operator no longer gains a power of E over the source, which is the analytic face of the threshold $\mu = \frac{1}{2}$ of Theorem 4.9 (here $F \sim f^2$ with $f \sim e^{2\mu\xi}$, $\nu = \mu$).

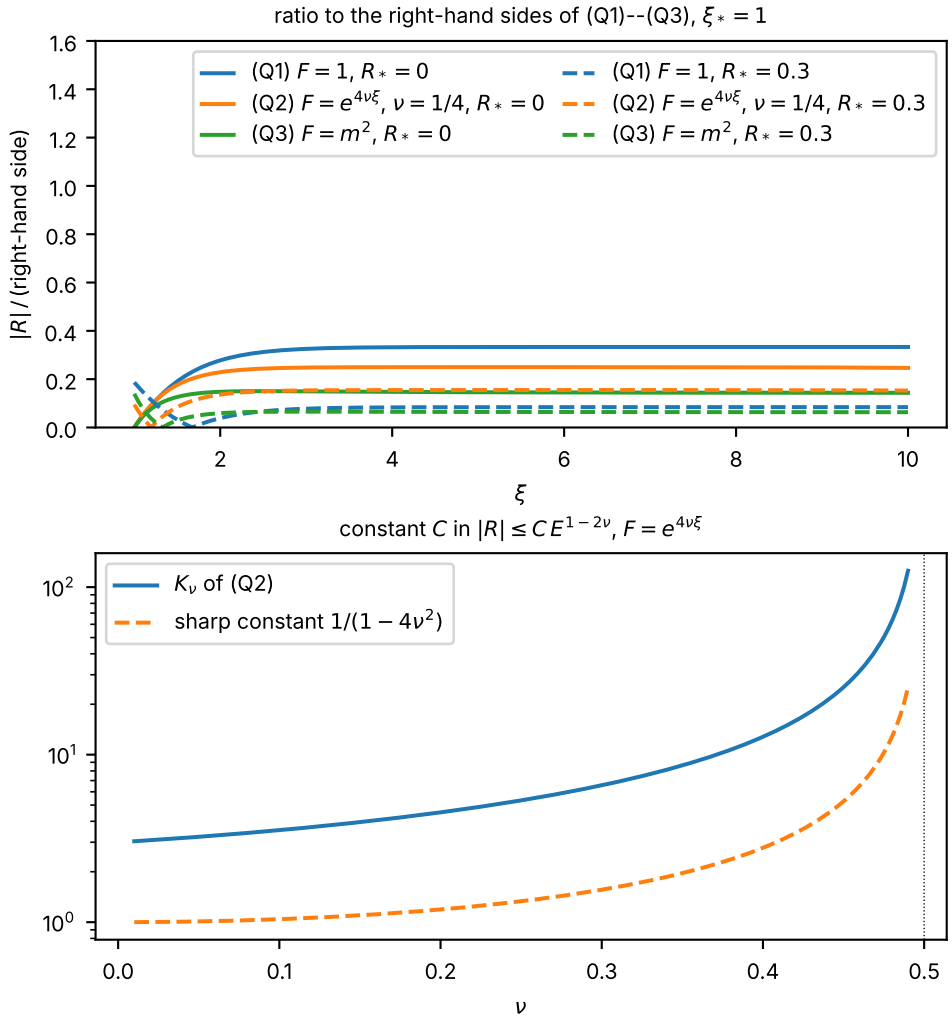


FIGURE 4.2. Numerical check of Lemma 4.13 (generated by `code/ch04_pole/make_a_green_check.py`; direct quadrature of (4.9)–(4.10), no solver). Left: $|\mathfrak{r}_{R_*} F|$ divided by the right-hand side of (Q1) ($F = 1$), (Q2) ($F = e^{\xi}$, $\nu = \frac{1}{4}$) and (Q3) ($F = m^2$), for $\xi_* = 1$, $R_* = 0$ (solid) and $R_* = 0.3$ (dashed): all ratios stay below 0.34. Right: the constant K_ν of (Q2) and the sharp constant $1/(1-4\nu^2)$ of Example 4.14 (the ratio at $\nu = \frac{1}{4}$ is $\frac{4/3}{16/3} = 0.25$, the plateau of the orange solid curve on the left).

Remark 4.15 (relation to the Green operator of Chapter 2)**.** Lemma 2.23 solves $\mathsf{L}[R] = F$ with $R = o(E^2)$ at the pole, i.e. it kills *both* kernel elements at the pole, which needs $\int |F|E^{-1} < \infty$ and is the right operator for effective coordinates. The operator $\mathfrak{r}_{R_*}F$ above only imposes $R \to 0$ (it kills x but keeps t^2) and fixes the remaining freedom by a Dirichlet datum at ξ_*; it accepts sources F that do not decay and gains exactly one power of E. The two differ by a multiple of t^2.

4.5. The four strata and the a priori asymptotics

In words. *Theorem 4.5 sorts every local solution with genuine swirl by three data: the limit $q = Q(1) \geq -2$ (equivalently μ), the sign of $\lim h = \pm\mu$ (dominant or recessive), and, when $q = -2$, whether $f(1)$ is finite. This gives four* types*: G (generic), R (recessive, $0 < \mu < \frac{1}{2}$), L_1 and L_2 (logarithmic). Each has a precise asymptotic behaviour of z, f and $R = Q - q$, proved by the same bootstrap: a rough bound on h gives a rough bound on z, hence on f and F; the Green operator turns that into an exponentially small bound for R; then $w = \frac{1}{2}R$ is integrable and asymptotic integration (Lemma 3.16) gives the precise form of z.*

Definition 4.16 (pole types; strata)**.** Let $\kappa \neq 0$ and let q, μ be as in Theorem 4.5. The local solution at $x = 1$ is of type

- G (*generic*) if $q > -2$ and $h \to \mu$ (so $U_\theta(1) = -2 + 2\mu > -2$);
- R (*recessive*) if $q > -2$ and $h \to -\mu$ (so $0 < \mu < \frac{1}{2}$ and $U_\theta(1) = -2 - 2\mu \in (-3, -2)$, by Theorem 4.9);
- $\mathsf{L}_1, \mathsf{L}_2$ (*logarithmic*) if $q = -2$ (so $\mu = 0$, $h \to 0$, $U_\theta(1) = -2$); of type L_1 if moreover $f(1)$ is finite, of type L_2 otherwise.

At $x = -1$ the types are defined through σ_2: G means $U_\theta(-1) = 2 - 2\mu < 2$, R means $U_\theta(-1) \in (2, 3)$, $\mathsf{L}_1, \mathsf{L}_2$ mean $U_\theta(-1) = 2$. The set of local solutions of a given type is called a *stratum*.

Convention for $\kappa = 0$. If $\kappa = 0$ then $f \equiv A$ is constant and Q is a polynomial of degree ≤ 2 in x (see Lemma 4.20). We say that such a local solution is *of type* G *at* $x = 1$ if $q = Q(1) > -2$ and $h \to +\mu$, $\mu = \sqrt{1 + q/2} > 0$, in the same words as for $\kappa \neq 0$. The types $\mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$ are defined only for $\kappa \neq 0$. At $x = -1$ the definition is transported by σ_2 as above. Type G in this extended sense is the stratum on which the openness and the coordinates of Corollaries 4.42 and 4.43 are stated.

The types are exhaustive and mutually exclusive. Indeed by Theorem 4.5(a) either $q > -2$ (then $\mu > 0$ and by (b) exactly one of $h \to \mu$, $h \to -\mu$ holds, since $\mu \neq -\mu$) or $q = -2$ (then $\mu = 0$ and $h \to 0$); in the latter case $f(1)$ is finite or not. The ranges of $U_\theta(1)$ in the definition follow from $U_\theta(1) = -2 \pm 2\mu$, and for type R Theorem 4.9 gives $\mu < \frac{1}{2}$. Thus $U_\theta(1)$ itself separates R (< -2), $\mathsf{L}_1 \cup \mathsf{L}_2$ ($= -2$) and G (> -2). In types $\mathsf{L}_1, \mathsf{L}_2$ the dominant/recessive distinction degenerates (both exponents are 0), and the model solutions are $a + b\xi$; Proposition 4.19(c) shows that L_1 corresponds to $b > 0$ and L_2 to $b = 0$.

Remark 4.17 (why R has codimension one; the ordering picture)**.** In the model of Toy example 4.3 the positive solutions with $z(\xi_1) = 1$ and slope $\beta = z'(\xi_1)$ are positive and dominant for $\beta > -\mu$, recessive for $\beta = -\mu$, and have a zero for $\beta < -\mu$. By Lemma 3.18(b) the same is true for $z'' = (\mu^2 + w)z$, $\int |w| < \infty$: the recessive solution is the *lowest* positive solution through a given point, and type R is the

lower wall of the region of type G. This is the elementary reason why, in the global picture (Chapter 5), the lower edge of a family of global solutions is of type R, L_1 or L_2 and the interior is of type G.

Remark 4.18 (the pole in the variable x; the velocity near the axis)**.** In the original variable $t = 1 - x \to 0$ the linear equation $2(1-x^2)^2 y_{xx} = Qy$ of Lemma 2.9 is, near the pole, an Euler equation: $1 - x^2 = t(2-t) \approx 2t$ and $Q \to q$ give $8t^2 y_{tt} \approx qy$, so $y \approx t^r$ with the indicial equation $8r(r-1) = q$, i.e. $r = \frac{1 \pm \mu}{2}$. Since $t \approx 2e^{-2\xi}$ and $\cosh\xi \approx \frac{1}{2}e^{\xi}$, one has $e^{\pm\mu\xi}/\cosh\xi \approx 2(t/2)^{(1\mp\mu)/2}$, so

$$z \sim e^{\mu\xi} \iff y \sim Ct^{(1-\mu)/2} \quad \text{(dominant)},$$
$$z \sim e^{-\mu\xi} \iff y \sim Ct^{(1+\mu)/2} \quad \text{(recessive)},$$

the two indicial roots $r = \frac{1 \mp \mu}{2}$ of Example 2.31. With $U = 2(1-x^2)y_x/y \approx 2 \cdot 2t \cdot (-r/t) = -4r$ (note $y_x = -y_t$) one recovers $U(1) = -2 + 2\mu$ (dominant) and $U(1) = -2 - 2\mu$ (recessive). The recessive branch is the *larger* root's solution: it is the exceptional one. For $\mu = 0$ the indicial root is double ($r = \frac{1}{2}$) and the model solutions $t^{1/2}$, $t^{1/2}\ln t$ correspond to $z \sim 1$ and $z \sim \xi$. Finally, the velocity component $u_\theta = U/\sin\theta$ behaves like $U(1)/\theta$ at the axis ($\sin\theta \approx \theta$): the flow is singular on the axis unless $U(1) = 0$, i.e. $\mu = 1$ on the dominant branch ($q = 0$, $c_2' = 0$: the Landau value). Types R and $\mathsf{L}_1, \mathsf{L}_2$ have $U(1) < 0$ and $U(1) = -2$, respectively, so the axis carries a genuine line singularity of the corresponding strength; this is the physical meaning of the strata.

Table 4.1 lists the four types at $x = 1$.

type	$q = Q(1)$	μ	$\lim h$	z	$U_\theta(1)$	$f(1)$	$R = Q - q$
G	> -2	> 0	$+\mu$	$\sim a\, e^{\mu\xi}$	$-2 + 2\mu > -2$	finite	$O(E)$
R	$\in (-2, -\frac{3}{2})$	$(0, \frac{1}{2})$	$-\mu$	$\sim a\, e^{-\mu\xi}$	$-2 - 2\mu \in (-3, -2)$	$\pm\infty$, $\sim \frac{\kappa e^{2\mu\xi}}{2\mu a^2}$	$O(E^{1-2\mu})$
L_1	-2	0	0	$\sim b\,\xi$, $b > 0$	-2	finite	$O(E)$
L_2	-2	0	0	$\to a > 0$	-2	$\pm\infty$, $\sim \frac{\kappa}{a^2}\xi$	$O(E\, m^2)$

TABLE 4.1. The four types at $x = 1$ ($\kappa \neq 0$); $E = e^{-2\xi}$, $m = 1 + \xi$. The last column and the asymptotics of z and f are those of Proposition 4.19; the value $q \in (-2, -\frac{3}{2})$ for type R is $q = 2(\mu^2 - 1)$ with $\mu \in (0, \frac{1}{2})$.

Proposition 4.19 (a priori asymptotics)**.** *Let (U, f, Q) be a local solution at the north pole with $\kappa \neq 0$, $R = Q - q$, $w = R/2$, $m = 1 + \xi - \xi_*$ for a fixed $\xi_* \geq \max(\xi_0, 0)$; the constants C below depend on the solution.*

(a) (Type G.) *$a := \lim_{\xi\to\infty} z e^{-\mu\xi} \in (0, \infty)$ exists, $f(1)$ is finite, and $|R| \leq C\,E$.*

(b) (Type R.) *$a := \lim z e^{\mu\xi} \in (0, \infty)$ exists, $|f| \leq Ce^{2\mu\xi}$, $f \sim \frac{\kappa}{2\mu a^2} e^{2\mu\xi}$, and $|R| \leq C\,E^{1-2\mu}$.*

(c) (Types $\mathsf{L}_1, \mathsf{L}_2$.) *$z = a + b\xi + o(1)$ and $z' \to b \geq 0$. If $b > 0$ (type L_1) then f is bounded and $|R| \leq C\,E$. If $b = 0$ (type L_2) then $a > 0$, $f = \frac{\kappa}{a^2}\xi + O(1)$ and $|R| \leq C\,E\,m^2$.*

PROOF. Throughout $W = \mu^2 + w$ with $w = \frac{1}{2}R$ and (4.4) holds. For every $\xi_* \geq \xi_0$, $R = \mathfrak{r}_{R(\xi_*)}F$ as soon as F satisfies the integrability hypothesis of Lemma 4.13, because $R \to 0$ and $\mathsf{L}[R] = F$ (the uniqueness part). This is the only way the Green operator is used.

(a) Type G. Fix $\varepsilon \in (0, \mu)$; for large ξ, $h \geq \mu - \varepsilon$, so $z \geq ce^{(\mu-\varepsilon)\xi}$ and $\int^\infty z^{-2} < \infty$. By Theorem 4.5(c), $f(1)$ is finite; f is monotone, hence bounded near the pole, and $F = \Lambda_{\mathrm{p}} - f^2$ is bounded. By the uniqueness part of Lemma 4.13 and (Q1), $|R| \leq CE$. Thus $\int^\infty |w| < \infty$ and Lemma 3.16(a) gives $z = Az_+ + Bz_-$; $h \to \mu$ forces $A \neq 0$, and then $e^{-\mu\xi}z \to A$, with $A > 0$ because $z > 0$. So $a = A$.

(b) Type R. Here $h \to -\mu$ with $0 < \mu < \frac{1}{2}$ (Theorem 4.9). Fix $\varepsilon > 0$ with $\mu + \varepsilon < \frac{1}{2}$ and $\varepsilon < \mu$. For large ξ, $-\mu - \varepsilon \leq h \leq -\mu + \varepsilon$, so $ce^{-(\mu+\varepsilon)\xi} \leq z \leq Ce^{-(\mu-\varepsilon)\xi}$. Hence $|f'| \leq |\kappa|c^{-2}e^{2(\mu+\varepsilon)\xi}$, so $|f| \leq Ce^{2(\mu+\varepsilon)\xi}$ and $|F| \leq CE^{-2(\mu+\varepsilon)}$ (recall $E^{-2\nu} = e^{4\nu\xi}$). (Q2) with $\nu = \mu + \varepsilon < \frac{1}{2}$ gives $|R| \leq CE^{1-2\mu-2\varepsilon}$, so w is integrable and, by Lemma 3.16(a), $z = Az_+ + Bz_-$ with $A = 0$ (else $h \to \mu$). Hence $ze^{\mu\xi} \to B =: a$, and $a > 0$ because $z > 0$. Then $f' = \kappa z^{-2} = \kappa a^{-2}e^{2\mu\xi}(1 + o(1))$; since $\int^\xi e^{2\mu\eta}\mathrm{d}\eta \to \infty$, integration gives $f = \frac{\kappa}{2\mu a^2}e^{2\mu\xi}(1 + o(1))$. In particular $|f| \leq Ce^{2\mu\xi}$, $|F| \leq CE^{-2\mu}$, and (Q2) with $\nu = \mu$ gives $|R| \leq CE^{1-2\mu}$.

(c) Types $\mathsf{L}_1, \mathsf{L}_2$. Here $\mu = 0$ and $h \to 0$. For every $\varepsilon > 0$, $|h| \leq \varepsilon$ for large ξ, so $ce^{-\varepsilon\xi} \leq z \leq Ce^{\varepsilon\xi}$ ("$z = e^{o(\xi)}$"). Hence $|f| \leq C_\varepsilon e^{2\varepsilon\xi}$, $|F| \leq C_\varepsilon E^{-2\varepsilon}$, and (Q2) with $\nu = \varepsilon \in (0, \frac{1}{2})$ gives $|R| \leq C_\varepsilon E^{1-2\varepsilon}$. Thus $\int^\infty \xi^2|w| < \infty$ and Lemma 3.16(b) gives $z = a + b\xi + o(1)$, $z' \to b$; $b \geq 0$ because $b < 0$ would make z negative. If $b > 0$, then $z \geq \frac{b}{2}\xi$ for large ξ, so $\int^\infty z^{-2} < \infty$, f is bounded (Theorem 4.5(c)), and (Q1) gives $|R| \leq CE$. If $b = 0$, then $z = az_1$ with $z_1 \to 1$ (Lemma 3.16(b)), so $a > 0$ because $z > 0$; since $\int|z_1 - 1| < \infty$ and $|z_1^{-2} - 1| \leq 12|z_1 - 1|$ where $z_1 \in [\frac{1}{2}, 2]$, also $\int^\infty |z^{-2} - a^{-2}| < \infty$; hence $f = f(\xi_*) + \kappa\int_{\xi_*}^\xi z^{-2} = \frac{\kappa}{a^2}\xi + O(1)$. Then $|F| \leq C(1 + \xi)^2 \leq C'm^2$ for $\xi \geq \xi_*$, and (Q3) gives $|R| \leq CEm^2$. □

Lemma 4.20 (a priori asymptotics of type G for $\kappa = 0$)**.** *Let (U, f, Q) be a local solution at the north pole with $\kappa = 0$ which is of type G in the sense of the convention after Definition 4.16, $R = Q - q$, $w = R/2$. Then $f \equiv A$ is constant, $a := \lim_{\xi\to\infty} ze^{-\mu\xi} \in (0, \infty)$ exists, and $|R| \leq C\,E$.*

PROOF. $f' = \kappa z^{-2} = 0$. Then $\mathsf{L}[Q] = \lambda - A^2$ is constant and by Lemma 2.18(b) $Q = -\frac{1}{2}(\lambda - A^2) + \alpha x + \beta(1 - x)^2$ is a polynomial of degree ≤ 2 in x; hence $R = Q - Q(1)$ is a polynomial vanishing at $x = 1$, so $|R| \leq C(1 - x) = Ct \leq 2CE$. Thus $\int^\infty |w| < \infty$, and Lemma 3.16(a) gives $z = Az_+ + Bz_-$; $h \to \mu > 0$ forces $A \neq 0$ (otherwise $h \to -\mu$), and $A > 0$ because $z > 0$. So $e^{-\mu\xi}z \to A = a$. □

Corollary 4.21 (ranges at a pole)**.** *Let (U, f, Q) be a local solution at $x = 1$ with $\kappa \neq 0$, and let $P_{\mathbf{c}}$ be the constants of Q at some base point, $\psi_{\mathbf{c}} = P_{\mathbf{c}} - Q$. Then $\psi_{\mathbf{c}}(1) = 2c_2 - q$ and: type G has $q > -2$, $U_\theta(1) > -2$ and $\psi_{\mathbf{c}}(1) < 2c_2 + 2$; type R has $q \in (-2, -\frac{3}{2})$, $U_\theta(1) \in (-3, -2)$ and $\psi_{\mathbf{c}}(1) \in (2c_2 + \frac{3}{2}, 2c_2 + 2)$; types $\mathsf{L}_1, \mathsf{L}_2$ have $q = -2$, $U_\theta(1) = -2$, $\psi_{\mathbf{c}}(1) = 2c_2 + 2$. In particular $U_\theta(1) > -3$ and the swirl contribution to $Q(1)$ is, at an R, L_1 or L_2 pole, $\psi_{\mathbf{c}}(1) > 2c_2 + \frac{3}{2}$. The identities hold with $c_2 \mapsto c_1$, $U_\theta(1) \mapsto -U_\theta(-1)$ at the south pole.*

PROOF. $P_{\mathbf{c}}(1) = c_2(1 + 1) = 2c_2$, so $\psi_{\mathbf{c}}(1) = 2c_2 - q$, finite by Theorem 4.5(a). The ranges of $q = 2(\mu^2 - 1)$: type G, $\mu > 0$; type R, $\mu \in (0, \frac{1}{2})$ by Theorem 4.9, so $q \in (-2, -\frac{3}{2})$; types $\mathsf{L}_1, \mathsf{L}_2$, $\mu = 0$. Then $\psi_{\mathbf{c}}(1) = 2c_2 - q$ gives the stated intervals;

the values of $U_\theta(1)$ are those of Definition 4.16. At the south pole use σ_2, which exchanges c_1 and c_2 (Lemma 2.8). □

Proposition 4.22 (the exponent measured from a zero of the swirl)**.** *Let (U, f, Q) be a local solution at the north pole on $[x_0, 1)$ with $f(x_0) = 0$ and $\kappa \neq 0$ (a class-Z base point, Chapter 5), let $\boldsymbol{c}$ be the constants of Q at x_0 (the Taylor polynomial $P_{\boldsymbol{c}}$ of degree two of Q at x_0) and $\psi_{\boldsymbol{c}} = P_{\boldsymbol{c}} - Q$. Then $\psi_{\boldsymbol{c}}$ is strictly increasing on $[x_0, 1)$, $\psi_{\boldsymbol{c}} > 0$ on $(x_0, 1)$, $\psi_{\boldsymbol{c}}(1) = 2c_2 - q \in (0, \infty)$, and*

$$\mu^2 = 1 + c_2 - \tfrac{1}{2}\psi_{\boldsymbol{c}}(1) < 1 + c_2. \tag{4.13}$$

Consequently $c_2 > -1$ and, with $k_2 = \sqrt{1 + c_2}$, $0 \le \mu < k_2$; the pole is of type L_1 or L_2 ($\mu = 0$) iff $\psi_{\boldsymbol{c}}(1) = 2k_2^2$, and $\psi_{\boldsymbol{c}}(1) < 2k_2^2$ in types G, R. Mirror statement at a south pole, with σ_2 ($c_2 \mapsto c_1$).

Proof. By Proposition 2.20, $\lambda = \mathsf{L}[P_{\boldsymbol{c}}] + f(x_0)^2 = \mathsf{L}[P_{\boldsymbol{c}}]$, and $\mathsf{L}[P_{\boldsymbol{c}}]$ is the constant $-2c_1 - 2c_2 - 4c_3$; hence $\mathsf{L}[\psi_{\boldsymbol{c}}] = \mathsf{L}[P_{\boldsymbol{c}}] - \mathsf{L}[Q] = \lambda - (\lambda - f^2) = f^2$. Q and $P_{\boldsymbol{c}}$ have the same 2-jet at x_0, so $\psi_{\boldsymbol{c}}(x_0) = \partial_x\psi_{\boldsymbol{c}}(x_0) = 0$ and the Cauchy formula (2.34) (Lemma 2.18(d), $g = \psi_{\boldsymbol{c}}$, data zero) gives

$$\psi_{\boldsymbol{c}}(x) = \int_{x_0}^{x} K_{\mathsf{L}}(x, s) f(s)^2 \, \mathrm{d}s, \qquad \partial_x\psi_{\boldsymbol{c}}(x) = \int_{x_0}^{x} \partial_x K_{\mathsf{L}}(x, s) f(s)^2 \, \mathrm{d}s$$

(differentiate under the integral, $K_{\mathsf{L}}(x, x) = 0$). By Lemma 2.18(c), $K_{\mathsf{L}} > 0$ and $\partial_x K_{\mathsf{L}} > 0$ for $-1 < s < x < 1$, and $f^2 > 0$ on $(x_0, 1)$ (f is strictly monotone with its only zero at x_0, Corollary 2.10). So $\partial_x\psi_{\boldsymbol{c}} > 0$ and $\psi_{\boldsymbol{c}} > 0$ on $(x_0, 1)$. By Theorem 4.5(a) the limit q exists and is finite, and $\psi_{\boldsymbol{c}}(1) = P_{\boldsymbol{c}}(1) - q = 2c_2 - q$; being the limit of a strictly increasing positive function, $\psi_{\boldsymbol{c}}(1) > 0$. Then $\mu^2 = 1 + \frac{q}{2} = 1 + c_2 - \frac{1}{2}\psi_{\boldsymbol{c}}(1)$ gives (4.13); since $\mu^2 \ge 0$, $1 + c_2 > \frac{1}{2}\psi_{\boldsymbol{c}}(1) > 0$. Finally $\mu = 0$ iff $\frac{1}{2}\psi_{\boldsymbol{c}}(1) = 1 + c_2 = k_2^2$. □

Example 4.23 (the exponent drop in the numerical runs)**.** In Numerical observation 4.28 the base point is $x_0 = 0$ with $f(0) = 0$, so (4.13) applies. First row ($c = 0$, $k_2 = 1$): $\psi_{\boldsymbol{c}}(1) = -q = 1.2334 < 2k_2^2 = 2$ and $\mu^2 = 1 - \frac{1}{2}(1.2334) = 0.3833$. Third row ($c = (0, -0.9, 0)$, $k_2^2 = 0.1$): $\psi_{\boldsymbol{c}}(1) = 2c_2 - q = -1.8 + 1.9002 = 0.1002 < 0.2$ and $\mu^2 = 0.1 - 0.0501 = 0.0499$, $\mu = 0.2234$. The swirl lowers the exponent measured from the base point; in the second row, $c = 0$ but $\kappa > 0$, the exponent has dropped from $k_2 = 1$ to $\mu = 0.006$, i.e. to the logarithmic stratum.

Corollary 4.24 (bounded swirl; effective constants)**.** *Let (U, f, Q) be a local solution at $x = 1$ with $\kappa \neq 0$.*

(a) *$f(1)$ is finite if and only if the type is G or L_1; $|f(1)| = \infty$ if and only if the type is R or L_2. In the second case $f \sim \frac{\kappa}{2\mu a^2} e^{2\mu\xi}$ (type R), resp. $f \sim \frac{\kappa}{a^2}\xi$ (type L_2), and $f(1) = \operatorname{sgn}\kappa \cdot \infty$.*

(b) *Let the solution be global. Then f is bounded on $(-1, 1)$ if and only if the type is G or L_1 at both poles (always so if $\kappa = 0$). In that case $f(\pm 1)$ and $Q(\pm 1)$ are finite, i.e. hypothesis (2.37) of Proposition 2.27 holds, and the effective constants $\boldsymbol{c}'$, the centrifugal part Q_{sw} and W_{e} of Chapter 2 are defined.*

(c) *In the situation of* (b) *the pole exponents are the effective exponents: $\mu_+ = k_2' := \sqrt{1 + c_2'}$ at the north pole and $\mu_- = k_1' := \sqrt{1 + c_1'}$ at the south pole. A pole of type G has $c_i' > -1$ and a pole of type L_1 has $c_i' = -1$. In particular a global solution of type G at both poles has $c_1', c_2' > -1$.*

PROOF. (a) By Proposition 4.19: type G has $f(1)$ finite, R has $f \sim \frac{\kappa}{2\mu a^2} e^{2\mu\xi} \to \operatorname{sgn}\kappa \cdot \infty$, L_1 has f bounded (hence $f(1)$ finite, by monotonicity), L_2 has $f = \frac{\kappa}{a^2}\xi + O(1)$. The four types exhaust all cases (Definition 4.16). (b) If $\kappa = 0$, f is constant and $Q = P_{\boldsymbol{c}}$ (Example 2.30), so (2.37) holds. Let $\kappa \neq 0$. f is strictly monotone on $(-1,1)$ (Corollary 2.10), hence bounded iff both limits $f(\pm 1)$ are finite, i.e. by (a) (and its mirror image under σ_2) iff both poles are of type G or L_1. The limits $Q(\pm 1)$ are finite for every global solution by Theorem 4.5(a) at both poles. (c) By (2.38), $c_2' = \frac{1}{2}Q(1) = \frac{q}{2}$, so $\mu^2 = 1 + \frac{q}{2} = 1 + c_2' = k_2'^2$; at the south pole use σ_2, which exchanges c_1' and c_2'. Type G has $\mu > 0$, i.e. $c_2' > -1$, and type L_1 has $q = -2$, i.e. $c_2' = -1$. □

Corollary 4.25 (finite swirl energy; the logarithmic strata)**.** *Let (U, f, Q) be a local solution at $x = 1$ with $\kappa \neq 0$.*

(a) *The swirl energy is finite in every type: $\int^1 f^2 \,\mathrm{d}x < \infty$.*

(b) *Type L_1: $U_\theta(x) + 2 = \dfrac{4}{\ln \frac{1}{1-x}}\,(1 + o(1))$ as $x \to 1$.*

(c) *Type L_2: $U_\theta(x) + 2 = O\big((1-x)\ln^2 \frac{1}{1-x}\big)$: the approach to -2 is almost linear in $1 - x$, not logarithmic.*

PROOF. (a) By (4.6) it suffices to show $\int^\infty f^2 e^{-2\xi}\mathrm{d}\xi < \infty$. Types G, L_1: f is bounded. Type R: $f^2 \le Ce^{4\mu\xi}$ and $4\mu - 2 < 0$ by $\mu < \frac{1}{2}$ (Theorem 4.9). Type L_2: $f^2 \le C\xi^2$. (b), (c) $U = 2(h - \tanh\xi)$, so $U + 2 = 2h + 2t$ with $t = 1 - \tanh\xi \le 2E$. In type L_1, $z = a + b\xi + o(1)$ and $z' \to b > 0$ (Proposition 4.19(c)), so $h = z'/z = (b + o(1))/(b\xi + O(1)) = (1 + o(1))/\xi$; and $-\ln(1-x) = 2\xi - \ln 2 + o(1)$ (because $1 - x = t = 2E/(1+E)$), so $\frac{2}{\xi} = \frac{4}{\ln(1/(1-x))}(1 + o(1))$ while $2t = O(E)$ is negligible. In type L_2, $z = az_1$ with $z_1 \to 1$, $z_1' \to 0$ (Lemma 3.16(b)); $z_1'' = wz_1$ with $|w| \le CEm^2$ and z_1 bounded give $|z_1'(\xi)| = |\int_\xi^\infty wz_1| \le C\,Em^2$ (integrating by parts the elementary integral $\int_\xi^\infty \eta^2 e^{-2\eta}\mathrm{d}\eta = O(\xi^2 e^{-2\xi})$), hence $h = z_1'/z_1 = O(Em^2)$ and $U + 2 = O(Em^2)$. Finally $E \approx \frac{1}{2}(1 - x)$ and $m \le C \ln \frac{1}{1-x}$. □

Lemma 4.26 (types are intrinsic)**.** *The type of a local solution at $x = 1$ is determined by the germ of the solution at the pole: it does not depend on the choice of the cut x_0, it is unchanged by the symmetry σ_1 ($f \mapsto -f$) and by the rescaling $(z, \kappa) \mapsto (sz, s^2\kappa)$, $s > 0$ (Lemma 2.14); the exponent μ and the limits q, $\lim h$, $U(1)$ do not depend on the normalisation of z, whereas the constants a, b of Proposition 4.19 scale as $a \mapsto sa$, $b \mapsto sb$.*

PROOF. The type is read off from $q = \lim Q$, $\lim h = \lim z'/z$ and the finiteness of $\lim f$ (Definition 4.16); these are limits at $\xi = \infty$ of Q, of h and of f, which do not see a change of x_0. σ_1 does not change U, Q and changes f to $-f$. Under $(z, \kappa) \mapsto (sz, s^2\kappa)$ the functions U, f, Q, h are unchanged (Lemma 2.14), while $z \mapsto sz$ gives the stated scaling of a, b. □

Remark 4.27 (bookkeeping of the bootstrap; where $\mu < \frac{1}{2}$ enters again)**.** The proof of Proposition 4.19 runs a chain of implications in which each link has a fixed exponent loss, and it *closes* exactly when the exponent stays below the integrability threshold:

$$|h| \le \mu + \varepsilon \;\Rightarrow\; z \gtrsim e^{-(\mu+\varepsilon)\xi} \;\Rightarrow\; |f| \lesssim e^{2(\mu+\varepsilon)\xi} \;\Rightarrow\; |F| \lesssim E^{-2(\mu+\varepsilon)}$$

$$\xrightarrow{\text{(Q2)}} |R| \lesssim E^{1-2(\mu+\varepsilon)} \Rightarrow \int |w| < \infty.$$

The last step needs $1-2(\mu+\varepsilon)>0$, i.e. $\mu<\frac{1}{2}$: for $\mu\geq\frac{1}{2}$ the loop cannot be closed, and Theorem 4.9 shows that this is not a defect of the method but a true obstruction. On the dominant branch no exponent is lost (f stays bounded, so (Q1) applies), which is why type G exists for every $\mu>0$. In type L_2 the source F grows like ξ^2, the weight m^2 of (Q3) absorbs it, and the loss is only the polynomial factor m^2.

Numerical observation 4.28 (the three types in real solutions)**.** *Figure 4.3 shows the meridional circulation $U_\theta(x)$ of three solutions of the class-Z shooting problem (Chapter 5; base point $x_0=0$, $f(0)=0$) in the variable $s=-\ln(1-x)=2\xi-\ln 2+o(1)$. Method: shooting (explicit Runge–Kutta of order 8, tolerances $10^{-12}/10^{-14}$, integration to $s=14$; the edge value γ^- is found by bisection to 10^{-13}; the edge run uses $\gamma=\gamma^-+10^{-11}$). Results at $s=14$:*

run	q	$\mu=\sqrt{1+q/2}$	U_θ	$-2\pm 2\mu$
$c=0,\ \kappa=1$, *interior* ($\gamma=\gamma^-+0.1$)	-1.2334	0.6191	-0.7609	-0.7618
$c=0,\ \kappa=1$, *edge* ($\gamma^-=-1.7806$)	-1.99993	0.0060	-1.6769	-2
$c=(0,-0.9,0),\ \kappa=0.3$, *edge* ($\gamma^-=-1.0186$)	-1.9002	0.2234	-2.4461	-2.4467

The interior solution is of type G*, the first edge is of type* L_1 *($q+2=7\cdot 10^{-5}$; $U_\theta+2=0.323$ at $s=14$, to be compared with $4/s=0.286$: the logarithmic approach $U_\theta+2\sim 4/s$ proved in Corollary 4.25(b), with a slowly decaying relative error $O(1/s)$), the second edge is of type* R *with $\mu<\frac{1}{2}$, as Theorem 4.9 requires. This is an illustration, not a proof: which types occur on edges is the subject of Chapter 7 (Chapter 5 for the lens).*

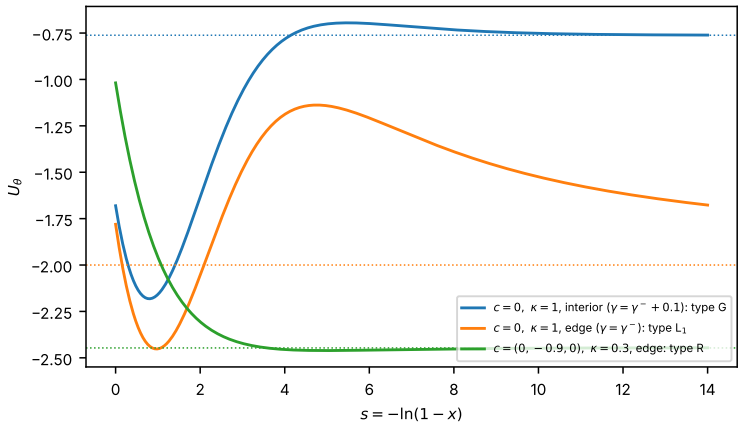


FIGURE 4.3. Numerical observation 4.28 (generated by `code/ch04_pole/make_a_pole_types.py`). U_θ against $s=-\ln(1-x)$ for three class-Z solutions; the dotted lines are the predicted limits $-2+2\mu$ (type G), -2 (type L_1, approached only logarithmically) and $-2-2\mu$ (type R).

Example 4.29 (type changes along the Landau-type family $\boldsymbol{c} = (0, c_2, 0)$)**.** Take $x_0 = 0$, $f(0) = 0$ and $\boldsymbol{c} = (0, c_2, 0)$, $c_2 \in (-1, \infty)$: the potential $W_{\boldsymbol{c}} = 1 + \frac{1}{2}P_{\boldsymbol{c}}$ of the line equation at $\kappa = 0$ satisfies $W_{\boldsymbol{c}}(-\infty) = 1$ and $W_{\boldsymbol{c}}(+\infty) = 1 + c_2 = k_2^2$. For $\kappa = 0$ the pole data are read off from the polynomial $P_{\boldsymbol{c}}$: $q = 2c_2$, $\mu = k_2$, and the lower edge solution U_c^- is recessive with $U_c^-(1) = -2 - 2k_2$. For $\kappa \neq 0$ the lower edge $\gamma^-(\kappa)$ of the class-Z shooting problem (Chapter 5) has a pole type, and Theorems 4.5 and 4.9 constrain it. They give a dichotomy:

(i) $k_2 < \frac{1}{2}$, i.e. $c_2 < -\frac{3}{4}$: the recessive value $-2-2k_2 \in (-3, -2)$ is admissible, and the edge can remain of type R with μ close to k_2.

(ii) $k_2 \geq \frac{1}{2}$, i.e. $c_2 \geq -\frac{3}{4}$: the recessive value $-2 - 2k_2 \leq -3$ is excluded for $\kappa \neq 0$ by Theorem 4.9. The edge must therefore be dominant or logarithmic, and the numerics below show that it is logarithmic (L_1).

Figure 4.4 and the table show this, for $\kappa = 0.05$ and $\kappa = 0.3$. Method as in Numerical observation 4.28, integration to $s = 30$ (the script is `code/ch04_pole/make_a_landau_family.py`). Values of $\mu = \sqrt{1 + q/2}$ at $s = 30$:

c_2	k_2	μ ($\kappa = 0.05$)	μ ($\kappa = 0.3$)	type of the edge
-0.95	0.224	0.221	0.136	R
-0.90	0.316	0.313	0.223	R
-0.80	0.447	0.439	0.296	R
-0.70	0.548	0.480	0.275	transition
-0.60	0.632	0.330	0.188	transition
-0.40	0.775	0.017	$< 10^{-4}$	L_1
0.00	1	10^{-4}	$< 10^{-4}$	L_1
0.50	1.225	10^{-4}	$< 10^{-4}$	L_1

Three observations. First, μ on the edge never exceeds $\frac{1}{2}$ (the maximum in the table is 0.480, at $c_2 = -0.70$; the script gives 0.484 at $c_2 = -0.74$, not shown), as Theorem 4.9 requires. Second, for $c_2 < -\frac{3}{4}$ the edge exponent is close to k_2 for small κ and is lowered by the swirl for larger κ, in the manner of Proposition 4.22. Third, for $c_2 > -\frac{3}{4}$ the exponent collapses to the logarithmic value: $q + 2 \to 0$ and the edge is of type L_1 (compare Corollary 4.25(b)). The transition region near $c_2 = -\frac{3}{4}$ is a boundary layer in κ whose width shrinks with κ; at fixed $s = 30$ it is blurred, and the table is an illustration of the dichotomy, not a determination of where the type changes.

Remark 4.30 (where the hypothesis $\kappa \neq 0$ is used; what has been machine-checked)**.** For $\kappa = 0$ nothing in this chapter is false, but nothing interesting happens: $Q = P_{\boldsymbol{c}}$ is a polynomial, $q = 2c_2$, $\mu = k_2$, and the no-swirl theory of Chapter 3 describes everything, including the recessive branch with any $\mu > 0$ (Example 4.4). The hypothesis $\kappa \neq 0$ is used in exactly four places: (i) f is strictly monotone, which fixes the sign of Q_{xxx} near the pole (Theorem 4.5); (ii) $|f| \to \infty$ on the recessive branch (Lemma 4.7(a)), the source of the energy argument; (iii) $f' \neq 0$, used for $\rho = f/f'$ in the borderline case of Theorem 4.9; (iv) the dichotomy "$f(1)$ finite or infinite" of Corollary 4.24. All elementary identities used in Sections 4.3 and 4.4 ((4.5), the kernel elements $\tanh\xi$ and t^2, the identity $\mathsf{L}[\mathcal{A}\tanh + \mathcal{B}t^2] = F$ for $\mathcal{A}' = FE$, $\mathcal{B}' = -F\sinh\xi\cosh\xi$, the constants K_ν and $\frac{5}{4}$ of (Q2)–(Q3), the deviation equation (4.8) and $\rho' = 1 - (1 - 2\eta)\rho$) were re-derived by computer

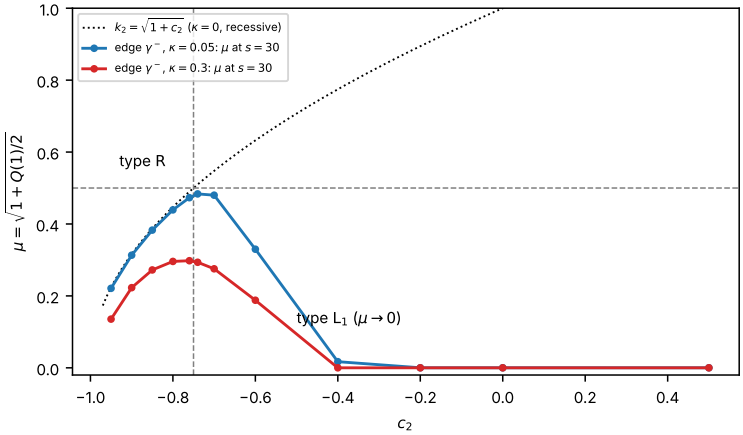


FIGURE 4.4. Example 4.29 (generated by `code/ch04_pole/make_a_landau_family.py`). The pole exponent μ of the lower-edge solution at $s = 30$ against c_2, for $\kappa = 0.05$ and $\kappa = 0.3$; dotted: $k_2 = \sqrt{1+c_2}$, the exponent at $\kappa = 0$; dashed: the wall $\mu = \frac{1}{2}$ and $c_2 = -\frac{3}{4}$.

algebra in `code/ch04_pole/verify_a_ch04.py`; the same script tests the bounds (Q1)–(Q3) numerically (for $\xi_* = 1$, $R_* \in \{0, \pm 0.3\}$ and the sources $F = 1$, e^{ξ}, m^2: $|\mathfrak{r}_{R_*} F|$ never exceeds 0.34 times the right-hand side).

Table 4.2 lists what the first half of the chapter hands to the rest of the book.

result of Section 4.2–Section 4.5	used for
Theorem 4.5 (limits q, μ, $U(\pm 1)$)	definition of $\boldsymbol{c}'$ and of the pole values compared with $U_c^{\pm}(\pm 1)$ in Chapter 5; the exponent μ in all charts of the second half
Theorem 4.9 ($U_\theta(1) > -3$)	the window of type R (Chapter 7); non-attachment to U_c^- for $k_2 > \frac{1}{2}$ (Chapter 5)
Lemma 4.13 (weights $E, E^{1-2\nu}, Em^2$)	contraction constants of the pole charts (second half); the weighted spaces of the existence theory in Chapter 6
Definition 4.16, Proposition 4.19	the four strata of edge solutions (Chapter 5, Chapter 7); germs at a pole (Chapter 6)
Corollary 4.24 (bounded swirl)	hypotheses of the centrifugal decomposition, hence the effective data of Chapter 8

TABLE 4.2. What the first half of the chapter hands to the rest of the book.

What we have, what comes next. Every local solution at a pole has a limit $q \geq -2$ of the potential, a limit $h \to \pm\mu$ with $\mu = \sqrt{1+q/2}$ and a limit of the swirl in $[-\infty, \infty]$ (Theorem 4.5); the recessive branch is possible only for $\mu < \frac{1}{2}$, so $U_\theta(1) > -3$ (Theorem 4.9); and the four types have the precise asymptotics of Proposition 4.19, with the weights E, $E^{1-2\mu}$, Em^2 of Lemma 4.13. What is missing is a *parametrization*: we know how a solution of a given type behaves, but not how the set of solutions of that type is shaped, nor how it depends on parameters. This is the content of the second half of the chapter: the types sit inside the Jost plane (σ, ω) (Proposition 4.32), and the weights E, $E^{1-2\mu}$, Em^2 are exactly those in which the map $w \mapsto \frac{1}{2}\mathfrak{r}_{R_*}(\Lambda_{\mathrm{p}} - f[w]^2)$ is a contraction (Lemma 4.35, Theorem 4.38, Theorem 4.40).

4.6. The Jost plane

In words. *Near a pole the equation is $z'' = (\sigma + w)z$ with $\sigma = \mu^2$ and w exponentially small. Its solution plane has a basis C (cosine-like), S (sine-like) that depends* analytically on σ through $\sigma = 0$ *(Lemma 3.38), and every solution is a multiple of $z = C + \omega S$ (ω is "slope over height" at the pole) or a multiple of S. Which pairs (σ, ω) give a solution that stays positive? A closed convex region $\widehat{\Gamma}$ of the (ω, σ)-plane: its interior is type* G*, its parabolic boundary arc is type* R*, its straight boundary ray is type* L_1*, and its corner is type* L_2*. The picture explains in one stroke why the four strata of Definition 4.16 are not four unrelated objects:* R *and* L_1 *are two faces of one wall and* L_2 *is where they meet.*

Throughout this section we use the hypotheses (J) of Lemma 3.38: a number $\mu_* \in (0, \frac{1}{160}]$, a cut $\xi_* \geq 1$, a constant $M > 0$ with $(M+1)e^{-\xi_*} \leq \frac{1}{16}$, and a continuous w on $[\xi_*, \infty)$ with $|w(\xi)| \leq Me^{-\xi}$ (note $e^{-\xi} = E^{1/2}$). For such w and $\sigma \in [0, \mu_*^2]$ the functions $C = C(\cdot;\sigma), S = S(\cdot;\sigma)$ of Lemma 3.38 solve $u'' = (\sigma + w)u$ and $\mathrm{Wr}(C, S) = 1$. For $\sigma > 0$ we write $\mu = \sqrt{\sigma}$ and $F_\pm = C \pm \mu S = e^{\pm\mu\xi}\hat{F}_\pm$ with $\hat{F}_\pm \in [\frac{2}{3}, \frac{4}{3}]$, $\hat{F}_\pm \to 1$ (Lemma 3.38(ii)); F_- is the recessive solution.

Definition 4.31 (the Jost region)**.** $\widehat{\Gamma} := \{(\sigma, \omega) \in \mathbb{R}^2 :\ \sigma \geq 0,\ \omega \geq -\sqrt{\sigma}\} = \{\sigma \geq (\omega_-)^2\}$, where $\omega_- := \max(-\omega, 0)$.

$\widehat{\Gamma}$ is the epigraph of the convex function $\omega \mapsto (\omega_-)^2$, hence closed and convex. Its boundary consists of the *parabolic arc* $\{\sigma = \omega^2,\ \omega < 0\}$, the *ray* $\{\sigma = 0,\ \omega > 0\}$ and the *corner* $(\sigma, \omega) = (0, 0)$; the interior is $\{\sigma > (\omega_-)^2\}$. The boundary curve is $C^{1,1}$ and not C^2 at the corner: its second derivative is 2 on the left and 0 on the right (Figure 4.5).

Proposition 4.32 (the Jost plane)**.** *Assume* (J)*, let $0 \leq \sigma \leq \mu_*^2$, $\omega \in \mathbb{R}$ and $z := C + \omega S$, a solution of $z'' = (\sigma + w)z$.*

(a) *$z > 0$ on $[\xi_*, \infty)$ if and only if $(\sigma, \omega) \in \widehat{\Gamma}$. If $(\sigma, \omega) \notin \widehat{\Gamma}$, then $z < 0$ for all large ξ.*

(b) *For $(\sigma, \omega) \in \widehat{\Gamma}$ the solution z has the following behaviour at $+\infty$:*

- *interior, $\sigma > 0$, $\omega > -\mu$ (type* G*): $z = \alpha F_+ + \beta F_-$ with $\alpha = \frac{1}{2}(1 + \omega/\mu) > 0$, $\beta = \frac{1}{2}(1 - \omega/\mu) > 0$; so $e^{-\mu\xi}z \to \alpha$ and $z'/z \to \mu$;*
- *parabolic arc, $\sigma > 0$, $\omega = -\mu$ (type* R*, $\mu \leq \mu_* < \frac{1}{2}$): $z = F_-$, $e^{\mu\xi}z \to 1$ and $z'/z \to -\mu$;*
- *ray, $\sigma = 0 < \omega$ (type* L_1*): $z' \to \omega > 0$ and $z/\xi \to \omega$;*

- *corner,* $(\sigma,\omega)=(0,0)$ *(type* L_2*):* $z\to 1$, $z'\to 0$.

(c) *If* w *is the potential perturbation of a solution of the* Q*-system,* $Q=2\sigma-2+2w$ *(so that* $W=1+\frac12 Q=\sigma+w$*), then* $Q(+\infty)=2\sigma-2$*: the vertical coordinate of the Jost plane is* $\sigma=\mu^2=1+\frac12 Q(1)$.

Proof. (a),(b) *Case* $\sigma>0$. Write $z=\alpha F_+ +\beta F_-$ with α,β as above (insert $C=\frac12(F_++F_-)$, $S=\frac{1}{2\mu}(F_+-F_-)$). If $\omega>-\mu$, then $\alpha>0$ and $\beta>0$ if $\omega<\mu$; if $\omega\ge\mu$ then $\beta\le 0$ but $z=C+\omega S\ge\frac23(C_0+\omega S_0)>0$ by Lemma 3.38(iii) (here $\omega\ge 0$, $C_0=\cosh\mu\xi$, $S_0=\mu^{-1}\sinh\mu\xi$). Thus: for $0\le\omega$, $z>0$ by (iii); for $-\mu<\omega<0$, both $\alpha,\beta>0$ and $z>0$; for $\omega=-\mu$, $z=F_-\!>0$. For $\omega<-\mu$ we have $\alpha<0<\beta$ and $z/F_-=\beta+\alpha F_+/F_-$ with $F_+/F_-=e^{2\mu\xi}\hat F_+/\hat F_-\ge\frac12 e^{2\mu\xi}\to\infty$, so $z<0$ for large ξ. Since $\hat F_\pm\to 1$: if $\alpha>0$ then $e^{-\mu\xi}z\to\alpha$ and $z'/z\to\mu$ (because $\hat F_+'/\hat F_+\to 0$, Lemma 3.38(ii)); if $z=F_-$, then $e^{\mu\xi}z\to 1$ and $z'/z\to-\mu$. *Case* $\sigma=0$. Here $C_0=1$, $S_0=\xi$ and (iii) gives $\frac23\le C\le\frac43$, $S\ge\frac23\xi$. If $\omega\ge 0$ then $z\ge\frac23>0$. If $\omega<0$ then $z\le\frac43-\frac23|\omega|\xi\to-\infty$. By (iv), $z'=C'+\omega S'\to\omega$, $z/\xi\to\omega$ for $\omega>0$, and $z=C\to 1$, $z'\to 0$ for $\omega=0$. (c) $W=1+\frac12 Q=\sigma+w$ and $w\to 0$ give $Q\to 2\sigma-2$. □

Remark 4.33 (types of the model versus types of solutions)**.** The proposition classifies the model solution $C+\omega S$ of the linear equation; it is the type of Definition 4.16 as soon as $w=\frac12(Q-q)$ comes from a solution of the Q-system (for $\kappa\neq 0$): the type is read off from q, $\lim z'/z$ and, for $q=-2$, from $b=\lim z'$ (Proposition 4.19(c)). For $\sigma>0$ the dichotomy "dominant/recessive" holds for every σ, not only $\sigma\le\mu_*^2$ (Lemma 3.16(a)); the Jost basis is needed only to cross $\sigma=0$. The coordinates (α,μ) of the dominant/recessive splitting are *not* smooth at $\sigma=0$ ($\alpha=\frac12(1+\omega/\sqrt{\sigma})$), whereas $z=C+\omega S$ is analytic in (σ,ω): this is why (σ,ω) and not (α,μ) are the coordinates of the corner chart (Theorem 4.45).

Toy example 4.34 ($w\equiv 0$: the Jost plane in closed form)**.** For $w\equiv 0$, $C=\cosh\mu\xi$, $S=\mu^{-1}\sinh\mu\xi$ and $z=\cosh\mu\xi+\frac{\omega}{\mu}\sinh\mu\xi$. This has a zero iff $\omega<-\mu$ (then $\tanh\mu\xi=\mu/|\omega|<1$ has exactly one solution, $\xi_0=\mu^{-1}\operatorname{artanh}(\mu/|\omega|)$) and never for $\omega\ge-\mu$; for $\sigma=0$, $z=1+\omega\xi$ vanishes at $\xi=-1/\omega$ iff $\omega<0$. For $\mu=0.1$, $\omega=-0.2$ this is $\xi_0=10\operatorname{artanh}\frac12\approx 5.49$: the solution looks positive on a long stretch and then turns negative; the further (σ,ω) lies below the parabola, the earlier. The region where $z>0$ for all ξ is $\widehat\Gamma$ exactly.

4.7. Weighted Volterra equations and the charts of G, R, L_1, L_2

In words. *A solution at the pole is the fixed point of a loop of four easy steps (the* chart loop*): given the potential perturbation* w*, (1) solve the* linear *equation* $z''=(\mu^2+w)z$ *with the normalisation of the stratum; (2) integrate* $f'=\kappa z^{-2}$*; (3) form the source* $F=\Lambda_{\mathrm p}-f^2$*; (4) solve* $\mathsf{L}[R]=F$ *with the Green operator of Lemma 4.13 and put* $w=\frac12 R$*. Each step is a Volterra-type integral equation (the integrals run towards the pole), so each step gains a small factor* $E_*=e^{-2\xi_*}$ *if* w *is small* in the weight natural to the stratum *(*E *for* G *and* L_1*,* $E^{1-2\varsigma}$ *for* R*,* Em^2 *for* L_2*). Hence for a cut* ξ_* *far enough out the loop is a contraction. Its fixed point depends smoothly on the finitely many free constants (the* coordinates *of the stratum), because a contraction whose constants vary smoothly has a fixed point*

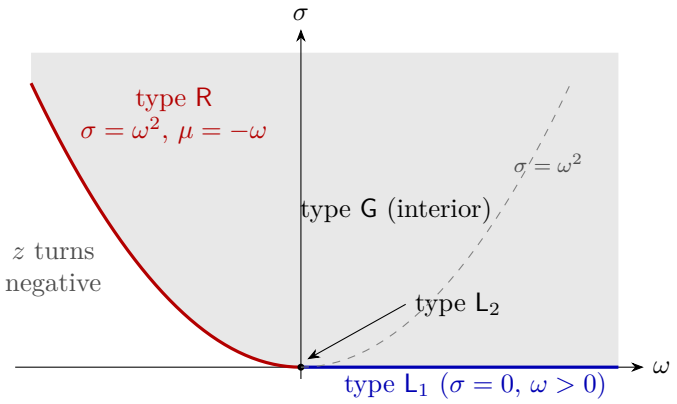


FIGURE 4.5. The Jost region $\widehat{\Gamma}$ (grey) in the (ω,σ)-plane, $\sigma=\mu^2=1+\frac12 Q(1)$ (Proposition 4.32). Left of the corner the boundary is the parabola $\sigma=\omega^2$ (recessive solutions, type R); right of it the ray $\sigma=0$ (type L_1); the corner is L_2. The two boundary pieces have the same tangent at the corner and different curvature. Dashed: the continuation $\sigma=\omega^2$, $\omega>0$ (pure dominant Jost solution F_+), which is *not* a boundary.

that varies smoothly (Theorem A.11). Everything that is special to a stratum is the weight, the normalisation of z, and the bookkeeping of which constants are free.

4.7.1. The linear steps. Fix $\xi_*\ge 0$ and write $I=[\xi_*,\infty)$, $E_*=e^{-2\xi_*}$, $m(\xi)=1+\xi-\xi_*$. For a continuous weight $\varpi>0$ on I let $C_\varpi=\{h\in C(I):\|h\|_\varpi=\sup_I|h|/\varpi<\infty\}$; C_ϖ is a Banach space (multiplication by ϖ is an isometry onto the bounded continuous functions), and $\|h_1h_2\|_{\varpi_1\varpi_2}\le\|h_1\|_{\varpi_1}\|h_2\|_{\varpi_2}$. We write C_b for $\varpi\equiv1$. The potential perturbation w always lies in a ball $\{\|w\|_\varpi\le r\}$ of such a space, $\varpi\in\{E,\ E^{1-2\varsigma},\ Em^2\}$ with $0<\varsigma<\frac12$ a number fixed once and for all. The lemma collects the three linear steps; each part has the same proof pattern: bound the integral operator by the weight, apply the Neumann series (Lemma A.1), identify the unique solution.

Lemma 4.35 (weighted Volterra equations)**.** *Let $\xi_*\ge0$ and let $w\in C(I)$.*
(A) (Dominant, $\mu>0$.) *Let $|w|\le rE$, $\chi_\mu:=\frac{1}{2e\min(\mu,1)}$, $Y_\mu:=C_{e^{\mu\xi}}$, and*

$$(Kh)(\xi)=\int_I g(\xi,\eta)w(\eta)h(\eta)\mathrm{d}\eta,$$
$$g(\xi,\eta)=-\tfrac1\mu\sinh\big(\mu(\min(\xi,\eta)-\xi_*)\big)e^{-\mu(\max(\xi,\eta)-\xi_*)}.$$

Then K maps Y_μ to itself, $\|K\|_{Y_\mu\to Y_\mu}\le\theta_{\mathrm D}:=\frac{rE_*}{2\mu}(\frac12+\chi_\mu)$, and Kh satisfies $(Kh)''=\mu^2Kh+wh$, $(Kh)(\xi_*)=0$, $e^{-\mu\xi}Kh\to0$. Let $\zeta>0$, $n_0=\min(1,\zeta)$, $n_1=\max(1,\zeta)$, $z_0(\xi)=e^{\mu\xi}+(\zeta-1)e^{2\mu\xi_*}e^{-\mu\xi}$ and assume $\theta_{\mathrm D}\le\min(\frac14,\frac{n_0}{4n_1})$. Then $z=(1-K)^{-1}z_0$ is the unique solution of $z''=(\mu^2+w)z$ with $z(\xi_*)=\zeta e^{\mu\xi_*}$ and $e^{-\mu\xi}z\to1$; it satisfies $\frac12 n_0\le e^{-\mu\xi}z\le 2n_1$ and $z'(\xi_*)=e^{\mu\xi_*}\big[\mu(2-\zeta)-\int_I w\,e^{-\mu\eta}z\,\mathrm{d}\eta\big]$.

(B) (Recessive.) *Let* $\mu > 0$, $|w| \le rE^{1-2\varsigma}$, $\theta_{\mathrm{R}} := \frac{rE_*^{1-2\varsigma}}{4\mu(1-2\varsigma)} \le \frac14$, *and* $(V\varphi)(\xi) = \int_\xi^\infty k_\mu(\eta-\xi)w(\eta)\varphi(\eta)\mathrm{d}\eta$ *with* $k_\mu(s) = \frac{1-e^{-2\mu s}}{2\mu} \in [0, \frac{1}{2\mu}]$. *Then* $\|V\|_{C_b\to C_b} \le \theta_{\mathrm{R}}$, $\hat z = (1-V)^{-1}1 \in [\frac23, \frac43]$, *and* $z = e^{-\mu\xi}\hat z$ *solves* $z'' = (\mu^2 + w)z$; *it is the unique solution with* $e^{\mu\xi}z \to 1$. *If* w_1, w_2 *are two such potentials (same* r*), then* $\sup_I|\hat z_1 - \hat z_2|E^{-(1-2\varsigma)} \le \frac{4}{9\mu(1-2\varsigma)}\|w_1 - w_2\|_{E^{1-2\varsigma}}$. *Also* $\hat z'(\xi_*) = -\int_I e^{-2\mu(\eta-\xi_*)}w\hat z\,\mathrm{d}\eta$ *and* $z' = e^{-\mu\xi}(\hat z' - \mu\hat z)$.

(C) (Logarithmic, $\mu = 0$.) *Let* $(K_0h)(\xi) = \int_I(\min(\xi,\eta) - \xi_*)\,w(\eta)h(\eta)\mathrm{d}\eta$. *Then* $(K_0h)(\xi_*) = 0$, $(K_0h)' = \int_\xi^\infty wh$, $(K_0h)'' = -wh$. (L1) *If* $|w| \le rE$, *then* $\|K_0\|_{C_m\to C_m} \le \frac34 rE_*$. *Let* $\zeta > 0$, $b_0 = \min(1,\zeta)$, $b_1 = \max(1,\zeta)$ *and assume* $\frac34 rE_* \le \min(\frac14, \frac{b_0}{4b_1})$. *Then* $z = (1+K_0)^{-1}(\zeta - 1 + m)$ *is the unique* $z \in C_m$ *with* $z'' = wz$, $z(\xi_*) = \zeta$, $z'(\infty) = 1$, *and* $\frac23 b_0 m \le z \le \frac43 b_1 m$, $z'(\xi_*) = 1 - \int_I wz\mathrm{d}\eta$. (L2) *If* $|w| \le rEm^2$, *then* $\|K_0\|_{C_b\to C_b} \le \frac98 rE_*$. *If* $\frac98 rE_* \le \frac14$, *then* $z = (1+K_0)^{-1}1$ *is the unique bounded* z *with* $z'' = wz$, $z(\xi_*) = 1$, $z'(\infty) = 0$, *and* $\frac23 \le z \le \frac43$, $z'(\xi_*) = -\int_I wz\mathrm{d}\eta$.

Proof. (A) *The operator.* Splitting at $\eta = \xi$,

$$(Kh)(\xi) = -\frac{1}{\mu}\Big[e^{-\mu(\xi-\xi_*)}\int_{\xi_*}^{\xi}\sinh(\mu(\eta-\xi_*))wh\mathrm{d}\eta + \sinh(\mu(\xi-\xi_*))\int_\xi^\infty e^{-\mu(\eta-\xi_*)}wh\mathrm{d}\eta\Big]$$

(the second integral converges since $|wh| \le r\|h\|_{Y_\mu}Ee^{\mu\eta}$). Differentiating, the terms from the variable limits cancel and $(Kh)' = -\frac{1}{\mu}\big[-\mu e^{-\mu(\xi-\xi_*)}\int_{\xi_*}^{\xi}\sinh(\cdot)wh + \mu\cosh(\mu(\xi-\xi_*))\int_\xi^\infty e^{-\mu(\eta-\xi_*)}wh\big]$; differentiating once more, the limit terms give $e^{-\mu(\xi-\xi_*)}(\sinh+\cosh)(\mu(\xi-\xi_*))\,wh = wh$, so $(Kh)'' = \mu^2Kh + wh$; clearly $(Kh)(\xi_*) = 0$. *The norm.* $|g|(\xi,\eta) \le \frac{1}{2\mu}e^{-\mu|\xi-\eta|}$. For $h \in Y_\mu$, $|h(\eta)| \le \|h\|e^{\mu\eta}$, and $e^{-\mu\xi}e^{-\mu|\xi-\eta|}e^{\mu\eta}$ equals 1 for $\eta \ge \xi$ and $e^{-2\mu(\xi-\eta)}$ for $\eta < \xi$. Hence $e^{-\mu\xi}|Kh(\xi)| \le \frac{\|h\|}{2\mu}\big[\int_\xi^\infty|w| + \int_{\xi_*}^{\xi}e^{-2\mu(\xi-\eta)}|w|\big]$. The first integral is $\le rE(\xi)/2 \le rE_*/2$. In the second put $s = \eta - \xi_*$, $\rho = \xi - \xi_*$, $\nu = \min(\mu,1)$; since $-2\mu(\rho - s) - 2s \le -2\nu(\rho - s) - 2\nu s = -2\nu\rho$, it is $\le rE_*\int_0^\rho e^{-2\mu(\rho-s)-2s}\mathrm{d}s \le rE_*\rho e^{-2\nu\rho} \le rE_*\chi_\mu$. This gives $\|K\| \le \theta_{\mathrm{D}}$, and the same computation shows $e^{-\mu\xi}Kh \to 0$ (tail of a convergent integral, and $\rho e^{-2\nu\rho} \to 0$). *The solution.* $z_0'' = \mu^2 z_0$, $z_0(\xi_*) = \zeta e^{\mu\xi_*}$ and $e^{-\mu\xi}z_0 = 1 + (\zeta-1)e^{-2\mu(\xi-\xi_*)}$ is a convex combination of 1 and ζ, so it lies in $[n_0, n_1]$. Since $\theta_{\mathrm{D}} \le \frac14$, $z = (1-K)^{-1}z_0 \in Y_\mu$ exists, solves $z = z_0 + Kz$, hence $z'' = \mu^2 z_0 + \mu^2 Kz + wz = (\mu^2+w)z$, $z(\xi_*) = z_0(\xi_*)$, $e^{-\mu\xi}z \to 1$. By Lemma A.1, $\|z - z_0\|_{Y_\mu} \le \frac43\theta_{\mathrm{D}}\|z_0\|_{Y_\mu} \le \frac43 \cdot \frac{n_0}{4n_1}n_1 = \frac{n_0}{3}$, so $e^{-\mu\xi}z \in [\frac23 n_0, n_1 + \frac13 n_0] \subset [\frac12 n_0, 2n_1]$. *Uniqueness.* If $\tilde z$ is any solution with the same data, then $\tilde z \in Y_\mu$ and $v = \tilde z - z_0 - K\tilde z$ satisfies $v'' = \mu^2 v$, $v(\xi_*) = 0$, $e^{-\mu\xi}v \to 1 - 1 - 0 = 0$; so $v = \alpha e^{\mu\xi} + \beta e^{-\mu\xi}$ with $\alpha = 0$ and then $\beta = 0$; thus $\tilde z = z_0 + K\tilde z$ and $\tilde z = z$. *The derivative at the cut.* $z_0'(\xi_*) = \mu(2-\zeta)e^{\mu\xi_*}$ and, from the formula for $(Kh)'$ at $\xi = \xi_*$, $(Kz)'(\xi_*) = -\int_I e^{-\mu(\eta-\xi_*)}wz\mathrm{d}\eta$.

(B) For bounded φ, $|V\varphi(\xi)| \le \frac{r}{2\mu}\|\varphi\|_\infty\int_\xi^\infty E^{1-2\varsigma} = \frac{r}{4\mu(1-2\varsigma)}E(\xi)^{1-2\varsigma}\|\varphi\|_\infty \le \theta_{\mathrm{R}}\|\varphi\|_\infty$, using $\int_\xi^\infty e^{-2(1-2\varsigma)\eta}\mathrm{d}\eta = \frac{E(\xi)^{1-2\varsigma}}{2(1-2\varsigma)}$; so $\|V\| \le \theta_{\mathrm{R}} \le \frac14$, $\hat z = (1-V)^{-1}1$ exists in C_b, $\|\hat z - 1\|_\infty \le \frac43\theta_{\mathrm{R}} \le \frac13$. For $G(\xi) = \int_\xi^\infty k_\mu(\eta-\xi)g\mathrm{d}\eta$ one has $G' =$

$-\int_\xi^\infty e^{-2\mu(\eta-\xi)}g$ (because $k_\mu(0)=0$, $\partial_\xi k_\mu(\eta-\xi)=-e^{-2\mu(\eta-\xi)}$) and $G''=g+2\mu G'$; so $\hat z''-2\mu\hat z'=w\hat z$, which is the equation for $z=e^{-\mu\xi}\hat z$. Also $\hat z-1=V\hat z\to 0$. Uniqueness: by Lemma 3.16(a) (as $\int|w|<\infty$) every solution is $Az_+ + Bz_-$, and $e^{\mu\xi}z$ bounded forces $A=0$, then $e^{\mu\xi}z\to 1$ forces $z=z_-$, which is the solution above (the bounded solution of the Volterra equation is unique). *Difference.* $\hat z_1-\hat z_2 = V_1(\hat z_1-\hat z_2)+(V_1-V_2)\hat z_2$. Put $\Delta=\|w_1-w_2\|_{E^{1-2\varsigma}}$ and $D=\sup|\hat z_1-\hat z_2|/E^{1-2\varsigma}$ ($<\infty$ because $|\hat z_i-1|\le\frac43\frac{r}{4\mu(1-2\varsigma)}E^{1-2\varsigma}$). Then $|(V_1-V_2)\hat z_2|\le\frac{1}{2\mu}\cdot\frac43\Delta\frac{E^{1-2\varsigma}}{2(1-2\varsigma)}$ and $|V_1(\hat z_1-\hat z_2)|\le\frac{r}{2\mu}\int_\xi^\infty E^{1-2\varsigma}\,DE^{1-2\varsigma}\le\theta_{\mathrm R}DE^{1-2\varsigma}$ (as $E(\eta)^{1-2\varsigma}\le E_*^{1-2\varsigma}$). Hence $D\le\frac14 D+\frac{\Delta}{3\mu(1-2\varsigma)}$, i.e. $D\le\frac{4\Delta}{9\mu(1-2\varsigma)}$. The formula for $\hat z'(\xi_*)$ is $G'(\xi_*)$ above.

(C) The three formulas for K_0h follow by differentiating $\int_{\xi_*}^{\xi}(\eta-\xi_*)wh\mathrm d\eta+(\xi-\xi_*)\int_\xi^\infty wh\mathrm d\eta$. (L1) For $\|h\|_m<\infty$, $\min(\xi,\eta)-\xi_*\le\xi-\xi_*\le m(\xi)$ and $|wh|\le r\|h\|_m Em$, so $|K_0h|/m\le r\|h\|_m\int_I Em\mathrm d\eta = rE_*\|h\|_m\int_0^\infty(1+s)e^{-2s}\mathrm ds = \frac34 rE_*\|h\|_m$. The function $z_{\mathrm{lin}}=\zeta-1+m=\zeta+\xi-\xi_*$ satisfies $b_0m\le z_{\mathrm{lin}}\le b_1m$ (if $\zeta\ge1$: $m\le\zeta-1+m\le\zeta m$ since $m\ge1$; if $\zeta<1$: $\zeta m\le m-(1-\zeta)\le m$). With $\theta=\frac34 rE_*\le\min(\frac14,\frac{b_0}{4b_1})$, $z=(1+K_0)^{-1}z_{\mathrm{lin}}$ satisfies $\|z-z_{\mathrm{lin}}\|_m\le\frac43\theta b_1\le\frac13 b_0$ (Neumann series), which gives the two-sided bound. Then $z=z_{\mathrm{lin}}-K_0z$ gives $z''=wz$, $z(\xi_*)=\zeta$, $z'(\infty)=1-\lim\int_\xi^\infty wz=1$. Uniqueness: if $\tilde z\in C_m$ has these properties, $v=\tilde z+K_0\tilde z-z_{\mathrm{lin}}$ has $v''=0$, $v(\xi_*)=0$, $v'(\infty)=0$, so $v\equiv0$ and $\tilde z=(1+K_0)^{-1}z_{\mathrm{lin}}$. (L2) For bounded h, $|K_0h|\le\|h\|_\infty\int_I(\eta-\xi_*)rEm^2\mathrm d\eta = rE_*\|h\|_\infty\int_0^\infty s(1+s)^2e^{-2s}\mathrm ds=\frac98 rE_*\|h\|_\infty$; the rest is identical ($\|z-1\|\le\frac43\cdot\frac14$). □

Toy example 4.36 (a Volterra equation solved by Bessel functions)**.** Take $w(\xi)=\varepsilon e^{-2\xi}$ ($\varepsilon>0$). With $t=\sqrt\varepsilon e^{-\xi}$ one has (ξ-derivative) $=-t\,\partial_t$, so $z''=(\mu^2+\varepsilon e^{-2\xi})z$ becomes $(t\partial_t)^2z=(\mu^2+t^2)z$, the modified Bessel equation of order μ. The solution tending to 0 at the pole is $I_\mu(t)$, and the normalisation of Lemma 4.35(B) is

$$\hat z(\xi)=e^{\mu\xi}z=\Gamma(1+\mu)\Big(\frac{2}{t}\Big)^{\mu}I_\mu(t),\qquad t=\sqrt\varepsilon\,e^{-\xi},$$

which tends to 1 as $t\to0$ because $I_\mu(t)\sim(t/2)^\mu/\Gamma(1+\mu)$. The Volterra iteration $\hat z_{n+1}=1+V\hat z_n$ converges geometrically. For $\varepsilon=1$, $\mu=0.3$ the crude bound $\|V\|\le\int_{\xi_*}^\infty\frac{e^{-2\eta}}{2\mu}\mathrm d\eta$ is 0.83 at $\xi_*=0$ and 0.002 at $\xi_*=3$; the iterates agree with the Bessel expression, and settle to 10^{-10} in 7 steps at $\xi_*=0$ and in 3 steps at $\xi_*=3$ (`verify_ch04b.py` compares them with I_μ). Figure 4.6 shows z, $\hat z=e^{\mu\xi}z$ and the swirl $1+f$, $f=\kappa\int_0^\xi z^{-2}$, for $\mu=0.3,0.1,0.03,0$: the family is smooth through $\mu=0$, where $z\to I_0(e^{-\xi})/I_0(0)$ is bounded and f grows only linearly — the passage from type $\mathsf R$ to type $\mathsf L_2$ that the corner chart of Section 4.8 describes.

4.7.2. The chart loop and the normalisations κ_{ch}. Fix a stratum and a cut $\xi_*\ge0$. A *normalised local solution* of type $\mathsf G,\mathsf R,\mathsf L_1,\mathsf L_2$ is a local solution (z,f,Q) with $\kappa\neq0$ in which z has been rescaled ($z\mapsto cz$, $\kappa\mapsto c^2\kappa$, which does not change f, Q, U) so that the normalisation of Table 4.3 holds; the resulting value of κ is denoted κ_{ch} (*chart normalisation*). The coordinates of the stratum are κ_{ch}, the constants of z not fixed by the normalisation, and the three constants $\varphi,\Lambda_{\mathrm p},\rho$ which describe f and Q. Put $q=2\mu^2-2$ ($\mu=0$ for $\mathsf L_1,\mathsf L_2$), $w=\frac12 R$, $R=Q-q$, $F=\Lambda_{\mathrm p}-f^2$; then Lemma 4.13 and Proposition 4.19 give

$$z''=(\mu^2+w)z,\qquad f'=\kappa_{\mathrm{ch}}z^{-2},\qquad w=\tfrac12\mathfrak t_{R_*}\big(\Lambda_{\mathrm p}-f^2\big),\qquad \lambda=\Lambda_{\mathrm p}-2q,\tag{4.14}$$

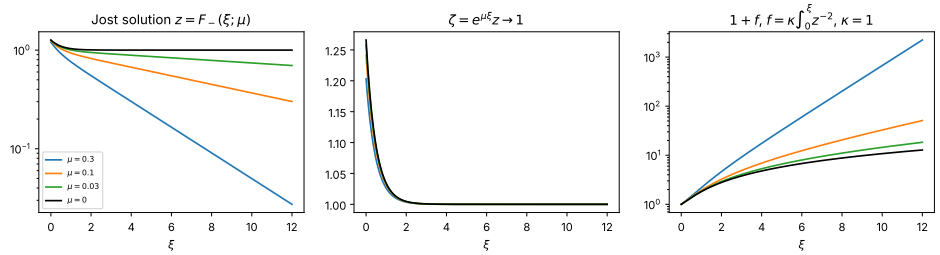


FIGURE 4.6. Toy example 4.36: the recessive solution $z = F_-(\xi;\mu)$ of $z'' = (\mu^2 + e^{-2\xi})z$ for $\mu = 0.3, 0.1, 0.03, 0$ (left, log scale), its normalisation $\hat{z} = e^{\mu\xi} z \to 1$ (middle), and $1 + f$, $f = \int_0^\xi z^{-2}$ (right). Produced by `code/ch04_pole/make_toy_rchart.py`.

and conversely a solution of (4.14) with $w \to 0$ is a solution of the system $\mathsf{L}[Q] = \lambda - f^2$, $z'' = (1 + \frac{1}{2}Q)z$, $f' = \kappa_{\mathrm{ch}} z^{-2}$ of Chapter 2 (because $\mathsf{L}[q] = -2q$, so $\mathsf{L}[R] = \mathsf{L}[Q] + 2q$).

stratum	normalisation κ_{ch}	coordinates p	f	weight of w	dim
G	$ze^{-\mu\xi} \to 1$	$(\mu, \zeta, \kappa_{\mathrm{ch}}, \varphi, \Lambda_{\mathrm{p}}, \rho)$, $z(\xi_*) = \zeta e^{\mu\xi_*}$	$f = \varphi - \kappa_{\mathrm{ch}} \int_\xi^\infty z^{-2}$	E	6
R	$ze^{\mu\xi} \to 1$	$(\mu, \kappa_{\mathrm{ch}}, \varphi, \Lambda_{\mathrm{p}}, \rho)$, $f(\xi_*) = e^{2\mu\xi_*}\varphi$	$f = f(\xi_*) + \kappa_{\mathrm{ch}} \int_{\xi_*}^\xi z^{-2}$	$E^{1-2\varsigma}$, $\varsigma > \mu$	5
L_1	$z' \to 1$	$(\zeta, \kappa_{\mathrm{ch}}, \varphi, \Lambda_{\mathrm{p}}, \rho)$, $z(\xi_*) = \zeta$	$f = \varphi - \kappa_{\mathrm{ch}} \int_\xi^\infty z^{-2}$	E	5
L_2	$z(\xi_*) = 1$	$(\kappa_{\mathrm{ch}}, \varphi, \Lambda_{\mathrm{p}}, \rho)$, $f(\xi_*) = \varphi$	$f = \varphi + \kappa_{\mathrm{ch}} \int_{\xi_*}^\xi z^{-2}$	Em^2	4

TABLE 4.3. The four stratum charts. In all cases $R_* = R(\xi_*) = \rho E_*$, except for R where $R_* = \rho E_*^{1-2\varsigma}$. The normalisation of z is the one of Lemma 4.35: (A) for G, (B) for R, (C)(L1) for L_1 and (C)(L2) for L_2. The dimension is that of the stratum in the space of scale-free states (see Corollary 4.42); it drops by one from G to R and L_1 (the dominant coefficient ζ or the exponent μ is lost) and by two to L_2.

Remark 4.37 (why the scaled coordinates)**.** The coordinates ρ, φ are chosen so that they stay in a compact set when the cut ξ_* moves out along a fixed solution: $\rho = R(\xi_*)/E_*$ is bounded by Proposition 4.19, $\zeta = ze^{-\mu\xi} \to 1$ at G, and in the recessive chart $f(\xi_*)e^{-2\mu\xi_*} \to \kappa_{\mathrm{ch}}/(2\mu)$. At L_2 one has $f(\xi_*) = \kappa_{\mathrm{ch}}\xi_* + O(1)$, which grows linearly; this is why the smallness condition for L_2 contains $|\varphi|$ (Theorem 4.40(d)).

The loop. For a parameter p in the chart space and w in the ball $\mathbb{B}_r = \{\|w\|_\varpi \le r\} \subset C_\varpi$ let $z = z[p, w]$ be the solution of Lemma 4.35 (A), (B), (C)(L1) or (C)(L2) with the data of Table 4.3, $f = f[p, w]$ the function in the table, and

$$T_p(w) = \tfrac{1}{2}\,\mathfrak{r}_{R_*}\big(\Lambda_{\mathrm{p}} - f^2\big). \tag{4.15}$$

A fixed point $w = T_p(w)$ is exactly a normalised local solution of the stratum with coordinates p and cut ξ_*: indeed then $z'' = (\mu^2 + w)z$, $f' = \kappa_{\mathrm{ch}} z^{-2}$ by construction

and $\mathsf{L}[2w] = \Lambda_{\mathrm{p}} - f^2$, $2w \to 0$ (Lemma 4.13), i.e. $Q := q + 2w$ solves $\mathsf{L}[Q] = \lambda - f^2$. Conversely, a normalised local solution of the stratum on $[\xi_*, \infty)$ whose w lies in $\mathbb{B}_r$ is a fixed point, by the uniqueness statements of Lemma 4.35 (for z) and Lemma 4.13 (for R).

4.7.3. The charts of G and of R, L_1, L_2. For $B \geq 1$ let K_B^{G} be the compact set $\{p \in \mathbb{R}^6 : \ B^{-1} \leq \mu, \zeta \leq B, \ |\kappa_{\mathrm{ch}}|, |\varphi|, |\Lambda_{\mathrm{p}}|, |\rho| \leq B\}$ and define $K_B^{\mathsf{R}}, K_B^{\mathsf{L}_1}, K_B^{\mathsf{L}_2}$ analogously (with $\mu \in [B^{-1}, \varsigma - B^{-1}]$ for R, where $\varsigma \in (0, \frac{1}{2})$ is the fixed number of Theorem 4.40, and the entries of p of Table 4.3 bounded by B; $\zeta \in [B^{-1}, B]$ for L_1). Every compact subset of the (open) parameter space is contained in one of these.

THEOREM 4.38 (the chart of G). *Let $B \geq 1$ and $r_0 > 0$. There is $\xi_G = \xi_G(B, r_0) \geq 0$ such that for every cut $\xi_* \geq \xi_G$ and every $p \in K_B^{\mathsf{G}}$ (including $\kappa_{\mathrm{ch}} = 0$, the case without swirl):*

(a) (Existence, uniqueness.) *The loop T_p has a unique fixed point w_p in $\mathbb{B}_{r_0} \subset C_E$ (for $r_0 \geq r_B := \frac{1}{2}(3\Phi_B + 4B)$, Φ_B below); w_p is the potential perturbation of a normalised local solution u_p of type G on $[\xi_*, \infty)$ with coordinates p and $Q(+\infty) = 2\mu^2 - 2$; $|w_p| \leq r_B E$.*

(b) (Smoothness.) *$p \mapsto w_p \in C_E$ is of class C^∞ on the interior of K_B^{G} (more precisely, on the open set where the above holds); hence $(p, \xi) \mapsto (z, f, Q)$ and all their ξ-derivatives are C^∞, and $\partial_p^\alpha w_p = O(E)$ uniformly.*

(c) (Completeness.) *Let u be a normalised local solution of type G (with $\kappa \neq 0$, or with $\kappa = 0$ in the sense of the convention after Definition 4.16). Then there are B, r_0 and ξ_u such that for every cut $\xi_* \geq \xi_u$ the coordinates $p(\xi_*)$ of u at the cut lie in K_B^{G}, $\xi_* \geq \xi_G(B, r_0)$, and $u|_{[\xi_*, \infty)} = u_{p(\xi_*)}$.*

(d) (Injectivity, immersion.) *The state map $\mathsf{S}_{\xi_*} : p \mapsto s(u_p)(\xi_*) \in \mathbb{R}^6$ (scale-free state of Corollary 4.42) is injective and a C^∞ immersion, hence a local C^∞ diffeomorphism.*

PROOF. *Setup.* Put $\hat{z} = e^{-\mu\xi} z$, so that $\hat{z}$ is the solution of the rescaled Volterra equation $\hat{z} = \hat{z}_0 + \hat{K}_{\mu,w}\hat{z}$, where $\hat{z}_0 = 1 + (\zeta - 1)e^{-2\mu(\xi - \xi_*)}$ and $(\hat{K}_{\mu,w}\hat{h})(\xi) = \int_I \hat{g}_\mu(\xi, \eta) w(\eta) \hat{h}(\eta) \mathrm{d}\eta$ with, by conjugating the kernel of Lemma 4.35(A) with $e^{\mu\xi}$,

$$\hat{g}_\mu(\xi, \eta) = -\int_0^{\xi - \xi_*} e^{-2\mu v} \mathrm{d}v \qquad (\eta \geq \xi),$$

$$\hat{g}_\mu(\xi, \eta) = -e^{-2\mu(\xi - \eta)} \int_0^{\eta - \xi_*} e^{-2\mu v} \mathrm{d}v \qquad (\eta < \xi).$$

(Check: $-\frac{1}{\mu}\sinh(\mu a) e^{-\mu b}\, e^{-\mu\xi} e^{\mu\eta}$ with $a = \min -\xi_*$, $b = \max -\xi_*$; for $\eta \geq \xi$ this is $-\frac{1}{2\mu}(1 - e^{-2\mu(\xi - \xi_*)})$ and for $\eta < \xi$ it is $-\frac{1}{2\mu}(1 - e^{-2\mu(\eta - \xi_*)})e^{-2\mu(\xi - \eta)}$.) The point of the rescaling is that

$$|\partial_\mu^j \hat{g}_\mu(\xi, \eta)| \leq c_j(\mu) := C_j\, \mu^{-j-1} \qquad (j \geq 0), \tag{4.16}$$

with C_j depending only on j, because $\partial_\mu^j \int_0^a e^{-2\mu v}\mathrm{d}v = \int_0^a (-2v)^j e^{-2\mu v}\mathrm{d}v$ has modulus $\leq \int_0^\infty (2v)^j e^{-2\mu v}\mathrm{d}v = \frac{j!}{2\mu^{j+1}}$, $\partial_\mu^i e^{-2\mu s} = (-2s)^i e^{-2\mu s}$ with $(2s)^i e^{-2\mu s} \leq (i/(e\mu))^i$, and Leibniz' rule. So the family $\mu \mapsto \hat{K}_{\mu,\cdot} \in \mathcal{L}(C_E, \mathcal{L}(C_b))$ is C^∞ (differentiate under the integral; $\int_I E \mathrm{d}\eta = E_*/2$), with $\|\partial_\mu^j \hat{K}\| \leq c_j(\mu) E_*/2$. On K_B^{G}, $\mu \geq B^{-1}$, so all constants below depend only on B.

Step 1: the linear step. For $w \in \mathbb{B}_r$, $r \le 2r_B$ say, Lemma 4.35(A) applies as soon as $\theta_{\mathrm{D}} = \frac{rE_*}{2\mu}(\frac12 + \chi_\mu) \le \min(\frac14, \frac{n_0}{4n_1})$; since $n_0 \ge B^{-1}$, $n_1 \le B$ this holds for $E_* \le c(B)/r$. Then $\frac12 n_0 \le \hat z \le 2n_1$ and $\hat z$ depends analytically on (μ, ζ, w) (Neumann series, Lemma A.1, and the smoothness of $\hat K$; it is the only place where μ enters through a kernel). Hence $z^{-2} \le 4n_0^{-2} e^{-2\mu\xi}$ and

$$|f| \le |\varphi| + |\kappa_{\mathrm{ch}}| \int_\xi^\infty 4n_0^{-2} e^{-2\mu\eta} \mathrm{d}\eta \le c_f := B + 2B^3 \cdot B = B + 2B^4.$$

(here $n_0^{-2} \le B^2$ and $\int_\xi^\infty e^{-2\mu\eta} \le \frac{1}{2\mu} \le \frac{B}{2}$.) Consequently $|F| = |\Lambda_{\mathrm{p}} - f^2| \le \Phi_B := B + c_f^2$ *independently of* ξ_*, r.

Step 2: invariance. By (Q1) of Lemma 4.13 with $R_* = \rho E_*$ ($|R_*|/E_* \le B$), $|T_p(w)| \le \frac12(3\Phi_B + 4B)E = r_B E$. Take $r \ge 2r_B$; then $T_p(\mathbb{B}_r) \subset \mathbb{B}_{r/2} \subset \operatorname{int}\mathbb{B}_r$.

Step 3: contraction. Let $w_1, w_2 \in \mathbb{B}_r$, $\Delta = \|w_1 - w_2\|_E$, $\hat z_i$ the corresponding solutions. From $\hat z_1 - \hat z_2 = \hat K_1(\hat z_1 - \hat z_2) + \hat K_{w_1 - w_2}\hat z_2$ and $\|\hat K_1\| \le \frac14$: $\|\hat z_1 - \hat z_2\|_\infty \le \frac43 \cdot \frac{E_*}{2} c_0(\mu)\,\Delta \cdot 2n_1 \le c_1 E_* \Delta$. Then, pointwise,

$$|z_1^{-2} - z_2^{-2}| = e^{-2\mu\xi}|\hat z_1^{-2} - \hat z_2^{-2}| \le e^{-2\mu\xi} \frac{|\hat z_1 + \hat z_2|}{\hat z_1^2 \hat z_2^2} |\hat z_1 - \hat z_2| \le c_2 E_* \Delta\, e^{-2\mu\xi},$$

so $|f_1 - f_2| \le |\kappa_{\mathrm{ch}}| c_2 E_* \Delta/(2\mu) \le c_3 E_* \Delta$, $|F_1 - F_2| = |f_1 + f_2||f_1 - f_2| \le 2c_f c_3 E_* \Delta$, and by (Q1) with R_*-datum 0, $\|T_p(w_1) - T_p(w_2)\|_E \le \frac12 \cdot 3 \cdot 2c_f c_3 E_* \Delta =: C_B E_* \Delta$. Choose ξ_G with $C_B E_* \le \frac12$ and $E_* \le c(B)/r$ for $r = \max(2r_B, r_0)$: then T_p is a $\frac12$-contraction of $\mathbb{B}_r$ to itself, with a unique fixed point w_p, and $|w_p| \le r_B E$ because $T_p(\mathbb{B}_r) \subset \mathbb{B}_{r_B}$ (Step 2). This is (a): the identification with a local solution of type G is the remark after (4.15); that $\mu^2 = 1 + q/2$, $Q(\infty) = q$, and that the type is G ($q > -2$, $h \to \mu$) is Lemma 4.35(A) ($e^{-\mu\xi} z \to 1$).

Step 4: smoothness, (b). We apply Theorem A.11 with parameter set $P = \operatorname{int} K_{2B}^{\mathsf{G}} \supset K_B^{\mathsf{G}}$ (the estimates of Steps 1–3 hold on K_{2B}^{G} with constants depending on B only; enlarge ξ_G accordingly), $X = C_E$, $O = \operatorname{int}\mathbb{B}_{2r}$, closed set $\mathbb{B}_r$ ((iii) holds since $T_p(\mathbb{B}_r) \subset \mathbb{B}_{r/2}$) and $\theta = \frac12$. It remains to see that $T : P \times O \to C_E$ is C^k for every k. The map $(p, w) \mapsto \hat z \in C_b$ is C^∞ (composition of the smooth $\hat K$ with the inverse in $\mathcal{L}(C_b)$). The functions $z^{-2} = e^{-2\mu\xi}\hat z^{-2}$ and $f = \varphi - \kappa_{\mathrm{ch}} \int_\xi^\infty z^{-2}$ have (p, w)-derivatives that are finite sums of products of $\partial_\mu^i(e^{-2\mu\eta}) = (-2\eta)^i e^{-2\mu\eta}$, derivatives of $\hat z^{-2}$ in C_b, and integrals of these: each is bounded uniformly in ξ by $C_i(B)$, since $\eta^i e^{-2\mu\eta} \le (i/(2e\mu))^i$. Hence $F = \Lambda_{\mathrm{p}} - f^2$ has all pointwise derivatives of order $\le k+1$ bounded by constants K_j times $\|h\|^j$. Lemma A.15 (with the weight $\varpi \equiv 1$ on the convex set $P \times O'$, O' a ball) gives $F \in C^k(P \times O'; C_b)$ for every k. Finally, by the joint linearity in (R_*, F) (Lemma 4.13) and since the solution of $\mathsf{L}[g] = 0$, $g \to 0$, $g(\xi_*) = R_*$ is $R_* t^2/t_*^2$, one has $\frac12 \mathfrak{r}_{\rho E_*} F = \frac12 \mathfrak{r}_0 F + \frac12 \rho E_* t^2/t_*^2$; the first term is bounded linear $C_b \to C_E$ by (Q1), the second is affine in ρ with values of size $\le 2|\rho|E$ (as $t \le 2E$, $t_* \ge E_*$). Thus $T \in C^k$ for every k and Theorem A.11 gives $w \in C^\infty(P; C_E)$; its derivative is $(I - \mathrm{D}_w T)^{-1}\mathrm{D}_p T$ with norm $\le 2\|\mathrm{D}_p T\|$. Pointwise derivatives in ξ follow from the equation (w'' is expressed by the Green operator).

Step 5: completeness, (c). Let u be given. By Proposition 4.19(a) if $\kappa \ne 0$, and by Lemma 4.20 if $\kappa = 0$, $|R| \le C_u E$ on $[\xi_u, \infty)$, $a = \lim z e^{-\mu\xi} = 1$ (normalised) and $f(\infty) = \varphi$ exists. For a cut $\xi_* \ge \xi_u$ the coordinates are μ, $\zeta(\xi_*) = e^{-\mu\xi_*} z(\xi_*) \to 1$, $\kappa_{\mathrm{ch}} = \kappa$, φ, $\Lambda_{\mathrm{p}} = \lambda + 2q$, $\rho(\xi_*) = R(\xi_*)/E_*$ with $|\rho| \le C_u$. So there is B with

$p(\xi_*) \in K_B^{\mathsf{G}}$ for all $\xi_* \geq \xi_u'$, and $r_0 := \frac{1}{2}C_u$ bounds $\|w\|_E$ on $[\xi_*, \infty)$. The function $w = \frac{1}{2}R$ is a fixed point of $T_{p(\xi_*)}$ in $\mathbb{B}_{r_0}$ for every such cut by the converse in the remark after (4.15); for $\xi_* \geq \max(\xi_u', \xi_G(B, r_0))$ the fixed point is unique, so it equals $w_{p(\xi_*)}$ and $u = u_{p(\xi_*)}$.

Step 6: injectivity and immersion, (d). The scale-free state is $s = (h, k, f, Q, \dot{Q}, \lambda)$ with $h = z'/z$, $k = \kappa/z^2$, $\dot{Q} = Q'$; it satisfies the autonomous-in-form system $h' = 1 + \frac{1}{2}Q - h^2$, $k' = -2hk$, $f' = k$, $Q'' = (\lambda - f^2 + 2Q - 2\sinh 2\xi\, Q')/\cosh^2\xi$, $\lambda' = 0$, which has a unique solution for any initial state (Theorem A.4), so S_{ξ_*} is injective because p is recovered from the solution (μ from $Q(\infty) = 2\mu^2 - 2$, $\kappa_{\mathrm{ch}} = \kappa$ from $k/\hat{z}^{-2}$, and so on: the six coordinates are functionals of the solution, as in (c)). For the immersion, let $\delta p \in \mathbb{R}^6$ with $\mathsf{DS}\,\delta p = 0$. By (b) the p-derivatives $\delta s(\xi) = \partial_p s(u_p)(\xi)\delta p$ exist, are continuous in ξ, and satisfy the linearised system (differentiate the system in p); $\delta s(\xi_*) = 0$ gives $\delta s \equiv 0$ on $[\xi_*, \infty)$ by uniqueness for linear systems. Now $\delta Q \equiv 0$ and $Q = q + 2w_p$ with $\partial_p w_p \in C_E$, so $\delta q = \lim \delta Q = 0$, hence $\delta\mu = 0$ and $\delta w \equiv 0$; then $\delta h \equiv 0$ means $\delta z = cz$. Since $\delta\mu = 0$, $e^{-\mu\xi}\delta z = \delta\hat{z}$, and $\hat{z}_p - \hat{z}_0 \in C_0$ (functions tending to 0; $\hat{K}$ maps C_b into C_0 by Lemma 4.35(A)) is a C^∞ map into the closed subspace C_0, so its derivative lies in C_0 and $\delta\hat{z} \to 0$ (the derivative of $\hat{z}_0 = 1 + (\zeta - 1)e^{-2\mu(\xi - \xi_*)}$ also tends to 0); on the other hand $e^{-\mu\xi}\delta z = c\hat{z} \to c$. Hence $c = 0$; so $0 = \delta z(\xi_*) = e^{\mu\xi_*}\delta\zeta$ gives $\delta\zeta = 0$, $\delta k \equiv 0$ gives $\delta\kappa = 0$, $\delta f \equiv 0$ gives $\delta\varphi = 0$, $\delta\lambda = 0$ gives $\delta\Lambda_{\mathrm{p}} = 0$ (as $\delta q = 0$), and $\delta Q(\xi_*) = E_*\delta\rho = 0$. So $\delta p = 0$. A C^∞ injective immersion between manifolds of the same dimension (6) is a local diffeomorphism. □

Toy example 4.39 (the chart of G for w constant in the weight)**.** Take $\kappa_{\mathrm{ch}} = 0$ (no swirl): then $f \equiv \varphi$, $F \equiv \Lambda_{\mathrm{p}} - \varphi^2$ is constant and $T_p(w) = \frac{1}{2}\mathfrak{r}_{\rho E_*}(\Lambda_{\mathrm{p}} - \varphi^2)$ does not depend on w at all: the fixed point is explicit, $w_p = \frac{1}{2}\mathfrak{r}_{\rho E_*}(\Lambda_{\mathrm{p}} - \varphi^2)$, of size $\leq (\frac{3}{2}|\Lambda_{\mathrm{p}} - \varphi^2| + 2|\rho|)E$ (this is (Q1)), and z is then the solution of the linear Volterra equation of Lemma 4.35(A). The contraction constant is 0; all of the nonlinearity of the problem is in the factor κ_{ch}, which multiplies the E_*-small terms in Step 3.

For the other three strata the proof is the same loop with other weights. We state the theorem and give the changes.

THEOREM 4.40 (the charts of R, L_1, L_2). *Let $\varsigma \in (0, \frac{1}{2})$ be fixed. For $\mathsf{c} \in \{\mathsf{R}, \mathsf{L}_1, \mathsf{L}_2\}$ let ϖ_{c} be the weight of Table 4.3, and let $B \geq 1$ (for $\mathsf{c} = \mathsf{R}$ also $B \geq (\frac{1}{2} - \varsigma)^{-1}$ and $\mu \leq \varsigma - B^{-1}$ in K_B^{R}). For every $r_0 > 0$ there is $\xi_{\mathsf{c}}(B, r_0)$ (explicitly, $2000(1 + B)^2(1 + r_0)E_* \leq 1$ for L_2 and $1000B^4(1 + r_0)E_* \leq 1$ for L_1, in addition to $\xi_* \geq \xi_{\mathsf{c}}$) so that for all cuts $\xi_* \geq \xi_{\mathsf{c}}$ and all $p \in K_B^{\mathsf{c}}$ the assertions* (a)–(d) *of Theorem 4.38 hold with K_B^{G} replaced by K_B^{c}, C_E by $C_{\varpi_{\mathsf{c}}}$, G by c and the state-map dimension 6 by $5, 5, 4$. In particular:*

(a) *the fixed point w_p of the loop in $\mathbb{B}_{r_0} \subset C_{\varpi_{\mathsf{c}}}$ is unique and defines a normalised local solution of type c with $Q(+\infty) = 2\mu^2 - 2$ ($\mu = 0$ for $\mathsf{L}_1, \mathsf{L}_2$);*

(b) *$p \mapsto w_p \in C_{\varpi_{\mathsf{c}}}$ is C^∞;*

(c) *every normalised local solution of type c is, at all sufficiently late cuts, the fixed point of the loop with its own coordinates (for R: with any $\varsigma \in (\mu, \frac{1}{2})$); for L_1, L_2 the coordinate bound B grows linearly with the cut, see the proof);*

(d) *S_{ξ_*} is an injective C^∞ immersion of the open parameter set into the scale-free state space $\mathbb{R}^6$.*

Proof. Set up exactly as in Theorem 4.38: for each stratum we list the linear step, the bound for f and F, the (Q)-estimate that replaces (Q1), and the smallness of the Lipschitz constant.

Type R. *Linear step:* Lemma 4.35(B); $\hat z=(1-V)^{-1}1\in[\frac23,\frac43]$, with the kernel $k_\mu(s)=\int_0^s e^{-2\mu v}\mathrm{d}v$, for which $|\partial_\mu^j k_\mu(s)|\le j!\,\mu^{-j-1}$ (as $\int_0^\infty (2v)^j e^{-2\mu v}\mathrm{d}v=j!\mu^{-j-1}$); thus $\mu\mapsto V_{\mu,\cdot}\in\mathcal{L}(C_{E^{1-2\varsigma}},\mathcal{L}(C_b))$ is C^∞ and $\|V\|\le\theta_{\mathrm R}=\frac{rE_*^{1-2\varsigma}}{4\mu(1-2\varsigma)}$. *Bounds:* $z^{-2}=e^{2\mu\xi}\hat z^{-2}\le\frac94 e^{2\mu\xi}$; with $f=e^{2\mu\xi_*}\varphi+\kappa_{\mathrm{ch}}\int_{\xi_*}^{\xi}z^{-2}$ we get $|f|\le(B+\frac{9B}{8\mu})e^{2\mu\xi}\le c_f e^{2\mu\xi}$ and $|F|\le|\Lambda_{\mathrm p}|+c_f^2e^{4\mu\xi}\le\Phi_B E^{-2\varsigma}$ ($e^{4\mu\xi}\le e^{4\varsigma\xi}=E^{-2\varsigma}$). Every p-derivative of order $j\le k+1$ of f carries at most a factor ξ^j and $\xi^j e^{4\mu\xi}\le C_j(\varsigma-\mu)^{-j}e^{4\varsigma\xi}$: this is why $\varsigma>\mu$ *strictly* and the weight $E^{-2\varsigma}$ on F is the correct choice for all derivatives at once; no k-dependence arises. *Estimate:* (Q2) with $\nu=\varsigma$: $|\frac12\mathfrak{r}_{R_*}F|\le\frac12(\Phi_BK_\varsigma+4|\rho|)E^{1-2\varsigma}=:r_BE^{1-2\varsigma}$ for $R_*=\rho E_*^{1-2\varsigma}$ (since $4|R_*|E_*^{2\varsigma-1}=4|\rho|$); the ball is invariant and $T(\mathbb{B}_r)\subset\mathbb{B}_{r/2}$ for $r\ge2r_B$. *Contraction:* by Lemma 4.35(B) (difference estimate) $|\hat z_1-\hat z_2|\le c\,\Delta\,E^{1-2\varsigma}$, so $|z_1^{-2}-z_2^{-2}|\le c'\Delta e^{2\mu\xi}E^{1-2\varsigma}$ and $|f_1-f_2|\le c''\Delta\int_{\xi_*}^{\xi}e^{\gamma_0\eta}\mathrm{d}\eta$, $\gamma_0:=2\mu+4\varsigma-2$. Then $|F_1-F_2|\le|f_1+f_2||f_1-f_2|\le c_fe^{2\mu\xi}|f_1-f_2|$, to be compared with $E^{-2\varsigma}=e^{4\varsigma\xi}$. Uniformly in γ_0: $\int_{\xi_*}^{\xi}e^{\gamma_0\eta}\mathrm{d}\eta\le(1+\xi)e^{\gamma_0^+\xi}$ with $\gamma_0^+=\max(\gamma_0,0)$ (the integrand is $\le e^{\gamma_0^+\eta}$ and the interval has length $\le1+\xi$), so there is no factor $1/|\gamma_0|$. Hence $|F_1-F_2|\le C\Delta\,e^{4\varsigma\xi}(1+\xi)e^{-\delta\xi}$ with $\delta:=\min(2-4\mu,\,4\varsigma-2\mu)$: indeed $2\mu+\gamma_0^+-4\varsigma$ equals $-(2-4\mu)$ if $\gamma_0\ge0$ and $2\mu-4\varsigma$ if $\gamma_0<0$. On $K_B^{\mathsf R}$ we have $\mu\le\varsigma-B^{-1}$ and $\mu\ge B^{-1}$, so $2-4\mu\ge2-4\varsigma+4B^{-1}\ge4B^{-1}$ and $4\varsigma-2\mu\ge2\varsigma+2B^{-1}\ge4B^{-1}$: $\delta\ge4B^{-1}$ uniformly. Since $\sup_{\xi\ge\xi_*}(1+\xi)e^{-\delta\xi}\le C_BE_*^{\delta/4}$, we get $|F_1-F_2|\le C\Delta\,E_*^{1/B}E^{-2\varsigma}$, and (Q2) with $\nu=\varsigma$ and R_*-datum 0 bounds the Lipschitz constant by $\frac12K_\varsigma CE_*^{1/B}\le\frac12$ for late cuts. *Completeness:* Proposition 4.19(b): $|R|\le CE^{1-2\mu}\le CE^{1-2\varsigma}$ ($\mu<\varsigma$), $\varphi=e^{-2\mu\xi_*}f(\xi_*)\to\kappa_{\mathrm{ch}}/(2\mu)$ (Remark 4.37), and $\rho=R(\xi_*)E_*^{2\varsigma-1}$ is bounded. *Immersion:* as in Step 6; the coordinate list is $(\mu,\kappa_{\mathrm{ch}},\varphi,\Lambda_{\mathrm p},\rho)$ and now $\delta z=cz$ from $\delta h\equiv0$, and $c=0$ because $e^{\mu\xi}\delta z=\delta\hat z\to0$ (V has range in C_0, so the p-derivative of $\hat z=(1-V)^{-1}1$ lies in C_0) while $e^{\mu\xi}\delta z=c\hat z\to c$.

Type $\mathsf L_1$. *Linear step:* Lemma 4.35(C)(L1), $z\in C_m$, $\frac23b_0m\le z\le\frac43b_1m$ with $b_0=\min(1,\zeta)$, $b_1=\max(1,\zeta)$; $\hat K$ is replaced by $-K_0$ which does not depend on any parameter, and $z_{\mathrm{lin}}=\zeta-1+m$ is affine in ζ: smoothness is immediate from the Neumann series. *Bounds:* $z^{-2}\le\frac94b_0^{-2}m^{-2}$, $f=\varphi-\kappa_{\mathrm{ch}}\int_\xi^\infty z^{-2}$ is bounded by $B+\frac94B^3$ ($\int_\xi^\infty m^{-2}\le1$), $|F|\le\Phi_B$ and (Q1) gives the ball $\mathbb{B}_{r_B}\subset C_E$ as for $\mathsf G$. *Contraction, with the dependence on B.* Take $\zeta\ge1$ (the case occurring in completeness), so $b_0=1$, $b_1=\zeta\le B$, $\frac23m\le z\le\frac43Bm$, $|f|\le4B$, $|F|\le B+16B^2=:\Phi_B$, $r_B=\frac12(3\Phi_B+4B)\le25B^2$; the ball radius is $r=\max(2r_B,r_0)\le50B^2+r_0$. Lemma (C)(L1) needs $\frac34rE_*\le\frac{1}{4B}$. Next, $\|z_1-z_2\|_m\le\frac34\cdot\frac34E_*\Delta\cdot\|z_2\|_m\le2BE_*\Delta$ (with $\|z_2\|_m\le\frac43B$), and

$$|z_1^{-2}-z_2^{-2}|\le|z_1-z_2|\,|z_1+z_2|\,z_1^{-2}z_2^{-2}\le2BE_*\Delta\,m\cdot\tfrac83Bm\cdot\tfrac{81}{16m^4}\le36B^2E_*\Delta\,m^{-2}$$

(the power is m^{-2}: numerator $\sim m\cdot m$, denominator $\sim m^4$), so $|f_1-f_2|\le36B^3E_*\Delta$ ($\int m^{-2}\le1$), $|F_1-F_2|\le8B\cdot36B^3E_*\Delta=288B^4E_*\Delta$ and by (Q1) the Lipschitz constant is $\le\frac12\cdot3\cdot288B^4E_*=432B^4E_*$. Both conditions hold if $1000B^4(1+r_0)E_*\le1$. *Completeness:* Proposition 4.19(c), $b>0$ normalised to $b=1$: $z=a+\xi+o(1)$, $z'\to1$, $|R|\le C_uE$, $f(\infty)=\varphi$. At the cut the coordinate $\zeta=z(\xi_*)=\xi_*+a+o(1)$ is

not bounded as the cut moves out, so $p(\xi_*) \in K_B^{\mathsf{L}_1}$ only with $B = B(\xi_*) = C(1+\xi_*)$ ($\kappa_{\rm ch}, \varphi, \Lambda_{\rm p}$ are fixed and $|\rho| \le C_u$, $r_0 = \frac{1}{2}C_u$); as for L_2 the explicit condition $1000B^4(1+r_0)E_* \le 1$ holds for all late cuts because $\xi_*^4 e^{-2\xi_*} \to 0$. *Immersion:* $\delta w \equiv 0$ from $\delta Q \equiv 0$ (here $q \equiv -2$ so $\delta q = 0$ automatically), $\delta h \equiv 0$ gives $\delta z = cz$; $z' \to 1$ for every p and $(K_0 h)' = \int_\xi^\infty wh \to 0$ show, as in the G case, that $\delta z' \to 0$, while $cz' \to c$, so $c = 0$; then $\delta\zeta = 0, \delta\kappa_{\rm ch} = 0, \delta\varphi = 0, \delta\Lambda_{\rm p} = 0, \delta\rho = 0$ exactly as before.

Type L_2. *Linear step:* (L2); $z \in [\frac{2}{3}, \frac{4}{3}]$ bounded, $z(\xi_*) = 1$. *Bounds:* $|f| \le |\varphi| + \frac{9}{4}|\kappa_{\rm ch}|(\xi - \xi_*) \le 4Bm$ (for $B \ge 1$), $|F| \le B + 16B^2m^2 \le 17B^2m^2$. (Q3) gives $|\frac{1}{2}\mathfrak{r}_{R_*}F| \le \frac{1}{2}(7 \cdot 17B^2 + 4B)Em^2 \le 62B^2Em^2$, so $r_B = 62B^2$. *Contraction:* $\|z_1 - z_2\|_\infty \le \frac{4}{3} \cdot \frac{9}{8}E_*\Delta \cdot \frac{4}{3} = 2E_*\Delta$; $|z_1^{-2} - z_2^{-2}| \le \frac{27}{2} \cdot 2E_*\Delta = 27E_*\Delta$ (since $z_i \ge \frac{2}{3}$); $|f_1 - f_2| \le 27BE_*\Delta m$; $|F_1 - F_2| \le 8Bm \cdot 27BE_*\Delta m = 216B^2E_*\Delta m^2$; (Q3): the Lipschitz constant is $\le \frac{1}{2} \cdot 7 \cdot 216B^2E_* = 756B^2E_*$. Both this and $\frac{9}{8}\max(2r_B, r_0)E_* \le \frac{1}{4}$ hold when $2000(1+B)^2(1+r_0)E_* \le 1$. *Completeness:* Proposition 4.19(c), $b = 0$: $|R| \le CEm^2$, $f = \frac{\kappa}{a^2}\xi + O(1)$; here $\varphi = f(\xi_*) = O(\xi_*)$ and $\kappa_{\rm ch} = \kappa/z(\xi_*)^2$ stay in K_B with $B = B(\xi_*) = C(1+\xi_*)$, and the condition $2000(1+B)^2(1+r_0)E_* \le 1$ holds for all late cuts because $\xi_*^2 e^{-2\xi_*} \to 0$. *Immersion:* $\delta w \equiv 0$, $\delta h \equiv 0$ so $\delta z = cz$ and $z(\xi_*) = 1$ forces $c = 0$; then $\delta\kappa_{\rm ch} = \delta\varphi = \delta\Lambda_{\rm p} = \delta\rho = 0$. □

Remark 4.41 (sharper lower bounds)**.** The Neumann-series estimates of Lemma 4.35 give the lower bounds $\frac{2}{3}n_0$ for $e^{-\mu\xi}z$ in the dominant step and $\frac{2}{3}b_0 m$ for z in the L_1 step, slightly sharper than the factors $\frac{1}{2}n_0$, $\frac{1}{2}b_0$ that already suffice for the contraction argument. Nothing below depends on the difference.

Corollary 4.42 (type G is open; family-wise openness)**.** *For $\xi_c \ge 0$ let $\mathcal{S}_{\xi_c} \subset \mathbb{R}^6$ be the set of* all *scale-free states $s = (h, k, f, Q, \dot{Q}, \lambda)$ at ξ_c ($k = \kappa/z^2$ may vanish) whose forward solution exists on $[\xi_c, \infty)$ with $z > 0$ (z is any positive solution of $z'/z = h$), and let $\mathcal{G}_{\xi_c} \subset \mathcal{S}_{\xi_c}$ be the subset of those of type G (for $k \ne 0$ by Definition 4.16, for $k = 0$ by the convention after it).*

(a) *$\mathcal{G}_{\xi_c}$ is open in $\mathbb{R}^6$. The same holds at the south pole: with $\mathcal{G}^-_{\xi_c}$ defined by the backward solution on $(-\infty, -\xi_c]$, $\mathcal{G}^-_{\xi_c}$ is open, because σ_2 (Lemma 2.8) maps it diffeomorphically onto $\mathcal{G}_{\xi_c}$ (compare Remark 5.14).*

(b) (Family-wise.) *Let Π be a topological space and $\pi \mapsto s_\pi \in \mathbb{R}^6$ the scale-free state at some ξ_a of a family of solutions of the system depending continuously on π. Then $\{\pi :$ the forward solution exists on $[\xi_a, \infty)$ and is of type $\mathsf{G}\}$ is open in Π.*

(c) (Uniform constants.) *For every compact $N \subset \mathcal{G}_{\xi_c}$ there are B, r_0 and $\xi_* \ge \max(\xi_c, \xi_G(B, r_0))$ such that every solution with initial state in N is, on $[\xi_*, \infty)$, the chart solution u_p with $p \in K_B^{\mathsf{G}}$ and $|w_p| \le r_B E$; thus $m_\pm$, $f_{\max}$ and all constants of Theorem 4.38 are bounded on N.*

Proof. (a) Let $s_0 \in \mathcal{G}_{\xi_c}$ and let u_0 be its solution; by Theorem 4.38(c) (valid for $\kappa = 0$ as well, Lemma 4.20) there are B, r_0 and a late cut $\xi_* \ge \xi_c$, $\xi_* \ge \xi_G(B, r_0)$, with $u_0|_{[\xi_*,\infty)} = u_{p_0}$, $p_0 \in \operatorname{int} K_B^{\mathsf{G}}$ (if $\kappa_{\rm ch} = 0$ the point p_0 is still interior: K_B^{G} contains a neighbourhood of $\kappa_{\rm ch} = 0$). By Theorem 4.38(d), S_{ξ_*} is a local diffeomorphism at p_0, so its image contains an open ball $U \ni s(u_0)(\xi_*)$; every state in U is the state at ξ_* of a solution u_p of type G on $[\xi_*, \infty)$. The flow $\Phi_{\xi_c}^{\xi_*}$ of the state system

from ξ_c to ξ_* is a C^∞ diffeomorphism of an open neighbourhood of s_0 onto an open neighbourhood of $s(u_0)(\xi_*)$ (Theorem A.4; u_0 exists on the compact interval $[\xi_c,\xi_*]$). Shrinking so that the image lies in U and $z>0$ persists on $[\xi_c,\xi_*]$ (an open condition on a compact interval), we get an open neighbourhood of s_0 in $\mathcal{G}_{\xi_c}$. The south pole follows from σ_2. (b) The map $\pi\mapsto s_\pi$ composed with the flow from ξ_a to ξ_c is continuous where the solution exists on $[\xi_a,\xi_c]$; if for π_0 the forward solution exists on $[\xi_a,\infty)$ and is of type G, its state at ξ_c is in the open set $\mathcal{G}_{\xi_c}$ by (a), and by continuous dependence on a compact interval the same holds for π near π_0. (c) By Theorem 4.38(c) each point of N has a neighbourhood with such B,r_0,ξ_* (by (a) and Theorem 4.38(d), the late cut and the ball U can be kept fixed for nearby states); cover the compact N by finitely many. □

Corollary 4.43 (interior dependence: the coordinates of type G)**.** *Let $\xi_c\ge 0$.*

(a) *In a neighbourhood of every point of $\mathcal{G}_{\xi_c}$ the map $s\mapsto p=\mathsf{S}_{\xi_*}^{-1}(\Phi_{\xi_c}^{\xi_*}s)$ (late cut ξ_*) is a C^∞ diffeomorphism onto an open set of $\mathbb{R}^6$; its components $(\mu,\zeta,\kappa_{\mathrm{ch}},\varphi,\Lambda_{\mathrm{p}},\rho)$ are thus C^∞ coordinates, valid also at $\kappa=0$. The functions $\mu=\sqrt{1+Q(\infty)/2}$, $\varphi=f(\infty)$, $\Lambda_{\mathrm{p}}=\lambda+2Q(\infty)$ do not depend on the cut; they are C^∞ on all of $\mathcal{G}_{\xi_c}$, and*

$$\mathrm{d}\mu\neq 0 \quad \textit{and} \quad \mathrm{d}\varphi\neq 0 \quad \textit{everywhere on } \mathcal{G}_{\xi_c}$$

(a coordinate function of a local diffeomorphism is a submersion). In particular $\sigma=\mu^2=1+\frac12 Q(\infty)>0$ is C^∞ on type G and $\partial_s Q(\xi)\to\partial_s q$ as $\xi\to\infty$, with an $O(E)$ error uniform on compact sets of states.

(b) (Dictionary with the coordinates of the global chapters.) *The state s contains λ and $k=\kappa/z^2$ at ξ_c, i.e. the swirl constant $\tilde\kappa=k$ of the solution $\tilde z$ scaled by $\tilde z(\xi_c)=1$. Put $a:=\lim_{\xi\to\infty}\tilde z\,e^{-\mu(\xi-\xi_c)}$ and $\rho_c=R(\xi_c)/E(\xi_c)$. Then $(\mu,\lambda,\rho_c,\varphi,\tilde\kappa,a)$, with $\lambda=\Lambda_{\mathrm{p}}-2q=\Lambda_{\mathrm{p}}+4-4\mu^2$, are C^∞ functions on $\mathcal{G}_{\xi_c}$, and they are a C^∞ coordinate system near every state whose cut can be taken equal to ξ_c (i.e. $\xi_c\ge\xi_G(B,r_0)$ at that state). The relation to the chart coordinates at the cut ξ_c is explicit: $a=1/\zeta_c$ and $\tilde\kappa=\kappa_{\mathrm{ch}}e^{-2\mu\xi_c}/\zeta_c^2$ (here $\zeta_c=e^{-\mu\xi_c}z_{\mathrm{chart}}(\xi_c)$), $\rho_c=\rho$.*

(c) *The limit of R/E is $\lim_{\xi\to\infty}R/E=\varphi^2-\Lambda_{\mathrm{p}}$, a C^∞ function.*

Proof. (a) By Theorem 4.38(d) and the inverse function theorem, with the flow of Corollary 4.42; the coordinates $\mu,\varphi,\Lambda_{\mathrm{p}}$ are functionals of the solution (limits at the pole), hence independent of the cut, and equal to components of p at every late cut. $\partial_s Q=\partial_s q+2\partial_s w_p$ and $\partial_p w_p\in C_E$ (Theorem 4.38(b)). (b) At a state with ξ_c a valid cut, p are the chart coordinates at ξ_c. If z_{chart} is the chart-normalised z, then $\tilde z=z_{\mathrm{chart}}/z_{\mathrm{chart}}(\xi_c)$, so $\tilde\kappa=\kappa_{\mathrm{ch}}/z_{\mathrm{chart}}(\xi_c)^2=\kappa_{\mathrm{ch}}e^{-2\mu\xi_c}/\zeta_c^2$ and $a=\lim z_{\mathrm{chart}}e^{-\mu\xi}\cdot e^{\mu\xi_c}/z_{\mathrm{chart}}(\xi_c)=e^{\mu\xi_c}/z_{\mathrm{chart}}(\xi_c)=1/\zeta_c$. Thus $(\zeta_c,\kappa_{\mathrm{ch}})\leftrightarrow(a,\tilde\kappa)$ is a diffeomorphism for $\zeta_c>0$ (for fixed μ,ξ_c), and the other four coordinates are unchanged up to $\lambda\leftrightarrow\Lambda_{\mathrm{p}}$ (shift by the function $2q(\mu)$). For general ξ_c, a is a C^∞ function of s: $a=e^{\mu\xi_c}/z_{\mathrm{chart}}(\xi_c)$ with $z_{\mathrm{chart}}(\xi_c)=z_p(\xi_*)\exp(-\int_{\xi_c}^{\xi_*}h)$, smooth in s by the flow. (c) Put $g=R+\frac12 F_\infty t$ with $F_\infty=\Lambda_{\mathrm{p}}-\varphi^2$. Since $\mathsf{L}[t]=-2$ (Toy example 4.12), $\mathsf{L}[g]=F-F_\infty=\varphi^2-f^2=O(e^{-2\mu\xi})=O(E^\mu)$ ($f-\varphi=-\kappa_{\mathrm{ch}}\int_\xi^\infty z^{-2}$, $z^{-2}\le Ce^{-2\mu\xi}$) and $g\to 0$. By Lemma 4.13, eq. (4.11) with $\Phi=CE^\mu$: $|g|\le CE^{1+\mu}$ (first term $\int_\xi^\infty E^{1+\mu}$, second $E\int_{\xi_*}^{\xi}e^{-2(\xi-\eta)}Ce^{-2\mu\eta}\mathrm{d}\eta\le CE\,e^{-2\mu\xi}/(2-2\mu)$, third $O(E^2)$). Hence $R/E\to-\frac12 F_\infty\cdot\lim t/E=-F_\infty$ ($t/E\to 2$). □

Remark 4.44 (R and L are not open; what the charts do not give)**.** For $\mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$ the same argument shows only that the images of S_{ξ_*} are immersed submanifolds of dimension $5, 5, 4$ of the six-dimensional state space; they are the boundary strata of G in the sense of Remark 4.17 but *this chapter does not prove that they are closed or embedded*, nor that G is dense near them (the corner chart below shows the local picture near the corner point, where the strata are pieces of the boundary of the convex region $\widehat{\Gamma}$). Part (iii) of the corresponding statement in the source paper (uniform constants relative to the stratum) is contained in Theorem 4.40(a),(b). The charts also say nothing about the global continuation of a local solution to the other pole.

4.8. The corner chart (local supplement)

In words. *The four charts of Section 4.7.3 are separate: each is built with the weights and normalisation natural to its stratum, and their smoothness constants blow up as $\mu \to 0$ (the factor $1/\mu$ in Lemma 4.35(A), (B)). Near the corner $(\omega, \sigma) = (0, 0)$ of the Jost region all four strata meet, and one may ask for a single parametrisation. Because the Jost basis is analytic through $\sigma = 0$, this exists: the loop is run with $z = C + \omega S$ and one weight, and the resulting map is C^k up to the boundary of the closed convex region $\widehat{\Gamma}$ — but only in the small neighbourhood $\sigma \le \mu_*^2$ with $\mu_* = \frac{1}{40(k+3)}$, because every derivative costs a power of $e^{\mu_* \xi}$. It is a* local supplement*: the charts of record for all pole behaviours are those of Theorems 4.38 and 4.40, which are valid for all μ in a compact set and have no k-dependent restriction. The corner chart is what is needed to show that the wall of L_2-edges is $C^{1,1}$ and not C^2 (Chapter 7).*

Setting. Fix $k \ge 1$, put $\mu_* = \frac{1}{40(k+3)}$ $(\le \frac{1}{160})$, $\nu' = \frac{1}{16}$, and let $\xi_* \ge 1$. The hypotheses (J) of Lemma 3.38 are in force with a constant M fixed below. The parameter is $p = (\beta, \omega, \sigma)$ with $\beta = (\lambda, \rho, f_0, \kappa_{\mathrm{c}}) \in \mathbb{R}^4$, and κ_{c} is the corner normalisation of the swirl constant (the coefficient of C in z is 1). For w as in (J) put

$$z = C + \omega S, \quad f = f_0 + \kappa_{\mathrm{c}} \int_{\xi_*}^{\xi} z^{-2}, \quad q = 2\sigma - 2, \quad \Lambda_{\mathrm{p}} = \lambda + 2q, \quad R_* = \rho E_*^{1-2\nu'},$$
$$T_p(w) = \tfrac{1}{2} \mathfrak{r}_{R_*}\big(\Lambda_{\mathrm{p}} - f^2\big). \tag{4.17}$$

A fixed point $w = T_p(w)$ gives, as in (4.15), a solution of the system on $[\xi_*, \infty)$ with $Q = 2\sigma - 2 + 2w$. With $|\beta|$ the maximum norm and a number $B_0 \ge 1$, let

$$\overline{\mathcal{D}} = \{(\beta, \omega, \sigma) : \ |\beta| \le B_0, \ 0 \le \sigma \le \mu_*^2, \ |\omega| \le \mu_*, \ \omega \ge -\sqrt{\sigma}\},$$

which is closed and convex with nonempty interior ($\widehat{\Gamma} \cap \{\sigma \le \mu_*^2, |\omega| \le \mu_*\}$ times a box). Put $F_0 = B_0 + 4 + 100 B_0^2$ (so that $|\Lambda_{\mathrm{p}}| + (|f_0| + 9|\kappa_{\mathrm{c}}|)^2 \le F_0$ on $\overline{\mathcal{D}}$: $\Lambda_{\mathrm{p}} = \lambda + 4\sigma - 4$ and $0 \le \sigma \le 1$ give $|\Lambda_{\mathrm{p}}| \le B_0 + 4$), $M = 1 + 4\mu_*^{-2} F_0 + 4B_0$, and let (H) be

$$(M+1) e^{-\xi_*} \le \tfrac{1}{16}, \qquad C_k (1 + B_0)^2 e^{-\xi_*} \le \tfrac{1}{2}, \tag{4.18}$$

where C_k is the constant of Step 2 below (it depends on k only, not on B_0).

THEOREM 4.45 (local corner chart). *Let $k \ge 1$, $B_0 \ge 1$ and let ξ_* satisfy* (H).

(a) *For every $p \in \overline{\mathcal{D}}$, T_p maps $\mathbb{B} = \{w \in C[\xi_*, \infty) : \ |w| \le M e^{-\xi}\}$ into itself and is a $\frac{1}{2}$-contraction in $C_{e^{-\xi}}$; its fixed point w_p is unique in $\mathbb{B}$.*

(b) *The solution u_p built from w_p is positive on $[\xi_*,\infty)$ (so regular), has $Q(+\infty)=2\sigma-2$, and, for $\kappa_{\mathrm c}\neq 0$, is of type* G *if $\omega>-\sqrt{\sigma}$,* R *if $\omega=-\sqrt{\sigma}>-\mu_*$ with $\mu=-\omega$,* L_1 *if $\sigma=0<\omega$, and* L_2 *at $(\omega,\sigma)=(0,0)$ (for $\kappa_{\mathrm c}=0$ read: the model solution $z=C+\omega S$ is dominant, recessive, linear, bounded respectively).*

(c) *The state map $\Phi: p\mapsto s(u_p)(\xi_*)\in\mathbb{R}^6$ is C^k on $\overline{\mathcal{D}}$ up to the boundary, i.e. all derivatives up to order k exist in the interior and extend continuously; by Corollary A.18 it is the restriction of a C^k map $\widetilde{\Phi}$ defined on an open set of $\mathbb{R}^6$.*

(d) *Φ is injective, and $\mathrm{D}\widetilde{\Phi}$ is invertible at every $p\in\overline{\mathcal{D}}$ with $|\beta|<B_0$, $|\omega|<\mu_*$, $\sigma<\mu_*^2$; so $\widetilde{\Phi}$ is a local C^k diffeomorphism there.*

(e) *For every C^1 curve $p(\tau)$ in $\overline{\mathcal{D}}$ one has $\partial_\tau Q(\xi)\to 2\partial_\tau\sigma$ as $\xi\to\infty$.*

(f) *Every normalised local solution of type* L_2 *($z\to a>0$, $\kappa_{\mathrm c}=\kappa/a^2$) is, at every sufficiently late cut, the point $(\omega,\sigma)=(0,0)$ of the chart, with $B_0=C(1+\xi_*)$ in* (H)*. Solutions of type* G, R, L_1 *near the corner are* not *claimed to lie in the chart for late cuts.*

Proof. *Step 1: size of z and of the source.* By Lemma 3.38 and Proposition 4.32, $z=C+\omega S$ lies in $B=\{\varphi:\sup e^{-3\mu_*\xi}|\varphi|<\infty\}$ and is analytic in (σ,ω,w) (we complexify below). On $\overline{\mathcal{D}}$, $z\geq\frac{2}{3}z_0$ with $z_0=C_0+\omega S_0$ (Lemma 3.38(iii), $\omega\geq-\mu$ if $\sigma>0$, and for $\omega<0$ write $z_0=\frac12(1+\omega/\mu)e^{\mu\xi}+\frac12(1-\omega/\mu)e^{-\mu\xi}\geq\frac12 e^{-\mu\xi}$; for $\omega\geq0$, $z_0\geq1$). Hence $z^{-1}\leq 3e^{\mu_*\xi}$, $z^{-2}\leq 9e^{2\mu_*\xi}$, $|f|\leq\varphi_*(1+\xi-\xi_*)e^{2\mu_*\xi}$ with $\varphi_*=|f_0|+9|\kappa_{\mathrm c}|\leq 10B_0$, and $|F|=|\Lambda_{\mathrm p}-f^2|\leq F_0(1+\xi)^2e^{4\mu_*\xi}\leq F_0\mu_*^{-2}e^{5\mu_*\xi}\leq F_0\mu_*^{-2}E^{-2\nu'}$, using $(1+\xi)^2\leq\mu_*^{-2}e^{\mu_*\xi}$ (checked in `verify_ch04b.py`) and $5\mu_*\leq\frac14=4\nu'$. *Step 2: invariance and contraction.* With $K_{\nu'}\leq 3.3$ and $|R_*|E_*^{2\nu'-1}=|\rho|$, (Q2) gives $|T_pw|\leq\frac12(3.3F_0\mu_*^{-2}+4B_0)E^{1-2\nu'}\leq\frac12(\cdots)e^{-3\xi_*/4}e^{-\xi}\leq(M-1)e^{-\xi}$ by the choice of M (here $E^{7/8}=e^{-\xi}E^{3/8}$): $T_p(\mathbb{B})\subset\mathbb{B}$. Let $w_1,w_2\in\mathbb{B}$ and $\Delta=\sup|w_1-w_2|e^{\xi}$. The operator $\mathcal{V}_\sigma[w_1-w_2]$ of Lemma 3.38 has norm $\leq(1-5\mu_*)^{-2}\Delta e^{-\xi_*}\leq 1.1\Delta e^{-\xi_*}$ on B (Step 1 of the proof of Lemma 3.38), and $C_1-C_2=(1-\mathcal{V}_1)^{-1}\mathcal{V}[w_1-w_2]C_2$, likewise for S; so $|z_1-z_2|\leq K\Delta e^{-\xi_*}e^{3\mu_*\xi}$ (K depends on μ_* only). Then $|z_1^{-2}-z_2^{-2}|\leq|z_1-z_2|z_1^{-1}z_2^{-1}(z_1^{-1}+z_2^{-1})\leq K'\Delta e^{-\xi_*}e^{6\mu_*\xi}$, $|f_1-f_2|\leq B_0K'\Delta e^{-\xi_*}(1+\xi)e^{6\mu_*\xi}$, $|F_1-F_2|\leq|f_1+f_2||f_1-f_2|\leq K''(1+B_0)^2\Delta e^{-\xi_*}(1+\xi)^2e^{8\mu_*\xi}\leq K''\mu_*^{-2}(1+B_0)^2\Delta e^{-\xi_*}E^{-2\nu'}$ (since $9\mu_*\leq\frac14$), and (Q2) with R_*-datum 0 together with $E^{1-2\nu'}e^{\xi}\leq 1$ gives $\sup|T_pw_1-T_pw_2|e^{\xi}\leq C_k(1+B_0)^2e^{-\xi_*}\Delta\leq\frac12\Delta$. This is (a) (the unique fixed point of the contraction of the closed set $\mathbb{B}$ of the Banach space $C_{e^{-\xi}}$). *Step 3: analytic dependence and Cauchy bounds.* Fix $p_0\in\overline{\mathcal{D}}$, $w_0\in\mathbb{B}$ and a continuous h with $\sup|h|e^{\xi}=1$. The map $(\sigma,\omega,t)\mapsto z(\sigma,\omega,w_0+th)\in B$ is analytic on the polydisc $|\sigma-\sigma_0|<\mu_*^2$, $|\omega-\omega_0|<\mu_*$, $|t|<1$ (Lemma 3.38 with $M+1$ in place of M: this is why (J) is assumed with $(M+1)e^{-\xi_*}\leq\frac{1}{16}$; note $|\sigma|<2\mu_*^2<4\mu_*^2$) and bounded there by a constant K_0'. Cauchy's inequality for Banach-valued analytic maps gives for $j\leq k+1$ and every direction $h=(\delta\sigma,\delta\omega,\delta w)$ with $|h|:=|\delta\sigma|/\mu_*^2+|\delta\omega|/\mu_*+\sup|\delta w|e^{\xi}$: $\|\mathrm{D}^jz[h^j]\|_B\leq K_j|h|^j$, with $K_j=j!\,K_0'$. No derivative is lost here. *Step 4: the nonlinearity; derivatives cost weight.* Pointwise, z^{-2} is a smooth function of $z>0$ and by Faà di Bruno $\mathrm{D}^j(z^{-2})[h_1,\dots,h_j]=\sum_\pi(-1)^{|\pi|}(|\pi|+1)!\,z^{-2-|\pi|}\prod_{P\in\pi}\mathrm{D}^{|P|}z[h_P]$ (sum over partitions of $\{1..j\}$). By Steps 1 and 3 each term is $\leq C\,e^{(2+|\pi|)\mu_*\xi}e^{3|\pi|\mu_*\xi}|h|^j$, so $|\mathrm{D}^j(z^{-2})[h^j]|\leq C_j|h|^je^{(2+4j)\mu_*\xi}$. Integrating, $|\mathrm{D}^jf[h^j]|\leq C_j(1+B_0)|h|^j(1+\xi)e^{(2+4j)\mu_*\xi}$; and $\mathrm{D}^jF[h^j]=-\sum_i\binom{j}{i}\mathrm{D}^if\,\mathrm{D}^{j-i}f$ (plus $\mathrm{D}\Lambda_{\mathrm p}$ for $j\leq1$) satisfies

$|\mathrm{D}^j F[h^j]| \le C_j(1+B_0)^2|h|^j(1+\xi)^2 e^{(4+4j)\mu_*\xi} \le C_j'|h|^j e^{\xi/4}$ for $j \le k+1$, because $(1+\xi)^2 \le \mu_*^{-2}e^{\mu_*\xi}$ and $(5+4(k+1))\mu_* = \frac{4k+9}{40(k+3)} \le \frac14$. *This* is where the k-dependence of μ_* comes from: a C^k statement needs $(5+4(k+1))\mu_* \le \frac14$. *Step 5: T is C^k.* On the convex open set $\operatorname{int}\overline{\mathcal{D}} \times \operatorname{int}\mathbb{B}'$ ($\mathbb{B}'$ a slightly larger ball in $C_{e^{-\xi}}$ on which (J) with $M+1$ holds) all the bounds of Steps 1–4 are uniform, so Lemma A.15 with $\varpi = e^{\xi/4}$ shows that $(p,w) \mapsto F \in C_{E^{-2\nu'}}$ is C^k with $\mathrm{D}^k F$ Lipschitz. The Green operator $F \mapsto \mathfrak{r}_{R_*}F$ is bounded $C_{E^{-2\nu'}} \to C_{E^{1-2\nu'}} \subset C_{e^{-\xi}}$ by (Q2), affine in ρ; so $T \in C^k$. *Step 6: the fixed point.* With $\theta = \frac12$ and the closed set $\mathbb{B}$ (with $T(p,\mathbb{B}) \subset \{|w| \le (M-1)e^{-\xi}\} \subset \operatorname{int}\mathbb{B}$), Theorem A.11 gives $p \mapsto w_p \in C^k(\operatorname{int}\overline{\mathcal{D}}; C_{e^{-\xi}})$ with derivatives $(I - \mathrm{D}_w T)^{-1}\mathrm{D}_p T$ ($\|(I-\mathrm{D}_wT)^{-1}\| \le 2$); all derivatives up to order k are bounded and $\mathrm{D}^k w_p$ is Lipschitz: by induction on j, $\mathrm{D}^j w_p$ is a polynomial expression in $\mathrm{D}^i T$ ($i \le j$) and $(I-\mathrm{D}_wT)^{-1}$, with Lipschitz factors for $j=k$ (by the Lipschitz bound of D^kT and $\mathrm{D}^iT \in C^1$ with bounded derivative for $i<k$); so all derivatives extend continuously (uniformly) to $\overline{\mathcal{D}}$. *Step 7: the state map; (c).* $s = (h,k,f,Q,\dot Q,\lambda)$ at ξ_* consists of $h = z'/z$, $k = \kappa_{\mathrm{c}}/z^2$, f_0, $Q(\xi_*) = 2\sigma - 2 + \rho E_*^{1-2\nu'}$, $\dot Q(\xi_*) = R'(\xi_*) = \operatorname{sech}^2\xi_*\,(\mathcal{A}(\xi_*) - 2t_*\mathcal{B}_*)$ and $\lambda = \Lambda_{\mathrm{p}} - 2q$. Here $z(\xi_*)$ and $z'(\xi_*) = C_0' + \omega S_0' - \mathcal{V}^{\mathrm{c}}[w]z$ with $\mathcal{V}^{\mathrm{c}}\varphi = \int_\xi^\infty C_0(\eta-\xi)w\varphi$ (differentiate $z = z_0 + \mathcal{V}z$; $S_0(0)=0$) are evaluations of B-valued analytic maps composed with $p \mapsto (\sigma,\omega,w_p)$; $\mathcal{A}(\xi_*) = -\int FE$ is a bounded linear functional on $C_{E^{-2\nu'}}$ ($E\cdot E^{-2\nu'}$ is integrable) and $\mathcal{B}_*$ is affine in $(R_*,\mathcal{A})$. So all components are C^k up to the boundary, and Corollary A.18 (convex closed set with nonempty interior) gives $\widetilde\Phi$. *Step 8: (b), (e).* (b) is Proposition 4.32 for $\xi \ge \xi_*$ with $w = w_p$ (the type is read off from $z = C + \omega S$ and the relation $w \to 0$, Remark 4.33); positivity of z makes the system regular. (e): $Q = 2\sigma - 2 + 2w_p$ and $\partial_p w_p \in C_{e^{-\xi}}$ tends to 0. *Step 9: injectivity and invertibility; (d). Injectivity:* $s(u_p)(\xi_*)$ determines the solution on $[\xi_*,\infty)$ (Theorem A.4), hence $q = Q(\infty)$ and $\sigma = (q+2)/2$, w, C, S; h determines z up to a scalar and the condition "coefficient of C equals 1" fixes it, so $\omega = \mathrm{Wr}(C,z)$ ($\mathrm{Wr}(C,S)=1$); $\kappa_{\mathrm{c}} = kz^2$, then f_0, ρ (from $Q(\xi_*)$) and λ follow. *Invertibility:* let $\delta p \in \mathbb{R}^6$ with $\mathrm{D}\Phi\,\delta p = 0$. In the interior every direction is admissible, the p-derivatives of $(z,f,Q,\dots)$ satisfy the linearised system, and by continuity of derivatives (Step 6) also on $\overline{\mathcal{D}}$. Their value at ξ_* vanishes, so $\delta s \equiv 0$ on $[\xi_*,\infty)$ (linear uniqueness). From $\delta Q = 2\delta\sigma + 2\delta w \equiv 0$ and $\delta w \to 0$ we get $\delta\sigma = 0$, $\delta w \equiv 0$; from $\delta(z'/z) \equiv 0$, $\delta z = c\,z$; but with $\delta\sigma = \delta w = 0$, $\delta z = \delta\omega\, S$, so $\delta\omega\, S = c(C+\omega S)$, and independence of C, S ($\mathrm{Wr}=1$) gives $c = 0 = \delta\omega$ (this replaces the argument through $\delta q = 4\mu\delta\mu$, which degenerates at $\mu = 0$). Then $\delta k \equiv 0$ gives $\delta\kappa_{\mathrm{c}} = 0$, $\delta f(\xi_*) = 0$ gives $\delta f_0 = 0$, $\delta Q(\xi_*) = E_*^{1-2\nu'}\delta\rho = 0$, and $\delta\lambda = 0$. So $\mathrm{D}\Phi$ is injective, hence invertible (6×6) and so is $\mathrm{D}\widetilde\Phi$ at the points where the one-sided derivatives are the derivatives of $\widetilde\Phi$; the inverse function theorem gives (d). *Step 10: (f).* Let u be a normalised local solution of type L_2. By Proposition 4.19(c), $|R| \le CEm^2$ and $f = \frac{\kappa}{a^2}\xi + O(1)$. The function z/a solves $y'' = wy$, $y \to 1$, $y' \to 0$, hence $y = C$ (a bounded solution is a multiple of C since $S \to \infty$, Lemma 3.38(iv)): so u corresponds to $(\omega,\sigma) = (0,0)$ with $\kappa_{\mathrm{c}} = \kappa/a^2$, $f_0 = f(\xi_*) = O(\xi_*)$, $\rho = R(\xi_*)E_*^{2\nu'-1} = O(E_*^{2\nu'}) = O(e^{-\xi_*/4})$. So $|\beta| \le B_0 := C(1+\xi_*)$, (H) holds for large ξ_* (both left sides are $O(\xi_*^2 e^{-\xi_*})$), $|w| \le CEm^2 \le C(1+\xi_*)^2 e^{-\xi_*}e^{-\xi} \le Me^{-\xi}$ on $[\xi_*,\infty)$ for large ξ_*, and by the uniqueness in (a) $w = w_p$. □

Remark 4.46 (how the corner chart restricts to the charts of record; κ-normalisations)**.** The chart normalisations of Table 4.3 differ from the corner normalisation $z = C + \omega S$. If u is a normalised solution of the corner chart with parameter (ω, σ) and swirl constant κ_{c}, then the chart-of-record constants are

$$\mathsf{G}:\ \kappa_{\mathrm{ch}} = \kappa_{\mathrm{c}}/\alpha^2,\ \ \alpha = \tfrac{1}{2}(1+\omega/\mu); \qquad \mathsf{R}:\ \kappa_{\mathrm{ch}} = \kappa_{\mathrm{c}}\ \ (z = F_-,\ e^{\mu\xi}z \to 1);$$
$$\mathsf{L}_1:\ \kappa_{\mathrm{ch}} = \kappa_{\mathrm{c}}/\omega^2\ \ (\omega > 0,\ z' \to \omega); \qquad \mathsf{L}_2:\ \kappa_{\mathrm{ch}} = \kappa_{\mathrm{c}}/C(\xi_*)^2\ \ (z(\xi_*) = 1),$$

by the rescaling $z \mapsto z/\sqrt{(\text{normalising quantity})}$, $\kappa \mapsto \kappa/(\cdot)$ of Section 4.7.2; for L_2 also $\kappa_{\mathrm{c}} = \kappa/a^2$ with $a = \lim z$. Since $\alpha \to \infty$ as $\omega/\mu \to \infty$ and $\alpha \to 0$ on the parabola, the chart-of-record constants of the dominant and the recessive strata are *not* continuous functions on $\overline{\mathcal{D}}$ at the wall; κ_{c} is, which is the point of the corner normalisation. The corner chart does not use ς or ζ; the chart-of-record coordinates ζ for G and L_1 are recovered as $\zeta = e^{-\mu\xi_*}z(\xi_*)/\alpha$ and $\zeta = z(\xi_*)/\omega$.

Remark 4.47 (k-dependence and why the corner chart is only local)**.** In Theorem 4.45 the neighbourhood $\sigma \le \mu_*^2$, $|\omega| \le \mu_*$ shrinks like $1/k$ as the differentiability order k grows (Step 4), and C^∞ up to the wall is *not* claimed. The charts of record do not have this restriction: in G every derivative in μ costs a factor ξ against the decay $e^{-2\mu\xi}$, which is harmless for $\mu \ge B^{-1}$; in R it costs $\xi^j e^{4\mu\xi} \le C_j e^{4\varsigma\xi}$ with $\varsigma > \mu$ (Theorem 4.40). What the corner chart provides, and the charts of record do not, is a single C^k parametrisation of a neighbourhood of an L_2 solution that contains G, R, L_1 solutions as the pieces $\sigma > (\omega_-)^2$, $\sigma = \omega^2$, $\sigma = 0 < \omega$. Its use is therefore confined to Chapter 7, where the $C^{1,1}$ regularity of the wall of L_2 edges is derived from it; nothing else in the book depends on the corner chart.

4.9. Worked examples and the numerical atlas of the chapter

In words. *Everything in this section is a* check*, not a proof: it runs the chart loop on the computer for sample parameters, compares the toys of this chapter with closed forms, and collects the one piece of numerical evidence about the threshold $U_\theta(1) = -3$ that the chapter's theorems predict but do not prove sharp. Status labels are strict:* Theorems *are proved above; the* Numerical observations *below are computed and not proved.*

4.9.1. Toys, and what each one tests. Toy example 4.34 ($w \equiv 0$) is the Jost plane in closed form: $z = \cosh\mu\xi + \frac{\omega}{\mu}\sinh\mu\xi$ has a zero iff $\omega < -\mu$, at $\xi_0 = \mu^{-1}\operatorname{artanh}(\mu/|\omega|)$. Toy example 4.36 ($w = \varepsilon e^{-2\xi}$) is a Volterra equation whose solution is a modified Bessel function; the iterates converge to it at the rate predicted by Lemma 4.35(B). Toy example 4.39 ($\kappa_{\mathrm{ch}} = 0$) is the degenerate loop in which T_p is constant in w. Together they exercise the three ingredients of every chart: the linear step (the first two), the Green step (the third), and the smallness of the Lipschitz constant (the second).

Example 4.48 (a sample G solution, the size of the correction)**.** Take $\mu = 0.3$, $\zeta = 1.5$, $\kappa_{\mathrm{ch}} = 1$, $\varphi = 0.5$, $\Lambda_{\mathrm{p}} = 1$, $\rho = 1$. Then $\hat{z}_0 = 1 + 0.5e^{-0.6(\xi-\xi_*)}$, $f = 0.5 - \int_\xi^\infty e^{-0.6\eta}\hat{z}^{-2}\mathrm{d}\eta$, $F = 1 - f^2$. At $\xi_* = 3$ the loop gives $\sup|w|/E = 0.55$: the potential perturbation is of size $\frac{1}{2}\rho E + \dots$, dominated by the datum $\frac{1}{2}R(\xi_*) = \frac{1}{2}\rho E_*$ propagated by $t^2/t_*^2 \approx (E/E_*)^2 \cdot 4/\dots$ and the source F. This is the size predicted by (Q1): $|w| \le \frac{1}{2}(3\Phi_0 + 4|\rho|)E$ with $\Phi_0 = \sup|F| \approx 1.0$, i.e. $\le 3.5E$ — an overestimate by a factor ≈ 6, so the constants of Theorem 4.38 are far from sharp.

Toy example 4.49 (the logarithmic charts: first Picard iterate in closed form)**.** Start the loop at $w = 0$. *Type* L_1*:* $z = \zeta + (\xi - \xi_*)$ exactly (so $z' = 1$), $f = \varphi - \kappa_{\mathrm{ch}}/z$, and $F = \Lambda_{\mathrm{p}} - (\varphi - \kappa_{\mathrm{ch}}/z)^2$ is bounded with limit $\Lambda_{\mathrm{p}} - \varphi^2$ at the pole; (Q1) gives $|w^{(1)}| \le \frac{1}{2}(3\sup|F| + 4|\rho|)E$, and $w^{(1)}$ is already of the size of the fixed point, whose correction to z is the E_*-small term $K_0 w^{(1)}$. *Type* L_2*:* $z \equiv 1$, $f = \varphi + \kappa_{\mathrm{ch}}(\xi - \xi_*)$, $F = \Lambda_{\mathrm{p}} - (\varphi + \kappa_{\mathrm{ch}}(m-1))^2$ is a quadratic polynomial in m, so $|F| \le F_0 m^2$ and (Q3) gives $w^{(1)} = O(Em^2)$: this is the weight Em^2 of Table 4.3, and the logarithmic growth of the swirl $f \sim \kappa_{\mathrm{ch}}\xi$ is exactly what forces the factor m^2 (for L_1 the swirl is bounded and the weight is E). In both cases the factor $\frac{3}{4}$ resp. $\frac{9}{8}$ of Lemma 4.35(C) is the $\int(1+s)e^{-2s}$ resp. $\int s(1+s)^2 e^{-2s}$ that appears when K_0 is applied to $w^{(1)}$.

4.9.2. The numerical atlas.

Atlas 4.50 (the pole charts: loops, contraction rates and the exponent near the threshold)**.** *All numbers are produced by* `code/ch04_pole/verify_ch04b.py` *(trapezoid quadrature on* $N = 900$ *points over a window of length* 16 *after the cut; double precision) and by stored shooting data. Status: numerical, not proved.*

(1) Constants. $\int_0^\infty (1+s)e^{-2s}\mathrm{d}s = \frac{3}{4}$, $\int_0^\infty s(1+s)^2 e^{-2s}\mathrm{d}s = \frac{9}{8}$ *(computer algebra);* $K_{1/16} = 3.3016$*;* $\max_\xi (1+\xi)^2 e^{-\mu_*\xi}\mu_*^2 = 0.545$ *for* $k = 1, \dots, 10$ *(*≤ 1 *as claimed in Step 1 of Theorem 4.45);* $(1-5\mu_*)^{-2} \le 1.07 \le 1.1$*;* $(5 + 4(k+1))\mu_* \le \frac{1}{4}$ *for all* $k \ge 1$*. The kernel conjugation* $\hat{g}_\mu$ *and the formula* $R' = \operatorname{sech}^2(\mathcal{A} - 2t\mathcal{B})$ *are verified by computer algebra.*

(2) The chart loop for G *(Example 4.48), iterated from* $w = 0$*:*

ξ_*	E_*	$\sup\lvert w\rvert/E$	*first-step ratio*	*steps*
1	$1.4 \cdot 10^{-1}$	0.500	$4.0 \cdot 10^{-4}$	3
2	$1.8 \cdot 10^{-2}$	0.500	$6.7 \cdot 10^{-5}$	3
3	$2.5 \cdot 10^{-3}$	0.553	$5.6 \cdot 10^{-6}$	2
4	$3.4 \cdot 10^{-4}$	0.716	$3.4 \cdot 10^{-7}$	2

(Columns: the cut, E_**, the size of the fixed point in the weight* E*, the observed contraction ratio* $\|w_2 - w_1\|/\|w_1 - w_0\|$*, and the number of iterations to reach* 10^{-10}*.) The observed ratio decreases roughly like* $E_*^{1.5}$*, faster than the first power of* E_* *guaranteed by Step 3 of Theorem 4.38: the proof loses by estimating* $|f_1 + f_2|$ *and* $|\hat{z}_1 + \hat{z}_2|$ *at their worst values.*

(3) The recessive loop *(Toy example 4.36): the Picard iterates of Lemma 4.35(B) reproduce* $\Gamma(1+\mu)(2/t)^\mu I_\mu(t)$ *(*$t = \sqrt{\varepsilon}e^{-\xi}$*) to the quadrature error (*$\le 10^{-5}$ *at* $\xi_* = 0$*,* 10^{-8} *at* $\xi_* = 3$ *with* $\mu = 0.3$*), in* 7 *and* 3 *steps respectively. Figure 4.6 shows the family* $\mu \downarrow 0$ *passing smoothly from* R *to* L_2*.*

(4) The Jost toy: *for* $\mu = 0.1$*,* $\omega = -0.2$*:* $\xi_0 = 5.4931$ *(zero of* z*), agreeing with* $10\,\mathrm{artanh}\,\frac{1}{2}$*.*

Numerical observation 4.51 (approach of the recessive exponent to the threshold)**.** *The theorem* $U_\theta(1) > -3$ *(Theorem 4.9) says* $q = Q(1) > -\frac{3}{2}$ *on the recessive branch. Figure 4.7 (stored shooting data on the class-*Z *lower edge, base point* $x_0 = 0$*,* $\boldsymbol{c} = (0, c_2, 0)$*,* $\kappa = 0.003, \dots, 0.3$*) shows* $Q(1)$ *along the edge as a function of* κ*. For* $c_2 = -\frac{3}{4}$ *one finds* $-(Q(1) + \frac{3}{2}) = 2.4 \cdot 10^{-4}$*,* $2.7 \cdot 10^{-3}$*,* $2.1 \cdot 10^{-2}$ *at* $\kappa = 0.003, 0.01, 0.03$*, i.e.* $\approx 27\kappa^2$ *for* $\kappa \le 0.03$ *and* 0.32 *at* $\kappa = 0.3$*:* $Q(1) \to -\frac{3}{2}$ *from below like* κ^2 *as* $\kappa \to 0$*, so the threshold of Theorem 4.9 is approached*

but not crossed; for $c_2 = -0.95, -0.9, -0.8$ *the values of* $Q(1)$ *at* $\kappa = 0.003$ *are* $-1.900, -1.800, -1.600$ *(i.e.* $U_\theta(1) = -2-2\mu$ *with* $\mu^2 = (Q(1)+2)/2 = 0.05, 0.1, 0.2$*), and* $Q(1) \to -2$ *for* $c_2 = 0$ *(type* L_1 *or* L_2*). For* $c_2 = -0.7, -0.75$ *at the smallest* κ *the stored values (*$\approx -1.40, -1.50$*) are read at the finite window* $s_{\max} = 40$*, before the layer is crossed, and are not values of* $Q(1)$*. The identification of the limit* $-\frac{3}{2}$ *at* $c_2 = -\frac{3}{4}$ *with the threshold, and the* κ^2*-rate, are numerical observations; the identification of the three types along the edge is that of Numerical observation 4.28.*

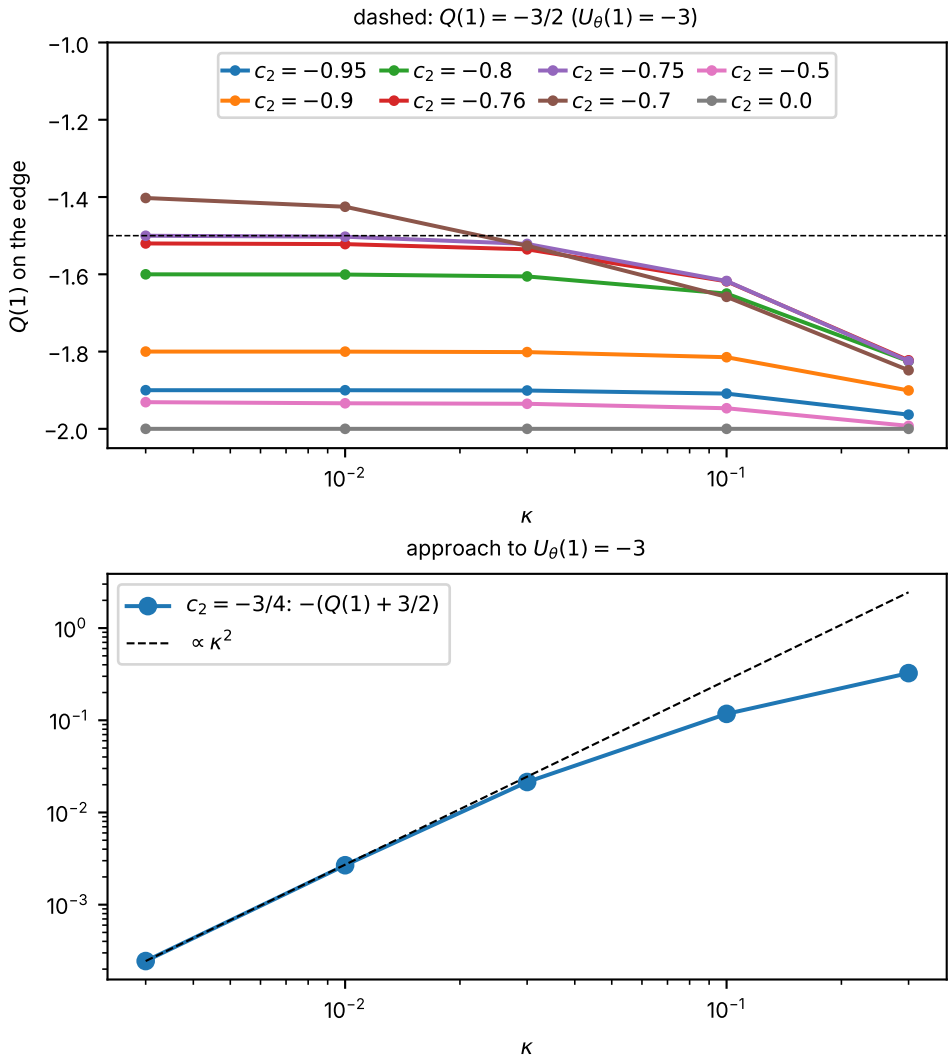


FIGURE 4.7. Numerical observation 4.51. Left: $Q(1)$ on the lower edge against κ for $c_2 = -0.95, \dots, 0$ (dashed: the threshold $Q(1) = -\frac{3}{2}$, $U_\theta(1) = -3$). Right: $-(Q(1) + \frac{3}{2})$ against κ for $c_2 = -\frac{3}{4}$ on log–log axes, with a κ^2 reference line. For $c_2 \geq -\frac{3}{4}$ and small κ the plotted value of Q at the end of the integration window ($s_{\max} = 40$) is still close to the no-swirl value $2c_2$: the boundary layer of width $\sim \ln(1/\kappa)$ has not been crossed, so these points do not contradict $Q(1) < -\frac{3}{2}$ (Theorem 4.9), which concerns the limit $s \to \infty$. Produced by `code/ch04_pole/make_qedge_small.py` from stored shooting data.

CHAPTER 5

Global solutions, the region Ω, comparison, and class Z (the lens)

In words. *A global solution with genuine swirl has a swirl f that is strictly monotone in x (Corollary 2.10); so either f vanishes exactly once, and the fluid turns in opposite directions on the two sides of one cone (*Z*, the* counter-rotating *solutions), or f never vanishes (*N*, one direction of rotation everywhere). The reflections $f \mapsto -f$ and $x \mapsto -x$ generate a group $\mathbb{Z}_2^2$ that acts* freely *on* N*, which is therefore $\mathsf{N}_0 \times \mathbb{Z}_2^2$ with $\mathsf{N}_0 = \{\kappa > 0,\ f > 0\}$. Collecting* all *global solutions in one set $\Omega \subset \mathbb{R}^6$ of Cauchy data at a base point, we prove that Ω is* closed*: a solution of the Q-system cannot stop existing in a limit, because an upper bound on the potential, coming for free from the sign of the swirl term, makes a very negative Riccati variable blow up persistently. The part Ω_{G} where both poles are generic is* open*, so the boundary of Ω consists of solutions with a non-generic pole. In the variables (U, g, f, p_2, p_1, p_0) the system is* cooperative *— every component is pushed up by its predecessor in the cycle $U \to g \to f \to p_2 \to p_1 \to p_0 \to U$ — as long as f grows in the direction of integration; ordered data then give ordered solutions (Kamke). One comparison lemma, with one exact "midpoint" linearisation, is the engine of the whole book: it makes the forward-safe sets half-lines, the edge of Ω a graph, the shifts of Chapter 6 monotone, and, applied to a geometric-arithmetic mean of two solutions, gives the log-convexity of the lens (Chapter 12). The second half of the chapter then parametrises class* Z *completely by the* lens *of admissible (κ, γ).*

Roadmap. Section 5.1 fixes the symmetries, the classes $\mathsf{Z}, \mathsf{N}, \mathsf{N}_0$ and proves the dichotomy $\mathsf{Z} \sqcup \mathsf{N}$ (Lemma 5.1, Definition 5.3, and Proposition 5.4). Section 5.2 defines Ω with its forward and backward parts and proves that it is in bijection with the set of global solutions (Lemma 5.7), that it is closed (Theorem 5.12), that Ω_{G} is open (Proposition 5.15), that Z is open in $\Omega \cap \{\kappa \neq 0\}$ and that the Z/N interface is $\{f(\pm 1) = 0\}$ (Proposition 5.16); it ends with the first consequence of the pole theory for global swirl solutions, the *threshold* 3 of Li–Li–Yan (Corollary 5.17). Section 5.3 is the comparison toolbox: cooperative systems, the cone lemma and Kamke's theorem (cited from Appendix A), the cooperative form of the Q-system with its Jacobian and the six-cycle, forward invariance of $\{g \geq 0, f \geq 0\}$, the *comparison lemma* for the Q-system (Lemma 5.28), ordering of data at a zero of the swirl (Corollary 5.29) and the directions in which the system is cooperative (Lemma 5.31). Section 5.4 proves the *midpoint rule* in logarithmic variables (Lemma 5.36). Part b of the chapter (Sections 5.5–5.8) uses these tools to prove that the forward-safe set at a base point with $A\kappa \geq 0$ is a half-line (Proposition 5.42), classifies the edge types (Lemma 5.46), parametrises class Z by the lens (Theorem 5.62), and studies the wedge at a Li–Li–Yan point and the closure of Ω_{G} on the faces (Propositions 5.67

and 5.69); the lens at $\boldsymbol{c} = 0$ is computed in Example 5.71 and the numerical atlas of the chapter is Atlas 5.73. *Conventions.* From the line variable $\xi = \operatorname{artanh} x$ on, a prime is $\mathrm{d}/\mathrm{d}\xi$; in this chapter the cooperative system of Sections 5.3 and 5.4 is written in the variable x with ∂_x explicit. A bare κ is the normalisation $\kappa_x = f_x(x_b)$ at the base point x_b under consideration (Lemma 2.15, item (1)); its sign is intrinsic. Solutions are never quotiented by the symmetry group.

5.1. The classes: symmetries and the dichotomy

In words. *A global solution (U, f, Q) is a point of one set, and the book studies the geometry of that set. Two reflections act on it: reversing the swirl ($f \mapsto -f$) and exchanging the poles ($x \mapsto -x$). Because f is strictly monotone, a solution with genuine swirl has only four possible sign patterns $(\operatorname{sgn}\kappa, \operatorname{sgn} f)$ when f does not vanish, and these four patterns are permuted* simply transitively *by the two reflections; this is why it is enough to study one quarter of class* N *and why we never lose anything by assuming $\kappa > 0$, $f > 0$ there. If f vanishes (once, at x_0) the solution is a counter-rotating one and we call it class* Z*.*

Recall that a *solution* means a solution (U, f, Q) of the Q-system (2.19)–(2.21) on its maximal interval and that it is *global* if this interval is $(-1, 1)$ (Corollary 2.25: $z > 0$ on $\mathbb{R}$). Its swirl constant κ is defined by $y^2 f_x \equiv \kappa$ (Lemma 2.9); the normalisation of y is free, and the sign of κ and the question whether $\kappa = 0$ are intrinsic.

Lemma 5.1 (symmetries act on global solutions)**.** *Let $S = (U, f, Q)$ be a solution with swirl constant κ on $(-1, 1)$ and put*

$$\sigma_1 S := (U, -f, Q), \qquad \sigma_2 S\,(x) := \big(-U(-x),\, f(-x),\, Q(-x)\big),$$
$$\sigma_1\sigma_2 S\,(x) = \big(-U(-x),\, -f(-x),\, Q(-x)\big).$$

(a) *$\sigma_1 S$, $\sigma_2 S$, $\sigma_1\sigma_2 S$ are solutions on $(-1, 1)$; $\sigma_1^2 = \sigma_2^2 = \mathrm{id}$ and $\sigma_1\sigma_2 = \sigma_2\sigma_1$. In particular they map global solutions to global solutions, and σ_2 maps the pole $x = -1$ to $x = +1$.*

(b) *The swirl constants are $\kappa(\sigma_1 S) = -\kappa$, $\kappa(\sigma_2 S) = -\kappa$, $\kappa(\sigma_1\sigma_2 S) = +\kappa$.*

(c) *If $f(x_0) = 0$, then the swirl of $\sigma_1 S$ vanishes at x_0, and the swirl of $\sigma_2 S$ and of $\sigma_1\sigma_2 S$ vanishes at $-x_0$. If $\kappa = 0$ then f is constant and the same holds for the three images.*

(d) *(Action on Cauchy data.) Let $(\boldsymbol{c}, \gamma, A, \kappa)$ be the Cauchy data of S at the base point x_b (Section 5.2). Then the Cauchy data of $\sigma_1 S$ at x_b are $(\boldsymbol{c}, \gamma, -A, -\kappa)$; those of $\sigma_2 S$ at $-x_b$ are $(\boldsymbol{c}_m, -\gamma, A, -\kappa)$ and those of $\sigma_1\sigma_2 S$ at $-x_b$ are $(\boldsymbol{c}_m, -\gamma, -A, \kappa)$, where $\boldsymbol{c}_m := (c_2, c_1, c_3)$.*

Proof. (a) Invariance of the Q-system under σ_1, σ_2 is Lemma 2.8; the composition rules follow from the formulas. A solution lives on $(-1, 1)$ iff its image does, since $x \mapsto -x$ preserves $(-1, 1)$. (b) σ_1 does not change U, hence not y, and changes the sign of f_x; so $\kappa = y^2 f_x$ changes sign. For σ_2 put $\tilde{y}(x) := y(-x) > 0$. Then $2(1 - x^2)\tilde{y}_x/\tilde{y} = -2(1 - x^2)y_x(-x)/y(-x) = -U(-x) = \tilde{U}(x)$, so $\tilde{y}$ is an admissible function y for $\sigma_2 S$, and with $\tilde{f}(x) = f(-x)$, $\tilde{f}_x(x) = -f_x(-x)$: $\tilde{y}^2 \tilde{f}_x(x) = -y(-x)^2 f_x(-x) = -\kappa$. For $\sigma_1\sigma_2$ combine the two. (c) is read off from the formulas. (d) At x_b the data of $\sigma_1 S$ are obtained by changing the sign of f and f_x only. For $\sigma_2 S$ the base point moves to $-x_b$; there $\tilde{U}(-x_b) = -\gamma$,

$\tilde f(-x_b) = A$, $\tilde f_x(-x_b) = -f_x(x_b) = -\kappa$, the gauge $\tilde y(-x_b) = y(x_b) = 1$ is preserved, and the degree-two Taylor polynomial of $\tilde Q(s) = Q(-s)$ at $-x_b$ is $P_{\boldsymbol c}(-s) = c_1(1+s) + c_2(1-s) + c_3(1-s^2) = P_{\boldsymbol c_m}(s)$. Compose for $\sigma_1\sigma_2$. □

Toy example 5.2 (the dichotomy for a fixed potential)**.** Freeze the potential at the rest state, $Q \equiv 0$, so $W \equiv 1$ and $z = \cosh\xi$, $y \equiv 1$, $f = A + \kappa x$. (This is *not* a solution of the system when $\kappa \neq 0$, because (2.21) forces Q to react to the swirl; it only shows the mechanism.) Then f is a strictly monotone function of x with $f(\mp 1) = A \mp \kappa$, and it has a zero in $(-1,1)$ iff $|A| < |\kappa|$. If $|A| \geq |\kappa|$ the sign of f is constant, and the four patterns $(\operatorname{sgn}\kappa, \operatorname{sgn} f) \in \{\pm\}^2$ are reached: $(A,\kappa) = (2,1), (-2,-1), (-2,1), (2,-1)$, exchanged by $\sigma_1 : (A,\kappa) \mapsto (-A,-\kappa)$ and by $\sigma_2 : (A,\kappa) \mapsto (A,-\kappa)$ (from Lemma 5.1(d)). The counter-rotating pattern is $|A| < |\kappa|$, e.g. $(A,\kappa) = (0,1)$.

Definition 5.3 (genuine swirl; the classes $\mathsf{Z}, \mathsf{N}, \mathsf{N}_0$)**.** A global solution has *genuine swirl* if $\kappa \neq 0$; it has *no swirl* (trivial swirl) if $\kappa = 0$, i.e. $f \equiv A$ is constant (Corollary 2.10). Let Sol be the set of global solutions with genuine swirl. Since $f_x = \kappa y^{-2}$ has the sign of κ, f is strictly monotone and has at most one zero. A solution $S \in \mathrm{Sol}$ belongs to *class* Z if f vanishes at some (necessarily unique) $x_0 \in (-1,1)$, and to *class* N otherwise. Put

$$\mathsf{N}_0 := \{S \in \mathsf{N} :\ \kappa > 0,\ f > 0 \text{ on } (-1,1)\} = \{f > 0,\ f_x > 0\}.$$

(Here $f = U_\varphi = u_\varphi \sin\theta$, and $2\pi f(\cos\theta)$ is the circulation of the velocity around the axis on the circle of polar angle θ, independent of the radius; so f is monotone in the polar angle. In class Z the fluid rotates in opposite directions on the two sides of the cone $\{x = x_0\}$.)

Proposition 5.4 (the dichotomy)**.** (a) $\mathrm{Sol} = \mathsf{Z} \sqcup \mathsf{N}$ *(disjoint union).*

(b) *The group* $\mathrm{Sym} = \langle\sigma_1,\sigma_2\rangle \cong \mathbb{Z}_2^2$ *preserves* Sol, Z *and* N. *It acts* freely *on* N, *and* N_0 *is a fundamental domain:* $(S,\sigma) \mapsto \sigma S$ *is a bijection* $\mathsf{N}_0 \times \mathrm{Sym} \to \mathsf{N}$, *i.e.* $\mathsf{N} = \mathsf{N}_0 \times \mathbb{Z}_2^2$.

(c) *On* Z *the elements* σ_1, σ_2 *act freely, and* $\sigma_1\sigma_2$ *fixes exactly the* odd *solutions, those with* $x_0 = 0$, $U(0) = 0$ *and* $c_1 = c_2$ *(*$\boldsymbol c$ *the Taylor constants of* Q *at* 0*); on* Z *the element* $\sigma_1\sigma_2$ *acts freely elsewhere.*

(d) *For* $S \in \mathsf{N}_0$ *the limit* $f_- := f(-1) = \lim_{x\downarrow -1} f(x)$ *exists, is finite, and* $0 \leq f_- < f(x)$ *for all* $x \in (-1,1)$.

Proof. (a) By Corollary 2.10 f has at most one zero; so a solution in Sol either has one (class Z) or none (class N).

(b) By Lemma 5.1 $\mathrm{Sym} = \{\mathrm{id}, \sigma_1, \sigma_2, \sigma_1\sigma_2\}$, its elements are commuting involutions, they map Sol to Sol (by (b) there, $\kappa \neq 0$ is preserved) and, by (c) there, a zero of f to a zero of f (at $\pm x_0$): the classes are preserved. For $S \in \mathsf{N}$ the function f has no zero, so $\operatorname{sgn} f$ is constant $= \pm 1$; also $\operatorname{sgn}\kappa = \pm 1$. By Lemma 5.1(b),(c) the pair $(\operatorname{sgn}\kappa, \operatorname{sgn} f)$ transforms as

	id	σ_1	σ_2	$\sigma_1\sigma_2$
$(\operatorname{sgn}\kappa, \operatorname{sgn} f)$ from $(+,+)$	$(+,+)$	$(-,-)$	$(-,+)$	$(+,-)$

(for σ_2: $\kappa \mapsto -\kappa$ and $f \mapsto f(-x)$ keeps its sign), so the four elements of Sym send $(+,+)$ to four different patterns: the action of Sym on the four patterns is simply transitive. If $\sigma S = S$ then σ fixes the pattern of S, hence $\sigma = \mathrm{id}$: the action is free.

Every orbit Sym$S$ contains exactly one element with pattern $(+,+)$, i.e. exactly one element of N_0. This is the bijection $\mathsf{N}_0 \times \mathrm{Sym} \to \mathsf{N}$.

(c) If $\sigma_1 S = S$ then $f = -f$, so $f \equiv 0$, impossible for $\kappa \neq 0$; if $\sigma_2 S = S$ then $f(x) = f(-x)$ contradicts injectivity of f ($\kappa \neq 0$). Now let $\sigma_1\sigma_2 S = S$: $U(x) = -U(-x)$, $f(x) = -f(-x)$, $Q(x) = Q(-x)$. Then $f(0) = 0$, so $x_0 = 0$; $U(0) = 0$; and Q even means that its Taylor polynomial $P_{\boldsymbol{c}}$ at 0 is even; since $P_{\boldsymbol{c}}(-x) = P_{\boldsymbol{c}_m}(x)$ this is $c_1 = c_2$. Conversely, let $S \in \mathsf{Z}$ have $x_0 = 0$, $U(0) = 0$ and $c_1 = c_2$. By Lemma 5.1(d) the Cauchy data at 0 of $\sigma_1\sigma_2 S$ are $(\boldsymbol{c}_m, -\gamma, -A, \kappa) = (\boldsymbol{c}, 0, 0, \kappa)$, the data of S itself; by uniqueness for the regular system (2.22) (Remark 2.7) $\sigma_1\sigma_2 S = S$.

(d) For $S \in \mathsf{N}_0$, f is strictly increasing and positive on $(-1,1)$; as $x \downarrow -1$ it decreases and is bounded below by 0, so $f_- = f(-1) \in [0, f(x))$ exists and is finite. □

Example 5.5 (odd solutions at $\boldsymbol{c} = 0$)**.** Take $\boldsymbol{c} = 0$, $A = 0$, $\gamma = 0$ and any $\kappa > 0$ (base point 0). By Proposition 5.4(c) the corresponding solution, if global, is fixed by $\sigma_1\sigma_2$: it is an odd solution, $U(-x) = -U(x)$, $f(-x) = -f(x)$, Q even. Numerically (Example 5.71) the set of κ for which it is global is $(0, \kappa_{\max}(0,0)]$ with $\kappa_{\max}(0,0) = 4.1608$, and the point $\kappa = \kappa_{\max}$ is the tip of the lens, $\gamma_* = 0$. For $\kappa = 0$ the solution with $\gamma = 0$ is the rest state $U \equiv 0$ (Example 2.29).

Remark 5.6 (N is not an artefact of the base point)**.** The dichotomy is a statement about f alone and uses no base point; all four sign sectors of N and the counter-rotating class Z appear in the slices of Ω of Figure 5.1.

5.2. The region $\Omega \subset \mathbb{R}^6$: closed, with Ω_{G} open

In words. *By Remark 2.7 a solution of the Q-system is determined by its state at one point, and every state in $\mathbb{R}^6$ is allowed. So the set of global solutions* is *a subset Ω of $\mathbb{R}^6$, and the monograph studies it as a set: we show it is closed (a limit of global solutions is global), that the part where both poles are generic is open, and that the swirl classes are open/closed in the way the pictures suggest. Closedness has a one-line reason: in the Riccati equation $h' = W - h^2$ for $h = z'/z$ the potential W is bounded above, uniformly on bounded sets of data,* because $-f^2 \leq 0$*; and a Riccati variable below $-\sqrt{2\sup W}$ blows up to $-\infty$ in finite time, which is an open condition on the data.*

5.2.1. Definition of Ω and the bijection with global solutions. Fix a base point $x_b \in (-1,1)$, $\xi_b = \operatorname{artanh} x_b$. By Lemma 2.5 and Remark 2.6 the data $\omega = (\boldsymbol{c}, \gamma, A, \kappa) \in \mathbb{R}^6$ — $\boldsymbol{c}$ the Taylor constants of Q at x_b, $\gamma = U(x_b)$, $A = f(x_b)$, $\kappa = \kappa_x = f_x(x_b)$ (gauge $y(x_b) = 1$) — correspond bijectively and linearly to the state $Y_b = (U, f, f_x, Q, Q_x, Q_{xx})(x_b) \in \mathbb{R}^6$ of the regular system (2.22); write $L_{x_b} : \omega \mapsto Y_b$ for this linear isomorphism. Let $(a^*(\omega), b^*(\omega)) \ni x_b$ be the maximal interval of the solution with data ω and put

$$\Omega^+(x_b) := \{\omega :\ b^*(\omega) = 1\}, \qquad \Omega^-(x_b) := \{\omega :\ a^*(\omega) = -1\},$$
$$\Omega(x_b) := \Omega^+(x_b) \cap \Omega^-(x_b).$$

$\Omega^+(x_b)$ (resp. $\Omega^-(x_b)$) is the set of data whose solution exists on $[x_b, 1)$ (resp. $(-1, x_b]$); its slices at fixed $(\boldsymbol{c}, A, \kappa)$ are the *forward-* and *backward-safe sets* $\mathrm{FS}, \mathrm{BS} \subset \mathbb{R}$ of γ's (Proposition 5.42), and the fibre of Ω is $\mathrm{FS} \cap \mathrm{BS}$. When no confusion is possible we write Ω for $\Omega(x_b)$; the default base point is $x_b = 0$.

Lemma 5.7 (Ω and global solutions; base-point independence)**.** (a) $\omega \mapsto$ *(the solution through ω) is a bijection from $\Omega(x_b)$ onto the set of all global solutions (with or without swirl). No quotient by the symmetry group is taken.*

(b) *For $x_b, x_b' \in (-1,1)$ let $D_{b,b'} \subset \mathbb{R}^6$ be the set of data at x_b whose solution exists on the closed interval between x_b and x_b'. Then $D_{b,b'}$ is open and the map $\Phi_{b\to b'}$, sending data at x_b to the data of the* same *solution at x_b', is a C^∞ diffeomorphism $D_{b,b'} \to D_{b',b}$ whose inverse is $\Phi_{b'\to b}$. It maps $\Omega(x_b)$ onto $\Omega(x_b')$, and $\Omega^{\pm}(x_b) \cap D_{b,b'}$ onto $\Omega^{\pm}(x_b') \cap D_{b',b}$.*

(c) *σ_1 acts on $\Omega(x_b)$ by $(\boldsymbol{c},\gamma,A,\kappa) \mapsto (\boldsymbol{c},\gamma,-A,-\kappa)$; at the base point $x_b = 0$ also $\sigma_2 : (\boldsymbol{c},\gamma,A,\kappa) \mapsto (\boldsymbol{c}_m,-\gamma,A,-\kappa)$ and $\sigma_1\sigma_2 : (\boldsymbol{c},\gamma,A,\kappa) \mapsto (\boldsymbol{c}_m,-\gamma,-A,\kappa)$; these are linear involutions of $\mathbb{R}^6$ that preserve $\Omega(0)$.*

Proof. (a) The right-hand side of (2.22) is smooth on $(-1,1)\times\mathbb{R}^6$, so by Remark 2.7(i) a solution is determined by its state at x_b, and $\omega \leftrightarrow Y_b$ is a linear isomorphism (Remark 2.6); every global solution has a state at x_b. The normalisation of y does not affect (U,f,Q) (Lemma 2.9), so fixing $y(x_b) = 1$ loses nothing. (b) By Remark 2.7(iii) the set of data whose solution exists on a given compact interval is open and the solution depends continuously (indeed smoothly, the field being smooth) on the data; evaluating at x_b' gives the smooth map $\Phi_{b\to b'}$ on $D_{b,b'}$, with image in $D_{b',b}$ (the same solution exists on the same interval), and $\Phi_{b'\to b}\circ\Phi_{b\to b'} = \mathrm{id}$ by uniqueness; the same with b, b' exchanged. A global solution exists on $(-1,1) \supset [x_b, x_b']$, so $\Omega(x_b) \subset D_{b,b'}$, and the solution is the same; the statement on $\Omega^{\pm}$ is the same observation for one-sided intervals. (c) is Lemma 5.1(d) at x_b, resp. $-x_b = x_b = 0$; σ_j preserves global solutions by part (a) of that lemma. □

Remark 5.8 (the Wronskian identity)**.** The identity that expresses $\gamma_{\mathrm{F}} - U_c^-$ as the integral $\int \psi_{\boldsymbol{c}} z_0 z$ of the swirl integral against the recessive no-swirl solution is not needed in this chapter and is proved once in Proposition 3.51 (identity (E)).

Example 5.9 (Ω at the rest state)**.** Let $\boldsymbol{c} = 0$ and $\kappa = 0$, $A \in \mathbb{R}$ arbitrary. Then $f \equiv A$, $Q = P_{\mathbf{0}} = 0$, $W \equiv 1$, and the global solutions are the positive solutions $z = ae^{\xi} + be^{-\xi}$ of $z'' = z$, $a, b \geq 0$ not both 0 (Proposition 3.43); $\gamma = U(0) = 2z'(0)/z(0) = 2(a-b)/(a+b) \in [-2,2]$. Hence $\Omega \cap \{\boldsymbol{c} = 0, \kappa = 0\} = [-2,2]\times\mathbb{R} \ni (\gamma, A)$, a closed set that is *not* open: for $\gamma > 2$ the solution of the Riccati equation $h' = 1 - h^2$, $h(0) = \gamma/2 > 1$, blows up to $+\infty$ at the *left* end. At $\gamma = \pm 2$ the solution is $z = e^{\pm\xi}$, recessive at one pole, so $\Omega_{\mathsf{G}} \cap \{\boldsymbol{c} = 0, \kappa = 0\} = (-2,2)\times\mathbb{R}$ (both $a, b > 0$, i.e. dominant at both poles, see Proposition 5.15); the two lines $\gamma = \pm 2$ lie in $\Omega \setminus \Omega_{\mathsf{G}}$.

5.2.2. Ω is closed.

Toy example 5.10 (a scalar Riccati equation: blow-up is persistent)**.** Let $h' = M - h^2$ with constant $M > 0$. The solutions with $|h| < \sqrt{M}$ are $h = \sqrt{M}\tanh(\sqrt{M}(\xi - \xi_0))$; those with $h < -\sqrt{M}$ are $h = -\sqrt{M}\coth(\sqrt{M}(\xi_0 - \xi))$ for $\xi < \xi_0$, which blow up to $-\infty$ at ξ_0; those with $h > \sqrt{M}$ are $h = \sqrt{M}\coth(\sqrt{M}(\xi - \xi_0))$ for $\xi > \xi_0$; and $h \equiv \pm\sqrt{M}$. (Check: $(\sqrt{M}\tanh)' = M\,\mathrm{sech}^2 = M - M\tanh^2$, and similarly for coth.) The set of initial values $h(0)$ for which the solution exists on $[0,\infty)$ is $[-\sqrt{M}, \infty)$, a *closed* half-line: the complement $\{h(0) < -\sqrt{M}\}$ is open, and $h(0) = -\sqrt{M}$ is the constant solution that just survives. The proof of Theorem 5.12

is this observation for $h = z'/z$, where M is replaced by an upper bound for W that holds uniformly for data in a compact set.

Lemma 5.11 (escape lemma)**.** *Let $M \ge 0$ and $H > 0$ with $H^2 \ge 2M$, and $\xi_2 \in \mathbb{R}$. There is no C^1 function $h : [\xi_2, \infty) \to \mathbb{R}$ with $h' \le M - h^2$ and $h(\xi_2) < -H$.*

Proof. Suppose h exists. We claim $h < -H$ on $[\xi_2, \infty)$. Otherwise let $\xi_1 > \xi_2$ be the first point with $h(\xi_1) = -H$; then $h < -H$ on $[\xi_2, \xi_1)$, so $h'(\xi_1) \ge 0$, while $h'(\xi_1) \le M - H^2 \le -H^2/2 < 0$: contradiction. Hence $h^2 > H^2 \ge 2M$ and $h' \le M - h^2 \le -\frac{1}{2}h^2$ on $[\xi_2, \infty)$. As $h < 0$, $(1/h)' = -h'/h^2 \ge \frac{1}{2}$, so $1/h(\xi) \ge 1/h(\xi_2) + \frac{1}{2}(\xi - \xi_2)$, and for $\xi \ge \xi_2 + 2/|h(\xi_2)|$ the right side is ≥ 0, contradicting $1/h(\xi) < 0$. □

Theorem 5.12 (Ω is closed). *For every base point $x_b \in (-1, 1)$ the sets $\Omega^+(x_b)$, $\Omega^-(x_b)$ and $\Omega(x_b) = \Omega^+(x_b) \cap \Omega^-(x_b)$ are closed subsets of $\mathbb{R}^6$.*

Proof. *Step 0 (reduction to Ω^+).* Let $R = \mathrm{diag}(-1, 1, -1, 1, -1, 1)$ act on states $(U, f, f_x, Q, Q_x, Q_{xx})$. By Lemma 5.1(d) (or directly from $\tilde{U}(s) = -U(-s)$, $\tilde{f}(s) = f(-s)$, $\tilde{Q}(s) = Q(-s)$) the solution $\sigma_2 S$ has at $-x_b$ the state RY_b, and it exists on $[-x_b, 1)$ iff S exists on $(-1, x_b]$. In terms of Cauchy data this reads

$$\Omega^-(x_b) = L_{x_b}^{-1}\, R^{-1}\, L_{-x_b}\, \Omega^+(-x_b),$$

a composition of linear isomorphisms of $\mathbb{R}^6$ (R, L_{x_b}, L_{-x_b}), hence a homeomorphism. So it suffices to prove that $\Omega^+(x_b)$ is closed for *every* x_b, and then $\Omega = \Omega^+ \cap \Omega^-$ is closed.

Step 1 (a bound for Q from above, with explicit dependence on the data). Let $\omega \in \mathbb{R}^6$, with state Y_b, and let the solution exist on $[x_b, x)$, $x \le 1$. By Proposition 2.20 $\mathsf{L}[Q] = \lambda - f^2$ with

$$\lambda = \lambda(\omega) := (1 - x_b^2)Q_{xx}(x_b) + 2x_bQ_x(x_b) - 2Q(x_b) + f(x_b)^2,$$

a continuous function of ω. The Cauchy formula (Lemma 2.18(d)) with base point x_b gives, for $x_b \le x < b^*(\omega)$,

$$Q(x) = Q(x_b)A_1(x) + Q_x(x_b)A_2(x) + \int_{x_b}^{x} K_{\mathsf{L}}(x, s)\big(\lambda - f(s)^2\big)\mathrm{d}s,$$

where $A_1 = 1 + \frac{(x-x_b)^2}{1-x_b^2}$, $A_2 = \frac{(x-x_b)(1-x_bx)}{1-x_b^2}$, and $K_{\mathsf{L}}(x, s) \ge 0$ for $s \le x$ (Lemma 2.18(c)). Since $-f^2 \le 0$ (this is the only property of f that is used: *no sign of f or of κ enters*), $K_{\mathsf{L}} \ge 0$ and $\int_{x_b}^{x} K_{\mathsf{L}}(x, s)\mathrm{d}s = \frac{(x-x_b)^2}{2(1-x_b^2)}$,

$$Q(x) \le |Q(x_b)|A_1 + |Q_x(x_b)|A_2 + \max(\lambda, 0)\tfrac{(x-x_b)^2}{2(1-x_b^2)}.$$

For $x_b \le x < 1$ we have $0 < x - x_b \le 1 - x_b$ and $0 < 1 - x_bx < 2$, hence $1 \le A_1 \le 1 + \frac{(1-x_b)^2}{1-x_b^2} = \frac{2}{1+x_b}$, $0 \le A_2 \le \frac{(1-x_b)\cdot 2}{1-x_b^2} = \frac{2}{1+x_b}$ and $\int_{x_b}^{x} K_{\mathsf{L}} \le \frac{1-x_b}{2(1+x_b)}$. Therefore

$$\begin{aligned} Q(x) \le M_Q(\omega) &:= \frac{2}{1+x_b}\big(|Q(x_b)| + |Q_x(x_b)|\big) \\ &\quad + \frac{1-x_b}{2(1+x_b)}\max(\lambda(\omega), 0), \end{aligned} \tag{5.1}$$

$$W = 1 + \tfrac{1}{2}Q \le M(\omega) := 1 + \tfrac{1}{2}M_Q(\omega)$$

for *every* $x \in [x_b, b^*(\omega))$; $M(\omega) \ge 1$ is a continuous function of $\omega \in \mathbb{R}^6$.

Step 2 (the Riccati inequality). In the line variable put $h = z'/z = \frac{1}{2}U + x$ (Lemma 2.14); it satisfies $h' = W - h^2$, so by (5.1), $h' \le M(\omega) - h^2$ on $[\xi_b, \xi^*(\omega))$ ($\xi^* = \operatorname{artanh} b^*$).

Step 3 (limits). Let $\omega_n \to \omega$ with $\omega_n \in \Omega^+(x_b)$ and suppose $\omega \notin \Omega^+(x_b)$, i.e. $b^*(\omega) < 1$, $\xi^* < \infty$. By Lemma 2.24 (the maximal intervals of the Y-system and of its line form coincide, since U is finite exactly where $z > 0$; Lemma 2.14), $U \to -\infty$ as $\xi \uparrow \xi^*$; hence $h = \frac{1}{2}U + \tanh\xi \to -\infty$. Put $M := M(\omega) + 1$ and $H := \sqrt{2M}$ (then $H > 0$, $H^2 = 2M$) and choose $\xi_2 \in (\xi_b, \xi^*)$ with $h(\xi_2) < -H - 1$. By continuous dependence (Remark 2.7(iii)) the states at ξ_2 of the solutions through ω_n converge to the state of the solution through ω, so $h_n(\xi_2) \to h(\xi_2)$ and there is n_1 with $h_n(\xi_2) < -H$ for $n \ge n_1$; by continuity of $M(\cdot)$ there is n_2 with $M(\omega_n) \le M$ for $n \ge n_2$. For $n \ge \max(n_1, n_2)$ the function h_n is C^1 on $[\xi_2, \infty)$, because $\omega_n \in \Omega^+(x_b)$ means that the solution exists on $[x_b, 1)$, i.e. $z_n > 0$ on $[\xi_b, \infty)$; by Step 2 $h_n' \le M(\omega_n) - h_n^2 \le M - h_n^2$ and $h_n(\xi_2) < -H$. This contradicts Lemma 5.11. Hence $\omega \in \Omega^+(x_b)$: $\Omega^+(x_b)$ is closed. □

Remark 5.13 (on the proof of closedness)**.** The argument holds at an arbitrary base point and for *all* data in $\mathbb{R}^6$ (any sign of A and κ, $\kappa = 0$ included). Three points are worth recording: (i) the Cauchy formula is based at x_b itself, so that M is an explicit continuous function of the data (5.1); (ii) the limit $h \to -\infty$ is taken from Lemma 2.24; (iii) the blow-up of h_n is the separate Lemma 5.11. The same argument shows that the set $\mathcal{S}$ of germs at a pole that are global is closed (Lemma 6.25). Closedness holds for Ω^+ alone; Ω^+ and Ω are *not* open (Example 5.9).

5.2.3. Ω_{G} is open. Recall (Definition 4.16) that a pole of a global solution is of *type* G if $\mu > 0$ and z is dominant there ($h \to +\mu$ at $+\infty$, i.e. $U(1) = -2 + 2\mu$; and, by σ_2, $h \to -\mu$ at $-\infty$). For $\kappa = 0$ this is the convention stated after Definition 4.16: $q = Q(1) > -2$ and $h \to +\mu$, $\mu = \sqrt{1 + q/2} > 0$; the types $\mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$ exist only for $\kappa \neq 0$. A pole that is not of type G is *non-generic*. Let

$$\Omega_{\mathsf{G}} := \{\omega \in \Omega : \text{ both poles are of type } \mathsf{G}\} \qquad (\kappa = 0 \text{ allowed}).$$

Chapter 4 describes type G in terms of a scale-free state at a point $\xi_c \ge 0$, so we first translate the Cauchy data of this chapter into it.

Remark 5.14 (from Cauchy data to the scale-free state)**.** The scale-free state of Chapter 4 at $\xi_c = \operatorname{artanh} x_c$, $x_c \in [0, 1)$, is $s = (h, k, f, Q, \dot{Q}, \lambda) \in \mathbb{R}^6$. In terms of the state $Y_c = (U, f, f_x, Q, Q_x, Q_{xx})(x_c)$ of (2.22) it is

$$h = \tfrac{U}{2} + x_c, \qquad k = (1 - x_c^2)\, f_x \ \big(= \kappa/z^2\big), \qquad f, \qquad Q,$$
$$\dot{Q} = (1 - x_c^2)\, Q_x, \qquad \lambda = (1 - x_c^2)\, Q_{xx} + 2x_c Q_x - 2Q + f^2,$$

the last being the first integral $\mathsf{L}[Q] = \lambda - f^2$ of Proposition 2.20. This map $T_{x_c} : \mathbb{R}^6 \to \mathbb{R}^6$ is affine and triangular, with diagonal entries $\frac{1}{2}$, $1 - x_c^2$, 1, 1, $1 - x_c^2$, $1 - x_c^2$ (the coefficient of Q_{xx} in λ is $\operatorname{sech}^2 \xi_c = 1 - x_c^2$), all nonzero; so it is an affine isomorphism, in particular a C^∞ diffeomorphism of $\mathbb{R}^6$, and it sends the states with $f_x = 0$ to the states with $k = 0$. Nothing here depends on $\kappa \neq 0$.

Transport to a base point x_b. Put $x_c := \max(x_b, 0)$ and let $D_{b,c}$, $\Phi_{b\to c}$ be as in Lemma 5.7(b) (for $x_b \ge 0$, $\Phi_{b\to c} = \mathrm{id}$). For $\omega \in D_{b,c}$ let $\tau_b(\omega) := T_{x_c}\big(L_{x_c} \Phi_{b\to c}(\omega)\big)$ be the scale-free state at ξ_c of the solution through ω. By Lemma 5.7(b), τ_b is a C^∞ diffeomorphism of the open set $D_{b,c}$ onto an open subset of $\mathbb{R}^6$. This handles

the north pole for every base point, including $x_b < 0$, where $\xi_c \geq 0$ is not available at x_b itself: the behaviour at $x = 1$ depends only on the solution on $[x_c, 1)$, and $\Phi_{b\to 0}$ transports the data from $x_b < 0$ to the equator. For the south pole apply the same construction to $\sigma_2 S$, whose state at $-x_b$ is RY_b (Lemma 5.1(d)); here $x_c = \max(-x_b, 0)$, and for $-x_b < 0$ one uses $\Phi_{-b\to 0}$.

Let $\Omega_G^+(x_b) \subset \mathbb{R}^6$ be the set of data whose solution exists on $[x_b, 1)$ and is of type G at $x = 1$, and $\Omega_G^-(x_b)$ the set of data whose solution exists on $(-1, x_b]$ and is of type G at $x = -1$; thus $\Omega_{\mathsf{G}}(x_b) = \Omega_G^+(x_b) \cap \Omega_G^-(x_b)$, and by σ_2, $\Omega_G^-(x_b) = L_{x_b}^{-1} R^{-1} L_{-x_b} \Omega_G^+(-x_b)$. We use two facts of Chapter 4, with the notation $\mathcal{G}_{\xi_c}$ for the set of scale-free states at $\xi_c \geq 0$ whose forward solution exists on $[\xi_c, \infty)$ and is of type G.

(i) (Corollary 4.42.) $\mathcal{G}_{\xi_c}$ is open in $\mathbb{R}^6$; this is proved there with uniform constants, for every $\kappa \in \mathbb{R}$ including $\kappa = 0$ (Theorem 4.38 holds for $\kappa_{\mathrm{ch}} = 0$, and Lemma 4.20 supplies the a priori asymptotics at $\kappa = 0$). By Remark 5.14, $\Omega_G^+(x_b) = \tau_b^{-1}(\mathcal{G}_{\xi_c})$ (a solution of type G at $x = 1$ that exists on $[x_b, 1)$ exists on $[x_b, x_c]$, so its data lie in $D_{b,c}$), hence $\Omega_G^+(x_b)$ *is open in* $\mathbb{R}^6$, and so is $\Omega_G^-(x_b)$.

(ii) (Theorem 4.38 and Corollary 4.43(a).) Near every point of $\mathcal{G}_{\xi_c}$ the map from the state to the chart coordinates $(\mu, \zeta, \kappa_{\mathrm{ch}}, \varphi, \Lambda_{\mathrm{p}}, \rho)$ at a late cut is a *local* C^∞ diffeomorphism (the cut depends on the state). Three of these coordinates are functionals of the solution that do not depend on the cut: $\mu = \sqrt{1 + Q(\infty)/2}$, $\varphi = f(\infty) = f(1)$ and Λ_{p}. Hence they are C^∞ on all of $\mathcal{G}_{\xi_c}$, and, being coordinate functions of a local diffeomorphism, *they have nowhere vanishing differential*; in particular $\mathrm{d}\varphi \neq 0$. (The full coordinate system of Corollary 4.43(b) holds only near states for which ξ_c itself is a valid cut; we never use it.) Composing with τ_b and, for $f(-1)$, with σ_2: $f(1)$ *is a* C^∞ *function with nowhere vanishing differential on* $\Omega_G^+(x_b)$*, and* $f(-1)$ *on* $\Omega_G^-(x_b)$.

Proposition 5.15 (Ω_{G} is open)**.** (a) Ω_{G} *is open in* $\mathbb{R}^6$ *and* $\Omega_{\mathsf{G}} \subset \operatorname{int}\Omega$*; consequently* $\partial\Omega$ *is disjoint from* Ω_{G}*.*

(b) *If* $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ *and* γ *lies strictly between* $U_{\boldsymbol{c}}^-(x_b)$ *and* $U_{\boldsymbol{c}}^+(x_b)$ *(*$\boldsymbol{c}$ *and* $U_{\boldsymbol{c}}^\pm$ *as in Theorem 3.44), then*

$$(\boldsymbol{c}, \gamma, A, 0) \in \Omega_{\mathsf{G}} \qquad \text{for every } A \in \mathbb{R};$$

in particular these points are interior points of Ω*.*

Proof. (a) $\Omega_{\mathsf{G}}(x_b) = \Omega_G^+(x_b) \cap \Omega_G^-(x_b)$ is the intersection of two open sets, by fact (i) above (with the transport of Remark 5.14 for the base points $x_b < 0$ and for the south pole). So Ω_{G} is open, hence contained in the interior of Ω. As Ω is closed (Theorem 5.12), $\partial\Omega = \Omega \setminus \operatorname{int}\Omega \subset \Omega \setminus \Omega_{\mathsf{G}}$. (b) Since $\kappa = 0$, $f \equiv A$, $f_x \equiv 0$, so $\psi \equiv 0$ and $Q = P_{\boldsymbol{c}}$ (Lemma 2.5; here $\boldsymbol{c}$ is the Taylor constant vector at *every* base point). The solution is the no-swirl solution $U_{\boldsymbol{c},\gamma}$ with $U(x_b) = \gamma$ of Theorem 3.44(ii). As γ is strictly between $U_{\boldsymbol{c}}^\mp(x_b)$, $U \neq U_{\boldsymbol{c}}^\pm$, and by Theorem 3.44(iii) $U(-1) = 2 - 2k_1$, $U(1) = -2 + 2k_2$, with $k_i = \sqrt{1 + c_i} > 0$, $Q(\mp 1) = 2c_{1,2}$ and $\mu = k_i$ at the two poles: $h \to +k_2$ at $+\infty$ and $h \to -k_1$ at $-\infty$ (Theorem 4.5). Both poles are dominant with $\mu > 0$: type G. A does not enter. □

5.2.4. The classes inside Ω; the interface.

Proposition 5.16 (Z is open; the Z/N interface)**.** *Work at a fixed base point x_b and identify solutions with their data in $\Omega \subset \mathbb{R}^6$.*

(a) Z *is open relative to* $\Omega \cap \{\kappa \neq 0\}$*; equivalently* N *is relatively closed in* $\Omega \cap \{\kappa \neq 0\}$.

(b) *Let* $\omega \in \Omega_{\mathsf{G}}$ *with* $\kappa > 0$. *Then* $f(\pm 1) := \lim_{x \to \pm 1} f(x)$ *exist, are finite,* $f(-1) < f(1)$*, and*

$$\omega \in \mathsf{N}_0 \iff f(-1) \geq 0; \qquad \omega \in \mathsf{Z} \iff f(-1) < 0 < f(1);$$
$$\omega \in \sigma_1\sigma_2\mathsf{N}_0 \iff f(1) \leq 0.$$

(c) *The functions* $f(\pm 1)$ *are* C^∞ *on* Ω_{G} *with nowhere vanishing differential. Hence, inside* $\Omega_{\mathsf{G}} \cap \{\kappa > 0\}$*, the interface between* Z *and the two outer sectors of* N *is* $\{f(-1) = 0\} \cup \{f(1) = 0\}$*, a union of* C^∞ *hypersurfaces, and it* belongs to N.

(For $\kappa < 0$ *apply* σ_1 *or* σ_2*.)*

Proof. (a) Let $\omega \in \mathsf{Z}$ with zero x_1. Then $f_x(x_1) = \kappa y^{-2} \neq 0$ (Lemma 2.9), so f changes sign at x_1: there are $x_\pm = x_1 \pm \delta$ in the interval with $f(x_-)f(x_+) < 0$. For $\omega_n \to \omega$, $\omega_n \in \Omega \cap \{\kappa \neq 0\}$, the solutions converge locally uniformly (Remark 2.7(iii)), so $f_n(x_-)f_n(x_+) < 0$ for large n and f_n has a zero between: $\omega_n \in \mathsf{Z}$. (b) Type G at $x = 1$ gives $f(1)$ finite (Proposition 4.19: $\int^\infty z^{-2} < \infty$ for $z \sim ae^{\mu\xi}$, $\mu > 0$), and likewise at -1 by σ_2. f is strictly increasing ($\kappa > 0$), so $f(-1) < f(x) < f(1)$. Then $f > 0$ on $(-1, 1)$ iff $f(-1) \geq 0$ (if $f(-1) \geq 0$ then $f > f(-1) \geq 0$; if $f(-1) < 0$ then $f < 0$ near -1); f has a zero iff $f(-1) < 0 < f(1)$ (intermediate value theorem for the strictly increasing f); and $f < 0$ on $(-1, 1)$ iff $f(1) \leq 0$. For $\kappa > 0$ the class $\sigma_1\sigma_2\mathsf{N}_0 = \{\kappa > 0,\ f < 0\}$ by Lemma 5.1. (c) By fact (ii) above (Corollary 4.43(a), valid at $\kappa = 0$ as well), $f(1) = \varphi \circ \tau_b$ is C^∞ on the open set $\Omega_G^+(x_b) \supset \Omega_{\mathsf{G}}$ with nowhere vanishing differential, and by σ_2 the same holds for $f(-1)$. So the zero sets are C^∞ hypersurfaces. At a point with $f(1) = 0$, $\kappa > 0$: $f < 0$ on $(-1, 1)$, so the solution is in $\sigma_1\sigma_2\mathsf{N}_0 \subset \mathsf{N}$, and it is a limit of points of Z ($f(1) > 0 > f(-1)$) and of $\sigma_1\sigma_2\mathsf{N}_0$ ($f(1) < 0$); similarly for $f(-1) = 0$, where $f > 0$ on $(-1, 1)$ and the point is in N_0. □

5.2.5. The threshold 3 of Li–Li–Yan. In words. *Swirl cannot make a solution approach the extreme no-swirl solution $U_{\boldsymbol{c}}^-$ at the pole when the pole exponent of that extreme solution is at least $\frac{1}{2}$. The reason is that a swirling solution can only be recessive (the branch $U(1) = -2 - 2\mu$) if $\mu < \frac{1}{2}$, so that $U(1) > -3$, whereas $U_{\boldsymbol{c}}^-(1) = -2 - 2k_2 \leq -3$ as soon as $k_2 \geq \frac{1}{2}$. This is the number* 3 *of the third paper of Li–Li–Yan [26]: it is the condition $\mu = k = \frac{1}{2}$ on the pole exponent.*

Corollary 5.17 (threshold 3)**.** *Let S be a global solution with $\kappa \neq 0$. Then*

$$U(1) > -3 \qquad \text{and} \qquad U(-1) < 3.$$

Consequently, if $\tilde{\boldsymbol{c}} \in \mathcal{J}$ has $k_2(\tilde{\boldsymbol{c}}) = \sqrt{1 + \tilde{c}_2} \geq \frac{1}{2}$, then

$$U(1) - U_{\tilde{\boldsymbol{c}}}^-(1) \ > \ 2k_2(\tilde{\boldsymbol{c}}) - 1 \ \geq \ 0,$$

and symmetrically $U_{\tilde{\boldsymbol{c}}}^+(-1) - U(-1) > 2k_1(\tilde{\boldsymbol{c}}) - 1$ if $k_1(\tilde{\boldsymbol{c}}) \geq \frac{1}{2}$. For $k_2 > \frac{1}{2}$ this is a gap that is uniform over all *global swirl solutions; for $k_2 = \frac{1}{2}$ it says that no swirl solution has $U(1) = U_{\tilde{\boldsymbol{c}}}^-(1) = -3$ (no uniform gap is claimed). Moreover, if the Taylor constants of S at the base point are $\boldsymbol{c}$ and $\psi = \psi_{\boldsymbol{c}}$, a pole of type* R *at $+1$ has $\psi(1) > 2c_2 + \frac{3}{2}$, and a pole of type* L_1 *or* L_2 *has $\psi(1) = 2c_2 + 2$.*

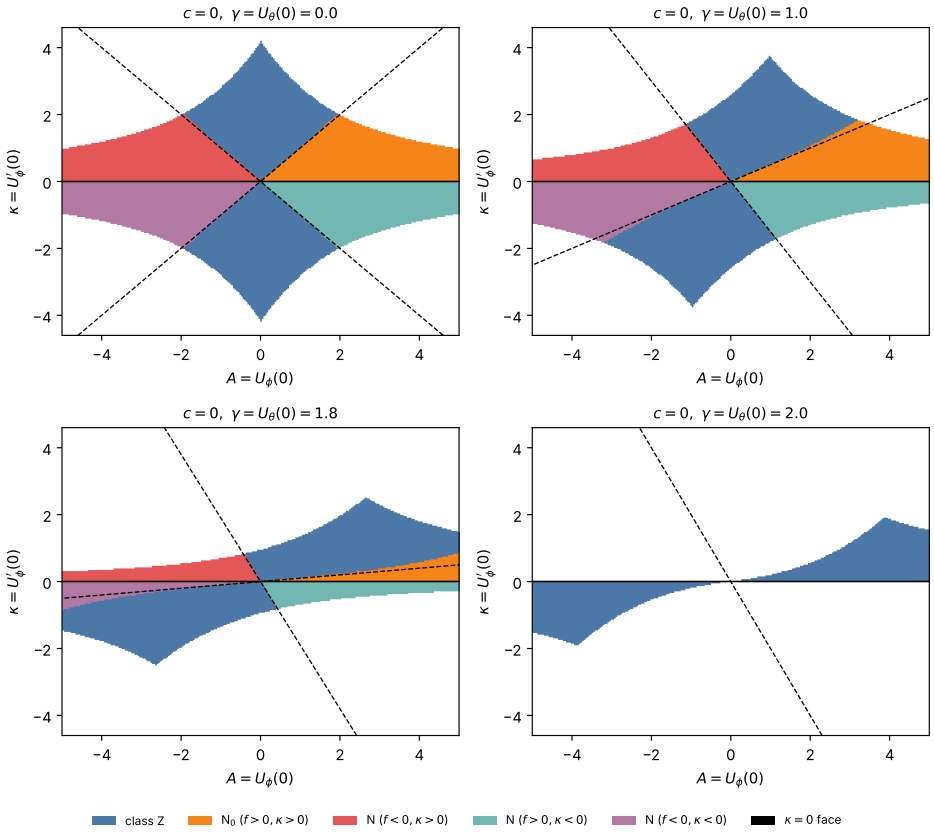


FIGURE 5.1. *Numerical illustration* (not used in any proof): the slices $\Omega \cap \{\boldsymbol{c} = 0,\ \gamma = \gamma_0\}$ in the (A, κ)-plane, $A = f(0)$, $\kappa = f_x(0)$, for $\gamma_0 = 0, 1, 1.8, 2$, coloured by class: Z (blue), N_0 (orange, $f > 0, \kappa > 0$), and the other three sectors of N, namely $f < 0, \kappa > 0$ (red), $f > 0, \kappa < 0$ (green) and $f < 0, \kappa < 0$ (purple); the black line is the face $\kappa = 0$ ($f \equiv A$; all of it belongs to Ω for $|\gamma_0| < 2$, Example 5.9). The dashed lines are the tangents $A = -F_0(\pm 1)\kappa$ at $\kappa = 0$ to the two interface hypersurfaces $\{f(\pm 1) = 0\}$ of Proposition 5.16, $F_0(\pm 1) = \int_0^{\pm 1} y_0^{-2}$ for the no-swirl solution ($F_0(1) = 2/(\gamma_0 + 2)$, $F_0(-1) = 2/(\gamma_0 - 2)$ at $\boldsymbol{c} = 0$; Proposition 5.67). Note the symmetry under $\sigma_1\sigma_2$ at $\gamma_0 = 0$ and the exchange of the four sectors by σ_1, σ_2. Code: `code/ch05_global/make_a_slice.py`.

PROOF. By Theorem 4.5, $Q(1) \geq -2$ exists and $U(1) = -2 \pm 2\mu$ with $\mu = \sqrt{1 + Q(1)/2} \geq 0$ (the two signs being the dominant and the recessive branch; for $\mu = 0$ only $U(1) = -2$). On the dominant branch $U(1) = -2 + 2\mu \geq -2 > -3$. On the recessive branch $h \to -\mu$, and since $\kappa \neq 0$, Theorem 4.9 gives $\mu < \frac{1}{2}$, hence

$U(1) = -2 - 2\mu > -3$ (the borderline $\mu = \frac{1}{2}$ is excluded there). So $U(1) > -3$; applying this to $\sigma_2 S$, $U(-1) < 3$. By Theorem 3.44(iii),(iv), $U^-_{\tilde{\boldsymbol{c}}}(1) = -2 - 2k_2(\tilde{\boldsymbol{c}})$ (the edge values $U^- = U^*$ if $\tilde{c}_3 = \bar{c}_3$), and $-2 - 2k_2 \le -3$ iff $k_2 \ge \frac{1}{2}$. Then $U(1) - U^-_{\tilde{\boldsymbol{c}}}(1) > -3 + 2 + 2k_2 = 2k_2 - 1$; the mirror statement follows by σ_2 ($U^+_{\tilde{\boldsymbol{c}}}(-1) = 2 + 2k_1$). For the last claim write $Q = P_{\boldsymbol{c}} - \psi$, $P_{\boldsymbol{c}}(1) = 2c_2$, so $\psi(1) = 2c_2 - Q(1)$; type R means $0 < \mu < \frac{1}{2}$ (Definition 4.16, Theorem 4.9), i.e. $Q(1) = 2\mu^2 - 2 \in (-2, -\frac{3}{2})$ and $\psi(1) = 2c_2 - Q(1) > 2c_2 + \frac{3}{2}$; types $\mathsf{L}_1, \mathsf{L}_2$ have $\mu = 0$, $Q(1) = -2$ and $\psi(1) = 2c_2 + 2$. □

Example 5.18 (the rest state)**.** For $\tilde{\boldsymbol{c}} = 0$: $k_2 = 1$, $U^-_0(1) = -4$. Every global swirl solution, at any $\boldsymbol{c}$ whatever, has $U(1) > -3$: it stays more than 1 above the lowest no-swirl solution at the pole. (Landau solutions have $U(1) = -2 + 2\mu \ge -2$ of the dominant branch.)

Remark 5.19 (relation to LLY III)**.** The condition "3" of [26] on the endpoint solutions is the condition $U^-(1) = -2 - 2k = -3$, i.e. $k = \frac{1}{2}$; Corollary 5.17 shows independently, in the pole-weighted sense, that for $k_2 > \frac{1}{2}$ no sequence of swirl solutions converges to $U^-_{\tilde{\boldsymbol{c}}}$ at the pole. The exact norm in which [26] prove non-existence near the endpoint solutions is not used here.

5.3. Cooperative systems and the comparison lemma

In words. *A comparison principle says: if two solutions start in the order $Y \le \hat{Y}$ they stay in that order. For one first-order equation this is automatic (graphs of solutions cannot cross). For a system it is true when every component is pushed* up *— never down — by every other component: a* cooperative *(Kamke) system; its linearisation has nonnegative off-diagonal entries (a* Metzler *matrix), which keep the positive orthant invariant. The Q-system is not cooperative as written, but after* reversing the order of every component except U *it is, on the set where the swirl f and its slope $g = f_x$ are ≥ 0: a larger U damps g (since $g_x = -Ug/(1-x^2)$), hence f, hence the swirl integrals p_2, p_1, p_0, and smaller p_0 raises U again. We prove the comparison lemma (with the strict version, which uses that the six couplings form a cycle) from the cone lemma of Appendix A, and determine in which directions of integration the system is cooperative: forward when $fg > 0$, backward when $fg < 0$.*

5.3.1. Metzler systems, the cone lemma and Kamke's theorem. A real matrix B is *Metzler* if $B_{ij} \ge 0$ for $i \ne j$. For $\Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_n)$, $\sigma_i = \pm 1$, write $Y \le_\Sigma \hat{Y}$ iff $\Sigma(\hat{Y} - Y) \ge 0$ componentwise; a system $\partial_t Y = F(t, Y)$ is Σ-*cooperative* on a set D if $\Sigma\, \partial_Y F\, \Sigma$ is Metzler on D (note $(\Sigma B \Sigma)_{ij} = \sigma_i \sigma_j B_{ij}$).

Toy example 5.20 (ordered flows; mutual inhibition)**.** (1) The pair $\partial_t x = -x + y$, $\partial_t y = x - y^2$ has Jacobian $\left(\begin{smallmatrix} -1 & 1 \\ 1 & -2y \end{smallmatrix}\right)$, off-diagonal entries $1 \ge 0$: cooperative. Solutions with componentwise ordered data stay ordered (Figure 5.2, left). (2) The pair $\partial_t p = -q$, $\partial_t q = -p$ (mutual inhibition) is not cooperative: the data $(0,0) \le (1,0)$ lead to $p \equiv q \equiv 0$ and $\hat{p} = \cosh t$, $\hat{q} = -\sinh t < 0 = q$; the componentwise order is lost, but the order $\Sigma = \operatorname{diag}(1, -1)$ ("larger p, smaller q") is kept, because $w = (\hat{p} - p, q - \hat{q})$ solves $\partial_t w = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) w$, which is Metzler (Figure 5.2, right). Exactly this sign change is what happens in the Q-system, with $\Sigma = \operatorname{diag}(+, -, -, -, -, -)$.

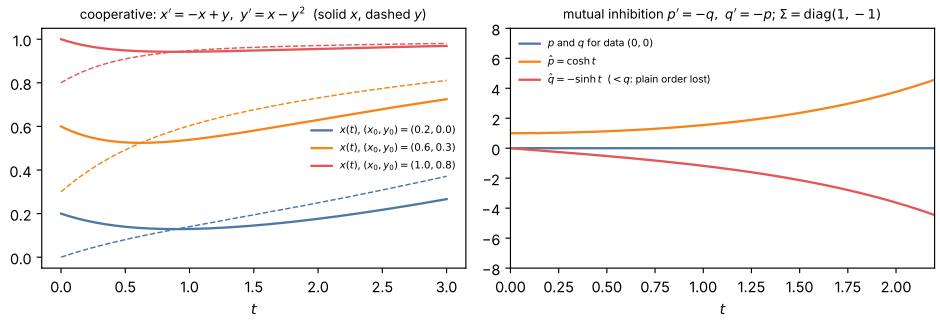


FIGURE 5.2. Toy example 5.20. Left: three solutions of the cooperative system $x' = -x + y$, $y' = x - y^2$ with componentwise ordered initial data stay ordered (x solid, y dashed). Right: for the non-cooperative pair $p' = -q$, $q' = -p$ the data $(0,0) \le_\Sigma (1,0)$ give $\hat{p} \ge p$ and $\hat{q} \le q$ for all t, i.e. order is preserved for $\Sigma = \mathrm{diag}(1,-1)$ only. Code: `code/ch05_global/make_a_toy_kamke.py`.

The two results of Appendix A that carry the whole chapter are the following (their proofs are in Appendix A.4; we restate the parts we use).

Lemma 5.21 (cone lemma; strict version)**.** *Let $B : [a,b] \to \mathbb{R}^{n\times n}$ be continuous and Metzler, $\rho : [a,b] \to \mathbb{R}^n$ continuous with $\rho \ge 0$, and $W \in C^1([a,b];\mathbb{R}^n)$ with $\partial_x W = BW + \rho$. (a) If $W(a) \ge 0$ then $W \ge 0$ on $[a,b]$. (b) If moreover the directed graph with an arrow $j \to i$ whenever $B_{ij} > 0$ on $(a,b]$ is strongly connected, and either $W(a) \ne 0$ or ρ does not vanish identically on any $[a,x]$, $x > a$, then $W_i > 0$ on $(a,b]$ for every i.*

PROOF. This is Lemma A.21 (Picard iteration after removing the diagonal by integrating factors; strict version by propagation of zeros around the cycle). □

THEOREM 5.22 (Kamke–Müller comparison with forcing). *Let $D \subset \mathbb{R}^n$ be convex, $F, \hat{F}$ defined and C^1 on a neighbourhood of $I \times D$ ($I \subset \mathbb{R}$ an interval), Σ a sign matrix with $\Sigma\, \partial_Y F\, \Sigma$ Metzler on $I \times D$ and $\Sigma F \le \Sigma \hat{F}$ there. If $Y, \hat{Y} : [a,b] \to D$ are C^1 with $\Sigma \partial_t Y \le \Sigma F(t, Y)$, $\Sigma \partial_t \hat{Y} \ge \Sigma \hat{F}(t, \hat{Y})$ and $Y(a) \le_\Sigma \hat{Y}(a)$, then $Y \le_\Sigma \hat{Y}$ on $[a,b]$.*

PROOF. For $\Sigma = \mathrm{id}$ this is Theorem A.24, which is stated for convex D (open or not) and F of class C^1 on an open set containing D; the signed version is Corollary A.25, which applies it to ΣY. □

In this chapter we shall *not* apply Theorem 5.22 as a black box: the Q-system is quadratic, so the integral mean-value matrix in its proof can be replaced by the exact *midpoint matrix* (Lemma 5.27), the proof reduces to Lemma 5.21, and the strict version is visible. (Classical references: [16, 18, 32, 43, 50].)

5.3.2. The cooperative form of the Q-system. Fix a vector of constants $\boldsymbol{c} = (c_1, c_2, c_3) \in \mathbb{R}^3$ and write $d = d(x) = 1 - x^2$. On $(-1,1) \times \mathbb{R}^6$ consider the

polynomial vector field $F_{\boldsymbol{c}}(x,Y)$, $Y=(U,g,f,p_2,p_1,p_0)$,

$$\partial_x Y = F_{\boldsymbol{c}}(x,Y),\qquad \begin{aligned} \partial_x U &= \frac{P_{\boldsymbol{c}}(x)-p_0-2xU-\frac12 U^2}{d}, & \partial_x g &= -\frac{Ug}{d},\\ \partial_x f &= g, & \partial_x p_2 &= \frac{2fg}{d},\\ \partial_x p_1 &= p_2, & \partial_x p_0 &= p_1. \end{aligned} \tag{5.2}$$

Lemma 5.23 (the cooperative form is the Q-system)**.** *(a) If Y solves (5.2) on an interval I, then $g=f_x$, $p_k=\partial_x^k p_0$ and, with $Q:=P_{\boldsymbol{c}}-p_0$, (U,f,Q) solves the Q-system on I. (b) Conversely let (U,f,Q) solve the Q-system on $I\ni x_0$, let $\boldsymbol{c}$ be the Taylor constants of Q at x_0 and $\psi:=P_{\boldsymbol{c}}-Q$ $(=\psi_{\boldsymbol{c}})$. Then $Y=(U,f_x,f,\psi_{xx},\psi_x,\psi)$ solves (5.2) with $Y(x_0)=(\gamma,\kappa,A,0,0,0)$, $\gamma=U(x_0)$, $A=f(x_0)$, $\kappa=f_x(x_0)$. In particular the solution with Cauchy data $(\boldsymbol{c},\gamma,A,\kappa)$ at x_0 is the solution of (5.2) with data $(\gamma,\kappa,A,0,0,0)$, and for data with $p_k(x_0)\neq 0$ the system (5.2) describes the same solutions with a different, redundant parametrisation.*

Proof. (a) $\partial_x f=g$ and $p_1=\partial_x p_0$, $p_2=\partial_x p_1$ give $g=f_x$ and $p_k=\partial_x^k p_0$. Then $Q_x=P_{\boldsymbol{c}}'-p_1$, $Q_{xx}=P_{\boldsymbol{c}}''-p_2$, $Q_{xxx}=-\partial_x p_2=-2ff_x/d$ since $P_{\boldsymbol{c}}'''=0$: this is (2.21); the equation for U is (2.19) with $Q=P_{\boldsymbol{c}}-p_0$; the equation for $g=f_x$ is (2.20). (b) As $P_{\boldsymbol{c}}$ is the Taylor polynomial of Q at x_0, $\psi(x_0)=\psi_x(x_0)=\psi_{xx}(x_0)=0$; and $\psi_{xxx}=-Q_{xxx}=2ff_x/d$. The remaining equations are (2.19),(2.20), and the definitions. □

Lemma 5.24 (reflection in the cooperative form)**.** *Let $\boldsymbol{c}_m=(c_2,c_1,c_3)$ and let R be the diagonal matrix with diagonal entries $-1,1,-1,1,-1,1$. If Y solves (5.2) for $\boldsymbol{c}$ on $[\alpha,\beta]\subset(-1,1)$, then*

$$Y^\sharp(s):=RY(-s)$$

solves (5.2) for $\boldsymbol{c}_m$ on $[-\beta,-\alpha]$.

Proof. By Lemma 5.23(a), $(U,f,Q=P_{\boldsymbol{c}}-p_0)$ is a solution of the Q-system, so $(\tilde U,\tilde f,\tilde Q)(s):=(-U(-s),-f(-s),Q(-s))$ is one as well (Lemma 5.1). Since $P_{\boldsymbol{c}}(-s)=P_{\boldsymbol{c}_m}(s)$, $\tilde p_0:=P_{\boldsymbol{c}_m}-\tilde Q=p_0(-s)$; by Lemma 5.23(b)–(a) (the variables built from $(\tilde U,\tilde f,\tilde Q)$ and $\boldsymbol{c}_m$ solve (5.2) for $\boldsymbol{c}_m$, whatever the value of the p's at the base point) they are $\tilde g=\tilde f_s=g(-s)$, $\tilde p_2=p_2(-s)$, $\tilde p_1=-p_1(-s)$, $\tilde p_0=p_0(-s)$, i.e. $RY(-s)$. (Direct substitution is the identity $-RF_{\boldsymbol{c}}(-s,RZ)=F_{\boldsymbol{c}_m}(s,Z)$, verified by computer algebra in `verify_ch05a.py`, block V3.) □

The Jacobian and the six-cycle. Differentiating (5.2), the nonzero entries of $J=\partial_Y F_{\boldsymbol{c}}$ are, in the order (U,g,f,p_2,p_1,p_0):

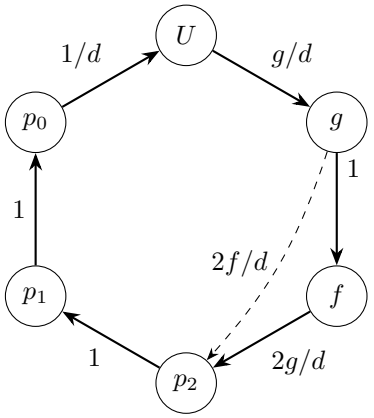
$$J_{UU}=-\tfrac{2x+U}{d},\quad J_{U,p_0}=-\tfrac1d;\qquad J_{gU}=-\tfrac{g}{d},\quad J_{gg}=-\tfrac{U}{d};\qquad J_{fg}=1;$$
$$J_{p_2,g}=\tfrac{2f}{d},\quad J_{p_2,f}=\tfrac{2g}{d};\qquad J_{p_1,p_2}=1;\qquad J_{p_0,p_1}=1.$$

Conjugating with $\Sigma=\operatorname{diag}(+,-,-,-,-,-)$ multiplies by $\sigma_i\sigma_j$: the entries J_{U,p_0}, J_{gU} change sign, and

$$(\Sigma J\Sigma)_{U,p_0}=\tfrac1d,\qquad (\Sigma J\Sigma)_{gU}=\tfrac{g}{d},\qquad (\Sigma J\Sigma)_{fg}=1,$$
$$(\Sigma J\Sigma)_{p_2,g}=\tfrac{2f}{d},\qquad (\Sigma J\Sigma)_{p_2,f}=\tfrac{2g}{d},\qquad (\Sigma J\Sigma)_{p_1,p_2}=(\Sigma J\Sigma)_{p_0,p_1}=1. \tag{5.3}$$

So $\Sigma J\Sigma$ is Metzler exactly where $g\ge 0$ and $f\ge 0$, i.e. on the closed convex set $D_+:=\{g\ge 0,\ f\ge 0\}\subset\mathbb{R}^6$. The coupling graph is the cycle $U\to g\to f\to p_2\to p_1\to p_0\to U$ (an arrow $j\to i$ for each nonzero entry (i,j)) with the extra

chord $g \to p_2$ (Figure 5.3). The parameters enter only through $P_{\mathbf{c}}$ in F_U, and $\partial_{c_i} F_U = \frac{1-x}{d}, \frac{1+x}{d}, 1$ are all > 0, with $\sigma_U = +1$: *raising any c_i moves a solution up in the order Σ.*



entries of $\Sigma J \Sigma$ (5.3);
all ≥ 0 iff $g \geq 0,\ f \geq 0$.
larger $U \Rightarrow$ smaller g, f, p_k
$\Rightarrow$ larger U.

FIGURE 5.3. The coupling graph of the Q-system in the variables (U, g, f, p_2, p_1, p_0): a cycle through all six components (solid arrows, with the entry of $\Sigma J \Sigma$ on each arrow) and the chord $g \to p_2$ (dashed). Positivity of one component is transmitted around the cycle; this is why the comparison lemma has a strict version (Lemma 5.28(b)).

Lemma 5.25 (forward invariance of D_+)**.** *Let Y solve (5.2) on $[a, b) \subset (-1, 1)$ with $g(a) \geq 0$, $f(a) \geq 0$. Then for all $x \in [a, b)$: (a) $g(x) \geq 0$; $g(x) > 0$ if $g(a) > 0$ and $g \equiv 0$ if $g(a) = 0$; (b) f is nondecreasing, $f(x) \geq f(a) \geq 0$; (c) if $p_k(a) \geq 0$ for $k = 0, 1, 2$ then p_2, p_1, p_0 are nondecreasing and ≥ 0 on $[a, b)$. This applies in particular to the normalised data $p_k(x_0) = 0$ of Lemma 5.23(b). Thus D_+, and $D_+ \cap \{p_k \geq 0\}$, are forward invariant. By σ_1 (that is, $(g, f) \mapsto (-g, -f)$ in (5.2)) the set $\{g \leq 0, f \leq 0\}$ is also forward invariant, and $|f|$ is nondecreasing there.*

PROOF. (a) $\partial_x g = -Ug/d$ is linear in g with a continuous coefficient on $[a, b)$, so $g(x) = g(a) \exp\big(-\int_a^x U/d\big)$ has the sign of $g(a)$. (b) $\partial_x f = g \geq 0$. (c) By (a),(b), $\partial_x p_2 = 2fg/d \geq 0$, so $p_2 \geq p_2(a) \geq 0$; then $\partial_x p_1 = p_2 \geq 0$ gives $p_1 \geq p_1(a) \geq 0$, and $\partial_x p_0 = p_1 \geq 0$ gives $p_0 \geq p_0(a) \geq 0$. The last statement: $(U, g, f, p_2, p_1, p_0) \mapsto (U, -g, -f, p_2, p_1, p_0)$ maps solutions to solutions (σ_1 with $f \mapsto -f$: $\partial_x p_2 = 2fg/d$ is unchanged). □

Remark 5.26 (why only forward)**.** D_+ is the region where the swirl is nonnegative and growing; once a solution is in it, it stays as x increases. Going *backward* from a point with $f = 0$, $g > 0$ the swirl becomes negative and $fg < 0$; the system is then cooperative for a *different* sign matrix (Lemma 5.31). Note that $p_k \geq 0$ in the forward direction means $\psi \geq 0$, $Q \leq P_{\mathbf{c}}$.

5.3.3. The comparison lemma.

Lemma 5.27 (exact linearisation at the midpoint)**.** *Let $Y, \hat{Y} \in \mathbb{R}^6$ be two states, $\boldsymbol{c}, \hat{\boldsymbol{c}} \in \mathbb{R}^3$, $x \in (-1,1)$, $\bar{Y} = \frac{1}{2}(Y+\hat{Y})$ with components $\bar{U}, \bar{g}, \bar{f}$, and $W := \Sigma(\hat{Y}-Y) = (\hat{U}-U,\ g-\hat{g},\ f-\hat{f},\ p_2-\hat{p}_2,\ p_1-\hat{p}_1,\ p_0-\hat{p}_0)$. Then*

$$\Sigma\big[F_{\hat{\boldsymbol{c}}}(x,\hat{Y}) - F_{\boldsymbol{c}}(x,Y)\big] = \mathfrak{A}(x;\bar{Y})\,W + \frac{P_{\hat{\boldsymbol{c}}}(x) - P_{\boldsymbol{c}}(x)}{d}\,e_U, \tag{5.4}$$

where $e_U = (1,0,0,0,0,0)^{\mathsf{T}}$ and $\mathfrak{A}(x;\bar{Y}) = \Sigma\,\partial_Y F_{\boldsymbol{c}}(x,\bar{Y})\,\Sigma$ has the nonzero entries

$$\mathfrak{A}_{UU} = -\tfrac{2x+\bar{U}}{d},\quad \mathfrak{A}_{U,p_0} = \tfrac{1}{d},\qquad \mathfrak{A}_{gU} = \tfrac{\bar{g}}{d},\qquad \mathfrak{A}_{gg} = -\tfrac{\bar{U}}{d},\quad \mathfrak{A}_{fg} = 1,$$
$$\mathfrak{A}_{p_2,f} = \tfrac{2\bar{g}}{d},\qquad \mathfrak{A}_{p_2,g} = \tfrac{2\bar{f}}{d},\quad \mathfrak{A}_{p_1,p_2} = \mathfrak{A}_{p_0,p_1} = 1.$$

Proof. $F_{\boldsymbol{c}}$ is a polynomial of degree ≤ 2 in Y whose quadratic terms are $-\frac{U^2}{2d}$, $-\frac{Ug}{d}$, $\frac{2fg}{d}$. For products, $\hat{a}\hat{b} - ab = \frac{1}{2}(\hat{a}-a)(\hat{b}+b) + \frac{1}{2}(a+\hat{a})(\hat{b}-b)$ (expand both sides), and $\hat{a}^2 - a^2 = (\hat{a}-a)(\hat{a}+a)$. Applying this to the six components, multiplying by σ_i and writing $\hat{Y} - Y = \Sigma W$, one finds for instance

$$W_g' = -\partial_x(\hat{g}-g) = \frac{\hat{U}\hat{g} - Ug}{d} = \frac{\bar{g}}{d}W_U - \frac{\bar{U}}{d}W_g,$$
$$W_{p_2}' = -\frac{2(\hat{f}\hat{g} - fg)}{d} = \frac{2\bar{g}}{d}W_f + \frac{2\bar{f}}{d}W_g,$$

and similarly $W_U' = \frac{1}{d}\big[P_{\hat{\boldsymbol{c}}} - P_{\boldsymbol{c}} + W_{p_0} - (2x+\bar{U})W_U\big]$, $W_f' = W_g$, $W_{p_1}' = W_{p_2}$, $W_{p_0}' = W_{p_1}$ (here $'$ means the right-hand side of ∂_x). These are the rows of (5.4); $\mathfrak{A}(x;\bar{Y}) = \Sigma\partial_Y F_{\boldsymbol{c}}(x,\bar{Y})\Sigma$ because the Jacobian of a quadratic polynomial is affine, so $\partial_Y F(\bar{Y})$ is the mean value of $\partial_Y F$ along the segment. (All six rows, and the table, are verified by computer algebra in `verify_ch05a.py`, block V1.) □

Lemma 5.28 (comparison for the Q-system)**.** *Let $\boldsymbol{c} \le \hat{\boldsymbol{c}}$ (componentwise) and let $Y, \hat{Y} \in C^1([a,b];\mathbb{R}^6)$, $[a,b] \subset (-1,1)$, solve $\partial_x Y = F_{\boldsymbol{c}}(x,Y)$, $\partial_x \hat{Y} = F_{\hat{\boldsymbol{c}}}(x,\hat{Y})$ with $g, \hat{g}, f, \hat{f} \ge 0$ on $[a,b]$. Put $W = \Sigma(\hat{Y} - Y)$ as above.*

(a) *If $W(a) \ge 0$ then $W \ge 0$ on $[a,b]$; that is $\hat{U} \ge U$, $\hat{g} \le g$, $\hat{f} \le f$, $\hat{p}_k \le p_k$ $(k = 0,1,2)$.*

(b) *If, in addition, $g + \hat{g} > 0$ on $(a,b]$ and ($W(a) \ne 0$ or $\hat{\boldsymbol{c}} \ne \boldsymbol{c}$), then every component of W is > 0 on $(a,b]$.*

Proof. By Lemma 5.27, $\partial_x W = \mathfrak{A}(x)W + \rho$ with $\mathfrak{A}(x) = \mathfrak{A}(x;\bar{Y}(x))$, $\bar{Y} = \frac{1}{2}(Y+\hat{Y})$ and $\rho = \frac{P_{\hat{\boldsymbol{c}}} - P_{\boldsymbol{c}}}{d}\,e_U$. $\mathfrak{A}$ is continuous on $[a,b]$; its off-diagonal entries are $\frac{1}{d}, \frac{\bar{g}}{d}, 1, \frac{2\bar{g}}{d}, \frac{2\bar{f}}{d}, 1, 1$ with $\bar{g} = \frac{1}{2}(g+\hat{g}) \ge 0$, $\bar{f} = \frac{1}{2}(f+\hat{f}) \ge 0$: $\mathfrak{A}$ is Metzler. And $P_{\hat{\boldsymbol{c}}} - P_{\boldsymbol{c}} = (\hat{c}_1 - c_1)(1-x) + (\hat{c}_2 - c_2)(1+x) + (\hat{c}_3 - c_3)(1-x^2) \ge 0$ on $(-1,1)$, so $\rho \ge 0$. (a) is Lemma 5.21(a). (b) The arrows $U \to g$, $g \to f$, $f \to p_2$, $p_2 \to p_1$, $p_1 \to p_0$, $p_0 \to U$ correspond to the entries $\mathfrak{A}_{gU} = \bar{g}/d$, $\mathfrak{A}_{fg} = 1$, $\mathfrak{A}_{p_2,f} = 2\bar{g}/d$, $\mathfrak{A}_{p_1,p_2} = 1$, $\mathfrak{A}_{p_0,p_1} = 1$, $\mathfrak{A}_{U,p_0} = 1/d$, all > 0 on $(a,b]$ because $g + \hat{g} > 0$ there: the graph contains the cycle through all six indices, hence is strongly connected. If $\hat{\boldsymbol{c}} \ne \boldsymbol{c}$, $P_{\hat{\boldsymbol{c}}} - P_{\boldsymbol{c}}$ is a nonzero polynomial (since $1-x, 1+x, 1-x^2$ are linearly independent), so it has finitely many zeros and ρ does not vanish identically on any $[a,x]$, $x > a$. Lemma 5.21(b) applies. □

Corollary 5.29 (ordering of data at a zero of the swirl)**.** *Let $x_0 \in (-1,1)$, $\boldsymbol{c} \le \hat{\boldsymbol{c}}$, $\gamma \le \hat{\gamma}$ and $\kappa \ge \hat{\kappa} \ge 0$, and let $Y, \hat{Y}$ be the solutions of (5.2) for $\boldsymbol{c}, \hat{\boldsymbol{c}}$ with $Y(x_0) =$*

$(\gamma,\kappa,0,0,0,0)$, $\hat{Y}(x_0)=(\hat{\gamma},\hat{\kappa},0,0,0,0)$ *(solutions of the Q-system with a zero of f at x_0 and Taylor constants $\boldsymbol{c},\hat{\boldsymbol{c}}$ there).*

(a) (forward) *On every $[x_0,b]$ on which both solutions exist: $\hat{U}\ge U$, $\hat{g}\le g$, $\hat{f}\le f$, $\hat{\psi}_{xx}\le\psi_{xx}$, $\hat{\psi}_x\le\psi_x$, $\hat{\psi}\le\psi$. If moreover $\kappa>0$ and $(\hat{\gamma},\hat{\kappa},\hat{\boldsymbol{c}})\neq(\gamma,\kappa,\boldsymbol{c})$, all six inequalities are strict on $(x_0,b]$.*

(b) (backward) *If instead $\boldsymbol{c}\le\hat{\boldsymbol{c}}$, $\hat{\gamma}\le\gamma$ and $\kappa\ge\hat{\kappa}\ge0$, then on every $[b,x_0]$ on which both solutions exist: $\hat{U}\le U$, $\hat{g}\le g$, $f\le\hat{f}$ (≤0) and $\hat{\psi}\le\psi$ (and $\hat{\psi}_{xx}\le\psi_{xx}$, $\hat{\psi}_x\ge\psi_x$).*

Proof. (a) Both solutions have $g,\hat{g}\ge0$, $f,\hat{f}\ge0$ and $p_k,\hat{p}_k\ge0$ on the common interval (Lemma 5.25). The order vector at x_0 is $W(x_0)=(\hat{\gamma}-\gamma,\ \kappa-\hat{\kappa},0,0,0,0)\ge0$, so Lemma 5.28(a) gives the inequalities. If $\kappa>0$ then $g=\kappa\exp(-\int_{x_0}U/d)>0$ so $g+\hat{g}>0$, and $(\hat{\gamma},\hat{\kappa},\hat{\boldsymbol{c}})\neq(\gamma,\kappa,\boldsymbol{c})$ means $W(x_0)\neq0$ or $\hat{\boldsymbol{c}}\neq\boldsymbol{c}$; Lemma 5.28(b) gives strictness. (b) Apply (a) to the reflected solutions $Y^\sharp,\hat{Y}^\sharp$ of Lemma 5.24, which solve (5.2) for $\boldsymbol{c}_m\le\hat{\boldsymbol{c}}_m$ on $[-x_0,-b]$ with data $RY(x_0)=(-\gamma,\kappa,0,0,0,0)$, $(-\hat{\gamma},\hat{\kappa},0,0,0,0)$; the hypotheses of (a) hold ($-\gamma\le-\hat{\gamma}$, $\kappa\ge\hat{\kappa}\ge0$). Hence $\hat{U}^\sharp\ge U^\sharp$, i.e. $-\hat{U}(-s)\ge-U(-s)$, which is $\hat{U}\le U$ at $x=-s$; likewise $\hat{g}\le g$ ($g^\sharp=g(-s)$), $-\hat{f}\le-f$, i.e. $f\le\hat{f}$ ($f\le0$ on $[b,x_0]$ since $f(x_0)=0$, $g\ge0$), $\hat{p}_0\le p_0$, $\hat{p}_2\le p_2$ and $-\hat{p}_1\le-p_1$. □

Remark 5.30 (direction of integration)**.** Larger γ, smaller $\kappa\ge0$ and larger $\boldsymbol{c}$ push a solution *up* (larger U) when integrating from a zero of f in the forward direction, and *down* in the backward direction: the sign of the effect of γ on U is the same, but the roles of "up" and "larger γ" are those of the reflected problem. This is what makes FS an up-set and BS a down-set in γ (Proposition 5.42).

5.3.4. In which direction is the system cooperative?

Lemma 5.31 (directions of cooperativity)**.** *Let $x\in(-1,1)$ and let $Y=(U,g,f,p_2,p_1,p_0)$ be a state with $fg\neq0$ and $J=\partial_Y F_{\boldsymbol{c}}(x,Y)$ as above. Call a sign matrix $\Sigma'=\operatorname{diag}(\sigma_i)$* forward-cooperative *at Y if $\sigma_i\sigma_j J_{ij}\ge0$ for all $i\neq j$, and* backward-cooperative *if $\sigma_i\sigma_j(-J_{ij})\ge0$ for all $i\neq j$ (cooperativity of the system with x reversed).*

(a) *If $f>0$, $g>0$ the forward-cooperative sign matrices are exactly $\pm\Sigma$, $\Sigma=\operatorname{diag}(+,-,-,-,-,-)$, and there is no backward-cooperative one. If $f<0$, $g<0$ they are exactly $\pm\operatorname{diag}(+,+,+,-,-,-)$, and none is backward-cooperative.*

(b) *If $f>0>g$ the backward-cooperative sign matrices are exactly $\pm\operatorname{diag}(+,-,+,+,-,+)$; if $f<0<g$ they are exactly $\pm\operatorname{diag}(+,+,-,+,-,+)$; in both cases none is forward-cooperative.*

(c) *In general, $\Sigma J\Sigma$ is Metzler iff $g\ge0$ and $f\ge0$ (also when $fg=0$).*

In words: the forward flow is cooperative exactly when $fg>0$ (the swirl grows in absolute value), the backward flow exactly when $fg<0$.

Proof. All off-diagonal entries of J listed above are nonzero when $fg\neq0$, so each condition fixes a product of signs. Let $s_f=\operatorname{sgn}f$, $s_g=\operatorname{sgn}g$. Forward conditions, edge by edge: (U,p_0): $-\sigma_U\sigma_{p_0}\ge0$, i.e. $\sigma_{p_0}=-\sigma_U$; (g,U): $-s_g\sigma_g\sigma_U\ge0$; (f,g): $\sigma_f\sigma_g=+1$; (p_2,g): $s_f\sigma_{p_2}\sigma_g=+1$; (p_2,f): $s_g\sigma_{p_2}\sigma_f=+1$; (p_1,p_2): $\sigma_{p_1}=\sigma_{p_2}$; (p_0,p_1): $\sigma_{p_0}=\sigma_{p_1}$. Multiply the fourth and fifth conditions: $s_fs_g\,\sigma_g\sigma_f=1$ (as $\sigma_{p_2}^2=1$), and use the third, $\sigma_f\sigma_g=1$: $s_fs_g=1$. So a forward-cooperative

Σ' exists only if $fg > 0$. For $f, g > 0$ the conditions read $\sigma_{p_0} = \sigma_{p_1} = \sigma_{p_2} = -\sigma_U$, $\sigma_g = \sigma_f = \sigma_{p_2}$, $\sigma_g\sigma_U = -1$: the solutions are $\pm(+,-,-,-,-,-)$ (the last condition is then automatic). For $f, g < 0$: $\sigma_f = \sigma_g = -\sigma_{p_2}$ and $\sigma_g\sigma_U = +1$ give $\pm(+,+,+,-,-,-)$. Backward, all five sign conditions with J_{ij} replaced by $-J_{ij}$: $\sigma_{p_0} = \sigma_U$, $\sigma_g\sigma_U = s_g$, $\sigma_f\sigma_g = -1$, $\sigma_{p_2}\sigma_g = -s_f$, $\sigma_{p_2}\sigma_f = -s_g$, $\sigma_{p_1} = -\sigma_{p_2}$, $\sigma_{p_0} = -\sigma_{p_1}$; multiplying the fourth and fifth, $s_f s_g = \sigma_f\sigma_g = -1$: only $fg < 0$ is possible, and then $\sigma_U = \sigma_{p_0} = \sigma_{p_2} =: \sigma$, $\sigma_g = -\sigma s_f$, $\sigma_f = -\sigma s_g$, $\sigma_{p_1} = -\sigma$ solve all conditions. For $f > 0 > g$ this is $\pm(+,-,+,+,-,+)$, for $f < 0 < g$ it is $\pm(+,+,-,+,-,+)$. (c) is read off from (5.3). The enumeration of all 2^6 sign matrices was also done by computer (`verify_ch05a.py`, block V5). □

Remark 5.32 (the triangle)**.** The obstruction in the backward direction at $fg > 0$ is the triangle $g \to f \to p_2 \leftarrow g$ (the entries $J_{fg}, J_{p_2,f}, J_{p_2,g}$ all of the same sign): reversing the direction negates all three edges, and a triangle with three negative edges admits no sign assignment. (The reader can check on the 3×3 model $\partial g = 0$, $\partial f = g$, $\partial p = 2fg$.)

At a base point x_b with data (A, κ), $f = A + \kappa F$ and $g = \kappa y^{-2}$ with $y(x_b) = 1$ and $F(x) = \int_{x_b}^{x} y^{-2}$, which is strictly increasing with $F(x_b) = 0$: so

$$fg = \kappa^2 y^{-2}\Big(\frac{A}{\kappa} + F(x)\Big) \qquad (\kappa \neq 0). \tag{5.5}$$

Proposition 5.33 (directions at a base point)**.** *Let $\kappa \neq 0$ and let the solution exist on the interval considered.*

(a) *Forward: $fg > 0$ on all of $(x_b, 1)$ (where the solution exists) iff $A\kappa \geq 0$.*

(b) *Backward: $fg < 0$ on all of $(-1, x_b)$ iff $A\kappa \leq 0$.*

(c) *If $A\kappa > 0$, the backward flow starts in $\{fg > 0\}$ (not cooperative) and enters $\{fg < 0\}$ only at the zero $x_1 < x_b$ of f (if f has one); symmetrically forward if $A\kappa < 0$. For $A = 0$ both halves are cooperative, with Σ (forward) and the backward matrices of Lemma 5.31(b).*

Proof. Since F is strictly increasing with $F(x_b) = 0$, $F > 0$ for $x > x_b$ and $F < 0$ for $x < x_b$. (a) If $A/\kappa \geq 0$ then $A/\kappa + F(x) > 0$ for $x > x_b$. If $A/\kappa < 0$ then $A/\kappa + F < 0$ for x close to x_b. (b) If $A/\kappa \leq 0$ then $A/\kappa + F < 0$ for $x < x_b$; if $A/\kappa > 0$ it is positive for x near x_b. (c) follows from (5.5): for $A\kappa > 0$, $A/\kappa + F$ changes sign exactly where $F(x_1) = -A/\kappa < 0$, i.e. where $f = 0$. □

Comparison is therefore available on the forward half for $A\kappa \geq 0$ and on the backward half for $A\kappa \leq 0$, and *not* on the other half: pointwise monotonicity of U in γ is in fact false there.

Numerical observation 5.34 (pointwise failure of monotonicity for $A\kappa < 0$)**.** *At the base point $x_b = 0$, for* 600 *random $(\boldsymbol{c}, A, \kappa)$ with $A\kappa < 0$ and pairs $(\gamma, \gamma + 0.05)$ both forward-global (at most three pairs per sample; all* 600 *samples were used and* 1800 *pairs were tested), $U_{\gamma+0.05} < U_\gamma$ at some $x \in (0,1)$ in* 1153 *of the* 1800 *pairs, with worst violation* 0.0274 *(tolerance 10^{-7}; script `code/ch05_global/pointwise.py`, which prints these counts). So whatever replaces the order on γ for $A\kappa < 0$ must be a statement about the* set FS*, not about pointwise order of solutions; compare Atlas 5.73.*

5.4. The midpoint rule in logarithmic variables

In words. *The comparison lemma compares a solution with another* solution. *If we compare a solution with a* sub-solution *— a function V with $\partial_x V = F(x,V)+r$ and defect r of the right sign — the same lemma still works (the defect is a forcing term, Lemma 5.21). The useful sub-solutions are means of two solutions. The arithmetic mean of two solutions fails: the defect of the g-equation has the wrong sign at the very first point. The* geometric *mean for g, f (a linear equation in $\ln g$) and the arithmetic mean for the other variables works, because the defects are squares by AM–GM at three places. This is the whole mechanism behind the log-convexity of the lens (Theorem 12.48).*

Toy example 5.35 ($y' = y^2$: the geometric mean is a supersolution)**.** The solutions of $y' = y^2$ are $y(t;a) = a/(1-at)$, $at < 1$. For $y_i = y(\cdot;a_i)$ the geometric mean $V = \sqrt{y_1 y_2}$ has $V' = \frac12 V(y_1+y_2) \ge V\sqrt{y_1y_2} = V^2$ (AM–GM), a supersolution with $V(0) = \sqrt{a_1a_2}$; comparison (Theorem 5.22, $n=1$) with the true solution through $\sqrt{a_1a_2}$ gives $y(t;\sqrt{a_1a_2}) \le \sqrt{y(t;a_1)y(t;a_2)}$, i.e. $s \mapsto \ln y(t;e^s)$ is midpoint convex. This is Toy example A.27; below the same mechanism is carried out for six components, with the signs of Σ deciding which of V and the true solution is above.

Lemma 5.36 (the midpoint rule: the geometric–arithmetic mean)**.** *Let $x_0 \in (-1,1)$ and for $i = 1,2$ let $\boldsymbol{c}^i \in \mathbb{R}^3$, $\gamma_i \in \mathbb{R}$, $\kappa_i > 0$ and let $Y^i = (U^i, g^i, f^i, p_2^i, p_1^i, p_0^i)$ solve $\partial_x Y^i = F_{\boldsymbol{c}^i}(x, Y^i)$ on $[x_0, b] \subset (-1,1)$ with $Y^i(x_0) = (\gamma_i, \kappa_i, 0, 0, 0, 0)$. Put $\bar{\boldsymbol{c}} = \frac12(\boldsymbol{c}^1 + \boldsymbol{c}^2)$, $\bar\gamma = \frac12(\gamma_1+\gamma_2)$ and*

$$V := \Big(\tfrac{U^1+U^2}{2},\ \sqrt{g^1g^2},\ \sqrt{f^1f^2},\ \tfrac{p_2^1+p_2^2}{2},\ \tfrac{p_1^1+p_1^2}{2},\ \tfrac{p_0^1+p_0^2}{2}\Big).$$

(a) *$V \in C^1([x_0,b])$, $V(x_0) = (\bar\gamma, \sqrt{\kappa_1\kappa_2}, 0,0,0,0)$, and $\partial_x V = F_{\bar{\boldsymbol{c}}}(x,V) + r$ with*

$$r_U = -\frac{(U^1-U^2)^2}{8d} \le 0, \qquad r_g = 0, \qquad r_{p_1} = r_{p_0} = 0,$$

$$r_f = \tfrac12 V_f\Big(\sqrt{\tfrac{g^1}{f^1}} - \sqrt{\tfrac{g^2}{f^2}}\Big)^2 \ge 0, \qquad r_{p_2} = \frac{(\sqrt{f^1g^1} - \sqrt{f^2g^2})^2}{d} \ge 0$$

(for $x > x_0$; r_f extends continuously to x_0 with $r_f(x_0) = 0$).

(b) *Let $\hat Y$ be the maximal solution of $\partial_x \hat Y = F_{\bar{\boldsymbol{c}}}(x, \hat Y)$ with $\hat Y(x_0) = V(x_0)$, defined on $[x_0, \hat b)$. Then on $[x_0, \min(b, \hat b))$*

$$\hat U \ge V_U, \qquad \hat g \le V_g, \quad \hat f \le V_f, \quad \hat p_k \le V_{p_k} \quad (k = 0,1,2).$$

Proof. For $x > x_0$: $g^i = \kappa_i \exp(-\int_{x_0}^x U^i/d) > 0$ and $f^i = \int_{x_0}^x g^i > 0$ (Lemma 5.25). Moreover $f^i = (x-x_0)\varphi_i$ with $\varphi_i = \int_0^1 g^i(x_0 + t(x-x_0))\mathrm{d}t$ smooth and $\varphi_i(x_0) = \kappa_i > 0$, so $\sqrt{f^1f^2} = (x-x_0)\sqrt{\varphi_1\varphi_2}$ is C^1 on $[x_0,b]$; $V \in C^1$ and $V(x_0)$ is as stated. $P_{\boldsymbol{c}}$ is affine in $\boldsymbol{c}$, so $\frac12(P_{\boldsymbol{c}^1} + P_{\boldsymbol{c}^2}) = P_{\bar{\boldsymbol{c}}}$: the parameter enters only through the average. *U.* The terms of F_U other than $-\frac{U^2}{2d}$ are affine in $(U, p_0, P_{\boldsymbol{c}})$ and commute with averaging; and $\frac12\big(\frac{(U^1)^2}{2} + \frac{(U^2)^2}{2}\big) - \frac12\big(\frac{U^1+U^2}{2}\big)^2 = \frac{(U^1-U^2)^2}{8}$. So $\partial_x V_U = \frac12\sum_i F_U(Y^i) = F_U(x,V) - \frac{(U^1-U^2)^2}{8d}$. *$g$.* $\partial_x \ln V_g = \frac12\sum_i \partial_x \ln g^i = -\frac{1}{2d}(U^1+U^2) = -V_U/d$: the equation for g is linear in $\ln g$, and the geometric mean satisfies it exactly: $r_g = 0$. *f.* $\partial_x V_f = V_f \cdot \frac12\big(\frac{g^1}{f^1} + \frac{g^2}{f^2}\big)$, while $F_f(V) = V_g = V_f\sqrt{\frac{g^1g^2}{f^1f^2}}$. With $a_i = \sqrt{g^i/f^i}$: $r_f = V_f \cdot \frac12(a_1^2 + a_2^2 - 2a_1a_2) = \frac12 V_f(a_1-a_2)^2 \ge 0$ (AM–GM).

At x_0: $r_f = \frac{1}{2}\sqrt{\varphi_1\varphi_2}\big(\sqrt{g^1/\varphi_1} - \sqrt{g^2/\varphi_2}\big)^2 \to 0$ since $g^i(x_0) = \varphi_i(x_0) = \kappa_i$. p_2. $\partial_x V_{p_2} = \frac{f^1 g^1 + f^2 g^2}{d}$ while $F_{p_2}(V) = \frac{2V_f V_g}{d} = \frac{2\sqrt{f^1 g^1 f^2 g^2}}{d}$; the difference is $\frac{(\sqrt{f^1 g^1} - \sqrt{f^2 g^2})^2}{d} \geq 0$ (AM–GM again). p_1, p_0. Linear equations: averaging commutes. (All defects are verified by computer algebra in `verify_ch05a.py`, block V2.) This proves (a).

(b) For $x > x_0$ also $\hat{g} > 0$, $\hat{f} > 0$ ($\hat{g}(x_0) = \sqrt{\kappa_1\kappa_2} > 0$, $\hat{f}(x_0) = 0$, $\partial_x \hat{f} = \hat{g}$) and $V_g, V_f > 0$; at x_0, f-components are 0. So $\bar{Y} := \frac{1}{2}(\hat{Y} + V)$ has $\bar{g} \geq 0$, $\bar{f} \geq 0$ on $[x_0, \min(b, \hat{b}))$. Let $W = \Sigma(\hat{Y} - V) \in C^1$. By Lemma 5.27 (same constants $\bar{\boldsymbol{c}}$) and (a),

$$\partial_x W = \Sigma\big(F_{\bar{\boldsymbol{c}}}(\hat{Y}) - F_{\bar{\boldsymbol{c}}}(V)\big) - \Sigma r = \mathfrak{A}(x; \bar{Y})\, W + \rho,$$
$$\rho := -\Sigma r = (-r_U,\ r_g,\ r_f,\ r_{p_2},\ r_{p_1},\ r_{p_0}) = (-r_U, 0, r_f, r_{p_2}, 0, 0) \geq 0,$$

with $\mathfrak{A}$ Metzler and continuous, and $W(x_0) = 0$. By Lemma 5.21(a) on each $[x_0, b']$, $b' < \min(b, \hat{b})$: $W \geq 0$, which is the claim (Σ reverses the sign of all components but U). □

Remark 5.37 (why not the arithmetic mean)**.** Replace $\sqrt{g^1 g^2}$ by $\frac{1}{2}(g^1 + g^2)$ in V. Then $V(x_0)$ has the data $(\bar{\gamma}, \bar{\kappa})$ with $\bar{\kappa} = \frac{1}{2}(\kappa_1 + \kappa_2)$, which is what convexity in κ would need, but the defect of the g-equation is $r_g = -\frac{1}{d}\big(\overline{Ug} - \bar{U}\bar{g}\big) = -\frac{(U^1 - U^2)(g^1 - g^2)}{4d}$, and the comparison needs $r_g \geq 0$ (in the order Σ). At x_0 this equals $-\frac{(\gamma_1 - \gamma_2)(\kappa_1 - \kappa_2)}{4d} < 0$ whenever the pair of data is "increasing", as it is for two points of the lower wall of the lens (γ^- increases with κ): the sign is wrong at the very first point. The geometric mean is the cure; what is lost is the Jensen gap between $\sqrt{\kappa_1\kappa_2}$ and $\frac{1}{2}(\kappa_1 + \kappa_2)$. This is why Theorem 12.48 is a convexity statement in $\ln\kappa$, not in κ.

Remark 5.38 (use in the book)**.** Lemma 5.36 is the engine of Theorem 12.48: if $(\boldsymbol{c}^i, \kappa_i, \gamma_i)$ are *forward admissible* (the solutions exist up to $x = 1$), then so is the data $(\bar{\boldsymbol{c}}, \sqrt{\kappa_1\kappa_2}, \bar{\gamma})$, because $\hat{U} \geq V_U$ and $V_U = \frac{1}{2}(U^1 + U^2)$ stays bounded below, while U of a solution that stops existing tends to $-\infty$ (Lemma 2.24); the passage to the backward half is the reflection of Lemma 5.24. We stop here because the argument needs the admissible sets of the lens, which are introduced in part b of this chapter.

5.5. Safe sets: the forward solution as a half-line in γ

In words. *Fix a base point x_b and the swirl data (A, κ) there, both ≥ 0, and the constants $\boldsymbol{c}$. Which values $\gamma = U(x_b)$ allow the solution to reach the north pole? The only way to fail is for U to run to $-\infty$ at some $x < 1$ (a pole of the Riccati equation), and a larger U at the start is never worse: by the comparison lemma the solution with the larger γ stays above, with less swirl integral ψ. So the* forward-safe set FS *is a closed half-line $[\gamma^-, \infty)$; it is helped by smaller swirl and by larger constants. The endpoint γ^- is the* edge*: strictly inside the half-line the pole $x = 1$ is generic (type* G*), at the edge it is not (type* R *or* $\mathsf{L}_1, \mathsf{L}_2$*), because type* G *is an open condition (Chapter 4). Everything backward follows by the reflection $\sigma_1\sigma_2$. This section proves the half-line principle for every base point with $A \geq 0$, $\kappa \geq 0$; the book uses it for $A = 0$ (a zero of the swirl, class* Z*) and for $x_b = 0$ (the equator chart).*

Conventions. A *base datum* is $p = (x_b, \boldsymbol{c}, A, \kappa) \in (-1,1) \times \mathbb{R}^3 \times [0,\infty) \times [0,\infty)$; $\kappa = \kappa_x = f_x(x_b)$ in the gauge $y(x_b) = 1$. For $\gamma \in \mathbb{R}$ let $Y(\cdot; p, \gamma) = (U, g, f, p_2, p_1, p_0)$ be the maximal solution of the cooperative system (5.2) with $Y(x_b) = (\gamma, \kappa, A, 0, 0, 0)$; by Lemma 5.23 it is the solution of the Q-system with Cauchy data $(\boldsymbol{c}, \gamma, A, \kappa)$ at x_b, with $g = f_x$ and $\psi := p_0 = \psi_{\boldsymbol{c}}$. The linear isomorphism between the state Y and the state $(U, f, f_x, Q, Q_x, Q_{xx})$ of (2.22) maps solutions to solutions, so the two systems have the same maximal intervals. Write $[x_b, X)$, $X = X(p,\gamma) \in (x_b, 1]$, for the forward part of the maximal interval, and

$$\mathrm{FS}(p) := \{\gamma :\ X(p,\gamma) = 1\},$$
$$\mathrm{BS}(p) := \{\gamma :\ \text{solution exists on } (-1, x_b]\}, \qquad \mathrm{GS}(p) := \mathrm{FS}(p) \cap \mathrm{BS}(p),$$

so $\mathrm{GS}(p)$ is the fibre of $\Omega(x_b)$ over $(\boldsymbol{c}, A, \kappa)$. Let $G^+ := \{(p,\gamma) : \gamma \in \mathrm{FS}(p)\}$ and $G^- := \{(p,\gamma) : \gamma \in \mathrm{BS}(p)\}$. Put $P_{\max}(\boldsymbol{c}) := \max_{[-1,1]} |P_{\boldsymbol{c}}|$ and $K_P(\boldsymbol{c}) := 4 + \sqrt{16 + 4P_{\max}(\boldsymbol{c})}$; K_P is continuous in $\boldsymbol{c}$ because $|P_{\max}(\boldsymbol{c}) - P_{\max}(\boldsymbol{c}')| \le 2|c_1 - c_1'| + 2|c_2 - c_2'| + |c_3 - c_3'|$, and $K_P(\boldsymbol{c}_m) = K_P(\boldsymbol{c})$ since $P_{\boldsymbol{c}_m}(x) = P_{\boldsymbol{c}}(-x)$. Continuous dependence (Remark 2.7(iii)) is used for the data $(x_b, \boldsymbol{c}, A, \kappa, \gamma)$ together: $\boldsymbol{c}$ is adjoined to the state as a constant component, and the starting point x_b is allowed to vary.

Toy example 5.39 (the Landau family: FS and BS are half-lines, the edge is the one-ray solution)**.** Take $\boldsymbol{c} = 0$, $\kappa = 0$ (so $f \equiv A$, $Q \equiv 0$, and A plays no role). Then (2.24) reads $y_{xx} = 0$: with $y(x_b) = 1$, $y_x(x_b) = \ell$ we have $y = 1 + \ell(x - x_b)$, and

$$U = \frac{2(1-x^2)\ell}{1 + \ell(x - x_b)}, \qquad \gamma = U(x_b) = 2(1 - x_b^2)\ell$$

(one checks directly that this solves $(1-x^2)U_x + 2xU + \frac{1}{2}U^2 = 0$). The solution exists on $[x_b, 1)$ iff $y > 0$ there, i.e. iff $\ell \ge -\frac{1}{1-x_b}$, i.e. $\gamma \ge -2(1 + x_b)$; and on $(-1, x_b]$ iff $\ell \le \frac{1}{1+x_b}$, i.e. $\gamma \le 2(1 - x_b)$. Hence

$$\mathrm{FS} = [-2(1+x_b), \infty), \qquad \mathrm{BS} = (-\infty, 2(1-x_b)],$$
$$\mathrm{GS} = [-2(1+x_b),\ 2(1-x_b)] \ \ (= [-2, 2] \text{ for } x_b = 0).$$

The edge $\gamma^- = -2(1 + x_b)$ is $y = (1-x)/(1-x_b)$, $U = -2(1+x)$: it vanishes at $x = 1$ (the *recessive* solution, $U(1) = -4 = -2 - 2k_2$, $k_2 = 1$), whereas every solution with $\gamma > \gamma^-$ has $U(1) = 0 = -2 + 2k_2$ (dominant). This is the whole picture of this section in one line; the swirl will move the edge to the right (Proposition 5.42(c)) and change its nature (Lemma 5.46(b)). Cf. Example 5.9.

Lemma 5.40 (reflection: BS from FS)**.** *For every datum $(x_b, \boldsymbol{c}, A, \kappa)$ (any signs of A, κ) and every γ:*

$$\gamma \in \mathrm{BS}(x_b, \boldsymbol{c}, A, \kappa) \iff -\gamma \in \mathrm{FS}(-x_b, \boldsymbol{c}_m, -A, \kappa),$$
$$\text{and} \qquad \mathrm{FS}(x_b, \boldsymbol{c}, A, \kappa) = \mathrm{FS}(x_b, \boldsymbol{c}, -A, -\kappa).$$

Consequently $\mathrm{GS}(x_b, \boldsymbol{c}, A, \kappa) = \mathrm{FS}(x_b, \boldsymbol{c}, A, \kappa) \cap \big(-\mathrm{FS}(-x_b, \boldsymbol{c}_m, -A, \kappa)\big)$, *and G^- is the image of G^+ under the linear homeomorphism $(x_b, \boldsymbol{c}, A, \kappa, \gamma) \mapsto (-x_b, \boldsymbol{c}_m, -A, \kappa, -\gamma)$, which is its own inverse.*

Proof. By Lemma 5.1(d) the solution $\sigma_1\sigma_2 S$ has the data $(\boldsymbol{c}_m, -\gamma, -A, \kappa)$ at $-x_b$ when S has $(\boldsymbol{c}, \gamma, A, \kappa)$ at x_b, and $\sigma_1\sigma_2 S(x) = (-U(-x), -f(-x), Q(-x))$ exists on an interval I iff S exists on $-I$. Taking $I = [-x_b, 1)$ gives the first claim.

The second: $\sigma_1 S$ has the data $(\boldsymbol{c}, \gamma, -A, -\kappa)$ at x_b and the same domain as S. The last statement is the first one read as a map of the data. □

Lemma 5.41 (a priori bounds for the forward solution)**.** *Let $p = (x_b, \boldsymbol{c}, A, \kappa)$ be a base datum, $\gamma \in \mathbb{R}$, $Y = Y(\cdot; p, \gamma)$ with forward interval $[x_b, X)$.*

(a) *$g \geq 0$; $f \geq A \geq 0$ is nondecreasing; $p_2, p_1, p_0 \geq 0$ are nondecreasing. Hence $\psi = p_0 \geq 0$ and $Q = P_{\boldsymbol{c}} - \psi \leq P_{\boldsymbol{c}} \leq P_{\max}$ on $[x_b, X)$.*

(b) *$U \leq \max(\gamma, K_P)$ on $[x_b, X)$.*

(c) *If $U(x_1) < -K_P$ for some $x_1 \in [x_b, X)$, then U is decreasing on $[x_1, X)$, $\operatorname{artanh} X \leq \operatorname{artanh} x_1 + 4/|U(x_1)|$, and in particular $X < 1$.*

(d) *If $X < 1$, then $U(x) \to -\infty$ as $x \uparrow X$.*

(e) *If $X = 1$ and $\kappa > 0$, then $\psi(1) := \lim_{x\to 1} p_0$ exists, $0 < \psi(1) \leq 2 + 2c_2$ (so $c_2 > -1$), and $\psi(1) = \int_{x_b}^1 \frac{1-s}{1+s} f(s) g(s) \mathrm{d}s$.*

Proof. (a) $g(x_b) = \kappa \geq 0$, $f(x_b) = A \geq 0$ and $p_k(x_b) = 0$, so Lemma 5.25(a)–(c) applies. $Q = P_{\boldsymbol{c}} - p_0$ by Lemma 5.23.

Write $N := Q - 2xU - \frac{1}{2}U^2$, so $U_x = N/(1-x^2)$, and by (a) $N \leq P_{\max} + 2|U| - \frac{1}{2}U^2$ ($|x| < 1$).

(b) The positive root of $\frac{1}{2}r^2 - 2r - P_{\max} = 0$ is $2 + \sqrt{4 + 2P_{\max}} < K_P$ (as $\sqrt{16 + 4P_{\max}} \geq \sqrt{4 + 2P_{\max}}$), so $N < 0$, hence $U_x < 0$, wherever $U \geq K_P$. Let $M = \max(\gamma, K_P)$ and suppose $U(x_1) > M$. Then $x_2 := \sup\{x \in [x_b, x_1] : U(x) \leq M\}$ satisfies $U(x_2) = M \geq K_P$ and $U > M$ on $(x_2, x_1]$, whereas $U_x(x_2) < 0$: contradiction.

(c) K_P is also the positive root of $\frac{1}{4}r^2 - 2r - P_{\max} = 0$ (roots $4 \pm \sqrt{16 + 4P_{\max}}$), so for $U \leq -K_P$ we have $\frac{1}{4}U^2 - 2|U| - P_{\max} \geq 0$, i.e. $N \leq -\frac{1}{4}U^2$ and

$$U_x \leq -\frac{U^2}{4(1-x^2)} < 0 \qquad \text{wherever } U \leq -K_P. \tag{5.6}$$

Let $x_3 := \sup\{x \in [x_1, X) : U < -K_P \text{ on } [x_1, x]\}$. On $[x_1, x_3)$, U is decreasing by (5.6); if $x_3 < X$ then $U(x_3) \leq U(x_1) < -K_P$ and by continuity $U < -K_P$ slightly beyond x_3, contradicting the definition. So $x_3 = X$: $U < -K_P$ and $U_x < 0$ on $[x_1, X)$. Then $V := 1/U < 0$ satisfies $V_x = -U_x/U^2 \geq \frac{1}{4(1-x^2)}$ on $[x_1, X)$, and integrating ($\int \frac{\mathrm{d}s}{1-s^2} = \operatorname{artanh}$),

$$0 > V(x) \geq -\frac{1}{|U(x_1)|} + \frac{1}{4}\big(\operatorname{artanh} x - \operatorname{artanh} x_1\big) \qquad (x_1 \leq x < X).$$

So $\operatorname{artanh} x - \operatorname{artanh} x_1 < 4/|U(x_1)|$ for all $x < X$, which is the claim, and $X < 1$ since $\operatorname{artanh} 1 = \infty$.

(d) By Lemma 2.24 (a finite right endpoint $\xi^* = \operatorname{artanh} X$ of the maximal interval) $U \to -\infty$.

(e) By (a), p_0 is nondecreasing, so $\psi(1) = \lim p_0 \in [0, \infty]$ exists. The restriction of Y to $[x_b, 1)$ is a local solution at the north pole (Definition 4.1), so by Theorem 4.5(a) $q = \lim Q$ exists, is finite and ≥ -2; as $P_{\boldsymbol{c}}(1) = 2c_2$ and $Q = P_{\boldsymbol{c}} - p_0$, $\psi(1) = 2c_2 - q \leq 2c_2 + 2$. For $x > x_b$ we have $g > 0$ ($\kappa > 0$) and $f > 0$, so $\partial_x p_2 = 2fg/(1-x^2) > 0$ and then $p_2, p_1, p_0 > 0$ on $(x_b, 1)$: $\psi(1) \geq \psi(x) > 0$. Thus $0 < \psi(1) \leq 2 + 2c_2$, which forces $c_2 > -1$. Finally, Taylor's formula with integral remainder at x_b (where ψ, ψ_x, ψ_{xx} vanish and $\psi_{xxx} = 2fg/(1-s^2)$) gives, for every

datum and every x in the interval,

$$\psi(x) = \int_{x_b}^{x} (x-s)^2 \frac{f(s)g(s)}{1-s^2} \mathrm{d}s; \tag{5.7}$$

the integrand $\mathbf{1}_{s<x}(x-s)^2 \frac{fg}{1-s^2}$ is ≥ 0 and increases with x to $(1-s)^2 \frac{fg}{1-s^2} = \frac{1-s}{1+s} fg$, so monotone convergence gives the formula. □

Strategy. Lemma 5.41 says that a solution which has not died is bounded above and, once U is below $-K_P$, is doomed: this makes "safe" a closed condition. The comparison lemma of Section 5.3 (applied to two solutions with the same swirl data and different γ) makes it an up-set. Strictness of the comparison (the cooperative graph is strongly connected as soon as the swirl is positive) then gives the strict monotonicity of the edge in all parameters, which is what is needed later to turn edges into graphs.

Proposition 5.42 (the forward-safe set is a closed half-line)**.** *Let $p = (x_b, \boldsymbol{c}, A, \kappa)$ be a base datum (so $A \geq 0$, $\kappa \geq 0$).*

(a) *G^+ is closed in $(-1,1) \times \mathbb{R}^3 \times [0,\infty)^2 \times \mathbb{R}$.*

(b) $\mathrm{FS}(p)$ *is empty or a closed half-line $[\gamma^-(p), \infty)$, and $\gamma^-(p) \geq -K_P(\boldsymbol{c})$. Put $\gamma^-(p) := +\infty$ if $\mathrm{FS}(p) = \emptyset$.*

(c) (Monotonicity.) *If $\boldsymbol{c} \leq \boldsymbol{c}'$, $0 \leq A' \leq A$ and $0 \leq \kappa' \leq \kappa$, then*

$$\mathrm{FS}(x_b, \boldsymbol{c}, A, \kappa) \subset \mathrm{FS}(x_b, \boldsymbol{c}', A', \kappa');$$

so γ^- is nonincreasing in $\boldsymbol{c}$ and nondecreasing in A and in κ. Strictly: *if $\mathrm{FS}(p) \neq \emptyset$ and $(\boldsymbol{c}', A', \kappa') \neq (\boldsymbol{c}, A, \kappa)$, then $\gamma^-(x_b, \boldsymbol{c}', A', \kappa') < \gamma^-(x_b, \boldsymbol{c}, A, \kappa)$, provided $\kappa > 0$, or $\kappa = 0$ and $\boldsymbol{c}' \neq \boldsymbol{c}$.*

(d) (Mirror.) *If $A \leq 0$ and $\kappa \geq 0$, then $\mathrm{BS}(x_b, \boldsymbol{c}, A, \kappa) = -\mathrm{FS}(-x_b, \boldsymbol{c}_m, -A, \kappa)$ is empty or a closed half-line $(-\infty, \gamma^+]$, where*

$$\gamma^+(x_b, \boldsymbol{c}, A, \kappa) = -\gamma^-(-x_b, \boldsymbol{c}_m, -A, \kappa) \leq K_P(\boldsymbol{c});$$

the monotonicities of (c) hold with the inequalities for γ^+ reversed (γ^+ is nondecreasing in $\boldsymbol{c}$ and nonincreasing in κ).

By σ_1 (Lemma 5.40) the half-line statement for FS *holds whenever $A\kappa \geq 0$, and the one for* BS *whenever $A\kappa \leq 0$:* FS is an up-set if $A\kappa \geq 0$, BS is a down-set if $A\kappa \leq 0$.

Proof. (a) We show that the complement is open. Let $(p_0, \gamma_0) \notin G^+$, $X_0 := X(p_0, \gamma_0) < 1$. By Lemma 5.41(d), $U \to -\infty$ as $x \uparrow X_0$, so we can choose $x_1 \in (x_b, X_0)$ with $U(x_1) < -K_P(\boldsymbol{c}_0) - 1$. The solution exists on $[x_b - \delta, x_1]$ for some $\delta > 0$ (the maximal interval is open). By continuous dependence, for (p, γ) close to (p_0, γ_0) (with $A, \kappa \geq 0$) the solution $Y(\cdot; p, \gamma)$ exists on $[x_b - \delta, x_1]$, in particular is defined at $x_1 > x_b(p)$, and its first component satisfies $|U^\sharp(x_1) - U(x_1)| < \frac{1}{2}$; and $K_P(\boldsymbol{c}) < K_P(\boldsymbol{c}_0) + \frac{1}{2}$ by continuity of K_P. Then $U^\sharp(x_1) < -K_P(\boldsymbol{c}_0) - \frac{1}{2} < -K_P(\boldsymbol{c})$, and Lemma 5.41(c), applied to the forward solution from the base point $x_b(p) < x_1$, gives $X(p, \gamma) < 1$. So $(p, \gamma) \notin G^+$. (The case $\kappa = 0$ or $\kappa_0 = 0$ is included.)

(b) *Up-closedness.* Let $\gamma \in \mathrm{FS}(p)$, $\gamma' > \gamma$, $Y, \hat{Y}$ the solutions with these γ's and the same p. Both have $g, \hat{g}, f, \hat{f} \geq 0$ (Lemma 5.41(a)), $\boldsymbol{c} = \hat{\boldsymbol{c}}$ and $W(x_b) = \Sigma(\hat{Y} - Y)(x_b) = (\gamma' - \gamma, 0, 0, 0, 0, 0) \geq 0$. By Lemma 5.28(a) on $[x_b, b]$ for every $b < X(p, \gamma')$: $\hat{U} \geq U$. If $X(p, \gamma') < 1$, then $\hat{U} \to -\infty$ there (Lemma 5.41(d)), whereas $\hat{U} \geq U \geq \min_{[x_b, X(p,\gamma')]} U > -\infty$ (U is continuous on $[x_b, 1)$): contradiction.

So $\gamma' \in \mathrm{FS}(p)$. *Closedness* of $\mathrm{FS}(p)$: it is the preimage of G^+ under $\gamma \mapsto (p,\gamma)$. *Lower bound:* if $\gamma < -K_P$ then $U(x_b) < -K_P$ and Lemma 5.41(c) with $x_1 = x_b$ gives $X < 1$. A nonempty closed up-set of $\mathbb{R}$ that is bounded below is $[\min,\infty)$.

(c) *Inclusions.* Let $\gamma \in \mathrm{FS}(p)$ and $p' = (x_b, \boldsymbol{c}', A', \kappa')$ as stated; $Y = Y(\cdot; p, \gamma)$, $\hat{Y} = Y(\cdot; p', \gamma)$, both with $g, f \geq 0$. Then $W(x_b) = (0, \kappa - \kappa', A - A', 0, 0, 0) \geq 0$ and $\boldsymbol{c} \leq \boldsymbol{c}'$, so exactly as in (b) $\hat{U} \geq U$ on $[x_b, X(p', \gamma))$ and $X(p', \gamma) = 1$. *Strictness.* Let $\mathrm{FS}(p) \neq \emptyset$, $\gamma_0 := \gamma^-(p)$, $Y = Y(\cdot; p, \gamma_0)$ (forward global), $Y' = Y(\cdot; p', \gamma_0)$ (forward global by the inclusion). Fix $x_1 \in (x_b, 1)$. *Claim: there is $\delta_0 > 0$ such that for $0 < \delta \leq \delta_0$ the solution $Y'_\delta := Y(\cdot; p', \gamma_0 - \delta)$ exists on $[x_b, x_1]$ and $W(Y, Y'_\delta)(x_1) \geq 0$.* If $\kappa > 0$: $g > 0$ on $[x_b, 1)$ (as $g = \kappa \exp(-\int U/(1-r^2))$), so $g + g' > 0$ on $(x_b, x_1]$, and $W(x_b) \neq 0$ or $\boldsymbol{c}' \neq \boldsymbol{c}$ (this is exactly $(\boldsymbol{c}', A', \kappa') \neq (\boldsymbol{c}, A, \kappa)$); Lemma 5.28(b) shows that all six components of $W(Y, Y')(x_1)$ are > 0. By continuous dependence $Y'_\delta(x_1) \to Y'(x_1)$ as $\delta \to 0$, and strict inequalities survive small perturbations. If $\kappa = 0$, $\boldsymbol{c}' \neq \boldsymbol{c}$: then A is irrelevant (here $g \equiv 0$, $f \equiv A$, $p_k \equiv 0$, and A does not enter the equation of U), so we may take $A' = A$; Y and Y'_δ have $g = \hat{g} = 0$, $f = \hat{f}$, $p_k = \hat{p}_k = 0$, hence $W = (d, 0, 0, 0, 0, 0)$ with $d = \hat{U} - U$ for all δ; by Lemma 5.27 $d_x = a(x)d + (P_{\boldsymbol{c}'} - P_{\boldsymbol{c}})/(1 - x^2)$ with a continuous coefficient a and $P_{\boldsymbol{c}'} - P_{\boldsymbol{c}} = \sum_i (c'_i - c_i)\,\phi_i > 0$ on $(-1, 1)$ ($\phi_i = 1 - x, 1 + x, 1 - x^2$; all $c'_i - c_i \geq 0$, not all 0); with $\alpha(x) = \exp\int_{x_b}^{x} a$, variation of constants gives $d(x_1) = \alpha(x_1)\big[d(x_b) + \int_{x_b}^{x_1} \alpha(s)^{-1} \frac{(P_{\boldsymbol{c}'} - P_{\boldsymbol{c}})(s)}{1 - s^2}\mathrm{d}s\big]$, where $d(x_b) = 0$ and the integrand is positive: $d(x_1) > 0$ for $\delta = 0$, and by continuous dependence $d(x_1) > 0$ persists for small δ. *Conclusion.* Apply Lemma 5.28(a) on $[x_1, b]$, $b < X(p', \gamma_0 - \delta)$, to Y and Y'_δ (both have $g, f \geq 0$; $\boldsymbol{c} \leq \boldsymbol{c}'$; $W(x_1) \geq 0$): $\hat{U}_\delta \geq U$, and as in (b) $X(p', \gamma_0 - \delta) = 1$. So $\gamma_0 - \delta \in \mathrm{FS}(p')$ and $\gamma^-(p') \leq \gamma_0 - \delta < \gamma^-(p)$.

(d) By Lemma 5.40, $\mathrm{BS}(x_b, \boldsymbol{c}, A, \kappa) = -\mathrm{FS}(-x_b, \boldsymbol{c}_m, -A, \kappa)$, and the datum $(-x_b, \boldsymbol{c}_m, -A, \kappa)$ is a base datum iff $A \leq 0$. Apply (b), (c) to it. $K_P(\boldsymbol{c}_m) = K_P(\boldsymbol{c})$ and $\boldsymbol{c} \leq \boldsymbol{c}' \Rightarrow \boldsymbol{c}_m \leq \boldsymbol{c}'_m$. The final sentence: if $A\kappa \geq 0$ with $A, \kappa \leq 0$ use $\mathrm{FS}(x_b, \boldsymbol{c}, A, \kappa) = \mathrm{FS}(x_b, \boldsymbol{c}, -A, -\kappa)$; for BS with $A\kappa \leq 0$ and $A \geq 0 \geq \kappa$ the same. □

Remark 5.43 (which half-line, when)**.** Collecting Proposition 5.33, Lemma 5.40 and Proposition 5.42 for a datum $(x_b, \boldsymbol{c}, A, \kappa)$ (all signs):

signs of (A, κ)	forward set FS	backward set BS
$A \geq 0$, $\kappa \geq 0$	up-set $[\gamma^-, \infty)$	no statement
$A \leq 0$, $\kappa \leq 0$	up-set $[\gamma^-, \infty)$ (via σ_1)	no statement
$A \leq 0$, $\kappa \geq 0$	no statement	down-set $(-\infty, \gamma^+]$
$A \geq 0$, $\kappa \leq 0$	no statement	down-set (via σ_1)

At $A = 0$ both halves are covered, which is why the class-Z fibre is the closed interval $[\gamma^-, \gamma^+]$ (Definition 5.55); at $A\kappa \neq 0$ only one half of GS is controlled by comparison.

Remark 5.44 (reading Proposition 5.42; what is and is not claimed)**.** In words: forward existence is helped by a larger U at the start, by smaller swirl (κ or A) and by larger constants $\boldsymbol{c}$ — all three have the same cause: the right-hand side of the U-equation is increasing in Q, swirl lowers Q ($\psi \geq 0$), larger $\boldsymbol{c}$ raises $P_{\boldsymbol{c}}$, and the only danger is $U \to -\infty$. The proposition is stated for *every* base point with $A \geq 0$, $\kappa \geq 0$; the book uses it only for $A = 0$ (any $x_b = x_0$: the lens of Section 5.6) and

for $x_b = 0$ with $A \geq 0$ (the equator chart); the proof is the same. For $A\kappa < 0$ *no* such statement is available: pointwise monotonicity of U in γ fails there (Numerical observation 5.34), and whether FS is a half-line is not known (Chapter 13). In Landau's example ($K_P = 8$) it is far from sharp.

Toy example 5.45 (the edge for a fixed potential: the lowest positive solution is the recessive one)**.** Freeze the potential at its value at the pole, $W \equiv \mu^2 > 0$, and consider positive solutions of $z'' = \mu^2 z$ through $z(\xi_b) = 1$ with slope β. Then $z = \cosh\mu(\xi - \xi_b) + \frac{\beta}{\mu}\sinh\mu(\xi - \xi_b)$ is positive on $[\xi_b, \infty)$ iff $\beta \geq -\mu$; for $\beta > -\mu$ it grows like $e^{\mu\xi}$ (dominant), and for $\beta = -\mu$ it is $e^{-\mu(\xi-\xi_b)}$ (recessive), the lowest positive solution through the point. Since $U = 2(h - x)$ with $h = z'/z$, the safe set in γ is $[\gamma^-, \infty)$ with γ^- the recessive value, and the edge has $h \to -\mu$, $U(1) = -2 - 2\mu$; every interior point has $h \to +\mu$. A swirl adds $f' = \kappa z^{-2}$, and for the recessive solution $\int^\infty z^{-2} = \infty$: the swirl diverges at the edge ($f(1) = +\infty$, type R), while it stays bounded in the interior. This is the content of Lemma 5.46 (and, for $\kappa = 0$, part (c) is exactly this picture with $W_{\boldsymbol{c}}$ in place of μ^2).

Lemma 5.46 (the edge of the safe set: generic inside, not generic on the edge)**.** *Let $p = (x_b, \boldsymbol{c}, A, \kappa)$ be a base datum with $\mathrm{FS}(p) \neq \emptyset$, $\gamma^- = \gamma^-(p)$, and for $\gamma \in \mathrm{FS}(p)$ let S_γ be the forward solution.*

(a) (Interior.) *Let $\kappa > 0$ and $\gamma > \gamma^-$. Then S_γ is of type G at $x = 1$ (so $c_2 > -1$ and $U(1) = -2 + 2\mu > -2$, $\mu > 0$).*

(b) (Edge.) *Let $\kappa > 0$. Then S_{γ^-} is* not *of type G; it is of type R (then $U(1) \in (-3, -2)$, $\mu < \frac{1}{2}$, $f(1) = +\infty$) or of type $\mathsf{L}_1, \mathsf{L}_2$ (then $Q(1) = -2$, $U(1) = -2$ and $\psi(1) = 2 + 2c_2$).*

(c) (No swirl.) *Let $\kappa = 0$ and $c_2 > -1$. Then $\mathrm{FS}(p) = [\gamma^-, \infty)$ with $\gamma^- = \gamma^-(x_b, \boldsymbol{c})$ independent of A; S_γ is the no-swirl solution with $U(1) = -2 + 2k_2 > -2$ for $\gamma > \gamma^-$ and $U(1) = -2 - 2k_2$ for $\gamma = \gamma^-$. If moreover $\boldsymbol{c} \in \mathcal{J}$ and $c_1 > -1$, then $\gamma^- = U^-_{\boldsymbol{c}}(x_b)$ and (by Lemma 5.40) $\gamma^+(x_b, \boldsymbol{c}, A, 0) = U^+_{\boldsymbol{c}}(x_b)$, so $\mathrm{GS} = [U^-_{\boldsymbol{c}}(x_b), U^+_{\boldsymbol{c}}(x_b)]$ (the window of Theorem 3.44).*

Proof. (a) Take $\gamma_1 \in (\gamma^-, \gamma)$, let Y_1 and $\hat{Y}$ be the solutions with γ_1 and γ (both global forward, $\gamma_1 \in \mathrm{FS}(p)$ by Proposition 5.42(b)), and $W = \Sigma(\hat{Y} - Y_1)$. They have the same $\boldsymbol{c}, A, \kappa$, $g, \hat{g} > 0$ and $W(x_b) = (\gamma - \gamma_1, 0, \dots, 0) \neq 0$, so by Lemma 5.28(b) all components of W are positive on $(x_b, 1)$. Fix $x_1 \in (x_b, 1)$ and put $\delta := W_{p_0}(x_1) = \psi_1(x_1) - \hat{\psi}(x_1) > 0$. Since $\partial_x W_{p_0} = W_{p_1} > 0$, W_{p_0} is increasing, so $\psi_1(1) - \hat{\psi}(1) \geq \delta$. By Lemma 5.41(e) both limits exist and, with $q = 2c_2 - \psi(1)$, $\hat{q} \geq q_1 + \delta \geq -2 + \delta$. Hence $\hat{q} > -2$, $\hat{\mu} = \sqrt{1 + \hat{q}/2} > 0$, and $\hat{\mu} > \mu_1$. Moreover $W_U > 0$ gives $\hat{U} \geq U_1$ on $[x_b, 1)$, so $\hat{U}(1) \geq U_1(1)$. By Theorem 4.5(b), $U_1(1) \geq -2 - 2\mu_1 > -2 - 2\hat{\mu}$; if $\hat{U}(1) = -2 - 2\hat{\mu}$ this contradicts $\hat{U}(1) \geq U_1(1)$. Hence $\hat{U}(1) = -2 + 2\hat{\mu}$, i.e. $h \to \hat{\mu} > 0$ and $\hat{q} > -2$: type G (Definition 4.16).

(b) Suppose S_{γ^-} were of type G. Choose $x_c \geq \max(x_b, 0)$ and consider the scale-free state of S_{γ^-} at $\xi_c = \operatorname{artanh} x_c$; it lies in the open set $\mathcal{G}_{\xi_c}$ of Corollary 4.42. The state at x_c depends continuously on γ (continuous dependence on $[x_b, x_c]$), and the passage from the data at x_c to the scale-free state is an affine isomorphism (Remark 5.14); so for γ slightly below γ^- the state at ξ_c is in $\mathcal{G}_{\xi_c}$: the solution exists on $[x_c, 1)$ and, as it exists on $[x_b, x_c]$ by the same continuity, $\gamma \in \mathrm{FS}(p)$, contradicting the definition of γ^-. Hence S_{γ^-} is not of type G; the types are exhaustive (Definition 4.16), so it is R, L_1 or L_2. The values for R are those

of Corollary 4.21, $f(1) = +\infty$ by Proposition 4.19(b); for $\mathsf{L}_1, \mathsf{L}_2$, $q = -2$ and $\psi(1) = 2c_2 - q = 2c_2 + 2$.

(c) Let $\kappa = 0$: $g \equiv 0$, $f \equiv A$, $p_k \equiv 0$, so $Q = P_{\boldsymbol{c}}$ and S_γ is the solution of (3.6) with $U(x_b) = \gamma$ (here A plays no role, so γ^- does not depend on it). With $z = y\cosh\xi$, $z(\xi_b) = 1$, it corresponds to the positive solution of $z'' = W_{\boldsymbol{c}} z$, $W_{\boldsymbol{c}} = 1 + \frac{1}{2}P_{\boldsymbol{c}}$, with $z'(\xi_b) = \frac{\gamma}{2} + x_b$ (Lemma 3.40); it exists on $[x_b, 1)$ iff $z > 0$ on $[\xi_b, \infty)$. As $\xi \to \infty$, $W_{\boldsymbol{c}} \to k_2^2 > 0$ with $W_{\boldsymbol{c}} - k_2^2 = O(e^{-2\xi})$, so by Lemma 3.16(a) there are solutions $z_\pm \sim e^{\pm k_2 \xi}$, $\mathrm{Wr}(z_-, z_+) =: w > 0$ (constant, computed at ∞: $\to 2k_2$), and z_- is, up to a factor, the only recessive solution. Every solution is $z = \alpha z_+ + \beta z_-$, and $z > 0$ near ∞ forces $\alpha \geq 0$, with $\alpha = 0$ iff z is a multiple of z_-. *Dominance for $\gamma > \gamma^-$.* Let $\gamma > \gamma^-$ and z_1, z_2 the solutions with $\gamma_1 = \gamma^-$, $\gamma_2 = \gamma$; both are positive on $[\xi_b, \infty)$ and $\mathrm{Wr}(z_1, z_2) = z_2'(\xi_b) - z_1'(\xi_b) = \frac{1}{2}(\gamma - \gamma^-) > 0$. In terms of coefficients, $\mathrm{Wr}(z_1, z_2) = (\alpha_1\beta_2 - \beta_1\alpha_2)\,\mathrm{Wr}(z_+, z_-) = (\beta_1\alpha_2 - \alpha_1\beta_2)\,w$. If $\alpha_2 = 0$ we would get $-\alpha_1\beta_2 w > 0$, so $\alpha_1\beta_2 < 0$; as $\alpha_1 \geq 0$ and $\beta_2 > 0$ ($z_2 = \beta_2 z_-$ is positive) this is impossible. So $\alpha_2 > 0$: S_γ is dominant, $U(1) = -2 + 2k_2$. *The edge is recessive.* Suppose $\alpha_1 > 0$ for $z_1 = z_{\gamma^-}$. Let v be the solution with $v(\xi_b) = 0$, $v'(\xi_b) = 1$. There is ξ_2 with $z_1 \geq \frac{\alpha_1}{2} z_+ \geq c e^{k_2\xi}$ on $[\xi_2, \infty)$ ($c > 0$), while $|v| \leq C e^{k_2\xi}$ there; and $z_1 \geq m > 0$, $|v| \leq C$ on $[\xi_b, \xi_2]$. For $\varepsilon < \min(c/C, m/C)$, $z_1 - \varepsilon v > 0$ on $[\xi_b, \infty)$, a positive solution with $z(\xi_b) = 1$ and slope $z_1'(\xi_b) - \varepsilon$, i.e. $\gamma^- - 2\varepsilon \in \mathrm{FS}(p)$: contradicting minimality. So $\alpha_1 = 0$ and $U(1) = -2 - 2k_2$ at the edge. *The window.* Let $\boldsymbol{c} \in \mathcal{J}$, $c_1 > -1$. Then $p_+ > 0$ on all of $\mathbb{R}$ is recessive at $+\infty$ (Lemma 3.42, Lemma 3.21) and $U^-_{\boldsymbol{c}} = 2(\zeta_+ - \tanh\xi)$ (Theorem 3.44). Its solution $z = p_+/p_+(\xi_b)$ is positive on $[\xi_b, \infty)$, hence $U^-_{\boldsymbol{c}}(x_b) \in \mathrm{FS}(p)$. The edge solution is recessive and positive on $[\xi_b, \infty)$, hence a multiple of z_- and of p_+, so it has the same normalised slope: $\gamma^- = U^-_{\boldsymbol{c}}(x_b)$. The statement for γ^+ is the mirror (Lemma 5.40, $c_1 \leftrightarrow c_2$). □

5.5.1. Small swirl is harmless.

Lemma 5.47 (a dominant no-swirl solution survives a small perturbation of $\boldsymbol{c}$)**.** *Let $x_b \in (-1, 1)$, $\boldsymbol{c} \in \mathbb{R}^3$ with $c_2 > -1$ and $\gamma > \gamma^-(x_b, \boldsymbol{c}, 0, 0)$ (so the no-swirl solution $U_{\boldsymbol{c},\gamma}$ with $U(x_b) = \gamma$ is forward global and dominant at $x = 1$, Lemma 5.46(c)). Then for all small $e > 0$, $\tilde{\boldsymbol{c}} := \boldsymbol{c} - \frac{e}{2}(1, 1, 0)$ has $\tilde{c}_2 > -1$ and the no-swirl solution for $\tilde{\boldsymbol{c}}$ with the same value γ at x_b is forward global and dominant: $\tilde{U}(1) = -2 + 2\tilde{k}_2$.*

Proof. Let $z > 0$ solve $z'' = Wz$, $W = 1 + \frac{1}{2}P_{\boldsymbol{c}}$, and $h = z'/z$, so $h' = W - h^2$, $h(\xi_b) = \frac{\gamma}{2} + x_b$ (Lemma 3.40); dominance means $h \to k_2 := \sqrt{1 + c_2} > 0$. Hence $M := \sup_{[\xi_b, \infty)} |h| < \infty$ and there is ξ_1 with $h \geq \frac{3}{4}k_2$ on $[\xi_1, \infty)$. Since $P_{\boldsymbol{c}} - P_{\tilde{\boldsymbol{c}}} = e$, the solution $\tilde{h}$ of $\tilde{h}' = W - \frac{e}{2} - \tilde{h}^2$ with the same initial value satisfies, for $d := h - \tilde{h}$ on its interval of existence, $d(\xi_b) = 0$ and

$$d' = \tfrac{e}{2} - (2h - d)d = \tfrac{e}{2} - 2hd + d^2.$$

At $d = 0$, $d' = \frac{e}{2} > 0$, so $d \geq 0$. While $d \leq 1$ on $[\xi_b, \xi_1]$: $d' \leq \frac{e}{2} + (2M + 1)d$, hence $d \leq \frac{e}{2}(\xi_1 - \xi_b)e^{(2M+1)(\xi_1 - \xi_b)} =: C_1 e$. On $[\xi_1, \infty)$, while $d \leq \frac{k_2}{4}$: $d' \leq \frac{e}{2} - \frac{3}{2}k_2 d + \frac{k_2}{4}d \leq \frac{e}{2} - k_2 d$, so d cannot exceed $\max(C_1 e, \frac{e}{2k_2})$. For e so small that this maximum is $< \frac{k_2}{4}$, a continuation argument gives $0 \leq d \leq \frac{k_2}{4}$ on the whole interval of existence of $\tilde{h}$. Thus $\tilde{h} \geq h - \frac{k_2}{4} \geq -M - 1$ is bounded below, so $\tilde{h}$ does not blow down and exists on $[\xi_b, \infty)$, and $\tilde{h} \geq \frac{1}{2}k_2 > 0$ on $[\xi_1, \infty)$. By Theorem 4.5(b) (for the no-swirl solution $\tilde{z} = \exp\int \tilde{h}$), $\tilde{h} \to \pm\tilde{k}_2$, $\tilde{k}_2 = \sqrt{1 + \tilde{c}_2} > 0$ for e small; positivity of the limit forces $+\tilde{k}_2$: dominant. □

Lemma 5.48 (small swirl is harmless)**.** *Let $x_b \in (-1,1)$, $\tilde{\boldsymbol{c}} \in \mathbb{R}^3$ with $\tilde{c}_2 > -1$, $\boldsymbol{c} \in \mathbb{R}^3$ and $e > 0$ with $P_{\boldsymbol{c}} - P_{\tilde{\boldsymbol{c}}} \geq e$ on $(-1,1)$, and let $\tilde{\gamma} \in \mathbb{R}$ be such that the no-swirl solution $\tilde{U}$ for $\tilde{\boldsymbol{c}}$ with $\tilde{U}(x_b) = \tilde{\gamma}$ is forward global and dominant at $x = 1$ ($\tilde{\gamma} > \gamma^-(x_b, \tilde{\boldsymbol{c}}, 0, 0)$; for $\tilde{\boldsymbol{c}} \in \operatorname{int} \mathcal{J}$ this holds for every $\tilde{\gamma} \in (U^-_{\tilde{\boldsymbol{c}}}(x_b), U^+_{\tilde{\boldsymbol{c}}}(x_b))$). Let*

$$\tilde{g}(x) := \exp\Big(-\int_{x_b}^{x} \frac{\tilde{U}}{1-r^2}\mathrm{d}r\Big), \qquad \tilde{f}(x) := \int_{x_b}^{x} \tilde{g},$$

$$\tilde{F} := \tilde{f}(1), \qquad B_+ := \int_{x_b}^{1} \frac{1-s}{1+s}\, \tilde{f}\, \tilde{g}\mathrm{d}s.$$

Then $\tilde{F}$ is finite, $B_+ \leq \frac{1}{2}\tilde{F}^2 < \infty$, and the following holds for every datum $\omega = (\boldsymbol{c}, \gamma, A, \kappa)$ at x_b with $\gamma \geq \tilde{\gamma}$ and $\mathcal{E}(\omega) := |\kappa|\big(|A|\tilde{F} + |\kappa| B_+\big) < e$: the forward solution exists on $[x_b, 1)$ (so $\gamma \in \mathrm{FS}(x_b, \boldsymbol{c}, A, \kappa)$), and on it

$$U \geq \tilde{U}, \qquad |g| \leq |\kappa|\, \tilde{g}, \qquad |f| \leq |A| + |\kappa| \tilde{f}, \qquad |\psi| \leq \mathcal{E}(\omega) < e. \tag{5.8}$$

At $x = 1$ it is of type G with $U(1) = -2 + 2\mu$, $\mu \geq \tilde{k}_2 := \sqrt{1 + \tilde{c}_2} > 0$ (for $\kappa \neq 0$; for $\kappa = 0$ it is the no-swirl solution).

Proof. *Finiteness.* By dominance $\tilde{U}(1) = -2 + 2\tilde{k}_2$, i.e. $z = y\cosh\xi \sim a e^{\tilde{k}_2 \xi}$ with $a > 0$ (Theorem 4.5(b), Lemma 3.16). With $\mathrm{d}x = \operatorname{sech}^2 \xi \mathrm{d}\xi$ and $\tilde{g} = y^{-2}$ ($y(x_b) = 1$, so $z(\xi_b) = \cosh \xi_b$), $\tilde{g}\mathrm{d}x = y^{-2}\operatorname{sech}^2 \xi \mathrm{d}\xi = \mathrm{d}\xi / z^2$, so $\tilde{F} = \int_{\xi_b}^{\infty} \mathrm{d}\xi / z^2 < \infty$. Further $\tilde{f}\tilde{g} = \frac{1}{2}(\tilde{f}^2)'$ and $0 < \frac{1-s}{1+s} \leq 1$ give $B_+ \leq \frac{1}{2}\tilde{F}^2$.

Bootstrap. Let $[x_b, X)$ be the forward interval of the solution through ω and $X_* := \sup\{x < X : |\psi| < e \text{ on } [x_b, x]\} > x_b$ ($\psi(x_b) = 0$). On $[x_b, X_*)$ we have $Q = P_{\boldsymbol{c}} - \psi \geq P_{\boldsymbol{c}} - e \geq P_{\tilde{\boldsymbol{c}}}$. The difference $D = U - \tilde{U}$ satisfies $D_x = a(x)D + (Q - P_{\tilde{\boldsymbol{c}}})/(1-x^2)$ with continuous $a = -(2x + \frac{1}{2}(U + \tilde{U}))/(1-x^2)$, $D(x_b) = \gamma - \tilde{\gamma} \geq 0$ and nonnegative forcing; by variation of constants $D \geq 0$: $U \geq \tilde{U}$. Then $|g| = |\kappa| \exp(-\int_{x_b}^{x} U/(1-r^2)) \leq |\kappa|\tilde{g}$, and $|f| \leq |A| + \int |g| \leq |A| + |\kappa|\tilde{f}$. By Taylor's formula (5.7) (valid for any datum), for $x < X_*$,

$$\begin{aligned} |\psi(x)| &\leq \int_{x_b}^{x} (x-s)^2 \frac{|f||g|}{1-s^2}\mathrm{d}s \leq \int_{x_b}^{1} \frac{1-s}{1+s}\big(|A| + |\kappa|\tilde{f}\big)|\kappa|\tilde{g}\mathrm{d}s \\ &\leq |\kappa||A|\tilde{F} + \kappa^2 B_+ = \mathcal{E}(\omega), \end{aligned}$$

using $(x-s)^2 \leq (1-s)^2$ and $\int \frac{1-s}{1+s}\tilde{g} \leq \tilde{F}$. If $X_* < X$, then by continuity $|\psi(X_*)| \leq \mathcal{E}(\omega) < e$, so $|\psi| < e$ slightly beyond X_*: contradiction. Hence $X_* = X$ and (5.8) holds on $[x_b, X)$. If $X < 1$, Lemma 2.24 gives $U \to -\infty$ as $x \uparrow X$ (this holds for data of any sign), impossible since $U \geq \tilde{U} \geq \min_{[x_b, X]} \tilde{U}$. So $X = 1$.

Type. $U(1) \geq \tilde{U}(1) = -2 + 2\tilde{k}_2 > -2$, and $U(1) \in \{-2 \pm 2\mu\}$ (Theorem 4.5(b)), so $U(1) = -2 + 2\mu$ with $\mu \geq \tilde{k}_2 > 0$: type G. □

Remark 5.49 (the same estimate, two uses)**.** With $A = 0$, $\kappa > 0$: $\mathcal{E} = \kappa^2 B_+$, so the lens contains $\{\gamma\} \times (0, \sqrt{e/B_+})$ for every admissible γ of $\tilde{\boldsymbol{c}}$ — this is Proposition 5.58(c). With κ, A small and arbitrary sign, (5.8) gives the *uniform* domination $g/\kappa = y^{-2} \leq \tilde{g}$ used in the proof of Proposition 5.67. The point of the proof is that no cooperativity is needed: only the scalar Riccati comparison, because the smallness of ψ is closed by continuation.

Proposition 5.50 (semicontinuity and continuity of the thresholds)**.** *Let $\mathcal{D} := \{p = (x_b, \boldsymbol{c}, A, \kappa) : \ A \geq 0, \kappa \geq 0, \ \mathrm{FS}(p) \neq \emptyset\}$ and $\gamma^- : \mathcal{D} \to [-K_P, \infty)$.*

(a) *$\mathcal{D}$ is closed in $(-1,1)\times\mathbb{R}^3\times[0,\infty)^2$ and γ^- is lower semicontinuous on $\mathcal{D}$ in all variables $(x_b,\boldsymbol{c},A,\kappa)$.*

(b) *γ^- is continuous at every $p\in\mathcal{D}$ with $\kappa>0$, and at every $p\in\mathcal{D}$ with $\kappa=0$ and $c_2>-1$; here continuity is for $p'\to p$ inside $\mathcal{D}$, and the* points of a neighbourhood of p in $\{A\ge 0,\kappa\ge 0\}$ all belong to $\mathcal{D}$.

(c) *Consequently, for $\boldsymbol{c}\in\mathcal{J}$ with $c_1,c_2>-1$ and any $A\ge 0$,*

$$\lim_{\kappa\downarrow 0}\gamma^-(x_b,\boldsymbol{c},A,\kappa)=U^-_{\boldsymbol{c}}(x_b),\qquad \lim_{\kappa\downarrow 0}\gamma^+(x_b,\boldsymbol{c},-A,\kappa)=U^+_{\boldsymbol{c}}(x_b).$$

Proof. (a) $\mathcal{D}$ is the projection of the closed set G^+ (Proposition 5.42(a)) along γ; since $\gamma^-\ge -K_P(\boldsymbol{c})$ and K_P is locally bounded, the fibre-points $\gamma\in[\gamma^-,\gamma^-+1]$ stay in a compact set locally, so the projection is closed. If $p_n\to p$, $p_n\in\mathcal{D}$, and $\ell:=\liminf\gamma^-(p_n)<\infty$, choose a subsequence with $\gamma^-(p_n)\to\ell\in\mathbb{R}$ ($\ell\ge -K_P$); $(p_n,\gamma^-(p_n))\in G^+$ closed gives $(p,\ell)\in G^+$, so $p\in\mathcal{D}$ and $\gamma^-(p)\le\ell$.

(b) By (a) it remains to prove $\limsup\gamma^-(p')\le\gamma^-(p)$ as $p'\to p$ in $\mathcal{D}$ (and, for the last clause, that all p' near p lie in $\mathcal{D}$). Let $\gamma>\gamma^-(p)$. *Case* $\kappa>0$. By Lemma 5.46(a) $S_\gamma(p)$ is of type G. Take $x_c\ge\max(x_b,0)$, $x_c>x_b$, and let $\Phi(p',\gamma')$ be the scale-free state at $\xi_c=\operatorname{artanh}x_c$ of the solution with data (p',γ') at x_b'; Φ is defined and continuous near (p,γ) by continuous dependence on $(x_b,\boldsymbol{c},A,\kappa,\gamma)$ jointly (Remark 2.7(iii)), and $\Phi(p,\gamma)\in\mathcal{G}_{\xi_c}$ (Remark 5.14). Since $\mathcal{G}_{\xi_c}$ is open (Corollary 4.42) and $\kappa'>0$ near p, for p' near p the solution with data (p',γ) exists on $[x_b',x_c]$ and on $[x_c,1)$: $\gamma\in\mathrm{FS}(p')$, so $\gamma^-(p')\le\gamma$. *Case* $\kappa=0$, $c_2>-1$. Then $\gamma>\gamma^-(p)=\gamma^-(x_b,\boldsymbol{c},0,0)$ (γ^- does not depend on A, Lemma 5.46(c)). Fix $\tilde\gamma\in(\gamma^-(p),\gamma)$ and, by Lemma 5.47, $e>0$ such that $\tilde{\boldsymbol{c}}=\boldsymbol{c}-\frac{e}{2}(1,1,0)$ has $\tilde c_2>-1$ and the no-swirl solution $\tilde U$ for $\tilde{\boldsymbol{c}}$ with $\tilde U(x_b)=\tilde\gamma$ is forward global and dominant; let $\tilde F,B_+$ be as in Lemma 5.48. For $p'=(x_b',\boldsymbol{c}',A',\kappa')$ close to p let $\omega'=(\boldsymbol{c}'',\gamma'',A'',\kappa'')$ be the data at x_b of the solution with data (p',γ) at x_b'; it exists near x_b and $\omega'\to(\boldsymbol{c},\gamma,A,0)$ by continuous dependence, and the solution is forward global iff the one from ω' is. For p' close enough, $P_{\boldsymbol{c}''}-P_{\tilde{\boldsymbol{c}}}\ge\frac{e}{2}$, $\gamma''\ge\tilde\gamma$ and $\mathcal{E}(\omega')=|\kappa''|(|A''|\tilde F+|\kappa''|B_+)<\frac{e}{2}$, so Lemma 5.48 (with $e/2$) shows that the solution exists on $[x_b,1)$: $\gamma\in\mathrm{FS}(p')$ and $\gamma^-(p')\le\gamma$. In particular all p' near p belong to $\mathcal{D}$. Since $\gamma>\gamma^-(p)$ was arbitrary, $\limsup\gamma^-(p')\le\gamma^-(p)$.

(c) At $\kappa=0$, $c_2>-1$, $\gamma^-(x_b,\boldsymbol{c},A,0)=U^-_{\boldsymbol{c}}(x_b)$ by Lemma 5.46(c), and $(x_b,\boldsymbol{c},A,0)\in\mathcal{D}$. Apply (b). For γ^+ use Lemma 5.40: $\gamma^+(x_b,\boldsymbol{c},-A,\kappa)=-\gamma^-(-x_b,\boldsymbol{c}_m,A,\kappa)$, and $\boldsymbol{c}_m$ has second entry $c_1>-1$, $U^-_{\boldsymbol{c}_m}(-x_b)=-U^+_{\boldsymbol{c}}(x_b)$ by the reflection. □

Corollary 5.51 (the edge solutions do not converge uniformly as $\kappa\to 0$)**.** *Let $\boldsymbol{c}\in\operatorname{int}\mathcal{J}$ with $c_2>-\frac34$ ($k_2>\frac12$), $A\ge 0$, and for small $\kappa>0$ (so that* FS $\neq\emptyset$*, Lemma 5.48) let S_κ be the forward edge solution $S_{\gamma^-(x_b,\boldsymbol{c},A,\kappa)}$. Then $S_\kappa\to S_0:=U^-_{\boldsymbol{c}}$* locally uniformly on $[x_b,1)$ *as $\kappa\downarrow 0$, but*

$$\liminf_{\kappa\downarrow 0}\ \big(U_\kappa(1)-U^-_{\boldsymbol{c}}(1)\big)\ \ge\ 2k_2-1>0,$$

so the convergence is not uniform on $[x_b,1]$, and the limit and the pole values do not commute: $\lim_{\kappa\to 0}U_\kappa(1)\neq U^-_{\boldsymbol{c}}(1)$ *(if the limit exists).*

Proof. By Proposition 5.50(c) the data $(x_b,\boldsymbol{c},A,\kappa,\gamma^-(\kappa))\to(x_b,\boldsymbol{c},A,0,U^-_{\boldsymbol{c}}(x_b))$, and continuous dependence gives local uniform convergence on $[x_b,1)$ (the limit solution exists there). At $x=1$: by Lemma 5.46(b),

S_κ is of type R or $\mathsf{L}_1, \mathsf{L}_2$, so $U_\kappa(1) > -3$ (for R by Theorem 4.9, for $\mathsf{L}_1, \mathsf{L}_2$ since $U_\kappa(1) = -2$). By Theorem 3.44(iii), $U_{\boldsymbol{c}}^-(1) = -2 - 2k_2$. Hence $U_\kappa(1) - U_{\boldsymbol{c}}^-(1) > -3 + 2 + 2k_2 = 2k_2 - 1$. □

Numerical observation 5.52 (the non-uniform limit)**.** *At* $\boldsymbol{c} = (0, \frac{1}{2}, 0)$, $x_b = 0.3$, $A = 0$ *one finds* $\gamma^-(\kappa) = -2.959796, -2.959777, -2.959603, -2.957072, -2.923432, -2.646553$ *at* $\kappa = 0, 0.005, 0.02, 0.1, 0.5, 2$*: the thresholds tend to* $U_{\boldsymbol{c}}^-(0.3)$ *continuously, while* $Q(1) = -2.0000$ *(within* 10^{-5}*) at all* $\kappa > 0$ *tested, i.e. the edge is of type* $\mathsf{L}_1, \mathsf{L}_2$ *with* $U(1) = -2$ *instead of* $U_{\boldsymbol{c}}^-(1) = -2 - 2\sqrt{1.5} = -4.449$*. Method and code in Atlas 5.73.*

Numerical observation 5.53 (which edge type occurs)**.** *At* $x_b = 0$, $A = 0$ *the lower edge* $\gamma = \gamma^-(\kappa)$ *was integrated to* $\xi = 30$ *and* $Q(1) = 2c_2 - \psi(1)$ *read off:*

κ		0.01	0.05	0.2	1	3
$\boldsymbol{c} = (0, -0.9, 0)$*:* γ^-		-1.0575	-1.0566	-1.0409	-0.2187	4.3473
	$Q(1)$	-1.8002	-1.8038	-1.8524	-2.0000	-2.0000
$\boldsymbol{c} = 0$*:* γ^-		-1.99994	-1.9989	-1.9865	-1.7806	-0.7548
	$Q(1)$	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000

For $c_2 = -0.9$ *(*$k_2 = 0.316$*) the edge is of type* R *for small* κ *(*$q \in (-2, -\frac{3}{2})$*,* $\mu = 0.316, 0.313, 0.272$*,* $U(1) = -2 - 2\mu \in (-3, -2)$*, as Corollary 4.21 requires) and becomes logarithmic (*$q = -2$*) between* $\kappa = 0.2$ *and* 1*; for* $c_2 = 0$ *(*$k_2 = 1 > \frac{1}{2}$*) it is logarithmic for every* $\kappa > 0$ *tested, in accordance with Corollary 5.51. The type* G *never occurs on the edge, as proved.*

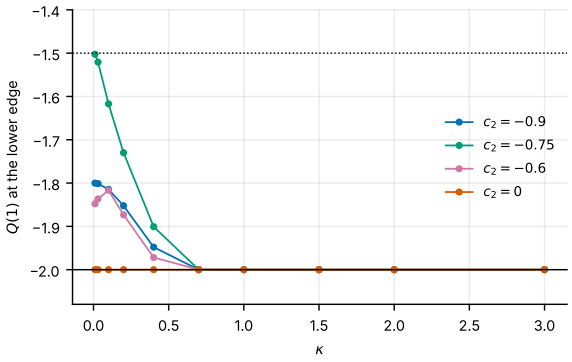


FIGURE 5.4. *Numerical illustration* (not used in any proof): $Q(1) = 2c_2 - \psi(1)$ at the lower edge $\gamma = \gamma^-(\kappa)$, $\boldsymbol{c} = (0, c_2, 0)$, $x_b = 0$, $A = 0$, for $c_2 = -0.9, -0.75, -0.6, 0$. Type R requires $Q(1) \in (-2, -\frac{3}{2})$ (dotted line: $-\frac{3}{2}$), type $\mathsf{L}_1, \mathsf{L}_2$ has $Q(1) = -2$ (solid line). As $\kappa \downarrow 0$, $Q(1) \to 2k_2^2 - 2$ if $k_2 < \frac{1}{2}$ (the no-swirl value, $= -1.8$ at $c_2 = -0.9$; -1.5 at $c_2 = -0.75$), whereas for $k_2 \geq \frac{1}{2}$ the edge is $\mathsf{L}_1, \mathsf{L}_2$ for every $\kappa > 0$ ($c_2 = 0$). Code: `code/ch05_global/make_b_extra.py`.

Remark 5.54 (the mechanism of the non-uniform limit)**.** Let $A = 0$ and $\kappa \downarrow 0$. For every fixed $x < 1$ the swirl integral at the edge solution tends to 0: on $[x_b, x]$ the functions $g/\kappa = y^{-2}$ converge (Proposition 5.50), so $f = O(\kappa)$, $g = O(\kappa)$ and $\psi(x) = O(\kappa^2)$ by (5.7). But at the pole, $\psi(1) = 2c_2 - Q(1) \geq 2c_2 + \frac{3}{2}$ for *every* $\kappa > 0$ by Corollary 4.21. So $\psi(x) \to 0$ for each $x < 1$ while $\psi(1) \not\to 0$: the swirl integral accumulates in a boundary layer at the pole whose position moves to 1 as $\kappa \to 0$. This is the non-uniformity of Corollary 5.51; the quantitative version is the subject of Chapter 7. Numerically, for $k_2 < \frac{1}{2}$ the edge keeps $q \to 2k_2^2 - 2$ as $\kappa \downarrow 0$ (Figure 5.4), i.e. $\psi(1) \to 0$ and the limit is continuous.

5.6. Class Z and the lens

In words. *A counter-rotating solution has one zero x_0 of the swirl; use that point as the base point. Its Cauchy data there are $(\boldsymbol{c}, \gamma, 0, \kappa)$ with $\kappa > 0$ (by σ_1), and two sets of numbers matter: γ, the Riccati variable, and κ, the strength of the swirl. For a fixed $\boldsymbol{c}$ and x_0 the global solutions fill a planar region* $\mathrm{Lens}(x_0, \boldsymbol{c}) = \{(\kappa, \gamma)\}$*, and we show that it has the shape of a* lens*: at $\kappa \to 0$ it is the Li–Li–Yan window $[U^-_{\boldsymbol{c}}(x_0), U^+_{\boldsymbol{c}}(x_0)]$ (swirl absent), and it closes up at a single tip $\kappa = \kappa_{\max}$ as the swirl grows; the lower boundary $\gamma^-(\kappa)$ increases, the upper boundary $\gamma^+(\kappa)$ decreases, and at the tip they meet. The lens is non-empty exactly for $\boldsymbol{c}$ in the interior of the admissible set, and $\kappa_{\max}$ is a continuous, strictly monotone function of $\boldsymbol{c}$. Inside the lens both poles are generic; on the lower edge the* north *pole is recessive or logarithmic; on the upper edge the south pole; at the tip both. This is the structure on which the global theory of the monograph rests (Chapters 7 and 12).*

5.6.1. The lens and its symmetry.

Definition 5.55 (the lens)**.** For $x_0 \in (-1, 1)$ and $\boldsymbol{c} \in \mathbb{R}^3$ put

$$\mathrm{Lens}(x_0, \boldsymbol{c}) := \{(\kappa, \gamma) \in (0, \infty) \times \mathbb{R} : \ (\boldsymbol{c}, \gamma, 0, \kappa) \in \Omega(x_0)\},$$
$$\gamma^-(\kappa) = \gamma^-(x_0, \boldsymbol{c}, \kappa) := \gamma^-(x_0, \boldsymbol{c}, 0, \kappa),$$

and $\gamma^+(\kappa) = \gamma^+(x_0, \boldsymbol{c}, 0, \kappa)$ (Proposition 5.42). By definition of Ω, $\mathrm{Lens}(x_0, \boldsymbol{c}) = \{(\kappa, \gamma) : \ \kappa > 0, \ \gamma^-(\kappa) \leq \gamma \leq \gamma^+(\kappa)\}$, the fibre over κ being empty if FS or BS is empty or $\gamma^- > \gamma^+$.

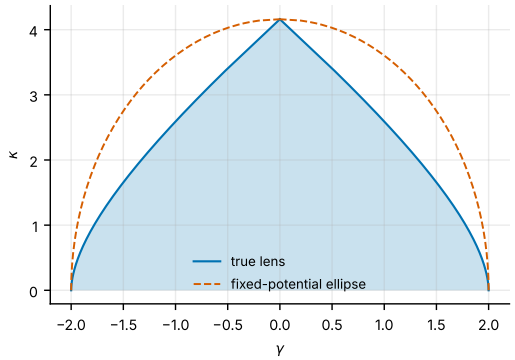
By Lemma 5.7(a), a class-Z solution with $\kappa > 0$ is the same thing as a point $(x_0, \boldsymbol{c}, \kappa, \gamma)$ with $(\kappa, \gamma) \in \mathrm{Lens}(x_0, \boldsymbol{c})$ (x_0 the zero of f, $\boldsymbol{c}$ the Taylor constants of Q at x_0, $\kappa = f_x(x_0)$ in the gauge $y(x_0) = 1$, $\gamma = U(x_0)$); the solutions with $\kappa < 0$ are their images under σ_1.

Proposition 5.56 (reflection of the lens)**.** $(\kappa, \gamma) \in \mathrm{Lens}(x_0, \boldsymbol{c}) \iff (\kappa, -\gamma) \in \mathrm{Lens}(-x_0, \boldsymbol{c}_m)$*. Hence $\gamma^+(x_0, \boldsymbol{c}, \kappa) = -\gamma^-(-x_0, \boldsymbol{c}_m, \kappa)$, and if $x_0 = 0$ and $c_1 = c_2$, then* $\mathrm{Lens}(0, \boldsymbol{c})$ *is symmetric under $\gamma \mapsto -\gamma$: $\gamma^+(\kappa) = -\gamma^-(\kappa)$.*

Proof. By Lemma 5.1(d) the solution $\sigma_1\sigma_2 S$ has data $(\boldsymbol{c}_m, -\gamma, -A, \kappa) = (\boldsymbol{c}_m, -\gamma, 0, \kappa)$ at $-x_0$ when S has $(\boldsymbol{c}, \gamma, 0, \kappa)$ at x_0, and S is global iff $\sigma_1\sigma_2 S$ is. The formula for γ^+ is Proposition 5.42(d). For $x_0 = 0$, $\boldsymbol{c}_m = \boldsymbol{c}$ and the two lenses coincide. □

Toy example 5.57 (a lens from a fixed potential)**.** Let $x_0 = 0$, $\boldsymbol{c} = 0$ and freeze the effect of the swirl as a uniform lowering of the potential: replace Q by the constant $-s$, $s \geq 0$ (this is *not* a solution of the system; it only shows the mechanism).

Then $W = 1 - \frac{s}{2} =: k^2$, the positive solutions of $z'' = k^2 z$ with $z(0) = 1$ are $z = \cosh k\xi + \beta \sinh k\xi$ with $|\beta| \le 1$ (those with $z'(0) = \beta k$, $|\beta| \le 1$ are positive on $\mathbb{R}$), and $\gamma = 2z'(0) \in [-2k, 2k]$. Now the swirl enters through $s = s(\kappa)$, increasing from 0 (the swirl integral $\psi \ge 0$ grows with κ^2), and the window of admissible γ is $|\gamma| \le 2\sqrt{1 - s(\kappa)/2}$: it *shrinks* from $[-2, 2]$ as κ grows and closes up when $s = 2$, i.e. at a finite $\kappa_{\max}$, with the tip at $\gamma_* = 0$ by symmetry. For $s = b\kappa^2$ this is the ellipse $\frac{\gamma^2}{4} + \frac{b}{2}\kappa^2 = 1$. The true lens (Figure 5.5) has the same topology but is not an ellipse; the monotone dependence of the two edges on κ and the existence of a tip are what Proposition 5.59 proves.



Numerical illustration (not used in any proof): the true lens at $\boldsymbol{c} = 0$, $x_0 = 0$ (shaded) and the ellipse of the fixed-potential model with the same tip $\kappa_{\max} = 4.1608$ (dashed). The true lens is thinner at small κ, as the swirl integral acts where the potential is largest. Code: `code/ch05_global/make_b_extra.py`.

5.6.2. Admissible constants, and bounds on κ.

Proposition 5.58 (admissible constants; bounds on the swirl)**.** *Let $x_0 \in (-1, 1)$ and $\boldsymbol{c} \in \mathbb{R}^3$.*

(a) *If* $\mathrm{Lens}(x_0, \boldsymbol{c}) \neq \emptyset$, *then* $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$.

(b) *For* $(\kappa, \gamma) \in \mathrm{Lens}(x_0, \boldsymbol{c})$: $|\gamma| \le K_P(\boldsymbol{c})$ *and*

$$\kappa \;\le\; \kappa_{\mathrm{up}}(x_0, \boldsymbol{c}) := \frac{\sqrt{384\,(1 + c_2)}}{m\,(1 - x_0)^2}, \qquad m = \Big(\frac{1 + x_0}{4}\Big)^{K_P(\boldsymbol{c})/2},$$

and also $\kappa \le \kappa_{\mathrm{up}}(-x_0, \boldsymbol{c}_m)$.

(c) *If* $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$, *there are* $\kappa_{\mathrm{lo}}(x_0, \boldsymbol{c}) > 0$ *and* $\tilde\gamma \in \mathbb{R}$ *such that* $(\boldsymbol{c}, \tilde\gamma, 0, \kappa) \in \Omega_{\mathsf{G}}$ *for all* $0 < \kappa < \kappa_{\mathrm{lo}}$; *in particular* $\mathrm{Lens}(x_0, \boldsymbol{c}) \neq \emptyset$. *One may take*

$$\kappa_{\mathrm{lo}} = \sqrt{e/\max(B_+, B_-)}, \qquad B_+ = B_+(x_0, \tilde{\boldsymbol{c}}, \tilde\gamma),\ B_- = B_+(-x_0, \tilde{\boldsymbol{c}}_m, -\tilde\gamma)$$

with B_+ of Lemma 5.48, for any $e > 0$ with $\tilde{\boldsymbol{c}} := \boldsymbol{c} - \frac{e}{2}(1, 1, 0) \in \operatorname{int} \mathcal{J}$ and any $\tilde\gamma \in (U^-_{\tilde{\boldsymbol{c}}}(x_0), U^+_{\tilde{\boldsymbol{c}}}(x_0))$.

Proof. (a) Let $(\kappa, \gamma) \in \mathrm{Lens}(x_0, \boldsymbol{c})$. By Proposition 5.42(c),(d) with $\kappa' = 0$: $\mathrm{FS}(x_0, \boldsymbol{c}, 0, \kappa) \subset \mathrm{FS}(x_0, \boldsymbol{c}, 0, 0)$ and $\mathrm{BS}(x_0, \boldsymbol{c}, 0, \kappa) \subset \mathrm{BS}(x_0, \boldsymbol{c}, 0, 0)$ (the latter by the mirror, $\mathrm{BS}(\kappa) = -\mathrm{FS}(-x_0, \boldsymbol{c}_m, 0, \kappa)$). So the no-swirl solution with $U(x_0) = \gamma$ is global and $\boldsymbol{c} \in \mathcal{J}$ by Theorem 3.44(i). Since $\kappa > 0$ the solution is a global solution with genuine swirl and zero at x_0, so Proposition 4.22 gives $c_2 > -1$, and its mirror (applied to $\sigma_1\sigma_2 S$, whose Taylor constants at $-x_0$ are $\boldsymbol{c}_m$) gives $c_1 > -1$.

By Lemma 5.46(c), $\gamma^-(x_0,\boldsymbol{c},0,0)=U^-_{\boldsymbol{c}}(x_0)$ and $\gamma^+(x_0,\boldsymbol{c},0,0)=U^+_{\boldsymbol{c}}(x_0)$. The strictness in Proposition 5.42(c) (with $\kappa>0$ for the larger datum, $\mathrm{FS}(x_0,\boldsymbol{c},0,\kappa)\neq\emptyset$) gives $U^-_{\boldsymbol{c}}(x_0)=\gamma^-(0)<\gamma^-(\kappa)\le\gamma\le\gamma^+(\kappa)<\gamma^+(0)=U^+_{\boldsymbol{c}}(x_0)$. Thus $U^-_{\boldsymbol{c}}<U^+_{\boldsymbol{c}}$, i.e. $c_3\neq\bar c_3$ (Theorem 3.44(ii)), and $\boldsymbol{c}\in\operatorname{int}\mathcal{J}$.

(b) $\gamma\le\gamma^+(x_0,\boldsymbol{c},0,\kappa)\le K_P$ by Proposition 5.42(d), and $\gamma\ge\gamma^-\ge -K_P$ by (b) there. The solution is forward global, so by Lemma 5.41(b) $U\le\max(\gamma,K_P)=K_P$ on $[x_0,1)$. Hence for $x\in[x_0,x_m]$, $x_m:=\frac{1+x_0}{2}$,

$$\begin{aligned} g(x)&=\kappa\exp\Big(-\int_{x_0}^{x}\frac{U}{1-r^2}\mathrm{d}r\Big)\ge\kappa\,e^{-K_P(\operatorname{artanh}x-\operatorname{artanh}x_0)}\\ &\ge\kappa\Big(\frac{1+x_0}{3+x_0}\Big)^{K_P/2}\ge\kappa\,m, \end{aligned}$$

because $e^{2(\operatorname{artanh}x_m-\operatorname{artanh}x_0)}=\frac{(1+x_m)(1-x_0)}{(1-x_m)(1+x_0)}=\frac{3+x_0}{1+x_0}$ and $3+x_0\le4$. Then $f(x)=\int_{x_0}^{x}g\ge\kappa m(x-x_0)$ on $[x_0,x_m]$, and by (5.7), with $1-s^2\le1$ and $L:=x_m-x_0=\frac{1-x_0}{2}$,

$$\begin{aligned} \psi(x_m)&\ge\int_{x_0}^{x_m}(x_m-s)^2\,\kappa m(s-x_0)\,\kappa m\mathrm{d}s=\kappa^2m^2\int_0^L(L-t)^2t\mathrm{d}t\\ &=\kappa^2m^2\frac{L^4}{12}=\kappa^2m^2\frac{(1-x_0)^4}{192}. \end{aligned}$$

By Lemma 5.41(a),(e): $\psi(x_m)\le\psi(1)\le2+2c_2$, so $\kappa^2\le384(1+c_2)/(m^2(1-x_0)^4)$. The second bound is the first one for $\sigma_1\sigma_2S$ (Proposition 5.56).

(c) Choose $e>0$ with $\tilde{\boldsymbol{c}}\in\operatorname{int}\mathcal{J}$ (possible: $\operatorname{int}\mathcal{J}$ is open) and $\tilde\gamma$ in the open window; $P_{\boldsymbol{c}}-P_{\tilde{\boldsymbol{c}}}=\frac{e}{2}\big((1-x)+(1+x)\big)=e$. Lemma 5.48 at $x_b=x_0$ with $\gamma=\tilde\gamma$, $A=0$ gives for $0<\kappa<\sqrt{e/B_+}$ a forward solution, global and of type G at $x=1$. Applied to $\sigma_1\sigma_2S$ (data $(\boldsymbol{c}_m,-\tilde\gamma,0,\kappa)$ at $-x_0$, with $\tilde{\boldsymbol{c}}_m=\boldsymbol{c}_m-\frac{e}{2}(1,1,0)$ and $-\tilde\gamma\in(U^-_{\tilde{\boldsymbol{c}}_m}(-x_0),U^+_{\tilde{\boldsymbol{c}}_m}(-x_0))$ because $U^{\pm}_{\tilde{\boldsymbol{c}}_m}(-x)=-U^{\mp}_{\tilde{\boldsymbol{c}}}(x)$) it gives a solution on $(-1,x_0]$, of type G at $x=-1$, for $\kappa<\sqrt{e/B_-}$. Hence for $0<\kappa<\kappa_{\mathrm{lo}}$ the data $(\boldsymbol{c},\tilde\gamma,0,\kappa)$ lie in Ω with both poles generic. □

5.6.3. The lens theorem.

Strategy. The lens is the intersection of the forward-safe half-line $[\gamma^-,\infty)$ and the backward-safe half-line $(-\infty,\gamma^+]$ over $\kappa>0$. Strict monotonicity of both edges in κ gives the nested shape; openness of Ω_{G} forces the interior of each fibre to be generic and the fibre at $\kappa_{\max}$ to be a point; the dependence on $(x_0,\boldsymbol{c})$ follows from closedness of the safe sets (upper semicontinuity of $\kappa_{\max}$) and openness of Ω_{G} (lower semicontinuity).

Proposition 5.59 (the lens)**.** *Let $\boldsymbol{c}\in\operatorname{int}\mathcal{J}$, $x_0\in(-1,1)$, and* $\mathrm{Lens}=\mathrm{Lens}(x_0,\boldsymbol{c})$.

(a) (Shape.) $\kappa_{\max}=\kappa_{\max}(x_0,\boldsymbol{c}):=\sup\{\kappa:\ \exists\gamma,\ (\kappa,\gamma)\in\mathrm{Lens}\}$ *is finite, positive and attained,* $\kappa_{\mathrm{lo}}\le\kappa_{\max}\le\kappa_{\mathrm{up}}$, *and*

$$\mathrm{Lens}=\{(\kappa,\gamma):\ 0<\kappa\le\kappa_{\max},\ \gamma^-(\kappa)\le\gamma\le\gamma^+(\kappa)\},$$

with γ^- continuous and strictly increasing, γ^+ continuous and strictly decreasing on $(0,\kappa_{\max}]$, $\gamma^-(\kappa)<\gamma^+(\kappa)$ for $\kappa<\kappa_{\max}$.

(b) (Tip.) $\gamma^-(\kappa_{\max})=\gamma^+(\kappa_{\max})=:\gamma_*(x_0,\boldsymbol{c})$*: the fibre over $\kappa_{\max}$ is a single point. Inside, i.e. for $\gamma^-(\kappa)<\gamma<\gamma^+(\kappa)$, $(\boldsymbol{c},\gamma,0,\kappa)\in\Omega_{\mathsf{G}}$; the tip is not in Ω_{G}.*

(c) (Edges.) *For $\gamma = \gamma^-(\kappa)$ the north pole is of type R, L_1 or L_2 and, if $\kappa < \kappa_{\max}$, the south pole is of type G; symmetrically for $\gamma = \gamma^+(\kappa)$. At the tip* both *poles are of type R, L_1 or L_2.*

(d) (Dependence.) *$\kappa_{\max}$ and γ_* are continuous on $\{(x_0, \boldsymbol{c}) : \boldsymbol{c} \in \operatorname{int} \mathcal{J}\}$; $\kappa_{\max}$ is strictly increasing in each c_i; $\kappa_{\max}(x_0, \boldsymbol{c}) = \kappa_{\max}(-x_0, \boldsymbol{c}_m)$ and $\gamma_*(-x_0, \boldsymbol{c}_m) = -\gamma_*(x_0, \boldsymbol{c})$; and $\kappa_{\max}(x_0, \boldsymbol{c}) \to 0$ as $\boldsymbol{c} \to \boldsymbol{c}^0 \in \partial\mathcal{J}$ (with $\kappa_{\max} := 0$ for $\boldsymbol{c} \notin \operatorname{int} \mathcal{J}$ the function is upper semicontinuous everywhere).*

(e) (Floor.) *$\gamma^-(\kappa) \to U^-_{\boldsymbol{c}}(x_0)$ and $\gamma^+(\kappa) \to U^+_{\boldsymbol{c}}(x_0)$ as $\kappa \downarrow 0$.*

Proof. *(a), first part.* The fibre $\{\gamma : (\kappa, \gamma) \in \text{Lens}\} = [\gamma^-(\kappa), \gamma^+(\kappa)]$ if nonempty. $\kappa_{\max} \geq \kappa_{\text{lo}} > 0$ and $\leq \kappa_{\text{up}} < \infty$ by Proposition 5.58. Take $\kappa_n \uparrow \kappa_{\max}$ and γ_n in the fibres; $|\gamma_n| \leq K_P$, so a subsequence converges to some γ, and $(\boldsymbol{c}, \gamma_n, 0, \kappa_n) \in \Omega(x_0)$ closed (Theorem 5.12) gives $(\kappa_{\max}, \gamma) \in \text{Lens}$: the supremum is attained. *Monotonicity.* Let $0 < \kappa < \kappa' \leq \kappa_{\max}$ with κ' in the set of κ's with nonempty fibre, and γ' in its fibre. By Proposition 5.42(c) (strict form, $\kappa' > 0$ and $\text{FS}(x_0, \boldsymbol{c}, 0, \kappa') \neq \emptyset$) $\gamma^-(\kappa) < \gamma^-(\kappa') \leq \gamma' \leq \gamma^+(\kappa') < \gamma^+(\kappa)$ (the last by the mirror). So the fibre over κ is nonempty with nonempty interior, γ^- is strictly increasing and γ^+ strictly decreasing wherever defined, and the fibres exist for all $\kappa \in (0, \kappa_{\max}]$. Continuity of γ^- on $(0, \kappa_{\max}]$ is Proposition 5.50(b) (the data have $\kappa > 0$), and of γ^+ by the mirror. *(b)* Suppose $\gamma^-(\kappa_{\max}) < \gamma^+(\kappa_{\max})$ and take γ strictly between. By Lemma 5.46(a) and its mirror, both poles of the solution $(\boldsymbol{c}, \gamma, 0, \kappa_{\max})$ are generic, so the point lies in Ω_{G}, which is open (Proposition 5.15): $(\boldsymbol{c}, \gamma, 0, \kappa_{\max} + \varepsilon) \in \Omega$ for small $\varepsilon > 0$, contradicting the maximality of $\kappa_{\max}$. So the fibre over $\kappa_{\max}$ is the single point γ_*. For $\gamma^-(\kappa) < \gamma < \gamma^+(\kappa)$ the same argument shows Ω_{G}. The tip is not in Ω_{G} by the same argument (an interior point would allow $\kappa > \kappa_{\max}$). *(c)* At $\gamma = \gamma^-(\kappa)$ the north pole is not generic by Lemma 5.46(b), so it is R or $\mathsf{L}_1, \mathsf{L}_2$; the south pole is generic iff $\gamma < \gamma^+(\kappa)$ (mirror of Lemma 5.46(a)), which holds for $\kappa < \kappa_{\max}$. At the tip $\gamma_* = \gamma^- = \gamma^+$: the north pole is an edge pole and the south pole an edge pole. *(e)* Is Proposition 5.50(c) ($c_1, c_2 > -1$ as $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$). *(d) Monotone.* Let $\boldsymbol{c} \leq \boldsymbol{c}'$, $\boldsymbol{c} \neq \boldsymbol{c}'$, both in $\operatorname{int} \mathcal{J}$ (the strict monotonicity below shows $\text{Lens}(x_0, \boldsymbol{c}') \neq \emptyset$, so $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$ by Proposition 5.58(a)). At $\kappa_0 := \kappa_{\max}(\boldsymbol{c})$, the strict monotonicity of Proposition 5.42(c),(d) gives $\gamma^-(\boldsymbol{c}', \kappa_0) < \gamma^-(\boldsymbol{c}, \kappa_0) = \gamma_*(\boldsymbol{c}) = \gamma^+(\boldsymbol{c}, \kappa_0) < \gamma^+(\boldsymbol{c}', \kappa_0)$: the fibre of $\text{Lens}(\boldsymbol{c}')$ over κ_0 has nonempty interior. By (b) for $\boldsymbol{c}'$, $\kappa_0 < \kappa_{\max}(\boldsymbol{c}')$. *Mirror.* By Proposition 5.56 the lens of $(-x_0, \boldsymbol{c}_m)$ is the image of that of $(x_0, \boldsymbol{c})$ under $\gamma \mapsto -\gamma$. *Upper semicontinuity.* Let $(x_n, \boldsymbol{c}_n) \to (x, \boldsymbol{c})$, $\kappa_n := \kappa_{\max}(x_n, \boldsymbol{c}_n)$, $\gamma_n := \gamma_*(x_n, \boldsymbol{c}_n)$ (or $\kappa_n = 0$ if $\boldsymbol{c}_n \notin \operatorname{int} \mathcal{J}$). The bounds $\kappa_n \leq \kappa_{\text{up}}(x_n, \boldsymbol{c}_n)$ and $|\gamma_n| \leq K_P(\boldsymbol{c}_n)$ are locally uniform (continuity of K_P and of κ_{up} in $(x_0, \boldsymbol{c})$), so a subsequence converges to $(\bar\kappa, \bar\gamma)$. If $\bar\kappa > 0$, closedness of G^+ and G^- in the variables $(x_b, \boldsymbol{c}, A, \kappa, \gamma)$ (Proposition 5.42(a), Lemma 5.40) gives $(\bar\kappa, \bar\gamma) \in \text{Lens}(x, \boldsymbol{c})$, so $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$ and $\bar\kappa \leq \kappa_{\max}(x, \boldsymbol{c})$. If $\bar\kappa = 0$ the inequality is trivial. In particular for $\boldsymbol{c} \in \partial\mathcal{J}$ every limit is 0. *Lower semicontinuity.* Let $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$, $0 < \kappa < \kappa_{\max}(x_0, \boldsymbol{c})$ and $\gamma^-(\kappa) < \gamma < \gamma^+(\kappa)$. Then $\omega := (\boldsymbol{c}, \gamma, 0, \kappa) \in \Omega_{\mathsf{G}}(x_0)$. For $(x', \boldsymbol{c}')$ close to $(x_0, \boldsymbol{c})$ let ω' be the state at x_0 of the solution with data $(\boldsymbol{c}', \gamma, 0, \kappa)$ at x'; $\omega' \to \omega$ by continuous dependence (the base point is allowed to vary, Remark 2.7(iii)), so $\omega' \in \Omega_{\mathsf{G}}(x_0)$ for $(x', \boldsymbol{c}')$ close enough, because $\Omega_{\mathsf{G}}(x_0)$ is open; by Lemma 5.7(b) the solution is global, hence $(\kappa, \gamma) \in \text{Lens}(x', \boldsymbol{c}')$ and $\kappa_{\max}(x', \boldsymbol{c}') \geq \kappa$. So $\liminf \kappa_{\max} \geq \kappa$ for every $\kappa < \kappa_{\max}(x_0, \boldsymbol{c})$. *Continuity of γ_*.* $\gamma_*(x_0, \boldsymbol{c}) = \gamma^-(x_0, \boldsymbol{c}, 0, \kappa_{\max}(x_0, \boldsymbol{c}))$; γ^- is continuous in all its arguments at points

with $\kappa > 0$ (Proposition 5.50(b)) and $\kappa_{\max}$ is continuous. The reflection formula follows from Proposition 5.56. □

Numerical observation 5.60 ($\kappa_{\max}$ along c_3: strict monotonicity and collapse at the boundary of $\mathcal{J}$)**.** *At $x_0 = 0$, $\boldsymbol{c} = (0, 0, c_3)$ one has $\bar{c}_3(0, 0) = -4$, so* $\operatorname{int}\mathcal{J}$ *is $c_3 > -4$, and*

c_3	-3.99	-3.95	-3.7	-3	-2	0	5
$\kappa_{\max}$	0.2102	0.5008	1.2992	2.3269	3.1379	4.1608	5.7586

increasing in c_3 (Proposition 5.59(d)) and tending to 0 as $c_3 \downarrow -4$ (roughly like $\sqrt{c_3 + 4}$: $0.21/\sqrt{0.01} = 2.1$, $0.50/\sqrt{0.05} = 2.2$). The entry at $c_3 = -2.9$ is 2.4281 in Atlas 5.73.

Remark 5.61 (the lens in words)**.** For κ small, the swirl is a small perturbation and the lens is the no-swirl window $[U^-_{\boldsymbol{c}}(x_0), U^+_{\boldsymbol{c}}(x_0)]$ thinned by $O(\kappa^2)$ (the values in Example 5.71 give $\gamma^-(\kappa) + 2 = 0.0011,\ 0.0039,\ 0.0135$ at $\kappa = 0.05,\ 0.1,\ 0.2$, roughly quadratic in κ). The lower edge is R or $\mathsf{L}_1, \mathsf{L}_2$ at the north pole because pushing γ down until the solution is just recessive is the *last* thing the safe set tolerates (Lemma 5.46); by the reflection the upper edge is the same at the south pole. The tip is the largest swirl for which a global solution exists at all, and its two poles are both non-generic. What is *not* said here: whether γ^- is convex or $\kappa_{\max}$ is concave (log-convexity of the lens is Theorem 12.48) and how γ^- behaves analytically; Chapter 12 proves the quantitative statements.

THEOREM 5.62 (class Z is parametrised by the lens). *Let $\mathsf{Z}_+ := \mathsf{Z} \cap \{\kappa > 0\}$. The map $S \mapsto (x_0, \boldsymbol{c}, \kappa, \gamma)$ (zero of f, Taylor constants of Q and $\gamma = U$ at x_0, $\kappa = \kappa_x$ at x_0) is a bijection from Z_+ onto*

$$\mathcal{L} := \left\{ (x_0, \boldsymbol{c}, \kappa, \gamma) : \begin{array}{c} x_0 \in (-1, 1),\ \boldsymbol{c} \in \operatorname{int}\mathcal{J},\ 0 < \kappa \le \kappa_{\max}(x_0, \boldsymbol{c}), \\ \gamma^-(x_0, \boldsymbol{c}, \kappa) \le \gamma \le \gamma^+(x_0, \boldsymbol{c}, \kappa) \end{array} \right\},$$

and $\mathsf{Z} = \mathsf{Z}_+ \sqcup \sigma_1 \mathsf{Z}_+$. The pole types are given by the position in the lens:

position in $\operatorname{Lens}(x_0, \boldsymbol{c})$	*north pole*	*south pole*
interior: $\gamma^- < \gamma < \gamma^+$	G	G
lower edge $\gamma = \gamma^-$, $\kappa < \kappa_{\max}$	$\mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$	G
upper edge $\gamma = \gamma^+$, $\kappa < \kappa_{\max}$	G	$\mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$
tip $(\kappa_{\max}, \gamma_*)$	$\mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$	$\mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$

Moreover: $\mathcal{L}$ is a closed subset of $\{(x_0, \boldsymbol{c}, \kappa, \gamma)\}$ ($\kappa > 0$); the interior is contained in Ω_{G}; the lens of each $(x_0, \boldsymbol{c})$ is the region between the graph of γ^- (strictly increasing) and the graph of γ^+ (strictly decreasing) over $(0, \kappa_{\max}]$, with limits $U^{\mp}_{\boldsymbol{c}}(x_0)$ at $\kappa \downarrow 0$, closing up at one tip; $\kappa_{\max}$ is continuous and strictly increasing in each c_i, mirror-symmetric, and tends to 0 at $\partial\mathcal{J}$. The parameter space has dimension $6 = 1{+}3{+}1{+}1$ (the numbers $x_0, \boldsymbol{c}, \kappa, \gamma$), the dimension of $\Omega \subset \mathbb{R}^6$ (x_0 is determined by the solution as the zero of f, so it plays the role of a base point and not of an extra parameter); every class-Z solution is $\sigma_1^j S_+$ for exactly one $j \in \{0, 1\}$ and exactly one point of $\mathcal{L}$.

PROOF. The bijection: a solution with $\kappa > 0$ and a zero x_0 of f has data $(\boldsymbol{c}, \gamma, 0, \kappa)$ at x_0, which belong to $\Omega(x_0)$ by Lemma 5.7(a); conversely each point of $\Omega(x_0)$ with $A = 0$, $\kappa > 0$ is a global solution with a zero of f at x_0 (class Z). So $\mathsf{Z}_+ \leftrightarrow \{(x_0, \boldsymbol{c}, \kappa, \gamma) : (\kappa, \gamma) \in \operatorname{Lens}(x_0, \boldsymbol{c})\}$. By Proposition 5.58(a) $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$, and

the description of the fibres is Proposition 5.59(a); the table is Proposition 5.59(b),(c) together with Lemma 5.46(a),(b) and its mirror; the rest is Proposition 5.59(a),(d),(e). Closedness of $\mathcal{L}$: if $(x_n, \boldsymbol{c}_n, \kappa_n, \gamma_n) \to (x, \boldsymbol{c}, \kappa, \gamma)$ with $\kappa > 0$ then $(\boldsymbol{c}_n, \gamma_n, 0, \kappa_n)$ at x_n converge to the data at x, and $G^{\pm}$ are closed (Proposition 5.42(a), Lemma 5.40). Finally $\mathsf{Z} = \mathsf{Z}_+ \sqcup \sigma_1\mathsf{Z}_+$ since $\kappa \neq 0$ for genuine swirl and σ_1 exchanges the signs (Lemma 5.1). □

Example 5.63 (an asymmetric lens)**.** Take $\boldsymbol{c} = (2, -\frac{1}{2}, 0)$ and $x_0 = \frac{1}{2}$. Here $c_1 \neq c_2$ and $x_0 \neq 0$, so Proposition 5.56 does not make the lens symmetric: the numerical tip is $\kappa_{\max} = 6.2919$ at $\gamma_* = -1.1538$ (third panel of Figure 5.5). The reflection says that the lens of $(-\frac{1}{2}, (-\frac{1}{2}, 2, 0))$ is its mirror image: the same $\kappa_{\max} = 6.2919$, with $\gamma_* = +1.1538$. By Theorem 5.62 every interior point of the first lens is a solution with both poles generic, the lower edge has an edge pole at $x = 1$ only, the upper edge at $x = -1$ only, and the tip has edge poles at both.

Remark 5.64 (what the next chapters take from this section)**.** Chapter 7 uses Theorem 5.62 as follows: the edges γ^-, γ^+ of the lens are the graphs along which the north, resp. south, pole is non-generic, and their regularity and the behaviour of the tip are studied there; Chapter 12 proves the quantitative statements (convexity of the lens in suitable variables, log-convexity Theorem 12.48, and the sharp ranges of $\kappa_{\max}$) from the comparison principle of Proposition 5.42. Everything here is qualitative and uses only the half-line structure, the strictness of the comparison, and the openness of Ω_{G}.

5.7. The wedge at a no-swirl point, and the faces of Ω

In words. *The face $\kappa = 0$ of trivial swirl is described in effective coordinates in Theorem 8.53 (Chapter 8), and the global pictures of Ω are collected in Chapter 7; here we show, with the tools of this chapter, that every no-swirl solution with γ strictly inside the window is an interior point of Ω (Proposition 5.15) and describe what surrounds it. Which of the nearby swirling solutions are counter-rotating? The swirl at the two poles is $f(\pm 1) = A \pm \kappa$(positive numbers), so f changes sign iff $-\kappa F_+ < A < \kappa F_-$: in the (A, κ)-plane, class Z is a* wedge *with the no-swirl point $(0, 0)$ as its vertex and two edges of slopes given by the integrals $F_0(\pm 1) = \int y_0^{-2}$ of the no-swirl solution. The same positivity gives the second result of the section: on the two "faces" $A = 0$ and $\kappa = 0$, every point of Ω is a limit of points where both poles are generic — raising the constants $\boldsymbol{c}$ pushes the solution into the interior.*

Lemma 5.65 (the face $\kappa = 0$ of Ω)**.** $\Omega(x_b) \cap \{\kappa = 0\} = \{(\boldsymbol{c}, \gamma, A, 0) : \ \boldsymbol{c} \in \mathcal{J},\ U^-_{\boldsymbol{c}}(x_b) \le \gamma \le U^+_{\boldsymbol{c}}(x_b),\ A \in \mathbb{R}\}$.

Proof. If $\kappa = 0$ then $f \equiv A$ (Corollary 2.10), $\psi \equiv 0$ and $Q = P_{\boldsymbol{c}}$ (Lemma 5.23, $Y = (\gamma, 0, A, 0, 0, 0)$ is a stationary state of the last four components), so the solution is the no-swirl solution with $U(x_b) = \gamma$ for (3.6); it is global iff $\boldsymbol{c} \in \mathcal{J}$ and γ is in the window (Theorem 3.44). □

Toy example 5.66 (the wedge at the rest state)**.** At $\boldsymbol{c} = 0$, $x_b = 0$, $\kappa = 0$ every $\gamma \in (-2, 2)$ is admissible (Example 5.9), with $z = ae^{\xi} + be^{-\xi}$, $a = \frac{\gamma+2}{4}$, $b = \frac{2-\gamma}{4}$ ($a + b = 1$, $a - b = \frac{\gamma}{2}$). The swirl of a nearby solution is $f = A + \kappa \int_0^x y^{-2}$, with

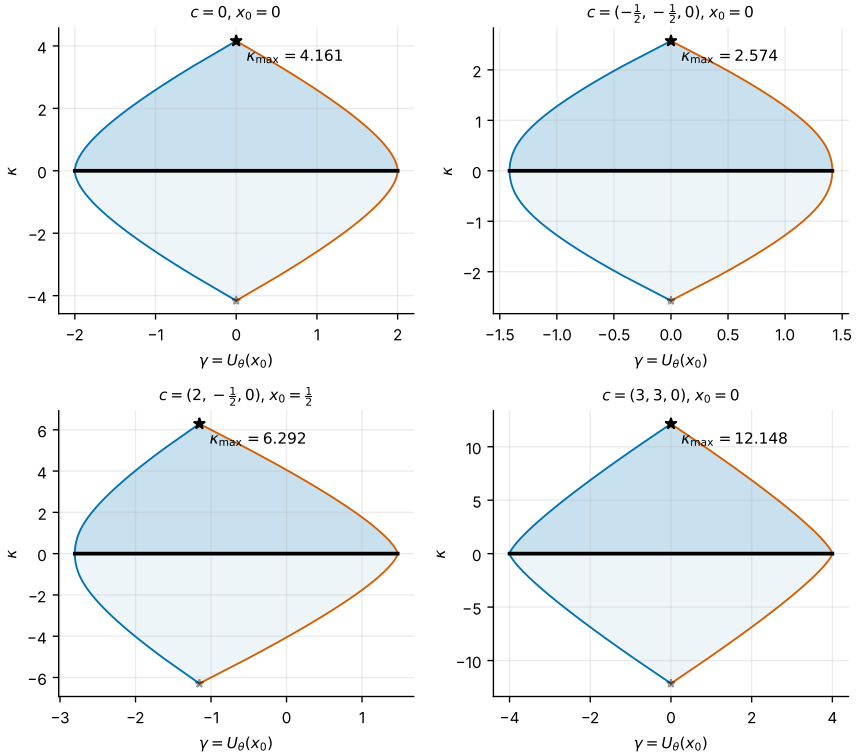


FIGURE 5.5. *Numerical illustration* (not used in any proof): the lens $\mathrm{Lens}(x_0, \boldsymbol{c})$ in the (γ, κ)-plane (γ horizontal) for four constants. Top left, top right, bottom left, bottom right: $\boldsymbol{c} = 0$, $x_0 = 0$ (tip $\kappa_{\max} = 4.1608$ at $\gamma_* = 0$; symmetric); $\boldsymbol{c} = (-\frac{1}{2}, -\frac{1}{2}, 0)$, $x_0 = 0$ (tip 2.5737, $\gamma_* = 0$); $\boldsymbol{c} = (2, -\frac{1}{2}, 0)$, $x_0 = \frac{1}{2}$ (tip 6.2919 at $\gamma_* = -1.1538$); $\boldsymbol{c} = (3, 3, 0)$, $x_0 = 0$ (tip 12.148, $\gamma_* = 0$). Shaded: the lens for $\kappa > 0$; faded: its image for $\kappa < 0$ under σ_1; black: the floor $[U_{\boldsymbol{c}}^-(x_0), U_{\boldsymbol{c}}^+(x_0)]$ at $\kappa = 0$. The lower edge is γ^-, the upper edge γ^+; they meet at the tip. Data: shooting with bisection in γ at each κ (Atlas 5.73). Code: `code/ch05_global/make_b_lens.py` (reads the stored table `data_lens.json`; Appendix E).

$y^{-2}\mathrm{d}x = \mathrm{d}\xi/z^2$ in the line variable; the reduction of order $(e^{-\xi}/z)' = -2a/z^2$, $(e^{\xi}/z)' = 2b/z^2$ gives

$$\int_0^\infty \frac{\mathrm{d}\xi}{z^2} = \frac{1}{2a} = \frac{2}{\gamma + 2}, \qquad \int_{-\infty}^0 \frac{\mathrm{d}\xi}{z^2} = \frac{1}{2b} = \frac{2}{2 - \gamma}.$$

So $f(1) = A + \frac{2\kappa}{\gamma+2}$, $f(-1) = A - \frac{2\kappa}{2-\gamma}$ to first order in κ: class Z is $\{-\frac{2\kappa}{\gamma+2} < A < \frac{2\kappa}{2-\gamma}\}$ ($\kappa > 0$), a wedge of total slope-width $\frac{2}{\gamma+2} + \frac{2}{2-\gamma} = \frac{8}{4-\gamma^2}$: narrow at $\gamma = 0$ ($= 2$), widening to infinity as $\gamma \to \pm 2$ (the window ends).

Proposition 5.67 (the wedge)**.** *Let $x_b \in (-1,1)$, $\boldsymbol{c}^0 \in \operatorname{int}\mathcal{J}$, $U^-_{\boldsymbol{c}^0}(x_b) < \gamma^0 < U^+_{\boldsymbol{c}^0}(x_b)$ and $\omega^0 = (\boldsymbol{c}^0, \gamma^0, 0, 0)$. There is an open neighbourhood $V \subset \Omega_{\mathsf{G}}(x_b)$ of ω^0 and continuous functions $F_+, F_- : V \to (0,\infty)$ such that for all $\omega = (\boldsymbol{c}, \gamma, A, \kappa) \in V$ (any signs of A, κ)*

$$\begin{aligned} f(1) &= A + \kappa F_+(\omega), \qquad f(-1) = A - \kappa F_-(\omega), \\ F_+ &= \int_{x_b}^{1} y^{-2}\mathrm{d}x, \qquad F_- = \int_{-1}^{x_b} y^{-2}\mathrm{d}x, \end{aligned} \tag{5.9}$$

where $y^{-2} = \exp(-\int_{x_b}^{x} U/(1-r^2))$. Consequently

$$\begin{aligned} \mathsf{Z} \cap V &= \{\omega \in V :\ f(1)f(-1) < 0\} \\ &= \{\kappa > 0,\ -\kappa F_+ < A < \kappa F_-\} \ \cup\ \{\kappa < 0,\ -|\kappa| F_- < A < |\kappa| F_+\}. \end{aligned}$$

At $\kappa = 0$: $F_\pm(\boldsymbol{c}, \gamma, A, 0) = \pm F_0(\pm 1)$, with the signed integrals $F_0(\pm 1) = \int_{x_b}^{\pm 1} y_0^{-2}$ of the no-swirl solution y_0; the hypersurfaces $\{f(1) = 0\}$, $\{f(-1) = 0\}$ of Proposition 5.16 are C^∞ near ω^0 and tangent there to

$$A = -F_0(1)\,\kappa \qquad \text{and} \qquad A = -F_0(-1)\,\kappa, \tag{5.10}$$

that is, $\partial_\kappa f(\pm 1)(\omega^0) = F_0(\pm 1)$, $\partial_A f(\pm 1)(\omega^0) = 1$, and the derivatives in $\boldsymbol{c}, \gamma$ vanish at $\kappa = 0$. At $\boldsymbol{c} = 0$, $x_b = 0$: $F_0(1) = \frac{2}{\gamma+2}$, $F_0(-1) = \frac{2}{\gamma-2}$.

Proof. *Choice of V.* Pick $e > 0$ with $\tilde{\boldsymbol{c}} := \boldsymbol{c}^0 - \frac{e}{2}(1,1,0) \in \operatorname{int}\mathcal{J}$ (open) and so small that $U^-_{\tilde{\boldsymbol{c}}}(x_b) < \gamma^0 < U^+_{\tilde{\boldsymbol{c}}}(x_b)$ (the thresholds $U^\pm_{\boldsymbol{c}}(x_b)$ depend continuously on $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ by Lemma 3.25); fix $\tilde\gamma_f < \gamma^0 < \tilde\gamma_b$ in this window. Let B_+ be the constant of Lemma 5.48 for $(x_b, \tilde{\boldsymbol{c}}, \tilde\gamma_f)$, with $\tilde F_f, \tilde g_f$, and let $\tilde F_b, B_b, \tilde g_b$ be the corresponding objects for the mirrored problem $(-x_b, \tilde{\boldsymbol{c}}_m, -\tilde\gamma_b)$ (window by $U^\pm_{\tilde{\boldsymbol{c}}_m}(-x) = -U^\mp_{\tilde{\boldsymbol{c}}}(x)$). Let V be the open set of $\omega = (\boldsymbol{c}, \gamma, A, \kappa)$ with $P_{\boldsymbol{c}} - P_{\tilde{\boldsymbol{c}}} > \frac{e}{2}$, $\tilde\gamma_f < \gamma < \tilde\gamma_b$, and $|\kappa|(|A|\max(\tilde F_f, \tilde F_b) + |\kappa|\max(B_+, B_b)) < \frac{e}{2}$ (an open neighbourhood of ω^0: $P_{\boldsymbol{c}} - P_{\tilde{\boldsymbol{c}}} \to e$ as $\boldsymbol{c} \to \boldsymbol{c}^0$). Applying Lemma 5.48 (with $e/2$) to the forward half, and to the reflected backward half ($\sigma_1\sigma_2$ maps the data to $(\boldsymbol{c}_m, -\gamma, -A, \kappa)$ at $-x_b$, and $P_{\boldsymbol{c}}(-x)$ vs. $P_{\boldsymbol{c}_m}$), each $\omega \in V$ has a solution on $(-1,1)$, generic at both poles, so $V \subset \Omega_{\mathsf{G}}$, with

$$y^{-2}(x) = e^{-\int_{x_b}^{x} U/(1-r^2)} \le \tilde g_f(x)\ \ (x \ge x_b), \qquad y^{-2}(x) \le \tilde g_b^{\mathrm{refl}}(x)\ \ (x \le x_b),$$

the second being the reflected bound $|g(-s)| \le |\kappa|\tilde g_b(s)$; both majorants are integrable on their intervals ($\tilde F_f, \tilde F_b < \infty$).

The formulas. $f(1) = A + \int_{x_b}^{1} f_x\mathrm{d}x = A + \kappa\int_{x_b}^{1} y^{-2}\mathrm{d}x$, because $f_x = g = \kappa y^{-2}$ (Lemma 2.9; $g = \kappa\exp(-\int U/(1-r^2))$ from the g-equation of (5.2), also for $\kappa = 0$ where both sides vanish). Likewise $f(-1) = A - \kappa\int_{-1}^{x_b} y^{-2}$. Both integrals are positive and finite by the majorants. *Continuity of $F_\pm$*: if $\omega_n \to \omega$ in V, $U_{\omega_n} \to U_\omega$ locally uniformly on $(-1,1)$ (Remark 2.7(iii)), so $y_n^{-2} \to y^{-2}$ pointwise, and dominated convergence applies.

The class Z. For $\kappa > 0$, Proposition 5.16(b) gives $\omega \in \mathsf{Z} \iff f(-1) < 0 < f(1)$; by (5.9) this reads $A < \kappa F_-$ and $A > -\kappa F_+$. For $\kappa < 0$, f is strictly decreasing (Corollary 2.10) with finite limits $f(\pm 1)$, so it has a zero iff $f(-1) > 0 > f(1)$, i.e. $A + |\kappa|F_- > 0$ and $A - |\kappa|F_+ < 0$. For $\kappa = 0$, $f \equiv A$ is constant, there is no genuine swirl, and $f(1)f(-1) = A^2 \ge 0$. So $\mathsf{Z} \cap V = \{f(1)f(-1) < 0\}$.

Tangents. At $\kappa = 0$ the formulas (5.9) give $f(\pm 1) = A$ identically in $(\boldsymbol{c}, \gamma)$, hence $\partial_A f(\pm 1) = 1$ and $\partial_{\boldsymbol{c}} f(\pm 1) = \partial_\gamma f(\pm 1) = 0$ there. The functions $f(\pm 1)$ are C^∞ on Ω_{G} (Proposition 5.16(c)); $\partial_\kappa f(1)(\omega^0) = \lim_{\kappa\to 0} f(1)(\boldsymbol{c}^0, \gamma^0, 0, \kappa)/\kappa = \lim F_+ = F_+(\omega^0) = F_0(1)$ by continuity of F_+; similarly $\partial_\kappa f(-1)(\omega^0) = -F_-(\omega^0) = F_0(-1)$. Since $df(\pm 1)(\omega^0) = \mathrm{d}A + F_0(\pm 1)\mathrm{d}\kappa \neq 0$, the zero sets are C^∞ hypersurfaces tangent to $\{\mathrm{d}A + F_0(\pm 1)\mathrm{d}\kappa = 0\}$, which is (5.10) in the (A, κ)-plane. The values at $\boldsymbol{c} = 0$, $x_b = 0$ were computed in Toy example 5.66. □

Remark 5.68 (reading the wedge)**.** At the vertex the whole segment $\{\kappa = 0 = A\}$ (the no-swirl solutions without rotation) is approached by Z through the open cone $\{-F_0(1) < A/\kappa < -F_0(-1)\}$, and by the rotating classes N through its complement. The slopes depend on γ and blow up at the ends of the window ($F_0(1) \to \infty$ as $\gamma \to -2$ at $\boldsymbol{c} = 0$): the cone opens up to a half-plane exactly where the edge of the window is reached (Figure 5.1).

Proposition 5.69 (the faces $A = 0$ and $\kappa = 0$ lie in the closure of Ω_{G})**.** *Let $x_b \in (-1, 1)$ and $\omega = (\boldsymbol{c}, \gamma, A, \kappa) \in \Omega(x_b)$ with $A = 0$ or $\kappa = 0$. Then $\omega_\varepsilon := (\boldsymbol{c} + \varepsilon(1,1,1), \gamma, A, \kappa) \in \Omega_{\mathsf{G}}(x_b)$ for every $\varepsilon > 0$. In particular*

$$\Omega \cap (\{A = 0\} \cup \{\kappa = 0\}) \ \subset\ \overline{\Omega_{\mathsf{G}}}.$$

Proof. Put $\boldsymbol{c}' = \boldsymbol{c} + \varepsilon(1,1,1)$; $\boldsymbol{c}' \geq \boldsymbol{c}$, $\boldsymbol{c}' \neq \boldsymbol{c}$, and $\boldsymbol{c}'_1, \boldsymbol{c}'_2 > c_1, c_2$.

Case $\kappa = 0$. For $\kappa = 0$ the sign of A is irrelevant: $f \equiv A$ does not enter the equation of U, so FS, BS and the solution of the Q-system do not depend on A (Lemma 5.46(c) and its proof); we apply Proposition 5.42, which is stated for $A \geq 0$ (resp. $A \leq 0$ for BS), with A replaced by 0. By Lemma 5.65, $\boldsymbol{c} \in \mathcal{J}$, so $c_1, c_2 \geq -1$ and $c'_1, c'_2 > -1$. The solution is the no-swirl solution and $\gamma \in \mathrm{FS}(x_b, \boldsymbol{c}, A, 0) \cap \mathrm{BS}(x_b, \boldsymbol{c}, A, 0)$. By Proposition 5.42(c),(d) (strict form for $\kappa = 0$, $\boldsymbol{c}' \neq \boldsymbol{c}$): $\gamma^-(x_b, \boldsymbol{c}', A, 0) < \gamma^-(x_b, \boldsymbol{c}, A, 0) \leq \gamma \leq \gamma^+(x_b, \boldsymbol{c}, A, 0) < \gamma^+(x_b, \boldsymbol{c}', A, 0)$ (for BS the mirror gives the strict form likewise). Thus $(\boldsymbol{c}', \gamma, A, 0) \in \Omega$, so $\boldsymbol{c}' \in \mathcal{J}$ (Lemma 5.65); with $c'_1, c'_2 > -1$, Lemma 5.46(c) identifies $\gamma^-(\boldsymbol{c}') = U^-_{\boldsymbol{c}'}(x_b)$, $\gamma^+(\boldsymbol{c}') = U^+_{\boldsymbol{c}'}(x_b)$, hence $U^-_{\boldsymbol{c}'}(x_b) < \gamma < U^+_{\boldsymbol{c}'}(x_b)$, so $U^-_{\boldsymbol{c}'} < U^+_{\boldsymbol{c}'}$, $c'_3 \neq \bar{c}_3$, $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$, and $\omega_\varepsilon \in \Omega_{\mathsf{G}}$ by Proposition 5.15(b).

Case $A = 0$, $\kappa \neq 0$. By σ_1 ($(\boldsymbol{c}, \gamma, 0, \kappa) \mapsto (\boldsymbol{c}, \gamma, 0, -\kappa)$, which preserves Ω and Ω_{G}) we may take $\kappa > 0$. Since $\omega \in \Omega$, $\gamma \in \mathrm{FS} \cap \mathrm{BS}$ for $p = (x_b, \boldsymbol{c}, 0, \kappa)$. By Proposition 5.42(c),(d), strictly ($\kappa > 0$): $\gamma^-(x_b, \boldsymbol{c}', 0, \kappa) < \gamma^-(x_b, \boldsymbol{c}, 0, \kappa) \leq \gamma \leq \gamma^+(x_b, \boldsymbol{c}, 0, \kappa) < \gamma^+(x_b, \boldsymbol{c}', 0, \kappa)$. So γ is strictly inside the safe set of $\boldsymbol{c}'$ in both directions; by Lemma 5.46(a) and its mirror both poles of the solution with data ω_ε are generic: $\omega_\varepsilon \in \Omega_{\mathsf{G}}$.

Letting $\varepsilon \downarrow 0$ gives $\omega \in \overline{\Omega_{\mathsf{G}}}$. □

Remark 5.70 (what is not proved)**.** Away from the two faces ($A\kappa \neq 0$), Proposition 5.42 is not available (Numerical observation 5.34), and the argument above does not show that Ω is the closure of Ω_{G}, nor that $\Omega \setminus \Omega_{\mathsf{G}}$ is nowhere dense there. This is recorded in Chapter 13.

5.8. The lens at the rest state, and the numerical atlas

In words. *The theory of the chapter fixes the* shape *of the lens but not its size. For the rest state ($\boldsymbol{c} = 0$, base point 0) the lens can be computed: the floor is the window $[-2, 2]$, the two edges leave the floor at $\kappa = 0$ and meet at a tip $\kappa_{\max} = 4.1608$ on the axis $\gamma = 0$. This section records which parts of that picture are* proved *(the floor,*

the symmetry, the monotonicity, and a wide bracket $2.91 \le \kappa_{\max} \le 5017$ *for the tip) and which are* numerical *(the table of the edge and the values of* $\kappa_{\max}$*), and then collects in one atlas the numerical evidence of the whole chapter together with the method that produced it.*

Example 5.71 (the lens at $\boldsymbol{c} = 0$, $x_0 = 0$, and the bounds $\kappa_{\mathrm{lo}}, \kappa_{\mathrm{up}}$)**.** Throughout $\kappa = \kappa_x$ (equal to $\varkappa$ at $x_0 = 0$). At $c = 0$, $x_0 = 0$: window $[-2, 2]$ at $\kappa = 0$; $\gamma^-(\kappa) = -2, -1.9989, -1.9961, -1.9865, -1.9542, -1.9077, -1.8492, -1.7806, -1.7033$ at $\kappa = 0, 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1, 1.2$, $\gamma^+ = -\gamma^-$; $\kappa_{\max}(0, 0) = 4.1608$ with $\gamma_* = 0$; $\kappa_{\max}(\pm 0.5, 0) = 4.6310$, $\kappa_{\max}(0, (2, -0.5, 0)) = 4.7071$, $\kappa_{\max}(0.5, (2, -0.5, 0)) = \kappa_{\max}(-0.5, (-0.5, 2, 0)) = 6.2919$, $\kappa_{\max}(0, (0, 0, -2.9)) = 2.4281$, $\kappa_{\max}(0, (0, 0, 5)) = 5.7586$, $\kappa_{\max}(0, (1, 1, \bar{c}_3 + 0.3)) = 1.3846$;

What is proved here, and what is numerical. The following parts of the example are consequences of the theory of this chapter and are used as checks. (i) *The floor.* At $\kappa = 0$, $\boldsymbol{c} = 0$, $x_0 = 0$ the window is $[-2, 2]$: Toy example 5.39 and Example 5.9 ($U_{\mathbf{0}}^{\pm}(0) = \pm 2$), and $\gamma^-(\kappa) \to -2$ (Proposition 5.59(e)). (ii) *Symmetry.* $\boldsymbol{c}_m = \boldsymbol{c}$ and $x_0 = 0$, so $\gamma^+ = -\gamma^-$ and the tip is at $\gamma_* = 0$: Proposition 5.56, Proposition 5.59(b) ($\gamma^-(\kappa_{\max}) = \gamma^+(\kappa_{\max}) = -\gamma^-(\kappa_{\max})$). The same reflection gives $\kappa_{\max}(\pm\frac{1}{2}, 0)$ equal, and $\kappa_{\max}(\frac{1}{2}, (2, -\frac{1}{2}, 0)) = \kappa_{\max}(-\frac{1}{2}, (-\frac{1}{2}, 2, 0))$. (iii) *Monotonicity.* $\gamma^-(\kappa)$ increases in κ (Proposition 5.59(a)): the table is increasing; $\kappa_{\max}$ is strictly increasing in each c_i (Proposition 5.59(d)): $2.4281 < 4.1608 < 5.7586$ along $c_3 = -2.9, 0, 5$. (iv) *The bounds.* $\kappa_{\mathrm{up}}(0, \mathbf{0}) = \sqrt{384} \cdot 4^4 \approx 5016.6$ (Proposition 5.58(b) with $K_P(\mathbf{0}) = 8$, $m = 4^{-4}$), and from Proposition 5.58(c) with $x_0 = 0$, $\boldsymbol{c} = \mathbf{0}$, $\tilde{\boldsymbol{c}} = (-\frac{e}{2}, -\frac{e}{2}, 0)$: the potential of $\tilde{\boldsymbol{c}}$ is the constant $W = 1 - \frac{e}{2} = k^2$, $z = \cosh k\xi$ ($\tilde{\gamma} = 0$), $\tilde{g} = \cosh^2 \xi / \cosh^2 k\xi$, $\tilde{f} = \tanh(k\xi)/k$, and

$$B_+ = B_- = \frac{1}{k} \int_0^\infty e^{-2\xi} \frac{\tanh k\xi}{\cosh^2 k\xi} \mathrm{d}\xi, \qquad \kappa_{\mathrm{lo}}(e) = \sqrt{e / B_+(e)}.$$

Maximising over $e \in (0, 2)$ gives $\kappa_{\mathrm{lo}} \approx 2.912$ at $e \approx 1.711$. So $2.91 \le \kappa_{\max}(0, \mathbf{0}) \le 5017$; the true value 4.1608 is inside the (wide) proved bracket. The numerical entries — the table of $\gamma^-(\kappa)$ and the values of $\kappa_{\max}$ — are observations and are recorded, with the recomputation, in Atlas 5.73.

Figure 5.6 compares the proved lower bound with the numerical tip of the lens.

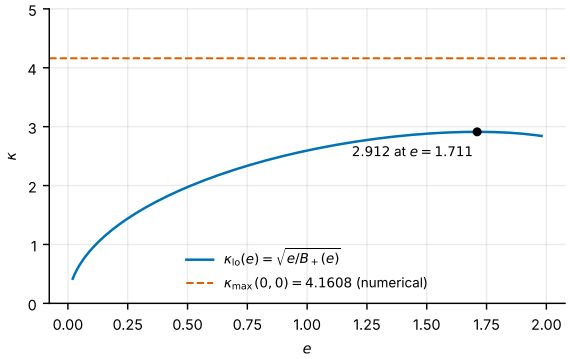


FIGURE 5.6. *Numerical illustration* (not used in any proof): the proved lower bound $\kappa_{\mathrm{lo}}(e) = \sqrt{e/B_+(e)}$ of Proposition 5.58(c) at $\boldsymbol{c} = 0$, $x_0 = 0$ with $\tilde{\boldsymbol{c}} = (-\frac{e}{2}, -\frac{e}{2}, 0)$, $\tilde{\gamma} = 0$, as a function of $e \in (0, 2)$; its maximum 2.912 at $e = 1.711$ is below the numerical tip $\kappa_{\max}(0, \mathbf{0}) = 4.1608$ (dashed). The upper bound $\kappa_{\mathrm{up}} = 5016.6$ is off the scale. Code: `code/ch05_global/make_b_bounds.py`.

Remark 5.72 (Chapter 5 at a glance)**.**

object	statement	where
Ω, Ω_{G}	closed; open; $\partial\Omega \subset \Omega \setminus \Omega_{\mathsf{G}}$	Theorem 5.12 and Proposition 5.15
comparison	$\hat{U} \geq U$ for ordered data (and strictly)	Lemma 5.28 and Corollary 5.29
FS, BS	half-lines for $A\kappa \geq 0$, resp. ≤ 0; edge monotone	Proposition 5.42
edge of FS	type R or $\mathsf{L}_1, \mathsf{L}_2$ ($\kappa > 0$); recessive no-swirl ($\kappa = 0$)	Lemma 5.46
thresholds	l.s.c.; continuous where $\kappa > 0$ or $c_2 > -1$	Proposition 5.50
class Z	$\leftrightarrow$ lens $\{\gamma^- \leq \gamma \leq \gamma^+,\ \kappa \leq \kappa_{\max}\}$	Theorem 5.62
wedge	$\mathsf{Z} \cap V = \{f(1)f(-1) < 0\}$, slopes $F_0(\pm 1)$	Proposition 5.67
faces	$A = 0$ or $\kappa = 0$: in $\overline{\Omega_{\mathsf{G}}}$	Proposition 5.69

Atlas 5.73 (the global picture: lens, wedge slopes, non-uniform limit and sign orders)**.** *All numbers are produced by* `code/ch05_global/verify_ch05b.py` *(python, double precision; the shooting solver is described in Appendix E). Method: the system* (5.2) *in* $\xi = \operatorname{artanh} x$ *by RK4 with step* $0.02/(1 + |h| + \sqrt{|W|})$ *up to* $\xi_0 + 14$ *(+30 for the non-uniform limit), where* $h = z'/z$ *and* $W = 1 + \frac{1}{2}Q$*. A trajectory is declared* unsafe *as soon as* $U < -K_P - 1$ *(then it blows up, Lemma 5.41(c)). A trajectory that survives to the end of the window* ξ_{end} *is tested once more, with the quantities* $q := 2c_2 - \psi(\xi_{\mathrm{end}})$ *(the numerical value of* $Q(1)$*),* $D := 1 + \frac{q}{2}$ *and* $h(\xi_{\mathrm{end}})$*: it is declared* unsafe *if* $D < 0$ *(then* $Q < -2$*, which Theorem 4.5(a) forbids at a pole, so*

the solution cannot reach the pole) or if $h \le -\sqrt{D}$ *(the recessive branch* $h \to -\mu$*), and* safe *if* $D \ge 0$ *and* $h > -\sqrt{D}$*. This terminal test is the classification of the pole by* (q, h) *from Theorem 4.5 and Lemma 5.46: it separates type* G *(*$h \to +\mu$*) from the edge types* $\mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$ *and from solutions that fail later. Without it the survivors of a finite window include solutions that die after* ξ_{end}*, and the method returns wrong values (for the tips of Atlas 5.73(1),(2) it would give about* 4.27*,* 2.70*,* 12.24 *in place of* 4.1608*,* 2.5737*,* 12.148*).* $\gamma^-(\kappa)$ *by bisection in* γ *to* 10^{-11}*,* $\kappa_{\max}$ *by bisection on the sign of* $\gamma^+ - \gamma^-$ *to relative* 10^{-9}*. Status: numerical, not proved.*

(1) The lens at $\boldsymbol{c} = 0$, $x_0 = 0$ *(the table of Example 5.71). The tip is* $\kappa_{\max} = 4.1608149$*,* $\gamma_* = 0$ *to within* 10^{-9}*; mirror symmetry* $\gamma^+ = -\gamma^-$ *and monotonicity in* c_i *hold in all runs.*

(2) Values of $\kappa_{\max}$ *(the values used in Example 5.71, rounded to four places):*

$(x_0; \boldsymbol{c})$	$\kappa_{\max}$
$(\pm 0.5;\, 0, 0, 0)$	4.63097
$(0;\, 2, -0.5, 0)$	4.70711
$(0.5;\, 2, -0.5, 0) \;=\; (-0.5;\, -0.5, 2, 0)$	6.29191
$(0;\, 0, 0, -2.9)$	2.42808
$(0;\, 0, 0, 5)$	5.75860
$(0;\, 1, 1, \bar{c}_3 + 0.3)$	1.38459

(3) The bounds of Example 5.71. $\kappa_{\mathrm{up}}(0, \mathbf{0}) = \sqrt{384}\, 4^4 = 5016.55$ *(*$K_P = 8$*);* $\max_e \kappa_{\mathrm{lo}}(e) = 2.912$ *at* $e = 1.711$ *(with* $e = 1.7$*:* 2.9120*), where* B_+ *computed by the ODE agrees with the closed one-dimensional integral to five digits (*$B_+ = 0.20048$ *at* $e = 1.7$*). The proved bracket* $[2.91, 5.02 \cdot 10^3]$ *contains* 4.1608*; neither end is sharp.*

(4) Wedge slopes *(Proposition 5.67). Shooting at* $\boldsymbol{c} = 0$*,* $x_0 = 0$*,* $A = 0$*:*

γ	κ	$f(1)/\kappa$	$\frac{2}{\gamma+2}$	$f(-1)/\kappa$	*vs* $\frac{2}{\gamma-2}$
0	$10^{-3}, 10^{-2}$	1.00000	1.00000	−1.00000	−1.00000
0.5	$10^{-3}, 10^{-2}$	0.80000	0.80000	−1.33333	−1.33333
−1	$10^{-3}, 10^{-2}$	2.00000	2.00000	−0.66667	−0.66667

(evaluation of $F_0(\pm 1)$ *by computer algebra and shooting agree to five digits).*

(5) The non-uniform limit *(Numerical observation 5.52).* $\boldsymbol{c} = (0, \frac{1}{2}, 0)$*,* $x_b = 0.3$*,* $A = 0$*:* $\gamma^-(\kappa) = -2.959796, -2.959777, -2.959603, -2.957072, -2.923432, -2.646553$ *at* $\kappa = 0,\ 0.005,\ 0.02,\ 0.1,\ 0.5,\ 2$*;* $Q(1) + 2$ *is of order* 10^{-5} *at* $\kappa = 0.005$ *and smaller for larger* κ*: the edge is of type* $\mathsf{L}_1, \mathsf{L}_2$ *(*$U(1) = -2$*) for every* $\kappa > 0$ *tested, while the no-swirl edge has* $U^-_{\boldsymbol{c}}(1) = -2 - 2\sqrt{1.5} = -4.449$*.*

(6) Edge types *(Numerical observation 5.53). Along the lower edge at* $\boldsymbol{c} = (0, -0.9, 0)$ *and* $\boldsymbol{c} = 0$ *(*$x_b = 0$*,* $\xi_{\mathrm{end}} = 30$*,* $\mathrm{d}\xi = 0.01$*):* $Q(1) = -1.8002, -1.8038, -1.8524, -2.0000, -2.0000$ *at* $\kappa = 0.01,\ 0.05,\ 0.2,\ 1,\ 3$*, resp.* $Q(1) = -2.0000$ *for all five* κ *at* $\boldsymbol{c} = 0$*.*

(7) $\kappa_{\max}$ along c_3 *(Numerical observation 5.60),* $x_0 = 0$*,* $\boldsymbol{c} = (0, 0, c_3)$*:* 0.2102*,* 0.5008*,* 1.2992*,* 2.3269*,* 3.1379*,* 4.1608*,* 5.7586 *at* $c_3 = -3.99, -3.95, -3.7, -3, -2, 0, 5$ *(strictly increasing, as proved).*

(8) Sign orders of the cooperative form *(Lemma 5.31; enumerated by* `verify_ch05a.py`*, block V5, over all* 2^6 *sign matrices):*

region	*direction*	*cooperative sign matrices*
$f > 0,\ g > 0$	*forward*	$\pm(+,-,-,-,-,-)$
$f < 0,\ g < 0$	*forward*	$\pm(+,+,+,-,-,-)$
$f > 0 > g$	*backward*	$\pm(+,-,+,+,-,+)$
$f < 0 < g$	*backward*	$\pm(+,+,-,+,-,+)$
other combinations	—	*none*

The pointwise failure of monotonicity for $A\kappa < 0$ is recorded in Numerical observation 5.34 and not repeated here.

Part 3

Classification and the global picture

CHAPTER 6

Class N: the shifts and the half-cylinder

In words. *Class* N_0 *consists of the global solutions whose swirl is positive and increasing. For them the Q-system is cooperative in the direction* away from the south pole*, where the swirl is smallest (Section 5.3), so the natural place to compare two solutions is the pole* $x = -1$ *itself. But a pole is a singular point of the equations: there is no Cauchy problem there, and a solution near the pole is described by its* germ*, the local solution together with its asymptotic data (the exponent* μ*, the constant* λ*, the limit* f_- *of the swirl, the swirl constant, and two more numbers). The two missing numbers are the coefficient of the recessive mode in* z *and the coefficient of the free mode* $(1+x)^2$ *in* Q*, and they cannot be used as coordinates, because the formal expansions that define them divide by numbers that vanish at* resonant *values of* μ *(Toy example 6.1). We replace them by two canonical* flows *on germs: the* t^2*-shift* T_c*, which adds* $c(1+x)^2$ *to* Q*, and the recessive shift* $\mathcal{R}_C$*, which adds the recessive solution with Wronskian* C *to* z*. Each is the unique fixed point of an explicit Volterra contraction, and every constant in it is an explicit moment, with no resonant denominator. This part of the chapter builds the germs, proves that their bounds are uniform in a neighbourhood* also through $\mu = 0$*, where the pole changes from type* G *to type* L_1*, constructs the two flows, shows that both are order preserving and that the set of germs that are global solutions is closed, and proves that the flows are continuous in the germ, including across* $\mu = 0$*.*

Roadmap. Section 6.1 defines the germ space $\mathcal{G}$ with its topology and records what Chapter 4 gives (types, openness of type G, continuity of the frame). Section 6.2 collects the tools (exponential moments, the weight $Z_\mu = \sinh(\mu(1-\xi))/\mu$, a bound on $Q - Q(-1)$ that does not use $\mu > 0$) and proves the *uniform envelopes* for z near a germ of type G (Lemma 6.11(a), weight $e^{-\mu\xi}$) and near a germ of type L_1 (Lemma 6.11(b), weight Z_μ), and the continuity of the frame (Proposition 6.13). Section 6.3 constructs T_c for $0 \le \mu < 2$ and $\mathcal{R}_C$ for $\mu > \frac{1}{2}$ as fixed points, proves uniqueness and the group laws (Theorems 6.19 and 6.22). Section 6.4 proves that positive shifts of germs in N_0 are order preserving (Proposition 6.23), that the set $\mathcal{S}$ of global germs is closed (Lemma 6.25) and that a positive shift of a global germ is of type G at the north pole (Lemma 6.26). Section 6.5 proves that the flows are jointly continuous in the germ and the shift parameter (Lemma 6.28) and, with the uniform envelope, across $\mu = 0$ (Theorem 6.29). The second half of the chapter uses these results to show that the set of shifts c for which $T_c S$ is global is a half-line $[\alpha_-(S), \infty)$ (Theorem 6.32), to build the charts $\mathsf{N}_0 \cong \Gamma_T \times [0, \infty)$ (Theorem 6.34) and $\mathsf{N}_0 \cap \{\mu > 1\} \cong B \times [0, \infty)$ (Theorem 6.50), to show that the two flows commute (Theorem 6.54), and to assemble the structure theorem (Theorem 6.61) and the numerical atlas (Atlas 6.65). *Conventions.* A prime is $\mathrm{d}/\mathrm{d}\xi$, $\xi = \operatorname{artanh} x$; at the south pole $\xi \to -\infty$ and $E(\xi) = e^{2\xi}$ (written E in formulas of this chapter),

$t = 1+x = 2E/(1+E) \in [E, 2E]$ for $\xi \le 0$. The statements of Chapter 4 are proved at the north pole; we use them at the south pole through σ_2 (Lemma 5.1), which exchanges the poles and changes the sign of κ and nothing we use. A bare κ in this chapter is the chart constant κ_{ch} (normalisation 6 of Lemma 2.15); its value in the cut frame $z(\xi_1) = 1$ is $\kappa_{S,\xi_1} = \kappa_{\mathrm{ch}}\, z(\xi_1)^{-2}$ (normalisation 7). The constants κ_{env} of Lemma 6.11 are bounds for κ_{S',ξ_1} and have nothing to do with the lens quantity $\kappa_{\max}(x_0, c)$ of Chapter 5.

Toy example 6.1 (why Frobenius coefficients are not coordinates)**.** Consider the linear equation $z'' = (\mu^2 + e^{2\xi})z$ as $\xi \to -\infty$, the simplest potential of the form "constant $+\,O(E)$". A formal solution $z = e^{-\mu\xi}\sum_k a_k e^{2k\xi}$ requires $a_k\big((2k-\mu)^2 - \mu^2\big) = a_{k-1}$, i.e.

$$a_k = \frac{a_{k-1}}{4k(k-\mu)} \qquad (k \ge 1,\ a_0 = 1),$$

so $a_1 = \frac{1}{4(1-\mu)}$ blows up as $\mu \to 1$, while the recessive series $e^{\mu\xi}\sum_k b_k e^{2k\xi}$, $b_k = b_{k-1}/(4k(k+\mu))$, is harmless. At $\mu = 1$ the term $a_1 e^{(2-\mu)\xi}$ and the recessive mode $e^{\mu\xi}$ have the same exponent and a logarithm is needed: $z_1 = e^{-\xi} + \frac{1}{2}\xi e^{\xi}$ satisfies $z_1'' - (1+e^{2\xi})z_1 = -\frac{1}{2}\xi e^{3\xi}$ (the three terms of $z_1'' - (1+e^{2\xi})z_1$ are $e^{\xi} - e^{\xi} - \frac{1}{2}\xi e^{3\xi}$: $z_1'' = e^{-\xi} + \frac{1}{2}\xi e^{\xi} + e^{\xi}$, and $e^{2\xi}z_1 = e^{\xi} + \frac{1}{2}\xi e^{3\xi}$), two orders smaller, and the true solutions are $e^{-\xi} + \frac{1}{2}\xi e^{\xi} + \beta e^{\xi} + O(\xi e^{3\xi})$ with a free β. Write a solution as $A(D_\mu + \alpha R_\mu)$ with D_μ, R_μ the dominant and recessive series; its terms beyond $e^{-\mu\xi}$ are $e^{\mu\xi}\big(\alpha + \frac{1}{4(1-\mu)} + \frac{\xi}{2} + O((1-\mu)\xi^2)\big)$ (use $e^{2(1-\mu)\xi} = 1 + 2(1-\mu)\xi + \cdots$). If the solutions depend continuously on μ and converge to the $\mu = 1$ solution, then $\alpha + \frac{1}{4(1-\mu)} \to \beta$: the *label* α tends to $-\infty$ as $\mu \uparrow 1$ and to $+\infty$ as $\mu \downarrow 1$ although nothing happens to the solutions. *Differences are regular*: for two solutions of the same equation $\mathrm{Wr}(z_A, z_B)$ is constant, equal to $2\mu A_A A_B(\alpha_B - \alpha_A)$ at the pole and to a continuous function of μ if computed at $\xi = 0$. The toy shows only what happens at $\mu = 1$; we make no claim about other resonant exponents or about the coefficients of the true Q-expansion, because none is needed: the two flows below are built from *differences* measured by Wronskians and by $(1+x)^2$-increments, never from such labels.

6.1. Germs at the south pole

In words. *A germ is a solution of the Q-system near $x = -1$ together with its maximal continuation to the right. Two germs are close when their six-dimensional states are close at one interior point. Within* N_0 *every germ has pole type* G *or* L_1*, and the stratum* G *is open; the only new point is that type* L_1 *is the boundary of type* G *($\mu \to 0$), which is why uniform bounds must be proved in a neighbourhood that contains both types.*

Types at the south pole. For a local solution at $x = -1$ (the image under σ_2 of Definition 4.1) let $q_0 = Q(-1) \ge -2$ and $\mu = \sqrt{1 + q_0/2}$ (Theorem 4.5). Through σ_2, Definition 4.16 reads: type G means $q_0 > -2$ and $h = z'/z \to -\mu$ (so z grows like $e^{-\mu\xi}$ as $\xi \to -\infty$ and $U_\theta(-1) = 2 - 2\mu < 2$); type R means $q_0 > -2$ and $h \to +\mu$; type L_1 means $q_0 = -2$ and $f(-1)$ is finite, in which case $z = a + b|\xi| + o(1)$, $z' \to -b < 0$ (Proposition 4.19(c) mirrored); type L_2 means $q_0 = -2$ and $f(-1)$ infinite. We use the *pole normalisation* $ze^{\mu\xi} \to 1$ in type G and $z' \to -1$ in type L_1 (so $z/|\xi| \to 1$), and write κ for the swirl constant in it; since $(z, \kappa) \mapsto (sz, s^2\kappa)$

changes nothing else (Lemma 2.14), the normalisation is a choice of representative. The *frame* of a solution is $(\mu, \lambda, f_-, \kappa)$ with $f_- := f(-1)$. For a cut $\xi_1 \le 0$ the *cut-normalised* representative is

$$\begin{aligned} z_{S,\xi_1}(\xi) &= \exp \int_{\xi_1}^{\xi} h_S, \quad h_S = \tfrac{1}{2}U + \tanh \xi, \\ \kappa_{S,\xi_1} &:= f'(\xi_1) = \cosh^{-2}\xi_1 \, f_x(\tanh \xi_1) \end{aligned} \tag{6.1}$$

(normalisation 7 of Lemma 2.15); both depend continuously on the state at ξ_1, and $\kappa_{S,\xi_1} = \kappa \, z(\xi_1)^{-2}$ in the pole normalisation.

Definition 6.2 (germs)**.** A *germ* is a maximal solution $S = (U, f, Q)$ of the Q-system (2.22) on an interval $(-1, x_S^*)$, $x_S^* \in (-1, 1]$, with $\kappa \neq 0$, which is of type G or L_1 at $x = -1$. Its state is $Y_S = (U, f, f_x, Q, Q_x, Q_{xx})$. Let $\mathcal{G}$ be the set of germs, $\mathcal{G}_{\mathsf{G}}, \mathcal{G}_{\mathsf{L}_1}$ the subsets of the two types, and

$$\mathcal{G}_{\mathsf{N}_0} = \{S \in \mathcal{G} : \ \kappa > 0, \ f_- \ge 0\}, \qquad \mathcal{S} = \{S \in \mathcal{G}_{\mathsf{N}_0} : \ x_S^* = 1\}.$$

For $S \in \mathcal{G}$, $x_* < x_S^*$ and $\varepsilon > 0$ put $N(S, x_*, \varepsilon) = \{S' \in \mathcal{G} : \ x_{S'}^* > x_*, \ |Y_{S'}(x_*) - Y_S(x_*)| < \varepsilon\}$.

Lemma 6.3 (germs; the topology)**.** (a) *A local solution at $x = -1$ with $\kappa \neq 0$ and $f(-1)$ finite is of type G or L_1. Every $S \in \mathsf{N}_0$ (Definition 5.3) is a germ of $\mathcal{G}_{\mathsf{N}_0}$, hence of type G or L_1 at the south pole, and of type G, R (with $\mu_+ < \frac{1}{2}$), L_1 or L_2 at the north pole.*

(b) *The sets $N(S, x_*, \varepsilon)$ form a base of a Hausdorff topology on $\mathcal{G}$; for fixed S and $x_* < x_S^*$ the sets $N(S, x_*, 1/k)$, $k \in \mathbb{N}$, form a neighbourhood base of S. Thus $\mathcal{G}$ is first countable, and $S_n \to S$ iff for one (equivalently every) $x_* < x_S^*$ we have $x_{S_n}^* > x_*$ for large n and $Y_{S_n}(x_*) \to Y_S(x_*)$. Continuity of maps on $\mathcal{G}$ (and on its products with $\mathbb{R}$) can be tested on sequences.*

(c) *$\mathcal{G}_{\mathsf{N}_0}$ is closed in $\mathcal{G}$, and $\mathcal{S}$ is exactly the set of solutions of class N_0.*

Proof. (a) Types are exhaustive and exclusive (Definition 4.16). By Proposition 4.19(b),(c), mirrored, $|f| \to \infty$ at $x = -1$ in types R and L_2; so $f(-1)$ finite leaves G, L_1. For $S \in \mathsf{N}_0$, f_- exists, is finite and ≥ 0 (Proposition 5.4(d)), and $\kappa > 0$. At $x = +1$ the type is one of $\mathsf{G}, \mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$ (Definition 4.16) and $\mu_+ < \frac{1}{2}$ in type R (Theorem 4.9).

(b) We use continuous dependence (Remark 2.7(iii)): if the solution through Y_a at x_a exists on a compact interval $I \ni x_a$, then for Y_a' near Y_a the solution exists on I and converges uniformly on I as $Y_a' \to Y_a$. *Base.* Let $S \in N_1 \cap N_2$, $N_i = N(S_i, x_i, \varepsilon_i)$, put $\eta_i = \varepsilon_i - |Y_S(x_i) - Y_{S_i}(x_i)| > 0$ and $\bar{x} = \max(x_1, x_2) < x_S^*$. There is $\delta > 0$ such that every S' with $x_{S'}^* > \bar{x}$ and $|Y_{S'}(\bar{x}) - Y_S(\bar{x})| < \delta$ exists on $[\min(x_1, x_2), \bar{x}]$ with $|Y_{S'}(x_i) - Y_S(x_i)| < \eta_i$; by the triangle inequality $N(S, \bar{x}, \delta) \subset N_1 \cap N_2$. Hence the sets form a base. *Hausdorff.* If $S_1 \neq S_2$ choose $x_* < \min(x_{S_1}^*, x_{S_2}^*)$; by uniqueness $Y_{S_1}(x_*) \neq Y_{S_2}(x_*)$, and the basic sets of radius half the distance are disjoint. *Neighbourhood base.* If $S \in N(S', x', \varepsilon')$ then, as in the first step, $N(S, x', \delta) \subset N(S', x', \varepsilon')$ for some δ, and by continuous dependence on the compact interval between x_* and x' there is k with $N(S, x_*, 1/k) \subset N(S, x', \delta)$. The description of convergent sequences follows in the same way. In a first countable space continuity and closedness are tested on sequences.

(c) Let $S_n \to S$, $S_n \in \mathcal{G}_{\mathsf{N}_0}$. The sign of κ is the sign of f_x, a component of the state, so $f_{x,S}(x_*) = \lim f_{x,S_n}(x_*) \ge 0$ and $\kappa_S \neq 0$ gives $\kappa_S > 0$. Moreover

$f_{S_n} \geq f_{S_n,-} \geq 0$ (increasing functions), so $f_S(x) = \lim f_{S_n}(x) \geq 0$ and $f_{S,-} \geq 0$. If $S \in \mathcal{S}$ then f is strictly increasing with $f > f_- \geq 0$ on $(-1,1)$, so $S \in \mathsf{N}_0$; conversely an element of N_0 is a global germ with $\kappa > 0$, $f_- \geq 0$ by (a). □

Proposition 6.4 (type G is open; the frame is continuous on it)**.** *Let $S \in \mathcal{G}_{\mathsf{G}}$ and $x_* < x_S^*$. There is $\varepsilon > 0$ such that every solution of the Q-system whose state at x_* is ε-close to $Y_S(x_*)$ exists on $(-1, x_*]$ and is of type G at $x = -1$; in particular $\mathcal{G}_{\mathsf{G}}$ is open in $\mathcal{G}$. On $\mathcal{G}_{\mathsf{G}}$ the maps $S \mapsto \mu(S)$, $\lambda(S)$, $f_-(S)$ are continuous, and so is $S \mapsto a_{\xi_1}(S) := \lim_{\xi \to -\infty} z_{S,\xi_1}(\xi)\, e^{\mu(\xi - \xi_1)} \in (0, \infty)$ for each fixed cut $\xi_1 < \operatorname{artanh} x_S^*$.*

Proof. Apply σ_2: $\sigma_2 S$ is a local solution at the north pole of type G with $\kappa \neq 0$, and its state at ξ is the reflected state of S at $-\xi$. The scale-free state $s = (h, k, f, Q, \dot{Q}, \lambda)$ of Corollary 4.42 and the state Y at the same point are related by a diffeomorphism of $\mathbb{R}^6$ ($h = \frac{1}{2}U + x$, $k = (1-x^2)f_x$, $\dot{Q} = (1-x^2)Q_x$, and $\lambda = (1-x^2)Q_{xx} + 2xQ_x - 2Q + f^2$, by Proposition 2.20). Choose $\xi_c \geq 0$ with $-\xi_c \leq \operatorname{artanh} x_*$. By continuous dependence (Remark 2.7(iii)) there is $\varepsilon > 0$ such that every solution with $|Y(x_*) - Y_S(x_*)| < \varepsilon$ exists on $[x_c, x_*]$, $x_c := -\tanh \xi_c$, and its state at x_c is close to that of S; hence the scale-free state at ξ_c of its σ_2-image lies in the open set $\mathcal{G}_{\xi_c} \ni s(\sigma_2 S)(\xi_c)$ of Corollary 4.42. So the image exists on $[\xi_c, \infty)$ with $z > 0$ and is of type G, i.e. the solution exists on $(-1, x_c]$, hence on $(-1, x_*]$, and is of type G at -1 ($\kappa \neq 0$ because $k \neq 0$ on $\mathcal{G}_{\xi_c}$). This is the first claim. By Corollary 4.43, on $\mathcal{G}_{\xi_c}$ the exponent μ, the limit $\varphi = f(\infty)$ and the dominant coefficient a (relative to the value at the cut) are smooth functions of s; composed with the continuous maps $Y(x_*) \mapsto s$ above they are continuous functions of S, and λ is a function of the state. The coefficient a of Corollary 4.43 is, after reflection, $a_{-\xi_c}$; for another cut ξ_1 use $z_{S,\xi_1} = z_{S,-\xi_c} / z_{S,-\xi_c}(\xi_1)$, where $z_{S,-\xi_c}(\xi_1) = \exp \int_{-\xi_c}^{\xi_1} h_S$ is continuous in S by continuous dependence on a compact interval. □

6.2. Tools and uniform envelopes

In words. *Everything in this chapter is a Volterra equation on a half line $(-\infty, \xi_1]$ whose coefficients are built from $z^{\pm 2}$ and from the swirl. What makes it solvable is a pair of two-sided bounds $m_- Z \leq z \leq m_+ Z$ with a* weight *Z whose growth is controlled; we call such a pair an* envelope*. Near a germ of type G the weight is $e^{-\mu\xi}$ and the envelope is a by-product of the chart of Chapter 4. Near a germ of type L_1 the solution grows linearly, $z \sim b|\xi|$, while nearby germs of type G with tiny μ look like $e^{-\mu\xi}$ only for $|\xi| \gg 1/\mu$; no single power or exponential serves. The weight $Z_\mu(\xi) = \sinh(\mu(1-\xi))/\mu$ (the solution of $y'' = \mu^2 y$ that vanishes at $\xi = 1$) is linear for $\mu = 0$, exponential for $\mu|\xi| \gg 1$, and depends continuously on μ; with it one gets an envelope that does not degenerate as $\mu \to 0$.*

6.2.1. Four classical facts and the moment lemma. We use the following facts, all standard. **(F1)** Banach's fixed point theorem (Theorem A.2), in a weighted sup-norm space $\{u \in C(I) : \sup|u|/\varpi < \infty\}$ ($\varpi > 0$ continuous), which is complete because $u \mapsto u/\varpi$ is an isometry onto the bounded continuous functions; closed subsets are complete. **(F2)** Continuous dependence of solutions of the Q-system on the state at one point, uniformly on compact intervals of existence (Remark 2.7(iii)). **(F3)** Dominated convergence. **(F4)** If $g_n \to g$ pointwise on a compact interval I and $|g_n'| \leq L$ for all n, then $g_n \to g$ uniformly on I. (Given $\eta > 0$ pick a grid of mesh

$\le \eta/(3L)$ and n_0 with $|g_n - g| \le \eta/3$ on the grid for $n \ge n_0$; g is also L-Lipschitz, so $|g_n - g| \le \eta$ on I.) Toy: $g_n(\xi) = \sin(n\xi)/n$ has $|g_n'| \le 1$ and tends to 0 uniformly, whereas $\max(0, 1 - n|\xi|) \to \mathbf{1}_{\{0\}}$ pointwise but not uniformly; the Lipschitz bound is what excludes the latter.

We say that a function $Z : (-\infty, \xi_1] \to [1, \infty)$ has *rate* $\bar\mu \ge 0$ if it is non-increasing and $Z(\eta) \le (1 + \xi - \eta) e^{\bar\mu(\xi-\eta)} Z(\xi)$ for $\eta \le \xi \le \xi_1$. Examples: $Z = e^{-\mu(\xi-\xi_1)}$ has rate μ (as $1 \le 1 + s$), and $Z = 1 + \xi_1 - \xi$ has rate 0 (as $\frac{1+\xi_1-\eta}{1+\xi_1-\xi} = 1 + \frac{\xi-\eta}{1+\xi_1-\xi} \le 1 + (\xi - \eta)$).

Lemma 6.5 (exponential moments)**.** *Let Z have rate $\bar\mu$ on $(-\infty, \xi_1]$ and $\mathcal{M}(r) := \int_0^\infty (1+s)^2 e^{-rs} \mathrm{d}s = \frac{1}{r} + \frac{2}{r^2} + \frac{2}{r^3}$ ($r > 0$; decreasing in r). For $\xi \le \xi_1$:* (a) *if $p > 2\bar\mu$, then $\int_{-\infty}^{\xi} Z^2 e^{p\eta} \mathrm{d}\eta \le \mathcal{M}(p - 2\bar\mu)\, Z(\xi)^2 e^{p\xi}$;* (b) *if $p > 0$, then $\int_{-\infty}^{\xi} Z^{-2} e^{p\eta} \mathrm{d}\eta \le Z(\xi)^{-2} e^{p\xi}/p$;* (c) *if $\bar\mu < 2$, then $Z(\xi)^2 E(\xi)^2 \to 0$ as $\xi \to -\infty$.*

Proof. (a) Put $\eta = \xi - s$: $Z(\eta)^2 \le Z(\xi)^2 (1+s)^2 e^{2\bar\mu s}$ and $e^{p\eta} = e^{p\xi} e^{-ps}$, so the integral is at most $Z(\xi)^2 e^{p\xi} \int_0^\infty (1+s)^2 e^{-(p-2\bar\mu)s} \mathrm{d}s$, and $\int_0^\infty s^k e^{-rs} \mathrm{d}s = k!/r^{k+1}$. (b) Z is non-increasing, so $Z(\eta) \ge Z(\xi)$ for $\eta \le \xi$; then $\int_{-\infty}^{\xi} e^{p\eta} = e^{p\xi}/p$. (c) By the rate condition between ξ and ξ_1, $Z(\xi)^2 E(\xi)^2 \le Z(\xi_1)^2 (1 + \xi_1 - \xi)^2 e^{2\bar\mu(\xi_1 - \xi)} e^{4\xi} \to 0$ since $4 > 2\bar\mu$. □

6.2.2. The weight Z_μ.

Lemma 6.6 (the weight Z_μ)**.** *Let $\mu \ge 0$, $Z_\mu(\xi) = \sinh(\mu(1-\xi))/\mu$ for $\mu > 0$ and $Z_0(\xi) = 1 - \xi$; let $\xi \le 0$.*

(i) *$Z_\mu \ge 1 - \xi \ge 1$; Z_μ is decreasing in ξ and non-decreasing in μ; $Z_\mu'' = \mu^2 Z_\mu$, $Z_\mu(1) = 0$, $Z_\mu'(1) = -1$.*

(ii) *For $\eta \le \xi \le 0$ and $s = \xi - \eta$:* $\dfrac{Z_\mu(\eta)}{Z_\mu(\xi)} = \cosh \mu s + \coth(\mu(1-\xi)) \sinh \mu s \le (1+s) e^{\mu s}$*; so Z_μ has rate μ.*

(iii) $\int_{-\infty}^{\xi} Z_\mu^{-2} \mathrm{d}\eta \le \dfrac{1}{1-\xi}$.

(iv) *With $k_\mu(\xi, \eta) = \sinh(\mu(\eta - \xi))/\mu$ ($= \eta - \xi$ if $\mu = 0$), for $\xi \le \eta \le 1$,*

$$k_\mu(\xi, \eta)\, Z_\mu(\eta) \le Z_\mu(\xi)\, \min\{\eta - \xi,\ 1 - \eta\}. \tag{6.2}$$

(v) *Let $\xi_1 \le 0$, $a_1 = 1 - \xi_1$, $\beta > 0$ and $y_\beta(\xi) = \cosh \mu s + \frac{\beta}{\mu} \sinh \mu s$, $s = \xi_1 - \xi \ge 0$ (the solution of $y'' = \mu^2 y$ with $y(\xi_1) = 1$, $y'(\xi_1) = -\beta$). Put $p = \mu / \sinh(\mu a_1)$ ($= 1/a_1$ if $\mu = 0$), $q = \beta / \cosh(\mu a_1)$, $\ell = \min(p, q)$, $L = \max(p, q)$. Then $\ell Z_\mu \le y_\beta \le L Z_\mu$ on $(-\infty, \xi_1]$. If $\mu \le 1$ then $\frac{1}{a_1 \cosh a_1} \le p \le \frac{1}{a_1}$.*

Proof. (i) $\sinh y \ge y$ gives $Z_\mu \ge 1 - \xi$; $\partial_\xi Z_\mu = -\cosh(\mu(1-\xi)) < 0$; $\mu \mapsto \sinh(\mu y)/\mu$ has derivative $(\mu y \cosh \mu y - \sinh \mu y)/\mu^2 \ge 0$; the ODE and the values at $\xi = 1$ are immediate. (ii) With $a = 1 - \xi$, $\sinh(\mu(a+s)) = \sinh \mu a \cosh \mu s + \cosh \mu a \sinh \mu s$ gives the identity. Using $\coth y \le 1 + \frac{1}{y}$ ($e^{2y} - 1 \ge 2y$), $\sinh x \le x \cosh x$ and $a \ge 1$: $\coth(\mu a) \sinh \mu s \le \sinh \mu s + \frac{\sinh \mu s}{\mu a} \le \sinh \mu s + s \cosh \mu s$, so the ratio is $\le e^{\mu s} + s \cosh \mu s \le (1+s) e^{\mu s}$. For $\mu = 0$ the ratio is $1 + s/a \le 1 + s$. (iii) $Z_\mu(\eta) \ge 1 - \eta$ and $\int_{-\infty}^{\xi} (1-\eta)^{-2} \mathrm{d}\eta = \frac{1}{1-\xi}$. (iv) Let $r_1 = \eta - \xi$, $r_2 = 1 - \eta$. For $\mu > 0$, $\sinh(a+b) = \sinh a \sinh b (\coth a + \coth b)$ with $a = \mu r_1$, $b = \mu r_2$ gives

$$\frac{k_\mu(\xi,\eta)Z_\mu(\eta)}{Z_\mu(\xi)} = \frac{\sinh\mu r_1 \sinh\mu r_2}{\mu\sinh\mu(r_1+r_2)} = \frac{1}{\mu(\coth\mu r_1+\coth\mu r_2)} \le \frac{\tanh(\mu\min(r_1,r_2))}{\mu} \le \min(r_1,r_2),$$

because coth is decreasing, so the sum is $\ge \coth(\mu\min(r_1,r_2))$. For $\mu=0$ the left side is $\frac{r_1r_2}{r_1+r_2} \le \min(r_1,r_2)$. (v) Let $\mathrm{ch}=\cosh\mu s$ and $\mathrm{sh}=\sinh(\mu s)/\mu$ (both positive). Then $y_\beta = \mathrm{ch}+\beta\,\mathrm{sh}$ and, by (ii)'s addition formula with $\mu(a_1+s)$, $Z_\mu(\xi)=\varrho\,\mathrm{ch}+\varsigma\,\mathrm{sh}$ with $\varrho=\sinh(\mu a_1)/\mu$, $\varsigma=\cosh\mu a_1$. The ratio of two positive combinations lies between the ratios $1/\varrho = p$ and $\beta/\varsigma = q$ of the coefficients (mediant inequality). Finally for $\mu\le 1$, $x=\mu a_1 \le a_1$: $x/\sinh x \le 1$ and, since $\sinh x \le x\cosh x$, $x/\sinh x \ge 1/\cosh x \ge 1/\cosh a_1$; and $p = \frac{\mu}{\sinh x} = \frac{1}{a_1}\frac{x}{\sinh x}$. □

The weights are plotted in Figure 6.1.

Toy example 6.7 (the normalisation degenerates at $\mu=0$, the cut normalisation does not)**.** Take the frozen equation $z''=\mu^2 z$ with $z(\xi_1)=1$, $z'(\xi_1)=-\beta<0$. Then $z=A_\mu e^{-\mu\xi}+B_\mu e^{\mu\xi}$ with $A_\mu = \frac{(\beta+\mu)e^{\mu\xi_1}}{2\mu}$ (solve the two conditions: $z\mp z'/\mu = 2A_\mu e^{-\mu\xi}$, $2B_\mu e^{\mu\xi}$ at ξ_1), so $2\mu A_\mu \to \beta$ as $\mu\to 0$, while $z\to 1+\beta(\xi_1-\xi)$ is linear. The pole normalisation $ze^{\mu\xi}\to 1$ divides z by A_μ and therefore *divides* κ *by* A_μ^2: $\kappa_{\mathsf G} = \kappa_{\xi_1}/A_\mu^2 \approx 4\mu^2\kappa_{\xi_1}/\beta^2 \to 0$. A family of germs converging to a germ of type $\mathsf L_1$ with $\kappa_{\xi_1}>0$ therefore has pole-normalised swirl constants tending to 0, whereas the limit germ has a positive constant κ_{ξ_1}/b^2: *the pole-normalised* κ *is discontinuous at* $\mathsf L_1$, and so is the constant $\sup z/e^{-\mu\xi} \sim 1/\mu$. The fixed-cut quantities $z_{S,\xi_1}, \kappa_{S,\xi_1}$ are continuous and z/Z_μ stays between two constants; this is why the chapter works with the cut normalisation (Figure 6.2), and why the shifts, which are invariant under $(z,\kappa)\mapsto(sz,s^2\kappa)$, may be built in it.

6.2.3. A bound for $Q-Q(-1)$ that does not use $\mu>0$.

Lemma 6.8 (Q is Lipschitz in $1+x$ at the pole)**.** *Let $S\in\mathcal{G}_{\mathsf{N}_0}$ be a germ (type $\mathsf G$ or $\mathsf L_1$), $x_B\le 0$, $x_B < x_S^*$ and $t_x = 1+x$. Then $0\le f\le f(x_B)$ on $(-1,x_B]$ and*

$$|Q(x)-Q(-1)| \le M' t_x \quad (-1<x\le x_B),$$
$$M' = M'(S,x_B) := 2|Q(x_B)| + 2|Q_x(x_B)| + 3\big(|\lambda| + f(x_B)^2\big).$$

Consequently $w := \frac12(Q-Q(-1))$ satisfies $|w|\le M'E$ on $(-\infty,\operatorname{artanh} x_B]$, and $1+\frac12 Q = \mu^2 + w$. No assumption on $Q(-1)$ other than finiteness is made.

Proof. Since $\kappa>0$, f is increasing and $f(-1)=f_-\ge 0$, so $0\le f\le f(x_B)$ and $F:=\lambda - f^2$ satisfies $|F|\le F_0 := |\lambda| + f(x_B)^2$. The Cauchy formula (Lemma 2.18(d)) with base point x_B and $\mathsf L[Q]=F$ gives, for $x<x_B$, $Q(x) = Q(x_B)A_1(x) + Q_x(x_B)A_2(x) - \int_x^{x_B} K_{\mathsf L}(x,s)F(s)\mathrm{d}s$. Write $t_s = 1+s$, $t_B = 1+x_B \in (0,1]$. With $x=t_x-1$, $s=t_s-1$ the following (from $K_{\mathsf L}(x,s) = (x-s)(1-xs)/(1-s^2)^2$ of Lemma 2.18(c): the numerator of $K_{\mathsf L}(x,s)-K_{\mathsf L}(-1,s)$ is $-sx^2+(1+s^2)x+1+s+s^2$, which vanishes at $x=-1$ and equals $(x+1)(-sx+1+s+s^2) = t_x\big(t_s^2+t_x(1-t_s)\big)$ in the variables $t=1+x$, $t_s=1+s$)

$$K_{\mathsf L}(x,s) - K_{\mathsf L}(-1,s) = \frac{t_x\big(t_s^2+t_x(1-t_s)\big)}{(1-s^2)^2}, \qquad K_{\mathsf L}(-1,s) = -(1-s)^{-2}.$$

As $(1-s^2)^2 = t_s^2(2-t_s)^2 \ge t_s^2$, $0<t_x\le t_s\le 1$ for $x\le s\le x_B$: $0\le K_{\mathsf L}(x,s) - K_{\mathsf L}(-1,s) \le t_x(1+t_x/t_s^2)$ and $\int_x^{x_B} t_x(1+t_x/t_s^2)\mathrm{d}s = t_x(t_B-t_x) + t_x^2(\frac{1}{t_x}-\frac{1}{t_B}) \le$

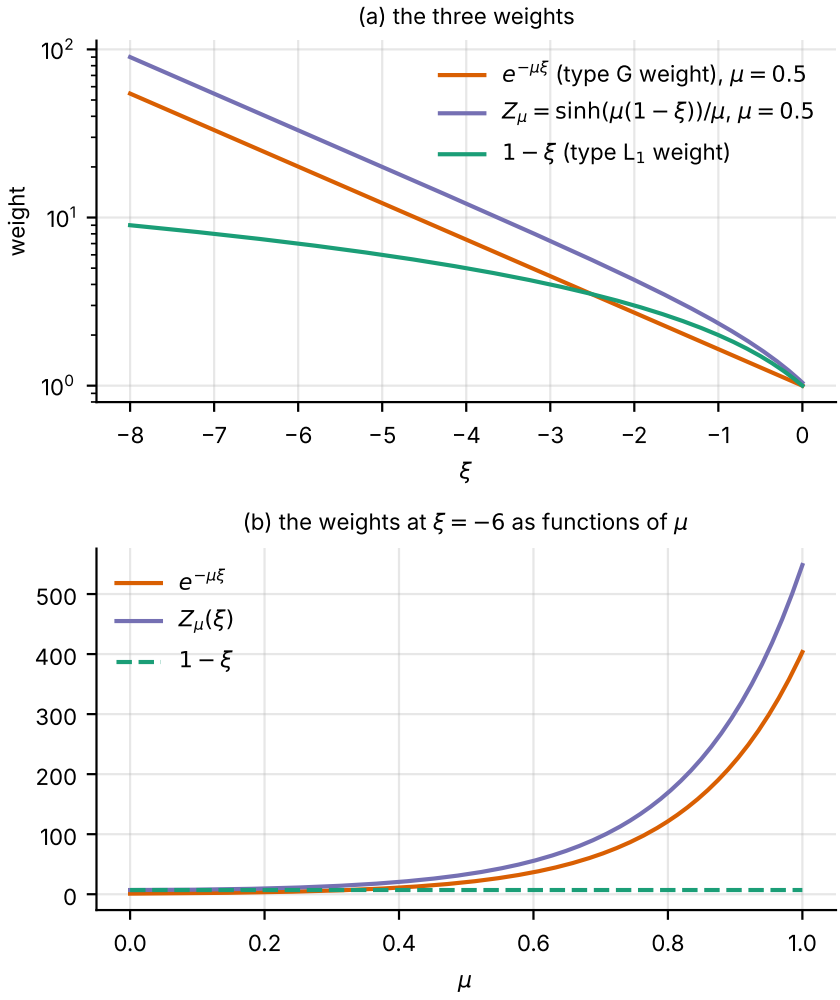


FIGURE 6.1. (a) The type-G weight $e^{-\mu\xi}$, the type-L_1 weight $1-\xi$ and Z_μ ($\mu = 0.5$). (b) At $\xi = -6$ the weight $Z_\mu(\xi)$ is a continuous function of $\mu \in [0,1]$ that joins $1-\xi$ at $\mu = 0$; $e^{-\mu\xi}$ does not. Code: `code/ch06_classN/make_a_envelope.py`.

$t_x t_B + t_x \le 2t_x$. Also $|K_L(-1,s)| \le 1$ for $s \le 0$. Hence, letting $x \to -1$ in the formula (the integral converges by this bound), $Q(-1) = Q(x_B)A_1(-1) + Q_x(x_B)A_2(-1) - \int_{-1}^{x_B} K_L(-1,s)F\,ds$, and

$$Q(x) - Q(-1) = Q(x_B)\big(A_1(x) - A_1(-1)\big) + Q_x(x_B)\big(A_2(x) - A_2(-1)\big) - \int_x^{x_B} \big(K_L(x,s) - K_L(-1,s)\big)F\,ds + \int_{-1}^{x} K_L(-1,s)F\,ds.$$

The last two terms are bounded by $2t_xF_0$ and t_xF_0. A direct computation gives $A_1(x)-A_1(-1)=-\frac{t_x(2t_B-t_x)}{t_B(2-t_B)}$ and $A_2(x)-A_2(-1)=\frac{t_x(t_B^2+t_x(1-t_B))}{t_B(2-t_B)}$; for $0<t_x\le t_B\le 1$ their moduli are at most $\frac{2t_x}{2-t_B}\le 2t_x$ and $\frac{t_x}{2-t_B}\le t_x$. Summing gives M'. Finally $t\le 2E$ and $\mu^2=1+Q(-1)/2$. □

Lemma 6.9 (the exponent is small near L_1)**.** *Let $S\in\mathcal{G}_{\mathsf{N}_0}$ be of type L_1 and $\nu>0$. There is a neighbourhood V of S in $\mathcal{G}_{\mathsf{N}_0}$ with $\mu(S')\le\nu$ for all $S'\in V$.*

Proof. Fix $x_B\le 0$, $x_B<x_S^*$, and $\varepsilon_0>0$ such that on $V_0=N(S,x_B,\varepsilon_0)\cap\mathcal{G}_{\mathsf{N}_0}$ the state at x_B stays in a compact set; λ is a function of the state, so $M_B:=\sup_{V_0}M'(\cdot,x_B)<\infty$. For $x_C\le x_B$ and $S'\in V_0$, Lemma 6.8 gives $|Q_{S'}(x_C)-q_0'|\le M_Bt_C$, and likewise $|Q_S(x_C)+2|\le M_Bt_C$ ($q_0(S)=-2$). If $S'\in N(S,x_C,\delta)\cap V_0$ then $q_0'+2\le|q_0'-Q_{S'}(x_C)|+|Q_{S'}(x_C)-Q_S(x_C)|+|Q_S(x_C)+2|<2M_Bt_C+\delta$, i.e. $\mu'^2=\frac12(q_0'+2)<M_Bt_C+\frac{\delta}{2}$. Choose x_C with $M_Bt_C<\nu^2/4$ and $\delta<\nu^2/2$; then $\mu'<\nu$. By Lemma 6.3(b) $N(S,x_C,\delta')\subset N(S,x_B,\varepsilon_0)$ for small $\delta'\le\delta$, so $V=N(S,x_C,\delta')\cap\mathcal{G}_{\mathsf{N}_0}$ works. □

6.2.4. Envelopes.

Definition 6.10 (envelope)**.** Let $S\in\mathcal{G}$ with a representative (z,κ) (any positive z with $f'=\kappa z^{-2}$), let $\xi_1\le 0$ with $\tanh\xi_1<x_S^*$, and let $\bar\mu\ge 0$. An *envelope of S on $(-\infty,\xi_1]$ with rate $\bar\mu$ and data $\mathfrak{d}=(m_-,m_+,k_0,f_{\max},\bar\mu)$* is a function $Z:(-\infty,\xi_1]\to[1,\infty)$ of rate $\bar\mu$ together with constants $0<m_-\le m_+$, $f_{\max}$ and $k_0=|\kappa|m_-^{-2}$ such that

$$m_-Z\le z\le m_+Z,\qquad |f|\le f_{\max}\qquad\text{on }(-\infty,\xi_1];$$
$$\text{in particular}\quad |\kappa|z^{-2}\le k_0Z^{-2}.$$

Lemma 6.11 (uniform envelopes)**.** (a) (Near type G.) *Let $S\in\mathcal{G}_{\mathsf{G}}$. There are $\xi_1\le 0$, a neighbourhood $V\subset\mathcal{G}_{\mathsf{G}}$ of S and constants $0<m_-\le m_+$, $f_{\max}$, $\kappa_{\rm env}$, $0<\mu_{\min}\le\mu_{\max}$ such that every $S'\in V$ exists on $(-\infty,\xi_1]$, has $\mu'=\mu(S')\in[\mu_{\min},\mu_{\max}]$, $|\kappa_{S',\xi_1}|\le\kappa_{\rm env}$, and its cut-normalised z_{S',ξ_1} has the envelope $Z(\xi)=e^{-\mu'(\xi-\xi_1)}$ with these constants. If $\mu(S)<2$ one may take $\mu_{\max}<2$, and for every $m<\mu(S)$ one may take $\mu_{\min}>m$ (in particular $\mu_{\min}>\frac12$ if $\mu(S)>\frac12$, and $\mu_{\min}>1$ if $\mu(S)>1$).*

(b) (Near type L_1.) *Let $S\in\mathcal{G}_{\mathsf{N}_0}$ be of type L_1 and $\nu\in(0,\frac12]$. There are $\xi_1\le 0$, a neighbourhood V of S in $\mathcal{G}_{\mathsf{N}_0}$ and $f_{\max},\kappa_{\rm env}$ such that every $S'\in V$ (of type G or L_1, with $\mu'\le\nu$) exists on $(-\infty,\xi_1]$, has $0\le f\le f_{\max}$, $0<\kappa_{S',\xi_1}\le\kappa_{\rm env}$, $h_{S'}(\xi_1)<0$, and*

$$m_-Z_{\mu'}\le z_{S',\xi_1}\le m_+Z_{\mu'}\quad\text{on }(-\infty,\xi_1],$$
$$m_-=\frac{1}{6a_1\cosh a_1},\quad m_+=\frac{4}{a_1},\quad a_1=1-\xi_1.$$

In particular $Z_{\mu'}$ is an envelope with rate $\mu'\le\frac12$ and $Z_{\mu'}\le Z_{1/2}$ (Lemma 6.6(i),(ii)).

Proof. (a) Apply σ_2 and use Chapter 4 at the north pole for σ_2S; we write the argument there and read it back at $x=-1$ by reflection. By Theorem 4.38(c) there are B,r_0 and a cut $\xi_*\ge\xi_G(B,r_0)$ at which the chart coordinates p_0 of σ_2S lie in the interior of the compact set K_B^{G} ($B^{-1}\le\mu,\zeta\le B$, other coordinates $\le B$ in modulus), and by Theorem 4.38(d) the state map S_{ξ_*} is a diffeomorphism from a neighbourhood $P_0\subset K_B^{\mathsf{G}}$ of p_0 onto a neighbourhood $\mathcal{U}$ of $s_0=s(\sigma_2S)(\xi_*)$.

Every solution of the Q-system whose scale-free state at ξ_* lies in $\mathcal{U}$ coincides on $[\xi_*,\infty)$ with u_p, $p=\mathsf{S}_{\xi_*}^{-1}(s)$ (it is a solution with the same state, and solutions are determined by the state). By Lemma 4.35(A) for u_p (this is the bound that Theorem 4.38(a) provides), the normalised z_p ($z_pe^{-\mu\xi}\to1$, $z_p(\xi_*)=\zeta e^{\mu\xi_*}$) satisfies $\frac12 n_0\le e^{-\mu\xi}z_p\le 2n_1$ with $n_0=\min(1,\zeta)\ge B^{-1}$, $n_1=\max(1,\zeta)\le B$. The cut-normalised solution is $z=z_p/z_p(\xi_*)$, and $z_p(\xi_*)=\zeta e^{\mu\xi_*}\in[B^{-1},B]e^{\mu\xi_*}$; hence

$$(4B^2)^{-1}e^{\mu(\xi-\xi_*)}\le z(\xi)\le 4B^2e^{\mu(\xi-\xi_*)}\qquad(\xi\ge\xi_*).$$

The swirl f is monotone with limit $\varphi\in[-B,B]$ at the pole and with $f(\xi_*)$ and $f'(\xi_*)=\kappa_{S',\xi_*}$ continuous functions of s, hence bounded on a compact neighbourhood of s_0 inside $\mathcal{U}$; so $|f|\le f_{\max}$ and $|\kappa|\le\kappa_{\mathrm{env}}$ there. Reading back: put $\xi_1=-\xi_*$ and $V=N(S,x_1,\varepsilon)$, $x_1=\tanh\xi_1$, with ε so small that $Y_{S'}(x_1)\mapsto s(\sigma_2S')(\xi_*)$ maps V into that compact neighbourhood (the map is a homeomorphism, see the proof of Proposition 6.4). By Proposition 6.4 the elements of V are of type G, and $\mu(S')\in[B^{-1},B]$; by continuity of μ we may shrink V so that $\mu_{\max}<2$ if $\mu(S)<2$ and $\mu_{\min}>m$ for any prescribed $m<\mu(S)$.

(b) *Step 1: a base point and the bound for w.* Fix $x_B\le0$, $x_B<x_S^*$, $\xi_B=\operatorname{artanh}x_B$, $V_0=N(S,x_B,\varepsilon_0)\cap\mathcal{G}_{\mathsf{N}_0}$ as in Lemma 6.9, $M_B=\sup_{V_0}M'$ and $f_{\max}=\sup_{V_0}f(x_B)$. By Lemma 6.8, every $S'\in V_0$ satisfies $1+\frac12Q'=\mu'^2+w'$ with $|w'|\le M_BE$ on $(-\infty,\xi_B]$ and $0\le f\le f_{\max}$. *Step 2: choice of the cut.* Since S is of type L_1, $z_S=a+b|\xi|+o(1)$, $z_S'\to-b<0$, so $h_S<0$ near $-\infty$ and $(1-\xi_1)|h_S(\xi_1)|\to1$ as $\xi_1\to-\infty$. Choose $\xi_1\le\xi_B$, $a_1=1-\xi_1$, with

$$\tfrac12\le a_1|h_S(\xi_1)|\le2,\qquad 12\,M_B(2a_1+1)e^{2-a_1}\le1;\tag{6.3}$$

both hold for all large a_1 (M_B is already fixed). *Step 3: the neighbourhood.* By Lemmas 6.3 and 6.9(b) choose $V=N(S,x_1,\delta)\cap\mathcal{G}_{\mathsf{N}_0}\subset V_0$ so small that $\mu'\le\nu$, $|h_{S'}(\xi_1)-h_S(\xi_1)|\le\frac12|h_S(\xi_1)|$ (h is a component of the state) and $\kappa_{S',\xi_1}\le\kappa_{\mathrm{env}}$. Then $\beta':=-h_{S'}(\xi_1)\in[\frac{1}{4a_1},\frac{3}{a_1}]$. *Step 4: Volterra equation.* Fix $S'\in V$, drop primes, and put $z=z_{S',\xi_1}$, so $z(\xi_1)=1$, $z'(\xi_1)=-\beta'$. From $z''=(\mu^2+w)z$, variation of constants backwards from ξ_1 (the kernel $k_\mu(\xi,\eta)=\sinh(\mu(\eta-\xi))/\mu$ solves $v''=\mu^2v+g$, $v(\xi_1)=v'(\xi_1)=0$ for $v=\int_\xi^{\xi_1}k_\mu g$) gives

$$z=y_{\beta'}+\mathcal{V}_wz,\qquad(\mathcal{V}_wg)(\xi)=\int_\xi^{\xi_1}k_\mu(\xi,\eta)w(\eta)g(\eta)\mathrm{d}\eta,$$

with $y_{\beta'}$ as in Lemma 6.6(v). For g with $\|g\|_Z:=\sup_{[\xi_T,\xi_1]}|g|/Z_\mu<\infty$, (6.2) and $\min\{\cdot,\cdot\}\le1-\eta$ give

$$\begin{aligned}|\mathcal{V}_wg(\xi)|&\le M_B\|g\|_Z\int_\xi^{\xi_1}k_\mu Z_\mu\,E\mathrm{d}\eta\\&\le M_B\|g\|_ZZ_\mu(\xi)\int_{-\infty}^{\xi_1}(1-\eta)e^{2\eta}\mathrm{d}\eta=M_B\|g\|_ZZ_\mu(\xi)\tfrac14(2a_1+1)E_1,\end{aligned}$$

$E_1=e^{2\xi_1}=e^{2-2a_1}$. The bound is independent of ξ_T *and of* μ': $\|\mathcal{V}_w\|_{Z\to Z}\le\theta:=\frac14M_B(2a_1+1)E_1$. *Step 5: Neumann series.* By Lemma 6.6(v) with $\mu'\le1$ and $\beta'\in[\frac{1}{4a_1},\frac{3}{a_1}]$: $\ell\ge\frac{1}{4a_1\cosh a_1}$, $L\le\frac{3}{a_1}$, so $\ell/L\ge\frac{1}{12\cosh a_1}$. Since $E_1\cosh a_1\le e^{2-2a_1}e^{a_1}=e^{2-a_1}$, (6.3) gives $\theta\le\frac{1}{48\cosh a_1}\le\frac{\ell}{4L}\le\frac14$. On each $[\xi_T,\xi_1]$ the operator $\mathcal{V}_w$ is a θ-contraction for $\|\cdot\|_Z$ and z solves $z=y+\mathcal{V}_wz$, hence $z=\sum_k\mathcal{V}_w^ky$ and

$\|z\|_Z \le \frac{4}{3}\|y\|_Z \le \frac{4}{3}L$, uniformly in ξ_T. Then $|z-y| \le \theta\|z\|_Z Z_\mu \le \frac{\ell}{3}Z_\mu$, and with $\ell Z_\mu \le y \le L Z_\mu$,

$$\tfrac{2}{3}\ell\, Z_\mu \le z \le \tfrac{4}{3}L\, Z_\mu, \qquad \tfrac{2}{3}\ell \ge \frac{1}{6a_1\cosh a_1} = m_-, \quad \tfrac{4}{3}L \le \frac{4}{a_1} = m_+.$$

Finally $0 \le f \le f_{\max}$ by Step 1, and S' exists on $(-\infty, \xi_1] \subset (-1, x_B]$. □

Remark 6.12 (on the constants)**.** Every ingredient of Lemma 6.11(b) is μ-free: Step 1 is Lemma 6.8 (no use of $\mu > 0$), Step 4 uses only (6.2), which holds for all $\mu \ge 0$ with the same constant, and Step 5 uses $\mu \le 1$ only through $\cosh(\mu a_1) \le \cosh a_1$. Nothing compares μ with an integer or a half-integer. The constants are explicit but astronomically bad ($M_B = 1$ forces $a_1 \ge 7.2$, hence $m_- \approx 3\cdot 10^{-5}$); this is irrelevant for qualitative continuity, which needs only their existence. We state (b) for $\nu \le \frac{1}{2}$, the range in which Theorem 6.29 uses it; for $\frac{1}{2} < \mu < 2$ near a germ of type G the weight $e^{-\mu\xi}$ of part (a) is used instead.

6.2.5. The frame is continuous, also at L_1.

Proposition 6.13 (continuity of the frame)**.** *For S of type L_1 and every $\nu > 0$ there is a neighbourhood V of S in $\mathcal{N}$ with $\mu(S') \le \nu$ on V. On $\mathcal{N}$ (germs at $x = -1$ with $\kappa > 0$, $f(-1) \ge 0$ finite, of type G or L_1) the maps $S \mapsto \mu(S), \lambda(S), f_-(S)$ and $S \mapsto \kappa_{S,\xi_1}$ (fixed cut) are continuous, also at germs of type L_1. Here $\mathcal{N} = \mathcal{G}_{\mathsf{N}_0}$ (Definition 6.2).*

Proof. The first sentence is Lemma 6.9. For the second: at a germ of type G, μ and f_- are continuous by Proposition 6.4 ($\mathcal{G}_{\mathsf{G}}$ is open, so nearby germs are of type G). At a germ S of type L_1: $\mu \ge 0 = \mu(S)$ and $\mu \le \nu$ near S by the first sentence, for every ν. The function λ is a function of the state, and so is $\kappa_{S,\xi_1} = f'(\xi_1)$. For f_- take ξ_1, V from Lemma 6.11(b) ($\nu = \frac{1}{2}$): $f_-(S') = f_{S'}(\xi_1) - \kappa_{S',\xi_1}\int_{-\infty}^{\xi_1} z_{S',\xi_1}^{-2}\,\mathrm{d}\xi$ with $z_{S',\xi_1} \ge m_- Z_{\mu'}$. For $\xi_T < \xi_1$ the part of the integral over $[\xi_T, \xi_1]$ is continuous in S' by (F2), and the tail is at most $\kappa_{\mathrm{env}} m_-^{-2}\int_{-\infty}^{\xi_T} Z_{\mu'}^{-2} \le \kappa_{\mathrm{env}} m_-^{-2}/(1-\xi_T)$ by Lemma 6.6(iii), uniformly in $S' \in V$. So f_- is a uniform limit (as $\xi_T \to -\infty$) of functions that are continuous along sequences (Lemma 6.3(b)), hence continuous. □

Remark 6.14 (μ and f_- at L_1)**.** Proposition 6.13 shows that f_- is *continuous* at germs of type L_1, where $\mu = 0$; nothing more is claimed. Numerically, along the family B of Atlas 6.65, f_- increases linearly in μ; this is an observation, not a statement proved here.

6.3. The two shift flows

In words. *Near the pole a germ is determined by its frame $(\mu, \lambda, f_-, \kappa)$ together with two further numbers: the coefficient of $t^2 = (1+x)^2$ in Q and the admixture of the second solution of $z'' = Wz$. The t^2-*shift *T_c changes the first by c and the second not at all; the* recessive shift *$\mathcal{R}_C$ does the opposite. Neither is defined by an expansion (whose coefficients have poles at resonant μ, Toy example 6.1); each is defined as the unique germ with the same frame that differs from S by exactly that much, and is constructed as the fixed point of a Volterra contraction whose constants are explicit in the envelope of Section 6.2.*

Toy example 6.15 (the amplitude of the t^2-shift)**.** Take no swirl, a constant potential $W \equiv \mu^2$ and $z = e^{-\mu\xi}$. Adding ct^2 to Q adds $\varepsilon = \frac{1}{2}ct^2 \approx 2cE^2$ to W ($t = 2E/(1+E)$), and $\tilde z = z(1+u)$ satisfies $u'' - 2\mu u' \approx 2cE^2 = 2ce^{4\xi}$. With $u = aE^2$: $(16 - 8\mu)a = 2c$, so

$$u \approx \frac{c\,E^2}{4(2-\mu)}.$$

The only singular factor is $(2-\mu)^{-1}$; the amplitude is regular at $\mu = 0$ ($c/8$; `verify_ch06a.py`), and for $\mu \geq 2$ the perturbation $e^{-\mu\xi}E^2$ is not smaller than the recessive mode $e^{\mu\xi}$, so "$\mathrm{Wr}(z,\tilde z) \to 0$" no longer singles out one germ. Similarly the recessive shift is a perturbation $\approx \frac{C}{2\mu}e^{\mu\xi}$ whose effect on Q is $O(E^{2\mu+1})$, which is $o(E^2)$ iff $\mu > \frac{1}{2}$.

Definition 6.16 (shift classes)**.** Let $S \in \mathcal{G}_{\mathsf{N}_0}$ be of type G or L_1 with a representative z, $f' = \kappa z^{-2}$, λ its first-integral constant, $c, C \in \mathbb{R}$. A germ $\tilde S \in \mathcal{G}_{\mathsf{N}_0}$ *has the frame of* S if it has a representative $\tilde z$ with $\tilde z/z \to 1$, $\tilde f' = \kappa\tilde z^{-2}$ (the same κ), $\tilde f - f \to 0$ and $\tilde\lambda = \lambda$ (limits at $\xi \to -\infty$). Put

$$C_T(c)(S) = \{\tilde S \text{ with the frame of } S:\ \mathrm{Wr}(z,\tilde z) \to 0,\ \tilde Q - Q - ct^2 = o(E^2)\},$$
$$C_A(C)(S) = \{\tilde S \text{ with the frame of } S:\ \mathrm{Wr}(z,\tilde z) \to C,\ \tilde Q - Q = o(E^2)\}.$$

Both classes are unchanged if (z,κ) is replaced by $(sz, s^2\kappa)$, $s > 0$.

The Volterra equation. For a representative z of S, $u \in C((-\infty,\xi_2])$ with $u > -1$ and $c, C \in \mathbb{R}$ put

$$\begin{aligned} \tilde f &= f + \kappa\int_{-\infty}^{\xi} z^{-2}\big((1+u)^{-2} - 1\big), \\ F &= f^2 - \tilde f^2, \qquad R = \mathsf{G}_{\mathsf{L}}[F], \qquad \varepsilon = \tfrac{1}{2}(ct^2 + R), \\ \mathcal{F}_{c,C}(u)(\xi) &= \int_{-\infty}^{\xi} z^{-2}(\eta)\Big(C + \int_{-\infty}^{\eta} z^2\,\varepsilon\,(1+u)\,\mathrm{d}\zeta\Big)\mathrm{d}\eta, \end{aligned} \tag{6.4}$$

(G_{L} the Green operator of Lemma 2.23). We write $\mathcal{F} = \mathcal{F}_{c,0}$ for the t^2-shift and $\mathcal{F}_A = \mathcal{F}_{0,C}$ for the recessive shift. Only κz^{-2} and $z^{-2}\cdot z^2$ occur, which is the scale invariance.

Lemma 6.17 (fixed points are shifted germs)**.** *Let $u \in C((-\infty,\xi_2])$ with $u > -1$, $u \to 0$ at $-\infty$, such that $\int_{-\infty}|F|E^{-1} < \infty$ and $u = \mathcal{F}_{c,C}(u)$. Then $\tilde z = z(1+u)$, $\tilde f$, $\tilde Q = Q + ct^2 + R$ satisfy*

$$\tilde z'' = (1 + \tfrac{1}{2}\tilde Q)\tilde z, \quad \tilde f' = \kappa\tilde z^{-2}, \quad \mathsf{L}[\tilde Q] + \tilde f^2 = \lambda,$$
$$\mathrm{Wr}(z,\tilde z) = z^2u' = C + \int_{-\infty}^{\xi} z^2\varepsilon(1+u),$$

so $(U,\tilde f,\tilde Q)$ with $U = 2(\tilde z'/\tilde z - \tanh\xi)$ solves the Q-system on $(-\infty,\xi_2]$ and has the frame of S; $\tilde Q - Q - ct^2 = R = o(E^2)$.

Proof. Differentiating $u = \mathcal{F}_{c,C}(u)$ twice gives $(z^2u')' = z^2\varepsilon(1+u)$ and $z^2u' \to C$. With $g = z$, $V_g = 1 + \frac{1}{2}Q$, $V = 1 + \frac{1}{2}\tilde Q$ we have $V - V_g = \varepsilon$, so Corollary 3.6 (equation (3.1)) is equivalent to $\tilde z'' = V\tilde z$. Next $\tilde f' = f' + \kappa z^{-2}((1+u)^{-2} - 1) = \kappa\tilde z^{-2}$. Finally $\mathsf{L}[t^2] = 0$ (Lemma 2.18(b)), $\mathsf{L}[R] = F$ (Lemma 2.23) and $\mathsf{L}[Q] = \lambda - f^2$ (Proposition 2.20) give $\mathsf{L}[\tilde Q] = \lambda - f^2 + f^2 - \tilde f^2$. By Proposition 2.20 the third

relation is the sixth equation of (2.22), the first relation is the first (Riccati) equation for U, and the second relation gives the second and third. The estimate $R = o(E^2)$ is the last assertion of Lemma 2.23. Since $\tilde f - f = \kappa \int z^{-2}((1+u)^{-2}-1) \to 0$, $\tilde z/z \to 1$, $\tilde\lambda = \lambda$, the frame is that of S. □

6.3.1. The t^2-shift. Fix an envelope datum for a family $\mathfrak{F} \subset \mathcal{G}_{\mathsf{N}_0}$: every $S \in \mathfrak{F}$ has on $(-\infty, \xi_1]$ the envelope $Z = Z_S$ of rate $\le \bar\mu < 2$ with common $m_\pm, k_0, f_{\max}$ for $z = z_{S,\xi_1}$ (Lemma 6.11). Let $\bar c > 0$ and put

$$\rho_0 = (m_+/m_-)^2, \quad \bar f = 2f_{\max} + 2, \quad \delta = \min\{\tfrac14, \tfrac{2-\bar\mu}{2}\},$$
$$a_0 = \rho_0 \mathcal{M}(4 - 2\bar\mu), \quad a_1 = \rho_0 \mathcal{M}(4 - 2\delta - 2\bar\mu),$$
$$\hat B = \tfrac32 a_0 \bar c, \quad \hat\Phi = \tfrac52 k_0 \bar f \hat B,$$

so that $4 - 2\bar\mu > 0$, $4 - 2\delta - 2\bar\mu \ge 2 - \bar\mu > 0$ and $2 - \delta > 1$. For $\xi_2 \le \xi_1$, $E_2 = e^{2\xi_2}$, assume

$$E_2 \le 1, \quad \hat B E_2^2 \le \tfrac12, \quad \tfrac52 k_0 \hat B E_2^2 \le 1, \quad a_0 k_0 \bar f E_2 \le \tfrac14, \quad \frac{a_1 \Theta}{4 - 2\delta} \le \frac12,$$
$$\Theta = \frac{24 k_0 \bar f E_2}{(1-\delta)(4-2\delta)} + 2\bar c E_2^2 + \hat\Phi E_2^3. \tag{6.5}$$

All five hold for $\xi_2 \le \hat\xi_2$, where $\hat\xi_2$ depends only on $(m_\pm, k_0, f_{\max}, \bar\mu, \bar c)$ (each left side tends to 0 or is $\le$ its bound already). Let $X = \{u \in C((-\infty, \xi_2]) : |u| \le \hat B E^2\}$ and $\|u\|_w = \sup |u|\, E^{-(2-\delta)}$.

Proposition 6.18 (uniform contraction for T_c)**.** *For every $S \in \mathfrak{F}$, $|c| \le \bar c$ and $\xi_2 \le \hat\xi_2$, the map $\mathcal{F} = \mathcal{F}_{c,0}$ of (6.4) is defined on X, maps X into X, and is a $\frac12$-contraction for $\|\cdot\|_w$; X is complete for $\|\cdot\|_w$. Its fixed point $u_{S,c}$ satisfies the hypotheses of Lemma 6.17. Moreover, for $B \ge 0$ with $BE_2^2 \le \frac12$ and $\frac52 k_0 B E_2^2 \le 1$ and any continuous u with $|u| \le BE^2$,*

$$|\mathcal{F}(u)| \le \Big[\tfrac34 a_0 \bar c + \tfrac{15}{16} a_0 k_0 \bar f B\, E_2\Big] E^2. \tag{6.6}$$

Proof. Write $Z = Z_S$, $z = z_{S,\xi_1}$, so $m_- Z \le z \le m_+ Z$, $|\kappa| z^{-2} \le k_0 Z^{-2}$, $|f| \le f_{\max}$, and Z has rate $\bar\mu$ (Lemma 6.5 applies). Let $|u| \le BE^2$ with $BE_2^2 \le \frac12$, $\frac52 k_0 B E_2^2 \le 1$.

Step 1. $1 + u \in [\frac12, \frac32]$; by the mean value theorem $|(1+u)^{-2} - 1| \le 10|u|$ and $|(1+u_1)^{-2} - (1+u_2)^{-2}| \le 16|u_1 - u_2|$. *Step 2 (swirl).* By Lemma 6.5(b) with $p = 4$: $|\tilde f - f| \le 10 k_0 B \int_{-\infty}^{\xi} Z^{-2} e^{4\eta} \mathrm{d}\eta \le \frac52 k_0 B Z^{-2} E^2 \le 1$. Hence $|\tilde f| \le f_{\max} + 1$, $|f + \tilde f| \le \bar f$ and $|F| \le \Phi_B Z^{-2} E^2$, $\Phi_B = \frac52 k_0 \bar f B$. *Step 3.* Lemma 2.23 ($q = 2$, Z non-increasing) gives $|R| \le 2\Phi_B Z^{-2} E^3$, and with $t^2 \le 4E^2$, $Z \ge 1$: $|\varepsilon| \le 2\bar c E^2 + \Phi_B E^3$. *Step 4.* Using $E^3 \le E_2 E^2$ and Lemma 6.5(a) with $p = 4$: $|\int_{-\infty}^{\eta} z^2 \varepsilon (1+u)| \le \frac32 m_+^2 \mathcal{M}(4 - 2\bar\mu)(2\bar c + \Phi_B E_2) Z^2 E^2$. Multiplying by $z^{-2} \le m_-^{-2} Z^{-2}$ and integrating ($\int_{-\infty}^{\xi} E^2 = E^2/4$) gives $|\mathcal{F}(u)| \le \frac38 a_0 (2\bar c + \Phi_B E_2) E^2$, which is (6.6). For $B = \hat B$: $\frac34 a_0 \bar c = \frac12 \hat B$ and $\frac{15}{16} a_0 k_0 \bar f E_2 \le \frac12$ by (6.5), so $\mathcal{F}(X) \subset X$. *Step 5 (weak Lipschitz bound).* Let $u_1, u_2 \in X$, $d = \|u_1 - u_2\|_w$. By Step 1 and Lemma 6.5(b) with $p = 4 - 2\delta$: $|\Delta \tilde f| \le \frac{16 k_0 d}{4 - 2\delta} Z^{-2} E^{2-2\delta}$, $|\Delta F| \le \bar f |\Delta \tilde f| =: \Phi_w Z^{-2} E^{2-2\delta}$ and by Lemma 2.23 ($q = 2 - \delta > 1$) $|\Delta R| \le \frac{2\Phi_w}{1-\delta} Z^{-2} E^{3-\delta}$. As $\Delta(\varepsilon(1+u)) = \frac12 \Delta R (1 + u_1) + \varepsilon_2 \Delta u$,

$$|\Delta(\varepsilon(1+u))| \le \Theta\, d\, E^{2-\delta} \qquad (E \le E_2,\ Z \ge 1).$$

Repeating Step 4 with $p = 4 - 2\delta > 2\bar\mu$ gives $|\Delta\mathcal{F}| \le \frac{a_1\Theta}{4-2\delta} d\, E^{2-\delta} \le \frac12 dE^{2-\delta}$. *Completeness.* If $u_n \in X$ is $\|\cdot\|_w$-Cauchy it converges locally uniformly and the limit lies in X. Banach's theorem (Theorem A.2) gives the fixed point. The hypotheses of Lemma 6.17 hold: $u_{S,c} \to 0$, $|F| \le \Phi Z^{-2}E^2$. □

THEOREM 6.19 (the t^2-shift). *Let $S \in \mathcal{G}_{\mathsf{N}_0}$ be of type G with $\mu < 2$, or of type L_1, and $c \in \mathbb{R}$.*

(a) (Existence and estimates.) *There are ξ_2 and B such that (6.4) with $C = 0$ has a unique solution u with $|u| \le BE^2$ on $(-\infty, \xi_2]$. The corresponding $(\tilde z, \tilde f, \tilde Q)$ continued as a maximal solution is a germ $T_cS \in \mathcal{G}_{\mathsf{N}_0}$ with $|\tilde f - f| \le \frac52 k_0 BZ^{-2}E^2$ and $|\tilde Q - Q - ct^2| \le 5k_0\bar f BZ^{-2}E^3$.*

(b) *T_cS is the unique element of $C_T(c)(S)$; it has the type, exponent μ and frame (λ, f_-, κ) of S.*

(c) *$T_0 = \mathrm{id}$, $T_cT_{c'} = T_{c+c'}$, and $T_cS = S$ only for $c = 0$.*

PROOF. By Lemma 6.11 (with $\mathfrak{F} = \{S\}$) S has an envelope of rate $\bar\mu < 2$ ($\bar\mu = \mu$ for type G, $\frac12$ for L_1). Take $\bar c = |c| + 1$.

(a) Proposition 6.18 gives $u \in X$ and Lemma 6.17 the solution on $(-\infty, \xi_2]$. The state at ξ_2 determines the maximal solution (Remark 2.7(i)). The estimates are Steps 2 and 3 with $B = \hat B$ ($|R| \le 2\Phi_B Z^{-2}E^3$, $\Phi_B = \frac52 k_0\bar f B$). The frame statement follows from Lemma 6.17; $\tilde\kappa = \kappa > 0$ and $\tilde f_- = f_- \ge 0$, so $T_cS \in \mathcal{G}_{\mathsf{N}_0}$, and it has the type and exponent of S because $\tilde z/z \to 1$ and $\tilde Q(-1) = Q(-1)$. Uniqueness in the ball is Banach.

(b) *Membership* is Lemma 6.17: $\mathrm{Wr}(z, \tilde z) = z^2u' = \int_{-\infty}^{\xi} z^2\varepsilon(1+u) = O(Z^2E^2) \to 0$ by Lemma 6.5(c). *Uniqueness.* Let $\tilde S \in C_T(c)(S)$, $u = \tilde z/z - 1 \to 0$, $R := \tilde Q - Q - ct^2 = o(E^2)$. As in the proof of Lemma 6.17, Corollary 3.6 gives $(z^2u')' = z^2\varepsilon(1+u)$ with $\varepsilon = \frac12(ct^2 + R) = O(E^2)$; since $z^2u' \to 0$, $z^2u'(\xi) = \int_{-\infty}^{\xi} z^2\varepsilon(1+u) = O(Z^2E^2)$ (Lemma 6.5(a), $|1+u| \le 2$ near the pole), hence $|u'| \le m_-^{-2}Z^{-2}O(Z^2E^2) = O(E^2)$ and $u = O(E^2)$. Then $\tilde f - f = \kappa\int z^{-2}((1+u)^{-2} - 1) = O(Z^{-2}E^2)$ (here $\tilde f - f \to 0$ is used) and $F = O(E^2)$. Both first integrals have the same λ, so $\mathsf{L}[R] = F$ ($\mathsf{L}[t^2] = 0$); as $R = o(E^2)$, Lemma 2.23 gives $R = \mathsf{G}_{\mathsf{L}}[F]$. So $u = \mathcal{F}(u)$. Let $C''(\xi_*) = \sup_{\xi\le\xi_*}|u|E^{-2} < \infty$ for ξ_* close to $-\infty$. For $\xi_* \le \hat\xi_2$ so small that $C''(\xi_*)E_*^2 \le \frac12$, $\frac52 k_0 C''(\xi_*)E_*^2 \le 1$, (6.6) with $B = C''(\xi_*)$ and $a_0k_0\bar f E_* \le \frac14$ gives $C''(\xi_*) \le \frac34 a_0\bar c + \frac{15}{64}C''(\xi_*)$, i.e. $C''(\xi_*) \le \frac{64}{49}\cdot\frac34 a_0\bar c \le \hat B$. Thus $u \in X$ (on $(-\infty, \xi_*]$) and by uniqueness in X, $u = u_{S,c}$. The germs $\tilde S$ and T_cS then have the same state at ξ_* and coincide.

(c) $\mathcal{F}_{0,0}(0) = 0$, so $u = 0$ and $T_0S = S$. Let $S_1 = T_cS$, $S_2 = T_{c'}S_1$ (the hypotheses hold for S_1: same type and μ, and $z_1 = z(1+u)$ with $1+u \in [\frac12, \frac32]$ near the pole has the envelope Z with constants $\frac12 m_-, \frac32 m_+$). Put $z_2 = z_1(1+u_2)$ and $1+u_{12} = (1+u)(1+u_2)$. Then $z_2/z \to 1$; $z^2u'_{12} = (1+u_2)\,z^2u' + (1+u)\,\frac{z^2}{z_1^2}\,z_1^2u'_2 \to 0$ because $z^2u' \to 0$, $z_1^2u'_2 = \mathrm{Wr}(z_1, z_2) \to 0$ and $z^2/z_1^2 \to 1$; $Q_2 - Q = (c+c')t^2 + o(E^2)$; the frame is common. So $S_2 \in C_T(c+c')(S)$, and $S_2 = T_{c+c'}S$ by (b). If $T_cS = S$ then $ct^2 + R = 0$ with $R = o(E^2)$, so $c = 0$. □

Remark 6.20 (μ enters only through moment constants)**.** In Proposition 6.18 the exponent of S enters only through $\mathcal{M}(4-2\bar\mu)$, $\mathcal{M}(4-2\delta-2\bar\mu)$ and the rate of Z (Lemma 6.6(ii)). No factor $(\mu - n)^{-1}$ or $(n\mu - k)^{-1}$ occurs, because the proof

integrates exponentials $\int_{-\infty}^{\xi} e^{p\eta} = e^{p\xi}/p$ with p bounded away from 0 and never compares coefficients of an expansion; this is why the construction is uniform across the resonant values of Toy example 6.1. The singular factor $(2-\mu)^{-1}$ of Toy example 6.15 is genuine.

6.3.2. The recessive shift. Let now $\mathfrak{F} \subset \mathcal{G}_{\mathsf{N}_0} \cap \mathcal{G}_{\mathsf{G}}$ have, on $(-\infty, \xi_1]$, the envelope $Z = e^{-\mu'(\xi-\xi_1)}$ ($\mu' = \mu(S')$) with common $m_\pm, k_0, f_{\max}$ and $\mu' \in [\nu, \mu_{\max}]$, $\nu > \frac{1}{2}$ (Lemma 6.11(a)). Put $\varpi_1 = e^{-2\mu'\xi_1} \le \varpi_{\max} := e^{-2\mu_{\max}\xi_1}$ so that $Z^{-2} = \varpi_1 E^{\mu'}$. With $\varsigma = \nu - \frac{1}{2}$, $\rho_0, \bar f$ as above and $\bar C > 0$, put

$$\hat B_A = \frac{\bar C \varpi_{\max}}{\nu m_-^2}, \qquad \Lambda_0(E_*) = \frac{15\rho_0 k_0 \bar f\, E_*}{32\,\nu\varsigma(\nu+1)(2\nu+1)},$$

$$\Lambda_1 = \frac{8\rho_0 k_0 \bar f E_*}{\varsigma(2\nu+1)(2\nu+3)}, \qquad \Lambda_2 = \frac{5\rho_0 k_0 \bar f \hat B_A E_*^{\nu+1}}{4\nu\varsigma(2\nu+3)(4\nu+3)}.$$

Choose $\hat\xi_2 \le \xi_1$ such that for $\xi_* \le \hat\xi_2$ (with $E_* = e^{2\xi_*}$)

$$\hat B_A E_*^\nu \le \tfrac12, \qquad \tfrac{5}{2\nu} k_0 \hat B_A E_*^\nu \le 1, \qquad \Lambda_0(E_*) \le \tfrac12, \qquad \Lambda_1 + \Lambda_2 \le \tfrac12; \tag{6.7}$$

possible because all four left sides tend to 0. Let $X_A = \{|u| \le \hat B_A E^\nu\}$ on $(-\infty, \xi_2]$, $\xi_2 \le \hat\xi_2$, and $\|u\|_A = \sup |u| E^{-1/2}$.

Proposition 6.21 (uniform contraction for $\mathcal{R}_C$)**.** *For every $S \in \mathfrak{F}$ and $|C| \le \bar C$, $\mathcal{F}_A = \mathcal{F}_{0,C}$ maps X_A into itself and is a $\frac12$-contraction for $\|\cdot\|_A$; X_A is complete for it. For $|u| \le BE^\nu$ (with $BE_*^\nu \le \frac12$, $\frac{5}{2\nu}k_0 BE_*^\nu \le 1$) one has $|\mathcal{F}_A(u)| \le [\frac12 \hat B_A + \Lambda_0(E_*)B]E^\nu$. The fixed point satisfies the hypotheses of Lemma 6.17 ($c = 0$).*

Proof. Let $|u| \le BE^\nu$, $\nu \le \mu'$, $q := \mu' + \nu > 1$. *Zeroth term.* $|C\int_{-\infty}^{\xi} z^{-2}| \le \bar C m_-^{-2}\varpi_1 E^{\mu'}/(2\mu') \le \frac12 \hat B_A E^\nu$. *Swirl.* $|\kappa z^{-2}| \le k_0 \varpi_1 E^{\mu'}$ and $|(1+u)^{-2} - 1| \le 10|u|$, so $|\tilde f - f| \le 10 k_0 B\varpi_1 \int_{-\infty}^{\xi} e^{2q\eta}\mathrm{d}\eta \le \frac{5}{2\nu}k_0 B\varpi_1 E^q \le 1$ and $|F| \le \Phi E^q$, $\Phi = \frac{5}{2\nu}k_0 \bar f B \varpi_1$. Lemma 2.23 with $Z \equiv 1$ (note $q > 1$, $q - 1 \ge 2\varsigma$) gives $|R| \le \frac{\Phi}{\varsigma} E^{q+1}$, $|\varepsilon| \le \frac{\Phi}{2\varsigma} E^{q+1}$. *Self-map.* Since $z^2 \le m_+^2 \varpi_1^{-1} E^{-\mu'}$, $|\int_{-\infty}^{\eta} z^2 \varepsilon(1+u)| \le \frac32 m_+^2 \varpi_1^{-1} \frac{\Phi}{2\varsigma} \int_{-\infty}^{\eta} e^{2(\nu+1)\zeta} = \frac{3m_+^2\Phi}{8\varsigma(\nu+1)\varpi_1} E^{\nu+1}$. Multiplying by $z^{-2} \le m_-^{-2}\varpi_1 E^{\mu'}$ and integrating, $\frac{1}{2(\mu'+\nu+1)} \le \frac{1}{2(2\nu+1)}$: the second part of $\mathcal{F}_A(u)$ is at most $\frac{3\rho_0\Phi}{16\varsigma(\nu+1)(2\nu+1)} E^{q+1}$, and $\varpi_1 E^{q+1} = E^\nu(\varpi_1 E^{\mu'})E \le E^\nu E_*$ (using $\varpi_1 E^{\mu'} = Z^{-2} \le 1$). Inserting Φ: $\le \Lambda_0(E_*) BE^\nu$. With $B = \hat B_A$ and $\Lambda_0 \le \frac12$ we get $\mathcal{F}_A(X_A) \subset X_A$. *Weak Lipschitz bound.* Let $|\Delta u| \le dE^{1/2}$ and $u_i \in X_A$ (so $|\Delta|((1+u)^{-2}) \le 16|\Delta u|$). Then $|\Delta\tilde f| \le 16 k_0 \varpi_1 d \int e^{2(\mu'+\frac12)\eta} \le \frac{16k_0\varpi_1 d}{2\nu+1} E^{\mu'+1/2}$, $|\Delta F| \le \Phi_w E^{\mu'+1/2}$ with $\Phi_w = \frac{16k_0\bar f\varpi_1 d}{2\nu+1}$, and Lemma 2.23 ($q_w = \mu' + \frac12 > 1$, $q_w - 1 \ge \varsigma$): $|\Delta R| \le \frac{2\Phi_w}{\varsigma} E^{\mu'+3/2}$. From $\Delta(\varepsilon(1+u)) = \frac12 \Delta R(1+u_1) + \varepsilon_2 \Delta u$ and the bound for ε_2 above (with $B = \hat B_A$), the same two integrations give $|\Delta\mathcal{F}_A| \le (\Lambda_1 + \Lambda_2) dE^{1/2}$: the first term contributes $\frac32 \frac{\Phi_w}{\varsigma} E^{\mu'+3/2} \to \frac{\rho_0\Phi_w}{2\varsigma(2\nu+3)} E^{\mu'+3/2}$ and $\varpi_1 E^{\mu'+1} \le E_*$; the second contributes $\le \frac{\rho_0\Phi_B}{2\varsigma(2\nu+3)(4\nu+3)} dE^{\mu'+\nu+3/2}$ with $\varpi_1 E^{\mu'+\nu+1} \le E_*^{\nu+1}$. Completeness and the hypotheses of Lemma 6.17 are as in Proposition 6.18. □

Theorem 6.22 (the recessive shift). *Let $S \in \mathcal{G}_{\mathsf{N}_0}$ be of type G with $\mu > \frac12$ and $C \in \mathbb{R}$.*

(a) *The equation* $u = \mathcal{F}_{0,C}(u)$ *has a unique solution with* $|u| \le BE^{\mu}$ *near the pole; it defines a germ* $\mathcal{R}_C S \in \mathcal{G}_{\mathsf{N}_0}$ *with* $\tilde z - z = \frac{C}{2\mu} e^{\mu\xi}(1+o(1))$ ($z \sim ae^{-\mu\xi}$ *normalised as in* (b)), $|\tilde f - f| = O(E^{2\mu})$ *and* $\tilde Q - Q = O(E^{2\mu+1})$.

(b) $\mathcal{R}_C S$ *is the unique element of* $C_A(C)(S)$*; it has the type and frame of* S.

(c) $\mathcal{R}_0 = \mathrm{id}$, $\mathcal{R}_C\mathcal{R}_{C'} = \mathcal{R}_{C+C'}$, *and* $\mathcal{R}_C S = S$ *only for* $C = 0$.

Proof. (a) Lemma 6.11(a) with $\mathfrak{F} = \{S\}$, $\bar C = |C| + 1$, and Proposition 6.21 (with $\nu = \mu$ and $B \le \hat B_A$) give u and, by Lemma 6.17, the solution; the estimates are those in the proof (the exponent $\mu = \nu$ is used for the unique ball). For the first-order term: $z^2u' \to C$ and $z^2 \sim a^2e^{-2\mu\xi}$ give $u' \sim Ca^{-2}e^{2\mu\xi}$, so $u \sim \frac{C}{2\mu a^2}e^{2\mu\xi}$ and $\tilde z - z = zu \sim \frac{C}{2\mu a}e^{\mu\xi}$ with $a = \lim ze^{\mu\xi}$. (b) Membership: Lemma 6.17 and $\tilde Q - Q = R = O(E^{2\mu+1}) = o(E^2)$ because $\mu > \frac12$. Uniqueness: let $\tilde S \in C_A(C)(S)$, $u = \tilde z/z - 1 \to 0$; then $z^2u' \to C$, so $|u'| \le (|C|+1)z^{-2} = O(E^{\mu})$ and $u = O(E^{\mu})$. Hence $\tilde f - f = O(E^{2\mu})$, $F = O(E^{2\mu})$ with $2\mu > 1$, $\mathsf{L}[R] = F$ for $R = \tilde Q - Q = o(E^2)$, and Lemma 2.23 gives $R = \mathsf{G}_{\mathsf{L}}[F] = O(E^{2\mu+1})$, $\varepsilon = \frac12 R$. Then $z^2u'(\xi) = C + \int_{-\infty}^{\xi} z^2\varepsilon(1+u)$ (the integrand is $O(E^{\mu+1})$, integrable), i.e. $u = \mathcal{F}_{0,C}(u)$. With $C''(\xi_*) = \sup_{\xi\le\xi_*}|u|E^{-\mu}$ the a priori bound of Proposition 6.21 gives $C'' \le \frac12\hat B_A + \Lambda_0 C'' \le \frac12\hat B_A + \frac12 C''$, so $u \in X_A$ and equals the fixed point. (c) A_0: $u = 0$. If $S_1 = \mathcal{R}_C S$, $S_2 = \mathcal{R}_{C'}S_1$ then $z_2/z \to 1$, $z^2u'_{12} \to C + C'$ (as in Theorem 6.19(c)), $Q_2 - Q = o(E^2)$, same frame: $S_2 \in C_A(C+C')(S)$. If $\mathcal{R}_C S = S$ then $C = \lim \mathrm{Wr}(z,z) = 0$. □

6.4. Order, closedness and the interior property

In words. *A positive shift moves a germ up in the Kamke order of Lemma 5.28: z, z'/z and Q increase and the swirl decreases. The mechanism is a chain of four implications: smaller swirl ⇒ larger potential (positivity of the Green kernel) ⇒ the Wronskian z^2u' grows ⇒ $\tilde z/z$ is increasing ⇒ the swirl is smaller still. Two consequences are used throughout the sequel: a positive shift of a global solution is global, and it is of generic type* G *at the north pole.*

Proposition 6.23 (the shifts are order preserving in N_0)**.** *Let* $S \in \mathcal{G}_{\mathsf{N}_0}$ *have maximal interval* $(-\infty, \xi_S^*)$ *in* ξ, *and let* $\tilde S = T_c S$ *with* $c > 0$ ($\mu < 2$ *or type* L_1), *or* $\tilde S = \mathcal{R}_C S$ *with* $C > 0$ (*type* G, $\mu > \frac12$)*;* $u = \tilde z/z - 1$. *Then* $\tilde S$ *exists at least on* $(-\infty, \xi_S^*)$ *and there*

$$u > 0, \quad u' > 0, \quad \tilde f < f, \quad \tilde Q - Q \ge ct^2 \ \ (\textit{resp.} > 0), \quad \partial_x(\tilde Q - Q) \ge 2ct \ \ (\textit{resp.} > 0),$$

and $z^2u' = \mathrm{Wr}(z,\tilde z)$ *is positive and increasing. If* S *is global, so is* $\tilde S$, *and* $\tilde Q(1) \ge Q(1) + 4c$ (*resp.* $\tilde Q(1) > Q(1)$).

Proof. Write $\mathcal{F} = \mathcal{F}_{c,C}$ with $(c,C) = (c,0)$, $c > 0$, or $(0,C)$, $C > 0$, and X the ball of Proposition 6.18 or Proposition 6.21.

Step 1 (near the pole: Picard iteration). Put $u_0 = 0$, $u_{n+1} = \mathcal{F}(u_n)$. The iterates lie in X and converge in $\|\cdot\|_w$, hence pointwise, to the fixed point u. Claim: $u_n \ge 0$, and $u_{n+1} \ge Cl$ in the $\mathcal{R}$ case, $l = \int_{-\infty}^{\xi} z^{-2}$. If $u_n \ge 0$ then $(1+u_n)^{-2} \le 1$, so $\tilde f_n \le f$; and $\tilde f_n = f_- + \kappa\int z^{-2}(1+u_n)^{-2} \ge f_- \ge 0$ ($\kappa > 0$). Hence $F_n = (f - \tilde f_n)(f + \tilde f_n) \ge 0$, $R_n = \mathsf{G}_{\mathsf{L}}[F_n] \ge 0$ (as $K_{\mathsf{L}} \ge 0$ for $s < x$), $\varepsilon_n \ge \frac12 ct^2 \ge 0$ (resp. ≥ 0), and $u_{n+1} \ge 0$, $\ge Cl > 0$ for the recessive shift. So $u \ge 0$. Repeating the signs with u: $\varepsilon \ge \frac12 ct^2 > 0$

(resp. $z^2u' = C + \int z^2\varepsilon(1+u) \geq C > 0$), $z^2u' > 0$ and increasing, $u' > 0$, and $u > 0$ as $u(-\infty) = 0$.

Step 2 (continuation). Let $\xi_m = \sup\{\xi < \xi_S^* : \tilde{S} \text{ exists and } u > 0 \text{ on } (-\infty, \xi]\} \geq \xi_2$. On $(-\infty, \xi_m)$: $u > 0$ gives $\tilde{f} < f$ (the integral in $\tilde{f} - f$ has a negative integrand), $\tilde{f} \geq f_- \geq 0$, so $F = f^2 - \tilde{f}^2 > 0$. Both germs satisfy their first integrals with the same λ, so $\mathsf{L}[R] = F$ with $R = \tilde{Q} - Q - ct^2$ on $(-1, \tanh \xi_m)$; as R and $\mathsf{G}_\mathsf{L}[F]$ are $o(E^2)$ at the pole, Lemma 2.23 gives $R = \mathsf{G}_\mathsf{L}[F] > 0$ and $\partial_x R = \int_{-1}^{x} \partial_x K_\mathsf{L}\, F \mathrm{d}s > 0$ (Lemma 2.18(c)). So $\varepsilon > 0$ and $(z^2u')' = z^2\varepsilon(1+u) > 0$: z^2u' is positive and increasing, u is increasing. Suppose $\xi_m < \xi_S^*$. On $[\xi_2, \xi_m)$, z is bounded above and below by positive constants and S has bounded state there, so $0 < R \leq \int K_\mathsf{L} f^2$ and $\partial_x R$ are bounded; $f_- \leq \tilde{f} < f$; the linear equation $\tilde{z}'' = (1 + \tilde{Q}/2)\tilde{z}$ with bounded coefficient has bounded $\tilde{z}, \tilde{z}'$; $\tilde{z} > z \geq z_{\min} > 0$ gives $\tilde{h} = \tilde{z}'/\tilde{z}$ bounded, $\tilde{f}_x = \kappa \tilde{z}^{-2} \cosh^2 \xi$ bounded and $\tilde{Q}_{xx}$ bounded by the first integral. So the state of $\tilde{S}$ stays in a compact set as $\xi \uparrow \xi_m < \xi_S^*$ and $\tilde{S}$ extends beyond ξ_m (Remark 2.7(ii)); since $u(\xi_m) \geq u(\xi_2) > 0$, $u > 0$ slightly beyond ξ_m, contradicting the definition of ξ_m. Hence $\xi_m = \xi_S^*$, and all inequalities hold: $\tilde{Q} - Q = ct^2 + R \geq ct^2$, $\partial_x(\tilde{Q} - Q) = 2ct + \partial_x R \geq 2ct$ ($R > 0$, $\partial_x R > 0$ in the recessive case, where $\tilde{Q} - Q = R$).

Step 3 (global germs). If $\xi_S^* = \infty$ then $\tilde{S}$ is global and $\tilde{Q}(1) - Q(1) = \lim_{x\to 1}(ct^2 + R) \geq 4c$ (both limits exist, Theorem 4.5), resp. $= \lim R(x) \geq R(x_0) > 0$ since R is increasing. □

Corollary 6.24 (negative shifts)**.** *For $c < 0$ ($C < 0$) and $S \in \mathcal{G}_{\mathsf{N}_0}$ in the domain of the flow, $\tilde{S} = T_c S$ ($\mathcal{R}_C S$) exists at most as long as S, and the inequalities of Proposition 6.23 reverse on the interval of existence of $\tilde{S}$; in particular $T_c S$ global implies S global.*

Proof. $\tilde{S} \in \mathcal{G}_{\mathsf{N}_0}$ is of the same type and exponent (Theorem 6.19(b)), and $S = T_{-c}\tilde{S}$ by Theorem 6.19(c) (resp. $S = \mathcal{R}_{-C}\tilde{S}$). Apply Proposition 6.23 to $\tilde{S}$ and the positive shift $-c$ ($-C$). □

Lemma 6.25 ($\mathcal{S}$ is closed; $\mathcal{S}_\mathsf{G}$ is relatively open)**.** *$\mathcal{S}$ is closed in $\mathcal{G}$. The set $\mathcal{S}_\mathsf{G} = \{S \in \mathcal{S} : S \text{ of type } \mathsf{G} \text{ at } x = 1\}$ is open relative to $\mathcal{G}_{\mathsf{N}_0}$.*

Proof. *Closedness.* By Lemma 6.3(c), $\mathcal{G}_{\mathsf{N}_0}$ is closed, so a limit S of germs $S_n \in \mathcal{S}$ lies in $\mathcal{G}_{\mathsf{N}_0}$. Fix $x_a < x_S^*$ (so $x_a < x_{S_n}^*$ for large n and $Y_{S_n}(x_a) \to \omega := Y_S(x_a)$, Lemma 6.3(b)). Each S_n is global, so $Y_{S_n}(x_a) \in \Omega^+(x_a)$, the set of states whose forward solution exists on $[x_a, 1)$. This set is closed (Theorem 5.12), hence $\omega \in \Omega^+(x_a)$, i.e. the solution through ω, which is S, exists on $[x_a, 1)$. So $x_S^* = 1$ and $S \in \mathcal{S}$.

Openness. Let $S \in \mathcal{S}_\mathsf{G}$, $x_a < 1$, $\xi_a = \operatorname{artanh} x_a$. The scale-free state at ξ_a is a continuous function of the state $Y(x_a)$, and $S' \mapsto Y_{S'}(x_a)$ is continuous on $N(S, x_a, 1)$ (Lemma 6.3(b)). By Corollary 4.42(b) the set of $S' \in N(S, x_a, 1)$ whose forward solution exists on $[\xi_a, \infty)$ and is of type G is open, contains S, and consists of germs that are global (they exist on $(-1, x_a]$ anyway) and of type G at $x = 1$. Intersected with $\mathcal{G}_{\mathsf{N}_0}$ it lies in $\mathcal{S}_\mathsf{G}$. □

Lemma 6.26 (interior property)**.** *Let $S \in \mathcal{S}$. If $\mu < 2$ or S is of type L_1 and $c > 0$, then $T_c S \in \mathcal{S}_\mathsf{G}$. If S is of type G with $\mu > \frac{1}{2}$ and $C > 0$, then $\mathcal{R}_C S \in \mathcal{S}_\mathsf{G}$.*

PROOF. Let $\tilde{S}$ be the shifted germ. By Proposition 6.23 it is global with $\tilde{Q}(1) > Q(1) \geq -2$ (Theorem 4.5); by Definition 4.16 at the north pole, type L_1 or L_2 means $Q(1) = -2$, so $\tilde{S}$ is of type G or R at $x = 1$. Suppose type R: then $\tilde{h} = \tilde{z}'/\tilde{z} \to -\tilde{\mu}_+ < 0$, hence $\tilde{z} \to 0$ as $\xi \to \infty$. Now $\Omega := \mathrm{Wr}(z, \tilde{z}) = z^2 u' > 0$ is increasing (Proposition 6.23), so $u' \geq \Omega_0 z^{-2}$ on $[\xi_0, \infty)$, $\Omega_0 = \Omega(\xi_0) > 0$, and u is increasing with $u > 0$. By the classification at the north pole (Definition 4.16, Proposition 4.19) S is of one of the types $\mathsf{G}, \mathsf{R}, \mathsf{L}_1, \mathsf{L}_2$ there ($\mu_+ < \frac{1}{2}$ for R, Lemma 6.3(a)):

- type G: $z \to \infty$ and $\tilde{z} = z(1 + u) \geq z$;
- type L_1 or L_2: $z = a + b\xi + o(1)$ with $b > 0$, or $b = 0 < a$ (Proposition 4.19(c)), so $z \geq z_0 > 0$ on $[\xi_0, \infty)$ and $\tilde{z} \geq z \geq z_0$;
- type R: $z \asymp e^{-\mu_+ \xi}$, hence $u' \geq c e^{2\mu_+ \xi}$, $u \geq c' e^{2\mu_+ \xi}$ for large ξ and $\tilde{z} \geq zu \geq c'' e^{\mu_+ \xi} \to \infty$.

In every case $\tilde{z} \not\to 0$, contradicting type R. So $\tilde{S}$ is of type G at $x = 1$. □

Remark 6.27 (the edge)**.** Consequently the set Γ of those $S \in \mathcal{S}$ that are not of type G at $x = 1$ (types R, L_1, L_2 at the north pole) can lie only at the lower end of a shift orbit: if $S \in \Gamma$ then $S \neq T_c S'$ and $S \neq \mathcal{R}_C S'$ for any $S' \in \mathcal{S}$ and $c, C > 0$. This is the starting point of Chapter 7.

6.5. Continuity of the flows, also through $\mu = 0$

In words. *The shift T_c is built at each germ by a contraction, and on each stratum separately its continuity in (S, c) is routine. The point is that at a germ of type L_1 the stratum G is not enough: nearby germs of type G with tiny μ must be handled with the* same *constants. The envelope with weight Z_μ provides them, and the proof of continuity then does not distinguish the two types.*

Lemma 6.28 (continuity of the flows)**.** *Let $\mathcal{D}_T$ be the set of germs in $\mathcal{G}_{\mathsf{N}_0}$ that are of type L_1, or of type G with $\mu < 2$, and $\mathcal{D}_A$ the set of germs in $\mathcal{G}_{\mathsf{N}_0} \cap \mathcal{G}_{\mathsf{G}}$ with $\mu > \frac{1}{2}$. These are open in $\mathcal{G}_{\mathsf{N}_0}$, and*

$$(S, c) \mapsto T_c S : \ \mathcal{D}_T \times \mathbb{R} \to \mathcal{G}_{\mathsf{N}_0}, \qquad (S, C) \mapsto \mathcal{R}_C S : \ \mathcal{D}_A \times \mathbb{R} \to \mathcal{G}_{\mathsf{N}_0}$$

are continuous, including at germs of type L_1.

PROOF. *Openness.* Type G is open and μ is continuous on it (Proposition 6.4); near a germ of type L_1 every germ has $\mu \leq \nu$ (Lemma 6.9). Since $\mathcal{G}_{\mathsf{N}_0}$ is first countable (Lemma 6.3(b)) it suffices to take sequences $(S_n, c_n) \to (S_0, c_0)$.

The t^2-shift. Step 0 (uniform constants). By Lemma 6.11, with $\mathfrak{F} = \{S_n : \ n \geq n_0\} \cup \{S_0\}$ (for large n_0, $S_n \in V$): if S_0 is of type G, $Z_n = e^{-\mu_n(\xi - \xi_1)}$ and $\bar{\mu} = \mu_{\max} < 2$; if of type L_1, $Z_n = Z_{\mu_n}$ and $\bar{\mu} = \frac{1}{2}$. In both cases $Z_n(\zeta) \leq K_0(1 + \xi_1 - \zeta) e^{\bar{\mu}(\xi_1 - \zeta)}$, $K_0 = \sup_n Z_n(\xi_1) < \infty$ (Lemma 6.6(i),(ii)). Take $\bar{c} = |c_0| + 1$, $\xi_2 \leq \hat{\xi}_2$ from (6.5), and $|c_n - c_0| \leq 1$. Write $z_n = z_{S_n, \xi_1}$, κ_n, $\mathcal{F}_n = \mathcal{F}_{c_n, 0}$ for S_n, with fixed points $u_n \in X$, and $\mathcal{F}_0$, u_0 for (S_0, c_0) (Proposition 6.18).

Step 1 (reduction). $u_n - u_0 = [\mathcal{F}_n(u_n) - \mathcal{F}_n(u_0)] + [\mathcal{F}_n(u_0) - \mathcal{F}_0(u_0)]$ and the $\frac{1}{2}$-contraction give $\|u_n - u_0\|_w \leq 2 \|\mathcal{F}_n(u_0) - \mathcal{F}_0(u_0)\|_w$.

Step 2 (data). $Y_{S_n}(x_*) \to Y_{S_0}(x_*)$ at $x_* = \tanh \xi_1$ implies (Remark 2.7(iii)) convergence of the states, uniformly on every $[\xi_T, \xi_1]$: $h_n \to h_0$, $z_n \to z_0$ and $f_n \to f_0$ locally uniformly, $\kappa_n = f_n'(\xi_1) \to \kappa_0$.

Step 3 (pointwise convergence of $\mathcal{F}_n(u_0)$). Fix $\xi \le \xi_2$. (a) $\tilde f_n - f_n = \kappa_n \int_{-\infty}^{\xi} z_n^{-2}((1+u_0)^{-2}-1)$ has pointwise convergent integrand dominated by $10k_0\hat B e^{4\eta}$ (as $Z_n \ge 1$), so $\tilde f_n \to \tilde f_0$ pointwise (Remark 2.7 gives $f_n \to f_0$ pointwise), and $F_n \to F_0$ with $|F_n| \le \hat\Phi E^2 \le \hat\Phi t^2$. (b) $R_n(x) = \int_{-1}^{x} K_{\mathsf L}(x,s)F_n(s)\mathrm{d}s$ with $0 \le K_{\mathsf L} \le (t_x/t_s)^2$ (proof of Lemma 2.23): the integrand is dominated by $\hat\Phi t_x^2$, so $R_n \to R_0$ pointwise and $\varepsilon_n \to \varepsilon_0$, $|\varepsilon_n| \le 2\bar c E^2 + \hat\Phi E^3$. (c) The inner integrand $z_n^2\varepsilon_n(1+u_0)$ converges pointwise and is bounded by $C(1+\xi_1-\zeta)^2 e^{2\bar\mu(\xi_1-\zeta)}e^{4\zeta}$, integrable on $(-\infty,\xi_1]$ and independent of n. So the inner integrals converge for each η, $|z_n^{-2}\cdot\text{inner}| \le C_1E^2$ uniformly in n (Step 4 of Proposition 6.18), and dominated convergence in the outer integral gives $\mathcal{F}_n(u_0)(\xi) \to \mathcal{F}_0(u_0)(\xi)$.

Step 4 (norm convergence). Both functions lie in X, so for $\xi \le \xi_T$ their difference weighted by $E^{-(2-\delta)}$ is at most $2\hat B E_T^{\delta}$, small for ξ_T negative. On $[\xi_T,\xi_2]$ the functions $\mathcal{F}_n(u_0)$ converge pointwise and have derivatives $z_n^{-2}\cdot\text{inner}$ bounded by C_1, so the convergence is uniform (F4). Hence $\|\mathcal{F}_n(u_0)-\mathcal{F}_0(u_0)\|_w \to 0$ and $u_n \to u_0$ uniformly on $(-\infty,\xi_2]$.

Step 5 (the state at ξ_2). All germs $T_{c_n}S_n$, $T_{c_0}S_0$ exist on $(-1,x_2]$, $x_2=\tanh\xi_2$, so by Lemma 6.3(b) it suffices to show $Y(x_2)$ converges. With $\xi=\xi_2$: $\tilde z = z(1+u)$, $\tilde z' = z'(1+u)+zu'$, $\tilde U = 2(\tilde z'/\tilde z - x_2)$, $\tilde f = f + \kappa\int z^{-2}((1+u)^{-2}-1)$, $\tilde f_x = \kappa\tilde z^{-2}\cosh^2\xi$, $\tilde Q = Q + ct^2 + R$, $\tilde Q_x = Q_x + 2ct + R_x$, $(1-x^2)\tilde Q_{xx} = \lambda - \tilde f^2 - 2x\tilde Q_x + 2\tilde Q$ (λ is a frame constant, equal for all shifts). So Y is a continuous function of z, z', Q, Q_x at ξ_2 (Step 2), of κ_n, c_n, and of the four integrals $u_n'(\xi_2)$, $\tilde f_n(\xi_2)$, $R_n(x_2)$, $R_{n,x}(x_2)$ evaluated on u_n. These converge by dominated convergence: $\tilde f_n - f_n$ as in Step 3(a) ($u_n \to u_0$ pointwise, $u_n \in X$); $R_n(x_2)$ as in Step 3(b); $R_{n,x}(x_2) = \int_{-1}^{x_2}\partial_x K_{\mathsf L}(x_2,s)F_n\mathrm{d}s$ with $0 < \partial_x K_{\mathsf L} \le 3t_x/t_s^2$ (for $-1 < s < x \le 0$: $(x-s)^2 + 1 - x^2 \le t_x^2 + 2t_x \le 3t_x$ and $(1-s^2)^2 \ge t_s^2$), dominated by $3\hat\Phi t_x$; and $u_n'(\xi_2)$ as in Step 3(c). Hence $T_{c_n}S_n \to T_{c_0}S_0$.

The recessive shift. Let $S_0 \in \mathcal{D}_A$. By Lemma 6.11(a) and Proposition 6.21 the family $\{S_n\}$ has uniform constants with $\nu = \mu_{\min} > \frac12$, X_A, $\|\cdot\|_A$. Steps 1–5 are repeated with $\mathcal{F}_A$, $\|\cdot\|_A$, and the following changes: in Step 3 the zeroth term $C_n\int_{-\infty}^{\xi} z_n^{-2}$ converges (integrand dominated by $m_-^{-2}\varpi_{\max}E^{\nu}$), the swirl integrand is dominated by $10k_0\varpi_{\max}\hat B_A e^{2(\mu_n+\nu)\eta} \le 10k_0\varpi_{\max}\hat B_A e^{4\nu\eta}$ and $|F_n| \le \Phi E^{2\nu}$ with $2\nu > 1$, so $|K_{\mathsf L}F_n| \le \Phi\, t_x^2\, t_s^{2\nu-2}$, integrable in s (as $2\nu-2 > -1$); the inner integrand $z_n^2\varepsilon_n(1+u_0)$ is dominated by $CE^{\nu+1}\cdot\varpi_{\max}^{-1}E^{-\mu_n} \le C'e^{2\zeta}$ (since $Z_n^2 \le e^{2\mu_{\max}(\xi_1-\zeta)}$ and $\varepsilon_n = O(E^{q+1})$ with $q+1 \ge \mu_n+\nu+1$), integrable. In Step 4 the weight is $E^{-1/2}$ and the tail is $\le 2\hat B_A E_T^{\nu-1/2}$. Step 5 is unchanged ($c=0$, $u_n'(\xi_2) = z_n^{-2}(C_n+\dots)$). □

THEOREM 6.29 (uniform envelope and continuity across $\mu=0$). *Let $S_0 \in \mathcal{G}_{\mathsf{N}_0}$ be of type L_1.*

(A) (Envelope.) *There are ξ_1, a neighbourhood V of S_0 and constants such that every $S \in V$ (of type $\mathsf G$ or $\mathsf L_1$) has $m_-Z_\mu \le z_{S,\xi_1} \le m_+Z_\mu$ with $m_\pm$ independent of S (Lemma 6.11(b)), and $\mu, \lambda, f_-, \kappa_{S,\xi_1}$ are continuous at S_0 (Proposition 6.13).*

(B) (Joint continuity.) *$(S,c) \mapsto T_cS$ is continuous at every (S_0,c_0), $c_0 \in \mathbb{R}$. In particular if $S_n \to S_0$ with S_n of type $\mathsf G$, $\mu_n \to 0$, then $T_{c_n}S_n \to T_{c_0}S_0$ whenever $c_n \to c_0$. The frame functions μ, λ, f_- of T_cS equal those of S, hence are continuous in (S,c) there as well.*

PROOF. (A) is Lemma 6.11(b) with $\nu = \frac{1}{2}$ and Proposition 6.13. (B) is Lemma 6.28 at a germ of type L_1: its proof uses only the uniform constants of (A) and Proposition 6.18 with $\bar{\mu} = \frac{1}{2}$. The frame statement follows from Theorem 6.19(b). □

Remark 6.30 (what is not claimed)**.** The frame function κ_{S,ξ_1} of the *pole* normalisation is not continuous at L_1 (Toy example 6.7); Lemma 6.28 uses the cut normalisation, in which it is. Continuity of the threshold of the shift orbit across $\mu = 0$ is proved in the second half of this chapter (Theorem 6.32).

6.6. The t^2-chart: $N_0 \cap \{0 \le \mu < 2\} \cong \Gamma_T \times [0,\infty)$

In words. *Fix a germ S at the south pole and push it along its t^2-orbit $\{T_c S\}$. Two facts are already in hand: a positive shift of a global germ is global (Proposition 6.23), and the set of global germs is closed (Lemma 6.25). So the shifts c for which $T_c S$ is global form a closed up-set of $\mathbb{R}$. This section shows that this set is non-empty and bounded below, hence a half-line $[\alpha_-(S),\infty)$, that the threshold $\alpha_-(S)$ depends continuously on S* also through $\mu = 0$*, and that therefore every orbit meets the set Γ_T of edge solutions exactly once. The result is a chart: the global solutions of class N_0 with $0 \le \mu < 2$ are a half-cylinder $\Gamma_T \times [0,\infty)$, the second factor being the distance, measured by the t^2-shift, from the edge.*

Toy example 6.31 (why a large shift makes the solution global: the circle closes)**.** Take a constant potential $z'' = s^2 z$ on $[0,\infty)$ with $z(0) = 1$, $z'(0) = -\rho < 0$ (the solution may start out decaying). If $s \ge 2\rho$, then $z = \cosh s\tau - \frac{\rho}{s}\sinh s\tau \ge \frac{1}{4}e^{s\tau}$ (because $\cosh - \frac{1}{2}\sinh = \frac{1}{4}e^{s\tau} + \frac{3}{4}e^{-s\tau}$), so z grows exponentially, and the swirl increment is bounded: $\int_0^\infty z^{-2}\mathrm{d}\tau \le 16\int_0^\infty e^{-2s\tau}\mathrm{d}\tau = 8/s$ (the exact values are 2, $\frac{1}{2}$, $\frac{1}{5}$ against the bound $8/s = 8$, 2.67, 0.8 for $(s,\rho) = (1,\frac{1}{2}),(3,1),(10,5)$; `verify_ch06b.py`, Part 2). In the real problem a swirl f with $f' = \kappa z^{-2}$ feeds back into the potential through $\mathsf{L}[Q] = \lambda - f^2$ and lowers it; but a bounded swirl lowers it by a bounded amount, and the shift ct^2 raises it by an amount proportional to c. Take c so large that the raised potential is $\ge s^2$ with s so large that $8\kappa/s < 1$: then the swirl increment stays below 1, the potential stays high, z keeps growing, and nothing can go wrong. The proof below is this argument, closed by a continuity argument.

Notation. Let $\mathcal{D}_T = \{S \in \mathcal{G}_{\mathsf{N}_0} :\ S \text{ of type } \mathsf{L}_1, \text{ or of type } \mathsf{G} \text{ with } \mu < 2\}$ be the domain of the t^2-shift (Lemma 6.28); it is open in $\mathcal{G}_{\mathsf{N}_0}$ and invariant under T_c (the frame is preserved, Theorem 6.19(b)). Put $\mathcal{S}_T = \mathcal{S} \cap \mathcal{D}_T$ (this is the set of solutions of class N_0 with $0 \le \mu < 2$, Lemma 6.3(c)), and

$$I(S) = \{c \in \mathbb{R} :\ T_c S \in \mathcal{S}\}, \qquad \Gamma_T = \{S \in \mathcal{S}_T :\ S \text{ not of type } \mathsf{G} \text{ at } x = 1\}.$$

Γ_T is the part of the edge Γ_{N} of class N (Chapter 7) that has $\mu < 2$ at the south pole.

THEOREM 6.32 (the threshold; finite, a half-line, continuous across $\mu = 0$). *Let $S \in \mathcal{D}_T$.*

(a) (Finiteness.) *$T_c S$ is global and of type G at $x = 1$ for all sufficiently large c.*

(b) (Half-line.) *$I(S) = [\alpha_-(S),\infty)$ with $\alpha_-(S) \in \mathbb{R}$; and $\alpha_-(T_a S) = \alpha_-(S) - a$ for $a \in \mathbb{R}$.*

(c) $T_cS \in \mathcal{S}_{\mathsf{G}}$ *(global, of type* G *at* $x=1$*) for* $c > \alpha_-(S)$*, and* $T_{\alpha_-(S)}S \in \Gamma_T$*. Hence* $\Gamma_T = \{S \in \mathcal{D}_T :\ \alpha_-(S) = 0\} \cap \mathcal{S}$*.*

(d) (Continuity.) $\alpha_- : \mathcal{D}_T \to \mathbb{R}$ *is continuous; in particular it is continuous across* $\mu = 0$*, i.e. along every family of germs of type* G *that converges to a germ of type* L_1*.*

Proof. (a) If S is global, $T_cS \in \mathcal{S}_{\mathsf{G}}$ for every $c > 0$ by Lemma 6.26. Let S be non-global, with maximal ξ-interval $(-\infty, \xi_S^*)$, $\xi_S^* < \infty$, and let z be a representative. Fix $\xi_a < \xi_S^*$, put $x_a = \tanh\xi_a$, $t_a = 1 + x_a$, $\tau = \xi - \xi_a$, $z_a = z(\xi_a)$, $r_a = z'(\xi_a)/z_a$, $\rho_a = \max(0, -r_a)$, $f_a = f(\xi_a)$, $F_a = f_a + 1$, $c_a = \frac{1-x_a}{2(1+x_a)}$. For $c > 0$ let $S_c = T_cS$ with functions z_c, f_c, Q_c; λ and κ do not depend on c.

Step 1 (comparison at ξ_a*).* By Proposition 6.23, S_c exists on $(-\infty, \xi_S^*) \ni \xi_a$ and there $z_c > z$, $z_c'/z_c > z'/z \geq -\rho_a$ (as $u' > 0$), $f_c < f$, $Q_c \geq Q + ct^2$, $\partial_x Q_c \geq \partial_x Q + 2ct$. Hence at x_a

$$Q_c \geq q_a := Q(x_a) + ct_a^2, \qquad \partial_x Q_c \geq q_a' := Q_x(x_a) + 2ct_a, \qquad 0 \leq f_- \leq f_c \leq f_a.$$

Step 2 (a lower bound for Q_c *forward of* x_a*).* Suppose $q_a, q_a' > 0$. As long as S_c exists on $[x_a, x]$ and $f_c \leq F_a$ on it, the Cauchy formula (2.34) for $\mathsf{L}[Q_c] = \lambda - f_c^2$ gives

$$Q_c(x) = Q_c(x_a)A_1(x) + \partial_x Q_c(x_a)A_2(x) + \int_{x_a}^{x} K_{\mathsf{L}}(x,s)(\lambda - f_c^2)\mathrm{d}s \geq\ q_a - (|\lambda| + F_a^2)c_a =: q_*,$$

because $A_1 \geq 1$, $A_2 \geq 0$, $Q_c(x_a) \geq q_a > 0$, $\partial_x Q_c(x_a) \geq q_a' > 0$, $K_{\mathsf{L}} \geq 0$ and $\int_{x_a}^{x} K_{\mathsf{L}} \leq c_a$ (Lemma 2.18(d)); note $0 \leq f_c \leq F_a$.

Step 3 (Sturm comparison). Suppose in addition $q_* > -2$, put $s = \sqrt{1 + q_*/2}$ and assume $s \geq 2\rho_a$. Where Step 2 applies, $z_c'' = (1 + \frac{1}{2}Q_c)z_c \geq s^2 z_c$. Let $\zeta = z_c(\xi_a)(\cosh s\tau - \frac{\rho_a}{s}\sinh s\tau)$ and $y = z_c - \zeta$. Then $g := y'' - s^2 y = z_c'' - s^2 z_c \geq 0$, $y(\xi_a) = 0$ and $y'(\xi_a) = z_c'(\xi_a) + \rho_a z_c(\xi_a) \geq 0$, so $y(\xi) = \frac{y'(\xi_a)}{s}\sinh s\tau + \frac{1}{s}\int_{\xi_a}^{\xi} \sinh(s(\xi - \eta))g(\eta)\mathrm{d}\eta \geq 0$. With $s \geq 2\rho_a$, $\cosh s\tau - \frac{\rho}{s}\sinh s\tau \geq \frac{1}{4}e^{s\tau}$ (Toy example 6.31); therefore $z_c \geq \zeta \geq \frac{1}{4}z_c(\xi_a)e^{s\tau} \geq \frac{1}{4}z_a e^{s\tau}$.

Step 4 (closing the circle). Where Step 3 applies, $f_c(\xi) - f_c(\xi_a) = \kappa\int_{\xi_a}^{\xi} z_c^{-2} \leq 16\kappa z_a^{-2}\int_0^{\infty} e^{-2s\tau}\mathrm{d}\tau = \frac{8\kappa}{sz_a^2}$. Assume $8\kappa < sz_a^2$; then $f_c < f_a + 1 = F_a$ with a strict inequality. Let $J = \{\xi \geq \xi_a :\ S_c \text{ exists on } (-1, \tanh\xi],\ f_c \leq F_a \text{ on } [\xi_a, \xi]\}$, an interval containing ξ_a. It is closed in $[\xi_a, \xi_c^*)$ by continuity of f_c and open there because Steps 2–4 give $f_c < F_a$ strictly on J. So $J = [\xi_a, \xi_c^*)$, $z_c \geq \frac{1}{4}z_a e^{s\tau}$ is bounded below on bounded intervals, and $\xi_c^* = \infty$ by Lemma 2.24. Thus S_c is global, $Q_c(1) = \lim Q_c \geq q_* > -2$, and $z_c \to \infty$: at $x = 1$ the type is G or R (Definition 4.16), and R has $z \to 0$. So S_c is of type G.

Step 5 (the conditions hold for large c*).* q_a, q_a' grow linearly in c, $q_* = ct_a^2 + O(1)$ and $s^2 = 1 + q_*/2 = \frac{1}{2}ct_a^2 + O(1)$, while $\rho_a, z_a, \kappa, F_a, c_a$ do not depend on c. So $q_a, q_a' > 0$, $q_* > -2$, $s \geq 2\rho_a$ and $8\kappa < sz_a^2$ for all large c.

(b) $I(S)$ is *closed*: if $c_n \in I(S)$, $c_n \to c$, then $T_{c_n}S \to T_cS$ by Lemma 6.28 (at a fixed germ; also at L_1 by Theorem 6.29), and $\mathcal{S}$ is closed (Lemma 6.25). It is an *up-set*: if $c \in I(S)$ and $c' > c$, then $T_{c'}S = T_{c'-c}T_cS$ (Theorem 6.19(c)) is a positive shift of a global germ, hence global (Proposition 6.23). It is *non-empty* by (a). It is *bounded below*: if $I(S) \supset (-\infty, c_1]$, then for $c < c_1$ the germ T_cS is global

and, by Proposition 6.23 with the shift $c_1 - c > 0$, $Q_{T_{c_1}S}(1) \ge Q_{T_cS}(1) + 4(c_1 - c)$, i.e. $Q_{T_cS}(1) \le Q_{T_{c_1}S}(1) - 4(c_1 - c) \to -\infty$ as $c \to -\infty$, contradicting $Q(1) \ge -2$ (Theorem 4.5). A closed, non-empty up-set of $\mathbb{R}$ that is bounded below is $[\alpha_-, \infty)$. Finally $I(T_aS) = \{c : T_{c+a}S \in \mathcal{S}\} = I(S) - a$.

(c) For $c > \alpha_-(S)$ apply Lemma 6.26 to the global germ $T_{\alpha_-}S$ and the shift $c - \alpha_- > 0$. If $T_{\alpha_-}S$ were of type G at $x = 1$, then $T_{\alpha_-}S \in \mathcal{S}_{\mathsf{G}}$, which is open in $\mathcal{G}_{\mathsf{N}_0}$ (Lemma 6.25); since $T_{\alpha_- - \epsilon}S \to T_{\alpha_-}S$ as $\epsilon \downarrow 0$ (Lemma 6.28) we would get $\alpha_- - \epsilon \in I(S)$, contradicting minimality. The last assertion: if $S \in \mathcal{S} \cap \mathcal{D}_T$ then $0 \in I(S)$; $\alpha_-(S) < 0$ would make $S = T_0S$ of type G by what was just proved; conversely $S \in \Gamma_T$ has $\alpha_-(S) \le 0$ and $\alpha_-(S) < 0$ is excluded for the same reason.

(d) By first countability (Lemma 6.3(b)) we may use sequences $S_n \to S$ in $\mathcal{D}_T$. *Upper semicontinuity.* Let $\epsilon > 0$. By (c), $T_{\alpha_-(S)+\epsilon}S \in \mathcal{S}_{\mathsf{G}}$, which is open relative to $\mathcal{G}_{\mathsf{N}_0}$. By Lemma 6.28 (jointly continuous in the germ, also when S is of type L_1 and S_n of type G with $\mu_n \to 0$, Theorem 6.29) $T_{\alpha_-(S)+\epsilon}S_n \to T_{\alpha_-(S)+\epsilon}S$ and these germs lie in $\mathcal{G}_{\mathsf{N}_0}$; so they lie in $\mathcal{S}_{\mathsf{G}} \subset \mathcal{S}$ for large n, i.e. $\alpha_-(S_n) \le \alpha_-(S) + \epsilon$. *Lower semicontinuity.* Suppose a subsequence has $\alpha_-(S_n) \to L < \alpha_-(S)$, $L = -\infty$ allowed. If $L > -\infty$, then $T_{\alpha_-(S_n)}S_n \in \mathcal{S}$ converges to T_LS (joint continuity), which is global because $\mathcal{S}$ is closed; so $L \in I(S)$, contradicting $L < \alpha_-(S)$. If $L = -\infty$, then for every m and large n, $-m \in I(S_n)$ (up-set), so $T_{-m}S_n \in \mathcal{S}$ and the limit $T_{-m}S \in \mathcal{S}$: $I(S) \supset (-\infty, 0]$, contradicting (b). The value $+\infty$ is excluded by the upper bound. □

Remark 6.33 (what had to be new at $\mu = 0$)**.** Everything in this proof is a monotonicity or closedness argument and is the same on the two strata G and L_1; the only input that is not, is the *joint* continuity of $(S, c) \mapsto T_cS$ at germs of type L_1 (Theorem 6.29), which rests on the uniform envelope Z_μ of Lemma 6.11(b). Without it, continuity of α_- would be proved on each stratum separately and the two halves of the chart would not be known to fit. The restriction $\mu < 2$ enters only through the existence of T_c (Toy example 6.15); the amplitude $\frac{1}{4(2-\mu)}$ is a true singularity, and the chart is not claimed up to $\mu = 2$. For $\mu > 1$ the recessive shift gives a second chart (Section 6.7) that has no restriction on μ.

THEOREM 6.34 (the t^2-chart). *Let $\Psi : \Gamma_T \times [0,\infty) \to \mathcal{S}_T$, $\Psi(\tilde{S}, s) = T_s\tilde{S}$. Then Ψ is a homeomorphism, with inverse $S \mapsto (T_{\alpha_-(S)}S, -\alpha_-(S))$. Moreover $\mathcal{D}_T \cong \Gamma_T \times \mathbb{R}$ by the same formulas, every T-orbit in $\mathcal{D}_T$ meets Γ_T exactly once, and $\Gamma_T \to \mathcal{D}_T/T$ is a homeomorphism. In particular $\mathsf{N}_0 \cap \{0 \le \mu < 2\} = \mathcal{S}_T \cong \Gamma_T \times [0,\infty)$, the type of $\Psi(\tilde{S}, s)$ at $x = 1$ being G iff $s > 0$.*

PROOF. *Well defined.* For $S \in \mathcal{S}_T$, $0 \in I(S)$ so $s(S) := -\alpha_-(S) \ge 0$; $\tilde{S} := T_{\alpha_-(S)}S$ is global (definition of α_-), lies in $\mathcal{D}_T$ (the frame is preserved) and is not of type G at $x = 1$ (Theorem 6.32(c)), so $\tilde{S} \in \Gamma_T$. For $(\tilde{S}, s) \in \Gamma_T \times [0,\infty)$: $\alpha_-(\tilde{S}) = 0$ (if $\tilde{S} \in \Gamma_T$ then $0 \in I$, so $\alpha_- \le 0$, and $\alpha_- < 0$ would make $\tilde{S}$ of type G), hence $\alpha_-(T_s\tilde{S}) = -s \le 0$, i.e. $T_s\tilde{S} \in \mathcal{S}_T$. *Inverse maps.* By the group law, $T_{-\alpha_-(S)}T_{\alpha_-(S)}S = S$ and $\alpha_-(T_s\tilde{S}) = -s$, $T_{\alpha_-(T_s\tilde{S})}T_s\tilde{S} = T_{-s}T_s\tilde{S} = \tilde{S}$. *Continuity.* $S \mapsto \alpha_-(S)$ is continuous (Theorem 6.32(d)) and $(S, c) \mapsto T_cS$ is continuous (Lemma 6.28, Theorem 6.29); so the inverse is continuous, and so is Ψ. *Orbits.* The same formulas, with $s \in \mathbb{R}$, give $\mathcal{D}_T \cong \Gamma_T \times \mathbb{R}$. Since $\alpha_-(T_cS) = \alpha_-(S) - c$ vanishes only at $c = \alpha_-(S)$, every orbit meets Γ_T exactly once. The map $S \mapsto T_{\alpha_-(S)}S$ is continuous and constant on orbits, so it induces a continuous map

$\mathcal{D}_T/T \to \Gamma_T$, inverse to the continuous bijection $\Gamma_T \to \mathcal{D}_T/T$. The type statement is Theorem 6.32(c). □

Remark 6.35 (topology; the picture; Frobenius labels)**.** The topology on $\mathcal{S}_T$ is the topology of $\Omega \subset \mathbb{R}^6$: for a global germ the neighbourhoods $N(S, x_*, \epsilon)$ of Definition 6.2 are neighbourhoods of the state at one (equivalently every) point, Lemma 5.7. Figure 6.5(a) draws the chart: the edge Γ_T is the floor, the T-orbits are the half-lines $s \mapsto \Psi(\tilde{S}, s)$, and s is the distance from the edge. For $1 < \mu < 2$, where the number p_2 is defined (Proposition 6.37) and $p_2(T_c S) = p_2(S) + c$, put $p_2^-(S) := p_2(S) + \alpha_-(S)$; it is constant along T-orbits, S is global iff $p_2(S) \geq p_2^-(S)$, and $\alpha_- = p_2^- - p_2$ (the T-coordinate $s = -\alpha_- = p_2 - p_2^-$ of Remark 6.58). The chart is thus a half-space in p_2 over the frame. For $\mu \leq 1$ the book uses no coordinate of this kind: the quantity α_- is defined through the shift flow and is continuous (Theorem 6.32), whatever the behaviour of any particular normalisation of the expansion near the resonances (Toy example 6.1).

6.7. The recessive chart for $\mu > 1$: the explicit base

In words. *For $\mu \geq 2$ the t^2-shift does not exist, but the recessive shift does, and for $\mu > 1$ it can be analysed completely. The reason is that the swirl correction to Q is then smaller than t^2, so Q has a well defined* polynomial part $P_d = q_0 + p_1 t + p_2 t^2$ *at the pole, and the whole effect of the swirl is a correction $-\Psi$ that is invisible to the recessive shift. A global solution has a potential below that of the swirl-free potential P_d, which forces P_d into the interior of the Li–Li–Yan set (necessity); and conversely a large recessive shift makes the swirl negligible, so that P_d in the interior suffices (sufficiency). The result is an* explicit *base B and the chart $\mathsf{N}_0 \cap \{\mu > 1\} \cong B \times [0, \infty)$.*

Setting. $\mathcal{G}_A = \mathcal{G}_{\mathsf{N}_0} \cap \mathcal{G}_{\mathsf{G}} \cap \{\mu > 1\}$ and $\mathcal{S}_A = \mathcal{S} \cap \mathcal{G}_A$. Germs of type G are normalised by $z e^{\mu\xi} \to 1$, $z' e^{\mu\xi} \to -\mu$ (Definition 6.2); κ is the swirl constant in this normalisation. On $\mathcal{G}_A$ the recessive shift $\mathcal{R}_C$ exists for all $C \in \mathbb{R}$ (Theorem 6.22); $E = e^{2\xi}$, $t = 1 + x$.

Toy example 6.36 (the polynomial part; a swirl that is constant)**.** If $f \equiv f_-$ were constant ($\kappa = 0$), $\mathsf{L}[Q] = \lambda - f_-^2$ would say that Q is a quadratic polynomial, $Q = P_{\boldsymbol{c}}$ with $\boldsymbol{c} = (c_1, c_2, c_3)$, and $z'' = W_{\boldsymbol{c}} z$ would be a no-swirl problem of Chapter 3. A genuine swirl $f = f_- + \kappa \int z^{-2}$ differs from f_- by $O(E^\mu)$, and the correction to Q is $\mathsf{G}_{\mathsf{L}}[f^2 - f_-^2] = O(E^{1+\mu})$. For $\mu > 1$ this is $o(t^2)$, so the three numbers (q_0, p_1, p_2) read off from Q at the pole are exactly the data of a quadratic. For $\mu \leq 1$ the correction is larger than t^2 and no such statement is possible; this is where the threshold $\mu > 1$ comes from. For example, $\mu = 1.5$, $\lambda = 0$, $f_- = 0.5$, $p_2 = 0$ give $q_0 = 2.5$, $p_1 = \frac{1}{2}(f_-^2 - \lambda) - q_0 = -2.375$, and $\boldsymbol{c}^d = (1.25, -1.125, 0)$ (Proposition 6.37; `verify_ch06b.py`, Part 1).

Proposition 6.37 (Green representation; the polynomial part)**.** *Let $S \in \mathcal{G}_A$ and $g_S := f^2 - f_-^2 \geq 0$, $\Psi_S := \mathsf{G}_{\mathsf{L}}[g_S]$. Then $g_S = O(E^\mu)$, $\Psi_S \geq 0$, $\partial_x \Psi_S > 0$, $\Psi_S = O(E^{1+\mu}) = o(t^2)$, and on the whole interval of existence*

$$Q = P_d - \Psi_S, \qquad P_d = q_0 + p_1 t + p_2 t^2,$$

$$q_0 = 2(\mu^2 - 1), \quad p_1 = \tfrac{1}{2}(f_-^2 - \lambda) - q_0, \quad p_2 = \lim_{t \to 0} \frac{Q - q_0 - p_1 t}{t^2}.$$

The number p_2 is invariant under $\mathcal{R}_C$ and shifted by c under T_c ($\mu < 2$). Moreover $P_d = P_{\boldsymbol{c}^d}$ with $\boldsymbol{c}^d = \big(\mu^2 - 1,\ \mu^2 - 1 + p_1 + 2p_2,\ -p_2\big)$, and we put $\mu_+^2 := 1 + c_2^d = \mu^2 + p_1 + 2p_2$.

PROOF. $f - f_- = \kappa \int_{-\infty}^{\xi} z^{-2} = O(E^\mu)$ by the envelope $z \geq m_- e^{-\mu\xi}$ (Lemma 6.11(a)), and $f + f_-$ is bounded, so $g_S = O(E^\mu)$; $g_S \geq 0$ because $f \geq f_- \geq 0$, and $g_S > 0$ on $(-1, x_S^*)$ since f is strictly increasing. Put $Y = Q - q_0 - p_1 t$. By the first integral, $\mathsf{L}[Q] = \lambda - f^2$, and $\mathsf{L}[q_0 + p_1 t] = -2q_0 - 2p_1 = \lambda - f_-^2$ (Lemma 2.18(b), definition of p_1), so $\mathsf{L}[Y] = -g_S$; and $Y \to 0$ at the pole. Since $\int g_S E^{-1} < \infty$ for $\mu > 1$, Lemma 2.23 gives $\Psi_S = O(E^{1+\mu})$ (use $q = \mu$) and $-Y = \Psi_S + \beta t^2$ for a unique β. Dividing by t^2 and using $\Psi_S / t^2 = O(E^{\mu-1}) \to 0$ shows that the limit defining p_2 exists, and $p_2 = -\beta$; so $Y = -\Psi_S + p_2 t^2$, i.e. $Q = P_d - \Psi_S$ near the pole, hence on the whole interval (both sides solve $\mathsf{L}[\cdot] = \lambda - f^2$ and the difference lies in $\ker \mathsf{L}$ and is $o(t^2)$, hence is 0: $\ker \mathsf{L}$ is spanned by x and $(1+x)^2$, and if $\alpha x + \beta(1+x)^2 = o(t^2)$ then $\alpha = 0$ (let $x \to -1$), and then $\beta = 0$). $\Psi_S \geq 0$ and $\partial_x \Psi_S > 0$ by Lemma 2.18(c). A recessive shift changes Q by $O(E^{2\mu+1}) = o(E^2)$ and keeps the frame $(q_0, \lambda, f_-, \kappa)$, hence p_1 and p_2 (Theorem 6.22); T_c keeps the frame and changes Q by $ct^2 + o(E^2)$ (Theorem 6.19), hence $p_2 \mapsto p_2 + c$. Finally $P_{\boldsymbol{c}}(-1 + t) = c_1(2 - t) + c_2 t + c_3(2t - t^2)$ (because $1 - x = 2 - t$, $1 + x = t$, $1 - x^2 = 2t - t^2$); matching coefficients with $q_0 + p_1 t + p_2 t^2$ gives $2c_1 = q_0$, $c_3 = -p_2$, $c_2 = p_1 + c_1 - 2c_3$. □

The map $\Pi : \mathcal{G}_A \to \mathcal{Y}_0 := (1,\infty) \times \mathbb{R} \times [0,\infty) \times (0,\infty) \times \mathbb{R}$, $\Pi(S) = (\mu, \lambda, f_-, \kappa, p_2)$, is therefore invariant under $\mathcal{R}_C$. It depends on p_2 only through the potential $w_P := 1 + \frac{1}{2} P_d = \mu^2 + \hat{w}$, $\hat{w} = \frac{1}{2}(p_1 t + p_2 t^2)$, $|\hat{w}| \leq \pi_y E$ for $\xi \leq 0$, $\pi_y := |p_1| + 2|p_2|$ (use $t \leq 2E$). Let u_P be the solution of $u'' = w_P u$ that is recessive at the south pole, $u_P e^{-\mu\xi} \to \frac{1}{2\mu}$.

Lemma 6.38 (the no-swirl solutions u_P and z_d)**.** *Let $y = \Pi(S) \in \mathcal{Y}_0$.*

(a) *u_P exists, is unique, and $u_P = e^{\mu\xi}\omega$ with $\omega = \frac{1}{2\mu} + \int_{-\infty}^{\xi} k_\mu(\xi - \eta)\hat{w}\omega \mathrm{d}\eta$, $k_\mu(s) = \frac{1 - e^{-2\mu s}}{2\mu} \in [0, \frac{1}{2\mu}]$. If $\xi_b \leq 0$ and $\pi_y E_b \leq \mu/5$, then on $(-\infty, \xi_b]$: $|2\mu\omega - 1| \leq \frac{\pi_y E}{3\mu} \leq \frac{1}{15}$, so $\frac{1}{4\mu} e^{\mu\xi} \leq u_P \leq \frac{1}{\mu} e^{\mu\xi}$, $|u_P'| \leq e^{\mu\xi}$, $u_P' = \frac{1}{2} e^{\mu\xi}(1 + O(E))$. Also $\mathrm{Wr}(u_P, z) \to -1$ for every germ with $z e^{\mu\xi} \to 1$, $z' e^{\mu\xi} \to -\mu$.*

(b) *$u_P(\xi)$, $u_P'(\xi)$ depend continuously on y for fixed ξ, and the bounds in* (a) *are uniform for y in compact subsets of $\mathcal{Y}_0$.*

(c) *For $\xi \leq \xi_b$ put $z_d(\xi) = u_P(\xi) \int_{\xi}^{\xi_b} u_P^{-2}$. Then $z_d'' = w_P z_d$, $\mathrm{Wr}(u_P, z_d) = -1$, $z_d e^{\mu\xi} \to 1$, $z_d' e^{\mu\xi} \to -\mu$; and every solution v of $v'' = w_P v$ is $v = a\, z_d + b\, u_P$ with $a = -\mathrm{Wr}(u_P, v)$, $b = v(\xi_b)/u_P(\xi_b)$.*

PROOF. (a) Substituting $u = e^{\mu\xi}\omega$ gives $(e^{2\mu\xi}\omega')' = e^{2\mu\xi}\hat{w}\omega$; the solution with $\omega \to \frac{1}{2\mu}$, $e^{2\mu\xi}\omega' \to 0$ satisfies $\omega' = \int_{-\infty}^{\xi} e^{-2\mu(\xi-\eta)}\hat{w}\omega$, and integrating once more (exchange the order) gives the Volterra equation. On bounded functions the operator has norm $\leq \frac{\pi_y}{2\mu} \int_{-\infty}^{\xi_b} e^{2\eta}\mathrm{d}\eta = \frac{\pi_y E_b}{4\mu} \leq \frac{1}{20}$, so there is a unique bounded solution, $\|\omega\|_\infty \leq \frac{20}{19} \cdot \frac{1}{2\mu}$ and $|\omega - \frac{1}{2\mu}| \leq \frac{\pi_y E}{4\mu}\|\omega\|_\infty$, which gives the bound on $2\mu\omega - 1$ and on u_P; for $\omega' = \int_{-\infty}^{\xi} e^{-2\mu(\xi-\eta)}\hat{w}\omega \mathrm{d}\eta$ one has $|\omega'| \leq \pi_y \|\omega\|_\infty \int_{-\infty}^{\xi} e^{-2\mu(\xi-\eta)} e^{2\eta}\mathrm{d}\eta = \frac{\pi_y E}{2\mu+2}\|\omega\|_\infty$, which gives $|u_P' - \mu u_P| = e^{\mu\xi}|\omega'|$ and with it the stated bounds on u_P'. Uniqueness: two recessive solutions differ by $o(e^{\mu\xi})$, and by (c) every solution is

$az_d + bu_P$ with $z_d \sim e^{-\mu\xi}$. Finally $\mathrm{Wr}(u_P, z) \to \frac{1}{2\mu}(-\mu) - \frac{1}{2}\cdot 1 = -1$. (b) For $y_n \to y$ in a compact set the difference $\omega_n - \omega$ satisfies the linear Volterra equation with source $(k_{\mu_n}\hat{w}_n - k_\mu \hat{w})\omega$, which is $o(1)$ uniformly because $|\partial_\mu k_\mu| \le \frac{1}{\mu} + \frac{1}{2\mu^2}$ and $|\hat{w}_n - \hat{w}| \le (|p_{1,n} - p_1| + 2|p_{2,n} - p_2|)E$; the operator bound gives $\sup|\omega_n - \omega| \to 0$ (also for ω'). Continuous dependence of the linear ODE transports this to every fixed ξ. (c) $(z_d/u_P)' = -u_P^{-2}$ gives $\mathrm{Wr}(u_P, z_d) = u_P^2(z_d/u_P)' = -1$; z_d solves the equation by reduction of order (Corollary 3.6). By (a), $u_P^{-2} = 4\mu^2 e^{-2\mu\xi}(1+O(E))$, so $\int_\xi^{\xi_b} u_P^{-2} = 2\mu e^{-2\mu\xi}(1+o(1))$ and $z_d e^{\mu\xi} \to \frac{1}{2\mu}\cdot 2\mu = 1$; from $z_d' = u_P' \int_\xi^{\xi_b} u_P^{-2} - u_P^{-1}$ we get $z_d' e^{\mu\xi} \to \frac{1}{2}\cdot 2\mu - 2\mu = -\mu$. For a solution v, $\mathrm{Wr}(u_P, v - az_d) = 0$, so $v - az_d = bu_P$, and $z_d(\xi_b) = 0$ gives b. □

Definition 6.39 (the base set B)**.** For $y = (\mu, \lambda, f_-, \kappa, p_2) \in \mathcal{Y}_0$ let p_1 and $\mu_+^2 = \mu^2 + p_1 + 2p_2$ be as in Proposition 6.37, and

$$B = \{y \in \mathcal{Y}_0 :\ \mu_+^2 > 0,\ \ 2(\mu+1)(\mu+\mu_+) + p_1 > 0\}, \qquad \mu_+ = \sqrt{\mu_+^2}.$$

Lemma 6.40 (the algebra behind B)**.** *For $y \in \mathcal{Y}_0$ the following are equivalent:* (i) $y \in B$*;* (ii) $\boldsymbol{c}^d \in \operatorname{int}\mathcal{J}$*;* (iii) $u_P > 0$ *on $\mathbb{R}$ and $u_P'/u_P \to \mu_+ > 0$ as $\xi \to +\infty$ (u_P is dominant at $+\infty$). For fixed $(\mu, \lambda, f_-, \kappa)$, B is the half-line $p_2 > p_2^{\mathrm{crit}} := \frac{1}{2}(m^2 - \mu^2 - p_1)$, $m = \max(0, -\mu - \frac{p_1}{2(1+\mu)})$. B is open in $\mathcal{Y}_0$.*

Proof. Recall $\operatorname{int}\mathcal{J} = \{c_1, c_2 > -1,\ c_3 > \bar{c}_3\}$, $\bar{c}_3 = -\frac{1}{2}(k_1+k_2)(k_1+k_2+2)$, $k_i = \sqrt{1+c_i}$ (Chapter 3). Here $c_1^d = \mu^2 - 1 > -1$ with $k_1 = \mu$; $c_2^d > -1$ iff $\mu_+^2 > 0$, and then $k_2 = \mu_+$. The condition $c_3^d > \bar{c}_3$ reads $-p_2 > -\frac{1}{2}(\mu+\mu_+)(\mu+\mu_++2)$; inserting $p_2 = \frac{1}{2}(\mu_+^2 - \mu^2 - p_1)$ and multiplying by 2: $\mu_+^2 - \mu^2 - p_1 < (\mu+\mu_+)^2 + 2(\mu+\mu_+)$, i.e. $0 < 2\mu^2 + 2\mu\mu_+ + 2\mu + 2\mu_+ + p_1 = 2(\mu+1)(\mu+\mu_+) + p_1$ (expand $(\mu+\mu_+)^2$: the terms μ_+^2 cancel). So (i)⇔(ii). (ii)⇔(iii): u_P solves $z'' = W_{\boldsymbol{c}^d} z$ and is recessive at $-\infty$, so it is a positive multiple of the principal solution p_- whenever it is positive and bounded relative to g (Lemma 3.42(b),(d)). If $c_2^d < -1$, $W_{\boldsymbol{c}^d} \to 1 + c_2^d < 0$ at $+\infty$ and no positive solution exists (Proposition 3.43(a)); if $c_2^d = -1$ then $\mu_+ = 0$, excluded in (iii). If $c_2^d > -1$: for $\varepsilon := c_3^d - \bar{c}_3 > 0$ the principal solution p_- is positive with p_-/g increasing to ∞, i.e. dominant at $+\infty$ (Proposition 3.43(c)); for $\varepsilon = 0$ it equals g, recessive at $+\infty$; for $\varepsilon < 0$ there is no positive solution (Proposition 3.43(a)). Hence $u_P > 0$ and dominant iff $\varepsilon > 0$. For the half-line: with μ, p_1 fixed, (i) says $\mu_+ > 0$ and $\mu_+ > -\mu - \frac{p_1}{2(\mu+1)}$, i.e. $\mu_+ > m$ (as $\mu_+ \ge 0$, equivalently $\mu_+^2 > m^2$), i.e. $p_2 > \frac{1}{2}(m^2 - \mu^2 - p_1)$. Openness: (i) consists of strict inequalities between continuous functions of y. □

Numerical observation 6.41 (the criterion $\Pi \in B$ against direct integration)**.** *For* 300 *random points $(\mu, p_1, p_2) \in [1.05, 3] \times [-6, 6] \times [-4, 4]$ we integrated $u'' = w_P u$ from the recessive start at $\xi = -18$ to $\xi = 18$ and recorded whether u_P stays positive and its log-derivative at $+18$ is positive. This happened for* 189 *points, and the criterion of Lemma 6.40 (i) gave the same answer in all* 300 *cases (no mismatches;* `verify_ch06b.py`*, Part 3). Figure 6.3 shows the set B in the (p_1, p_2)-plane for $\mu = 1.5$.*

Lemma 6.42 (necessity)**.** *If $S \in \mathcal{S}_A$, then $\Pi(S) \in B$.*

Proof. $\sigma := \mathrm{Wr}(u_P, z)$ satisfies $\sigma' = u_P z(W - w_P) = -\frac{1}{2}\Psi_S u_P z \le 0$ wherever $u_P > 0$ ($W - w_P = \frac{1}{2}(Q - P_d) = -\frac{1}{2}\Psi_S$, Proposition 6.37; $z > 0$ because S is

global), and $\sigma \to -1$ at $-\infty$ (Lemma 6.38(a)); so $\sigma \le -1$. If u_P had a first zero ξ_0, then $u_P'(\xi_0) < 0$ and $\sigma(\xi_0) = -u_P'(\xi_0)z(\xi_0) > 0$: impossible. So $u_P > 0$ on $\mathbb{R}$. Next $-2 \le Q(1) = P_d(1) - \Psi_S(1) < P_d(1) = 2c_2^d$ (Theorem 4.5; Ψ_S is positive and increasing), so $c_2^d > -1$ and $\mu_+ > 0$. Finally $\tilde{\mu}_+ := \sqrt{1 + Q(1)/2} < \mu_+$ and, S being global, $|z| + |z'| \le C' e^{(\tilde{\mu}_+ + \varepsilon)\xi}$ for every $\varepsilon > 0$ (Proposition 4.19). If u_P were recessive at $+\infty$, $u_P, u_P' = O(e^{-\mu_+ \xi})$, we would get $|\sigma| \le |u_P||z'| + |u_P'||z| \to 0$ for $\varepsilon < \mu_+ - \tilde{\mu}_+$, contradicting $\sigma \le -1$. Hence u_P is positive and dominant at $+\infty$; by Lemma 6.40, $\Pi(S) \in B$. □

Lemma 6.43 (pole-region estimate for large recessive shifts)**.** *Let $S \in \mathcal{G}_A$ (global or not), $\theta := \frac{\mu-1}{\mu}$, and let ξ_b be such that $u_P > 0$ on $(-\infty, \xi_b]$; $S_C := \mathcal{R}_C S$, $z_C = u_P \omega_C$. Then there are Λ_* and, for every $\varepsilon > 0$, a C_0 such that for all $C \ge C_0$:*

(i) *S_C exists on $(-\infty, \xi_b]$ and $z_C \ge \frac{1}{3} C u_P$ there;*
(ii) *$0 \le f_C - f_- \le 3\kappa/C$ on $(-\infty, \xi_b]$;*
(iii) *$0 \le \Psi_C := P_d - Q_C \le \Lambda_* C^{-\theta}$ and $|\partial_x \Psi_C| \le \Lambda_* C^{-\theta}$ on $(-1, x_b]$;*
(iv) *$|z_C'/z_C - u_P'/u_P| \le \varepsilon$ at ξ_b (indeed on $[\xi_1, \xi_b]$ for a suitable ξ_1).*

Toy example 6.44 (why a large C makes the swirl small)**.** Write $z_C = u_P \omega_C$. The swirl obeys $f_C' = \kappa u_P^{-2} \omega_C^{-2}$, and the Wronskian with u_P makes ω_C decrease at least like $\omega_C' \le -u_P^{-2}$ (this is the content of Step 1 below). Hence $f_C' \le \kappa(-\omega_C')\omega_C^{-2}$ and $f_C - f_- \le \kappa/\omega_C$ *whatever* u_P is: the swirl increment is at most the reciprocal of the amplitude of z_C relative to u_P. A shift by C makes this amplitude $\approx C$ already near the pole, so the swirl is $O(1/C)$. The lemma shows that this survives the feedback of the swirl on the potential, up to any fixed point ξ_b.

Proof. *Step 1 (Wronskian).* $\Psi_C = \mathsf{G}_{\mathsf{L}}[f_C^2 - f_-^2] \ge 0$ because $f_C \ge f_- \ge 0$, and $W_C = w_P - \frac{1}{2}\Psi_C$ (Proposition 6.37 for S_C, which has $\Pi(S_C) = \Pi(S)$). Hence $\sigma_C := \mathrm{Wr}(u_P, z_C) = u_P^2 \omega_C'$ satisfies $\sigma_C' = -\frac{1}{2}\Psi_C u_P z_C \le 0$ and $\sigma_C(-\infty) = -1$: indeed $\mathrm{Wr}(u_P, z) \to -1$, and $z_C - z = zu$ with $zu, (zu)' = O(e^{\mu\xi})$ [$u = O(E^\mu)$ by Theorem 6.22(a) and $z^2 u' = C + \int z^2 \varepsilon(1+u) = O(1)$ by reduction of order, whence $zu' = O(e^{\mu\xi})$ and $z'u = O(e^{-\mu\xi}E^\mu) = O(e^{\mu\xi})$], so $\mathrm{Wr}(u_P, zu) = O(e^{2\mu\xi}) \to 0$. Thus $\sigma_C = -(1 + m_C)$, $m_C(\xi) = \frac{1}{2}\int_{-\infty}^{\xi} \Psi_C u_P z_C \ge 0$ nondecreasing (the integrand is $O(E^{1+\mu}) \cdot O(1)$), and, wherever S_C exists and $u_P > 0$,

$$\omega_C' = -(1 + m_C) u_P^{-2} \le -u_P^{-2},$$
$$f_C - f_- = \kappa \int_{-\infty}^{\xi} u_P^{-2} \omega_C^{-2} \le \kappa \int_{-\infty}^{\xi} (-\omega_C') \omega_C^{-2} \le \frac{\kappa}{\omega_C}. \quad (*)$$

Step 2 (the pole region; constants depending on S only). Let c_S, k_S, m_+ and a first cut $\xi_1^{(0)} < \min(\xi_b, 0, x_S^*)$ be such that on $(-\infty, \xi_1^{(0)}]$: $f - f_- \le c_S E^\mu \le 1$, $z \le m_+ e^{-\mu\xi}$, $\pi_y E \le \mu/5$ (so Lemma 6.38(a) applies). These inequalities persist on $(-\infty, \xi_1]$ for every $\xi_1 \le \xi_1^{(0)}$, and the cut $\xi_1 \le \xi_1^{(0)}$ used below is chosen in Step 3 *after* c_S, k_S, m_+ are fixed; so Λ_* does not depend on ξ_1. By Proposition 6.23, S_C exists on $(-\infty, \xi_1]$ for $C > 0$ and $z_C \ge z$, $f_C \le f$; hence $f_C - f_- \le c_S E^\mu$, and by Lemma 2.23 with $q = \mu$, $\Psi_C \le k_S E^{1+\mu}$ for all $C \ge 0$, $k_S = 2(2f_- + 1)c_S/(\mu - 1)$. Put $\omega_1 = \omega_C(\xi_1)$, $m_1 = m_C(\xi_1)$. For $\xi \le \xi_1$, $\omega_C(\xi) \le \omega_1 + (1 + m_1)U(\xi)$ with

$U(\xi)=\int_\xi^{\xi_1}u_P^{-2}\le 8\mu e^{-2\mu\xi}$ and, since $u_P^2\le\mu^{-2}e^{2\mu\xi}$,

$$m_1=\tfrac12\int_{-\infty}^{\xi_1}\Psi_C u_P^2\omega_C\le a_1\omega_1+b_1(1+m_1),\quad a_1=\frac{k_S e^{(2+4\mu)\xi_1}}{2\mu^2(2+4\mu)},\ b_1=\frac{4k_S e^{(2+2\mu)\xi_1}}{\mu(2+2\mu)},$$

so $m_1\le 2a_1\omega_1+2b_1$ if $b_1\le\frac12$. Moreover $z_C=z(1+u)\ge z(1+Cl)\ge Czl$ with $l=\int_{-\infty}^{\xi}z^{-2}$ (Proposition 6.23, Step 1), and $zl/u_P\to 1$ at $-\infty$; so $\omega_1\ge\nu_1 C$ with $\nu_1\to 1$ as $\xi_1\to-\infty$. *Step 3 (the order of choices).* Put $U_b=\int_{\xi_1}^{\xi_b}u_P^{-2}$, $V_b=\int_{\xi_1}^{\xi_b}u_P^2$, $\underline{u}=\min_{[\xi_1,\xi_b]}u_P^2$, $M_b=4/(1-x_b)^2$. *First* choose ξ_1 so negative that $b_1\le\frac12$, $\nu_1\ge\frac34$, $2a_1U_b\le\frac18$ and $a_1/\underline{u}\le\varepsilon'$ (possible: $a_1U_b=O(e^{(2+2\mu)\xi_1})$, $a_1/\underline{u}=O(e^{(2+2\mu)\xi_1})$ because $u_P^2\ge\min(e^{2\mu\xi}/(16\mu^2),\text{const})$ on $[\xi_1,\xi_b]$). *Then* choose C_0 so large that for $C\ge C_0$: $\delta_C:=4\kappa/C\le 1$, $t_c:=(\delta_C/c_S)^{1/\mu}\le t(\xi_1)$, $\frac12\Lambda_*C^{-\theta}V_bU_b\le\frac18$ and $2U_b\le C/8$. *Bootstrap.* Let $J=\{\xi\in[\xi_1,\xi_b]:\ S_C$ exists on $(-\infty,\xi]$, $\omega_C>C/4$ on $[\xi_1,\xi]\}\ni\xi_1$. Fix $\xi\in J$. By $(*)$, $f_C-f_-<\delta_C$ on $[\xi_1,\xi]$, and on $(-\infty,\xi_1]$ also $f_C-f_-\le\min(c_SE^\mu,\delta_C)$ ($\omega_C\ge\omega_1>C/4$ there, ω_C being decreasing). Since $E\le t$ and $c_St^\mu\ge\delta_C$ for $t\ge t_c$, $f_C-f_-\le\min(c_St^\mu,\delta_C)$ for all $s\le x(\xi)$. For $s<x\le x_b$, $K_{\mathsf L}(x,s)\le M_bt_s^{-2}$ and $\partial_xK_{\mathsf L}(x,s)\le\frac54M_bt_s^{-2}$ ($|x-s|\le 2$, $1-xs\le 2$, $(x-s)^2+1-x^2\le 5$, $(1-s^2)^2=t_s^2(1-s)^2\ge t_s^2(1-x_b)^2$), so with $f_C+f_-\le 2f_-+1$,

$$\Psi_C,\ |\partial_x\Psi_C|\le 2M_b(2f_-+1)\Big[\int_0^{t_c}c_St^{\mu-2}\mathrm dt+\delta_C\int_{t_c}^{\infty}t^{-2}\mathrm dt\Big]$$
$$=2M_b(2f_-+1)\Big(\tfrac{1}{\mu-1}+1\Big)c_St_c^{\mu-1}=\Lambda_*C^{-\theta}$$

(using $c_St_c^\mu=\delta_C$, $t_c^{\mu-1}=(\delta_C/c_S)^\theta$), with $\Lambda_*=2M_b(2f_-+1)(\frac{1}{\mu-1}+1)c_S^{1-\theta}(4\kappa)^\theta$ independent of C, ξ. Hence $m_C(\xi)\le m_1+\frac12\Lambda_*C^{-\theta}V_b\omega_1$ (as $\omega_C\le\omega_1$ on $[\xi_1,\xi]$), and integrating $\omega_C'=-(1+m_C)u_P^{-2}$ from ξ_1,

$$\omega_C(\xi)\ge\omega_1\big[1-U_b(2a_1+\tfrac12\Lambda_*C^{-\theta}V_b)\big]-(1+2b_1)U_b$$
$$\ge\tfrac34\omega_1-2U_b\ge\tfrac{9}{16}C-\tfrac18C=\tfrac{7}{16}C>\tfrac{C}{4}.$$

So J is open in $[\xi_1,\xi_b]$; it is closed because $z_C=u_P\omega_C\ge\frac{7}{16}Cu_P>0$ on bounded intervals and Lemma 2.24 continues S_C; hence $J=[\xi_1,\xi_b]$. This proves (i), (ii) ($\kappa/\omega_C\le\frac{16\kappa}{7C}\le\frac{3\kappa}{C}$) and (iii) (the bound was proved for every $x\le x(\xi)$, $\xi\in J$; $z_C\ge\frac13Cu_P$ since $\frac{7}{16}>\frac13$ on $[\xi_1,\xi_b]$, and on $(-\infty,\xi_1]$ because $\omega_C\ge\omega_1\ge\frac34C$). For (iv): $z_C'/z_C-u_P'/u_P=\omega_C'/\omega_C=-(1+m_C)/(u_P^2\omega_C)$, and with $\omega_C\ge\frac{7}{12}\omega_1$, $m_C\le 1+(2a_1+\frac12\Lambda_*C^{-\theta}V_b)\omega_1$, $\omega_1\ge\frac34C$ this is at most $2\big(\frac{8}{3C\underline{u}}+\frac{2a_1}{\underline{u}}+\frac{\Lambda_*C^{-\theta}V_b}{2\underline{u}}\big)$, which is $\le\varepsilon$ after the choices above (ε' small, then C large). □

Lemma 6.45 (sufficiency)**.** *If $S\in\mathcal G_A$ and $\Pi(S)\in B$, then $\mathcal R_CS$ is global and of type* G *at $x=1$ for all large C.*

Proof. By Lemma 6.40, $u_P>0$ on $\mathbb R$, $r_P:=u_P'/u_P\to\mu_+>0$, and $P_d(1)=2c_2^d>-2$. Choose $\delta>0$ and $x_a\in[0,1)$ with $P_d\ge-2+3\delta$ on $[x_a,1]$, $r_P(\xi_a)>0$ and $(2f_-+1)c_a\le\delta$, $c_a=\frac{1-x_a}{2(1+x_a)}$ ($\delta=\frac16(P_d(1)+2)$ and x_a close to 1 do it). Apply Lemma 6.43 with $\xi_b=\xi_a$ and $\varepsilon=\frac12r_P(\xi_a)$: for C large, $z_C(\xi_a)\ge\frac13Cu_P(\xi_a)$, $r_C(\xi_a):=z_C'/z_C\ge\rho_a:=\frac12r_P(\xi_a)$, $f_C(\xi_a)-f_-\le 3\kappa/C$ and $\Psi_C,|\partial_x\Psi_C|\le\Lambda_*C^{-\theta}$ at x_a. Since $\mathsf L[\Psi_C]=f_C^2-f_-^2$, Lemma 2.18(d) gives for $x\ge x_a$, as long as $f_C-f_-\le 1$ (so $f_C^2-f_-^2\le 2f_-+1$),

$$Q_C(x) = P_d(x) - \Psi_C(x_a)A_1 - \partial_x\Psi_C(x_a)A_2 - \int_{x_a}^{x} K_{\mathsf{L}}(f_C^2 - f_-^2)$$
$$\geq\ -2 + 3\delta - 3\Lambda_* C^{-\theta} - \delta\ \geq\ -2 + \delta$$

(using $A_1 \leq 2$, $0 \leq A_2 \leq 1 - x_a \leq 1$ for $x_a \geq 0$, $\int_{x_a}^{x} K_{\mathsf{L}} \leq c_a$, and C so large that $3\Lambda_* C^{-\theta} \leq \delta$). Then $W_C = 1 + \frac{1}{2}Q_C \geq \frac{\delta}{2}$, and $r_C' = W_C - r_C^2 \geq \frac{\delta}{2} - r_C^2$ keeps $r_C \geq r_* := \min(\rho_a, \frac{1}{2}\sqrt{\delta/2})$: at a first point where $r_C = r_*$ one has $r_C' \geq \frac{\delta}{2} - \frac{\delta}{8} > 0$. Hence $z_C \geq z_C(\xi_a)e^{r_*(\xi-\xi_a)}$ and $f_C(\xi) - f_C(\xi_a) \leq \frac{\kappa}{2r_* z_C(\xi_a)^2} \leq \frac{9\kappa}{2r_* C^2 u_P(\xi_a)^2}$, so $f_C - f_- \leq \frac{3\kappa}{C} + \frac{9\kappa}{2r_* C^2 u_P(\xi_a)^2} \leq \frac{1}{2}$ for large C (the first term is the bound $f_C(\xi_a) - f_- \leq 3\kappa/C$ of Lemma 6.43(ii)) and the continuity argument closes ($\{\xi \geq \xi_a :\ S_C$ exists, $f_C - f_- \leq 1\}$ is closed and, by the bound just proved, open; z_C is bounded below on bounded intervals, so S_C exists on $[\xi_a, \infty)$ by Lemma 2.24; on $(-\infty, \xi_a]$ it exists by Lemma 6.43(i)). Thus S_C is global with $Q_C(1) \geq -2 + \delta > -2$ and $z_C'/z_C \geq r_* > 0$; by Theorem 4.5 and Definition 4.16 it is of type G at $x = 1$. □

THEOREM 6.46 (necessity and sufficiency). *Let $S \in \mathcal{G}_A$.* (a) *If S is global, then $\Pi(S) \in B$.* (b) *If $\Pi(S) \in B$, then $\mathcal{R}_C S$ is global and of type G at $x = 1$ for all large C. Consequently the $\mathcal{R}$-orbit through S contains a global solution if and only if $\Pi(S) \in B$.*

PROOF. (a) is Lemma 6.42, (b) is Lemma 6.45; $\Pi(\mathcal{R}_C S) = \Pi(S)$ gives the last sentence. □

Coordinates on the orbits. By Proposition 6.37, Π is constant on $\mathcal{R}$-orbits; to locate a germ inside its orbit we need one additive coordinate. Let v_y solve $v'' = w_P v$ with $v(0) = u_P'(0)$, $v'(0) = -u_P(0)$, and $a(y) = u_P(0)^2 + u_P'(0)^2 > 0$. Then $\mathrm{Wr}(u_P, v_y) = u_P v' - u_P' v = -a(y)$ at $\xi = 0$, hence constantly, and by Lemma 6.38(c), $v_y = a(y)z_d + bu_P = a(y)e^{-\mu\xi}(1 + o(1))$, $v_y' = -\mu a(y)e^{-\mu\xi}(1 + o(1))$.

Lemma 6.47 (the section and the coordinate $\mathfrak{a}$)**.** (a) *$\Pi : \mathcal{G}_A \to \mathcal{Y}_0$ is continuous.*

(b) *For every $y \in \mathcal{Y}_0$ there is a germ $S^y \in \mathcal{G}_A$ with $\Pi(S^y) = y$, depending continuously on y, with $z_{S^y} = a(y)^{-1}v_y + O(e^{(2+\mu)\xi})$ (so $\mathrm{Wr}(v_y, z_{S^y}) \to 0$).*

(c) *For $S \in \mathcal{G}_A$ the limit $\mathfrak{a}(S) := a(\Pi(S))^{-1} \lim_{\xi\to-\infty} \mathrm{Wr}(v_{\Pi(S)}, z_S)$ exists, $\mathfrak{a}$ is continuous on $\mathcal{G}_A$, $\mathfrak{a}(S^y) = 0$ and $\mathfrak{a}(\mathcal{R}_C S) = \mathfrak{a}(S) + C$.*

(d) *If $\Pi(S) = \Pi(\hat{S})$, then $\hat{S} = \mathcal{R}_C S$ with $C = \mathfrak{a}(\hat{S}) - \mathfrak{a}(S)$.*

PROOF. (a) For a cut $\xi_* \leq 0$ Proposition 6.37 gives $p_2 = t_*^{-2}(Q(\xi_*) - q_0 - p_1 t_* + \Psi_S(\xi_*))$. The frame and $Q(\xi_*)$ are continuous in the germ (Proposition 6.4), and $\Psi_S(\xi_*) = \int_{-1}^{x_*} K_{\mathsf{L}} g_S \mathrm{d}s$ is continuous by dominated convergence: near a given S the uniform envelope of Lemma 6.11(a) with exponent $\mu_{\min} > 1$ gives $|K_{\mathsf{L}}(x_*, s)g_{S'}(s)| \leq (t_*/t_s)^2 C t_s^{\mu_{\min}}$ (integrable since $\mu_{\min} > 1$), and $g_{S'}(s) \to g_S(s)$ pointwise because $f_{S'}(s) \to f_S(s)$ and $f_{S',-} \to f_{S,-}$. (b) *Setting.* Let $z_0 = v_y/a(y)$; then $\mathrm{Wr}(u_P, z_0) = -1$ and $z_0 = z_d + \beta(y)u_P$. Fix a compact neighbourhood $\hat{Y}$ of y^0 in $\mathcal{Y}_0$ on which $\mu \in [\mu_{\min}, \mu_{\max}] \subset (1, \infty)$ and $\pi_y, |\beta|, \kappa, f_-$ are bounded. By Lemma 6.38 there is $\xi_1 \leq 0$, uniform on $\hat{Y}$, with $z_0 \in [\frac{1}{2}, 2]e^{-\mu\xi}$, $|z_0'| \leq 2\mu e^{-\mu\xi}$, $u_P \in [\frac{1}{4\mu}, \frac{1}{\mu}]e^{\mu\xi}$, $|u_P'| \leq e^{\mu\xi}$ on $(-\infty, \xi_1]$. *The fixed-point problem.* We look for $z = z_0 + \delta$ with $z'' = (w_P - \frac{1}{2}\Psi)z$, $\Psi = \mathsf{G}_{\mathsf{L}}[f^2 - f_-^2]$, $f = f_- + \kappa\int_{-\infty}^{\xi} z^{-2}$, i.e. $\delta'' - w_P\delta = H := -\frac{1}{2}\Psi z$. Variation of constants with $\mathrm{Wr}(u_P, z_0) = -1$ gives $\delta = u_P\int_{-\infty}^{\xi} z_0 H - z_0\int_{-\infty}^{\xi} u_P H$ (indeed $\delta'' = w_P\delta + (u_P' z_0 - z_0' u_P)H = w_P\delta + H$). Work in $X = \{\delta :$

$|\delta| + |\delta'| \le Be^{(2+\mu)\xi}$ on $(-\infty, \xi_1]\}$ (the weight $e^{(2+\mu)\xi} = e^{-\mu\xi}E^{1+\mu}$). *Self-map.* If $BE_1^{1+\mu} \le \frac{1}{4}$ and $8\kappa E_1^{\mu}/\mu \le 1$, then $z \in [\frac{1}{4}, 3]e^{-\mu\xi}$, $0 \le f - f_- \le 8\kappa E^{\mu}/\mu \le 1$, $f^2 - f_-^2 \le g_y E^{\mu}$ with $g_y = 8\kappa(2f_- + 1)/\mu$, $\Psi \le \frac{2g_y}{\mu-1}E^{1+\mu}$ (Lemma 2.23, $q = \mu > 1$), $|H| \le h\, e^{(2+\mu)\xi}$ with $h = \frac{3g_y}{\mu-1}$. Since $|z_0 H| \le 2he^{2\xi}$ and $|u_P H| \le \frac{h}{\mu}e^{(2+2\mu)\xi}$, the two integrals give $|\delta_{\text{new}}| + |\delta'_{\text{new}}| \le B_0(y)e^{(2+\mu)\xi}$ with B_0 continuous in y and independent of B. Take $B = \max_{\hat{Y}} B_0$. *Contraction.* For $\delta_1, \delta_2 \in X$ with distance $d = \sup e^{-(2+\mu)\xi}(|\Delta\delta| + |\Delta\delta'|)$ one finds successively $|\Delta z| \le de^{(2+\mu)\xi}$, $|\Delta z^{-2}| \le Cde^{(2+4\mu)\xi}$, $|\Delta f| \le CdE^{1+2\mu}$, $|\Delta\Psi| \le CdE^{2+2\mu}$ (Lemma 2.23, $q = 1+2\mu$), $|\Delta H| \le Cde^{(4+3\mu)\xi}$; the same two integrals give $|\Delta\delta_{\text{new}}| + |\Delta\delta'_{\text{new}}| \le Cde^{(4+3\mu)\xi} = Cd\,E^{1+\mu}e^{(2+\mu)\xi}$, i.e. the Lipschitz constant is $\le C(y)E_1^{1+\mu} \le \frac{1}{2}$ for ξ_1 small, with $C(y)$ continuous. Banach's theorem gives δ. *The germ.* $z = z_0 + \delta$ satisfies $z'' = (1 + \frac{1}{2}Q)z$ with $Q := P_d - \Psi$, $\mathsf{L}[Q] = \lambda - f^2$ ($\mathsf{L}[P_d] = \lambda - f_-^2$, $\mathsf{L}[t^2] = 0$, $\mathsf{L}[\Psi] = f^2 - f_-^2$), $ze^{\mu\xi} \to 1$, $f(-1) = f_-$, $f' = \kappa z^{-2}$, and $Q - P_d = O(E^{1+\mu}) = o(t^2)$. So z, f, Q define a germ of type G with $\Pi = y$ (the exponent from $Q(-1) = q_0$, κ from the normalisation, λ from the first integral, p_2 from the t^2 coefficient); it lies in $\mathcal{G}_A$, and $\operatorname{Wr}(v_y, z) = a\operatorname{Wr}(z_0, z_0 + \delta) = a\operatorname{Wr}(z_0, \delta) = O(e^{-\mu\xi}e^{(2+\mu)\xi}) \to 0$. *Continuity in y.* Let $y_n \to y$ in $\hat{Y}$, and write T_y for the self-map above (with data y), δ^y for its fixed point. Put $\gamma = 2 + \mu_{\min} - \epsilon_0$ with $0 < \epsilon_0 \le \mu_{\min} - 1$ and $\|\delta\|_\gamma = \sup_{\xi \le \xi_1} e^{-\gamma\xi}(|\delta| + |\delta'|)$. Since $e^{(2+\mu)\xi} \le e^{\gamma\xi}$ for $\xi \le 0$, all the $X_{y'}$, $y' \in \hat{Y}$, lie in $X' = \{\|\delta\|_\gamma \le B\}$, which is complete. *(1) Reduction.* Repeating the contraction step with weight γ in place of $2 + \mu$ (the same chain: $|\Delta z| \le de^{\gamma\xi}$, $|\Delta z^{-2}| \le Cde^{(\gamma+3\mu)\xi}$ because $z \ge \frac{1}{4}e^{-\mu\xi}$, $|\Delta f| \le Cde^{(\gamma+3\mu)\xi}$, $|\Delta\Psi| \le Cde^{(\gamma+3\mu+2)\xi}$, hence $|\Delta H| \le Cde^{(\gamma+2+2\mu)\xi}$ using $\Psi \le CE^{1+\mu}$ and $|\Delta z| \le de^{\gamma\xi}$, and finally $|\Delta\delta_{\text{new}}| + |\Delta\delta'_{\text{new}}| \le Cde^{(\gamma+2+2\mu)\xi} \le Cd\,E_1^{1+\mu}e^{\gamma\xi}$), every $T_{y'}$ is a $\frac{1}{2}$-contraction of X' for $\|\cdot\|_\gamma$, after shrinking ξ_1 uniformly on $\hat{Y}$ (the same shrinking keeps $z = z_0 + \delta \ge \frac{1}{4}e^{-\mu\xi}$ on X'). Hence $\|\delta^{y_n} - \delta^y\|_\gamma = \|T_{y_n}\delta^{y_n} - T_y\delta^y\|_\gamma \le \frac{1}{2}\|\delta^{y_n} - \delta^y\|_\gamma + \|T_{y_n}\delta^y - T_y\delta^y\|_\gamma$, so $\|\delta^{y_n} - \delta^y\|_\gamma \le 2\|T_{y_n}\delta^y - T_y\delta^y\|_\gamma$. *(2) Pointwise convergence.* Fix $\delta = \delta^y$ and $\xi \le \xi_1$. By Lemma 6.38(b), $u_P^{y_n}, z_0^{y_n}$ and their derivatives converge to those of y locally uniformly. With $z^{(n)} = z_0^{y_n} + \delta$ we have $f^{(n)} = f_-^{(n)} + \kappa_n \int_{-\infty}^{\xi}(z^{(n)})^{-2} \to f$ pointwise by dominated convergence (majorant $16e^{2\mu_{\min}\eta}$, from $z^{(n)} \ge \frac{1}{4}e^{-\mu_n\eta}$), and then $\Psi^{(n)}(\eta) = \int_{-1}^{x} K_{\mathsf{L}}\,((f^{(n)})^2 - (f_-^{(n)})^2)\mathrm{d}s \to \Psi(\eta)$ by dominated convergence again (majorant $|K_{\mathsf{L}}(x,s)|g_y t_s^{\mu_{\min}}$, integrable as $\mu_{\min} > 1$). So the integrands $z_0^{y_n}H^{(n)}$, $u_P^{y_n}H^{(n)}$ converge pointwise, and they are dominated, uniformly in n, by $Ce^{2\eta}$ and $Ce^{(2+2\mu_{\min})\eta}$ respectively (the bounds $|z_0 H| \le 2he^{2\eta}$, $|u_P H| \le \frac{h}{\mu}e^{(2+2\mu)\eta}$ of the self-map step, with h, μ bounded on $\hat{Y}$). Hence $(T_{y_n}\delta)(\xi) \to (T_y\delta)(\xi)$ together with the derivative, for every $\xi \le \xi_1$. *(3) Tails.* By the self-map step, $|T_{y'}\delta| + |(T_{y'}\delta)'| \le B_0 e^{(2+\mu')\xi} \le B_0 e^{(2+\mu_{\min})\xi}$ for all $y' \in \hat{Y}$; so for $\xi \le \xi_T$ the weighted quantity $e^{-\gamma\xi}(\cdots)$ is at most $2B_0 e^{\epsilon_0\xi_T}$, uniformly in n. *(4) The middle.* On the compact interval $[\xi_T, \xi_1]$ the functions $T_{y_n}\delta$ are C^1 with derivatives bounded uniformly (by the same domination, since $(T\delta)'' = w_P T\delta + H$ with w_P, H bounded there), so the pointwise convergence of (2) is uniform by Arzelà–Ascoli (every subsequence has a uniformly convergent subsequence, whose limit is the pointwise limit); the same holds for the derivative (the second derivative is bounded likewise). Given ε, choose ξ_T by (3) and then n by (4): $\|T_{y_n}\delta^y - T_y\delta^y\|_\gamma \to 0$, hence $\delta^{y_n} \to \delta^y$ with derivatives, uniformly on $(-\infty, \xi_1]$ up to the factor $e^{\gamma\xi}$, by (1). *(5) The state*

at ξ_1. So $z^{y_n}(\xi_1), z^{y_n\prime}(\xi_1)$ converge; $f^{y_n}(\xi_1) = f_- + \kappa_n \int_{-\infty}^{\xi_1} (z^{y_n})^{-2}$ converges by dominated convergence (majorant $16e^{2\mu_{\min}\eta}$); $Q = P_d - \Psi$ converges at x_1 and $\partial_x Q$ converges, because $\partial_x K_{\mathsf{L}}(x,s) \le 5t_s^{-2}(1-x)^{-2}$ and $g_n \le Ct_s^{\mu_{\min}}$ give the majorant $Ct_s^{\mu_{\min}-2}$, integrable, and $g_n \to g$ pointwise; Q_{xx} then follows from the first integral $\mathsf{L}[Q] = \lambda - f^2$. Thus the states of S^{y_n} at ξ_1 converge to that of S^y, i.e. $S^{y_n} \to S^y$ (Lemma 6.3(b)). (c) $\mathrm{Wr}(v,z)' = vz(W - w_P) = -\frac{1}{2}\Psi_S vz$ with $|\Psi_S vz| \le CE^{1+\mu}e^{-2\mu\xi} = CE$, which is integrable, so the limit exists: $\lim \mathrm{Wr}(v,z) = \mathrm{Wr}(v,z)(\xi_*) + \frac{1}{2}\int_{-\infty}^{\xi_*} \Psi_S vz$. If $S_n \to S$ then $\Pi(S_n) \to \Pi(S)$ by (a), $v_{\Pi(S_n)} \to v_{\Pi(S)}$ pointwise with uniform bounds $|v| \le Ce^{-\mu\xi}$, $z_{S_n} \to z_S$ with uniform envelopes, $\Psi_{S_n} \le kE^{1+\mu_{\min}}$ and $\Psi_{S_n} \to \Psi_S$ pointwise; explicitly, the integrand $\Psi_{S_n} v_n z_n$ is bounded by $kE^{1+\mu_n} \cdot Ce^{-\mu_n\xi} \cdot m_+ Ce^{-\mu_n\xi} = C'e^{2\xi}$ (with μ_n the exponent of S_n, and $|v| \le Ce^{-\mu\xi}$, $z \le m_+ Z_\mu \le Ce^{-\mu\xi}$ by the envelope), an integrable function independent of n, and it converges pointwise; dominated convergence gives continuity of the integral, and $a(\Pi(S_n)) \to a(\Pi(S)) > 0$. For S^y, $\mathrm{Wr}(v_y, z) \to 0$ by (b). For $\mathcal{R}_C S$ (Π unchanged, $z_C - z = zu$) we have $zu = \frac{C}{2\mu}e^{\mu\xi}(1 + o(1))$ (Theorem 6.22(a)) and $(zu)' = z'u + zu' = -\frac{C}{2}e^{\mu\xi} + Ce^{\mu\xi} + o(e^{\mu\xi}) = \frac{C}{2}e^{\mu\xi}(1 + o(1))$ ($z^2u' \to C$); with $v = ae^{-\mu\xi}(1+o(1))$, $v' = -\mu a e^{-\mu\xi}(1+o(1))$ one gets $\mathrm{Wr}(v, zu) = v(zu)' - v'(zu) \to a\frac{C}{2} + \mu a\frac{C}{2\mu} = aC$. By linearity of $\mathrm{Wr}(v,\cdot)$, $\mathfrak{a}(\mathcal{R}_C S) = \mathfrak{a}(S) + C$. (d) Let $(\hat z, \hat f, \hat Q)$ belong to $\hat S$. Then $\mathrm{Wr}(z,\hat z)' = \frac{1}{2}z\hat z(\hat Q - Q)$ with $\hat Q - Q = \Psi_S - \Psi_{\hat S} = O(E^{1+\mu})$ (same P_d), so $z\hat z(\hat Q - Q) = O(E)$ is integrable and $C := \lim \mathrm{Wr}(z,\hat z)$ exists. With $S_1 = \mathcal{R}_C S$ the three-function identity (Lemma 6.53) gives $\mathrm{Wr}(z_1, \hat z) \to -C + C = 0$; moreover $\hat z / z_1 \to 1$ (same normalisation) and $\hat Q - Q_1 = \Psi_{S_1} - \Psi_{\hat S} = O(E^{1+\mu}) = o(E^2)$ since $\mu > 1$. So $\hat S \in C_A(0)(S_1) = \{S_1\}$ (Theorem 6.22(b)). Finally $C = \mathfrak{a}(\hat S) - \mathfrak{a}(S)$ by (c). □

Remark 6.48 (the Wronskian identity used above)**.** Lemma 6.53 is proved in Section 6.8; it is the elementary identity $\mathrm{Wr}(z_1, z_3) = \frac{z_3}{z_2}\mathrm{Wr}(z_1, z_2) + \frac{z_1}{z_2}\mathrm{Wr}(z_2, z_3)$.

Proposition 6.49 (the lower bound; coordinates; the recessive chart)**.** (i) Lower bound. *If* $S \in \mathcal{G}$ *has* $\mu > \frac{1}{2}$ *and* $\ell_\infty := \int_{\mathbb{R}} z^{-2} < \infty$ *(e.g.* S *of type* G *at* $x = 1$*), then* $\mathsf{A}_C S \notin \mathcal{S}$ *for* $C < -1/\ell_\infty$ *(*A_C *the recessive shift;* $\mathcal{G}$ *the germ space,* $\mathcal{S}$ *the global solutions).* (ii) Coordinates, $\mu > 1$. *Let* $\Pi : \mathcal{G}_{\mathsf{A}} \to \mathcal{Y}_0 = (1,\infty) \times \mathbb{R} \times [0,\infty) \times (0,\infty) \times \mathbb{R}$ *be the frame map. (a)* Π *is continuous. (b) For* $y \in \mathcal{Y}_0$ *there is a germ* $S^y \in \mathcal{G}_{\mathsf{A}}$*, depending continuously on* y*, with* $\Pi(S^y) = y$ *(*$v_y = a(y)e^{-\mu\xi}(1 + o(1))$*,* $a(y) = u_P(0)^2 + u_P'(0)^2$*). (c)* $\mathfrak{a}(S) := a(\Pi(S))^{-1} \lim_{\xi\to-\infty} \mathrm{Wr}(v_{\Pi(S)}, z_S)$ *exists and is continuous on* $\mathcal{G}_{\mathsf{A}}$*,* $\mathfrak{a}(S^y) = 0$*,* $\mathfrak{a}(\mathsf{A}_C S) = \mathfrak{a}(S) + C$*. (d) If* $\Pi(S) = \Pi(\hat S)$ *then* $\hat S = \mathsf{A}_C S$ *with* $C = \mathfrak{a}(\hat S) - \mathfrak{a}(S)$. (iii) Homeomorphism. *Hence* $\Phi = (\Pi, \mathfrak{a})$ *is a homeomorphism* $\mathcal{G}_{\mathsf{A}} \to \mathcal{Y}_0 \times \mathbb{R}$ *with inverse* $(y, s) \mapsto \mathsf{A}_s S^y$.

Proof. *The lower bound.* Let $C < -1/\ell_\infty$ and $l(\xi) = \int_{-\infty}^{\xi} z^{-2} \in [0, \ell_\infty)$, increasing continuously to ℓ_∞, so $1 + Cl$ decreases from 1 to $1 + C\ell_\infty < 0$ and vanishes at some ξ_0. If $\tilde S = \mathcal{R}_C S$ exists on $(-\infty, \xi]$, then by Corollary 6.24 (the case $C < 0$ of Proposition 6.23) $u = \tilde z/z - 1$ satisfies $\varepsilon := \frac{1}{2}(\tilde Q - Q) \le 0$, and $z^2u' = C + \int_{-\infty}^{\xi} z^2\varepsilon(1+u) \le C$ as long as $1 + u > 0$ (Corollary 3.6); so $u \le Cl$, i.e. $\tilde z/z = 1 + u \le 1 + Cl$. If $\tilde S$ were global, $\tilde z > 0$ on $\mathbb{R}$ would give $0 < 1 + u(\xi_0) \le 1 + Cl(\xi_0) = 0$, a contradiction. For S of type G at $x = 1$, z grows exponentially at both ends, so $\ell_\infty < \infty$. *The coordinates* (a)–(d) are Lemma 6.47.

The homeomorphism. Φ is continuous by (a),(c); injective by (d) (if $\Phi(S)=\Phi(\hat S)$ then $\hat S=\mathcal{R}_0 S$); surjective with the stated inverse because $\Pi(\mathcal{R}_s S^y)=y$ and $\mathfrak{a}(\mathcal{R}_s S^y)=s$ ($\mathcal{R}_s$ preserves frame and type, so $\mathcal{R}_s S^y\in\mathcal{G}_A$); the inverse $(y,s)\mapsto\mathcal{R}_s S^y$ is continuous by Lemma 6.47(b) and Lemma 6.28. □

THEOREM 6.50 (the recessive chart, $\mu>1$ including $\mu\geq 2$). (a) $\Phi=(\Pi,\mathfrak{a}):\mathcal{G}_A\to\mathcal{Y}_0\times\mathbb{R}$ *is a homeomorphism and* $\Phi(\mathcal{R}_C S)=\Phi(S)+(0,C)$.

(b) *There is a continuous function* $\Theta^-:B\to\mathbb{R}$ *with* $\Phi(\mathcal{S}_A)=\{(y,s):\ y\in B,\ s\geq\Theta^-(y)\}$, *and* $\Theta^-(y)\to+\infty$ *as* $y\to\partial B$ *within* $\mathcal{Y}_0$.

(c) $S\mapsto(\Pi(S),\ \mathfrak{a}(S)-\Theta^-(\Pi(S)))$ *is a homeomorphism* $\mathcal{S}_A\to B\times[0,\infty)$. *Its zero set* $\{\mathfrak{a}=\Theta^-\circ\Pi\}$ *is* $\Gamma_{\mathsf{N}}\cap\mathcal{G}_A$ (*global solutions not of type* G *at* $x=1$), *homeomorphic to* B; *above it all solutions are of type* G *at* $x=1$.

PROOF. (a) is Proposition 6.49. (b) Fix y. If $y\notin B$, the orbit $\{\mathcal{R}_s S^y\}=\Phi^{-1}(\{y\}\times\mathbb{R})$ contains no global solution (Theorem 6.46(a)). Let $y\in B$ and $I_y=\{s:\Phi^{-1}(y,s)\in\mathcal{S}\}$. It is non-empty (sufficiency: $s\geq C_0$ for some C_0), *closed* (Lemma 6.25 and continuity of Φ^{-1}), an *up-set* (a positive recessive shift of a global germ is global, Proposition 6.23), and *bounded below*: if $s_1\in I_y$ and $\Phi^{-1}(y,s_1+C_1)=\mathcal{R}_{C_1}\Phi^{-1}(y,s_1)\in\mathcal{S}_{\mathsf{G}}$ for some $C_1>0$ (Lemma 6.26), then $\ell_\infty<\infty$ for this germ and Proposition 6.49 excludes $s<s_1+C_1-1/\ell_\infty$. So $I_y=[\Theta^-(y),\infty)$. *Upper semicontinuity.* Let $y_0\in B$, $\varepsilon>0$. $\Phi^{-1}(y_0,\Theta^-(y_0)+\varepsilon)=\mathcal{R}_\varepsilon\Phi^{-1}(y_0,\Theta^-(y_0))\in\mathcal{S}_{\mathsf{G}}$ (Lemma 6.26), which is open in $\mathcal{G}_{\mathsf{N}_0}\supset\mathcal{G}_A$ (Lemma 6.25); by (a), $\Phi^{-1}(y,\Theta^-(y_0)+\varepsilon)\in\mathcal{S}$ for y near y_0. So such y lie in B (necessity) and $\Theta^-(y)\leq\Theta^-(y_0)+\varepsilon$. *Lower semicontinuity.* If $y_n\to y_0\in B$ and $\Theta^-(y_n)\to L<\Theta^-(y_0)$ ($L=-\infty$ allowed), fix $L'\in(L,\Theta^-(y_0))$. For large n, $\Theta^-(y_n)<L'$, so $\Phi^{-1}(y_n,L')\in\mathcal{S}$ (up-set), and $\Phi^{-1}(y_n,L')\to\Phi^{-1}(y_0,L')$ is global by closedness, contradicting $L'<\Theta^-(y_0)$. *Properness.* If $y_n\to y_0\in\partial B$ ($y_0\notin B$, B being open) with $\Theta^-(y_n)\leq M$, then $\Phi^{-1}(y_n,M)\in\mathcal{S}$ converge to $\Phi^{-1}(y_0,M)$, which is global, contradicting necessity. (c) The map is well defined (necessity and the half-line), continuous, with the continuous inverse $(y,r)\mapsto\Phi^{-1}(y,\Theta^-(y)+r)$. If $\mathfrak{a}(S)>\Theta^-(\Pi(S))$, S is a positive recessive shift of the global germ $\Phi^{-1}(\Pi(S),\Theta^-(\Pi(S)))$, hence of type G at $x=1$ (Lemma 6.26); if $\mathfrak{a}(S)=\Theta^-(\Pi(S))$ and S were of type G at $x=1$, openness of $\mathcal{S}_{\mathsf{G}}$ would make $\mathcal{R}_{-\varepsilon}S$ global for small ε (Lemma 6.28), contradicting minimality. So the zero set is the graph of Θ^- over B, equal to the set of global germs not of type G at $x=1$. □

Remark 6.51 ($\mu=2$ is not special here; $B\to\Gamma_T$ on $1<\mu<2$)**.** Nothing in Theorem 6.50 changes at $\mu=2$: p_2, B, u_P, v_y and all constants depend continuously on $\mu\in(1,\infty)$, with no factor $(\mu-2)^{-1}$; only the t^2-shift degenerates there (Toy example 6.15). On $1<\mu<2$ both charts describe the same set $\mathsf{N}_0\cap\{1<\mu<2\}$ and both edges are the intrinsically defined set Γ_{N} of global solutions not of type G at $x=1$; hence $B\cap\{\mu<2\}\cong\Gamma_T\cap\{\mu>1\}$ (by $y\mapsto\Phi^{-1}(y,\Theta^-(y))$). The finer relation between the two actions is Theorem 6.54.

6.8. Commuting shifts, the group $\mathbb{R}^2$, and the inverse relation

In words. *On* $\frac12<\mu<2$ *both shifts exist. Each is the unique germ with the same frame that differs from* S *by exactly one number: the* t^2*-content for* T_c, *the Wronskian for* $\mathcal{R}_C$. *A germ that differs from* S *by* both *numbers is therefore unique, and since* $T_c\mathcal{R}_C S$ *and* $\mathcal{R}_C T_c S$ *both have both numbers, they coincide. The consequence is a*

simple dictionary between the two charts: "how far below the edge, in the recessive direction, is T_cS?" and "how large must the t^2-shift be to survive a recessive shift by $-d$?" are the same question, answered by mutually inverse functions.

Toy example 6.52 (two corrections that do not interact)**.** At constant potential μ^2 with $z = e^{-\mu\xi}$, the recessive shift is $\tilde{z} = z(1 + Cl)$, $l = \frac{e^{2\mu\xi}}{2\mu}$, and the potential change $\varepsilon = \frac{1}{2}ct^2 \approx 2cE^2$ of the t^2-shift adds a correction of relative size $\approx \frac{2c}{16-8\mu}E^2$ to z (Toy example 6.15). The first correction is of relative order E^μ, the second of order E^2; the effect of ε on the first is of order $E^{\mu+2}$, smaller than both. So up to $o(E^2)$, which is the precision to which the shifts are defined (Definition 6.16), the two corrections just add, in either order. Theorem 6.54 makes this exact.

Lemma 6.53 (reduction of order; the three-function identity)**.** (a) *If $z > 0$, $z'' = Wz$ and $\tilde{z} = z(1 + u)$ satisfies $\tilde{z}'' = (W + \varepsilon)\tilde{z}$, then $(z^2u')' = z^2\varepsilon(1 + u)$ and $\mathrm{Wr}(z, \tilde{z}) = z^2u'$.*

(b) *For positive z_1, z_2, z_3: $\mathrm{Wr}(z_1, z_3) = \frac{z_3}{z_2}\mathrm{Wr}(z_1, z_2) + \frac{z_1}{z_2}\mathrm{Wr}(z_2, z_3)$. In particular, if $z_3/z_2 \to 1$, $z_1/z_2 \to 1$ and two of the three Wronskians have limits, so has the third, and $\lim \mathrm{Wr}(z_1, z_3) = \lim \mathrm{Wr}(z_1, z_2) + \lim \mathrm{Wr}(z_2, z_3)$.*

Proof. (a) $\tilde{z}'' = Wz(1+u)+2z'u'+zu''$ and $= (W+\varepsilon)z(1+u)$ give $zu''+2z'u' = \varepsilon z(1 + u)$; multiply by z. Also $\mathrm{Wr}(z, \tilde{z}) = z(z'(1 + u) + zu') - z'z(1 + u) = z^2u'$. (b) $\frac{z_3}{z_2}(z_1z_2' - z_1'z_2) + \frac{z_1}{z_2}(z_2z_3' - z_2'z_3) = z_1z_3' - z_1'z_3$; the terms with z_2' cancel. □

Theorem 6.54 (the shifts commute). *Let $S \in \mathcal{G}_{\mathsf{N}_0}$ be of type G with $\frac{1}{2} < \mu < 2$. For all $c, C \in \mathbb{R}$, $T_c\mathcal{R}_CS = \mathcal{R}_CT_cS$. More precisely, the* joint class

$$C_{TA}(c, C)(S) = \{\hat{S} \text{ with the frame of } S : \ \mathrm{Wr}(z, \hat{z}) \to C, \ \hat{Q} - Q - ct^2 = o(E^2)\}$$

contains exactly one germ, and it is $T_c\mathcal{R}_CS = \mathcal{R}_CT_cS$.

Proof. Let $S_1 = \mathcal{R}_CS$ (it exists since $\mu > \frac{1}{2}$ and has the frame, type and exponent of S, Theorem 6.22), $S_2 = T_cS_1$ (it exists since $\mu < 2$), $S_3 = T_cS$, $S_4 = \mathcal{R}_CS_3$. All have the frame of S; in particular all ratios of the corresponding z's tend to 1. *(1) $S_2 \in C_{TA}(c, C)(S)$.* By Theorem 6.19(b), $\mathrm{Wr}(z_1, z_2) \to 0$ and $Q_2 - Q_1 - ct^2 = o(E^2)$; by Theorem 6.22(b), $\mathrm{Wr}(z, z_1) \to C$ and $Q_1 - Q = o(E^2)$. Lemma 6.53(b) gives $\mathrm{Wr}(z, z_2) \to C + 0$, and $Q_2 - Q - ct^2 = (Q_2 - Q_1 - ct^2) + (Q_1 - Q) = o(E^2)$. $S_4 \in C_{TA}(c, C)(S)$ likewise: $\mathrm{Wr}(z, z_3) \to 0$, $\mathrm{Wr}(z_3, z_4) \to C$, $Q_3 - Q - ct^2 = o(E^2)$, $Q_4 - Q_3 = o(E^2)$. *(2) Uniqueness.* Let $\hat{S} \in C_{TA}(c, C)(S)$. Then $\mathrm{Wr}(z_1, \hat{z}) \to -C + C = 0$ (Lemma 6.53(b) with $z_1, z, \hat{z}$) and $\hat{Q} - Q_1 - ct^2 = (\hat{Q} - Q - ct^2) - (Q_1 - Q) = o(E^2)$, so $\hat{S} \in C_T(c)(S_1) = \{T_cS_1\}$ (Theorem 6.19(b)): $\hat{S} = S_2$. Also $\mathrm{Wr}(z_3, \hat{z}) \to -0+C$ and $\hat{Q}-Q_3 = (\hat{Q}-Q-ct^2)-(Q_3-Q-ct^2) = o(E^2)$, so $\hat{S} \in C_A(C)(S_3) = \{\mathcal{R}_CS_3\}$ (Theorem 6.22(b)): $\hat{S} = S_4$. Combined with (1), $S_2 = S_4$ is the unique element of the joint class. □

Corollary 6.55 (the action of $\mathbb{R}^2$)**.** *On $\mathcal{G}_{\frac{1}{2},2} := \{S \in \mathcal{G}_{\mathsf{N}_0}$ of type G, $\frac{1}{2} < \mu < 2\}$ the rule $(c, C) \cdot S = T_c\mathcal{R}_CS$ is a continuous, free action of $\mathbb{R}^2$ which preserves the frame $(\mu, \lambda, f_-, \kappa)$ and, where Π is defined ($\mu > 1$), acts on Π by $p_2 \mapsto p_2 + c$. On $1 < \mu < 2$ it is simply transitive on each fibre of the frame map $(\mu, \lambda, f_-, \kappa)$.*

Proof. Action: Theorem 6.19(c), Theorem 6.22(c) and Theorem 6.54; continuity: Lemma 6.28. Free: if $(c, C) \cdot S = S$ then $p_2 + c = p_2$ (for $\mu > 1$; for $\mu \le 1$ use $\mathrm{Wr}(z, \tilde{z}) = C$ and $\tilde{Q} - Q - ct^2 = o(E^2)$, which for $\tilde{S} = S$ gives $C = 0$ and then

$ct^2 = o(E^2)$, i.e. $c = 0$). Transitive on $1 < \mu < 2$: two germs $S, \hat{S}$ with the same frame have $\Pi(T_c S) = \Pi(\hat{S})$ for $c = p_2(\hat{S}) - p_2(S)$ (Proposition 6.37), so $\hat{S} = \mathcal{R}_C T_c S$ by Lemma 6.47(d). □

For $S \in \mathcal{G}_{\frac{1}{2},2}$ the function $d \mapsto \alpha_-(\mathcal{R}_{-d}S)$ is defined (all $\mathcal{R}_{-d}S$ lie in the domain of T) and continuous on $\mathbb{R}$, by Lemma 6.28 and Theorem 6.32(d). Let $\ell_\infty(S) = \int_{\mathbb{R}} z^{-2} \mathrm{d}\xi \in (0, \infty]$ ($< \infty$ if S is of type G at $x = 1$, Proposition 6.49).

Definition 6.56 (depth)**.** For $S \in \mathcal{G}_{\frac{1}{2},2}$ and $c \geq \alpha_-(S)$ let $\Lambda(S, c) = \sup\{d \geq 0 : \mathcal{R}_{-d} T_c S \in \mathcal{S}\}$ be the *depth* of $T_c S$ below the edge in the recessive direction.

Proposition 6.57 (inverse relation)**.** *Let $S \in \mathcal{G}_{\frac{1}{2},2}$.*

(i) *For $c \in \mathbb{R}$, $d \geq 0$: $\mathcal{R}_{-d} T_c S \in \mathcal{S} \iff c \geq \alpha_-(\mathcal{R}_{-d}S)$.*

(ii) *$d \mapsto \alpha_-(\mathcal{R}_{-d}S)$ is continuous and strictly increasing on $[0, \infty)$, with limit $+\infty$ as $d \to \infty$.*

(iii) *$c \mapsto \Lambda(S, c)$ is finite on $[\alpha_-(S), \infty)$ and is the inverse function of $d \mapsto \alpha_-(\mathcal{R}_{-d}S)$: $\alpha_-(\mathcal{R}_{-\Lambda(S,c)}S) = c$. In particular $\Lambda(S, \cdot)$ is continuous and strictly increasing.*

(iv) *$\Lambda(S, c) \leq 1/\ell_\infty(T_c S)$ for $c > \alpha_-(S)$.*

Proof. (i) By Theorem 6.54, $\mathcal{R}_{-d} T_c S = T_c \mathcal{R}_{-d} S$, which is global iff $c \geq \alpha_-(\mathcal{R}_{-d}S)$ (Theorem 6.32(b)). (ii) Continuity was noted above. *Monotone:* if $0 \leq d' < d$ and $c \geq \alpha_-(\mathcal{R}_{-d}S)$, then $\mathcal{R}_{-d'} T_c S = \mathcal{R}_{d-d'} \mathcal{R}_{-d} T_c S$ is a positive recessive shift of a global germ, hence global (Proposition 6.23); so $c \geq \alpha_-(\mathcal{R}_{-d'}S)$, i.e. $\alpha_-(\mathcal{R}_{-d'}S) \leq \alpha_-(\mathcal{R}_{-d}S)$. *Strict:* with $c = \alpha_-(\mathcal{R}_{-d}S)$, $T_c \mathcal{R}_{-d} S \in \Gamma_T$ is global and not of type G at $x = 1$ (Theorem 6.32(c)); then $T_c \mathcal{R}_{-d'} S = \mathcal{R}_{d-d'} T_c \mathcal{R}_{-d} S$ is of type G at $x = 1$ (Lemma 6.26, $\mu > \frac{1}{2}$), hence $\alpha_-(\mathcal{R}_{-d'}S) < c$ by Theorem 6.32(c). *Unbounded:* (this uses no hypothesis on S) let $c > \alpha_-(S)$, which exists since α_- is finite; $T_c S \in \mathcal{S}_{\mathsf{G}}$ has $\ell_\infty < \infty$, and by Proposition 6.49, $\mathcal{R}_{-d} T_c S \notin \mathcal{S}$ for $d > 1/\ell_\infty(T_c S)$; by (i), $\alpha_-(\mathcal{R}_{-d}S) > c$ for these d. (iii)–(iv) By (i), $\{d \geq 0 : \mathcal{R}_{-d} T_c S \in \mathcal{S}\} = \{d \geq 0 : \alpha_-(\mathcal{R}_{-d}S) \leq c\}$; by (ii) it is a closed interval $[0, \Lambda]$ containing 0 (for $c \geq \alpha_-(S)$), bounded by the bound just proved, which is (iv); at its right end continuity and strict monotonicity give $\alpha_-(\mathcal{R}_{-\Lambda}S) = c$ (if it were $< c$, slightly larger d would still satisfy the inequality). So $\Lambda(S, \cdot)$ inverts $d \mapsto \alpha_-(\mathcal{R}_{-d}S)$, which is a continuous strictly increasing bijection $[0, \infty) \to [\alpha_-(S), \infty)$. □

Remark 6.58 (the closed form for $\mu > 1$; the two charts agree)**.** For $1 < \mu < 2$: $\Pi(T_c S) = \Pi(S) + c\, e_{p_2}$, $\mathfrak{a}(\mathcal{R}_C S) = \mathfrak{a}(S) + C$, and $T_c S$ is global iff $\mathfrak{a}(T_c S) \geq \Theta^-(\Pi(S) + c\, e_{p_2})$ (Theorem 6.50). By Proposition 6.57(i), $\Lambda(S, c) = \mathfrak{a}(T_c S) - \Theta^-(\Pi(S) + c\, e_{p_2})$, the recessive depth of $T_c S$ in the coordinates of Lemma 6.47; and $c \mapsto \Lambda(S, c)$ inverts $d \mapsto \alpha_-(\mathcal{R}_{-d}S)$. Thus the two charts $\mathsf{N}_0 \cap \{\mu < 2\} \cong \Gamma_T \times [0, \infty)$ (Theorem 6.34) and $\mathsf{N}_0 \cap \{\mu > 1\} \cong B \times [0, \infty)$ (Theorem 6.50) are related, on $1 < \mu < 2$, by the explicit $\mathbb{R}^2$-action: the T-coordinate of a global solution is read off from the $\mathcal{R}$-chart as $p_2 - p_2^{\mathrm{edge}}$, and vice versa. A quantitative form of the relation, $\alpha_-(\mathcal{R}_C S)/|C|^{2/\mu} \to \gamma_*(\mu)$ as $C \to -\infty$, is proved in Chapter 12; the two exactly solvable cases are in Example 6.64.

6.9. Gluing with class Z, the structure theorem for class N, and the atlas

In words. *Class* N *has swirl of one sign; class* Z *has swirl with one zero. At the south pole both appear in the same space of germs, and the single number $f_- = f(-1)$*

decides between them: $f_- \geq 0$ is class N_0, $f_- < 0 < f(1)$ is class Z. The zero of the swirl runs into the pole as $f_- \to 0^-$. This closes the description of the germ space; the section then collects the chapter into one theorem (the structure of class N), works out the one exactly solvable example (a rest state, with the Bessel model as its asymptotic cousin), and reports the numerical atlas.

Proposition 6.59 (class N_0 against class Z at the south pole)**.** *Let S be a global solution of type G at $x = -1$ with $\kappa > 0$ (pole normalisation), and $f_+ := f(1) \in (f_-, \infty]$.*

(a) *$S \in \mathsf{N}_0$ iff $f_- \geq 0$; $S \in \mathsf{Z}$ iff $f_- < 0 < f_+$; $S \in \sigma_1\sigma_2(\mathsf{N}_0)$ iff $f_+ \leq 0$. The set $\{f_+ > 0\}$ is open, since it is the condition $f(x_0) > 0$ at one interior point x_0 (and f is continuous in the state there); $\{f_- < 0\}$ and $\{f_- \geq 0\}$ are separated by the continuous function f_-.*

(b) *If $S \in \mathsf{Z}$ with zero $x_0 = \tanh\xi_0$ of the swirl, then $|f_-| = \kappa \int_{-\infty}^{\xi_0} z^{-2}$. If $m_- e^{-\mu\xi} \leq z \leq m_+ e^{-\mu\xi}$ on $(-\infty, \xi_1]$ and $|f_-| < \kappa m_+^{-2} e^{2\mu\xi_1}/(2\mu)$, then $\xi_0 \leq \xi_1$ and $E_0 := e^{2\xi_0}$ satisfies $E_0^\mu \in [2\mu m_-^2,\, 2\mu m_+^2] \cdot |f_-|/\kappa$. In particular $1 + x_0 \to 0$ as $f_- \to 0^-$, locally uniformly in the remaining coordinates of the germ chart, and $1 + x_0 \sim 2(2\mu|f_-|/\kappa)^{1/\mu}$ when $z \approx e^{-\mu\xi}$.*

(c) *The slope of the swirl at its zero, $f_x(x_0) = \kappa \cosh^2\xi_0 / z(\xi_0)^2 \asymp \kappa E_0^{\mu-1}$, tends to ∞ as $f_- \to 0^-$ if $\mu < 1$. Hence the natural coordinates of class Z (the slope at the zero, Chapter 5) do not extend continuously to $f_- = 0$ when $\mu < 1$.*

Proof. (a) f is strictly increasing from f_- to f_+. If $f_- \geq 0$ then $f > 0$ on $(-1, 1)$ and $S \in \mathsf{N}_0$; if $f_- < 0 < f_+$ then f has exactly one zero, so $S \in \mathsf{Z}$; if $f_+ \leq 0$ then $f < 0$ and $\sigma_1\sigma_2 S$ (Lemma 5.1) has $\hat{f}(x) = -f(-x) > 0$, $\hat{f}_x(x) = f_x(-x) > 0$, so lies in N_0. The function f_- is a chart coordinate at $x = -1$ and f_+ one at $x = 1$ where it is finite (Corollary 4.42), hence they are continuous. (b) $0 = f(\xi_0) = f_- + \kappa \int_{-\infty}^{\xi_0} z^{-2}$ and $\int_{-\infty}^{\xi} z^{-2} \in [m_+^{-2}, m_-^{-2}] \frac{e^{2\mu\xi}}{2\mu}$ for $\xi \leq \xi_1$. If $|f_-| < \kappa m_+^{-2} e^{2\mu\xi_1}/(2\mu)$ then $\xi_0 \leq \xi_1$ (otherwise $|f_-| \geq \kappa \int_{-\infty}^{\xi_1} z^{-2} \geq \kappa m_+^{-2} e^{2\mu\xi_1}/(2\mu)$), and then $|f_-|/\kappa \in [m_+^{-2}, m_-^{-2}] E_0^\mu/(2\mu)$, which is the claim. Since $1 + x_0 = t_0 \leq 2E_0$ the pole is approached; for $z \approx e^{-\mu\xi}$, $E_0^\mu \approx 2\mu|f_-|/\kappa$ and $t_0 \approx 2E_0$. (c) $f_x = f' \cosh^2\xi$ ($\mathrm{d}x/\mathrm{d}\xi = \operatorname{sech}^2\xi$), $\cosh^2\xi_0 \sim \frac{1}{4} E_0^{-1}$ and $z(\xi_0)^{-2} \asymp E_0^\mu$. □

Numerical observation 6.60 (the zero of the swirl runs into the pole)**.** *Fix a type-G germ chart with frame $\mu \approx 0.3$ and let $f_- \to 0^-$ by moving one coordinate of the germ ($f_a = f_a^* - s$ with $f_-(f_a^*) = 0$; frame data and code in `ch06b_germs.py`). The computed zero $1 + x_0$ of the swirl against the leading-order prediction of Proposition 6.59(b), with κ in the pole normalisation, is*

f_-	μ	$1 + x_0$	*prediction*	*ratio*
-0.2756	0.3277	$5.25 \cdot 10^{-3}$	$6.75 \cdot 10^{-3}$	0.778
-0.0905	0.3127	$1.60 \cdot 10^{-4}$	$1.75 \cdot 10^{-4}$	0.914
-0.0270	0.3063	$2.95 \cdot 10^{-6}$	$3.00 \cdot 10^{-6}$	0.983
-0.0090	0.3043	$7.64 \cdot 10^{-8}$	$7.59 \cdot 10^{-8}$	1.006

so the leading-order law is accurate to 2% for $|f_-| \leq 0.03$ and to 1% at $|f_-| = 0.009$ (`verify_ch06b.py`, Part 4; Figure 6.4(a)).

THEOREM 6.61 (the structure of class N). $N = N_0 \times (\mathbb{Z}_2 \times \mathbb{Z}_2)$. *Every* $S \in N_0$ *is of type* G *or* L_1 *at the south pole and of type* G, R *(*$\mu_+ < \frac{1}{2}$*),* L_1 *or* L_2 *at the north pole. (i)* $0 \le \mu < 2$*:* $N_0 \cap \{\mu < 2\} \cong \Gamma_T \times [0,\infty)$ *via the* t^2*-shift, where* $\Gamma_T = \{S \in N_0 \cap \{\mu < 2\} : S$ *not of type* G *at* $x = 1\}$*;* c^- *is finite and continuous across* $\mu = 0$*. (ii)* $\mu > 1$*:* $N_0 \cap \{\mu > 1\} \cong B \times [0,\infty)$ *via* $(\Pi, \mathfrak{a} - \Theta^-)$*,* $B = \{\Pi : P_d \in \operatorname{int} \mathcal{J}\}$*,* $\Theta^- \to \infty$ *at* ∂B*. (iii) The boundary* $\Gamma_T \times \{0\}$ *(resp.* $B \times \{0\}$*) is the edge* Γ_{N}*, the solutions that are not of type* G *at the north pole; all other solutions are of type* G *there.*

PROOF. $\mathsf{N} = \mathsf{N}_0 \times \mathbb{Z}_2^2$ is Proposition 5.4(b); the possible types at the two poles are Lemma 6.3(a). (i) is Theorem 6.34 together with Theorem 6.32(a),(b),(d) (finiteness and continuity of $c^- = \alpha_-$, across $\mu = 0$ included); note $\alpha_-(T_c S) = \alpha_-(S) - c$. (ii) is Theorem 6.50(c); the identification $B = \{\Pi : P_d \in \operatorname{int} \mathcal{J}\}$ is Lemma 6.40, and $\Theta^- \to \infty$ at ∂B is Theorem 6.50(b). (iii): by Theorem 6.32(c) and Theorem 6.50(c) the zero sets $\{\alpha_- = 0\}$, resp. $\{\mathfrak{a} = \Theta^- \circ \Pi\}$, are the global germs that are not of type G at $x = 1$, and all other global germs of the chart are of type G there. Remark 6.51 and Proposition 6.57 compare the two charts on $1 < \mu < 2$. □

Remark 6.62 (what the structure theorem rests on)**.** Finiteness and continuity of the threshold on each of the strata $\{\mu > 0\}$ and $\{\mu = 0\}$ separately would not give continuity across $\mu = 0$; this is Theorem 6.32(d), which uses the uniform envelope of Theorem 6.29. The pole-region estimate Lemma 6.43 is stated with an explicit order of choices (first the cut ξ_1, then C_0) because Lemma 6.45 needs Λ_* independent of the cut. The commutation theorem and the inverse relation are used in Chapter 12, where the threshold law is derived.

Corollary 6.63 (flows, chart and monotonicity of the threshold in N_0)**.** (a) *The shift flows* $(S,c) \mapsto T_c S$ *(*$\mu < 2$ *or type* L_1*) and* $(S,C) \mapsto \mathcal{R}_C S$ *(type* G*,* $\mu > \frac{1}{2}$*) are continuous, also at germs of type* L_1 *for the* t^2*-shift.* (b) *For* $0 \le \mu < 2$*,* $\mathsf{N}_0 \cap \{\mu < 2\} \cong \Gamma_T \times [0,\infty)$ *by the* t^2*-shift, where* Γ_T *includes the germs of type* L_1*.* (c) *The threshold is monotone:* $\alpha_-(T_a S) = \alpha_-(S) - a$*, and* $d \mapsto \alpha_-(\mathcal{R}_{-d} S)$ *is continuous and strictly increasing on* $\mathsf{N}_0 \cap \{\frac{1}{2} < \mu < 2\}$*.*

PROOF. (a) is Lemma 6.28; (b) is Theorem 6.34; (c) is Theorem 6.32(b) and Proposition 6.57(ii). □

Example 6.64 (the rest state: an exact threshold at $\mu = 1$; the Bessel model)**.** Take no swirl ($\kappa = 0$, $f \equiv 0$, $\lambda = 0$) and $Q \equiv 0$, $\mu = 1$, $z = e^{-\xi}$: the rest state. Since $\mathsf{L}[t^2] = 0$, T_c adds ct^2 and generates no swirl, so $z'' = (1 + \frac{c}{2}t^2)z$. This is not a class N_0 germ ($\kappa = 0$) but the limit $\kappa \to 0$ of the germs studied here, and all statements of this chapter that do not use $\kappa > 0$ hold verbatim. With $r = \sqrt{1+2c}$ and $x = \tanh \xi$ the solution with $z_c = e^{-\xi}(1 + O(E^2))$ is

$$z_c = 2^{(1-r)/2}(1+x)^{-1/2}(1-x)^{r/2}\Big(1 + \tfrac{r-1}{2r}\big(2^r(1-x)^{-r} - 1\big)\Big),$$

(an ODE residual of 0 to rounding for a symbolic r, `verify_ch06b.py`, Part 1), and

$$\ell_\infty(c) = \int_{\mathbb{R}} z_c^{-2} \mathrm{d}\xi = \frac{1}{r-1}$$

(quadrature against $\frac{1}{r-1}$: $c = 0.5,\ 2,\ 12$ gives $2.41421356,\ 0.80901699,\ 0.25000000$). By Proposition 6.49's mechanism, $\tilde{z} = z_c(1 + C l_c)$, $l_c = \int_{-\infty}^{\xi} z_c^{-2}$, is positive on $\mathbb{R}$

for $C<0$ iff $|C|\le 1/\ell_\infty = r-1$; solving, *the threshold of the recessive shift is exact*:

$$\alpha_-(\mathcal{R}_C S_0)=\tfrac12|C|(|C|+2),\qquad C<0\qquad (7.5 \text{ at } |C|=3,\ 60 \text{ at } |C|=10).$$

Equivalently, the global region of the (c,C)-plane is bounded below by $C=1-\sqrt{1+2c}$ for $c\ge-\frac12$. A direct numerical integration of $z''=(1+\frac{c}{2}t^2)z$ with the recessive start reproduces this boundary to $3\cdot10^{-5}$ (the largest deviation of the six values below is $2.7\cdot10^{-5}$, at $c=-0.4$; `verify_ch06b.py`, Part 4c; Figure 6.4(b)): for $c=-0.4,-0.2,0,0.5,2,12$ the computed critical values are 0.55276, 0.22541, -0.000005, -0.41421, -1.23606, -4.00000 against $1-\sqrt{1+2c}=0.55279$, 0.22540, 0, -0.41421, -1.23607, -4. For general $\mu\in(0,2)$ the pole zone is modelled by $\zeta''=(\mu^2+2cE^2)\zeta$, solved by modified Bessel functions: with $\nu=\mu/2$, $\ell_\infty=\ell_*(\mu)c^{-\mu/2}$, $\ell_*(\mu)=\dfrac{8^\nu\Gamma(\nu)}{4\Gamma(1-\nu)}$, and $\alpha_-=\gamma_*(\mu)|C|^{2/\mu}$, $\gamma_*=\ell_*^{2/\mu}$. For $\mu=0.6,1,1.3,1.6,1.9$: $\ell_*=1.0752,\ 0.70711,\ 0.52535,\ 0.33462,\ 0.09549$ and $\gamma_*=1.2732,\ 0.5,\ 0.37147,\ 0.25451,\ 0.08439$, and numerical quadrature of $\int\zeta_c^{-2}$ agrees with $\ell_*c^{-\mu/2}$ to the printed accuracy (ratio 1.0000; Part 2a). At $\mu=1$ this reproduces $\gamma_*=\frac12$ of the rest state, whose correction $|C|^{-1}$ is the relative size of the first deviation from the Bessel law. Chapter 12 proves that the Bessel law is the correct limit for genuine germs.

Atlas 6.65 (class N: thresholds through $\mu=0$, resonances, the criterion for B, the zero of the swirl)**.** Method. *A germ is specified by its state $(z,z',f,Q,Q')=(1,-0.4,f_a,Q_a,0.5)$ at $\xi_a=-1$ with $\kappa=1$, $\lambda=0$; it is integrated backward to $\xi=-9$ (Dormand–Prince of order 8, relative tolerance 10^{-13}), where μ and f_- are read off with the pure-exponential tail (error $O(e^{-18})\approx10^{-8}$). The t^2-shift is computed by integrating the exact difference equations from $\xi=-9$ with initial datum $\Delta Q=ct^2$, and α_- by bisection on c (tolerance 10^{-10}) with a globality test on the forward continuation. All numbers below are produced by* `code/ch06_classN/verify_ch06b.py` *(full run ≈3 minutes;* `-quick` *skips Part 4); the figures by the* `make_b_*.py` *scripts; the raw output is* `ch06b_numbers.json`*.*

Numerical observation 6.66 (the threshold is continuous through $\mu=0$ on two one-parameter families)**.** Wall. *The state at which $\mu=0$ (type L_1 at the south pole) with $f_a=2$ is $Q_a^{\mathrm{wall}}=-1.45431$, with $f_-=0.213726$ and threshold $\alpha_-=6.584684$.* Family B *moves Q_a off the wall by σ (f_a fixed):*

σ	μ	f_-	α_-
10^{-4}	0.007515	0.233697	6.584455
10^{-3}	0.023763	0.275465	6.582393
10^{-2}	0.075147	0.396009	6.561774
0.1	0.237646	0.689555	6.355921
0.2	0.336098	0.821882	6.127882

with $\mu^2/\sigma\to0.5647$ and $\mathrm{d}\alpha_-/\mathrm{d}\sigma=-2.2910$. Family A *keeps the frame (λ,κ,f_-) fixed ($f_-=0.213726$) and re-solves (Q_a,f_a) so that μ takes prescribed values: $\alpha_-=6.501415,\ 6.178006,\ 5.794292,\ 5.082780$ at $\mu=0.01,0.05,0.1,0.2$, i.e. difference quotients $(\alpha_--\alpha_-^{\mathrm{wall}})/\mu=-8.327,\ -8.134,\ -7.904,\ -7.510$.*

Numerical observation 6.67 ($\mu\sim\sqrt{\sigma}$ along B; the linear rate along family A is a parametrisation effect)**.** *Along family B, $\mu^2/\sigma\to0.5647$, so $\mu\sim\sqrt{\sigma}$, while*

$\alpha_- - \alpha_-^{\mathrm{wall}}$ *is proportional to* σ*:* $\mathrm{d}\alpha_-/\mathrm{d}\sigma = -2.2910$ *at the wall, equivalently* $(\alpha_- - \alpha_-^{\mathrm{wall}})/\mu^2 = -4.057$ *at* $\sigma = 10^{-2}$ *(*$-2.2910/0.5647 = -4.057$*). Along family A the state moves linearly in* μ *(not in* μ^2*). At* $\mu = 0.01$ *the numerical gradient of* α_- *in the* (Q_a, f_a)*-plane is* $(-2.2898, 4.2068)$ *and the tangent of family A is* $(-0.4152, -2.2152)$*; their scalar product* -8.368 *is the slope predicted for family A, against the measured difference quotient* -8.327*. So the apparent linear rate in* μ *along A is what a differentiable function of the state gives along a curve that is linear in* μ*, while along B the same function is linear in* μ^2*. These are observations at the displayed parameters (*`verify_ch06b.py`*, Part 4a–4b); whether* α_- *is differentiable in the state at* $\mu = 0$ *is not decided here (see Chapter 13).*

Numerical observation 6.68 (no jump at the resonances $\mu = \frac{1}{2}, 1, \frac{3}{2}$)**.** *The coefficient* $a_1 = \frac{1}{4(1-\mu)}$ *of Toy example 6.1 is singular at the resonant exponent* $\mu = 1$*, but the threshold shows no singularity at* $\mu = \frac{1}{2}, 1, \frac{3}{2}$*: for base germs with prescribed* $\mu = \mu_0 + \{-0.01, -0.005, 0, 0.005, 0.01\}$ *(*$f_a = 2$*)*

μ_0	α_- *at the five exponents*
0.5	5.617275, 5.597566, 5.577663, 5.557565, 5.537272
1.0	2.706870, 2.668465, 2.629881, 2.591120, 2.552181
1.5	−1.960093, −2.014203, −2.068422, −2.122746, −2.177176

a smooth, nearly linear sequence; the second differences $\alpha_-(\mu_0 - 0.01) - 2\alpha_-(\mu_0) + \alpha_-(\mu_0 + 0.01)$ *are* $-7.8 \cdot 10^{-4}$*,* $-7.1 \cdot 10^{-4}$*,* $-4.3 \cdot 10^{-4}$*, of the size expected from the curvature of a smooth function and with no trace of a jump or kink.*

Numerical observation 6.69 (the criterion $\Pi \in B$ reproduces direct integration)**.** *This is Numerical observation 6.41:* 300 *random* (μ, p_1, p_2)*,* 189 *in B, no disagreement; Figure 6.3.*

Numerical observation 6.70 (rest state and gluing)**.** *Example 6.64:* $\ell_\infty(c) = 1/(r-1)$ *and* $\alpha_-(\mathcal{R}_C S_0) = \frac{1}{2}|C|(|C| + 2)$ *to* $3 \cdot 10^{-5}$ *(boundary* $C = 1 - \sqrt{1 + 2c}$*); Numerical observation 6.60: the zero of the swirl follows* $1 + x_0 \approx 2(2\mu|f_-|/\kappa)^{1/\mu}$ *to* 2% *for* $|f_-| \le 0.03$*.*

Figure 6.6 shows the threshold through $\mu = 0$*.*

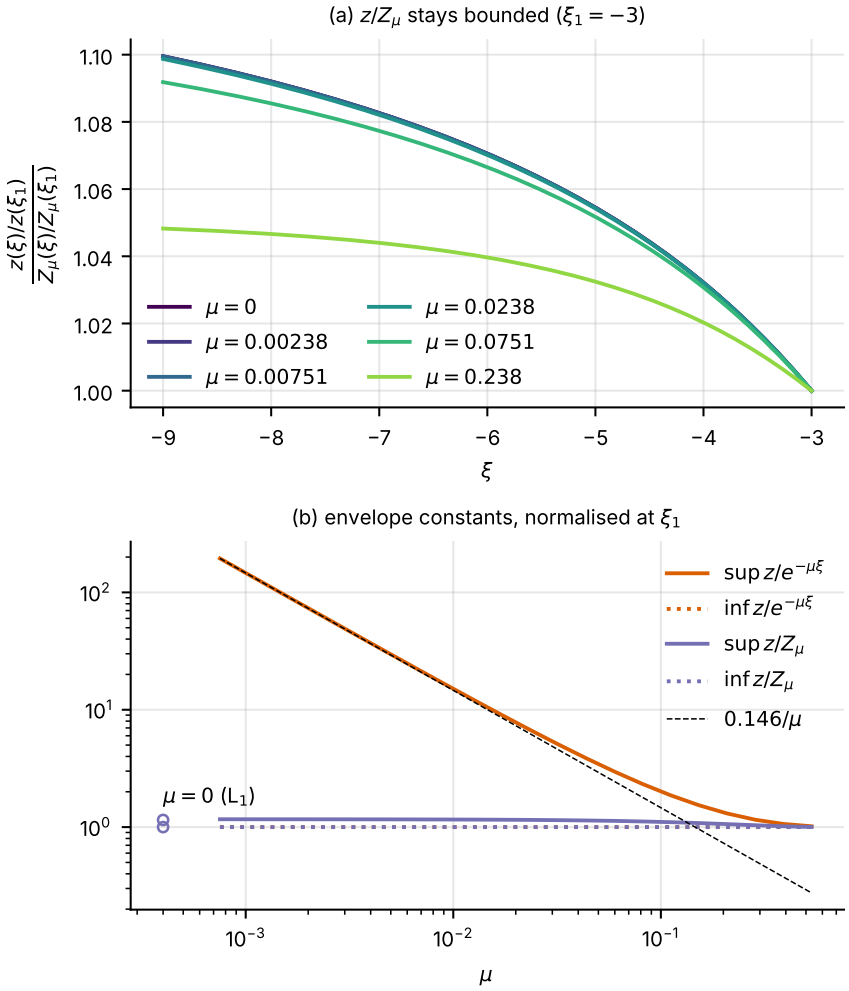


FIGURE 6.2. Numerically computed germs along a segment in state space through $\mu = 0$ (cut $\xi_1 = -3$, normalisation $z(\xi_1) = 1$; below the computational cut the germ is continued by its exact asymptotic form). (a) z/Z_μ, normalised to 1 at ξ_1, for $\mu \in \{0, 0.0024, 0.0075, 0.024, 0.075, 0.24\}$: the curves for $\mu \le 0.075$ nearly coincide. (b) sup and inf of z over the weight, for the weight $e^{-\mu\xi}$ and for Z_μ: the first constant behaves like $0.146/\mu$ for small μ ($\mu \sup z/e^{-\mu\xi}$ lies between 0.1463 and 0.1505 for $\mu \le 0.01$, and is 0.54 at $\mu = 0.53$), while $\sup z/Z_\mu \in [1.002, 1.167]$ and $\inf z/Z_\mu \ge 0.99999$ for all computed germs (open circles: the L_1 germ; `verify_ch06b.py`, Part 5). Code: `code/ch06_classN/make_a_envelope.py`.

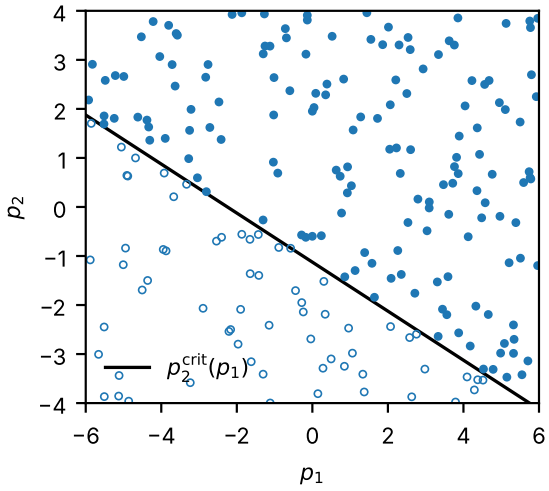


FIGURE 6.3. The base set B at fixed $\mu = 1.5$ in the plane of the two polynomial-part coordinates (p_1, p_2): the exact boundary $p_2 = p_2^{\mathrm{crit}}(p_1)$ of Lemma 6.40 (line) and a numerical test of positivity and dominance of u_P (dots; filled: in B). Code: `code/ch06_classN/make_b_Bset.py`.

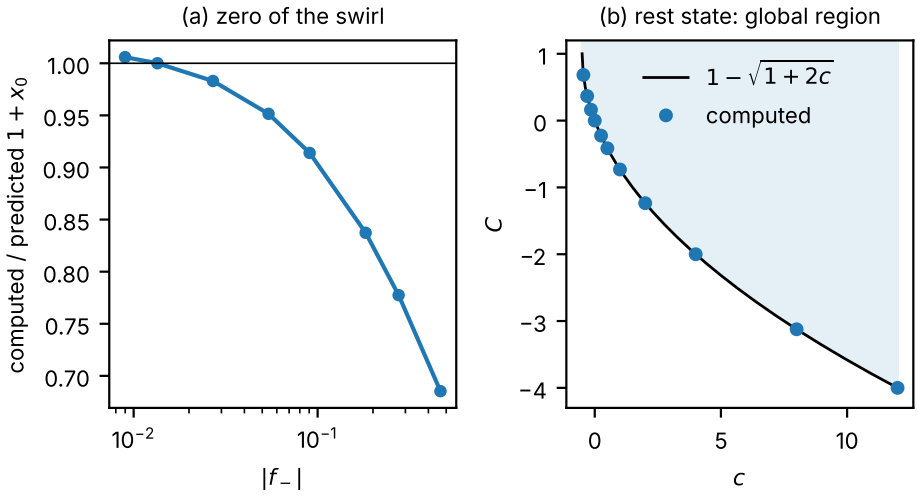


FIGURE 6.4. (a) Gluing with class Z (Numerical observation 6.60): ratio of the computed zero $1 + x_0$ of the swirl to the leading-order prediction $2(2\mu|f_-|/\kappa)^{1/\mu}$, against $|f_-|$ (log scale). (b) The rest state (Example 6.64): the numerically computed boundary of the set of (c, C) for which $\mathcal{R}_C$ of the rest state is global, and the exact curve $C = 1 - \sqrt{1 + 2c}$. Code: `code/ch06_classN/make_b_gluing.py`.

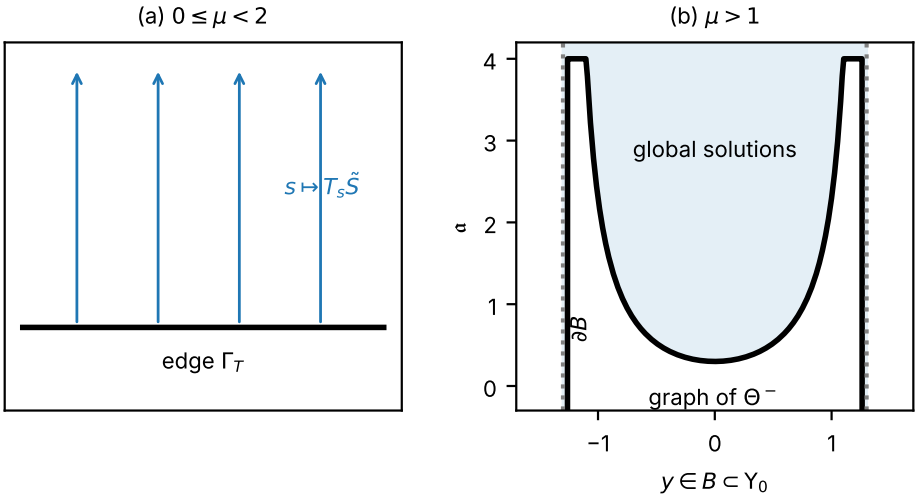


FIGURE 6.5. The two charts of the global solutions of class N_0. (a) $0 \le \mu < 2$: the edge Γ_T is the floor, the t^2-orbits are the half-lines $s \mapsto T_s\tilde{S}$. (b) $\mu > 1$: the base $B \subset \mathcal{Y}_0$ is an open set, Θ^- blows up at ∂B, and the solutions are the closed region above the graph of Θ^-. On $1 < \mu < 2$ the two edges coincide. Schematic; the data are in Atlas 6.65. Code: `code/ch06_classN/make_b_cylinder.py`.

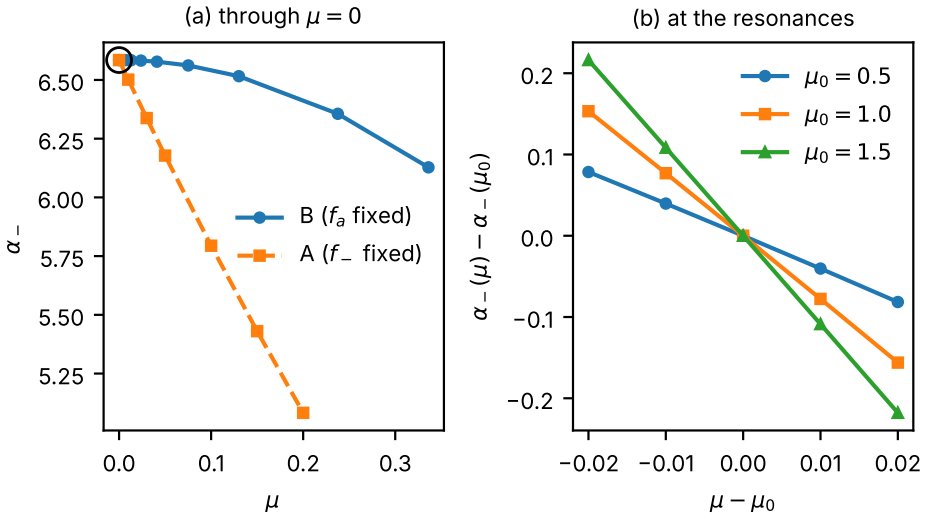


FIGURE 6.6. The threshold α_- through $\mu = 0$. (a) Family B and family A as functions of μ (Numerical observation 6.66); open circle: the type-L_1 germ at the wall, the common limit. (b) α_- near $\mu_0 = \frac{1}{2}, 1, \frac{3}{2}$ (Numerical observation 6.68), shifted by $\alpha_-(\mu_0)$. Code: `code/ch06_classN/make_b_threshold.py`.

CHAPTER 7

Edges, regularity, walls, and the global picture

In words. *A global swirl solution of class* Z *is determined by a point of the lens: a datum* $(x_0, \boldsymbol{c}, \kappa)$ *and a slope* $\gamma = U(x_0)$ *between the two thresholds* $\gamma^- \le \gamma \le \gamma^+$. *This chapter studies the* edge $\gamma = \gamma^-$, *the borderline solution that only just survives up to the north pole, and the global picture of the whole set* Ω *of global solutions. The edge solution ends at the pole in one of three ways,* R *(a recessive solution with* $\mu < \frac{1}{2}$*),* L_1 *or* L_2 *(the two logarithmic ones), and its dependence on the data has a different character for each. The mechanism is one idea used four times. Vary a parameter and you get a* Jacobi field, *a solution of the linearised system. Because the system is cooperative (Chapter 5), the Jacobi fields of* γ *and of* $\boldsymbol{c}$ *have all components of one sign, and that of* κ *the opposite sign. Two quantities then move monotonically: the variation* δQ *of the potential, and a Wronskian* $v = \frac{1}{2} z^2\, \delta U$ *with* $v' = \frac{1}{2} z^2 \delta Q$. *On the other hand a variation* within *the family of pole solutions of type* R *has* $v \to 0$, *and one within the family of type* L_1 *has* $\delta Q \to 0$. *A monotone positive quantity does not tend to zero, so* ∂_γ *is never tangent to the family of pole solutions: the solutions through the edge cross the hypersurface of edge solutions transversally. The implicit function theorem then makes* γ^- *smooth, with an explicit formula for its derivatives, and the same argument, with the shift of Chapter 6 in place of* γ, *makes the thresholds of class* N *smooth. Without any nondegeneracy, the same sign structure gives Lipschitz bounds, hence a Lipschitz tip* $\kappa_{\max}$. *Where* R *and* L_1 *meet, at the wall of* L_2 *edges, the unified corner chart of Chapter 4 shows exactly how smoothness fails:* γ^- *is* $C^{1,1}$ *and not* C^2, *with a Hessian jump of rank one.*

Roadmap. Section 7.1 sets up the Jacobi fields of the lower threshold, proves the three sign facts (Lemma 7.3), that tangent vectors of the pole charts have $v \to 0$ (R) or $\delta Q \to 0$ (L_1, L_2) (Lemmas 7.4 and 7.5), and transversality (Proposition 7.7). Section 7.2 proves the Lipschitz bounds, with no assumption on the type of the edge (Theorem 7.10). Section 7.3 proves that γ^- is C^∞ at edges of type R and L_1, with the variational formula for its derivatives and a linear law for $Q(1)$ at an L_1 edge (Theorem 7.12), and that the tip $\kappa_{\max}$ is smooth when its two edges are (Corollary 7.14); it ends with a numerical check of the formulas (Numerical observation 7.15). Section 7.4 runs the same scheme for the thresholds of class N (Theorem 7.19). Section 7.5 contains the small-swirl R edges, smooth in κ^2 (Theorem 7.21), the continuity of $Q(1)$ along the edge (Proposition 7.26), the structure of the set of edge states (Lemma 7.28), the interface between R and L_1 edges and the density of regular edges (Proposition 7.29, Corollary 7.35), and the wall theorem for L_2 edges (Theorem 7.32, Proposition 7.34), with the numerical wall of Numerical observation 7.37 and the numerical atlas of the part (Atlas 7.39). Part b of the chapter studies the edge hypersurface in the full space of data, the

interval property of the fibres of Ω where it is proved, attachment and the layers at large swirl. *Conventions.* Throughout, $x_0 \in (-1,1)$ is the zero of the swirl, $\kappa > 0$ is the swirl constant $\kappa_x = f_x(x_0)$ there, γ^-, γ^+ are the thresholds of Proposition 5.42, $d = 1 - x^2$, and a prime is $\mathrm{d}/\mathrm{d}\xi$ with $\xi = \operatorname{artanh} x$ unless ∂_x is written.

7.1. Jacobi fields of the threshold: tangent signs and transversality

In words. *Move a datum of the lens a little and you move the solution; the first-order motion is a* Jacobi field. *The cooperative structure of the system decides its signs: raising γ or $\boldsymbol{c}$ lifts every component of the state in the order of Chapter 5, raising κ lowers them. A solution that is just recessive at the pole cannot be moved by γ without leaving the family of recessive (or logarithmic) pole solutions, because the motion has a nonzero limit of a quantity that vanishes along that family.*

7.1.1. Setting. Fix $p = (x_0, \boldsymbol{c}, \kappa)$ with $\kappa > 0$ and write $F_{\boldsymbol{c}}$ for the vector field of (5.2) in the variables $Y = (U, g, f, p_2, p_1, p_0)$. For $\gamma \in \mathbb{R}$ let $Y(\cdot; p, \gamma)$ be the solution of $\partial_x Y = F_{\boldsymbol{c}}(x, Y)$ with $Y(x_0) = (\gamma, \kappa, 0, 0, 0, 0)$. By Lemma 5.23, (U, f, Q) with $Q = P_{\boldsymbol{c}} - p_0$ solves the Q-system, $f(x_0) = 0$, $f_x(x_0) = \kappa$, and $\boldsymbol{c}$ are the Taylor constants of Q at x_0; every class-Z solution with $\kappa > 0$ arises this way. Let

$$\mathrm{FS}(p) = \{\gamma : \ Y(\cdot; p, \gamma) \text{ exists on } [x_0, 1)\},$$
$$\mathcal{P} := \{p = (x_0, \boldsymbol{c}, \kappa) : \ \kappa > 0, \ \mathrm{FS}(p) \neq \emptyset\}.$$

By Proposition 5.42, $\mathrm{FS}(p) = [\gamma^-(p), \infty)$ for $p \in \mathcal{P}$, and by Lemma 5.46 the *edge solution* $Y_e := Y(\cdot; p, \gamma^-(p))$ is of type R, L_1 or L_2 at $x = 1$, while $Y(\cdot; p, \gamma)$ is of type G for $\gamma > \gamma^-(p)$. Along the forward solution $f > 0$ and $g > 0$ on $(x_0, 1)$ (Lemma 5.25). We write $\mathcal{P}_{\mathsf{R}}, \mathcal{P}_{\mathsf{L}_1}, \mathcal{P}_{\mathsf{L}_2}$ for the sets of $p \in \mathcal{P}$ whose edge solution has the indicated type; they partition $\mathcal{P}$.

The *pole variable* z and the Riccati variable are, as in Chapter 2, $h = \frac{1}{2}U + x = z'/z$, with

$$z'' = (1 + \tfrac{1}{2}Q)\, z, \qquad U' = Q - 2hU \quad \big(\text{i.e. } U_\xi + 2\tanh\xi\, U + \tfrac{1}{2}U^2 = Q\big), \tag{7.1}$$

where $z > 0$ is any positive solution of $z'/z = h$ on the interval of existence. For $\nu \in \{\gamma, \kappa, c_1, c_2, c_3, x_0\}$ the *Jacobi field* is $\mathsf{J}_\nu(x) := \partial_\nu Y(x; p, \gamma)$ evaluated at $\gamma = \gamma^-(p)$ (the derivative is taken at fixed x); it exists and is smooth on $[x_0, 1)$ because the flow of the polynomial field $F_{\boldsymbol{c}}$ is smooth in the data (Theorem A.4), and it solves the linearised system

$$\partial_x \mathsf{J} = \partial_Y F_{\boldsymbol{c}}(x, Y_e)\, \mathsf{J} + \partial_\nu F_{\boldsymbol{c}}, \tag{7.2}$$

where $\partial_\nu F_{\boldsymbol{c}} = 0$ unless $\nu = c_i$, and $\partial_{c_i} F_{\boldsymbol{c}} = \frac{b_i}{d} e_U$ with $b = (1-x,\ 1+x,\ 1-x^2)$, with $\mathsf{J}_\gamma(x_0) = e_U$, $\mathsf{J}_\kappa(x_0) = e_g$, $\mathsf{J}_{c_i}(x_0) = 0$, and $\mathsf{J}_{x_0}(x_0) = -F_{\boldsymbol{c}}(x_0, Y_e(x_0))$ (differentiate $Y(x; x_0) = Y(x; x_1, Y(x_1; x_0))$ in x_0). Write $\delta U, \delta Q, \ldots$ for components. The variation of the potential is $\delta Q = \delta P_{\boldsymbol{c}} - \delta p_0$, with $\delta P_{\boldsymbol{c}} = b_i$ for $\nu = c_i$ and 0 otherwise; the variation fields of the Q-*system* itself, $(\delta U, \delta f, \delta Q)$, are what the pole charts will see. The sign matrix is $\Sigma = \mathrm{diag}(+, -, -, -, -, -)$ and $\mathsf{W}_\nu := \Sigma \mathsf{J}_\nu$.

Lemma 7.1 (the Wronskian of a variation)**.** *Along any family of solutions, with δ the variation at fixed x and $v := \frac{1}{2} z^2\, \delta U$,*

$$v' = \tfrac{1}{2} z^2\, \delta Q, \qquad \textit{equivalently} \qquad \partial_x v = \frac{z^2\, \delta Q}{2(1 - x^2)}.$$

Moreover $v = \mathrm{Wr}(z, \delta z) = z\,\delta z' - z'\,\delta z$ *for the variation* δz *of any differentiable family* z_τ *of positive solutions with* $z'_\tau/z_\tau = h_\tau$.

Proof. At fixed ξ, $\delta h = \frac{1}{2}\delta U$ since $h = \frac{1}{2}U + x$. Varying the Riccati equation $h' = 1 + \frac{1}{2}Q - h^2$ gives $\delta h' = \frac{1}{2}\delta Q - 2h\,\delta h$, so

$$(z^2\delta h)' = 2zz'\,\delta h + z^2\delta h' = 2z^2 h\,\delta h + \tfrac{1}{2}z^2\delta Q - 2z^2 h\,\delta h = \tfrac{1}{2}z^2\delta Q,$$

and $z^2\delta h = \frac{1}{2}z^2\delta U = v$. Finally $\delta h = \delta(z'/z) = (z\,\delta z' - z'\,\delta z)/z^2$. □

Toy example 7.2 (no swirl: the Wronskian is constant)**.** Freeze the potential at its value at the pole, $W \equiv \mu^2$, as in Toy example 5.45. The solutions through $z(0) = 1$ with slope $z'(0) = \beta$ are $z_\beta = \cosh\mu s + \frac{\beta}{\mu}\sinh\mu s$ $(s = \xi - \xi_b)$, and $\partial_\beta z_\beta = \mu^{-1}\sinh\mu s$. At the recessive solution $\beta = -\mu$, $z = e^{-\mu s}$ and

$$v = \mathrm{Wr}(z, \partial_\beta z) = e^{-\mu s}\cosh\mu s + \mu e^{-\mu s}\,\mu^{-1}\sinh\mu s = e^{-\mu s}e^{\mu s} = 1.$$

Here $\delta Q \equiv 0$, so v is constant by Lemma 7.1, and it is a *nonzero* constant. This is exactly what Lemma 7.4 excludes for a tangent vector of the family of recessive solutions (there $v = O(e^{-2\mu\xi})$): moving γ leaves the family. For genuine swirl the constant is replaced by a nondecreasing positive function, and the same conclusion holds (Proposition 7.7).

7.1.2. The three sign facts.

Lemma 7.3 (tangent signs of the Jacobi fields)**.** *Let* $p \in \mathcal{P}$ *and* $\gamma \geq \gamma^-(p)$*, and let the Jacobi fields be taken along* $Y(\cdot; p, \gamma)$*. For every* $x \in (x_0, 1)$*:*

(a) $\mathsf{W}_\gamma > 0$, $\mathsf{W}_{c_i} > 0$ *and* $\Sigma(-\mathsf{J}_\kappa) > 0$ *componentwise; that is,* $\delta U > 0$, $\delta g < 0$, $\delta f < 0$, $\delta p_k < 0$ *for* $\nu = \gamma, c_i$ *and all signs reversed for* $\nu = \kappa$*.*

(b) $\delta U_\gamma, \delta U_{c_i} > 0 > \delta U_\kappa$*; hence* $v_\gamma, v_{c_i} > 0$ *are nondecreasing and* $v_\kappa < 0$ *is nonincreasing.*

(c) $\delta Q_\gamma > 0$ *is nondecreasing;* $\delta Q_{c_i} = b_i - \delta p_0 \geq -\delta p_0 > 0$ *with* $-\delta p_0$ *nondecreasing;* $\delta Q_\kappa < 0$ *is nonincreasing. Consequently, for every* $x_1 \in (x_0, 1)$ *there is* $m_1 > 0$ *(continuous in* p *and* γ*) with* $\delta Q_\gamma \geq m_1$, $\delta Q_{c_i} \geq m_1$, $-\delta Q_\kappa \geq m_1$, $v_\gamma \geq m_1$, $v_{c_i} \geq m_1$, $-v_\kappa \geq m_1$ *on* $[x_1, 1)$ *(for* v *the function* z *is fixed, and* m_1 *then depends on* $z(x_1)$*).*

Proof. (a) By Lemma 5.27 with $\hat{Y} = Y$ (so $\bar{Y} = Y$), $\Sigma\,\partial_Y F_{\mathbf{c}}(x, Y)\,\Sigma = \mathfrak{A}(x; Y)$ has the nonzero off-diagonal entries $\frac{1}{d}, \frac{g}{d}, 1, \frac{2g}{d}, \frac{2f}{d}, 1, 1$, all ≥ 0 since $f, g \geq 0$, and they contain the arrows $U \to g \to f \to p_2 \to p_1 \to p_0 \to U$ with positive entries on $(x_0, 1)$ because $g > 0$ there. Multiplying (7.2) by Σ, $\mathsf{W} = \Sigma\mathsf{J}$ solves $\partial_x\mathsf{W} = \mathfrak{A}\,\mathsf{W} + \rho$ with $\rho = \Sigma\partial_\nu F_{\mathbf{c}}$; $\rho = 0$ for γ, κ and $\rho = \frac{b_i}{d}e_U \geq 0$ for c_i $(\sigma_U = +)$. Initial values: $\mathsf{W}_\gamma(x_0) = e_U \neq 0$; for $-\mathsf{J}_\kappa$, $\Sigma(-e_g) = e_g \neq 0$; for c_i, $\mathsf{W}(x_0) = 0$ but ρ does not vanish identically on any $[x_0, x]$ ($b_i > 0$ on $(-1, 1)$). Lemma 5.21(b) gives > 0 on $(x_0, 1)$. (b) $\sigma_U = +1$ gives the signs of δU; then Lemma 7.1 and $z > 0$. (c) Recall Σ acts on p_0, p_1 by -1. For γ and κ, $\delta P_{\mathbf{c}} = 0$, so $\delta Q = -\delta p_0$ and, from $Q_x = P'_{\mathbf{c}} - p_1$, $\partial_x\delta Q = -\delta p_1$. For $\nu = \gamma$ these are W_{p_0} and W_{p_1}, both > 0 by (a): $\delta Q_\gamma > 0$ and nondecreasing. For $\nu = \kappa$ the positive vector is $\Sigma(-\mathsf{J}_\kappa)$, whose components p_0, p_1 are $+\delta p_0$, $+\delta p_1$, so $\delta p_0^\kappa, \delta p_1^\kappa > 0$ and $\delta Q_\kappa = -\delta p_0^\kappa < 0$ with $\partial_x\delta Q_\kappa = -\delta p_1^\kappa < 0$. For c_i: $\delta Q = b_i - \delta p_0 \geq -\delta p_0 = \mathsf{W}_{p_0} > 0$ ($b_i \geq 0$ on $(-1, 1)$), and $\partial_x\mathsf{W}_{p_0} = \mathfrak{A}_{p_0,p_1}\mathsf{W}_{p_1} = \mathsf{W}_{p_1} > 0$. The lower bounds follow by evaluating at x_1 (continuity of the Jacobi fields in (p, γ)), and for v from monotonicity: $v(x) \geq v(x_1)$ since $v' > 0$. □

7.1.3. What a tangent vector of a pole chart looks like. The pole charts parametrise the *solutions of a fixed type near the pole*: for a cut ξ_* (large) the state map S_{ξ_*} of Theorems 4.38 and 4.40 is an injective C^∞ immersion of the open set of coordinates p_{c} of the stratum $\mathsf{c} \in \{\mathsf{R}, \mathsf{L}_1, \mathsf{L}_2\}$ into the scale-free state space $\mathbb{R}^6$ ($6 = \dim$ of the space of local solutions of the Q-system with $\kappa \neq 0$); the image $\mathcal{M}_{\mathsf{c}}$ is an immersed hypersurface (R, L_1: 5 coordinates) resp. of codimension two (L_2: 4 coordinates). The scale-free state s and the state Y of Chapter 5 at $x_* = \tanh \xi_*$ determine each other through the affine isomorphism of Remark 5.14, which depends on $\boldsymbol{c}$ only through $P_{\boldsymbol{c}}$; for this reason it is cleaner to speak of *variation fields* $(\delta U, \delta f, \delta Q)$, the solutions of the linearised Q-system, which do not refer to any reference polynomial: a variation field is *tangent to* $\mathcal{M}_{\mathsf{c}}$ if its state at ξ_* is a tangent vector of $\mathcal{M}_{\mathsf{c}}$, equivalently (uniqueness for the linearised system) if it is ∂_τ of a C^1 curve $\tau \mapsto u_{p_{\mathsf{c}}(\tau)}$ of chart solutions on $[\xi_*, \infty)$.

Lemma 7.4 (R: tangent fields have $v \to 0$)**.** *Let Y be a chart solution of type R (a normalised local solution of type R in the sense of Theorem 4.40; in particular the edge solution of an R edge, or the no-swirl recessive solution), ξ_* as in Theorem 4.40, and let $\delta = \partial_\tau$ be a tangent vector of $\mathcal{M}_{\mathsf{R}}$ at the state of Y at ξ_*, propagated by the linearised flow. Then $v = \frac{1}{2} z^2 \delta U \to 0$ as $\xi \to \infty$.*

Proof. Let $p(\tau)$ be a C^1 curve of coordinates and u_τ the chart solutions, $z_\tau = e^{-\mu_\tau \xi} \zeta_\tau$ the recessive solution of $z'' = (\mu^2 + w) z$, normalised by $\zeta_\tau \to 1$ (Theorem 4.40: the potential is $Q = 2\mu^2 - 2 + 2w$, $w \in C_{E^{1-2\varsigma}}$, $\varsigma \in (\mu, \frac{1}{2})$ fixed). By Lemma 7.1, with $\delta = \partial_\tau$, $v = z\,\delta z' - z'\,\delta z$. Substituting $z = e^{-\mu\xi}\zeta$, $\delta z = e^{-\mu\xi}(\delta\zeta - \delta\mu\,\xi\zeta)$ (and the corresponding $\delta z'$) the terms with ξ cancel and

$$v = e^{-2\mu\xi}\big(\zeta\,\delta\zeta' - \zeta'\,\delta\zeta - \delta\mu\,\zeta^2\big)$$

(a direct expansion, confirmed by a computer-algebra check, `verify_ch07a.py`). Now ζ and $\delta\zeta = \partial_\tau\zeta$ are bounded: $\zeta = e^{\mu\xi} z = \hat{z}$ is the solution of the rescaled Volterra equation of Theorem 4.40, bounded between $\frac{2}{3}$ and $\frac{4}{3}$, and smooth in the coordinates as a map into C_b. The derivatives are bounded too: $\zeta'' - 2\mu\zeta' = w\zeta$ gives $(e^{-2\mu\xi}\zeta')' = e^{-2\mu\xi} w\zeta$, hence, as $e^{-2\mu\xi}\zeta' \to 0$,

$$\zeta' = -e^{2\mu\xi} \int_\xi^\infty e^{-2\mu\eta}\, w\zeta \,\mathrm{d}\eta,$$

$$|\zeta'| \le C\, e^{2\mu\xi} \int_\xi^\infty e^{-2\mu\eta} e^{-2(1-2\varsigma)\eta} \mathrm{d}\eta \le C' e^{-2(1-2\varsigma)\xi},$$

and the same bound, with an extra factor $1+\xi$, for $\delta\zeta'$ by differentiating this formula in τ ($\partial_\tau w \in C_{E^{1-2\varsigma}}$, Theorem 4.40(b)). So $v = O(e^{-2\mu\xi}) \to 0$ since $\mu > 0$. □

Lemma 7.5 ($\mathsf{L}_1, \mathsf{L}_2$: tangent fields have $\delta Q \to 0$)**.** *Let Y be a chart solution of type L_1 (L_2) (for instance an edge solution), ξ_* as in Theorem 4.40, and δ a tangent vector of $\mathcal{M}_{\mathsf{L}_1}$ ($\mathcal{M}_{\mathsf{L}_2}$), propagated by the linearised flow. Then $\delta Q(\xi) \to 0$ as $\xi \to \infty$.*

Proof. In the L charts $q = -2$ for every coordinate, so $Q_\tau = -2 + 2w_\tau$ with w_τ in the weighted space C_E (resp. C_{Em^2}) and $\tau \mapsto w_\tau$ is C^1 into the same space (Theorem 4.40(b)). Thus $\delta Q = 2\partial_\tau w_\tau = O(E)$ (resp. $O(Em^2)$) $\to 0$. □

Remark 7.6 (why two functionals)**.** In stratum R the exponent μ is a coordinate, so a tangent vector can change the limit $q = 2\mu^2 - 2$ of Q, and $\delta Q \not\to 0$ in general (for the γ-field at an R edge it tends to a positive constant, Figure 7.1); that is why

v is used there. In the L strata $q \equiv -2$, so a tangent field has $\delta Q \to 0$, while v tends to a finite limit depending on the tangent vector and carries no information.

7.1.4. Transversality.

Proposition 7.7 (transversality)**.** *Let $p \in \mathcal{P}_{\mathsf{R}} \cup \mathcal{P}_{\mathsf{L}_1}$ and let $\mathcal{M}_{\mathsf{c}}$ be the chart image through the state of the edge solution at ξ_* ($\mathsf{c} = \mathsf{R}$ or L_1). Then J_γ is not tangent to $\mathcal{M}_{\mathsf{c}}$; neither are $\mathsf{J}_\kappa, \mathsf{J}_{c_i}$. More precisely, for $\nu \in \{\kappa, c_1, c_2, c_3\}$ there is exactly one $s_\nu \in \mathbb{R}$ with $\mathsf{J}_\nu + s_\nu \mathsf{J}_\gamma$ tangent; it satisfies $s_\kappa > 0$ and $s_{c_i} < 0$, and*

$$s_\nu = -\lim_{x\to 1} \frac{\delta U_\nu}{\delta U_\gamma} \quad (\mathsf{c} = \mathsf{R}), \qquad s_\nu = -\lim_{x\to 1} \frac{\delta Q_\nu}{\delta Q_\gamma} \quad (\mathsf{c} = \mathsf{L}_1),$$

the limits of the ratios existing. The same is true for $\nu = x_0$, with no statement on the sign of s_{x_0}.

Proof. *Type* R. By Lemma 7.3(b), $v_\gamma \geq v_\gamma(x_1) > 0$ on $[x_1, 1)$, so $v_\gamma \not\to 0$, and by Lemma 7.4 J_γ is not tangent. Since the tangent space of the immersed hypersurface $\mathcal{M}_{\mathsf{R}}$ at the point has codimension one in the 6-dimensional space of variation fields and does not contain J_γ, for each ν there is exactly one s_ν with $\mathsf{J}_\nu + s_\nu \mathsf{J}_\gamma$ tangent. Then $v_\nu + s_\nu v_\gamma \to 0$ by Lemma 7.4. If $\nu = \kappa$: $v_\kappa \leq v_\kappa(x_1) < 0$ and $v_\gamma > 0$, so $s_\kappa \leq 0$ would give $v_\kappa + s_\kappa v_\gamma \leq v_\kappa(x_1) < 0$ on $[x_1, 1)$, contradicting convergence to 0: $s_\kappa > 0$. If $\nu = c_i$: $v_{c_i} \geq v_{c_i}(x_1) > 0$ forces $s_{c_i} < 0$ in the same way. Dividing $v_\nu + s_\nu v_\gamma \to 0$ by $v_\gamma \geq v_\gamma(x_1) > 0$ gives $s_\nu = -\lim v_\nu / v_\gamma = -\lim \delta U_\nu / \delta U_\gamma$ (z^2 cancels). *Type* L_1. By Lemma 7.3(c), $\delta Q_\gamma \geq m_1 > 0$, so $\delta Q_\gamma \not\to 0$ and, by Lemma 7.5, J_γ is not tangent. If $\mathsf{J}_\kappa + s\mathsf{J}_\gamma$ is tangent then $\delta Q_\kappa + s\,\delta Q_\gamma \to 0$; since $\delta Q_\kappa \leq -m_1 < 0 < \delta Q_\gamma$, $s \leq 0$ is impossible. Likewise $\delta Q_{c_i} \geq m_1$ forces $s < 0$. The ratio formula follows. For x_0 the argument gives existence of s_{x_0} and the formula, as only the existence of the limit of the ratio was used. □

This is the key point of the chapter: *transversality at* R *and* L_1 *edges is a consequence of the sign structure of the linearised flow: a monotone positive quantity (v_γ, resp. δQ_γ) against tangent vectors on which it tends to zero.* At L_2 edges the chart image has codimension two, and what replaces transversality is Theorem 7.32.

7.2. Lipschitz regularity: no nondegeneracy needed

In words. *The thresholds are the borderline of a monotone family: to survive, a solution must start high enough, and the order structure of Chapter 5 says how much higher a change of the data forces it to start. At a fixed interior station x_1 the variation of the state with γ is a vector with all components positive (an* order unit*), so a perturbation of the other data of size δ can be compensated by a change of γ of size $L\delta$. Nothing about the end of the edge solution at the pole enters, which is why this holds at* every *edge, in particular at the walls of L_2 edges.*

Lemma 7.8 (order criterion for survival)**.** *Let $\boldsymbol{c} \in \mathbb{R}^3$, let $Y, \hat{Y}$ solve $\partial_x Y = F_{\boldsymbol{c}}(x, Y)$ (same $\boldsymbol{c}$) near $x_1 \in (-1, 1)$ with $g, f, \hat{g}, \hat{f} \geq 0$ at x_1, and suppose $\Sigma(\hat{Y} - Y)(x_1) \geq 0$ componentwise. If Y exists on $[x_1, 1)$, then so does $\hat{Y}$, and $\hat{U} \geq U$ there. Equivalently: if $\hat{Y}$ ceases to exist at some $X < 1$, so does Y.*

Proof. By Lemma 5.25 both solutions keep $g, f \geq 0$ on their forward intervals. On a common interval $[x_1, b]$, Lemma 5.28(a) with $\hat{\boldsymbol{c}} = \boldsymbol{c}$ gives $\hat{U} \geq U$. If $\hat{Y}$

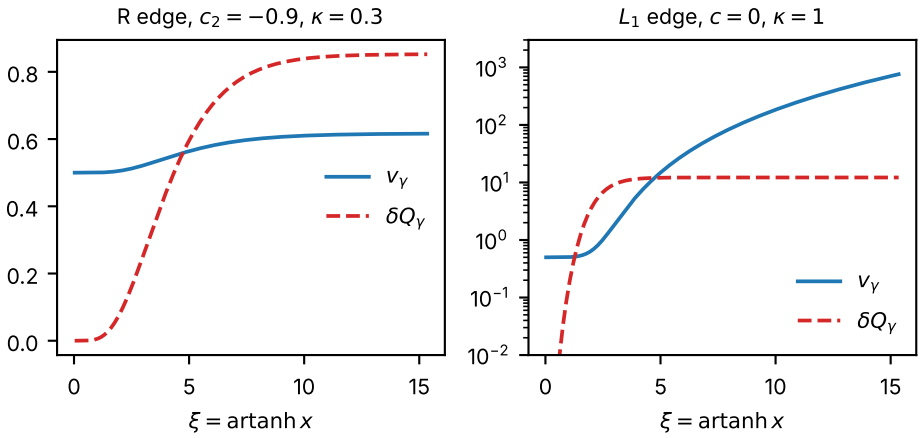


FIGURE 7.1. The mechanism of Section 7.1. Left: an R edge ($\boldsymbol{c} = (0, -0.9, 0)$, $x_0 = 0$, $\kappa = 0.3$): v_γ rises from 0.5 to 0.616 and δQ_γ increases to 0.852 (both tend to positive limits). Right: an L_1 edge ($\boldsymbol{c} = 0$, $\kappa = 1$): δQ_γ increases to $\ell_\gamma = 12.20$ while v_γ keeps growing (760 at the end of the window, $x = 1 - e^{-30}$); the abscissa is $\xi = \operatorname{artanh} x$. Code: `code/ch07_edges/make_a_mech.py`.

ceased to exist at $X < 1$, then $\hat{U} \to -\infty$ as $x \uparrow X$ (Lemma 2.24), whereas $\hat{U} \geq U \geq \min_{[x_1, X]} U > -\infty$ (Y exists on $[x_1, X]$): a contradiction. □

Remark 7.9 (every solution of the Q-system solves $F_{\boldsymbol{c}}$ for every $\boldsymbol{c}$)**.** By Lemma 5.23(a), if (U, f, Q) solves the Q-system then $Y = (U, f_x, f, \partial_x^2(P_{\boldsymbol{c}} - Q), \partial_x(P_{\boldsymbol{c}} - Q), P_{\boldsymbol{c}} - Q)$ solves $\partial_x Y = F_{\boldsymbol{c}}(x, Y)$ for *any* $\boldsymbol{c}$, the Taylor constants at the base point playing no role in the equations. Thus two solutions with different Taylor constants may be compared in the variables of one fixed reference polynomial $P_{\boldsymbol{c}}$; the state of $Y(\cdot; p', \gamma')$ at x_1 in the variables of $P_{\boldsymbol{c}}$ is an affine function of its own state, with coefficients depending smoothly on $\boldsymbol{c}' - \boldsymbol{c}$. We write $\Psi_{\boldsymbol{c}}(p', \gamma') \in \mathbb{R}^6$ for it and $\Psi_{\mathrm{own}}(p', \gamma') = Y(x_1; p', \gamma')$ for the state in the own variables.

THEOREM 7.10 (Lipschitz bounds). (i) *$\mathcal{P}$ is open in $(-1, 1) \times \mathbb{R}^3 \times (0, \infty)$ and γ^- is locally Lipschitz on it; so is γ^+ on the mirror set $\{(x_0, \boldsymbol{c}, \kappa) : (-x_0, \boldsymbol{c}_m, \kappa) \in \mathcal{P}\}$, where $\gamma^+(x_0, \boldsymbol{c}, \kappa) = -\gamma^-(-x_0, \boldsymbol{c}_m, \kappa)$.*

(ii) (bi-Lipschitz in κ and $\boldsymbol{c}$.) *For every compact $K \subset \mathcal{P}$ there are $0 < \ell \leq L$ and $\delta_0 > 0$ such that for $p = (x_0, \boldsymbol{c}, \kappa) \in K$ and p' with $|p' - p| < \delta_0$:*

$$\gamma^-(x_0, \boldsymbol{c}, \kappa') - \gamma^-(x_0, \boldsymbol{c}, \kappa) \geq \ell\,(\kappa' - \kappa) \quad (\kappa' > \kappa),$$
$$\gamma^-(x_0, \boldsymbol{c}, \kappa) - \gamma^-(x_0, \boldsymbol{c}', \kappa) \geq \ell\,|\boldsymbol{c}' - \boldsymbol{c}| \quad (\boldsymbol{c}' \geq \boldsymbol{c},\ \boldsymbol{c}' \neq \boldsymbol{c}),$$

and $|\gamma^-(p') - \gamma^-(p)| \leq L|p' - p|$.

(iii) *The tip function $\kappa_{\max}$ of Proposition 5.59 is locally Lipschitz on $\{(x_0, \boldsymbol{c}) : \boldsymbol{c} \in \operatorname{int} \mathcal{J}\}$.*

PROOF. *Openness.* If $p \in \mathcal{P}$ and $\gamma > \gamma^-(p)$, the solution is of type G at $x = 1$ (Lemma 5.46(a)), which is an open condition on the data (Corollary 4.42(b), as in the proof of Proposition 5.50(b), case $\kappa > 0$); so $\gamma \in \mathrm{FS}(p')$ for p' near p.

Step 1: an order unit. Fix a compact $K \subset \mathcal{P}$ and $x_1 \in (\max_K x_0, 1)$. For $p \in K$ the edge solution is regular on $[x_0, x_1]$, hence so is every solution with data (p', γ') in a neighbourhood $\mathcal{N}$ of the compact set $\{(p, \gamma^-(p)) : p \in K\}$, and the maps $\Psi_{\boldsymbol{c}}$, Ψ_{own} of Remark 7.9 are C^∞ on $\mathcal{N}$ (flow of a polynomial field composed with an affine change of variables). By Lemma 7.3(a), $w(p) := \Sigma\, \partial_\gamma \Psi_{\boldsymbol{c}}(p, \gamma^-(p)) = \mathsf{W}_\gamma(x_1)$ has all components > 0, and $w_\kappa := \Sigma(-\partial_\kappa \Psi_{\boldsymbol{c}})$ and $w_{c_i} := \Sigma \partial_{c_i} \Psi_{\mathrm{own}}$ as well; all are continuous in p. So $m := \min_{p \in K} \min\{\text{components of } w, w_\kappa, w_{c_i}\} > 0$ and $M := \max_K \max\{|w|, |w_\kappa|, |w_{c_i}|\} < \infty$; the C^2 norms of Ψ on $\mathcal{N}$ are bounded by some C_2; and $C_1 := \sup_{\mathcal{N}} |D_p \Psi_{\boldsymbol{c}}|$.

Step 2: upper bound. For $p \in K$, $|p' - p| = \delta$ small and $\eta = L\delta$ with $L := 2C_1/m + 1$,

$$\begin{aligned} \Sigma\big(\Psi_{\boldsymbol{c}}(p', \gamma^-(p) + \eta) - \Psi_{\boldsymbol{c}}(p, \gamma^-(p))\big) &= \eta\, w(p) + \Sigma D_p \Psi_{\boldsymbol{c}}\,(p' - p) + O(\eta^2 + \delta^2) \\ &\geq \big(Lm - C_1 - C_2(L^2 + 1)\delta\big)\, \delta\, \mathbf{1} \;\geq 0 \end{aligned}$$

for $\delta \leq \delta_0$. Both states are in the variables of the same $P_{\boldsymbol{c}}$ (Remark 7.9), have $g, f \geq 0$ (they are forward states of class-Z solutions), and the edge solution of p exists on $[x_1, 1)$, so Lemma 7.8 shows that the solution with data $(p', \gamma^-(p) + \eta)$ exists on $[x_1, 1)$, and on $[x_0', x_1]$ by smoothness: $\gamma^-(p') \leq \gamma^-(p) + L|p' - p|$.

Step 3: symmetry; (i). The constants are uniform on K. Exchanging the roles of p, p' (both in K) gives $\gamma^-(p) \leq \gamma^-(p') + L|p - p'|$; so γ^- is Lipschitz with constant L near each point of K. The mirror statement follows from the reflection formula (Lemma 5.40).

Step 4: lower bounds in κ and $\boldsymbol{c}$ (ii). κ: let $\kappa' > \kappa$, $p' = (x_0, \boldsymbol{c}, \kappa')$, $\varepsilon > 0$ small and $\gamma^* = \gamma^-(p) - \varepsilon$; the solution Y^* with data (p, γ^*) is not forward global, so it ceases to exist at some $X < 1$. Put $\gamma' = \gamma^* + \ell(\kappa' - \kappa)$ with $\ell := m/(2M)$. By Taylor expansion at (p, γ^*), whose Σ-derivatives are within $O(\varepsilon)$ of $w, -w_\kappa$,

$$\Sigma\big(\Psi_{\boldsymbol{c}}(p', \gamma') - \Psi_{\boldsymbol{c}}(p, \gamma^*)\big) \leq \big(\ell M - m + O(\varepsilon + \kappa' - \kappa)\big)(\kappa' - \kappa)\, \mathbf{1} \leq 0,$$

i.e. $\Sigma(\hat{Y} - Y)(x_1) \geq 0$ with $\hat{Y} = Y^*$ and $Y = Y(\cdot; p', \gamma')$. If Y existed on $[x_1, 1)$, Lemma 7.8 would give that $\hat{Y} = Y^*$ does; it does not, so $\gamma' \notin \mathrm{FS}(p')$: $\gamma^-(p') \geq \gamma' = \gamma^-(p) - \varepsilon + \ell(\kappa' - \kappa)$, and $\varepsilon \downarrow 0$. ***c**:* let $\boldsymbol{c}' \geq \boldsymbol{c}$, $\boldsymbol{c}' \neq \boldsymbol{c}$, $p' = (x_0, \boldsymbol{c}', \kappa)$, $\gamma'' = \gamma^-(p) - \ell|\boldsymbol{c}' - \boldsymbol{c}|$ ($|\cdot|$ the ℓ^1 norm, $\ell \leq m/(2M)$ again). Compare in their own variables the global edge solution Y_e of p and $\hat{Y} = Y(\cdot; p', \gamma'')$: $\Sigma(\hat{Y} - Y_e)(x_1) = \sum_i w_{c_i} \Delta c_i - \ell|\boldsymbol{c}' - \boldsymbol{c}|\, w + O(|\boldsymbol{c}' - \boldsymbol{c}|^2) \geq (m - \ell M - C_2|\boldsymbol{c}' - \boldsymbol{c}|)|\boldsymbol{c}' - \boldsymbol{c}|\, \mathbf{1} \geq 0$, and the *forced* comparison lemma (Lemma 5.28(a), $\hat{\boldsymbol{c}} \geq \boldsymbol{c}$, forcing $\rho = (P_{\hat{\boldsymbol{c}}} - P_{\boldsymbol{c}})/d \geq 0$) gives $\hat{U} \geq U_e$ on $[x_1, b]$, so $\hat{Y}$ is forward global as in Lemma 7.8: $\gamma^-(p') \leq \gamma^-(p) - \ell|\boldsymbol{c}' - \boldsymbol{c}|$.

Step 5: the tip (iii). Let $(x_*, \boldsymbol{c}_*)$ with $\boldsymbol{c}_* \in \operatorname{int} \mathcal{J}$ and $\kappa_* = \kappa_{\max}(x_*, \boldsymbol{c}_*)$. By Proposition 5.59(b) the tip $p_* = (x_*, \boldsymbol{c}_*, \kappa_*)$ has $\gamma^-(p_*) = \gamma^+(p_*) = \gamma_*$, so p_* and its mirror lie in $\mathcal{P}$, which is open; on a compact neighbourhood K' of p_* inside $\mathcal{P} \cap \{\text{mirror of } \mathcal{P}\}$ the function $\Delta := \gamma^- - \gamma^+$ is defined, Lipschitz in $(x_0, \boldsymbol{c})$ with constant $2L$ by (i), and strictly increasing in κ with $\Delta(\kappa') - \Delta(\kappa) \geq 2\ell(\kappa' - \kappa)$ by (ii) and its mirror. By Proposition 5.59(b),(d), for $(x_0, \boldsymbol{c})$ near $(x_*, \boldsymbol{c}_*)$ the tip $\kappa_{\max}(x_0, \boldsymbol{c})$ is the unique zero of $\Delta(x_0, \boldsymbol{c}, \cdot)$ in K'. For two such points q, q' with tips κ, κ': $2\ell|\kappa' - \kappa| \leq |\Delta(q, \kappa') - \Delta(q, \kappa)| = |\Delta(q, \kappa') - \Delta(q', \kappa')| \leq 2L|q - q'|$. □

Remark 7.11. The only inputs are the sign structure of the *linearised* flow at a single fixed station x_1 and the blow-up characterisation of the safe set; neither the type of the edge nor any property of the pole charts enters. This is why Theorem 7.10 holds at all edge points, including L_2. The constants ℓ, L degenerate at the boundary of $\mathcal{P}$, and $\kappa \downarrow 0$ with $c_2 > -\frac{3}{4}$ is such a boundary (Section 7.5).

7.3. Smoothness at edges of type R and L_1

In words. *Near a pole solution of type R or L_1 the set of all pole solutions is a smooth hypersurface in the six-dimensional state space at a fixed late cut (Chapter 4). The solutions of the family $\gamma \mapsto Y(\cdot; p, \gamma)$ cross it transversally at $\gamma = \gamma^-(p)$ (Proposition 7.7), so by the implicit function theorem the crossing parameter is a smooth function of p; and it is the threshold, because a solution that is of pole type R or L_1 is never in the interior of the safe set. The derivatives of γ^- are the slopes of the tangent vectors in the hyperplane spanned by J_ν and J_γ, computed from the Jacobi fields by the ratio formulas.*

THEOREM 7.12 (smoothness at R and L_1 edges). *Let $p_0 \in \mathcal{P}$ and suppose that the edge solution $Y_e = Y(\cdot; p_0, \gamma^-(p_0))$ is of type R or of type L_1 at $x = 1$. Then there is a neighbourhood $N \subset \mathcal{P}$ of p_0 such that:*

(i) *$\gamma^- \in C^\infty(N)$ and the edge solution has the same type for all $p \in N$; in particular $\mathcal{P}_{\mathsf{R}}$ and $\mathcal{P}_{\mathsf{L}_1}$ are open.*

(ii) *For $\nu \in \{x_0, c_1, c_2, c_3, \kappa\}$,*

$$\partial_\nu \gamma^- = -\lim_{x\to 1} \frac{\delta U_\nu}{\delta U_\gamma} \quad (\mathsf{R}), \qquad \partial_\nu \gamma^- = -\lim_{x\to 1} \frac{\delta Q_\nu}{\delta Q_\gamma} \quad (\mathsf{L}_1),$$

the Jacobi fields being those along the edge solution of p and the limits of the ratios existing.

(iii) *$\partial_\kappa \gamma^- > 0$ and $\partial_{c_i} \gamma^- < 0$ $(i = 1, 2, 3)$ on N, strictly.*

(iv) (linear law at L_1) *If the edge is of type L_1, there are $\bar\gamma > \gamma^-(p_0)$ and $m > 0$ such that for p near p_0 and $\gamma^-(p) < \gamma < \bar\gamma$ the limit $q(\gamma) = Q(1)$ of the solution $Y(\cdot; p, \gamma)$ satisfies $q(\gamma) + 2 \geq m\,(\gamma - \gamma^-(p))$; equivalently $\mu(\gamma) \geq \sqrt{m/2}\,\sqrt{\gamma - \gamma^-(p)}$.*

PROOF. Let $\mathsf{c} \in \{\mathsf{R}, \mathsf{L}_1\}$ be the type of Y_e. *Set-up.* By completeness of the charts (Theorem 4.40(c)) choose a cut ξ_* so late that the scale-free state of Y_e at ξ_* is $\mathsf{S}_{\xi_*}(p_{\mathrm{c}})$ for a coordinate p_{c} in the domain of the chart; let K be a small closed ball around p_{c} in it. Since S_{ξ_*} is an injective C^∞ immersion, $\mathsf{S}_{\xi_*}|_K$ is a C^∞ embedding of a manifold with boundary of dimension 5 into $\mathbb{R}^6$ (shrink K): there are a neighbourhood $\mathcal{V}$ of $s_e := \mathsf{S}_{\xi_*}(p_{\mathrm{c}})$ and a C^∞ function $s : \mathcal{V} \to \mathbb{R}$ with $\mathrm{d}s \neq 0$ and $\{s = 0\} = \mathsf{S}_{\xi_*}(\operatorname{int} K) \cap \mathcal{V}$. Let $\Theta(p, \gamma) \in \mathbb{R}^6$ be the scale-free state at ξ_* of the solution $Y(\cdot; p, \gamma)$, i.e. the composition of the flow from x_0 to $x_* = \tanh \xi_*$ with the affine isomorphism of Remark 5.14; Θ is C^∞ near $(p_0, \gamma^-(p_0))$ (the edge solution exists on $[x_0, x_*]$), and $\partial_\nu \Theta$ is the state of the variation field of J_ν at ξ_* (Remark 7.9). Put $H(p, \gamma) = s(\Theta(p, \gamma))$.

Implicit function theorem. $H(p_0, \gamma^-(p_0)) = 0$ and $\partial_\gamma H = \mathrm{d}s(\partial_\gamma \Theta) \neq 0$, because $\partial_\gamma \Theta$ is not tangent to $\mathsf{S}_{\xi_*}(K)$ (Proposition 7.7) and $\ker \mathrm{d}s = T\{s = 0\}$. So there are a neighbourhood N of p_0 and a C^∞ function $\hat g$ on N with $H(p, \hat g(p)) = 0$, $\hat g(p_0) = \gamma^-(p_0)$ (Theorem A.3).

Identification with the threshold. For $p \in N$ the state $\Theta(p, \hat{g}(p))$ lies on $\mathsf{S}_{\xi_*}(\operatorname{int} K)$, so it is the scale-free state at ξ_* of a chart solution of type c defined on $[\xi_*, \infty)$; by uniqueness (Theorem A.4: the scale-free state determines the solution) the solution $Y(\cdot; p, \hat{g}(p))$ coincides on $[x_*, 1)$ with it: it is forward global, so $\hat{g}(p) \in \mathrm{FS}(p)$, $\hat{g}(p) \geq \gamma^-(p)$, and it is of type c, hence not of type G, so $\hat{g}(p) \not> \gamma^-(p)$ by Lemma 5.46(a). Thus $\hat{g} = \gamma^-$ on N, the edge solution of every $p \in N$ has type c, and (i) holds. (Here $N \subset \mathcal{P}$ since $\hat{g}(p) \in \mathrm{FS}(p)$.)

(ii), (iii). Differentiate $H(p, \hat{g}(p)) = 0$ in ν: $\mathrm{d}s\big(\partial_\nu \Theta + \partial_\nu \hat{g}\, \partial_\gamma \Theta\big) = 0$, i.e. $\mathsf{J}_\nu + \partial_\nu \hat{g}\, \mathsf{J}_\gamma$ is tangent to $\mathsf{S}_{\xi_*}(K)$; by the uniqueness of s_ν in Proposition 7.7, $\partial_\nu \hat{g} = s_\nu$, and the formulas and signs of Proposition 7.7 are (ii) and (iii).

(iv). Let $\gamma > \gamma^- = \gamma^-(p)$. By Corollary 5.29(a) (strict form) the solutions Y_e and $\hat{Y} = Y(\cdot; p, \gamma)$, which have the same $\boldsymbol{c}$ and κ, satisfy $\hat{\psi} \leq \psi$ and $\hat{\psi}_x \leq \psi_x$ on $[x_0, 1)$ ($\psi = P_{\boldsymbol{c}} - Q$), so $\Delta Q := \hat{Q} - Q = \psi - \hat{\psi} \geq 0$ has $\partial_x \Delta Q \geq 0$ and $q(\gamma) - q(\gamma^-) = \lim_{x \to 1} \Delta Q \geq \Delta Q(x_1)$ for any $x_1 \in (x_0, 1)$. By Lemma 7.3(c), $\delta Q_\gamma(x_1; p, \gamma') \geq m_1 > 0$ uniformly for p near p_0, γ' near $\gamma^-(p_0)$; hence $\Delta Q(x_1) = \int_{\gamma^-}^{\gamma} \delta Q_\gamma(x_1; p, \gamma') \mathrm{d}\gamma' \geq m_1(\gamma - \gamma^-)$. For an L_1 edge $q(\gamma^-) = -2$ and $\mu = \sqrt{1 + q/2}$. □

Remark 7.13. (1) The linear lower bound in (iv) is rigorous; the matching upper bound $q + 2 \leq C(\gamma - \gamma^-)$ with $C = \ell_\gamma = \lim \delta Q_\gamma$ is only observed (Numerical observation 7.15), and at a wall it is exactly the content of Theorem 7.32(a). (2) The sign of $\partial_{x_0} \gamma^-$ is not constrained by the cooperative structure: the base point moves along a solution that is not monotone in x_0 (in Numerical observation 7.15 it is negative in both examples). (3) The proof is the same for the two types; what differs is only the functional (v or δQ) that detects tangency.

Corollary 7.14 (smoothness of the tip)**.** *Let $\kappa_* = \kappa_{\max}(x_0, \boldsymbol{c})$ be the tip, $p_* = (x_0, \boldsymbol{c}, \kappa_*)$, and suppose the lower edge solution at p_* and the lower edge solution at the mirror datum $(-x_0, \boldsymbol{c}_m, \kappa_*)$ are each of type R or L_1. Then $\kappa_{\max}$ is C^∞ near $(x_0, \boldsymbol{c})$, the two walls of the lens meet transversally, $\partial_\kappa \gamma^- > 0 > \partial_\kappa \gamma^+$, and for $i = 1, 2, 3$*

$$\partial_{c_i} \kappa_{\max} = -\frac{\partial_{c_i} \gamma^- - \partial_{c_i} \gamma^+}{\partial_\kappa \gamma^- - \partial_\kappa \gamma^+} > 0.$$

Proof. By Theorem 7.12 γ^- is C^∞ near p_* with $\partial_\kappa \gamma^- > 0$, $\partial_{c_i} \gamma^- < 0$; by the mirror relation $\gamma^+(x_0, \boldsymbol{c}, \kappa) = -\gamma^-(-x_0, \boldsymbol{c}_m, \kappa)$ (Lemma 5.40) so is γ^+, with $\partial_\kappa \gamma^+ < 0$ and $\partial_{c_i} \gamma^+ = -\partial_{c_{i'}} \gamma^-(-x_0, \boldsymbol{c}_m, \kappa) > 0$, i' the mirrored index ($1 \leftrightarrow 2$, $3 \leftrightarrow 3$). So $\Delta = \gamma^- - \gamma^+$ is C^∞ with $\partial_\kappa \Delta > 0$ and $\partial_{c_i} \Delta < 0$. The tip is the zero of Δ (Proposition 5.59(b)); the implicit function theorem gives smoothness and the formula, whose sign is that of $-\partial_{c_i} \Delta / \partial_\kappa \Delta > 0$. □

Numerical observation 7.15 (the variational formulas against finite differences; the linear law)**.** *All derivatives of this observation are computed by* `verify_ch07a.py` *from the Jacobi fields of the shooting code at $s_{\max} = 30$ ($s = -\ln(1 - x)$), and compared with central differences of γ^- computed by bisection ($h = 10^{-4}$). The L_1 datum is $\boldsymbol{c} = 0$, $\kappa = 1$, $x_0 = 0$; the R datum is $\boldsymbol{c} = (0, -0.9, 0)$, $\kappa = 0.3$, $x_0 = 0$.*

datum	γ^-	$\partial_\kappa \gamma^-$	$\partial_{c_1} \gamma^-$	$\partial_{c_2} \gamma^-$	$\partial_{c_3} \gamma^-$	$\partial_{x_0} \gamma^-$
L_1, *variational*	−1.78060	0.36573	−0.28800	−0.75292	−0.35800	−2.36012
L_1, *finite diff.*		0.36573	−0.28800	−0.75292	−0.35800	−2.36012
R, *variational*	−1.01859	0.27756	−0.44048	−2.36711	−0.59229	−1.54834
R, *finite diff.*		0.27813	−0.44044	−2.36843	−0.59222	−1.54857

The signs of Theorem 7.12(iii) are those observed. The variational values at the R *edge are read at* $s_{\max} = 30$*, where the ratios* $\delta U_\nu/\delta U_\gamma$ *differ from the finite-difference values by at most* 0.2%*. For the linear law of (iv): at* $\boldsymbol{c} = 0$*,* $\kappa = 1$ *the quotient* $(q+2)/(\gamma - \gamma^-)$ *equals* 12.128, 12.195, 12.201 *for* $\gamma - \gamma^- = 10^{-3}, 10^{-4}, 10^{-5}$*, so* $q + 2 \approx 12.20\,(\gamma - \gamma^-)$*: the lower bound is sharp up to a constant, and the observed limit is* $\ell_\gamma = \lim \delta Q_\gamma$ *(twice the* γ*-derivative of* μ^2*).*

7.4. Class N: smoothness of the shift thresholds

In words. *A solution of class* N_0 *is a germ* S *at the south pole pushed along its orbit by the* t^2*-shift* T_c *(Chapter 6); it is global once* c *is at least the threshold* $\alpha_-(S)$*, and the global solution at* $c = \alpha_-(S)$ *is an edge solution at the north pole. The north edge has exactly the same types as for class* Z*, and the shift plays the role of* γ*: a positive shift lifts the state in the order of Chapter 5. Its variation field has a Wronskian bounded below by an explicit integral and a potential variation bounded below by* t^2*, so it is transversal to the families of pole solutions, and the implicit function theorem gives* α_- *smooth wherever the edge is of type* R *or* L_1*. The same holds for the recessive shift, with the threshold measured along each orbit; smoothness of the base function* Θ^- *in the coordinates of the section is not proved (Remark 7.20).*

7.4.1. Smooth dependence of the shifted germ. Let $\mathfrak{P} \subset \mathbb{R}^m$ be open. A *chart family* is a family $p \mapsto S_p \in \mathcal{G}_{\mathsf{N}_0}$, $p \in \mathfrak{P}$, of germs at the south pole given by the coordinates p of a pole chart, of type G (Theorem 4.38) or of type L_1 (Theorem 4.40), mirrored by σ_2 to $x = -1$. Every germ of type G or L_1 belongs to such a family (Theorem 4.38(c),(d), Theorem 4.40(c),(d)). Write $E = E = e^{2\xi}$, fix $\xi_1 \le 0$, and let $z_p = z_{S_p,\xi_1}$ be the representative with $z_p(\xi_1) = 1$, with swirl constant κ_p. Put

$$k_p = \kappa_p z_p^{-2}\ (= f_p'), \qquad r_p(\eta,\zeta) = \frac{z_p(\zeta)^2}{z_p(\eta)^2} \quad (\zeta \le \eta \le \xi_1),$$

$$l_p(\xi) = \int_{-\infty}^{\xi} z_p^{-2}.$$

Writing out (6.4) (only κz^{-2} and $z^{-2}\cdot z^2$ occur),

$$\begin{aligned} \mathcal{F}_{c,C}(u)(\xi) &= C\, l_p(\xi) + \int_{-\infty}^{\xi}\int_{-\infty}^{\eta} r_p(\eta,\zeta)\,\varepsilon(\zeta)\,(1+u(\zeta))\mathrm{d}\zeta\mathrm{d}\eta, \\ \varepsilon &= \tfrac{1}{2}(ct^2 + R), \quad R = \mathsf{G}_{\mathsf{L}}[F], \quad F = f_p^2 - \tilde f^2, \quad \tilde f = f_p + \Delta f, \\ \Delta f(\xi) &= \int_{-\infty}^{\xi} k_p(\eta)\, n(u(\eta))\mathrm{d}\eta, \quad n(s) = (1+s)^{-2} - 1. \end{aligned} \tag{7.3}$$

A chart family has the following three properties, which are all that is used below.

(C1) *Envelope.* For every compact $K \subset \mathfrak{P}$ the germs S_p, $p \in K$, have a common envelope datum on $(-\infty, \xi_1]$ (Lemma 6.11, with a finite cover of K): constants $m_\pm, k_0, f_{\max}$ and a rate $\bar\mu < 2$ with $|k_p| \le k_0$, $|f_p| \le f_{\max}$ and $r_p(\eta,\zeta) \le \rho_0(1+\eta-\zeta)^2 e^{2\bar\mu(\eta-\zeta)}$, $\rho_0 = (m_+/m_-)^2$.

(C2) *Smoothness with polynomial loss.* The functions $k_p(\eta)$, $z_p(\eta)^{-2}$, $r_p(\eta,\zeta)$, $f_p(\xi)$ are C^∞ in p and, for every compact K, every $\hat\mu > \bar\mu$ and every multi-index α,

there is K_α with

$$|\partial_p^\alpha k_p(\eta)|,\ |\partial_p^\alpha z_p(\eta)^{-2}| \le K_\alpha(1+|\eta|)^{|\alpha|}E(\eta)^d,$$
$$|\partial_p^\alpha r_p(\eta,\zeta)| \le K_\alpha e^{2\hat\mu(\eta-\zeta)}, \qquad |\partial_p^\alpha f_p(\xi)| \le K_\alpha(1+|\xi|)^{|\alpha|},$$

with $d=0$; and with $d=\nu$ if $\mu(S_p)\ge\nu$ on K and the germs are of type G.

(C3) *State.* $p\mapsto(z_p,f_p,Q_p)$ is C^∞ together with all ξ-derivatives; in particular the scale-free state $(h,k,f,Q,\dot Q,\lambda)(S_p)(\xi_2)$ of Proposition 6.4 is C^∞ in p.

Verification. (C3) is Theorem 4.38(b) and Theorem 4.40(b). (C1) is Lemma 6.11 together with the uniform constants of Corollary 4.42(c). For (C2), type G: by Theorem 4.38(b) and the proof of its Step 4, $z_p = e^{-\mu(\xi-\xi_1)}\hat z_p(\xi)/\hat z_p(\xi_1)$ with $p\mapsto\hat z_p$ a C^∞ map into the bounded continuous functions, $\hat z_p\in[\frac12 n_0,2n_1]$ and all p-derivatives bounded. Hence $r_p(\eta,\zeta)=e^{2\mu(\eta-\zeta)}\hat z_p(\zeta)^2/\hat z_p(\eta)^2$ and $k_p=\kappa_{\mathrm{ch}}e^{2\mu\eta}\hat z_p(\eta)^{-2}$ (κ_{ch} the chart coordinate), and $\partial_\mu^j e^{2\mu s}=(2s)^j e^{2\mu s}$ gives $|\partial_p^\alpha r_p|\le K(1+\eta-\zeta)^{|\alpha|}e^{2\mu(\eta-\zeta)}\le K'e^{2\hat\mu(\eta-\zeta)}$ for $\hat\mu>\mu$, and $|\partial_p^\alpha k_p|\le K(1+|\eta|)^{|\alpha|}e^{2\mu\eta}\le K(1+|\eta|)^{|\alpha|}E^\nu$ (same for z_p^{-2}); $f_p=f_-+\int_{-\infty}^{\xi}k_p$ has bounded p-derivatives. Type L_1: by Theorem 4.40, $z_p\in C_m$ ($m=1+|\xi|$) depends on p through a Neumann series, so $|\partial_p^\alpha z_p|\le Km$, $z_p\ge cm$, and $|\partial_p^\alpha r_p|\le K(m(\zeta)/m(\eta))^2\le K(1+\eta-\zeta)^2\le K'e^{2\hat\mu(\eta-\zeta)}$, $|\partial_p^\alpha k_p|\le Km^{-2}\le K$; f_p is bounded with bounded derivatives.

Lemma 7.16 (smooth dependence of the shifts)**.** *Let $(S_p)_{p\in\mathfrak P}$ be a chart family and $p_0\in\mathfrak P$.*

(a) (t^2*-shift.*) *Let the germs be of type L_1 or of type G with $\mu<2$, and let $c_0\in\mathbb R$. There are $\xi_2\le\xi_1$, a ball $P\ni p_0$, an open interval $I\ni c_0$ and $\hat B>0$ such that, with $\mathcal X$ the Banach space of continuous functions on $J=(-\infty,\xi_2]$ with the norm $\|u\|_w=\sup_J|u|E^{-(2-\delta)}$ (δ of Proposition 6.18), the fixed point $u(p,c)\in\{|u|\le\hat BE^2\}$ of $\mathcal F_{c,0}$ is a C^∞ map $P\times I\to\mathcal X$.*

(b) (*Recessive shift.*) *Let the germs be of type G with $\mu>\frac12$ and $C_0\in\mathbb R$. The same holds for the fixed point $u(p,C)\in\{|u|\le\hat B_AE^\nu\}$ of $\mathcal F_{0,C}$, $\nu=\min_P\mu>\frac12$, with the norm $\sup|u|E^{-1/2}$.*

Consequently the scale-free state at ξ_2 of the shifted germ (T_cS_p, resp. $\mathcal R_CS_p$) is C^∞ in (p,c) (resp. (p,C)), and $(p,c,\xi)\mapsto(U,f,Q)(T_cS_p)(\xi)$, resp. $(p,C,\xi)\mapsto(U,f,Q)(\mathcal R_CS_p)(\xi)$, is C^∞ on the open set of parameters and ξ in the maximal interval of existence of the solution.

Proof. We write the proof for (a) and indicate at the end of each step what changes in (b).

Step 0: constants and domain. Let $P_1\ni p_0$ be a ball with $K=\bar P_1\subset\mathfrak P$ compact, $\bar c=|c_0|+2$. With the common envelope datum (C1) of K, Proposition 6.18 provides $\hat B$, δ, $a_1=\rho_0\mathrm M(4-2\delta-2\bar\mu)$ ($\mathrm M(r)=\int_0^\infty(1+s)^2e^{-rs}\mathrm ds$), $\hat\Phi=\frac52k_0\bar f\hat B$, $\bar f=2f_{\max}+2$, and $\hat\xi_2$ such that for $\xi_2\le\hat\xi_2$, all $p\in K$, $|c|\le\bar c$ the map $\mathcal F_{c,0}$ maps $X=\{|u|\le\hat BE^2\}$ into itself and is a $\frac12$-contraction for $\|\cdot\|_w$ with unique fixed point $u(p,c)\in X$; and (6.5) holds. Decrease ξ_2 (this keeps all of this) so that moreover

$$E_2^{2-\delta}\le\tfrac14, \qquad \hat BE_2^\delta\le\tfrac12 \qquad (E_2=e^{2\xi_2}), \tag{7.4}$$

possible because $\delta>0$. Put $P=P_1$, $I=(c_0-1,c_0+1)$, $\mathcal X=(\{u\in C(J):\|u\|_w<\infty\},\|\cdot\|_w)$, $\mathcal O=\{u\in\mathcal X:\|u\|_w<1\}$, $V=\mathbb R^m\times\mathbb R\times\mathcal X$ with $\|(h_p,h_c,h_u)\|=|h_p|+|h_c|+\|h_u\|_w$ and $D=P\times I\times\mathcal O$, a convex open subset of V. For $u\in\mathcal O$,

$|u(\xi)| < E^{2-\delta} \le \frac14$. And $X \subset \mathcal{O}$: $\|u\|_w \le \hat B \sup_J E^\delta \le \frac12$ for $u \in X$. This is the point that the set X itself has *empty interior* in $\mathcal{X}$; we shall use $\mathcal{O} \supset X$ as domain and never need the interior of X. The map to be studied is $T(q) = \mathcal{F}_{c,0}(u)$ of (7.3), $q = (p, c, u) \in D$.

Step 1: a calculus of classes. Fix $k \ge 0$. For $a \ge 0$ let $\mathfrak{C}_a$ be the set of maps $\Phi : D \to C(J)$ such that for every $\xi \in J$ the function $q \mapsto \Phi(q)(\xi)$ is of class C^{k+1} on D and

$$\begin{gathered} |\mathrm{D}^j\Phi(q)[h^j](\xi)| \le K_j(1+|\xi|)^j E(\xi)^a \|h\|^j \\ (q \in D,\ h \in V,\ 0 \le j \le k+1,\ \xi \in J). \end{gathered} \tag{7.5}$$

(For $j = 0$ this is $|\Phi(q)| \le K_0 E^a$.) The exponent j of $(1+|\xi|)$ is the number of derivatives: each derivative in a direction of p may cost one power of $|\xi|$, by (C2). Note that $(1+|\xi|)^j E^a \le C_{j,\beta} E^{a-\beta}$ for every $\beta > 0$. The following rules are used.

(R1) products: $\Phi \in \mathfrak{C}_a$, $\Psi \in \mathfrak{C}_b \Rightarrow \Phi\Psi \in \mathfrak{C}_{a+b}$; and $\mathfrak{C}_a \subset \mathfrak{C}_{a'}$ for $a \ge a'$ (as $E \le 1$). Indeed Leibniz' rule gives $\mathrm{D}^j(\Phi\Psi)[h^j] = \sum_i \binom{j}{i} \mathrm{D}^i\Phi[h^i]\,\mathrm{D}^{j-i}\Psi[h^{j-i}]$, bounded by $\sum_i \binom{j}{i} K_i K'_{j-i}(1+|\xi|)^j E^{a+b}\|h\|^j$.

(R2) integration: Let $\Gamma_p(\xi,\eta)$, $\eta \le \xi$, be C^{k+1} in p with $|\partial_p^\alpha \Gamma_p(\xi,\eta)| \le K_\alpha(1+|\eta|)^{|\alpha|} E(\eta)^d e^{b(\xi-\eta)}$ ($|\alpha| \le k+1$; $b, d \ge 0$), let $\Phi \in \mathfrak{C}_a$ and $2(a+d) > b$. Then $\Xi(q)(\xi) = \int_{-\infty}^{\xi} \Gamma_p(\xi,\eta)\Phi(q)(\eta)\mathrm{d}\eta$ belongs to $\mathfrak{C}_{a+d}$. Indeed, differentiating under the integral sign (the integrands of the derivatives of order $\le k+1$ have the integrable majorants below, uniformly in $q \in D$) gives $\mathrm{D}^j\Xi[h^j] = \sum_i \binom{j}{i} \int \mathrm{D}_p^i\Gamma[h_p^i]\,\mathrm{D}^{j-i}\Phi[h^{j-i}]\mathrm{d}\eta$, bounded by $C\|h\|^j \int_{-\infty}^{\xi}(1+|\eta|)^j E(\eta)^{a+d} e^{b(\xi-\eta)}\mathrm{d}\eta$. With $1+|\eta| \le (1+|\xi|)(1+\xi-\eta)$ and $E(\eta) = E(\xi)e^{-2(\xi-\eta)}$ the integral is at most $(1+|\xi|)^j E(\xi)^{a+d}\int_0^\infty (1+s)^j e^{-(2(a+d)-b)s}\mathrm{d}s$.

(R3) Green operator: If $\Phi \in \mathfrak{C}_a$ with $a > 1$, then $\mathsf{G}_\mathsf{L}[\Phi] : q \mapsto \mathsf{G}_\mathsf{L}[\Phi(q)]$ belongs to $\mathfrak{C}_{a+1}$, and so does $\dot R = (1-x^2)\partial_x \mathsf{G}_\mathsf{L}[\Phi]$. Indeed $\mathrm{D}^j \mathsf{G}_\mathsf{L}[\Phi][h^j] = \mathsf{G}_\mathsf{L}[\mathrm{D}^j\Phi[h^j]]$ (differentiation under the integral, majorant from (2.36)), and by (2.36) $|\mathsf{G}_\mathsf{L}[g](\xi)| \le 4E(\xi)^2 \int_{-\infty}^{\xi} |g|\, E^{-1}\mathrm{d}\eta$; for $\dot R$ the kernel $(1-x^2)\partial_x K_\mathsf{L}(x,s)(1-s^2) = (1-x^2)\frac{(x-s)^2+1-x^2}{1-s^2} \le \frac{6t_x^2}{t_s} \le 24\, E(\xi)^2 E(\eta)^{-1}$ ($t = 1+x \le 2E$, $t \ge E$) gives the same bound with 24 in place of 4. With $|g| \le K(1+|\eta|)^j E^a$ the integral is $\le K(1+|\xi|)^j E(\xi)^{a-1}\int_0^\infty (1+s)^j e^{-2(a-1)s}\mathrm{d}s$.

The ingredients. (i) $N(q) = n \circ u$ belongs to $\mathfrak{C}_{2-\delta}$: $\mathrm{D}^j N[h^j](\xi) = n^{(j)}(u(\xi))\, h_u(\xi)^j$, $|n^{(j)}| \le K_j$ on $|s| \le \frac14$, $|h_u(\xi)| \le \|h\| E^{2-\delta}$, so for $j \ge 1$ the bound is $K_j\|h\|^j E^{j(2-\delta)} \le K_j \|h\|^j E^{2-\delta}$, and for $j = 0$, $|n(u)| \le 4|u| \le 4E^{2-\delta}$. (ii) $f_p \in \mathfrak{C}_0$ and $1 \in \mathfrak{C}_0$, $1+u \in \mathfrak{C}_0$, $c \in \mathfrak{C}_0$ (by (C1), (C2) and (7.4)), $t^2 \in \mathfrak{C}_2$ ($t^2 \le 4E^2$ and it does not depend on q), so $ct^2 \in \mathfrak{C}_2$ by (R1). (iii) The kernels $k_p(\eta)$ [with $b = 0$, $d = 0$], $r_p(\eta,\zeta)$ [with $b = 2\hat\mu$, $d = 0$] and $z_p^{-2}(\eta)$ satisfy the hypothesis of (R2) by (C2).

Step 2: T is of class C^k. By (R2) with $\Gamma_p = k_p$ ($2(2-\delta) > 0$): $\Delta f \in \mathfrak{C}_{2-\delta}$. By (R1), $F = -\Delta f\,(2f_p + \Delta f) \in \mathfrak{C}_{2-\delta}$. Since $2-\delta > 1$, (R3) gives $R \in \mathfrak{C}_{3-\delta} \subset \mathfrak{C}_2$ ($\delta \le \frac14$), so $\varepsilon = \frac12(ct^2 + R) \in \mathfrak{C}_2$ and $g = \varepsilon(1+u) \in \mathfrak{C}_2$. Finally (R2) twice, first with $\Gamma = r_p$, $b = 2\hat\mu$, $a = 2$ (we choose $\hat\mu = \frac12(\bar\mu + 2) < 2$, so $2\hat\mu < 4 = 2a$), then with $\Gamma \equiv 1$, $b = 0$, gives $T \in \mathfrak{C}_2$. By (7.5) and $(1+|\xi|)^j E^2 \le C_j E^{2-\delta}$,

$$|\mathrm{D}^j T(q)[h^j](\xi)| \le K'_j\, E(\xi)^{2-\delta}\|h\|^j \qquad (0 \le j \le k+1).$$

Lemma A.15 with the weight $\mathfrak{w} = E^{2-\delta}$ (so that $\mathcal{X}_{\mathfrak{w}} = \mathcal{X}$) gives $T \in C^k(D;\mathcal{X})$, and the derivatives of T are the pointwise ones. Since k is arbitrary and D does not depend on k, T is C^∞. *In case* (b)*:* replace $\mathcal{X}$ by the space with the norm $\sup|u|E^{-1/2}$, δ by $\frac12$ in the definition of $\mathcal{O}$ ((7.4) becomes $E_2^{1/2} \le \frac14$, $\hat B_A E_2^{\nu-1/2} \le \frac12$), and use $d = \nu$ in (C2). Then $N \in \mathfrak{C}_{1/2}$, $\Delta f \in \mathfrak{C}_{\nu+1/2}$ (R2 with $a = \frac12$, $d = \nu$), $F \in \mathfrak{C}_{\nu+1/2}$, $R \in \mathfrak{C}_{\nu+3/2}$ (R3, $\nu + \frac12 > 1$), $\varepsilon(1+u) = \frac12 R(1+u) \in \mathfrak{C}_{\nu+3/2}$, and the double integral with $\hat\mu = \bar\mu + \frac14$ is in $\mathfrak{C}_{\nu+3/2}$ (need $2\hat\mu < 2\nu+3$; we shrink P so that $\bar\mu = \max_P \mu < \nu + \frac12$). Moreover $C\,l_p \in \mathfrak{C}_\nu$ (R2 with $\Phi \equiv 1$, $\Gamma = z_p^{-2}$, $d = \nu$, $b = 0$, and (R1)). So $T \in \mathfrak{C}_\nu$ and $(1+|\xi|)^j E^\nu \le C_j E^{1/2}$ gives $T \in C^\infty(D;\mathcal{X})$ as before.

Step 3: the linearised contraction. We claim $\|\mathrm{D}_u T(p,c,u^*)\|_{\mathcal{X}\to\mathcal{X}} \le \frac12$ for $(p,c) \in P \times I$ and $u^* = u(p,c) \in X$. (This replaces hypothesis (iii) of Theorem A.11, which does not hold here; the hypothesis (ii) of that theorem only bounds differences of elements of X, and $u^* + th \notin X$ in general.) Let $\|h\|_w \le 1$. We differentiate the chain of Step 2 in the direction $(0,0,h)$ and bound each link, using $|u^*| \le \hat B E^2 \le \frac12$ (so $1 + u^* \in [\frac12, \frac32]$) and the bounds of Steps 2–3 of the proof of Proposition 6.18 for $u^* \in X$: $|\tilde f| \le f_{\max} + 1$ (so $|2\tilde f| \le \bar f$), $|\varepsilon^*| \le 2\bar c E^2 + \hat\Phi E^3$. First $|n'(u^*)h| \le 16|h| \le 16E^{2-\delta}$ ($n'(s) = -2(1+s)^{-3}$). Then $|\mathrm{d}\Delta f| \le 16k_0 \int_{-\infty}^{\xi} E^{2-\delta} = \frac{16k_0}{4-2\delta}E^{2-\delta}$; $\mathrm{d}F = -2\tilde f\,\mathrm{d}\Delta f$, so $|\mathrm{d}F| \le \Phi_w E^{2-\delta}$ with $\Phi_w = \frac{16k_0\bar f}{4-2\delta}$; by Lemma 2.23 with $Z \equiv 1$, $q = 2-\delta > 1$: $|\mathrm{d}R| \le \frac{2\Phi_w}{1-\delta}E^{3-\delta}$. Now $\mathrm{d}(\varepsilon(1+u)) = \frac12 \mathrm{d}R\,(1+u^*) + \varepsilon^* h$, whence, with $E^{3-\delta} \le E_2 E^{2-\delta}$ and $|\varepsilon^* h| \le (2\bar c E_2^2 + \hat\Phi E_2^3)E^{2-\delta}$,

$$|\mathrm{d}(\varepsilon(1+u))| \le \Big[\frac{24k_0\bar f E_2}{(1-\delta)(4-2\delta)} + 2\bar c E_2^2 + \hat\Phi E_2^3\Big]E^{2-\delta} = \Theta\,E^{2-\delta},$$

the quantity Θ of (6.5). By (C1), for $|g| \le GE^{2-\delta}$: $|\int_{-\infty}^{\eta} r_p(\eta,\zeta)g(\zeta)\mathrm{d}\zeta| \le \rho_0 G\int_0^\infty (1+s)^2 e^{2\bar\mu s}e^{(4-2\delta)(\eta-s)}\mathrm{d}s = a_1 GE(\eta)^{2-\delta}$ ($4 - 2\delta - 2\bar\mu \ge 2 - \bar\mu > 0$), and then $\int_{-\infty}^{\xi} E^{2-\delta} = E^{2-\delta}/(4-2\delta)$. Hence $\|\mathrm{D}_u T\,h\|_w \le \frac{a_1\Theta}{4-2\delta} \le \frac12$ by the fifth condition of (6.5). *In case* (b)*:* the same chain is the linearisation of the "weak Lipschitz bound" in the proof of Proposition 6.21: there Δu is replaced by h ($|h| \le E^{1/2}$), the three facts used are $|n'(u^*)h| \le 16|h|$, $|\mathrm{d}F| \le \bar f|\mathrm{d}\Delta f|$ and the identity $\mathrm{d}(\varepsilon(1+u)) = \frac12 \mathrm{d}R(1+u^*) + \varepsilon^* h$ with $\varepsilon^* = \frac12 R^*$, $u^* \in X_A$, which are the linearisations of the corresponding facts there (with the same constants), and the two integrations give $\|\mathrm{D}_u T_A h\|_A \le (\Lambda_1 + \Lambda_2) \le \frac12$ by (6.7).

Step 4: the fixed point. (Continuity.) For $q = (p,c), q' = (p',c')$ in $P \times I$ and $u = u(q)$, $u' = u(q') \in X$, the contraction of Proposition 6.18 gives $\|u - u'\|_w \le \frac12\|u - u'\|_w + \|T(q,u') - T(q',u')\|_w$ (both $u, u' \in X$), so $\|u(q) - u(q')\|_w \le 2\|T(q,u(q')) - T(q',u(q'))\|_w \to 0$ as $q \to q'$, by the continuity of T (Step 2). (Implicit function.) Fix q_0 and $u_0 = u(q_0)$. The map $\mathcal{F}(q,u) = u - T(q,u)$ is C^∞ on $(P \times I) \times \mathcal{O}$, $\mathcal{F}(q_0,u_0) = 0$, and $\mathrm{D}_u\mathcal{F} = \mathrm{I} - \mathrm{D}_u T$ is invertible by Step 3 and Lemma A.1. Theorem A.3 gives neighbourhoods $Q_0 \ni q_0$, $\mathcal{O}_0 \ni u_0$ and a C^∞ map $\tilde u : Q_0 \to \mathcal{O}_0$ whose graph is $\{\mathcal{F} = 0\} \cap (Q_0 \times \mathcal{O}_0)$. By the continuity just proved, $u(q) \in \mathcal{O}_0$ for q near q_0, and $\mathcal{F}(q,u(q)) = 0$; so $u = \tilde u$ near q_0. Thus $u \in C^\infty(P \times I;\mathcal{X})$, which is (a), (b).

Step 5: the state at ξ_2. By Proposition 6.18 and Lemma 6.17, $T_c S_p$ is the germ with $\tilde z = z_p(1+u)$, $\tilde f = f_p + \Delta f$, $\tilde Q = Q_p + ct^2 + R$ and $z_p^2 u' = C + \int_{-\infty}^{\xi} z_p^2 \varepsilon(1+u)$, i.e. $u'(\xi) = C z_p(\xi)^{-2} + \Xi_1(\xi)$ with $\Xi_1(\eta) = \int_{-\infty}^{\eta} r_p(\eta,\zeta)g(\zeta)\mathrm{d}\zeta$ ($C = 0$ in (a)). The

scale-free state at ξ_2 is therefore

$$\tilde h = h_p + \frac{u'}{1+u}, \quad \tilde k = \frac{k_p}{(1+u)^2}, \quad \tilde f = f_p + \Delta f, \quad \tilde\lambda = \lambda_p,$$
$$\tilde Q = Q_p + ct^2 + R, \quad \dot{\tilde Q} = \dot Q_p + c\,\tfrac{\mathrm d}{\mathrm d\xi}t^2 + \dot R,$$

all evaluated at ξ_2 ($\dot{} = \mathrm d/\mathrm d\xi = (1-x^2)\partial_x$; $\tilde\lambda = \lambda_p$ because the shifted germ has the frame of S_p). The quantities $u(\xi_2)$, $u'(\xi_2)$, $\Delta f(\xi_2)$, $R(\xi_2)$, $\dot R(\xi_2)$ are values at the point ξ_2 of members of the classes $\mathfrak C_a$ (Steps 1–2; $\Xi_1 \in \mathfrak C_2$, $\dot R \in \mathfrak C_{3-\delta}$ by (R3)) evaluated at $q = (p, c, u(p,c))$; by definition of the classes, $q \mapsto \Phi(q)(\xi_2)$ is C^{k+1}, and $(p,c) \mapsto (p,c,u(p,c))$ is C^∞ by Step 4, and evaluation at ξ_2 is a bounded linear functional on $\mathcal X$. With (C3) all components are C^∞ in (p,c), for every k. The solution of the Q-system with this state at ξ_2 is the shifted germ (the scale-free state determines the solution, Theorem A.4), and it exists on $(-\infty, \xi_2]$; by the smooth dependence of ODE solutions on their data (Theorem A.4(c)) the map $(p,c,\xi) \mapsto (U,f,Q)(\xi)$ is C^∞ on the open set where the solution exists, which contains $(-\infty,\xi_2]$ and extends along the flow. □

Remark 7.17 (what was needed)**.** The domain of the implicit function theorem is $\mathcal O \supset X$, not X; the only information taken from the contraction estimates of Chapter 6 is that the fixed point lies in X and that the linearisation at it is a $\frac12$-contraction. Each p-derivative costs a factor $(1+|\xi|)$, which is absorbed by the gain E^δ between the weights E^2 and $E^{2-\delta}$ in (a), and by $E^{\nu-1/2}$ in (b); this is the reason for the margin δ in Proposition 6.18.

Lemma 7.18 (the shift Jacobi field)**.** *Let $S \in \mathcal G_{\mathsf N_0}$ be of type $\mathsf G$ with $\mu < 2$ or of type $\mathsf L_1$ and $c \in \mathbb R$, let (z_c, f_c, Q_c) be T_cS, and let $\delta = \partial_c$ be the variation field of the family $c \mapsto T_cS$ (it exists by Lemma 7.16, applied to a chart family through S). Then for $x \in (-1,1)$ on the existence interval of T_cS:*

$$\delta Q_c \;\ge\; t^2 \;(t = 1+x), \qquad v_c' = \tfrac12 z_c^2 \delta Q_c > 0,$$
$$v_c := \tfrac12 z_c^2\,\delta U_c \;\ge\; \tfrac12 \int_{-\infty}^{\xi} t^2 z_c^2 \,\mathrm d\eta > 0.$$

For the recessive shift $\mathcal R_C$ (type $\mathsf G$, $\mu > \frac12$), with $\delta = \partial_C$: $v_C \ge 1$, $\partial_x \delta Q_C \ge 0$ and $\delta Q_C > 0$.

Proof. By $T_{c+e}S = T_eT_cS$ (Theorem 6.19(c)) the variation at c is the variation at $e=0$ of $e \mapsto T_eS_c$, $S_c = T_cS$. For $e>0$ Proposition 6.23 gives $\tilde Q - Q_c \ge et^2$, hence $\delta Q_c = \lim_{e\downarrow 0} e^{-1}(\tilde Q - Q_c) \ge t^2$. Put $\tilde z = z_c(1+u_e)$ with u_e the shift function. By Lemma 7.1, $v_c = \mathrm{Wr}(z_c, \delta z) = \lim e^{-1}\,\mathrm{Wr}(z_c, \tilde z) = \lim e^{-1} z_c^2 u_e'$, and $z_c^2 u_e' = \int_{-\infty}^{\xi} z_c^2 \varepsilon (1+u_e)$ (Lemma 6.17) with $\varepsilon = \frac12(et^2 + R) \ge \frac12 et^2$ ($R \ge 0$, Proposition 6.23) and $1 + u_e \ge 1$: so $z_c^2 u_e' \ge \frac12 e \int_{-\infty}^{\xi} t^2 z_c^2$, finite by the envelope of z_c ($t^2 z_c^2 = O(e^{(4-2\mu)\eta})$, $\mu < 2$; for $\mathsf L_1$, $z \sim |\eta|$). The identity $v_c' = \frac12 z_c^2 \delta Q_c$ is Lemma 7.1. For the recessive shift, $z^2 u' = C + \int z^2 \varepsilon(1+u) \ge C$ with $\varepsilon = \frac12 R \ge 0$, $C = e$, gives $v_C \ge 1$; $\partial_x(\tilde Q - Q) = \partial_x R > 0$ and $\tilde Q - Q = R > 0$ (Proposition 6.23) give $\partial_x \delta Q_C \ge 0$ and $\delta Q_C \ge 0$, and $\delta Q_C > 0$ at one point (hence beyond) because $R_e = \mathsf G_{\mathsf L}[F_e]$ with $F_e = f^2 - \tilde f_e^2 > 0$ to first order in e ($\partial_e f_e < 0$, $f > 0$). □

7.4.2. The thresholds.

THEOREM 7.19 (smoothness of the shift thresholds). *Let $(S_p)_{p\in\mathfrak{P}}$ be a chart family (Section 7.4).*

(a) *Let the germs be of type G with $\mu<2$ or of type L_1, and suppose that the global germ $T_{\alpha_-(S_{p_0})}S_{p_0}$ is of type R or L_1 at $x=1$. Then $\alpha_-(S_p)$ is a C^∞ function of p near p_0, the north pole type of $T_{\alpha_-}S_p$ is constant, and for every coordinate p_j, $\partial_{p_j}\alpha_- = -\lim_{x\to1}\delta U_{p_j}/\delta U_c$ (R) resp. $-\lim_{x\to1}\delta Q_{p_j}/\delta Q_c$ (L_1), with variation fields along $T_{\alpha_-}S_p$. The set of edge germs of type G, $\mu<2$, is near $T_{\alpha_-}S_{p_0}$ a C^∞ hypersurface in the chart coordinates.*

(b) *Let the germs be of type G with $\mu>1$ (so $S_p\in\mathcal{G}_A$), let $\theta(S)=\Theta^-(\Pi(S))-\mathfrak{a}(S)$ be the least C for which $\mathcal{R}_CS$ is global (Theorem 6.50, defined for $\Pi(S)\in B$), and suppose that $\Pi(S_{p_0})\in B$ and $\mathcal{R}_\theta S_{p_0}$, $\theta=\theta(S_{p_0})$, is of type R or L_1 at $x=1$. Then $\Pi(S_p)\in B$ and $\theta(S_p)$ is a C^∞ function of p near p_0, the north pole type of $\mathcal{R}_\theta S_p$ is constant, $\partial_{p_j}\theta$ is given by the same ratio formulas with $\delta=\partial_C$ in place of ∂_c, and $\Gamma_{\mathsf{N}}\cap\mathcal{G}_A=\{\theta=0\}$ is near S_{p_0} a C^∞ hypersurface in the chart coordinates.*

PROOF. (a) Let $S_0^*=T_{\alpha_-(S_{p_0})}S_{p_0}$ and let $\mathsf{c}\in\{\mathsf{R},\mathsf{L}_1\}$ be its type at the north pole. Choose a late north cut ξ_* and an embedded chart piece $\mathsf{S}_{\xi_*}(K)$ with defining function s as in the proof of Theorem 7.12. Let $\Theta(p,c)$ be the scale-free state at ξ_* of T_cS_p; by Lemma 7.16(a) it is C^∞ near $(p_0,\alpha_-(S_{p_0}))$ (the germ S_0^* is global, so it exists up to ξ_*, and existence on a compact interval is an open condition). Put $H(p,c)=s(\Theta(p,c))$. The variation field ∂_c is not tangent to $\mathsf{S}_{\xi_*}(K)$: for $\mathsf{c}=\mathsf{R}$ Lemma 7.18 gives $v_c\ge v_c(\xi_1)>0$ (v_c nondecreasing) whereas $v\to0$ on tangent fields (Lemma 7.4); for $\mathsf{c}=\mathsf{L}_1$ it gives $\delta Q_c\ge t_1^2>0$ on $[x_1,1)$ whereas $\delta Q\to0$ on tangent fields (Lemma 7.5). So $\partial_cH\neq0$ and the implicit function theorem gives $\hat c(p)$, C^∞, with $\Theta(p,\hat c(p))\in\mathsf{S}_{\xi_*}(\operatorname{int}K)$. Then $T_{\hat c(p)}S_p$ is global of type c at $x=1$; hence $\hat c(p)\in I(S_p)=[\alpha_-(S_p),\infty)$ (Theorem 6.32(b)), and since T_cS_p is of type G at $x=1$ for $c>\alpha_-(S_p)$ (Theorem 6.32(c)), $\hat c(p)=\alpha_-(S_p)$. The formula for $\partial_{p_j}\alpha_-$ follows as in Theorem 7.12(ii): $\partial_{p_j}\Theta+\partial_{p_j}\hat c\,\partial_c\Theta$ is tangent, and the ratio of the limits exists because v_c, resp. δQ_c, is bounded below. For the last claim: $\alpha_-(T_aS)=\alpha_-(S)-a$ (Theorem 6.32(b)), so along the curve $a\mapsto T_aS_{p_0}$, which is C^∞ in the chart coordinates (Lemma 7.16(a) and the local diffeomorphism S_{ξ_*} of Theorem 4.38(d)), $\mathrm{d}\alpha_-\neq0$, and $\{\alpha_-=0\}$ is the set of edge germs.

(b) Put $\theta_0=\theta(S_{p_0})$ and let $S_0^*=\mathcal{R}_{\theta_0}S_{p_0}$, of type $\mathsf{c}\in\{\mathsf{R},\mathsf{L}_1\}$ at the north pole. Choose ξ_*, K, s as in (a), and let $\Theta(p,C)$ be the scale-free state at ξ_* of $\mathcal{R}_CS_p$; by Lemma 7.16(b) it is C^∞ near (p_0,θ_0). By Lemma 7.18 for the recessive shift, $v_C\ge1$ and δQ_C is nondecreasing and positive; for $\mathsf{c}=\mathsf{R}$ this contradicts tangency ($v\to0$, Lemma 7.4), and for $\mathsf{c}=\mathsf{L}_1$ it gives $\delta Q_C\ge\delta Q_C(x_1)>0$ on $[x_1,1)$, contradicting $\delta Q\to0$ on tangent fields (Lemma 7.5). So $\partial_CH\neq0$ for $H=s\circ\Theta$, and the implicit function theorem gives a C^∞ function $\hat C(p)$ with $\Theta(p,\hat C(p))\in\mathsf{S}_{\xi_*}(\operatorname{int}K)$. Thus $\mathcal{R}_{\hat C(p)}S_p$ is global of type c at $x=1$. By necessity (Theorem 6.46(a)) $\Pi(S_p)\in B$; by Theorem 6.50(b) the set of C with $\mathcal{R}_CS_p$ global is $[\theta(S_p),\infty)$, so $\hat C(p)\ge\theta(S_p)$, and by Theorem 6.50(c) $\mathcal{R}_CS_p$ is of type G at $x=1$ for $C>\theta(S_p)$, so $\hat C(p)=\theta(S_p)$. The ratio formulas follow as in (a), and $\{\theta=0\}=\Gamma_{\mathsf{N}}\cap\mathcal{G}_A$ by Theorem 6.50(c); it is a hypersurface because $\theta(\mathcal{R}_CS)=\theta(S)-C$ (Theorem 6.50(a)), so $\mathrm{d}\theta\neq0$ along the C^∞ orbit curve $C\mapsto\mathcal{R}_CS_{p_0}$. □

Remark 7.20. (1) Smoothness is claimed on each of the strata G and L_1 *at the south pole* separately: the south-pole charts are not unified across $\mu = 0$ (only continuity of α_- through $\mu = 0$ is proved, Theorem 6.32(d)). (2) Orbits and thresholds: $\partial_c \alpha_- = -1$ along an orbit, and the content of (a) is smoothness transversally to the orbits. (3) The north-pole type is a property of the *edge* germ; (a), (b) say that the set of edge germs of type R or L_1 is open in the set of edge germs. (4) *What is not proved.* The function $\Theta^- : B \to \mathbb{R}$ is continuous (Theorem 6.50) and equals $\theta \circ S$ for the section $y \mapsto S^y$ of Lemma 6.47; by (b) it is C^∞ at y_0 as soon as the section is a C^∞ family of germs in chart coordinates near y_0. Lemma 6.47(b) gives only the continuity of $y \mapsto S^y$, and smoothness of Θ^- as a function on B is not asserted; (b) is the statement about the edge as a hypersurface in the chart coordinates of the germs.

7.5. Small swirl, the continuity of $Q(1)$, and the wall of L_2 edges

In words. *Three further facts about the edge. (i) For $-1 < c_2 < -\frac{3}{4}$ and small κ the edge is of type R, because the no-swirl edge already is (with $\mu_0 = \sqrt{1+c_2} < \frac{1}{2}$) and the transversality of the γ-direction survives at $\kappa = 0$; the swirl enters only through κ^2, so γ^- is a smooth function of κ^2. (ii) The exponent at the pole, $q = Q(1)$, depends continuously on the data along the edge: it is smooth at R and L_1 edges, and at an L_2 edge it is squeezed between -2 and the Fatou bound. (iii) The only places where smoothness can fail are the walls of L_2 edges, where R turns into L_1. The unified corner chart shows that γ^- is then $C^{1,1}$ but not C^2 and computes the jump of the Hessian; in particular the wall cannot create concavity.*

7.5.1. Small swirl: R edges, smooth in κ^2.

THEOREM 7.21 (small swirl, R edges). *Let $x_0 \in (-1,1)$ and $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ with $-1 < c_2 < -\frac{3}{4}$, and put $\mu_0 = \sqrt{1+c_2} \in (0, \frac{1}{2})$. There are a neighbourhood N of $(x_0, \boldsymbol{c})$, a number $\kappa_0 > 0$ and a C^∞ function g on $N \times (-\kappa_0, \kappa_0)$ such that: for $0 < \kappa < \kappa_0$ and $(x_0', \boldsymbol{c}') \in N$ the lower edge of $(x_0', \boldsymbol{c}', \kappa)$ is of type R and $\gamma^- = g$; $g(\cdot, -\kappa) = g(\cdot, \kappa)$, so g is a C^∞ function of κ^2; $g(x_0', \boldsymbol{c}', 0) = U^-_{\boldsymbol{c}'}(x_0')$; and $Q(1) = 2\mu(\kappa)^2 - 2$ with μ a C^∞ function of κ^2, $\mu(0) = \mu_0$. In particular*

$$\gamma^- = U^-_{\boldsymbol{c}}(x_0) + a\,\kappa^2 + O(\kappa^4), \qquad Q(1) = 2c_2 + O(\kappa^2).$$

PROOF. *Step 1: the no-swirl edge is a point of the R chart.* At $\kappa = 0$: $f \equiv 0$, $Q = P_{\boldsymbol{c}}$, and $\gamma^-(x_0, \boldsymbol{c}, 0) = U^-_{\boldsymbol{c}}(x_0)$, the recessive no-swirl solution, with $U^-_{\boldsymbol{c}}(1) = -2 - 2\mu_0$ (Lemma 5.46(c), $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$). Its z is the recessive solution of $z'' = (\mu_0^2 + w)z$ with $w = \frac{1}{2}(P_{\boldsymbol{c}} - 2c_2) = \frac{1}{2}\big((c_1 - c_2)(1-x) + c_3(1-x^2)\big) = O(E)$; since $E \le E^{1-2\varsigma}$ for $\varsigma \in (\mu_0, \frac{1}{2})$, this is a normalised local solution of type R with swirl constant $\kappa_{\mathrm{ch}} = 0$, i.e. it lies in the chart domain (the compact sets K_B^{R} bound $|\kappa_{\mathrm{ch}}|$ and contain $\kappa_{\mathrm{ch}} = 0$) and at a late cut its state is $\mathsf{S}_{\xi_*}(p_0)$ for a coordinate p_0, by completeness (Theorem 4.40(c)). *Step 2: transversality at $\kappa = 0$.* Take a defining function s of the embedded chart piece near this state, and put $H(x_0', \boldsymbol{c}', \kappa, \gamma) = s(\Theta(x_0', \boldsymbol{c}', \kappa, \gamma))$, which is C^∞ for κ of *either* sign. Along J_γ at $\kappa = 0$, $g = f = 0$ gives $\delta g' = -U\delta g/d$ with $\delta g(x_0) = 0$, so $\delta g \equiv 0$, $\delta f \equiv 0$, the forcing $2(f\,\delta g + g\,\delta f)/d$ of δp_2 vanishes and $\delta Q \equiv 0$. By Lemma 7.1, v_γ is constant and equals $\frac{1}{2}z(x_0)^2 > 0$ ($\delta U_\gamma(x_0) = 1$), as in Toy example 7.2. By Lemma 7.4 tangent vectors have $v \to 0$, so $\partial_\gamma H = \mathrm{d}s(\partial_\gamma \Theta) \neq 0$. *Step 3: the implicit function theorem and the identification.* There is a C^∞ function g on $N \times (-\kappa_0, \kappa_0)$ with $H(\cdot, g) = 0$ and $g = U^-_{\boldsymbol{c}}(x_0)$ at $(x_0, \boldsymbol{c}, 0)$ (Theorem A.3).

For $\kappa > 0$ the state $\Theta(\cdot, g)$ lies on the chart piece, so the solution is of type R on $[x_*, 1)$, forward global, with $f' = \kappa z^{-2} > 0$; thus $g \geq \gamma^-$, and $g > \gamma^-$ is excluded because the solution is not of type G (Lemma 5.46(a)): $g = \gamma^-$. *Step 4: evenness.* By Lemma 5.1, $\sigma_1(U, f, Q) = (U, -f, Q)$ is a solution with $\kappa \mapsto -\kappa$ and the same Taylor constants and γ; so $\Theta(x_0', \boldsymbol{c}', -\kappa, \gamma) = \mathsf{R}\,\Theta(x_0', \boldsymbol{c}', \kappa, \gamma)$ with R the linear map $(f, k) \mapsto (-f, -k)$ of the scale-free state. The chart piece is R-invariant ($(\varphi, \kappa_{\mathrm{ch}}) \mapsto (-\varphi, -\kappa_{\mathrm{ch}})$ maps chart solutions to chart solutions, the loop depending on f through f^2 and $z^{-2}\kappa$). Hence the zero set $\{H = 0\}$, a graph over $(x_0', \boldsymbol{c}', \kappa)$ near $(x_0, \boldsymbol{c}, 0)$, is symmetric under $\kappa \mapsto -\kappa$, i.e. $g(\cdot, -\kappa) = g(\cdot, \kappa)$ (it is not needed that $s \circ \mathsf{R} = s$). By Whitney's theorem on even functions (Theorem A.20, with parameters $p = (x_0', \boldsymbol{c}')$), g is a C^∞ function of $(x_0', \boldsymbol{c}', \kappa^2)$. *Step 5: μ and $Q(1)$.* The coordinate μ of $(\mathsf{S}_{\xi_*}|_K)^{-1}(\Theta(\cdot, g))$ is C^∞ (the inverse of an embedding), even in κ (μ is R-invariant), equals μ_0 at $\kappa = 0$, and $q = 2\mu^2 - 2$ at the pole. By Whitney's theorem again it is smooth in κ^2. The expansions follow. □

Corollary 7.22 (small swirl at the boundary of $\mathcal{J}$)**.** *Let $x_0 \in (-1, 1)$ and $\boldsymbol{c} \in \mathcal{J}$ with $c_1 > -1$ and $-1 < c_2 < -\frac{3}{4}$; thus $c_3 \geq \bar{c}_3$ and $\boldsymbol{c}$ may lie on $\partial\mathcal{J}$ ($c_3 = \bar{c}_3$). Put $\mu_0 = \sqrt{1 + c_2} \in (0, \frac{1}{2})$. There are a neighbourhood N of $(x_0, \boldsymbol{c})$ in $(-1, 1) \times \mathbb{R}^3$, with $c_1' > -1$ and $-1 < c_2' < -\frac{3}{4}$ on N, a number $\kappa_0 > 0$ and a C^∞ function g on $N \times (-\kappa_0, \kappa_0)$ such that: $g(\cdot, -\kappa) = g(\cdot, \kappa)$, so g is a C^∞ function of κ^2; $g(x_0', \boldsymbol{c}', 0) = U^-_{\boldsymbol{c}'}(x_0')$ for $(x_0', \boldsymbol{c}') \in N$ with $\boldsymbol{c}' \in \mathcal{J}$; and, for $0 < \kappa < \kappa_0$ and $(x_0', \boldsymbol{c}') \in N$ with $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$, the lower edge of $(x_0', \boldsymbol{c}', \kappa)$ is of type R, $\gamma^- = g$ and $Q(1) = 2\mu(\kappa)^2 - 2$ with μ a C^∞ function of κ^2, $\mu(0) = \mu_0$. In particular the expansions of Theorem 7.21 hold at $\boldsymbol{c}$.*

Proof. *Step 1: the no-swirl edge is a point of the R chart, also for $c_3 = \bar{c}_3$.* Let $\boldsymbol{c}' \in \mathcal{J}$ with $c_1' > -1$, $-1 < c_2' < -\frac{3}{4}$. The solution $U^-_{\boldsymbol{c}'}$ exists on $(-1, 1)$ and corresponds to $z_+ = p_+$ if $c_3' > \bar{c}_3$, and to $z_+ = g_{\mathrm{crit}}$, the doubly recessive solution, if $c_3' = \bar{c}_3$ (Theorem 3.44(ii),(iv), Lemma 3.40); in both cases $z_+ > 0$ on $\mathbb{R}$ and $z_+'/z_+ \to -k_2'$ at $+\infty$, i.e. z_+ is recessive and $U^-_{\boldsymbol{c}'}(1) = -2 - 2\mu_0'$, $\mu_0' = k_2'$ (Theorem 3.44(iii),(iv)). At $\kappa = 0$ the solution with data $(x_0', \boldsymbol{c}', 0, U^-_{\boldsymbol{c}'}(x_0'))$ has $f \equiv 0$, $Q = P_{\boldsymbol{c}'}$ (Lemma 5.46(c) for $c_1' > -1$, which covers $c_3' = \bar{c}_3$) and z_+ solves $z'' = (\mu_0'^2 + w)z$ with $w = \frac{1}{2}(P_{\boldsymbol{c}'} - 2c_2') = O(E)$. Only the behaviour at the north pole enters, so exactly as in Theorem 7.21 it is a normalised local solution of type R with $\kappa_{\mathrm{ch}} = 0$ ($E \leq E^{1-2\varsigma}$ for $\varsigma \in (\mu_0', \frac{1}{2})$), and at a late cut its state is $\mathsf{S}_{\xi_*}(p')$ for a chart coordinate p', depending continuously on $(x_0', \boldsymbol{c}')$ (Theorem 4.40(c)). *Step 2: transversality at $\kappa = 0$.* At $(x_0, \boldsymbol{c})$ take a defining function s of the chart piece near the state of Step 1 and $H = s \circ \Theta$, where $\Theta(x_0', \boldsymbol{c}', \kappa, \gamma)$ is the scale-free state at ξ_* of the solution with these Cauchy data at x_0'. Θ is C^∞ for *all* $\boldsymbol{c}' \in \mathbb{R}^3$, κ of either sign and γ near $U^-_{\boldsymbol{c}}(x_0)$, because the vector field is polynomial in the data and the solution through $(x_0, \boldsymbol{c}, 0, U^-_{\boldsymbol{c}}(x_0))$ exists on $[x_0, x_*]$ (continuous dependence); no membership $\boldsymbol{c}' \in \mathcal{J}$ is used. At $\kappa = 0$, $\boldsymbol{c}' = \boldsymbol{c}$ the argument of Theorem 7.21, Step 2, uses only $z = z_+ > 0$ on $[x_0, 1)$ and Lemma 7.1, Lemma 7.4: $\delta g \equiv \delta f \equiv \delta Q \equiv 0$, $v_\gamma = \frac{1}{2}z_+(x_0)^2$ is constant and positive, tangent vectors of the chart piece have $v \to 0$, so $\partial_\gamma H \neq 0$. *Step 3: the implicit function theorem.* Theorem A.3 gives $N \ni (x_0, \boldsymbol{c})$, κ_0 and a C^∞ function g on $N \times (-\kappa_0, \kappa_0)$ with $H(\cdot, g) = 0$, $g(x_0, \boldsymbol{c}, 0) = U^-_{\boldsymbol{c}}(x_0)$, and uniqueness of the zero near $U^-_{\boldsymbol{c}}(x_0)$. Shrink N so that $c_1' > -1$, $-1 < c_2' < -\frac{3}{4}$ on N and Step 1 applies at every $(x_0', \boldsymbol{c}') \in N$ with $\boldsymbol{c}' \in \mathcal{J}$: its state at ξ_* is on the chart piece, so $H(x_0', \boldsymbol{c}', 0, U^-_{\boldsymbol{c}'}(x_0')) = 0$. The map $\boldsymbol{c}' \mapsto U^-_{\boldsymbol{c}'}(x_0')$ is continuous on

$N \cap \mathcal{J}$ up to $c_3' = \bar{c}_3(c_1', c_2')$, because $U^-_{\boldsymbol{c}'}$ is built from the Jost solution u_+ of $W_{\boldsymbol{c}'}$, which depends continuously on $\boldsymbol{c}'$ in the weighted topology (Lemma 3.25); hence, after shrinking N, $U^-_{\boldsymbol{c}'}(x_0')$ lies in the interval on which the zero of $H(x_0', \boldsymbol{c}', 0, \cdot)$ is unique, and by uniqueness $g(x_0', \boldsymbol{c}', 0) = U^-_{\boldsymbol{c}'}(x_0')$. Shrinking N and κ_0 once more, the state $\Theta(\cdot, g)$ lies on the chart piece for *all* $(x_0', \boldsymbol{c}', \kappa) \in N \times (-\kappa_0, \kappa_0)$ (not only for $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$, $\kappa > 0$), since it is continuous and lies in the open chart piece at $\kappa = 0$; the exponent μ used below is read off there. The base state has $f = k = 0$, so it is fixed by the reflection σ_1 ($\kappa \mapsto -\kappa$), and the neighbourhood may be taken invariant under it. Now let $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$ and $\kappa > 0$, so that the lens and $\gamma^-(x_0', \boldsymbol{c}', \kappa)$ are defined. The state $\Theta(\cdot, g)$ lies on the chart piece, the solution is forward global of type R with $f' = \kappa z^{-2} > 0$, so $g \in \mathrm{FS}$ and $g \geq \gamma^-$; $g > \gamma^-$ is excluded because the solution is not of type G (Lemma 5.46(a)). Hence $\gamma^- = g$ and the edge is of type R. (On $\partial\mathcal{J}$ with $\kappa > 0$ we make no claim about γ^-.) *Steps 4–5.* Evenness of g, its smoothness in κ^2, and the statements on μ and $Q(1) = 2\mu^2 - 2$ use only the function H, the R-invariance of the chart piece and of the coordinate μ, and Whitney's theorem on even functions; none of them refers to $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$, so Steps 4 and 5 of the proof of Theorem 7.21 apply verbatim on $N \times (-\kappa_0, \kappa_0)$ with the g and H above. The value $\mu(0) = \mu_0$ is that of the point p_0 of Step 1. For $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$, $\kappa > 0$ the exponent μ is that of the edge solution because $\gamma^- = g$ (Step 3). □

Remark 7.23 (range)**.** The neighbourhood N and κ_0 degenerate as $c_2 \uparrow -\frac{3}{4}$ ($\mu_0 \uparrow \frac{1}{2}$): the weight margin $E_*^{2(\varsigma-\mu_0)}$ of the chart tends to 1. The numbers show it: $(q - 2c_2)/\kappa^2$ at $\kappa = 0.01$ equals -0.907, -1.52, -6.32 for $c_2 = -0.95, -0.9, -0.8$ (Numerical observation 7.24). For $c_2 > -\frac{3}{4}$ the no-swirl edge has $U(1) = -2 - 2\mu_0 \leq -3$, which swirl forbids at an R pole (Theorem 4.9): the limit $\kappa \to 0$ is singular and Theorem 7.21 says nothing; this is the boundary of $\mathcal{P}$ in the sense of Remark 7.11.

Numerical observation 7.24 (small swirl: $Q(1)$ along the lower edge, $x_0 = 0$, $\boldsymbol{c} = (0, c_2, 0)$)**.** *Computed by* `verify_ch07a.py` *(edge by bisection, $Q(1)$ read at $s_{\max} = 60$, where for $c_2 = -0.8$ the layer exponent $1 - 2\mu_0 = 0.106$ requires the large window). The last quantity $(Q(1) - 2c_2)/\kappa^2$ at $\kappa = 0.01$ is -0.907, -1.523, -6.322 for the three values of c_2 (a coefficient that grows as $c_2 \uparrow -\frac{3}{4}$):*

$c_2 \backslash \kappa$	0.003	0.01	0.03	0.1	0.3
-0.95: $Q(1)$	-1.90001	-1.90009	-1.90081	-1.90881	-1.96331
-0.90: $Q(1)$	-1.80001	-1.80015	-1.80137	-1.81462	-1.90064
-0.80: $Q(1)$	-1.60006	-1.60063	-1.60555	-1.65002	-1.82492

The values tend to $2c_2$ as $\kappa \downarrow 0$, like $2c_2 + O(\kappa^2)$, as Theorem 7.21 predicts, and move to -2 as κ grows (type R with $\mu \downarrow$; at $c_2 = -0.9$ the edge is L_1 beyond the wall at $\kappa_t \approx 0.593$, Numerical observation 7.37).

7.5.2. Continuity of $Q(1)$ along the edge.

Lemma 7.25 (energy formula)**.** *Let (U, f, Q) be a solution on $[x_a, 1)$ with first integral $\lambda = \mathsf{L}[Q] + f^2$, and suppose $q = Q(1)$ exists. Then*

$$q = \frac{2Q(x_a)}{1 + x_a} + \frac{(1 - x_a)\, Q_x(x_a)}{1 + x_a} + \frac{\lambda\,(1 - x_a)}{2(1 + x_a)} - \int_{x_a}^{1} \frac{f^2}{(1 + s)^2}\, \mathrm{d}s.$$

PROOF. Let $I(x)$ be the right-hand side with x_a replaced by x. Using $\mathsf{L}[Q] = (1-x^2)Q_{xx} + 2xQ_x - 2Q = \lambda - f^2$ one finds

$$(1+x)^2\, I'(x) = 2xQ_x + (1-x^2)Q_{xx} - 2Q - (\lambda - f^2) = 0,$$

so I is constant. As $x \to 1$, $\frac{2Q}{1+x} \to q$, $\frac{\lambda(1-x)}{2(1+x)} \to 0$ and $\int_x^1 \to 0$; since I is constant, $\frac{(1-x)Q_x}{1+x}$ converges to a number L and $I \equiv q + L$. If $L \neq 0$ then $Q_x \sim 2L/(1-x)$ near 1, which is not integrable and contradicts the boundedness of Q. So $L = 0$ and $I \equiv q$. □

Proposition 7.26 (continuity of $Q(1)$)**.** *On $\mathcal{P}$ the function $q(p) = Q(1)$ of the lower edge solution is continuous; it is C^∞ on $\mathcal{P}_{\mathsf{R}} \cup \mathcal{P}_{\mathsf{L}_1}$ and equals -2 on $\mathcal{P}_{\mathsf{L}_1} \cup \mathcal{P}_{\mathsf{L}_2}$. The same holds for $Q(-1)$ along the upper edge. Consequently, if $p_n \to p_0 \in \mathcal{P}$ and $x_a \in (x_0, 1)$, the swirl energies $\int_{x_a}^1 f_n^2 (1+s)^{-2} \mathrm{d}s$ of the edge solutions converge to that of the edge solution of p_0: no energy concentrates at the pole.*

PROOF. R *points.* By the proof of Theorem 7.12 the edge state is $\Theta(p, \gamma^-(p)) = \mathsf{S}_{\xi_*}(\beta(p))$ with $\beta = (\mathsf{S}_{\xi_*}|_K)^{-1} \circ \Theta(\cdot, \gamma^-(\cdot))$ a C^∞ function; $q = 2\mu^2 - 2$ with $\mu = \beta_1$ a coordinate. L_1 *points:* $\mathcal{P}_{\mathsf{L}_1}$ is open and $q \equiv -2$ there. L_2 *points.* Let $p_n \to p_0 \in \mathcal{P}_{\mathsf{L}_2}$. The edge solutions are continuous in p (γ^- is Lipschitz, Theorem 7.10), so their states at x_a converge, and so do Q, Q_x, λ at x_a and $f_n \to f$ pointwise on $[x_a, 1)$. By Lemma 7.25 and Fatou's lemma, $\limsup q(p_n) \le q(p_0)$: the three boundary terms converge, and $\liminf \int f_n^2 (1+s)^{-2} \ge \int f^2 (1+s)^{-2}$. Since $q(p_0) = -2$ and $q \ge -2$ (Theorem 4.5), $q(p_n) \to -2$. The energy statement follows from Lemma 7.25: all terms except the integral converge, as does $q(p_n)$. □

7.5.3. The set of edge states and the interface between R and L_1. Proposition 7.26 and Theorem 7.12 say that in $\mathcal{P}$ the open sets $\mathcal{P}_{\mathsf{R}} = \{q > -2\}$ and $\mathcal{P}_{\mathsf{L}_1}$ are separated only by $\mathcal{P}_{\mathsf{L}_2}$. To know *what* $\mathcal{P}_{\mathsf{L}_2}$ is we need a topological description of the set of forward-global non-generic states. Fix a station $x_* \in (-1, 1)$ and a reference polynomial P_0 and write states $Y = (U, g, f, p_2, p_1, p_0) \in \mathbb{R}^6$ relative to it; let $\mathcal{D}_+ = \{g > 0, f > 0\}$ and

$$\mathcal{G} = \{Y \in \mathcal{D}_+ : \text{ the forward solution from } Y \text{ at } x_* \text{ exists on } [x_*, 1)\},$$
$$\mathcal{N} = \{Y \in \mathcal{G} : \text{ not of type } \mathsf{G} \text{ at } x = 1\}.$$

Lemma 7.27 (states and class-Z data)**.** *Let Y_0 be the state at x_* of a solution with $f = 0$ at some $x_0 < x_*$, Taylor constants $\boldsymbol{c}$ and $\kappa = g(x_0) > 0$, $\gamma = U(x_0)$. Near Y_0 the map $Y \mapsto (x_0, \boldsymbol{c}, \kappa, \gamma)(Y)$ is a C^∞ diffeomorphism onto a neighbourhood of $(x_0, \boldsymbol{c}, \kappa, \gamma)$, with inverse $(x_0, \boldsymbol{c}, \kappa, \gamma) \mapsto Y(x_*; x_0, \boldsymbol{c}, \kappa, \gamma)$.*

PROOF. Let $f(x; Y)$ be the swirl of the solution with state Y at x_*; it is C^∞ in (x, Y) and has at (x_0, Y_0) a simple zero ($\partial_x f = g(x_0) = \kappa > 0$), so by the implicit function theorem there is a C^∞ function $x_0(Y)$ with $f(x_0(Y); Y) = 0$. Then $\kappa(Y) = g(x_0(Y); Y)$, $\gamma(Y) = U(x_0(Y); Y)$ and $\boldsymbol{c}(Y)$, the solution of the linear system $(P_{\boldsymbol{c}}, P'_{\boldsymbol{c}}, P''_{\boldsymbol{c}})(x_0) = (Q, Q_x, Q_{xx})(x_0; Y)$ ($1 - x$, $1 + x$, $1 - x^2$ is a basis of the quadratics), are C^∞. The flow from x_0 to x_* inverts the map. □

Lemma 7.28 (the edge states form a Lipschitz graph)**.** (a) *$\mathcal{G}$ is closed in $\mathcal{D}_+$ and an up-set for the order $Y \preceq \hat{Y} \iff \Sigma(\hat{Y} - Y) \ge 0$.*

(b) *The interior of $\mathcal{G}$ is the set of states of type G; hence $\mathcal{N} = \partial\mathcal{G} \cap \mathcal{D}_+$.*

(c) *$\mathcal{N}$ is a locally Lipschitz graph over a hyperplane, in particular a topological manifold of dimension* 5.

(d) (Local product structure.) *For $y_0 \in \mathcal{N}$ there are a direction $e = \Sigma\mathbf{1}$, a hyperplane Π complementary to e, a ball $B_\rho \subset \Pi$, a Lipschitz function τ on B_ρ and $r' > 0$ with $\tau(B_\rho) \subset (-\frac{r'}{2}, \frac{r'}{2})$ such that, in the cylinder $V' = \{\pi + te : \pi \in B_\rho,\ |t| < r'\}$, $\mathcal{N} \cap V' = \{t = \tau(\pi)\}$ is connected, $A = \{t > \tau\} \cap V'$ is a connected subset of $\operatorname{int}\mathcal{G}$ and $B = \{t < \tau\} \cap V'$ is connected with $B \cap \mathcal{G} = \emptyset$.*

Proof. (a) Closedness: write a state $Y \in \mathcal{D}_+$ as Cauchy data $(\boldsymbol{c}, \gamma, A, \kappa) =$ (Taylor constants of Q, U, f, g) at $x_b = x_*$ (an affine isomorphism); $A, \kappa > 0$, and $\mathcal{G}$ corresponds to G^+, closed by Proposition 5.42(a). Up-set: Lemma 7.8 (the two states have $g, f > 0$; any solution that is not forward global is not forward global when pushed down in $\preceq$). (b) *Type* G *is interior*: $\mathcal{G}_\mathsf{G}$ is open in the space of states (Corollary 4.42) and contained in $\mathcal{G}$. *Interior is type* G*:* let $Y \in \operatorname{int}\mathcal{G}$. For small $\varepsilon > 0$, $Y_1 = Y - \varepsilon e_U \in \mathcal{G}$ and $Y_1 \preceq Y$, $Y_1 \neq Y$. By Lemma 5.28(b) (strict form; $g + g_1 > 0$, the data differ) all components of $W = \Sigma(Y - Y_1)$ are > 0 on $(x_*, 1)$; as in the proof of Lemma 5.46(a), with $\delta = W_{p_0}(x_1) > 0$ and $\partial_x W_{p_0} = W_{p_1} > 0$, one gets $q \geq q_1 + \delta > -2$, $\mu > \mu_1 \geq 0$, $U \geq U_1$, and $U(1) = -2 - 2\mu$ would contradict $U(1) \geq U_1(1) \geq -2 - 2\mu_1 > -2 - 2\mu$; so $U(1) = -2 + 2\mu$ with $\mu > 0$: type G. (c) Choose $v = \mathbf{1}$ in the Σ-coordinates ΣY, an interior vector of the cone $K = \{\Sigma y \geq 0\}$, and $r > 0$ with $B(v, r) \subset K$. If $Y \in \partial\mathcal{G}$ and $t > 0$, then $Y + tB(\Sigma v, r) \subset \mathcal{G}$ by (a), so $Y + t\Sigma v \in \operatorname{int}\mathcal{G}$. Hence every line $Y + \mathbb{R}\Sigma v$ meets $\partial\mathcal{G}$ at most once, and two boundary points are never strictly comparable, which gives, with π the projection along Σv onto a hyperplane, $|t(Y) - t(Y')| \leq r^{-1}|\pi Y - \pi Y'|$ for the parameter t of the boundary point on its line. Each line meets $\partial\mathcal{G}$ at least once near any boundary point (the line leaves $\mathcal{G}$ in the direction $-\Sigma v$ and enters in $+\Sigma v$), so $\partial\mathcal{G}$ is the graph of a Lipschitz function over an open subset of the hyperplane: a topological 5-manifold.

(d) Take $e = \Sigma\mathbf{1}$ ($= \Sigma v$ of (c)), Π a complementary hyperplane through y_0 and the Lipschitz function τ of (c), so that $\mathcal{N}$ is the graph $\{t = \tau(\pi)\}$ near y_0; choose ρ, r' with $\tau(B_\rho) \subset (-\frac{r'}{2}, \frac{r'}{2})$ and $V' \cap \partial\mathcal{G} = \mathcal{N} \cap V'$. The graph of a continuous function over a ball is connected, and so are the two regions $\{t > \tau\}$ and $\{t < \tau\}$ of the cylinder (each is homeomorphic to $B_\rho \times (0,1)$ by $(\pi, t) \mapsto (\pi, (t - \tau(\pi))/(r' - \tau(\pi)))$, resp. the analogue below the graph). $A \subset \operatorname{int}\mathcal{G}$*:* for $Y = \pi + te \in A$ the point $\pi + \tau(\pi)e$ is in $\partial\mathcal{G}$ and $Y = \pi + \tau e + (t - \tau)e$ with $t - \tau > 0$, so $Y \in \operatorname{int}\mathcal{G}$ by the statement proved in (c). $B \cap \mathcal{G} = \emptyset$*:* if $Y = \pi + te \in B \cap \mathcal{G}$, then by (a) $Y + B((\tau - t)\Sigma v, (\tau - t)r) \subset \mathcal{G}$, so $\pi + \tau(\pi)e = Y + (\tau - t)e \in \operatorname{int}\mathcal{G}$, contradicting $\pi + \tau e \in \partial\mathcal{G}$. □

Proposition 7.29 (interface)**.** (a) *$\mathcal{P}_\mathsf{R} = \{q > -2\}$ is open in $\mathcal{P}$ and $\mathcal{P}_{\mathsf{L}_1} \cup \mathcal{P}_{\mathsf{L}_2} = \{q = -2\}$ is closed. If $p_0 \in \overline{\mathcal{P}_\mathsf{R}} \cap \overline{\mathcal{P}_{\mathsf{L}_1}} \cap \mathcal{P}$ then $p_0 \in \mathcal{P}_{\mathsf{L}_2}$: L_2 is the only way to pass from R to L_1.*

(b) *If a connected $C \subset \mathcal{P}$ meets $\mathcal{P}_\mathsf{R}$ and $\mathcal{P}_{\mathsf{L}_1}$, it meets $\mathcal{P}_{\mathsf{L}_2}$.*

Proof. (a) R edges have $q > -2$ and L edges $q = -2$ by definition of the types; the openness of $\mathcal{P}_\mathsf{R}$ is Theorem 7.12(i), and $\{q = -2\}$ is its complement; if $p_0 \in \mathcal{P}_\mathsf{R}$ the open set $\mathcal{P}_\mathsf{R}$ is a neighbourhood of p_0 missing $\mathcal{P}_{\mathsf{L}_1}$, and similarly for $\mathcal{P}_{\mathsf{L}_1}$. (b) $C \cap \mathcal{P}_\mathsf{R}$ and $C \cap \mathcal{P}_{\mathsf{L}_1}$ are disjoint, nonempty and relatively open; they cannot cover the connected C. □

7.5.4. The wall of L_2 edges. In words. *Between the region of the parameters where the edge is of type* R *(the pole rate* $\mu > 0$*) and the region where it is of type* L_1 *(*$\mu = 0$*, a pole that is only logarithmically attracting) there is a wall of edges of type* L_2*. In a neighbourhood of an* L_2 *edge the local corner chart of Chapter 4 puts all solutions that are forward global into one picture: the dominant solutions of the pole equation in the interior, the recessive ones on a parabola* $\sigma = \omega^2$*, the logarithmic ones on a half-line* $\sigma = 0 < \omega$*. Pulling the chart back by the flow turns the threshold* γ^- *into the root of a function that is* C^3 *except for a square* $(\omega_-)^2$*. The result is a precise description of the wall:* γ^- *is* $C^{1,1}$ *but* not C^2 *there, the second derivative jumps by an explicit positive multiple of the square of the normal derivative, and the walls form a closed, nowhere dense family of hypersurfaces separating* R *from* L_1*. Everything in this section is* local *in the sense of the corner chart: a neighbourhood that depends on the order of differentiability, and no claim of* C^∞ *up to the wall.*

Throughout this section $x_* = \tanh \xi_*$ is a late cut, $\Theta(p, \gamma) \in \mathbb{R}^6$ is the scale-free state at ξ_* of the solution with data (p, γ) (as in the proof of Theorem 7.12) and $\mathcal{G} = \mathcal{G}(x_*)$ is the set of states at x_* whose forward solution is global. Lemma 7.27 says that, near any state, Θ is a C^∞ diffeomorphism from the space of data $(x_0, \boldsymbol{c}, \kappa, \gamma)$. The proof of Theorem 7.32 needs one more consequence of the charts.

Lemma 7.30 (L_1 edges are stable)**.** *$\mathcal{P}_{\mathsf{L}_1}$ is open in $\mathcal{P}$. Hence $\mathcal{P}_{\mathsf{L}_2} = \mathcal{P} \setminus (\mathcal{P}_{\mathsf{R}} \cup \mathcal{P}_{\mathsf{L}_1})$ is closed in $\mathcal{P}$.*

Proof. Let $p_0 \in \mathcal{P}_{\mathsf{L}_1}$, Y_e its edge solution and $y_e = \Theta(p_0, \gamma^-(p_0))$ its state at a cut ξ_* so late that $y_e = \mathsf{S}_{\xi_*}(p_{\mathrm{c}})$ lies on the chart of L_1 (Theorem 4.40(c)). For a small closed ball K around p_{c}, $\mathsf{S}_{\xi_*}|_{\operatorname{int} K}$ is an embedding of a 5-dimensional open set; its image D consists of states of solutions of type L_1 which are forward global and not of type G, hence belong to $\mathcal{N} = \partial\mathcal{G} \cap \mathcal{D}_+$ (Lemma 7.28(b)). Since $\mathcal{N}$ is a topological 5-manifold (Lemma 7.28(c)) and $\mathsf{S}_{\xi_*}|_{\operatorname{int} K} : \operatorname{int} K \to \mathcal{N}$ is continuous and injective, invariance of domain says that D is open in $\mathcal{N}$: there is a neighbourhood $\mathcal{V}$ of y_e with $\mathcal{N} \cap \mathcal{V} \subset D$. The map $p \mapsto \Theta(p, \gamma^-(p)) \in \mathcal{N}$ is continuous (Theorem 7.10(i)), so for p near p_0 the edge state of p lies in D and the edge is of type L_1. The second statement follows from the openness of $\mathcal{P}_{\mathsf{R}}$ (Theorem 7.12(i)) and the trichotomy of Lemma 5.46. □

Lemma 7.31 (Hessian of a function vanishing to second order on a hypersurface)**.** *Let $U \subset \mathbb{R}^n$ be open, $\varphi \in C^3(U)$ with $\nabla\varphi \neq 0$ on $Z = \{\varphi = 0\}$, and $\Psi \in C^2(U)$ with $\Psi = 0$ and $\nabla\Psi = 0$ on Z. Then:*

(a) *On Z one has $\nabla^2\Psi = a\,\nabla\varphi \otimes \nabla\varphi$ with $a = \nabla^2\Psi(\nabla\varphi, \nabla\varphi)/|\nabla\varphi|^4$.*

(b) *Every point of Z has a neighbourhood on which $|\nabla\Psi(p)| \le C|\varphi(p)|$.*

(c) *If moreover $\Psi(p) = A\varphi(p)^2 + o(\varphi(p)^2)$ as $p \to p_* \in Z$, uniformly, then $a(p_*) = 2A$.*

Proof. (a) Let τ be tangent to Z at $p \in Z$ and $t \mapsto r(t)$ a C^1 curve in Z with $r(0) = p$, $r'(0) = \tau$ (Z is a C^3 hypersurface by the implicit function theorem). Then $\nabla\Psi(r(t)) = 0$, and differentiating at $t = 0$ gives $\nabla^2\Psi(p)\tau = 0$. So the symmetric matrix $H = \nabla^2\Psi(p)$ has the hyperplane $(\nabla\varphi)^\perp = T_pZ$ in its kernel; being symmetric, its range lies in $(\ker H)^\perp \subset \mathbb{R}\nabla\varphi$, so $H = a\,\nabla\varphi \otimes \nabla\varphi$, and evaluating on $(\nabla\varphi, \nabla\varphi)$ gives a. (b) By the implicit function theorem there are C^3 coordinates (u, t) near p_* with $t = \varphi$ (take φ as one coordinate), in which $Z = \{t = 0\}$ and the Euclidean distance of p to Z is at most $C|t| = C|\varphi(p)|$. If

$m \in Z$ is a nearest point, $\nabla\Psi(m) = 0$ and the mean value theorem for the C^1 map $\nabla\Psi$ gives $|\nabla\Psi(p)| \le \sup|\nabla^2\Psi|\,|p-m| \le C'|\varphi(p)|$. (c) Let $n = \nabla\varphi(p_*)/|\nabla\varphi(p_*)|^2$. Then $\varphi(p_* + tn) = t + O(t^2)$ and so $\Psi(p_* + tn) = At^2 + o(t^2)$; since $\Psi \in C^2$ with $\Psi(p_*) = 0$, $\nabla\Psi(p_*) = 0$, $\frac{d^2}{dt^2}\Psi(p_* + tn)|_{t=0} = \nabla^2\Psi(n,n) = 2A$. By (a) the left-hand side is $a(\nabla\varphi \cdot n)^2 = a$. □

THEOREM 7.32 (the wall of L_2 edges: local structure, order $k = 3$). *Let $p_0 \in \mathcal{P}_{\mathsf{L}_2}$ (so $\kappa_0 > 0$) and $\gamma_0 = \gamma^-(p_0)$. There are a neighbourhood U of p_0 in $\mathcal{P}$ and an interval $I \ni \gamma_0$ (depending on p_0 and on the order $k = 3$ of the corner chart of Theorem 4.45) such that, with $\ell_\gamma(p,\gamma) = \lim_{x\to 1}\delta Q_\gamma(x;p,\gamma)$:*

(a) (coordinates) *On $N = U \times I$ the map $\Xi = \widetilde{\Phi}^{-1} \circ \Theta = (\beta,\omega,\sigma) : N \to \mathbb{R}^4 \times \mathbb{R} \times \mathbb{R}$ is a C^3 diffeomorphism onto an open set containing $(\beta_0, 0, 0)$; a pair $(p,\gamma) \in N$ gives a forward global solution if and only if $\sigma \ge (\omega_-)^2$, $\omega_- = \max(-\omega, 0)$; $Q(1) + 2 = 2\sigma$ for these solutions; and $\partial_\gamma\sigma = \frac{1}{2}\ell_\gamma > 0$ on N (where ℓ_γ is read at the points with $\sigma > (\omega_-)^2$ and extended by continuity). In particular on the L_1 side $Q(1) + 2 = \ell_\gamma\,(\gamma - \gamma^-) + O\big((\gamma - \gamma^-)^2\big)$ with uniform constants.*

(b) (the wall) *Let $\gamma_L(p)$ be the root of $\sigma(p,\cdot)$ in I (a C^3 function on U) and $\omega_L(p) = \omega(p, \gamma_L(p))$. Then $\omega_L \in C^3(U)$, $\nabla\omega_L \ne 0$ and*

$$\mathcal{P}_{\mathsf{L}_2} \cap U = \{\omega_L = 0\}, \qquad \mathcal{P}_{\mathsf{R}} \cap U = \{\omega_L < 0\}, \qquad \mathcal{P}_{\mathsf{L}_1} \cap U = \{\omega_L > 0\},$$

in particular $\mathcal{P}_{\mathsf{L}_2} \cap U$ is a C^3 hypersurface and both R and L_1 edges occur on its two sides.

(c) (regularity) *$\gamma^- \in C^{1,1}(U)$ ($\nabla\gamma^-$ is Lipschitz), $\gamma^- \in C^\infty(U \setminus \{\omega_L = 0\})$, but $\gamma^- \notin C^2$ at any point of the wall: for $p \in \mathcal{P}_{\mathsf{L}_2} \cap U$ the limits of $\nabla^2\gamma^-$ from the R side and from the L_1 side exist and*

$$\big[\nabla^2\gamma^-\big]_{\mathsf{R}-\mathsf{L}_1}(p) = \frac{4}{\ell_\gamma(p,\gamma^-(p))}\,\nabla\omega_L(p) \otimes \nabla\omega_L(p).$$

(d) (profile) *$\gamma^- = \gamma_L$ on $\{\omega_L \ge 0\}$ and $\gamma^- = \gamma_L + \Psi$ on $\{\omega_L < 0\}$, where $\Psi \in C^3(U)$, $\Psi \ge 0$ and $\Psi = \dfrac{2}{\ell_\gamma}\omega_L^2 + O(|\omega_L|^3)$. On the R side $\omega_L = -\mu$ with μ the pole rate, and $Q(1) + 2 = 2\mu^2$.*

The neighbourhood U shrinks with k and no regularity beyond what is stated is claimed up to the wall.

PROOF. *Step 1: the chart near the edge state.* By Theorem 4.45(f) the state $y_0 = \Theta(p_0,\gamma_0)$ of the L_2 solution at a sufficiently late cut ξ_* is $\Phi(\beta_0,0,0)$ for a point $c_0 = (\beta_0,0,0)$ of the chart with $|\beta_0| < B_0$; here $k = 3$, so $\mu_* = \frac{1}{40(k+3)} = \frac{1}{240}$ and $\sigma \le \mu_*^2$, $|\omega| \le \mu_*$. By Theorem 4.45(d), $\widetilde{\Phi}$ is a C^3 diffeomorphism from an open ball $W \ni c_0$ (contained in $\{|\beta| < B_0,\ |\omega| < \mu_*,\ |\sigma| < \mu_*^2\}$) onto an open set $V \ni y_0$. Put $K_W = \overline{\mathcal{D}} \cap W = \{\sigma \ge (\omega_-)^2\} \cap W$ and $S_W = \{\sigma = (\omega_-)^2\} \cap W$; note $\sigma \ge 0$, $\omega \ge -\sqrt{\sigma}$ is the same as $\sigma \ge (\omega_-)^2$. S_W is the graph of the continuous function $(\beta,\omega) \mapsto (\omega_-)^2$, a topological 5-manifold, and by Theorem 4.45(b) its points are of type R ($\omega < 0$), L_1 ($\omega > 0$) or L_2 ($\omega = 0$), while the points with $\sigma > (\omega_-)^2$ are of type G. All of them are forward global, so $\Phi(K_W) \subset \mathcal{G}$, and $\Phi(S_W) \subset \mathcal{G} \setminus \operatorname{int}\mathcal{G} = \mathcal{N}$ by Lemma 7.28(b).

Step 2: the chart is complete near y_0. We claim there is a neighbourhood $V' \subset V$ of y_0 with $\mathcal{G} \cap V' = \widetilde{\Phi}(K_W) \cap V'$. By Lemma 7.28(d) there is a cylinder

$V' = \{\pi + te : \ \pi \in B_\rho,\ |t| < r'\} \subset V$ (shrink ρ, r') with $\mathcal{N} \cap V' = \{t = \tau(\pi)\}$ connected, $A = \{t > \tau\}$ a connected subset of $\operatorname{int}\mathcal{G}$ and $B = \{t < \tau\}$ connected with $B \cap \mathcal{G} = \emptyset$. Put $S = \widetilde{\Phi}(S_W) \cap V'$ and $K = \widetilde{\Phi}(K_W) \cap V'$. S is closed in V' ($\widetilde{\Phi}$ is a homeomorphism $W \to V$), is a topological 5-manifold, and $S \subset \mathcal{N} \cap V'$. By invariance of domain S is open in the 5-manifold $\mathcal{N} \cap V'$; it is nonempty ($y_0 \in S$) and $\mathcal{N} \cap V'$ is connected, so $S = \mathcal{N} \cap V'$. Next $K \setminus S = \widetilde{\Phi}(\{\sigma > (\omega_-)^2\} \cap W) \cap V'$ is open in V', and $K \setminus S = K \cap (V' \setminus \mathcal{N})$ is closed in $V' \setminus \mathcal{N} = A \sqcup B$. It lies in $\mathcal{G}$, so it lies in A, and it is not empty because points with $\sigma > (\omega_-)^2$ accumulate at c_0. As A is connected, $K \setminus S = A$, whence $K = A \cup S = \mathcal{G} \cap V'$. Choose a neighbourhood $N = U \times I$ of (p_0, γ_0) with $\Theta(N) \subset V'$ and put $\Xi = \widetilde{\Phi}^{-1} \circ \Theta : N \to W$. It is a C^3 diffeomorphism onto an open set (Θ is a diffeomorphism by Lemma 7.27), and for $(p, \gamma) \in N$: $Y(\cdot; p, \gamma)$ is forward global $\iff \Theta(p, \gamma) \in \mathcal{G} \iff \Xi(p, \gamma) \in K_W \iff \sigma \geq (\omega_-)^2$. By Theorem 4.45(b), $Q(+\infty) = 2\sigma - 2$, i.e. $Q(1) + 2 = 2\sigma$.

Step 3: $\partial_\gamma \sigma > 0$. Let $(p, \gamma) \in N$ with $\sigma > (\omega_-)^2$ (type G). The curve $\gamma \mapsto \Xi(p, \gamma)$ stays in the interior of K_W, and $\partial_\gamma Q = \delta Q_\gamma$, so Theorem 4.45(e) gives $\delta Q_\gamma(x) \to 2\partial_\gamma \sigma$ as $x \to 1$. By Lemma 7.3, $\delta Q_\gamma > 0$ is nondecreasing, so $\ell_\gamma = 2\partial_\gamma \sigma \geq \delta Q_\gamma(x_1; p, \gamma)$ for any fixed station x_1. Fix a station $x_1 \in (x_0, 1)$; by continuous dependence $\delta Q_\gamma(x_1; p, \gamma) \geq c_1 := \frac{1}{2}\delta Q_\gamma(x_1; p_0, \gamma_0) > 0$ near (p_0, γ_0). So $\partial_\gamma \sigma \geq c_1/2$ on the part of N where $\sigma > (\omega_-)^2$, a set that accumulates at (p_0, γ_0); by continuity of $\partial_\gamma \sigma$ and after shrinking N, $\partial_\gamma \sigma \geq c_1/4$ on all of N. Moreover, shrinking N so that $|\omega| \leq \varepsilon_0$ and $|\partial_\gamma \omega| \leq M$ with $2\varepsilon_0 M < c_1/8$, the functions

$$K = \sigma - \omega^2 \ (C^3), \qquad H = \sigma - (\omega_-)^2 \ (C^1,\ \nabla H \text{ Lipschitz})$$

have $\partial_\gamma K, \partial_\gamma H \geq c_1/8 > 0$. This is (a) except the last assertion: for $p \in U$ with $\gamma^-(p) = \gamma_L(p)$ one has $\sigma(p, \gamma) = \frac{1}{2}\ell(\gamma - \gamma_L) + O((\gamma - \gamma_L)^2)$ by Taylor's formula, where $\ell = 2\partial_\gamma \sigma(p, \gamma_L) = \ell_\gamma$.

Step 4: the roots. Shrinking U and I (so that K and H change sign on I for every $p \in U$) and using $\partial_\gamma K, \partial_\gamma H > 0$, there are unique roots in I: $\gamma_L(p)$ of $\sigma(p, \cdot)$ and $\gamma_R(p)$ of $K(p, \cdot)$, both C^3 by the implicit function theorem, and $\gamma_H(p)$ of $H(p, \cdot)$. Since γ^- is continuous (Theorem 7.10), $\gamma^-(p) \in I$ for p near p_0, and as $\mathrm{FS}(p) = [\gamma^-(p), \infty)$ is the set of forward global γ, Step 2 gives that $\gamma^-(p)$ is the smallest $\gamma \in I$ with $H \geq 0$, i.e. $\gamma^- = \gamma_H$. Put $\omega_L(p) = \omega(p, \gamma_L(p))$. Now:

- $\omega_L \geq 0$: $H(p, \gamma_L) = -\big((\omega_L)_-\big)^2 = 0$, so $\gamma^- = \gamma_L$.
- $\omega_L < 0$: $H(p, \gamma_L) = -\omega_L^2 < 0$, so $\gamma^- > \gamma_L$; at $\gamma' = \gamma^-$ one has $\sigma = (\omega_-)^2 > 0$ and so $\omega(p, \gamma') < 0$ (if $\omega \geq 0$ then $\sigma = 0$, but $\sigma > 0$ above γ_L); hence $K(p, \gamma') = 0$ and $\gamma^- = \gamma_R$.

Since $K(p, \gamma_L) = -\omega_L^2 \leq 0 = K(p, \gamma_R)$ and K is increasing, $\Psi := \gamma_R - \gamma_L \geq 0$ on U, with $\Psi = 0 \iff \omega_L = 0$. This gives $\gamma^- = \gamma_L + \mathbf{1}_{\omega_L < 0}\Psi$ with $\gamma_L, \gamma_R, \Psi \in C^3(U)$.

Step 5: the types and $\nabla \omega_L \neq 0$. The map $p \mapsto \Xi(p, \gamma_L(p)) = (\beta(p), \omega_L(p), 0)$ is a C^3 diffeomorphism of U onto an open subset of the hyperplane $\{\sigma = 0\}$ (the restriction of the diffeomorphism Ξ to the C^3 graph $\{\gamma = \gamma_L(p)\} = \{\sigma \circ \Xi = 0\}$, which Ξ maps onto $\{\sigma = 0\} \cap \Xi(N)$). Hence $(\beta, \omega_L) : U \to \mathbb{R}^5$ is a local C^3 diffeomorphism, and $\nabla \omega_L \neq 0$. If $\omega_L > 0$ the edge state is $(\beta, \omega_L, 0)$, of type L_1; if $\omega_L = 0$ it is $(\beta, 0, 0)$, of type L_2; if $\omega_L < 0$ it is $\Xi(p, \gamma_R) = (\beta, \omega, \omega^2)$ with $\omega = \omega(p, \gamma_R) < 0$, of type R with $\mu = -\omega$ and $Q(1) + 2 = 2\sigma = 2\mu^2$ (all by Theorem 4.45(b)). This is (b) and the last sentence of (d).

Step 6: the profile of Ψ. Put $\ell(p) = 2\partial_\gamma\sigma(p,\gamma_L(p))$, a positive C^2 function equal to ℓ_γ on $\{\omega_L \geq 0\}$. Taylor expansion at (p,γ_L) gives, uniformly in $p \in U$ and small $\psi \geq 0$, $\sigma(p,\gamma_L+\psi) = \frac{1}{2}\ell\psi + O(\psi^2)$ and $\omega(p,\gamma_L+\psi) = \omega_L + O(\psi)$. The equation $K(p,\gamma_R) = 0$ reads $\frac{1}{2}\ell\Psi + O(\Psi^2) = (\omega_L + O(\Psi))^2$. Hence $\Psi\big(\frac{1}{2}\ell - C\Psi\big) \leq 2\omega_L^2$, so (for U small) $\Psi \leq (8/\ell)\omega_L^2$, and then $\frac{1}{2}\ell\Psi = \omega_L^2 + O(|\omega_L|^3)$, i.e. $\Psi = \frac{2}{\ell}\omega_L^2 + O(|\omega_L|^3)$. This is (d).

Step 7: regularity. $\Psi \geq 0$ and $\Psi = 0$ on $Z = \{\omega_L = 0\}$, so Ψ has a minimum there and $\nabla\Psi = 0$ on Z. Lemma 7.31(b) gives $|\nabla\Psi| \leq C|\omega_L|$ near Z. Therefore $\nabla\gamma^- = \nabla\gamma_L + \mathbf{1}_{\omega_L<0}\nabla\Psi$ is continuous. It is Lipschitz on a ball $B \subset U$: let $p,p' \in B$; if both lie in $\{\omega_L \leq 0\}$ or both in $\{\omega_L \geq 0\}$ this follows from the C^2 regularity of γ_L,γ_R; if $\omega_L(p) < 0 < \omega_L(p')$ the segment meets Z at some m and $|\nabla\gamma^-(p) - \nabla\gamma^-(p')| \leq |\nabla\gamma_L(p) - \nabla\gamma_L(p')| + |\nabla\Psi(p)| \leq C|p-p'| + C'|p-m| \leq C''|p-p'|$. So $\gamma^- \in C^{1,1}(U)$. On $\{\omega_L < 0\}$, $\nabla^2\gamma^- = \nabla^2\gamma_R$ and on $\{\omega_L > 0\}$, $\nabla^2\gamma^- = \nabla^2\gamma_L$, both continuous up to Z, so the one-sided limits at $p \in Z$ exist and their difference is $\nabla^2\Psi(p)$. By Lemma 7.31(c) with $\varphi = \omega_L$, $A = 2/\ell$ (Step 6), $\nabla^2\Psi(p) = \frac{4}{\ell}\nabla\omega_L \otimes \nabla\omega_L$. As $\nabla\omega_L \neq 0$ and $\ell > 0$, for any n with $n\cdot\nabla\omega_L \neq 0$ the second derivative of $t \mapsto \gamma^-(p+tn)$ has different one-sided limits at $t=0$ (the jump is $\frac{4}{\ell}(n\cdot\nabla\omega_L)^2 \neq 0$), so γ^- is not twice differentiable there. Smoothness off the wall is Theorem 7.12(i). □

Remark 7.33 (what is claimed and what is not)**.** The theorem is exactly as local as the corner chart. The neighbourhood U depends on the point p_0 and shrinks as the order k of differentiability grows ($\mu_* = 1/(40(k+3))$, Remark 4.47); the argument is written for $k=3$ so that γ_L,γ_R are C^3 (a C^2 Hessian with a cubic Taylor remainder in Step 6, and a C^2 function ℓ). *No statement that γ^- is C^∞ up to the wall, or that γ_L and γ_R are C^∞, is made or used.* Away from the walls the regularity is that of Theorem 7.12. Note that what the theorem produces is a $C^{1,1}$ function whose Hessian is bounded and jumps; Proposition 7.34 shows what this means for convexity along a path.

Proposition 7.34 (the wall cannot create concavity)**.** *Let p_0, U be as in Theorem 7.32, and $t \mapsto p(t)$ a C^2 path in U with $p(t_0) \in \{\omega_L = 0\}$ and $\frac{d}{dt}\omega_L(p(t))|_{t_0} \neq 0$. Put $g(t) = \gamma^-(p(t))$. Then:*

(a) *$g \in C^{1,1}$ near t_0, and g'' has one-sided limits at t_0 whose difference* (R *side minus* L_1 *side*) *is* $\dfrac{4}{\ell_\gamma}\big(\frac{d}{dt}\omega_L(p(t))\big)^2 > 0$.

(b) *If g is convex on $[t_0-\varepsilon,t_0]$ and on $[t_0,t_0+\varepsilon]$ then it is convex on $[t_0-\varepsilon,t_0+\varepsilon]$.*

Proof. (a) $g' = \nabla\gamma^-\cdot p'$ is Lipschitz, since $\nabla\gamma^-$ is Lipschitz and $p \in C^2$. Off t_0, $g'' = p'^{\,T}\nabla^2\gamma^-\,p' + \nabla\gamma^-\cdot p''$; the second term is continuous at t_0 and the first jumps by $p'^{\,T}\big[\nabla^2\gamma^-\big]p' = \frac{4}{\ell_\gamma}(\nabla\omega_L\cdot p')^2$ by Theorem 7.32(c). (b) g' is continuous at t_0 and nondecreasing on each side, hence nondecreasing on the whole interval; this is convexity. □

Corollary 7.35 (L_2 edges are walls; density of regular edges)**.** (a) *$\mathcal{P}_{\mathsf{L}_2}$ is closed in $\mathcal{P}$ and nowhere dense; it is locally a C^3 hypersurface. Every point of $\mathcal{P}_{\mathsf{L}_2}$ is a limit of points of $\mathcal{P}_{\mathsf{R}}$ and of points of $\mathcal{P}_{\mathsf{L}_1}$.*

(b) *The set $\mathcal{P}_{\mathrm{reg}} = \mathcal{P}_{\mathsf{R}} \cup \mathcal{P}_{\mathsf{L}_1}$ is open and dense in $\mathcal{P}$, and $\gamma^- \in C^\infty(\mathcal{P}_{\mathrm{reg}}) \cap C^{1,1}_{\mathrm{loc}}(\mathcal{P})$.*

PROOF. Closedness is Lemma 7.30. If $p_0 \in \mathcal{P}_{\mathsf{L}_2}$, Theorem 7.32(b) shows that near p_0 the set $\mathcal{P}_{\mathsf{L}_2}$ is the hypersurface $\{\omega_L = 0\}$ with $\nabla\omega_L \neq 0$; a hypersurface has empty interior and ω_L takes both signs in every neighbourhood of p_0. So $\mathcal{P}_{\mathsf{L}_2}$ is a closed set with empty interior, $\mathcal{P}_{\mathrm{reg}}$ is open and dense, and by Theorem 7.12(i) and Theorem 7.32(c) the regularity of γ^- is as stated at all points. □

Remark 7.36 (what the wall shows for the structure)**.** In the language of the chapter: the edge hypersurface $\{\gamma = \gamma^-(p)\}$ is a $C^{1,1}$ graph with a codimension-one crease (a jump of the second fundamental form) along the walls, and the pole type changes there from R to L_1; this is the only way to change type (Proposition 7.29). The jump formula says that the crease has the sign of $+\nabla\omega_L \otimes \nabla\omega_L$ on the R side, so a graph that is convex on both sides stays convex across; whether the *lens* boundaries inherit global convexity from this local statement is a question for later chapters.

7.5.5. Numerical observations: the wall, the tips, the type map.

Numerical observation 7.37 (the wall at $\boldsymbol{c} = (0, -0.9, 0)$, $x_0 = 0$)**.** *Along $\kappa \mapsto \gamma^-(0, \boldsymbol{c}, \kappa)$ the edge is of type R for $\kappa < \kappa_t$ and of type L_1 for $\kappa > \kappa_t$, with $\kappa_t = 0.59263$ obtained by a cubic fit of $\mu = \sqrt{(Q(1)+2)/2}$ on the R side to its zero (`verify_ch07a.py`, 106 values of $\kappa \in [0.540, 0.645]$, edges by bisection to 10^{-14}, $Q(1)$ read at $\gamma^- + 10^{-12}$, $s_{\max} = 40$). In the notation of Theorem 7.32 the one-variable data are $\partial_\kappa\omega_L = -d\mu/d\kappa = 1.1170$ (cubic fit on the R side, evaluated at κ_t) and $\ell_\gamma = 0.57734$ (the limit of δQ_γ, computed from the variational equations at nine values of $\kappa \in [0.575, 0.61]$ and interpolated to κ_t). The theorem then predicts, along this line,*

$$\left[\partial_\kappa^2\gamma^-\right]_{\mathsf{R}-\mathsf{L}_1} = \frac{4(\partial_\kappa\omega_L)^2}{\ell_\gamma} = 8.644, \qquad \frac{\gamma^- - \gamma_L}{\mu^2} \longrightarrow \frac{2}{\ell_\gamma} = 3.4642 \quad (\mu \to 0).$$

The one-sided curvatures of γ^- at κ_t, from least-squares polynomials of degree $4, 5, 6, 7$ fitted on $\kappa < \kappa_t - 0.004$ and on $\kappa > \kappa_t + 0.004$, are

degree	4	5	6	7
$(\gamma^-)''$, R *side*	10.719	11.005	11.084	11.105
$(\gamma^-)''$, L_1 *side*	2.397	2.399	2.399	2.399
jump	8.322	8.606	8.685	8.706

The L_1 side is stable in the degree; the R side is not yet converged (the kink limits the polynomial approximation), and the jump increases towards the prediction 8.644 (within 4% for degree 4, 0.7% for degree 7). Second differences with step 10^{-3} show the jump directly:

κ	0.56	0.57	0.58	0.585	0.59	0.595	0.60	0.61	0.62	0.63
$\Delta^2\gamma^-/h^2$	6.34	7.36	8.70	9.54	10.52	2.38	2.34	2.26	2.19	2.12

Finally, with γ_L the degree-6 polynomial continuation of the L_1 branch, $(\gamma^- - \gamma_L)/\mu^2$ equals 2.85, 3.04, 3.25, 3.40 at $\mu = 0.053, 0.036, 0.018, 0.005$ and extrapolates linearly in μ (six points nearest the wall) to 3.4639, against $2/\ell_\gamma = 3.4642$ (Figure 7.2). The statements of Theorem 7.32, Proposition 7.34 are thus observed: γ^- has a corner in the second derivative with the predicted jump and no first-derivative kink.

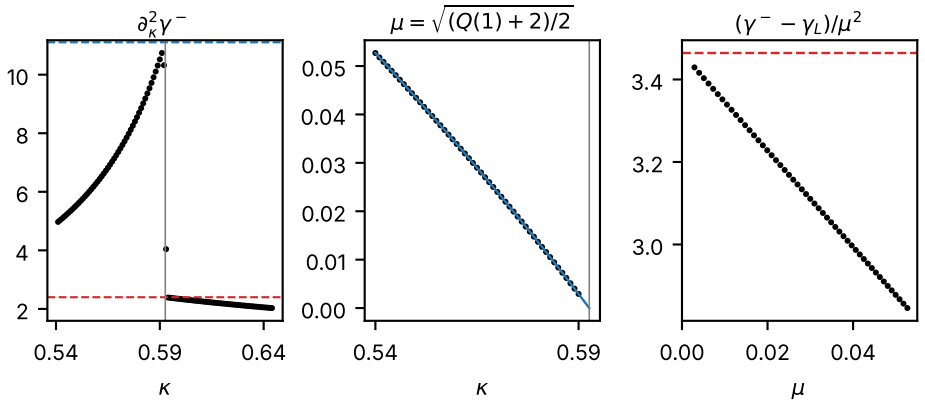


FIGURE 7.2. The wall at $\boldsymbol{c}=(0,-0.9,0)$, $x_0=0$ (Numerical observation 7.37). Left: second difference of γ^- in κ (step 10^{-3}); dashed: the one-sided curvatures of the degree-7 fits; the isolated point is the difference whose stencil straddles κ_t. Middle: $\mu=\sqrt{(Q(1)+2)/2}$ on the R side with its cubic fit, vanishing linearly at κ_t (vertical line). Right: $(\gamma^- - \gamma_L)/\mu^2$ against μ, with the predicted limit $2/\ell_\gamma$ (dashed). Code: `code/ch07_edges/make_a_wall.py`.

Numerical observation 7.38 (the pole types at the tips and the type along an edge)**.** *At the tip and on the edges the north pole is generically of type* L_1 *($Q(1)=-2$ to 10^{-5}, $U+2\sim 4/\ln\frac{1}{1-x}$, $U_\varphi(1)$ finite). Type* R *occurs when some c_i is close to -1: for $c=(-0.9,-0.9,-0.782)$ the tip has $Q(\pm1)=-1.857$, $U_\theta(\pm1)=\mp2.535$ and unbounded swirl; for $c=(-0.9,0,-2.132)$ the tip is* R *at $x=-1$ and* L_1 *at $x=1$. Along a lower edge with $c_2=-0.9$ the* R*-value of $Q(1)$ moves from -1.80 to -2.00 as κ grows from 0.01 to 2. Type* L_2 *was never observed.*
Recomputation *(`verify_ch07a.py`, part `tip` and `edges`; $x_0=0$, $s_{\max}=30$, $Q(1)$ at γ^-+10^{-10} resp. at the two poles at $\kappa_{\max}-10^{-5}$):*

data $\boldsymbol{c}$	$\kappa_{\max}$	$Q(1)$	$Q(-1)$	$U(1)$	$U(-1)$
$(0,0,0)$	4.16081	-2.00000	-2.00000	-1.8625	-1.8625
$(-0.9,-0.9,-0.782)$	0.35511	-1.85717	-1.85717	-2.5345	-2.5345
$(-0.9,0,-2.132)$	0.41236	-2.00000	-1.84186	-1.8481	-2.5624

The pole types follow the pattern of the statement: L_1 *at both poles for $\boldsymbol{c}=0$ ($Q+2=1.4\cdot10^{-10}$; $U(1)+2=0.1375$ at $s_{\max}=30$, where the logarithmic law $U+2\sim4/s$ predicts 0.133),* R *at both poles for $(-0.9,-0.9,-0.782)$ and* R *at the south,* L_1 *at the north pole for $(-0.9,0,-2.132)$. The value $U(\pm1)=\mp2.535$ of the statement is the value at $\kappa_{\max}-10^{-3}$ (∓2.53498); at $\kappa_{\max}-10^{-5}$ it is ∓2.53447, so that the limit at the tip is ∓2.5345 (the two readings differ by the slow approach to the tip). Along $\boldsymbol{c}=(0,-0.9,0)$ the values of $Q(1)$ at $\kappa=0.01,\,0.3,\,0.55,\,0.65,\,2$ are -1.80015, -1.90064, -1.99623, -2.00000, -2.00000: the type is* R *with $Q(1)$ decreasing from -1.80 to -2 up to the wall $\kappa_t=0.5926$ of Numerical observation 7.37, and* L_1 *beyond. An* L_2 *edge occurs* exactly *on the wall, a set of codimension one in $\mathcal{P}$*

(Theorem 7.32) that a scan on a grid does not meet; what the scan sees is the passage from R *to* L_1 *with* $Q(1) \to -2$ *continuously, in agreement with Proposition 7.26.*

Atlas 7.39 (the type of the edge in the (κ, c_2)-plane)**.** ***Definition of the map.*** *For* $x_0 = 0$ *and* $\boldsymbol{c} = (0, c_2, 0)$ *the number* $Q(1) + 2$ *of the edge solution is computed on a* 26×26 *grid,* $c_2 = -0.95, -0.90, \ldots, 0.30$ *and* κ *geometric from* 0.02 *to* 1.6 *(ratio* 1.19*), with edges by bisection and* $Q(1)$ *read at* $\gamma^- + 10^{-10}$*,* $s_{\max} = 40$ *(`verify_ch07a.py`, part `tmap`); Figure 7.3 shows* $\log_{10}(Q(1)+2)$ *with a floor at* -6*.*

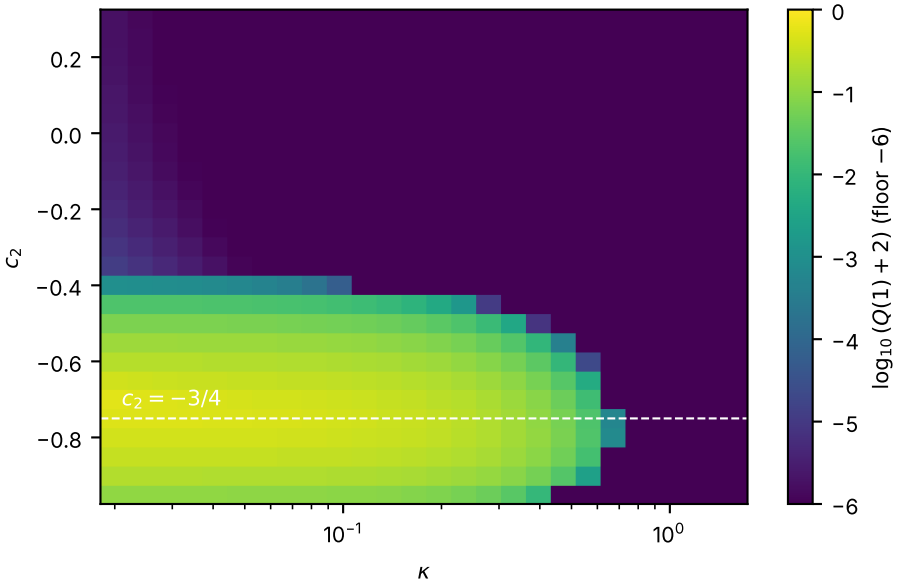


FIGURE 7.3. $\log_{10}(Q(1)+2)$ along the lower edge, $x_0 = 0$, $\boldsymbol{c} = (0, c_2, 0)$ (floor -6; values below 10^{-6} are of type L_1 to the accuracy of the computation). Dashed: $c_2 = -\frac{3}{4}$, to the left of which the small-swirl edge is of type R (Theorem 7.21). For $-\frac{3}{4} < c_2 < -0.4$ and small κ the entries are readings at $s_{\max} = 40$ of a slowly relaxing layer, not edge values. Code: `code/ch07_edges/make_a_tmap.py`.

What is seen. *(1) For* $c_2 \le -0.8$ *the picture is that of Numerical observation 7.24 and Numerical observation 7.37:* R *for small* κ *(at* $\kappa = 0.02$*,* $Q(1)+2 =$ 0.0996, 0.199, 0.299, 0.398*, close to* $2(1+c_2)$*, as Theorem 7.21 predicts), then a wall, then* L_1*. The first grid value of* κ *with* $Q(1)+2 < 10^{-6}$ *is* 0.469, 0.666, 0.666, 0.794 *for* $c_2 = -0.95, -0.90, -0.85, -0.80$*; the true wall* $\kappa_t(-0.9) = 0.5926$ *lies between the grid points* 0.559 *and* 0.666*. (2) For* $c_2 \ge -0.35$ *the edge is of type* L_1 *from the smallest* κ *of the grid on (first value with* $Q(1)+2 < 10^{-6}$ *at* $\kappa \le 0.057$*). (3) For* $-0.75 < c_2 < -0.4$ *the no-swirl edge has* $U(1) = -2 - 2\mu_0 \le -3$ *and the readings at* $s_{\max} = 40$ *near* $\kappa \to 0$ *are not at the pole limit; the part of the plane near the dashed line to the right of it is outside the scope of Theorem 7.21, and the numbers*

there are not used anywhere in this chapter. (4) The L_1 *region is, on the grid, the complement of a bounded region of the* (κ, c_2)*-plane in which the edge is of type* R*; for* $\kappa \geq 1$ *the grid values satisfy* $Q(1) + 2 < 1.5 \cdot 10^{-9}$ *for every* c_2*.*

Remark 7.40 (what this part does not cover)**.** The analytic results of this part are local at L_2 edges and rely on the sign structure of the linearised flow for R and L_1. The questions that these results leave open — global information on the hypersurface of edges at large κ, the matching upper bound for $Q(1) + 2$ away from the walls, the regularity of γ^- up to the wall beyond $C^{1,1}$, and the layer at $c_2 > -\frac{3}{4}$ — are collected in Chapter 13. Part b of this chapter continues with the edge in the full space of data and the interval property of the fibres.

7.6. The edge hypersurface in the space of data

In words. *Part a studied the threshold* $\gamma^-(p)$ *as a function of the parameters* $p = (x_0, \boldsymbol{c}, \kappa)$ *of the lens. The set* $\Omega \subset \mathbb{R}^6$ *of all global solutions is described by Cauchy data* $(\boldsymbol{c}, \gamma, A, \kappa)$ *at the base point* 0*, not by lens parameters, and the two descriptions are related by a change of base point. This section reads the results of part a as a statement about* Ω *itself. Away from the face* $\kappa = 0$ *the boundary of* Ω *in the swirl part consists exactly of the solutions that are not generic at one of the two poles; in the coordinates of the lens it is the graph of* γ^- *together with the graph of* γ^+*, and the change of base point is a diffeomorphism. Hence the* edge hypersurface *of* Ω *is a topological hypersurface, a smooth one where the pole type is* R *or* L_1*, and the surfaces on which the type changes are walls. The section ends with a precise list of what is proved about this hypersurface, a numerical scan for edges of type* R *at large swirl, and a heuristic explaining why the scan sees none.*

Throughout this section $\Omega = \Omega(0) \subset \mathbb{R}^6$ is the set of Cauchy data $\omega = (\boldsymbol{c}, \gamma, A, \kappa)$ at the base point $x_b = 0$ of global solutions (Lemma 5.7), Z and N are the classes of Definition 5.3, and $\Omega^+_{\mathsf{Z}} := \Omega \cap \mathsf{Z} \cap \{\kappa > 0\}$. The sign of κ is the sign of f_x on all of $(-1, 1)$, so it does not depend on the base point. We write $\mathcal{L}$ for the lens set of Theorem 5.62.

7.6.1. Change of base point.

Lemma 7.41 (from lens parameters to Cauchy data at 0)**.** *Let* Λ *be the set of* $(x_0, \boldsymbol{c}, \kappa, \gamma) \in (-1, 1) \times \mathbb{R}^3 \times (0, \infty) \times \mathbb{R}$ *for which the solution* $Y(\cdot; x_0, \boldsymbol{c}, \kappa, \gamma)$ *of* (5.2) *with* $Y(x_0) = (\gamma, \kappa, 0, 0, 0, 0)$ *exists on the closed interval between* x_0 *and* 0*, and let* $\Phi : \Lambda \to \mathbb{R}^6$ *send such a point to the Cauchy data at* $x_b = 0$ *of the same solution. Then* Λ *is open,* Φ *is an injective* C^∞ *local diffeomorphism, hence a* C^∞ *diffeomorphism of* Λ *onto an open subset of* $\mathbb{R}^6$*, and* $\Phi(\mathcal{L}) = \Omega^+_{\mathsf{Z}}$*.*

Proof. *Smooth map.* The field $F_{\boldsymbol{c}}$ is polynomial in $(Y, \boldsymbol{c})$ and rational in $x \in (-1, 1)$, so the solution depends smoothly on $(x_0, \boldsymbol{c}, \kappa, \gamma)$ and exists, for nearby parameters, on a neighbourhood of the compact interval between x_0 and 0; this gives openness of Λ and smoothness of the state at 0. The Cauchy data at 0 are the affine image of this state: $(U, g, f)(0) = (\gamma_0, \kappa_0, A)$ and $\boldsymbol{c}_0$ is the solution of the linear system $(P_{\boldsymbol{c}_0}, P'_{\boldsymbol{c}_0}, P''_{\boldsymbol{c}_0})(0) = (Q, Q_x, Q_{xx})(0)$, whose matrix is invertible (Lemma 2.18(b): $1-x, 1+x, 1-x^2$ is a basis of the quadratics). So Φ is C^∞. *Inverse.* Let $\omega^* = \Phi(x_0^*, \boldsymbol{c}^*, \kappa^*, \gamma^*)$. For ω near ω^* the solution with Cauchy data ω exists on a neighbourhood of the interval between 0 and x_0^*, and its swirl $f(x; \omega)$ is C^∞ in (x, ω) with $f(x_0^*; \omega^*) = 0$, $\partial_x f(x_0^*; \omega^*) = \kappa^* > 0$ (here $g = f_x$). By the implicit function

theorem there is a C^∞ function $x_0(\omega)$ with $f(x_0(\omega);\omega)=0$; and $\kappa(\omega)=g(x_0(\omega);\omega)$, $\gamma(\omega)=U(x_0(\omega);\omega)$ and $\boldsymbol{c}(\omega)$ (the solution of $(P_{\boldsymbol{c}},P'_{\boldsymbol{c}},P''_{\boldsymbol{c}})(x_0)=(Q,Q_x,Q_{xx})(x_0)$) are C^∞. Flowing from $x_0(\omega)$ to 0 returns ω, and flowing from 0 to x_0 returns the lens parameters, so the two maps are inverse to each other near ω^*. Hence Φ is a local diffeomorphism. *Injectivity and image.* f is strictly monotone (Corollary 2.10), so a solution with swirl has at most one zero x_0, and the solution determines $(x_0,\boldsymbol{c},\kappa,\gamma)$ by the recipe above; thus Φ is injective. A point of Λ lies in $\mathcal{L}$ iff its solution is global; by Theorem 5.62 and Lemma 5.7(a) these are exactly the global solutions of class Z with $\kappa>0$. □

7.6.2. The boundary of Ω in class Z. For the lens $\mathrm{Lens}(x_0,\boldsymbol{c})$ of Definition 5.55 write $P_{\mathcal{L}}=\{(x_0,\boldsymbol{c},\kappa):\ \exists\gamma,\ (x_0,\boldsymbol{c},\kappa,\gamma)\in\mathcal{L}\}$; by Proposition 5.59 this is $\{\boldsymbol{c}\in\operatorname{int}\mathcal{J},\ 0<\kappa\le\kappa_{\max}(x_0,\boldsymbol{c})\}$. Put

$$\mathfrak{E}^-:=\Phi\big\{(p,\gamma^-(p)):\ p\in P_{\mathcal{L}}\big\},\qquad \mathfrak{E}^+:=\Phi\big\{(p,\gamma^+(p)):\ p\in P_{\mathcal{L}}\big\},$$
$$\mathfrak{E}:=\mathfrak{E}^-\cup\mathfrak{E}^+.$$

Proposition 7.42 (the edge hypersurface)**.** (a) (Edge = boundary.) $\mathfrak{E}=\Omega^+_{\mathsf{Z}}\setminus\Omega_{\mathsf{G}}$ *is the set of global solutions of class* Z *with* $\kappa>0$ *that are not of type* G *at both poles, and* $\mathfrak{E}=\partial\Omega\cap\mathsf{Z}\cap\{\kappa>0\}$. *The solutions in* $\mathfrak{E}^-\setminus\mathfrak{E}^+$ *are of type* R, L_1 *or* L_2 *at the north pole and of type* G *at the south pole; those in* $\mathfrak{E}^+\setminus\mathfrak{E}^-$ *the reverse;* $\mathfrak{E}^-\cap\mathfrak{E}^+$ *consists of the tips. The reflection* $\sigma_1\sigma_2$ *maps* $\mathfrak{E}^-$ *onto* $\mathfrak{E}^+$.

(b) (Regularity.) *In the coordinates* (p,γ) *of* Λ, $\mathfrak{E}^-$ *is the graph of the locally Lipschitz function* γ^- *over the open set* $\mathcal{P}$ *(restricted to* $P_{\mathcal{L}}$*, the set where* $\kappa\le\kappa_{\max}$*), hence* $\mathfrak{E}^-$ *is a topological hypersurface* with boundary *of* $\mathbb{R}^6$*, the boundary being the set of tips (*$\kappa=\kappa_{\max}$*); its interior, the graph over the open set* $\{\kappa<\kappa_{\max}\}$*, is a topological hypersurface. The union* $\mathfrak{E}=\mathfrak{E}^-\cup\mathfrak{E}^+$ *is obtained by gluing the two sheets along the tips. Over* $\mathcal{P}_{\mathsf{R}}\cup\mathcal{P}_{\mathsf{L}_1}$ *the function* γ^- *is* C^∞ *and* $\mathfrak{E}^-$ *is a* C^∞ *embedded hypersurface; over* $\mathcal{P}_{\mathsf{L}_2}$ *it is* $C^{1,1}$*. The set of edge points of type* R *or* L_1 *is open and dense in* $\mathfrak{E}^-$*, and its complement, the set of* L_2 *edge points, is closed, nowhere dense, and locally a* C^3 *hypersurface of* $\mathfrak{E}^-$.

(c) (Orientation.) *Near an edge point of type* R *or* L_1 *the set* Ω *lies on one side of* $\mathfrak{E}^-$*: in the coordinates of* Λ *it is* $\{\gamma\ge\gamma^-(p)\}$*, and* $\partial_\kappa\gamma^->0>\partial_{c_i}\gamma^-$ (*Theorem 7.12(iii)*).

Proof. (a) By Lemma 7.41 and Theorem 5.62, $\Omega^+_{\mathsf{Z}}=\Phi(\mathcal{L})$, and the position of a lens point decides the pole types (table of Theorem 5.62): interior points are of type G at both poles and lie in Ω_{G}; the points on the graph of γ^- or of γ^+ are not. So $\Omega^+_{\mathsf{Z}}\setminus\Omega_{\mathsf{G}}=\mathfrak{E}$ and the statements on the pole types follow from the same table. By Proposition 5.15(a), Ω_{G} is open and $\partial\Omega\subset\Omega\setminus\Omega_{\mathsf{G}}$ (Ω is closed by Theorem 5.12), so $\partial\Omega\cap\mathsf{Z}\cap\{\kappa>0\}\subset\mathfrak{E}$. Conversely let $\omega=\Phi(p,\gamma^-(p))\in\mathfrak{E}^-$. For $\varepsilon>0$ the point $\Phi(p,\gamma^-(p)-\varepsilon)$ belongs to $\Phi(\Lambda)$ for small ε (openness of Λ), tends to ω as $\varepsilon\to0$, and is not in Ω because $\gamma^-(p)-\varepsilon\notin\mathrm{FS}(p)$; so $\omega\in\partial\Omega$. For $\mathfrak{E}^+$ use $\Phi(p,\gamma^+(p)+\varepsilon)$ and $\gamma^+(p)+\varepsilon\notin\mathrm{BS}(p)$. The reflection $\sigma_1\sigma_2:(\boldsymbol{c},\gamma,A,\kappa)\mapsto(\boldsymbol{c}_m,-\gamma,-A,\kappa)$ preserves Ω, κ and the class Z (Lemma 5.7(c)), and exchanges the two poles (Lemma 5.1); so it maps the non-generic north pole to a non-generic south pole, that is $\mathfrak{E}^-$ to $\mathfrak{E}^+$. (b) The first two statements are Theorem 7.10(i), Theorem 7.12(i), Theorem 7.32(c), and the graph of a locally Lipschitz (resp. C^∞, $C^{1,1}$) function over an open set is a topological (resp. smooth, $C^{1,1}$) hypersurface (over the closed set $P_{\mathcal{L}}$ it is one

with boundary, the boundary lying over $\{\kappa = \kappa_{\max}\}$, Theorem 7.10(iii)); Φ is a diffeomorphism (Lemma 7.41), so it preserves these properties. The last statement is Corollary 7.35 and Theorem 7.32(b). (c) By Lemma 7.41, the intersection of Ω_{Z}^{+} with a neighbourhood of an edge point is Φ of $\{(p,\gamma) : \gamma^{-}(p) \le \gamma \le \gamma^{+}(p)\}$; near a point of $\mathfrak{E}^{-} \setminus \mathfrak{E}^{+}$ the upper bound is not active ($\gamma^{-} < \gamma^{+}$ there by Proposition 5.59(a),(b)). The signs are those of Theorem 7.12(iii). □

Remark 7.43 (class N and the face $\kappa = 0$)**.** In class N the same picture holds with the shift in place of the lens: by Theorem 6.61(iii) the edge is Γ_{N}, the set of N_0 solutions that are not of type G at the north pole, it is the boundary of $\mathsf{N}_0 \cong \Gamma_T \times [0,\infty)$ (resp. $B \times [0,\infty)$), and Theorem 7.19 gives its smoothness at edges of type R and L_1 as a hypersurface in the chart coordinates of the germs; smoothness of the threshold Θ^{-} as a function on B is not proved (Remark 7.20). On the face $\kappa = 0$ there is no edge hypersurface in this sense: the fibre is the interval $[U_{\boldsymbol{c}}^{-}(0), U_{\boldsymbol{c}}^{+}(0)]$ for every A (Lemma 5.46(c)), and the points of the face with γ strictly inside belong to Ω_{G} (Proposition 5.15(b)). How the hypersurface $\mathfrak{E}$ ends on the face is the subject of Sections 7.8 and 7.8.4.

7.6.3. What is known about the whole hypersurface.

Proposition 7.44 (what is proved about the edge hypersurface)**.** *Let $\mathcal{P}_{\mathsf{c}}$ be as in part a.*

(a) (Types.) $\mathfrak{E}^{-} = E_{\mathsf{R}} \sqcup E_{\mathsf{L}_1} \sqcup E_{\mathsf{L}_2}$ *with $E_{\mathsf{R}}, E_{\mathsf{L}_1}$ relatively open, E_{L_2} relatively closed and nowhere dense. A connected subset of $\mathfrak{E}^{-}$ that meets E_{R} and E_{L_1} meets E_{L_2}. Hence the boundary of E_{R} in $\mathfrak{E}^{-}$ lies in E_{L_2}.*

(b) (Pole value.) *The value $Q(1)$ of the edge solution is continuous along $\mathfrak{E}^{-}$ and $E_{\mathsf{R}} = \{Q(1) > -2\}$ (Proposition 7.26).*

(c) (Small swirl, $c_2 < -\frac{3}{4}$.) *For $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ with $c_2 < -\frac{3}{4}$ and $x_0 = 0$ the edge point over $(\boldsymbol{c},\kappa)$ is of type R for all sufficiently small $\kappa > 0$, and γ^{-} is smooth in κ^2 there (Theorem 7.21).*

(d) (Floor.) *As $\kappa \downarrow 0$ the edge value $\gamma^{-}(x_0,\boldsymbol{c},\kappa)$ tends to $U_{\boldsymbol{c}}^{-}(x_0)$, $\gamma^{+} \to U_{\boldsymbol{c}}^{+}(x_0)$ (Proposition 5.59(e)).*

Proof. (a) is Lemma 7.30, Theorem 7.12(i), Corollary 7.35, Proposition 7.29 in the coordinates of Lemma 7.41. (b) is Proposition 7.26 together with the definition of the types ($q > -2$ for R, $q = -2$ for $\mathsf{L}_1, \mathsf{L}_2$, Definition 4.16). (c) is Theorem 7.21. (d) is Proposition 5.59(e). □

Not proved (and not claimed): whether edges of type R exist at large swirl, whether E_{R} and E_{L_1} are connected, what the type of the edge is for small κ when $c_2 > -\frac{3}{4}$ (this is the layer problem of Section 7.8.4), and the behaviour of $\mathfrak{E}$ at the tips and as $x_0 \to \pm 1$. The first of these questions is the reason for the following scan.

Numerical observation 7.45 (the type of the lower edge at large swirl)**.** *Take $x_0 = 0$ and $\boldsymbol{c} = (0, -0.8, c_3)$, so that $k_2 = \sqrt{0.2} < \frac{1}{2}$ and Theorem 7.21 makes the lower edge of type R for κ below some $\kappa_0(\boldsymbol{c})$. For $c_3 = 3, 30, 300$ the largest swirl constant of the lens is $\kappa_{\max} = 3.0059,\ 6.0469,\ 16.348$ (Proposition 5.59(a)). For $\kappa = f\kappa_{\max}$ with $f = 0.01, \dots, 0.9$ the table gives the number $Q(1)+2$ of the edge solution, read at $\gamma = \gamma^{-} + 10^{-9}$ and $s_{\max} = 30$; type R has $Q(1)+2 > 0$ (Definition 4.16), type L_1 has $Q(1)+2 = 0$.*

c_3	$\kappa_{\max}$	$f = 0.01$	0.03	0.1	0.3	0.6	0.9
3	3.0059	0.360	0.203	$4.6 \cdot 10^{-9}$	$7.3 \cdot 10^{-10}$	$2.9 \cdot 10^{-10}$	$1.8 \cdot 10^{-10}$
30	6.0469	$1.2 \cdot 10^{-8}$	$3.4 \cdot 10^{-9}$	$8.5 \cdot 10^{-10}$	$2.5 \cdot 10^{-10}$	$1.2 \cdot 10^{-10}$	$7.5 \cdot 10^{-11}$
300	16.3477	$2.7 \cdot 10^{-9}$	$8.6 \cdot 10^{-10}$	$2.5 \cdot 10^{-10}$	$7.9 \cdot 10^{-11}$	$3.9 \cdot 10^{-11}$	$2.6 \cdot 10^{-11}$

Of the 18 *edges only two, both at* $c_3 = 3$ *and* $\kappa = 0.030, 0.090$*, have* $Q(1) + 2$ *of order one and are of type* R*. The other* 16 *have* $Q(1) + 2 \le 1.3 \cdot 10^{-8}$ *and are of type* L_1 *to the accuracy of the computation, up to* $\kappa = 14.7$*; in particular no* R *edge is seen at* $c_3 = 30$ *and* 300*, not even at the smallest swirl* $\kappa = 0.06$ *and* 0.16 *of the scan, so the window* $\kappa < \kappa_0(\boldsymbol{c})$ *of Theorem 7.21 shrinks as* c_3 *grows. (*`code/ch07_edges/verify_ch07b.py`*, part* `scan`*.)*

Heuristic 7.46 (why the scan sees no large-swirl edge of type R)**.** *For large* κ *the solutions of class* Z *with the swirl bounded in the effective coordinates of Chapter 10 have a swirl step of height* $J \asymp \sqrt{\kappa}$ *with a thin logistic layer and a centrifugal plateau* $W \approx J^2/16$ *around the equator, and the layer position moves towards the pole at which* z *becomes recessive as the mixing parameter tends to* 0 *or* ∞*; an edge of type* R *is the end point of this process. The chart of type* R *(Theorem 4.40) requires the cut* ξ_* *to satisfy* $\kappa^2 E_*^{1-2\varsigma} \ll 1$*, that is* $1 - x_* \lesssim \kappa^{-2/(1-2\varsigma)}$*, far outside the first layer. An* R *edge at large* κ *would therefore need a second, pole, layer in addition to the equatorial step. This is a scaling argument, not a proof, and the scan of Numerical observation 7.45, which does not reach the regime* $\kappa \sim 10^3$*, neither supports nor contradicts it beyond* $\kappa \approx 15$*.*

7.7. The interval property of the fibres of Ω

In words. *Fix* $(\boldsymbol{c}, A, \kappa)$ *and shoot from the base point* 0 *with initial slope* γ*. Some slopes survive up to the north pole, some up to the south pole, and the fibre of* Ω *is the set of slopes that survive both ways. If* $A\kappa = 0$ *the forward survivors form a half-line pointing up and the backward survivors a half-line pointing down, so the fibre is an interval and* Ω *is the region between two graphs. If* $A\kappa > 0$ *the forward survivors are still a half-line, but for the backward survivors the comparison argument stops: going backwards from* 0 *the swirl first moves in the "wrong" direction until it vanishes. What we can prove is a chain of statements about the* edge points *of the backward survivors: a reduction to a sign condition at the pole, a theorem saying that near an edge point of type* R *or* L_1 *the backward survivors are the set* $\{\varphi \ge 0\}$ *for a smooth function whose derivative is a pole limit, a finite check on a compact interval that implies the sign condition, and a theorem showing that the check holds when* A/κ *is small. The general interval property is not proved here; it is stated as a conjecture in Chapter 13.*

7.7.1. Setting, the mirror problem, and two toys. For $\omega = (\boldsymbol{c}, \gamma, A, \kappa) \in \mathbb{R}^6$ let S_γ be the maximal solution with Cauchy data ω at $x_b = 0$. For fixed $(\boldsymbol{c}, A, \kappa)$ put

$$\mathrm{FS} = \{\gamma : \ S_\gamma \text{ exists on } [0,1)\}, \qquad \mathrm{BS} = \{\gamma : \ S_\gamma \text{ exists on } (-1,0]\},$$

so that the fibre is $\Omega_{\boldsymbol{c},A,\kappa} = \mathrm{FS} \cap \mathrm{BS}$ (Lemma 5.7). Proposition 5.42 and Lemma 5.40 give: FS is empty or a half-line $[\gamma_{\mathrm{F}}, \infty)$ if $A\kappa \ge 0$, and BS is empty or a half-line $(-\infty, \gamma_{\mathrm{B}}]$ if $A\kappa \le 0$; we write $\gamma_{\mathrm{F}} = \min \mathrm{FS}$, $\gamma_{\mathrm{B}} = \max \mathrm{BS}$ and put $\gamma_{\mathrm{F}} = +\infty$ (resp. $\gamma_{\mathrm{B}} = -\infty$) if the set is empty. When the fibre is non-empty, closed and bounded,

we write $\Gamma^- = \min \Omega_{\boldsymbol{c},A,\kappa}$ and $\Gamma^+ = \max \Omega_{\boldsymbol{c},A,\kappa}$. The reflection σ_1 acts on solutions by $f \mapsto -f$ and leaves (U, Q) unchanged, so it preserves the interval of existence in both directions: FS and BS are unchanged under $(A, \kappa) \mapsto (-A, -\kappa)$. *For $A\kappa > 0$ we may and do assume $A > 0$ and $\kappa > 0$.*

The mirror problem. By Lemma 5.7(c) the reflection $\sigma_1\sigma_2$ maps S_γ to the solution $\widetilde{S}_{\tilde\gamma}$ with data $(\tilde{\boldsymbol{c}}, \tilde\gamma, \tilde A, \kappa) = (\boldsymbol{c}_m, -\gamma, -A, \kappa)$, and

$$(\widetilde{U}, \tilde f, \widetilde{Q})(x) = (-U(-x), -f(-x), Q(-x)),$$
$$\text{so} \qquad \gamma \in \mathrm{BS}(\boldsymbol{c}, A, \kappa) \iff \tilde\gamma = -\gamma \in \mathrm{FS}(\boldsymbol{c}_m, -A, \kappa)$$

(Lemma 5.40). For $A > 0$ the mirror swirl starts at $\tilde f(0) = -A < 0$ and is increasing ($\tilde g = g(-x) > 0$): the mirror problem is a *forward* problem in the mixed quadrant, and it is cooperative only after the zero $\tilde x_z = -x_z$ of $\tilde f$. Tilded variables are the mirror variables, $\tilde\xi = -\xi$, $\tilde z(\tilde\xi) = z(-\tilde\xi)$. The Jacobi fields of the mirror problem, $\tilde J = \partial_{\tilde\gamma}\widetilde{S}$ (at fixed x), are related to those of the original by

$$\begin{gathered}\partial_{\tilde\gamma}\widetilde{U}(x) = \partial_\gamma U(-x), \qquad \partial_{\tilde\gamma}\widetilde{Q}(x) = -\partial_\gamma Q(-x) = \partial_\gamma\psi(-x), \\ \tilde v(\tilde\xi) = \tfrac12 \tilde z^2\, \partial_{\tilde\gamma}\widetilde{U} = v(-\tilde\xi),\end{gathered} \tag{7.6}$$

with $v = \frac12 z^2 \partial_\gamma U$ computed along the backward solution; here $\psi = P_{\boldsymbol{c}} - Q$ is the nonlocal term, which has vanishing 2-jet at 0 (Lemma 2.5).

Variables along the backward solution. Let $\xi = \operatorname{artanh} x \in (\xi_*, 0]$, $z > 0$ with $z(0) = 1$, $z'(0) = \gamma/2$, $W = 1 + \frac12 Q$ and $z'' = Wz$, $f' = \kappa z^{-2}$, $U = 2(z'/z - \tanh\xi)$ (Lemma 2.14). Differentiating in γ at fixed ξ, with $w = \partial_\gamma z$, $v = zw' - z'w$, $\psi_\gamma = \partial_\gamma\psi$, $f_\gamma = \partial_\gamma f$, $\delta Q = -\psi_\gamma$:

$$\begin{gathered}w(0) = 0,\ w'(0) = \tfrac12,\ v(0) = \tfrac12, \qquad v' = -\tfrac12 z^2 \psi_\gamma, \\ \partial_\gamma U = \frac{2v}{z^2}, \qquad f_\gamma' = -\frac{2\kappa w}{z^3},\ f_\gamma(0) = 0.\end{gathered} \tag{7.7}$$

The first identities are Lemma 7.1 (the sign convention $v' = \frac12 z^2 \delta Q$) with $\delta Q = -\psi_\gamma$; the last follows from $f' = \kappa z^{-2}$. Finally $(w/z)' = v/z^2$ and $w(0) = 0$ give

$$w(\xi) = -z(\xi) \int_\xi^0 \frac{v(\eta)}{z(\eta)^2}\, \mathrm{d}\eta. \tag{7.8}$$

A point $\gamma_0 \in \partial\mathrm{BS}$ is called an *edge point*; if f has a zero $x_z \in (-1, 0)$ along S_{γ_0} we say that *(Z) holds*, and we write $\xi_z = \operatorname{artanh} x_z$. For $A > 0$ and $\kappa > 0$ the swirl f is increasing in x, so along the backward solution $f > 0$ on $(x_z, 0]$ and $f < 0$ on $(-1, x_z)$.

Toy example 7.47 (a constant potential)**.** Freeze $W \equiv 1$. The solution through $z(0) = 1$, $z'(0) = \gamma/2$ is $z = \cosh\xi + \frac{\gamma}{2}\sinh\xi = \frac12\big[(1 + \frac{\gamma}{2})e^{\xi} + (1 - \frac{\gamma}{2})e^{-\xi}\big]$. As $\xi \to -\infty$ the term $e^{-\xi}$ dominates, so $z > 0$ on $(-\infty, 0]$ iff $\gamma \le 2$: the backward survivors are $\mathrm{BS} = (-\infty, 2]$, a half-line. For $\gamma < 2$ the solution is dominant ($z \sim e^{|\xi|}$); at $\gamma = 2$ it is $z = e^{\xi}$, recessive. So *the edge point of* BS *is the recessive solution*, as it is at the pole of the real problem. In the real problem $W = 1 + \frac12 Q$ depends on γ through ψ, and this is the whole difficulty.

Toy example 7.48 (cooperativity in dimension two)**.** Let $u' = -u + v$, $v' = u - 2v$. The off-diagonal entries are ≥ 0. If $u(0), v(0) \ge 0$, then $u, v \ge 0$ for all $t \ge 0$: at a first time at which $u = 0$ one has $u' = v \ge 0$, so u cannot become negative, and the same for v. The quadrant is invariant. For the Q-system the difference

of two solutions solves a linear system with such a matrix in the sign convention $\Sigma = \mathrm{diag}(+,-,-,-,-,-)$, on the region $\{f, g \geq 0\}$ (Lemma 5.28); the quadrant becomes the orthant $\{\Sigma W \geq 0\}$ of $\mathbb{R}^6$. The catch, for $A > 0$ and the *backward* problem, is that the mirror swirl $\tilde{f}$ starts negative: the system is not cooperative on $[0, \tilde{x}_z]$.

7.7.2. The nonlocal term and the Jacobi field.

Lemma 7.49 (the equation of ψ and of ψ_γ)**.** *Let $\gamma \in \mathrm{BS}$, $A \in \mathbb{R}$, $\kappa \neq 0$, and $\xi \leq 0$.*

(a) $\mathsf{L}\psi = f^2 - A^2$, $\psi(0) = \psi_x(0) = \psi_{xx}(0) = 0$*; and* $\Psi := \cosh^2\xi\,\psi$ *satisfies* $\Psi'' - 4\Psi = f^2 - A^2$ *with* $\Psi(0) = \Psi'(0) = 0$.

(b) $\Psi_\gamma := \cosh^2\xi\,\psi_\gamma$ *satisfies* $\Psi_\gamma'' - 4\Psi_\gamma = h$, $h := 2ff_\gamma$, $\Psi_\gamma(0) = \Psi_\gamma'(0) = 0$, *and*

$$\Psi_\gamma(\xi) = \int_\xi^0 \tfrac{1}{2}\sinh\big(2(\eta-\xi)\big)\,h(\eta)\,\mathrm{d}\eta,$$

$$\psi_\gamma(\xi) = \int_\xi^0 \mathsf{K}(\xi,\eta)\,h(\eta)\,\mathrm{d}\eta, \quad \mathsf{K} = \frac{\sinh(2(\eta-\xi))}{2\cosh^2\xi},$$

with $0 \leq \mathsf{K}(\xi,\eta) \leq e^{2\eta} \leq 1$ *for* $\xi \leq \eta \leq 0$.

(c) *With* $n := \Psi_\gamma' - 2\Psi_\gamma$ *one has* $(e^{2\xi}n)' = e^{2\xi}h$.

Proof. (a) By Lemma 2.5 at the base point 0, $\psi = P_{\boldsymbol{c}} - Q$ has vanishing 2-jet at 0 and $\psi_{xxx} = 2ff_x/(1-x^2)$ (this is (2.21) with $Q = P_{\boldsymbol{c}} - \psi$). By Lemma 2.18(a), $\partial_x\mathsf{L}\psi = (1-x^2)\psi_{xxx} = 2ff_x = \partial_x(f^2)$, so $\mathsf{L}\psi - f^2$ is constant, equal to its value $\mathsf{L}\psi(0) - A^2 = -A^2$ ($\mathsf{L}\psi(0) = \psi_{xx}(0) - 2\psi(0) = 0$). In the variable ξ, $\mathsf{L}\psi = \cosh^2\xi\,\psi'' + 4\sinh\xi\cosh\xi\,\psi' - 2\psi$ (Lemma 2.18(a)), and $\Psi'' - 4\Psi = \mathsf{L}\psi$ by direct differentiation of $\Psi = \cosh^2\xi\,\psi$ (a computer-algebra check: `code/ch07_edges/verify_ch07b.py`, S1). (b) The data $(\boldsymbol{c}, A, \kappa)$ do not depend on γ, so $\psi(0) = \psi_x(0) = \psi_{xx}(0) = 0$ for all γ and $f(0) = A$; differentiating (a) in γ at fixed ξ gives $\Psi_\gamma'' - 4\Psi_\gamma = 2ff_\gamma$ with zero Cauchy data at 0. The function $\Psi(\xi) = \int_\xi^0 \tfrac{1}{2}\sinh(2(\eta-\xi))h(\eta)\mathrm{d}\eta$ has $\Psi' = -\int_\xi^0 \cosh(2(\eta-\xi))h\,\mathrm{d}\eta$ and $\Psi'' = h + 2\int_\xi^0 \sinh(2(\eta-\xi))h\,\mathrm{d}\eta = h + 4\Psi$, and zero Cauchy data at 0; by uniqueness it is Ψ_γ (S2). For $\xi \leq \eta \leq 0$, $\sinh(2(\eta-\xi)) \geq 0$; and $\sinh(2(\eta-\xi)) \leq \tfrac{1}{2}e^{2(\eta-\xi)}$, $\cosh^2\xi \geq \tfrac{1}{4}e^{-2\xi}$ give $\mathsf{K} \leq e^{2\eta}$. (c) $(e^{2\xi}(\Psi' - 2\Psi))' = e^{2\xi}(\Psi'' - 4\Psi)$ (S3). □

7.7.3. The case $A\kappa = 0$, closedness, and a local criterion.

Proposition 7.50 (the fibre for $A\kappa = 0$)**.** *Let $A\kappa = 0$. Then* $\mathrm{FS} = [\gamma_{\mathrm{F}}, \infty)$ *or* $\emptyset$, $\mathrm{BS} = (-\infty, \gamma_{\mathrm{B}}]$ *or* $\emptyset$, *so $\Omega_{\boldsymbol{c},A,\kappa} = [\gamma_{\mathrm{F}}, \gamma_{\mathrm{B}}] \cap \mathbb{R}$ is a closed interval (possibly empty) with $\Gamma^- = \gamma_{\mathrm{F}}$, $\Gamma^+ = \gamma_{\mathrm{B}}$. If $(\boldsymbol{c}, \gamma, A, \kappa) \in \Omega$ and $\boldsymbol{c}' \geq \boldsymbol{c}$ then $(\boldsymbol{c}', \gamma, A, \kappa) \in \Omega$. For $A = 0$, $\kappa > 0$ the fibre is the lens fibre $[\gamma^-(0, \boldsymbol{c}, \kappa), \gamma^+(0, \boldsymbol{c}, \kappa)]$ of Definition 5.55; for $\kappa = 0$ it is $[U_{\boldsymbol{c}}^-(0), U_{\boldsymbol{c}}^+(0)]$ for every A if $\boldsymbol{c} \in \mathcal{J}$, $c_1, c_2 > -1$.*

Proof. If $A\kappa = 0$ then $A = 0$ or $\kappa = 0$, and by σ_1 (which does not change FS or BS) we may assume $A \geq 0$ and $\kappa \geq 0$. Then FS is empty or the half-line $[\gamma_{\mathrm{F}}, \infty)$ by Proposition 5.42(b). By Lemma 5.40, $\mathrm{BS}(\boldsymbol{c}, A, \kappa) = -\mathrm{FS}(\boldsymbol{c}_m, -A, \kappa)$. If $A = 0$ this is $-\mathrm{FS}(\boldsymbol{c}_m, A, \kappa)$; if $\kappa = 0$ the solution is the no-swirl solution and FS does not depend on the sign of A (Lemma 5.46(c)), so again $\mathrm{BS}(\boldsymbol{c}, A, \kappa) = -\mathrm{FS}(\boldsymbol{c}_m, A, \kappa)$. This is empty or a half-line $(-\infty, \gamma_{\mathrm{B}}]$ by Proposition 5.42(b) applied to $\boldsymbol{c}_m$ (this is part (d) of that proposition). The fibre is the intersection. For the monotonicity

in $\boldsymbol{c}$: if $\boldsymbol{c} \le \boldsymbol{c}'$, Proposition 5.42(c) gives $\mathrm{FS}(\boldsymbol{c}, A, \kappa) \subset \mathrm{FS}(\boldsymbol{c}', A, \kappa)$; and $\boldsymbol{c}_m \le \boldsymbol{c}'_m$, so $\mathrm{BS}(\boldsymbol{c}, A, \kappa) = -\mathrm{FS}(\boldsymbol{c}_m, A, \kappa) \subset -\mathrm{FS}(\boldsymbol{c}'_m, A, \kappa) = \mathrm{BS}(\boldsymbol{c}', A, \kappa)$. The last two statements are Definition 5.55 and Lemma 5.46(c). □

Lemma 7.51 (BS is closed and bounded above)**.** *For every* $(\boldsymbol{c}, A, \kappa)$ *the set* BS *is closed. If* $A > 0$ *and* $\kappa > 0$ *then* $\mathrm{BS} \subset (-\infty, 2\sqrt{W^*}\,]$ *with* $W^* = 1 + \frac{1}{2}\big(\max_{[-1,1]} P_{\boldsymbol{c}} + \frac{1}{2}A^2\big)$.

Proof. Closedness: $\Omega^-(0)$ is closed in $\mathbb{R}^6$ (Theorem 5.12) and BS is its slice at fixed $(\boldsymbol{c}, A, \kappa)$. Upper bound: by Lemma 2.18(d) with $x_a = 0$, zero Cauchy data and $F = f^2 - A^2 \ge -A^2$ (see Lemma 7.49(a)), $\psi(x) = \int_0^x K_{\mathsf{L}}(x,s)(f^2 - A^2)\mathrm{d}s$. For $x < 0$ one has $K_{\mathsf{L}}(x,s) < 0$ for $x < s \le 0$ (Lemma 2.18(c)), so $\psi(x) = \int_x^0 |K_{\mathsf{L}}(x,s)|(f^2 - A^2)\mathrm{d}s \ge -A^2 \int_x^0 |K_{\mathsf{L}}|\mathrm{d}s = -A^2x^2/2$, using $\int_0^x K_{\mathsf{L}}\,\mathrm{d}s = x^2/2$. Hence $Q = P_{\boldsymbol{c}} - \psi \le \max P_{\boldsymbol{c}} + A^2/2$ and $W \le W^*$ along the backward solution. The Riccati variable $h = z'/z$ satisfies $h' = W - h^2$; in the backward variable $\tau = -\xi$, $\mathrm{d}h/\mathrm{d}\tau = h^2 - W \ge h^2 - W^*$. If $h(0) = \gamma/2 > \sqrt{W^*}$ then h increases and, comparing with $\mathrm{d}k/\mathrm{d}\tau = k^2 - W^*$, blows up at a finite τ: $z \to 0$ at a finite $\xi < 0$, so the solution does not exist on $(-1, 0]$ (Lemma 2.14: U exists on an interval iff $z > 0$ there). So $\gamma \le 2\sqrt{W^*}$ on BS. □

Lemma 7.52 (local criterion)**.** *Say* $\gamma_0 \in \mathrm{BS}$ *is* left-open *if* $(\gamma_0 - \varepsilon, \gamma_0] \subset \mathrm{BS}$ *for some* $\varepsilon > 0$. *If every point of* BS *is left-open then* BS *is empty,* $\mathbb{R}$*, or* $(-\infty, \gamma_{\mathrm{B}}]$. *Points of* $\mathrm{BS} \setminus \partial\mathrm{BS}$ *are left-open.*

Proof. Let $\gamma_1 \in \mathrm{BS}$ and $m = \inf\{t : [t, \gamma_1] \subset \mathrm{BS}\}$. If $m > -\infty$, then $m \in \mathrm{BS}$ (BS is closed, Lemma 7.51) and left-openness at m contradicts the definition of m. Hence $(-\infty, \gamma_1] \subset \mathrm{BS}$ for every $\gamma_1 \in \mathrm{BS}$, so BS is a down-set; a closed down-set is $\emptyset$, $\mathbb{R}$ or $(-\infty, \max \mathrm{BS}]$. The last statement holds because points of $\mathrm{BS} \setminus \partial\mathrm{BS}$ are interior. □

7.7.4. Cooperative comparison and the reduction theorem. Let $A > 0$, $\kappa > 0$, $\gamma_0 \in \mathrm{BS}$, and suppose (Z) holds along S_{γ_0}. Let $\tilde{J} = \partial_{\tilde{\gamma}}\widetilde{S}$ be the Jacobi field of the mirror problem at $\tilde{\gamma}_0 = -\gamma_0$, a solution of the linearised Q-system on $[0, 1)$ (Lemma 2.5, Lemma 7.1). A *cut* is a point $x_e \in (-1, x_z)$, and we put $\tilde{x}_e = -x_e \in (\tilde{x}_z, 1)$. We say that *(E) holds at* γ_0 if there is a cut x_e with

$$\Sigma\tilde{J}(\tilde{x}_e) > 0 \qquad (\text{all six components, } \Sigma = \mathrm{diag}(+,-,-,-,-,-)),$$

i.e. $\partial_{\tilde{\gamma}}\widetilde{U} > 0$ and $\partial_{\tilde{\gamma}}\tilde{g}, \partial_{\tilde{\gamma}}\tilde{f}, \partial_{\tilde{\gamma}}\tilde{p}_k < 0$ at $\tilde{x}_e$: the Jacobi field is in the open orthant of Lemma 7.3(a), which it would be in for a problem started at a zero of the swirl.

Proposition 7.53 (orthant entry implies left-openness)**.** *Let* $A, \kappa > 0$, $\gamma_0 \in \mathrm{BS}$, *and let (E) hold at* γ_0. *Then* γ_0 *is left-open.*

Proof. We work in the mirror problem: put $\tilde{\gamma}_0 = -\gamma_0 \in \mathrm{FS}(\boldsymbol{c}_m, -A, \kappa)$, $Y = \widetilde{S}_{\tilde{\gamma}_0}$ and, for $\tilde{\gamma} > \tilde{\gamma}_0$ close to $\tilde{\gamma}_0$, $\hat{Y} = \widetilde{S}_{\tilde{\gamma}}$; both are solutions of (5.2) for the same constants $\boldsymbol{c}_m$ (Lemma 5.23, base point 0). We must show $\tilde{\gamma} \in \mathrm{FS}$ for $\tilde{\gamma} - \tilde{\gamma}_0$ small. *Entry at the cut.* $\hat{Y}$ exists on $[0, \tilde{x}_e]$ for $\tilde{\gamma}$ near $\tilde{\gamma}_0$ and depends smoothly on $\tilde{\gamma}$ there, so $\hat{Y}(\tilde{x}_e) - Y(\tilde{x}_e) = (\tilde{\gamma} - \tilde{\gamma}_0)\tilde{J}(\tilde{x}_e) + o(\tilde{\gamma} - \tilde{\gamma}_0)$. The set $\{\Sigma J > 0\}$ is open, so $\Sigma(\hat{Y} - Y)(\tilde{x}_e) > 0$ for $0 < \tilde{\gamma} - \tilde{\gamma}_0 < \varepsilon$. Moreover $\tilde{f}(\tilde{x}_e) > 0$ for Y because $\tilde{x}_e > \tilde{x}_z$ and $\tilde{f}$ is increasing, hence also for $\hat{Y}$ if ε is small; and $\tilde{g} > 0$ always (Lemma 5.25(a)). By Lemma 5.25(b), $f \ge f(\tilde{x}_e) > 0$ on $[\tilde{x}_e, \cdot)$ for both. *Comparison.* Y is forward global.

Let $\hat{X} \le 1$ be the end of the forward interval of $\hat{Y}$. For every $b < \hat{X}$, Lemma 5.28(a) on $[\tilde{x}_e, b]$ (same constants, $g, \hat{g}, f, \hat{f} \ge 0$, $\Sigma(\hat{Y} - Y)(\tilde{x}_e) \ge 0$) gives $\hat{U} \ge U$ on $[\tilde{x}_e, \hat{X})$. If $\hat{X} < 1$, Lemma 5.41(d), applied to $\hat{Y}$ re-based at $\tilde{x}_e$ (Lemma 5.7(b): Cauchy data at $\tilde{x}_e$ with $A'' = \hat{f}(\tilde{x}_e) > 0$, $\kappa'' = \hat{g}(\tilde{x}_e) > 0$), says $\hat{U} \to -\infty$ as $x \uparrow \hat{X}$, whereas $\hat{U} \ge U \ge \min_{[\tilde{x}_e, \hat{X}]} U > -\infty$: a contradiction. So $\hat{X} = 1$, $\tilde{\gamma} \in \mathrm{FS}$, that is $-\tilde{\gamma} \in \mathrm{BS}$ for $-\tilde{\gamma} \in (\gamma_0 - \varepsilon, \gamma_0)$. □

THEOREM 7.54 (reduction to edge points). *Let $A > 0$, $\kappa > 0$, and suppose (E) holds at every $\gamma_0 \in \partial\mathrm{BS}$. Then* BS *is empty, $\mathbb{R}$, or $(-\infty, \gamma_{\mathrm{B}}]$.*

PROOF. BS is closed, so $\partial\mathrm{BS} \subset \mathrm{BS}$. Points of $\mathrm{BS} \setminus \partial\mathrm{BS}$ are left-open (Lemma 7.52); points of $\partial\mathrm{BS}$ are left-open by Proposition 7.53. Apply Lemma 7.52. □

This is a proved implication, not the interval property: (E) is only a sufficient condition and is checked below only in a regime. The sharp form of the condition is a sign at the pole.

7.7.5. The pole chart at edges of type R and L_1: the sign condition (E′). In words. *Near the south pole a global solution is described by a few numbers (the pole chart of Chapter 4), and the solutions of a fixed pole type* R *or* L_1 *form a smooth hypersurface in the space of solutions. The family $\gamma \mapsto S_\gamma$ crosses it at an edge point, and it crosses it in one of two directions. Left-open means that it crosses from the non-surviving side (larger γ) to the surviving side. The direction is the sign of a single number, the derivative φ' below, and this number is a pole limit: the limit of a Wronskian (*R*) or of the variation of the nonlocal term (*L_1*). Positive means left-open; negative means that the edge point is the* lower *end of a bubble of survivors; zero is undecided.*

Re-basing at the zero of the swirl. Let $A, \kappa > 0$ and let $\gamma_0 \in \partial\mathrm{BS}$ with (Z). In the mirror problem $\tilde{f}$ has the simple zero $\tilde{x}_z > 0$ with $\tilde{g} = \kappa_z > 0$ there. For $\tilde{\gamma}$ near $\tilde{\gamma}_0 = -\gamma_0$ the mirror solution exists on a neighbourhood of $[0, \tilde{x}_z]$ and, by the implicit function theorem at the simple zero (as in the proof of Lemma 7.41), the zero $\tilde{x}_z(\tilde{\gamma})$, the Taylor constants $\boldsymbol{c}'(\tilde{\gamma})$ of $\widetilde{Q}$ at $\tilde{x}_z(\tilde{\gamma})$, $\kappa'(\tilde{\gamma}) = \tilde{g}(\tilde{x}_z)$ and $\gamma'(\tilde{\gamma}) = \widetilde{U}(\tilde{x}_z)$ are C^∞ functions of $\tilde{\gamma}$. Put $p(\tilde{\gamma}) = (\tilde{x}_z, \boldsymbol{c}', \kappa')$. On $[\tilde{x}_z, 1)$ the mirror solution is the class-Z forward solution $Y(\cdot; p(\tilde{\gamma}), \gamma'(\tilde{\gamma}))$ of part a, and the part on $[0, \tilde{x}_z]$ is regular. Since $\mathcal{P}$ is open (Theorem 7.10(i)) and $p(\tilde{\gamma}_0) \in \mathcal{P}$ (because $\tilde{\gamma}_0 \in \mathrm{FS}$, i.e. $\gamma'(\tilde{\gamma}_0) \in \mathrm{FS}(p(\tilde{\gamma}_0)) \neq \emptyset$), we get, for $\tilde{\gamma}$ near $\tilde{\gamma}_0$,

$$\begin{aligned} \tilde{\gamma} \in \mathrm{FS}(\boldsymbol{c}_m, -A, \kappa) &\iff \gamma'(\tilde{\gamma}) \in \mathrm{FS}(p(\tilde{\gamma})) = [\gamma^-(p(\tilde{\gamma})), \infty) \\ &\iff \varphi(\tilde{\gamma}) := \gamma'(\tilde{\gamma}) - \gamma^-\big(p(\tilde{\gamma})\big) \ge 0. \end{aligned} \tag{7.9}$$

THEOREM 7.55 (local structure of BS at edges of type R and L_1). *Let $A > 0$, $\kappa > 0$, let $\gamma_0 \in \partial\mathrm{BS}$ satisfy (Z), and assume that S_{γ_0} is of type* R *or* L_1 *at $x = -1$. Then φ is C^∞ near $\tilde{\gamma}_0 = -\gamma_0$, $\varphi(\tilde{\gamma}_0) = 0$ and, for small $\varepsilon > 0$,*

$$\mathrm{BS} \cap (\gamma_0 - \varepsilon, \gamma_0 + \varepsilon) = \{\gamma : \ \varphi(-\gamma) \ge 0\} \cap (\gamma_0 - \varepsilon, \gamma_0 + \varepsilon).$$

The derivative $\varphi'(\tilde{\gamma}_0)$ is a pole limit,

$$\varphi'(\tilde{\gamma}_0) = \lim_{\tilde{x} \to 1} \frac{\delta \widetilde{U}_{\tilde{J}}}{\delta \widetilde{U}_{\gamma'}} \quad (\mathsf{R}), \qquad \varphi'(\tilde{\gamma}_0) = \lim_{\tilde{x} \to 1} \frac{\delta \widetilde{Q}_{\tilde{J}}}{\delta \widetilde{Q}_{\gamma'}} \quad (\mathsf{L}_1),$$

where $\tilde J = \partial_{\tilde\gamma}\widetilde S$ and $J_{\gamma'} = \partial_\gamma Y$ is the Jacobi field of the class-$\mathsf Z$ *family, and its sign is decided as follows, with $v = \frac12 z^2\partial_\gamma U$ and $\psi_\gamma = \partial_\gamma\psi$ computed along the backward solution S_{γ_0}:*

(i) *type* $\mathsf R$*:* $\varphi'(\tilde\gamma_0) > 0 \iff \liminf_{\xi\to-\infty} v > 0$*;* $\varphi'(\tilde\gamma_0) < 0 \iff \limsup_{\xi\to-\infty} v < 0$*;* $\varphi'(\tilde\gamma_0) = 0 \iff v \to 0$*;*

(ii) *type* $\mathsf L_1$*: the same with v replaced by ψ_γ.*

Consequently: if $\varphi'(\tilde\gamma_0) > 0$ then $\mathrm{BS}\cap(\gamma_0-\varepsilon,\gamma_0+\varepsilon) = (\gamma_0-\varepsilon,\gamma_0]$, and γ_0 is left-open; if $\varphi'(\tilde\gamma_0) < 0$ then $\mathrm{BS}\cap(\gamma_0-\varepsilon,\gamma_0+\varepsilon) = [\gamma_0,\gamma_0+\varepsilon)$ and γ_0 is the lower end of a component of BS*; if $\varphi'(\tilde\gamma_0) = 0$ nothing is decided.*

Proof. *The function φ.* By (7.9), $\tilde\gamma \in \mathrm{FS}$ iff $\varphi(\tilde\gamma) \ge 0$ near $\tilde\gamma_0$, and $\varphi(\tilde\gamma_0) \ge 0$ since $\tilde\gamma_0 \in \mathrm{FS}$. If $\varphi(\tilde\gamma_0) > 0$ then, γ^- being continuous (Theorem 7.10(i)), $\varphi > 0$ in a neighbourhood, which would lie in FS and make $\tilde\gamma_0$ an interior point of FS, i.e. γ_0 an interior point of BS: contradiction. So $\varphi(\tilde\gamma_0) = 0$, i.e. $\gamma'(\tilde\gamma_0) = \gamma^-(p_0)$, $p_0 = p(\tilde\gamma_0)$: the forward solution from $\tilde x_z$ is the *edge solution* Y_e of p_0, and its type at $\tilde x = 1$ is the type of S_{γ_0} at $x = -1$, $\mathsf R$ or $\mathsf L_1$ by hypothesis. By Theorem 7.12(i), $\gamma^- \in C^\infty$ near p_0 and φ is C^∞ near $\tilde\gamma_0$. Since $\gamma \in \mathrm{BS} \iff -\gamma \in \mathrm{FS}$, the displayed description of BS follows. *The derivative.* For $x > \tilde x_z(\tilde\gamma_0)$ the mirror family is $\widetilde S_{\tilde\gamma}(x) = Y(x; p(\tilde\gamma), \gamma'(\tilde\gamma))$, so by the chain rule, with $\nu \in \{x_0, c_1, c_2, c_3, \kappa\}$ the components of p and $\mathsf J_\nu, \mathsf J_\gamma$ the Jacobi fields of part a along Y_e,

$$\tilde J = \sum_\nu \nu'(\tilde\gamma_0)\,\mathsf J_\nu + \gamma'_{\tilde\gamma}\,\mathsf J_\gamma.$$

By Proposition 7.7 and Theorem 7.12(ii) there is, for each ν, exactly one number s_ν with $\mathsf J_\nu + s_\nu\mathsf J_\gamma$ tangent to the chart hypersurface, and $s_\nu = \partial_\nu\gamma^-(p_0)$. Hence

$$\begin{aligned} \tilde J = T + \varphi'(\tilde\gamma_0)\,\mathsf J_\gamma, \qquad T := \sum_\nu \nu'(\tilde\gamma_0)\big(\mathsf J_\nu + \partial_\nu\gamma^-\,\mathsf J_\gamma\big) \text{ tangent},\\ \varphi' = \gamma'_{\tilde\gamma} - \sum_\nu \nu'\,\partial_\nu\gamma^-, \end{aligned} \tag{7.10}$$

the last identity being the chain rule for $\varphi = \gamma' - \gamma^-\circ p$. *Type* $\mathsf R$. With $\tilde v_{(\cdot)} = \frac12\tilde z^2\,\delta\widetilde U_{(\cdot)}$, (7.10) gives $\tilde v_{\tilde J} = \tilde v_T + \varphi'\,v_\gamma$. By Lemma 7.4, $\tilde v_T \to 0$; by Lemma 7.3(b), $v_\gamma \ge m_1 > 0$ on $[x_1, 1)$. Dividing by v_γ and using $\tilde v_T/v_\gamma \to 0$ gives $\tilde v_{\tilde J}/v_\gamma \to \varphi'$, which is the limit formula (z^2 cancels). If $\varphi' > 0$ then $\tilde v_{\tilde J} \ge \tilde v_T + \varphi' m_1 \ge \frac12\varphi' m_1$ for large $\tilde\xi$, so $\liminf \tilde v_{\tilde J} > 0$; if $\varphi' < 0$, $\limsup \tilde v_{\tilde J} \le \frac12\varphi' m_1 < 0$; if $\varphi' = 0$, $\tilde v_{\tilde J} = \tilde v_T \to 0$. The three alternatives for φ' are exhaustive and the three conclusions about $\tilde v_{\tilde J}$ are mutually exclusive, so the implications reverse. Finally $\tilde v_{\tilde J}(\tilde\xi) = v(-\tilde\xi)$ by (7.6), so the conditions are those on v as $\xi \to -\infty$. *Type* $\mathsf L_1$. Apply $\delta\widetilde Q$ to (7.10): $\delta\widetilde Q_{\tilde J} = \delta\widetilde Q_T + \varphi'\,\delta Q_\gamma$ with $\delta\widetilde Q_T \to 0$ (Lemma 7.5) and $\delta Q_\gamma \ge m_1 > 0$ (Lemma 7.3(c)); the same argument applies, and $\delta\widetilde Q_{\tilde J}(\tilde x) = \psi_\gamma(-\tilde x)$ by (7.6). *Consequences.* If $\varphi'(\tilde\gamma_0) > 0$, then $\varphi \ge 0$ exactly for $\tilde\gamma \ge \tilde\gamma_0$ near $\tilde\gamma_0$, i.e. $\gamma \le \gamma_0$; if $\varphi' < 0$, exactly for $\gamma \ge \gamma_0$. □

Corollary 7.56 (bubbles)**.** *Let $A, \kappa > 0$ and suppose that every point of* ∂BS *satisfies (Z) and is of type* $\mathsf R$ *or* $\mathsf L_1$*. (a) If some $\gamma_0 \in \partial\mathrm{BS}$ has $\varphi'(-\gamma_0) < 0$, then* BS *is not of the form $\emptyset$, $\mathbb R$, $(-\infty, \gamma_{\mathrm B}]$. (b) If* BS *is not of that form, then some $\gamma_0 \in \partial\mathrm{BS}$ has $\varphi'(-\gamma_0) \le 0$. (c) $\max\mathrm{BS}$ exists (if $\mathrm{BS} \ne \emptyset$) and $\varphi'(-\max\mathrm{BS}) \ge 0$.*

PROOF. (a) By Theorem 7.55, $\mathrm{BS} \cap (\gamma_0 - \varepsilon, \gamma_0 + \varepsilon) = [\gamma_0, \gamma_0 + \varepsilon)$: $\gamma_0 \in \mathrm{BS}$ but the points just below it are not, so BS is not a down-set. (b) If all $\varphi' > 0$, all edge points are left-open (Theorem 7.55), and Lemma 7.52 concludes. (c) BS is closed and bounded above (Lemma 7.51), so $\gamma_{\mathrm{B}} = \max \mathrm{BS}$ exists; $\gamma_{\mathrm{B}} \in \partial\mathrm{BS}$ and $\mathrm{BS} \cap (\gamma_{\mathrm{B}}, \gamma_{\mathrm{B}} + \varepsilon) = \emptyset$, which excludes $\varphi' < 0$ (Theorem 7.55: then $[\gamma_{\mathrm{B}}, \gamma_{\mathrm{B}} + \varepsilon) \subset \mathrm{BS}$). □

Remark 7.57 (what the chart theorem does not cover)**.** Three things are outside the theorem. (1) Edges of type L_2: there γ^- is only $C^{1,1}$ (Theorem 7.32) and the corner chart replaces the smooth chart; the identification of φ' with a pole limit would need the corner-chart analogue of Lemma 7.5. (2) Edges at which f has no zero on $(-1, 0)$: then the mirror swirl is negative on $[0, 1)$, the mirror solution is not of class Z after any base change, and Lemma 7.41 does not apply. At an edge of type R a zero always exists, because $\tilde{f}(0) = -A < 0$ while $\tilde{f} \to +\infty$ at an R pole (Corollary 4.24(a)); at an L_1 edge f stays bounded at the pole and may keep one sign when A/κ is large. (3) The degenerate case $\varphi' = 0$.

7.7.6. A compact criterion for the sign condition. In words. *The sign condition says that a signed integral is positive. Its integrand has the good sign far from the base point and the bad sign on a short stretch near $x = 0$, between the base point and the zero of the swirl. A maximum-principle argument shows that if the good sign wins at one point ξ_c of the regular interval $[\xi_c, 0]$ then it wins forever. So the sign condition at the pole reduces to a finite check, and the check holds whenever the bad stretch is short, that is when A/κ is small.*

Lemma 7.58 (one sign change)**.** *Let $A, \kappa > 0$, $\gamma_0 \in \mathrm{BS}$ with* (Z)*, and let $\xi_* < \xi_z$ be such that $w < 0$ on $(\xi_*, 0)$. Put $\Psi = \Psi_\gamma = \cosh^2 \xi\, \psi_\gamma$ and $L = \partial_\xi^2 - 4$. On $(\xi_*, 0)$: $L\Psi = h$ with $h > 0$ on (ξ_*, ξ_z) and $h < 0$ on $(\xi_z, 0)$; $\Psi < 0$ on $[\xi_z, 0)$; and either $\Psi < 0$ on all of $(\xi_*, 0)$, or there is $\xi_1 \in (\xi_*, \xi_z)$ with $\Psi \geq 0$ on $(\xi_*, \xi_1]$, $\Psi(\xi_1) = 0$, $\Psi'(\xi_1) < 0$ and $\Psi < 0$ on $(\xi_1, 0)$. In the second case $v \geq v(\xi_1)$ on $(\xi_*, 0]$.*

PROOF. *Signs of h.* By (7.7), $f_\gamma' = -2\kappa w/z^3 > 0$ on $(\xi_*, 0)$ and $f_\gamma(0) = 0$, so $f_\gamma < 0$ on $(\xi_*, 0)$. Since $f > 0$ on $(\xi_z, 0)$ and $f < 0$ on (ξ_*, ξ_z) we get $h = 2ff_\gamma < 0$ on $(\xi_z, 0)$ and $h > 0$ on (ξ_*, ξ_z). $\Psi < 0$ *on* $[\xi_z, 0)$ by the kernel formula of Lemma 7.49(b), whose integrand is negative there. *Principle (P): Ψ has no local maximum with value ≥ 0 in (ξ_*, ξ_z).* At such a point $\Psi'' \leq 0$ and $\Psi \geq 0$ give $L\Psi = \Psi'' - 4\Psi \leq 0 < h$, a contradiction. *Structure of $\{\Psi \geq 0\}$.* Suppose $\Psi(\xi_a) \geq 0$ for some $\xi_a \in (\xi_*, \xi_z)$ and $\Psi(\xi_b) < 0$ for some $\xi_b \in (\xi_*, \xi_a)$. On $[\xi_b, \xi_z]$ the values at the end points are negative and the maximum is $\geq \Psi(\xi_a) \geq 0$, so it is attained at an interior point, which is a local maximum with value ≥ 0: this contradicts (P). Hence $\{\Psi \geq 0\} \cap (\xi_*, \xi_z)$ is an initial segment. It is not all of (ξ_*, ξ_z), since $\Psi(\xi_z) < 0$. If it is nonempty, let $\xi_1 < \xi_z$ be its supremum: $\Psi(\xi_1) = 0$ by continuity ($\Psi < 0$ on $(\xi_1, \xi_z]$) and $\Psi'(\xi_1) \leq 0$. If $\Psi'(\xi_1) = 0$ then $\Psi''(\xi_1) = h(\xi_1) + 4\Psi(\xi_1) = h(\xi_1) > 0$, so ξ_1 is a strict local minimum, incompatible with $\Psi < 0$ to its right; hence $\Psi'(\xi_1) < 0$. *Minimum of v.* By (7.7), $v' = -\frac{1}{2} z^2 \psi_\gamma$ has the sign of $-\Psi$: $v' \leq 0$ on $(\xi_*, \xi_1]$ and $v' > 0$ on $(\xi_1, 0)$, so $v \geq v(\xi_1)$ on $(\xi_*, 0]$. □

THEOREM 7.59 (compact criterion for the sign condition). *Let $A, \kappa > 0$, $\gamma_0 \in \mathrm{BS}$, and suppose there is $\xi_c < \xi_z$ with*

(C1) *$w < 0$ on $[\xi_c, 0)$;* (C2) *$v > 0$ on $[\xi_c, 0]$;* (C3) *$\psi_\gamma(\xi_c) > 0$.*

Then $w<0$ on $(-\infty,0)$; there is $\xi_1\in(\xi_c,\xi_z)$ with $\psi_\gamma\ge 0$ on $(-\infty,\xi_1]$ and $\psi_\gamma<0$ on $(\xi_1,0)$ (exactly one sign change); $v\ge v(\xi_1)>0$ on $(-\infty,0]$, so $\liminf_{\xi\to-\infty}v\ge v(\xi_1)>0$; and, with $m:=e^{2\xi_1}|\Psi'(\xi_1)|>0$, $\liminf_{\xi\to-\infty}\psi_\gamma\ge m/4>0$. Consequently, if γ_0 is an edge point of type R *or* L_1 *then $\varphi'(-\gamma_0)>0$ and γ_0 is left-open (Theorem 7.55).*

Proof. *Continuation.* Let $E=\{\xi'\le\xi_c:\ w<0 \text{ on } [\xi',0),\ v>0 \text{ on } [\xi',0]\}$ and $\xi_*=\inf E$. By (C1), (C2) and continuity, $\xi_*<\xi_c$. Suppose $\xi_*>-\infty$. Lemma 7.58 applies on $(\xi_*,0)$. By (C3), $\Psi(\xi_c)>0$, so the second alternative holds, with $\xi_1\in(\xi_c,\xi_z)$ ($\xi_1\neq\xi_c$ since $\Psi(\xi_1)=0$), and $v\ge v(\xi_1)>0$ on $(\xi_*,0]$ by (C2) ($\xi_1\in[\xi_c,0]$). Since $\gamma_0\in$ BS, the solution exists at ξ_*, and $v(\xi_*)\ge v(\xi_1)>0$ by continuity. By (7.8), $w(\xi_*)=-z(\xi_*)\int_{\xi_*}^0 v/z^2<0$. By continuity $w<0$ and $v>0$ on a neighbourhood of ξ_* to the left, so ξ_* is not the infimum of E: a contradiction. Hence $\xi_*=-\infty$, $w<0$ on $(-\infty,0)$ and Lemma 7.58 applies on $(-\infty,0)$: $\psi_\gamma\ge0$ on $(-\infty,\xi_1]$, $\psi_\gamma<0$ on $(\xi_1,0)$, and $v\ge v(\xi_1)>0$. *The bound on ψ_γ.* On $(-\infty,\xi_z)$, $h>0$, so $e^{2\xi}n$ is increasing (Lemma 7.49(c)). For $\xi\le\xi_1$: $e^{2\xi}n(\xi)\le e^{2\xi_1}n(\xi_1)=e^{2\xi_1}\Psi'(\xi_1)=-m<0$ ($\Psi(\xi_1)=0$). Since $(e^{-2\xi}\Psi)'=e^{-2\xi}n\le -me^{-4\xi}$, integrating from ξ to ξ_1 and using $\Psi(\xi_1)=0$,

$$e^{-2\xi}\Psi(\xi)\ \ge\ m\int_\xi^{\xi_1}e^{-4\eta}\mathrm{d}\eta=\tfrac{m}{4}\big(e^{-4\xi}-e^{-4\xi_1}\big),$$

$$\psi_\gamma=\frac{\Psi}{\cosh^2\xi}\ge e^{2\xi}\Psi\ge\tfrac{m}{4}\big(1-e^{4(\xi-\xi_1)}\big),$$

using $\cosh^2\xi\le e^{-2\xi}$ for $\xi\le0$. So $\liminf_{\xi\to-\infty}\psi_\gamma\ge m/4$. *Conclusion.* At an edge point of type R, $\liminf v>0$ gives $\varphi'>0$ by Theorem 7.55(i); at type L_1, $\liminf\psi_\gamma>0$ gives $\varphi'>0$ by (ii). (Z) holds because $\xi_c<\xi_z$ requires f to have a zero. □

Conditions (C1)–(C3) involve only the regular flow on $[\xi_c,0]$ and are *open* in all parameters $(\boldsymbol{c},A,\kappa,\gamma_0)$ (strict inequalities on a compact interval). The criterion is not circular: it concerns an arbitrary $\gamma_0\in$ BS.

7.7.7. Small A/κ.

Theorem 7.60 (small A/κ). *Let $\mathcal{K}$ be a compact set of $(\boldsymbol{c},\kappa)$ with $\kappa>0$ and let $\Gamma>0$. There is $a_0=a_0(\mathcal{K},\Gamma)>0$ such that for $(\boldsymbol{c},\kappa)\in\mathcal{K}$ and $0<A/\kappa<a_0$, every edge point $\gamma_0\in\partial$BS with $\gamma_0\ge-\Gamma$ satisfies* (Z) *and* (C1)–(C3) *with $\xi_c=-s_0$, for a number $s_0=s_0(\mathcal{K},\Gamma)>0$; so the conclusions of Theorem 7.59 hold. In particular every such edge point of type* R *or* L_1 *is left-open: $\varphi'(-\gamma_0)>0$. The number a_0 can be chosen nonincreasing in Γ, and the explicit choice below tends to 0 as $\Gamma\to\infty$.*

Proof. Put $a=A/\kappa$ (so $A=\kappa a$) and let $\Gamma_{\mathcal{K}}$ be the supremum of $2\sqrt{W^*}$ over $\mathcal{K}\times(0,1]$ (Lemma 7.51; W^* is continuous in $(\boldsymbol{c},\kappa,a)$). Replacing Γ by $\max(\Gamma,\Gamma_{\mathcal{K}})$ we may assume BS $\subset(-\infty,\Gamma]$, so every $\gamma_0\in$ BS with $\gamma_0\ge-\Gamma$ has $|\gamma_0|\le\Gamma$. *Uniform regular flow.* The system for $(z,z',f,Q,\dots)$ together with the variational equations for $(w,w',f_\gamma,Q_\gamma,\dots)$ is smooth in ξ and in the parameters as long as $z>0$. On the compact parameter set $\{(\boldsymbol{c},\kappa)\in\mathcal{K},\ \gamma\in[-\Gamma,\Gamma],\ a\in[0,1]\}$ the solution exists on a common interval $[-s_1,0]$ and all components, with their first two derivatives, are bounded by a constant M. Choose $s_0\le\min(s_1,\frac12)$ so small, in

terms of M, that on $[-s_0,0]$

$$z\in[\tfrac34,\tfrac54],\qquad w'\in[\tfrac14,\tfrac34]\qquad (z(0)=1,\ w(0)=0,\ w'(0)=\tfrac12).$$

Then $-\frac34|\xi|\le w\le-\frac14|\xi|<0$ on $[-s_0,0)$: this is (C1). *The swirl.* From $f'=\kappa z^{-2}$ and $\frac{16}{25}\le z^{-2}\le\frac{16}{9}$: $f(\xi)\le\kappa a-\frac{16}{25}\kappa|\xi|$, so f has a zero ξ_z with $|\xi_z|\le\frac{25}{16}a\le 2a$, provided $2a<s_0$; this is (Z). If moreover $a<s_0/16$ then $\xi_z>-s_0/8$. On $[-s_0,-s_0/2]$, $a/|\xi|\le\frac18$, so $f\le-\kappa|\xi|\big(\frac{16}{25}-\frac18\big)\le-\frac12\kappa|\xi|$, i.e. $|f|\ge\frac12\kappa|\xi|$. From (7.7), $f_\gamma'=-2\kappa w/z^3$ and the bounds on w,z:

$$\tfrac{32}{125}\kappa|\xi|\le f_\gamma'\le\tfrac{32}{9}\kappa|\xi|\ \Longrightarrow\ \tfrac{16}{125}\kappa\xi^2\le|f_\gamma|\le\tfrac{16}{9}\kappa\xi^2\ \text{ on }[-s_0,0],\quad f_\gamma<0$$

(the constants $2\cdot\frac14\cdot\frac{64}{125}=\frac{32}{125}$ and $2\cdot\frac34\cdot\frac{64}{27}=\frac{32}{9}$ and their halves, and $\frac{16}{25}-\frac18\ge\frac12$, are checked in `verify_ch07b.py`, S8). *The good part of $\psi_\gamma(-s_0)$.* For $\eta\in[-\frac34 s_0,-\frac12 s_0]$ we have $\eta\in[-s_0,-s_0/2]$ and $\eta<\xi_z$, so $h=2ff_\gamma=2|f||f_\gamma|\ge2\cdot\frac12\kappa|\eta|\cdot\frac{16}{125}\kappa\eta^2\ge\frac{16}{125}\kappa^2(s_0/2)^3$. For $\xi=-s_0$ and such η: $\eta-\xi\in[\frac{s_0}{4},\frac{s_0}{2}]$, $\sinh(2(\eta-\xi))\ge2(\eta-\xi)\ge\frac{s_0}{2}$ and $\cosh^2\xi\le\cosh^2\frac12<\frac32$, so $\mathsf K(\xi,\eta)\ge s_0/6$. The interval has length $s_0/4$, and $h>0$ on all of $(-s_0,\xi_z)$ and $\mathsf K\ge0$, whence

$$\int_{-s_0}^{\xi_z}\mathsf K\,h\,\mathrm d\eta\ \ge\ \frac{s_0}{4}\cdot\frac{s_0}{6}\cdot\frac{16}{125}\kappa^2\frac{s_0^3}{8}=\frac{\kappa^2s_0^5}{1500}.$$

The bad part. On $(\xi_z,0)$: $0<f\le A=\kappa a$ (f is increasing with $f(0)=A$), so $|h|\le2\kappa a\cdot\frac{16}{9}\kappa\eta^2=\frac{32}{9}\kappa^2a\eta^2$, and with $\mathsf K\le1$ and $|\xi_z|\le2a$,

$$\int_{\xi_z}^{0}\mathsf K\,|h|\,\mathrm d\eta\le\frac{32}{27}\kappa^2a|\xi_z|^3\le\frac{256}{27}\kappa^2a^4=:H_0.$$

Hence, by Lemma 7.49(b), $\psi_\gamma(-s_0)\ge\kappa^2\big(\frac{s_0^5}{1500}-\frac{256}{27}a^4\big)>0$ if $a^4<\frac{27s_0^5}{256\cdot1500}$: this is (C3). Moreover $\psi_\gamma\ge-H_0$ on $[-s_0,0]$ (the part over $(-\infty,\xi_z)$ is ≥0), so by (7.7) and $z^2\le\frac{25}{16}$,

$$v(\xi)=\tfrac12+\tfrac12\int_{\xi}^{0}z^2\psi_\gamma\,\mathrm d\eta\ \ge\ \tfrac12-\tfrac12\cdot\tfrac{25}{16}H_0s_0>0\qquad\text{on }[-s_0,0]$$

if $\frac{25}{32}H_0s_0<\frac12$, which is (C2). So the conditions on a are $a<s_0/16$, $a^4<\frac{27s_0^5}{256\cdot1500}$ and $\frac{25}{32}\cdot\frac{256}{27}(\max_{\mathcal K}\kappa)^2a^4s_0<\frac12$; each depends on $(\mathcal K,\Gamma)$ only through s_0 and $\max_{\mathcal K}\kappa$. Take a_0 the smallest of the three bounds (and 1). Since M grows with Γ, $s_0\to0$ and so $a_0\to0$ as $\Gamma\to\infty$. Theorem 7.59 applies, and gives left-openness at edges of type $\mathsf R$ and $\mathsf L_1$. □

Remark 7.61 (what the theorem gives and what it does not)**.** For $A/\kappa<a_0(\Gamma)$ the half-line property of BS is *not* proved by this theorem, for three reasons. (a) The constant $a_0(\Gamma)$ degenerates as $\Gamma\to\infty$ and we have no a-priori lower bound for the location of edge points, so edges below $-\Gamma$ are not excluded (if $W\ge0$ along the solution then $\gamma<0$ is safe, since $h'=-W\le0$ forbids h to cross 0 upward in the backward direction; but $Q\ge-2$ on the whole interval is not available a priori). (b) Edges of type $\mathsf L_2$ are not covered by Theorem 7.55. (c) Edge points without a zero of f are excluded by the use of the re-basing. What is proved is the sign condition (E′), $\varphi'>0$, at all edges of type $\mathsf R$ and $\mathsf L_1$ with $\gamma_0\ge-\Gamma$.

Remark 7.62 (continuation in A: what a proof would need)**.** One would like to continue from $A=0$, where $\varphi'=1$ (re-basing is trivial: $\tilde{x}_z=0$ and $p(\tilde{\gamma})=(0,\boldsymbol{c}_m,\kappa)$, $\gamma'(\tilde{\gamma})=\tilde{\gamma}$, so $\varphi=\tilde{\gamma}-\gamma^-(p)$ with p independent of $\tilde{\gamma}$), to all $A>0$. The set of A for which (C1)–(C3) hold at all edge points is open (the conditions are open, and edge points move continuously in A as the threshold does, Theorem 7.10). It could cease to be closed only if the cut ξ_c has to move to $-\infty$ (the sign change ξ_1 of ψ_γ escapes to the pole, equivalently $v(\xi_1)\to 0$), or an edge enters from $\gamma=-\infty$, or an L_2 event occurs. So a proof by continuation reduces to a uniform bound $\xi_1\geq -\Xi$ on edge solutions; the numerical values of ξ_1 and of $\min v$ are in Numerical observation 7.70. We do not have such a bound.

7.7.8. The leading order at the zero of the swirl. It is tempting to evaluate the Jacobi field at the zero ξ_z itself, where the swirl vanishes and the mirror problem becomes cooperative. The following computation shows that this fails by a small amount, and explains why one has to wait until the positive source $2ff_\gamma$ after x_z has overcome the small negative contribution from $(x_z,0)$.

Formal computation 7.63 (the Jacobi field at the zero of the swirl for small A/κ)**.** *Formal (remainder not estimated). Let $a=A/\kappa_x\to 0$ ($A=f(x_b)$, κ_x the swirl constant at the base point), and let $\delta=\partial_\gamma$ denote the derivative with respect to the data γ: $w=\partial_\gamma z$, $\psi_\gamma=\partial_\gamma\psi$, $p_k=\partial_x^k\psi$. Near $x=0$: $w\approx x/2$, $\delta f\approx -\kappa_x x^2/2$, $\psi_\gamma''\approx -\kappa_x x^2(A+\kappa_x x)$; at $x_z=-a+O(a^2)$: $\delta p_1\approx\frac{1}{12}\kappa_x^2a^4>0$, $\delta p_0\approx-\frac{1}{30}\kappa_x^2a^5<0$, $\delta p_2\approx\frac{1}{10}\kappa_x^2a^5>0$.*

Derivation. Near $x=0$: $w\approx x/2$, so $\delta g=-2\kappa_x\cosh^2\xi\, w/z^3\approx-\kappa_x x$ and $\delta f\approx-\kappa_x x^2/2$; also $f\approx A+\kappa_x x$. Since $\mathsf{L}\approx\partial_x^2$ at leading order, $\psi_\gamma''\approx 2f\delta f=-\kappa_x x^2(A+\kappa_x x)$, hence $\psi_\gamma'\approx-\frac{1}{3}\kappa_x Ax^3-\frac{1}{4}\kappa_x^2x^4$ and $\psi_\gamma\approx-\frac{1}{12}\kappa_x Ax^4-\frac{1}{20}\kappa_x^2x^5$. At $x_z=-a+O(a^2)$, with $A=\kappa_x a$: $\delta p_1=\psi_\gamma'(x_z)\approx\kappa_x^2a^4(\frac{1}{3}-\frac{1}{4})=\frac{1}{12}\kappa_x^2a^4$ and $\delta p_0=\psi_\gamma(x_z)\approx-\kappa_x^2a^5(\frac{1}{12}-\frac{1}{20})=-\frac{1}{30}\kappa_x^2a^5$; finally $(1-x^2)\psi_\gamma''=h-2x\psi_\gamma'+2\psi_\gamma$ with $h=0$ at x_z gives $\delta p_2\approx 2a\cdot\frac{1}{12}\kappa_x^2a^4-\frac{2}{30}\kappa_x^2a^5=\frac{1}{10}\kappa_x^2a^5$. A series computation of the Jacobi field to order x^6, with the zero x_z solved perturbatively, reproduces the three leading coefficients exactly (a computer-algebra check, `code/ch07_edges/verify_ch07b.py`, S9), and a numerical integration of the variational system confirms them: the ratios $\delta p_1/(\kappa_x^2a^4/12)$ at $a=0.4,0.2,0.1,0.05,0.025$ are 0.9424, 0.9571, 0.9745, 0.9862, 0.9928, and $\delta p_0/(-\kappa_x^2a^5/30)$ are 0.9019, 0.9167, 0.9487, 0.9717, 0.9851 (`verify_ch07b.py`, part `lead`). In the original variables the orthant condition $\Sigma\tilde{J}>0$ at a cut reads $\partial_\gamma U>0$, $\partial_\gamma g>0$, $\partial_\gamma f<0$, $\partial_\gamma p_2>0$, $\partial_\gamma p_1<0$, $\partial_\gamma p_0>0$ (the mirror conditions of Lemma 7.3(a), translated by $\partial_{\tilde{\gamma}}=-\partial_\gamma$ and $\tilde{p}_k(x)=\pm p_k(-x)$ with the signs of $R=\mathrm{diag}(-,+,-,+,-,+)$, Lemma 5.24). So at x_z the two components $\delta p_1=\partial_\gamma p_1$ and $\delta p_0=\partial_\gamma p_0$ have the *wrong* sign, by amounts of order a^4 and a^5, and δp_2 the right one. This is why (E) is evaluated at a cut *after* the zero, and why the compact criterion needs the integral over $(-s_0,\xi_z)$ to win over $H_0\asymp a^4$.

7.7.9. What is proved: the interval property.

Proposition 7.64 (the fibre near the forward edge)**.** *Let $A,\kappa>0$ and $\gamma_{\mathrm{F}}=\min\mathrm{FS}$ (so $\mathrm{FS}\neq\emptyset$), and suppose that $S_{\gamma_{\mathrm{F}}}$ is of type G at $x=-1$ (it exists on $(-1,0]$ and the south pole is generic). Then there is $\varepsilon>0$ with $\Omega_{\boldsymbol{c},A,\kappa}\cap(\gamma_{\mathrm{F}}-\varepsilon,\gamma_{\mathrm{F}}+\varepsilon)=[\gamma_{\mathrm{F}},\gamma_{\mathrm{F}}+\varepsilon)$.*

PROOF. The set of Cauchy data whose backward solution is global and of type G at $x=-1$ is open in $\mathbb{R}^6$: it is one of the two open factors of Ω_{G} in the proof of Proposition 5.15(a) (Corollary 4.42, transported to the base point 0 by Lemma 5.7(b)). So $(\gamma_{\mathrm{F}}-\varepsilon,\gamma_{\mathrm{F}}+\varepsilon)\subset \mathrm{BS}$ for small ε, and the fibre near γ_{F} is $\mathrm{FS}\cap(\gamma_{\mathrm{F}}-\varepsilon,\gamma_{\mathrm{F}}+\varepsilon)=[\gamma_{\mathrm{F}},\gamma_{\mathrm{F}}+\varepsilon)$ (Proposition 5.42). □

Type G at the south pole is an open condition, and it holds at γ_{F} whenever $S_{\gamma_{\mathrm{F}}}$ is not the recessive solution at -1; this is the sense in which BS is a down-set "locally, near γ_{F}".

THEOREM 7.65 (the interval property: what is proved). *For $A\kappa=0$ ($\kappa=\kappa_x$ at the base point) the fibre $\Omega_{c,A,\kappa}=[\gamma_{\mathrm{F}},\gamma_{\mathrm{B}}]\cap\mathbb{R}$ is a closed interval with $\Gamma^-=\gamma_{\mathrm{F}}$, $\Gamma^+=\gamma_{\mathrm{B}}$, and Ω is an up-set in c. For $A\kappa>0$ the forward-safe set* FS *is a half-line and the backward-safe set* BS *is a down-set only locally, near points where it contains a neighbourhood of γ_{F};* the interval property (H1) for $A\kappa>0$ is not proved. *Reduction (*conditional*): if the cut-orthant hypothesis (E) holds for every $\gamma_0\in\partial\mathrm{BS}$ then, for $A\kappa>0$,* BS *is empty, $\mathbb{R}$ or $(-\infty,\gamma_{\mathrm{B}}]$. Local structure: at an edge point of type R or L_1 at which f has a zero,* $\mathrm{BS}=\{\varphi\ge 0\}$ *locally with φ' given by a pole limit; $\varphi'>0$ gives left-openness, $\varphi'<0$ a bubble. Small A/κ: for compact $\mathcal{K}\ni(c,\kappa_x)$ and $\Gamma>0$ there is $a_0(\mathcal{K},\Gamma)>0$ ($a_0\to 0$ as $\Gamma\to\infty$) such that for $0<A/\kappa_x<a_0$ every edge point of type R or L_1 with $\gamma_0\ge-\Gamma$ is left-open (property (E′)); this does not prove the half-line property of* BS*, and does not cover L_2 edges or edges below $-\Gamma$. The case $A=0$ is the case $A\kappa=0$ above.*

PROOF. The statements are assembled from the results of this section. *$A\kappa=0$:* Proposition 7.50, which also gives that Ω is an up-set in $\boldsymbol{c}$ there and, with $A=0$, identifies the case $A=0$ with the case $A\kappa=0$ (γ then runs over the lens fibre). *$A\kappa>0$,* FS*:* Proposition 5.42(b) with σ_1. BS *locally:* Proposition 7.64. *Reduction:* Theorem 7.54 (the hypothesis (E) is the cut-orthant hypothesis of Section 7.7.4). *Local structure:* Theorem 7.55; $\varphi'>0$ gives left-openness and $\varphi'<0$ a bubble. *Small A/κ:* Theorem 7.60; (E′) is $\varphi'>0$, and the qualifications in the statement (the half-line property is not proved, L_2 edges and edges below $-\Gamma$ are not covered) are those of Remark 7.61. The negative clauses of the statement assert no result and need no proof. □

Remark 7.66 (the general statement)**.** In the statement above, (H1) is the *half-line property*: for $A\kappa>0$, FS is empty or a half-line $[\gamma_{\mathrm{F}},\infty)$ and BS is empty or a half-line $(-\infty,\gamma_{\mathrm{B}}]$, so that the fibre of Ω is an interval; it is stated precisely in Chapter 13. The general statement for $A\kappa>0$, namely that BS is a half-line $(-\infty,\gamma_{\mathrm{B}}]$ for all $(\boldsymbol{c},A,\kappa)$, together with the monotonicity of $\Gamma^{\mp}$ in c and in A, is not proved in this monograph; the precise statement is in Chapter 13, and the numerical evidence is summarised in Numerical observation 7.92.

7.7.10. Numerical observations on the fibres. The following observations are the evidence behind the general statement of Remark 7.66; they prove nothing. All numbers come from `code/ch07_edges/verify_ch07b.py` (parts `slices`, `fine`, `down`).

Numerical observation 7.67 (the fibres are intervals; half-lines in γ)**.** *(1)* Random data. *For* 400 *random data with $c_1,c_2\in[-0.9,2]$, $c_3=\bar{c}_3+10^u$ ($u\in[-1,1.2]$), κ of either sign with $|\kappa|\in[0.05,6.3]$ and A of either sign ($|A|\le 3|\kappa|^{1/2}$ or $3|\kappa|$),*

membership in FS *and* BS *was tested on* 2001 *values of* $\gamma \in [-14, 14]$ *(half the data with* $A\kappa < 0$*, half with* $A\kappa > 0$*). For* $A\kappa < 0$*,* BS *is a down-set in* 190 *cases and empty in* 10*, and* FS *is an up-set in all* 200*. For* $A\kappa > 0$*,* FS *is empty in* 11 *and an up-set in* 189 *cases, and* BS *is a down-set in all* 200*: no gap, no second run. (2)* Adversarial data. *A further* 899 *data from six regimes (*$\boldsymbol{c}$ *close to* $\partial\mathcal{J}$*;* c_1 *or* c_2 *close to* -1*;* $|A| \in [5, 50]$*;* $|\kappa| \in [10^{-3.5}, 10^{-2}]$*;* $|\kappa| \in [5, 100]$*; a moderate regime with both signs of* κ*), tested on* 1601 *values of* $\gamma \in [-R, R]$*,* $R = 18 + 3\sqrt{|c_3|} + \frac{1}{2}\sqrt{|A|}$*, gave* 431 *cases with* $A\kappa > 0$*, in all of which* BS *is a down-set, and* 468 *with* $A\kappa < 0$*; in none of the* 899 *cases is* FS *other than an up-set or* BS *other than a down-set or empty. (3)* Blow-up distance. *For* 1900 *data with* $c_1, c_2 \in [-1, 4]$*,* $c_3 \in [-8, 8]$*,* $|A| \in [10^{-2}, 10^{1.5}]$ *and* $\kappa \in [10^{-2}, 10]$ *(*944 *of them with* $A\kappa > 0$*),* BS *was computed on* 600 *values of* $\gamma \in [-K, K]$*. It is non-empty for* 1867 *of them; in none of these is there a member of* BS *above a non-member (no gap), and in none does the distance from* $x = 0$ *at which the backward solution of a non-member blows up fail to be non-increasing in* γ*.*

Numerical observation 7.68 (a fine rescan of 1148 fibres)**.** *A set of* 1148 *data* $(\boldsymbol{c}, A, \kappa)$ *in eight families (the lens* $A = 0$*; generic* $A = a\kappa$*,* $|a| < 3$*; large* $|A|$*;* κ *near* $\kappa_{\max}$*;* $\boldsymbol{c}$ *near* $\partial\mathcal{J}$*;* c_i *near* -1*; large* κ*; data obtained from points of the lens by shifting A) was scanned with a grid of step* 0.0125 *in* $\gamma \in [-25, 25]$*. On* 869 *of them the fibre meets the grid; on none is it more than one run (the earlier, coarser scan of the same data found* 786*). An independent bisection test,* $\gamma_{\mathrm{F}} \le \gamma_{\mathrm{B}}$*, gives* 882 *non-empty fibres; the* 13 *in which the two disagree are fibres of width between* 0.0008 *and* 0.0081*, smaller than the grid step, and in all* 13 *the midpoint of* $[\gamma_{\mathrm{F}}, \gamma_{\mathrm{B}}]$ *is a member of* Ω*. The families* $A = 0$ *and generic* $A = a\kappa$ *have* 37 *of* 40 *and* 379 *of* 500 *non-empty fibres; the family of data with large* κ *has none.*

Numerical observation 7.69 (monotonicity of Ω)**.** (a) In $\boldsymbol{c}$*. For* 200 *sampled fibres, the two edge points and the midpoint of each were tested after increasing one of* c_1, c_2, c_3 *by* 0.02 *or* 0.3*: of* 3600 *tests,* 0 *leave* Ω*. This is the up-set property in* $\boldsymbol{c}$*, proved for* $A\kappa = 0$ *in Proposition 7.50 and observed here for both signs of* $A\kappa$*.* (b) In A*. For* 60 *random* $(\boldsymbol{c}, \kappa)$ *and* 25 *values of* $A \in [-6, 6]$ *the edges* γ_{F} *and* γ_{B} *are monotone in* A *in all* 120 *curves (both non-decreasing in* A *for* $\kappa > 0$*, and non-increasing for* $\kappa < 0$*, by* σ_1*).* (c) In κ: no. *For* 3000 *tests of the implication* $(\boldsymbol{c}, \gamma, A, \kappa) \in \Omega \Rightarrow (\boldsymbol{c}, \gamma, A, s\kappa) \in \Omega$ *with* $s \in \{0.9, 0.5, 0.1, 0\}$ *(edges and midpoints of sampled fibres),* 200 *fail:* 100 *with* $A\kappa < 0$ *and* 100 *with* $A\kappa > 0$*, of which* 65 *with* $s = 0$*, i.e. at the face. So* Ω *is not star-shaped in* κ *(compare Formal computation 7.80).*

Numerical observation 7.70 (the sign condition at the edge: N1, N2, N4)**.** *Data with* $A\kappa > 0$ *(so that the interval property is the unproved one) were sampled with* $c_1, c_2 \in [-1, 3]$*,* $c_3 \in [-4, 6]$*,* $\kappa \in [10^{-1.3}, 10^{0.7}]$ *and* $A = a\kappa$*,* $a \in [10^{-2}, 10]$*;* γ_{B} *was found by bisection to* 10^{-10}*.* Orthant entry (hypothesis (E))*. For* 300 *data and three members each, the Jacobi field of the mirror problem, followed backwards to* $\xi = -6$*, had a zero of* f *in the window in* 569 *of* 900 *samples, and in* 552 *of these all six components of* $\Sigma\tilde{J}$ *became positive at some cut beyond the zero (*$\xi_e < \xi_z$*) and stayed positive. At the edge itself, for* 150 *edges and* $\gamma = \gamma_{\mathrm{B}} - d$*,* $d = 10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}$ (600 *samples, window* $\xi \ge -9$*): a zero of* f *was found in all* 600 *and the orthant was entered and kept in all* 600*; no member of* BS *was found in* $(\gamma_{\mathrm{B}}, \gamma_{\mathrm{B}} + 0.3]$ *on a grid of* 150 *points, for any of the* 150 *edges.* Pole-side

quantities ((E′)). *For* 150 *edges of generic data and* 150 *edges of data biased towards type* R ($c_1 \in [-0.95, -0.8]$, $\kappa \in [10^{-2}, 10^{-0.7}]$)*, taken at* $\gamma_B - 10^{-9}$ *and followed to* $\xi = -8$*: in all* 300*,* $w < 0$ *and* $v > 0$ *on the whole window,* $\psi_\gamma > 0$ *at the end,* ψ_γ *has exactly one sign change* ξ_1*, and* $\xi_1 < \xi_z$*. The smallest value of* $v(\xi_1)$ *is* 0.469 *(generic) and* 0.4999 *(biased towards* R*); the end values* $Q(\xi_{\text{end}})$ *show* 141 *of the generic and* 0 *of the biased edges of type* L_1 *(*$Q < -1.99$*), and* 7 *and* 147 *of type* R *(*$Q > -1.9$*). The positions are* $\xi_1 \in [-2.55, -0.017]$ *(generic) and* $[-1.76, -0.017]$*. This is the quantity that Remark 7.62 would need to bound from below; in these samples it stays in* $[-2.6, -0.017]$*.*

Figure 7.4 shows one fibre of Ω and the Jacobi field at its upper edge.

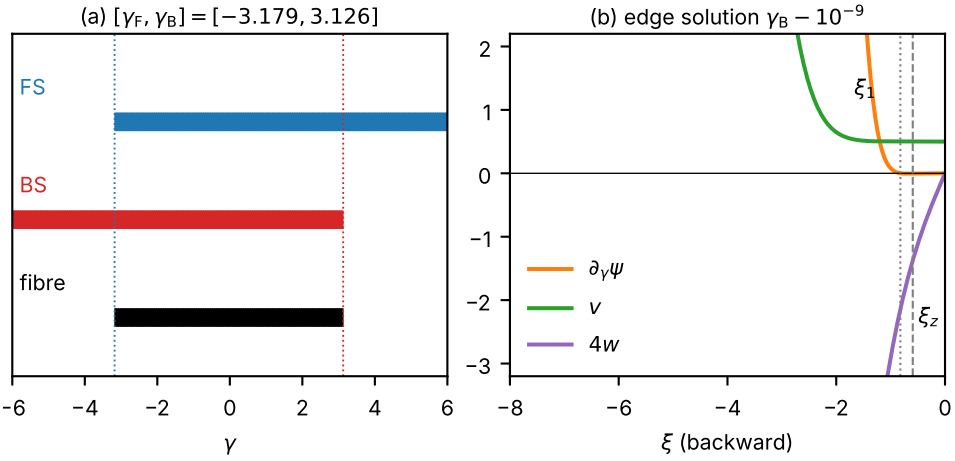


FIGURE 7.4. One fibre of Ω and the Jacobi field at its upper edge, for $\boldsymbol{c} = (0.5, 1, 2)$, $A = 0.8$, $\kappa = 0.5$. (a) The sets FS (top), BS (middle) and their intersection, the fibre (bottom), on a grid of 241 values of $\gamma \in [-6, 6]$; the dotted lines mark $\gamma_F = -3.179$ and $\gamma_B = 3.126$. (b) Along the backward edge solution $\gamma = \gamma_B - 10^{-9}$: $\partial_\gamma\psi$, $v = zw' - z'w$ and $4w$ against ξ (backward in $\xi \le 0$). $w < 0$ throughout, $v \ge 0.4998$ (it starts at $\frac{1}{2}$), $\partial_\gamma\psi$ changes sign once, at $\xi_1 = -0.821$, after the zero $\xi_z = -0.593$ of f, and ends at the value 75.6 (off scale). Code: `code/ch07_edges/make_b_interval.py`.

7.8. Attachment of the swirl solutions to the Li–Li–Yan solutions

In words. *The solutions without swirl, classified in [24, 25] and [26], form the face* $\kappa = 0$ *of* Ω*: for each admissible* $\boldsymbol{c}$ *a segment of* γ *between the two Li–Li–Yan endpoints* $U_{\boldsymbol{c}}^-(0)$ *and* $U_{\boldsymbol{c}}^+(0)$*, times a free line of values of the constant swirl* A*. Swirl solutions come out of this face in two ways. At an interior point of the segment the face lies inside* Ω *and the swirl solutions form a wedge around it (Proposition 5.67). At an endpoint the wedge degenerates, because the integral* F *that measures how far the swirl can grow becomes infinite exactly when the no-swirl solution is the*

recessive *one at a pole. What happens next depends on one number, the exponent* $k_2=\sqrt{1+c_2}$ *of the endpoint solution at the pole. If* $k_2<\frac12$ *the swirl solutions attach smoothly: the edge of* Ω *is a smooth hypersurface through the endpoint, and the solutions on it have a pole of type* R *(Theorem 7.74). If* $k_2>\frac12$ *this is impossible, because a swirl solution cannot have* $U(1)\le -3$*; the edge solution then leaves the endpoint solution in a boundary layer at the pole (Section 7.8.4). The threshold* 3 *of [26] is the value* $\mu=\frac12$ *of the exponent at the pole.*

7.8.1. The dictionary: finite F is dominant, infinite F is recessive. For $\boldsymbol{c}\in\operatorname{int}\mathcal{J}$ and $\gamma\in[U^-_{\boldsymbol{c}}(0),U^+_{\boldsymbol{c}}(0)]$ let z be the positive solution of $z''=W_{\boldsymbol{c}}z$, $W_{\boldsymbol{c}}=1+\frac12 P_{\boldsymbol{c}}$, with $z(0)=1$, $z'(0)=\gamma/2$ (Lemma 3.40), and let y be the linearising function of Proposition 5.67, so that $z^{-2}\mathrm{d}\xi=y^{-2}\mathrm{d}x$. The numbers

$$F_+(\gamma)=\int_0^1 y^{-2}\mathrm{d}x=\int_0^\infty z^{-2}\mathrm{d}\xi,$$
$$F_-(\gamma)=\int_{-1}^0 y^{-2}\mathrm{d}x=\int_{-\infty}^0 z^{-2}\mathrm{d}\xi\ \in(0,\infty]$$

are the constants of Proposition 5.67: if f is a swirl with $f_x=\kappa y^{-2}$ and $f(0)=A$ then $f(1)=A+\kappa F_+$ and $f(-1)=A-\kappa F_-$.

Proposition 7.71 (when is F infinite)**.** *Let* $\boldsymbol{c}\in\operatorname{int}\mathcal{J}$*, so that* $k_i=\sqrt{1+c_i}>0$ *and* $U^-_{\boldsymbol{c}}(0)<U^+_{\boldsymbol{c}}(0)$.

(a) $F_+(\gamma)<\infty$ *if and only if* $\gamma\neq U^-_{\boldsymbol{c}}(0)$*, and* $F_+(U^-_{\boldsymbol{c}}(0))=+\infty$*;* $F_-(\gamma)<\infty$ *if and only if* $\gamma\neq U^+_{\boldsymbol{c}}(0)$*. Moreover* $F_+(\gamma)\to+\infty$ *as* $\gamma\downarrow U^-_{\boldsymbol{c}}(0)$*, and* $F_-(\gamma)\to+\infty$ *as* $\gamma\uparrow U^+_{\boldsymbol{c}}(0)$.

(b) *Let* (U,f,Q) *be a local solution at the north pole with* $\kappa\neq0$ *and* $\mu>0$ *(Theorem 4.5). Then* $f(1)$ *is finite if and only if* $h=z'/z\to+\mu$*, i.e. the solution is of type* G*, and* $|f(1)|=\infty$ *if and only if* $h\to-\mu$*, i.e. type* R.

Proof. (a) By Theorem 3.44(iii) and $U=2(z'/z-\tanh\xi)$, $z'/z\to-k_2$ if $\gamma=U^-_{\boldsymbol{c}}(0)$ and $z'/z\to+k_2$ if $\gamma>U^-_{\boldsymbol{c}}(0)$ as $\xi\to\infty$ ($U(1)=-2\mp2k_2$ corresponds to $z'/z\to\mp k_2$). If $z'/z\to-k_2<0$ then, for $\xi\ge\xi_\varepsilon$, $(\ln z)'\le-k_2+\varepsilon$ and $z^{-2}\ge ce^{2(k_2-\varepsilon)\xi}$, so the integral diverges. If $z'/z\to k_2>0$ then $z^{-2}\le Ce^{-2(k_2-\varepsilon)\xi}$ is integrable. The statement for F_- is the mirror (σ_2, Lemma 3.40 with $\xi\mapsto-\xi$). For the limit let $\gamma_n\downarrow U^-_{\boldsymbol{c}}(0)$. The solutions z_n of $z''=W_{\boldsymbol{c}}z$ with $z_n(0)=1$, $z_n'(0)=\gamma_n/2$ converge to z_0 locally uniformly (linear equation, continuous dependence), so by Fatou's lemma $\liminf F_+(\gamma_n)\ge\int_0^\infty z_0^{-2}\mathrm{d}\xi=\infty$. (b) Here $f'=\kappa z^{-2}$ in ξ, so $f(1)$ is finite iff $\int^\infty z^{-2}\mathrm{d}\xi<\infty$ (Theorem 4.5(c)), and the argument of (a) applies with μ in place of k_2. □

Remark 7.72 (the dictionary)**.** Proposition 7.71 is the dictionary between the no-swirl solutions of [24–26] and the pole types of this monograph:

no swirl	F_+	swirl solution	$U(1)$
dominant, $\gamma>U^-_{\boldsymbol{c}}(0)$	finite	type G, $\mu>0$	$-2+2\mu>-2$
recessive, $k_2<\frac12$	∞	type R, $\mu<\frac12$	$-2-2\mu\in(-3,-2)$
recessive, $k_2\ge\frac12$	∞	does not exist ($U(1)>-3$)	≤-3

In words: finite $F \Leftrightarrow$ dominant $\Leftrightarrow$ type G; $F = \infty \Leftrightarrow$ recessive $\Leftrightarrow$ the endpoint of the Li–Li–Yan interval. The pole value μ of a swirl solution is the pole exponent of z, and the threshold 3 of [26] is $U(1) = -2 - 2\mu = -3$, i.e. $\mu = \frac{1}{2}$. The logarithmic types ($\mu = 0$, $f(1)$ finite for L_1 and infinite for L_2, Corollary 4.24(a)) have no no-swirl counterpart with $k_2 > 0$; they occur at the edge when the pole exponent of the swirl solution has dropped to 0.

Corollary 7.73 (the threshold 3 is $\mu = \frac{1}{2}$)**.** *Let $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$. The recessive endpoint solution has $U^-_{\boldsymbol{c}}(1) = -2 - 2k_2$, and $U^-_{\boldsymbol{c}}(1) \le -3$ if and only if $k_2 \ge \frac{1}{2}$. Every global solution with $\kappa \neq 0$ has $U(1) > -3$ and $U(-1) < 3$; in particular, for $k_2 \ge \frac{1}{2}$, no sequence of swirl solutions converges to $U^-_{\boldsymbol{c}}$ at the pole, and for $k_2 > \frac{1}{2}$ one has $U(1) - U^-_{\boldsymbol{c}}(1) > 2k_2 - 1 > 0$ uniformly in all swirl solutions (Corollary 5.17).*

Proof. $U^-_{\boldsymbol{c}}(1) = -2 - 2k_2$ is Theorem 3.44(iii), and $-2 - 2k_2 \le -3$ iff $k_2 \ge \frac{1}{2}$. The bounds on $U(\pm 1)$ and the uniform gap are Corollary 5.17; they rest on Theorem 4.9: on the recessive branch $\mu < \frac{1}{2}$, so $U(1) = -2 - 2\mu > -3$, and on the dominant branch $U(1) \ge -2$. □

The solutions with $\kappa = 0$ are not subject to this bound: the recessive endpoint solution itself has $U(1) = -2 - 2k_2$ for every k_2 (it has no swirl). The face $\kappa = 0$ reaches the endpoint for every $k_2 > 0$, but the swirl solutions can only approach it in the way described in the next subsection for $k_2 < \frac{1}{2}$ and in Section 7.8.4 for $k_2 > \frac{1}{2}$.

7.8.2. Attachment for $k_2 < \frac{1}{2}$.

Theorem 7.74 (attachment at an endpoint with $k_2 < \frac{1}{2}$). *Let $\boldsymbol{c}^0 \in \operatorname{int} \mathcal{J}$ with $c_2^0 < -\frac{3}{4}$ (equivalently $k_2 < \frac{1}{2}$) and $A_0 \ge 0$. There are a neighbourhood N of $(\boldsymbol{c}^0, A_0, 0)$ in $\mathbb{R}^5$, a number $\varepsilon > 0$ and a C^∞ function $g : N \to \mathbb{R}$ such that:*

(i) *$g(\boldsymbol{c}, A, 0) = U^-_{\boldsymbol{c}}(0)$ for all $(\boldsymbol{c}, A, 0) \in N$;*

(ii) *for $(\boldsymbol{c}, A, \kappa) \in N$ with $A \ge 0$, $\kappa \ge 0$ one has $\gamma_{\mathrm{F}} = g(\boldsymbol{c}, A, \kappa)$; for $\kappa > 0$ the solution with $\gamma = g$ is of type R at the north pole; and*

$$\Omega \cap \big\{(\boldsymbol{c}, \gamma, A, \kappa) :\ (\boldsymbol{c}, A, \kappa) \in N,\ A \ge 0,\ \kappa \ge 0,\ |\gamma - U^-_{\boldsymbol{c}^0}(0)| < \varepsilon\big\}$$
$$= \big\{\gamma \ge g(\boldsymbol{c}, A, \kappa),\ |\gamma - U^-_{\boldsymbol{c}^0}(0)| < \varepsilon\big\},$$

that is, locally Ω is the half-space above the smooth hypersurface $\gamma = g$, with no upper edge in the window;

(iii) *if $A_0 > 0$ then, after shrinking N, the solution of (ii) with $\gamma = g(\boldsymbol{c}, A, \kappa)$ and $\kappa > 0$ is of class N_0 ($f > 0$ on $(-1, 1)$ and $f(1) = +\infty$).*

By σ_1 the same holds near $(\boldsymbol{c}^0, -A_0, 0)$ in the quadrant $\{A \le 0, \kappa \le 0\}$.

Proof. *Step 1: the hypersurface of R solutions.* Fix a late cut ξ_* as in Theorem 4.40 for the type R with $\varsigma \in (k_2(\boldsymbol{c}^0), \frac{1}{2})$, and let $\Theta(\boldsymbol{c}, \gamma, A, \kappa)$ be the scale-free state at ξ_* of the solution with Cauchy data $(\boldsymbol{c}, \gamma, A, \kappa)$ at $x_b = 0$; Θ is C^∞ wherever the solution exists on $[0, x_*]$, for either sign of κ. At $(\boldsymbol{c}^0, \gamma^0, A_0, 0)$, $\gamma^0 = U^-_{\boldsymbol{c}^0}(0)$, the solution is the recessive no-swirl solution: $f \equiv A_0$, $Q = P_{\boldsymbol{c}^0}$, $z = z_0$ recessive with exponent $k_2 < \frac{1}{2}$. It is a normalised local solution of type R with $\kappa_{\mathrm{ch}} = 0$ (the compact sets K^{R}_B contain $\kappa_{\mathrm{ch}} = 0$ and put no restriction on the constant φ of f, Table 4.3; the argument is Step 1 of Theorem 7.21), so by Theorem 4.40(c),(d) its state at ξ_* lies on the C^∞ embedded 5-dimensional chart piece $\mathcal{M}_{\mathsf{R}} \subset \mathbb{R}^6$. Let s be a

defining function of $\mathcal{M}_{\mathsf{R}}$ near this state and $H = s \circ \Theta$. *Transversality.* At $\kappa = 0$ the data $(\boldsymbol{c}, A)$ fix $f \equiv A$ and $Q = P_{\boldsymbol{c}}$ (the trivial swirl, Lemma 5.46(c)), so the variation $\delta = \partial_\gamma$ of the solution has $\delta f \equiv 0$, $\delta g \equiv 0$, $\delta Q \equiv 0$ and only δU is non-zero; by Lemma 7.1, $v = \frac{1}{2} z^2 \delta U$ is constant, and equal to $\frac{1}{2}$ at $x = 0$ ($z(0) = 1$, $\delta U(0) = 1$). By Lemma 7.4 the tangent vectors of $\mathcal{M}_{\mathsf{R}}$ have $v \to 0$. So $\partial_\gamma \Theta \notin T\mathcal{M}_{\mathsf{R}} = \ker \mathrm{d}s$ and $\partial_\gamma H \neq 0$ at the point. The implicit function theorem gives a neighbourhood N and a C^∞ function g on N (κ of either sign) with $H(\boldsymbol{c}, g, A, \kappa) = 0$ and $g(\boldsymbol{c}^0, A_0, 0) = \gamma^0$. For $(\boldsymbol{c}, A, 0) \in N$ the recessive solution $\gamma = U^-_{\boldsymbol{c}}(0)$ is again a chart point of type R (the same argument), so $H(\boldsymbol{c}, U^-_{\boldsymbol{c}}(0), A, 0) = 0$, and uniqueness in the implicit function theorem gives (i). For $\kappa > 0$ the state $\Theta(\boldsymbol{c}, g, A, \kappa)$ lies on $\mathcal{M}_{\mathsf{R}}$, so the solution continues from x_* to the pole as a solution of type R, and it exists on $[0, x_*]$ by continuous dependence: it is forward global of type R. *Step 2: $g = \gamma_{\mathrm{F}}$ in the quadrant.* Let $A \geq 0$, $\kappa > 0$. By Lemma 5.46(a),(b), every $\gamma > \gamma_{\mathrm{F}}$ is of type G and $S_{\gamma_{\mathrm{F}}}$ is not. The solution at g is forward global, so $g \in \mathrm{FS}$, $g \geq \gamma_{\mathrm{F}}$; it is of type R, hence not of type G, so $g > \gamma_{\mathrm{F}}$ is excluded: $g = \gamma_{\mathrm{F}}$. For $\kappa = 0$, $\gamma_{\mathrm{F}} = U^-_{\boldsymbol{c}}(0) = g$ by Lemma 5.46(c). *Step 3: the backward half is open.* $U^-_{\boldsymbol{c}^0}$ is not the recessive solution at -1, because $U^-_{\boldsymbol{c}^0}(0) < U^+_{\boldsymbol{c}^0}(0)$; by Theorem 3.44(iii) $U^-_{\boldsymbol{c}^0}(-1) = 2 - 2k_1 < 2$, and $k_1 > 0$ since $\boldsymbol{c}^0 \in \operatorname{int} \mathcal{J}$: the backward half of $(\boldsymbol{c}^0, \gamma^0, A_0, 0)$ exists up to -1 and is of type G at the south pole. As in the proof of Proposition 7.64 (Corollary 4.42, transported by Lemma 5.7(b)), the same holds for all data in a neighbourhood, whatever the signs of A and κ. Shrinking N and choosing ε, this gives $\mathrm{BS} \supset \{|\gamma - \gamma^0| < \varepsilon\}$ for all $(\boldsymbol{c}, A, \kappa) \in N$, so $\Omega = \mathrm{FS}$ in the window. Since $\mathrm{FS} = [\gamma_{\mathrm{F}}, \infty)$ for $A \geq 0$, $\kappa \geq 0$ (Proposition 5.42(b)), this is (ii). *Step 4: class.* Let $A_0 > 0$. On the open set of data of type G at the south pole, $f(-1)$ is the coordinate φ of the chart of type G (Table 4.3, at the south pole through σ_2), a continuous function of the state at a late cut (Theorem 4.38(d)), hence of the data. At $\kappa = 0$ it equals $A_0 > 0$ at the base datum, so $f(-1) > 0$ on a neighbourhood. For $\kappa > 0$, f is increasing, so $f > 0$ on $(-1, 1)$: the solution is in N_0, and $f(1) = +\infty$ at the R pole (Corollary 4.24(a)). □

Remark 7.75 (what is not proved)**.** (a) *Mixed quadrants.* In the quadrants $A_0 > 0$, $\kappa < 0$ and $A_0 = 0$ with κ of either sign, the function g is still defined and smooth and still describes the hypersurface of forward solutions of type R, but $\mathrm{FS} = [g, \infty)$ is *not* proved there. The monotonicity of Proposition 5.42 requires $A\kappa \geq 0$; for $A_0 > 0 > \kappa$ the swirl vanishes at a point x_1 with $F(x_1) = -A/\kappa$ that tends to the pole as $\kappa \to 0$, so the estimates are not uniform in κ, and excluding forward solutions below g would need a comparison across the point where f changes sign. (b) *Other cases.* Nothing is claimed at endpoints with $k_2 \geq \frac{1}{2}$ (Corollary 7.73 and Section 7.8.4), at edges of type L_2, or at edge points without a zero of f other than the part of (iii) that is stated. The statement concerns the lower edge at the north pole; the upper edge at the south pole is the image under $\sigma_1\sigma_2$.

Corollary 7.76 (the exact-endpoint slice)**.** *Let $\boldsymbol{c} \in \mathcal{J}$ with $c_1, c_2 > -1$. Then $\Omega \cap \{\gamma = U^-_{\boldsymbol{c}}(0),\ \kappa \neq 0\} \subset \{A\kappa < 0\}$.*

Proof. Let $\kappa > 0$ and $A \geq 0$; the case $\kappa < 0$, $A \leq 0$ follows by σ_1, which maps Ω to itself and leaves γ unchanged. If $\mathrm{FS}(\boldsymbol{c}, A, \kappa) = \emptyset$ there is nothing to prove. Otherwise Proposition 5.42(c), applied with $(\boldsymbol{c}, A', \kappa') = (\boldsymbol{c}, 0, 0) \neq (\boldsymbol{c}, A, \kappa)$, gives $\gamma^-(\boldsymbol{c}, A, \kappa) > \gamma^-(\boldsymbol{c}, 0, 0) = U^-_{\boldsymbol{c}}(0)$, the last equality by Lemma 5.46(c). So $U^-_{\boldsymbol{c}}(0) \notin \mathrm{FS}$, and the point is not in Ω. Hence no point with $A\kappa \geq 0$, $\kappa \neq 0$ lies in Ω at $\gamma = U^-_{\boldsymbol{c}}(0)$. □

So on the exact-endpoint slice the face $\kappa = 0$ is the whole set in the quadrants $A\kappa \geq 0$, and the swirl members lie in the open quadrants $A\kappa < 0$.

Numerical observation 7.77 (the exact-endpoint slice and the double wedge)**.** *For* $\boldsymbol{c} = 0$, $(0, -0.5, 0)$ *and* $(1.25, 1.25, -7)$ *and* $\gamma = U_{\boldsymbol{c}}^-(0) = -2$, -1.5948, -0.2843, *the members of* Ω *in the* (A, κ)*-plane were counted on a grid of* 121×121 *points,* $A \in [-3, 3]$, $\kappa \in [-1.5, 1.5]$ *with* $\kappa = 0$ *left out: there are* 0, 0, 0 *members with* $A\kappa > 0$ *and* 2040, 3824, 3904 *members with* $A\kappa < 0$*. This is Corollary 7.76, which excludes the first number, together with the observation that the second is large. The members form a double wedge* $\{A < 0 < \kappa\} \cup \{\kappa < 0 < A\}$ *touching the face and, in the figure, all of them are of class* Z *(this is not proved). As* γ *rises above* $U_{\boldsymbol{c}}^-(0)$ *the wedge widens and class* N *appears in the quadrants* $A\kappa > 0$*, until at interior* γ *the four wedges of Proposition 5.67 are present (Example 7.89 below makes this explicit for* $\boldsymbol{c} = 0$*). The first three panels of Figure 7.5 are the case* $\boldsymbol{c} = 0$*.*

7.8.3. The shift of the edge, exactly and to first order. The distance of the edge from the endpoint is given by an exact identity.

Proposition 7.78 (the edge shift identity)**.** *Let* $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$*,* $A \geq 0$*,* $\kappa > 0$*, and let* S *be a forward global solution with data* $(\boldsymbol{c}, \gamma, A, \kappa)$ *at* $x_b = 0$ *that is of type* R*,* L_1 *or* L_2 *at the north pole. Let* z *be its Schrödinger function with* $z(0) = 1$ *and* z_0 *the recessive solution of* $z_0'' = W_{\boldsymbol{c}} z_0$ *with* $z_0(0) = 1$*, and let* $\psi = P_{\boldsymbol{c}} - Q \geq 0$ *be its swirl part. Then* $\psi z_0 z$ *is integrable on* $(0, \infty)$ *and*

$$\gamma - U_{\boldsymbol{c}}^-(0) = \int_0^\infty \psi \, z_0 z \, \mathrm{d}\xi > 0.$$

Proof. *Positivity of* ψ*.* ψ has vanishing 2-jet at 0 and $\psi_{xxx} = 2ff_x/(1-x^2)$ (Lemma 7.49(a)). For $A \geq 0$, $\kappa > 0$ the swirl f is increasing with $f(0) = A \geq 0$, so $f \geq 0$ and $f_x > 0$ on $[0, 1)$ and $\psi_{xxx} \geq 0$; hence $\psi_{xx}, \psi_x, \psi \geq 0$ and ψ is nondecreasing; by Theorem 4.5(a) it is bounded by $\psi(1) = 2c_2 - Q(1) < \infty$. *The Wronskian.* $z'' = (1 + \frac{1}{2}Q)z$ and $z_0'' = (1 + \frac{1}{2}P_{\boldsymbol{c}})z_0$, so $\mathrm{Wr} := z_0 z' - z_0' z$ satisfies $\mathrm{Wr}' = \frac{1}{2}(Q - P_{\boldsymbol{c}})z_0 z = -\frac{1}{2}\psi z_0 z$. At 0, $z(0) = z_0(0) = 1$, $z'(0) = \gamma/2$ and $z_0'(0) = U_{\boldsymbol{c}}^-(0)/2$ (Lemma 3.40, $\tanh 0 = 0$), so $\mathrm{Wr}(0) = \frac{1}{2}(\gamma - U_{\boldsymbol{c}}^-(0))$. *The end at the pole.* $z_0'/z_0 \to -k_2 < 0$ (Theorem 3.44(iii)) and, by Theorem 4.5(b) and the types, $z'/z \to -\mu \leq 0$ with $\mu \geq 0$ ($h \to -\mu$ at R; $h \to 0$ at $\mathsf{L}_1, \mathsf{L}_2$). So $z_0 z = \exp \int (z_0'/z_0 + z'/z) \leq e^{-k_2 \xi/2}$ for large ξ, which gives the integrability of $\psi z_0 z$ and $\mathrm{Wr} = z_0 z\,(z'/z - z_0'/z_0) \to 0$. Integrating Wr' over $(0, \infty)$ gives $-\mathrm{Wr}(0) = -\frac{1}{2}\int_0^\infty \psi z_0 z \, \mathrm{d}\xi$. Positivity: the integrand is ≥ 0 and not identically zero ($\kappa > 0$). □

At the edge, $\gamma = \gamma_{\mathrm{F}}$, the identity gives $\gamma_{\mathrm{F}} > U_{\boldsymbol{c}}^-(0)$, the strict case of Proposition 5.42(c), now with the size of the shift as an integral; this integral is what the following computations estimate.

Formal computation 7.79 (the first-order coefficient a_1)**.** *By Theorem 7.74, near an endpoint with* $k_2 < \frac{1}{2}$*,* $g(\boldsymbol{c}, A, \kappa) = U_{\boldsymbol{c}}^-(0) + a_1(\boldsymbol{c})A\kappa + b(\boldsymbol{c})\kappa^2 + \cdots$ *with* $a_1 = \partial_A \partial_\kappa g$ *(there is no term linear in* κ *alone, since* $g(\boldsymbol{c}, 0, \cdot)$ *is even by* σ_1*). To first order in* κ *the swirl is* $f \approx A$*,* $f_x \approx \kappa y_0^{-2}$*, so that* $\psi_{xxx} = 2ff_x/(1-x^2) \approx 2A\kappa y_0^{-2}/(1-x^2)$ *and* $\psi \approx A\kappa\Psi_1$ *with*

$$\Psi_1(x) = \int_0^x (x - s)^2 \, \frac{y_0(s)^{-2}}{1 - s^2} \, \mathrm{d}s, \qquad a_1 = \int_0^\infty \Psi_1(\tanh \xi)\, z_0(\xi)^2 \, \mathrm{d}\xi,$$

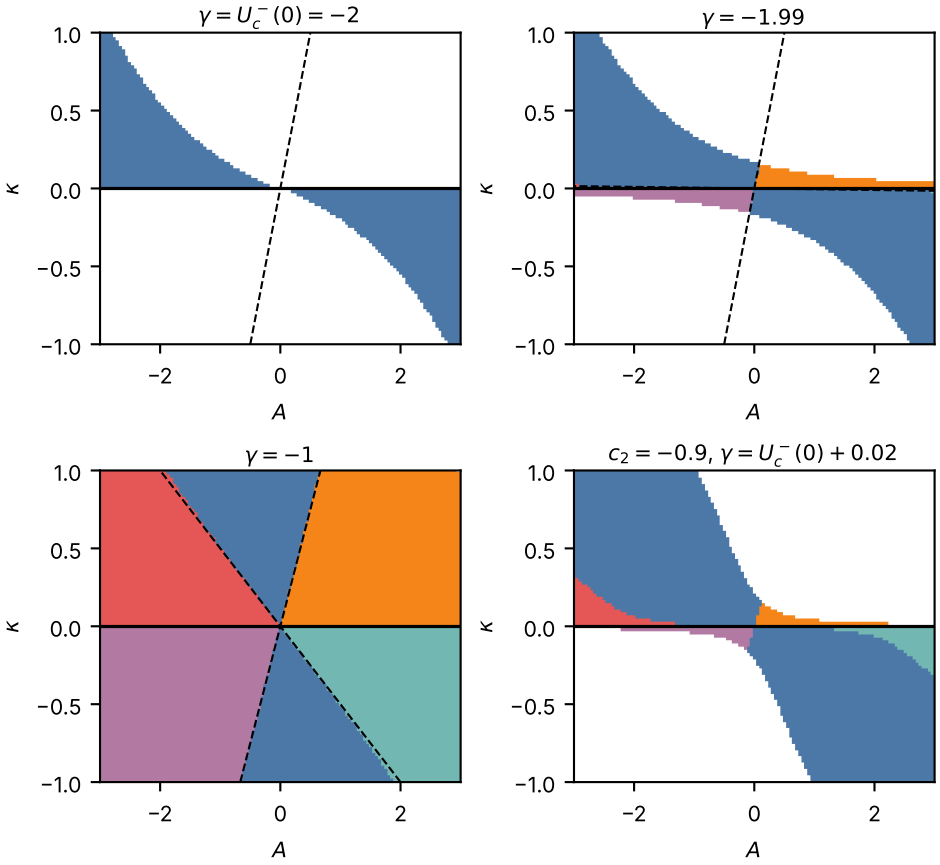


FIGURE 7.5. Sections of Ω in the (A, κ)-plane near the endpoint $U_{\boldsymbol{c}}^-(0)$, $|A| \le 3$, $|\kappa| \le 1$ (121×101 points). Top left, top right, bottom left: $\boldsymbol{c} = 0$ and $\gamma = -2, -1.99, -1$. Bottom right: $\boldsymbol{c} = (0, -0.9, 0)$, $k_2 = \sqrt{0.1} < \frac{1}{2}$, and $\gamma = U_{\boldsymbol{c}}^-(0) + 0.02$. White: not in Ω; blue: class Z; orange: N_0 ($\kappa > 0$, $f > 0$); red, teal, purple: the other three sign sectors of class N; black: the face $\kappa = 0$. Dashed: the lines $A = -\kappa F_+$ and $A = \kappa F_-$ with the no-swirl values $F_+ = 2/(\gamma + 2)$, $F_- = 2/(2 - \gamma)$ of Example 7.89 (case $\boldsymbol{c} = 0$ only; the Z/N interface of Proposition 5.67 is tangent to them at $\kappa = 0$; at $\gamma = -2$ only the second line exists). As $\gamma \downarrow -2$ the blue region thins into the double wedge $A\kappa < 0$ of Corollary 7.76; in the last panel ($k_2 < \frac{1}{2}$) the picture near the face is the smooth one of Theorem 7.74. Code: `code/ch07_edges/make_b_attach.py`.

where the second formula is Proposition 7.78 with $z \approx z_0$. At $\boldsymbol{c} = 0$: $y_0 = 1 - s$, $z_0 = e^{-\xi}$ ($k_2 = 1$, so this is outside Theorem 7.74, but the integral converges because $(x - s)^2/(1 - s)^2 \le 1$ gives $\Psi_1(x) \le \int_0^x \mathrm{d}s/(1 - s^2) = \xi$); numerically

$a_1(0) = 0.068147$, *and the quotient* $(\gamma_{\mathrm{F}}(A{=}1) - \gamma_{\mathrm{F}}(A{=}-1))/(2\kappa)$ *gives* 0.068126, 0.068140, 0.068145 *at* $\kappa = 0.004,\ 0.002,\ 0.001$. *At* $\boldsymbol{c} = (0, -0.9, 0)$ *(*$k_2 = 0.316 < \frac{1}{2}$*) the quotients* $(\gamma_{\mathrm{F}} - U^-_{\boldsymbol{c}}(0))/\kappa$ *at* $\kappa = 0.005$ *are* 0.42538 *for* $A = 1$, -0.41699 *for* $A = -1$ *and* 0.00202 *for* $A = 0$, *so* $a_1 \approx 0.42$ *there; the quotient for* $A = 0$ *is* 0.00807, 0.00403, 0.00202 *at* $\kappa = 0.02,\ 0.01,\ 0.005$, *i.e. it is proportional to* κ *and the shift is* $\approx 0.4\kappa^2$, *as the evenness predicts (*`verify_ch07b.py`, *part* `layer`*).*

Formal computation 7.80 (Ω is not star-shaped in κ)**.** *The slice* $\{\kappa : (\boldsymbol{c}, \gamma, A, \kappa) \in \Omega\}$ *at fixed* $(\boldsymbol{c}, \gamma, A)$ *is not star-shaped around* $\kappa = 0$ *in general. Mechanism: by the first-order term the whole fibre moves with* $A\kappa$. *The lower edge is* $\gamma_{\mathrm{F}} \simeq U^-_{\boldsymbol{c}}(0) + a_1(\boldsymbol{c})A\kappa$; *by the mirror (Lemma 5.40) the upper edge is* $\gamma_{\mathrm{B}} \simeq U^+_{\boldsymbol{c}}(0) + a_1(\boldsymbol{c}_m)A\kappa$ *with the appropriate sign. For* $A\kappa < 0$ *the lower edge sinks below* $U^-_{\boldsymbol{c}}(0)$, *so that a point with* $\gamma < U^-_{\boldsymbol{c}}(0)$ *is in* Ω *at* κ *and not at* $s\kappa$ *for small* s *(and not on the face, which needs* $\gamma \geq U^-_{\boldsymbol{c}}(0)$*); for* $A\kappa > 0$ *the upper edge rises above* $U^+_{\boldsymbol{c}}(0)$ *and points near the top are lost when* κ *is shrunk. The first statement uses* $\gamma_{\mathrm{F}} = g$ *in the mixed quadrant, which is not proved (Remark 7.75), so the claim is formal. What is proved is that* FS *is star-shaped in* κ *for* $A\kappa \geq 0$ *(Proposition 5.42(c)), and that the fibre* $\mathrm{FS} \cap \mathrm{BS}$ *is not, because* BS *is not monotone in* κ *for* $A\kappa > 0$ *(mirror image of the statement for* FS *and* $A\kappa < 0$*). The counts of Numerical observation 7.69(c) —* 200 *failures of the implication in* 3000 *tests,* 100 *with* $A\kappa < 0$ *and* 100 *with* $A\kappa > 0$ *— are consistent with this mechanism, as is the measured* a_1 *above.*

7.8.4. The boundary layer at an endpoint with $k_2 > \frac{1}{2}$. In words. *If* $k_2 > \frac{1}{2}$ *the recessive endpoint solution* $U^-_{\boldsymbol{c}}$ *has* $U^-_{\boldsymbol{c}}(1) \leq -3$, *and no swirl solution can have such a value at the pole (Corollary 7.73). So the lower edge* $\gamma = \gamma^-(\kappa)$, *which converges to the endpoint as* $\kappa \to 0$ *at every fixed* $x < 1$, *cannot converge to it* at *the pole: its pole value stays in* $(-3, -2]$. *The two statements are reconciled by a boundary layer, a region next to the pole where the swirl part* ψ *of* Q *grows from almost* 0 *to a value of order one. What can be proved is that the layer exists and is pushed into the pole as* $\kappa \to 0$. *Its width, the logarithmic law at* $k_2 = 1$ *and the exponents of the shift of the edge are computations that we carry out formally and check numerically.*

Throughout this subsection $x_b = 0$, $A = 0$ and $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$ with $c_2 > -\frac{3}{4}$ (so $k = k_2 > \frac{1}{2}$), and Y_κ is the edge solution with data $(\boldsymbol{c}, \gamma^-(\kappa), 0, \kappa)$ for $0 < \kappa \leq \kappa_{\max}(\boldsymbol{c})$, where $\gamma^- = \gamma^-(0, \boldsymbol{c}, \kappa)$ is the lower edge of the lens of Definition 5.55 at $x_0 = 0$. Write U_κ, $\psi_\kappa = P_{\boldsymbol{c}} - Q_\kappa$ for its components.

7.8.4.1. *What is proved.*

Corollary 7.81 (the layer)**.** (e1) (Local convergence, not uniform at the pole.) *For each* $x \in [0, 1)$, $\psi_\kappa(x) \to 0$ *and* $U_\kappa(x) \to U^-_{\boldsymbol{c}}(x)$ *as* $\kappa \downarrow 0$. *But* $U_\kappa(1) \in (-3, -2]$ *for every* $\kappa > 0$, *whereas* $U^-_{\boldsymbol{c}}(1) = -2 - 2k < -3$. *Thus*

$$\lim_{x \to 1} \lim_{\kappa \to 0} U_\kappa(x) = U^-_{\boldsymbol{c}}(1) < -3 \leq \liminf_{\kappa \to 0} \lim_{x \to 1} U_\kappa(x),$$

and the convergence is not uniform on $[0, 1)$.

(e2) (The swirl energy sits at the pole.) $\psi_\kappa(1) > 2c_2 + \frac{3}{2} > 0$ *for every* $\kappa > 0$, *while* $\psi_\kappa(x) \to 0$ *for every fixed* $x < 1$. *In particular* $\sup_x \psi_\kappa \not\to 0$.

(e3) (It is a boundary layer.) ψ_κ *is nondecreasing on* $[0, 1)$, *and for every* $\eta \in (0, 2c_2 + \frac{3}{2})$ *the point* $x_\eta(\kappa) = \min\{x : \psi_\kappa(x) \geq \eta\}$ *exists and* $x_\eta(\kappa) \to 1$ *as* $\kappa \downarrow 0$.

PROOF. (e1) By Proposition 5.59(e), $\gamma^-(\kappa) \to U_{\boldsymbol{c}}^-(0)$ as $\kappa \downarrow 0$, so the data of Y_κ tend to $(\boldsymbol{c}, U_{\boldsymbol{c}}^-(0), 0, 0)$, whose solution is the global no-swirl solution $U_{\boldsymbol{c}}^-$, with $Q = P_{\boldsymbol{c}}$. For fixed $x < 1$ it exists on $[0, x]$, so by continuous dependence on a compact interval $U_\kappa(x) \to U_{\boldsymbol{c}}^-(x)$ and $Q_\kappa(x) \to P_{\boldsymbol{c}}(x)$, i.e. $\psi_\kappa(x) \to 0$. The edge solution is of type R or $\mathsf{L}_1, \mathsf{L}_2$ (Lemma 5.46(b)), so $U_\kappa(1) \in (-3, -2)$ or $U_\kappa(1) = -2$ (Definition 4.16); and $U_{\boldsymbol{c}}^-(1) = -2 - 2k < -3$ since $k > \frac{1}{2}$ (Theorem 3.44(iii)). (e2) $\psi_\kappa(1) = 2c_2 - q$ with $q = Q_\kappa(1) \in [-2, -\frac{3}{2})$ (Corollary 4.21), which gives $\psi_\kappa(1) > 2c_2 + \frac{3}{2}$, and $2c_2 + \frac{3}{2} > 0$ iff $c_2 > -\frac{3}{4}$. The second statement is (e1). (e3) With $A = 0$, $\kappa > 0$ and $f(0) = 0$, f is increasing, so $f \geq 0$ and $f_x > 0$ on $[0, 1)$, and $\psi_{xxx} = 2ff_x/(1 - x^2) \geq 0$; as ψ has vanishing 2-jet at 0, ψ is nondecreasing. Since $\psi_\kappa(1) > \eta$ by (e2) the point $x_\eta(\kappa)$ exists. If $x_\eta(\kappa_n) \leq x_* < 1$ along some sequence $\kappa_n \to 0$, then $\psi_{\kappa_n}(x_*) \geq \eta$ by monotonicity, contradicting $\psi_{\kappa_n}(x_*) \to 0$ by (e1). □

In particular, any expansion of γ^-, U_κ or ψ_κ in powers of κ has to be non-uniform at the pole.

7.8.4.2. *The width of the layer.*

Formal computation 7.82 (the position of the layer)**.** *Outside the layer the edge solution is close to $U_{\boldsymbol{c}}^-$, so $z \approx a\, z_0$ with $z_0 \sim e^{-k\xi}$, and $f_\xi = \kappa z^{-2} \approx \kappa a^{-2} e^{2k\xi}$, $f \sim \kappa e^{2k\xi}$. Since $\psi_{xxx} = 2ff_x/(1 - x^2)$ and $t = 1 - x \approx 2e^{-2\xi}$,*

$$\psi \approx C\kappa^2 e^{(4k-2)\xi} \qquad (\textit{outside the layer, } k > \tfrac{1}{2}),$$

and the layer begins where $\psi = O(1)$: $\kappa^2 e^{(4k-2)\xi_L} = O(1)$, that is

$$\xi_L = \frac{\ln(1/\kappa)}{2k-1}, \qquad t_L = 2e^{-2\xi_L} \asymp \kappa^{2/(2k-1)}. \tag{7.11}$$

The prediction is that the level $\xi_\delta(\kappa) = \min\{\xi : \psi_\kappa \geq \delta\}$ satisfies $\mathrm{d}\xi_\delta/\mathrm{d}\ln(1/\kappa) \to 1/(2k-1)$ for each fixed $\delta \in (0, 2c_2 + \frac{3}{2})$.

Numerical observation 7.83 (the width of the layer)**.** *For $\boldsymbol{c} = (0, c_2, 0)$ with $c_2 =$ 0, 0.5, 2 ($k = 1$, 1.225, 1.732), $\delta = 1$, 1.25, 1 and $\kappa = 0.2$, 0.1, 0.05, 0.025, 0.0125 the levels $\xi_\delta(\kappa)$ and the slopes between consecutive values of $\ln(1/\kappa)$ are*

c_2	k	ξ_δ at $\kappa = 0.2, \ldots, 0.0125$					slopes				$\frac{1}{2k-1}$	$\psi(1)$
0	1	3.376	4.059	4.751	5.444	6.138	0.985	0.998	1.000	1.001	1.000	2
0.5	1.225	2.831	3.303	3.780	4.258	4.736	0.681	0.688	0.689	0.690	0.690	3
2	1.732	1.935	2.214	2.496	2.778	3.059	0.403	0.406	0.407	0.406	0.406	6

The slopes converge to the prediction $1/(2k-1)$ of Formal computation 7.82. The pole values $\psi(1) = 2, 3, 6$ are the type-L_1 value $2c_2 + 2$ (Corollary 4.21). (`verify_ch07b.py`, part `layer`.)

Figure 7.6 shows the layer of ψ for decreasing κ.

7.8.4.3. *The logarithmic law at $k = 1$.* Let $\boldsymbol{c} = 0$, $A = 0$. The outer solution at $\gamma = -2$ is $z_0 = e^{-\xi}$, $y_0 = 1 - x$, $F = x/(1 - x)$ and $f = \kappa x/(1 - x)$, and the triple integral for ψ can be done exactly: a computer-algebra check (`code/ch07_edges/verify_ch07b.py`, S7) gives

$$\psi(1 - t) = \frac{\kappa^2}{6t} + \frac{\kappa^2}{4}\ln\frac{t}{2} - \frac{\kappa^2}{8} + O(t), \qquad \text{that is } \psi \simeq \frac{\kappa^2 e^{2\xi}}{12} \quad (t = 2e^{-2\xi}).$$

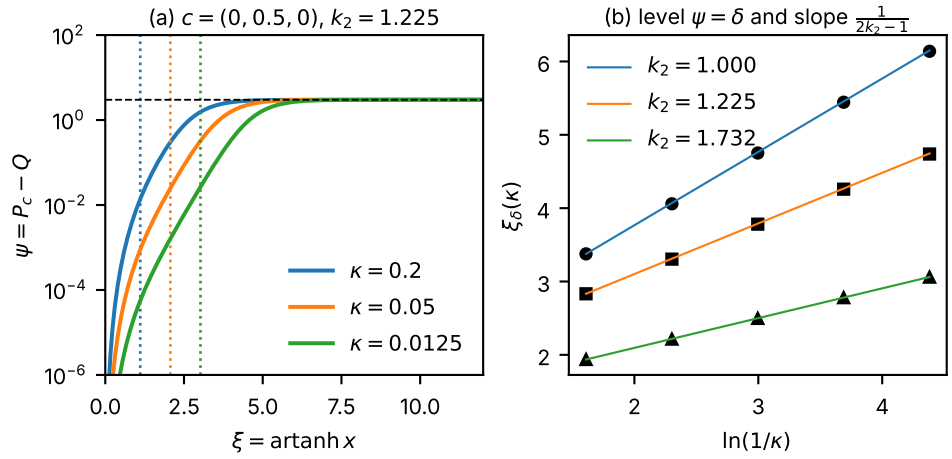


FIGURE 7.6. (a) $\psi = P_c - Q$ of the edge solution at $\boldsymbol{c} = (0, 0.5, 0)$ ($k = 1.225$) against ξ for $\kappa = 0.2$, 0.05, 0.0125 (log scale). Dotted: the predicted layer positions $\xi_L = \ln(1/\kappa)/(2k-1)$; dashed: the pole value $2c_2 + 2 = 3$. Before the layer ψ grows like $\kappa^2 e^{(4k-2)\xi}$; the layer moves out as κ decreases. (b) The level $\xi_\delta(\kappa)$ against $\ln(1/\kappa)$ for $k = 1$, 1.225, 1.732 (markers) and lines of the predicted slope $1/(2k-1)$. Code: `code/ch07_edges/make_b_layer.py`.

Insert this in Proposition 7.78 with $z_0 z \approx e^{-2\xi}$ in the outer region: $\int_0^{\xi_L} \psi z_0 z \, \mathrm{d}\xi \approx \kappa^2 \xi_L/12 + O(\kappa^2)$, and the layer contributes $O(\kappa^2)$. With $\xi_L = \ln(1/\kappa)$ ($k = 1$ in (7.11)):

Formal computation 7.84 (the logarithmic law)**.** *At $\boldsymbol{c} = 0$, $A = 0$, $x_b = 0$: $\gamma^-(\kappa) + 2 \simeq \kappa^2\big[D + \frac{1}{12}\ln\frac{1}{\kappa}\big]$ with a constant D that depends on the exact layer solution. For general $\boldsymbol{c}$ with $k_2 = 1$ the coefficient of the logarithm is $1/(12a^2)$, $a = \lim_{\xi\to\infty} e^{\xi} z_0$.*

The numerical statement concerns the *increments* of $(\gamma^- + 2)/\kappa^2$ per halving of κ, which must tend to $\frac{1}{12}\ln 2 = 0.057762$ (the slope $\frac{1}{12}$ is for the increment per unit of $\ln\frac{1}{\kappa}$).

Numerical observation 7.85 (the logarithmic law)**.** *For $\kappa = 2^{-j}$, $j = 0, \dots, 9$, the increments of $(\gamma^- + 2)/\kappa^2$ between j and $j+1$ are*

$j \to j{+}1$	$0 \to 1$	$1 \to 2$	$2 \to 3$	$3 \to 4$	$4 \to 5$	$5 \to 6$	$6 \to 7$	$7 \to 8$	$8 \to 9$
increment	0.05031	0.05142	0.05387	0.05583	0.05693	0.05743	0.05764	0.05772	0.05774

against $\frac{1}{12}\ln 2 = 0.057762$. The identity of Proposition 7.78 itself holds to $5 \cdot 10^{-4}$ or better: the ratio of the two sides is 0.99995, 0.99986, 0.99954 at $\kappa = 0.5$, 0.1, 0.02 (the integral being cut at $\xi = 13.9$; $\psi(\mathrm{end}) = 2.0000$). (`verify_ch07b.py`, part `layer`.)

7.8.4.4. *The exponents of the shift.* By Proposition 7.78 the integrand is $\psi z_0 z \sim \kappa^2 e^{(4k-2)\xi} e^{-2k\xi} = \kappa^2 e^{(2k-2)\xi}$ outside the layer. Hence:

Formal computation 7.86 (the exponents)**.**

k	*the integral is dominated by*	$\gamma^- - U_{\boldsymbol{c}}^-(0) \sim$	*measured local slope*
$\frac{1}{2} < k < 1$	$\xi = O(1)$ *(regular term)*	κ^2	$2.022 \to 1.995$ *($k = 0.707$)*
$k = 1$	ξ *up to* ξ_L	$\kappa^2 \ln(1/\kappa)$	$1.758 \to 1.845$ *(slowly to 2)*
$k > 1$	$\xi \approx \xi_L$	$\kappa^{2k/(2k-1)}$	$1.594 \to 1.651$ *($k = 1.225$, predicted 1.690);* $1.390 \to 1.403$ *($k = \sqrt{3}$, predicted 1.406)*

For $k > 1$: $\kappa^2 e^{(2k-2)\xi_L} = \kappa^{2-(2k-2)/(2k-1)} = \kappa^{2k/(2k-1)}$.

The measured slopes are local slopes of $\ln(\gamma^- - U_{\boldsymbol{c}}^-(0))$ against $\ln\kappa$ over $\kappa \in [0.0125, 0.4]$; they move monotonically towards the prediction as κ decreases (`verify_ch07b.py`, part `layer`, items for $c_2 = -0.5$, 0, 0.5, 2).

Remark 7.87 (the three laws are proved later)**.** The three laws of Formal computation 7.86, with the exact coefficient $1/(12a^2)$ of the logarithm in the case $k = 1$ (Formal computation 7.84) and as two-sided bounds for $k > 1$, are proved in Chapter 11 (Theorem 11.44). The computation above is the heuristic that leads to them; the numerical observations are consistent with the theorem.

Remark 7.88 (the resonance at $\mu = \frac{1}{2}$)**.** Let $f \sim \kappa\,\mathrm{const}\cdot t^{-k}$ near the pole, k the exponent of the no-swirl profile (k_2, not the exponent μ of the swirl solution). A particular solution of $\mathsf{L}Q = \lambda - f^2$ has the form $\sim Ct^{1-2k}$, and

$$\mathsf{L}(t^{1-2k}) = (8k^2 - 2)\,t^{-2k}\big(1 + O(t)\big) = 2(2k-1)(2k+1)\,t^{-2k}\big(1 + O(t)\big)$$

(S6 of `verify_ch07b.py`). So $C \propto \kappa^2/((2k-1)(2k+1))$: it is finite, giving a bounded contribution to $Q(1)$, iff $k < \frac{1}{2}$; at $k = \frac{1}{2}$ the right-hand side resonates with the exponent 0 of the homogeneous solutions and Q diverges logarithmically; for $k > \frac{1}{2}$ the term t^{1-2k} blows up, which is the outer growth of ψ above. This is the same borderline $\frac{1}{2}$ as in Corollary 7.73, seen from the equation for Q rather than from the values at the pole. For $k < \frac{1}{2}$ one has $\psi(1) = O(\kappa^2)$ (Theorem 7.21); for $k > \frac{1}{2}$ the bracket $\psi(1) \in (2c_2 + \frac{3}{2}, 2c_2 + 2]$ of Corollary 4.21 holds instead, confirmed in Numerical observation 7.83 ($\psi(1) = 2c_2 + 2$).

7.9. The global picture of Ω and an atlas of the edges

In words. *Every global solution of the reduced Navier–Stokes system with conical symmetry is one point of the six-dimensional space of Cauchy data at the base point, and the set Ω of all such points is* one *closed set. This section draws it. The set has a trivial part, the face $\kappa = 0$ of solutions without swirl; a part of class Z (the swirl changes sign), which is a lens-shaped region over the space of lens parameters; and a part of class N (the swirl has one sign), which comes in four copies. The three parts are separated by the interface $\{f(\pm 1) = 0\}$ inside Ω and have a common boundary, the edge hypersurface, on which the solutions fail to be generic at a pole; the edge has its own sub-strata, the walls where the pole type changes, and the tips where the lens closes. We state all this as a single theorem and show a two-dimensional and a*

three-dimensional slice at $\boldsymbol{c}=0$. The section ends with an atlas that says, for each region of Ω and the edge, what is proved, what is only computed, and where.

7.9.1. A toy: the Landau wedge at $\boldsymbol{c}=0$.

Example 7.89 (the wedge around the rest-state interval)**.** Take $\boldsymbol{c}=0$, so that $P_{\boldsymbol{c}}\equiv 0$, $W\equiv 1$, $z''=z$. The no-swirl solutions are $z=\cosh\xi+\frac{\gamma}{2}\sinh\xi$, $y=1+\frac{\gamma}{2}x$, and $U=\gamma(1-x^2)/(1+\gamma x/2)$; they are global iff $|\gamma|\le 2$, the Landau interval $[U_0^-(0),U_0^+(0)]=[-2,2]$. For $|\gamma|<2$

$$F_+=\int_0^1\frac{\mathrm{d}x}{(1+\gamma x/2)^2}=\frac{2}{2+\gamma},\qquad F_-=\int_{-1}^0\frac{\mathrm{d}x}{(1+\gamma x/2)^2}=\frac{2}{2-\gamma},$$

and by Proposition 5.67, for small κ, $f(1)=A+\kappa F_+$ and $f(-1)=A-\kappa F_-$ with $F_\pm$ equal to these values up to $O(\kappa^2)$. For $\kappa>0$ the swirl changes sign (class Z) iff $f(-1)<0<f(1)$, i.e. $-\kappa F_+<A<\kappa F_-$: a wedge in the (A,κ)-plane bounded by the two lines $A=-\kappa F_+$ and $A=\kappa F_-$. At $\gamma=0$ both slopes are 1 and the wedge is $|A|<|\kappa|$ (for κ of either sign); at $\gamma=1$, $F_+=\frac{2}{3}$, $F_-=2$ and the wedge is $-\frac{2}{3}\kappa<A<2\kappa$. Outside the wedge the solution is of class N: for $A\ge\kappa F_-$ it is in N_0, for $A\le-\kappa F_+$ in $\sigma_1\sigma_2\mathsf{N}_0$. As $\gamma\downarrow-2$, $F_+\to+\infty$ and the line $A=-\kappa F_+$ tends to the A-axis $\kappa=0$, so the wedge collapses onto the face; the other line stays at $A=\kappa/2$. At $\gamma=-2$ itself $F_+=\infty$ and only the line $A=\kappa/2$ remains. This is the mechanism behind Proposition 7.71 and Numerical observation 7.77.

7.9.2. The stratification. Recall $\Omega=\Omega(0)\subset\mathbb{R}^6$, $\omega=(\boldsymbol{c},\gamma,A,\kappa)$, and write σ_1,σ_2 for the symmetries of Lemma 5.7(c). Let $\Omega_0=\Omega\cap\{\kappa=0\}$ be the face, and N_0, N_0 be as in Definition 5.3; put $\mathsf{N}^{\kappa+,f+}:=\mathsf{N}_0$ and let $\mathsf{N}^{\kappa-,f-}=\sigma_1\mathsf{N}_0$, $\mathsf{N}^{\kappa-,f+}=\sigma_2\mathsf{N}_0$, $\mathsf{N}^{\kappa+,f-}=\sigma_1\sigma_2\mathsf{N}_0$ be its three images.

THEOREM 7.90 (Ω as one stratified set). (a) (One set.) $\Omega\subset\mathbb{R}^6$ *is closed, and* $\omega\mapsto$ (*the solution through* ω) *is a bijection from* Ω *onto the set of* all *global solutions, with and without swirl. As a set,* $\Omega=\Omega_0\sqcup\mathsf{Z}\sqcup\mathsf{N}$, $\mathsf{Z}=\mathsf{Z}_+\sqcup\sigma_1\mathsf{Z}_+$, $\mathsf{N}=\mathsf{N}_0\sqcup\sigma_1\mathsf{N}_0\sqcup\sigma_2\mathsf{N}_0\sqcup\sigma_1\sigma_2\mathsf{N}_0$*, and the sign of* κ *and, in* N*, of* f *is constant on each piece.*

(b) (The face.) $\Omega_0=\{(\boldsymbol{c},\gamma,A,0):\ \boldsymbol{c}\in\mathcal{J},\ U_{\boldsymbol{c}}^-(0)\le\gamma\le U_{\boldsymbol{c}}^+(0),\ A\in\mathbb{R}\}$*, a closed set of dimension* 5 *that is the product of a set in* $(\boldsymbol{c},\gamma)$*-space with the free line of values of* A*. Its points with* $\boldsymbol{c}\in\operatorname{int}\mathcal{J}$ *and* $U_{\boldsymbol{c}}^-(0)<\gamma<U_{\boldsymbol{c}}^+(0)$ *are interior points of* Ω*.*

(c) (Class Z.) $\mathsf{Z}_+=\Phi(\mathcal{L})$*, where* Φ *is the diffeomorphism of Lemma 7.41 and* $\mathcal{L}$ *the lens set of Theorem 5.62. Its interior in* $\mathbb{R}^6$ *is the generic part* $\Phi\{\gamma^-<\gamma<\gamma^+,\ \kappa<\kappa_{\max}\}\subset\Omega_{\mathsf{G}}$*, and its boundary inside the open set* $\Phi(\Lambda)$ *is the edge hypersurface* $\mathfrak{E}$ *of Proposition 7.42.*

(d) (Class N.) $\mathsf{N}_0\cong\Gamma_T\times[0,\infty)$ *for the part with* $\mu<2$ *and* $\cong B\times[0,\infty)$ *for the part with* $\mu>1$ (*Theorem 6.61*)*; the solutions of* N_0 *that are not of type* G *at the north pole are exactly the boundary* Γ_{N} *of this product.*

(e) (Generic part and boundary.) Ω_{G} *is open and contained in the interior of* Ω*, and* $\partial\Omega\cap\{\kappa\neq0\}$ *consists of solutions not of type* G *at both poles; in* Z_+ *it equals* $\mathfrak{E}$ (*in* $\sigma_1\mathsf{Z}_+$ *it is* $\sigma_1\mathfrak{E}$).

(f) (The interface.) *Inside* $\Omega_{\mathsf{G}}\cap\{\kappa>0\}$, $\mathsf{Z}=\{f(-1)<0<f(1)\}$ *and the interface between* Z *and* N *is* $\{f(-1)=0\}\cup\{f(1)=0\}$ (*Proposition 5.16*).

(g) (Sub-strata of the edge.) $\mathfrak{E}^- = E_{\mathsf{R}} \sqcup E_{\mathsf{L}_1} \sqcup E_{\mathsf{L}_2}$ *with* $E_{\mathsf{R}}, E_{\mathsf{L}_1}$ *relatively open and* E_{L_2} (*the* wall) *relatively closed and nowhere dense, locally a* C^3 *hypersurface of* $\mathfrak{E}^-$ (*dimension* 4)*;* $\mathfrak{E}^- \cap \mathfrak{E}^+$ *is the set of* tips $\kappa = \kappa_{\max}(x_0, \boldsymbol{c})$*, parametrised continuously by the four numbers* $(x_0, \boldsymbol{c})$*. The reflection* $\sigma_1\sigma_2$ *exchanges* $\mathfrak{E}^-$ *and* $\mathfrak{E}^+$ *and the two poles.*

Proof. (a) Closedness is Theorem 5.12, the bijection is Lemma 5.7(a). A global solution has $\kappa = 0$ (no swirl), or $\kappa \neq 0$ and then f is strictly monotone with at most one zero (Definition 5.3): class Z if there is a zero and N otherwise. $\mathsf{Z} = \mathsf{Z}_+ \sqcup \sigma_1\mathsf{Z}_+$ is Theorem 5.62, and $\mathsf{N} = \mathsf{N}_0 \times (\mathbb{Z}_2 \times \mathbb{Z}_2)$ is Theorem 6.61, the four copies being the images under σ_1, σ_2 (Lemma 5.7(c)): σ_1 changes the signs of κ and f, σ_2 that of κ only (it fixes $A = f(0)$). On each copy κ and f have constant sign. (b) The solutions with $\kappa = 0$ are the no-swirl solutions with $f \equiv A$; they are global iff $\boldsymbol{c} \in \mathcal{J}$ and γ lies in the window of Theorem 3.44 (Lemma 5.46(c)), and A is arbitrary since it does not enter the U-equation. The interior statement is Proposition 5.15(b). (c) is Theorem 5.62, Lemma 7.41 and Proposition 7.42(a): interior lens points are generic at both poles and lie in the open set $\Omega_{\mathsf{G}} \subset \operatorname{int}\Omega$, and the boundary points are the edge. (d) is Theorem 6.61(i)–(iii). (e) is Proposition 5.15(a), and $\partial\Omega \cap \{\kappa \neq 0\} \cap \mathsf{Z} = \mathfrak{E}$ is Proposition 7.42(a). (f) is Proposition 5.16. (g) is Proposition 7.44(a) and Proposition 5.59(b),(d) for the tips; the statement on the wall is Theorem 7.32 and Proposition 7.42(b); the reflection is Proposition 7.42(a). □

Remark 7.91 (what the theorem does not say)**.** Five things are not part of the statement. (1) How $\mathfrak{E}$ meets the face $\kappa = 0$: for $c_2 < -\frac{3}{4}$ at the endpoint it passes smoothly through it in the quadrants $A, \kappa \geq 0$ (Theorem 7.74), for $c_2 > -\frac{3}{4}$ through a pole layer (Corollary 7.81), and in the mixed quadrants the passage is not proved (Remark 7.75). (2) Whether the fibres over $(\boldsymbol{c}, A, \kappa)$ with $A\kappa > 0$ are intervals, so that Ω is the region between two graphs $\gamma = \Gamma^{\mp}(\boldsymbol{c}, A, \kappa)$ over the whole parameter space; this is proved for $A\kappa = 0$ and locally near edge points of type R and L_1 (Theorem 7.65). (3) Whether Ω is the closure of Ω_{G}; this holds on the faces $A = 0$ and $\kappa = 0$ (Proposition 5.69) and is not known in general. (4) The decomposition of the edge Γ_{N} in class N into pole types and the regularity of its sub-strata are not described here: only the smoothness at edges of type R and L_1 is proved (Theorem 7.19, Remark 7.43), not the structure at walls or the smoothness of Θ^- on B. (5) The closure and boundary structure of the face Ω_0 at $\gamma = U^{\pm}_{\boldsymbol{c}}(0)$ and at the boundary of $\mathcal{J}$.

7.9.3. Slices. Figure 7.7 shows Ω at $\boldsymbol{c} = 0$. The left panel is the three-dimensional slice $\{\boldsymbol{c} = 0,\ \kappa \geq 0\}$ in the coordinates (A, κ, γ), drawn as the region between the two sheets $\gamma = \Gamma^-(A, \kappa)$ (blue) and $\gamma = \Gamma^+(A, \kappa)$ (red); at the face $\kappa = 0$ the sheets are the planes $\gamma = \mp 2$, the Landau interval, drawn as a black rectangle. The right panel is the two-dimensional section $\gamma = 0$ of the whole set, both signs of κ, with the classes. The sheets are computed by bisection for FS and BS *assuming* that both are half-lines (Numerical observation 7.67); the right panel needs no such assumption, each point is tested for membership in Ω.

7.9.4. The atlas.

Numerical observation 7.92 (evidence for the interval property: 1900 fibres, 450 edges, no counterexample)**.** *The general interval property for* $A\kappa > 0$ *(Remark 7.66) is supported by the following computations, all described in Section 7.7.10.* (i) 400 +

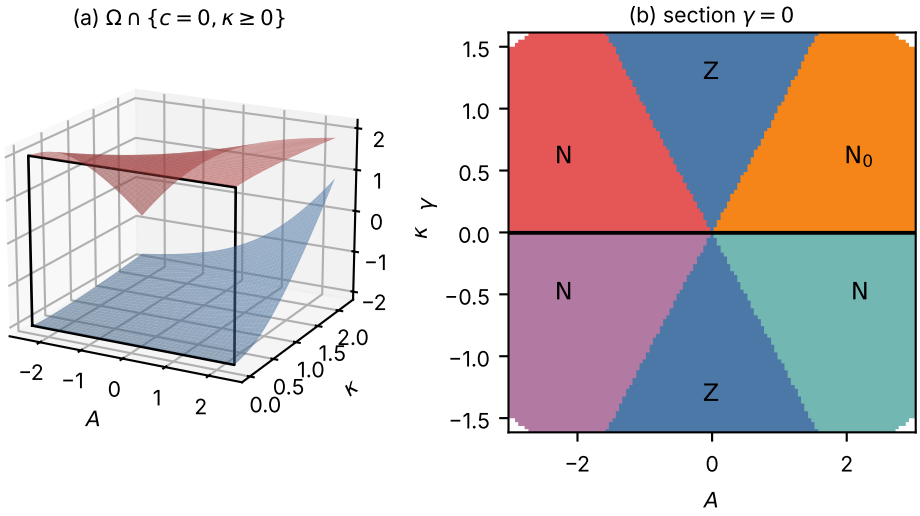


FIGURE 7.7. Ω at $\boldsymbol{c} = 0$. (a) The part $\kappa \geq 0$ as the region between the sheets $\gamma = \Gamma^{-}(A, \kappa)$ and $\gamma = \Gamma^{+}(A, \kappa)$ over 49×36 points, $A \in [-2.4, 2.4]$, $\kappa \in [0.02, 2.2]$; the black rectangle is the face $\kappa = 0$, $\gamma \in [-2, 2]$. (b) The section $\gamma = 0$ over 121×109 points, $A \in [-3, 3]$, $\kappa \in [-1.6, 1.6]$: white is outside Ω, blue the class Z (a double wedge around the κ-axis, Example 7.89), orange N_0, the other three colours the other sign sectors of N, black the face $\kappa = 0$. Code: `code/ch07_edges/make_b_omega.py`.

899 *random and adversarial data:* BS *is a down-set and* FS *an up-set in every case (Numerical observation 7.67(1),(2)).* (ii) 1900 *fibres in which* BS *was computed on a grid of* 600 *values of* γ, 944 *of them with* $A\kappa > 0$*: no gap and no violation of the monotonicity of the blow-up distance (Numerical observation 7.67(3)).* (iii) *A fine rescan of* 1148 *fibres: none has two components (Numerical observation 7.68).* (iv) *At* 450 *edge points of* BS *with* $A\kappa > 0$ *(*150 *in the test of hypothesis (E) with four offsets each, and* $150 + 150$ *in the test of the sign condition, generic and biased towards type* R*): the orthant condition (E) holds in all of the* 600 *samples of the first test, and in the second all* 300 *edges have* $\psi_\gamma > 0$ *at the pole, exactly one sign change,* $w < 0$ *and* $v > 0$ *(Numerical observation 7.70). No counterexample to the half-line property, to the monotonicity in* $\boldsymbol{c}$ *(*3600 *tests) or in* A *(*120 *curves) was found; the monotonicity in* κ *fails (*200 *of* 3000*), as expected (Numerical observation 7.69).*

Atlas 7.93 (where the edges of Ω are understood)**.**

region	*proved*	*computed or formal*
face $\kappa = 0$	Ω_0 *explicit, interior points for generic* γ *(Theorem 7.90(b))*	–
class Z*, edge* $\mathfrak{E}$	*Lipschitz hypersurface, smooth at* R, L_1*; walls* E_{L_2} *of class* $C^{1,1}$ *(Proposition 7.42)*	*wall position, Atlas 7.39*
endpoint $k_2 < \frac{1}{2}$ *(i.e.* $c_2 < -\frac{3}{4}$*)*	R *edge smooth in* κ^2 *at* $A = 0$*; attachment in the quadrants* $A, \kappa \geq 0$ *(Theorem 7.74)*	*first-order shift* $a_1(\boldsymbol{c})A\kappa$*, Formal computation 7.79*
endpoint $k_2 > \frac{1}{2}$	*layer exists, is pushed into the pole (Corollary 7.81); threshold* $\mu = \frac{1}{2}$ *(Corollary 7.73)*	*width* $\xi_L = \ln \frac{1}{\kappa}/(2k-1)$*, log law, exponents (Formal computations 7.82, 7.84 and 7.86, Numerical observations 7.83 and 7.85)*
exact endpoint slice	$A\kappa < 0$ *only (Corollary 7.76)*	*double wedge (Numerical observation 7.77)*
large κ	–	*no* R *edge up to* $\kappa = 14.7$ *(Numerical observation 7.45); two-layer heuristic (Heuristic 7.46)*
fibres, $A\kappa = 0$	*interval,* Ω *up-set in* $\boldsymbol{c}$ *(Proposition 7.50)*	–
fibres, $A\kappa > 0$	FS *half-line;* BS *half-line locally; reduction, chart, small* A/κ *(Theorem 7.65)*	BS *half-line in* 1900 *fibres and* 1299 *data, no counterexample (Numerical observation 7.92)*
monotonicity	*in* $\boldsymbol{c}$ *for* $A\kappa = 0$*; in* A *for* FS	*in* $\boldsymbol{c}$*,* A*: no violation; in* κ*: violated (Numerical observation 7.69)*

Part 4

Effective coordinates: existence and uniqueness

CHAPTER 8

Effective data, existence for every swirl strength, the index criterion

In words. *Chapters 2 and 3 turned the swirl problem into a statement about one linear equation: a global solution is a positive solution z of $z'' = Wz$ on $\mathbb{R}$ together with $f' = \kappa_{\mathrm{D}} z^{-2}$, and the swirl enters the potential only through $W = W_{\mathrm{e}} + \frac{1}{2} Q_{\mathrm{sw}}[f]$, where $Q_{\mathrm{sw}}[f] \geq 0$ is an explicit positive integral of f^2 (Proposition 2.27). The* effective data *are the six numbers $D = (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$ made of the three effective constants (the values of Q at the poles and the first-integral constant), the mixing ratio ϑ (how z is composed of the two Jost solutions), the swirl $f_- = f(-\infty)$ at the south pole, and κ_{D}. Given D and a guess f for the swirl, the potential W_f is known, $z_f = u_- + \vartheta u_+$ is known, and a new swirl $T_D f = f_- + \kappa_{\mathrm{D}} \int_{-\infty}^{\xi} z_f^{-2}$ is computed by a single quadrature; a global solution of type* G *at both poles* is *a fixed point of T_D. This chapter studies that fixed-point problem. The map T_D is compact on a bounded convex set of monotone profiles, and the exact integral $\int z^{-2} = 1/(\vartheta w_0)$ gives an a priori bound; Schauder's theorem then gives existence for every swirl strength once the unperturbed potential is admissible. The derivative of T_D is an explicit operator with positive kernels, compact, but* not *positive; its sign structure gives an antichain property of fixed points and, with an exact L^1 identity, a quantitative moderate-swirl uniqueness theorem.*

Roadmap. Section 8.1 fixes the effective data, the centrifugal operator $\mathcal{Q}$ (with the identity $\mathsf{L}[\mathcal{Q}g] = -g$) and the potentials W_f; Section 8.2 proves the exact integrals (Lemma 8.10), defines T_D and proves that its fixed points are exactly the solutions of type G at both poles (Proposition 8.13). Section 8.3 proves existence by Schauder's theorem on a convex set of monotone profiles built from a reference potential (Lemma 8.15, Theorem 8.16), the variant below the Li–Li–Yan set with its necessary condition (Proposition 8.18), and obtains existence for every κ_{D} and every c_3' as a corollary of the degree theorem proved in the second half of the chapter (Theorem 8.20). Section 8.4 computes the kernels $\Phi_\pm$ and the derivative of T_D (Lemma 8.23, Lemma 8.25, Lemma 8.27), proves the antichain property and the contraction criterion (Proposition 8.32) and the explicit moderate-swirl theorem (Theorem 8.34) with the L^1 identity of Lemma 8.30. The second half of the chapter (Theorem 8.42, Lemma 8.39, Proposition 8.48, Theorem 8.53, Example 8.57) proves that the Leray–Schauder degree of $I - T_D$ is $+1$ for *all* data, the index criterion, the no-swirl face, and collects examples; Chapters 9, 10, 11 use only the objects defined here.

Conventions of the chapter. $' = \mathrm{d}/\mathrm{d}\xi$; $\mathrm{BC} = \mathrm{BC}(\mathbb{R})$ is the Banach space of bounded continuous functions with $\|g\|_\infty = \sup|g|$; $\lg(s) = (1 + e^{-s})^{-1}$ (so $\lg' = \lg(1-\lg), 0 < \lg < 1, \lg' \leq \frac{1}{4}$). *The swirl constant is always* κ_{D} (normalisation 4

of Table 2.1): $\kappa_{\mathrm{D}} = \kappa_W[z]$ for $z = u_- + \vartheta u_+$ with $u_\pm(0) = 1$, so $f' = \kappa_{\mathrm{D}} z^{-2}$; in particular statements of this chapter never use $\kappa = \kappa_x$. In Sections 8.1–8.4 a bare κ occurs only for its sign, or where it is declared to be κ_{D}; *in Sections 8.5–8.8 we write* κ *for* κ_{D} *and* k_i *for* k_i', the abbreviations used in the statements of Theorem 8.42 and Lemma 8.39. Throughout, $c_1', c_2' > -1$ and $k_i' := \sqrt{1 + c_i'}$; for a solution of type G these are the pole exponents (Proposition 8.13), and if some $c_i' \le -1$ no solution of type G exists (Proposition 8.18). For a potential W with Jost solutions we write $\zeta_\pm^W = (u_\pm^W)'/u_\pm^W$, $\rho_W = \zeta_-^W - \zeta_+^W = w_0(W)/(u_-^W u_+^W)$ and $\ell_W(\xi) = \int_0^\xi \rho_W = \ln(u_-^W/u_+^W)$ (Lemma 3.21); the superscript is omitted when W is clear. The sign convention is that of Chapter 3: $w_0 = \mathrm{Wr}(u_+, u_-) = \zeta_-(0) - \zeta_+(0) > 0$.

8.1. Effective data, the centrifugal operator, admissible potentials

In words. *Given D we must be able to write down the potential of a candidate swirl f* without knowing the solution. *The reason this is possible is the decomposition $Q = P_{\boldsymbol{c}'} + Q_{\mathrm{sw}}[f]$ of Proposition 2.27: the polynomial $P_{\boldsymbol{c}'}$ carries the values of Q at the poles and the first integral, and the remainder is a positive, short-range functional of f^2. This section isolates the operator $f^2 \mapsto Q_{\mathrm{sw}}$ for arbitrary bounded f (not only solutions), proves the identity $\mathsf{L}[\mathcal{Q}g] = -g$ that makes fixed points solutions, and collects the elementary bounds on the potentials W_f.*

Recall (Proposition 2.27, (2.38)) that for a solution with f bounded and $Q(\pm 1)$ finite the effective constants are $c_1' = \frac{1}{2}Q(-1)$, $c_2' = \frac{1}{2}Q(1)$, $c_3' = -\frac{1}{4}(\lambda + 2c_1' + 2c_2')$, and $Q = P_{\boldsymbol{c}'} + Q_{\mathrm{sw}}$, $W = W_{\mathrm{e}} + \frac{1}{2}Q_{\mathrm{sw}}$. We now *prescribe* $\boldsymbol{c}'$ and construct solutions.

Definition 8.1 (effective data)**.** The *effective data space* is $\mathbb{D} = \mathbb{R}^3 \times (0,\infty) \times \mathbb{R} \times (0,\infty)$ with elements $D = (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$. Its dimension $3 + 1 + 1 + 1 = 6$ is the dimension of the set of global solutions (Chapter 5). For $D \in \mathbb{D}$ with $c_1', c_2' > -1$ put $k_i' = \sqrt{1 + c_i'}$, $W_{\mathrm{e}} = 1 + \frac{1}{2}P_{\boldsymbol{c}'}$, and $\mathbb{D}_+ = \{D \in \mathbb{D} : c_1', c_2' > -1\}$.

Lemma 8.2 (the centrifugal operator)**.** *For $g \in \mathrm{BC}$ put*

$$\mathcal{Q}[g](\xi) := \frac{1}{4\cosh^2\xi} \int_{\mathbb{R}} e^{-2|\xi-\eta|} g(\eta)\, \mathrm{d}\eta, \qquad \textit{so that} \qquad Q_{\mathrm{sw}}[f] = \mathcal{Q}[f^2]$$

by (2.39). Then:

(a) *$\mathcal{Q}$ is linear and monotone ($g_1 \le g_2 \Rightarrow \mathcal{Q}g_1 \le \mathcal{Q}g_2$); $\mathcal{Q}[a] = \frac{a}{4}\operatorname{sech}^2\xi$ for constants a; $0 \le \mathcal{Q}[g] \le \frac{1}{4}\sup g \operatorname{sech}^2\xi$ if $g \ge 0$; and $\mathcal{Q}[g] > 0$ everywhere if $g \ge 0$ is continuous and $\not\equiv 0$. In the weighted norm $\|q\|_w := \sup_{\mathbb{R}} \cosh^2\xi\, |q(\xi)|$: $\|\mathcal{Q}g\|_w \le \frac{1}{4}\|g\|_\infty$.*

(b) *If $g \in C^k$ then $\mathcal{Q}[g] \in C^{k+2}$.*

(c) *$\mathsf{L}[\mathcal{Q}g] = -g$, where L is the operator of Lemma 2.18 in its line form $\mathsf{L}[F] = \cosh^2\xi\, F'' + 4\sinh\xi\cosh\xi\, F' - 2F$. Moreover $\mathcal{Q}g(\xi) \to 0$ as $\xi \to \pm\infty$, and $\mathcal{Q}[g]$ is the only solution of $\mathsf{L}[F] = -g$ with this property.*

Proof. (a) Linearity and monotonicity are clear because the kernel is positive. $\int_{\mathbb{R}} e^{-2|\sigma|}\,\mathrm{d}\sigma = 1$ gives $\mathcal{Q}[a] = \frac{a}{4}\operatorname{sech}^2\xi$, hence $\cosh^2\xi\,|\mathcal{Q}g| \le \frac{1}{4}\|g\|_\infty$, and $0 \le \mathcal{Q}g \le \mathcal{Q}[\sup g]$ for $g \ge 0$. If $g \ge 0$ is continuous and $g(\eta_0) > 0$ then $g > \frac{1}{2}g(\eta_0)$ on an interval, and the kernel is strictly positive. (b) Put $G(\xi) = \int e^{-2|\xi-\eta|}g(\eta)\,\mathrm{d}\eta = e^{-2\xi}\int_{-\infty}^{\xi} e^{2\eta}g + e^{2\xi}\int_{\xi}^{\infty} e^{-2\eta}g$. Differentiating, the two terms $\pm g(\xi)$ cancel and $G' = -2e^{-2\xi}\int_{-\infty}^{\xi} e^{2\eta}g + 2e^{2\xi}\int_{\xi}^{\infty} e^{-2\eta}g$; differentiating again, $G'' = 4G - 4g$. So

$G \in C^2$ and, inductively, $G \in C^{k+2}$ if $g \in C^k$; $\mathcal{Q}g = G/(4\cosh^2\xi)$. (c) Write $\mathcal{Q}g = H/\cosh^2\xi$ with $H = G/4$ and put $T = \tanh\xi$. Then $(\cosh^{-2})' = -2T\cosh^{-2}$ and

$$\cosh^2\xi\,(\mathcal{Q}g)'' = H'' - 4TH' - 2H + 6T^2H,$$
$$4\sinh\xi\cosh\xi\,(\mathcal{Q}g)' = 4TH' - 8T^2H,$$
$$-2\mathcal{Q}g = -2H + 2T^2H,$$

(using $\cosh^{-2} = 1 - T^2$), whose sum is $H'' - 4H = \frac{1}{4}(G'' - 4G) = -g$. (A computer-algebra check is in `verify_ch08a.py`.) The decay follows from (a). If F is another solution tending to 0 at $\pm\infty$ then $F - \mathcal{Q}g$ solves $\mathsf{L}[F] = 0$ and tends to 0 at both poles; in the variable x both $\pm(F - \mathcal{Q}g)$ satisfy the hypotheses of Lemma 2.19 ($\mathsf{L}[\cdot] \le 0$, boundary values ≥ 0), so $F = \mathcal{Q}g$. □

Toy example 8.3 (the operator at constant source)**.** For $g \equiv a$: $\mathcal{Q}[a] = \frac{a}{4}(1 - x^2)$ and indeed $\mathsf{L}[\frac{a}{4}(1 - x^2)] = -a$ (Toy example 2.21); this is the swirl $f \equiv A$, $a = A^2$, whose centrifugal part is $\frac{1}{4}A^2(1-x^2)$. For a general f the potential $W = W_{\mathrm{e}} + \frac{1}{2}\mathcal{Q}[f^2]$ is a short-range perturbation of W_{e}, of size at most $\frac{1}{8}\|f\|_\infty^2\operatorname{sech}^2\xi$.

Definition 8.4 (admissible potentials; the potentials W_f)**.** A continuous $W : \mathbb{R} \to \mathbb{R}$ is *admissible* if $W \in \mathcal{W}(k_1, k_2)$ for some $k_1, k_2 > 0$ (Definition 3.20: $W - k_1^2 = O(e^{2\xi})$ at $-\infty$, $W - k_2^2 = O(e^{-2\xi})$ at $+\infty$) and the Jost solution φ_-^W is positive on $\mathbb{R}$ and unbounded at $+\infty$; equivalently W is subcritical (Definition 3.31). For admissible W the normalised Jost solutions $u_\pm = u_\pm^W$ ($u_\pm(0) = 1$), the logarithmic derivatives $\zeta_\pm$ and $w_0(W) = \zeta_-(0) - \zeta_+(0) > 0$ are defined and have the properties of Lemma 3.21. For $D \in \mathbb{D}_+$ and $f \in \mathrm{BC}$ put

$$W_f := W_{\mathrm{e}} + \tfrac{1}{2}\mathcal{Q}[f^2], \qquad \mathcal{A}_D := \{f \in \mathrm{BC} :\ W_f \text{ is admissible}\}.$$

Lemma 8.5 (the potentials W_f)**.** *Let $D \in \mathbb{D}_+$.*

(a) *$W_{\mathrm{e}} \in \mathcal{W}(k_1', k_2'; C_{\boldsymbol{c}'})$ with $C_{\boldsymbol{c}'} = |c_1' - c_2'| + 2|c_3'|$, and for every $f \in \mathrm{BC}$, $W_f \in \mathcal{W}(k_1', k_2'; C_{\boldsymbol{c}'} + \frac{1}{2}\|f\|_\infty^2)$: all potentials W_f have the limits $k_1'^2, k_2'^2$ at the exponential rates of Definition 3.20.*

(b) *W_{e} is admissible iff $c_3' > \bar{c}_3(c_1', c_2')$, i.e. $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$; it is critical iff $c_3' = \bar{c}_3$ and supercritical iff $c_3' < \bar{c}_3$.*

(c) *(Swirl raises the potential, controlled.) For $m \ge 0$ put $\widetilde{W}_m := W_{\mathrm{e}} + \frac{1}{8}m^2\operatorname{sech}^2\xi = 1 + \frac{1}{2}P_{(c_1', c_2', c_3' + m^2/4)}$. If $m^2 \le g^2 \le S^2$ on $\mathbb{R}$ then $\widetilde{W}_m \le W_g \le \widetilde{W}_S$. The potential $\widetilde{W}_m$ is admissible iff $c_3' + \frac{1}{4}m^2 > \bar{c}_3(c_1', c_2')$.*

(d) *$\mathcal{A}_D$ is open in BC; more precisely, if $f_0 \in \mathcal{A}_D$, $W_0 = W_{f_0}$, $g_0 = \sup_{\mathbb{R}} 1/\rho_{W_0}$ and $\|f - f_0\|_\infty(\|f\|_\infty + \|f_0\|_\infty) \le 1/g_0$, then $f \in \mathcal{A}_D$.*

Proof. (a) Lemma 3.35(a),(b) gives $W_{\mathrm{e}} \in \mathcal{W}(k_1', k_2'; C_{\boldsymbol{c}'})$, and $0 \le \frac{1}{2}\mathcal{Q}[f^2] \le \frac{1}{8}\|f\|^2\operatorname{sech}^2\xi \le \frac{1}{2}\|f\|^2 e^{-2|\xi|}$ by Lemma 8.2(a) and $\operatorname{sech}\xi \le 2e^{-|\xi|}$. (b) is Lemma 3.35(b). (c) $\mathcal{Q}[g^2]$ is monotone in g^2 and $\mathcal{Q}[m^2] = \frac{m^2}{4}\operatorname{sech}^2$, so $\frac{1}{2}\mathcal{Q}[m^2] = \frac{1}{8}m^2\operatorname{sech}^2\xi$. Since $P_{\boldsymbol{c}} - P_{\boldsymbol{c}'} = (c_3 - c_3')(1 - x^2)$ for $\boldsymbol{c} = (c_1', c_2', c_3)$, $\widetilde{W}_m = 1 + \frac{1}{2}P_{(c_1', c_2', c_3' + m^2/4)}$; admissibility is (b) with c_3' replaced by $c_3' + \frac{1}{4}m^2$. (d) $q := W_f - W_0 = \frac{1}{2}\mathcal{Q}[f^2 - f_0^2]$ has $\|q\|_w \le \frac{1}{8}\|f^2 - f_0^2\|_\infty \le \frac{1}{8}\|f - f_0\|(\|f\| + \|f_0\|) \le \frac{1}{8g_0}$. Now Lemma 8.25 (below; it uses only Lemma 3.21) shows that $W_0 + q$ is admissible. □

Remark 8.6 (why the effective constants are the right ones for existence)**.** (i) $P_{\boldsymbol{c}'}$ is the unique quadratic polynomial with the boundary values $Q(\pm1)$ and the first-integral constant $\mathsf{L}[P_{\boldsymbol{c}'}]=\lambda$, so the remainder Q_{sw} solves $\mathsf{L}[Q_{\mathrm{sw}}]=-f^2$ with zero boundary values and is given by a positive kernel. (ii) c_1', c_2' are read off at the poles: they fix the exponents k_i' of *every* potential W_f, independently of f. The Taylor constants $\boldsymbol{c}$ at the equator are different and depend on the whole profile: $c_i = c_i' + \frac{1}{2}\psi_{\mp}$ (Proposition 2.27(iii)). (iii) Swirl only raises the potential, and only at short range, so a lower bound $W_f \geq W_{\mathrm{ref}}$ can be fixed *before* f is known; positivity of solutions (Lemma 3.32) is inherited by larger potentials.

Example 8.7 (trivial swirl)**.** Let $f \equiv A$ be constant (so $\kappa_{\mathrm{D}}=0$, the boundary case of the chart, $\psi\equiv0$ and $Q=P_{\boldsymbol{c}}$). Then $c_1'=c_1$, $c_2'=c_2$ and $\lambda=\mathsf{L}[P_{\boldsymbol{c}}]+A^2 = -2c_1-2c_2-4c_3+A^2$ (Proposition 2.20), whence

$$c_3' = -\tfrac{1}{4}(\lambda+2c_1+2c_2) = c_3 - \tfrac{1}{4}A^2, \qquad Q_{\mathrm{sw}} = P_{\boldsymbol{c}} - P_{\boldsymbol{c}'} = \tfrac{1}{4}A^2(1-x^2) = \mathcal{Q}[A^2].$$

A constant swirl A lowers the effective constant c_3' by $A^2/4$ and compensates this exactly by the centrifugal term. In particular $c_3'+\frac{1}{4}A^2=c_3$: the Li–Li–Yan constants of the trivial-swirl solution are recovered from $c_3'+\frac{1}{4}S^2$ with $S=|A|$, which is the combination in Proposition 8.18. (Checked in `verify_ch08a.py`.)

Example 8.8 (sample dependence of the pole exponents on $\boldsymbol{c}'$)**.** The exponents $k_i'=\sqrt{1+c_i'}$ and the potential $W_{\mathrm{e}}=1+\frac{1}{2}P_{\boldsymbol{c}'}$ at the poles are read off directly ($\bar c_3=-2\bar k(\bar k+1)$ with $\bar k=\frac{1}{2}(k_1'+k_2')$, Lemma 3.35): $W_{\mathrm{e}}(-\infty)=1+c_1'\cdot\frac{1}{2}\cdot2 = 1+c_1'=k_1'^2$ and $W_{\mathrm{e}}(+\infty)=1+c_2'=k_2'^2$ (as $P_{\boldsymbol{c}'}\to 2c_1', 2c_2'$ at $x\to\mp1$).

$\boldsymbol{c}'$	(k_1',k_2')	$\bar c_3(c_1',c_2')$	W_{e}
$(0,0,0)$	$(1,1)$	-4	$\equiv 1$ (the rest state)
$(3,0,0)$	$(2,1)$	$-\frac{15}{2}$	$\to 4$ at $-\infty$, $\to 1$ at $+\infty$
$(0,-\frac{3}{4},0)$	$(1,\frac{1}{2})$	$-\frac{21}{8}$	$\to 1$ at $-\infty$, $\to \frac{1}{4}$ at $+\infty$

A pole with $c_i'\downarrow -1$ has $k_i'\downarrow 0$, the boundary of type G (Proposition 8.18): the Jost solution loses its exponential rate and the constants of Lemma 8.23(b) degenerate.

8.2. Exact integrals, the map T_D, and solutions as fixed points

In words. *If W is admissible and $z=u_-+\vartheta u_+$, then z^{-2} can be integrated in closed form: the quotient $R=u_-/u_+$ increases from 0 to ∞ and $z^{-2}\,\mathrm{d}\xi = \mathrm{d}R/(w_0(R+\vartheta)^2)$. So the swirl is a logistic function of $\ln R$ whose total height $J=\kappa_{\mathrm{D}}/(\vartheta w_0)$ is known in terms of w_0 alone. The map T_D sends a profile f to the swirl $f_- + \kappa_{\mathrm{D}}\int z_f^{-2}$ computed from the potential W_f; a fixed point of T_D is exactly a global solution of type G at both poles.*

Toy example 8.9 (the logistic step at a constant potential)**.** Let $W\equiv k^2$ (Example 3.22): $u_\pm = e^{\pm k\xi}$, $w_0=2k$, $R=e^{2k\xi}$, $z=e^{k\xi}+\vartheta e^{-k\xi}$. Then $u_-/z = 1/(1+\vartheta e^{-2k\xi}) = \lg(2k\xi-\ln\vartheta)$ and $\int_{-\infty}^{\xi} z^{-2} = \frac{1}{2k\vartheta}\lg(2k\xi-\ln\vartheta)$, so $f=f_-+J\lg(2k\xi-\ln\vartheta)$ is a *logistic step* of height $J=\kappa_{\mathrm{D}}/(2k\vartheta)$ centred at $\xi_c=\ln\vartheta/(2k)$: the mixing ratio ϑ fixes the position of the swirl layer and κ_{D} its height. The lemma below shows that this is the general picture, with $2k\xi$ replaced by $\ell=\ln R$. (For $k=1$ this is the rest state $\boldsymbol{c}'=0$, $f\equiv0$ before the swirl is added.)

Lemma 8.10 (exact integrals)**.** *Let W be admissible and $\vartheta > 0$, $z = u_- + \vartheta u_+$, $f(\xi) := f_- + \kappa_{\mathrm{D}} \int_{-\infty}^{\xi} z^{-2}$ (we show that the integral converges). For admissible W, $\vartheta > 0$, $z = u_- + \vartheta u_+$, $f' = \kappa_{\mathrm{D}} z^{-2}$ ($\kappa = \kappa_{\mathrm{D}}$), $f(-\infty) = f_-$: $R := u_-/u_+$ increases from 0 to ∞, $R' = w_0 u_+^{-2}$, $f = f_- + J\, u_-/z$ with $J = f_+ - f_- = \kappa_{\mathrm{D}}/(\vartheta w_0)$ and $(u_-/z)(0) = 1/(1+\vartheta)$. In addition:* (i) $\int_{-\infty}^{\xi} z^{-2} = \frac{1}{\vartheta w_0} \frac{u_-}{z}(\xi)$ *and* $\int_{\mathbb{R}} z^{-2} = \frac{1}{\vartheta w_0}$; (ii) *with $\ell = \ell_W = \int_0^{\xi} \rho_W = \ln R$,*

$$f = f_- + J \, \lg(\ell - \ln \vartheta), \qquad J = \frac{\kappa_{\mathrm{D}}}{\vartheta \, \rho_W(0)} = \frac{\kappa_{\mathrm{D}}}{\vartheta w_0}. \tag{8.1}$$

Proof. Since W is admissible, $u_\pm > 0$ and by Lemma 3.21(b): $u_- \to 0$ and $u_+ \to \infty$ at $-\infty$, $u_- \to \infty$ and $u_+ \to 0$ at $+\infty$. Hence $R = u_-/u_+$ is positive with $R \to 0$ at $-\infty$, $R \to \infty$ at $+\infty$, and

$$R' = \frac{u_-' u_+ - u_- u_+'}{u_+^2} = \frac{\mathrm{Wr}(u_+, u_-)}{u_+^2} = w_0 \, u_+^{-2} > 0.$$

So R is a strictly increasing C^1 bijection of $\mathbb{R}$ onto $(0, \infty)$. Write $z = u_+(R + \vartheta)$; then $z^{-2} = u_+^{-2}(R+\vartheta)^{-2} = R'/(w_0(R+\vartheta)^2)$ and

$$\frac{\mathrm{d}}{\mathrm{d}\xi} \frac{R}{R+\vartheta} = \frac{\vartheta R'}{(R+\vartheta)^2} = \vartheta w_0 \, z^{-2}, \qquad \frac{R}{R+\vartheta} = \frac{u_-}{z}.$$

As $u_-/z \to 0$ at $-\infty$ and $\to 1$ at $+\infty$ we obtain (i), so $z^{-2} \in L^1(\mathbb{R})$, f is well defined, $f(-\infty) = f_-$, $f = f_- + \frac{\kappa_{\mathrm{D}}}{\vartheta w_0} \frac{u_-}{z}$ and $f(+\infty) - f(-\infty) = \kappa_{\mathrm{D}}/(\vartheta w_0) = J$. Finally $u_\pm(0) = 1$ gives $(u_-/z)(0) = 1/(1+\vartheta)$. For (ii), $R/(R+\vartheta) = 1/(1+\vartheta/R) = \lg(\ln R - \ln \vartheta)$, $\ln R = \int_0^{\xi} \rho_W$ because $(\ln R)' = \zeta_- - \zeta_+$ and $R(0) = 1$, and $\rho_W(0) = w_0$. □

Definition 8.11 (the map T_D)**.** Let $D \in \mathbb{D}_+$. For $f \in \mathcal{A}_D$ let $z_f := u_-^{W_f} + \vartheta u_+^{W_f}$ and

$$T_D f := f_- + \kappa_{\mathrm{D}} \int_{-\infty}^{\xi} z_f^{-2} \, \mathrm{d}\eta.$$

By Lemma 8.10, for $f \in \mathcal{A}_D$ the function $T_D f$ is C^1, increasing, $T_D f(-\infty) = f_-$, $T_D f(+\infty) - f_- = J(f) := \kappa_{\mathrm{D}}/(\vartheta \, w_0(W_f))$ and

$$T_D f = f_- + J(f) \, \lg\big(\ell_{W_f} - \ln \vartheta\big), \qquad f_- \le T_D f \le f_- + J(f). \tag{8.2}$$

In particular $T_D(\mathcal{A}_D) \subset \mathrm{BC}$. No fixed-point relation is used here: (8.2) holds for every $f \in \mathcal{A}_D$, and $\mathcal{A}_D$ is open (Lemma 8.5(d)).

Toy example 8.12 (the first iterate at the rest state)**.** Take $\boldsymbol{c}' = 0$, $f_- = 0$, $\vartheta = 1$ and start from $f_0 \equiv 0$. Then $W_{f_0} = W_{\mathrm{e}} = 1$ and Toy example 8.9 gives $f_1 = T_D f_0 = \frac{\kappa_{\mathrm{D}}}{2} \lg(2\xi)$, a logistic step of height $\kappa_{\mathrm{D}}/2$. Now $f_1^2 > 0$, so $W_{f_1} = 1 + \frac{1}{2}\mathcal{Q}[f_1^2] > 1$: the potential has been *raised* near the layer, hence (Lemma 3.23) w_0 has increased and the height $J(f_1) = \kappa_{\mathrm{D}}/(\vartheta w_0(W_{f_1})) < \kappa_{\mathrm{D}}/2$ of the next step is smaller. The iteration $f_n = T_D^n f_0$ converges geometrically for small κ_{D}, with a rate bounded by Theorem 8.34; a numerical instance is Numerical observation 8.36.

Proposition 8.13 (solutions ↔ fixed points)**.** *Let $D \in \mathbb{D}_+$.*

(a) *If $f \in \mathcal{A}_D$ and $f = T_D f$, then with $z = z_f$, $Q := 2(W_f - 1) = P_{\boldsymbol{c}'} + \mathcal{Q}[f^2]$ and $U := 2(z'/z - \tanh \xi)$, the triple (U, f, Q) is a C^∞ global solution of the Q-system, of type* G *at both poles, with effective constants $\boldsymbol{c}'$, $f(-1) = f_-$,*

$U(-1) = 2 - 2k_1'$, $U(1) = -2 + 2k_2'$ *and* $f(1) = f_- + \kappa_{\mathrm{D}}/(\vartheta w_0)$. *Its swirl constant in the normalisation* $z = u_- + \vartheta u_+$ *is* κ_{D}.

(b) *Conversely every global solution with* $\kappa > 0$ (*in any normalisation*) *that is of type* G *at both poles arises in this way from exactly one* $D \in \mathbb{D}_+$. *Solutions with* $\kappa < 0$ *are obtained from these by* σ_1 ($f \mapsto -f$), *which maps* $(f_-, \kappa_{\mathrm{D}})$ *to* $(-f_-, -\kappa_{\mathrm{D}})$ *and does not change* $\boldsymbol{c}'$, ϑ.

PROOF. We use three facts about type G (Definition 4.16, Theorem 4.5, Proposition 4.19): the pole $x = 1$ is of type G iff $Q(1) > -2$ exists, $\mu_+ = \sqrt{1 + Q(1)/2} > 0$ and $z\, e^{-\mu_+ \xi} \to a_+ > 0$ at $+\infty$ (*dominant*); then $f(1)$ is finite. Mirror at $x = -1$: $\mu_- = \sqrt{1 + Q(-1)/2}$ and $z e^{\mu_- \xi} \to a_- > 0$ at $-\infty$.

(a) *Equations.* $f = T_D f$ is C^1 and $f' = \kappa_{\mathrm{D}} z^{-2}$. By Lemma 8.2(b),(c) $\mathcal{Q}[f^2]$ is C^3 if $f \in C^1$; so $W_f \in C^3$, $z \in C^5$ and $f' = \kappa_{\mathrm{D}} z^{-2} \in C^5$, $f \in C^6$. If $f \in C^m$ ($m \geq 6$) then $W_f \in C^{m+2}$, $z \in C^{m+4}$, $f \in C^{m+5}$: f, W_f, z are C^∞. $\mathsf{L}[P_{\boldsymbol{c}'}] = -2c_1' - 2c_2' - 4c_3' =: \lambda$ (Lemma 2.18(b)) and $\mathsf{L}[\mathcal{Q}[f^2]] = -f^2$ (Lemma 8.2(c)), so $\mathsf{L}[Q] = \lambda - f^2$. Further $z'' = W_f z = (1 + \frac{1}{2}Q)z$, $z > 0$, $f' = \kappa_{\mathrm{D}} z^{-2}$. With $y = z/\cosh\xi$, Lemma 2.14 (whose computation holds for every z) gives (2.24): $U = 2(1-x^2)y_x/y$, $2(1-x^2)^2 y_{xx} = Qy$, $y^2 f_x = \kappa_{\mathrm{D}}$. By Lemma 2.9 (converse parts) (U, f, Q) satisfies (2.19), (2.20); by Proposition 2.20, $\mathsf{L}[Q] + f^2 \equiv \lambda$ is equivalent to (2.21). By Lemma 2.5, (U, f) solves the original system with $\boldsymbol{c}$ the Taylor constants of Q at 0. It is global because $z > 0$ on $\mathbb{R}$ (Corollary 2.25). *Pole behaviour.* By Lemma 3.21, at $-\infty$ $u_- \to 0$, $u_+ \to \infty$, $\zeta_+ \to -k_1'$ and ζ_- is bounded, so $z = u_- + \vartheta u_+ \sim \vartheta u_+$ and $h = z'/z = \dfrac{\zeta_- u_-/u_+ + \vartheta \zeta_+}{u_-/u_+ + \vartheta} \to -k_1'$; likewise $h \to k_2'$ at $+\infty$. Hence $U = 2(h - \tanh\xi) \to 2 - 2k_1'$ at $\xi = -\infty$ and $\to -2 + 2k_2'$ at $+\infty$. Moreover $\mathcal{Q}[f^2] \to 0$ at both poles and $P_{\boldsymbol{c}'}(\mp 1) = 2c_{1,2}'$, so $Q(\mp 1) = 2c_{1,2}' > -2$, $\mu_\pm = k_{2,1}' > 0$ (indices: $\mu_+ = k_2'$, $\mu_- = k_1'$), and z is dominant at both ends: $z e^{-k_2'\xi} \to u_- e^{-k_2'\xi} \to a > 0$ at $+\infty$ (Lemma 3.21(b)(i); the u_+-term decays), and mirror. So both poles are of type G. f is bounded, $Q(\pm 1)$ finite, so the effective constants of the solution are $(\frac{1}{2}Q(-1), \frac{1}{2}Q(1), -\frac{1}{4}(\lambda + 2c_1' + 2c_2')) = \boldsymbol{c}'$ (Proposition 2.27). $f(-1) = f_-$ and $f(1) = f_- + \kappa_{\mathrm{D}}/(\vartheta w_0)$ by Lemma 8.10.

(b) Let (U, f, Q) be global with $\kappa > 0$, of type G at both poles. Then f is monotone with finite limits at ± 1 and $Q(\pm 1) > -2$ finite, so $\mu_\pm$ as above, $c_1' = \frac{1}{2}Q(-1) > -1$, $c_2' > -1$, and $Q = P_{\boldsymbol{c}'} + \mathcal{Q}[f^2]$ (Proposition 2.27, (2.39)), i.e. $W = W_f$ in the sense of Definition 8.4, with $k_i' = \mu_\mp$. Let $z > 0$ be the solution of Lemma 2.14; z is dominant: $z e^{k_1'\xi} \to A_- > 0$ at $-\infty$ and $z e^{-k_2'\xi} \to A_+ > 0$ at $+\infty$. *W_f is admissible.* Put $u = \varphi_-^{W_f}$. As $\xi \to -\infty$: $z'/z \to -k_1'$, $u'/u \to k_1'$, so $\mathrm{Wr}(z, u) = zu' - z'u = zu(u'/u - z'/z) \to A_- \cdot 2k_1' > 0$ (using $zu \to A_-$). $\mathrm{Wr}(z, u)$ is constant (z, u solve the same equation), so $(u/z)' = \mathrm{Wr}(z, u)/z^2 > 0$ on $\mathbb{R}$. Since $u/z \to 0$ at $-\infty$, $u/z > 0$ on $\mathbb{R}$, hence $u > 0$, and $u \geq (u/z)(0)\, z$ on $[0, \infty)$ is unbounded at $+\infty$. So W_f is admissible. *Coordinates.* Then $u_\pm$ is a basis, $z = \alpha u_- + \beta u_+$. At $+\infty$, $u_+/u_- \to 0$ and $z/u_- \to \alpha$; as z and u_- are both positive and $\sim e^{k_2'\xi}$ we get $\alpha = \lim z/u_- = A_+/\lim e^{-k_2'\xi} u_- > 0$; likewise $\beta > 0$ from $-\infty$. Replace (z, κ) by $(z/\alpha, \kappa/\alpha^2)$ (this changes neither U, f nor Q) and set $\vartheta = \beta/\alpha$, $\kappa_{\mathrm{D}} = \kappa/\alpha^2$, $f_- = f(-1)$. Then $z = u_- + \vartheta u_+$, $f' = \kappa_{\mathrm{D}} z^{-2}$, so $f = f_- + \kappa_{\mathrm{D}} \int_{-\infty}^{\xi} z^{-2} = T_D f$ and $f \in \mathcal{A}_D$. *Uniqueness of D.* $\boldsymbol{c}'$ and f_- are determined by the solution; U determines z'/z and hence z up to a positive factor, and the requirement that the coefficient of u_- be 1 fixes the factor, hence ϑ and

$\kappa_{\mathrm{D}} = f' z^2$. For $\kappa < 0$ apply σ_1 (Lemma 2.8), which preserves Q, λ, z and the pole types. □

Thus $D \mapsto \{\text{fixed points of } T_D\}$ is a "chart" of the six-dimensional set of global solutions of type G at both poles, provided it is single-valued; whether it is, is the uniqueness question of Section 8.4 and of Chapter 9. Inside this chart $f_- \geq 0$ means class N_0 and $f_- < 0 < f(1)$ means class Z (Definition 5.3, Proposition 5.16): if $f_- \geq 0$ and $\kappa_{\mathrm{D}} > 0$ then $f > f_- \geq 0$ has no zero.

Lemma 8.14 (Arzelà–Ascoli on the line)**.** *A set $\mathcal{F} \subset \mathrm{BC}$ is relatively compact if (i) it is bounded, (ii) it is equicontinuous and (iii) it has uniform tails: for every $\epsilon > 0$ there is L with $|g(\xi) - g(\eta)| \leq \epsilon$ for all $g \in \mathcal{F}$ and all $\xi, \eta \geq L$, and also for all $\xi, \eta \leq -L$.*

Proof. This is Lemma A.29. □

8.3. Existence by Schauder's theorem; below the Li–Li–Yan set

In words. *Schauder's theorem needs a closed convex set mapped into itself by a compact map. The exact integral supplies both: every image $T_D f$ is a nondecreasing profile with $f_- \leq T_D f \leq f_- + \kappa_{\mathrm{D}}/(\vartheta w_0(W_{\mathrm{ref}}))$ as soon as the potentials W_f stay above a reference potential W_{ref} (larger potentials have larger w_0), and the same reference gives a lower barrier $z_{\mathrm{low}} \leq z_f$ independent of f, which makes the images equicontinuous with uniform tails. For $\mathbf{c}' \in \operatorname{int} \mathcal{J}$ the reference is W_{e} itself and no smallness is needed; for $f_- \geq 0$ the swirl is bounded below by f_- and the reference $W_{\mathrm{e}} + \frac{1}{8} f_-^2 \operatorname{sech}^2$ is admissible already below the Li–Li–Yan set. Conversely a solution can exist only if the largest possible swirl would make the potential admissible.*

Lemma 8.15 (the reference-potential principle)**.** *Let $D \in \mathbb{D}_+$ and let $W_{\mathrm{ref}} \in \mathcal{W}(k_1', k_2')$ be admissible; write $u_\pm^{\mathrm{ref}}$, $w_0^{\mathrm{ref}} = w_0(W_{\mathrm{ref}})$ and*

$$B_{\mathrm{ref}} := \frac{\kappa_{\mathrm{D}}}{\vartheta\, w_0^{\mathrm{ref}}},$$

$$K_{\mathrm{ref}} := \{ f \in C(\mathbb{R}) : \ f \text{ nondecreasing},\ f(-\infty) = f_-,\ f \leq f_- + B_{\mathrm{ref}} \},$$

$z_{\mathrm{low}} := u_-^{\mathrm{ref}}$ on $[0, \infty)$ and $z_{\mathrm{low}} := \vartheta u_+^{\mathrm{ref}}$ on $(-\infty, 0)$. Assume

$$W_f \geq W_{\mathrm{ref}} \quad \text{for every } f \in K_{\mathrm{ref}}. \tag{8.3}$$

Then:

(a) *K_{ref} is a nonempty, closed, convex, bounded subset of* BC *($f_- \leq f \leq f_- + B_{\mathrm{ref}}$) and $K_{\mathrm{ref}} \subset \mathcal{A}_D$.*

(b) *For $f \in K_{\mathrm{ref}}$: $w_0(W_f) \geq w_0^{\mathrm{ref}}$, $\rho_{W_f} \geq \rho_{\mathrm{ref}}$, $z_f \geq z_{\mathrm{low}}$, and $T_D f \in K_{\mathrm{ref}}$.*

(c) *z_{low} is positive, bounded below by a positive constant, and there are $c_* > 0$ with $z_{\mathrm{low}} \geq c_* e^{k_2' \xi}$ on $[0, \infty)$ and $z_{\mathrm{low}} \geq c_* e^{-k_1' \xi}$ on $(-\infty, 0)$. Hence $\Lambda_{\mathrm{low}} := \sup z_{\mathrm{low}}^{-2} < \infty$, $C_* := \int_{\mathbb{R}} z_{\mathrm{low}}^{-2} < \infty$ and $\int_{|\xi| > L} z_{\mathrm{low}}^{-2} \to 0$ as $L \to \infty$.*

(d) *$T_D : K_{\mathrm{ref}} \to K_{\mathrm{ref}}$ is continuous and $T_D(K_{\mathrm{ref}})$ is relatively compact in* BC*. Hence T_D has a fixed point in K_{ref} (Theorem A.31).*

(e) *Every $f \in \mathcal{A}_D$ with $f = T_D f$ and $W_f \geq W_{\mathrm{ref}}$ lies in K_{ref}; in particular $0 \leq f - f_- \leq B_{\mathrm{ref}}$.*

PROOF. (a) $K_{\mathrm{ref}} \ni$ the constant f_-. A nondecreasing f with $f(-\infty) = f_-$ and $f \le f_- + B_{\mathrm{ref}}$ satisfies $f_- \le f \le f_- + B_{\mathrm{ref}}$. Convex combinations of nondecreasing functions with the same limit at $-\infty$ and the same upper bound have these properties. If $f_n \in K_{\mathrm{ref}}$, $f_n \to f$ uniformly, then f is nondecreasing, $f \le f_- + B_{\mathrm{ref}}$ and $|f(\xi) - f_-| \le \|f - f_n\|_\infty + |f_n(\xi) - f_-|$ shows $f(-\infty) = f_-$. Finally $W_f \ge W_{\mathrm{ref}}$ and $W_f \in \mathcal{W}(k_1', k_2')$ (Lemma 8.5(a)); Lemma 3.23 makes W_f admissible. (b) Lemma 3.23 with $W_{\mathrm{ref}} \le W_f$ gives $\zeta_-^f \ge \zeta_-^{\mathrm{ref}}$, $\zeta_+^f \le \zeta_+^{\mathrm{ref}}$ (so $\rho_f \ge \rho_{\mathrm{ref}}$), $w_0(W_f) \ge w_0^{\mathrm{ref}}$, $u_-^f \ge u_-^{\mathrm{ref}}$ on $[0,\infty)$ and $u_+^f \ge u_+^{\mathrm{ref}}$ on $(-\infty, 0]$. Hence for $\xi \ge 0$, $z_f \ge u_-^f \ge u_-^{\mathrm{ref}}$ (as $u_+^f > 0$), and for $\xi < 0$, $z_f \ge \vartheta u_+^f \ge \vartheta u_+^{\mathrm{ref}}$: $z_f \ge z_{\mathrm{low}}$. By (8.2), $T_D f$ is C^1, increasing, $T_D f(-\infty) = f_-$ and $T_D f \le f_- + \kappa_{\mathrm{D}}/(\vartheta w_0(W_f)) \le f_- + B_{\mathrm{ref}}$. (c) z_{low} is continuous and positive on each half-line. By Lemma 3.21(b)(i), $e^{-k_2'\xi} u_-^{\mathrm{ref}} \to a > 0$ at $+\infty$; so $e^{-k_2'\xi} u_-^{\mathrm{ref}}$ is continuous, positive and has a positive limit, hence is bounded below by some $c_1 > 0$ on $[0,\infty)$. Likewise $e^{k_1'\xi} u_+^{\mathrm{ref}} \ge c_2 > 0$ on $(-\infty, 0]$. Take $c_* = \min(c_1, \vartheta c_2)$ (and $c_* \le 1$): then $z_{\mathrm{low}} \ge c_*$ on $\mathbb{R}$, and $z_{\mathrm{low}}^{-2} \le c_*^{-2} e^{-2k_2'\xi}$ resp. $c_*^{-2} e^{2k_1'\xi}$ is integrable. (d) *Compactness.* Let $g = T_D f$, $f \in K_{\mathrm{ref}}$. Then $f_- \le g \le f_- + B_{\mathrm{ref}}$, $|g(\xi) - g(\eta)| = \kappa_{\mathrm{D}} \int_\eta^\xi z_f^{-2} \le \kappa_{\mathrm{D}} \Lambda_{\mathrm{low}} |\xi - \eta|$ for $\eta < \xi$ (a Lipschitz bound independent of f), and, for $\xi, \eta \ge L$, $|g(\xi) - g(\eta)| \le \kappa_{\mathrm{D}} \int_L^\infty z_{\mathrm{low}}^{-2}$, for $\xi, \eta \le -L$ similarly; both tend to 0 with L uniformly in f by (c). Lemma 8.14 applies. *Continuity.* Let $f_n \to f$ in K_{ref}. By Lemma 8.2(a), $\cosh^2\xi\, |W_{f_n} - W_f| = \frac{1}{2}\cosh^2\xi\, |\mathcal{Q}[f_n^2 - f^2]| \le \frac{1}{8}\|f_n^2 - f^2\|_\infty \to 0$, and all W_{f_n}, W_f lie in one class $\mathcal{W}(k_1', k_2'; C)$ (Lemma 8.5(a)). By Lemma 3.25(b) the Jost solutions $u_\pm^{f_n} \to u_\pm^f$ locally uniformly, hence $z_{f_n} \to z_f$ pointwise. Since $z_{f_n}^{-2}, z_f^{-2} \le z_{\mathrm{low}}^{-2} \in L^1$, dominated convergence gives $\|T_D f_n - T_D f\|_\infty \le \kappa_{\mathrm{D}} \int_{\mathbb{R}} |z_{f_n}^{-2} - z_f^{-2}| \to 0$. By (b) T_D maps K_{ref} into itself; Theorem A.31 gives the fixed point. (e) $f = T_D f$ is nondecreasing with $f(-\infty) = f_-$ and, by (8.2), $f \le f_- + \kappa_{\mathrm{D}}/(\vartheta w_0(W_f)) \le f_- + B_{\mathrm{ref}}$ because $w_0(W_f) \ge w_0^{\mathrm{ref}}$ (Lemma 3.23). □

THEOREM 8.16 (existence for $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$). *Let $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$ ($c_1', c_2' > -1$, $c_3' > \bar{c}_3(c_1', c_2')$), $\vartheta > 0$, $f_- \in \mathbb{R}$ and $\kappa_{\mathrm{D}} > 0$. Then $\mathcal{A}_D = \mathrm{BC}$ and T_D has a fixed point; hence there is a global solution with effective data D, of type G at both poles. Every such solution satisfies*

$$0 \le f - f_- \le \frac{\kappa_{\mathrm{D}}}{\vartheta\, w_0(W_{\mathrm{e}})},$$

and $T_D(\mathrm{BC}) \subset K := \{f \in C(\mathbb{R}) :$ f nondecreasing, $f(-\infty) = f_-$, $f \le f_- + \kappa_{\mathrm{D}}/(\vartheta w_0(W_{\mathrm{e}}))\}$. No smallness of κ_{D}, f_- or $\boldsymbol{c}'$ is assumed.

PROOF. By Lemma 8.5(b), W_{e} is admissible. Since $\mathcal{Q}[f^2] \ge 0$, $W_f \ge W_{\mathrm{e}}$ for every $f \in \mathrm{BC}$, so (8.3) holds with $W_{\mathrm{ref}} = W_{\mathrm{e}}$ (then $K_{\mathrm{ref}} = K$, z_{low} and C_* are the lower barrier and constant of the construction). By Lemma 8.5(a) and Lemma 3.23, W_f is admissible for *every* $f \in \mathrm{BC}$: $\mathcal{A}_D = \mathrm{BC}$. For every $f \in \mathrm{BC}$, (8.2) and $w_0(W_f) \ge w_0(W_{\mathrm{e}})$ give $T_D f \in K$, i.e. $T_D(\mathrm{BC}) \subset K$ (this is stronger than (b) of the lemma, which only treated $f \in K$). Lemma 8.15(d) gives a fixed point $f \in K$; by Proposition 8.13(a) it is a global solution of type G at both poles with effective data D. Every fixed point lies in $T_D(\mathrm{BC}) \subset K$, which is the stated bound; every solution with data D is a fixed point (Proposition 8.13(b)). □

For $\boldsymbol{c}' = 0$ (Example 3.22, $W_{\mathrm{e}} \equiv 1$, $w_0 = 2$) the bound reads $0 \le f - f_- \le \kappa_{\mathrm{D}}/(2\vartheta)$; it is linear in κ_{D} and far from sharp for large κ_{D} (Chapter 10: $J \approx \sqrt{2\kappa_{\mathrm{D}}/\vartheta}$), but it holds with no restriction on the data.

Toy example 8.17 (Schauder's two ingredients at the rest state)**.** Take $\boldsymbol{c}' = 0$, $f_- = 0$, $\vartheta = 1$, so $W_{\mathrm{e}} \equiv 1$, $u_\pm = e^{\pm\xi}$, $w_0 = 2$ and $K = \{f \text{ nondecreasing},\ f(-\infty) = 0,\ f \le \kappa_{\mathrm{D}}/2\}$. The barrier is $z_{\mathrm{low}} = e^{\xi}$ on $\xi \ge 0$ and $e^{-\xi}$ on $\xi < 0$; $z_{\mathrm{low}}^{-2} = e^{-2|\xi|}$ and $C_* = \int z_{\mathrm{low}}^{-2} = 1$. For *every* $f \in K$ (not only for fixed points) the image $g = T_D f$ satisfies $0 \le g \le \kappa_{\mathrm{D}}/2$ (it stays in K) and $0 \le g' = \kappa_{\mathrm{D}} z_f^{-2} \le \kappa_{\mathrm{D}} e^{-2|\xi|}$: the images are uniformly Lipschitz (constant κ_{D}) and the increments beyond $|\xi| \ge L$ are at most $\frac{\kappa_{\mathrm{D}}}{2} e^{-2L}$, uniformly in f. These are exactly the hypotheses (bounded, equicontinuous, uniform tails) of Lemma 8.14; the point of Lemma 8.15 is that the same two estimates hold for a general admissible reference potential, with z_{low} in the place of $e^{-|\xi|}$. The example also shows what is *not* proved by Schauder: whether the fixed point is unique (Theorem 8.34 for $\kappa_{\mathrm{D}} < 4/\sqrt{3}$, and Chapter 9 beyond).

Proposition 8.18 (sign-definite swirl below $\mathcal{J}$; a necessary condition)**.** *If $c_1', c_2' > -1$, $f_- \ge 0$, $\kappa_{\mathrm{D}} > 0$ and $c_3' + \frac{1}{4} f_-^2 > \bar{c}_3(c_1', c_2')$ then T_D has a fixed point. Conversely a global solution of type* G *at both poles with effective constants $\boldsymbol{c}'$ and $S = \max(|f_-|, |f_+|)$ has $c_1', c_2' > -1$ and $c_3' + \frac{1}{4} S^2 \ge \bar{c}_3(c_1', c_2')$. Moreover, in the first statement every fixed point lies in $\widetilde{K} := \{f \in C(\mathbb{R}) :\ f \text{ nondecreasing},\ f(-\infty) = f_-,\ f \le f_- + \kappa_{\mathrm{D}}/(\vartheta w_0(\widetilde{W}))\}$, $\widetilde{W} = W_{\mathrm{e}} + \frac{1}{8} f_-^2 \operatorname{sech}^2 \xi$, and belongs to class* N_0.

Proof. *Existence.* Let $f_- \ge 0$ and $c_3' + \frac{1}{4} f_-^2 > \bar{c}_3(c_1', c_2')$. By Lemma 8.5(c) with $m = f_-$, $\widetilde{W} = \widetilde{W}_{f_-}$ is admissible. Let $f \in K_{\mathrm{ref}}$ with $W_{\mathrm{ref}} = \widetilde{W}$. Then $f \ge f_- \ge 0$, so $f^2 \ge f_-^2$ and $W_f \ge \widetilde{W}_{f_-} = \widetilde{W}$ (Lemma 8.5(c)): (8.3) holds. Lemma 8.15(d) gives a fixed point in $\widetilde{K} = K_{\mathrm{ref}}$ (note that T_D need not be defined on all of BC now: it is only defined on $\mathcal{A}_D \supset \widetilde{K}$). Every fixed point $f = T_D f$ is nondecreasing with $f(-\infty) = f_- \ge 0$, so $f^2 \ge f_-^2$, $W_f \ge \widetilde{W}$, and Lemma 8.15(e) puts it in $\widetilde{K}$. As $f = T_D f > f_- \ge 0$ and $f' = \kappa_{\mathrm{D}} z^{-2} > 0$, f has no zero: the solution is in N_0 (Definition 5.3).

Necessary condition. Let (U, f, Q) be global, of type G at both poles, with effective constants $\boldsymbol{c}'$ and $\kappa \ne 0$ (the case without swirl is separate: for a global solution without swirl, $f \equiv A$, $S = |A|$, $z'' = W_{\boldsymbol{c}} z$ has a positive solution, so $W_{\boldsymbol{c}}$ is not supercritical, Lemma 3.32(b), i.e. $c_3 \ge \bar{c}_3$, and $c_3' + \frac{1}{4} A^2 = c_3$ by Example 8.7). Type G means $Q(\pm 1) > -2$, i.e. $c_1', c_2' > -1$. By monotonicity $\sup|f| = \max(|f_-|, |f_+|) = S$, and Lemma 8.5(c) gives $W = W_f \le W^{\mathrm{up}} := \widetilde{W}_S = 1 + \frac{1}{2} P_{(c_1', c_2', c_3' + S^2/4)}$. Suppose $c_3' + \frac{1}{4} S^2 < \bar{c}_3$. Then W^{up} is supercritical (Lemma 3.35(b)): $u := \varphi_-^{W^{\mathrm{up}}}$ has a zero. Let $z > 0$ solve $z'' = Wz$ with $U = 2(z'/z - \tanh\xi)$; as in Proposition 8.13(b), $z \sim A_- e^{-k_1'\xi}$ with $A_- > 0$, $z'/z \to -k_1'$ at $-\infty$, so $\mathrm{Wr}(z, u) = zu' - z'u \to 2k_1' A_- > 0$. Let ξ_1 be the first zero of u. On $(-\infty, \xi_1)$ we have $z, u > 0$ and $\mathrm{Wr}(z, u)' = zu'' - z''u = (W^{\mathrm{up}} - W) zu \ge 0$, so $\mathrm{Wr}(z, u) \ge 2k_1' A_- > 0$ and $(u/z)' = \mathrm{Wr}(z, u)/z^2 > 0$; since $u/z \to 0$ at $-\infty$, u/z is positive and increasing on $(-\infty, \xi_1)$, so $u(\xi_1)/z(\xi_1) \ge (u/z)(\xi_a) > 0$ for any $\xi_a < \xi_1$, contradicting $u(\xi_1) = 0$. Hence $c_3' + \frac{1}{4} S^2 \ge \bar{c}_3$. □

Example 8.19 (a datum below $\mathcal{J}$: the swirl is forced to be large)**.** Let $\boldsymbol{c}' = (0, 0, -5)$. Then $W_{\mathrm{e}} = 1 + \frac{1}{2} c_3' \operatorname{sech}^2 \xi = 1 - \frac{5}{2} \operatorname{sech}^2 \xi$, which is supercritical ($\bar{c}_3(0, 0) = -4$; the critical potential is $1 - 2\operatorname{sech}^2 \xi$ with zero mode $\operatorname{sech}\xi$, Toy example 3.34); so no

solution without swirl exists and W_{e} is not admissible. Take $f_- \geq 0$. Proposition 8.18 gives a fixed point whenever $-5 + \frac{1}{4}f_-^2 > -4$, i.e. $f_- > 2$; and, together with Theorem 8.20, *every* solution with these effective constants has $S = \max(|f_-|, |f_+|)$ with $-5 + \frac{1}{4}S^2 \geq -4$, i.e. $S \geq 2$. In particular for $f_- = 0$ (with $\vartheta = 1$, any $\kappa_{\mathrm{D}} > 0$) a solution exists by Theorem 8.20, and it has $f_+ = J \geq 2$ *however small* κ_{D} is: the swirl cannot be switched off continuously at this datum. (For $\kappa_{\mathrm{D}} \to 0$ the profile tends to a step of height ≥ 2; Theorem 8.53 describes this limit.)

THEOREM 8.20 (Thm E: existence for every swirl strength). *Let $c_1', c_2' > -1$, $c_3' \in \mathbb{R}$ arbitrary, $\vartheta > 0$, $f_- \in \mathbb{R}$. Then for* every *$\kappa_{\mathrm{D}} > 0$ there is a global solution of type* G *at both poles with effective data $(\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$.*

PROOF. By Theorem 8.42(b) the map T_D has a fixed point for every $\kappa_{\mathrm{D}} > 0$ (the degree of $I - H$ on $\mathcal{O}$ is $+1$, hence $\neq 0$, and $H = T_D \circ r$ with r the identity near every fixed point), and by Proposition 8.13(a) a fixed point is a global solution of type G at both poles with these data. □

Remark 8.21 (what this corollary replaces)**.** For $c_3' > \bar{c}_3$ the result is Theorem 8.16. For $c_3' \leq \bar{c}_3$ no solution without swirl exists and W_{e} is not admissible, so T_D is defined only on the open set $\mathcal{A}_D$ and Theorem 8.16 is unavailable: a large value of f on a half-line produces, through $\mathcal{Q}$, a centrifugal barrier that makes W_f admissible. The proof of Theorem 8.42 controls this uniformly (a priori bounds, Lemma 8.39) and does not need the large-κ_{D} continuation of the original argument; in particular no threshold $\kappa_1(\boldsymbol{c}', \vartheta, f_-)$ appears, and the statement holds for all $\kappa_{\mathrm{D}} > 0$.

8.4. The derivative of T_D, the antichain property, moderate swirl

In words. *Varying the swirl f varies the potential W_f by $\delta W = \mathcal{Q}[f\,\delta f]$, a short-range function whose sign is the sign of $f\,\delta f$. The first variation of the Jost logarithmic derivatives is then an explicit integral against a* positive *kernel (the kernels $\Phi_\pm$), and inserting it into the exact formula $T_D f = f_- + J\lg(\ell - \ln\vartheta)$ gives a closed formula for the derivative $\mathrm{D}T_D(f)$. Three consequences follow. First, $\mathrm{D}T_D(f)$ is compact and T_D is C^1. Second, the kernels have an* exact *L^1 norm, $\int \Phi_\pm = \int \operatorname{sech}^2/\rho$, which turns bounds on the derivative into a one-line computation. Third, the sign of $\mathrm{D}T_D(f)v$ on the half line $\xi \leq 0$ is that of $-fv$; hence $\mathrm{D}T_D$ is* not *a positive operator, increasing f* lowers *$T_D f$ on the left half line (the antichain property), and, when the feedback is weak enough, T_D is a contraction.*

The kernels.

Definition 8.22 (kernel operators)**.** Let W be admissible with Jost solutions $u_\pm$ and $\rho = w_0/(u_- u_+)$. For a bounded continuous (or measurable) h put

$$(A_W^- h)(\xi) = \frac{1}{u_-(\xi)^2}\int_{-\infty}^{\xi} u_-^2 h,$$

$$(A_W^+ h)(\xi) = \frac{1}{u_+(\xi)^2}\int_{\xi}^{\infty} u_+^2 h,$$

$$\Phi_h^W := A_W^- h + A_W^+ h,$$

and $\Phi_\mp^W := A_W^\mp[\operatorname{sech}^2]$, $\Psi_W := \Phi_-^W + \Phi_+^W$ (so Φ_h^W is Φ_h^W with h in the subscript, and $\Phi_\mp^W$ are the two functions with the sign as subscript; Φ_- and Φ_+ are the functions occurring in the statement of Proposition 8.32). Put $\Gamma_{\ker}(W) := \int_{\mathbb{R}} \operatorname{sech}^2\xi/\rho_W(\xi)\,\mathrm{d}\xi$.

Lemma 8.23 (the kernels $\Phi_\pm$)**.** *Let $W \in \mathcal{W}(k_1, k_2)$ be admissible.*

(a) Positivity. *$h \geq 0 \Rightarrow A_W^\pm h \geq 0$, and $|A_W^\pm h| \leq A_W^\pm |h|$; if $|h| \leq m \operatorname{sech}^2$ then $|A_W^\mp h| \leq m\Phi_\mp^W$. If $h \geq 0$, $h \not\equiv 0$ is continuous then $A_W^\pm h > 0$ everywhere.*

(b) Decay. *There is C with $\Phi_-^W(\eta) \leq Ce^{2\eta}$ for $\eta \leq 0$ and $\Phi_-^W(\eta) \leq C(1+\eta)e^{-2\min(1,k_2)\eta}$ for $\eta \geq 0$; $\Phi_+^W(\eta) = \Phi_-^{\check{W}}(-\eta)$ for the reflected potential $\check{W}(\xi) = W(-\xi)$. In particular $\Phi_\pm^W \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$.*

(c) Exact L^1 identity. *For $h \geq 0$ measurable, $\int_{\mathbb{R}} A_W^- h = \int_{\mathbb{R}} A_W^+ h = \int_{\mathbb{R}} h/\rho_W$; hence $\|\Phi_h^W\|_{L^1} = 2\int h/\rho_W$ and $\int \Phi_-^W = \int \Phi_+^W = \Gamma_{\ker}(W) < \infty$.*

(d) Value at the origin. *For $h \geq 0$: $\Phi_h^W(0) \leq w_0(W) \int_{\mathbb{R}} h/\rho_W$.*

(e) Monotonicity in W. *If $W' \geq W$ (both admissible, same limits) then $\rho_{W'} \geq \rho_W$ and hence $\Gamma_{\ker}(W') \leq \Gamma_{\ker}(W)$ and $\int h/\rho_{W'} \leq \int h/\rho_W$ for $h \geq 0$.*

Proof. (a) The integrands are positive; the strict claim holds because $u_\pm^2 > 0$. (b) By Lemma 3.21(a),(b)(i) the functions $e^{-k_1\xi}u_-$ and $e^{-k_2\xi}u_-$ are continuous and positive with positive limits at $-\infty$, $+\infty$ respectively; hence there are $0 < c \leq C_0$ with $c\,e^{k_1\xi} \leq u_- \leq C_0 e^{k_1\xi}$ on $(-\infty, 0]$ and $c\,e^{k_2\xi} \leq u_- \leq C_0 e^{k_2\xi}$ on $[0, \infty)$. Using $\operatorname{sech}^2\eta \leq 4e^{-2|\eta|}$: for $\eta \leq 0$, $\Phi_-(\eta) \leq \frac{4C_0^2}{c^2} e^{-2k_1\eta} \int_{-\infty}^{\eta} e^{(2+2k_1)r}\mathrm{d}r = \frac{2C_0^2}{(1+k_1)c^2} e^{2\eta}$. For $\eta \geq 0$, splitting the integral at $r = 0$ and using $u_-(\eta) \geq c\,e^{k_2\eta}$, $u_-(r)^2 \operatorname{sech}^2 r \leq 4C_0^2 e^{2(1+k_1)r}$ ($r \leq 0$) and $\leq 4C_0^2 e^{2(k_2-1)r}$ ($r \geq 0$),

$$\Phi_-(\eta) \leq \frac{4C_0^2}{c^2} e^{-2k_2\eta}\Big(\int_{-\infty}^{0} e^{2(1+k_1)r}\mathrm{d}r + \int_0^{\eta} e^{2(k_2-1)r}\mathrm{d}r\Big) \leq C(1+\eta)e^{-2\min(1,k_2)\eta},$$

because $\int_0^\eta e^{2(k_2-1)r}\mathrm{d}r \leq (1+\eta)e^{2(k_2-1)^+\eta}$. The statement for Φ_+ is the mirror image ($u_+^W(\xi) = u_-^{\check{W}}(-\xi)$ up to normalisation, which cancels in A^+). (c) By Tonelli, $\int_{\mathbb{R}} A^- h = \int h(\eta) u_-(\eta)^2 \int_\eta^\infty u_-^{-2}(\xi)\mathrm{d}\xi\,\mathrm{d}\eta$, and by Lemma 3.21(b)(iii) $\int_\eta^\infty u_-^{-2} = u_+(\eta)/(w_0 u_-(\eta))$, so the inner factor is $u_-u_+/w_0 = 1/\rho$. Likewise $\int_{\mathbb{R}} A^+ h = \int h\,u_+^2 \int_{-\infty}^\eta u_+^{-2} = \int h\,u_+^2\,u_-/(w_0 u_+) = \int h/\rho$. Finiteness: $1/\rho \leq g_0 := \sup 1/\rho < \infty$ (Lemma 3.21(b)(iv)) and $\operatorname{sech}^2 \in L^1$. (d) Since $u_\pm(0) = 1$, $\Phi_h(0) = \int_{-\infty}^0 u_-^2 h + \int_0^\infty u_+^2 h$. Now $u_-^2 = \frac{u_-}{u_+}\,u_-u_+ = R\,w_0/\rho$ with $R = u_-/u_+ \leq 1$ on $(-\infty, 0]$ (R is increasing and $R(0) = 1$, Lemma 8.10), and $u_+^2 = R^{-1}w_0/\rho \leq w_0/\rho$ on $[0, \infty)$. (e) is Lemma 3.23: $\zeta_-^{W'} \geq \zeta_-^W$, $\zeta_+^{W'} \leq \zeta_+^W$. □

Toy example 8.24 (the kernels at the rest state)**.** For $W \equiv 1$ ($\boldsymbol{c}' = 0$, $k_1' = k_2' = 1$): $u_\pm = e^{\pm\xi}$, $\rho \equiv 2$, $w_0 = 2$ and

$$\Phi_-(\eta) = \int_{-\infty}^{\eta} \operatorname{sech}^2(r)\, e^{2(r-\eta)}\,\mathrm{d}r,$$
$$\Phi_+(\eta) = \Phi_-(-\eta),$$
$$\int \Phi_\pm = \frac{1}{2}\int \operatorname{sech}^2 = 1 = \Gamma_{\ker}(1).$$

The decay is $\Phi_-(\eta) \sim e^{2\eta}$ as $\eta \to -\infty$ and $\Phi_-(\eta) \sim 4\eta e^{-2\eta}$ as $\eta \to +\infty$ (the borderline case $k_2 = 1$ of Lemma 8.23(b); the factor $1+\eta$ is sharp). The right panel of Figure 8.1 shows both kernels.

Differentiating the Jost logarithmic derivatives.

Lemma 8.25 (stability of the Jost solutions under small weighted perturbations)**.** *Let $W_0 \in \mathcal{W}(k_1,k_2;C)$ be admissible, $g_0 = \sup_{\mathbb{R}} 1/\rho_{W_0}$, and let q be continuous with $\varepsilon := \|q\|_w = \sup \cosh^2\xi|q|$ satisfying $g_0\varepsilon \le \frac18$. Put $W_1 = W_0 + q$ (which lies in $\mathcal{W}(k_1,k_2;C+4\varepsilon)$) and write $\Psi = \Psi_{W_0}$, $\Phi^{\mp} = \Phi^{W_0}_{\mp}$. Then W_1 is admissible and, for the logarithmic derivatives,*

(a) $|\zeta_-^{W_1} - \zeta_-^{W_0}| \le 2\varepsilon\Phi^-$, $|\zeta_+^{W_1} - \zeta_+^{W_0}| \le 2\varepsilon\Phi^+$; *hence* $|\rho_{W_1} - \rho_{W_0}| \le 2\varepsilon\Psi$ *and* $\frac23 \le u_-^{W_1}/u_-^{W_0} \le \frac43$ *(for the normalisation of φ_-);*

(b) $|\zeta_-^{W_1} - \zeta_-^{W_0} - A^-_{W_0}q| \le 8g_0\varepsilon^2\Phi^-$, $|\zeta_+^{W_1} - \zeta_+^{W_0} + A^+_{W_0}q| \le 8g_0\varepsilon^2\Phi^+$; *hence* $|\rho_{W_1} - \rho_{W_0} - \Phi^{W_0}_q| \le 8g_0\varepsilon^2\Psi$;

(c) *for every continuous p with $\|p\|_w < \infty$:* $|A^{\mp}_{W_1}p - A^{\mp}_{W_0}p| \le 24g_0\varepsilon\|p\|_w\Phi^{\mp}$.

Proof. We treat the left Jost solution; the right one is the mirror image, with $\check W$. Let $u_0 = \varphi_-^{W_0}$ and look for $u_1 = m\,u_0$. Then $u_1'' = (W_0+q)u_1$ is equivalent to $(u_0^2m')' = q\,u_0^2m$, which with $u_0^2m' \to 0$ at $-\infty$ becomes the Volterra equation

$$m(\xi) = 1 + \int_{-\infty}^{\xi} K(\xi,\eta)q(\eta)m(\eta)\,\mathrm{d}\eta, \qquad K(\xi,\eta) = u_0(\eta)^2\int_{\eta}^{\xi} u_0^{-2} \quad (\eta<\xi). \tag{8.4}$$

By Lemma 3.21(b)(iii) $\int_\eta^\xi u_0^{-2} \le \int_\eta^\infty u_0^{-2} = u_+(\eta)/(w_0u_0(\eta))$, so $0 \le K(\xi,\eta) \le u_0u_+/w_0 = 1/\rho_{W_0}(\eta) \le g_0$ ($u_\pm$ normalised; the quotient does not depend on the normalisation). Put $h = |q|/\rho_{W_0} \ge 0$; $\int_{\mathbb{R}} h \le g_0\varepsilon\int\operatorname{sech}^2 = 2g_0\varepsilon \le \frac14$. The Volterra iteration for (8.4) converges, with $|m| \le \exp\int h \le e^{1/4}$ (Gronwall, Remark 3.17), $|m-1| \le \int_{-\infty}^{\xi} h|m| \le 2g_0\varepsilon e^{1/4} < \frac13$. So

$$\tfrac23 \le m \le \tfrac43, \qquad |m(\eta)-1| \le 2e^{1/4}g_0\varepsilon \quad \text{for all } \eta, \tag{8.5}$$

and $m \to 1$ at $-\infty$. Thus $u_1 = mu_0$ solves $u_1'' = W_1u_1$, $u_1e^{-k_1\xi} \to \lim u_0e^{-k_1\xi}$: it is a multiple of $\varphi_-^{W_1}$; it is positive and $u_1 \ge \frac23 u_0 \to \infty$ at $+\infty$. So $\varphi_-^{W_1} > 0$ is unbounded: W_1 is admissible (Definition 8.4). (a) $\zeta_-^{W_1} - \zeta_-^{W_0} = m'/m = \dfrac{1}{m(\xi)u_0(\xi)^2}\displaystyle\int_{-\infty}^{\xi} qu_0^2m\,\mathrm{d}\eta$, and $m(\eta)/m(\xi) \le 2$ by (8.5). Hence $|\cdot| \le 2A^-_{W_0}|q| \le 2\varepsilon\Phi^-$. (b) $\zeta_-^{W_1} - \zeta_-^{W_0} - A^-q = u_0^{-2}\int_{-\infty}^{\xi} qu_0^2\big(\frac{m(\eta)}{m(\xi)} - 1\big)$ and, by (8.5), $|\frac{m(\eta)}{m(\xi)} - 1| \le \frac32(|m(\eta)-1| + |m(\xi)-1|) \le 6e^{1/4}g_0\varepsilon \le 8g_0\varepsilon$. (c) $u_1/u_0 = m$ gives $A^-_{W_1}p - A^-_{W_0}p = u_0^{-2}\int_{-\infty}^{\xi} u_0^2p\,\big(\frac{m(\eta)^2}{m(\xi)^2} - 1\big)$ and, with $\delta = \frac{m(\eta)}{m(\xi)} - 1$, $|\delta| \le 8g_0\varepsilon \le 1$, we have $|(1+\delta)^2 - 1| = |\delta||2+\delta| \le 3|\delta| \le 24g_0\varepsilon$; finally $|p| \le \|p\|_w\operatorname{sech}^2$. □

Toy example 8.26 (the first-order kernel by hand)**.** For $W_0 \equiv 1$ and $q = \varepsilon\operatorname{sech}^2\xi$: $u_0 = e^\xi$, $\rho_0 \equiv 2$, $g_0 = \frac12$, and the exact first-order formula is $\zeta_-^{W_1} - \zeta_-^{W_0} \approx A^-q = \varepsilon e^{-2\xi}\int_{-\infty}^{\xi} e^{2r}\operatorname{sech}^2 r\,\mathrm{d}r = \varepsilon\Phi_-(\xi)$. The lemma asserts that the true difference is within $8g_0\varepsilon^2\Phi_- = 4\varepsilon^2\Phi_-$ of this, and never more than $2\varepsilon\Phi_-$; at $\xi \to -\infty$ both sides are $\approx \varepsilon e^{2\xi}$.

The derivative of T_D.

Lemma 8.27 (the derivative of T_D and its sign)**.** *Under the existence hypotheses (effective data $D = (c',\vartheta,f_-,\kappa_{\mathrm{D}})$): (a) $T_D : BC(\mathbb{R}) \to BC(\mathbb{R})$ is continuously Fréchet differentiable; (b) $DT_D(f)$ is compact; (c) if $f \ge 0$, $DT_D(f)$ has no eigenvector*

$v \geq 0$, $v \neq 0$ with eigenvalue $\nu > 0$. DT_D is not a positive operator and its spectrum can be complex: at the datum D^* of Chapter 9 the lowest of the three certified fixed points ($J = 19.0407$) has the leading complex pair $0.3504 \pm 0.1247\,\mathrm{i}$ (the middle fixed point has the real eigenvalue 1.2818, the upper one 0.6296); these eigenvalues are computed numerically and are not certified. Here the existence hypotheses *are those under which T_D is defined: T_D is considered on the open set $\mathcal{A}_D$ (Lemma 8.5(d)), which is all of* BC *for $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$ (Theorem 8.16). Moreover, for $f \in \mathcal{A}_D$, $v \in$ BC, with $W = W_f$, $\rho = \rho_W$, $\varphi = \lg(\ell_W - \ln\vartheta)$ and $\delta\rho = \Phi^W_{\mathcal{Q}[fv]}$,*

$$\boxed{\mathrm{D}T_D(f)v \;=\; J(f)\,\varphi\Big[-\frac{\delta\rho(0)}{\rho(0)} + (1-\varphi)\int_0^\xi \delta\rho\Big]\,, \qquad J(f) = \frac{\kappa_{\mathrm{D}}}{\vartheta\,\rho(0)}.} \tag{8.6}$$

For $f \geq 0$ and $v \geq 0$ one has $\delta\rho \geq 0$ and $\mathrm{D}T_D(f)v \leq 0$ on $\xi \leq 0$, with $\mathrm{D}T_D(f)v < 0$ on $\xi \leq 0$ if $fv \not\equiv 0$.

Proof. *Step 1: first variation of ρ.* Fix $f \in \mathcal{A}_D$, $W = W_f$, $g_0 = \sup 1/\rho_W$. For $v \in$ BC put $q = W_{f+v} - W = \mathcal{Q}[fv] + \frac12\mathcal{Q}[v^2] =: q_1 + q_2$. By Lemma 8.2(a), $\|q_1\|_w \leq \frac14\|f\|_\infty\|v\|_\infty$ and $\|q_2\|_w \leq \frac18\|v\|_\infty^2$, so $\varepsilon = \|q\|_w = O(\|v\|)$ and for $\|v\|$ small $g_0\varepsilon \leq \frac18$. Lemma 8.25 shows that $f + v \in \mathcal{A}_D$ and

$$\big|\rho_{f+v} - \rho_f - \Phi^W_{q_1}\big| \leq |\Phi^W_{q_2}| + 8g_0\varepsilon^2\Psi \leq \big(\tfrac18\|v\|^2 + 8g_0\varepsilon^2\big)\Psi = O(\|v\|^2)\,\Psi,$$

with $\Psi = \Psi_W \in L^1 \cap L^\infty$ (Lemma 8.23(b)). So $\delta\rho = \Phi^W_{q_1} = \Phi^W_{\mathcal{Q}[fv]}$ is the first variation of ρ, linear and bounded in v with $|\delta\rho| \leq \frac14\|f\|\|v\|\Psi$, and the remainder is $O(\|v\|^2)$ both in L^∞ and, after integration, uniformly in ξ: $|\int_0^\xi(\dots)| \leq O(\|v\|^2)\|\Psi\|_{L^1}$. *Step 2: Fréchet differentiability.* $T_Df = f_- + J\varphi$ with $J = \kappa_{\mathrm{D}}/(\vartheta\rho(0))$, $\varphi = \lg(s)$, $s = \ell_W - \ln\vartheta = \int_0^\xi \rho - \ln\vartheta$ ((8.2)). Step 1 gives $J_{f+v} = J - J\,\delta\rho(0)/\rho(0) + O(\|v\|^2)$ and $s_{f+v} = s + \int_0^\xi\delta\rho + O(\|v\|^2)$, uniformly in ξ, with $\sup_\xi|s_{f+v} - s| \leq \|\Psi\|_1 \cdot O(\|v\|)$. Since $\lg' = \lg(1-\lg)$ and $|\lg''| \leq \frac{1}{10}$, Taylor's formula gives $\varphi_{f+v} = \varphi + \varphi(1-\varphi)\int_0^\xi\delta\rho + O(\|v\|^2)$ uniformly in ξ. Multiplying out, $T_D(f+v) - T_Df = J\varphi\big[-\delta\rho(0)/\rho(0)\big] + J\varphi(1-\varphi)\int_0^\xi\delta\rho + O(\|v\|^2)$, which is (8.6); the right side is linear and bounded in v ($|\delta\rho(0)|/\rho(0) \leq \frac14\|f\|\|v\|g_0\Psi(0)$, $|\int_0^\xi\delta\rho| \leq \frac14\|f\|\|v\|\|\Psi\|_1$). This is the Fréchet differentiability. *Step 3: C^1.* Let $f_n \to f$ in $\mathcal{A}_D$ and $\varepsilon_n = \|W_{f_n} - W_f\|_w \leq \frac14(\|f_n\| + \|f\|)\|f_n - f\| \to 0$. By Lemma 8.25(a) $\rho_{f_n} \to \rho_f$ uniformly and $\ell_{f_n} \to \ell_f$ uniformly (as $|\ell_{f_n} - \ell_f| \leq 2\varepsilon_n\|\Psi\|_1$), so $J(f_n) \to J(f)$ and $\varphi_n \to \varphi$ uniformly (lg is $\frac14$-Lipschitz). For $\delta\rho_n = \Phi^{W_n}_{\mathcal{Q}[f_nv]}$: $\delta\rho_n - \delta\rho = (\Phi^{W_n} - \Phi^W)\mathcal{Q}[f_nv] + \Phi^W\mathcal{Q}[(f_n - f)v]$ and Lemma 8.25(c) together with Lemma 8.23(a) bound this by $\big(24g_0\varepsilon_n\frac14\|f_n\| + \frac14\|f_n - f\|\big)\|v\|\,\Psi$, i.e. $\leq C\|f_n - f\|\|v\|\Psi$. So the three ingredients of (8.6) converge, in L^∞ and after integration uniformly, at a rate $O(\|f_n - f\|)\|v\|$: $\|\mathrm{D}T_D(f_n) - \mathrm{D}T_D(f)\|_{\mathrm{op}} \to 0$. This proves (a). *Step 4: compactness (b).* Let $\mathcal{F} = \{\mathrm{D}T_D(f)v : \|v\|_\infty \leq 1\}$. By (8.6) and Step 1, with $a_v = \delta\rho(0)/\rho(0)$ and $b_v(\xi) = \int_0^\xi\delta\rho$: $|a_v| \leq C$, $|b_v| \leq C$, $|b_v'| = |\delta\rho| \leq C$, uniformly in $\|v\| \leq 1$. $\mathcal{F}$ is bounded. Equicontinuous: φ is Lipschitz ($\varphi' = \varphi(1-\varphi)\rho \leq \frac14\sup\rho$), b_v is uniformly Lipschitz, so the product $J\varphi[-a_v + (1-\varphi)b_v]$ is uniformly Lipschitz in v. Uniform tails: $\varphi(\xi) \to 0$ at $-\infty$ and $1-\varphi \to 0$ at $+\infty$ (Lemma 8.10); for $\xi, \eta \leq -L$: $|\mathrm{D}T_Dv(\xi) - \mathrm{D}T_Dv(\eta)| \leq J\,[(\varphi(\xi) + \varphi(\eta))C + (\varphi(\xi) + \varphi(\eta))C] \leq 4JC\varphi(-L)$, and for $\xi, \eta \geq L$: $|\varphi(\xi) - \varphi(\eta)| \leq 1 - \varphi(L)$ and $(1-\varphi) \leq 1 - \varphi(L)$, giving $\leq JC(1-\varphi(L))\cdot 3$; both $\to 0$ as $L \to \infty$ uniformly in v. Lemma 8.14 shows that $\mathcal{F}$ is relatively compact:

$\mathrm{D}T_D(f)$ is compact. *Step 5: sign (c).* Let $f \geq 0$, $v \geq 0$; then $fv \geq 0$, $\mathcal{Q}[fv] \geq 0$ (Lemma 8.2(a)), and $\delta\rho = \Phi^W_{\mathcal{Q}[fv]} \geq 0$ by Lemma 8.23(a), with $\delta\rho > 0$ everywhere if $fv \not\equiv 0$ (then $\mathcal{Q}[fv] > 0$). For $\xi \leq 0$, $\int_0^\xi \delta\rho = -\int_\xi^0 \delta\rho \leq 0$ and $0 \leq 1-\varphi \leq 1$, so the bracket in (8.6) is $\leq -\delta\rho(0)/\rho(0) \leq 0$, and < 0 if $fv \not\equiv 0$. Since $J\varphi > 0$, $\mathrm{D}T_D(f)v \leq 0$, resp. < 0, on $\xi \leq 0$. Now suppose $\mathrm{D}T_D(f)v = \nu v$ with $v \geq 0$, $v \not\equiv 0$, $\nu > 0$. Then $\nu v \geq 0$ on $\xi \leq 0$, so $\mathrm{D}T_D(f)v \geq 0$ there, which forces $fv \equiv 0$ (otherwise it is < 0); but then $\mathcal{Q}[fv] = 0$, $\delta\rho = 0$ and $\mathrm{D}T_D(f)v \equiv 0 = \nu v$, contradicting $\nu > 0$, $v \not\equiv 0$. □

Remark 8.28 (range of validity; homotopies in c_3')**.** Lemma 8.23, Lemma 8.25, Lemma 8.27 and Lemma 8.30 are proved for *every* $D \in \mathbb{D}_+$ ($c_1', c_2' > -1$, $c_3' \in \mathbb{R}$ arbitrary), for every admissible W in the relevant class, and for f in the open set $\mathcal{A}_D$; nothing uses $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$. They therefore apply verbatim to the deformed data $D_s = (c_1', c_2', c_3' + s, \vartheta, f_-, \kappa_\mathrm{D})$, $s \geq 0$, with $W_{f,s} = W_\mathrm{e} + \frac{s}{2}\operatorname{sech}^2\xi + \frac{1}{2}\mathcal{Q}[f^2]$ (the potential W_e for $c_3' + s$ is $W_\mathrm{e} + \frac{s}{2}\operatorname{sech}^2\xi$).

Remark 8.29 ($\mathrm{D}T_D$ is not positive; the spectrum)**.** By Step 5, for $f, v \geq 0$ the image $\mathrm{D}T_D(f)v$ is *negative* on $\xi \leq 0$ although $v \geq 0$: $\mathrm{D}T_D(f)$ does not map the cone of nonnegative functions into itself, so Krein–Rutman theory does not apply and the leading part of the spectrum need not be real. The last sentence of the statement of Lemma 8.27, which quotes the spectrum at one specific datum, is a *numerical* fact established with the solver of Chapter 9 (it is not used in this chapter); here only the sign structure matters. The finer sign structure (Lemma 9.2, including the case $fv \leq 0$) is used in Chapter 9. A numerical illustration of the sign structure is Figure 8.1.

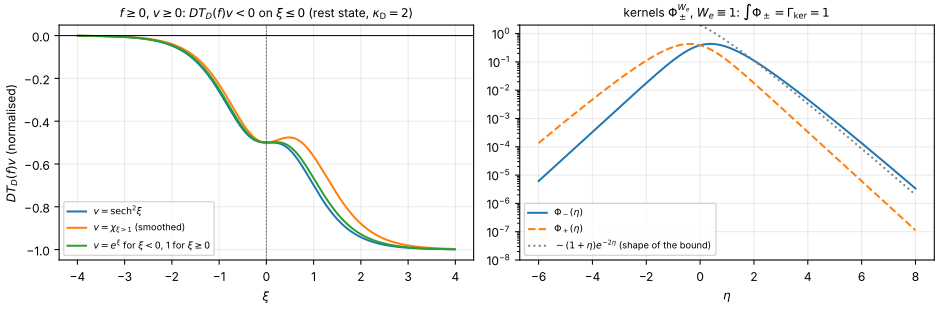


FIGURE 8.1. Left: $\mathrm{D}T_D(f)v$ for three nonnegative directions v at the fixed point $f \geq 0$ of the rest state ($\boldsymbol{c}' = 0$, $\vartheta = 1$, $f_- = 0$, $\kappa_\mathrm{D} = 2$), normalised by their sup norm: the image is negative on $\xi \leq 0$ (Lemma 8.27, Step 5) although $v \geq 0$. Right: the kernels Φ_-, Φ_+ for $W_e \equiv 1$ (Toy example 8.24); each has integral 1. Code: `code/ch08_effective/make_DTsign.py` (see Appendix E).

Lemma 8.30 (bound on $\mathrm{D}T_D$ by the exact L^1 identity)**.** *For $f \in \mathcal{A}_D$, with $W = W_f$,*

$$\|\mathrm{D}T_D(f)\|_{\mathrm{op}} \;\leq\; \frac{3}{8}\,J(f)\,\|f\|_\infty\,\Gamma_{\ker}(W_f).$$

PROOF. Let $\|v\|_\infty \le 1$, $g = \mathcal{Q}[fv]$, so $|g| \le \frac14\|f\|\operatorname{sech}^2$ and $|\delta\rho| = |\Phi_g| \le \Phi_{|g|}$. By Lemma 8.23(d), $|\delta\rho(0)|/\rho(0) \le \Phi_{|g|}(0)/w_0 \le \int |g|/\rho \le \frac14\|f\|\Gamma_{\ker}$; by (c), $|\int_0^\xi \delta\rho| \le \|\Phi_{|g|}\|_{L^1} = 2\int |g|/\rho \le \frac12\|f\|\Gamma_{\ker}$. In (8.6), $0<\varphi<1$ and $\varphi(1-\varphi)\le\frac14$, so $|\mathrm{D}T_D(f)v| \le J\big(\varphi|a| + \varphi(1-\varphi)|b|\big) \le J\|f\|\Gamma_{\ker}(\frac14 + \frac14\cdot\frac12)$. □

Toy example 8.31 (small swirl: $\mathrm{D}T_D$ vanishes at $f=0$)**.** If $f\equiv 0$ then $\mathcal{Q}[fv]=0$, $\delta\rho=0$ and (8.6) gives $\mathrm{D}T_D(0)=0$ (T_D depends on f through f^2). Quantitatively Lemma 8.30 gives $\|\mathrm{D}T_D(f)\| \le \frac38 J\|f\|_\infty\Gamma_{\ker}$, linear in $\|f\|_\infty$. Since the solution has $\|f\|_\infty \approx |f_-| + O(\kappa_{\mathrm{D}})$, the feedback is weak when both f_- and κ_{D} are small: this is the mechanism behind Theorem 8.34.

The antichain property and the contraction criterion.

Proposition 8.32 (contraction criterion and antichain property)**.** *(1) Under the hypotheses of Theorem 8.16 ($c' \in \operatorname{int}\mathcal{J}$, so W_e is admissible; $\kappa = \kappa_{\mathrm{D}}$) let z_{low} be the lower barrier of Lemma 8.15, $C_* = \int_{\mathbb{R}} z_{\mathrm{low}}^{-2}$, $S_0 = |f_-| + \kappa_{\mathrm{D}}/(\vartheta w_0(W_e))$, $\Phi_-(\eta) = \int_{-\infty}^{\eta} \operatorname{sech}^2(r)\big(u_-^{W_e}(r)/u_-^{W_e}(\eta)\big)^2 dr$, Φ_+ its mirror, $\Gamma_U = \max\big(\int_{\mathbb{R}}\Phi_-, \int_{\mathbb{R}}\Phi_+\big) < \infty$. If $\kappa_{\mathrm{D}} S_0 \Gamma_U C_* < 2$ then T_D is a contraction on K and the solution with effective data D is unique. (2) For $f_- \ge 0$: (a) $f_1 \ge f_2 \ge 0 \Rightarrow T_D f_1 \le T_D f_2$ on $(-\infty, 0]$, strictly at 0 unless $f_1 = f_2$; (b) two distinct fixed points are never pointwise ordered.*

Reading of the statement. Here z_{low} is the function of Lemma 8.15 with $W_{\mathrm{ref}} = W_e$, and $K = K_{\mathrm{ref}}$; $C_* = \int z_{\mathrm{low}}^{-2} < \infty$ by Lemma 8.15(c). With $\Gamma_{\ker}$ of Definition 8.22, $\Phi_\mp = \Phi_\mp^{W_e}$ and by Lemma 8.23(c) $\Gamma_U = \Gamma_{\ker}(W_e) = \int \operatorname{sech}^2/\rho_{W_e}$: the two integrals in the maximum coincide. We write Γ_U only in the statement above. In (2) the maps are evaluated on $\mathcal{A}_D$; note that $f_2 \in \mathcal{A}_D$ implies $f_1 \in \mathcal{A}_D$ when $f_1 \ge f_2 \ge 0$, since $W_{f_1} \ge W_{f_2}$ and Lemma 3.23 applies.

PROOF. *(1).* Let $f_1, f_2 \in K$, $\Delta = \|f_1 - f_2\|_\infty$, $W_i = W_{f_i} \ge W_e$, $u_i = u_-^{W_i}$, $v_i = u_+^{W_i}$, $\zeta_\pm^i$. Since $f_i \in [f_-, f_- + B_e]$ with $B_e = \kappa_{\mathrm{D}}/(\vartheta w_0(W_e))$, $\|f_i\|_\infty \le S_0$ and $\|f_1^2 - f_2^2\| \le 2S_0\Delta$, so by Lemma 8.2(a)

$$|W_1 - W_2| = \tfrac12|\mathcal{Q}[f_1^2 - f_2^2]| \le \tfrac18 \cdot 2S_0\Delta\operatorname{sech}^2 = \tfrac14 S_0\Delta\operatorname{sech}^2.$$

By the exact difference formula of Lemma 3.23, for all η,

$$|\zeta_-^1 - \zeta_-^2|(\eta) \le \int_{-\infty}^{\eta} |W_1 - W_2|(r)\frac{u_1(r)u_2(r)}{u_1(\eta)u_2(\eta)}\mathrm{d}r.$$

Because $W_i \ge W_e$ we have $\zeta_-^i \ge \zeta_-^e$ (Lemma 3.23), so for $r<\eta$, $u_i(r)/u_i(\eta) = \exp(-\int_r^\eta \zeta_-^i) \le u_e(r)/u_e(\eta)$, $u_e = u_-^{W_e}$. Hence $|\zeta_-^1 - \zeta_-^2| \le \frac14 S_0\Delta\,\Phi_-$ with $\Phi_- = \Phi_-^{W_e}$ (exactly the function of the statement); mirror: $|\zeta_+^1 - \zeta_+^2| \le \frac14 S_0\Delta\,\Phi_+$. Integrating from 0 (all u, v are normalised at 0),

$$\Big|\ln\frac{u_1}{u_2}(\xi)\Big| \le \tfrac14 S_0\Delta\int\Phi_- \le \tfrac14 S_0\Delta\Gamma_U, \qquad \Big|\ln\frac{v_1}{v_2}(\xi)\Big| \le \tfrac14 S_0\Delta\Gamma_U.$$

Now $z_i = u_i + \vartheta v_i$ and $\frac{z_1}{z_2} = \frac{u_1+\vartheta v_1}{u_2+\vartheta v_2}$ lies between $\frac{u_1}{u_2}$ and $\frac{v_1}{v_2}$ (mediant inequality): $|\ln z_1 - \ln z_2| \le \frac14 S_0\Gamma_U\Delta$. By the mean value theorem, $|z_1^{-2} - z_2^{-2}| \le 2|\ln z_1 - \ln z_2|\max(z_1^{-2}, z_2^{-2}) \le 2|\ln z_1 - \ln z_2|\, z_{\mathrm{low}}^{-2}$ since $z_i \ge z_{\mathrm{low}}$ (Lemma 8.15(b)). Therefore

$$\|T_Df_1 - T_Df_2\|_\infty \le \kappa_{\mathrm{D}}\int_{\mathbb{R}} |z_1^{-2} - z_2^{-2}| \le \tfrac12\,\kappa_{\mathrm{D}} S_0\Gamma_U C_*\,\Delta.$$

If $\kappa_{\mathrm{D}} S_0 \Gamma_U C_* < 2$ the Lipschitz constant is < 1; K is closed and $T_D(K) \subset K$ (Lemma 8.15(b)), so Banach's theorem gives exactly one fixed point in K. Every fixed point lies in K (Theorem 8.16); hence the solution with data D is unique (Proposition 8.13). *(2)(a).* Let $f_- \geq 0$, $f_1 \geq f_2 \geq 0$, $f_i \in \mathcal{A}_D$. Then $f_1^2 \geq f_2^2$ and $W_1 \geq W_2$ (monotonicity of $\mathcal{Q}$), so by Lemma 3.23 $\rho_1 \geq \rho_2$ pointwise and $J_1 := J(f_1) \leq J_2$. For $\xi \leq 0$, $\ell_i(\xi) = -\int_\xi^0 \rho_i$, so $\ell_1(\xi) \leq \ell_2(\xi)$ and $\varphi_1 \leq \varphi_2$ (lg is increasing). Since all factors are positive, $T_D f_1 = f_- + J_1\varphi_1 \leq f_- + J_2\varphi_2 = T_D f_2$ on $(-\infty, 0]$. At $\xi = 0$, $T_D f_i(0) = f_- + J_i/(1+\vartheta)$ (Lemma 8.10), so the inequality is strict iff $J_1 < J_2$ iff $\rho_1(0) > \rho_2(0)$. If $f_1 \neq f_2$ then $f_1 > f_2 \geq 0$ on an interval, so $f_1^2 - f_2^2 \geq 0$ is continuous and $\not\equiv 0$, and $\Delta W = W_1 - W_2 = \frac{1}{2}\mathcal{Q}[f_1^2 - f_2^2] > 0$ everywhere (Lemma 8.2(a)). By Lemma 3.23, $\rho_1(0) - \rho_2(0) = (\zeta_-^1 - \zeta_-^2)(0) - (\zeta_+^1 - \zeta_+^2)(0) = \int_{-\infty}^0 \Delta W\, u_1 u_2 + \int_0^\infty \Delta W\, v_1 v_2 > 0$. *(2)(b).* Let $f_1 \neq f_2$ be fixed points with, say, $f_1 \geq f_2$. As $f_i \geq f_- \geq 0$, (a) applies: $f_1(0) = T_D f_1(0) < T_D f_2(0) = f_2(0)$, contradicting $f_1 \geq f_2$. □

Toy example 8.33 (the criterion at the rest state)**.** Take $\boldsymbol{c}' = 0$, $f_- = 0$, $\vartheta = 1$ (the rest state; $W_{\mathrm{e}} \equiv 1$, $\bar{c}_3(0,0) = -4 < 0$ so $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$). Then $w_0 = 2$, $\rho \equiv 2$, $B_e = \kappa_{\mathrm{D}}/2$, $S_0 = \kappa_{\mathrm{D}}/2$, $\Gamma_U = 1$ (Toy example 8.24), and $z_{\mathrm{low}} = e^{\xi}$ for $\xi \geq 0$, $e^{-\xi}$ for $\xi < 0$, so $C_* = \int z_{\mathrm{low}}^{-2} = \frac{1}{2} + \frac{1}{2} = 1$. The criterion $\kappa_{\mathrm{D}} S_0 \Gamma_U C_* < 2$ reads $\kappa_{\mathrm{D}}^2 < 4$. For general ϑ, $C_* = \frac{1}{2}(1 + \vartheta^{-2})$, $S_0 = \kappa_{\mathrm{D}}/(2\vartheta)$ and the criterion is $\kappa_{\mathrm{D}}^2(1 + \vartheta^{-2}) < 8\vartheta$ (checked in `verify_ch08a.py`).

Explicit moderate-swirl uniqueness.

THEOREM 8.34 (moderate swirl: uniqueness and nondegeneracy). *Let $c_1', c_2' > -1$, $\vartheta > 0$, $f_- \in \mathbb{R}$, $\kappa_{\mathrm{D}} > 0$, $m_- := \max(f_-, 0)$ and assume $c_3' + \frac{1}{4}m_-^2 > \bar{c}_3(c_1', c_2')$ (automatic if $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$). Let $\widetilde{W} = W_{\mathrm{e}} + \frac{1}{8}m_-^2 \operatorname{sech}^2 \xi$ (admissible, Lemma 8.5(c)), $\widetilde{w}_0 = w_0(\widetilde{W})$, $\widetilde{\Gamma} = \Gamma_{\ker}(\widetilde{W})$, $B = \kappa_{\mathrm{D}}/(\vartheta \widetilde{w}_0)$, $S_1 := \max(|f_-|, f_- + B)$ and*

$$\Lambda_{\mathrm{mod}} := \frac{3\,\kappa_{\mathrm{D}}\, S_1\, \widetilde{\Gamma}}{4\,\vartheta\, \widetilde{w}_0}\ .$$

If $\Lambda_{\mathrm{mod}} < 1$, then T_D has exactly one fixed point f^* in $\mathcal{A}_D$, hence exactly one global solution of type G *at both poles with effective data D; the Picard iteration $f_{n+1} = T_D f_n$ converges to f^* from every $f_0 \in \widetilde{K}$ with $\|f_n - f^*\|_\infty \leq (\Lambda_{\mathrm{mod}}/2)^n \|f_0 - f^*\|_\infty$; and $\|\mathrm{D}T_D(f^*)\|_{\mathrm{op}} \leq \frac{1}{2}\Lambda_{\mathrm{mod}} < \frac{1}{2}$, so $\operatorname{spec} \mathrm{D}T_D(f^*)$ lies in the disc $|\nu| \leq \frac{1}{2}\Lambda_{\mathrm{mod}}$; in particular $1 \notin \operatorname{spec} \mathrm{D}T_D(f^*)$ (f^* is nondegenerate).*

PROOF. *Reference set.* Let $\widetilde{K} = K_{\mathrm{ref}}$ for $W_{\mathrm{ref}} = \widetilde{W}$ (so $B_{\mathrm{ref}} = B$). For $f \in \widetilde{K}$, $f \geq f_-$ gives $f^2 \geq m_-^2$ (if $f_- \geq 0$; if $f_- < 0$, $m_- = 0$ and $f^2 \geq 0$), so $W_f \geq \widetilde{W}$ by Lemma 8.2(a) and Lemma 8.5(c): hypothesis (8.3) of Lemma 8.15 holds. Hence $\widetilde{K} \subset \mathcal{A}_D$ is closed, convex, $T_D(\widetilde{K}) \subset \widetilde{K}$, and a fixed point exists; moreover every fixed point of T_D in $\mathcal{A}_D$ is nondecreasing with $f(-\infty) = f_-$, so $f^2 \geq m_-^2$, $W_f \geq \widetilde{W}$ and by Lemma 8.15(e) lies in $\widetilde{K}$. *Bounds on $\widetilde{K}$.* For $f \in \widetilde{K}$: $f_- \leq f \leq f_- + B$, so $\|f\|_\infty \leq S_1$; $\rho_f \geq \widetilde{\rho}$ gives $w_0(W_f) \geq \widetilde{w}_0$, $J(f) \leq B$, and $\Gamma_{\ker}(W_f) \leq \widetilde{\Gamma}$ (Lemma 8.23(e)). Lemma 8.30 yields

$$\|\mathrm{D}T_D(f)\|_{\mathrm{op}} \leq \tfrac{3}{8}\, B\, S_1\, \widetilde{\Gamma} = \tfrac{1}{2}\Lambda_{\mathrm{mod}} \qquad (f \in \widetilde{K}).$$

Contraction. $\widetilde{K}$ is convex and contained in the open set $\mathcal{A}_D$ on which T_D is C^1 (Lemma 8.27); for $f_1, f_2 \in \widetilde{K}$ the segment lies in $\widetilde{K}$, and $\|T_D f_1 - T_D f_2\| \leq$

$\sup\|\mathrm{D}T_D\|\,\|f_1-f_2\| \le \frac{1}{2}\Lambda_{\mathrm{mod}}\|f_1-f_2\|$. Banach's theorem gives a unique fixed point $f^*\in\widetilde{K}$ and the stated Picard rate; as every fixed point lies in $\widetilde{K}$, it is the only one in $\mathcal{A}_D$, and Proposition 8.13 translates this into uniqueness of solutions. The spectral radius of an operator is at most its norm. □

Remark 8.35 (the constants)**.** The proof gives the contraction constant $\frac{1}{2}\Lambda_{\mathrm{mod}}$, so $\Lambda_{\mathrm{mod}}<2$ already suffices for uniqueness and the Picard rate; the normalisation $\Lambda_{\mathrm{mod}}<1$ of the original statement is kept, since it additionally gives $\|\mathrm{D}T_D\|<\frac{1}{2}$ and is the form used later. S_1 is $S_0=|f_-|+B$ of Proposition 8.32 when $f_-\ge 0$; we use the name S_1 to avoid a clash with the S_0 of Proposition 8.32 (which is defined with W_e, not $\widetilde{W}$). At the rest state ($\boldsymbol{c}'=0$, $f_-=0$): $\widetilde{w}_0=2$, $\widetilde{\Gamma}=1$, $S_1=B=\kappa_{\mathrm{D}}/(2\vartheta)$ and $\Lambda_{\mathrm{mod}}=3\kappa_{\mathrm{D}}^2/(16\vartheta^2)$; $\Lambda_{\mathrm{mod}}<1$ iff $\kappa_{\mathrm{D}}<4\vartheta/\sqrt{3}$, which admits larger κ_{D} than Proposition 8.32 ($\kappa_{\mathrm{D}}^2(1+\vartheta^{-2})<8\vartheta$) for every ϑ, because $2\vartheta^2-3\vartheta+2>0$. The theorem is the quantitative special case of the index criterion (Theorem 8.42(d), Proposition 8.48): all of $\operatorname{spec}\mathrm{D}T_D(f^*)$ lies in a disc of radius $<\frac{1}{2}$.

Numerical observation 8.36 (the Picard rate at the rest state)**.** *For $\boldsymbol{c}'=0$, $f_-=0$, $\vartheta=1$ and $\kappa_{\mathrm{D}}=2$ (the top of the range of the criteria) the iteration $f_{n+1}=T_Df_n$ started at $f_0=0$ converges with observed error ratios* 0.0254, 0.0255, 0.0255, …*; the leading eigenvalue of $\mathrm{D}T_D(f^*)$ is* -0.0255 (*real, negative*)*, the total swirl is $J=0.985$, to be compared with the proved bound $\frac{1}{2}\Lambda_{\mathrm{mod}}=0.375$ ($\Lambda_{\mathrm{mod}}=0.75$). The bounds of this section are therefore not sharp by a factor of about* 15 *at this datum. Code:* `code/ch08_effective/verify_ch08a.py` *(part 6, the same solver); numerical, not part of any proof.*

Handing over to the second half of the chapter. Theorem 8.16 and Proposition 8.18 give existence on the sets K, $\widetilde{K}$ and Theorem 8.20 for every data; Lemmas 8.27 and 8.30 and Theorem 8.34 give the derivative and quantitative uniqueness. What is missing is an existence and counting statement for *all* data that does not rest on smallness: the Leray–Schauder degree of $I-T_D$ (Theorem 8.42), with the a priori bounds of Lemma 8.39, the index criterion Proposition 8.48 (which uses $\mathrm{D}T_D$ of Lemma 8.27), the no-swirl face Theorem 8.53 and examples Example 8.57.

8.5. The degree theorem: a solution for every swirl strength

In words. *Fix the effective data $D=(\boldsymbol{c}',\vartheta,f_-,\kappa_{\mathrm{D}})$. The fixed points of T_D are the solutions. We want to count them, with signs, without knowing anything about them: the* Leray–Schauder degree *is a signed count that does not change when the map is deformed, as long as no fixed point reaches the boundary of the region in which we count. The deformation is $c_3'\mapsto c_3'+s$ at fixed $(\vartheta,f_-,\kappa_{\mathrm{D}})$: for s huge the potential carries a huge bump, the Wronskian w_0 is huge, the jump $J=\kappa_{\mathrm{D}}/(\vartheta w_0)$ is tiny, and the problem is trivial (one fixed point, index $+1$); for $s=0$ it is the problem we want. The whole proof is the statement that* along the deformation, all fixed points stay in a bounded region where w_0 is bounded below *(Lemma 8.39), uniformly in s. The result is that the degree is $+1$ for* every *datum: a solution exists for every $\kappa_{\mathrm{D}}>0$, the number of solutions is odd when they are nondegenerate, and uniqueness is a spectral question about $\mathrm{D}T_D$ (Proposition 8.48).*

8.5.1. Two toys: why the boundary condition matters, and the shape of the homotopy.

Toy example 8.37 (a degree that jumps)**.** On $E=\mathbb{R}$ let $F_t(x)=tx$ and $\mathcal{O}=(-1,1)$. For $t\neq 1$ the only fixed point is 0, and $\deg_{\mathrm{LS}}(I-F_t,\mathcal{O})=\operatorname{sgn}(1-t)$: it equals $+1$ for $t<1$ and -1 for $t>1$. At $t=1$ every point is fixed, in particular the boundary points ±1: the hypothesis "no fixed point on $\partial\mathcal{O}$ for all t" of homotopy invariance (D2) fails exactly where the degree jumps. In our problem the role of t is played by s, and the a priori bounds of Lemma 8.39 are the statement that no fixed point ever reaches $\partial\mathcal{O}$.

Toy example 8.38 (a scalar model of the homotopy)**.** Let $T_s(x)=\kappa/(1+s+x^2)$ on $\mathbb{R}$, $\kappa>0$, $s\geq 0$. Here x plays the role of J and $1+s+x^2$ the role of ϑw_0 (swirl raises w_0, hence lowers J; the term s is the bump added by $c_3'\mapsto c_3'+s$). A fixed point solves $x(1+s+x^2)=\kappa$: it is unique, $0<x_s\leq\kappa/(1+s)\leq\kappa$ (a bound *uniform in* s), and $T_s'(x_s)<0$, so its index is $+1$. With $\mathcal{O}=(-\kappa-1,\kappa+1)$ we have $T_s(\overline{\mathcal{O}})\subset(0,\kappa]\subset\mathcal{O}$, so $\deg_{\mathrm{LS}}(I-T_s,\mathcal{O})=1$ by Corollary A.37 for every $s\geq 0$, e.g. $x_0=2$ for $\kappa=10$. In the real problem $T_{D,s}$ is not a monotone scalar map and its degree at $s=0$ cannot be read off; one computes it at $s=s_{\max}$ and transports it along s by homotopy invariance, which needs the uniform bound.

8.5.2. What is used.

Conventions. Throughout this chapter $\kappa:=\kappa_{\mathrm{D}}$ (Lemma 2.15), $k_i:=\sqrt{1+c_i'}$ ($=k_i'$ of the notation table; the exponents of W_{e}), $k:=\min(k_1,k_2)$, $\bar c_3=\bar c_3(c_1',c_2')$ and $f_-^+=\max(f_-,0)$. We always assume $c_1',c_2'>-1$, $\vartheta>0$, $\kappa>0$ and $f_-\in\mathbb{R}$; $c_3'\in\mathbb{R}$ is arbitrary. The centrifugal operator $\mathcal{Q}[h](\xi)=\frac{1}{4\cosh^2\xi}\int_{\mathbb{R}}e^{-2|\xi-\eta|}h(\eta)\,d\eta$ is that of Lemma 8.2, so $Q_{\mathrm{sw}}=\mathcal{Q}[f^2]$; the symbol $Q[\,\cdot\,]$ of the quoted statements below means $\mathcal{Q}[\,\cdot\,]$. By Lemma 8.2(a), $\mathcal{Q}$ is linear and order preserving, $\mathcal{Q}[1]=\frac14\operatorname{sech}^2$, $|\mathcal{Q}[h]|\leq\frac14\|h\|_\infty\operatorname{sech}^2$, hence $\mathcal{Q}[h]\geq 0$ for $h\geq 0$, and in the weighted distance $d_w(V,\widetilde V)=\sup_\xi\cosh^2\xi\,|V-\widetilde V|$ of Lemma 3.25

$$d_w\big(\mathcal{Q}[h],\mathcal{Q}[\tilde h]\big)\leq\tfrac14\|h-\tilde h\|_\infty. \tag{8.7}$$

For $c\in\mathrm{BC}(\mathbb{R})$ and $s\geq 0$ put

$$W_{c,s}:=W_{\mathrm{e}}+\tfrac12\mathcal{Q}[c^2+4s]=W_{\mathrm{e}}+\tfrac12 s\operatorname{sech}^2\xi+\tfrac12\mathcal{Q}[c^2],\qquad \widetilde{W}^{\,s}:=W_{f_-^+,s}, \tag{8.8}$$

where we used $\mathcal{Q}[4s]=s\operatorname{sech}^2$. Thus $W_{c,s}$ is the potential W_c of the datum with c_3' replaced by $c_3'+s$, because $W_{\mathrm{e}}+\frac12 s\operatorname{sech}^2=1+\frac12 P_{(c_1',c_2',c_3'+s)}$; *this is the sense in which s raises c_3'.*

(U1) The class of potentials. By Lemma 3.35(a), $W_{\mathrm{e}}\in\mathcal{W}(k_1,k_2;C_e)$, $C_e=|c_1'-c_2'|+2|c_3'|$; since $0\leq\frac12\mathcal{Q}[c^2+4s]\leq\frac18(\|c\|_\infty^2+4s)\operatorname{sech}^2\xi\leq\frac12(\|c\|_\infty^2+4s)e^{-2|\xi|}$ we get $W_{c,s}\in\mathcal{W}\big(k_1,k_2;C_e+\frac12(\|c\|_\infty^2+4s)\big)$ (Definition 3.20). By Equation (8.7),

$$d_w(W_{c,s},W_{\tilde c,\tilde s})\leq\tfrac18\big(\|c^2-\tilde c^2\|_\infty+4|s-\tilde s|\big). \tag{8.9}$$

(U2) The map. Let $\mathcal{A}_s=\{c\in\mathrm{BC}(\mathbb{R}):W_{c,s}\text{ admissible}\}$; it is open in $\mathrm{BC}(\mathbb{R})$ by Equation (8.9) and Lemma 3.32(c). For $c\in\mathcal{A}_s$ let $u_\pm$ be the Jost solutions of $W_{c,s}$, $R=u_-/u_+$,

$$\Sigma_c:=\frac{u_-}{u_-+\vartheta u_+}=\frac{R}{R+\vartheta},\qquad J_c:=\frac{\kappa}{\vartheta\,w_0(W_{c,s})},\qquad T_{D,s}\,c:=f_-+J_c\,\Sigma_c.$$

By Lemma 8.10, $T_{D,s}c = f_- + \kappa\int_{-\infty}^{\xi}(u_- + \vartheta u_+)^{-2}$, Σ_c increases strictly from 0 to 1, $\Sigma_c(0) = \frac{1}{1+\vartheta}$, and $\int_{\mathbb{R}}(u_- + \vartheta u_+)^{-2} = 1/(\vartheta w_0)$. $T_{D,0} = T_D$, and $T_{D,s} = T_{D_s}$ for the datum $D_s = (c_1', c_2', c_3' + s, \vartheta, f_-, \kappa)$. By Proposition 8.13 the fixed points of T_D are exactly the global solutions of type G at both poles with effective data D. By Lemma 8.27(a),(b) applied to D_s (its proof is local, near any c with W_c admissible), $T_{D,s}$ is C^1 on the open set $\mathcal{A}_s$ with compact derivative.

(U3) The Jost toolbox of Chapter 3. Comparison: Lemma 3.23; continuity in d_w, closedness of $\{w_0 > 0\}$ on positive pairs: Lemma 3.25; the trichotomy: Lemma 3.32; compactness of the profiles Σ: Lemma 3.28; explicit admissible and critical potentials: Lemma 3.35; the Riccati barrier: Lemma 3.10(b)(iii).

(U4) Degree theory (Appendix A). Theorem A.35 (D1)–(D5), Corollary A.36, and the homotopy invariance on sets that move with the parameter, Lemma A.39.

8.5.3. A priori bounds, uniform in the deformation parameter.

Lemma 8.39 (a priori bounds, uniform in s)**.** *Notation: $\kappa = \kappa_{\mathrm{D}}$, $k = \min(k_1, k_2)$, $W_{c,s} = W_e + \frac{1}{2}Q[c^2 + 4s]$ for $c \in BC(\mathbb{R})$, $s \geq 0$, $T_{D,s}$ the corresponding map, $\widetilde{W}^s = W_e + \frac{1}{2}Q[(f_-^+)^2 + 4s]$ with $\tilde{w}_0(s) = w_0(\widetilde{W}^s)$, $\tilde{\Gamma}(s)$ the kernel integral $\Gamma_{\ker}$ of $\widetilde{W}^s$, $L_s(\kappa_{\mathrm{D}}) := \Lambda_{\mathrm{mod}}(s)$, the moderate-swirl number of Theorem 8.34 for the data D_s (with c_3' replaced by $c_3' + s$), i.e. $L_s(\kappa_{\mathrm{D}}) = \frac{3}{4}\,\kappa_{\mathrm{D}}\,S_1(s)\,\tilde{\Gamma}(s)/(\vartheta\,\tilde{w}_0(s))$ with $S_1(s)$ the sup bound of Theorem 8.34. There are $m_*, w_* > 0$ depending on (c_1', c_2', c_3') only such that with $J_1 = (1+\vartheta)(m_* - f_-)^+$ and $J_{\max} = \max(J_1, \kappa_{\mathrm{D}}/(\vartheta w_*))$: for nondecreasing $c \in BC$ with $c(0) \geq m_*$, $W_{c,s} \geq W_* := W_e + \frac{1}{2}Q[m_*^2\mathbf{1}_{[0,\infty)}]$ for all $s \geq 0$, $W_* \geq k^2/2$, $W_{c,s}$ is admissible and $w_0 \geq w_* = w_0(W_*)$. Every fixed point of $T_{D,s}$ has $f_- \leq f \leq f_- + J$, $J \leq J_{\max}$, $w_0 = \kappa_{\mathrm{D}}/(\vartheta J) \geq \kappa_{\mathrm{D}}/(\vartheta J_{\max})$. For $c_3' + s > \bar{c}_3$: $\tilde{w}_0(s) \uparrow \infty$, $\tilde{\Gamma}(s) \downarrow$, so $L_s(\kappa_{\mathrm{D}}) \to 0$ as $s \to \infty$.*

Reading of the last sentence. $\tilde{\Gamma}(s) = \Gamma_{\ker}(\widetilde{W}^s) = \int_{\mathbb{R}} \operatorname{sech}^2\xi/\rho_{\widetilde{W}^s}(\xi)\,d\xi$ (finite because $\sup 1/\rho < \infty$, Lemma 3.21(b)(iv)), the kernel integral of Lemma 8.30. Put $B_s := \kappa/(\vartheta\tilde{w}_0(s))$ and $S_1(s) := \max(|f_-|, f_- + B_s)$. The potential $\widetilde{W}^s = W_{\mathrm{e}} + \frac{1}{2}s\operatorname{sech}^2 + \frac{1}{8}(f_-^+)^2\operatorname{sech}^2$ is the reference potential $\widetilde{W}$ of Theorem 8.34 for the datum D_s (c_3' replaced by $c_3' + s$; $m_- = f_-^+$), B_s and $S_1(s)$ are its B and S_1, and therefore $L_s = \Lambda_{\mathrm{mod}}(s) = \frac{3}{4}\kappa S_1(s)\tilde{\Gamma}(s)/(\vartheta\tilde{w}_0(s))$ is the number Λ_{mod} of that theorem for D_s. (The constant C_* of Lemma 8.15 does not occur here.) The three assertions are proved in order.

Proof of the first assertion (the barrier W_*). *Choice of m_*.* $W_{\mathrm{e}} \to k_1^2, k_2^2 \geq k^2$ at $\mp\infty$, hence $W_{\mathrm{e}} \geq k^2/2$ outside a compact interval $[-X, X]$. The function $q(\xi) = \mathcal{Q}[\mathbf{1}_{[0,\infty)}](\xi) = (4\cosh^2\xi)^{-1}\int_0^\infty e^{-2|\xi-\eta|}d\eta$ is continuous and positive, so $q_0 := \min_{[-X,X]} q > 0$. Let $b = \sup|W_{\mathrm{e}}|$ and choose $m_* > 0$ with $\frac{1}{2}m_*^2 q_0 \geq \frac{1}{2}k^2 + b$. Then $W_* = W_{\mathrm{e}} + \frac{1}{2}m_*^2 q \geq \frac{1}{2}k^2$: on $[-X, X]$ because $W_{\mathrm{e}} \geq -b$, outside because $q \geq 0$. Note that m_* depends on c' only, and so does w_* below. W_* is admissible.* Let φ_- be its Jost solution, $\varphi_- \sim e^{k_1\xi}$ at $-\infty$ (Lemma 3.21(a); $W_* \in \mathcal{W}(k_1, k_2)$ by (U1)). Near $-\infty$, $\varphi_- > 0$ and $\varphi_-'/\varphi_- \to k_1 > 0$. As long as $\varphi_- > 0$ and $\varphi_-' > 0$ we have $\varphi_-'' = W_*\varphi_- > 0$, so φ_-' increases and stays positive: by continuation $\varphi_- > 0$ and $\varphi_-' > 0$ on $\mathbb{R}$. Then $\varphi_-'' \geq \frac{1}{2}k^2\varphi_-$ and $\varphi_-' \geq \varphi_-'(0) > 0$ on $[0, \infty)$, so $\varphi_- \to \infty$: W_* is admissible, and $w_* := w_0(W_*) > 0$. (No hypothesis on c_3' was used: W_* is admissible whether W_{e} is subcritical or supercritical.) *The comparison.* Let $c \in \mathrm{BC}(\mathbb{R})$ be nondecreasing with $c(0) \geq m_*$ and $s \geq 0$. For $\xi \geq 0$,

$c(\xi) \geq c(0) \geq m_* > 0$, so $c^2 \geq m_*^2$; for $\xi < 0$, $m_*^2 \mathbf{1}_{[0,\infty)} = 0 \leq c^2 + 4s$. Hence $c^2 + 4s \geq m_*^2 \mathbf{1}_{[0,\infty)}$ and, $\mathcal{Q}$ being order preserving, $W_{c,s} \geq W_*$. Both potentials lie in $\mathcal{W}(k_1, k_2)$ by (U1), and W_* is admissible, so Lemma 3.23 gives that $W_{c,s}$ is admissible and $w_0(W_{c,s}) \geq w_0(W_*) = w_*$. □

Proof of the second assertion (fixed points). Let $f \in \mathcal{A}_s$ be a fixed point of $T_{D,s}$, $f = f_- + J\Sigma_f$, $J = J_f = \kappa/(\vartheta w_0)$. Since $0 < \Sigma_f < 1$ is increasing, $f_- \leq f \leq f_- + J$ and f is nondecreasing, with $f(0) = f_- + J/(1+\vartheta)$. If $J \geq J_1$ then $f(0) \geq f_- + (m_* - f_-)^+ \geq m_*$, so the first assertion applies to $c = f$: $w_0(W_{f,s}) \geq w_*$, i.e. $J = \kappa/(\vartheta w_0) \leq \kappa/(\vartheta w_*) \leq J_{\max}$. If $J < J_1$ then $J < J_1 \leq J_{\max}$. In both cases $J \leq J_{\max}$, and $w_0 = \kappa/(\vartheta J) \geq \kappa/(\vartheta J_{\max})$. □

Proof of the third assertion ($s \to \infty$). Let $s_0 \geq 0$ with $c_3' + s_0 > \bar{c}_3$ and let $s \geq s_0$. *Admissibility and monotonicity.* Since $\mathcal{Q}[(f_-^+)^2] \geq 0$, $\widetilde{W}^s \geq W_{\mathrm{e}} + \frac{1}{2} s \operatorname{sech}^2 = W_{(c_1', c_2', c_3'+s)}$, which is admissible by Lemma 3.35(b) ($c_3' + s > \bar{c}_3$); so $\widetilde{W}^s$ is admissible by Lemma 3.23. For $s_0 \leq s < s'$, $\widetilde{W}^{s'} - \widetilde{W}^s = \frac{1}{2}(s'-s)\operatorname{sech}^2 \geq 0$, so by Lemma 3.23 $\zeta_-^{s'} \geq \zeta_-^s$, $\zeta_+^{s'} \leq \zeta_+^s$ pointwise: $\rho_{s'} \geq \rho_s$, hence $\tilde{w}_0(s) = \rho_s(0)$ is nondecreasing and $\tilde{\Gamma}(s) = \int \operatorname{sech}^2 / \rho_s$ is nonincreasing. *Divergence of $\tilde{w}_0$.* Put $b_1 = \sup(W_{\mathrm{e}})^-$, i.e. $W_{\mathrm{e}} \geq -b_1$, and $\lambda_s^2 = \frac{s}{2\cosh^2 1} - b_1$. On $[-1,1]$, $\operatorname{sech}^2 \xi \geq \operatorname{sech}^2 1$, so $\widetilde{W}^s \geq -b_1 + \frac{1}{2} s \operatorname{sech}^2 \xi \geq \lambda_s^2$ there. Let $C_0 = \max(0, -\zeta_-^{s_0}(-1))$ and $C_0' = \max(0, \zeta_+^{s_0}(1))$. By the comparison just proved, $\zeta_-^s(-1) \geq \zeta_-^{s_0}(-1) \geq -C_0$ and $\zeta_+^s(1) \leq C_0'$. Let s be so large that $\lambda_s > \max(C_0, C_0')$. The tanh barrier Lemma 3.10(b)(iii) applied to $h = \zeta_-^s$ on $[-1,0]$ and to $h = \zeta_+^s$ on $[0,1]$ (both solve $h' = \widetilde{W}^s - h^2$) gives

$$\zeta_-^s(0) \geq \lambda_s \tanh\bigl(\lambda_s - \operatorname{artanh} \tfrac{C_0}{\lambda_s}\bigr), \qquad -\zeta_+^s(0) \geq \lambda_s \tanh\bigl(\lambda_s - \operatorname{artanh} \tfrac{C_0'}{\lambda_s}\bigr).$$

As $s \to \infty$, $\lambda_s \to \infty$, $\operatorname{artanh}(C_0/\lambda_s) \to 0$, and the right-hand sides tend to $+\infty$: $\tilde{w}_0(s) = \zeta_-^s(0) - \zeta_+^s(0) \to \infty$. *The number L_s.* $B_s = \kappa/(\vartheta \tilde{w}_0(s)) \downarrow 0$, so $S_1(s) \leq \max(|f_-|, f_- + B_{s_0})$ is bounded; $\tilde{\Gamma}(s) \leq \tilde{\Gamma}(s_0)$ (Lemma 8.23(e)); and $\tilde{w}_0(s) \to \infty$. Hence $L_s(\kappa) = \frac{3}{4}\kappa S_1 \tilde{\Gamma}/(\vartheta \tilde{w}_0) \to 0$. □

Remark 8.40 (a second ending of the homotopy)**.** Since $L_s = \Lambda_{\mathrm{mod}}(s) \to 0$, so that $\Lambda_{\mathrm{mod}}(s) < 1$ for large s, Theorem 8.34 (applied to D_s, whose datum satisfies $c_3' + s > \bar{c}_3$) shows that $T_{D,s}$ is a contraction on a convex set containing all its fixed points: the end of the deformation can be read from Banach's theorem. The proof below does not need this: the monotonicity $\widetilde{W}^s \uparrow$ and $\tilde{w}_0(s) \to \infty$ suffice (Step 3), which is why the explicit constants of Λ_{mod} do not enter the degree theorem.

Figure 8.2 shows the deformation for a supercritical datum.

8.5.4. The region in which fixed points are counted. Fix the data D and let $m_*, w_*, J_1, J_{\max}$ be as in Lemma 8.39; put $w_{\min} := \kappa/(\vartheta J_{\max})$. Since $J_{\max} \geq J_1 \geq (m_* - f_-)^+$ and $J_{\max} \geq \kappa/(\vartheta w_*)$,

$$J_{\max} \geq m_* - f_-, \qquad w_{\min} \leq w_*. \tag{8.10}$$

By the third assertion of Lemma 8.39 choose $s_{\max} \geq 0$ with $c_3' + s_{\max} > \bar{c}_3$ and $\tilde{w}_0(s_{\max}) > w_{\min}$; thus $B_{s_{\max}} = \kappa/(\vartheta \tilde{w}_0(s_{\max})) < J_{\max}$. Define

$$\begin{gathered} \hat{J} := J_{\max} + 1, \qquad R_0 := |f_-| + \hat{J} + 2, \qquad S' := |f_-| + \hat{J} + 1, \\ r(g) := \min\bigl(\max(g, f_- - 1),\, f_- + \hat{J}\bigr) \end{gathered} \tag{8.11}$$

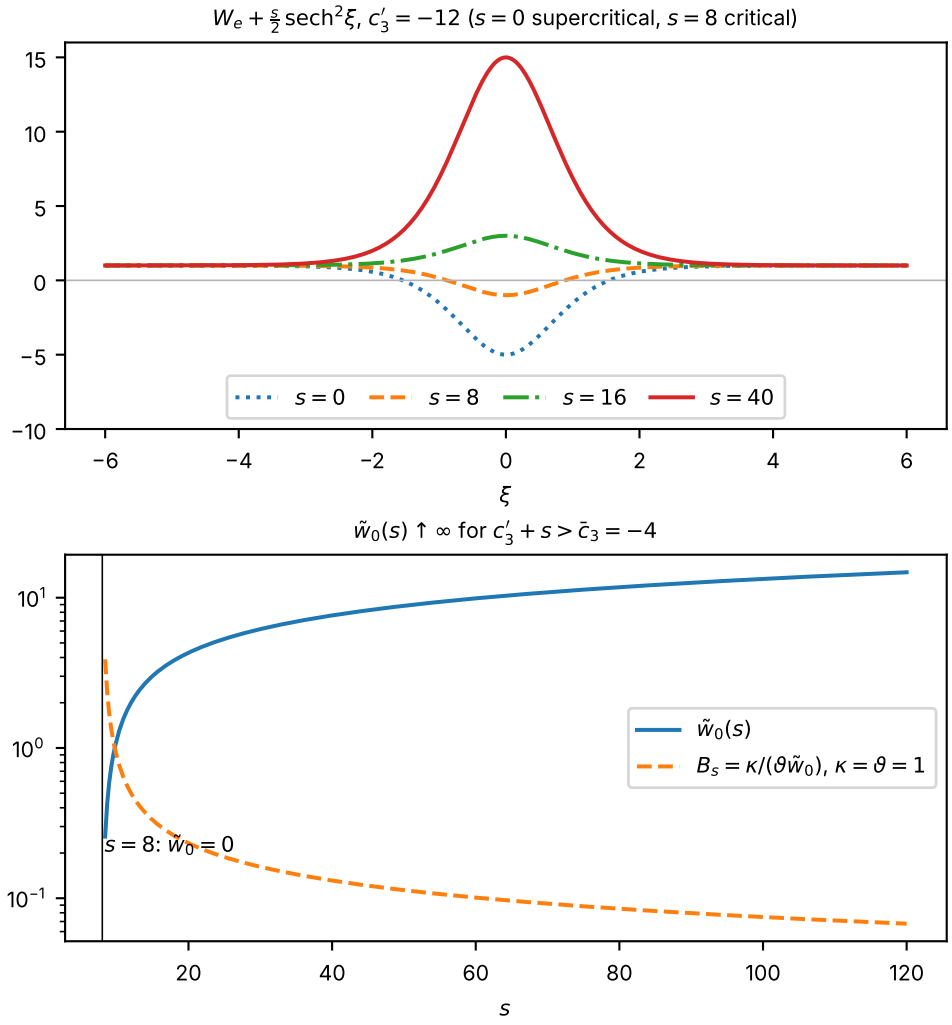


FIGURE 8.2. The deformation $c_3' \mapsto c_3' + s$ for the supercritical datum $c' = (0, 0, -12)$ ($\bar{c}_3 = -4$). Left: $W_e + \frac{s}{2}\operatorname{sech}^2\xi = 1 - (6 - \frac{s}{2})\operatorname{sech}^2\xi$ for $s = 0$ (Pöschl–Teller with $\ell = 2$: supercritical, T_D is defined on no constant), $s = 8$ (critical) and $s = 16, 40$ (admissible, with an ever higher bump). Right: $\tilde{w}_0(s)$ (log scale), computed by Riccati integration and checked against the closed form of Proposition 3.47 (maximal difference $3 \cdot 10^{-11}$ at $s = 9, 16, 40, 100$), and the bound $B_s = \kappa/(\vartheta\tilde{w}_0)$ for $\kappa = \vartheta = 1$: $\tilde{w}_0 \to 0$ as $s \downarrow 8$ and $\tilde{w}_0 \uparrow \infty$ as $s \to \infty$. Code: `code/ch08_effective/make_wzero_s.py`.

(r acts pointwise: it clamps g into $[f_- - 1, f_- + \hat{J}]$). Then $r : \mathrm{BC}(\mathbb{R}) \to \mathrm{BC}(\mathbb{R})$ is 1-Lipschitz, $|r(g)| \le S'$, and $r(g) = g$ whenever $f_- - 1 \le g \le f_- + J_{\max} + 1$, in particular whenever $\|g - g_0\|_\infty < 1$ for some g_0 with $f_- \le g_0 \le f_- + J_{\max}$ (see

Step 5). For $s \in [0, s_{\max}]$ and $g \in \mathrm{BC}(\mathbb{R})$ put

$$V_{g,s} := W_{r(g),s}, \qquad H(s,g) := T_{D,s}\big(r(g)\big) \ \text{ whenever } V_{g,s} \text{ is admissible,}$$

so $H(0,\cdot) = T_D \circ r$. Let

$$\begin{aligned}\mathcal{C} := \big\{(s,g) \in [0,s_{\max}] \times \mathrm{BC}(\mathbb{R}) :\ & \|g\|_\infty \le R_0,\\ & V_{g,s} \text{ admissible},\ w_0(V_{g,s}) \ge \tfrac{1}{2} w_{\min}\big\},\\ \mathcal{U} := \big\{(s,g) \in [0,s_{\max}] \times \mathrm{BC}(\mathbb{R}) :\ & \|g\|_\infty < R_0,\\ & V_{g,s} \text{ admissible},\ w_0(V_{g,s}) > \tfrac{1}{2} w_{\min}\big\},\end{aligned}$$

with slices $\mathcal{U}_s = \{g : (s,g) \in \mathcal{U}\}$. The set $\mathcal{O}$ of the theorem will be $\mathcal{U}_0$.

Lemma 8.41 (the region and the map)**.** *(a) $\mathcal{U}$ is relatively open and bounded in $[0,s_{\max}] \times \mathrm{BC}(\mathbb{R})$, the slices $\mathcal{U}_s$ are open, $\mathcal{U} \subset \mathcal{C}$, and $\mathcal{C}$ is closed. Hence $\overline{\mathcal{U}} \subset \mathcal{C}$, and every point of $\overline{\mathcal{U}} \setminus \mathcal{U}$ has $\|g\|_\infty = R_0$ or $w_0(V_{g,s}) = \frac{1}{2} w_{\min}$. (b) H is continuous on $\mathcal{C}$ and $H(\mathcal{C})$ is relatively compact in $\mathrm{BC}(\mathbb{R})$. In particular H is compact on $\overline{\mathcal{U}}$.*

Proof. *Uniform class.* For (s,g) with $s \le s_{\max}$ the potential $V_{g,s}$ equals $W_e + \frac{1}{2}\mathcal{Q}[h]$ with $h = r(g)^2 + 4s \in [0, S'^2 + 4s_{\max}]$. By (U1) and Lemma 3.35, all these potentials belong to one class $\mathcal{K}$ of Lemma 3.28: they lie in $\mathcal{W}(k_1,k_2;C)$ with $C = C_e + \frac{1}{2}(S'^2 + 4s_{\max})$; $V \le B^2 := \max\big(\sup W_{\mathrm{e}} + \frac{1}{8}(S'^2 + 4s_{\max}), k_1^2, k_2^2\big)$; and $V \ge W_{\mathrm{e}} \ge k^2/2 =: m^2$ outside $[-X,X]$, with X of the proof of Lemma 8.39. (a) By Equation (8.9), $(s,g) \mapsto V_{g,s}$ is Lipschitz from $[0,s_{\max}] \times \mathrm{BC}(\mathbb{R})$ into $(\mathcal{W}, d_w)$ (r is 1-Lipschitz and $|r(g)^2 - r(\tilde g)^2| \le 2S'|r(g) - r(\tilde g)|$). Admissibility is open (Lemma 3.32(c)) and w_0 is continuous on admissible potentials (Lemma 3.25(b)); so the three defining conditions of $\mathcal{U}$ are open conditions. $\mathcal{U} \subset \mathcal{C}$ and the bound $\|g\| < R_0$ give boundedness. *$\mathcal{C}$ is closed:* let $(s_n,g_n) \in \mathcal{C}$ converge to (s,g). Then $V_n := V_{g_n,s_n} \to V := V_{g,s}$ in d_w, hence in L^1_{loc}; each V_n is admissible, so $V_n \in \mathcal{P}$ (Lemma 3.25(c)), and by that lemma $V \in \mathcal{P}$ and $w_0(V) = \lim w_0(V_n) \ge \frac{1}{2} w_{\min} > 0$. A positive pair with $w_0 > 0$ is not critical, hence admissible (Lemma 3.25(c)). So $(s,g) \in \mathcal{C}$. The last sentence of (a) is the definition of $\mathcal{U}$ inside $\mathcal{C}$. (b) *Compactness.* For $(s,g) \in \mathcal{C}$, $H(s,g) = f_- + J\Sigma$ with $\Sigma = \Sigma_{r(g)}$ the profile of $V_{g,s} \in \mathcal{K}$ and $J = \kappa/(\vartheta w_0) \in (0, 2J_{\max}]$ because $w_0 \ge \frac{1}{2} w_{\min}$. By Lemma 3.28(f) the profiles Σ of admissible $W \in \mathcal{K}$ with $w_0 \ge \frac{1}{2} w_{\min}$ form a relatively compact set $\mathcal{Y} \subset \mathrm{BC}(\mathbb{R})$. So $H(\mathcal{C}) \subset f_- + [0, 2J_{\max}] \cdot \mathcal{Y}$, the image of a compact set under a continuous map, which is relatively compact. *Continuity.* Let $(s_n,g_n) \to (s,g)$ in $\mathcal{C}$. By Lemma 3.25(b) (all V_n, V admissible, $V_n \to V$ in d_w), $u_\pm^{(n)} \to u_\pm$ and $w_0(V_n) \to w_0(V)$ locally uniformly, so $J_n \to J$ and $\Sigma_n = R_n/(R_n + \vartheta) \to \Sigma$ locally uniformly. Every subsequence of (Σ_n) has, by relative compactness of $\mathcal{Y}$, a sub-subsequence converging in $\mathrm{BC}(\mathbb{R})$; its limit coincides with Σ (local uniform convergence). So $\Sigma_n \to \Sigma$ in $\mathrm{BC}(\mathbb{R})$ and $H(s_n,g_n) \to H(s,g)$. □

8.5.5. The theorem.

Theorem 8.42 (degree one for all data). *For all $c_1', c_2' > -1$, c_3', f_-, $\vartheta, \kappa_{\mathrm{D}} > 0$ ($\kappa = \kappa_{\mathrm{D}}$) there are $J_{\max}, R_0, \hat J$ and an open bounded $\mathcal{O} \subset BC(\mathbb{R})$ with compact $H = T_D \circ r$ such that (a) all fixed points lie in $\mathcal{O}$, none on $\partial\mathcal{O}$; (b) $\deg_{\mathrm{LS}}(I - H, \mathcal{O}) = +1$, hence a solution exists for every $\kappa_{\mathrm{D}} > 0$; (c) if all fixed points are nondegenerate there are finitely many and their number is odd; (d) if at no fixed point DT_D has a real eigenvalue > 1 or the eigenvalue 1 (i.e. $1 \notin \operatorname{spec} DT_D$ and no real eigenvalue*

exceeds 1*), the fixed point is unique; conversely, if there are two or more fixed points then some fixed point has a real eigenvalue* > 1 *or* $1 \in \operatorname{spec} DT_D$ *(index criterion).*

Here $J_{\max}$, $\hat{J}$, R_0 and r are the constants and the clamp of Lemma 8.39 and Equation (8.11), $\mathcal{O} = \mathcal{U}_0$, and $H = T_D \circ r$ on $\overline{\mathcal{O}}$. Nondegenerate means $1 \notin \operatorname{spec} \mathrm{D}T_D(f)$ (the spectrum of the complexification of the compact operator $\mathrm{D}T_D(f)$); the *number of real eigenvalues* > 1 is counted with algebraic multiplicity, as in (D5).

Proof. All constants are those of the previous subsection. The hypotheses of homotopy invariance on moving sets (Lemma A.39) are verified in Steps 1–2, the degree at the end $s = s_{\max}$ is computed in Step 3, (a),(b) follow in Step 4 and (c),(d) in Step 5.

Step 1: the map is compact on $\overline{\mathcal{U}}$*.* This is Lemma 8.41.

Step 2: fixed points of H *never lie on* $\partial\mathcal{U}$ *and are fixed points of* $T_{D,s}$*.* Let $(s,g) \in \mathcal{C}$ with $g = H(s,g)$ and put $c = r(g)$, $J_c = \kappa/(\vartheta w_0(V_{g,s}))$, Σ_c as in (U2). Then $g = f_- + J_c\Sigma_c$ is nondecreasing and $\geq f_-$, so

$$c = r(g) = \min(g, f_- + \hat{J}) \quad \text{is nondecreasing, and}$$

$$c(0) = \min\Big(f_- + \frac{J_c}{1+\vartheta},\ f_- + \hat{J}\Big).$$

If $J_c \geq J_1$ then $f_- + J_c/(1+\vartheta) \geq f_- + (m_* - f_-)^+ \geq m_*$, and $f_- + \hat{J} = f_- + J_{\max} + 1 > m_*$ by Equation (8.10); so $c(0) \geq m_*$, and Lemma 8.39 (first assertion; $V_{g,s} = W_{c,s}$) gives $w_0 \geq w_*$, i.e. $J_c \leq \kappa/(\vartheta w_*) \leq J_{\max}$. If $J_c < J_1$ then $J_c < J_{\max}$. So always $J_c \leq J_{\max}$, and:

(i) $f_- \leq g \leq f_- + J_{\max} < f_- + \hat{J}$, hence $r(g) = g$ and $g = T_{D,s}(g)$: g is a fixed point of $T_{D,s}$;

(ii) $\|g\|_\infty \leq |f_-| + J_{\max} < R_0$;

(iii) $w_0(V_{g,s}) = \kappa/(\vartheta J_c) \geq w_{\min} > \frac{1}{2} w_{\min}$.

By (ii), (iii) the point (s,g) lies in $\mathcal{U}$ (the other defining condition, admissibility, holds on $\mathcal{C}$). Hence H has no fixed point in $\mathcal{C} \setminus \mathcal{U} \supset \overline{\mathcal{U}} \setminus \mathcal{U} = \partial\mathcal{U}$. By Lemma A.39 the degree $\deg_{\mathrm{LS}}(I - H(s,\cdot), \mathcal{U}_s)$ is independent of $s \in [0, s_{\max}]$.

Step 3: the degree at $s = s_{\max}$ *equals* 1*.* Write $s = s_{\max}$. We use the homotopy $H_t(g) = tH(s,g) + (1-t)f_-$, $t \in [0,1]$, on $\overline{\mathcal{U}_s} \subset \mathcal{C}_s$. It is compact (continuous, with image in the continuous image of $[0,1] \times \overline{H(\mathcal{C})}$). The constant f_- belongs to $\mathcal{U}_s$: $r(f_-) = f_-$ and $V_{f_-,s} = W_{f_-,s} \geq \widetilde{W}^s$ (because $f_-^2 \geq (f_-^+)^2$), so $V_{f_-,s}$ is admissible with $w_0 \geq \tilde{w}_0(s) > w_{\min}$ by Lemma 3.23, and $|f_-| < R_0$. Let $g \in \overline{\mathcal{U}_s}$ with $g = H_t(g)$. Then $g = f_- + tJ_c\Sigma_c$ with $c = r(g)$, so $g \geq f_-$ and $c = \min(g, f_- + \hat{J}) \geq f_-$, whence $c^2 \geq (f_-^+)^2$ (if $f_- \geq 0$ then $c \geq f_- \geq 0$; if $f_- < 0$ the right side is 0) and $V_{g,s} = W_{c,s} \geq \widetilde{W}^s$. By Lemma 3.23, $w_0(V_{g,s}) \geq \tilde{w}_0(s) > w_{\min} > \frac{1}{2} w_{\min}$ and $J_c \leq B_s < J_{\max}$. Therefore $\|g\|_\infty \leq |f_-| + B_s < R_0$, so $g \in \mathcal{U}_s$ (admissibility holds on $\mathcal{C}$). Thus H_t has no fixed point on $\partial\mathcal{U}_s \subset \overline{\mathcal{U}_s} \setminus \mathcal{U}_s$ for any t, and by (D2), (D1) of Theorem A.35,

$$\deg_{\mathrm{LS}}(I - H(s_{\max},\cdot), \mathcal{U}_{s_{\max}}) = \deg_{\mathrm{LS}}(I - H_0, \mathcal{U}_{s_{\max}}) = \deg_{\mathrm{LS}}(I - f_-, \mathcal{U}_{s_{\max}}) = 1.$$

Step 4: (a) and (b). By Step 2 and Step 3, $\deg_{\mathrm{LS}}(I - H(0,\cdot), \mathcal{U}_0) = \deg_{\mathrm{LS}}(I - H(s_{\max},\cdot), \mathcal{U}_{s_{\max}}) = 1$. Put $\mathcal{O} = \mathcal{U}_0$ (open and bounded, Lemma 8.41(a)); $H(0,\cdot) = T_D \circ r$ is compact on $\overline{\mathcal{O}}$. (b) is the value of the degree, and the existence of a fixed point of $H(0,\cdot)$ follows from (D3); by Step 2(i) (with $s = 0$) it is a fixed point of

T_D, i.e. by Proposition 8.13 a solution with effective data D (this holds for every $\kappa > 0$ since $\kappa = \kappa_{\mathrm{D}}$ was arbitrary). (a): let f be a fixed point of T_D, $W_f = W_{f,0}$ admissible. By Lemma 8.39 (second assertion, $s = 0$) $f_- \le f \le f_- + J_{\max}$ and $w_0(W_f) \ge w_{\min}$. So $r(f) = f$, $\|f\|_\infty \le |f_-| + J_{\max} < R_0$, and $(0, f) \in \mathcal{U}$; i.e. $f \in \mathcal{O}$, and f is not in $\overline{\mathcal{O}} \setminus \mathcal{O} \supset \partial\mathcal{O}$ by Step 2. Conversely every fixed point of $H(0,\cdot)$ in $\overline{\mathcal{O}}$ is a fixed point of T_D (Step 2(i)), so the fixed-point sets coincide.

Step 5: (c) and (d). Let f be a fixed point of T_D. If $\|g - f\|_\infty < 1$ then $g \ge f - 1 \ge f_- - 1$ and $g \le f + 1 \le f_- + J_{\max} + 1 = f_- + \hat{J}$, so $r(g) = g$: near f, $H(0,\cdot) = T_D$, which is C^1 with compact derivative $\mathrm{D}T_D(f)$ by (U2). Suppose $1 \notin \operatorname{spec} \mathrm{D}T_D(f)$. By (D5), f is an isolated fixed point and its index is $(-1)^{\beta(f)}$, $\beta(f)$ the number of real eigenvalues of $\mathrm{D}T_D(f)$ above 1 (with algebraic multiplicity). Now assume this for all fixed points. The fixed-point set Fix is closed and contained in $\overline{H(\mathcal{C})}$ (compact), hence compact; a compact set of isolated points is finite. By Corollary A.36 and (b),

$$\sum_{f \in \mathrm{Fix}} (-1)^{\beta(f)} = 1, \qquad \text{so } N_{\mathrm{even}} - N_{\mathrm{odd}} = 1, \quad N = \#\mathrm{Fix} = 1 + 2N_{\mathrm{odd}} \ \text{ is odd.}$$

(c) is proved. For (d), first suppose $\operatorname{spec} \mathrm{D}T_D(f) \cap [1, \infty) = \emptyset$ for every fixed point. Then all fixed points are nondegenerate with $\beta = 0$, so by the display $N = N_{\mathrm{even}} = 1$. Conversely, let $\#\mathrm{Fix} \ge 2$ and suppose, for contradiction, that no fixed point has a real eigenvalue > 1 or the eigenvalue 1; this is the situation just treated, so $\#\mathrm{Fix} = 1$: contradiction. □

Remark 8.43 (how the proof meets the hypotheses of the degree theory; for the independent reader)**.** For a quick audit, the hypotheses used are located as follows.

hypothesis	where it is verified
$\mathcal{U}$ relatively open, bounded; slices open	Lemma 8.41(a)
H compact on $\overline{\mathcal{U}}$	Lemma 8.41(b): uniform class $\mathcal{K}$, Lemma 3.28(f), closedness of $\mathcal{C}$ (Lemma 3.25(c))
no fixed point of H on $\partial\mathcal{U}$ (for Lemma A.39)	Step 2: $J_c \le J_{\max}$ from Lemma 8.39
degree at the end slice	Step 3: homotopy to the constant f_-, boundary condition from $\widetilde{W}^{s_{\max}}$ and $\tilde{w}_0(s_{\max}) > w_{\min}$
$H = T_D$ near fixed points, C^1, compact derivative	Step 5, Lemma 8.27
finiteness of Fix and index formula	Theorem A.35(D5), Corollary A.36

The three places where $c_3' < \bar{c}_3$ is delicate (see Remark 8.44) are the admissibility of W_* (proved without reference to c_3'), the closedness of $\mathcal{C}$, and the compactness of $\mathcal{Y}$.

Remark 8.44 (the case $c_3' < \bar{c}_3$: where the Jost compactness lemmas enter)**.** If $c_3' \ge \bar{c}_3$ the potential W_{e} is admissible or critical and the lower bound $W_{f,s} \ge W_{\mathrm{e}}$ is a useful reference; if $c_3' < \bar{c}_3$ then W_{e} is supercritical (Lemma 3.35(b)) and T_D is defined only on the open set $\mathcal{A}_0$ of profiles whose swirl lifts the potential above criticality; $0 \notin \mathcal{A}_0$, so the classical homotopy $f \mapsto sT_Df$ (see Remark 8.50) is not

available. Three things replace it. (1) The barrier W_* is built from the swirl itself, $W_* = W_{\mathrm{e}} + \frac{1}{2}m_*^2\mathcal{Q}[\mathbf{1}_{[0,\infty)}]$: once $f(0) \geq m_*$ and f is nondecreasing, the swirl on $[0,\infty)$ is $\geq m_*$, and this makes the potential admissible with $w_0 \geq w_*$, *whatever* c_3' is (first assertion of Lemma 8.39). (2) A fixed point with small J has $w_0 = \kappa/(\vartheta J)$ *large*: the two cases $J \geq J_1$ and $J < J_1$ of Step 2 together bound w_0 below. (3) A lower bound on w_0 along a family of potentials is exactly what the weighted-topology lemmas need: it makes the limit of admissible potentials admissible rather than critical (Lemma 3.25(c)) and makes the profiles Σ precompact (Lemma 3.28(f)). This is why the lemmas of Chapter 3 on closedness of w_0 and compactness of Σ are used here and not only the comparison lemma. The same proof covers class Z ($f_- < 0$): nothing in it uses the sign of f_-.

Corollary 8.45 (existence below the no-swirl set; $J \geq M - f_-$)**.** *Let* $c_1', c_2' > -1$, $c_3' < \bar{c}_3$ *(so* $M = 2\sqrt{\bar{c}_3 - c_3'} > 0$*),* $0 \leq f_- < M$, $\vartheta > 0$. *Then for every* $\kappa > 0$ *there is a global solution of type* G *at both poles with effective data* D*; it lies in class* N_0 *and* $f(+\infty) - f_- = J \geq M - f_-$.

Proof. Existence is Theorem 8.42(b) with Proposition 8.13. Let f be a solution. By Proposition 8.18 (necessary condition), $c_3' + \frac{1}{4}S^2 \geq \bar{c}_3$ for $S = \max(|f_-|, |f_+|)$, i.e. $S \geq M$ since $M^2 = 4(\bar{c}_3 - c_3')$. As $|f_-| = f_- < M$ and $f_+ \geq f_- \geq 0$ (f is nondecreasing) we get $f_+ \geq M$, so $J = f_+ - f_- \geq M - f_-$. Moreover $f' = \kappa z^{-2} > 0$, so $f > f_- \geq 0$ on $\mathbb{R}$: f has no zero and the solution is of class N (Definition 5.3); with $f > 0$, $\kappa > 0$ it is in the fundamental domain N_0 (Proposition 5.4). □

Remark 8.46 (existence for every $\kappa > 0$)**.** A continuation in the total jump J, along which $\kappa = J\vartheta w_0$ is continuous, would give existence only for $\kappa \geq \kappa_1(c', \vartheta, f_-)$; for $0 \leq f_- < M$ such a continuation needs a starting slice with $J \geq M - f_-$. Theorem 8.42 gives existence for every $\kappa > 0$ at once: the starting slice is the end of an explicit deformation and no non-existence statement is used. Existence for $c_1', c_2' > -1$, any c_3', any $\kappa > 0$ is Theorem 8.20, a one-line corollary of Theorem 8.42; for $0 \leq f_- < M$ its existence part is contained in Corollary 8.45. The bound $J \geq M - f_-$ is not a consequence of the degree argument: it comes from the necessary condition Proposition 8.18.

8.6. The index criterion

In words. *The degree is* $+1$*; the index of a nondegenerate fixed point is* $+1$ *if the number* β *of real eigenvalues of* $\mathrm{D}T_D$ *above* 1 *is even and* -1 *if it is odd. Hence a second solution can appear only together with a third one and with a fixed point that has a real eigenvalue above* 1*. Conversely, a* spectral *hypothesis (no real eigenvalue* ≥ 1 *at any fixed point) gives uniqueness. This is a criterion, not a theorem about all data: it holds at small swirl, at large swirl, and fails at an explicit datum (Chapter 9). Complex eigenvalues, however large, never matter.*

Toy example 8.47 (complex eigenvalues have index $+1$)**.** Let $F(v) = Av$ on $\mathbb{R}^2$ with $A = \left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$, eigenvalues $a \pm ib$. The fixed point 0 has $\det(I - A) = (1-a)^2 + b^2 > 0$, so $\deg_{\mathrm{LS}}(I - F, \mathcal{O}) = \operatorname{sgn}\det(I - A) = +1$ on every ball, even if $|a \pm ib| = 10$: no real eigenvalue exceeds 1, so $\beta = 0$. The same happens for $\mathrm{D}T_D$ at the lowest fixed point of the datum D^* of Chapter 9, whose leading eigenvalues are $0.3504 \pm 0.1247\,\mathrm{i}$ (Lemma 8.27); and an operator with a real eigenvalue -10 also has index $+1$. Only real eigenvalues > 1 change the sign.

Proposition 8.48 (index criterion)**.** *Let D be arbitrary data as in Theorem 8.42 and* Fix *the set of fixed points of T_D. Then* Fix *is compact,* $\mathrm{Fix} \subset \{f : f_- \le f \le f_- + J_{\max}\}$*, and:*

(a) *(*parity*) if every $f \in \mathrm{Fix}$ is nondegenerate, then* Fix *is finite, $\sum_f (-1)^{\beta(f)} = 1$, hence $\#\mathrm{Fix}$ is odd;*

(b) *(*uniqueness criterion*) if $\operatorname{spec} \mathrm{D}T_D(f) \cap [1,\infty) = \emptyset$ for every $f \in \mathrm{Fix}$ (for instance if $\|\mathrm{D}T_D(f)\| < 1$, or the spectral radius is < 1, at every fixed point), then $\#\mathrm{Fix} = 1$;*

(c) *(*converse*) if $\#\mathrm{Fix} \ge 2$ then some $f \in \mathrm{Fix}$ has $\operatorname{spec} \mathrm{D}T_D(f) \cap [1,\infty) \ne \emptyset$; if all fixed points are nondegenerate and $\#\mathrm{Fix} \ge 2$ then $\#\mathrm{Fix} \ge 3$ and some fixed point has an odd number $\beta \ge 1$ of real eigenvalues above 1 (index -1);*

(d) *(*a unique solution is stable in the index sense*) if $\mathrm{Fix} = \{f\}$ and f is nondegenerate, then $\beta(f)$ is even.*

Proof. Compactness and the bound: Theorem 8.42(a) and Lemma 8.39; Fix is closed and contained in $\overline{H(\mathcal{C})}$. (a) is Step 5 of the proof of Theorem 8.42. (b) and the first half of (c) are Theorem 8.42(d); if $\|\mathrm{D}T_D(f)\| < 1$ or the spectral radius is < 1, every point of $\operatorname{spec} \mathrm{D}T_D(f)$ has modulus < 1, so no real eigenvalue is ≥ 1. Second half of (c): by (a), $N_{\mathrm{odd}} = (N-1)/2 \ge 1$ when $N \ge 3$; if $N \ge 2$ and all fixed points are nondegenerate then N is odd, so $N \ge 3$. (d): by (a), $(-1)^{\beta(f)} = 1$. □

Remark 8.49 (what the criterion does and does not say)**.** (i) Uniqueness is *not* equivalent to the spectral hypothesis: a unique, nondegenerate fixed point may have $\beta = 2$. (ii) The hypothesis of (b) holds for small κ ($\|\mathrm{D}T_D\| < 1$ by Theorem 8.34) and, numerically, for κ in an intermediate range; for large κ the eigenvalue -1 is approached (the leading eigenvalue is -0.83 at $\kappa = 64$ and -0.94 at $\kappa = 256$ at the rest state, Numerical observation 8.58), a heuristic that Chapter 10 makes precise (Theorem 10.11) and that is not used here. (iii) The spectral hypothesis does not hold for all data with $f_- \ge 0$: there are data with several solutions (Corollary 9.69, a computer-assisted proof). The sign structure of $\mathrm{D}T_D$ (Lemma 8.27(c), Chapter 9) shows what *can* be said: for $f \ge 0$ there is no nonnegative eigenvector with positive eigenvalue, so a real eigenvalue ≥ 1 has a sign-changing eigenvector. (iv) For $c' \in \operatorname{int} \mathcal{J}$ and κ moderate, Theorem 9.51 pushes the range of the criterion up to the first fold: the number of solutions can change only where $1 \in \operatorname{spec} \mathrm{D}T_D$.

Remark 8.50 (a direct proof of $\deg_{\mathrm{LS}} = 1$ for $c' \in \operatorname{int} \mathcal{J}$)**.** If $c_3' > \bar{c}_3$ then W_{e} is admissible, and $W_f \ge W_{\mathrm{e}}$ for *every* $f \in \mathrm{BC}(\mathbb{R})$, so T_D is defined on all of $\mathrm{BC}(\mathbb{R})$; by Lemma 3.23, $w_0(W_f) \ge w_0(W_{\mathrm{e}})$, and $\|T_D f\|_\infty \le |f_-| + B_0$, $B_0 = \kappa/(\vartheta w_0(W_{\mathrm{e}}))$. Let $\mathcal{B} = \{\|g\|_\infty < |f_-| + B_0 + 1\}$. For $s \in [0,1]$ the map $(s,g) \mapsto sT_D g$ is compact (Lemma 8.41-type argument with $s = 0$ fixed) and has no fixed point on $\partial\mathcal{B}$ ($\|sT_D g\| \le |f_-| + B_0$), so $\deg_{\mathrm{LS}}(I - T_D, \mathcal{B}) = \deg_{\mathrm{LS}}(I, \mathcal{B}) = 1$ by (D1), (D2). For such data this reproves $\deg_{\mathrm{LS}} = 1$ without the clamp, with a convex region and a deformation $T_D \to 0$. It breaks down when $c_3' < \bar{c}_3$, because $0 \notin \mathcal{A}_0$; the homotopy in c_3' of Theorem 8.42 is the replacement that works for all data.

Numerical observation 8.51 (degree and spectra at the certified datum)**.** *For $D^* = \big(c' = (\frac{5}{4}, -\frac{15}{16}, -\frac{517}{160}), \vartheta = 40, f_- = 0, \kappa_{\mathrm{D}} = 1400\big)$ (Theorem 9.68), the damped iteration of the fixed-point solver (Appendix E) finds the lowest fixed point, $J = 19.0407$, with leading eigenvalues $0.35042 \pm 0.12475\,\mathrm{i}$ (computed and*

asserted by `code/ch08_effective/verify_ch08b.py`*, part 6); the other two (*$J = 33.5892, 46.3185$*) are unstable under iteration and are found by Newton's method in the computations of Atlas 9.76, not reproduced here, with leading eigenvalues* 1.2818 *(real,* $\beta = 1$*) and* 0.6296*. The indices are therefore* $+1, -1, +1$ *and add up to* 1*, as Theorem 8.42(c) requires.*

8.7. The no-swirl face

In words. *At* $\kappa_{\mathrm{D}} = 0$ *the equation* $f = T_D f$ *reads* $f = f_-$*: the trivial swirl, and the solution is one of the explicit no-swirl solutions of Chapter 3. Where the potential* $W_{f_-} = W_{\mathrm{e}} + \frac{1}{8} f_-^2 \operatorname{sech}^2$ *of this solution is admissible, the implicit function theorem continues it to small* κ_{D}*, and for small* κ_{D} *it is the* only *solution: every solution is close to the face. Where* $\kappa_{\mathrm{D}} \to 0$ *forces a nontrivial swirl (the potential of the trivial swirl is supercritical) nothing converges to the face, and the limits are the bubbles of Chapter 11. At the rest state one computes the first correction by hand.*

Toy example 8.52 (the implicit function theorem at a face)**.** Scalar model of $f = f_- + \kappa G(f)$: $x = \kappa/(1+x^2)$. $F(\kappa, x) = x - \kappa/(1+x^2)$ is smooth, $F(0,0) = 0$, $\partial_x F(0,0) = 1 \neq 0$. Hence there is a unique smooth branch $x(\kappa)$ near $\kappa = 0$, and $x = \kappa(1 - x^2 + x^4 - \dots)$ gives $x(\kappa) = \kappa - \kappa^3 + 2\kappa^5 - \dots$. The corresponding statement for T_D is Theorem 8.53(a),(d): $J = j - \frac{1}{4} c_\vartheta j^3 + O(j^5)$.

For $\kappa \in \mathbb{R}$ write $T_{D,\kappa} f = f_- + \kappa G(f)$, $G(f) := \int_{-\infty}^{\xi} (u_- + \vartheta u_+)^{-2}$ with $u_\pm$ the Jost solutions of W_f (Lemma 8.10); G does not depend on κ. For the rest state $c' = 0$ write $\Sigma^0_\vartheta(\xi) := \lg(2\xi - \ln\vartheta) = 1/(1 + \vartheta e^{-2\xi})$ (the profile Σ of $W \equiv 1$, Example 8.57) and

$$c_\vartheta := \int_{\mathbb{R}} \mathcal{Q}\big[(\Sigma^0_\vartheta)^2\big](r)\, e^{-2|r|}\, dr \ \in (0,\infty), \qquad j := \frac{\kappa}{2\vartheta}.$$

THEOREM 8.53 (the no-swirl face). *Let* $c_1', c_2' > -1$*,* $c_3' \in \mathbb{R}$*,* $f_- \in \mathbb{R}$*,* $\vartheta > 0$*, and* $M = 2\sqrt{(\bar c_3 - c_3')^+}$*. Write* $W_{f_-} = W_{\mathrm{e}} + \frac{1}{8} f_-^2 \operatorname{sech}^2$ *for the potential of the constant swirl* $f \equiv f_-$*, i.e. of the datum* $(c_1', c_2', c_3' + \frac{1}{4} f_-^2)$*.*

(a) *(*the face and its collar*) Assume* (Adm′)*:* $c_3' + \frac{1}{4} f_-^2 > \bar c_3$*. Then the constant* f_- *is in* $\mathcal{A}_0$*, the no-swirl solution with* $z = u_- + \vartheta u_+$*,* $u_\pm$ *the Jost solutions of* W_{f_-}*, is the point* $\kappa = 0$ *of a family of effective data, and there are* $\kappa_1 > 0$*,* $\delta > 0$ *such that for* $|\kappa| < \kappa_1$ *the equation* $f = T_{D,\kappa} f$ *has exactly one solution* f^κ *in* $\{\|f - f_-\|_\infty < \delta\}$*;* $\kappa \mapsto f^\kappa$ *is* C^1*,* $f^0 = f_-$ *and* $\frac{d}{d\kappa} f^\kappa\big|_0 = f_1 := \int_{-\infty}^{\xi} (u_- + \vartheta u_+)^{-2} = \frac{1}{\vartheta w_0(W_{f_-})} \Sigma_{f_-}$*.*

(b) *(*every solution is near the face*) Assume* (Adm)*:* $c_3' + \frac{1}{4}(f_-^+)^2 > \bar c_3$ *(this is* (Adm′) *if* $f_- \geq 0$*, and means* $c_3' > \bar c_3$ *if* $f_- < 0$*). Then every fixed point of* T_D *satisfies* $0 \leq f - f_- \leq J \leq B_\kappa := \kappa/(\vartheta \tilde w_0)$*,* $\tilde w_0 = w_0(\widetilde{W}^0)$*. Hence if* $0 < \kappa < \kappa_2 := \min(\kappa_1, \delta\vartheta\tilde w_0)$ *the solution is unique and equals* f^κ*.*

(c) *(*the face is not approached when it is forbidden*) If* $|f_-| < M$ *(so* $c_3' < \bar c_3$*), every solution has* $J \geq M - f_-$ *(hence* $\|f - f_-\| \not\to 0$ *as* $\kappa \to 0$*).*

(d) *(*the rest state*) Let* $c' = 0$ *and* $f_- = 0$*. Then* $\widetilde{W}^0 = W_{\mathrm{e}} \equiv 1$ *and for* $\kappa \to 0$ *the unique solution satisfies* $f = j\,\Sigma^0_\vartheta + O(\kappa^3)$ *uniformly and*

$$J = j - \tfrac{1}{4} c_\vartheta\, j^3 + O(j^5), \qquad w_0 = 2 + \tfrac{1}{2} c_\vartheta J^2 + O(J^4).$$

PROOF. (a) $W_{f_-} = W_{(c_1',c_2',c_3'+f_-^2/4)}$ is admissible by Lemma 3.35(b) and (Adm′), so $f_- \in \mathcal{A}_0$ ($\mathcal{A}_0$ is open, it is the domain of T_D). By Lemma 8.27(a) applied to the datum with $\kappa = 1$, $G = T_{D,1} - f_-$ is C^1 on $\mathcal{A}_0$, so $\mathcal{F}(\kappa, f) = f - f_- - \kappa G(f)$ is C^1 on $\mathbb{R} \times \mathcal{A}_0$ with values in $\mathrm{BC}(\mathbb{R})$. $\mathcal{F}(0, f_-) = 0$ and $D_f\mathcal{F}(0, f_-) = I$ is invertible. The implicit function theorem (Theorem A.3) gives κ_1, δ and a C^1 map $\kappa \mapsto f^\kappa$ with $f^0 = f_-$ such that the zeros of $\mathcal{F}$ in $(-\kappa_1, \kappa_1) \times \{\|f - f_-\| < \delta\}$ are exactly (κ, f^κ). Differentiating $f^\kappa = f_- + \kappa G(f^\kappa)$ at $\kappa = 0$ gives $\frac{d}{d\kappa} f^\kappa|_0 = G(f_-) = f_1$, and $f_1 = \frac{1}{\vartheta w_0}\Sigma_{f_-}$ by Lemma 8.10. (b) Let f be a fixed point; W_f is admissible and $f = f_- + J\Sigma_f \geq f_-$, so $f^2 \geq (f_-^+)^2$ and $W_f \geq \widetilde{W}^0$. Now $\widetilde{W}^0 = W_{(c_1',c_2',c_3'+(f_-^+)^2/4)}$ is admissible by (Adm) and Lemma 3.35(b), so Lemma 3.23 gives $w_0(W_f) \geq \tilde{w}_0$, $J = \kappa/(\vartheta w_0(W_f)) \leq B_\kappa$ and $0 \leq f - f_- \leq J$. If $\kappa < \delta\vartheta\tilde{w}_0$ then $B_\kappa < \delta$, so f lies in the ball of (a); if also $\kappa < \kappa_1$, $f = f^\kappa$. (c) This is the necessary condition Proposition 8.18 used in Corollary 8.45: if $|f_-| < M$ then $\max(|f_-|, |f_+|) \geq M$ forces $f_+ \geq M$ (as $f_+ \geq f_- > -M$), so $J \geq M - f_-$ for all κ. (d) $W_{\mathrm{e}} \equiv 1$, $\widetilde{W}^0 \equiv 1$, so (Adm) holds ($0 > -4 = \bar{c}_3$), $u_\pm^{(0)} = e^{\pm\xi}$, $\rho^{(0)} \equiv 2$, $w_0^{(0)} = 2$, $R^{(0)} = e^{2\xi}$ and $\Sigma_\vartheta^0 = R^{(0)}/(R^{(0)} + \vartheta)$. Let $f = J\Sigma$ be a solution (κ small, hence unique by (b)); by (b) $J \leq \kappa/(2\vartheta) = j$ and $f = O(j)$. Put $\delta_W := W_f - 1 = \frac{1}{2}\mathcal{Q}[f^2]$, so $0 \leq \delta_W \leq \frac{1}{8}J^2\operatorname{sech}^2\xi \leq \frac{1}{2}J^2 e^{-2|\xi|}$.

Step 1: $\delta\rho$. By Lemma 3.23 (with $W_{\mathrm{ref}} \equiv 1$) $\zeta_- \geq 1$, $\zeta_+ \leq -1$. Lemma 7.7 in the form of Lemma 3.23 gives, with $u_1 = u_-^{W_f}$ (normalised) and $u_2 = e^r$,

$$\zeta_-^{W_f}(\xi) - 1 = \int_{-\infty}^{\xi} \delta_W(r)\frac{u_1(r)e^r}{u_1(\xi)e^\xi}dr \leq \int_{-\infty}^{\xi} \delta_W(r)e^{-2(\xi-r)}dr,$$

because $u_1(r)/u_1(\xi) = \exp(-\int_r^\xi \zeta_-^{W_f}) \leq e^{-(\xi-r)}$; mirror: $0 \leq -1 - \zeta_+^{W_f}(\xi) \leq \int_\xi^\infty \delta_W(r)e^{-2(r-\xi)}dr$. So with $\delta\rho := \rho - 2$,

$$\begin{aligned} 0 \leq \zeta_-^{W_f} - 1, \quad -1 - \zeta_+^{W_f} &\leq \int_{\mathbb{R}} \delta_W(r)e^{-2|\xi-r|}dr \\ &\leq \tfrac{1}{2}J^2 \int e^{-2|r|-2|\xi-r|}dr = \tfrac{1}{2}J^2(|\xi| + \tfrac{1}{2})e^{-2|\xi|}, \end{aligned}$$

and $0 \leq \delta\rho \leq J^2(|\xi| + \frac{1}{2})e^{-2|\xi|}$. In particular $\int_{\mathbb{R}} \delta\rho \leq C_1 J^2$ and $\int_{\mathbb{R}}(\zeta_-^{W_f} - 1) \leq C_1 J^2$.

Step 2: the Jost solutions. $u_1(r)e^{-r} = \exp\int_0^r(\zeta_-^{W_f} - 1) \in [e^{-C_1J^2}, e^{C_1J^2}]$ for all $r \in \mathbb{R}$, i.e. $u_-^{W_f}(r) = e^r(1 + O(J^2))$ uniformly; likewise $u_+^{W_f}(r) = e^{-r}(1 + O(J^2))$. Inserting this in the formulas of Step 1 at $\xi = 0$ (where $u_i(0) = 1$),

$$w_0 = \rho(0) = 2 + \int_{\mathbb{R}} \delta_W(r)e^{-2|r|}\big(1 + O(J^2)\big)dr = 2 + \int_{\mathbb{R}} \delta_W e^{-2|r|}dr + O(J^4),$$

using $\int \delta_W e^{-2|r|} \leq CJ^2$.

Step 3: the profile. $\ln R(\xi) = \int_0^\xi \rho = 2\xi + O(J^2)$ uniformly by Step 1 ($\ln R(0) = 0$), and $\Sigma = \lg(\ln R - \ln\vartheta)$ with $\lg' \leq \frac{1}{4}$, so $\|\Sigma - \Sigma_\vartheta^0\|_\infty \leq \frac{1}{4}C_1J^2$. Then $|\Sigma^2 - (\Sigma_\vartheta^0)^2| \leq 2|\Sigma - \Sigma_\vartheta^0|$ and $|\mathcal{Q}[\Sigma^2] - \mathcal{Q}[(\Sigma_\vartheta^0)^2]| \leq CJ^2\operatorname{sech}^2$. Since $\delta_W = \frac{1}{2}J^2\mathcal{Q}[\Sigma^2]$,

$$\int \delta_W e^{-2|r|}dr = \tfrac{1}{2}J^2\big(c_\vartheta + O(J^2)\big), \qquad w_0 = 2 + \tfrac{1}{2}c_\vartheta J^2 + O(J^4).$$

Step 4. $J = \kappa/(\vartheta w_0)$ means $J\,(2 + \frac{1}{2}c_\vartheta J^2 + O(J^4)) = \kappa/\vartheta = 2j$, i.e. $J + \frac{1}{4}c_\vartheta J^3 + O(J^5) = j$. Hence $J \leq j$ and $J = j + O(j^3)$, so $J^3 = j^3 + O(j^5)$ and $J =$

$j - \frac{1}{4}c_\vartheta j^3 + O(j^5)$. Finally $f = J\Sigma = (j + O(j^3))(\Sigma^0_\vartheta + O(j^2)) = j\Sigma^0_\vartheta + O(j^3)$. $c_\vartheta > 0$ because the integrand is positive, and $c_\vartheta < \infty$ since $\mathcal{Q}[\cdot] \le \frac{1}{4}\operatorname{sech}^2$. □

Numerical observation 8.54 (the constant c_ϑ and the expansion)**.** *Quadrature (scipy; tolerance 10^{-10}) gives $c_\vartheta = 0.0626935404$ for $\vartheta = 1$ and $c_\vartheta = 0.0254633439$ for $\vartheta = 4$. For $\vartheta = 1$ this agrees with $\frac{\pi^2}{12} + \frac{1}{2}\ln^2 2 - 1 = 0.06269354038321\ldots$ to 10^{-13} (the closed form was found by an integer-relation (PSLQ) search; it is a numerical identification, not a proof). The fixed-point solver gives $(J - j)/j^3 = -0.01567, -0.01563, -0.01549, -0.01496$ for $\vartheta = 1$, $\kappa = 0.2, 0.5, 1, 2$, against $-c_\vartheta/4 = -0.015673$, and $-0.006366, -0.006366, -0.006364, -0.006360$ for $\vartheta = 4$, against $-c_\vartheta/4 = -0.006366$. Code: `code/ch08_effective/verify_ch08b.py`, parts 2–3 (values asserted at the printed precision).*

Remark 8.55 (the critical bubbles)**.** For $0 \le f_- < M$ and $\kappa \downarrow 0$ one expects, and Chapter 11 proves (Theorem 11.6), that the solutions converge to the critical bubbles with total jump $J \to (1 + \vartheta)(M - f_-)$; part (c) above gives only the lower bound $J \ge M - f_-$.

Remark 8.56 (the face, the collar and the chart)**.** (i) In effective coordinates the face $\kappa = 0$ is $\{(c', \vartheta, f_-) : c'_3 + \frac{1}{4}f_-^2 > \bar{c}_3\}$ and each of its points is one explicit no-swirl solution (classified in Theorem 3.44: the potential has constants $c = (c'_1, c'_2, c'_3 + \frac{1}{4}f_-^2)$ by Proposition 2.27(iii) with $\psi_{\boldsymbol{c}} = 0$, and $z = u_- + \vartheta u_+$ is a point of the open cone of positive solutions); (a) says that near the face the effective data $(c', \vartheta, f_-, \kappa)$ are a C^1 chart of the set of solutions, which is a collar $\{\kappa \ge 0\}$ of the face. (ii) The sign-reversed range $\kappa < 0$ in (a) is the same solution set reflected by σ_1 (Lemma 5.1); only $\kappa > 0$ is physical. (iii) Beyond the collar the solution set is described by the degree theory and the fold analysis of Chapter 9; below the face ($c'_3 < \bar{c}_3$ and $0 \le f_- < M$ with $M = 2\sqrt{\bar{c}_3 - c'_3}$) by the bubbles of Chapter 11 (Theorem 11.6; the range $f_- < 0$ is not described there); the datum with $f_- < -M$, $c'_3 < \bar{c}_3$ has, for small κ, the trivial branch of (a) *and* bubble branches, with indices adding up to 1 (Atlas 9.76). (iv) That T is real-analytic in the data, so that the collar is analytic, is Proposition 9.16.

8.8. Examples, numerical illustrations and the tools

In words. *Three worked examples fix the vocabulary: a constant swirl (nothing happens except that c'_3 drops), the Landau family (the whole no-swirl cone at $c' = 0$), and the first-order solution at the rest state with the dictionary between the effective constant κ_{D} and the slope of the swirl at the equator. Then the fixed-point iteration is shown for one datum, and the tools of the chapter are collected with their places in Appendix A.*

Example 8.57 (effective data of the explicit solutions)**.** (a) *Trivial swirl.* Let $f \equiv A$, constants $c = (c_1, c_2, c_3)$. Then $\psi_{\boldsymbol{c}} = 0$, $Q = P_{\boldsymbol{c}}$, and $\lambda = \mathsf{L}[P_{\boldsymbol{c}}] + A^2 = -2c_1 - 2c_2 - 4c_3 + A^2$ gives $c'_i = c_i$ $(i = 1, 2)$, $c'_3 = -\frac{1}{4}(\lambda + 2c'_1 + 2c'_2) = c_3 - \frac{1}{4}A^2$, $f_- = A$, $J = 0$, $\kappa_{\mathrm{D}} = 0$ (Proposition 2.27(iii) with $\psi_{\boldsymbol{c}} = 0$). $Q_{\mathrm{sw}} = P_{\boldsymbol{c}} - P_{\boldsymbol{c}'} = \frac{1}{4}A^2(1 - x^2)$: a constant swirl lowers c'_3 by $\frac{1}{4}A^2$ and is compensated exactly by the centrifugal term. In particular the potential of a constant swirl is W_c, independent of A. (b) *Landau jets.* Take $c' = 0$, $f_- = 0$, $\kappa_{\mathrm{D}} = 0$. Then $f \equiv 0$, $W = W_{\mathrm{e}} \equiv 1$, $u_\pm = e^{\pm\xi}$,

$z = e^{\xi} + \vartheta e^{-\xi}$, and with $x = \tanh\xi$,

$$\frac{z'}{z} - x = \frac{(1-\vartheta)(1-x^2)}{(1+\vartheta)+(1-\vartheta)x},$$
$$U = 2\Big(\frac{z'}{z} - x\Big) = \frac{2(1-x^2)}{\lambda_L + x}, \quad \lambda_L = \frac{1+\vartheta}{1-\vartheta}\ (\vartheta \neq 1),$$

and $U \equiv 0$ for $\vartheta = 1$ (the rest state). These are Landau's solutions: $\vartheta \in (0,\infty)\setminus\{1\}$ corresponds to $|\lambda_L| > 1$, with $\vartheta = (\lambda_L - 1)/(\lambda_L + 1)$; the endpoints $\vartheta = 0, \infty$ ($z = u_-$ resp. u_+, recessive at one pole) are the endpoints of the no-swirl interval. (Verified by computer algebra: `verify_ch08b.py`.) (c) *The first-order solution and the dictionary of* κ. At the rest state, $T_D(0) = f_- + \frac{\kappa_D}{2\vartheta}\Sigma^0_\vartheta$ with $\Sigma^0_\vartheta = 1/(1+\vartheta e^{-2\xi})$ is the first Picard iterate; the integral $\int_{-\infty}^{\xi}(e^{\eta}+\vartheta e^{-\eta})^{-2}d\eta = \frac{1}{2\vartheta}\frac{e^{2\xi}}{e^{2\xi}+\vartheta}$ is the content of Lemma 8.10 for $W \equiv 1$. Since $z(0) = u_-(0) + \vartheta u_+(0) = 1 + \vartheta$, the slope of the swirl at the equator is $\kappa_0 = f'(0) = \kappa_D/(1+\vartheta)^2$, i.e. $\kappa_D = (1+\vartheta)^2\kappa_0$ (Lemma 2.15(4)), and $J = \kappa_D/(\vartheta w_0)$ with $w_0 = 2$ to first order. Example: $\vartheta = 3$, $\kappa_D = 8$ ($\kappa_0 = \frac{1}{2}$): first order $J \approx \frac{4}{3}$, $f(0) = J/(1+\vartheta) \approx \frac{1}{3}$, $U(0) \approx 2\frac{1-\vartheta}{1+\vartheta} = -1$; the fixed point of T_D (fixed-point solver, residual 10^{-13}) has $J = 1.3153$, $f(0) = 0.3288 = J/4$ (this relation is exact), $w_0 = 2.0273$, $U(0) = -1.0266$ (`verify_ch08b.py`, part 4).

Numerical observation 8.58 (Picard iteration of T_D and the profiles)**.** *For $c' = 0$, $f_- = 0$, $\vartheta = 1$ (`code/ch08_effective/make_iteration.py`, `make_profiles.py`, `verify_ch08b.py` part 5) the undamped iterates $f_{n+1} = T_D f_n$, $f_0 \equiv f_-$, converge rapidly for $\kappa = 1, 4, 16$ (fixed points $J = 0.4981, 1.8941, 5.5162$ against the bound $B = \kappa/(2\vartheta) = 0.5, 2, 8$; leading eigenvalues of $\mathrm{D}T_D$ at the fixed point -0.007, -0.089, -0.470). For $\kappa = 64$ (fixed point $J = 11.887$) they converge slowly, with alternating errors: $J_n = 32$, 5.09, 21.6, 7.15, 17.5, 8.54, 15.4, 9.53, 14.2, 10.2, 13.4, 10.7, ..., the error decreasing at the rate 0.83 of the leading eigenvalue of $\mathrm{D}T_D$ ($|J_n - J| < 4\cdot 10^{-6}$ at $n = 79$). The eigenvalue tends to -1 as κ grows (-0.94 at $\kappa = 256$), which is why the fixed points are computed with damping $\omega = 0.7$. The profiles (Figure 8.4) show the linear regime $J \approx j$ ($\kappa \lesssim 1$) turning into the logistic layer and $J \sim \sqrt{2\kappa/\vartheta}$ (Theorem 10.11).*

Figure 8.3 shows the Picard iteration for $\boldsymbol{c}' = 0$.

Remark 8.59 (the tools of the chapter, and where their statements are)**.** All general theorems used in Chapter 8 are stated precisely and referenced in Appendix A: the Arzelà–Ascoli criterion on the line (Lemma A.29) and Schauder's theorem (Theorem A.31) for the existence proof of Chapter 8 on the convex set K; the Leray–Schauder degree with its five properties (Theorem A.35), the parity corollary (Corollary A.36) and the degree of a compact self-map of a convex set (Corollary A.37); the homotopy invariance on moving sets (Lemma A.39), used for the deformation in c_3'; the implicit function theorem (Theorem A.3) for Theorem 8.53; and the Leray–Schauder continuation principle (Theorem A.42), which is not needed in this chapter but gives the connected solution branches of Theorem 9.51. The roles are: Schauder produces *a* fixed point on a convex set; degree theory counts fixed points with sign *without* such a set; the index formula translates the count into spectral information; the implicit function theorem describes the solution near an isolated nondegenerate one.

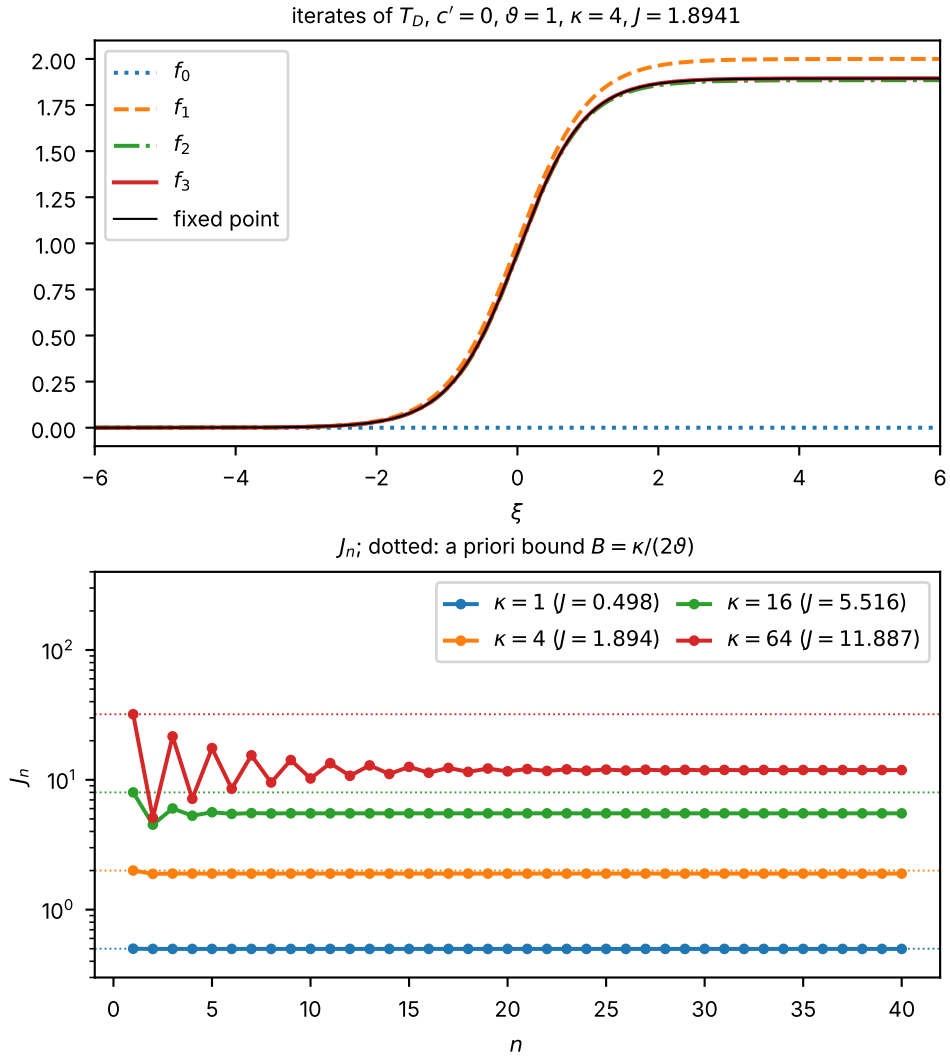


FIGURE 8.3. Picard iteration for $c' = 0$, $f_- = 0$, $\vartheta = 1$. Left: $f_0 \equiv 0$, $f_1 = \frac{\kappa}{2}\Sigma_1^0 = 2\Sigma_1^0$, f_2, f_3 and the fixed point ($\kappa = 4$, $J = 1.8941$). Right: $J_n = f_n(+\infty) - f_-$ for $\kappa = 1, 4, 16, 64$; dotted: the a priori bound $B = \kappa/(2\vartheta)$. The undamped iteration converges geometrically for $\kappa \le 16$; at $\kappa = 64$ the errors alternate in sign and decrease slowly (rate 0.83, the leading eigenvalue of $\mathrm{D}T_D$). Code: `code/ch08_effective/make_iteration.py`.

What we have, what comes next

We have proved that the Leray–Schauder degree of the fixed-point problem $f = T_D f$ is $+1$ for *every* effective datum (Theorem 8.42): a global solution of type

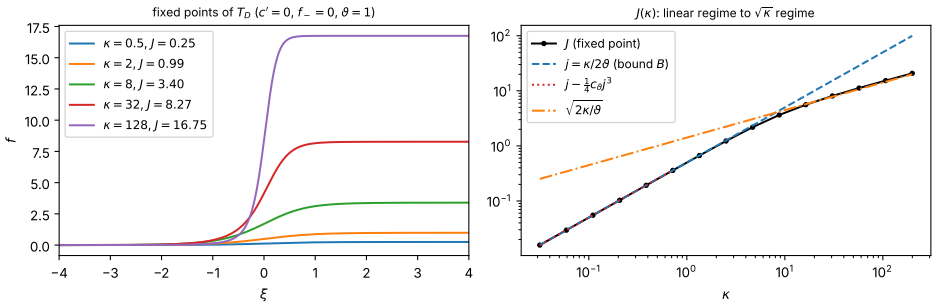


FIGURE 8.4. Left: the fixed points $f(\xi)$ for $c' = 0$, $f_- = 0$, $\vartheta = 1$ and $\kappa = 0.5, 2, 8, 32, 128$: a logistic step that sharpens and moves up with κ. Right: $J(\kappa)$ (dots) against the face asymptote $j = \kappa/2\vartheta$ (dashed), its first correction $j - \frac{1}{4}c_\vartheta j^3$ (dotted, Theorem 8.53(d)) and the large-swirl law $\sqrt{2\kappa/\vartheta}$ (dash-dotted, Theorem 10.11). Code: `code/ch08_effective/make_profiles.py`.

G at both poles exists for every $\kappa_{\mathrm{D}} > 0$ and every $\boldsymbol{c}'$ with $c_1', c_2' > -1$, including $c_3' < \bar{c}_3$ where no solution without swirl exists; the number of solutions is odd at nondegenerate data; the proof rests on the a priori bounds of Lemma 8.39 (uniform in the deformation $c_3' \mapsto c_3' + s$), which keep all fixed points in a bounded region where $w_0 \geq w_{\min} > 0$, and on the Jost lemmas of Chapter 3. The index criterion (Proposition 8.48) turns uniqueness into a spectral statement about $\mathrm{D}T_D$; the no-swirl face (Theorem 8.53) shows that for $c_3' + \frac{1}{4}f_-^2 > \bar{c}_3$ and small κ_{D} the only solution is the C^1 continuation of the trivial swirl, with $J = \frac{\kappa_{\mathrm{D}}}{2\vartheta} - \frac{1}{4}c_\vartheta(\frac{\kappa_{\mathrm{D}}}{2\vartheta})^3 + \dots$ at the rest state; examples and one iteration are in Section 8.8. Chapter 9 studies when the spectral hypothesis of the index criterion holds and how it fails (sign structure of $\mathrm{D}T_D$, folds and cusps, the first and last fold, the computer-assisted certificate at D^*); Chapter 10 uses the a priori bounds and the compactness of this chapter to prove the $\sqrt{\kappa}$ law, and Chapter 11 the same for the critical bubble as $\kappa_{\mathrm{D}} \to 0$ below the no-swirl set.

CHAPTER 9

Uniqueness and non-uniqueness: folds, the ridge, and the certificate

In words. *Chapter 8 turned the existence problem into a fixed-point problem $f = T_D f$ and proved that it always has a solution, that the number of solutions is odd when they are all nondegenerate, and that it is exactly one for moderate swirl. This chapter asks what happens in between. The answer has a geometric form. A global solution can be described by its Cauchy data ω at the equator, a point of an open subset of $\mathbb{R}^6$, or by its effective data D, a point of a six-dimensional space; the map $E : \omega \mapsto D$ is smooth, and "two solutions with the same D" means that E is not injective. Where E is a local diffeomorphism the solutions are locally unique and depend smoothly on D; where it is not, the linearisation $I - \mathrm{D}T_D$ is singular, and the generic singularity is a Whitney* fold*: a branch of solutions turns back, and on one side of a hypersurface in D-space two more solutions appear, with Leray–Schauder indices $+1$ and -1. Folds end at* cusps*. The linearisation has a sign structure that restricts where a fold can sit: if the swirl is non-negative, a fold cannot have a non-negative kernel vector. We prove all of this from scratch: the sign structure, the analyticity of T_D in (D, f) and smoothness of E, the translation of the Lyapunov–Schmidt reduction into folds and cusps of E, and the structure of the fold set, including the compactness (properness of E) which turns a local covering statement into a global one.*

Roadmap. Section 9.1 is the sign structure of $\mathrm{D}T_D$ (Lemma 9.2), a sharpening of Lemma 8.27(c) that includes the case of opposite signs. Section 9.2 fixes the Cauchy chart and the map E (Lemma 9.11), proves that the Jost data of a potential in the exponential class depend real-analytically on it (Lemma 9.15), and deduces Proposition 9.16: $T_D f$ is real-analytic in (D, f) and E is C^∞. Section 9.3 translates the fold and cusp lemmas of Appendix A into statements about E (Lemma 9.19). Section 9.4 defines the fold sets, proves that they coincide where a transversality condition holds (Lemma 9.21, Theorem 9.22), gives the local structure at a regular fold (Proposition 9.34, Proposition 9.40), proves the locally uniform a priori bounds and the properness of E (Lemma 9.37, Proposition 9.38) and collects everything in Theorem 9.42. The continuation in κ_{D} (first and last fold, Theorem 9.51), the mechanism (Heuristic 9.60), the computer-assisted three-solution theorem (Theorem 9.68, Corollary 9.69) and the numerical atlas (Atlas 9.76) are in the second half of the chapter.

Conventions. $' = \mathrm{d}/\mathrm{d}\xi$, $\mathrm{BC} = \mathrm{BC}(\mathbb{R})$, $\mathbb{D}_+ = \{D = (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}}) \in \mathbb{D} : \ c_1', c_2' > -1\}$ (an open subset of $\mathbb{R}^3 \times (0,\infty) \times \mathbb{R} \times (0,\infty) = \mathbb{D} \cong$ an open subset of $\mathbb{R}^6$; it is contractible, being homeomorphic to $(-1,\infty)^2 \times \mathbb{R} \times (0,\infty) \times \mathbb{R} \times (0,\infty)$). For $D \in \mathbb{D}_+$, W_{e}, $W_f = W_{\mathrm{e}} + \frac{1}{2}\mathcal{Q}[f^2]$, $\mathcal{A}_D = \{f \in \mathrm{BC} : W_f \text{ admissible}\}$, T_D, $\Gamma_{\ker}$ are those of Chapter 8; $k_i' = \sqrt{1 + c_i'}$. Sign convention: $w_0 = \mathrm{Wr}(u_+, u_-) = \zeta_-(0) - \zeta_+(0) > 0$.

"Real-analytic" for a map between Banach spaces means locally the sum of a norm-convergent series of bounded homogeneous polynomials ([8], [7, §4]); compositions of analytic maps are analytic, bounded multilinear maps are analytic, and inversion is analytic on the invertible elements of a Banach algebra. In this chapter a bare κ is the swirl constant $\kappa = \kappa_{\mathrm{D}}$ (normalisation 4 of Lemma 2.15); the Cauchy data at the equator $x_b = 0$ are expressed through $\kappa_0 = f'(0) = f_x(0)$ (normalisation 3), and $\kappa_{\mathrm{D}} = (1+\vartheta)^2\kappa_0$.

9.1. The sign structure of the derivative of T_D

In words. *Raising the swirl f raises the potential, and a higher potential makes the Jost solutions grow faster; this changes the height $J = \kappa_{\mathrm{D}}/(\vartheta w_0)$ and the position $\ell = \ln(u_-/u_+)$ of the layer. The first variation of the Jost logarithmic derivatives is the integral of the variation of the potential against a positive kernel, so it has the sign of δW. Inserted in the formula for $\mathrm{D}T_D$ this gives the following picture: if $f\,\delta f \geq 0$, then $\mathrm{D}T_D(f)\,\delta f$ is* negative *on the left half-line (the antichain property of Lemma 8.27), and if $f\,\delta f \leq 0$ it is* positive *there, bounded below by a multiple of the number $|\delta\rho(0)|/\rho(0)$. The consequence is a restriction on eigenvectors: for $f \geq 0$ a non-negative v can never be an eigenvector for a non-negative real eigenvalue, because $\mathrm{D}T_D v$ has the wrong sign on $\xi \leq 0$. In particular a fold with non-negative swirl has a kernel vector that changes sign.*

Toy example 9.1 (the sign structure at the rest state)**.** Take $\boldsymbol{c}' = 0$, $\vartheta = 1$, $f_- = 0$ and the potential $W \equiv 1$ (the potential of $f \equiv 0$; Example 3.22): $u_\pm = e^{\mp\xi}$ (u_+ recessive at $+\infty$), $\rho \equiv w_0 = 2$. A perturbation $\delta W = \varepsilon\,\operatorname{sech}^2\xi$ ($\varepsilon > 0$, this is $\delta W = \mathcal{Q}[g]$ with $g \equiv 4\varepsilon$ by Lemma 8.2(a)) gives

$$\delta\zeta_-(\xi) = e^{-2\xi}\int_{-\infty}^{\xi} e^{2\eta}\varepsilon\operatorname{sech}^2\eta\,\mathrm{d}\eta = \varepsilon\Phi_-(\xi),$$
$$\delta\zeta_+(\xi) = -\varepsilon\Phi_+(\xi), \qquad \delta\rho = \varepsilon(\Phi_- + \Phi_+) > 0,$$

with $\Phi_\mp$ as in Toy example 8.24; the substitution $t = e^{2r}$ gives $\Phi_-(0) = \int_0^1 \frac{2t}{(1+t)^2}\mathrm{d}t = 2\ln 2 - 1$, so $\delta\rho(0)/\rho(0) = \varepsilon(2\ln 2 - 1)$. By formula (8.6) of Lemma 8.27, with $\varphi(0) = \frac{1}{2}$ and $J = \kappa_{\mathrm{D}}/2$, at $\xi = 0$ the derivative equals $\mathrm{D}T_D\,\delta f(0) = -\frac{\kappa_{\mathrm{D}}}{4}(2\ln 2 - 1)\varepsilon < 0$, and for $\xi < 0$ it is more negative still, because $-(1-\varphi)\int_\xi^0 \delta\rho < 0$. If instead $g \equiv -4\varepsilon$ the signs of $\delta\rho$ and of $\mathrm{D}T_D\,\delta f$ are reversed on $\xi \leq 0$. (The two constants are checked in `verify_ch09a.py`.)

Lemma 9.2 (sign structure of $\mathrm{D}T_D$)**.** *$\delta\zeta_-(\xi) = u_-^{-2}\int_{-\infty}^{\xi} u_-^2\,\delta W$, $\delta\zeta_+(\xi) = -u_+^{-2}\int_{\xi}^{\infty} u_+^2\,\delta W$, so $\delta\rho = \delta\zeta_- - \delta\zeta_+$ has the sign of δW. For a fixed point f and $\delta f \geq 0$, $\delta f \not\equiv 0$, $g = f\,\delta f$: if $g \geq 0$ then $(DT_D\delta f)(\xi) < 0$ for $\xi \leq 0$; if $g \leq 0$ then $(DT_D\delta f) \geq J\varphi|\delta\rho(0)|/\rho(0) > 0$ for $\xi \leq 0$. Corollary: if $f \geq 0$, DT_D has no eigenvector $v \geq 0$ with real eigenvalue $\nu \geq 0$, so every fold ($\kappa = \kappa_{\mathrm{D}}$) with $f_- \geq 0$ has a sign-changing kernel vector; a non-negative kernel vector forces $f_- < 0$. Here, for $D \in \mathbb{D}_+$ and a fixed point $f = T_D f$ (so $f \in \mathcal{A}_D$), $W = W_f$, $u_\pm$ are its Jost solutions, $\zeta_\pm = u_\pm'/u_\pm$, $\rho = \zeta_- - \zeta_+$, $\varphi = \lg(\ell_W - \ln\vartheta)$, $J = \kappa_{\mathrm{D}}/(\vartheta\rho(0))$ and $DT_D = \mathrm{D}T_D(f)$; for $\delta f \in \mathrm{BC}$ the variation of the potential is $\delta W = \mathcal{Q}[f\,\delta f]$ (so $g = f\,\delta f$ and $\delta W = \mathcal{Q}[g]$), and $\delta\zeta_\mp$, $\delta\rho$ denote the first variations of $\zeta_\mp, \rho$ in the direction δW, that is, the linear parts of $\zeta_\mp^{W+\delta W} - \zeta_\mp^W$ and $\rho_{W+\delta W} - \rho_W$ (these*

are the Fréchet derivatives at W along δW in the weighted norm). "Fold" means a fixed point with $1 \in \operatorname{spec} DT_D$, and a kernel vector is a $v \neq 0$ with $DT_D v = v$.

PROOF. *The first variations.* Let $\delta W = q$ be continuous with $\|q\|_w = \sup \cosh^2 \xi\, |q| < \infty$, and let t be small. Lemma 8.25(b), applied to $W_0 = W$ and the perturbation tq (so $\varepsilon = |t|\|q\|_w$ and $g_0\varepsilon \le \frac{1}{8}$ for $|t|$ small), gives

$$\begin{aligned} \left|\zeta_-^{W+tq} - \zeta_-^W - tA_W^- q\right| &\le 8 g_0 t^2 \|q\|_w^2\, \Phi_-^W, \\ \left|\zeta_+^{W+tq} - \zeta_+^W + tA_W^+ q\right| &\le 8 g_0 t^2 \|q\|_w^2\, \Phi_+^W, \end{aligned}$$

with $\Phi_\mp^W \in L^1 \cap L^\infty$ (Lemma 8.23(b)). Hence the first variations exist and are

$$\delta\zeta_- = A_W^- q = u_-^{-2} \int_{-\infty}^{\xi} u_-^2 q, \qquad \delta\zeta_+ = -A_W^+ q = -u_+^{-2} \int_{\xi}^{\infty} u_+^2 q,$$

(Definition 8.22), which are the formulas of the statement. So $\delta\rho = A_W^- q + A_W^+ q = \Phi_q^W$. If $q \ge 0$ both integrands are non-negative and $\delta\rho \ge 0$, and $\delta\rho > 0$ everywhere if moreover $q \not\equiv 0$ (Lemma 8.23(a)); if $q \le 0$ the signs reverse. This is the first assertion; the proof of Lemma 8.27 (Step 1) shows that for $q = \mathcal{Q}[f\,\delta f]$ this $\delta\rho$ is the one that enters (8.6).

The sign of $\mathrm{D}T_D\,\delta f$. Let $\delta f \ge 0$, $\delta f \not\equiv 0$ and $g = f\,\delta f$. Because $f = T_D f$ with $\kappa_{\mathrm{D}} > 0$, $f' = \kappa_{\mathrm{D}} z^{-2} > 0$ (Lemma 8.10): f is strictly increasing, so it has at most one zero, and as δf is continuous and not identically zero, $g \not\equiv 0$. Then $\mathcal{Q}[g] \neq 0$. By (8.6), for $\xi \le 0$,

$$\mathrm{D}T_D(f)\delta f(\xi) = J\varphi\Big[-\frac{\delta\rho(0)}{\rho(0)} - (1-\varphi)\int_{\xi}^{0} \delta\rho \Big], \qquad J\varphi > 0.$$

If $g \ge 0$ then $\mathcal{Q}[g] \ge 0$, $\mathcal{Q}[g] \not\equiv 0$, so $\delta\rho > 0$ and both terms in the bracket are negative: $\mathrm{D}T_D\,\delta f < 0$ on $\xi \le 0$. If $g \le 0$ then $\delta\rho < 0$ everywhere, $-\delta\rho(0)/\rho(0) = |\delta\rho(0)|/\rho(0) > 0$ and $-(1-\varphi)\int_\xi^0 \delta\rho = (1-\varphi)\int_\xi^0 |\delta\rho| \ge 0$, so $\mathrm{D}T_D\,\delta f \ge J\varphi|\delta\rho(0)|/\rho(0) > 0$ on $\xi \le 0$.

The corollary. Let $f \ge 0$ and suppose $\mathrm{D}T_D(f)v = \nu v$ with $v \ge 0$, $v \neq 0$ (we may take $v \in \mathrm{BC}$ real, since the operator is real) and real $\nu \ge 0$. Since $f = T_D f$ is strictly increasing with $f(-\infty) = f_-$, $f \ge 0$ gives $f > 0$ everywhere. So $g = fv \ge 0$ and $g \not\equiv 0$ (because $v \not\equiv 0$ is continuous and $f > 0$), and by the previous step $\nu v = \mathrm{D}T_D\, v < 0$ on $\xi \le 0$. But $\nu v \ge 0$: a contradiction. Now let f be a fixed point with $f_- \ge 0$ ($f \ge 0$) and suppose $1 \in \operatorname{spec} \mathrm{D}T_D(f)$; the operator is compact, so 1 is an eigenvalue with a real eigenvector v. If $v \ge 0$ or $v \le 0$ the previous paragraph (with $\nu = 1$, applied to $\pm v$) is contradicted. So every kernel vector takes both signs. For the contrapositive: if $v \ge 0$ (or ≤ 0) is a kernel vector of $I - \mathrm{D}T_D(f)$ then $f \not\ge 0$, so $f(\xi_0) < 0$ for some ξ_0, hence $f_- < f(\xi_0) < 0$ (monotone). □

Remark 9.3 (what the lemma uses, and what it does not say)**.** The first assertion holds for every $f \in \mathcal{A}_D$, the second for every $f \in \mathcal{A}_D$ with $g \not\equiv 0$; only the strict monotonicity of f (hence $g \not\equiv 0$) uses that f is a fixed point. The lemma says nothing about complex eigenvalues of $\mathrm{D}T_D$: the spectrum can be genuinely complex (the lowest fixed point at the certified datum has the pair $0.3504 \pm 0.1247\,\mathrm{i}$, Numerical observation 8.51), so the corollary is a statement about *real* eigenvalues $\nu \ge 0$ only, and it cannot be improved to "$\rho(\mathrm{D}T_D) < 1$"; indeed that stronger statement is false, because the effective map is not injective (Theorem 9.68). A quantitative use of the sign structure is the mechanism of Heuristic 9.60.

Numerical observation 9.4 (the sign structure at two fixed points)**.** *For the rest state ($\boldsymbol{c}' = 0$, $\vartheta = 1$, $f_- = 0$, $\kappa_{\mathrm{D}} = 2$, total swirl $J = 0.9850$, so $f \geq 0$) and the non-negative directions $v = \operatorname{sech}^2 \xi$, $v = \lg(8(\xi-1))$ and $v = \lg(-8(\xi+3))$, the discretised operator $\mathrm{D}T_D(f)v$ computed with the fixed-point solver of Appendix E is negative on all grid points with $\xi \leq 0$, as Lemma 9.2 requires ($g = fv \geq 0$). For $f_- = -1.5$, $\kappa_{\mathrm{D}} = 6$ ($J = 2.909$, so f changes sign near $\xi = 0$) and $v = \exp(-(\xi+3)^2/4)$ restricted to the region $\{f < -0.3\}$ (so $g = fv \leq 0$, $v \geq 0$) the image is positive on all grid points with $\xi \leq 0$; the three smooth directions of the first example are of mixed type there and the lemma makes no claim. The same run checks by finite differences the formula $\partial_{\kappa_{\mathrm{D}}} T = (f - f_-)/\kappa_{\mathrm{D}}$ of Proposition 9.25 (maximal discrepancy $5 \cdot 10^{-5}$ for a step 10^{-6}). Code:* `code/ch09_uniq/verify_ch09a.py`*; numerical, not part of any proof.*

Remark 9.5 (reading the lemma on the half-lines)**.** The restriction to $\xi \leq 0$ is not a choice of convention: on $\xi \geq 0$ the term $(1-\varphi)\int_0^\xi \delta\rho$ in (8.6) is non-negative for $\delta\rho \geq 0$ and the term $-\delta\rho(0)/\rho(0)$ is negative, so the two compete and no sign is forced; on $\xi \leq 0$ the factor $\int_0^\xi \delta\rho = -\int_\xi^0 \delta\rho$ has the same sign as $-\delta\rho$.

9.1.1. The structure of $\mathrm{D}T_D$: a rank-one part and a remainder. Formula (8.6) has two terms of different nature, and the sign structure becomes sharper if they are separated. Write, for $f \in \mathcal{A}_D$ and $W = W_f$,

$$\ell(\delta f) := \frac{\delta\rho(0)}{\rho(0)}, \quad \delta\rho = \Phi^W_{\mathcal{Q}[f\delta f]}, \qquad u := J\varphi, \qquad R\,\delta f := J\varphi(1-\varphi)\int_0^\xi \delta\rho,$$

so that $\mathrm{D}T_D(f)\,\delta f = -\ell(\delta f)\,u + R\,\delta f$. Here ℓ is a bounded linear functional on BC ($|\ell(\delta f)| \leq \frac{1}{4}\|f\|_\infty \|\delta f\|_\infty\, g_0 \Psi_W(0)$ by Lemma 8.23), $u \in \mathrm{BC}$ is the logistic profile of the layer, and R is bounded. $R = \mathrm{D}T_D(f) + u \otimes \ell$ is compact (a compact operator plus an operator of rank one, Lemma 8.27(b)).

Toy example 9.6 (the matrix determinant lemma)**.** In $\mathbb{R}^n$, $I - A = (I-R) + u\ell^\top$ with $A = -u\ell^\top + R$ and $\det(I-A) = \det(I-R)\,(1 + \ell^\top (I-R)^{-1}u)$; for $n=1$, $R = r$, $u = \ell = 1$: $\det = 1 - r + 1$. The infinite-dimensional version below holds for compact R.

Proposition 9.7 (the secular equation)**.** *Let $f \in \mathcal{A}_D$ and $\nu \in \mathbb{C} \setminus \operatorname{spec} R$ (R complexified). Then $\nu \in \operatorname{spec} \mathrm{D}T_D(f)$ iff*

$$\mathcal{F}(\nu) := 1 + \ell\big((\nu - R)^{-1}u\big) = 0, \tag{9.1}$$

and then $\ker(\nu - \mathrm{D}T_D(f))$ is spanned by $(\nu - R)^{-1}u$. In particular, if $1 \notin \operatorname{spec} R$, then $1 \in \operatorname{spec} \mathrm{D}T_D(f)$ iff $\mathcal{F}(1) = 0$, and the kernel of $I - \mathrm{D}T_D(f)$ is then one-dimensional, spanned by $v = (I-R)^{-1}u$.

Proof. $\nu - \mathrm{D}T_D(f) = (\nu - R) + u \otimes \ell = (\nu - R)\big(I + (\nu-R)^{-1}u \otimes \ell\big)$. An operator $I + a \otimes \ell$ ($a \in \mathrm{BC}$, $\ell \in \mathrm{BC}^*$) has kernel $\{x : x = -\ell(x)a\}$, which is $\operatorname{span}(a)$ if $1 + \ell(a) = 0$ and $\{0\}$ otherwise (apply ℓ to $x = -\ell(x)a$: $\ell(x)(1+\ell(a)) = 0$). It is Fredholm of index 0, so it is invertible iff its kernel is 0. Take $a = (\nu - R)^{-1}u$. □

Lemma 9.8 (eigenvectors: values at the poles)**.** *If $\mathrm{D}T_D(f)\,\delta f = \nu\,\delta f$ with $\nu \neq 0$, then $\delta f(-\infty) = 0$ and $\delta f(+\infty) = -\frac{J}{\nu}\,\ell(\delta f)$ (both limits exist). If $f \geq 0$ is a fixed point, $\delta f \geq 0$, $\delta f \not\equiv 0$ and ν is real, then $\nu < 0$ and $\delta f(+\infty) > 0$.*

PROOF. By (8.6), $\mathrm{D}T_D(f)\delta f = -\ell(\delta f)J\varphi + J\varphi(1-\varphi)b$ with $b(\xi) = \int_0^\xi \delta\rho$ bounded ($|b| \le \|\delta\rho\|_{L^1} \le C\|\delta f\|$, Lemma 8.23(c)). As $\xi \to -\infty$, $\varphi \to 0$, so the right side tends to 0; as $\xi \to +\infty$, $\varphi \to 1$ and $1-\varphi \to 0$, so it tends to $-J\ell(\delta f)$ (Lemma 8.10). Dividing by ν gives the first claim. For the second, $\nu \ge 0$ is excluded by Lemma 9.2 (the corollary, which uses $f \ge 0$); $\nu \ne 0$ and real, so $\nu < 0$; and $g = f\delta f \ge 0$, $g \not\equiv 0$ gives $\delta\rho > 0$, $\ell(\delta f) > 0$, so $\delta f(+\infty) = -J\ell/\nu > 0$. □

Numerical observation 9.9 (the secular function at the rest state)**.** *For $\boldsymbol{c}' = 0$, $\vartheta = 1$, $f_- = 0$ the fixed point has $J = 0.9850$ at $\kappa_{\mathrm{D}} = 2$ and $J = 2.6962$ at $\kappa_{\mathrm{D}} = 6$. Computing ℓ, $u = J\varphi$ and $R = \mathrm{D}T_D(f) + u \otimes \ell$ with the fixed-point solver of Appendix E and solving $(\nu - R)y = u$ by GMRES, the function $\mathcal{F}(\nu) = 1 + \ell((\nu - R)^{-1}u)$ of (9.1) vanishes (to $5 \cdot 10^{-14}$) at the leading eigenvalue of $\mathrm{D}T_D(f)$, which is real and negative: $\nu = -0.02552$ at $\kappa_{\mathrm{D}} = 2$ and $\nu = -0.16701$ at $\kappa_{\mathrm{D}} = 6$, in accordance with Lemma 9.8 (a non-negative eigenvector can only have a negative eigenvalue). Around the zero $\mathcal{F}$ is monotone: $\mathcal{F}(\nu \pm 0.3) = 1.110,\ 0.910$ at $\kappa_{\mathrm{D}} = 2$ and $2.887,\ 0.608$ at $\kappa_{\mathrm{D}} = 6$. Code:* `code/ch09_uniq/secular_check.py`*; numerical, not part of any proof.*

Figure 9.1 plots the secular function at the rest state.

Remark 9.10 (what the secular equation says)**.** A fold is the vanishing of one scalar function, $\mathcal{F}(1) = 0$: the rank-one term $-\ell(\cdot)u$ feeds the value of the profile at the origin back into the height J and the position of the layer, and R propagates it. For $f \ge 0$ and $\nu > 0$ the sign structure shows that $(\nu - R)^{-1}u$ cannot be non-negative; in the language of (9.1) this is $\ell((\nu - R)^{-1}u) > -1$ whenever $(\nu - R)^{-1}u \ge 0$. The mechanism is discussed in Heuristic 9.60.

9.2. Charts, the map E, and analyticity of T_D

In words. *The set of global solutions has two coordinate systems: the Cauchy data at the equator, which are global and smooth but say nothing about the structure of the problem, and the effective data, read off at the poles, which turn the problem into a fixed-point equation. Fold theory needs both: the Cauchy chart provides an open subset of $\mathbb{R}^6$ on which solutions are points, the effective chart provides the Banach-space equation. We prove that the change of chart E is C^∞ and that the equation is real-analytic. The only analysis is a statement about Jost solutions: the Volterra integral operators that define them depend real-analytically on the potential, with no smallness, because Volterra operators have spectral radius zero.*

Let $\Omega = \Omega(0)$ and Ω_{G} be as in Chapter 5 (base point $x_b = 0$, Cauchy data $\omega = (\boldsymbol{c}, \gamma, A, \tilde\kappa) \in \mathbb{R}^6$ with $\boldsymbol{c}$ the Taylor constants of Q at 0, $\gamma = U(0)$, $A = f(0)$, $\tilde\kappa = \kappa_x(0) = f'(0)$; Lemma 2.15(3)), and put

$$\mathrm{Sol} := \Omega_{\mathsf{G}} \cap \{\tilde\kappa > 0\}.$$

By Proposition 5.15, Sol is open in $\mathbb{R}^6$; by σ_1 (Lemma 5.7(c)) the part $\tilde\kappa < 0$ of Ω_{G} is the mirror image of Sol and nothing is lost. For $\omega \in \mathrm{Sol}$ write $(U, f, Q) = (U_\omega, f_\omega, Q_\omega)$ for its solution and $W_\omega = 1 + \frac{1}{2}Q_\omega$.

Lemma 9.11 (change of chart)**.** (a) *(Definition of E.) For $\omega \in \mathrm{Sol}$ let $D = E(\omega) \in \mathbb{D}_+$ be the unique effective data of Proposition 8.13(b) of the*

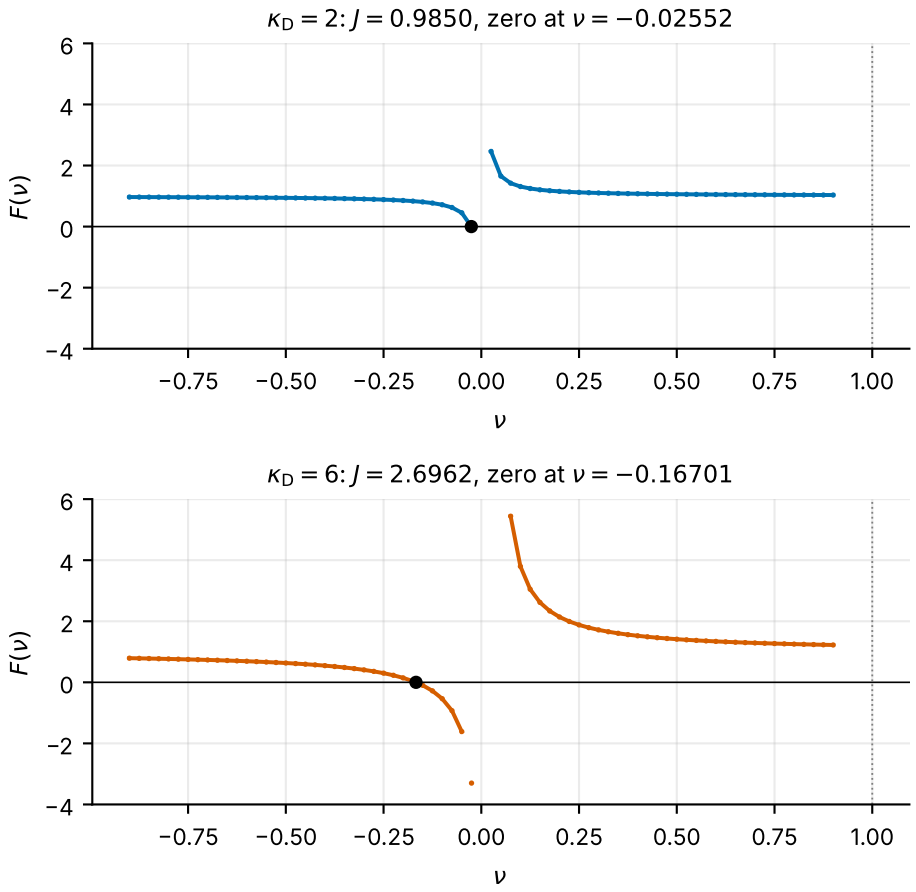


FIGURE 9.1. The secular function at the rest state (Numerical observation 9.9).

solution $(U_\omega, f_\omega, Q_\omega)$. *Then* $f_\omega = T_{E(\omega)} f_\omega$, $W_\omega = W_{E(\omega),f_\omega}$, *and, with* $u_\pm$ *the Jost solutions of* W_ω, $\rho_0 = \zeta_-(0) - \zeta_+(0) = w_0(W_\omega)$,

$$\begin{aligned} c_1' = \tfrac{1}{2} Q_\omega(-\infty), \quad c_2' = \tfrac{1}{2} Q_\omega(+\infty), \\ c_3' = -\tfrac{1}{4}(\lambda + 2c_1' + 2c_2'), \quad f_- = f_\omega(-\infty), \end{aligned} \tag{9.2}$$

$$\vartheta = \frac{1-a}{a}, \quad \kappa_{\mathrm{D}} = \frac{\tilde{\kappa}}{a^2} = (1+\vartheta)^2 \tilde{\kappa}, \qquad a = \frac{\gamma/2 - \zeta_+(0)}{\zeta_-(0) - \zeta_+(0)} \in (0,1), \tag{9.3}$$

where $\lambda = -2c_1 - 2c_2 - 4c_3 + A^2$ *is the first-integral constant of the solution.*

(b) *(*The inverse on fixed points.*) For* $(D, f) \in \mathbb{D}_+ \times \mathrm{BC}$ *with* $f \in \mathcal{A}_D$ *let* $Q = P_{\boldsymbol{c}'} + \mathcal{Q}[f^2]$ *and define* $\Sigma(D, f) = (\boldsymbol{c}, \gamma, A, \tilde{\kappa})$ *by*

$$\begin{aligned} c_3 = -\tfrac{1}{2} Q_{xx}(0), \quad c_{1,2} = \tfrac{1}{2} Q(0) \mp \tfrac{1}{2} Q_x(0) + \tfrac{1}{4} Q_{xx}(0), \\ \gamma = \frac{2(\zeta_-(0) + \vartheta\, \zeta_+(0))}{1+\vartheta}, \quad A = f(0), \quad \tilde{\kappa} = \frac{\kappa_{\mathrm{D}}}{(1+\vartheta)^2}, \end{aligned}$$

where $Q_x(0)=Q'(0)$ *and* $Q_{xx}(0)=Q''(0)$ *(at* $\xi=0$, $\partial_x=\partial_\xi$ *and* $\partial_x^2=\partial_\xi^2$ *because* $x=\tanh\xi$ *has* $x'(0)=1$, $x''(0)=0$*), and* $\zeta_\pm$ *belong to* W_f*. Then, if* $f=T_Df$*:* $\Sigma(D,f)\in\mathrm{Sol}$*,* $E(\Sigma(D,f))=D$ *and* $f_{\Sigma(D,f)}=f$*. Conversely* $\Sigma(E(\omega),f_\omega)=\omega$ *for every* $\omega\in\mathrm{Sol}$.

(c) *Consequently* $\omega\mapsto(E(\omega),f_\omega)$ *is a bijection from* Sol *onto the set* $\mathrm{Fix}:=\{(D,f):D\in\mathbb{D}_+,\ f\in\mathcal{A}_D,\ f=T_Df\}$*, with inverse* $\Sigma|_{\mathrm{Fix}}$.

Proof. (a) By Proposition 8.13(b) the solution is of type G at both poles and arises from exactly one $D\in\mathbb{D}_+$ with $f=T_Df$; the identification $W=W_f$, i.e. $Q=P_{\boldsymbol{c}'}+\mathcal{Q}[f^2]$, is stated there, and (9.2) is Proposition 2.27 together with Proposition 8.13(a) (effective constants, $f(-1)=f_-$). Let $z>0$ be the solution of $z''=Wz$ with the gauge $y=z/\cosh\xi$, $y(0)=1$, i.e. $z(0)=1$; then $U=2(z'/z-\tanh\xi)$ gives $z'(0)=\gamma/2$, and $\kappa_W[z]=f'z^2=\tilde\kappa$ at $\xi=0$ (Lemma 2.15(3)). As $u_\pm$ is a basis, $z=\alpha u_-+\beta u_+$ with $\alpha+\beta=z(0)=1$ and $\alpha\zeta_-(0)+\beta\zeta_+(0)=z'(0)=\gamma/2$; so $\alpha=a$ as in (9.3), $\beta=1-a$. By the construction in the proof of Proposition 8.13(b) ("coordinates"), $\alpha,\beta>0$, $\vartheta=\beta/\alpha$ and $\kappa_{\mathrm{D}}=\kappa_W[z/\alpha]=\tilde\kappa/\alpha^2$. (b) Let $f=T_Df$. By Proposition 8.13(a), (U,f,Q) with $z_f=u_-+\vartheta u_+$, $U=2(z_f'/z_f-\tanh\xi)$, is a global solution of type G at both poles with effective data D and swirl constant κ_{D} in the gauge $z=z_f$. The gauge $z(0)=1$ is $z=z_f/(1+\vartheta)$, in which the swirl constant at 0 is $\tilde\kappa=\kappa_{\mathrm{D}}/(1+\vartheta)^2$ and $\gamma=U(0)=2z'(0)/z(0)=2(\zeta_-(0)+\vartheta\zeta_+(0))/(1+\vartheta)$. The Taylor constants $\boldsymbol{c}$ of Q at 0 are determined by its 2-jet through the linear relations of Remark 2.6 ($Q(0)=c_1+c_2+c_3$, $Q_x(0)=c_2-c_1$, $Q_{xx}(0)=-2c_3$), which is how Σ is defined. Hence $\Sigma(D,f)$ are the Cauchy data of this solution; $\Sigma(D,f)\in\Omega_{\mathsf{G}}$ and $\tilde\kappa>0$. The uniqueness in Proposition 8.13(b) gives $E(\Sigma(D,f))=D$ and $f_{\Sigma(D,f)}=f$. The last identity of (b) is (a) read backwards. (c) is a restatement. □

Remark 9.12 (the third relation in (9.2))**.** λ is the first-integral constant of the solution (Proposition 2.20), $\lambda=\mathsf{L}[Q]+f^2$. Given the Cauchy data it is the polynomial $-2c_1-2c_2-4c_3+A^2$ of ω; this is the relation used in Example 8.7. In the definition of Σ we avoid λ and use the 2-jet of Q at 0, which is a bounded linear functional of f^2 on BC (Lemma 8.2(b): $\mathcal{Q}[g]\in C^2$ with $\mathcal{Q}[g],\mathcal{Q}[g]',\mathcal{Q}[g]''$ bounded by $C\|g\|_\infty$).

Example 9.13 (the conversion at the rest potential)**.** Let the potential be $W\equiv1$ ($\boldsymbol{c}'=0$, $f\equiv0$ would give this potential; for $\tilde\kappa\to0$ the solutions of Sol approach the no-swirl face, Example 5.9). Then $u_\pm=e^{\mp\xi}$, $\zeta_\pm(0)=\mp1$, $\rho_0=\zeta_-(0)-\zeta_+(0)=2$ and (9.3) reads

$$a=\frac{\gamma+2}{4},\qquad \vartheta=\frac{2-\gamma}{2+\gamma},\qquad \kappa_{\mathrm{D}}=\frac{16\,\tilde\kappa}{(\gamma+2)^2}\qquad(-2<\gamma<2).$$

As $\gamma\downarrow-2$ the mixing ratio $\vartheta\to\infty$ (the solution is almost purely u_+-dominated at $-\infty$) and as $\gamma\uparrow2$, $\vartheta\to0$; the endpoints $\gamma=\pm2$ are the points of $\Omega\setminus\Omega_{\mathsf{G}}$ (Example 5.9). So at the rest state E maps the interval of admissible γ diffeomorphically onto the half-line $\vartheta\in(0,\infty)$, and the Cauchy chart is a compactification of the effective chart in the directions $\vartheta\to0,\infty$. (Symbolic check in `verify_ch09a.py`.)

9.2.1. Jost data depend analytically on the potential. The potential of a swirl solution lies in $\mathcal{W}(k_1,k_2)$ with k_i and the profile of $W-k_i^2$ depending on parameters; we need an analytic dependence in the parameters of the Jost

logarithmic derivatives at 0 and of the profile $\varphi = \lg(\ell - \ln\vartheta)$. The tool is the Volterra inversion, and it is best seen in a toy.

Toy example 9.14 (the Volterra series to first order, and the first variation)**.** Let $k > 0$, $w(\eta) = a\,e^{-2\eta}$ on $[0,\infty)$ (so $\|w\| = a$ in the weighted norm below) and $K_k[w]h(\xi) = \int_\xi^\infty \frac{1-e^{-2k(\eta-\xi)}}{2k}\,w(\eta)h(\eta)\,\mathrm{d}\eta$. Then $K_k[w]1 = \frac{a\,e^{-2\xi}}{4(1+k)}$, hence $v = (1-K)^{-1}1 = 1 + \frac{a\,e^{-2\xi}}{4(1+k)} + O(a^2)$, and since $\zeta_+ = -k + v'/v$,

$$\zeta_+(0) = -k - \frac{a}{2(1+k)} + O(a^2).$$

This is the first variation of Lemma 9.2: $\delta\zeta_+(0) = -u_+(0)^{-2}\int_0^\infty u_+^2\,\delta W = -\int_0^\infty e^{-2k\eta}a\,e^{-2\eta}\mathrm{d}\eta = -\frac{a}{2(1+k)}$ for $u_+ = e^{-k\xi}$. The point of the lemma below is that the whole series $\sum_n K^n 1$ converges for *every* a (no smallness: $\|K^n\| \le (a/4k)^n/n!$), so that $a \mapsto \zeta_+(0)$ is real-analytic on the whole half-line $a > -\infty$ where $v(0) \neq 0$. (Checked by computer algebra in `verify_ch09a.py`.)

Let $\mathfrak{X}_+ = \{h \in C([0,\infty)) : \|h\|_{\mathfrak{X}_+} = \sup_{\xi\ge 0} e^{2\xi}|h(\xi)| < \infty\}$ and $\mathfrak{X}_- = \{h \in C((-\infty,0]) : \|h\|_{\mathfrak{X}_-} = \sup_{\xi\le 0} e^{-2\xi}|h(\xi)| < \infty\}$ (Banach spaces) and $\mathfrak{P} = (0,\infty)^2 \times \mathfrak{X}_- \times \mathfrak{X}_+ \ni p = (k_1,k_2,w_1,w_2)$. A continuous $W \in \mathcal{W}(k_1,k_2)$ defines $p(W) = (k_1,k_2,(W-k_1^2)|_{(-\infty,0]},(W-k_2^2)|_{[0,\infty)}) \in \mathfrak{P}$. For $p \in \mathfrak{P}$ put

$$(K_k^+[w]h)(\xi) = \int_\xi^\infty \frac{1-e^{-2k(\eta-\xi)}}{2k}w(\eta)h(\eta)\,\mathrm{d}\eta \quad (h \in C_b([0,\infty))),$$

and $K_k^-[w]$ the mirror operator on $C_b((-\infty,0])$, $(K_k^-[w]h)(\xi) = \int_{-\infty}^\xi \frac{1-e^{-2k(\xi-\eta)}}{2k}w(\eta)h(\eta)\mathrm{d}\eta$.

Lemma 9.15 (analytic dependence of the Jost data)**.** (a) *The maps $(k,w) \mapsto K_k^\pm[w] \in \mathcal{L}(C_b)$ are real-analytic on $(0,\infty)\times\mathfrak{X}_\pm$, $\|(K_k^\pm[w])^n\| \le (\|w\|/4k)^n/n!$, $1-K_k^\pm[w]$ is invertible for all (k,w), and $v_+ := (1-K_{k_2}^+[w_2])^{-1}1 \in C_b([0,\infty))$, $v_- := (1-K_{k_1}^-[w_1])^{-1}1 \in C_b((-\infty,0])$ are real-analytic functions of $p \in \mathfrak{P}$.*

(b) *Put $\zeta_+(0) = -k_2 + v_+'(0)/v_+(0)$ with $v_+'(0) = -\int_0^\infty e^{-2k_2\eta}w_2v_+\mathrm{d}\eta$, $\zeta_-(0) = k_1 + v_-'(0)/v_-(0)$ with $v_-'(0) = \int_{-\infty}^0 e^{2k_1\eta}w_1v_-\mathrm{d}\eta$, $\rho_0 = \zeta_-(0)-\zeta_+(0)$, and let $\mathfrak{P}_{\mathrm{adm}} = \{p : v_+ > 0$ on $[0,\infty)$, $v_- > 0$ on $(-\infty,0]$, $\rho_0 > 0\}$. Then $\mathfrak{P}_{\mathrm{adm}}$ is open in $\mathfrak{P}$ and $\zeta_\pm(0)$, ρ_0 are real-analytic on it.*

(c) *If $W \in \mathcal{W}(k_1,k_2)$ is continuous, then W is admissible iff $p(W) \in \mathfrak{P}_{\mathrm{adm}}$, and then $\zeta_\pm(0)$, $\rho_0 = w_0(W)$ are those of the Jost solutions of W (Definition 3.20).*

(d) *For $\vartheta > 0$ and $p \in \mathfrak{P}_{\mathrm{adm}}$ there are functions $\varphi^+ \in C_b([0,\infty))$, $\varphi^- \in C_b((-\infty,0])$, real-analytic in (ϑ,p), with $\varphi^+(0) = \varphi^-(0) = 1/(1+\vartheta)$, such that for $p = p(W)$ the function $\varphi = \lg(\ell_W - \ln\vartheta)$ of (8.2) equals $\varphi^\pm$ on $\{\pm\xi \ge 0\}$. Consequently*

$$\mathcal{T}(\vartheta, f_-, \kappa_{\mathrm{D}}; p) := f_- + \frac{\kappa_{\mathrm{D}}}{\vartheta\rho_0}\,\varphi \ \in \mathrm{BC}(\mathbb{R})$$

$$(\varphi = \varphi^- \text{ on } \xi \le 0,\ \varphi^+ \text{ on } \xi \ge 0)$$

is a real-analytic map $(0,\infty)\times\mathbb{R}\times(0,\infty)\times\mathfrak{P}_{\mathrm{adm}} \to \mathrm{BC}$, and $T_Df = \mathcal{T}(\vartheta, f_-, \kappa_{\mathrm{D}}; p(W_f))$ for $f \in \mathcal{A}_D$.

PROOF. (a) The kernel is $\kappa_k(\eta-\xi)$ with $\kappa_k(s)=\int_0^s e^{-2kr}\mathrm{d}r\in[0,\frac{1}{2k}]$. For $|k'-k|<k$, $\kappa_{k'}(s)=\sum_j \frac{(k'-k)^j}{j!}\int_0^s(-2r)^j e^{-2kr}\mathrm{d}r$ and $|\int_0^s(2r)^j e^{-2kr}\mathrm{d}r|\le j!/(2k^{j+1})$. Since $\int_0^\infty|w|\le\frac12\|w\|_{\mathfrak{X}_+}$, this expresses $K^+_{k'}[w']$ as $\sum_j(k'-k)^jB_j[w']$ with B_j linear in w' and $\|B_j[w']\|\le\|w'\|/(4k^{j+1})$; thus $(k',w')\mapsto K^+_{k'}[w']$ is a norm-convergent power series in $(k'-k,w')$, i.e. analytic (similarly at every w). Iterating the Volterra structure, $|(K^nh)(\xi)|\le\|h\|_\infty(2k)^{-n}\int_{\xi\le\eta_1\le\cdots\le\eta_n}\prod|w(\eta_j)|\le\|h\|_\infty(2k)^{-n}\frac{(\int_0^\infty|w|)^n}{n!}\le\|h\|_\infty\frac{(\|w\|/4k)^n}{n!}$, so the Neumann series converges, $1-K$ is invertible in the Banach algebra $\mathcal{L}(C_b)$ and $v_\pm=(1-K)^{-1}1$ is analytic as the composition of K with inversion ([8]). The mirror case is identical. (b) The functional $\lambda_k:g\mapsto\int_0^\infty e^{-2k\eta}g(\eta)\mathrm{d}\eta$ is bounded on $\mathfrak{X}_+$ ($|\lambda_kg|\le\|g\|/(2(k+1))$), analytic in $k>0$ because $|\int(2\eta)^je^{-2k\eta}g|\le\|g\|\,j!/(2(k+1)^{j+1})$, and $(w,v)\mapsto wv$ is bounded bilinear $\mathfrak{X}_+\times C_b\to\mathfrak{X}_+$; hence $v_+'(0)$ is analytic, as are $v_+(0)$ (evaluation is bounded linear), $\zeta_+(0)$ and, mirrored, $\zeta_-(0)$, ρ_0 (where $v_\pm(0)\neq0$). Openness: $v_+(\xi)\to1$ as $\xi\to\infty$ (from the Volterra equation and $\int_\xi^\infty|w|\to0$), so a positive continuous v_+ has $\inf v_+>0$ and stays positive under small perturbations in C_b; the same for v_-; $\rho_0>0$ is open. (c) Let $W\in\mathcal{W}(k_1,k_2)$. By Lemma 3.21(a) and its mirror, $\varphi^W_+=e^{-k_2\xi}v_+$ on $[0,\infty)$ and $\varphi^W_-=e^{k_1\xi}v_-$ on $(-\infty,0]$, where $v_\pm$ are the bounded solutions of the Volterra equations, i.e. the functions of (a) for $p(W)$ (uniqueness of the bounded solution). The logarithmic derivatives at 0 are $\zeta_+(0)=-k_2+v_+'(0)/v_+(0)$ and $\zeta_-(0)=k_1+v_-'(0)/v_-(0)$, with $v_\pm'(0)$ the integrals of (b) (differentiate the Volterra equation; the kernel vanishes on the diagonal). If W is admissible, $\varphi^W_\pm>0$ (Lemma 3.21(b)), so $v_\pm>0$ on the halves, and $\rho_0=w_0(W)>0$. Conversely let $p(W)\in\mathfrak{P}_{\mathrm{adm}}$. Then $\varphi^W_+>0$ on $[0,\infty)$ and $\varphi^W_->0$ on $(-\infty,0]$; continue both solutions by the ODE and normalise $u_\pm(0)=1$. The Wronskian $\mathrm{Wr}(u_+,u_-)$ is constant, equal to $\zeta_-(0)-\zeta_+(0)=\rho_0>0$ at 0. On $(-\infty,0]$: $(u_+/u_-)'=-\rho_0/u_-^2<0$, so $u_+/u_-\ge(u_+/u_-)(0)=1$ and $u_+>0$ there; on $[0,\infty)$: $(u_-/u_+)'=\rho_0/u_+^2>0$, so $u_-\ge u_+>0$ there. Thus $\varphi^W_->0$ on $\mathbb{R}$, and it is unbounded at $+\infty$: $u_-/u_+=1+\rho_0\int_0^\xi u_+^{-2}\to\infty$ since $u_+\le Ce^{-k_2\xi}$ gives $\int_0^\xi u_+^{-2}\ge c\,e^{2k_2\xi}$, and $u_-\ge\rho_0\,u_+\int_0^\xi u_+^{-2}\ge c'e^{k_2\xi}$ (using also $u_+\ge(\inf v_+)e^{-k_2\xi}/v_+(0)$). So W is admissible. (d) On $\xi\ge0$ set $E_+:=e^{-\ell}=u_+/u_-$ (here $\ell=\ln(u_-/u_+)=\int_0^\xi\rho$): from $(u_-/u_+)'=\rho_0u_+^{-2}$, $1/E_+=1+\rho_0\int_0^\xi u_+^{-2}=e^{2k_2\xi}\hat Y_+$ with

$$\hat Y_+(\xi)=e^{-2k_2\xi}+\rho_0\,v_+(0)^2\int_0^\xi e^{-2k_2(\xi-\eta)}v_+(\eta)^{-2}\,\mathrm{d}\eta\ \in C_b([0,\infty))$$

($u_+=e^{-k_2\xi}v_+/v_+(0)$). Now (i) $k\mapsto e^{-2k\xi}\in C_b([0,\infty))$ is analytic for $k>0$ ($\|(2\xi)^je^{-2k\xi}\|_\infty\le j!\,k^{-j}$); (ii) $h\mapsto\int_0^\xi e^{-2k(\xi-\eta)}h(\eta)\mathrm{d}\eta$ is bounded on C_b with norm $\le\frac{1}{2k}$, analytic in k by the same estimate as in (a); (iii) $v_+\mapsto v_+^{-2}$ is analytic on the open set of invertible elements of the Banach algebra $C_b([0,\infty))$ ($\inf v_+>0$). So $\hat Y_+$ is analytic in p. It is continuous, positive, and tends to $\rho_0v_+(0)^2/(2k_2)>0$ at ∞, so $\inf\hat Y_+>0$ and $\hat Y_+$ is invertible in C_b. Then $E_+=e^{-2k_2\xi}\hat Y_+^{-1}\in C_b$ is analytic, $0<E_+\le1$, and

$$\varphi^+:=\frac{1}{1+\vartheta E_+}\qquad(\xi\ge0)$$

(the denominator is ≥1, hence invertible in C_b) is analytic in (ϑ,p); and $\varphi^+=\frac{R}{R+\vartheta}=\lg(\ell-\ln\vartheta)$ with $R=u_-/u_+=1/E_+$. On $\xi\le0$ use instead $Y_-:=u_+/u_-=$

$1/R$, for which $(u_+/u_-)' = -\rho_0 u_-^{-2}$ gives $Y_- = 1 + \rho_0 \int_\xi^0 u_-^{-2} = e^{-2k_1\xi}\hat{Y}_-$ with $\hat{Y}_-(\xi) = e^{2k_1\xi} + \rho_0 v_-(0)^2 \int_\xi^0 e^{2k_1(\xi-\eta)} v_-(\eta)^{-2}\mathrm{d}\eta$, analytic into $C_b((-\infty,0])$ and bounded below by the same argument; then

$$\varphi^- := \frac{1}{1+\vartheta Y_-} = \frac{e^{2k_1\xi}}{e^{2k_1\xi} + \vartheta\hat{Y}_-}$$

is analytic (the denominator is $\geq \vartheta \inf \hat{Y}_- > 0$). Both equal $1/(1+\vartheta)$ at 0, where $R = 1$. The pair (φ^-, φ^+) lies in the closed subspace $\{(a,b) : a(0) = b(0)\} \cong \mathrm{BC}(\mathbb{R})$ of $C_b((-\infty,0]) \times C_b([0,\infty))$, so $\varphi \in \mathrm{BC}$ is analytic, and $\mathcal{T}$ is analytic since $\rho_0 > 0$. The last assertion is (8.2) together with (c). (Remark: a weaker fixed weight is not needed because $k \mapsto e^{\mp 2k\xi}$ is already analytic into C_b for $k > 0$.) □

Proposition 9.16 (smoothness of E, analyticity of T)**.** (R) *The set $\mathcal{A}$ of pairs $(D,f) \in \mathbb{D}_+ \times \mathrm{BC}$ with $f \in \mathcal{A}_D$ is open, the map $(D,f) \mapsto T_D f \in \mathrm{BC}$ is real-analytic on $\mathcal{A}$ (in particular C^k for every k), its partial derivative in f is the operator $\mathrm{D}T_D(f)$ of (8.6), and the map $\Sigma : \mathcal{A} \to \mathbb{R}^6$ of Lemma 9.11 is real-analytic.* (S) *$E : \mathrm{Sol} \to \mathbb{D}_+$ is C^∞, and $\mathrm{Sol} \ni \omega \mapsto f_\omega \in \mathrm{BC}(\mathbb{R})$ is C^∞ (sup norm).*

Proof. (R) The potential $W_{D,f} = W_\mathrm{e} + \frac{1}{2}\mathcal{Q}[f^2]$ lies in $\mathcal{W}(k_1', k_2')$ (Lemma 8.5(a)), and on the two halves

$$W - k_2'^2 = \tfrac{1}{2}\big[(c_1' - c_2')(1 - \tanh\xi) + c_3' \operatorname{sech}^2\xi + \mathcal{Q}[f^2]\big] \ (\xi \geq 0),$$
$$W - k_1'^2 = \tfrac{1}{2}\big[(c_2' - c_1')(1 + \tanh\xi) + c_3' \operatorname{sech}^2\xi + \mathcal{Q}[f^2]\big] \ (\xi \leq 0).$$

Since $1 \mp \tanh\xi \leq 2e^{-2|\xi|}$ on the relevant half, $\operatorname{sech}^2\xi \leq 4e^{-2|\xi|}$ and $\|\mathcal{Q}g\|_{\mathfrak{x}_\pm} \leq \|g\|_\infty$ (Lemma 8.2(a)), the map $\pi : (D,f) \mapsto p(W_{D,f}) = (k_1', k_2', w_1, w_2) \in \mathfrak{P}$ is affine in $\boldsymbol{c}'$ and quadratic in f except for the square roots $k_i' = \sqrt{1 + c_i'}$, which are analytic on $c_i' > -1$; hence π is real-analytic. By Lemma 9.15(c), $\mathcal{A} = \pi^{-1}(\mathfrak{P}_\mathrm{adm})$, which is open, and by Lemma 9.15(d), $T_D f = \mathcal{T}(\vartheta, f_-, \kappa_\mathrm{D}; \pi(D,f))$ is a composition of analytic maps. The Fréchet derivative in f is unique, so it is the operator computed in the proof of Lemma 8.27, formula (8.6). Finally Σ is built from the 2-jet of $Q = P_{\boldsymbol{c}'} + \mathcal{Q}[f^2]$ at 0 (affine in $\boldsymbol{c}'$, quadratic in f, bounded as noted in Remark 9.12) and from $\zeta_\pm(0)$ (analytic by Lemma 9.15(b)), $A = f(0)$ and $\tilde{\kappa} = \kappa_\mathrm{D}/(1+\vartheta)^2$.

(S) *Pole data.* Let $\omega_0 \in \mathrm{Sol}$. The scale-free state at $\xi_c = 0$ is $s(\omega) = (\gamma/2, \tilde{\kappa}, A, Q(0), Q'(0), \lambda)$ with $Q(0) = c_1 + c_2 + c_3$, $Q'(0) = c_2 - c_1$, $\lambda = -2c_1 - 2c_2 - 4c_3 + A^2$ (Remark 2.6); it is a polynomial diffeomorphism from $\mathbb{R}^6$ onto its image, and $s(\mathrm{Sol})$ is open and contained in the set $\mathcal{G}_0$ of states whose forward solution is of type G at $+\infty$ (Corollary 4.42). By Corollary 4.43(a) the quantities $\mu_+ = \sqrt{1 + \frac{1}{2}Q(+\infty)}$ and $f(+\infty)$ are C^∞ functions of the state, hence of ω. The involution σ_2 (Lemma 5.7(c)) is linear on $\mathbb{R}^6$, preserves Sol up to the sign of $\tilde{\kappa}$, exchanges the poles ($Q_{\sigma_2\omega}(x) = Q_\omega(-x)$, $f_{\sigma_2\omega}(x) = f_\omega(-x)$), and the same corollary applies to $\sigma_2\omega$ (the state may have either sign of k); thus $\mu_- = \sqrt{1 + \frac{1}{2}Q(-\infty)}$ and $f(-\infty)$ are C^∞ in ω. So $c_i' = \mu_\mp^2 - 1$, $f_- = f_\omega(-\infty)$ and $c_3' = -\frac{1}{4}(\lambda + 2c_1' + 2c_2')$ are C^∞. *Potential data.* Let $N \subset \mathrm{Sol}$ be a compact neighbourhood of ω_0. By Corollary 4.42(c) and Theorem 4.38(c) there is a cut $\xi_* > 0$ such that for all $\omega \in N$ the solution on $[\xi_*, \infty)$ is the chart solution $u_{p_+(\omega)}$, $Q_\omega = q_+(\omega) + 2w_{p_+(\omega)}$ with $q_+ = 2\mu_+^2 - 2$ (Theorem 4.38(a)), and $\omega \mapsto p_+(\omega) = \mathsf{S}_{\xi_*}^{-1}(\Phi_{0\to\xi_*} s(\omega))$ is C^∞ (Corollary 4.43(a) and the smooth flow of

Remark 2.7, Lemma 5.7(b)). By Theorem 4.38(b) $p \mapsto w_p \in C_E$ is C^∞, where C_E is the space with the weight $E = e^{-2\xi}$; so $\omega \mapsto \frac{1}{2}(Q_\omega - q_+)|_{[\xi_*,\infty)} = w_{p_+(\omega)}$ is C^∞ into C_E. On the compact interval $[0,\xi_*]$ the function $\frac{1}{2}(Q_\omega(\xi) - q_+(\omega))$ is C^∞ in (ω,ξ) jointly (the field of (2.22) is polynomial in the state), hence $\omega \mapsto$ it is C^∞ into $C([0,\xi_*])$ by Lemma A.15. The two pieces agree at ξ_*, and $\mathfrak{X}_+$ is isomorphic to the closed subspace of $C([0,\xi_*]) \times C_E$ of pairs with equal values at ξ_*; therefore $\omega \mapsto w_2(\omega) := (W_\omega - \mu_+^2)|_{[0,\infty)} = \frac{1}{2}(Q_\omega - q_+)|_{[0,\infty)} \in \mathfrak{X}_+$ is C^∞. Applying this to $\sigma_2\omega$ (for which $W_{\sigma_2\omega}(\xi) = W_\omega(-\xi)$) gives $w_1(\omega) \in \mathfrak{X}_-$. Hence $\omega \mapsto p(W_\omega) \in \mathfrak{P}$ is C^∞, with values in $\mathfrak{P}_{\mathrm{adm}}$ because W_ω is admissible (Lemma 9.11(a), Lemma 9.15(c)). *The map E.* By Lemma 9.15(b), $\zeta_\pm(0) = \zeta_\pm(0)(p(W_\omega))$ are C^∞ in ω, so a, ϑ and κ_{D} of (9.3) are C^∞ ($a \in (0,1)$). This proves that E is C^∞, and, since $\vartheta, f_-, \kappa_{\mathrm{D}}$ and $p(W_\omega)$ are C^∞ and $W_\omega = W_{E(\omega),f_\omega}$, the profile $f_\omega = T_{E(\omega)}f_\omega = \mathcal{T}(\vartheta, f_-, \kappa_{\mathrm{D}}; p(W_\omega))$ is C^∞ into BC by Lemma 9.15(d). □

Remark 9.17 (what is, and is not, used)**.** The proof of (R) needs nothing about the pole charts; it uses the formula $T_D f = f_- + J\varphi$ and the positivity of the Jost solutions. Pole charts enter (S) only because E is defined on Cauchy data, where the potential $W_\omega = 1 + \frac{1}{2}Q_\omega$ is known on a compact interval from the ODE and near the poles from the chart theorem. Part (R) is the regularity that the Lyapunov–Schmidt reduction needs: with C^1 only, the condition "$h_{aa} \neq 0$" would not make sense (Remark A.51).

9.3. Folds and cusps of the map E

In words. *A fixed-point equation $f = T_D f$ in a Banach space whose linearisation has a one-dimensional kernel behaves like* one *equation in* one *unknown a, the coordinate along the kernel vector (Lyapunov–Schmidt reduction). After the reduction the solution set is a graph $s = \chi(a,\lambda')$ over the unknown, and the projection to the parameters is a map of the scalar form $(a,\lambda') \mapsto (\chi,\lambda')$. If $\chi_{aa} \neq 0$ this is a Whitney fold: the solution branch turns back; if also $\chi_{aa} = 0$ and $\chi_{aaa} \neq 0$ with a nondegenerate unfolding, it is a cusp. In terms of the original data this says that E has a fold, resp. a cusp, and the coefficients are explicit: $\chi_{aa} = -\psi^* \mathrm{D}_f^2 T[v,v]/(\psi^* B\,\partial_s)$. The only extra input is that the solutions of the fixed-point problem* are *the points of the Cauchy chart (Lemma 9.11(c)), so the reduced coordinate a is a smooth coordinate on* Sol.

Toy example 9.18 (the cubic, as an E-like map)**.** On $X = C[0,1]$ let $T_{b,\tau}u = \frac{2}{3}(1+x)\big(b\,m(u) - m(u)^3 + \tau\big)$ with $m(u) = \int_0^1 u$ (Toy example A.44); parameters $D = (b,\tau) \in \mathbb{R}^2$. A fixed point is $u = c(1+x)$, $m = \frac{3}{2}c$, with $m^3 + (1-b)m = \tau$. Here the "Cauchy chart" is $\omega = (m,b) \in \mathbb{R}^2$ and "E" is $E(m,b) = (b,\ m^3 + (1-b)m)$, of the scalar form $(\varphi(m,b), b)$ with $\varphi = m^3 + (1-b)m$. $\mathrm{D}T(u)\delta u = \frac{2}{3}(1+x)(b - 3m^2)m(\delta u)$ has the single nonzero eigenvalue $\nu = b - 3m^2$, so $1 \in \operatorname{spec} \mathrm{D}T$ iff $\varphi_m = 1 - b + 3m^2 = 0$ (this is $\det \mathrm{D}E = \varphi_m$); the left functional is $\psi^* = m(\cdot)$ with $\psi^* \partial_\tau T = 1 \neq 0$ (transversality); $v = 1 + x$, $\psi^* \mathrm{D}_u^2 T[v,v] = -6m\,m(v)^2 \neq 0$ iff $m \neq 0$, and $\chi_{aa} = -\psi^* \mathrm{D}^2 T[v,v]/\psi^* B = 6m\,m(v)^2$ agrees with $\varphi_{mm} = 6m$ for the normalisation $m(v) = 1$ of the kernel vector. The origin $m = 0$, $b = 1$ is the cusp: $\varphi_{mmm} = 6 \neq 0$, $\varphi_{mb} = -1 \neq 0$. (Checked in `verify_ch09a.py`.)

For $\omega \in \mathrm{Sol}$ write $D = E(\omega)$, $f = f_\omega$ and

$$A_\omega := \mathrm{D}_f T(D, f_\omega) = \mathrm{D}T_D(f_\omega) \in \mathcal{L}(\mathrm{BC}),$$
$$B_\omega := \mathrm{D}_D T(D, f_\omega) \in \mathcal{L}(\mathbb{R}^6, \mathrm{BC}),$$
$$k(\omega) := \dim\ker(I - A_\omega).$$

By Lemma 8.27(b), A_ω is compact; so B_ω (finite rank) is compact, $I - A_\omega$ is Fredholm of index 0 (Theorem A.33) and $1 \in \operatorname{spec} A_\omega$ iff $k(\omega) \geq 1$.

Lemma 9.19 (fold and cusp of E: scalar form, Whitney fold, Whitney cusp, Lyapunov–Schmidt)**.** *Let $\omega_0 \in \mathrm{Sol}$ with $k(\omega_0) = 1$, $\ker(I - A_{\omega_0}) = \operatorname{span}(v)$, let $\psi^* \in \mathrm{BC}^*$ with $\ker\psi^* = \operatorname{ran}(I - A_{\omega_0})$, and assume transversality*

$$\text{(T)} \qquad \psi^* B_{\omega_0} \neq 0.$$

Let $\ell := \psi^* \circ B_{\omega_0} \in (\mathbb{R}^6)^*$; choose linear coordinates $D = (s, \lambda') \in \mathbb{R} \times \mathbb{R}^5$ on $\mathbb{D}$ centred at $D_0 = E(\omega_0)$ with $\ell(\partial_s) \neq 0$. Then:

(a) (Lyapunov–Schmidt reduction of E.) *There are a neighbourhood V of ω_0 in* Sol*, a C^∞ diffeomorphism $\nu = (a, \lambda') : V \to \nu(V) \subset \mathbb{R} \times \mathbb{R}^5$ onto a neighbourhood of 0, with $\nu(\omega_0) = 0$ and $\partial_a|_{\omega_0} f_\omega = v$ (the "kernel dictionary"), and a C^∞ function χ near 0 with $\chi(0) = 0$, $\chi_a(0) = 0$, such that*

$$s(E(\omega)) = \chi(\nu(\omega)), \qquad \lambda'(E(\omega)) = \lambda'(\omega) \qquad (\omega \in V),$$

i.e. E has the scalar form $(a, \lambda') \mapsto (\chi(a, \lambda'), \lambda')$. Moreover $\det \mathrm{D}E = j\,\chi_a$ with j smooth and nowhere zero, $\ker \mathrm{D}E(\omega_0) = \operatorname{span}(\partial_a)$, and

$$\chi_{aa}(0) = -\frac{\psi^* \mathrm{D}_f^2 T[v, v]}{\psi^* B_{\omega_0} \partial_s}, \tag{9.4}$$

where $\mathrm{D}_f^2 T$ is taken at (D_0, f_{ω_0}).

(b) (Fold.) *ω_0 is a regular fold of E (Definition A.47) iff $\psi^* \mathrm{D}_f^2 T[v, v] \neq 0$. In that case Lemma A.46 applies: there are coordinates in which $E(\eta, \lambda') = (\pm\eta^2, \lambda')$.*

(c) (Cusp.) *Suppose $\psi^* \mathrm{D}_f^2 T[v, v] = 0$, put $y_2 := (I - A_{\omega_0})|_Y^{-1} P\, \mathrm{D}_f^2 T[v, v]$ (Y a closed complement of $\operatorname{span}(v)$, $P = I - e\psi^*$ with $\psi^*(e) = 1$) and*

$$\chi_{aaa}(0) = -\frac{h_{aaa}}{h_s} = -\frac{\psi^*\big(\mathrm{D}_f^3 T[v, v, v] + 3\mathrm{D}_f^2 T[v, y_2]\big)}{\psi^* B_{\omega_0} \partial_s}. \tag{9.5}$$

If $\chi_{aaa}(0) \neq 0$ and $\nabla_{\lambda'} \chi_a(0) \neq 0$, then ω_0 is a cusp point of E (Definition A.49) and Lemma A.48 applies: near D_0 the number of preimages near ω_0 is $3, 2, 1$ according to the position of D relative to the two fold hypersurfaces, which approach each other like $|a(\lambda')|^{3/2}$.

Proof. (a) *The reduction.* By Proposition 9.16(R), $G(D, f) = f - T_D f$ is real-analytic on the open set $\mathcal{A} \subset \mathbb{R}^6 \times \mathrm{BC}$, and $A = A_{\omega_0}$ is compact (Lemma 8.27(b)). The hypotheses of Proposition A.50 hold with $\Pi = \mathbb{R}^6$, $d = 6$, $\lambda = D$, $X = \mathrm{BC}$, $L = I - A_{\omega_0}$, $B = B_{\omega_0}$, and (T) is $\ell^* B \neq 0$. Since T is C^∞, Remark A.51 applies: the map y and the reduced function $h(D, a) = \psi^* G(D, f_0 + av + y(D, a))$ are C^∞. By Proposition A.50(a), near (D_0, f_0) ($f_0 = f_{\omega_0}$) the set $\mathcal{N} = \{(D, f) \in \mathcal{A} :\ f = T_D f\}$ is a C^∞ embedded submanifold of dimension 6, parametrised by

$$\Psi : (a, \lambda') \longmapsto \Big(D = (\chi(a, \lambda'), \lambda'),\ f = f_0 + av + y(D, a)\Big),$$

where $s=\chi(a,\lambda')$ solves $h(s,\lambda',a)=0$ ($h_s=-\psi^*B\partial_s\neq 0$), $\chi(0)=0$, $\chi_a(0)=-h_a/h_s=0$. Let $\iota:\mathrm{Sol}\to\mathbb{R}^6\times\mathrm{BC}$, $\iota(\omega)=(E(\omega),f_\omega)$. It is C^∞ (Proposition 9.16(S)), takes values in $\mathrm{Fix}\supset\mathcal{N}$, and $\Sigma\circ\iota=\mathrm{id}$, $\iota\circ\Sigma|_{\mathrm{Fix}}=\mathrm{id}$ (Lemma 9.11(c)). Let $\mathcal{N}_0\subset\mathcal{N}$ be the part of $\mathcal{N}$ near (D_0,f_0) in the ambient space and $V:=\iota^{-1}(\mathcal{N}_0)$, open since ι is continuous and $\mathcal{N}_0=\mathrm{Fix}\cap\mathcal{O}$ for an open ambient set $\mathcal{O}$ ($\mathcal{A}$ is open and every fixed point in $\mathcal{O}$ lies on $\mathcal{N}$ by the local uniqueness in Proposition A.50). Then $\iota:V\to\mathcal{N}_0$ is a bijection with inverse $\Sigma|_{\mathcal{N}_0}$, both C^∞ (a smooth map into the ambient space with values in an embedded submanifold is smooth into it). Define $\nu:=\Psi^{-1}\circ\iota:V\to\mathbb{R}^6$; it is a C^∞ diffeomorphism onto a neighbourhood of 0 and, by construction, $E=\mathrm{pr}_D\circ\Psi\circ\nu$, which is the stated scalar form. The kernel dictionary follows from $\iota\circ\nu^{-1}(a,0)=\Psi(a,0)$ and $\partial_a f|_0=v+y_a=v$ ($y_a(0)=0$, Proposition A.50). *Jacobian.* In the scalar form the matrix of $\mathrm{D}E$ is $\left(\begin{smallmatrix}\chi_a & \nabla_{\lambda'}\chi\\ 0 & I\end{smallmatrix}\right)$, so $\det\mathrm{D}E=\chi_a$ in the coordinates (a,λ') and the coordinates (s,λ'), hence $\det\mathrm{D}E=j\chi_a$ with j the (nowhere zero) product of the Jacobian determinants of the two coordinate changes; at ω_0 the kernel is ∂_a (Lemma A.45). *Formula* (9.4). Proposition A.50: $h_{aa}(0)=-\psi^*\mathrm{D}_f^2T[v,v]$ and $h_s=-\psi^*B\partial_s$; differentiating $h(\chi(a,\lambda'),\lambda',a)=0$ twice in a at 0 gives $h_s\chi_{aa}+h_{aa}=0$ ($\chi_a=h_a=0$). (b) Definition A.47 and the invariance paragraph after it: ω_0 is a regular fold iff $\varphi_{xx}\neq 0$ for the scalar form, here $\chi_{aa}(0)\neq 0$, which by (9.4) is $\psi^*\mathrm{D}_f^2T[v,v]\neq 0$. Then Lemma A.46 gives the normal form. (c) Differentiate $H(a):=h(\chi(a,0),0,a)\equiv 0$ three times at $a=0$: $H'''=h_s\chi_{aaa}+3h_{sa}\chi_{aa}+h_{aaa}+(\text{terms containing }\chi_a=0)$. With $\chi_{aa}(0)=0$ (equivalently $h_{aa}=0$) this gives $\chi_{aaa}(0)=-h_{aaa}/h_s$. By Remark A.51, $h_{aaa}=\psi^*\big(\mathrm{D}_f^3G[v,v,v]+3\mathrm{D}_f^2G[v,y_{aa}]\big)$ with $y_{aa}=-(L|_Y)^{-1}P\mathrm{D}_f^2G[v,v]$; since $G=I-T$, $\mathrm{D}^kG=-\mathrm{D}^kT$ for $k\geq 2$, so $y_{aa}=(L|_Y)^{-1}P\mathrm{D}_f^2T[v,v]=y_2$ and $h_{aaa}=-\psi^*(\mathrm{D}_f^3T[v,v,v]+3\mathrm{D}_f^2T[v,y_2])$. With $h_s=-\psi^*B\partial_s$ this is (9.5). (The identity $\chi_{aaa}=-h_{aaa}/h_s$ and the formula for h_{aaa} are verified by computer algebra in `verify_ch09a.py`.) In the scalar-form coordinates, (C1) of Definition A.49 is $\chi_{aa}(0)=0$, (C3) is $\chi_{aaa}(0)\neq 0$ and (C2) is $\nabla_{\lambda'}\chi_a(0)\neq 0$; Lemma A.48 gives the count and the width. □

9.4. The structure of the fold set

In words. *There are two natural definitions of a fold: spectral (some fixed point of T_D has 1 in the spectrum of $\mathrm{D}T_D$) and geometric (the Jacobian of E vanishes). They are related by the identity $(I-A_\omega)\partial_w f_\omega=B_\omega\mathrm{D}E(\omega)w$ obtained by differentiating $f_\omega=T_{E(\omega)}f_\omega$: the kernel of $\mathrm{D}E$ embeds in the kernel of $I-A$, and the two coincide when the parameters D can push the equation in every direction of the cokernel (transversality). Once the fold set is a hypersurface, the number of solutions changes by two across it, and the two new solutions have opposite Leray–Schauder indices. To pass from this local picture to a global statement we need that E is* proper*: solutions with data in a compact set stay in a compact set. This is where the a priori bounds of Chapter 8 enter, in a version that is uniform in the data.*

Definition 9.20 (fold sets)**.** $\Phi_{\mathrm{fold}}:=\{\omega\in\mathrm{Sol}:\ 1\in\operatorname{spec}A_\omega\}=\{k\geq 1\}$ is the *spectral fold set*; $\Phi_{\mathrm{fold}}^{\det}:=\{\omega\in\mathrm{Sol}:\ \det\mathrm{D}E(\omega)=0\}$ is the *Jacobian fold set*. The *fold values* are $E(\Phi_{\mathrm{fold}})\subset\mathbb{D}_+$: the data D at which some fixed point of T_D is degenerate. Both sets are closed in Sol (k is upper semicontinuous by the argument of Proposition 9.40, and $\det\mathrm{D}E$ is continuous).

Lemma 9.21 (the differentiated fixed-point identity)**.** *For every* $\omega \in \mathrm{Sol}$ *and* $w \in \mathbb{R}^6$,

$$(I - A_\omega)\,\partial_w f_\omega = B_\omega\,\mathrm{D}E(\omega)w, \qquad \text{and} \qquad w = \mathrm{D}\Sigma\big(\mathrm{D}E(\omega)w,\ \partial_w f_\omega\big). \tag{9.6}$$

In particular the linear map $w \mapsto (\mathrm{D}E(\omega)w, \partial_w f_\omega) \in \mathbb{R}^6 \times \mathrm{BC}$ *is injective.*

Proof. By Lemma 9.11, $f_\omega = T_{E(\omega)} f_\omega$ for all ω, and both sides are C^1 in ω (Proposition 9.16); the chain rule gives $\partial_w f_\omega = B_\omega \mathrm{D}E(\omega)w + A_\omega \partial_w f_\omega$. The second identity is the chain rule applied to $\omega = \Sigma(E(\omega), f_\omega)$ (Lemma 9.11(b), Σ analytic). If $\mathrm{D}E(\omega)w = 0$ and $\partial_w f_\omega = 0$ then $w = \mathrm{D}\Sigma(0,0) = 0$. □

Theorem 9.22 (the fold set is the singular set of E where transversality holds). *Let* $\omega \in \mathrm{Sol}$ *and* $k = k(\omega)$.

(a) $w \mapsto \partial_w f_\omega$ *maps* $\ker \mathrm{D}E(\omega)$ *injectively into* $\ker(I - A_\omega)$. *Hence* $\dim\ker \mathrm{D}E(\omega) \le k(\omega)$*; in particular* $\Phi^{\det}_{\mathrm{fold}} \subset \Phi_{\mathrm{fold}}$, *and* $\mathrm{D}E(\omega)$ *is invertible whenever* $1 \notin \operatorname{spec} A_\omega$.

(b) *Let* $\beta_\omega : \mathbb{R}^6 \to \mathrm{BC}/\operatorname{ran}(I - A_\omega)$, $\beta_\omega(\delta D) = [B_\omega \delta D]$ (*a map into a* k*-dimensional space*). *If* β_ω *is onto* (*transversality* (T))*, then* $\dim\ker \mathrm{D}E(\omega) = k$ *and* $w \mapsto \partial_w f_\omega$ *is an isomorphism* $\ker \mathrm{D}E(\omega) \to \ker(I - A_\omega)$.

(c) *Consequently* $\Phi_{\mathrm{fold}} = \Phi^{\det}_{\mathrm{fold}}$ *at every point where* (T) *holds. For* $k = 1$, *with* $\ker(I - A_\omega) = \operatorname{span}(v)$ *and* ψ^* *the left null functional,* (T) *says* $\psi^* B_\omega \neq 0$.

Proof. (a) If $\mathrm{D}E(\omega)w = 0$, (9.6) gives $(I - A_\omega)\partial_w f_\omega = 0$, and injectivity is Lemma 9.21. If $1 \notin \operatorname{spec} A_\omega$ then $\ker(I - A_\omega) = 0$, so $\ker \mathrm{D}E(\omega) = 0$. (b) Let $\mathcal{K} := \{(\delta D, \delta f) \in \mathbb{R}^6 \times \mathrm{BC} : (I - A_\omega)\delta f = B_\omega \delta D\}$. By (9.6) the image $\mathcal{T}$ of the injective map $w \mapsto (\mathrm{D}E(\omega)w, \partial_w f_\omega)$ is a 6-dimensional subspace of $\mathcal{K}$. The projection $\mathcal{K} \to \mathbb{R}^6$, $(\delta D, \delta f) \mapsto \delta D$, has image $\ker \beta_\omega$ and kernel $\{0\} \times \ker(I - A_\omega)$, so $\dim \mathcal{K} = \dim\ker \beta_\omega + k$. If β_ω is onto, $\dim\ker \beta_\omega = 6 - k$ and $\dim \mathcal{K} = 6 = \dim \mathcal{T}$, so $\mathcal{K} = \mathcal{T}$. Then every $(0, v)$ with $v \in \ker(I - A_\omega)$ lies in $\mathcal{T}$: $v = \partial_w f_\omega$ with $\mathrm{D}E(\omega)w = 0$. So the injection of (a) is onto and $\dim\ker \mathrm{D}E(\omega) = k$. (c) follows; for $k = 1$, $\mathrm{BC}/\operatorname{ran}(I - A_\omega) \cong \mathbb{R}$ via ψ^*. □

Remark 9.23 (transversality is sufficient, not necessary, and cannot be dropped)**.** Two scalar toys with $\Sigma(\lambda, f) = f$ show both directions. (1) $T(\lambda, x) = x - x^3$ on $\mathbb{R}$ has the fixed-point set $\{x = 0\}$, a graph over λ, $E = \mathrm{id}$, $\det \mathrm{D}E = 1$, yet $\mathrm{D}_x T = 1$ at every fixed point; here $B = 0$ and $\dim\ker \mathrm{D}E = 0 < k = 1$: so (T) cannot be omitted in (b). (2) $T(\lambda, x) = x - \lambda^2$: the fixed-point set is $\{\lambda = 0\}$, $E \equiv 0$, $\ker \mathrm{D}E = \mathbb{R} = \ker(I - A)$, so $\dim\ker \mathrm{D}E = k = 1$ although $B = -2\lambda = 0$ and (T) fails. So (T) is sufficient for $\dim\ker \mathrm{D}E = k$ but not necessary, and only the sufficient direction is claimed. Part (a) and everything starting from $\Phi^{\det}_{\mathrm{fold}}$ need no transversality. The condition (T) is a genuine hypothesis: at the folds that are computed in Atlas 9.76 it is verified numerically (the number $\psi^* B_\omega \delta D$ is far from 0); no theorem says it holds at every fold.

9.4.1. Transversality in the parameters f_- and κ_{D}. Two of the six columns of $B_\omega = \mathrm{D}_D T$ are explicit, and this is enough to see that the transversality condition (T) is a mild one.

Toy example 9.24 (the columns of B in the scalar cubic)**.** In Toy example 9.18 the map is $T_{b,\tau} u = \frac{2}{3}(1 + x)(b\,m(u) - m(u)^3 + \tau)$; the column $\partial_\tau T = \frac{2}{3}(1 + x)$ is explicit

and the left null functional is $\psi^* = m(\cdot)$ with $\psi^* \partial_\tau T = 1$. The same happens for T_D: the parameter f_- shifts T by the constant function 1, and κ_{D} multiplies the quadrature $\int z_f^{-2}$.

Proposition 9.25 (two columns of B_ω)**.** *For $(D, f) \in \mathcal{A}$, in the coordinates $D = (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$,*

$$\partial_{f_-} T_D f = \mathbf{1} \quad (\textit{the constant function}),$$

$$\partial_{\kappa_{\mathrm{D}}} T_D f = \frac{T_D f - f_-}{\kappa_{\mathrm{D}}} = \int_{-\infty}^{\xi} z_f^{-2} = \frac{\varphi}{\vartheta\, w_0(W_f)}.$$

Consequently, if $k(\omega) = 1$ with left null functional ψ^*, then (T) *holds as soon as $\psi^*(\mathbf{1}) \neq 0$ or $\psi^*(f_\omega - f_-) \neq 0$; and if* (T) *fails then ψ^* vanishes on both $\mathbf{1}$ and $f_\omega - f_-$.*

Proof. By Definition 8.11 and Lemma 8.10, $T_D f = f_- + \kappa_{\mathrm{D}}\, G(\vartheta, \boldsymbol{c}', f)$ with $G = \int_{-\infty}^{\xi} (u_-^{W_f} + \vartheta u_+^{W_f})^{-2}$; the potential $W_f = W_{\mathrm{e}} + \frac{1}{2}\mathcal{Q}[f^2]$ and the Jost solutions depend on $(\boldsymbol{c}', f)$ only. Hence $T_D f$ is affine in $(f_-, \kappa_{\mathrm{D}})$ in the stated way, and differentiating (the derivatives exist by Proposition 9.16(R)) gives the two columns; the third form is Lemma 8.10(i)–(ii). For the consequence, $\psi^* B_\omega \delta D = \delta f_-\, \psi^*(\mathbf{1}) + \delta\kappa_{\mathrm{D}}\, \psi^*((f_\omega - f_-)/\kappa_{\mathrm{D}}) +$ (terms in $\delta\boldsymbol{c}', \delta\vartheta$), and (T) is the statement that this functional of δD is not zero. □

Corollary 9.26 (the continuation direction)**.** *Let $\omega_0 \in \mathrm{Sol}$ have $k(\omega_0) = 1$ and $\psi^*(f_{\omega_0} - f_-) \neq 0$. Then* (T) *holds, and in Lemma 9.19 one may take $s = \kappa_{\mathrm{D}}$, $\lambda' = (\boldsymbol{c}', \vartheta, f_-)$; the coefficient* (9.4) *becomes*

$$\chi_{aa}(0) = -\frac{\kappa_{\mathrm{D}}\, \psi^* \mathrm{D}_f^2 T[v, v]}{\psi^*(f_{\omega_0} - f_-)},$$

so ω_0 is a turning point of the solution branch in the variable κ_{D} (at fixed $\boldsymbol{c}', \vartheta, f_-$) iff $\psi^* \mathrm{D}_f^2 T[v, v] \neq 0$.

Proof. By Proposition 9.25, $\psi^* B_{\omega_0} \partial_{\kappa_{\mathrm{D}}} = \psi^*(f_{\omega_0} - f_-)/\kappa_{\mathrm{D}} \neq 0$, so (T) holds with $\ell(\partial_s) \neq 0$ for $s = \kappa_{\mathrm{D}}$; insert in (9.4). □

Corollary 9.27 (regular values are generic)**.** *The set $E(\Phi^{\det}_{\mathrm{fold}})$ of fold values is closed and has Lebesgue measure zero; hence $\mathbb{D}_+ \setminus E(\Phi^{\det}_{\mathrm{fold}})$ is open and dense, and the number $N(D)$ of solutions is finite and locally constant on it (Theorem 9.42(Ba)).*

Proof. E is C^∞ (Proposition 9.16(S)) between open subsets of $\mathbb{R}^6$; by Sard's theorem the set of critical values $E(\{\det \mathrm{D}E = 0\}) = E(\Phi^{\det}_{\mathrm{fold}})$ has measure zero. It is closed because E is proper (Proposition 9.38): a proper continuous map to a locally compact Hausdorff space is closed. A closed set of measure zero has empty interior. The remaining assertions are Step 1 of the proof of Theorem 9.42. □

Remark 9.28 (indices add up to one across a fold)**.** By Proposition 8.48(a), if all fixed points of T_D are nondegenerate then $\sum_f (-1)^{\beta(f)} = 1$. Crossing a regular fold at which (T) holds and $k = 1$, two solutions with indices $+1$ and -1 appear (Proposition 9.40), so the sum is unchanged, consistently with the constancy of the degree in Theorem 8.42(b). In particular a datum with exactly three nondegenerate solutions has indices $(+1, -1, +1)$ in the order of β of Numerical observation 8.51.

9.4.2. Nondegenerate fixed points: analytic continuation.

Toy example 9.29 (the implicit function theorem for $x = \kappa/(1+x^2)$)**.** For $F(\kappa, x) = x - \kappa/(1+x^2)$ one has $F_x = 1 + 2\kappa x/(1+x^2)^2$; at a solution $x > 0$ ($\kappa > 0$) this is > 1, so the branch $x(\kappa)$ is analytic for all $\kappa > 0$ and there is no fold: $T_\kappa(x) = \kappa/(1+x^2)$ has $T' < 0$, i.e. $1 \notin \operatorname{spec} T'$ (Toy example 8.38). In the Banach-space problem the same argument works wherever $1 \notin \operatorname{spec} \mathrm{D}T_D$.

Proposition 9.30 (continuation of nondegenerate fixed points)**.** *Let $\omega_0 \in \mathrm{Sol}$ with $1 \notin \operatorname{spec} A_{\omega_0}$ ($\omega_0 \notin \Phi_{\mathrm{fold}}$), $D_0 = E(\omega_0)$, $f_0 = f_{\omega_0}$. There are an open neighbourhood $\mathcal{N}$ of D_0 and a real-analytic map $F : \mathcal{N} \to \mathrm{BC}$ with $F(D_0) = f_0$, $F(D) = T_D F(D)$, such that $F(D)$ is the only fixed point of T_D in a fixed neighbourhood of f_0; E is a real-analytic diffeomorphism from the neighbourhood $\Sigma(\operatorname{graph} F)$ of ω_0 onto $\mathcal{N}$, with inverse $D \mapsto \Sigma(D, F(D))$.*

Proof. $G(D, f) = f - T_D f$ is real-analytic on $\mathcal{A}$ (Proposition 9.16(R)) and $\mathrm{D}_f G(D_0, f_0) = I - A_{\omega_0}$ is a bounded bijection of BC (A_{ω_0} compact and $1 \notin$ spec: the Riesz theory, Theorem A.33, gives a bounded inverse). Theorem A.3 (its analytic version) gives F and the local uniqueness. By Lemma 9.11(b), $\omega(D) := \Sigma(D, F(D)) \in \mathrm{Sol}$, with $E(\omega(D)) = D$ and $f_{\omega(D)} = F(D)$; ω is analytic as a composition of analytic maps (Proposition 9.16(R)). Conversely every ω near ω_0 has $(E(\omega), f_\omega)$ near (D_0, f_0), hence $f_\omega = F(E(\omega))$ and $\omega = \omega(E(\omega))$. So E is a bijection with analytic inverse $\omega(\cdot)$ between neighbourhoods; $\mathrm{D}E(\omega_0)$ is invertible, and E itself is analytic there (inverse of an analytic diffeomorphism). □

Corollary 9.31 (non-uniqueness is an open condition)**.** *If D_0 carries two distinct nondegenerate fixed points $f_1 \neq f_2$ of T_{D_0}, then there is a neighbourhood $\mathcal{N}$ of D_0 in $\mathbb{D}_+$ with $N(D) \geq 2$ for all $D \in \mathcal{N}$, and $N(D) \geq 3$ at every regular value $D \in \mathcal{N}$ of E.*

Proof. Let F_1, F_2 be the continuations of Proposition 9.30 on a common neighbourhood; $F_1(D_0) \neq F_2(D_0)$ and continuity give $F_1(D) \neq F_2(D)$ for D near D_0, and they are fixed points, so $N(D) \geq 2$ by Lemma 9.11(c). At a regular value $N(D)$ is odd (Theorem 9.42(Bb)), hence ≥ 3. □

Corollary 9.32 (width of the three-solution region at a cusp)**.** *In the situation of Lemma 9.19(c), let $a(\lambda')$ be the value of χ_a along the curve $\chi_{aa} = 0$ (so $\nabla a(0) = \nabla_{\lambda'} \chi_a(0) \neq 0$). For λ' near 0 with $a(\lambda')\chi_{aaa}(0) < 0$, the fibre of E over the points (s, λ') near $E(\omega_0)$ has three points near ω_0 for s strictly between the two critical values $s_{\min}(\lambda') < s_{\max}(\lambda')$ (the two fold hypersurfaces), and*

$$s_{\max}(\lambda') - s_{\min}(\lambda') = \frac{4\sqrt{2}}{3} \frac{|a(\lambda')|^{3/2}}{\sqrt{|\chi_{aaa}(0)|}} \big(1 + o(1)\big) \qquad (\lambda' \to 0);$$

for $a(\lambda')\chi_{aaa}(0) \geq 0$ the fibre has at most one point near ω_0.

Proof. In the scalar form of Lemma 9.19(a), $E = (\chi, \lambda')$ with $\varphi = \chi$, $x = a$, $h_0 = \chi_{aaa}(0) \neq 0$, $|H(0)| = |h_0|/2$; apply Lemma A.48(ii),(iii): the width is $\frac{4}{3}|a|^{3/2}/\sqrt{|h_0|/2} = \frac{4\sqrt{2}}{3}|a|^{3/2}/\sqrt{|h_0|}$. □

Toy example 9.33 (width in the cubic)**.** For $\varphi(m, b) = m^3 + (1-b)m$ the critical points are $m = \pm\sqrt{(b-1)/3}$ with critical values $\mp\frac{2}{3\sqrt{3}}(b-1)^{3/2}$, so the width is

$\frac{4}{3\sqrt{3}}(b-1)^{3/2}$; here $\varphi_{mmm}=6$ and $a(b)=\varphi_m(\tau(b),b)=1-b$, and Corollary 9.32 gives $\frac{4\sqrt{2}}{3}(b-1)^{3/2}/\sqrt{6}=\frac{4}{3\sqrt{3}}(b-1)^{3/2}$, the same number.

Figure 9.2 plots the fold normal form and the width of the window.

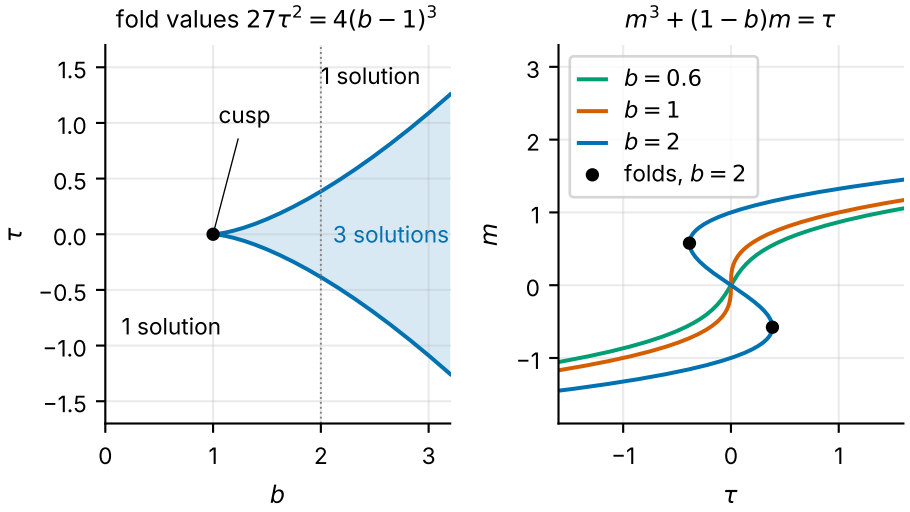


FIGURE 9.2. The fold normal form and the width of the three-solution window (Toy example 9.33).

Table 9.1 summarises the results of this half of the chapter.

Proposition 9.34 (the fold hypersurface and the count)**.** *Let* $\omega_0\in\Phi^{\det}_{\text{fold}}$ *with* $\dim\ker \mathrm{D}E(\omega_0)=1$ *be a regular fold of* E *(Definition A.47; no transversality is assumed). Then:*

(i) $\Phi^{\det}_{\text{fold}}$ *is, near* ω_0*, a smooth hypersurface of* Sol*, and* $E|_{\Phi^{\det}_{\text{fold}}}$ *is an embedding onto a smooth hypersurface* $\Gamma\ni E(\omega_0)$ *of* $\mathbb{D}_+$ *(a codimension-one set of fold values).*

(ii) *Each* D *near* $E(\omega_0)$ *has* 0 *or* 2 *preimages near* ω_0*, according to the side of* Γ*, and one on* Γ*.*

(iii) *Let* $D_0=E(\omega_0)$ *and suppose all other points of the fibre* $E^{-1}(D_0)$ *are regular points of* E*. Then the fibre is finite, and the number* $N(D)=\#E^{-1}(D)$ *equals* N_* *for* D *on one side of* Γ *near* D_0 *and* N_*+2 *on the other, where* $N_*=\#E^{-1}(D_0)-1$*.*

PROOF. E is C^∞ (Proposition 9.16(S)) between open subsets of $\mathbb{R}^6$ and has corank 1 at ω_0, so Lemma A.45 and Lemma A.46 apply and give (i), (ii) directly. (iii) The fibre is compact (Proposition 9.38 below) and discrete (near a regular point E is a local diffeomorphism, and ω_0 is isolated in its fibre by (ii)), hence finite. Choose pairwise disjoint neighbourhoods of the points of the fibre; by properness the preimages of all D close to D_0 lie in these neighbourhoods (Proposition 9.38); near a regular point there is exactly one preimage, and near ω_0 there are 0 or 2 by (ii). □

object	statement	where
sign structure of $\mathrm{D}T_D$	$\delta\rho \sim \delta W$; $g \geq 0 \Rightarrow \mathrm{D}T_D\delta f < 0$ on $\xi \leq 0$; no $v \geq 0$ with $\nu \geq 0$	Lemma 9.2, Lemma 9.8
secular equation	$1 \in \operatorname{spec}\mathrm{D}T_D \iff 1 + \ell((I-R)^{-1}u) = 0$	Proposition 9.7
analyticity	T analytic in (D,f), E smooth, Σ analytic	Proposition 9.16
nondegenerate points	analytic continuation, E local diffeomorphism	Proposition 9.30
kernel dictionary	$\ker \mathrm{D}E \hookrightarrow \ker(I-A)$, equality under (T)	Theorem 9.22, Proposition 9.25
fold, cusp	Lyapunov–Schmidt: χ_{aa}, χ_{aaa}	Lemma 9.19, Corollary 9.32
local structure	hypersurface, count, indices $(+1,-1)$	Proposition 9.34, Proposition 9.40
compactness	uniform a priori bounds, E proper	Lemma 9.37, Proposition 9.38
global structure	covering, parity, uniqueness criterion, non-emptiness	Theorem 9.42, Corollary 9.46

TABLE 9.1. The results of Sections 9.1–9.4.

9.4.3. Locally uniform a priori bounds and properness. The next lemma is the compactness input that makes E proper. It uses the exact formula $T_Df = f_- + J\Sigma_f$ and the *uniform* estimates for Jost solutions of Chapter 3; we spell out the constants and where each comes from, since the statement is that they can be chosen locally uniformly in D.

Toy example 9.35 (properness in the cubic model)**.** In Toy example 9.18, $E(m,b) = (b, m^3 + (1-b)m)$ is proper: if b and $\tau = m^3 + (1-b)m$ stay in a compact set then m does, because $|m|^3 \leq |\tau| + |1-b||m|$. The fibre over (b,τ) has one point for $b \leq 1$ and one or three for $b > 1$; the fold values form the curve $27\tau^2 = 4(b-1)^3$ with its cusp at $(1,0)$. Properness is what turns "locally 1 or 3" into "globally 1 or 3" (Theorem 9.42(B)).

Lemma 9.36 (compactness of the profiles)**.** *Let $K \subset \mathbb{D}_+$ be compact and $S, w_{\min} > 0$. The set*

$$\mathcal{P}(K,S,w_{\min}) := \{T_Df :\ D \in K,\ f \in \mathcal{A}_D,\ \|f\|_\infty \leq S,\ w_0(W_f) \geq w_{\min}\}$$

is relatively compact in $\mathrm{BC}(\mathbb{R})$*. Moreover, the potentials W_f occurring here lie in one class $\mathcal{K}(\underline{k}, \overline{\varpi}, C, m, x, x)$ of Section 3.4.1.*

PROOF. *The class.* Put $\underline{k} = \min_K k_i' > 0$, $\overline{k} = \max_K k_i'$, $C_K = \sup_K(|c_1' - c_2'| + 2|c_3'|)$, $b_K = \sup_{D\in K}\sup_\xi |W_{\mathrm{e}}| < \infty$ ($W_{\mathrm{e}} = 1 + \frac{1}{2}P_{\boldsymbol{c}'}$ is bounded by $1 + \frac{1}{2}(|c_1'| + |c_2'| + |c_3'|)\cdot 2$) and $\overline{\varpi}^{\,2} := b_K + S^2/8$; note $k_i'^2 = \lim W_{\mathrm{e}} \leq b_K$, so $\overline{k} \leq \overline{\varpi}$. For $D \in K$ and $\|f\|_\infty \leq S$: $W_f \in \mathcal{W}(k_1', k_2'; C_K + \frac{1}{2}S^2)$ (Lemma 8.5(a)) and $W_f \leq W_{\mathrm{e}} + \frac{1}{8}S^2 \leq \overline{\varpi}^{\,2}$ (Lemma 8.2(a)). Choose X with $C_K e^{-2X} \leq \underline{k}^2/2$; then $W_{\mathrm{e}} \geq k_i'^2 - C_K e^{-2|\xi|} \geq \underline{k}^2/2$ for $|\xi| \geq X$, and since $\mathcal{Q}[f^2] \geq 0$, $W_f \geq \underline{k}^2/2 =: m^2$ on $|\xi| \geq X$. So $W_f \in \mathcal{K}(\underline{k}, \overline{\varpi}, C, m, X, X)$ with $C = C_K + \frac{1}{2}S^2$. *Compactness.* Let

$D \in K$, f as in the statement, $W = W_f$ admissible with $w_0(W) \geq w_{\min}$, $u_\pm$ its Jost solutions and $\Sigma^1 = u_-/(u_- + u_+) = R/(R+1)$. By Lemma 3.28(f) (with $\vartheta = 1$), the set $\mathcal{S}$ of all such Σ^1 is relatively compact in BC. By Lemma 8.10, $T_D f = f_- + J\Sigma^\vartheta$ with $\Sigma^\vartheta = R/(R+\vartheta) = \sigma_\vartheta(\Sigma^1)$, $\sigma_\vartheta(s) = s/(s + \vartheta(1-s))$ and $J = \kappa_{\mathrm{D}}/(\vartheta w_0(W)) \in (0, \max_K \kappa_{\mathrm{D}}/(\vartheta w_{\min})]$. The map $(\vartheta, s) \mapsto \sigma_\vartheta(s)$ is Lipschitz on $[\vartheta_-, \vartheta_+] \times [0,1]$ ($\vartheta_\pm$ the extreme values on K) because the denominator is $\geq \min(1, \vartheta_-)$. So $\mathcal{P}(K, S, w_{\min})$ is contained in the image of the compact set $\{(f_-, J, \vartheta)\} \times \overline{\mathcal{S}}$ under a continuous map into BC, hence relatively compact. □

Lemma 9.37 (locally uniform a priori bounds)**.** *Let $K \subset \mathbb{D}_+$ be compact. There are constants $J_K, S_K < \infty$ and $w_K > 0$ such that every fixed point $f = T_D f$ with $D \in K$ satisfies*

(i) *$f_- \leq f \leq f_- + J$, f nondecreasing, with $J = \kappa_{\mathrm{D}}/(\vartheta w_0(W_f)) \leq J_K$, and $\|f\|_\infty \leq S_K$;*
(ii) *$w_0(W_f) \geq w_K$;*
(iii) *f lies in the compact set $\overline{\mathcal{P}(K, S_K, w_K)} \subset \mathrm{BC}$;*
(iv) *$0 \leq f' \leq L_K := J_K\, \overline{\vartheta}\, \overline{\varpi}_K/(2\min(1, \underline{\vartheta})^2)$, where $\underline{\vartheta}, \overline{\vartheta}$ are the extreme values of ϑ on K and $\overline{\varpi}_K^2 = b_K + S_K^2/8$.*

In particular $\mathrm{Fix}_K := \{(D, f) \in \mathrm{Fix} :\ D \in K\}$ *is a compact subset of* $\mathbb{D}_+ \times \mathrm{BC}$.

Proof. *Step 1: a uniform barrier.* With $\underline{k}, C_K, b_K, X$ as in Lemma 9.36 let $q := \mathcal{Q}[\mathbf{1}_{[0,\infty)}]$, that is (the defining integral of Lemma 8.2 makes sense for bounded measurable g, and the indicator is not continuous, so we use the formula directly, as in Lemma 8.39) $q(\xi) = (4\cosh^2\xi)^{-1} \int_0^\infty e^{-2|\xi-\eta|}\,\mathrm{d}\eta$, which is continuous and positive with $q \leq \frac{1}{4}\operatorname{sech}^2\xi$, $q_0 := \min_{[-X,X]} q > 0$, and choose $m_K > 0$ with $\frac{1}{2}m_K^2 q_0 \geq \frac{1}{2}\underline{k}^2 + b_K$. For $D \in K$ put $W_*^D := W_{\mathrm{e}} + \frac{1}{2}m_K^2 q$. Then $W_*^D \geq \frac{1}{2}\underline{k}^2$ on $\mathbb{R}$ (on $[-X, X]$ because $W_{\mathrm{e}} \geq -b_K$, outside because $q \geq 0$ and $W_{\mathrm{e}} \geq \underline{k}^2/2$), and $W_*^D \in \mathcal{W}(k_1', k_2'; C_K + \frac{1}{2}m_K^2)$ (since $0 \leq q \leq \frac{1}{4}\operatorname{sech}^2 \leq e^{-2|\xi|}$). W_*^D is admissible: its Jost solution $\varphi_- \sim e^{k_1'\xi}$ at $-\infty$ is positive with $\varphi_-'/\varphi_- \to k_1' > 0$; as long as $\varphi_- > 0$ and $\varphi_-' > 0$ we have $\varphi_-'' = W_*\varphi_- > 0$, so φ_-' stays positive, hence $\varphi_- > 0$, $\varphi_-' > 0$ on $\mathbb{R}$, and $\varphi_-'' \geq \frac{1}{2}\underline{k}^2\varphi_-$ on $[0,\infty)$ gives $\varphi_- \to \infty$. Let $w_*^D := w_0(W_*^D) > 0$. The map $\boldsymbol{c}' \mapsto W_*^D$ is continuous for locally uniform convergence in a fixed class $\mathcal{W}(\cdot,\cdot; C)$, so $D \mapsto w_*^D$ is continuous (Lemma 3.25(b)); put $w_* := \min_K w_*^D > 0$. Note that m_K depends only on $(\underline{k}, b_K, X)$, that is, on K through these three numbers and not on the individual datum D; and that Lemma 3.28(f) is used with the fixed value $\vartheta = 1$, the dependence on ϑ being carried by the Lipschitz map σ_ϑ. *Step 2: fixed points.* Let $f = T_D f$, $D \in K$. By Lemma 8.10, $f = f_- + J\Sigma_f$ with $0 < \Sigma_f < 1$ increasing, so f is nondecreasing, $f_- \leq f \leq f_- + J$, $J = \kappa_{\mathrm{D}}/(\vartheta w_0(W_f))$, $f(0) = f_- + J/(1+\vartheta)$. Put $J_1 := (1+\vartheta)(m_K - f_-)^+$. If $J \geq J_1$ then $f(0) \geq m_K$, hence $f \geq m_K$ on $[0,\infty)$ and $f^2 \geq m_K^2 \mathbf{1}_{[0,\infty)}$ on $\mathbb{R}$, so $W_f \geq W_*^D$ ($\mathcal{Q}$ is order preserving). Both potentials lie in $\mathcal{W}(k_1', k_2')$ and W_*^D is admissible, so Lemma 3.23 gives $w_0(W_f) \geq w_*^D \geq w_*$, i.e. $J \leq \kappa_{\mathrm{D}}/(\vartheta w_*)$. If $J < J_1$ nothing is to prove. Hence

$$J \leq J_K := \sup_{D \in K} \max\Big((1+\vartheta)(m_K - f_-)^+,\ \frac{\kappa_{\mathrm{D}}}{\vartheta w_*} \Big) < \infty,$$

(this is the bound $J \leq J_{\max}$ of Lemma 8.39 with constants chosen uniformly on K), $w_0(W_f) = \kappa_{\mathrm{D}}/(\vartheta J) \geq w_K := \min_K \kappa_{\mathrm{D}}/(\vartheta J_K) > 0$, and $\|f\|_\infty \leq S_K := \sup_K |f_-| + J_K$. This proves (i), (ii). For (iv): $f' = J\,\sigma_\vartheta'(\Sigma^1)\,(\Sigma^1)'$ with $0 <$

$(\Sigma^1)' \le \overline{\varpi}_K/2$ by Lemma 3.28(d) (the potentials lie in the class of Lemma 9.36) and $0 < \sigma'_\vartheta(s) = \vartheta/(s+\vartheta(1-s))^2 \le \overline{\vartheta}/\min(1,\underline{\vartheta})^2$. *Step 3.* Since $f = T_D f$, $f \in \mathcal{P}(K, S_K, w_K)$ by definition, and (iii) is Lemma 9.36. Fix_K is contained in $K \times \overline{\mathcal{P}}$, compact. It is closed: let $(D_n, f_n) \in \mathrm{Fix}_K$ converge to $(D, f) \in K \times \mathrm{BC}$. The potentials $W_{D_n,f_n} \to W_{D,f}$ locally uniformly, with $k_i'(D_n) \to k_i'(D)$, in one class $\mathcal{W}(\cdot,\cdot;C)$, and are admissible with $w_0 \ge w_K$. By Lemma 3.25(c) (positive pairs) the limit $W_{D,f}$ has $\varphi_-^W > 0$ and $w_0(W_{D,f}) = \lim w_0(W_{D_n,f_n}) \ge w_K > 0$; so it is not critical, i.e. it is admissible and $(D, f) \in \mathcal{A}$. By continuity of T on $\mathcal{A}$ (Proposition 9.16(R)), $f = \lim f_n = \lim T_{D_n} f_n = T_D f$. So $(D, f) \in \mathrm{Fix}_K$. □

Proposition 9.38 (properness of E)**.** *$E : \mathrm{Sol} \to \mathbb{D}_+$ is proper: preimages of compact sets are compact.*

Proof. Let $K \subset \mathbb{D}_+$ be compact and $\omega_n \in E^{-1}(K)$. By Lemma 9.11, $(E(\omega_n), f_{\omega_n}) \in \mathrm{Fix}_K$, which is compact (Lemma 9.37); pass to a subsequence with $(E(\omega_n), f_{\omega_n}) \to (D, f) \in \mathrm{Fix}_K$. Then $\omega := \Sigma(D, f) \in \mathrm{Sol}$, $E(\omega) = D \in K$, and by continuity of Σ (Proposition 9.16(R)), $\omega_n = \Sigma(E(\omega_n), f_{\omega_n}) \to \omega$. □

Remark 9.39 (on the use of the a priori estimates)**.** The bounds used are the exact integrals (Lemma 8.10), the comparison and continuity of Jost solutions (Lemma 3.23, Lemma 3.25), the uniform estimates of Lemma 3.28, and the barrier argument of Lemma 8.39 with constants chosen on a compact set of data; nothing from Chapters 10–11 is used, and the statements hold on all of $\mathbb{D}_+$, not only for $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$ (in the supercritical case W_e is not admissible, and it is the barrier W_*^D that supplies the lower bound for w_0).

9.4.4. Indices at a fold.

Proposition 9.40 (Leray–Schauder indices at a regular fold)**.** *Let $\omega_0 \in \mathrm{Sol}$ be a regular fold of E with $k(\omega_0) = 1$ at which* (T) *holds, and let D be near $E(\omega_0)$ on the two-solution side. Then the two nearby solutions ω_1, ω_2 are nondegenerate ($1 \notin \operatorname{spec} A_{\omega_i}$) and their Leray–Schauder indices $\mathrm{ind}(T_D, f_{\omega_i}) = (-1)^{\beta_i}$ (β_i the number of real eigenvalues > 1 of A_{ω_i}, with algebraic multiplicity) are opposite: $\mathrm{ind}_1 + \mathrm{ind}_2 = 0$. If moreover $\psi^* v \neq 0$ (1 is an algebraically simple eigenvalue), then along the branch through the fold a real eigenvalue $\nu(a)$ of A near 1 crosses 1 transversally, $\nu'(0) \neq 0$, so β changes by exactly one.*

Proof. *Nondegeneracy.* By Lemma 9.19(a) there are coordinates (a, λ') with $E = (\chi, \lambda')$, and $\Phi^{\det}_{\mathrm{fold}} = \{\chi_a = 0\}$ near ω_0; the regular fold has $\chi_{aa}(0) \neq 0$, so this is $\{a = a^*(\lambda')\}$. For ω near ω_0 one has $k(\omega) \le 1$: with $L_0 = I - A_{\omega_0}$ and a closed complement Y of $\mathrm{span}(v)$, $L_0|_Y$ is bounded below, hence so is $(I - A_\omega)|_Y$ for ω near ω_0 ($\omega \mapsto A_\omega$ is continuous in operator norm by Lemma 8.27(a)), so $\ker(I - A_\omega) \cap Y = 0$ and $\dim\ker(I - A_\omega) \le \operatorname{codim} Y = 1$. If $k(\omega) = 1$, then $\psi_\omega^* B_\omega \neq 0$ by continuity, so (T) holds at ω and $\omega \in \Phi^{\det}_{\mathrm{fold}}$ by Theorem 9.22(c). The two solutions $\omega_{1,2}$ near ω_0 with data D are off $\Phi^{\det}_{\mathrm{fold}}$ (Lemma A.46: they correspond to $\eta = \pm\sqrt{w/\epsilon} \neq 0$), so $k(\omega_i) = 0$. *Indices.* Let $D_0 = E(\omega_0)$, $f_0 = f_{\omega_0}$. The point f_0 is an isolated fixed point of T_{D_0}: a fixed point $g \in \mathcal{A}_{D_0}$ near f_0 gives $\omega = \Sigma(D_0, g)$ near ω_0 with $E(\omega) = D_0$, so $\omega = \omega_0$ by Proposition 9.34(ii), and $g = f_0$. Choose a ball $\mathcal{B} = B_r(f_0) \subset \mathrm{BC}$ with $\overline{\mathcal{B}} \subset \mathcal{A}_{D_0}$ and no other fixed point of T_{D_0} in $\overline{\mathcal{B}}$. There is a connected open neighbourhood $\mathcal{N}$ of D_0 with $\overline{\mathcal{B}} \subset \mathcal{A}_D$ for $D \in \mathcal{N}$ (openness of $\mathcal{A}$), such that T_D has no fixed point on $\partial\mathcal{B}$: otherwise

$D_n \to D_0$ and fixed points $g_n \in \partial\mathcal{B}$ of T_{D_n} exist; they lie in the compact set Fix_K (Lemma 9.37, $K = \{D_n\} \cup \{D_0\}$), a subsequence converges to a fixed point of T_{D_0} on $\partial\mathcal{B}$, a contradiction. The map $(D, g) \mapsto T_D g$ is compact on $\mathcal{N}' \times \overline{\mathcal{B}}$ for a compact neighbourhood $\mathcal{N}' \subset \mathcal{N}$ of D_0: its image lies in $\mathcal{P}(\mathcal{N}', S, w_{\min})$ for $S = \|f_0\|_\infty + 1$ and $w_{\min} = \frac{1}{2} w_0(W_{f_0})$ (shrink r and $\mathcal{N}'$ so that $w_0(W_g) \geq w_{\min}$ and $\|g\| \leq S$ on $\overline{\mathcal{B}}$, possible because $g \mapsto w_0(W_g)$ is continuous on $\mathcal{A}_D$ by Lemma 3.25(b)), which is relatively compact by Lemma 9.36. By homotopy invariance (D2) of Theorem A.35 along paths in $\mathcal{N}'$, $\deg_{\mathrm{LS}}(I - T_D, \mathcal{B})$ is independent of $D \in \mathcal{N}'$. On the zero-solution side of Γ there is no fixed point in $\mathcal{B}$ (Proposition 9.34(ii) and the correspondence Σ), so the degree is 0; on the two-solution side the fixed points in $\mathcal{B}$ are exactly $f_{\omega_1}, f_{\omega_2}$, nondegenerate, and by additivity (D4) and the index formula (D5), $0 = \mathrm{ind}_1 + \mathrm{ind}_2$ with $\mathrm{ind}_i = (-1)^{\beta_i} = \pm 1$. *Eigenvalue crossing.* Along the curve $a \mapsto \omega(a) = \nu^{-1}(a, 0)$ put $D(a) = (\chi(a, 0), 0)$, $f(a) = f_{\omega(a)}$, $A_a = A_{\omega(a)}$, $L_a = I - A_a$, $B_a = B_{\omega(a)}$. Differentiating $f(a) = T_{D(a)} f(a)$ gives $L_a f_a = B_a D_a$ with $D_a = (\chi_a, 0)$, and $f_a(0) = v$. If $\psi^* v \neq 0$, then $v \notin \operatorname{ran} L_0$, so $\ker L_0^2 = \ker L_0$ and 0 is an algebraically simple eigenvalue of L_0; hence for small a there is a C^1 simple eigenvalue $\mu(a)$ of L_a with $\mu(0) = 0$ and a left eigenfunctional ψ_a^* (C^1 in a). Applying ψ_a^*: $\mu(a)\psi_a^*(f_a) = \chi_a(a)\, \psi_a^*(B_a \partial_s)$. Differentiating at $a = 0$ (where $\mu = \chi_a = 0$) gives $\mu'(0)\psi^*(v) = \chi_{aa}(0)\, \psi^*(B_{\omega_0}\partial_s)$. The right side is nonzero by (T) and by regularity ($\chi_{aa}(0) \neq 0$), so $\mu'(0) \neq 0$; the eigenvalue of A is $\nu = 1 - \mu$. □

9.4.5. The structure theorem.

Toy example 9.41 (proper local diffeomorphisms are coverings)**.** $p(x) = e^x$ from $\mathbb{R}$ to $(0, \infty)$ is a local diffeomorphism and a homeomorphism; $q(x) = x^3 - x$ from $\mathbb{R}$ to $\mathbb{R}$ is proper, has two fold points $x = \pm 1/\sqrt{3}$ and $\#q^{-1}(y)$ equals 1 or 3, odd away from the fold values $\pm \frac{2}{3\sqrt{3}}$. A proper local diffeomorphism $F : M \to N$ onto a connected N is a finite covering; if moreover N is simply connected and locally path connected, each component of M maps homeomorphically onto N, so $\#F^{-1}(y)$ is the number of components. This is the mechanism of part (B) below.

THEOREM 9.42 (structure of the fold set and global structure of the solution set). (A) Local structure.

- (A1) (The fold set is the singular set of E where (T) holds.) $\Phi_{\mathrm{fold}}^{\mathrm{det}} \subset \Phi_{\mathrm{fold}}$, *with equality at the points where* (T) *holds; at such points* $\ker \mathrm{D}E \cong \ker(I - A)$ *via* $w \mapsto \partial_w f_\omega$ *(Theorem 9.22).*
- (A2) (Hypersurface and count.) *At a regular fold with* $\dim \ker \mathrm{D}E = 1$, $\Phi_{\mathrm{fold}}^{\mathrm{det}}$ *is a smooth hypersurface mapped by* E *diffeomorphically onto a hypersurface* Γ *of fold values, and the number of solutions with a given* D *changes by* 2 *across* Γ *(Proposition 9.34); regularity is* $\psi^* \mathrm{D}_f^2 T[v, v] \neq 0$ *when* (T) *holds (Lemma 9.19).*
- (A3) (Indices.) *The two solutions on the two-solution side of a regular fold at which* (T) *holds and* $k = 1$ *have Leray–Schauder indices* $+1$ *and* -1 *(Proposition 9.40).*
- (A4) (Locally uniform a priori bounds; properness.) *For every compact* $K \subset \mathbb{D}_+$ *the fixed points of* T_D, $D \in K$, *are bounded, nondecreasing, uniformly Lipschitz (Lemma 9.37(iv)) and have uniform tails: they form a compact subset of* BC*; and* $E : \mathrm{Sol} \to \mathbb{D}_+$ *is proper (Lemma 9.37, Proposition 9.38).*

(B) Global structure. *Let $N(D) := \#E^{-1}(D) \in \{1, 2, \dots\} \cup \{\infty\}$ be the number of global solutions of type* G *at both poles with $\tilde{\kappa} > 0$ and effective data D.*

- (Ba) (Finite covering off the fold values.) *$E(\Phi^{\det}_{\mathrm{fold}})$ is closed in $\mathbb{D}_+$; $N(D) < \infty$ for $D \in \mathbb{D}_+ \setminus E(\Phi^{\det}_{\mathrm{fold}})$ and N is locally constant there. Every D has $N(D) \geq 1$.*
- (Bb) (Parity of the count at regular values.) *If D is a regular value of E (no preimage in $\Phi^{\det}_{\mathrm{fold}}$), then $N(D)$ is odd.*
- (Bc) (Uniqueness while no eigenvalue is 1.) *Let $\mathcal{U} \subset \mathbb{D}_+$ be open, connected and simply connected, and suppose some $D_1 \in \mathcal{U}$ has exactly one solution. If $\Phi^{\det}_{\mathrm{fold}} \cap E^{-1}(\mathcal{U}) = \emptyset$ (in particular if $1 \notin \operatorname{spec} \mathrm{D}T_D(f)$ at every fixed point of every T_D, $D \in \mathcal{U}$), then $E : E^{-1}(\mathcal{U}) \to \mathcal{U}$ is a diffeomorphism: every $D \in \mathcal{U}$ has exactly one solution.*
- (Bd) (Non-emptiness criterion.) *If $\mathcal{U}$ as in* (Bc) *contains a D_1 with exactly one solution and a D_2 with at least two, then $\Phi^{\det}_{\mathrm{fold}} \cap E^{-1}(\mathcal{U}) \neq \emptyset$, hence $\Phi_{\mathrm{fold}} \cap E^{-1}(\mathcal{U}) \neq \emptyset$.*

Proof. (A1) is Theorem 9.22, (A2) is Proposition 9.34 and Lemma 9.19(b), (A3) is Proposition 9.40, (A4) is Lemma 9.37 (the Lipschitz bound is Lemma 9.37(iv); the tails are the compactness of $\overline{\mathcal{P}}$) and Proposition 9.38.

(Ba) *Step 1: finiteness and local structure.* $E(\Phi^{\det}_{\mathrm{fold}})$ is the image of a closed set under a proper map into a locally compact Hausdorff space, hence closed. Let $D \notin E(\Phi^{\det}_{\mathrm{fold}})$. The fibre $E^{-1}(D)$ is compact (Proposition 9.38), and near each of its points E is a local diffeomorphism; so the fibre is discrete, hence finite, $\{\omega_1, \dots, \omega_n\}$ ($n \geq 1$ by Theorem 8.20 and Proposition 8.13). Choose pairwise disjoint open $V_i \ni \omega_i$ mapped diffeomorphically onto open $W_i \ni D$. There is a neighbourhood $W \subset \bigcap W_i$ of D with $E^{-1}(W) \subset \bigcup V_i$: otherwise there are $\tilde{\omega}_m \notin \bigcup V_i$ with $E(\tilde{\omega}_m) \to D$; by properness a subsequence converges to some $\tilde{\omega}$ with $E(\tilde{\omega}) = D$ and $\tilde{\omega} \notin \bigcup V_i$ (V_i open), contradicting that the fibre is $\{\omega_i\}$. Thus for $D' \in W$ the fibre has exactly n points, one in each V_i: N is locally constant on $\mathbb{D}_+ \setminus E(\Phi^{\det}_{\mathrm{fold}})$. (Bc) and (Bd) *Step 2.* If $\Phi^{\det}_{\mathrm{fold}} \cap E^{-1}(\mathcal{U}) = \emptyset$, then $E : E^{-1}(\mathcal{U}) \to \mathcal{U}$ is a local diffeomorphism, proper (compact subsets of $\mathcal{U}$ have compact preimages, Proposition 9.38), hence by Step 1 a finite covering over $\mathcal{U}$ (every point of $\mathcal{U}$ has an evenly covered neighbourhood). As $\mathcal{U}$ is simply connected and locally path connected, each connected component of $E^{-1}(\mathcal{U})$ maps homeomorphically onto $\mathcal{U}$ (lifting criterion; e.g. Hatcher, *Algebraic Topology*, Prop. 1.33), so $N(D)$ is the number of components and is independent of $D \in \mathcal{U}$. If $N(D_1) = 1$, E is a bijective local diffeomorphism, i.e. a diffeomorphism. The parenthetical condition suffices by Theorem 9.22(a) ($\Phi^{\det}_{\mathrm{fold}} \subset \Phi_{\mathrm{fold}}$). (Bd) is the contrapositive: if $\Phi^{\det}_{\mathrm{fold}} \cap E^{-1}(\mathcal{U}) = \emptyset$, then $N \equiv N(D_1) = 1$ on $\mathcal{U}$, contradicting $N(D_2) \geq 2$. (Bb) $E : \mathrm{Sol} \to \mathbb{D}_+$ is a proper smooth map between manifolds of the same dimension with connected target; its mod-2 degree is defined by $N(D) \bmod 2$ at any regular value D and is independent of D (for proper maps between non-compact manifolds see Hirsch, *Differential Topology*, Ch. 5 [15]; Milnor, *Topology from the Differentiable Viewpoint*, §4, and Guillemin–Pollack, Ch. 2, treat compact domains, where the argument is the same). The datum $D_1 = (\boldsymbol{c}' = 0, \vartheta = 1, f_- = 0, \kappa_{\mathrm{D}} = 1)$ has $\Lambda_{\mathrm{mod}} = \frac{3}{16} < 1$ (Remark 8.35), so by Theorem 8.34 it has exactly one fixed point f^* and $\|\mathrm{D}T_{D_1}(f^*)\| \leq \frac{1}{2}\Lambda_{\mathrm{mod}} < 1$; thus $1 \notin \operatorname{spec} A$ and $\det \mathrm{D}E \neq 0$ there (Theorem 9.22(a)), so D_1 is a regular value with $N(D_1) = 1$. Hence $N(D)$ is odd at every regular value. □

Remark 9.43 (the two solutions born at a fold are not ordered)**.** Let ω_0 be a regular fold with $k = 1$, (T), and $f_- \geq 0$. In the coordinates of Lemma 9.19(a) the two solutions over $D = (\chi(a_1, 0), 0) = (\chi(a_2, 0), 0)$ near D_0 are $a_1 \neq a_2$ with $a_{1,2} = \pm\eta + O(\eta^2)$, and $f_{\omega(a_1)} - f_{\omega(a_2)} = (a_1 - a_2)\, v + O(\eta^2)$ in BC, where v is the kernel vector. By Lemma 9.2 (corollary) v takes both signs, so to first order the two profiles cross. This is consistent with, and is a first-order reflection of, the antichain property Proposition 8.32(b): two distinct fixed points with $f_- \geq 0$ are never pointwise ordered. Note however that the kernel vector changes sign for $f_- \geq 0$ *because* the eigenvalue is $1 \geq 0$; for negative eigenvalues sign-definite eigenvectors do occur (Numerical observation 9.9).

Remark 9.44 (codimension count)**.** Under the hypotheses of Proposition 9.34 the fold values form a hypersurface (dimension 5) in the six-dimensional space $\mathbb{D}_+$. Hence along a generic one-parameter family, e.g. the line $\kappa_{\mathrm{D}} \mapsto (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$, folds are isolated points (turning points of the branch in κ_{D}, Corollary 9.26); along a generic two-parameter family they form curves, which end at cusp points; the cusp set is of codimension 2 in Sol and E maps it to a set of dimension 4. These are statements about the *generic* situation, in which the conditions $\chi_{aa} \neq 0$, resp. $\chi_{aaa} \neq 0$ and $\nabla_{\lambda'}\chi_a \neq 0$, hold; which of these conditions holds at the folds of the solution set is a matter of computation and is reported in Atlas 9.76.

Remark 9.45 (relation to the index criterion of Chapter 8)**.** Parts (Bb)–(Bd) are not new as statements; they are the geometric form of Proposition 8.48: (a) parity holds there for nondegenerate fixed points, with a proof by degree theory for T_D that needs neither the smooth structure nor regular values of E, and the mod-2 argument of (Bb) only reproduces it (it applies at regular values of E, which by Theorem 9.22(a) includes all D with no degenerate fixed point); (c) and (d) are the family version of the criterion "a change in the number of solutions forces an eigenvalue 1" (Theorem 8.42(d)). What the geometric picture adds is the smooth structure: under the hypothesis of (Bc) the solutions form a smooth graph over the effective data, and at a regular fold the covering degenerates exactly as described by Proposition 9.34 and Proposition 9.40. Also, Proposition 8.48(b) is a pointwise criterion (no real eigenvalue ≥ 1), whereas (Bc) is a statement over a simply connected set of data; neither implies the other.

Corollary 9.46 (the fold set is non-empty)**.** $\Phi_{\mathrm{fold}}^{\det} \neq \emptyset$*, hence* $\Phi_{\mathrm{fold}} \neq \emptyset$*: there are data D and fixed points of T_D with* $1 \in \operatorname{spec} \mathrm{D}T_D(f)$*. More precisely, E is not a diffeomorphism of* Sol *onto* $\mathbb{D}_+$*.*

Proof. Take $\mathcal{U} = \mathbb{D}_+$ (contractible, hence connected and simply connected), $D_1 = (0, 1, 0, 1)$ as in the proof of (Bb), which has exactly one solution (Theorem 8.34), and $D_2 = D^* = (\boldsymbol{c}' = (\frac{5}{4}, -\frac{15}{16}, -\frac{517}{160}), \vartheta = 40, f_- = 0, \kappa_{\mathrm{D}} = 1400) \in \mathbb{D}_+$, which has at least three distinct solutions by Theorem 9.68. Theorem 9.42(Bd) applies. □

Remark 9.47 (where the folds are, and what follows)**.** Corollary 9.46 rests on the computer-assisted Theorem 9.68; the *existence* of non-unique data had been observed numerically long before (Atlas 9.76). The sign structure of Lemma 9.2 restricts what a fold looks like: with $f_- \geq 0$ every kernel vector changes sign, and a kernel vector without sign change forces $f_- < 0$. The second half of the chapter uses the structure theorem along the line $\kappa_{\mathrm{D}} \mapsto (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$ to define the first and

last fold and to prove uniqueness below the first and above the last (Theorem 9.51), explains the mechanism of the fold (Heuristic 9.60), and proves Theorem 9.68 and Corollary 9.69.

9.5. Continuation in κ_{D}: the first and the last fold

In words. *Freeze the data $(\boldsymbol{c}', \vartheta, f_-)$ and let the swirl strength κ_{D} run from 0 to ∞. For each κ_{D} the fixed points of T_D form a finite set (when no fixed point is degenerate), the set moves continuously with κ_{D}, and its cardinality can change only at a* fold, *a value of κ_{D} at which some fixed point has 1 in the spectrum of $\mathrm{D}T_D$. For small κ_{D} there is exactly one fixed point (Chapter 8), and for large κ_{D} too (Chapter 10). So the picture is that of an S-curve with a bounded window $[\kappa_{\mathrm{first}}, \kappa_{\mathrm{last}}]$ outside of which the solution is unique: the first fold marks the end of the small-swirl regime, the last fold the start of the large-swirl regime. The window may be empty (the typical case at moderate ϑ), and the theorem of this section says nothing about whether it is not; Sections 9.6–9.8 show that it is not.*

Toy example 9.48 (an S-curve in one dimension)**.** Let $T_\kappa(x) = x - \frac{1}{4}(x^3 - 3x - \kappa + 3)$, so that the fixed points of T_κ are the roots of $x^3 - 3x = \kappa - 3$ and $T_\kappa'(x) = 1 - \frac{1}{4}(3x^2 - 3)$. For $\kappa < 1$ and $\kappa > 5$ there is one root. At $\kappa = 3$ there are three, $x = -\sqrt{3}, 0, \sqrt{3}$, with $T' = -0.5,\ 1.75,\ -0.5$: the outer ones are attracting, the middle one repelling, and the *indices* $\operatorname{sgn}(1 - T')$ are $+1, -1, +1$ with sum $+1$, as for the degree of Theorem 8.42. The folds are at $x = \mp 1$, $\kappa = 1$ and $\kappa = 5$, where $T' = 1$ (the double root). So the *first fold* is $\kappa_{\mathrm{first}} = 1$, the *last* is $\kappa_{\mathrm{last}} = 5$, there is exactly one solution below and above them, and an odd number, 1 or 3, for every other κ (`verify_ch09b.py`, part 2). Theorem 9.51 is the same statement for T_D on the Banach space $\mathrm{BC}(\mathbb{R})$.

Fix $\boldsymbol{c}' = (c_1', c_2', c_3')$ with $c_1', c_2' > -1$, $\vartheta > 0$ and $f_- \in \mathbb{R}$, and write $D_\kappa = (\boldsymbol{c}', \vartheta, f_-, \kappa)$ for $\kappa = \kappa_{\mathrm{D}} > 0$, $T_\kappa := T_{D_\kappa}$, $\mathrm{Fix}_\kappa := \{f \in \mathcal{A}_{D_\kappa} : f = T_\kappa f\}$ and

$$\mathcal{K}_{\mathrm{deg}} := \{\kappa > 0 :\ 1 \in \operatorname{spec} \mathrm{D}T_\kappa(f) \text{ for some } f \in \mathrm{Fix}_\kappa\},$$
$$\kappa_{\mathrm{first}} := \inf \mathcal{K}_{\mathrm{deg}}, \quad \kappa_{\mathrm{last}} := \sup \mathcal{K}_{\mathrm{deg}},$$

with $\inf \emptyset = +\infty$, $\sup \emptyset = 0$ (the *first* and *last fold*; they are values of κ_{D} at which a fixed point is degenerate, whether or not it creates new solutions, see Remark 9.53). Since $\mathrm{D}T_\kappa(f)$ is compact (Lemma 8.27(b)), $1 \in \operatorname{spec} \mathrm{D}T_\kappa(f)$ means that 1 is an eigenvalue, i.e. $f \in \Phi_{\mathrm{fold}}$ in the notation of Definition 9.20. Let

$$\Lambda_{\mathrm{mod}}(\kappa) := \frac{3\,\kappa\, S_1(\kappa)\, \widetilde{\Gamma}}{4\,\vartheta\, \widetilde{w}_0}, \qquad S_1(\kappa) = \max\big(|f_-|,\ f_- + \kappa/(\vartheta \widetilde{w}_0)\big),$$

be the moderate-swirl number of Theorem 8.34 for D_κ (it exists when $c_3' + \frac{1}{4}(f_-^+)^2 > \bar{c}_3$, e.g. if $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$; $\widetilde{w}_0$ and $\widetilde{\Gamma}$ do not depend on κ). It is continuous and strictly increasing in κ, with $\Lambda_{\mathrm{mod}}(\kappa) \to 0$ as $\kappa \to 0$ and $\to \infty$ as $\kappa \to \infty$ ($S_1 \geq |f_-|$, $S_1 \geq f_- + \kappa/(\vartheta \widetilde{w}_0)$, and the factor κ is strictly increasing). So $\Lambda_{\mathrm{mod}}(\kappa) = 1$ has exactly one root $\kappa_{\mathrm{lo}} = \kappa_{\mathrm{lo}}(\boldsymbol{c}', \vartheta, f_-) > 0$; at the rest state $\kappa_{\mathrm{lo}} = 4\vartheta/\sqrt{3}$ (Remark 8.35).

Lemma 9.49 (compactness along a segment)**.** *For every compact $I \subset (0, \infty)$ the set $\mathcal{F}_I = \{(\kappa, f) : \kappa \in I,\ f \in \mathrm{Fix}_\kappa\}$ is a compact subset of $\mathbb{R} \times \mathrm{BC}$.*

PROOF. $K_I = \{D_\kappa : \kappa \in I\}$ is a compact subset of $\mathbb{D}_+$ (a continuous image of I), and $\mathcal{F}_I$ is the image of the compact set Fix_{K_I} of Lemma 9.37 under the continuous map $(D, f) \mapsto (\kappa_{\mathrm{D}}, f)$. □

Lemma 9.50 (constancy of the number of solutions between folds)**.** *Let $I \subset (0,\infty)$ be a compact interval with $I \cap \mathcal{K}_{\deg} = \emptyset$. Then $\#\mathrm{Fix}_\kappa$ is finite and independent of $\kappa \in I$, and every $f_0 \in \mathrm{Fix}_{\kappa_0}$ ($\kappa_0 \in I$) is the value at κ_0 of a real-analytic curve $\kappa \mapsto F(\kappa) \in \mathrm{Fix}_\kappa$, $\kappa \in I$. In particular $\#\mathrm{Fix}_\kappa$ is odd (Proposition 8.48(a)).*

PROOF. Let $\kappa_0 \in I$. Every $f \in \mathrm{Fix}_{\kappa_0}$ is nondegenerate, so by Proposition 9.30 it is the only fixed point of T_{κ_0} in a ball $B_r(f) \subset \mathcal{A}_{D_{\kappa_0}}$; hence Fix_{κ_0} is discrete, and compact (Lemma 9.49), so finite, $\{f_1, \dots, f_n\}$, with $n \geq 1$ by Theorem 8.42(b). Choose $r > 0$ so that the balls $B_i = B_r(f_i)$ are disjoint, $\overline{B_i} \subset \mathcal{A}_{D_{\kappa_0}}$, and Proposition 9.30 applies in B_i. The proposition gives analytic curves F_i on an interval $I_0 \ni \kappa_0$ (the data D_κ form a curve in the neighbourhood $\mathcal{N}$ of that proposition) with $F_i(\kappa) \in B_i$ and $F_i(\kappa)$ the only fixed point of T_κ in B_i. We claim that after shrinking I_0 there are no other fixed points. Otherwise there are $\kappa_m \to \kappa_0$ and $g_m \in \mathrm{Fix}_{\kappa_m} \setminus \bigcup B_i$; by Lemma 9.49 a subsequence converges to $g \in \mathrm{Fix}_{\kappa_0}$, and $g \notin \bigcup B_i$ because the union of the balls is open, contradicting $\mathrm{Fix}_{\kappa_0} \subset \bigcup B_i$. So $\#\mathrm{Fix}_\kappa = n$ on I_0; the function $\kappa \mapsto \#\mathrm{Fix}_\kappa$ is locally constant on the connected set I, hence constant. The curves F_i extend along I by the same argument (the set of $\kappa \in I$ up to which the curve through f_0 exists is open and closed). □

THEOREM 9.51 (continuation in κ_{D}: first and last fold). *Let $c_1', c_2' > -1$, $\vartheta > 0$, $f_- \in \mathbb{R}$ be fixed.*

(i) (Closedness.) *$\mathcal{K}_{\deg}$ is closed in $(0,\infty)$.*

(ii) (Small swirl.) *Assume $c_3' + \frac{1}{4}(f_-^+)^2 > \bar{c}_3$. Then $\kappa_{\mathrm{first}} \geq \kappa_{\mathrm{lo}} > 0$, $\#\mathrm{Fix}_\kappa = 1$ for all $\kappa < \kappa_{\mathrm{first}}$, and the fixed point is nondegenerate and depends analytically on κ there.*

(iii) (Large swirl, conditional.) *Suppose that*

(H_∞) *there is $\kappa_{**} < \infty$ such that for every $\kappa \geq \kappa_{**}$ the map T_κ has exactly one fixed point f and $1 \notin \operatorname{spec} \mathrm{D}T_\kappa(f)$ (for instance $\operatorname{spec} \mathrm{D}T_\kappa(f) \cap [\frac{1}{2}, \infty) = \emptyset$).*

Then $\mathcal{K}_{\deg} \subset (0, \kappa_{**})$ is bounded, $\kappa_{\mathrm{last}} \leq \kappa_{**} < \infty$, and $\#\mathrm{Fix}_\kappa = 1$ for all $\kappa > \kappa_{\mathrm{last}}$, with the fixed point nondegenerate and analytic in κ there.

(iv) (Non-uniqueness forces folds.) *Assume $c_3' + \frac{1}{4}(f_-^+)^2 > \bar{c}_3$ and $\#\mathrm{Fix}_{\kappa_1} \geq 2$ for some κ_1. Then $\mathcal{K}_{\deg} \neq \emptyset$, $\kappa_{\mathrm{first}} \in \mathcal{K}_{\deg}$ (a degenerate fixed point exists at κ_{first}), $\kappa_{\mathrm{lo}} \leq \kappa_{\mathrm{first}} \leq \kappa_1$, and $\#\mathrm{Fix}_\kappa$ is odd for every $\kappa \notin \mathcal{K}_{\deg}$. If in addition ($\mathrm{H}_\infty$) holds, then also $\kappa_{\mathrm{last}} \in \mathcal{K}_{\deg}$, $\kappa_1 \leq \kappa_{\mathrm{last}} < \infty$, and $\{\kappa : \#\mathrm{Fix}_\kappa \geq 2\} \subset [\kappa_{\mathrm{first}}, \kappa_{\mathrm{last}}]$.*

PROOF. (i) Let $\kappa_m \to \kappa \in (0,\infty)$ with $\kappa_m \in \mathcal{K}_{\deg}$, and $f_m \in \mathrm{Fix}_{\kappa_m}$ with $1 \in \operatorname{spec} \mathrm{D}T_{\kappa_m}(f_m)$. By Lemma 9.49 a subsequence converges to $f \in \mathrm{Fix}_\kappa$, and $\mathrm{D}T_{\kappa_m}(f_m) \to \mathrm{D}T_\kappa(f)$ in operator norm (Proposition 9.16(R): $\mathrm{D}_f T_D$ is real-analytic, in particular continuous, in (D, f)). If $I - \mathrm{D}T_\kappa(f)$ were invertible, so would be $I - \mathrm{D}T_{\kappa_m}(f_m)$ for large m (invertible operators form an open set), a contradiction. So $\kappa \in \mathcal{K}_{\deg}$.

(ii) For $\kappa < \kappa_{\mathrm{lo}}$, $\Lambda_{\mathrm{mod}}(\kappa) < 1$ and Theorem 8.34 gives $\#\mathrm{Fix}_\kappa = 1$ with $\|\mathrm{D}T_\kappa(f^*)\| \leq \frac{1}{2}\Lambda_{\mathrm{mod}} < \frac{1}{2}$, so $\kappa \notin \mathcal{K}_{\deg}$; hence $\kappa_{\mathrm{first}} \geq \kappa_{\mathrm{lo}}$. Let $\kappa < \kappa_{\mathrm{first}}$ and choose

$\kappa_a < \min(\kappa_{\mathrm{lo}}, \kappa)$. The compact interval $[\kappa_a, \kappa]$ misses $\mathcal{K}_{\mathrm{deg}}$, so by Lemma 9.50 #Fix on it equals $\#\mathrm{Fix}_{\kappa_a} = 1$. Analyticity is Lemma 9.50.

(iii) By (H_∞), $\mathcal{K}_{\mathrm{deg}} \subset (0, \kappa_{**})$, so $\kappa_{\mathrm{last}} \le \kappa_{**}$. For $\kappa > \kappa_{\mathrm{last}}$ the compact interval $[\kappa, \kappa_{**} + 1]$ misses $\mathcal{K}_{\mathrm{deg}}$ and ends in a point with #Fix = 1; apply Lemma 9.50.

(iv) If $\mathcal{K}_{\mathrm{deg}} = \emptyset$, then $\#\mathrm{Fix}_\kappa$ is constant on $(0, \infty)$ (apply Lemma 9.50 on every compact subinterval), equal to 1 by (ii); this contradicts $\#\mathrm{Fix}_{\kappa_1} \ge 2$. So $\mathcal{K}_{\mathrm{deg}} \neq \emptyset$; it is closed by (i) and contained in $[\kappa_{\mathrm{lo}}, \infty)$ by (ii), hence contains its infimum $\kappa_{\mathrm{first}} \ge \kappa_{\mathrm{lo}} > 0$. By (ii), $\kappa_1 \ge \kappa_{\mathrm{first}}$ (below κ_{first} there is one solution). Oddness off $\mathcal{K}_{\mathrm{deg}}$ is Proposition 8.48(a). Under (H_∞) the closed set $\mathcal{K}_{\mathrm{deg}}$ is also bounded, so it contains its supremum, and by (iii) $\kappa_1 \le \kappa_{\mathrm{last}}$; the same reasoning for every κ with $\#\mathrm{Fix}_\kappa \ge 2$ gives the inclusion. □

Remark 9.52 (the large-swirl hypothesis)**.** Hypothesis (H_∞) is proved in Chapter 10 for every $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$ (and every $\vartheta > 0$, $f_- \in \mathbb{R}$): this is Theorem 10.39, which supplies the constant κ_{**}. The unconditional statement of this theorem for $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$ (parts (iii), (iv) without the hypothesis) is Theorem 11.58 in Chapter 11. Nothing in the proofs of this chapter uses those results; they are quoted where the numerical sections and the discussion refer to the window. In particular, if $\mathcal{K}_{\mathrm{deg}} = \emptyset$ for such data, then by Theorem 9.51(ii) and Lemma 9.50 uniqueness holds for all κ_{D}; the converse fails (see Remark 9.53(1)), and the only way for the number of solutions to leave 1 is through a point of $\mathcal{K}_{\mathrm{deg}}$, that is, through a degenerate fixed point, outside the uniqueness regimes of Chapters 8 and 10.

Remark 9.53 (what the theorem does not say)**.** (1) A degenerate fixed point need not create a second solution: $T_\kappa(x) = x - (x-1)^3 + \kappa - \kappa_0$ has the single fixed point $x = 1 + (\kappa - \kappa_0)^{1/3}$ for every κ, with $T' = 1$ at $\kappa = \kappa_0$ (a cubic tangency has index $+1$ on both sides). So $\kappa_{\mathrm{first}} < \infty$ is necessary, not sufficient, for non-uniqueness; a *regular* fold (Lemma 9.19) does create two solutions, and by Corollary 9.26 a fixed point with $k = 1$ and $\psi^*(f - f_-) \neq 0$ is a turning point of the branch in κ_{D} exactly when $\psi^* \mathrm{D}_f^2 T[v, v] \neq 0$. (2) For $\boldsymbol{c}' \notin \operatorname{int} \mathcal{J}$ the hypothesis (H_∞) is not known to hold (see Remark 10.41 in Chapter 10), so (iii), (iv) apply only if it is assumed; (i), (ii) and the counting lemmas hold for all data. (3) The thresholds $\kappa_{\mathrm{first}}, \kappa_{\mathrm{last}}$ are not explicit, and the constant κ_{**} of (H_∞) (see Theorem 10.39 in Chapter 10) is not either; the theorem is a statement about where non-uniqueness *can* sit, not about whether it occurs. Whether the window is empty is a computation: for the rest state with $\vartheta = 1$ no fold is found for $0.01 \le \kappa_{\mathrm{D}} \le 3 \cdot 10^4$, while for the data of Sections 9.6 to 9.8 it is not empty.

Example 9.54 (the rest state)**.** For $\boldsymbol{c}' = 0$, $f_- = 0$ one has $\widetilde{w}_0 = 2$, $\widetilde{\Gamma} = 1$, $\Lambda_{\mathrm{mod}} = 3\kappa_{\mathrm{D}}^2/(16\vartheta^2)$ and $\kappa_{\mathrm{lo}} = 4\vartheta/\sqrt{3}$ (= 2.3094 for $\vartheta = 1$, 230.94 for $\vartheta = 100$, 23094 for $\vartheta = 10^4$). Theorem 9.51(ii) says that the first fold, if there is one, lies beyond these values; (iii) says that, under (H_∞), the last fold, if there is one, lies below the (non-explicit) $\kappa_{**}(\vartheta)$. Numerically no fold is found for $\vartheta = 1$ (at the 44 sampled values of κ_{D}; Numerical observation 9.62) and non-empty for every sampled $\vartheta \ge 2400$ (Numerical observation 9.63; the sampled values are 2400, 2500, 2700, 3000, 5000, 7000, 10^4, $2 \cdot 10^4$): the lower fold lies at $\kappa_{\mathrm{D}} = 2.357 \cdot 10^5$ for $\vartheta = 2400$ and $1.363 \cdot 10^6$ for $\vartheta = 10^4$, a factor 42.5 and 59.0 above $\kappa_{\mathrm{lo}} = 4\vartheta/\sqrt{3}$.

Corollary 9.55 (the three-solution window at the certified datum)**.** *Let* $D^* = (\boldsymbol{c}' = (\frac{5}{4}, -\frac{15}{16}, -\frac{517}{160}), \vartheta = 40, f_- = 0, \kappa_{\mathrm{D}} = 1400)$ *be the datum of Theorem 9.68,*

and let $\mathcal{K}_{\deg}$, κ_{first}, κ_{last} *refer to the line* $\kappa_{\text{D}} \mapsto (\boldsymbol{c}', \vartheta, f_-, \kappa_{\text{D}})$ *through it.* (a) $\kappa_{\text{lo}} \le \kappa_{\text{first}} \le 1400$*: there is a degenerate fixed point at some* $\kappa_{\text{D}} \le 1400$*, and* $\#\text{Fix}_\kappa = 1$ *for* $\kappa_{\text{D}} < \kappa_{\text{first}}$. (b) *If* (H_∞) *holds on this line (it does, Remark 9.52), then also* $1400 \le \kappa_{\text{last}} < \infty$*: there is a degenerate fixed point at some* $\kappa_{\text{D}} \ge 1400$*, and the set of* κ_{D} *with at least two solutions contains* 1400 *and lies in* $[\kappa_{\text{first}}, \kappa_{\text{last}}]$.

PROOF. $c_3' = \bar{c}_3 + \frac{1}{20} > \bar{c}_3$, so $c_3' + \frac{1}{4}(f_-^+)^2 > \bar{c}_3$, and $\#\text{Fix}_{1400} \ge 3$ by Theorem 9.68; apply Theorem 9.51(iv). □

Numerically the branch has exactly two turning points on this line, $\kappa_{\text{D}} = 1300.03$ and 1462.16 (Numerical observation 9.75).

9.6. Class-Z folds and the mechanism

In words. *The simplest non-uniqueness found for the effective-data problem sits in class* Z*, where the swirl changes sign. At the datum of Numerical observation 9.58 there are three fixed points, two of which are stable under the iteration* $f \mapsto T_D f$*; as* κ_{D} *or* c_3' *is varied the branch is S-shaped, with two folds, and the three-solution region is a wedge in the* $(c_3', \kappa_{\text{D}})$ *plane, closing in a cusp. There is a simple reason why a sign change is needed for the simplest mechanism: the swirl enters the potential through* f^2*, and the potential decides how large the swirl increment* J *is; if* f *vanishes near the equator, a larger* J *makes* f^2 *smaller there, which deepens the well and increases* J *further, a positive feedback that can overpower the contraction. If* $f > 0$ *the same chain closes with the opposite sign. This section proves the exact identities behind the loop, shows it in a toy, and records the numerics; the loop itself is a heuristic.*

Lemma 9.56 (the loop identities)**.** *Let* $D \in \mathbb{D}_+$, $f = T_D f$, $W = W_f$, $\rho = \rho_W$. *Then*

(a) $J := f_+ - f_- = \kappa_{\text{D}}/(\vartheta\,\rho(0))$ *and* $f(0) = f_- + J/(1+\vartheta)$*; in particular* $f(0) > f_-$ *and*

$$\rho(0) = \frac{\kappa_{\text{D}}}{\vartheta(1+\vartheta)\,(f(0) - f_-)}. \tag{9.7}$$

(b) *If* $g \in \mathcal{A}_D$ *satisfies* $g^2 \ge f^2$ *pointwise, then* $\rho_{W_g}(0) \ge \rho(0)$*: a pointwise larger* $|swirl|$ *gives a larger Wronskian at the equator.*

PROOF. (a) By (8.1) of Lemma 8.10, $f = f_- + J\,\lg(\ell - \ln\vartheta)$ with $\ell = \int_0^\xi \rho$, $J = \kappa_{\text{D}}/(\vartheta\rho(0))$; at $\xi = 0$, $\ell = 0$ and $\lg(-\ln\vartheta) = 1/(1+\vartheta)$. Eliminating J gives (9.7). (b) $\mathcal{Q}$ is order preserving (Lemma 8.2(a)), so $g^2 \ge f^2$ gives $W_g \ge W_f$; both are admissible, with the same limits at $\mp\infty$, and Lemma 3.23 gives $w_0(W_g) \ge w_0(W_f)$. □

Read (9.7) and (b) together as a loop. An increase of J moves $f(0) = f_- + J/(1+\vartheta)$ upwards. If $f_- < 0$ and $f(0) < 0$, then $f(0)^2$ *decreases*; if the Wronskian $\rho(0)$ followed $f(0)^2$ it would decrease, and $J = \kappa_{\text{D}}/(\vartheta\rho(0))$ would increase: the loop has positive sign. If $f \ge 0$ the sign is negative. The lemma proves that every link is an identity or a monotonicity, except that $\rho(0)$ depends on f^2 on the whole line, not only on $f(0)^2$; replacing this dependence by the simplest one gives a toy in which everything can be computed.

Toy example 9.57 (the ring: a one-variable caricature of the loop)**.** Take the class-Z values $\vartheta = 4.49$, $f_- = -2.64$ of Numerical observation 9.58 and replace $\rho(0)$ by $a + b\,f(0)^2$ with $a = 0.15$, $b = 0.10$. By (9.7) the fixed-point equation becomes $\kappa_{\mathrm{D}} = \vartheta J\,(a + b(f_- + J/(1+\vartheta))^2)$, i.e. κ_{D} as an explicit cubic function of J:

$$\kappa_{\mathrm{D}}(J) = \vartheta J\Big(a + b\big(f_- + \tfrac{J}{1+\vartheta}\big)^2\Big), \qquad \kappa_{\mathrm{D}}'(J) = \vartheta\Big(a + bf(0)^2 + \tfrac{2b\,f(0)\,J}{1+\vartheta}\Big).$$

The last term is negative exactly when $f(0) < 0$. For $f_- = -2.64$ the curve has two turning points, $(J, \kappa_{\mathrm{D}}) = (6.787,\ 10.576)$ and $(12.538,\ 9.159)$, and at $\kappa_{\mathrm{D}} = 10$ there are three roots, $J = 4.872,\ 9.301,\ 14.814$, with $f(0) = -1.753,\ -0.946,\ +0.058$; for $f_- \geq 0$ all three terms of κ_{D}' are non-negative and the first is positive, so $\kappa_{\mathrm{D}}(J)$ is strictly increasing ($\kappa_{\mathrm{D}}'(J) \geq \kappa_{\mathrm{D}}'(0) = 3.80$ for $f_- = +2.64$), and there is exactly one solution for each κ_{D} (Figure 9.3(c)). The toy has the shape of the phenomenon (an S-curve, three solutions of which the middle one is unstable, none for $f_- \geq 0$); it is not a model: its numbers differ from those of Numerical observation 9.58.

Numerical observation 9.58 (three fixed points in class Z at a moderate datum)**.** *Take the datum $\boldsymbol{c}' = (-0.15,\ 0.09,\ \bar{c}_3 + 0.04)$, $\vartheta = 4.49$, $f_- = -2.64$, $\kappa_{\mathrm{D}} = 10$ ($\bar{c}_3 = -3.89853$, so $c_3' = -3.85853 \in \operatorname{int}\mathcal{J}$ and Theorem 8.16 applies). Solving $f = T_D f$ by Newton's method with arclength continuation on three grids ($N = 336, 736, 1440$ nodes) gives three fixed points, all of class* Z *($f_- < 0 < f_- + J$):*

	lower	*middle*	*upper*
J	3.6466	8.7391	13.8803
$f(0) = f_- + J/(1+\vartheta)$	-1.9758	-1.0482	-0.1117
$\rho(0) = \kappa_{\mathrm{D}}/(\vartheta J)$	0.6108	0.2549	0.1605
leading real eigenvalue $\nu_{\max}$ *of* $\mathrm{D}T_D(f)$	0.5376	1.6063	0.0136

The outer two are stable under the iteration (leading eigenvalue below 1); the middle one has a real eigenvalue > 1 (index -1, indices $+1, -1, +1$). The two rows $f(0)$ and $\rho(0)$ are the two sides of (9.7): the numbers satisfy $\rho(0) = \kappa_{\mathrm{D}}/(\vartheta J)$ to the four digits shown. Code: `code/ch09_uniq/verify_ch09b.py` *(part 5). Numerical, not part of any proof.*

Numerical observation 9.59 (the class-Z folds)**.** *Along the line $\kappa_{\mathrm{D}} \mapsto (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$ through the datum of Numerical observation 9.58 the branch of fixed points is S-shaped (Figure 9.3(a)) with two turning points at*

$$\kappa_{\mathrm{D}} = 6.30477\ (J = 12.788) \qquad \textit{and} \qquad \kappa_{\mathrm{D}} = 11.35302\ (J = 5.983),$$

identical to the digits shown on all three grids; $\nu_{\max} > 1$ on the middle branch exactly between them, with maximum $\Lambda = 1.9022$. An independent computation in the shooting chart (the Cauchy data of Lemma 9.11, Newton's method on the fold conditions $E(\omega) = D$, $\det \mathrm{D}E = 0$, with Runge–Kutta step $h = 0.005,\ 0.0025,\ 0.00125$ and box $L = 14, 18$) gives 6.304766 *and* 11.353022*, equal to six digits for all steps and boxes. Continuing the fold in the second parameter $c_3' - \bar{c}_3$ (with ϑ and f_- fixed) gives a fold curve with two branches that meet in a cusp near $(c_3' - \bar{c}_3, \kappa_{\mathrm{D}}) = (0.16,\ 13.74)$ (Figure 9.3(b)); at $\kappa_{\mathrm{D}} = 10$ the two folds are at $c_3' - \bar{c}_3 = -0.0469$ and 0.0827, and at $c_3' - \bar{c}_3 = 0.04$ the interpolated folds are $\kappa_{\mathrm{D}} = 11.358$ and 6.302, within $5 \cdot 10^{-3}$ of the values above. Inside the wedge between the two branches there are three solutions. The cusp is a point where the singularity theory of Lemma 9.19 predicts a Whitney cusp; whether it is one is stated as a conjecture in Conjecture 13.18.*

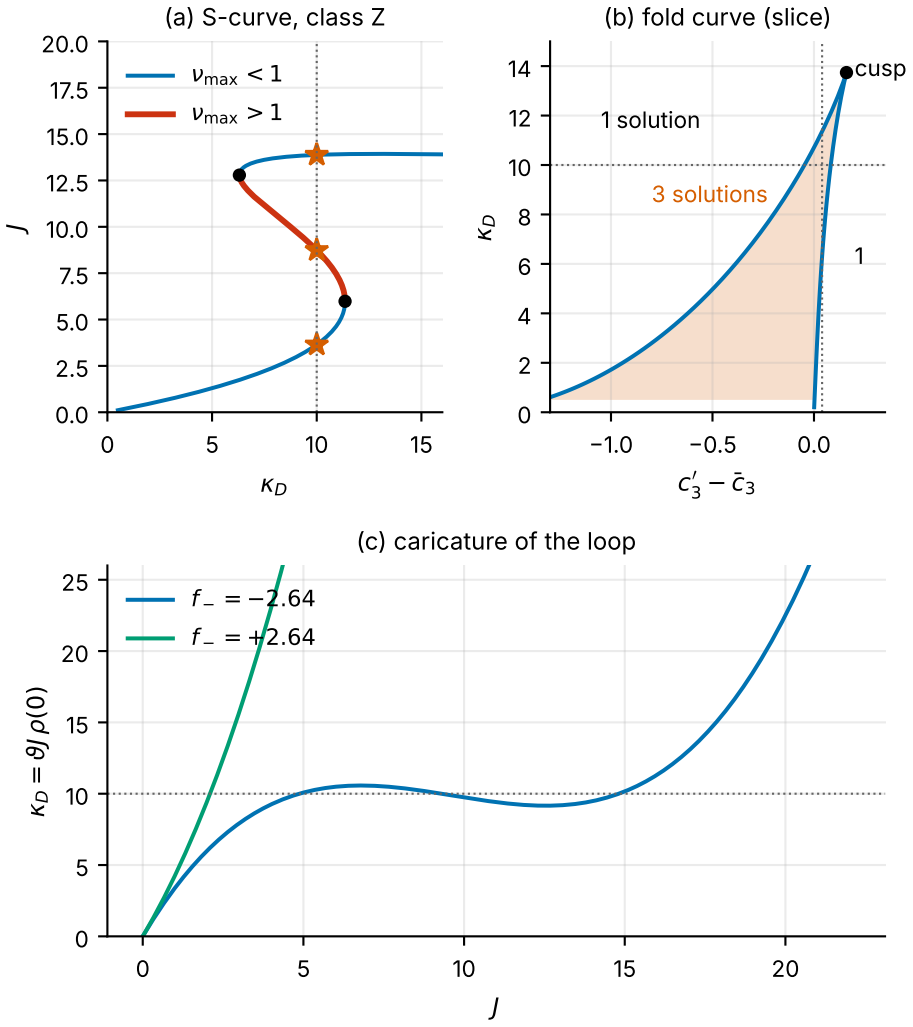


FIGURE 9.3. The class-Z fold picture at $\boldsymbol{c}' = (-0.15, 0.09, \bar{c}_3 + 0.04)$, $\vartheta = 4.49$, $f_- = -2.64$. (a) J against κ_D: S-curve with the folds $\kappa_D = 6.30477$, 11.35302 (dots) and the three fixed points at $\kappa_D = 10$ (stars); red: leading real eigenvalue > 1. (b) The fold curve in the plane $(c_3' - \bar{c}_3, \kappa_D)$; the shaded wedge has three solutions, outside one; the dotted lines are $\kappa_D = 10$ and $c_3' - \bar{c}_3 = 0.04$. (c) The caricature of Toy example 9.57: $\kappa_D(J)$ is S-shaped for $f_- < 0$ and monotone for $f_- > 0$. Code: `code/ch09_uniq/make_b_classZ.py`.

Heuristic 9.60 (mechanism of the folds, and the physical hysteresis)**.** *The potential is* $W_f = 1 + \frac{1}{2}P_{\boldsymbol{c}'} + \frac{1}{2}\mathcal{Q}[f^2]$. *The polynomial* $P_{\boldsymbol{c}'}$ *carries a well at the equator whose depth is governed by* c_3' *(it is deepest as* $c_3' \downarrow \bar{c}_3$*), and* $\mathcal{Q}[f^2]$ *fills the well with weight*

f^2. *By Lemma 9.56, the swirl enters through the loop* $J \to f(0) \to f(0)^2 \to \rho(0) \to J = \kappa_{\mathrm{D}}/(\vartheta\rho(0))$ *(the third arrow is Lemma 9.56(b), valid pointwise, and heuristic when used at the single point* $\xi = 0$*). When* $f(0) < 0$ *the loop has positive sign, a loop gain* > 1 *is a real eigenvalue* > 1 *of* $\mathrm{D}T_D$*, and the fold is where the gain passes through* 1*. The three fixed points of Numerical observation 9.58 are then the three ways the layer can sit relative to the well: the lower one has* $f(0) = -1.98$ *and* f^2 *fills the well (*$\rho(0) = 0.61$*), so* J *is small; the upper one has* $f(0) \approx -0.11$*, almost zero, so the well stays deep (*$\rho(0) = 0.16$*),* J *is large and the swirl layer is* pinned *so that* f *crosses* 0 *near the equator; the middle one is the unstable balance. When* $f > 0$ *near the well the chain has the opposite sign, and Lemma 9.2 (and Lemma 8.27(c)) excludes the positive eigenvectors that such a gain would require; this is the rigorous content, and it excludes only* one-signed *kernel vectors, not folds in class* N_0 *(Section 9.7).*

Dictionary with the physical hysteresis. *The* (-1)*-homogeneous swirling flows are the conically similar swirling jets studied by Goldshtik, Paull and Pillow, Sozou–Wilkinson–Shtern and Shtern–Hussain [10, 36, 40, 41, 44]; In the physical literature on these flows the term hysteresis (a jump between two flow states when a control parameter is raised and lowered) is the natural counterpart of an S-shaped branch; we have not checked the specific statements of these papers against the present results, and the following dictionary is an interpretation, not a comparison. In the present language the swirl intensity corresponds to a combination of the effective data (for instance* κ_{D} *or* $c_3' - \bar{c}_3$*), the two states of large and small swirl increment* J *to the upper and lower branches of the S-curve, the switching points to the folds, and the loop in the parameter plane (Figure 9.3(a), (b)) to the hysteresis loop; beyond the cusp there is no hysteresis. "Stable" means here that all eigenvalues of* $\mathrm{D}T_D$ *lie below* 1*, the analogue of linear stability within the class of conically similar flows, not stability for the time-dependent Navier–Stokes equations. A quantitative comparison with these papers is not attempted.*

Remark 9.61 (class Z and class N_0)**.** The observation above shows that, for $f_- = -2.64$, a fixed point with a real eigenvalue ≥ 1 and a second solution occur already at moderate data. Corollary 9.69 shows that three solutions also occur for $f_- = 0$, i.e. in class N_0, so no sign restriction on f_- gives uniqueness. The *sign structure* is what survives: $f_- \geq 0$ excludes non-negative kernel vectors at a fold (Lemma 9.2).

9.7. The ridge: folds at large ϑ in class N_0

In words. *In class* N_0 *(*$f_- \geq 0$*) the loop of the previous section has negative sign, and at generic data the leading eigenvalue of* $\mathrm{D}T_D$ *is far below* 1 *(Numerical observation 9.62); the worst cases found are on a narrow* ridge *on which it comes close to the threshold* 1 *(Numerical observation 9.65). The ridge is where the amplification inside* $\mathrm{D}T_D$ *is largest: for large* ϑ *the swirl layer sits at distance about* $\ln\vartheta$ *from the equator, the second term of* $\mathrm{D}T_D$ *in (8.6) accumulates the response of* ρ *over this distance, and the amplification grows with* $\ln\vartheta$*. Past a critical* ϑ_c *the leading eigenvalue exceeds* 1*, the branch in* κ_{D} *becomes S-shaped, and the three-solution window opens in a cusp. At the rest state* $\boldsymbol{c}' = 0$*,* $f_- = 0$ *no fold was found for* $\vartheta \leq 2300$ *and one was found at every sampled* $\vartheta \geq 2400$*, so* $\vartheta_c \approx 2.3 \cdot 10^3$*. Everything in this section is numerical or heuristic; the rigorous result for a class-*N_0 *datum is Theorem 9.68 in the next section.*

Numerical observation 9.62 (the rest state: no folds at moderate ϑ)**.** *At $\boldsymbol{c}' = 0$, $f_- = 0$, $\vartheta = 1$ the solution was followed for 44 values of $\kappa_{\mathrm{D}} \in [0.01, 3 \cdot 10^4]$ and the spectrum of $\mathrm{D}T_D$ computed. The leading real eigenvalue stays below 0.0047 (maximum at $\kappa_{\mathrm{D}} \approx 14.6$), the dominant eigenvalue by modulus reaches -0.992 (the rank-one regime of the large-swirl theorem, see Theorem 10.39 in Chapter 10), and no fold is found at these 44 sampled values: the proved range $\kappa_{\mathrm{D}} < \kappa_{\mathrm{lo}} = 2.309$ of Example 9.54 is conservative by a wide margin. For a generic datum in class N_0 ($\boldsymbol{c}' = (0.6, -0.4, \bar{c}_3 + 0.5)$, $\vartheta = 3$, $f_- = 1$) the corresponding numbers are $\kappa_{\mathrm{lo}} = 0.451$ and $\nu_{\max} \le 0.0613$ (at $\kappa_{\mathrm{D}} \approx 20.6$). Code: scan with the fixed-point solver of Chapter 8, run once and cached; read by* `code/ch09_uniq/verify_ch09b.py` *(part 5). Numerical, not part of any proof.*

Numerical observation 9.63 (the ridge at the rest state: three solutions and the birth of the fold pair)**.** (1) Three solutions at $\vartheta = 10^4$. *At $\boldsymbol{c}' = 0$, $f_- = 0$, $\vartheta = 10^4$ the branch in κ_{D} is S-shaped (Figure 9.4(c)) with turning points at $\kappa_{\mathrm{D}} = 1362776$ ($J = 44.71$) and $\kappa_{\mathrm{D}} = 1641832$ ($J = 75.15$); $\Lambda = \max \nu_{\max} = 1.3013$, attained near $\kappa_{\mathrm{D}} = 1.491 \cdot 10^6$. At $\kappa_{\mathrm{D}} = 1.5 \cdot 10^6$ there are three fixed points, with residual $\|f - T_D f\|_\infty \le 2 \cdot 10^{-13}$,*

$$J = 37.8731,\ 61.1612,\ 72.0626, \qquad \nu_{\max} = 0.4568,\ 1.3007,\ 0.5829$$

(indices $+1, -1, +1$); they have $f_- = 0$ and $f' > 0$, so they are in class N_0, and $\boldsymbol{c}' = 0 \in \operatorname{int} \mathcal{J}$. These are three solutions in class N_0 with the same effective data. (2) Birth of the fold pair. *Following the branch for other ϑ at $\boldsymbol{c}' = 0$, $f_- = 0$: no fold for $\vartheta = 1000, 2000, 2300$ ($\Lambda = 0.299,\ 0.969,\ 0.998$), a fold pair for $\vartheta = 2400$ ($\kappa_{\mathrm{D}} = 235656$ and 235817, $\Lambda = 1.0069$), 2500 (248253 and 248812, $\Lambda = 1.0153$), and $\vartheta = 2700, 3000, 5000, 7000, 10^4, 2{\cdot}10^4$ ($\Lambda = 1.031,\ 1.053,\ 1.158,\ 1.228,\ 1.301,\ 1.445$). The width $w(\vartheta)$ of the window is $162,\ 557,\ 1781,\ 4492$ for $\vartheta = 2400, 2500, 2700, 3000$. A law $w = c(\vartheta - \vartheta_c)^{3/2}$ fitted to $\vartheta = 2500, 2700$ gives $\vartheta_c = 2329$ and predicts $w(3000) = 4334$ against 4492 observed; equivalently $w^{2/3}$ is linear in ϑ (Figure 9.4(b)). This is the width law at a cusp (Corollary 9.32, Toy example 9.33), tested on four points and one fitted parameter pair: consistent with a Whitney cusp, not a demonstration of one (Conjecture 13.18, Open problem 13.19). Beyond the birth, Λ is well fitted for $\vartheta \ge 2500$ by $1 + 0.206 \ln(\vartheta/2320)$ (error $\le 4 \cdot 10^{-4}$), but the fitted constant 2320 is forced by the requirement $\Lambda = 1$ at birth and the range is a single decade.* (3) The position of the ridge. *Between $\vartheta = 2500$ and $2 \cdot 10^4$ the lower and upper fold at the rest state grow like $\vartheta^{1.22}$ and $\vartheta^{1.37}$ (local exponents of the cached scans; the window lies at finite κ_{D} for each ϑ, and the constant $\kappa_{**}(\vartheta)$ of Theorem 10.39 in Chapter 10 cannot be uniform in ϑ). The exponents found are below 2 at the rest state ($1.22,\ 1.37$) and at D^* (1.31 and 1.94 over $[100, 160]$, Numerical observation 9.64); a pure power law is not established by them: the lower fold at $\vartheta = 10^4$ is 1.23 times $2\vartheta(\ln \vartheta/1.236)^2$, which only shows that a form $\vartheta \ln^2 \vartheta$ is compatible with this one value. Code:* `code/ch09_uniq/verify_ch09b.py` *(part 5), figure* `make_b_ridge.py`*; the scans for $\vartheta \le 3000$ (other than 2300–2500) were run once with the fixed-point solver of Chapter 8 and cached, and $\vartheta = 10^4$ and $\vartheta = 2300, 2400, 2500$ were recomputed by* `_w_ridge.py`*. Numerical, not part of any proof.*

Numerical observation 9.64 (the ridge is not a peculiarity of $f_- = 0$ or of the rest state)**.** *Evidence for the robustness of the phenomenon, computed at*

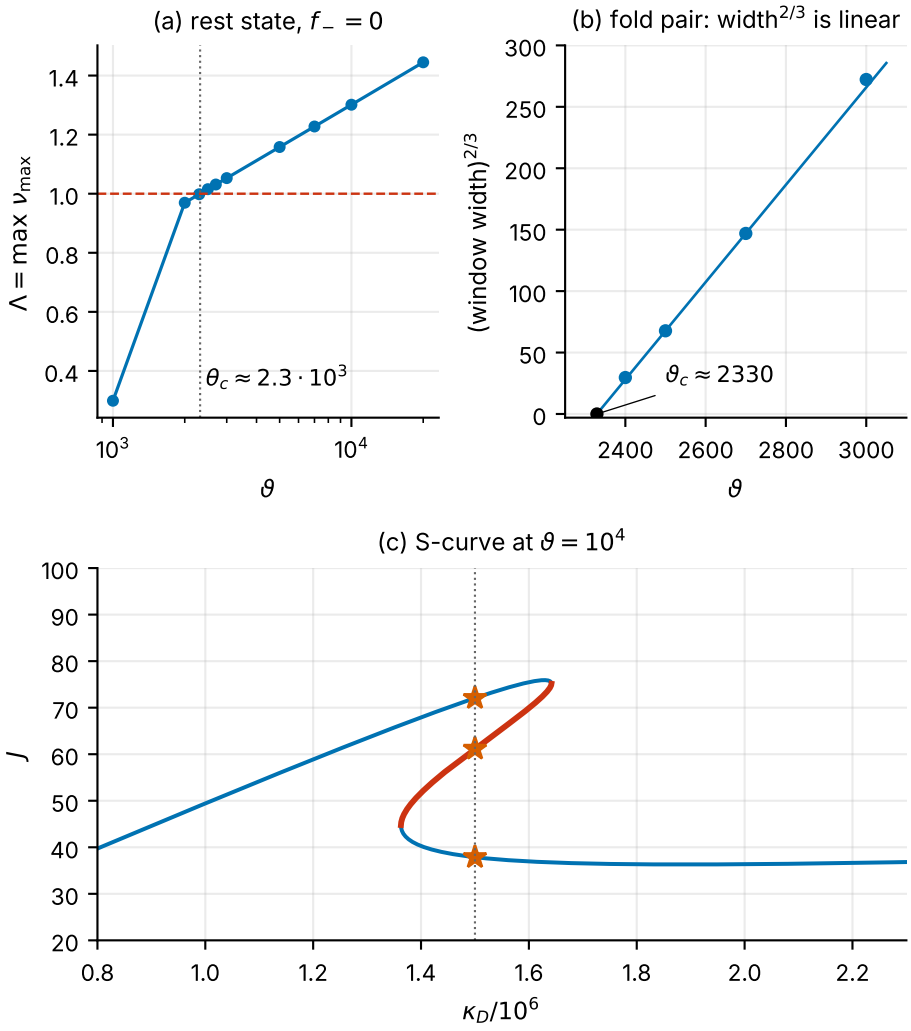


FIGURE 9.4. The ridge at the rest state ($\boldsymbol{c}' = 0$, $f_- = 0$). (a) $\Lambda(\vartheta) = \max_{\kappa_D} \nu_{\max}$ against ϑ; the fold pair exists once $\Lambda > 1$ (dashed). (b) (Width of the three-solution window)$^{2/3}$ against ϑ for $\vartheta = 2400, 2500, 2700, 3000$: a straight line through zero at $\vartheta_c \approx 2330$ is the 3/2-law of a cusp. (c) J against $\kappa_D/10^6$ at $\vartheta = 10^4$ with the three fixed points at $\kappa_D = 1.5 \cdot 10^6$ (stars); red: $\nu_{\max} > 1$. Code: `code/ch09_uniq/make_b_ridge.py`.

the datum D^* of Theorem 9.68 (the other data fixed): the fold pair exists for every sampled $f_- \in [-0.5, 1.0]$ ($\Lambda = 1.3848,\ 1.2921,\ 1.1834,\ 1.0619$ at $f_- = -0.5,\ 0,\ 0.5,\ 1$) and has disappeared at $f_- = 1.5$ ($\Lambda = 0.9321$); so the three solutions persist for strictly positive f_-, and $\Lambda(f_-)$ decreases smoothly through $f_- = 0$: so Λ itself has no peak at $f_- = 0$ at this datum (a different datum,

with a ridge in f_-, is in Numerical observation 9.65) With ϑ varied at D^*: $\Lambda =$ 0.384, 0.565, 0.718, 0.847, 0.985, 1.102, 1.292, 1.569, 1.927, 2.263 for $\vartheta =$ $8, 12, 16, 20, 25, 30, 40, 60, 100, 160$, so that Λ crosses 1 near $\vartheta \approx 25.6$; the fold positions have local exponents $\mathrm{d}\ln\kappa_\mathrm{D}/\mathrm{d}\ln\vartheta = 1.94$ (upper fold) and 1.31 (lower fold) over $[100, 160]$. The cached scan contains two further entries, $\vartheta = 250$ ($\Lambda = 2.586$, one fold found) and $\vartheta = 400$ ($\Lambda = 1.773$, one fold found); they are not monotone with the values above and one fold of the pair is missing, so the continuation did not complete there (the cause was not investigated) and these two entries are not used. Over the reliable range $8 \le \vartheta \le 160$ the growth of Λ shows no sign of saturating, but nothing is claimed beyond $\vartheta = 160$; the regime of larger ϑ is discussed in Open problem 13.21. Code: scans with the fixed-point solver of Chapter 8, run once and cached, read by `verify_ch09b.py` (part 5). Numerical.

Numerical observation 9.65 (a datum with a ridge close to 1 and no fold)**.** *Take $\boldsymbol{c}' = (1.288, -0.905, \bar{c}_3 + 0.11)$, $\vartheta = 28.5$, $f_- = 0$. Following the fixed point of T_D in κ_D from $0.01\vartheta^2$ to about $2.9\vartheta^2$ with the solver of the atlas (the grid of Numerical observation 9.63) gives a branch without turning points, with $\max\nu_\mathrm{max} = 0.9901$ near $\kappa_\mathrm{D} \approx 768$; for $f_- = \pm 0.01$ the maxima are 0.9886 and 0.9913 (at $\kappa_\mathrm{D} \approx 767$, 769), again without a fold. The branch was sampled at about 110 points; at the sample nearest to $\kappa_\mathrm{D} = 760$ the value is 0.881, so the ridge in κ_D is narrow and its height is not resolved to better than the sample spacing. At this datum the leading eigenvalue thus stays below 1 on the whole branch, but only by 1%; at D^* ($\vartheta = 40$, $c_3' - \bar{c}_3 = 0.05$) it reaches $\Lambda = 1.29$ (Numerical observation 9.75). Code: `_w_nearone.py`, `verify_ch09b.py` (part 5). Numerical, not part of any proof.*

Heuristic 9.66 (the plateau picture of the ridge)**.** *By (8.6), $\mathrm{D}T_D(f)v = J\varphi\big[-\delta\rho(0)/\rho(0) + (1-\varphi)\int_0^\xi \delta\rho\big]$ with $\varphi = \lg(\ell - \ln\vartheta)$. The factor $\varphi(1-\varphi)$ localises the second term in the layer $\xi \approx \xi_c$ with $\ell(\xi_c) = \ln\vartheta$, where $\int_0^\xi \delta\rho$ has accumulated the response of ρ over the length ξ_c. If ρ were constant, $\rho \approx 2q$, on $[0, \xi_c]$, then $\xi_c \approx \ln\vartheta/(2q)$ and the amplification would be proportional to ξ_c, i.e. to $\ln\vartheta$; a fold is where the amplification passes through 1. This predicts the* scaling *of Λ with $\ln\vartheta$, which is consistent with the fits of Numerical observation 9.63(2) and Numerical observation 9.64, and nothing more: a quantitative test of the plateau value of ρ against the computed profiles fails (the computed ρ on $[0.5, 4]$ is not constant and is well below the predicted plateau for the middle solution at D^*), and any mechanism in which an integral over a length $\propto \ln\vartheta$ enters would give the same scaling. The heuristic is not derived with controlled errors and no constant is claimed.*

Remark 9.67 (folds in class N_0 and the sign structure)**.** Lemma 9.2 says that at a fixed point with $f \ge 0$ a non-negative kernel vector with eigenvalue ≥ 1 cannot exist. So a fold in class N_0 *has a sign-changing kernel vector*; this is a restriction on what a fold looks like, not an obstruction to its existence. The folds of this section exist because the amplification in the second term of (8.6) can exceed 1 with a sign-changing eigenvector, although $\mathrm{D}T_D$ is not positive (its leading eigenvalues can even be complex, Numerical observation 8.51).

9.8. The computer-assisted proof of three solutions

In words. *The numerical evidence of the previous section is strong, but the question* is the effective data map injective on class N_0? *is a yes/no question, and we answer*

it with a proof. We exhibit one datum D^* and prove that T_{D^*} has three distinct fixed points. The proof is a computer-assisted existence proof for a finite-dimensional *problem: a solution is determined by four numbers at the equator, the four conditions that its effective data be D^* form four equations $\Phi(s) = 0$ in these four numbers, and for each of the three approximate solutions found numerically we prove, with interval arithmetic, that a true zero of Φ lies within 10^{-19} of it. Three ingredients make this rigorous: a validated integrator over the finite interval $[-12, 26]$; an explicit enclosure of the effects of the two infinite tails $\xi > 26$ and $\xi < -12$, which uses only comparison arguments for the Riccati equation; and Krawczyk's test, which turns an enclosure of a Jacobian into the existence of a zero. This section states the theorem, reduces it (in full) to a short list of certificate statements proved in Appendix D, explains the architecture on a one-dimensional toy, and says exactly what is and is not proved.*

9.8.1. The theorem.

THEOREM 9.68 (three solutions in class N_0: computer-assisted). *Let*

$$D^* = \Big(\boldsymbol{c}' = \big(\tfrac{5}{4}, -\tfrac{15}{16}, -\tfrac{517}{160}\big),\ \vartheta = 40,\ f_- = 0,\ \kappa_{\mathrm{D}} = 1400\Big),$$

so that $k_1' = \frac{3}{2}$, $k_2' = \frac{1}{4}$, $\bar{c}_3 = -\frac{105}{32}$ and $c_3' = \bar{c}_3 + \frac{1}{20}$, hence $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$ and T_D is defined on all of $\mathrm{BC}(\mathbb{R})$. *Then*

(a) *T_{D^*} has at least three distinct fixed points f_1, f_2, f_3; each is nondecreasing with $f_k(-\infty) = 0$ and $f_k' > 0$, and the jumps $J_k = f_k(+\infty)$ satisfy*

$$J_1 \in 19.040738660117701 \pm 6 \cdot 10^{-16},$$
$$J_2 \in 33.58919484747842 \pm 2 \cdot 10^{-15},$$
$$J_3 \in 46.31852825252179 \pm 5 \cdot 10^{-15}.$$

(b) *Equivalently there are at least three distinct global solutions (U, f, Q) of the reduced system, of type G at both poles, in class N_0, with effective data exactly D^*; their Cauchy data $s = (h_0, f_0, Q_0, Q_1) = (\frac{1}{2}U(0), f(0), Q(0), Q_x(0))$ lie in the boxes $\bar{s}_k + 10^{-19}[-1, 1]^4$ around the three centres of Table 9.2.*

(c) *Consequently the effective-data map E is not injective on class-N_0 solutions with $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$ (Corollary 9.69).*

Order of proof. Parts (a) and (b) are proved from the certificates alone (proof of Theorem 9.68 below); Corollary 9.55 and Corollary 9.69 are then deduced from (a), (b); part (c) merely restates Corollary 9.69 and is not used in any of these proofs, so there is no circularity.

Corollary 9.69 (the effective-data map is not injective; the spectral condition fails)**.** (a) *The map $E : \mathrm{Sol} \to \mathbb{D}_+$ is not injective on the set of class-N_0 solutions of type G at both poles with $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$: the three solutions $\omega_1, \omega_2, \omega_3$ of Theorem 9.68(b) have $E(\omega_k) = D^*$ and are pairwise different. So uniqueness of the global solution with given effective data fails in class N_0.* (b) *At least one of the fixed points of T_{D^*} has a real eigenvalue ≥ 1 in* $\operatorname{spec} \mathrm{D}T_{D^*}$ *(the condition "no real eigenvalue ≥ 1 at any fixed point" fails at D^*); and the number of solutions with data D^* is at least* 3*, so that by Corollary 9.55(a) the line through D^* contains a degenerate fixed point at some $\kappa_{\mathrm{D}} \leq 1400$ (and, with* (H_∞)*, at some $\kappa_{\mathrm{D}} \geq 1400$).*

PROOF. (a) $\omega_k \in \mathrm{Sol}$ are the points of Theorem 9.68(b); $E(\omega_k) = D^*$ and $f_- = 0$, $\kappa_{\mathrm{D}} > 0$, so they are in N_0 ($f_k > 0$ on $\mathbb{R}$ since $f_k' > 0$ and $f_k(-\infty) = 0$; Definition 5.3); they are pairwise different by Theorem 9.68(a) and Lemma 9.11(c). (b) By Proposition 8.48(c), $\#\mathrm{Fix} \geq 2$ forces some fixed point with $\operatorname{spec} \mathrm{D}T_D \cap [1,\infty) \neq \emptyset$; this is the contrapositive of Proposition 8.48(b). The last clause is Corollary 9.55. □

Remark 9.70 (what is, and what is not, proved)**.** The theorem proves the existence of *at least three* solutions at one datum, with the stated enclosures of J_k. It does *not* prove: that there are *exactly* three (this would need a global exclusion argument over a non-compact set of Cauchy data, Open problem 13.16); that each fixed point is nondegenerate, or any statement on the spectrum of $\mathrm{D}T_{D^*}$ (the eigenvalues quoted in Numerical observation 9.75 are not certified); or that three solutions persist for nearby data (this holds if the fixed points are nondegenerate, by Corollary 9.31, and is observed, Numerical observation 9.64). The S-curve in c_3' of class Z is in Numerical observation 9.58; Theorem 9.68 shows that the phenomenon is not confined to class Z.

9.8.2. Reduction to four equations in four unknowns. For $s = (h_0, f_0, Q_0, Q_1) \in \mathbb{R}^4$ consider the initial value problem for the polynomial system, in $\xi = \operatorname{artanh} x$, with $T = \tanh\xi$, $S = \operatorname{sech}^2\xi$ (so $T' = S$, $S' = -2ST$):

$$\begin{gathered} h' = 1 + \tfrac{1}{2}Q - h^2, \quad E' = -2hE, \quad f' = \tilde{\kappa}E, \\ Q' = P, \quad P' = S\,(\lambda - f^2 + 2Q) - 4TP, \\ h(0) = h_0, \quad E(0) = 1, \quad f(0) = f_0, \quad Q(0) = Q_0, \quad P(0) = Q_1, \\ \lambda = \lambda^* := \tfrac{123}{10}, \quad \tilde{\kappa} = \tilde{\kappa}^* := \tfrac{1400}{41^2}. \end{gathered} \tag{9.8}$$

This is the system on the line (2.29) in the variables $h = z'/z$, $E = z^{-2}$ with the normalisation $z(0) = 1$: indeed $z'' = (1 + \frac{1}{2}Q)z$ gives $h' = 1 + \frac{1}{2}Q - h^2$, $E' = -2hE$ and $f' = \tilde{\kappa}z^{-2} = \tilde{\kappa}E$; the last equation is $\mathsf{L}[Q] = \lambda - f^2$ solved for Q'' (with $\mathsf{L}[g] = \cosh^2\xi\, g'' + 4\sinh\xi\cosh\xi\, g' - 2g$, Proposition 2.20); $U = 2(h - T)$. (Symbolic check of the last equation: `verify_ch09b.py`, part 1.) Let $\mathcal{S} \subset \mathbb{R}^4$ be the set of s for which the solution exists on all of $\mathbb{R}$ with $E > 0$ (i.e. $z > 0$) and is of type G at both poles. For $s \in \mathcal{S}$ the limits $f(\pm\infty)$ and $Q(\pm\infty) := \lim_{\xi\to\pm\infty} Q$ exist, and we put

$$\Phi(s) = \Big(Q(-\infty) - \tfrac{5}{2},\ \ Q(+\infty) + \tfrac{15}{8},\ \ f(+\infty) + 40f(-\infty) - 41f_0,\ \ f(-\infty)\Big). \tag{9.9}$$

Lemma 9.71 (the reduction)**.** (a) *For $s \in \mathcal{S}$ the point $\omega(s) = (\boldsymbol{c}, \gamma, A, \tilde{\kappa}) \in \mathrm{Sol}$ with $\gamma = 2h_0$, $A = f_0$, $\tilde{\kappa} = \tilde{\kappa}^*$ and $\boldsymbol{c}$ given by $c_3 = -\frac{1}{2}Q_{xx}$, $c_{1,2} = \frac{1}{2}Q_0 \mp \frac{1}{2}Q_1 + \frac{1}{4}Q_{xx}$, $Q_{xx} = \lambda^* - f_0^2 + 2Q_0$, is the Cauchy datum at $x = 0$ of the solution (9.8).* (b) *For $s \in \mathcal{S}$: $E(\omega(s)) = D^*$ if and only if $\Phi(s) = 0$.* (c) *Every $\omega \in \mathrm{Sol}$ with $E(\omega) = D^*$ is of the form $\omega(s)$ for exactly one $s \in \mathcal{S}$, and this s satisfies $\Phi(s) = 0$.*

PROOF. (a) At $\xi = 0$, $T = 0$, $S = 1$, $\partial_x = \partial_\xi$; the second-order jet of Q at 0 is (Q_0, Q_1, Q_{xx}) with $Q_{xx}(0) = \mathsf{L}[Q](0) + 2Q_0 = \lambda - f_0^2 + 2Q_0$ (Proposition 2.20, $\mathsf{L}[g](0) = g_{xx}(0) - 2g(0)$), and the relations $Q(0) = c_1 + c_2 + c_3$, $Q_x(0) = c_2 - c_1$, $Q_{xx}(0) = -2c_3$ (Remark 2.6) give the stated $\boldsymbol{c}$ (computer-algebra check: part 1 of the script); $\lambda = -2c_1 - 2c_2 - 4c_3 + A^2$ (Proposition 2.20) is then λ^*. The state $z(0) = 1$, $z'(0) = h_0$, i.e. $\gamma = U(0) = 2h_0$ ($T(0) = 0$), $f(0) = A$, $\tilde{\kappa} = f'(0)$ since $E(0) = 1$. By definition of $\mathcal{S}$ the solution is global of type G at both poles with $\tilde{\kappa} > 0$,

i.e. $\omega(s) \in$ Sol. (b) By Lemma 9.11(a), $E(\omega) = D$ has $c_1' = \frac{1}{2}Q(-\infty)$, $c_2' = \frac{1}{2}Q(+\infty)$, $f_- = f(-\infty)$, $c_3' = -\frac{1}{4}(\lambda + 2c_1' + 2c_2')$ and $\kappa_{\mathrm{D}} = (1+\vartheta)^2\tilde{\kappa}$; moreover $E(\omega)$ is the datum of a fixed point $f = T_D f$, so by Lemma 9.56(a) $f(0) = f_- + J/(1+\vartheta)$ with $J = f(+\infty) - f(-\infty)$, i.e. $1 + \vartheta = J/(f_0 - f_-)$. Hence, given $\lambda = \lambda^*$ and $\tilde{\kappa} = \tilde{\kappa}^*$: $c_1' = \frac{5}{4} \iff Q(-\infty) = \frac{5}{2}$; $c_2' = -\frac{15}{16} \iff Q(+\infty) = -\frac{15}{8}$; with these, $c_3' = -\frac{517}{160} \iff \lambda^* = \frac{123}{10}$ (because $-4c_3' - 2c_1' - 2c_2' = \frac{517}{40} - \frac{5}{2} + \frac{15}{8} = \frac{123}{10}$, exact arithmetic in part 1); $f_- = 0 \iff f(-\infty) = 0$; $\vartheta = 40 \iff J = 41(f_0 - f_-) \iff f(+\infty) + 40f(-\infty) - 41f_0 = 0$; and then $\kappa_{\mathrm{D}} = 41^2\tilde{\kappa}^* = 1400$. (c) If $E(\omega) = D^*$ then $\tilde{\kappa} = \kappa_{\mathrm{D}}/(1+\vartheta)^2 = \tilde{\kappa}^*$ and $\lambda(\omega) = -4c_3' - 2c_1' - 2c_2' = \lambda^*$ by Lemma 9.11(a); the map $\omega \mapsto (\frac{1}{2}\gamma, A, Q(0), Q_x(0))$ is a bijection from $\{\tilde{\kappa} = \tilde{\kappa}^*,\ \lambda = \lambda^*\} \subset \mathbb{R}^6$ onto $\mathbb{R}^4$ (its inverse is (a)); $\omega \in$ Sol means $s \in \mathcal{S}$. □

So *to find solutions with data D^* is to find zeros of* $\Phi : \mathcal{S} \to \mathbb{R}^4$. Φ involves limits at $\xi = \pm\infty$, which a computer cannot evaluate; the plan is to integrate (9.8) to $\xi = X_R = 26$ and $\xi = -X_L = -12$ (validated), and to replace the limits by explicit formulas in the values at the cuts plus enclosed errors.

9.8.3. The three tools, on toys.

Toy example 9.72 (the architecture in one dimension: the free problem)**.** Let the swirl feed back nothing: $Q \equiv 0$ (the rest potential $W \equiv 1$, $\lambda = 0$). Then $h' = 1 - h^2$ gives $h = \tanh(\xi + c)$, $c = \operatorname{artanh} h_0$, $E = \cosh^2 c/\cosh^2(\xi + c)$, so $J = \tilde{\kappa}\int E = \dfrac{2\tilde{\kappa}}{1-h_0^2}$ and $f(0) - f_- = \tilde{\kappa}\int_{-\infty}^{0} E = \dfrac{\tilde{\kappa}(1+h_0)}{1-h_0^2}$, and $\Phi_3(h_0) = J - 41(f_0 - f_-) = -\tilde{\kappa}\dfrac{39 + 41h_0}{1-h_0^2}$ with the unique zero $h_0 = -\frac{39}{41}$ (that is $\vartheta = \frac{1-h_0}{1+h_0} = 40$, the relation $\vartheta = (2-\gamma)/(2+\gamma)$ of Example 9.13). Now run Krawczyk's test (Lemma A.57) on $X = [x_0 - r, x_0 + r]$, $x_0 = -\frac{39}{41}$, $r = 10^{-3}$, with $\tilde{\kappa} = \frac{1400}{1681}$: $\Phi_3'(h) = -\tilde{\kappa}(41 + 78h + 41h^2)/(1-h^2)^2$, $\Phi_3'(x_0) = -358.75$, $C = 1/\Phi_3'(x_0)$, $\Phi_3(x_0) = 0$, and ball arithmetic at 200 bits gives $\mathbf{K} = x_0 + \mathbf{M}(X - x_0)$ with $|\mathbf{M}| \le 0.0838$ and radius $8.4 \cdot 10^{-5} < r$: the inclusion $\mathbf{K} \subset \operatorname{int} X$ holds, so there is exactly one zero in X, which we knew. The *point* of the toy is what is not in it: for $Q \neq 0$ the function Φ has no closed form, the flow is only known on $[-12, 26]$ and the limits at the poles must be enclosed; yet the same three steps (enclose the flow, enclose the Jacobian, test $\mathbf{K} \subset \operatorname{int} X$) apply. The free value $x_0 = -0.95122$, $J = 17.5$ is also a reasonable first guess for the lowest solution of D^*: its certified centre has $h_0 = -0.95147$ and $J = 19.0407$ (the feedback is weak for the lowest solution). Code: `verify_ch09b.py`, part 2.

(T1) Enclosures and the mean value form. A *ball* $[x \pm \varrho]$ carries the true value of a quantity; interval arithmetic with outward rounding (Lemma A.52) produces balls for sums, products and quotients. Naive enclosure of a flow loses accuracy exponentially (the *wrapping effect*); the remedy is the *mean value form* (Lemma A.53): one encloses the flow of the midpoint of the box of initial values and an interval matrix for its derivative (the variational equation) and propagates $\varphi(\mathbf{Y}) \subset \varphi(\hat{y}) + \mathbf{A}(\mathbf{Y} - \hat{y})$. One step of size h is a Taylor series of degree 24 with a remainder bound whose validity is certified by an a priori enclosure (Lemma A.54); since (9.8) has polynomial right-hand side, the Taylor coefficients are computed exactly by the recursion of that lemma. This is what is called Lohner's method [28]: $\mathbf{Y}_{\mathrm{new}} = \hat{y}_{\mathrm{new}} + \mathbf{A}_{\mathrm{step}}(\mathbf{Y} - \hat{y})$, $\mathbf{A}_{\mathrm{new}} = \mathbf{A}_{\mathrm{step}}\mathbf{A}$. The statement used is Theorem D.8: if the box $\mathbf{Y}_j$ contains $\varphi_{t_j}(s)$

for all s in the parameter box and $\mathbf{A}_j \supset \partial_s \varphi_{t_j}(s)$, then $\mathbf{Y}_{j+1}, \mathbf{A}_{j+1}$ have the same property at t_{j+1}.

(T2) The tails. Beyond the cut $X = X_R$ the solution is not computed but enclosed. Three facts do the work. (i) The equation $\mathsf{L}[Q] = \lambda - f^2$ can be solved from a point $x_a = \tanh X$ out to the pole with a Cauchy kernel of known sign, so that $Q(x)$ lies in a tiny interval determined by $Q(X), Q'(X)$ and the a priori range of f (this is Lemma 2.23 in the form of the Cauchy formula, with $\varepsilon = e^{-2X}$). (ii) Hence $W = 1 + \frac{1}{2}Q$ is trapped between two constants $a^2 \le W \le b^2$ on $[X, \infty)$, and the Riccati equation $h' = W - h^2$ is trapped between the solutions of $\hat{h}' = a^2 - \hat{h}^2$ and $\hat{h}' = b^2 - \hat{h}^2$ with the same initial value (comparison, Toy example 9.73 below), so that z stays positive and the pole is of type G with $h \to \mu = \sqrt{1 + \frac{1}{2}Q(1)}$. (iii) The remaining increment of f is then $\int_X^\infty \tilde{\kappa} E$, trapped between $\tilde{\kappa} E_X/(b + h_X)$ and $\tilde{\kappa} E_X/(a + h_X)$.

Toy example 9.73 (the tail of f for a constant and for a squeezed potential)**.** If $W \equiv \mu^2$ from $\xi = X$ on, $z = z_X\big(\cosh \mu t + \frac{h_X}{\mu} \sinh \mu t\big)$ $(t = \xi - X)$ and $\int_0^\infty (\cosh \mu t + \frac{h_X}{\mu} \sinh \mu t)^{-2}\, \mathrm{d}t = \frac{1}{\mu + h_X}$; for $\mu = h_X = \frac{1}{4}$ the potential is $e^{t/4}$ and the integral is $\int_0^\infty e^{-t/2} \mathrm{d}t = 2 = \frac{1}{1/4 + 1/4}$. If W is only known to satisfy $a^2 \le W \le b^2$ with $a = \frac{1}{4}$, $b = 0.30$, $h_X = \frac{1}{4}$, the integral $\int_0^\infty z^{-2}$ (with $z(X) = 1$) is trapped in $[\frac{1}{b + h_X}, \frac{1}{a + h_X}] = [1.8182, 2]$. For 200 random potentials $W \in [a^2, b^2]$ the computed integral lies in $[1.843, 1.964]$, inside the enclosure, as it must (`verify_ch09b.py`, part 2). At D^* the right pole is $\mu_R = \frac{1}{4}$ and the left one $\mu_L = \frac{3}{2}$, i.e. $\mu_R = k_2'$, $\mu_L = k_1'$ (Table 9.2, certified values $\mu_R \in 0.2500 \pm 5 \cdot 10^{-17}$, $\mu_L \in 1.5000 \pm 1.6 \cdot 10^{-10}$).

(T3) Krawczyk. Let Φ be continuous on a box $\mathbf{X} = \bar{s} + r[-1, 1]^4$ and $\Phi = \Phi_0 + \mathsf{Z}$ with Φ_0 smooth and $\mathsf{Z}(s) \in \mathbf{T}$ for all $s \in \mathbf{X}$, $\mathbf{T}$ a (tiny) box. If C is a matrix, $\mathbf{J}$ an interval matrix containing $\mathrm{D}\Phi_0(\mathbf{X})$, $\mathbf{V} \ni \Phi_0(\bar{s})$ and

$$\mathbf{K} := \bar{s} - C(\mathbf{V} + \mathbf{T}) + (I - C\mathbf{J})(\mathbf{X} - \bar{s}) \;\subset\; \operatorname{int} \mathbf{X},$$

then C is invertible and Φ has a zero in $\mathbf{K}$ (Corollary A.59, proved in Appendix A by Brouwer's theorem; its proof does not use differentiability of Z). Equivalently (Lemma A.57) $|C(\mathbf{V}+\mathbf{T})| + |I - C\mathbf{J}|\, r < r$ componentwise, which is what is computed: write $Y = \max |C(\mathbf{V} + \mathbf{T})|$ and $Z_1 = \|I - C\mathbf{J}\|_\infty$; the test passes if $Y/r + Z_1 < 1$.

9.8.4. The certificate lemmas and the proof of Theorem 9.68. *The truncated map.* Let $\varepsilon_R = e^{-2X_R}$, $c_R = \cosh^2 X_R$ and write $(h_R, E_R, f_R, Q_R, P_R)$ for the state of (9.8) at $\xi = X_R = 26$, and $(h_L, \dots)$ at $\xi = -X_L = -12$ ($\varepsilon_L = e^{-2X_L}$, $c_L = \cosh^2 X_L$). Put

$$\begin{aligned}
Q_+^{\mathrm{tr}} &= Q_R(1 + \varepsilon_R) + c_R \varepsilon_R P_R + \tfrac{\varepsilon_R}{2}(\lambda^* - f_R^2), \\
f_+^{\mathrm{tr}} &= f_R + \frac{\tilde{\kappa}^* E_R}{h_R + a_R}, \quad a_R = \sqrt{1 + \tfrac{1}{2} Q_+^{\mathrm{tr}}}, \\
Q_-^{\mathrm{tr}} &= Q_L(1 + \varepsilon_L) - c_L \varepsilon_L P_L + \tfrac{\varepsilon_L}{2}(\lambda^* - f_L^2), \\
f_-^{\mathrm{tr}} &= f_L - \frac{\tilde{\kappa}^* E_L}{a_L - h_L}, \quad a_L = \sqrt{1 + \tfrac{1}{2} Q_-^{\mathrm{tr}}},
\end{aligned}$$

and $\Phi_0(s) = \big(Q_-^{\mathrm{tr}} - \frac{5}{2},\ Q_+^{\mathrm{tr}} + \frac{15}{8},\ f_+^{\mathrm{tr}} + 40 f_-^{\mathrm{tr}} - 41 f_0,\ f_-^{\mathrm{tr}}\big)$, a smooth function of s through the flow. These are the values the formulas would give if W were constant beyond the cuts; they are the centres of the tail enclosures.

Why the formula for $Q_\pm^{\mathrm{tr}}$ is the right centre. In the variable $x = \tanh\xi$ the equation $\cosh^2\xi\, Q_{\xi\xi} + 4\sinh\xi\cosh\xi\, Q_\xi - 2Q = g$, $g := \lambda - f^2$, reads $(1-x^2)Q_{xx} + 2xQ_x - 2Q = g$ (use $Q_\xi = (1-x^2)Q_x$ and $\cosh^2\xi = 1/(1-x^2)$). The homogeneous equation has the solutions x and $1+x^2$ (direct substitution), and a constant g has the particular solution $-g/2$. So if $g \equiv g_R$ were constant on $[x_a, 1]$, $x_a = \tanh X_R$, then $q := Q + g_R/2 = Ax + B(1+x^2)$, and eliminating A, B from the values $q(x_a)$, $q_x(x_a)$ gives $B(1-x_a^2) = q(x_a) - x_a q_x(x_a)$ and

$$q(1) = q(x_a) + (1-x_a)q_x(x_a) + B(1-x_a)^2 = (1+\varepsilon_R)\, q(x_a) + \varepsilon_R\, q_x(x_a),$$

because $(1-x_a)/(1+x_a) = e^{-2X_R} = \varepsilon_R$. With $q_x(x_a) = Q_x(x_a) = c_R P_R$ ($P = Q_\xi$, $Q_x = \cosh^2\xi\, Q_\xi$) this is exactly $Q(1) = Q_R(1+\varepsilon_R) + c_R\varepsilon_R P_R + \frac{\varepsilon_R}{2} g_R = Q_+^{\mathrm{tr}}$ with $g_R = \lambda^* - f_R^2$. The true g is not constant: f increases on $[X_R, \infty)$ by at most τ^+, so, if $f_R \geq 0$, $g - g_R \in [-\tau^+(2f_R + \tau^+), 0]$, and the difference $Q(1) - Q_+^{\mathrm{tr}}$ is the (sign-definite, $O(\varepsilon_R)$) response to this deviation, enclosed in Proposition D.10; this is also why the enclosure widths in Numerical observation 9.74(1) decay like $e^{-2.5X}$. The left formula is the mirror image ($x \to -x$, $P \to -P$). (The symbolic identity above is checked in `verify_ch09b.py`, part 2.)

The proof of Theorem 9.68 consists of the following four certificate statements. The first three are general lemmas proved on paper in Appendix D; the fourth is the numerical output of the computer for the three boxes.

(C1) Lemma D.6.: The polynomial system (9.8), the variational system for ∂_s of the state, and the definition of Φ_0 above; for $s \in \mathcal{S}$, $\Phi(s) = 0$ iff $E(\omega(s)) = D^*$ (this is Lemma 9.71, proved above).

(C2) Proposition D.10 (right and left tail).: Let the state at $\xi = X_R$ satisfy, with $f_R \geq 0$, with $F_{\max} > f_R$ any constant, $G = |\lambda^*| + F_{\max}^2$, $Q^{\mathrm{tr}} = Q_+^{\mathrm{tr}}$, $D = Q_R[0,\varepsilon_R] + c_R P_R[0,\varepsilon_R] + \varepsilon_R G[-1,1]$ and $a^2 = 1 + \frac{1}{2}(Q^{\mathrm{tr}} - \sup D)$, $b^2 = 1 + \frac{1}{2}(Q^{\mathrm{tr}} - \inf D)$, the hypotheses (H1) $a^2 > 0$, (H2) $h_R + a > 0$, (H3) $f_R + \tilde\kappa^* E_R/(a+h_R) < F_{\max}$. Then the solution extends to $[X_R, \infty)$ with $z > 0$, $Q(x) \in Q^{\mathrm{tr}} - D$ for $x \in [\tanh X_R, 1]$ (so $a^2 \leq W \leq b^2$), f is increasing with $f(\infty) - f_R \in \tilde\kappa^* E_R\big[\frac{1}{b+h_R}, \frac{1}{a+h_R}\big]$, $h \to \mu > 0$ and $Q(1) > -2$ (type G), and $Q(1) - Q^{\mathrm{tr}} \in -\frac{\varepsilon_R}{2}[0, \tau^+(2f_R + \tau^+)]$ with $\tau^+ = \tilde\kappa^* E_R/(a+h_R)$. At the left pole $\xi = -X_L$ the mirror statement holds with the sign reversed: if $f_L - \tau^- \geq 0$ (with τ^- the corresponding bound for the decrease of f on $(-\infty, -X_L]$), then $Q(-1) - Q_-^{\mathrm{tr}} \in +\frac{\varepsilon_L}{2}[0,\ f_L^2 - (f_L - \tau^-)^2]$; without that sign condition only $|Q(-1) - Q_-^{\mathrm{tr}}| \leq \frac{\varepsilon_L}{2} G$ is claimed. Consequently, for every s in a box on which (H1)–(H3) hold at both poles, $s \in \mathcal{S}$, the limits in Φ exist, Φ is continuous on the box, and $\Phi(s) - \Phi_0(s) \in \mathbf{T}$ for an explicit box $\mathbf{T}$.

(C3) Lemma D.5.: The Krawczyk–Brouwer statement of (T3) in the form needed for $\Phi = \Phi_0 + \mathsf{Z}$ (a restatement of Corollary A.59).

(C4) Theorem D.8 and Proposition D.11.: The validated integration of (9.8) and of its variational system over the box $\mathbf{X}_k = \bar s_k + 10^{-19}[-1,1]^4$ from $\xi = 0$ to $\xi = 26$ and to $\xi = -12$ (one-step enclosure theorem), and the output for $k = 1, 2, 3$: the hypotheses (H1)–(H3) of (C2) hold at both poles for every $s \in \mathbf{X}_k$, the enclosures $\mathbf{J}_k \supset \mathrm{D}\Phi_0(\mathbf{X}_k)$ and $\mathbf{T}_k$, $\mathbf{V}_k$ are computed with C the rounded midpoint inverse of $\mathrm{D}\Phi_0(\bar s_k)$, and the inclusion $\mathbf{K}_k \subset \operatorname{int}\mathbf{X}_k$ of (T3) is verified together with the enclosures of J

and of $f(-\infty)$ over $\mathbf{X}_k$ listed in Table 9.2. The three boxes are pairwise disjoint.

k	$J_k = f_k(+\infty) - f_k(-\infty)$	h_0 at the centre $\bar{s}_k$	Z_1	$\max Y/r$	$\operatorname{sgn}\det$
1	$19.040738660117701 \pm 5.7 \cdot 10^{-16}$	$-0.951474132\ldots$	$2.5 \cdot 10^{-14}$	$7.6 \cdot 10^{-6}$	$-$
2	$33.58919484747842 \pm 1.9 \cdot 10^{-15}$	$-0.306353\ldots$	$1.8 \cdot 10^{-14}$	$2.1 \cdot 10^{-5}$	$+$
3	$46.31852825252179 \pm 4.8 \cdot 10^{-15}$	$-0.034142\ldots$	$3.7 \cdot 10^{-15}$	$1.07 \cdot 10^{-4}$	$-$

TABLE 9.2. The certified data (computed with 160-bit arithmetic, Taylor degree 24, cuts $X_R = 26$, $X_L = 12$, box radius $r = 10^{-19}$). The Krawczyk test needs $Y/r + Z_1 < 1$ and passes with margin of order 10^4. $f_k(-\infty) \in \pm(1.4, 2.2, 3.2) \cdot 10^{-19}$, $\mu_R \in 0.25 \pm 5 \cdot 10^{-17}$, $\mu_L \in 1.5 \pm 1.6 \cdot 10^{-10}$. The tail corrections $\mathbf{T}_k$ are at most $1.2 \cdot 10^{-24}$, $2.5 \cdot 10^{-23}$, $3.7 \cdot 10^{-22}$. The signs of $\det \mathrm{D}\Phi_0$ are $(-, +, -)$, those of the Leray–Schauder indices $(+1, -1, +1)$ (a by-product, not used). Source: certificate of Appendix D (combined SHA-256 of the three result files, timing fields removed: `6839ccbd6dacea72489d00d237c267f4c06367e094b0039a60c030c94bcf3d69`).

PROOF OF THEOREM 9.68. Fix $k \in \{1, 2, 3\}$ and the box $\mathbf{X}_k$.

Step 1: the box lies in the domain. By (C4) the hypotheses (H1)–(H3) of (C2) hold at both poles for every $s \in \mathbf{X}_k$ (with $F_{\max} = \sup f_R + 1$ and the analogous constants at the left pole). By (C2) the solution of (9.8) through each such s exists on $\mathbb{R}$ with $z > 0$ (indeed $E = \exp(-2\int h) > 0$ as long as h is finite, so $z > 0$ on the cut intervals, and (C2) gives it beyond the cuts) and is of type G at both poles, i.e. $\mathbf{X}_k \subset \mathcal{S}$ (the solution is then a global solution of the reduced system, with $f' = \tilde{\kappa}^* E > 0$); Φ is defined and continuous on $\mathbf{X}_k$ and $\Phi = \Phi_0 + \mathsf{Z}$ with $\mathsf{Z}(s) \in \mathbf{T}_k$.

Step 2: a zero. By (C4) the inclusion $\mathbf{K}_k \subset \operatorname{int} \mathbf{X}_k$ holds, so (C3) gives a point $s_k^* \in \mathbf{K}_k \subset \mathbf{X}_k$ with $\Phi(s_k^*) = 0$.

Step 3: a solution with data D^*. By Lemma 9.71(b), $\omega_k := \omega(s_k^*) \in \mathrm{Sol}$ has $E(\omega_k) = D^*$; by Lemma 9.11(a) its swirl $f_k := f_{\omega_k}$ is a fixed point of T_{D^*}, nondecreasing with $f_k(-\infty) = 0$ and $f_k' = \tilde{\kappa}^* E > 0$. The jump $J_k = f_k(+\infty) - f_k(-\infty)$ lies in the enclosure of J over $\mathbf{X}_k$ listed in Table 9.2 (part of (C4)).

Step 4: three different solutions. The three enclosures of J_k are disjoint ($19.04\ldots$, $33.58\ldots$, $46.31\ldots$), so f_1, f_2, f_3 are pairwise different fixed points; equally the boxes are disjoint, so s_1^*, s_2^*, s_3^* and hence the ω_k are different (a point $\omega \in \mathrm{Sol}$ and its Cauchy datum s determine each other, Lemma 9.71(c)). This proves (a) and (b); the class is N_0 because $f_- = 0$, $\kappa_{\mathrm{D}} > 0$ (Definition 5.3), and $D^* \in \mathbb{D}_+$ with $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$. (c) is Corollary 9.69, whose proof uses only (a), (b) and the results of Chapter 8. □

9.8.5. Reliability: what the computer is trusted for. The trust base of the proof is: (1) the correctness of the ball arithmetic of the Arb library (called through python-flint 0.9.0, Python 3.13; [17]); (2) the correctness of the three programs that implement (C2)–(C4) (validated Taylor steps, tail enclosures, Krawczyk test), guarded by regression tests: an exact solution at $\tilde{\kappa} = 0$ (the free problem of Toy example 9.72), agreement with an independent double-precision integration, and

the tail enclosures against long integrations (Numerical observation 9.74); (3) the hand-written statements (C1)–(C3), re-derived in Appendix D and in this chapter. The certificate is reproduced by one command (Appendix D); the result files of the three runs have a canonical hash (above), and the reproduction on a fresh machine reproduced the same hashes.

Numerical observation 9.74 (sanity checks of the certificate)**.** (1) Tails against the cut. *The certified half-width of the enclosure of $f(+\infty)$ for solution* 1 *with cut $X = 8, 10, \dots, 20$ is* $2.73 \cdot 10^{-6}$, $1.84 \cdot 10^{-8}$, $1.24 \cdot 10^{-10}$, $8.4 \cdot 10^{-13}$, $5.6 \cdot 10^{-15}$, $3.8 \cdot 10^{-17}$, $2.6 \cdot 10^{-19}$*, always* 7.1 *times the true truncation error and decaying like $e^{-2.5X}$ ($2.5 = 2k_2' + 2$, $k_2' = \frac{1}{4}$) (Figure 9.5(b)).* (2) An independent double-precision calculation. *Starting from the three certified centres, Newton's method on the truncated map Φ_0 with a double-precision Dormand–Prince integrator (written afresh, sharing no code with the certificate) converges to $J =$* 19.0407386601, 33.5891948475, 46.3185282525*; the three values differ from the midpoints of the certified enclosures by* $4.6 \cdot 10^{-14}$, $6.4 \cdot 10^{-14}$, $1.2 \cdot 10^{-12}$*, and $f(0) - f_- = J/41$ holds to 10^{-9} in each case (the identity of Lemma 9.56(a), an independent test of the bookkeeping $\vartheta = 40$).* (3) An independent solver for T_{D^*}. *The fixed-point solver of the numerical atlas (collocation on panels, Newton's method with dense Jacobian; it works in the effective variables and shares nothing with the shooting code) finds the same three fixed points, with $|J_{\mathrm{solver}} - J_{\mathrm{certified}}| = 5.3 \cdot 10^{-14}$, $2.1 \cdot 10^{-13}$, $1.1 \cdot 10^{-13}$ and residuals $\|f - T_{D^*} f\|_\infty \le 2 \cdot 10^{-13}$ (Numerical observation 9.75). Code:* `code/ch09_uniq/verify_ch09b.py` *(parts 3 and 4;* `-full` *re-runs the validated Krawczyk test for solution* 1*, about one minute);* `make_b_cap.py`*. Numerical sanity checks, not part of the proof.*

9.8.6. The branch through D^* and the spectra.

Numerical observation 9.75 (three solutions at D^*: the S-curve, the spectra)**.** *Continuation of the fixed points of T_D in κ_{D} at the other data of D^* (pseudo-arclength in $\ln \kappa_{\mathrm{D}}$) gives the S-curve of Figure 9.6 with turning points $\kappa_{\mathrm{D}} =$* 1300.03 *($J =$* 23.29*) and* 1462.16 *($J =$* 41.56*): the window has width* 162.1 *and ratio* 1.125*, and $\nu_{\max} > 1$ exactly on the middle branch (maximum $\Lambda =$* 1.2921*). At $\kappa_{\mathrm{D}} =$* 1400 *the three fixed points have $J =$* 19.040739, 33.589195, 46.318528 *(agreeing with the certified enclosures to* $2.1 \cdot 10^{-13}$*), residuals $\le 2 \cdot 10^{-13}$ and leading eigenvalues of* $\mathrm{D}T_{D^*}(f_k)$

$$0.35043 \pm 0.12474\,\mathrm{i} \quad (\textit{third eigenvalue } 0.00954),$$

$$1.28184 \quad (\textit{next } 0.15042), \qquad 0.62962 \quad (\textit{next } 0.06433),$$

so the numbers of real eigenvalues above 1 *are $\beta = (0, 1, 0)$ and the indices $(+1, -1, +1)$ add up to* 1*, as Theorem 8.42(c) requires; the complex leading pair of the lowest fixed point is an instance of the remark that $\mathrm{D}T_D$ is not a positive operator (Lemma 8.27). The eigenvalues are computed, not certified; the middle one exceeds* 1 *as Corollary 9.69(b) demands. The numbers $\kappa_{\mathrm{D}} =$* 1300.03, 1462.16 *locate the first and last fold of Corollary 9.55 numerically. Code:* `verify_ch09b.py` *(part 5; worker* `_w_dstar.py`*). Numerical, not part of any proof.*

9.8.7. The numerical atlas of the chapter.

Atlas 9.76 (Chapter 9: where the fixed-point problem has several solutions)**.** *All numerical results of this chapter are collected in Table 9.3; none is used in a proof.*

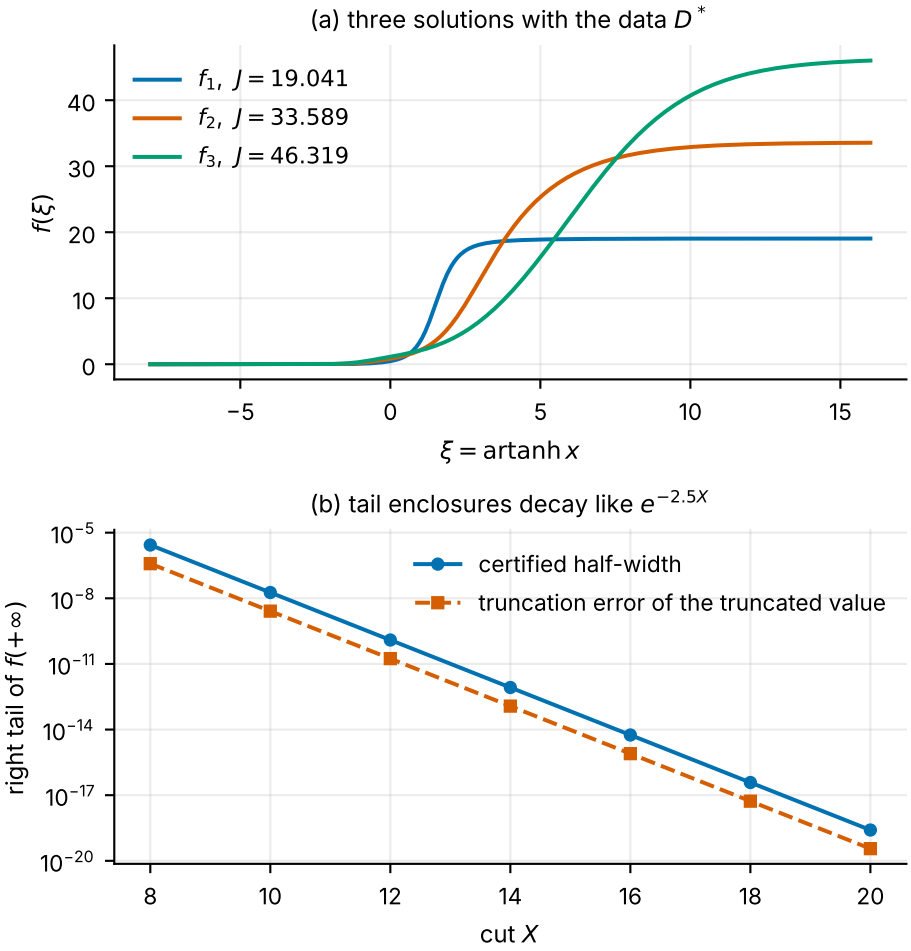


FIGURE 9.5. (a) The three solutions $f_k(\xi)$, $\xi = \operatorname{artanh} x$, with the data D^* (double-precision shooting from the certified centres; J_k in the legend). (b) The certified half-width of the right tail enclosure of $f(+\infty)$ against the cut X, and the true truncation error. Code: `code/ch09_uniq/make_b_cap.py`.

Method. *Fixed points of T_D are computed on a nonuniform grid on $[-L, L]$ ($L = 14$ to 24) made of panels with Gauss–Legendre nodes of order 6 to 10 (panel width $h_0 = 0.25$ to 0.6 near the equator, up to 1.5 far away); $\mathcal{Q}$ is applied exactly on each panel by a forward/backward recursion for the kernel $e^{-2|\xi-\eta|}$, the Jost solutions are obtained by a transfer matrix per panel from spectral collocation, with the Riccati logarithms started from the exact exponentials at $\pm L$; $\mathrm{D}T_D$ is assembled as a dense matrix by differentiating the whole chain (formula (8.6)) and its leading eigenvalues computed by a dense eigensolver; fixed points, including the unstable ones, are found by Newton's method and followed by pseudo-arclength continuation in $\ln \kappa_{\mathrm{D}}$,*

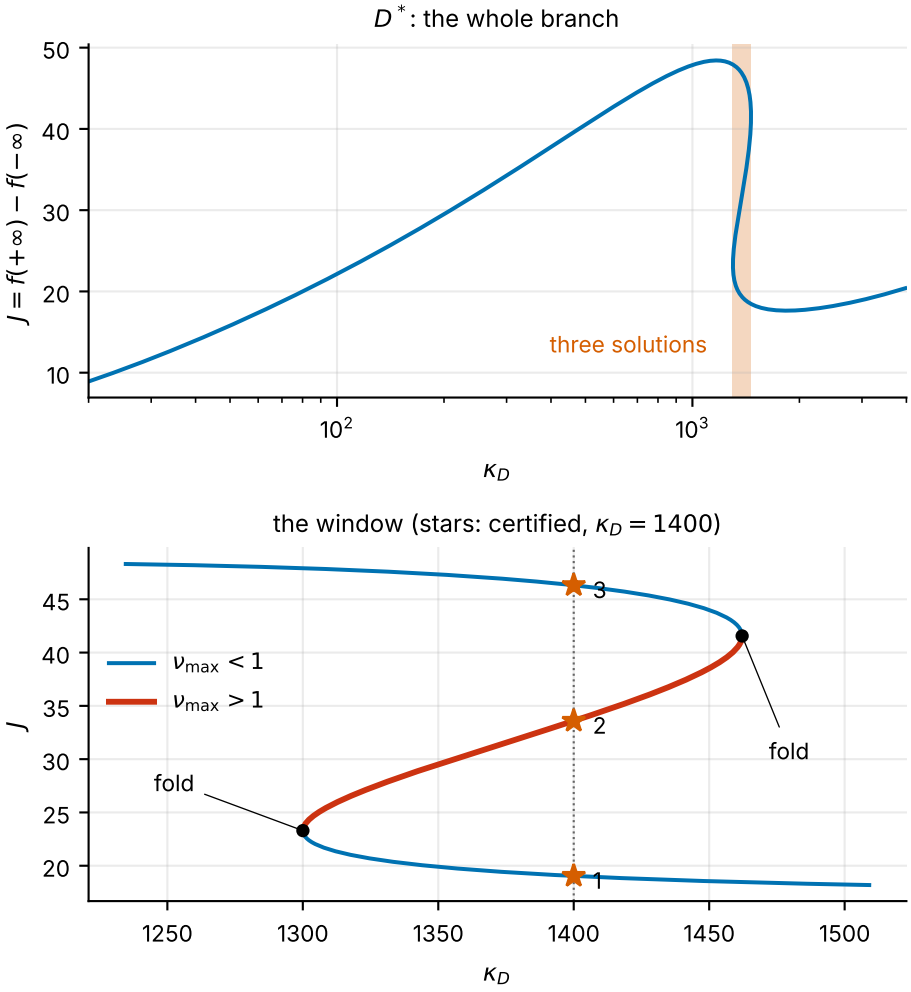


FIGURE 9.6. The solution branch of T_D in κ_{D} at the certified datum, the S-curve of Numerical observation 9.75. Left: J against κ_{D} (logarithmic axis): unique solution for small and large κ_{D}, with the three-solution window shaded. Right: zoom on the window; the three solutions of Theorem 9.68 at $\kappa_{\mathrm{D}} = 1400$ (stars); red: leading real eigenvalue > 1; dots: the two folds $\kappa_{\mathrm{D}} = 1300.03$ and 1462.16. Code: `code/ch09_uniq/make_b_scurve.py`.

with folds detected by sign changes of $\mathrm{d}\kappa_{\mathrm{D}}$ *along the arclength and refined by a cubic fit; the class-*Z *folds are in addition computed independently in the shooting chart by Newton's method on* $\{E(\omega) = D,\ \det \mathrm{D}E(\omega) = 0\}$. Tolerance. *Residuals* $\|f - T_D f\|_\infty \le 2 \cdot 10^{-13}$*; J is converged to* 10^{-9} *or better under refinement of the order, the panel size and L (agreement with the certificate to* $2 \cdot 10^{-13}$ *at* D^**);*

eigenvalues agree to 10^{-4} *under refinement; fold positions agree on three grids to the digits quoted.* Code. `code/ch09_uniq/` *(*`verify_ch09b.py` *prints every number of Sections 9.5–9.8.7;* `make figures` *regenerates the four figures;* `make tables` *recomputes the cached branches).*

family	*datum*	*result*	*where*
class Z	$\boldsymbol{c}' = (-0.15, 0.09, \bar{c}_3 + 0.04)$, $\vartheta = 4.49$, $f_- = -2.64$	*three solutions at* $\kappa_{\mathrm{D}} = 10$*:* $J = 3.6466, 8.7391, 13.8803$*; folds* $\kappa_{\mathrm{D}} = 6.30477,\ 11.35302$*; cusp* $\approx (0.16, 13.74)$ *in* $(c_3' - \bar{c}_3, \kappa_{\mathrm{D}})$	*Numerical observations 9.58 and 9.59*
class N_0 *at* D^*	$\boldsymbol{c}' = (\frac{5}{4}, -\frac{15}{16}, \bar{c}_3 + \frac{1}{20})$, $\vartheta = 40$, $f_- = 0$	*three solutions at* $\kappa_{\mathrm{D}} = 1400$ *(certified); folds* $\kappa_{\mathrm{D}} = 1300.03,\ 1462.16$*;* $\Lambda = 1.2921$*; fold pair for* $f_- \le 1$*, none at* $f_- = 1.5$	*Theorem 9.68 and Numerical observations 9.64 and 9.75*
D^* *with* ϑ *varied*	*as above,* $8 \le \vartheta \le 160$	Λ *crosses* 1 *near* $\vartheta = 25.6$*;* $\Lambda(160) = 2.263$	*Numerical observation 9.64*
rest state, ridge	$\boldsymbol{c}' = 0$, $f_- = 0$, $\vartheta = 10^4$	*folds* $\kappa_{\mathrm{D}} = 1362776,\ 1641832$*; three solutions at* $1.5 \cdot 10^6$*:* $J = 37.873, 61.161, 72.063$	*Numerical observation 9.63*
rest state, cusp	$\boldsymbol{c}' = 0$, $f_- = 0$, $2300 \le \vartheta \le 2 \cdot 10^4$	*no fold at* 2300*, folds at* 2400*; width*$^{2/3}$ *linear,* $\vartheta_c \approx 2330$	*Numerical observation 9.63*
rest state, no fold	$\boldsymbol{c}' = 0$, $f_- = 0$, $\vartheta = 1$, $0.01 \le \kappa_{\mathrm{D}} \le 3 \cdot 10^4$	$\nu_{\max} \le 0.0047$*, no fold*	*Numerical observation 9.62*
ridge close to 1	$\boldsymbol{c}' = (1.288, -0.905, \bar{c}_3 + 0.11)$, $\vartheta = 28.5$, $f_- = 0$	$\max \nu_{\max} = 0.9901$ *near* $\kappa_{\mathrm{D}} = 768$*, no fold*	*Numerical observation 9.65*

TABLE 9.3. The numerical atlas of Chapter 9.

Summary of the second half of the chapter. Along any line in κ_{D} the number of solutions is odd and changes only at a degenerate fixed point; there is a first and a last such κ_{D} (Theorem 9.51, with the large-swirl hypothesis of Remark 9.52). Folds exist: in class Z at moderate data, in class N_0 at large ϑ, and at the certified datum D^*, where non-uniqueness is a theorem (Theorem 9.68). The sign structure of Section 9.1 restricts what a fold looks like; it does not exclude folds. What remains open is collected in Chapter 13: the exact number of solutions at D^* (Open problem 13.16), the exact fold set and its cusps (Conjecture 13.18, Open problem 13.19, Open problem 13.17) and the large-ϑ regime (Open problem 13.21).

Part 5

Asymptotic regimes

CHAPTER 10

Large swirl: the $\sqrt{\kappa}$ law and the logistic layer

In words. *In the effective coordinates of Chapter 8 a global solution of type* G *at both poles is a fixed point $f = T_D f$, and its swirl rises from f_- to $f_- + J$ with a jump $J = \kappa_{\mathrm{D}}/(\vartheta w_0)$ that is* exactly *determined by one number, the Wronskian w_0 of the Jost solutions of the potential $W = W_{\mathrm{e}} + \frac{1}{2}\mathcal{Q}[f^2]$. For small swirl $W \approx W_{\mathrm{e}}$ and w_0 is a constant of the data, so J grows linearly with κ_{D}. For large swirl the potential feels the swirl: it is W_{e} plus $\frac{1}{8}f^2\operatorname{sech}^2\xi$, so near the equator $W \approx J^2/16$, the Jost solutions are $e^{\pm J\xi/4}$ to leading order, and $w_0 \approx 2\sqrt{W} \approx J/2$. Inserting this in $\kappa_{\mathrm{D}} = \vartheta J w_0$ gives $\kappa_{\mathrm{D}} \approx \frac{1}{2}\vartheta J^2$: the jump grows only like $\sqrt{\kappa_{\mathrm{D}}}$. This chapter makes the heuristic into a theorem* for every solution*: $J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta}\,(1 + O(\kappa_{\mathrm{D}}^{-1/2}))$ with explicit constants, and even $J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta} + 1 + \ln\vartheta - f_-/2 + O(\kappa_{\mathrm{D}}^{-1/2})$. The swirl itself is a logistic step of width $O(1/J)$ at the equator (the* layer*), the radial velocity is a jet of the same width, and the linearisation of T_D has a rank-one structure that forces uniqueness for large swirl.*

Roadmap. Section 10.1 collects the properties of the centrifugal operator $\mathcal{Q}$ that are needed (bounds on $\mathcal{Q}[g]$ and its first two derivatives, the *exact* potential of a step function) and the standing facts about a fixed point (Lemma 10.1, Lemma 10.3). Section 10.2 proves the a priori bounds for an arbitrary fixed point with J large: the barrier $W \geq B^2$ on $[-1,1]$, steepness $\rho \geq \rho_{\min}$ of the Jost gap, the step profile of f, $W(0) = J^2/16 + O(J)$ and the tracking $\zeta_\mp \approx \mp\sqrt{W}$ (Lemma 10.5). Section 10.3 closes the argument with the exact identity $\kappa_{\mathrm{D}} = \vartheta J w_0$ and proves the $\sqrt{\kappa_{\mathrm{D}}}$ law with explicit constants (Theorem 10.11), with the rest state as worked example and a numerical illustration. Section 10.4 computes the next order: second-order tracking, the exact integrals of the logistic layer, and the coefficient $J(\frac{f_-}{8} - \frac{1+\ln\vartheta}{4})$ of the potential, which gives Theorem 10.22. The second half of the chapter (Theorem 10.28, Lemma 10.33, Theorem 10.39, Corollary 10.43, Atlas 10.48) uses the lemmas proved here.

The logical structure of the first half of the chapter is shown in Figure 10.1.

Conventions of the chapter. $' = \mathrm{d}/\mathrm{d}\xi$. *The swirl constant is always* κ_{D} (normalisation 4 of Table 2.1); in the title and in verbal statements "the $\sqrt{\kappa}$ law" means the law in κ_{D}. The *standing hypotheses* are: $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$ (so W_{e} is admissible, Lemma 8.5(b)), $\vartheta > 0$, $f_- \in \mathbb{R}$, $\kappa_{\mathrm{D}} > 0$, and f is a fixed point of T_D — equivalently (Proposition 8.13) a global solution of type G at both poles with effective data $D = (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$ and $\kappa_{\mathrm{D}} > 0$ (solutions with $\kappa_{\mathrm{D}} < 0$ are obtained by σ_1). Existence of such an f for every $\kappa_{\mathrm{D}} > 0$ is Theorem 8.16; *uniqueness is never assumed* in this chapter. We write $W = W_f = W_{\mathrm{e}} + \frac{1}{2}\mathcal{Q}[f^2]$, $u_\pm, \zeta_\pm$ for the Jost solutions of W and their logarithmic derivatives ($u_\pm(0) = 1$), $\rho = \zeta_- - \zeta_+ = w_0/(u_- u_+)$, $\Sigma = \int_0^\xi \rho = \ln(u_-/u_+)$, $J = f(+\infty) - f(-\infty)$ and $\varpi = \sqrt{W}$. The sign convention

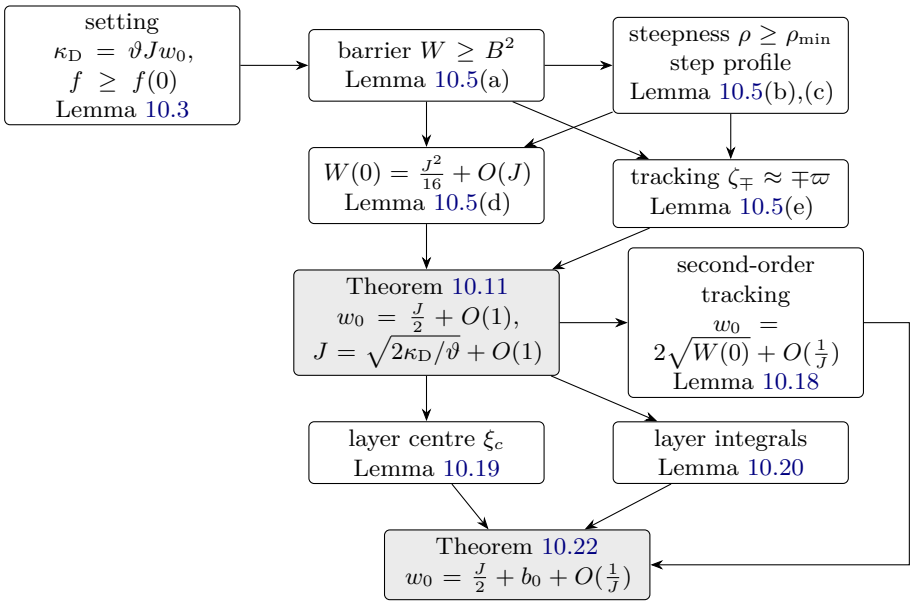


FIGURE 10.1. Logical structure of Section 10.2–Section 10.4. Every arrow is an application of a lemma to an *arbitrary* fixed point; the identity $\kappa_{\mathrm{D}} = \vartheta J w_0$ of the setting lemma closes the argument at both orders.

is that of Chapter 3: $w_0 = \mathrm{Wr}(u_+, u_-) = \rho(0) > 0$. The letters $A, B, S, \dots$ of Definition 10.4 are *chapter-local* constants (the base-point swirl A of Chapter 5 plays no role here), and E_w is not the pole weight E.

10.1. The operator $\mathcal{Q}$ and the standing facts

In words. *Everything in this chapter rests on two simple facts about the potential of a swirl profile: it is a* smooth *function with controlled first and second derivatives, and for a* step *profile it can be written down in closed form. The first fact lets the Riccati tracking of Chapter 3 work (it needs ϖ' and ϖ''); the second one computes $W(0)$ up to $O(J)$ by comparing the true profile, which is a smoothed step, with a sharp step.*

Lemma 10.1 (properties of $\mathcal{Q}$; the potential of a step)**.** *Let $\mathcal{Q}[g](\xi) = \frac{1}{4\cosh^2\xi}\int_{\mathbb{R}} e^{-2|\xi-\eta|} g(\eta)\,\mathrm{d}\eta$ be the operator of Lemma 8.2.*

(a) *For $g \in \mathrm{BC}(\mathbb{R})$, $\mathcal{Q}[g] \in C^2$ and*

$$|\mathcal{Q}[g]| \leq \tfrac{1}{4}\|g\|_\infty \operatorname{sech}^2\xi, \qquad |\mathcal{Q}[g]'| \leq \|g\|_\infty, \qquad |\mathcal{Q}[g]''| \leq 5\|g\|_\infty.$$

(b) (Exact potential of a step.) *Let $a, f_-, J \in \mathbb{R}$, $f_s = f_- + J\mathbf{1}_{(a,\infty)}$ and*

$$m_a(\xi) = \int_a^\infty e^{-2|\xi-\eta|}\mathrm{d}\eta = \begin{cases} 1 - \frac{1}{2}e^{-2(\xi-a)}, & \xi \geq a, \\ \frac{1}{2}e^{2(\xi-a)}, & \xi < a. \end{cases}$$

Then $\mathcal{Q}[f_s^2](\xi) = \frac{1}{4}\operatorname{sech}^2\xi\left[f_-^2 + (2f_-J + J^2)\,m_a(\xi)\right]$. *This function is* C^1, *tends to* 0 *at* $\pm\infty$, *and satisfies* $\mathsf{L}[\mathcal{Q}[f_s^2]] = -f_s^2$ *on* $\mathbb{R}\setminus\{a\}$.

(c) *For* $a = 0$ *put* $q_{\mathrm{lo}} := \mathcal{Q}[\mathbf{1}_{(0,\infty)}] = \frac{1}{4}\operatorname{sech}^2\xi\, m_0(\xi)$. *Then* $q_{\mathrm{lo}}(0) = \frac{1}{8}$, $q_{\mathrm{lo}}'(0) = \frac{1}{4}$, *and*

$$\min_{[-1,1]} q_{\mathrm{lo}} = q_{\mathrm{lo}}(-1) = q_1 := \frac{e^{-2}}{8\cosh^2 1} = 7.1046\ldots\cdot 10^{-3}.$$

PROOF. (a) Put $G(\xi) = \int e^{-2|\xi-\eta|}g(\eta)\mathrm{d}\eta$, so $\mathcal{Q}[g] = \frac{1}{4}\operatorname{sech}^2\xi\, G$. Since $\int e^{-2|s|}\mathrm{d}s = 1$, $|G| \le \|g\|_\infty$. Writing $G = e^{-2\xi}\int_{-\infty}^{\xi} e^{2\eta}g + e^{2\xi}\int_{\xi}^{\infty} e^{-2\eta}g$ and differentiating (the two boundary terms $\pm g(\xi)$ cancel),

$$G' = -2e^{-2\xi}\int_{-\infty}^{\xi} e^{2\eta}g + 2e^{2\xi}\int_{\xi}^{\infty} e^{-2\eta}g, \qquad G'' = 4G - 4g,$$

and each of the two terms of G' is at most $2\cdot\frac{1}{2}\|g\|_\infty$ in modulus, so $|G'| \le 2\|g\|_\infty$ and $|G''| \le 8\|g\|_\infty$. In particular $G \in C^2$. Now $(\operatorname{sech}^2)' = -2\operatorname{sech}^2\tanh$ and $(\operatorname{sech}^2)'' = \operatorname{sech}^2(4 - 6\operatorname{sech}^2)$, hence $|(\operatorname{sech}^2)'| \le 2\operatorname{sech}^2$ and $|(\operatorname{sech}^2)''| \le 4\operatorname{sech}^2$ (a symbolic check is in `verify_ch10a.py`). Then

$$|\mathcal{Q}[g]'| \le \tfrac{1}{4}\operatorname{sech}^2\xi\,(2|G| + |G'|) \le \tfrac{1}{4}(2+2)\|g\|_\infty = \|g\|_\infty,$$
$$|\mathcal{Q}[g]''| = \tfrac{1}{4}|(\operatorname{sech}^2)''G + 2(\operatorname{sech}^2)'G' + \operatorname{sech}^2 G''| \le \tfrac{1}{4}(4+8+8)\|g\|_\infty = 5\|g\|_\infty.$$

(b) Since $f_s^2 = f_-^2 + (2f_-J + J^2)\mathbf{1}_{(a,\infty)}$ and $\int e^{-2|\xi-\eta|}\mathrm{d}\eta = 1$, the formula follows from the definition of $\mathcal{Q}$ and from the value of m_a: for $\xi \ge a$, $\int_a^\xi e^{-2(\xi-\eta)}\mathrm{d}\eta + \int_\xi^\infty e^{-2(\eta-\xi)}\mathrm{d}\eta = \frac{1}{2}(1 - e^{-2(\xi-a)}) + \frac{1}{2}$, and for $\xi < a$ only the second kind of integral occurs and equals $\frac{1}{2}e^{-2(a-\xi)}$. The two expressions for m_a have the same value $\frac{1}{2}$ and the same derivative 1 at $\xi = a$, so $m_a \in C^1$; on each side $m_a'' = 4m_a - 4\cdot\mathbf{1}_{\xi>a}$, which is the identity $G'' = 4G - 4g$ of (a) and, by the computation in the proof of Lemma 8.2(c), gives $\mathsf{L}[\mathcal{Q}[f_s^2]] = -f_s^2$ off $\xi = a$. The decay is clear from the factor sech^2. (c) The values at 0 follow from $m_0(0) = \frac{1}{2}$, $m_0'(0) = 1$, $(\operatorname{sech}^2)'(0) = 0$. For $\xi \le 0$, $q_{\mathrm{lo}} = e^{2\xi}/(8\cosh^2\xi) = \frac{1}{2}(1 + e^{-2\xi})^{-2}$, which is increasing; hence $\min_{[-1,0]} q_{\mathrm{lo}} = q_{\mathrm{lo}}(-1) = q_1$. For $\xi \in [0,1]$, $m_0 \ge \frac{1}{2}$ and $\operatorname{sech}^2\xi \ge \operatorname{sech}^2 1$, so $q_{\mathrm{lo}} \ge \frac{1}{8}\operatorname{sech}^2 1 = 0.0525 > q_1$. □

Example 10.2 (the potential of a sharp step at the rest state)**.** For $\boldsymbol{c}' = 0$, $f_- = 0$ (so $W_{\mathrm{e}} \equiv 1$) and a sharp step of height J at 0, Lemma 10.1(b) gives

$$W = 1 + \tfrac{1}{2}\mathcal{Q}[J^2\mathbf{1}_{(0,\infty)}] = 1 + \tfrac{J^2}{2}q_{\mathrm{lo}} = 1 + \tfrac{J^2}{16}q_{\mathrm{step}},$$
$$q_{\mathrm{step}} := 8q_{\mathrm{lo}}, \quad q_{\mathrm{step}}(0) = 1,\ q_{\mathrm{step}}'(0) = 2.$$

So $W(0) = 1 + J^2/16$ and $W'(0) = J^2/8$; q_{step} decays like $2\operatorname{sech}^2\xi$ at $+\infty$ and like $e^{2\xi}$ at $-\infty$. The true potential of a fixed point differs from this one by $O(J)$ near 0: this difference is the whole content of Theorem 10.22.

Lemma 10.3 (standing facts about a fixed point)**.** *Under the standing hypotheses let* $S = |f_-| + J$, $A = (\sup W_{\mathrm{e}} + S^2/8)^{1/2}$. *Then:*

(a) $W \in C^2$ *is admissible,* $W \ge W_{\mathrm{e}}$, *and* $W \le A^2$ *on* $\mathbb{R}$. *The functions* $\zeta_\pm$ *solve* $\zeta' = W - \zeta^2$, $\rho' = -\rho(\zeta_- + \zeta_+)$, $\rho > 0$.

(b) *f is nondecreasing, $f_- \le f \le f_- + J$, $|f| \le S$, $f(0) = f_- + J/(1+\vartheta)$, and*

$$f = f_- + J\,\lg(\Sigma - \ln\vartheta), \qquad \kappa_{\mathrm{D}} = \vartheta\, J\, w_0. \tag{10.1}$$

(c) *$\zeta_- \le A$ and $-\zeta_+ \le A$ on $\mathbb{R}$; hence $w_0 \le 2A$ and $\kappa_{\mathrm{D}} \le 2\vartheta J A(J)$, where $J \mapsto A(J)$ is increasing. Consequently, if $\kappa_{\mathrm{D}} > 2\vartheta J_0 A(J_0)$ then every fixed point has $J > J_0$.*

(d) *$\zeta_-(-1) \ge \zeta_-^{W_{\mathrm{e}}}(-1)$ and $\zeta_+(1) \le \zeta_+^{W_{\mathrm{e}}}(1)$.*

(e) *$|W'| \le \|W_{\mathrm{e}}'\|_\infty + \frac{1}{2}S^2$ and $|W''| \le \|W_{\mathrm{e}}''\|_\infty + \frac{5}{2}S^2$.*

Proof. (a) $W = W_f = W_{\mathrm{e}} + \frac{1}{2}\mathcal{Q}[f^2]$ is admissible because $f \in \mathcal{A}_D$ (Proposition 8.13); W_{e} is smooth and $\mathcal{Q}[f^2] \in C^2$ (Lemma 8.2(b)). Since $\mathcal{Q}[f^2] \ge 0$ (Lemma 8.2(a)), $W \ge W_{\mathrm{e}}$. The bound $|f| \le S$ is proved in (b) and gives $\mathcal{Q}[f^2] \le \frac{1}{4}S^2\operatorname{sech}^2\xi \le \frac{1}{4}S^2$, whence $W \le \sup W_{\mathrm{e}} + \frac{1}{8}S^2 = A^2$. The Riccati equation is Lemma 3.21 and the definition of $\zeta_\pm$; $\rho' = \zeta_-' - \zeta_+' = -(\zeta_-^2 - \zeta_+^2) = -\rho(\zeta_- + \zeta_+)$. (b) By Lemma 8.10 (formula (8.1) with $f = T_D f$) $f = f_- + J\lg(\Sigma - \ln\vartheta)$ with $J = \kappa_{\mathrm{D}}/(\vartheta w_0)$, i.e. $\kappa_{\mathrm{D}} = \vartheta J w_0$; Σ is increasing ($\Sigma' = \rho > 0$) and $0 < \lg < 1$ is increasing, so f is nondecreasing with $f_- \le f \le f_- + J$ and $|f| \le |f_-| + J$. Finally $f(0) = f_- + J(u_-/z)(0) = f_- + J/(1+\vartheta)$ (Lemma 8.10). (c) The limit of W at $-\infty$ is $k_1'^2 = \lim W_{\mathrm{e}} \le \sup W_{\mathrm{e}} < A^2$ (as $S > 0$), and $\zeta_- \to k_1' < A$ at $-\infty$ (Lemma 3.21(b)(iv)). If $\zeta_-(\xi_1) > A$ for some ξ_1, let $\xi_0 < \xi_1$ be the last point with $\zeta_-(\xi_0) = A$; on $(\xi_0, \xi_1]$, $\zeta_- > A$, so $\zeta_-' = W - \zeta_-^2 < A^2 - A^2 = 0$ and $\zeta_-(\xi_1) < \zeta_-(\xi_0) = A$, a contradiction. Similarly $\zeta_+ \to -k_2' > -A$ at $+\infty$; if $\zeta_+(\xi_1) < -A$, let $\xi_0 > \xi_1$ be the first point with $\zeta_+(\xi_0) = -A$; on $[\xi_1, \xi_0)$, $\zeta_+ < -A$ so $\zeta_+' < 0$ and $\zeta_+(\xi_1) > \zeta_+(\xi_0) = -A$, again a contradiction. Hence $w_0 = \zeta_-(0) - \zeta_+(0) \le 2A$ and, by (10.1), $\kappa_{\mathrm{D}} = \vartheta J w_0 \le 2\vartheta J A(J)$. Since $J \mapsto JA(J)$ is increasing, $J \le J_0$ would give $\kappa_{\mathrm{D}} \le 2\vartheta J_0 A(J_0)$. (d) $W \ge W_{\mathrm{e}}$ with the same limits at the poles at the rates of Definition 3.20 (Lemma 8.5(a)), and W_{e} is admissible; Lemma 3.23 gives the two inequalities. (e) Lemma 10.1(a) with $g = f^2$, $\|g\|_\infty \le S^2$, and $W' = W_{\mathrm{e}}' + \frac{1}{2}\mathcal{Q}[f^2]'$, $W'' = W_{\mathrm{e}}'' + \frac{1}{2}\mathcal{Q}[f^2]''$. □

10.2. A priori bounds for large J

In words. *Fix a fixed point whose jump J is large; we do not know yet how large J is compared with κ_{D}. Because f is monotone, $f \ge f(0) \gtrsim J$ on the whole right half-line, so by* positivity *of $\mathcal{Q}$ the potential is $\gtrsim J^2$ on $[-1,1]$ (the* barrier*). A Riccati solution on an interval where $W \ge B^2$ with B large is attracted to $\pm B$ within a distance $O(1/B)$ (the tanh barrier of Lemma 3.10): hence ζ_- is large on the right of $-\frac{1}{2}$ and $-\zeta_+$ on its left, and the Jost gap ρ is large near 0 (*steepness*). Then (10.1) says that f is a logistic function of Σ with $\Sigma' = \rho$ large: a step of width $O(1/J)$. The potential of a step is explicit (Lemma 10.1), so $W(0) = J^2/16 + O(J)$, and the tracking lemma gives $\zeta_\mp = \mp\sqrt{W} + O(1)$. Nothing here uses uniqueness: every inequality is proved for an arbitrary fixed point from the monotonicity of f and the value $f(0)$, and J is the only parameter.*

Definition 10.4 (the constants of the chapter)**.** Fix $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$, $\vartheta > 0$, $f_- \in \mathbb{R}$, and let q_1 be as in Lemma 10.1(c). Put

$$C_- = \max\bigl(0, -\min_{[-1,1]} W_{\mathrm{e}}\bigr), \qquad C_0 = \max\bigl(0, -\zeta_-^{W_{\mathrm{e}}}(-1), \zeta_+^{W_{\mathrm{e}}}(1)\bigr)$$

($\zeta_\pm^{W_\mathrm{e}}$ the Riccati variables of W_e; for the rest state $\boldsymbol{c}'=0$, $W_\mathrm{e}\equiv 1$, $\zeta_\pm^{W_\mathrm{e}}=\pm1$ and $C_-=C_0=0$). For $J>0$ put

$$S=|f_-|+J,\quad m_0=f_-+\frac{J}{1+\vartheta},\quad A=\big(\sup W_\mathrm{e}+\tfrac18 S^2\big)^{1/2},$$
$$B=\big(\tfrac12 q_1 m_0^2-C_-\big)^{1/2},\quad B_1=B\tanh\Big(\frac{B}{2}-\operatorname{artanh}\frac{C_0}{B}\Big),\quad \rho_{\min}=2B_1,$$
$$C_{\mathrm{bd}}=\frac{\|W_\mathrm{e}'\|_\infty+\frac12 S^2}{2B},\quad E_w=\frac{C_{\mathrm{bd}}}{B}+Ae^{-B/4},$$
$$r_0=|W_\mathrm{e}(0)|+\frac{f_-^2}{8}+\frac{|f_-|J}{8}+\frac{SJ}{\rho_{\min}}\cdot\frac{|\ln\vartheta|+3}{4},$$
$$C_A=\frac{8r_0}{J}+2E_w$$

(whenever the square roots and artanh are defined). The three conditions

$$\text{(J1) } J\ge 2(1+\vartheta)|f_-|,\qquad \text{(J2) } B\ge\max(2C_0,6),\qquad \text{(J3) } |\ln\vartheta|\le B/9$$

are called the *largeness conditions*; (J2) includes $\frac12 q_1 m_0^2-C_-\ge 36$. These are numbers attached to the data and to J; they do not refer to a particular solution.

The largeness conditions are monotone in J: under (J1), $m_0\ge J/(2(1+\vartheta))>0$ increases with J, hence so does B. They hold for all large J, because by (J1) $B^2\ge\frac18 q_1 J^2/(1+\vartheta)^2-C_-\to\infty$. Let

$$J_*=J_*(\boldsymbol{c}',\vartheta,f_-):=\min\{J>0:\ \text{(J1)–(J3) hold}\}\tag{10.2}$$

(the minimum exists because the set is a closed up-set). On a compact set of (ϑ,f_-) the number J_* is bounded, since the lower bound for B^2 just given is continuous in (ϑ,f_-,J).

Lemma 10.5 (a priori bounds for a fixed point with large jump)**.** *Let f be a fixed point (standing hypotheses) whose jump J satisfies* (J1)–(J3)*. Then:*

(a) (Barrier.) $m_0=f(0)\ge J/(2(1+\vartheta))>0$ *and* $W\ge B^2$ *on* $[-1,1]$.

(b) (Steepness.) *With* $a_0'=\operatorname{artanh}(C_0/B)\le 0.55$*:* $B_1\ge 0.985B$, $\zeta_-\ge B_1$ *on* $[-\frac12,1]$, $-\zeta_+\ge B_1$ *on* $[-1,\frac12]$, *hence*

$$\rho\ge\rho_{\min}=2B_1\ge 1.97B\quad\text{on } [-\tfrac12,\tfrac12],\qquad |\Sigma(\xi)|\ge\rho_{\min}\min(|\xi|,\tfrac12);$$

moreover $\zeta_-\le A$, $-\zeta_+\le A$ *on* $\mathbb{R}$.

(c) (Step profile.) *Let ξ_c be the point with $\Sigma(\xi_c)=\ln\vartheta$ (the layer centre). Then $|\xi_c|\le|\ln\vartheta|/\rho_{\min}\le\frac{1}{16}$ and, for all $\eta\in\mathbb{R}$,*

$$\big|f(\eta)-f_--J\mathbf{1}_{\{\eta>\xi_c\}}\big|\le J\,e^{-\rho_{\min}\min(|\eta-\xi_c|,\,3/8)}.\tag{10.3}$$

(d) (Potential at the equator.) $W(0)=W_\mathrm{e}(0)+\frac{f_-^2}{8}+\frac{f_-J}{8}+\frac{J^2}{16}+E_1$ *with* $|E_1|\le\frac{SJ}{\rho_{\min}}\frac{|\ln\vartheta|+3}{4}$*; in particular* $|W(0)-J^2/16|\le r_0$.

(e) (Tracking.) *On* $[-1,1]$, $B\le\varpi\le A$ *and* $|\varpi'|\le C_{\mathrm{bd}}$. *For* $|\xi|\le\frac14$*:*

$$|\zeta_-(\xi)-\varpi(\xi)|\le E_w,\qquad |\zeta_+(\xi)+\varpi(\xi)|\le E_w,\qquad |\rho'(\xi)|\le 4AE_w.$$

Proof. *(a)* By (J1), $m_0=f_-+J/(1+\vartheta)\ge J/(1+\vartheta)-|f_-|\ge J/(2(1+\vartheta))>0$, and $f(0)=m_0$ by Lemma 10.3(b). As f is nondecreasing, $f\ge m_0>0$ on $[0,\infty)$, so $f^2\ge m_0^2\mathbf{1}_{(0,\infty)}$. Since $\mathcal{Q}$ is monotone, $\mathcal{Q}[f^2]\ge m_0^2 q_{\mathrm{lo}}\ge m_0^2 q_1$ on $[-1,1]$ (Lemma 10.1(c)), and $W_\mathrm{e}\ge -C_-$ there, so $W\ge -C_-+\frac12 q_1 m_0^2=B^2$.

(b) By Lemma 10.3(d) and the definition of C_0, $\zeta_-(-1) \ge -C_0$ and $\zeta_+(1) \le C_0$. Since $C_0 \le B/2 < B$ by (J2), the tanh barrier Lemma 3.10(b)(iii) applies on $[a,b] = [-1,1]$ with $m = B$ (by (a)) and $\lambda_0 = C_0$: for $\xi \in [-1,1]$

$$\zeta_-(\xi) \ge B\tanh\big(B(\xi+1) - a_0'\big), \qquad \zeta_+(\xi) \le -B\tanh\big(B(1-\xi) - a_0'\big).$$

For $\xi \ge -\frac12$ (resp. $\xi \le \frac12$) the right sides are $\ge B_1$ (resp. $\le -B_1$), because tanh is increasing and $B(\xi+1) - a_0' \ge B/2 - a_0'$. For $\xi \in [\frac12, 1]$ we only need $\zeta_- \ge B_1$, which holds because $\xi \mapsto B\tanh(B(\xi+1)-a_0')$ is increasing. Numerically $C_0/B \le \frac12$ gives $a_0' \le \operatorname{artanh}\frac12 = 0.5493$, and $B \ge 6$ gives $B/2 - a_0' \ge 2.45$ and $\tanh 2.45 = 0.9852 > 0.985$; so $B_1 \ge 0.985B$. On $[-\frac12, \frac12]$ both $\zeta_- \ge B_1$ and $-\zeta_+ \ge B_1$, so $\rho = \zeta_- - \zeta_+ \ge 2B_1$. As $\Sigma' = \rho > 0$ and $\Sigma(0) = 0$, $|\Sigma(\xi)| \ge \rho_{\min}|\xi|$ for $|\xi| \le \frac12$ and, by monotonicity, $|\Sigma(\xi)| \ge \rho_{\min}/2$ for $|\xi| \ge \frac12$. The upper bounds are Lemma 10.3(c).

(c) By (J3) and (b), $|\ln\vartheta| \le B/9 \le \rho_{\min}/(9\cdot 1.97) < \rho_{\min}/16 < \rho_{\min}/2$. Since Σ is increasing and $|\Sigma| \ge \rho_{\min}/2$ for $|\xi| \ge \frac12$, the solution ξ_c of $\Sigma(\xi_c) = \ln\vartheta$ lies in $[-\frac12, \frac12]$, and there $|\ln\vartheta| = |\Sigma(\xi_c)| \ge \rho_{\min}|\xi_c|$: $|\xi_c| \le |\ln\vartheta|/\rho_{\min} \le \frac{1}{16}$. Put $s(\eta) = \Sigma(\eta) - \ln\vartheta = \Sigma(\eta) - \Sigma(\xi_c)$. If $|\eta - \xi_c| \le \frac38$, then η and ξ_c lie in $[-\frac12, \frac12]$, where $\rho \ge \rho_{\min}$, so $|s(\eta)| \ge \rho_{\min}|\eta - \xi_c|$; if $|\eta - \xi_c| > \frac38$ the interval between ξ_c and η contains a subinterval of length $\frac38$ starting at ξ_c, so $|s(\eta)| \ge \frac38\rho_{\min}$. Thus $|s(\eta)| \ge \rho_{\min}\min(|\eta-\xi_c|, \frac38)$. By (10.1), $f - f_- - J\mathbf{1}_{\{\eta>\xi_c\}} = J(\lg(s) - \mathbf{1}_{\{s>0\}})$ (because $\eta > \xi_c \iff s > 0$), and $|\lg(s) - \mathbf{1}_{\{s>0\}}| \le e^{-|s|}$ (for $s > 0$ it is $e^{-s}/(1+e^{-s})$, for $s < 0$ it is $e^s/(1+e^s)$). This gives (10.3).

(d) Let $f_s = f_- + J\mathbf{1}_{\{\eta>\xi_c\}}$. Both f and f_s are bounded by S, so by (10.3), with $\epsilon(\eta) := e^{-\rho_{\min}\min(|\eta-\xi_c|, 3/8)}$ and $\varpi_0(\eta) := e^{-2|\eta|}$,

$$|f^2 - f_s^2| \le |f - f_s||f + f_s| \le 2SJ\epsilon,$$
$$\tfrac12|\mathcal{Q}[f^2 - f_s^2](0)| \le \tfrac18\int \varpi_0\, 2SJ\epsilon \le \frac{SJ}{4}\Big(\frac{2}{\rho_{\min}} + e^{-3\rho_{\min}/8}\Big) \le \frac{3SJ}{4\rho_{\min}},$$

where we split the integral at $|\eta-\xi_c| = \frac38$ (using $\varpi_0 \le 1$ inside and $\int\varpi_0 = 1$ outside) and used $e^{-3x/8} \le 8/(3ex) < 0.99/x$. By Lemma 10.1(b) with $a = \xi_c$ at $\xi = 0$: $\frac12\mathcal{Q}[f_s^2](0) = \frac18[f_-^2 + (2f_-J + J^2)\hat q]$ with $\hat q = m_{\xi_c}(0) = \int_{\xi_c}^\infty \varpi_0$. Now $\hat q(a) = \int_a^\infty \varpi_0$ has $\hat q(0) = \frac12$ and $|\hat q'| \le 1$, so $|\hat q - \frac12| \le |\xi_c| \le |\ln\vartheta|/\rho_{\min}$, and since f_- may have either sign, $|2f_-J + J^2| \le J(J + 2|f_-|) \le 2SJ$:

$$\Big|\tfrac18(2f_-J + J^2)(\hat q - \tfrac12)\Big| \le \frac{SJ}{4}\frac{|\ln\vartheta|}{\rho_{\min}}.$$

Adding up, $W(0) = W_{\mathrm{e}}(0) + \frac18[f_-^2 + (2f_-J + J^2)\frac12] + E_1 = W_{\mathrm{e}}(0) + \frac{f_-^2}{8} + \frac{f_-J}{8} + \frac{J^2}{16} + E_1$ with $|E_1| \le \frac{SJ(|\ln\vartheta|+3)}{4\rho_{\min}}$.

(e) By (a), $\varpi = \sqrt{W} \ge B$ on $[-1,1]$, and $\varpi \le A$ by Lemma 10.3(a). By Lemma 10.3(e), $|\varpi'| = |W'|/(2\varpi) \le (\|W_{\mathrm{e}}'\|_\infty + \frac12 S^2)/(2B) = C_{\mathrm{bd}}$. Apply Lemma 3.10(c) (first-order tracking) to $\zeta = \zeta_-$ on $[a_0, b] = [-\frac12, \frac12]$ with $m = B$: here $\zeta_-(-\frac12) \ge B_1 > 0$ by (b), and both $\zeta_-(-\frac12)$ and $\varpi(-\frac12)$ lie in $(0, A]$, so $|\zeta_- - \varpi|(-\frac12) \le A$ and, for $\xi \ge -\frac14$,

$$|\zeta_-(\xi) - \varpi(\xi)| \le \frac{C_{\mathrm{bd}}}{B} + Ae^{-B(\xi+1/2)} \le \frac{C_{\mathrm{bd}}}{B} + Ae^{-B/4} = E_w.$$

For ζ_+ reflect (Lemma 3.7): $\tilde\zeta(\eta) = -\zeta_+(-\eta)$ solves $\tilde\zeta' = \tilde W - \tilde\zeta^2$ with $\tilde W(\eta) = W(-\eta)$, $\tilde\varpi(\eta) = \varpi(-\eta)$, and $\tilde\zeta(-\frac12) = -\zeta_+(\frac12) \geq B_1 > 0$; the same argument gives $|\tilde\zeta - \tilde\varpi|(\eta) \leq E_w$ for $\eta \geq -\frac14$, i.e. $|\zeta_+(\xi) + \varpi(\xi)| \leq E_w$ for $\xi \leq \frac14$. Finally $|\zeta_- + \zeta_+| \leq 2E_w$ on $[-\frac14, \frac14]$ and $\rho \leq \zeta_- + |\zeta_+| \leq 2A$, so $|\rho'| = \rho|\zeta_- + \zeta_+| \leq 4AE_w$. □

Remark 10.6 (how to read the lemma)**.** Parts (a)–(c) are crude: the barrier uses only $f \geq f(0)$ on $[0,\infty)$, so $B \approx 0.06J/(1+\vartheta)$ is about 8.4 times smaller than the true $\varpi \approx J/4$ (and B^2 is about 4 times below the true $\min_{[-1,1]} W$, Numerical observation 10.7) (Example 10.12). This is why the constants of Theorem 10.11 are poor. They are not meant to be used numerically: their purpose is to place the solution in a regime ($\rho \gtrsim J$, $\xi_c = O(1/J)$, $W(0) = J^2/16 + O(J)$) in which the *asymptotic* statements of this chapter hold with constants depending only on the data. Part (c) is the first half of Theorem 10.28: the swirl is a step of width $O(1/J)$.

Figure 10.2 shows the potential and the tracking for the rest state.

Numerical observation 10.7 (how crude are the a priori bounds?)**.** *For the rest state the quantities of Lemma 10.5 can be compared with the computed fixed points (solver and method as in Numerical observation 10.16):*

κ_D	J	$\min_{[-1,1]} W$	B^2	$\min_{[-\frac12,\frac12]} \rho$	$\rho_{\min}$	$\lvert w_0 - J/2\rvert$ *vs.* C_A
10^3	*45.64*	*7.93*	*1.85*	*11.8*	*1.61*	0.91 *vs.* 756
10^5	*448.2*	*713.6*	*178.4*	*120.3*	*26.7*	0.99 *vs.* 675
10^6	*1415.2*	*7112*	*1779*	*380.2*	*84.4*	1.00 *vs.* 664

(At $\kappa_D = 10^3$ the largeness conditions of Definition 10.4 are not yet satisfied; the inequalities hold nevertheless.) The barrier is a factor 4 *below the true minimum of W and $\rho_{\min}$ is a factor* 4.5 *below the true* $\min\rho$*; the proved error constant C_A is about* 700 *times the true error. The a priori bounds are therefore* qualitatively *sharp (right powers of J) and quantitatively wasteful; this is the price of proving them for an arbitrary fixed point, with no information beyond monotonicity of f. Status: numerical observation;* `verify_ch10a.py`*, Part 3.*

Toy example 10.8 (tracking a slowly varying potential, again)**.** The mechanism of part (e) is the toy of Toy example 3.13 (Figure 3.1(d)): for $W = \Lambda^2 + 1 + 2\Lambda\tanh\xi$ every solution of $\zeta' = W - \zeta^2$ with $\zeta(0) > 0$ approaches $\Lambda + \tanh\xi$ at the rate 2Λ, while $\sqrt{W} = \Lambda + \tanh\xi + O(1/\Lambda)$ misses it by $O(1/\Lambda)$. In the present situation $\Lambda \approx J/4$, the distance to $\sqrt{W}$ is the lag $|\varpi'/(2\varpi)| = O(1)$ and the *initial* error, of size A at $\xi = -\frac12$, has decayed by the factor $e^{-B/4}$ at $\xi = -\frac14$. The constant E_w is exactly the sum of these two contributions.

10.3. The $\sqrt{\kappa_D}$ law

In words. *Now the pieces fit together. The exact identity $\kappa_D = \vartheta J w_0$ of Lemma 8.10 turns the question "how large is J compared with κ_D?" into "how large is $w_0 = \rho(0)$ compared with J?". By tracking, $w_0 = 2\sqrt{W(0)} + O(1)$, and by the step potential, $W(0) = J^2/16 + O(J)$, so $w_0 = J/2 + O(1)$ and $\kappa_D = \vartheta J(J/2 + O(1))$, i.e. $J = \sqrt{2\kappa_D/\vartheta} + O(1)$. The only circularity (steepness needs J large, J large needs κ_D large) is cut by the elementary bound $\kappa_D \leq 2\vartheta J A(J)$.*

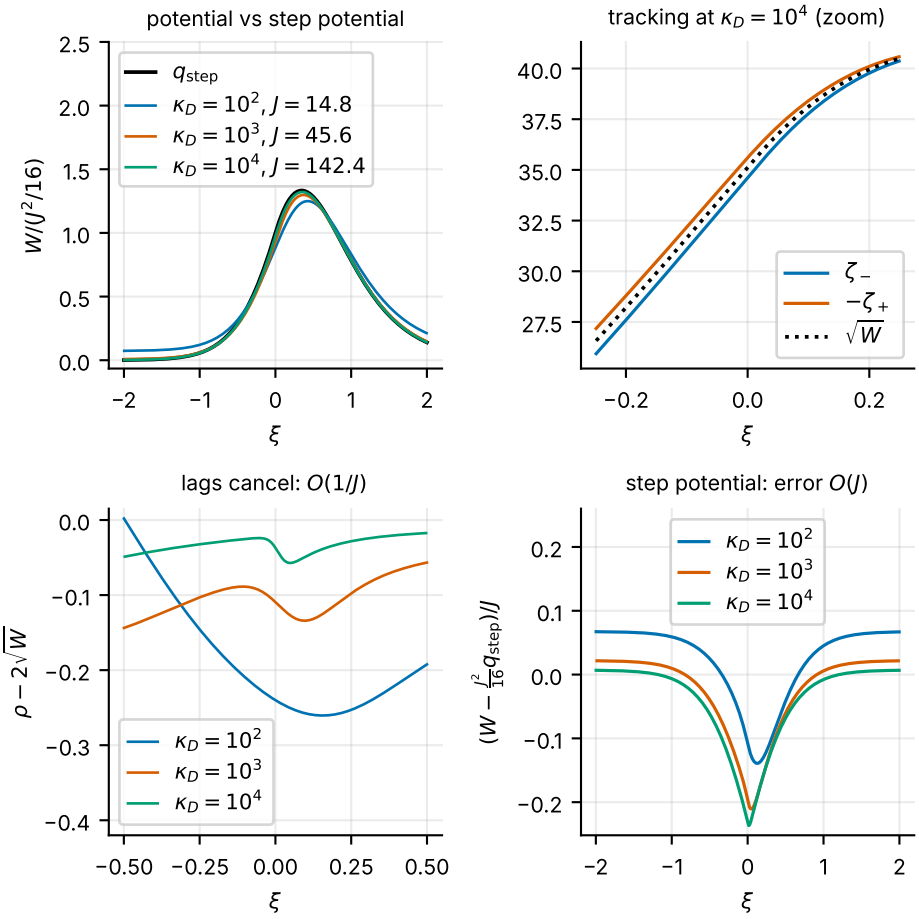


FIGURE 10.2. Potential and tracking for the rest state ($\boldsymbol{c}' = 0$, $\vartheta = 1$, $f_- = 0$). (a) $W/(J^2/16)$ for $\kappa_{\mathrm{D}} = 10^2, 10^3, 10^4$ compared with the potential q_{step} of a sharp step (Example 10.2); the two agree up to $O(1/J)$ near the layer, and $W(0) = J^2/16 + O(J)$ (Lemma 10.5(d)). (b) At $\kappa_{\mathrm{D}} = 10^4$, ζ_- and $-\zeta_+$ lie on the two sides of $\sqrt{W}$ at distance $O(1)$ (the lag; Lemma 10.5(e)). (c) The lags cancel in $\rho - 2\sqrt{W}$, which is $O(1/J)$ (Lemma 10.18). (d) The difference $(W - \frac{J^2}{16} q_{\text{step}})/J$ stays bounded (by 0.25 here, with value $\approx W_1 = -\frac{1}{4}$ at $\xi = 0$), as the $O(J)$ bound of Lemma 10.5(d) predicts. Code: `code/ch10_large/make_a_tracking.py`.

Toy example 10.9 (a self-consistent constant potential)**.** For a constant potential $W \equiv k^2$ one has $w_0 = 2k$ and, by Toy example 8.9, $J = \kappa_{\mathrm{D}}/(2k\vartheta)$. Let now the potential depend on the swirl through its size: a step of height J produces a potential of size $\approx J^2/16$ in the window where f varies (Example 10.2), so $k \approx J/4$,

$w_0 \approx J/2$ and the identity $\kappa_{\mathrm{D}} = \vartheta J w_0$ becomes $\kappa_{\mathrm{D}} \approx \frac{1}{2}\vartheta J^2$. The scalar map $J \mapsto \kappa_{\mathrm{D}}/(\vartheta w_0(J))$ with $w_0(J) = J/2$ has the fixed point $J_0 = \sqrt{2\kappa_{\mathrm{D}}/\vartheta}$. The theorem below says that this caricature gives the exact leading behaviour of *all* fixed points, and Theorem 10.22 computes the constant correction.

Lemma 10.10 (orders of magnitude)**.** *There are $c_* > 0$ and $C_\sharp < \infty$, depending only on $(\boldsymbol{c}', \vartheta, f_-)$, such that for all $J \geq J_*$:*

$$J \leq S \leq \tfrac{3}{2}J, \quad B \geq c_* J, \quad \rho_{\min} \geq 1.97 c_* J, \quad A \leq C_\sharp J,$$
$$C_{\mathrm{bd}} \leq C_\sharp J, \quad E_w \leq C_\sharp, \quad r_0 \leq C_\sharp J, \quad C_A \leq C_\sharp.$$

Proof. $S \geq J$ is clear and $S \leq \frac{3}{2}J$ by (J1) (as $|f_-| \leq J/(2(1+\vartheta))$). The function $J \mapsto B(J)/J$ is continuous and positive on $[J_*, \infty)$ (positive by (J2)) and tends to $\sqrt{q_1/2}/(1+\vartheta) > 0$ as $J \to \infty$ (since $m_0/J \to 1/(1+\vartheta)$ and $C_-/J^2 \to 0$); hence $c_* := \inf B/J > 0$. Then $\rho_{\min} \geq 1.97B$ (Lemma 10.5(b)), $A \leq (\sup W_{\mathrm{e}})^{1/2} + S/\sqrt{8} \lesssim J$, $C_{\mathrm{bd}} \leq (\|W_{\mathrm{e}}'\|_\infty + \frac{9}{8}J^2)/(2c_* J) \lesssim J$, and

$$E_w = \frac{C_{\mathrm{bd}}}{B} + Ae^{-B/4} \leq \frac{C_{\mathrm{bd}}}{c_* J} + Ce^{-c_* J/4} J \lesssim 1,$$
$$r_0 \leq |W_{\mathrm{e}}(0)| + \tfrac{f_-^2}{8} + \tfrac{|f_-|J}{8} + \frac{\frac{3}{2}J^2(|\ln\vartheta| + 3)}{4 \cdot 1.97 c_* J} \lesssim J,$$

so $C_A = 8r_0/J + 2E_w \lesssim 1$. □

Theorem 10.11 (the $\sqrt{\kappa_{\mathrm{D}}}$ law, explicit constants). *Let $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$, $\vartheta > 0$, $f_- \in \mathbb{R}$, $\kappa_{\mathrm{D}} > 0$, and let f be* any *fixed point of T_D (any global solution of type* G *at both poles with these effective data), $J = f(+\infty) - f(-\infty)$. If J satisfies the largeness conditions* (J1)–(J3)*, then*

$$|w_0 - J/2| \leq C_A, \qquad -C_A \leq J - \sqrt{2\kappa_{\mathrm{D}}/\vartheta} \leq 2C_A, \qquad \left|\frac{J}{\sqrt{2\kappa_{\mathrm{D}}/\vartheta}} - 1\right| \leq C_A\sqrt{\frac{2\vartheta}{\kappa_{\mathrm{D}}}}.$$

In "there exist" form: with J_* of (10.2) and $\kappa_* := 2\vartheta J_* A(J_*)$, every fixed point with $\kappa_{\mathrm{D}} > \kappa_*$ satisfies (J1)–(J3)*, and therefore*

$$\left|J - \sqrt{2\kappa_{\mathrm{D}}/\vartheta}\right| \leq C, \qquad C := 2\sup_{J \geq J_*} C_A(J) < \infty,$$

i.e. $J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta}\,(1 + O(\kappa_{\mathrm{D}}^{-1/2}))$ uniformly over all solutions. The numbers J_*, κ_*, C are bounded on compact sets of (ϑ, f_-).

Example 10.12 (the rest state: the proved constants)**.** For $\boldsymbol{c}' = 0$, $\vartheta = 1$, $f_- = 0$: $W_{\mathrm{e}} \equiv 1$, $C_- = C_0 = 0$, $\|W_{\mathrm{e}}'\|_\infty = 0$, $m_0 = J/2$, $B = \sqrt{q_1/8}\,J = 0.02980\,J$, $S = J$, $A = \sqrt{1 + J^2/8}$. Then (J2) reads $B \geq 6$, i.e. $J \geq J_* = 201.3$ (*not* an integer; the first integer is 202), $\kappa_* = 2J_* A(J_*) = 2.87 \cdot 10^4$ and, at J_*: $\rho_{\min} = 11.98$, $E_w = 297$, $r_0 = 2.55 \cdot 10^3$, $C_A = 696$; $C_A(J)$ decreases to $\approx 6.6 \cdot 10^2$ as $J \to \infty$. So Theorem 10.11 *proves* $J = \sqrt{2\kappa_{\mathrm{D}}} + O(1)$ for $\kappa_{\mathrm{D}} \geq 2.9 \cdot 10^4$ with $|J - \sqrt{2\kappa_{\mathrm{D}}}| \leq 1.4 \cdot 10^3$. In truth $J - \sqrt{2\kappa_{\mathrm{D}}} \to 1$ (Theorem 10.22, Numerical observation 10.16); the proved constants are poor because the barrier of Lemma 10.5(a) is a factor 8.4 too small in B (a factor 4 in B^2 against $\min_{[-1,1]} W$) and $E_w \approx S^2/(4B^2)$. (Computed in `verify_ch10a.py`.)

Proof of Theorem 10.11. *Step 1: w_0 is close to $J/2$.* By Lemma 10.5(e) at $\xi = 0$, $|\zeta_-(0) - \varpi(0)| \le E_w$ and $|\zeta_+(0) + \varpi(0)| \le E_w$, hence $|w_0 - 2\varpi(0)| \le 2E_w$. By Lemma 10.5(d), $W(0) = J^2/16 + R_0$ with $|R_0| \le r_0$ and $W(0) \ge B^2 > 0$, so

$$|\varpi(0) - \tfrac{J}{4}| = \frac{|W(0) - J^2/16|}{\varpi(0) + J/4} \le \frac{r_0}{J/4} = \frac{4r_0}{J}.$$

Therefore $|w_0 - J/2| \le |w_0 - 2\varpi(0)| + 2|\varpi(0) - J/4| \le 2E_w + 8r_0/J = C_A$.

Step 2: invert. By (10.1), $\kappa_{\mathrm{D}}/\vartheta = Jw_0$, so $|\kappa_{\mathrm{D}}/\vartheta - J^2/2| \le C_A J$. From $\kappa_{\mathrm{D}}/\vartheta \le J^2/2 + C_A J \le (J + C_A)^2/2$ we get $J \ge \sqrt{2\kappa_{\mathrm{D}}/\vartheta} - C_A$. From $\kappa_{\mathrm{D}}/\vartheta \ge J^2/2 - C_A J = \frac{1}{2}(J - C_A)^2 - \frac{1}{2}C_A^2$ we get $(J - C_A)^2 \le 2\kappa_{\mathrm{D}}/\vartheta + C_A^2$, hence $J \le C_A + \sqrt{2\kappa_{\mathrm{D}}/\vartheta + C_A^2} \le \sqrt{2\kappa_{\mathrm{D}}/\vartheta} + 2C_A$. Dividing by $\sqrt{2\kappa_{\mathrm{D}}/\vartheta}$ gives the relative bound, since $2C_A/\sqrt{2\kappa_{\mathrm{D}}/\vartheta} = C_A\sqrt{2\vartheta/\kappa_{\mathrm{D}}}$.

Step 3: every solution with $\kappa_{\mathrm{D}} > \kappa_*$ has $J \ge J_*$. By Lemma 10.3(c) with $J_0 = J_*$, $\kappa_{\mathrm{D}} > \kappa_* = 2\vartheta J_* A(J_*)$ forces $J > J_*$; and (J1)–(J3) hold for all $J \ge J_*$ by monotonicity. Then Steps 1–2 apply, and $C = 2\sup_{J \ge J_*} C_A(J) < \infty$ by Lemma 10.10. Finally J_* is bounded on compact sets, $A(J_*)$ is bounded there, and C_A is bounded by Lemma 10.10 with constants that are continuous in the data, which gives the last assertion. □

Example 10.13 (other data: the dependence on $\vartheta, f_-, \boldsymbol{c}'$)**.** The constants of Definition 10.4 for five data sets (computed in `verify_ch10a.py`, Part 5):

data $(\boldsymbol{c}', \vartheta, f_-)$	C_-	C_0	J_*	κ_*	$C_A(J_*)$
$(0,1,0)$ (rest state)	0	0	201.3	$2.87 \cdot 10^4$	696
$(0,1,-1)$	0	0	203.3	$2.94 \cdot 10^4$	716
$(0,4,\frac{1}{2})$	0	0	1044	$3.09 \cdot 10^6$	$3.9 \cdot 10^3$
$((-0.8,0.5,0),1,0)$	0	0	201.3	$2.87 \cdot 10^4$	696
$((0,0,-3),1,0)$	$\frac{1}{2}$	0	202.7	$2.91 \cdot 10^4$	705

Three observations. (i) For these $\boldsymbol{c}'$ the potential W_{e} is nowhere very negative and its Riccati variables are positive at ∓ 1, so C_-, C_0 are small and the constants hardly depend on $\boldsymbol{c}'$: the swirl dominates the potential, and the data $\boldsymbol{c}'$ only enter through the $O(1)$ constants $W_{\mathrm{e}}(0)$, $\|W_{\mathrm{e}}'\|_\infty$ (in r_0 and E_w). (ii) The data ϑ and f_- enter through $m_0 = f_- + J/(1+\vartheta)$ and the condition (J3) $|\ln\vartheta| \le B/9$: for $\vartheta = 4$ the factor $(1+\vartheta)^{-1}$ in B and $|\ln\vartheta| \le B/9$ together move the proved threshold by a factor 5. (iii) The proof needs far more than the truth: at $\kappa_{\mathrm{D}} = 2\cdot 10^5$ the computed fixed points for data sets of this kind satisfy $J - \sqrt{2\kappa_{\mathrm{D}}/\vartheta} = 1 + \ln\vartheta - f_-/2 + o(1)$ to within 0.03 (numerical observation; Figure 10.3 and `verify_ch10a.py`, Part 4), whereas the proved threshold κ_* is of order 10^4–10^6.

Corollary 10.14 (consequences of the $\sqrt{\kappa_{\mathrm{D}}}$ law)**.** *Let $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$, $\vartheta > 0$, $f_- \in \mathbb{R}$ and let $\kappa_{\mathrm{D}} > \kappa_*$ (notation of Theorem 10.11). For every global solution of type* G *at both poles with effective data $(\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$:*

(a) *$|J - \sqrt{2\kappa_{\mathrm{D}}/\vartheta}| \le C$; in particular $J \to \infty$ as $\kappa_{\mathrm{D}} \to \infty$, uniformly in the solution, and $J/\sqrt{\kappa_{\mathrm{D}}} \to \sqrt{2/\vartheta}$.*

(b) *The end values of the swirl are $f(+\infty) = f_- + \sqrt{2\kappa_{\mathrm{D}}/\vartheta} + O(1)$ and $f(0) = f_- + \frac{1}{1+\vartheta}\sqrt{2\kappa_{\mathrm{D}}/\vartheta} + O(1)$; the value at the equator is a fixed fraction $1/(1+\vartheta)$ of the total jump.*

(c) *$w_0 = \kappa_{\mathrm{D}}/(\vartheta J) = \sqrt{\kappa_{\mathrm{D}}/(2\vartheta)}\,(1 + O(\kappa_{\mathrm{D}}^{-1/2}))$.*

(d) *In terms of the equator slope* $\kappa_0 = f'(0)$ *for* $z(0) = 1$ (*normalisation 3;* $\kappa_{\mathrm{D}} = (1+\vartheta)^2\kappa_0$ *by Lemma 2.15(4)):* $J = (1+\vartheta)\sqrt{2\kappa_0/\vartheta}\,(1 + O(\kappa_0^{-1/2}))$.

Proof. (a) is Theorem 10.11 together with Lemma 10.3(c). (b) $f(+\infty) = f_- + J$ and $f(0) = f_- + J/(1+\vartheta)$ (Lemma 10.3(b)). (c) $w_0 = \kappa_{\mathrm{D}}/(\vartheta J)$ by (10.1) and $J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta}(1 + O(\kappa_{\mathrm{D}}^{-1/2}))$ by (a); $\kappa_{\mathrm{D}}/(\vartheta\sqrt{2\kappa_{\mathrm{D}}/\vartheta}) = \sqrt{\kappa_{\mathrm{D}}/(2\vartheta)}$. (d) substitute $\kappa_{\mathrm{D}} = (1+\vartheta)^2\kappa_0$ in (a): $\sqrt{2\kappa_{\mathrm{D}}/\vartheta} = (1+\vartheta)\sqrt{2\kappa_0/\vartheta}$, and $O(\kappa_{\mathrm{D}}^{-1/2}) = O(\kappa_0^{-1/2})$ for fixed ϑ. □

Remark 10.15 (what is and is not claimed)**.** (i) The theorem is a statement about *every* fixed point and uses neither uniqueness nor any smallness of ϑ or f_- beyond (J1)–(J3); (J1) and (J3) are artefacts of the crude barrier of Lemma 10.5(a): a fixed point with $f_- < 0$ may sit on a plateau $J \approx (1+\vartheta)|f_-|$ before the law takes over, and the regime $|\ln\vartheta| \sim J$ is not covered (Open problem Open problem 13.21 in Chapter 13). (ii) The hypothesis $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$ enters only through the admissibility of W_{e} and the constants C_-, C_0; for $\boldsymbol{c}' \notin \operatorname{int}\mathcal{J}$ the statement is not claimed (Open problem 13.20). (iii) Combined with Proposition 8.13, the theorem says: *in the chart of effective data, the jump* J *of the swirl is* $\sqrt{2\kappa_{\mathrm{D}}/\vartheta} + O(1)$ *for every global solution with large* κ_{D}; the physically relevant conclusion is that the swirl velocity $U_\varphi = f$ takes a total change of order $\sqrt{\kappa_{\mathrm{D}}}$, not of order κ_{D} as for small swirl.

Numerical observation 10.16 (the $\sqrt{\kappa_{\mathrm{D}}}$ law at the rest state)**.** *For* $\boldsymbol{c}' = 0$, $\vartheta = 1$, $f_- = 0$ *the fixed point was computed for* $\kappa_{\mathrm{D}} = 10, \dots, 10^7$ *(damped Picard iteration of* T_D*, residual* $< 10^{-11}$*, Riccati sweeps in the stable direction on a grid clustered at the layer; solver and scan scripts in* `code/ch10_large/`*, Appendix E).*

κ_{D}	J	$J - \sqrt{2\kappa_{\mathrm{D}}}$	$\sqrt{\kappa_{\mathrm{D}}}\,(J - \sqrt{2\kappa_{\mathrm{D}}} - 1)$	$w_0 - J/2$
10^1	*4.0139*	−0.4582	−4.611	0.4844
10^2	*14.8437*	0.7015	−2.985	−0.6849
10^3	*45.6419*	0.9206	−2.511	−0.9113
10^4	*142.3949*	0.9735	−2.646	−0.9702
10^5	*448.2050*	0.9914	−2.706	−0.9904
10^6	*1415.2108*	0.9973	−2.721	−0.9969
10^7	*4473.1351*	0.9991	−2.704	−0.9990

The law $J = \sqrt{2\kappa_{\mathrm{D}}} + O(1)$ *is visible from* $\kappa_{\mathrm{D}} \approx 10^2$*, i.e.* four orders of magnitude before *the proved threshold* $\kappa_* = 2.9 \cdot 10^4$ *of Example 10.12; the third column converges to the constant* 1 *predicted by Theorem 10.22 with error* $\approx -2.7/\sqrt{\kappa_{\mathrm{D}}}$*, and the last column to* $b_0 = -1$*. Other data (Figure 10.3):* $\vartheta = 4$, $f_- = \frac{1}{2}$, $\kappa_{\mathrm{D}} = 2 \cdot 10^5$ *gives* $J - \sqrt{2\kappa_{\mathrm{D}}/\vartheta} = 2.119$ *against the predicted limit* $1 + \ln\vartheta - f_-/2 = 2.136$*. The values at* $\kappa_{\mathrm{D}} = 10^3, 10^5, 10^6$ *were recomputed independently by* `verify_ch10a.py`*. Status: numerical observation; code* `code/ch10_large/`*, collected in Atlas 10.48.*

10.4. The next order: the potential at the equator and $J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta} + 1 + \ln\vartheta - f_-/2 + \dots$

In words. *Theorem 10.11 determines* w_0 *up to* $O(1)$*, and the* $O(1)$ *matters:* $\kappa_{\mathrm{D}} = \vartheta J w_0$ *turns an error* b *in* $w_0 = J/2 + b$ *into an error* b *in* J*. To find* b *we feed the result back. Now that we know* $\rho \gtrsim J$ *and the layer centre* $\xi_c = O(1/J)$*, the profile is a logistic step* $f = f_- + J\,\mathrm{lg}(\Sigma - \ln\vartheta)$ whose width and position are known,

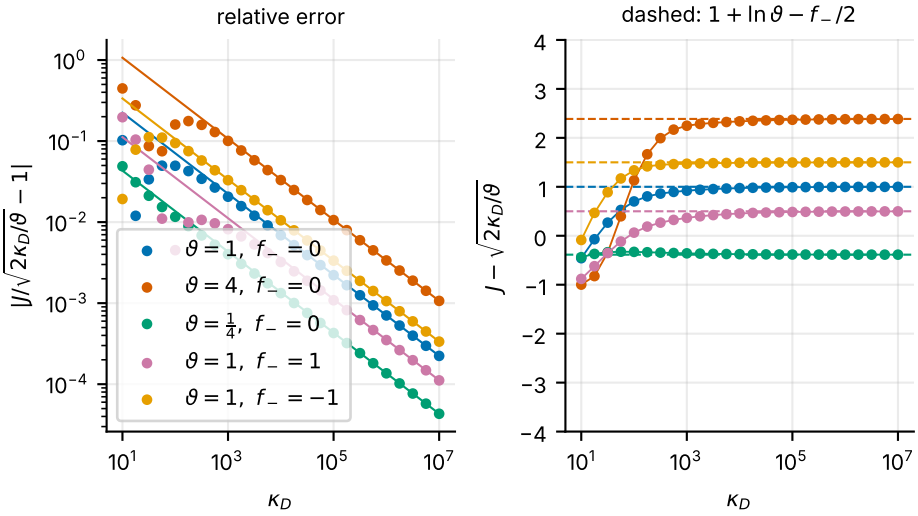


FIGURE 10.3. The $\sqrt{\kappa_{\mathrm{D}}}$ law and its next order for five data sets with $\boldsymbol{c}' = 0$ (legend). (a) Relative error $|J/\sqrt{2\kappa_{\mathrm{D}}/\vartheta} - 1|$ (dots) falls with slope $-\frac{1}{2}$ (Theorem 10.11) and agrees with $|1 + \ln\vartheta - f_-/2|\sqrt{\vartheta/(2\kappa_{\mathrm{D}})}$ (lines). (b) $J - \sqrt{2\kappa_{\mathrm{D}}/\vartheta}$ converges to $1 + \ln\vartheta - f_-/2$ (dashed; Theorem 10.22). Code: `code/ch10_large/make_a_sqrtlaw.py` (stored scan of the solver).

and the potential at the equator can be computed up to $O(1)$ by comparing with a sharp step. There are exactly three $O(J)$ contributions: the left state f_- (cross term $f_-J/8$), the displacement of the layer centre ($-\frac{J}{4}\ln\vartheta$) and the smoothing of the step ($-\frac{J}{4}$). A second-order WKB estimate (the lags of ζ_- and ζ_+ cancel in ρ) then gives $w_0 = 2\sqrt{W(0)} + O(1/J)$ instead of $O(1)$, and we read off $w_0 = \frac{J}{2} + 4W_1 + O(1/J)$.

Toy example 10.17 (sharp step versus smooth step at the rest state)**.** For the *sharp* step of Example 10.2 ($\boldsymbol{c}' = 0$, $f_- = 0$, step at 0, height J) the Jost gap at the origin, computed numerically by Riccati integration, is $w_0 = J/2 + b_{\mathrm{sharp}}$ with $b_{\mathrm{sharp}} = -0.046, -0.016, -0.0045$ for $J = 40, 160, 640$: it tends to 0 (consistent with $W(0) = 1 + J^2/16$ and $w_0 = 2\sqrt{W(0)} + O(1/J) = J/2 + O(1/J)$). For the *true* fixed point, $w_0 - J/2 \to -1$ (Numerical observation 10.16). The whole difference -1 comes from smoothing the step: the true profile is a logistic of width $2/J$, whose square has the same integral against the weight $e^{-2|\eta|}$ as a sharp step *except* for $\int(\lg^2 - \mathbf{1}_{\{s>0\}})\mathrm{d}s = -1$, which lowers $W(0)$ by $J^2/(8\rho) \approx J/4$ and w_0 by $2 \cdot \frac{1}{2} \cdot \frac{J/4}{J/4} = 1$. This is Lemma 10.20 and Remark 10.23 below. (Riccati integration with an adaptive Runge–Kutta scheme, tolerance 10^{-12}; `verify_ch10a.py`, Part 4.)

In this section "$X \lesssim Y$" means $X \le CY$ with C depending only on $(\boldsymbol{c}', \vartheta, f_-)$, for all fixed points with $J \ge J_*$; by Lemma 10.10 and Theorem 10.11, $S \asymp J \asymp B \asymp \rho_{\min}$,

$A, C_{\mathrm{bd}} \lesssim J$, $E_w \lesssim 1$ and $|w_0 - J/2| \lesssim 1$, and $\rho_{\min} \le w_0 \le 2A$ (so $w_0 \asymp J$). Put $L_\rho := 4AE_w \lesssim J$, the bound for $|\rho'|$ on $[-\frac14, \frac14]$ of Lemma 10.5(e).

For orientation, here are the sizes of the quantities that enter the next-order computation (all for a fixed point with $J \ge J_*$; "$\asymp$" with constants depending on the data only):

quantity	meaning	size	where
B, $\rho_{\min}$, w_0	lower bounds of ϖ, ρ near 0; the Jost gap	$\asymp J$	Lemma 10.10
A, $C_{\mathrm{bd}} = \sup\lvert\varpi'\rvert$	upper bound of ϖ; its slope	$\lesssim J$	Lemma 10.10
E_w	tracking error $\lvert\zeta_\mp \mp \varpi\rvert$	$O(1)$	Lemma 10.5(e)
$L_\rho = 4AE_w$	bound for $\lvert\rho'\rvert$ on $[-\frac14, \frac14]$	$O(J)$	Lemma 10.5(e)
a, F_0	lag $-\varpi'/(2\varpi)$; the quantity $a' + a^2$	$O(1)$	Lemma 10.18
ξ_c	layer centre	$2\ln\vartheta/J + O(J^{-2})$	Lemma 10.19
L_*	Lipschitz constant of ϖ_0/ρ in the variable s	$O(J^{-2})$	Lemma 10.20

Lemma 10.18 (second-order tracking: $w_0 = 2\sqrt{W(0)} + O(1/J)$)**.** *Let $a = -\varpi'/(2\varpi) = -W'/(4W)$ be the lag and $F = a' + a^2 = -\frac{W''}{4W} + \frac{5W'^2}{16W^2}$. Assume the fixed point satisfies* (J1)–(J3) *and $|a| \le B/2$ on $[-\frac12, \frac12]$. Then, with*

$$F_0 := \frac{\|W_{\mathrm{e}}''\|_\infty + \frac52 S^2}{4B^2} + \frac{5(\|W_{\mathrm{e}}'\|_\infty + \frac12 S^2)^2}{16B^4} \quad (\lesssim 1),$$

we have $|F| \le F_0$ on $[-\frac12, \frac12]$ and

$$|w_0 - 2\sqrt{W(0)}| \le \frac{4F_0}{B} + 2(A + \|a\|_\infty)e^{-B/4} \quad (\lesssim 1/J).$$

Proof. The formula for F is a direct computation ($a' = -\frac{W''}{4W} + \frac{W'^2}{4W^2}$, $a^2 = \frac{W'^2}{16W^2}$; checked by computer algebra). With $W \ge B^2$ (Lemma 10.5(a)) and the bounds of Lemma 10.3(e) we obtain $|F| \le F_0$; and $F_0 \lesssim 1$ because $S \asymp B \asymp J$. Apply Lemma 3.10(d) (second-order tracking) with $m = B$ to ζ_- on $[-\frac12, \frac12]$: the lag condition is the hypothesis $|a| \le B/2$, and $|r_-(-\frac12)| \le |\zeta_-(-\frac12) - \varpi(-\frac12)| + |a(-\frac12)| \le A + \|a\|_\infty$ for $r_- := \zeta_- - \varpi - a$, so

$$|r_-(0)| \le \frac{2F_0}{B} + (A + \|a\|_\infty)e^{-B/4}.$$

The symmetric statement of the same lemma for ζ_+, which is negative on $[-\frac12, \frac12]$ with $\zeta_+(\frac12) \le -B_1 < 0$, gives the same bound for $r_+ := \zeta_+ + \varpi - a$ at $\xi = 0$ (with the roles of the endpoints exchanged: the initial value is now taken at $\xi = \frac12$). Now $\zeta_- = \varpi + a + r_-$ and $\zeta_+ = -\varpi + a + r_+$, so the lags cancel in the difference (Remark 3.12): $w_0 = \rho(0) = 2\varpi(0) + r_-(0) - r_+(0)$. □

Lemma 10.19 (the layer centre)**.** *Under* (J1)–(J3)*: $|\xi_c - \ln\vartheta/w_0| \le L_\rho \xi_c^2/(2w_0) \lesssim J^{-2}$, hence $\xi_c = \frac{2\ln\vartheta}{J} + O(J^{-2})$ and $|\xi_c| \lesssim 1/J$.*

PROOF. Since $|\xi_c| \le \frac{1}{16}$ (Lemma 10.5(c)) and $|\rho'| \le L_\rho$ on $[-\frac14,\frac14]$ (Lemma 10.5(e)), $\ln\vartheta = \Sigma(\xi_c) = w_0\xi_c + \int_0^{\xi_c}(\rho - w_0)$ with $|\int_0^{\xi_c}(\rho-w_0)| \le L_\rho\xi_c^2/2$; divide by w_0. Now $|\xi_c| \le |\ln\vartheta|/\rho_{\min} \lesssim 1/J$, $L_\rho \lesssim J$ and $w_0 \ge \rho_{\min} \gtrsim J$ give the bound $\lesssim J\cdot J^{-2}/J$. Finally $|1/w_0 - 2/J| = |J-2w_0|/(Jw_0) \le 2C_A/(Jw_0) \lesssim J^{-2}$. □

Lemma 10.20 (the layer integrals)**.** *Let $\varpi_0(\eta) = e^{-2|\eta|}$, $s = \Sigma - \ln\vartheta$ and, for $k = 1,2$, $I_k := \int_{\mathbb{R}} \varpi_0(\eta)\big(\lg(s(\eta))^k - \mathbf{1}_{\{\eta>\xi_c\}}\big)\mathrm{d}\eta$. Under* (J1)–(J3)*, with $L_* := 2/\rho_{\min}^2 + L_\rho/\rho_{\min}^3$ ($\lesssim J^{-2}$),*

$$|I_1| \le 4L_* + 6e^{-3\rho_{\min}/16}, \qquad \Big|I_2 + \frac{\varpi_0(\xi_c)}{\rho(\xi_c)}\Big| \le 4L_* + 6e^{-3\rho_{\min}/16}.$$

PROOF. Put $D_k(s) = \lg(s)^k - \mathbf{1}_{\{s>0\}}$. Then $\int_{\mathbb{R}} D_1 = 0$ and $\int_{\mathbb{R}} D_2 = -1$: indeed $\lg(-s) = 1 - \lg(s)$ gives $\int_{-\infty}^0 \lg = \int_0^\infty (1-\lg)$, and

$$\int_{\mathbb{R}} D_2 = \int_0^\infty \big[(1-\lg)^2 + \lg^2 - 1\big]\mathrm{d}s = -2\int_0^\infty \lg(1-\lg)\mathrm{d}s = -2\big[\lg\big]_0^\infty = -1.$$

Moreover $|D_k(s)| \le 2e^{-|s|}$ (for $k=2$, $s>0$: $1-\lg^2 = (1-\lg)(1+\lg) \le 2e^{-s}$; for $s<0$: $\lg^2 \le \lg \le e^s$). Let $\Omega = \{\eta : |\eta - \xi_c| \le \frac{3}{16}\} \subset [-\frac14,\frac14]$ (as $|\xi_c| \le \frac{1}{16}$). *Outside Ω:* $|s(\eta)| \ge \frac{3}{16}\rho_{\min}$ by the proof of Lemma 10.5(c), so $\int_{\mathbb{R}\setminus\Omega} \varpi_0|D_k| \le 2e^{-3\rho_{\min}/16}\int \varpi_0 = 2e^{-3\rho_{\min}/16}$. *Inside Ω:* s is an increasing bijection of Ω onto an interval $I \ni 0$ containing $[-\frac{3}{16}\rho_{\min}, \frac{3}{16}\rho_{\min}]$, with $\mathrm{d}\eta = \mathrm{d}s/\rho$, so $\int_\Omega \varpi_0 D_k(s)\mathrm{d}\eta = \int_I g(s)D_k(s)\mathrm{d}s$, $g(s) := (\varpi_0/\rho)(\eta(s))$. Since $|\varpi_0'| = 2\varpi_0 \le 2$, $|\rho'| \le L_\rho$ and $\rho \ge \rho_{\min}$ on Ω, $|g'| = |(\varpi_0/\rho)'|/\rho \le (2/\rho_{\min} + L_\rho/\rho_{\min}^2)/\rho_{\min} = L_*$; and $g(0) = \varpi_0(\xi_c)/\rho(\xi_c) \le 1/\rho_{\min} \le 1$. Hence

$$\Big|\int_I g\,D_k - g(0)\int_{\mathbb{R}} D_k\Big| \le L_*\int_{\mathbb{R}} |s|\,2e^{-|s|}\mathrm{d}s + g(0)\int_{\mathbb{R}\setminus I} 2e^{-|s|}\mathrm{d}s \le 4L_* + 4e^{-3\rho_{\min}/16}.$$

Adding the two regions gives the claims, using $\int D_1 = 0$, $\int D_2 = -1$. □

Numerical observation 10.21 (the layer integrals)**.** *For the computed fixed points, $I_1 = \int \varpi_0(\lg(s) - \mathbf{1}_{\{\eta>\xi_c\}})$ and $I_2 = \int \varpi_0(\lg(s)^2 - \mathbf{1}_{\{\eta>\xi_c\}})$ compare with the predictions 0 and $-\varpi_0(\xi_c)/\rho(\xi_c)$ of Lemma 10.20 as follows (trapezoidal rule on the solver grid):*

data $(\boldsymbol{c}', \vartheta, f_-)$	κ_{D}	J	ξ_c $(2\ln\vartheta/J)$	I_1	I_2	*prediction*
$(0,1,0)$	10^3	*45.64*	0	$3.0\cdot10^{-3}$	-0.0378	-0.0456
$(0,1,0)$	10^5	*448.2*	0	$8.8\cdot10^{-5}$	-0.00434	-0.00448
$(0,4,\frac12)$	10^3	*24.28*	0.127 (0.114)	$2.0\cdot10^{-2}$	-0.0474	-0.0675
$(0,4,\frac12)$	10^5	*225.7*	0.01244 (0.01228)	$3.5\cdot10^{-4}$	-0.00829	-0.00870

The remainders (differences I_2−prediction: $7.8\cdot10^{-3}$ and $1.4\cdot10^{-4}$ at $\vartheta = 1$) fall by a factor ≈ 55 while J grows by ≈ 9.8, i.e. at least as fast as the $O(J^{-1})$ needed in the proof, and $\xi_c \approx 2\ln\vartheta/J$ as in Lemma 10.19. Status: numerical observation; ***verify_ch10a.py****, Part 5.*

We can now state and prove the next-order theorem; its statement follows.

THEOREM 10.22 (next order). *Under the hypotheses of the $\sqrt{\kappa_{\mathrm{D}}}$ theorem and for $J \ge J_B(c',\vartheta,f_-)$: $|w_0 - (J/2 + b_0)| \le C_B/J$ with $b_0 = f_-/2 - \ln\vartheta - 1$; $W(0) = J^2/16 + J(f_-/8 - (1+\ln\vartheta)/4) + O(1)$; hence for $\kappa_{\mathrm{D}} \ge \kappa_B$: $J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta} +$*

$1+\ln\vartheta - f_-/2 + O(\kappa_{\mathrm{D}}^{-1/2})$. *Here "the hypotheses of the $\sqrt{\kappa_{\mathrm{D}}}$ theorem" are those of Theorem 10.11 (the standing hypotheses and* (J1)–(J3)*); J_B is the least number $\ge J_*$ such that for every $J \ge J_B$:* (B1) $C_{\mathrm{bd}} \le B^2$ *(so that $|a| \le B/2$ on $[-\frac12,\frac12]$),* (B2) $r_0 \le J^2/32$, (B3) $J \ge 2|b_0|$*; and $\kappa_B := 2\vartheta J_B A(J_B)$. The constants C_B and the O-constants depend only on $(\boldsymbol{c}',\vartheta,f_-)$ and are bounded on compact sets of (ϑ,f_-).*

Proof. All conditions (B1)–(B3) hold for large J (Lemma 10.10: $C_{\mathrm{bd}} \lesssim J \ll B^2$, $r_0 \lesssim J$), so $J_B < \infty$. Let $J \ge J_B$ and $W_1 := \frac{f_-}{8} - \frac{1+\ln\vartheta}{4}$, so that $b_0 = 4W_1$.

Step 1: $\int \varpi_0 f^2$. By Lemma 8.2, $W(0) = W_{\mathrm{e}}(0) + \frac18\int \varpi_0 f^2$. By (10.1), $f = f_- + J\lg(s)$ and so

$$\int \varpi_0 f^2 = f_-^2 + 2f_-J\int \varpi_0 \lg(s) + J^2\int \varpi_0 \lg(s)^2, \qquad \int \varpi_0 \lg(s)^k = \hat q + I_k,$$

where $\hat q = \int_{\xi_c}^\infty \varpi_0$ and I_k is as in Lemma 10.20. Since $\hat q(0) = \frac12$, $\hat q'(0) = -1$, $|\hat q''| \le 2$: $\hat q = \frac12 - \xi_c + \theta\xi_c^2$ with $|\theta| \le 1$. Collecting,

$$\int \varpi_0 f^2 = \frac{J^2}{2} + f_-J - J^2\xi_c + J^2 I_2 + \Big[f_-^2 + 2f_-J\big(-\xi_c + \theta\xi_c^2 + I_1\big) + J^2\theta\xi_c^2\Big].$$

By Lemma 10.19, $|\xi_c| \lesssim 1/J$; by Lemma 10.20 and $L_* \lesssim J^{-2}$, $I_1 = O(J^{-2})$ and $J^2 I_2 = -J^2\varpi_0(\xi_c)/\rho(\xi_c) + O(1)$. So the bracket is $O(1)$, and since $0 \le 1-\varpi_0(\xi_c) \le 2|\xi_c|$ and $\rho(\xi_c) \ge \rho_{\min} \gtrsim J$ we may replace $\varpi_0(\xi_c)$ by 1 at the cost $J^2\cdot 2|\xi_c|/\rho(\xi_c) = O(1)$:

$$\int \varpi_0 f^2 = \frac{J^2}{2} + f_-J - J^2\xi_c - \frac{J^2}{\rho(\xi_c)} + O(1).$$

Step 2: the two remaining terms. $|\rho(\xi_c) - w_0| \le L_\rho|\xi_c| = O(1)$ and $w_0 = J/2 + O(1)$ (Theorem 10.11), so $\rho(\xi_c) = \frac{J}{2}(1+O(1/J))$ and $J^2/\rho(\xi_c) = 2J + O(1)$. By Lemma 10.19, $\xi_c = \ln\vartheta/w_0 + O(J^{-2}) = 2\ln\vartheta/J + O(J^{-2})$, so $J^2\xi_c = 2J\ln\vartheta + O(1)$. Hence

$$\begin{aligned} W(0) &= W_{\mathrm{e}}(0) + \frac18\Big[\frac{J^2}{2} + f_-J - 2J\ln\vartheta - 2J + O(1)\Big] \\ &= \frac{J^2}{16} + J\Big(\frac{f_-}{8} - \frac{\ln\vartheta}{4} - \frac14\Big) + O(1) = \frac{J^2}{16} + JW_1 + O(1), \end{aligned}$$

the second assertion of the theorem. (The coefficient W_1 and the bookkeeping were checked by computer algebra.)

Step 3: WKB. By (B2) and Lemma 10.5(d), $|W(0) - J^2/16| \le J^2/32$, i.e. $x := 16(W(0)-J^2/16)/J^2 = 16W_1/J + O(J^{-2})$ satisfies $|x| \le \frac12$; $\sqrt{1+x} = 1 + \frac{x}{2} + O(x^2)$ for $|x| \le \frac12$ gives

$$\sqrt{W(0)} = \frac{J}{4}\sqrt{1+x} = \frac{J}{4} + 2W_1 + O(1/J).$$

By (B1) the hypothesis of Lemma 10.18 holds ($|a| \le C_{\mathrm{bd}}/(2B) \le B/2$), so $w_0 = 2\sqrt{W(0)} + O(1/J) = \frac{J}{2} + 4W_1 + O(1/J) = \frac{J}{2} + b_0 + O(1/J)$: this is the first assertion, $|w_0 - (J/2 + b_0)| \le C_B/J$.

Step 4: invert. Write $w_0 = \frac{J}{2} + b_0 + \varepsilon$ with $|\varepsilon| \le C_B/J$. Then $\kappa_{\mathrm{D}}/\vartheta = Jw_0 = \frac12 J^2 + b_0 J + \varepsilon J$, i.e.

$$(J+b_0)^2 = \frac{2\kappa_{\mathrm{D}}}{\vartheta} + b_0^2 - 2\varepsilon J = \frac{2\kappa_{\mathrm{D}}}{\vartheta} + O(1),$$

and $J + b_0 \geq J/2 > 0$ by (B3). Hence $J + b_0 = \sqrt{2\kappa_{\mathrm{D}}/\vartheta}\,\sqrt{1 + O(1/\kappa_{\mathrm{D}})} = \sqrt{2\kappa_{\mathrm{D}}/\vartheta} + O(\kappa_{\mathrm{D}}^{-1/2})$, i.e. $J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta} - b_0 + O(\kappa_{\mathrm{D}}^{-1/2})$ with $-b_0 = 1 + \ln\vartheta - f_-/2$. By Lemma 10.3(c), $\kappa_{\mathrm{D}} \geq \kappa_B$ forces $J \geq J_B$, which makes Steps 1–3 available for every fixed point. □

Remark 10.23 (bookkeeping of the three $O(J)$ terms)**.** In $W(0) = \frac{J^2}{16} + J\big(\frac{f_-}{8} - \frac{\ln\vartheta}{4} - \frac{1}{4}\big) + O(1)$ the contributions are: $\frac{f_- J}{8}$ from a non-zero left state (the cross term $2f_- J\hat{q}$ of the square f^2); $-\frac{1}{4}\ln\vartheta\, J$ from the displacement $\xi_c \approx 2\ln\vartheta/J$ of the step (it costs $\frac{1}{8}J^2\xi_c$ in the mass $\hat{q}$ seen from the origin); $-\frac{J}{4}$ from the smoothing of the step, i.e. from $\int(\lg^2 - \mathbf{1}_{\{s>0\}})\mathrm{d}s = -1$ divided by the local rate $\rho \approx J/2$. The corresponding contributions to $J - \sqrt{2\kappa_{\mathrm{D}}/\vartheta}$ are $-f_-/2$, $+\ln\vartheta$ and $+1$. Numerically $16W(0)/J^2 \to 1$ with the predicted relative lag $4W_1/J$: for the rest state $(W(0) - J^2/16)/J = -0.197, -0.244, -0.248$ at $\kappa_{\mathrm{D}} = 10^3, 10^5, 10^6$, against $W_1 = -\frac{1}{4}$.

Remark 10.24 (what changes in the class-Z regime, and sources)**.** For $f_- < 0$ the profile crosses zero (class Z when $f_- + J > 0$), and the term $\frac{f_- J}{8} < 0$ lowers $W(0)$; the theorem is nevertheless valid for all $f_- \in \mathbb{R}$ once (J1)–(J3) hold. The statements Theorem 10.11 and Theorem 10.22 prove the $\sqrt{\kappa}$ law and strengthen it by the next term; the method is a four-step bootstrap (steepness, step potential, WKB, exact identity), with every inequality proved for an arbitrary fixed point.

Numerical observation 10.25 (the $O(1/J)$ remainder of Theorem 10.22)**.** *The theorem asserts $w_0 = J/2 + b_0 + O(1/J)$. The computed quantity $J\,(w_0 - J/2 - b_0)$ at $\kappa_{\mathrm{D}} = 2 \cdot 10^5$ is*

data $(\boldsymbol{c}', \vartheta, f_-)$	J	$J(w_0 - J/2 - b_0)$
$((0,0,0), 4, \frac{1}{2})$	318.3	7.80
$((-0.8, 0.5, 0), 1, 0)$	633.5	3.74
$((3,3,0), 1, 0)$	633.4	16.4
$((0,0,-3), 1, 0)$	633.5	−1.70

so the remainder is of order $10/J$ with a constant that depends on the data $\boldsymbol{c}'$ (through W_{e} and its derivatives, which enter F_0 of Lemma 10.18) and has either sign. The first-order quantities W_1 and b_0 do not depend on $\boldsymbol{c}'$: at the rest state $[W(0) - J^2/16]/J = -0.197, -0.244, -0.248$ for $\kappa_{\mathrm{D}} = 10^3, 10^5, 10^6$, against $W_1 = -\frac{1}{4}$. Status: numerical observation; `verify_ch10a.py`*, Parts 3 and 5.*

Remark 10.26 (how to read the proof of Theorem 10.22 after the first reading)**.** The proof has the same shape as that of Theorem 10.11, one order higher. At order J the potential W is the *step* $J^2/16$ with all lower-order information discarded; at order 1 one keeps (i) the left state f_-, (ii) the $O(1/J)$ displacement of the step, and (iii) the smoothing of the step into a logistic. None of these three contributions requires information that is not already in Lemma 10.5: (i) is the cross term in f^2, (ii) is Lemma 10.19, and (iii) is Lemma 10.20. The exact identity $\kappa_{\mathrm{D}} = \vartheta J w_0$ is used twice: once to convert w_0 into κ_{D} in (10.1), and once, in the form $\Sigma = \ln\vartheta + \cdots$, to locate the layer. The reader who wants to check the theorem independently can compare the output of `verify_ch10a.py`, Part 5, which evaluates each of the three terms numerically for the fixed points of Example 10.13.

Table 10.1 records the dependence of the results of this half of the chapter.

result	input from earlier chapters	used later for
Lemma 10.1	Lemma 8.2 (Ch. 8): definition and smoothing of $\mathcal{Q}$	every bound on W; step potential
Lemma 10.3	Lemma 8.10, Theorem 8.42 (Ch. 8): $\kappa_{\mathrm{D}} = \vartheta J w_0$	$\kappa_{\mathrm{D}} \leftrightarrow (J, w_0)$ at both orders
Lemma 10.5	Lemma 3.10, Lemma 3.14 (Ch. 3): barrier, tracking, reflection	Theorem 10.11, Lemma 10.18
Theorem 10.11	Lemma 10.5, Lemma 10.10	Corollary 10.14; uniqueness (second half)
Lemma 10.18–Lemma 10.20	second-order tracking (Ch. 3)	Theorem 10.22
Theorem 10.22	the three lemmas above	the layer theorem and Lemma 10.33 (second half)

TABLE 10.1. Dependence of the results of this half of the chapter.

Hand-over to the second half of the chapter. The step profile (10.3) and the location $\xi_c = 2\ln\vartheta/J + O(J^{-2})$ of Lemma 10.19 are parts (a),(b) of Theorem 10.28; the tracking Lemma 10.5(e), the bound $L_\rho = 4AE_w$ and the two lemmas Lemma 10.18, Lemma 10.20 are the inputs of the layer profile (Theorem 10.28(c),(d)) and of the window bounds and rank-one structure (Lemma 10.33); the orders of magnitude of Lemma 10.10 are used for uniqueness (Theorem 10.39) and for class Z (Corollary 10.43).

10.5. The layer

In words. *Section 10.4 ended with the facts that matter: $\rho \gtrsim J$ near the equator, the centre $\xi_c = 2\ln\vartheta/J + O(J^{-2})$ of the layer, the step profile (10.3) and the tracking $\zeta_\mp \approx \mp\varpi$. Here we read off the* shape *of the layer. In the stretched variable $t = J\xi/2 - \ln\vartheta$ the swirl is the logistic function, $(f - f_-)/J = \lg(t) + O(1/J)$; the polar velocity is a hyperbolic tangent, $U_\theta \approx \frac{J}{2}\tanh\frac{t}{2}$, and the radial velocity is a jet $u_r \approx \frac{J^2}{8}\operatorname{sech}^2\frac{t}{2}$. So the flow consists of two nearly uniform hemispheres in which fluid moves towards the equator, a thin conical shear layer of width $O(1/J)$ across which the swirl jumps by J, and a radial jet of the same width carrying away what the hemispheres feed in.*

Toy example 10.27 (the layer of a constant potential)**.** Let $W \equiv \omega^2$ be constant ($\omega > 0$ large). Then $u_\mp = e^{\pm\omega\xi}$, $\zeta_\mp = \pm\omega$, $\rho = 2\omega$, $\Sigma = 2\omega\xi$, $z = u_- + \vartheta u_+$, and the logistic variable is $\phi = u_-/z = \lg(2\omega\xi - \ln\vartheta)$. The swirl is $f = f_- + J\lg(t)$ with $t = 2\omega\xi - \ln\vartheta$, and for $h = z'/z = \zeta_+ + \rho\phi = \omega(2\phi - 1) = \omega\tanh\frac{t}{2}$ one finds $U = 2(h - \tanh\xi) \approx 2\omega\tanh\frac{t}{2}$ and, from $h' = W - h^2$,

$$W - h^2 = \omega^2\operatorname{sech}^2\tfrac{t}{2},$$

$$u_r = U_x = \cosh^2\xi\, U' = 2\cosh^2\xi\,(W - h^2) - 2 \approx 2\omega^2\operatorname{sech}^2\tfrac{t}{2}$$

(computer-algebra check: `verify_ch10b.py`, Part 1). With $\omega = J/4$ this is exactly the picture claimed above: $t = J\xi/2 - \ln\vartheta$, $U \approx \frac{J}{2}\tanh\frac{t}{2}$, $u_r \approx \frac{J^2}{8}\operatorname{sech}^2\frac{t}{2}$. The theorem below says that the *true* potential, which is not constant but only equal to $J^2/16$ at the equator up to $O(J)$, produces the same layer to the orders stated.

THEOREM 10.28 (the layer). *Notation ($\kappa = \kappa_{\mathrm{D}}$): $\rho = \zeta_- - \zeta_+$ the Riccati gap, $\Sigma(\xi) = \int_0^\xi \rho$, E_w the error in $\zeta_\mp \approx \mp\sqrt{W}$ of the $\sqrt{\kappa}$ theorem, $\Delta = C_A|\xi| + 2AE_w\xi^2$, A the bound $\sqrt{W} \le A$ there (so $\rho \le 2A$), u_r the radial velocity. Under the hypotheses of the $\sqrt{\kappa_{\mathrm{D}}}$ theorem ($c' \in \operatorname{int}\mathcal{J}$, $\vartheta > 0$, $\kappa_{\mathrm{D}} \ge \kappa_*$): (a) $\rho \ge \rho_{\min}$ on $[-\frac12,\frac12]$, the layer centre ξ_c defined by $\Sigma(\xi_c) = \ln\vartheta$ satisfies $|\xi_c| \le |\ln\vartheta|/\rho_{\min}$ and $\xi_c = 2\ln\vartheta/J + O(J^{-2})$; (b) $|f(\xi) - f_- - J\mathbf{1}_{\xi>\xi_c}| \le Je^{-\rho_{\min}\min(|\xi-\xi_c|,3/8)}$; (c) for $|\xi| \le \frac14$, $t = J\xi/2 - \ln\vartheta$: $|(f-f_-)/J - \lg(t)| \le \Delta e^{\Delta}e^{-|t|}$; (d) $|\zeta_\mp \mp \sqrt{W}| \le E_w$ on $[-\frac14,\frac14]$ and $U_\theta = \frac{J}{2}\tanh\frac{t}{2} + O_T(1)$, $u_r = \frac{J^2}{8}\operatorname{sech}^2\frac{t}{2} + O_T(J)$ for $|t| \le T$; here $\lg(s) = (1+e^{-s})^{-1}$.*

Reading of the statement. The data and the fixed point are those of Theorem 10.11: f is *any* global solution of type G at both poles with effective data $(\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$, $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$; the letters A, E_w, C_A, $\rho_{\min}$ are those of Definition 10.4 (so "A" in Δ is the sup-bound $\sup\varpi \le A$ of Lemma 10.3(a), and $L_\rho = 4AE_w$ of Lemma 10.5(e) is the Lipschitz constant of ρ used below); $U = U_\theta$ is the polar profile and $u_r = U_x = \cosh^2\xi\, U'$ the radial one (Chapter 2); $O_T(\cdot)$ has constants depending on the data $(\boldsymbol{c}', \vartheta, f_-)$ and on T only. "$\kappa_{\mathrm{D}} \ge \kappa_*$" (rather than $>$) is enough: if $\kappa_{\mathrm{D}} \ge \kappa_* = 2\vartheta J_* A(J_*)$ and $J < J_*$ then $\kappa_{\mathrm{D}} \le 2\vartheta JA(J) < \kappa_*$, since $J \mapsto JA(J)$ is strictly increasing; so $J \ge J_*$ and (J1)–(J3) hold.

PROOF. Let $J \ge J_*$ as just explained; Lemma 10.5, Lemma 10.10, Lemma 10.18, Lemma 10.19, Theorem 10.11 are available.

(a), (b). The lower bound $\rho \ge \rho_{\min}$ on $[-\frac12,\frac12]$ is Lemma 10.5(b); the bound $|\xi_c| \le |\ln\vartheta|/\rho_{\min}$ is Lemma 10.5(c) and $\xi_c = 2\ln\vartheta/J + O(J^{-2})$ is Lemma 10.19. Part (b) is (10.3) of Lemma 10.5(c).

(c). For $|\xi| \le \frac14$ we have $\Sigma(\xi) = w_0\xi + \int_0^\xi(\rho - w_0)$, and $|\rho'| \le L_\rho = 4AE_w$ on $[-\frac14,\frac14]$ (Lemma 10.5(e)), so $|\rho(\eta) - w_0| \le L_\rho|\eta|$ and

$$\left|\Sigma(\xi) - \frac{J\xi}{2}\right| \le |w_0 - \tfrac{J}{2}||\xi| + \tfrac12 L_\rho\xi^2 \le C_A|\xi| + 2AE_w\xi^2 = \Delta$$

by Theorem 10.11 ($|w_0 - J/2| \le C_A$). By (10.1), $(f-f_-)/J = \lg(s)$ with $s = \Sigma - \ln\vartheta$, and $s - t = \Sigma - J\xi/2$ has modulus $\le \Delta$. Since $\lg' = \lg(1-\lg) \le e^{-|r|}$, the mean value theorem gives $|\lg(s) - \lg(t)| \le \Delta\sup_{|r-t|\le\Delta} e^{-|r|} \le \Delta e^{\Delta}e^{-|t|}$.

(d), first assertion. This is Lemma 10.5(e).

(d), the profiles. Put $z = u_- + \vartheta u_+$ and $h = z'/z$. By Proposition 8.13, $U = 2(h - \tanh\xi)$; since $u_-/z =: \phi$ and $\vartheta u_+/z = 1 - \phi$,

$$h = \frac{\zeta_- u_- + \vartheta\zeta_+ u_+}{z} = \zeta_+ + \rho\,\phi, \qquad \phi = \frac{u_-}{z} = \lg(\Sigma - \ln\vartheta) = \lg(s)$$

(Lemma 8.10). Write $\zeta_- = \varpi + e_-$, $\zeta_+ = -\varpi + e_+$ with $|e_\pm| \le E_w$ on $[-\frac14,\frac14]$. Then

$$h = -\varpi + e_+ + (2\varpi + e_- - e_+)\phi = \varpi(2\phi - 1) + \epsilon, \qquad |\epsilon| \le 2E_w, \qquad 2\phi - 1 = \tanh\tfrac{s}{2},$$

so $U = 2\varpi\tanh\frac{s}{2} + O(E_w + 1) = 2\varpi\tanh\frac{s}{2} + O(1)$ on $[-\frac14,\frac14]$ (we used $|\tanh\xi| \le 1$ and $E_w \le C_\sharp$, Lemma 10.10). From $h' = W - h^2$ (because $z'' = Wz$) and

$U' = 2(h' - \operatorname{sech}^2\xi)$,

$$u_r = \cosh^2\xi\, U' = 2\cosh^2\xi\,(W - h^2) - 2,$$

$$W - h^2 = \varpi^2 - \varpi^2(2\phi-1)^2 - 2\varpi(2\phi-1)\epsilon - \epsilon^2 = \varpi^2 \operatorname{sech}^2\tfrac{s}{2} + O(AE_w + E_w^2),$$

using $1-(2\phi-1)^2 = 4\phi(1-\phi) = \operatorname{sech}^2\frac{s}{2}$ (computer-algebra check in Part 1 of `verify_ch10b.py`) and $\varpi \le A$. Now let $|t| \le T$. Then $\xi = 2(t + \ln\vartheta)/J$, so $|\xi| \le 2(T + |\ln\vartheta|)/J$; the following estimates hold with constants depending only on the data and on T. (i) $|\varpi(\xi) - \varpi(0)| \le C_{\mathrm{bd}}|\xi| \lesssim 1$ (Lemma 10.5(e), $C_{\mathrm{bd}} \lesssim J$) and $|\varpi(0) - \frac{J}{4}| = |W(0) - \frac{J^2}{16}|/(\varpi(0) + \frac{J}{4}) \le 4r_0/J \lesssim 1$ (Lemma 10.5(d)); hence $\varpi = \frac{J}{4} + O_T(1)$ and $\varpi^2 = \frac{J^2}{16} + O_T(J)$. (ii) $|s - t| \le \Delta(\xi) = C_A|\xi| + 2AE_w\xi^2 = O_T(1/J)$, because $C_A \lesssim 1$ and $AE_w\xi^2 \lesssim J\cdot J^{-2}$; as $\tanh\frac{\cdot}{2}$ and $\operatorname{sech}^2\frac{\cdot}{2}$ are Lipschitz, $\tanh\frac{s}{2} = \tanh\frac{t}{2} + O_T(1/J)$, $\operatorname{sech}^2\frac{s}{2} = \operatorname{sech}^2\frac{t}{2} + O_T(1/J)$. (iii) $\cosh^2\xi = 1 + O_T(J^{-2})$. Inserting: $U = 2(\frac{J}{4} + O_T(1))(\tanh\frac{t}{2} + O_T(\frac{1}{J})) + O(1) = \frac{J}{2}\tanh\frac{t}{2} + O_T(1)$, and $u_r = 2(\frac{J^2}{16} + O_T(J))(\operatorname{sech}^2\frac{t}{2} + O_T(\frac{1}{J})) + O(AE_w + E_w^2 + 1) = \frac{J^2}{8}\operatorname{sech}^2\frac{t}{2} + O_T(J)$ (here $AE_w \lesssim J$). □

Remark 10.29 (what the layer looks like in the original variables)**.** Recall $x = \cos\theta = \tanh\xi$, $u_\varphi = f/\sin\theta$, $u_\theta = U/\sin\theta$ (Chapter 2); for $|\xi| \ll 1$, $\theta = \frac{\pi}{2} - \xi + O(\xi^3)$. Theorem 10.28 says: *(i)* a thin conical shear layer around the cone $\theta_c = \frac{\pi}{2} - \frac{2\ln\vartheta}{J} + O(J^{-2})$, of angular width $O(1/J)$ (one unit of t is the angle $2/J$), across which u_φ jumps from $\approx f_-$ in the southern to $\approx f_- + J$ in the northern hemisphere; in particular the swirl is *not* spread over the sphere but concentrated at the equator when $\vartheta = 1$ and displaced by $2\ln\vartheta/J$ otherwise; *(ii)* a radial jet in the same layer, $u_r \approx \frac{J^2}{8}\operatorname{sech}^2\frac{t}{2}$; since $J^2/2 = \kappa_{\mathrm{D}}/\vartheta\,(1 + O(\kappa_{\mathrm{D}}^{-1/2}))$ its peak is $\approx \kappa_{\mathrm{D}}/(4\vartheta)$; *(iii)* away from the layer $U \approx \mp J/2$ in the northern/southern hemisphere ($U = 2(h - x)$, $h \approx \pm\varpi$): u_θ points *towards* the layer from both sides. The two pieces are linked by the exact identity $\int u_r\,\mathrm{d}x = [U]$, i.e. the total radial flux through the layer equals the total change of U across it, which by Theorem 10.28(d) is $J\tanh\frac{T}{2} + O_T(1)$ over $|t| \le T$. This is the swirling jet with a conical shear layer of the fluid-mechanics literature [36, 40, 41], here obtained for *every* solution.

Example 10.30 (the provable range of part (c))**.** For $\Delta \le 1$ we need $C_A|\xi| \le \frac{1}{2}$ and $2AE_w\xi^2 \le \frac{1}{2}$; then the error in (c) is $\le e\Delta e^{-|t|}$ and, since $\xi = 2(t + \ln\vartheta)/J$, equals $O((1+|t|)^2 e^{-|t|}/J)$. With the *provable* constants at the rest state this range is tiny: $\xi_1 = \min(\frac{1}{4}, \frac{1}{2C_A}, (4AE_w)^{-1/2})$ equals $1/1392$ at $J = J_* = 201.3$ ($C_A = 696$, $E_w = 297$) and $1/1327$ at $J = 1415$ ($C_A = 663.7$, $A = 500.4$, $E_w = 281.5$); in the variable $t = J\xi/2 - \ln\vartheta$ the provable range is $|t| \lesssim J\xi_1/2 = 0.07$ at $J = 201.3$ and 0.53 at $J = 1415$, a small fraction of one layer width, so (c) is informative only for $J \gg C_A$; for larger ξ only (b) is proved. The computed profile is much better (Figure 10.4): $J\sup_{|t|\le 6}|(f - f_-)/J - \lg(t)| = 0.71, 0.82, 0.83$ for $\kappa_{\mathrm{D}} = 10^2, 10^3, 10^4$ at the rest state (`verify_ch10b.py`, Part 2).

10.6. The window and the rank-one structure of $\mathrm{D}T_D$

In words. *Now linearise. A perturbation v of the swirl changes the potential by $N = \mathcal{Q}[fv]$, a short-range function of size $\lesssim J\|v\|$. Because $\varpi \approx J/4$ is large and slowly varying, the response of the Riccati variables is* quasi-static*: $\delta\zeta_\mp \approx N/(2\zeta_\mp)$, with an error one power of J smaller (the* window lemma*). Feeding this into the*

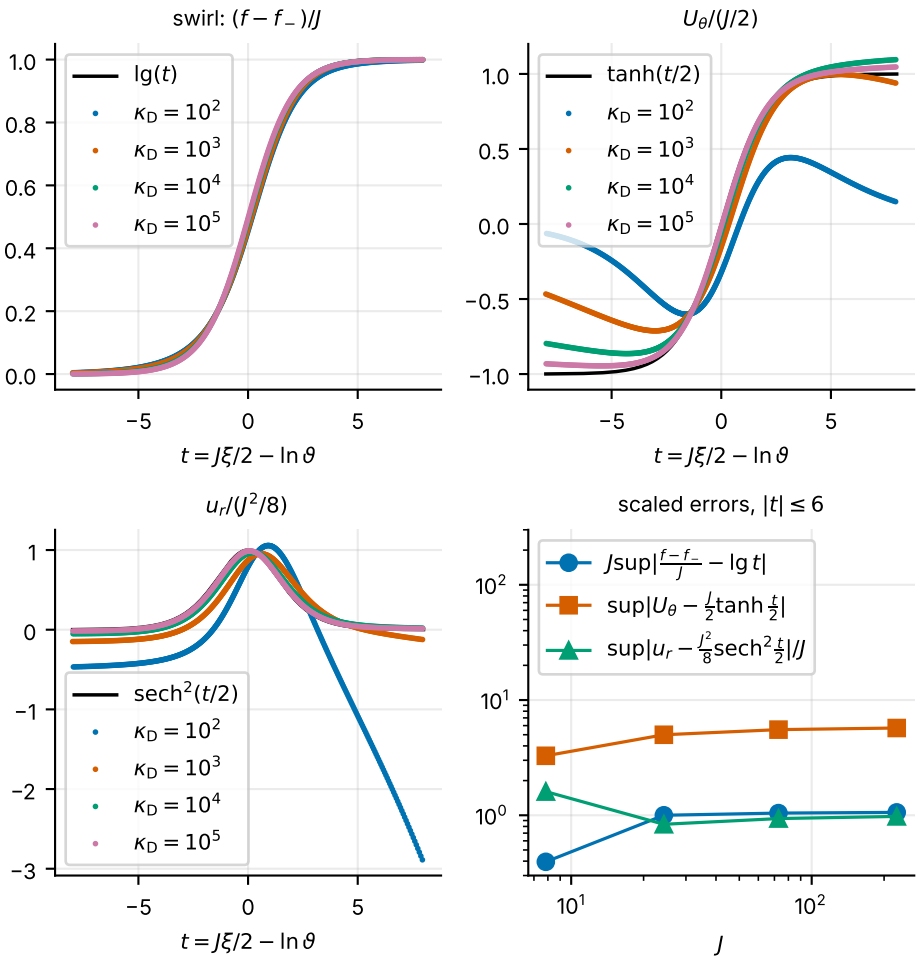


FIGURE 10.4. The layer for $\boldsymbol{c}' = 0$, $\vartheta = 4$, $f_- = \frac{1}{2}$ and $\kappa_D = 10^2, \dots, 10^5$ ($J = 7.8,\ 24.3,\ 72.8,\ 225.7$). (a)–(c) The swirl $(f - f_-)/J$, $U_\theta/(J/2)$ and $u_r/(J^2/8)$ as functions of $t = J\xi/2 - \ln\vartheta$ against $\lg(t)$, $\tanh\frac{t}{2}$, $\mathrm{sech}^2\frac{t}{2}$ (black); the collapse is visible for the swirl already at $J \approx 24$ and for U_θ, u_r for $J \gtrsim 70$, only on $|t| \lesssim T$ (the errors of Theorem 10.28(d) grow like $1 + |t|$; no uniformity in t is claimed). (d) The three errors on $|t| \le 6$, scaled by J, 1, $1/J$ respectively, stay bounded (Theorem 10.28(c),(d)). Code: `code/ch10_large/make_b_layer.py`.

formula (8.6) *for* DT_D *shows that everything that* v *can do to the swirl goes through* one number, *the change* $\delta\rho(0)$ *of the steepness at the equator, which moves the height* $J = \kappa_D/(\vartheta\rho(0))$ *and the position of the layer; the rest of the operator is* $O(1/J)$. *Thus* $\mathrm{DT}_D = \ell \otimes g + R$ *with* $\ell(g) = -1 + O(1/J)$ *and* $\|R\| = O(1/J)$*: one eigenvalue*

near -1 and all the others near 0. An eigenvalue near -1 is harmless for uniqueness, because the index criterion only fears real eigenvalues ≥ 1 (Section 10.7).

Toy example 10.31 (a rank-one operator and a small perturbation)**.** On $\mathbb{R}^2$ let $T_0 = \left(\begin{smallmatrix} -1 & -1 \\ 0 & 0 \end{smallmatrix}\right) = g \otimes \ell$ with $g = (1,0)^\top$ and $\ell(v) = -v_1 - v_2$, so $\ell(g) = -1$. Its eigenvalues are $\ell(g) = -1$ (eigenvector g) and 0 (kernel of ℓ). If R has norm $\delta \leq \frac{1}{16}$ the eigenvalues of $T_0 + R$ stay within $O(\delta)$ of -1 and 0. The point is that the *sign* of the large eigenvalue is read off from the single number $\ell(g)$ and survives every perturbation of size $\ll 1$: a rank-one structure with $\ell(g) < 0$ can never produce an eigenvalue near $+1$. Proposition 10.36 makes this precise for operators on $\mathrm{BC}(\mathbb{R})$.

Convention (uniform constants). Fix $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$ and a compact set $K \subset (0,\infty) \times \mathbb{R}$ of data (ϑ, f_-). "$X \lesssim Y$" means $X \leq CY$ with C depending only on $\boldsymbol{c}'$ and K. Put $\vartheta_{\max} = \max_K \vartheta$. For $J \geq J_*(\vartheta, f_-)$ (J1) gives $m_0 \geq J/(2(1+\vartheta))$, hence $B^2 \geq q_1 J^2/(8(1+\vartheta)^2) - C_-$, and C_- depends only on $\boldsymbol{c}'$; so, with $c_* := \sqrt{q_1}/(4(1+\vartheta_{\max}))$, $B \geq c_* J$ as soon as $J^2 \geq 16(1+\vartheta_{\max})^2 C_-/q_1$. Inspecting the proof of Lemma 10.10 one sees that, for $J \geq J_K$ with $J_K \geq \sup_K J_*$ chosen so large that this inequality and finitely many further inequalities of the form $J \geq \mathrm{const}(\boldsymbol{c}', K)$ used below hold (for instance $c_* J \geq 2C_\sharp$), all constants of Lemma 10.10 depend only on $(\boldsymbol{c}', K)$. By Lemma 10.3(c), every fixed point with $\kappa_{\mathrm{D}} > \kappa_K := \max_K 2\vartheta J_K A(J_K)$ has $J > J_K$. All fixed points below are such, with $(\vartheta, f_-) \in K$.

For a fixed point f and $v \in \mathrm{BC}(\mathbb{R})$ put $N := \mathcal{Q}[fv]$ (the first variation of $W = W_f$ in the direction v; Lemma 8.27), $\|v\| = \sup|v|$, and

$$\delta\zeta_- := A_W^- N, \qquad \delta\zeta_+ := -A_W^+ N,$$
$$\delta\rho := \delta\zeta_- - \delta\zeta_+ = \Phi_N^W, \qquad \Gamma_e := \Gamma_{\ker}(W_{\mathrm{e}}) < \infty$$

(Definition 8.22, Lemma 8.23(c)); $\delta\zeta_\mp$ are the decaying solutions of the linearised Riccati equations $\delta\zeta' = N - 2\zeta_\mp \delta\zeta$ (Lemma 8.25). By Lemma 10.1(a), $|N| \leq \frac{S}{4}\|v\| \operatorname{sech}^2 \xi$ and $|N'| \leq S\|v\|$, with $S \leq \frac{3}{2}J$. Let $I = [-\frac{1}{4}, \frac{1}{4}]$.

Toy example 10.32 (the quasi-static response)**.** Why is $\delta\zeta_\mp \approx N/(2\zeta_\mp)$? Take $\zeta \equiv c$ constant and a smooth source $N = \operatorname{sech}^2 \eta$. The decaying solution of $y' = N - 2cy$ is $y(\eta) = \int_{-\infty}^{\eta} N(r) e^{-2c(\eta - r)}\, dr$: the kernel has width $1/(2c)$, so for large c the source is seen as a constant and $y \approx N/(2c)$, with correction $-N'/(4c^2)$ from one integration by parts. Numerically (`verify_ch10b.py`, Part 8) $c|y - N/(2c)|$ equals 0.0093, 0.0004 and 0.0000 at $\eta = 0$ for $c = 5, 25, 100$; the response is slaved to the source. In the chapter $c \sim J/4$ and N varies on the scale $1/J$ (the layer width), which is why the window lemma needs the error equation below rather than this naive picture.

Lemma 10.33 (window bounds)**.** *There are constants C (depending only on $\boldsymbol{c}'$ and K) such that for every fixed point with $J > J_K$ and every $v \in \mathrm{BC}(\mathbb{R})$:*

(a) (Quasi-static response.) *For $\eta \in I$: $\delta\zeta_-(\eta) = \dfrac{N(\eta)}{2\zeta_-(\eta)} + \varepsilon_-(\eta)$ and $\delta\zeta_+(\eta) = \dfrac{N(\eta)}{2\zeta_+(\eta)} + \varepsilon_+(\eta)$, with $|\varepsilon_\pm| \leq C\|v\|/J$ and $|\delta\zeta_\pm'| \leq C\|v\|$. Hence $|\delta\rho| \leq C\|v\|$ and $|\delta\rho'| \leq C\|v\|$ on I.*

(b) (The value at the equator.) $\delta\rho(0) = \dfrac{N(0)}{\varpi(0)} + O\big(\dfrac{\|v\|}{J}\big)$, *and* $|\delta\rho(0)| \leq C\|v\|$.

(c) (Global L^1 bound.) $\int_{\mathbb{R}}|\delta\rho| \le \frac{1}{2}S\,\Gamma_e\|v\| \le C\,J\|v\|$.

Proof. *Preliminaries.* Lemma 10.5(b) gives $\zeta_- \ge B_1 \ge 0.985B \ge c\,J$ on $[-\frac12,1]$ and $\zeta_- \le A \lesssim J$ everywhere; the proof of Lemma 10.5(e) (first-order tracking from $a_0=-\frac12$) gives, for all $r\in[-\frac12,\frac12]$,

$$\begin{aligned} |\zeta_-(r)-\varpi(r)| &\le \frac{C_{\mathrm{bd}}}{B} + Ae^{-B(r+1/2)}, \\ \text{so that for } r\ge -\tfrac{7}{20}:\ |\zeta_-(r)-\varpi(r)| &\le \frac{C_{\mathrm{bd}}}{B} + Ae^{-3B/20} \lesssim 1. \end{aligned} \tag{10.4}$$

Write $E(r,\eta)=\exp(-2\int_r^\eta\zeta_-)=u_-(r)^2/u_-(\eta)^2$; on $[-\frac12,\eta]$, $E(r,\eta)\le e^{-2B_1(\eta-r)}$. By Lemma 3.23 the ratio $u_-^W/u_-^{W_{\mathrm e}}$ is nondecreasing, so $E(r,\eta)\le E_{W_{\mathrm e}}(r,\eta)$ for $r\le\eta$, whence

$$|\delta\zeta_-(\eta)| \le A^-_{W_{\mathrm e}}|N|(\eta) \le \tfrac{S}{4}\|v\|\,\Phi_-^{W_{\mathrm e}}(\eta) \le C\,J\|v\| \qquad (\eta\in\mathbb{R})$$

because $\Phi_-^{W_{\mathrm e}}$ is bounded (Lemma 8.23(b)). By the reflection $\xi\mapsto-\xi$ (Lemma 3.7), which maps $(W,N,\zeta_+,\delta\zeta_+)$ to $(\check W,\check N,-\zeta_-,\dots)$ and preserves all the hypotheses used below (the lower bound $\check W\ge B^2$ on $[-1,1]$, the bound on $\check W'$, the tracking estimate for $-\zeta_+$ on $[-\frac12,\frac12]$ and the bound $-\zeta_+\ge B_1$ on $[-1,\frac12]$), it is enough to treat $\delta\zeta_-$; for $\delta\zeta_+$ the same proof gives the stated bounds with $N/(2\zeta_+)$ (note $\zeta_+<0$).

(a). Put $m=N/(2\zeta_-)$ on $[-\frac12,\frac12]$ and $\varepsilon=\delta\zeta_- - m$. Since $\delta\zeta_-'=N-2\zeta_-\delta\zeta_-$ and $N=2\zeta_-m$,

$$\begin{aligned} \varepsilon' &= -2\zeta_-\varepsilon - m', \qquad \delta\zeta_-' = -2\zeta_-\varepsilon, \\ \varepsilon(\eta) &= \varepsilon(-\tfrac12)E(-\tfrac12,\eta) - \int_{-1/2}^{\eta} m'(r)E(r,\eta)\,\mathrm{d}r \end{aligned}$$

(the first identity is checked by computer algebra in Part 1 of `verify_ch10b.py`; the second is variation of constants; the third is $\delta\zeta_-'=N-2\zeta_-(\varepsilon+m)$). Now $m'=\dfrac{N'}{2\zeta_-}-\dfrac{N\zeta_-'}{2\zeta_-^2}$ and $\zeta_-'=W-\zeta_-^2=(\varpi-\zeta_-)(\varpi+\zeta_-)$ (Riccati equation). Hence, with $|N|\le\frac{S}{4}\|v\|$, $|N'|\le S\|v\|$, $\zeta_-\ge cJ$, $S\lesssim J$: (1) on $[-\frac12,\frac12]$ crudely $|\zeta_-'|\le A^2$ (as $0\le\zeta_-\le A$, $0\le W\le A^2$), so $|m'|\lesssim\|v\|(S/J+SJ^2/J^2)\lesssim J\|v\|$; (2) on $[-\frac{7}{20},\frac12]$, by (10.4), $|\zeta_-'|\le|\zeta_--\varpi|\cdot 2A\lesssim J$, so $|m'|\lesssim\|v\|(1+J\cdot J/J^2)\lesssim\|v\|$. Let $\eta\in I$ and split the r-integral at $r=\eta-\frac{1}{10}$ ($\ge-\frac{7}{20}$). For $r\le\eta-\frac{1}{10}$, $E\le e^{-B_1/5}e^{-B_1(\eta-r)}$, so by (1) this part is $\lesssim J\|v\|\,e^{-B_1/5}/B_1$. For $r\in[\eta-\frac{1}{10},\eta]$ use (2) and $\int_{-\infty}^{\eta}E(r,\eta)\,\mathrm{d}r\le\int_{-\infty}^{\eta}e^{-2B_1(\eta-r)}\mathrm{d}r=\frac{1}{2B_1}$ (for $r\ge-\frac12$): this part is $\lesssim\|v\|/J$. The boundary term is $\le(|\delta\zeta_-(-\frac12)|+|m(-\frac12)|)e^{-B_1/2}\lesssim J\|v\|\,e^{-cJ}$. Since $B_1\gtrsim J$, the exponentially small terms are $\lesssim\|v\|/J$: $|\varepsilon|\le C\|v\|/J$ on I, and $|\delta\zeta_-'|=2\zeta_-|\varepsilon|\le 2A\,C\|v\|/J\lesssim\|v\|$. Finally $|m|\le\frac{S\|v\|}{8B_1}\lesssim\|v\|$, so $|\delta\zeta_-|\lesssim\|v\|$, and the claims about $\delta\rho$ follow by subtraction.

(b). By (a) at $\eta=0$ and $\zeta_+<0$, $\delta\rho(0)=N(0)\big(\frac{1}{2\zeta_-}+\frac{1}{2|\zeta_+|}\big)+O(\|v\|/J)=N(0)\dfrac{\rho(0)}{2\zeta_-|\zeta_+|}+O(\|v\|/J)$. With $e_-=\zeta_--\varpi$, $e_+=\zeta_++\varpi$ ($|e_\pm|\le E_w$ at 0): $\rho=2\varpi+e_--e_+$, $\zeta_-|\zeta_+|=(\varpi+e_-)(\varpi-e_+)$ and a direct computation (checked

by computer algebra) gives

$$\frac{\rho}{2\zeta_-|\zeta_+|}-\frac{1}{\varpi}=\frac{-\varpi e_-+\varpi e_++2e_-e_+}{2\varpi\,\zeta_-|\zeta_+|},\qquad \Big|\cdot\Big|\le\frac{2\varpi E_w+2E_w^2}{2\varpi\cdot\frac14\varpi^2}\lesssim J^{-2}$$

(for $J\ge J_K$ we have $E_w\le\frac12 B\le\frac12\varpi$, so $\zeta_-,|\zeta_+|\ge\frac12\varpi$). As $|N(0)|\le\frac{S}{4}\|v\|\lesssim J\|v\|$, the replacement $\frac{\rho}{2\zeta_-|\zeta_+|}\to\frac{1}{\varpi}$ costs $O(\|v\|/J)$. The bound $|\delta\rho(0)|\lesssim\|v\|$ is (a).

(c). $|\delta\rho|\le\Phi^W_{|N|}$, and by Lemma 8.23(c) $\|\Phi^W_{|N|}\|_{L^1}=2\int|N|/\rho_W\le 2\cdot\frac{S}{4}\|v\|\,\Gamma_{\ker}(W)$ and $\Gamma_{\ker}(W)\le\Gamma_{\ker}(W_{\mathrm{e}})=\Gamma_e$ by Lemma 8.23(e). □

Toy example 10.34 (why the answer is rank one: the caricature)**.** Suppose the only dependence of the swirl on f were through the number $w_0=\rho(0)=J/2$ in $J=\kappa_{\mathrm{D}}/(\vartheta w_0)$ (the caricature of Toy example 10.9). Then T_D is the one-parameter family $f=f_-+J\lg(\Sigma-\ln\vartheta)$ with $J=\kappa_{\mathrm{D}}/(\vartheta\,w_0)$, and a perturbation changes f only through δJ and $\delta\Sigma$, i.e. along *two fixed profiles*: $\partial_J f=\phi$ and $\partial_\Sigma f=J\phi(1-\phi)$. Both are driven by the *same scalar* $\delta\rho(0)$ if ρ is steep: $\delta J=-J\,\delta\rho(0)/\rho(0)$ and $\delta\Sigma(\xi)=\int_0^\xi\delta\rho\approx\frac{\delta\rho(0)}{\rho(0)}\Sigma$. Together

$$\mathrm{D}T_D\,v\approx\ell(v)\,\phi\big(-1+(1-\phi)\Sigma\big)=\ell(v)\,g,\qquad \ell(v)=\frac{J\delta\rho(0)}{\rho(0)}.$$

In the scalar caricature $J\mapsto T(J)=2\kappa_{\mathrm{D}}/(\vartheta J)$ the derivative at the fixed point $J_0=\sqrt{2\kappa_{\mathrm{D}}/\vartheta}$ is $T'(J_0)=-2\kappa_{\mathrm{D}}/(\vartheta J_0^2)=-1$ (computer-algebra check): the caricature already has the eigenvalue -1, and Proposition 10.35(iv) says $\ell(g)=-1+O(1/J)$.

Proposition 10.35 (rank-one structure of $\mathrm{D}T_D$)**.** *For $J>J_K$ put, for $v\in\mathrm{BC}(\mathbb{R})$ ($\phi=\lg(\Sigma-\ln\vartheta)$),*

$$\ell(v):=\frac{J\,\delta\rho(0)}{\rho(0)},\qquad g:=\phi\big(-1+(1-\phi)\Sigma\big),$$
$$R:=\mathrm{D}T_D(f)-\ell\otimes g,\quad \textit{i.e.}\quad \mathrm{D}T_D(f)v=\ell(v)\,g+Rv.$$

Then, with constants depending only on $\boldsymbol{c}'$ and K:

(i) $\|\ell\|\le C$ *and* $\|g\|_\infty\le 1+e^{-1}+|\ln\vartheta|$;

(ii) $\ell(v)=\dfrac{2}{J}\displaystyle\int_{\mathbb{R}}e^{-2|\eta|}f(\eta)v(\eta)\,\mathrm{d}\eta+O\big(\frac{\|v\|}{J}\big)$;

(iii) $\|R\|_{\mathrm{BC}\to\mathrm{BC}}\le C/J$, *where* $Rv(\xi)=J\phi(1-\phi)\displaystyle\int_0^\xi\Big(\delta\rho(\eta)-\frac{\delta\rho(0)}{\rho(0)}\rho(\eta)\Big)\mathrm{d}\eta$;

(iv) $\ell(g)=-\dfrac{f_-+J}{J}+O\big(\dfrac{1}{J}\big)=-1+O\big(\dfrac{1}{J}\big)$.

Proof. By (8.6), $\mathrm{D}T_D(f)v=J\phi\big[-\frac{\delta\rho(0)}{\rho(0)}+(1-\phi)\int_0^\xi\delta\rho\big]$. Since $\Sigma=\int_0^\xi\rho$, $\ell(v)g=-J\phi\frac{\delta\rho(0)}{\rho(0)}+J\phi(1-\phi)\frac{\delta\rho(0)}{\rho(0)}\int_0^\xi\rho$; subtracting gives the formula for Rv (symbolic check, Part 1).

(ii) By Lemma 10.33(b), $\ell(v)=\frac{J}{\rho(0)}\big(\frac{N(0)}{\varpi(0)}+O(\frac{\|v\|}{J})\big)$. Now $\rho(0)=2\varpi(0)+O(E_w)$ and $\varpi(0)^2=W(0)=\frac{J^2}{16}+O(J)$, so $\rho(0)\varpi(0)=2\varpi(0)^2+O(J)=\frac{J^2}{8}\big(1+O(\frac1J)\big)$, i.e. $\frac{J}{\rho(0)\varpi(0)}=\frac{8}{J}(1+O(\frac1J))$. With $N(0)=\frac14\int e^{-2|\eta|}fv$ (definition of $\mathcal{Q}$) and $|\int e^{-2|\eta|}fv|\le S\|v\|\lesssim J\|v\|$ this is (ii). (i): (ii) gives $|\ell(v)|\le\frac{2}{J}S\|v\|+O(\|v\|/J)\lesssim\|v\|$; for g, $|\phi|\le1$ and, with $s=\Sigma-\ln\vartheta$, $\phi(1-\phi)\le e^{-|s|}$, $|\Sigma|\le|s|+|\ln\vartheta|$, so $|g|\le1+\sup_s(|s|+|\ln\vartheta|)e^{-|s|}\le1+e^{-1}+|\ln\vartheta|$.

(iii) *In the window* $|\xi| \le \frac14$: let $a = \delta\rho(0)/\rho(0)$, $H(\eta) = \delta\rho(\eta) - a\rho(\eta)$. Then $H(0) = 0$, $H' = \delta\rho' - a\rho'$ with $|\delta\rho'| \lesssim \|v\|$ (Lemma 10.33(a)), $|a| \lesssim \|v\|/J$ and $|\rho'| \le L_\rho \lesssim J$; so $|H'| \lesssim \|v\|$, $|H(\eta)| \lesssim |\eta|\|v\|$ and $|\int_0^\xi H| \lesssim \xi^2\|v\|$. Moreover $\phi(1-\phi) \le e^{-|s|}$ with $|s| \ge |\Sigma| - |\ln\vartheta| \ge \rho_{\min}|\xi| - |\ln\vartheta|$ (Lemma 10.5(b)), so

$$|Rv(\xi)| \lesssim J\xi^2 e^{|\ln\vartheta|}e^{-\rho_{\min}|\xi|}\|v\| \le e^{|\ln\vartheta|}\frac{4J}{e^2\rho_{\min}^2}\|v\| \lesssim \frac{\|v\|}{J},$$

using $\max_{\xi\ge0}\xi^2e^{-q\xi} = 4/(eq)^2$ and $\rho_{\min} \gtrsim J$. *Outside the window*, $|\xi| > \frac14$: $|\Sigma| \ge \rho_{\min}/4$, and since $|\ln\vartheta| < \rho_{\min}/16$ (proof of Lemma 10.5(c)), $|s| \ge \frac{3}{16}\rho_{\min}$. By Lemma 10.33(c) $|\int_0^\xi \delta\rho| \lesssim J\|v\|$ and $|a\Sigma| \lesssim \|v\||\Sigma|/J$, so $|Rv| \lesssim J^2(1+|\Sigma|)e^{-|s|}\|v\| \le J^2(1+|\ln\vartheta|+|s|)e^{-|s|}\|v\| \lesssim J^3e^{-3\rho_{\min}/16}\|v\| \lesssim \|v\|/J$ (the function $(c+y)e^{-y}$ decreases for $y \ge 1-c$, and $c \ge 1$).

(iv) By (ii), $\ell(g) = \frac{2}{J}\int e^{-2|\eta|}fg + O(\frac1J)$ (as $\|g\| \lesssim 1$). Put $\mathbf{1} = \mathbf{1}_{\{\eta>\xi_c\}}$, $f_s = f_- + J\mathbf{1}$ and $g_s = -\mathbf{1}$, so that $f_sg_s = -(f_-+J)\mathbf{1} = -f_+\mathbf{1}$ because $\mathbf{1}^2 = \mathbf{1}$. Then $fg + f_+\mathbf{1} = (f-f_s)g + f_s(g-g_s)$, and with $s = \Sigma - \ln\vartheta$: $|f-f_s| \le Je^{-|s|}$ (as in Lemma 10.5(c)), $|g-g_s| \le |\phi - \mathbf{1}_{s>0}| + \phi(1-\phi)|\Sigma| \le e^{-|s|}(1+|s|+|\ln\vartheta|)$ and $|f_s| \le S$. Thus $|fg + f_+\mathbf{1}| \le (J\|g\| + S)(2+|\ln\vartheta|+|s|)e^{-|s|}$. Integrate against $e^{-2|\eta|} \le 1$: on $\Omega = \{|\eta-\xi_c| \le \frac{3}{16}\}$ use $\mathrm{d}\eta = \mathrm{d}s/\rho \le \mathrm{d}s/\rho_{\min}$, which gives $\lesssim (J+S)/\rho_{\min} \lesssim 1$; outside Ω, $|s| \ge \frac{3}{16}\rho_{\min}$ and the integrand is exponentially small as in (iii). Since $\int e^{-2|\eta|}\mathbf{1} = \frac12 - \xi_c + O(\xi_c^2) = \frac12 + O(\frac1J)$ (Lemma 10.19),

$$\int e^{-2|\eta|}fg = -\frac{f_+}{2} + O(1), \qquad \ell(g) = -\frac{f_+}{J} + O\Big(\frac1J\Big) = -1 + O\Big(\frac1J\Big),$$

because $f_+ = f_- + J$. □

Proposition 10.36 (localisation of the spectrum)**.** *Let* $\delta := \|R\|$ *and* $M := \|\ell\|\,\|g\|$ *(both bounded as in Proposition 10.35,* $\delta \lesssim 1/J$*). If* $J > J_K$ *is so large that* $(10+8M)\,\delta \le \frac{1}{16}$ *and* $\ell(g) \le -\frac12$*, then*

$$\operatorname{spec}\mathrm{D}T_D(f) \;\subset\; \{|\nu| \le (2+8M)\delta\} \;\cup\; \{\lambda_*\}, \qquad \lambda_* \in \mathbb{R}, \quad |\lambda_* - \ell(g)| \le 8M\delta,$$

so $\lambda_* = -1 + O(1/J)$ *and every other point of the spectrum has modulus* $O(1/J)$*. In particular* $\operatorname{spec}\mathrm{D}T_D(f) \cap [\frac12, \infty) = \emptyset$ *and* $1 \notin \operatorname{spec}\mathrm{D}T_D(f)$.

Proof. We work on the complexification of $\mathrm{BC}(\mathbb{R})$; the operator $T = \mathrm{D}T_D(f) = g\otimes\ell + R$ ($v \mapsto \ell(v)g + Rv$) is compact and real, so $\operatorname{spec}T\setminus\{0\}$ consists of eigenvalues (Theorem A.33) and is symmetric under complex conjugation.

Step 1: reduction to a scalar equation. Let $|\nu| > \delta$. Then $\nu - R$ is invertible (Neumann series, Lemma A.1) and $\nu - T = (\nu-R)\big(I - a_\nu\otimes\ell\big)$ with $a_\nu := (\nu-R)^{-1}g \ne 0$. The operator $I - a\otimes\ell$ is invertible iff $\ell(a) \ne 1$ (its inverse is $I + a\otimes\ell/(1-\ell(a))$; if $\ell(a) = 1$ then a is in the kernel). Hence for $|\nu| > \delta$:

$$\nu \in \operatorname{spec}T \iff \Phi(\nu) = 1, \qquad \Phi(\nu) := \ell\big((\nu-R)^{-1}g\big) = \sum_{k\ge0}\frac{\ell(R^kg)}{\nu^{k+1}}.$$

For $|\nu| \ge 2\delta$, $|\Phi(\nu) - \ell(g)/\nu| \le M\sum_{k\ge1}\delta^k|\nu|^{-k-1} = \frac{M\delta}{|\nu|(|\nu|-\delta)} \le \frac{2M\delta}{|\nu|^2}$. Put $e(\nu) := \nu\Phi(\nu) - \ell(g)$, so $|e(\nu)| \le 2M\delta/|\nu|$ and an eigenvalue ν with $|\nu| \ge 2\delta$ satisfies $\nu = \ell(g) + e(\nu)$.

Step 2: where the eigenvalues are. Let $\nu \in \operatorname{spec}T$, $|\nu| \ge 2\delta$. If $|\nu| < \frac14$ then $|\nu - \ell(g)| \ge |\ell(g)| - |\nu| \ge \frac14$, while $|\nu - \ell(g)| = |e(\nu)| \le 2M\delta/|\nu|$, so $|\nu| \le 8M\delta$. If

$|\nu| \geq \frac{1}{4}$ then $|\nu - \ell(g)| \leq 8M\delta$. Thus every eigenvalue lies in $\{|\nu| \leq \max(2, 8M)\delta\}$ or in $D := \{|\nu - \ell(g)| \leq 8M\delta\}$.

Step 3: exactly one eigenvalue in D. On $D' := \{|\nu - \ell(g)| \leq 32M\delta\}$ we have $|\nu| \geq \frac{1}{2} - 32M\delta \geq \frac{1}{4}$, so $\Psi(\nu) := \ell(g) + e(\nu)$ is holomorphic there with $|\Psi - \ell(g)| \leq 8M\delta$; by Cauchy's estimate on the disc of radius $16M\delta$ around any point of $\{|\nu - \ell(g)| \leq 16M\delta\}$ (contained in D'), $|\Psi'| \leq 8M\delta/(16M\delta) = \frac{1}{2}$. So Ψ maps $\{|\nu - \ell(g)| \leq 16M\delta\}$ into itself and is a contraction there: it has exactly one fixed point λ_* in this disc, and eigenvalues in D are fixed points of Ψ. Hence $\operatorname{spec} T \cap D = \{\lambda_*\}$ (λ_* is an eigenvalue because $\Phi(\lambda_*) = \lambda_*^{-1}\Psi(\lambda_*) = 1$). Conjugation maps $\operatorname{spec} T \cap D$ to itself ($\ell(g)$ is real), so $\lambda_* = \bar{\lambda}_*$. The last claims follow from $\ell(g) = -1 + O(1/J)$ and $\delta \lesssim 1/J$. □

Remark 10.37 (how the proposition sharpens the numerical picture)**.** (i) The proposition says that $\mathrm{D}T_D(f)$ has, for large J, a *single* real eigenvalue $\lambda_* = -1 + O(1/J)$ and an $O(1/J)$ cluster; numerically the cluster is much smaller: the moduli $|\nu_k|$, $k \geq 2$, are 10^{-4}–10^{-5} at $J = 142$ (Figure 10.5, right; `verify_ch10b.py`, Part 4). (ii) The sign of $1 + \lambda_*$ depends on the data: at the rest state $1 + \nu_1 > 0$ and $J(1 + \nu_1) = 1.68, 1.52, 1.81, 1.93$ at $J = 14.8, 45.6, 142, 448$ (it stays between 1.5 and 2 on this range), but for $\vartheta = \frac{1}{4}$ (family C of Atlas 10.48) $\nu_1 = -1.0345, -1.0101, -1.0029, -1.0009$ at $J = 28, 89, 283, 894$: the eigenvalue approaches -1 *from below*. So the eigenvalue may approach -1 from either side, depending on the data; the theorem only needs $\lambda_* < 0$, i.e. $\ell(g) \approx -1$. (iii) Hypotheses on $\boldsymbol{c}'$: $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$ enters through W_{e} admissible (the constants of Definition 10.4) and $\Gamma_e < \infty$ only.

10.7. Uniqueness and class Z at large swirl

In words. *The index criterion of Chapter 8 (Proposition 8.48) says: if no fixed point of T_D has a real eigenvalue ≥ 1 of $\mathrm{D}T_D$, there is exactly one fixed point. At large swirl Proposition 10.36 verifies the hypothesis at* every *fixed point: the spectrum is one eigenvalue near -1 and a cluster near 0. Hence for large κ_{D} the solution with given effective data is unique and nondegenerate, whatever the sign of f_-; in particular in class* Z *the multiplicity found at moderate swirl disappears. In class* Z *one can say more, because the zero of the swirl sits at the foot of the layer and the lens parameter at the zero is $\asymp J \asymp \sqrt{\kappa_{\mathrm{D}}}$.*

Toy example 10.38 (the index in one variable, and why -1 is harmless)**.** Let $T : \mathbb{R} \to \mathbb{R}$ be continuous with $T(x) > x$ for $x \ll 0$ and $T(x) < x$ for $x \gg 0$. A fixed point x_* with $T'(x_*) < 1$ is a crossing of the graph of T through the diagonal from above to below (index $\operatorname{sgn}(1 - T') = +1$); one with $T'(x_*) > 1$ crosses from below (index -1); the indices add up to $+1$ (this is the degree, Theorem 8.42). If *every* fixed point has $T' < 1$ there is exactly one: two crossings of the first kind would need a crossing of the second kind between them. In the caricature $T(J) = 2\kappa_{\mathrm{D}}/(\vartheta J)$ of Toy example 10.34 one has $T'(J_0) = -1 < 1$, so the unique fixed point $J_0 = \sqrt{2\kappa_{\mathrm{D}}/\vartheta}$ is of index $+1$; a slope like -1 is as far from the dangerous value $+1$ as a slope can be. Complex eigenvalues of an operator never matter, and neither does a real eigenvalue < 1 (Toy example 8.47). Proposition 10.36 verifies exactly this for $\mathrm{D}T_D$.

THEOREM 10.39 (uniqueness for large swirl). *Let $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$ and let $K \subset (0,\infty)\times\mathbb{R}$ be compact. There is $\kappa_{**} = \kappa_{**}(\boldsymbol{c}', K) < \infty$ such that for $\kappa_{\mathrm{D}} \geq \kappa_{**}$ and $(\vartheta, f_-) \in K$:*

(a) *T_D has exactly one fixed point, i.e. there is exactly one global solution of type G at both poles with effective data $D = (\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$. (This includes class Z, $f_- < 0$.)*

(b) *It is nondegenerate in the strong sense $\operatorname{spec}\mathrm{D}T_D(f) \cap [\frac{1}{2}, \infty) = \emptyset$: $\mathrm{D}T_D(f) = \ell \otimes g + R$ with $\ell(g) = -1 + O(1/J)$, $\|R\| = O(1/J)$, one real eigenvalue $\lambda_* = -1 + O(1/J)$ and all other eigenvalues of modulus $O(1/J)$.*

(c) *With $\boldsymbol{c}'$ fixed, $(\vartheta, f_-, \kappa_{\mathrm{D}}) \mapsto f$ is real-analytic on $\operatorname{int} K \times (\kappa_{**}, \infty)$, with values in $\mathrm{BC}(\mathbb{R})$.*

PROOF. Let $J_{**} \geq J_K$ be so large that the hypotheses of Proposition 10.36 (including $\ell(g) \leq -\frac{1}{2}$, which follows from $|\ell(g)+1| \leq C/J$) hold for all $J \geq J_{**}$ and all $(\vartheta, f_-) \in K$ (possible because $\delta = \|R\| \leq C/J$, $M \leq C$ and $|\ell(g)+1| \leq C/J$ with $C = C(\boldsymbol{c}', K)$, Proposition 10.35), and put $\kappa_{**} := 1 + \max_{(\vartheta,f_-)\in K} 2\vartheta J_{**} A(J_{**})$ with $A = A(J;\vartheta,f_-)$ of Definition 10.4. Let $\kappa_{\mathrm{D}} \geq \kappa_{**}$ and let f be any fixed point of T_D (one exists by Theorem 8.16). By Lemma 10.3(c) its jump satisfies $J > J_{**}$, so Propositions 10.35 and 10.36 apply to f: $\operatorname{spec}\mathrm{D}T_D(f) \cap [\frac{1}{2}, \infty) = \emptyset$. This holds for *every* fixed point. Since $c' \in \operatorname{int}\mathcal{J}$ the data are covered by Proposition 8.48(b), which gives $\#\mathrm{Fix} = 1$; this is (a), and (b) is Proposition 10.36. For (c): $1 \notin \operatorname{spec}\mathrm{D}T_D(f)$, so Proposition 9.30 gives, near every $D_0 = (\boldsymbol{c}', \vartheta_0, f_{-,0}, \kappa_0)$ in the stated set, a real-analytic family $F(D)$ of fixed points, and by (a) $F(D)$ is *the* fixed point; so the unique fixed point is real-analytic in D. □

Corollary 10.40 (no folds at large swirl)**.** *For $(\vartheta, f_-) \in K$ and $\kappa_{\mathrm{D}} \geq \kappa_{**}$ the datum D is a regular value of the solution-to-data map E of Chapter 9 and carries exactly one solution. Consequently, for fixed $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$ and compact K, the set of κ_{D} for which $(\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$ carries more than one solution, or a fold, is bounded: it lies in $(0, \kappa_{**})$.*

PROOF. A fold is a solution with $1 \in \operatorname{spec}\mathrm{D}T_D$ (Theorem 9.22); none exists for $\kappa_{\mathrm{D}} \geq \kappa_{**}$ by Theorem 10.39(b), and uniqueness is Theorem 10.39(a). □

Remark 10.41 (what is and is not covered)**.** (i) κ_{**} is *not* explicit: J_{**} is where the $O(1/J)$ terms of Propositions 10.35 and 10.36 become small, and the constants $\delta = \|R\|$ come from Lemma 10.33. The numerical $\|R\|$ at the rest state is 0.032 and $7\cdot10^{-4}$ at $J = 14.8$ and 142 (Example 10.47; $J^2\|R\|$ between 7 and 14 there), and the multistart of Atlas 10.48(5) finds a single fixed point already at $\kappa_{\mathrm{D}} = 10^2$ ($J \approx 15$), while the proof needs $J \gtrsim 10^3$ or more. (ii) The proof needs $J \gtrsim (1+\vartheta)|f_-|$ (J1) and $|\ln\vartheta| \lesssim J$ (J3); both are real features and not only artefacts: a class-Z solution may sit on a plateau with $J \approx (1+\vartheta)|f_-|$ (Atlas 10.48(6)), and the regime $|\ln\vartheta| \sim J$ contains the ridge of Atlas 10.48(7), where the leading eigenvalue of $\mathrm{D}T_D$ comes within $1.02 \cdot 10^{-2}$ of 1. Both are outside Theorem 10.39 and are listed as Open problem 13.21 in Chapter 13. (iii) For $\boldsymbol{c}' \notin \operatorname{int}\mathcal{J}$ the potential W_{e} is not admissible, T_D is defined only on an open subset of $\mathrm{BC}(\mathbb{R})$ (Theorem 8.20), and the argument above is not claimed (Open problem 13.20).

Class Z at large swirl.

Toy example 10.42 (the zero of a logistic swirl and its slope)**.** Take $f_- < 0$ and $f = f_- + J\lg(t)$ with $J > |f_-|$. Its zero t_0 solves $\lg(t_0) = |f_-|/J$, i.e. $t_0 = -\ln(J/|f_-|-1)$: for large J the zero lies a distance $\ln(J/|f_-|)$ in t *below* the centre of the layer, in the left tail, where $f' = J\lg(1-\lg)\,\mathrm{d}t/\mathrm{d}\xi = J\lg(t_0)(1-\lg(t_0))\rho \approx |f_-|\rho$. With $\rho \approx J/2$ the slope at the zero is $\approx \frac{1}{2}|f_-|J \sim \sqrt{\kappa_{\mathrm{D}}}$: it *grows* with the swirl, while the lens of Theorem 5.62 bounds the slope by $\kappa_{\max}$ at the Cauchy data. The corollary makes this precise.

Corollary 10.43 (class Z at large swirl)**.** *Let $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$ and let $K \subset (0,\infty)\times(-\infty,0)$ be compact. There is $\kappa_{\mathsf{Z}} \geq \kappa_{**}$ (depending on $\boldsymbol{c}', K$) such that for $\kappa_{\mathrm{D}} \geq \kappa_{\mathsf{Z}}$ and $(\vartheta, f_-) \in K$ the unique solution f of Theorem 10.39 is of class* Z*, has a unique zero ξ_0 of f, and with J, ξ_c, ρ, $\rho_{\min}$ as above:*

(a) $\xi_0 < \xi_c$, $\xi_c - \xi_0 \leq \ln(J/|f_-|)/\rho_{\min}$ *and* $\xi_0 = \dfrac{2}{J}\big(\ln\vartheta - \ln\dfrac{J}{|f_-|}\big) + O\Big(\dfrac{\ln^2 J}{J^2}\Big)$; *in particular* $\xi_0 \to 0$.

(b) *The slope* $\varkappa := f'(\xi_0)$ (*normalisation* $z(\xi_0) = 1$ *of the swirl constant, Lemma 2.15(2)*) *equals exactly*

$$\varkappa = |f_-|\Big(1 - \frac{|f_-|}{J}\Big)\rho(\xi_0) = \frac{|f_-|}{2}\,J + O(\ln J), \qquad J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta} + O(1),$$

so $\varkappa \asymp \sqrt{\kappa_{\mathrm{D}}}$.

(c) *Let $(x_0, \tilde{\boldsymbol{c}}, \kappa, \gamma) \in \mathcal{L}$ be the point of the lens of Theorem 5.62 that corresponds to f ($x_0 = \tanh\xi_0$, $\tilde{\boldsymbol{c}} \in \operatorname{int}\mathcal{J}$ the Taylor constants of Q at x_0, $\kappa = \kappa_x = \cosh^2\xi_0\,\varkappa$). Then*

$$\kappa_{\max}(x_0, \tilde{\boldsymbol{c}}) \;\geq\; \kappa \;\geq\; \varkappa \;=\; \tfrac{1}{2}|f_-|J\,(1+o(1)),$$

$$\textit{i.e.} \qquad J \leq \frac{2+o(1)}{|f_-|}\,\kappa_{\max}(x_0, \tilde{\boldsymbol{c}}) \quad (\kappa_{\mathrm{D}} \to \infty).$$

*So the class-*Z *solutions of large swirl have J bounded by a constant times the size of the lens at their own zero; equivalently, along the large-swirl branch the lens at the zero grows like $\sqrt{\kappa_{\mathrm{D}}}$.*

(d) *The set of κ_{D} at which $(\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$ carries more than one solution of class* Z *(the S-shaped branch of the class-*Z *data below) is bounded, for $(\vartheta, f_-) \in K$.*

Proof. Let $\kappa_{\mathrm{D}} \geq \kappa_{**}$, so $J \geq J_K$, (J1)–(J3) hold and Theorem 10.39 gives the solution. By (J1), $J \geq 2(1+\vartheta)|f_-|$, so $f(\xi_c) = f_- + J\lg(0) = f_- + \frac{J}{2} \geq (1+\vartheta)|f_-| - |f_-| > 0$ (Lemma 10.5(c) and (10.1)); since f is increasing with $f(-\infty) = f_- < 0$, f has exactly one zero ξ_0, and $\xi_0 < \xi_c$. So f changes sign: class Z (Proposition 8.13, Definition 5.3).

(a) Apply (10.3) at $\eta = \xi_0 < \xi_c$: $|f_-| = |f(\xi_0) - f_-| \leq Je^{-\rho_{\min}\min(\xi_c - \xi_0, 3/8)}$. Since $\rho_{\min} \geq 1.97c_*J$ and $|f_-| \geq \delta_K > 0$ on K, the choice κ_{Z} large makes $Je^{-3\rho_{\min}/8} < \delta_K$, so the minimum is $\xi_c - \xi_0$ and $\xi_c - \xi_0 \leq \ln(J/|f_-|)/\rho_{\min}$, which is $\lesssim \ln J/J$ and in particular $\xi_0 \in [-\frac{1}{4}, \frac{1}{4}]$ for J large. Next, $\lg(\Sigma(\xi_0) - \ln\vartheta) = |f_-|/J$, i.e. $\Sigma(\xi_0) = \ln\vartheta - \ln\frac{J-|f_-|}{|f_-|} = \ln\vartheta - \ln\frac{J}{|f_-|} + O(\frac{1}{J})$ (toy). As in the proof of Lemma 10.19, $|\Sigma(\xi_0) - w_0\xi_0| \leq L_\rho\xi_0^2/2 \lesssim \ln^2 J/J$ and $|1/w_0 - 2/J| \lesssim J^{-2}$; dividing by $w_0 \asymp J$ gives the expansion.

(b) $f = f_- + J\phi$ with $\phi = \lg(\Sigma - \ln\vartheta)$, so $f' = J\phi(1-\phi)\rho$, and $\phi(\xi_0) = |f_-|/J$: this gives the exact formula. By the bound $|\rho'| \leq L_\rho$, $|\rho(\xi_0) - \rho(0)| \leq L_\rho|\xi_0| \lesssim \ln J$

and $\rho(0) = w_0 = J/2 + O(1)$ (Theorem 10.11), hence $\rho(\xi_0) = \frac{J}{2} + O(\ln J)$ and $\varkappa = |f_-|(1-\frac{|f_-|}{J})(\frac{J}{2}+O(\ln J)) = \frac{|f_-|}{2}J+O(\ln J)$. The relation $J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta}+O(1)$ is Theorem 10.11.

(c) By Theorem 5.62 the solution corresponds to a point of $\mathcal{L}$, whose constraint is $\kappa \le \kappa_{\max}(x_0, \tilde{\boldsymbol{c}})$; and by Lemma 2.15(2), $\kappa = \kappa_x = \cosh^2\xi_0\,\varkappa \ge \varkappa$ (here $\varkappa = (1-x_0^2)\kappa_x$). Insert (b).

(d) By Corollary 10.40 the number of solutions is 1 for $\kappa_{\mathrm{D}} \ge \kappa_{**}$; the solutions of class Z are among them. □

Remark 10.44 (reading of the corollary)**.** (i) Part (c) is a statement about the *pairing* of two pictures of the same solution: the effective data $(\boldsymbol{c}', \vartheta, f_-, \kappa_{\mathrm{D}})$ have $\kappa_{\mathrm{D}} \to \infty$ with everything else fixed, while the Cauchy constants $\tilde{\boldsymbol{c}}$ at the zero are forced to grow so that $\kappa_{\max}(x_0, \tilde{\boldsymbol{c}})$, which tends to 0 at $\partial\mathcal{J}$ and increases in each $\tilde{c}_i$ (Theorem 5.62), exceeds $\frac{1}{2}|f_-|J$. (ii) The corollary does *not* describe the S-shaped branch at moderate swirl, nor the plateau $J \approx (1+\vartheta)|f_-|$ that precedes the $\sqrt{\kappa_{\mathrm{D}}}$ law (Atlas 10.48(6)); these are Open problem 13.21.

10.8. The numerical atlas of the large-swirl regime

In words. *Everything in this section is numerical evidence, separated from the proofs: it shows how sharp the theorems of the chapter are, which regimes they do not cover and what the numerics say there. The picture is simple. The law* $J = \sqrt{2\kappa_{\mathrm{D}}/\vartheta} + 1 + \ln\vartheta - f_-/2$ *is accurate to a few percent from* $J \approx 15$*; the layer profile collapses on the logistic function;* $\mathrm{D}T_D$ *has one eigenvalue near* -1 *and numerically negligible others; class* Z *has a plateau before the law takes over; and for* $\ln\vartheta$ *comparable to* J *there is a near-fold, a ridge in* κ_{D}*, where the leading eigenvalue reaches* 0.99*.*

Numerical observation 10.45 (the eigenvector is the rank-one direction, and the iteration converges fast)**.** *At the rest state* ($\boldsymbol{c}' = 0$, $\vartheta = 1$, $f_- = 0$) *the eigenvector of* $\mathrm{D}T_D$ *for the eigenvalue* $\nu_1 \approx -1$*, normalised in* sup*, differs from the normalised* g *of Proposition 10.35 by* $\sup|v - g| = 0.0138,\ 0.0002,\ < 10^{-4}$ *at* $\kappa_{\mathrm{D}} = 10^2, 10^3, 10^4$ *(angle cosine* 0.99994*,* 1.00000*,* 1.00000*), while* $\nu_1 = -0.88689, -0.96672, -0.98730$*. Hence the damped iteration* $f \mapsto \frac{1}{2}(f + T_D f)$ *has multiplier* $|(1+\nu_1)/2| = 0.0566,\ 0.0166,\ 0.0064,\ 0.0022$ *at* $\kappa_{\mathrm{D}} = 10^2, \dots, 10^5$ *and needs* 36–37 *iterations to the tolerance* 10^{-11} *at every* $\kappa_{\mathrm{D}} = 10^2, \dots, 10^5$ *(the remaining work is the transient of the* $O(1/J)$ *cluster).* (Source: `verify_ch10b.py`, Part 8.) *This is evidence for, not part of, Proposition 10.36.*

Numerical observation 10.46 (the outer profile)**.** *Compare the computed* W *with the step profile* $\frac{J^2}{16}q_{\mathrm{step}}$ *of the caricature:* $\sup_{\mathbb{R}}|W - \frac{J^2}{16}q_{\mathrm{step}}|/J = 0.14,\ 0.21,\ 0.24,\ 0.25$ *at* $\kappa_{\mathrm{D}} = 10^2, \dots, 10^5$ *for the rest state, and* 0.21, 0.43, 0.50, 0.52 *for* $\vartheta = 4$, $f_- = \frac{1}{2}$*. The ratio is bounded and settles, as the* $O(J)$ *error claimed in the text predicts (Part 8). For* $\vartheta = 4$, $f_- = \frac{1}{2}$, $\kappa_{\mathrm{D}} = 10^4$ *(*$J = 72.775$*) the layer centre is at* $\xi_c = 0.03959$*, i.e. at colatitude* $87.73°$*; one unit of* t *is* $1.575°$ *of latitude.*

Example 10.47 (the rest state at large κ_{D})**.** For $\boldsymbol{c}' = 0$, $\vartheta = 1$, $f_- = 0$ the quantities of Theorem 10.11, Theorem 10.22, Theorem 10.28 and Proposition 10.35 are:

κ_{D}	J	$J-\sqrt{2\kappa_{\mathrm{D}}}$	$J\sup_{\lvert t\rvert\le 6}\lvert\frac{f}{J}-\lg(t)\rvert$	$\lVert R\rVert$	$\ell(g)$	ν_1
10^2	14.8437	0.7015	0.713	0.0316	-0.8892	-0.8869
10^3	45.6419	0.9206	0.823	0.0052	-0.9668	-0.9667
10^4	142.3949	0.9735	0.834	0.0007	-0.9873	-0.9873

The columns converge to the predicted values: $J-\sqrt{2\kappa_{\mathrm{D}}}\to 1$ (Theorem 10.22), the layer error is $O(1/J)$ (Theorem 10.28(c)), $\lVert R\rVert\to 0$ (proved: $O(J^{-1})$; observed: $J^2\lVert R\rVert$ between 7 and 14 for $J=15,\dots,142$, still rising), $\ell(g)\to -1$ and $\nu_1-\ell(g)\to 0$ (Proposition 10.36); $w_0-J/2=-0.685,-0.911,-0.970$ tends to $b_0=-1$. Status: numerical observation; `verify_ch10b.py`, Part 4.

Atlas 10.48 (large swirl: law, layer, spectrum, class Z and the ridge)**.** *All numbers are produced by* `code/ch10_large/verify_ch10b.py` *(and the figures by* `make_b_*.py`*); the solver and scan scripts are in* `code/ch10_large/` *(Appendix E).* Method. *Effective-coordinate map T_D on a non-uniform grid of $[-12,12]$ (spacing $\ge 0.1/J$ at the layer, $\le 2/J$ away from it, so that the explicit RK4 steps of the stable Riccati sweeps are stable), $\mathcal{Q}[f^2]$ by the exact two-sided recursion for piecewise linear f^2, T_Df from (10.1); damped Picard iteration $f\mapsto f+\frac12(T_Df-f)$ until $\sup\lvert T_Df-f\rvert<10^{-11}$ (more iterations are needed near the ridge); spectra by Arnoldi applied to the analytic derivative (8.6) (matrix-free, same grid) or, for $\lVert R\rVert$, by assembling the grid matrix. Only* stable *fixed points are found by damped iteration; the multistart of items (5) and uniqueness checks use ≥ 40 initial profiles $f_-+J_0\lg(J_0(\xi-\xi_0)/2)$ and are not exhaustive.* Validation. *(i) $J=0.49806,\ 4.01393,\ 14.84365,\ 45.64194,\ 448.20504$ at $\kappa_{\mathrm{D}}=1,10,10^2,10^3,10^5$ at the rest state; (ii) at $\kappa_{\mathrm{D}}=10^4$ halving the finest spacing and enlarging the box from 12 to 16 changes J by a relative $5\cdot10^{-8}$ (residual $4\cdot10^{-13}$); (iii) the computer-algebra identities of Part 1; (iv) the two class-*Z *solutions of item (6).*

(1) The law $J^2\vartheta/\kappa_{\mathrm{D}}\approx 2+4(1+\ln\vartheta)/J$ *(the origin of the $\sqrt{\kappa}$ law). In the J-formulation one fixes J, iterates $f\mapsto f_-+J\lg(\Sigma_f-\ln\vartheta)$ and reads off $\kappa_{\mathrm{D}}=J\vartheta\rho(0)$ ((10.1)); the ratio $J^2\vartheta/\kappa_{\mathrm{D}}$ for $f_-=0$ is*

data	$J=100$	*predicted*	$J=400$	*predicted*
$\boldsymbol{c}'=0$, $\vartheta=1$	*2.039*	*2.040*	*2.010*	*2.010*
$\vartheta=4$	*2.097*	*2.095*	*2.024*	*2.024*
$\vartheta=\frac14$	*1.986*	*1.985*	*1.996*	*1.996*
$\vartheta=16$	*2.160*	*2.151*	*2.038*	*2.038*
$\vartheta=10^{-3}$	*1.826*	*1.764*	*1.946*	*1.941*
$\boldsymbol{c}'=(-0.8,0.5,0)$, $\vartheta=1$	*2.039*	*2.040*	*2.010*	*2.010*
$\boldsymbol{c}'=(3,3,0)$, $\vartheta=1$	*2.034*	*2.040*	*2.010*	*2.010*

in agreement with the first-order prediction of Theorem 10.22: $J^2\vartheta/\kappa_{\mathrm{D}}=2/(1+2b_0/J+\dots)\approx 2-4b_0/J$ with $b_0=-1-\ln\vartheta$. The row $\vartheta=10^{-3}$ ($\lvert\ln\vartheta\rvert=6.9$, $J=100$) is the first sign of the regime $\lvert\ln\vartheta\rvert\sim J$ in which (J3) fails.

(2) The next-order law *for $f_-\neq 0$ and general $\boldsymbol{c}'$: see Numerical observation 10.16, Numerical observation 10.25, Figure 10.3; independent of $\boldsymbol{c}'$ at this order.*

(3) Layer profile *(Theorem 10.28):* $J\sup_{\lvert t\rvert\le 6}\lvert(f-f_-)/J-\lg(t)\rvert=$ 0.71, 0.82, 0.83, 0.84, 0.84 *at $\kappa_{\mathrm{D}}=10^2,\dots,10^6$ for the rest state, and* 0.40, 1.00, 1.05, 1.06, 1.065 *for $\vartheta=4$, $f_-=\frac12$;* $\sup_{\lvert t\rvert\le 6}\lvert U_\theta-\frac{J}{2}\tanh\frac{t}{2}\rvert$

grows slowly to 6.1 *(rest) and* 5.8 *($\vartheta=4$) and* $\sup|u_r-\frac{J^2}{8}\operatorname{sech}^2\frac{t}{2}|/J\to 0.88$ *and* 0.99 *respectively: $O(1)$ and $O(J)$ as in (d), Figure 10.4. The maximum of $|\zeta_\mp\mp\varpi|$ on $[-\frac14,\frac14]$ is* 0.27–0.67 *for $\kappa_{\mathrm D}\ge 10^2$ against the proved $E_w\approx 300$ of Example 10.12. For $\vartheta=4$, $f_-=\frac12$, $\kappa_{\mathrm D}=10^4$: the jet peak is $u_r=641.7$ against $\kappa_{\mathrm D}/(4\vartheta)=625$, and the layer centre is $\xi_c=0.0396$ against $2\ln\vartheta/J=0.0381$.*

(4) Window and rank one *(Lemma 10.33, Proposition 10.35). For the rest state and $\kappa_{\mathrm D}=10^3,\dots,10^6$ and four directions v (f, $\cos 3\xi$, a sign-changing function, 1): $J\sup_I|\varepsilon_\pm|/\|v\|\in[0.37,0.74]$ (bounded; proved $O(1)$), $\sup_I|\delta\zeta'_\pm|/\|v\|\le 0.26$, and $J|\delta\rho(0)-N(0)/\varpi(0)|/\|v\|\le 0.11$, decreasing to* 0.005 *at $\kappa_{\mathrm D}=10^6$ (faster than the proved $O(1)$). The assembled matrix gives $J^2\|R\|=10.8,12.3,13.7$ at $J=45.6,78.4,142.4$ and $J(1+\ell(g))=1.5,1.7,1.8$ at the rest state (same J); for $\vartheta=2$, $f_-=\frac12$: $\|R\|=0.0137,0.0053,0.0018$ at $J=32.9,56.1,101.4$.*

(5) Spectrum of $\mathrm DT_D$ *(Proposition 10.36, Figure 10.5). The families are $\boldsymbol{c}'=0$ with $(\vartheta,f_-)=A(1,0)$, $B(4,0)$, $C(\frac14,0)$, $D(1,1)$, $E(1,-1)$, and F: $\boldsymbol{c}'=(-0.8,0.5,0)$, G: $\boldsymbol{c}'=(3,3,0)$ with $(\vartheta,f_-)=(1,0)$:*

family	$\kappa_{\mathrm D}=10^2$	10^3	10^4	10^5
A	−0.8869	−0.9667	−0.9873	−0.9957
B	−0.2906	−0.8238	−0.9388	−0.9794
C	−1.0345	−1.0101	−1.0029	−1.0009
D	−0.8268	−0.9449	−0.9803	−0.9935
E	−0.9388	−0.9881	−0.9943	−0.9979
F	−0.9257	−0.9681	−0.9874	−0.9957
G	−0.6499	−0.9419	−0.9849	−0.9954

(ν_1 at the row's $\kappa_{\mathrm D}$.) In all cases the dominant eigenvalue is real and negative, $J(1+\nu_1)$ stays bounded, and the largest positive *real eigenvalue is $\le 3.3\cdot10^{-3}$ at $\kappa_{\mathrm D}=10^2$ and $\le 2.6\cdot10^{-7}$ at 10^5 (never near 1). Multistart (at least* 40 *starts) finds exactly one stable fixed point at $\kappa_{\mathrm D}=10^2,\dots,10^5$ for four data sets, including class* Z *($\boldsymbol{c}'=(3,3,0)$, $\vartheta=2$, $f_-=-1$): consistent with Theorem 10.39.*

(6) Class Z *(Corollary 10.43; data: $\boldsymbol{c}'=(-0.15,0.09,\bar c_3+0.04)$, $\vartheta=4.49$, $f_-=-2.64$). Stable fixed points found by multistart:*

$\kappa_{\mathrm D}$	*solutions J ($f(0)$)*	$\sqrt{2\kappa_{\mathrm D}/\vartheta}+1+\ln\vartheta-f_-/2$
7.77	2.338 (−2.214), 13.669 (−0.150)	*5.682*
9.39	3.208 (−2.056), 13.847 (−0.118)	*5.867*
10	3.647 (−1.976), 13.880 (−0.112)	*5.932*
12	13.929 (−0.103)	*6.134*
100	13.892 (−0.110)	*10.496*
300	17.227 (0.498)	*15.382*
10^3	25.807 (2.061)	*24.927*
10^4	70.769 (10.25)	*70.563*
10^6	671.248 (119.6)	*671.230*

Two stable solutions are found at $\kappa_{\mathrm D}=7.77,9.39,10$ and one at $\kappa_{\mathrm D}\ge 12$; its J stays pinned *near* 14 *($\approx(1+\vartheta)|f_-|=14.49$, where $f(0)=0$) up to $\kappa_{\mathrm D}\approx 100$ and then joins the $\sqrt{\kappa_{\mathrm D}}$ law, which it matches to* 0.21 *at $\kappa_{\mathrm D}=10^4$*

and to 0.02 *at* $\kappa_{\mathrm{D}} = 10^6$. *The condition (J1),* $J \geq 2(1+\vartheta)|f_-| = 28.99$, *is therefore a genuine feature of class* Z. *The exact formula of Corollary 10.43(b):* $\varkappa = |f_-|(1-|f_-|/J)\rho(\xi_0)$ *agrees with the numerical slope to* $5\cdot 10^{-4}$ *relative or better, and* $\varkappa/(\frac{1}{2}|f_-|J) = 0.807, 0.926, 0.973$ *at* $\kappa_{\mathrm{D}} = 10^4, 10^5, 10^6$ *(tending to* 1 *like* $\ln J/J$*);* $\xi_0 = -0.0572, -0.0282, -0.0122$ *against the prediction* $\frac{2}{J}(\ln\vartheta + \ln\frac{|f_-|}{J}) = -0.0505, -0.0270, -0.0120$.

(7) The ridge *(large* ϑ*;* $\boldsymbol{c}' = (1.288, -0.905, \bar{c}_3 + 0.11)$*,* $\vartheta = 28.5$*,* $f_- = 0$*). Here* $\ln\vartheta = 3.35$ *is not small against* $J \approx 20$–30. *A fine scan in* κ_{D} *(Figure 10.6) gives*

κ_{D}	*700*	*760*	*765*	*768*	*770*	*775*	*850*
J	*32.68*	*29.01*	*27.73*	*25.22*	*22.85*	*21.38*	*17.68*
$\nu_{\max}$	*0.558*	*0.881*	*0.939*	*0.9898*	*0.956*	*0.887*	*0.339*

and $\nu_{\max} < 10^{-4}$ *for* $\kappa_{\mathrm{D}} \geq 10^4$. *The jump* J *drops from* 29.0 *to* 21.4 *over* $\kappa_{\mathrm{D}} \in [760, 775]$ *while the largest real eigenvalue of* $\mathrm{D}T_D$ *peaks at* 0.9898 *(the ridge is extremely sensitive to* κ_{D}*): this is a near-fold (the margin to the fold value* 1 *is about* 1% *on this scan), no fold and a unique solution were found. For larger* κ_{D} *the law is* $J - \sqrt{2\kappa_{\mathrm{D}}/\vartheta} = 8.13,\ 4.92,\ 4.47$ *at* $\kappa_{\mathrm{D}} = 10^3, 10^4, 10^5$*, to be compared with* $1 + \ln\vartheta = 4.35$ *(Figure 10.6, right): the next-order constant is reached only slowly, since the correction is of relative size* $\ln\vartheta/J$.

Table 10.2 records the dependence of the results of the second half of the chapter.

result	input from earlier chapters	used for
Theorem 10.28	Lemma 10.5, Lemma 10.19, Theorem 10.11	profiles; Proposition 10.35(iv); Corollary 10.43
Lemma 10.33	Lemma 8.23, Lemma 8.25, Lemma 3.23, tracking	Proposition 10.35
Proposition 10.35, Proposition 10.36	Lemma 8.27 (formula (8.6)), Lemma 10.33	Theorem 10.39
Theorem 10.39	Proposition 8.48, Proposition 9.30, Lemma 10.3(c)	Corollary 10.40, Corollary 10.43(d), Theorem 11.58
Corollary 10.43	Theorem 5.62, Lemma 2.15, Theorem 10.28	Chapter 13 (Open problem 13.21)

TABLE 10.2. Dependence of the results of the second half of the chapter.

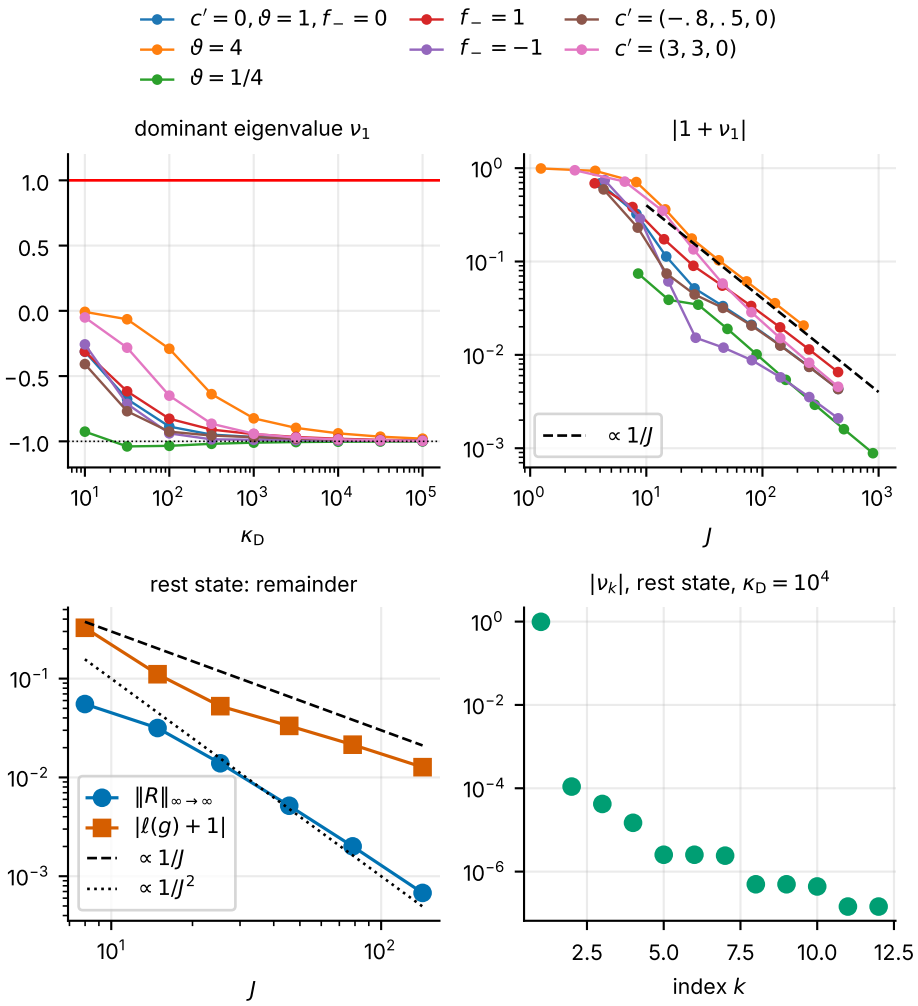


FIGURE 10.5. Spectrum of DT_D at large swirl. (a) The dominant eigenvalue ν_1 against κ_D for the seven families of Atlas 10.48 (red line: the critical value $+1$; dotted: -1). (b) $|1 + \nu_1|$ against J with a line $\propto 1/J$ (the dip is the sign change of $1 + \nu_1$ for family C). (c) At the rest state, from the assembled grid matrix: $\|R\|_{\infty\to\infty}$ and $|\ell(g) + 1|$ against J; $J^2\|R\|$ rises from 7 to 14 over $J = 15, \ldots, 142$, so on the computed range $\|R\|$ decays with an exponent between -1 and -2, consistent with the proved J^{-1} (Proposition 10.35(iii)). (d) Moduli of the twelve largest eigenvalues at $\kappa_D = 10^4$: one eigenvalue -0.987 and the rest below 10^{-3}. Code: `code/ch10_large/make_b_rank.py`.

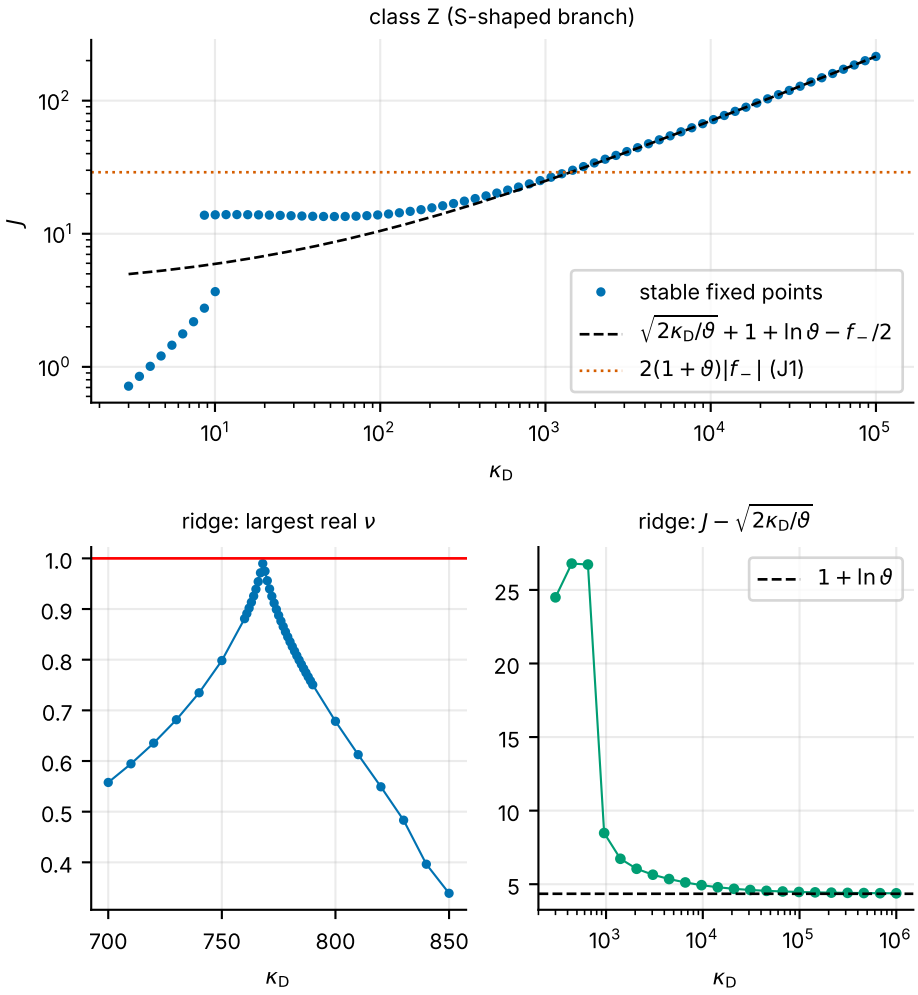


FIGURE 10.6. (a) Class Z (data as in the list above): all stable fixed points found by multistart against κ_D, the law $\sqrt{2\kappa_D/\vartheta} + 1 + \ln\vartheta - f_-/2$ (dashed) and the threshold $2(1+\vartheta)|f_-|$ of (J1) (dotted); the two lower branches at small κ_D are the S-curve. (b) The ridge, $\vartheta = 28.5$: leading real eigenvalue of DT_D (red: $+1$). (c) The same data, $J - \sqrt{2\kappa_D/\vartheta}$ against κ_D and the limit $1 + \ln\vartheta$. Code: `code/ch10_large/make_b_atlas.py`.

CHAPTER 11

Small swirl: the critical bubble and the laws near the no-swirl window

In words. *Below the no-swirl set ($c_3' < \bar{c}_3$) the swirl is* needed*: without it the potential W_{e} has no positive solution on the whole line, and one needs a large enough swirl to repair this. The cheapest repair is a constant swirl $f \equiv M$, which lifts c_3' exactly to the critical value $\bar{c}_3$; the potential then is the* critical *one, whose positive solution g decays at both poles. What happens when the swirl constant κ tends to 0? Since $f' = \kappa z^{-2}$ and z is bounded below on every compact set, the swirl cannot vary in the bulk: it sits on a plateau $f \approx M$. The plateau must then be critical, because $w_0 = \kappa/(\vartheta J) \to 0$ says exactly that the potential is critical. Near the poles z is exponentially large, so z^{-2} is exponentially small, and the swirl has to climb from f_- up to M and from M up to f_+ in two thin layers, at distance $\frac{1}{2k_i}\ln\frac{1}{\kappa}$ from the equator: a* critical bubble *with two logistic pole layers. This chapter proves this picture (Theorem 11.6, Theorem 11.10), derives from one exact identity (a Lyapunov–Schmidt condition) the rate at which the total jump J approaches its limit $J_* = (1+\vartheta)(M - f_-)$, and describes by formal computations what happens when the pole layers are not negligible (Section 11.4). The second half of the chapter studies the behaviour at the no-swirl edge and the combined regimes.*

Roadmap. Section 11.1 fixes the setting (hypothesis (H)) and proves the *critical bubble theorem* (Theorem 11.6): for every sequence of solutions with $\kappa \to 0$ one has $J \to J_*$, $f \to M$ locally uniformly, $W \to W_{\mathrm{crit}}$ and $u_\pm \to g$. The proof is a compactness argument that uses only the Jost lemmas of Chapter 3 and the a priori bounds of Chapter 8. Section 11.2 computes the two pole layers by matched asymptotics: the first part of Theorem 11.10. Section 11.3 proves the exact solvability identity $\int \delta\, g\, u_- = \gamma_{\mathrm{c}}\phi_+(0)w_0$ and from it the rate $|J - J_*| \le C\varepsilon_k(\kappa)$ and, for $\max k_i < 1$, the limit $(J - J_*)/\kappa \to \Gamma_{\mathrm{rate}}$ (second part of Theorem 11.10). Section 11.4 contains the *formal* (non-rigorous) computations for $\max k_i \ge 1$ and the formal law of the bubble eigenvalue, each checked against computations. The second half turns to the opposite end of the parameter range, data in the interior of the no-swirl set. Section 11.5 proves the cut lemma (Lemma 11.39) for the lower edge of the lens, Section 11.6 derives from it the three laws of the edge (Theorem 11.44) and Section 11.7 the exponent algebra and numerical convexity; none of these uses the bubble. Section 11.8 combines the two halves with the results of Chapters 9 and 10 into Theorem 11.58, where Theorem 11.6 enters in part (b); the final atlas (Atlas 11.62) collects the numerical observations.

The logical structure of the chapter is shown in Figure 11.1.

Conventions of the chapter. $' = \mathrm{d}/\mathrm{d}\xi$ and $x = \tanh\xi$. *The swirl constant is* κ_{D} (normalisation 4 of Table 2.1); in this chapter we write $\kappa := \kappa_{\mathrm{D}}$ (except in Sections 11.5 to 11.7, where $\kappa = \kappa_x$; see the setting stated at the beginning of

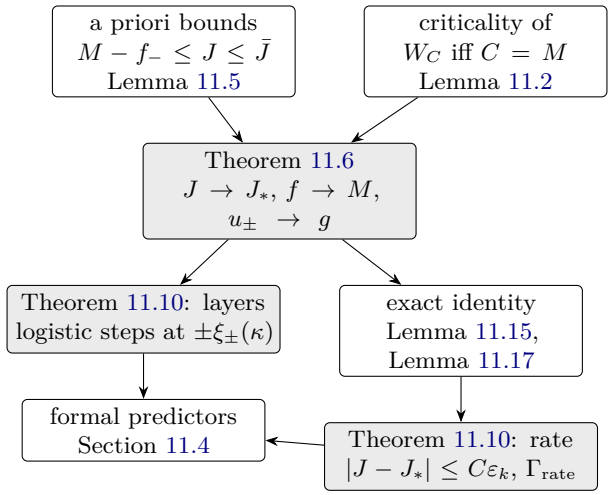


FIGURE 11.1. Logical structure of Section 11.1–Section 11.4. Every arrow applies a statement to an *arbitrary* solution; uniqueness is never assumed.

Section 11.5), and $k_i := k_i' = \sqrt{1 + c_i'}$ for the pole exponents, which are those of every potential W_f (Lemma 8.5(a)). A *solution* is a fixed point $f = T_D f$ of the map of Definition 8.11, equivalently (Proposition 8.13) a global solution of type G at both poles with effective data $D = (\boldsymbol{c}', \vartheta, f_-, \kappa)$; for such an f we write $W = W_f$, $u_\pm, \zeta_\pm$ for the Jost solutions of W normalised by $u_\pm(0) = 1$ and their logarithmic derivatives, $\rho = \zeta_- - \zeta_+ = w_0/(u_- u_+)$, $w_0 = \operatorname{Wr}(u_+, u_-) > 0$, $z = u_- + \vartheta u_+$, $R = u_-/u_+$, $\Sigma = R/(R + \vartheta)$ and $J = f(+\infty) - f_-$. We use the exact relations of Lemma 8.10:

$$f' = \kappa z^{-2}, \qquad R' = w_0 u_+^{-2}, \qquad f = f_- + J\,\Sigma, \\ J = \frac{\kappa}{\vartheta w_0}, \qquad f(0) = f_- + \frac{J}{1 + \vartheta}. \tag{11.1}$$

$\varphi_\pm$ denote the Jost solutions of W in the normalisation of Definition 3.20 ($e^{-k_1\xi}\varphi_- \to 1$ at $-\infty$, $e^{k_2\xi}\varphi_+ \to 1$ at $+\infty$), so $u_\pm = \varphi_\pm/\varphi_\pm(0)$; in the statement of Theorem 11.10, ϕ_+ is φ_+. $\lg(s) = (1 + e^{-s})^{-1}$. Uniqueness of the solution is *never assumed*: the numbers $J, w_0, \dots$ are those of an arbitrary solution, and statements about "solutions with $\kappa_n \to 0$" concern arbitrary sequences of solutions.

11.1. The critical bubble

In words. *When $\kappa \to 0$ the potential of a solution converges, along subsequences, to the potential of a constant swirl C, because the swirl cannot vary on compact sets; and since $w_0 \to 0$ this limit potential must be critical. Only the value $C = M$ makes $W_{\mathrm{e}} + \frac{1}{8}C^2 \operatorname{sech}^2$ critical, so the plateau is M, the Jost solutions both converge to the critical function g, and the total jump is $J_* = (1 + \vartheta)(M - f_-)$ because the equator value $f(0) = f_- + J/(1 + \vartheta)$ must equal M.*

Definition 11.1 (hypothesis (H), the critical objects)**.** *Hypothesis* (H): $c_1', c_2' > -1$, $c_3' < \bar{c}_3(c_1', c_2')$, $\vartheta > 0$ and $0 \le f_- < M := 2\sqrt{\bar{c}_3 - c_3'}$. Put $J_* := (1 + \vartheta)(M - f_-)$,

$\bar k = \frac12(k_1+k_2)$, $\gamma_c := 2^{\bar k}$ (so $\gamma_c^2 = 2^{k_1+k_2}$) and

$$W_{\rm crit} := W_e + \tfrac18 M^2 \operatorname{sech}^2\xi, \qquad g(\xi) := (1+x)^{k_1/2}(1-x)^{k_2/2},$$
$$\mathcal{G}_{\rm int}(\xi) = \int_0^\xi g^{-2}, \qquad \delta := W - W_{\rm crit} = \tfrac12 \mathcal{Q}[f^2 - M^2]$$

for a solution f (the second formula for δ is Lemma 8.2(a): $\mathcal{Q}[M^2] = \frac14 M^2 \operatorname{sech}^2$).

By Lemma 3.30 (with $c_3 = c_3'$, so $\varepsilon = c_3' - \bar c_3 = -M^2/4$), $W_{\rm crit} = W_{(c_1',c_2',\bar c_3)}$ is the critical potential of the explicit family, $g'' = W_{\rm crit} g$, $g(0)=1$, $g'/g = \frac{k_1-k_2}{2} - \bar k x$, $g \sim \gamma_c e^{k_1\xi}$ at $-\infty$, $g \sim \gamma_c e^{-k_2\xi}$ at $+\infty$ and $e^{-k_2\xi} \le g \le \gamma_c e^{-k_2\xi}$ ($\xi \ge 0$), $e^{k_1\xi} \le g \le \gamma_c e^{k_1\xi}$ ($\xi \le 0$).

Lemma 11.2 (constant swirl; the critical plateau)**.** *Assume* (H) *and let* $C \ge 0$. *The potential of the constant swirl* $f \equiv C$,

$$W_C := W_e + \tfrac18 C^2 \operatorname{sech}^2\xi = 1 + \tfrac12 P_{(c_1',c_2',c_3'+C^2/4)} \in \mathcal{W}(k_1,k_2),$$

is admissible iff $C > M$, *critical iff* $C = M$ *and supercritical iff* $C < M$. *Moreover* $W_M = W_{\rm crit}$ *and* $\varphi_-^{W_{\rm crit}} = \varphi_+^{W_{\rm crit}} = g/\gamma_c$.

Proof. $W_C = W_{(c_1',c_2',c_3'+C^2/4)}$ by Lemma 8.5(c), and $c_3' + \frac14 C^2 > \bar c_3$, $= \bar c_3$, $< \bar c_3$ iff $C > M$, $C = M$, $C < M$ (because $\bar c_3 = c_3' + \frac14 M^2$). The three cases are Lemma 3.35(b); there $\varphi_\pm$ are positive multiples of g in the critical case, and the normalisations $e^{-k_1\xi}\varphi_- \to 1$, $e^{k_2\xi}\varphi_+ \to 1$ together with the asymptotics of g give the factor $1/\gamma_c$. □

Toy example 11.3 (Pöschl–Teller: only $C = M$ is critical)**.** Take $\boldsymbol{c}' = (0,0,-8)$, so $k_1 = k_2 = 1$, $\bar c_3 = -4$, $M = 2\sqrt{-4+8} = 4$, $W_e = 1 - 4\operatorname{sech}^2\xi$ and $W_C = 1 - \ell(\ell+1)\operatorname{sech}^2\xi$ with $\ell(\ell+1) = 4 - \frac18 C^2$. The Pöschl–Teller potential with $\ell = 1$ is critical, with the zero mode $\operatorname{sech}\xi$ (Toy example 3.34): this happens iff $C = 4 = M$. For $C < 4$ one has $\ell > 1$ and a zero (no positive solution), for $C > 4$, $\ell < 1$ and W_C is admissible:

C	0	2	4	5
$\ell(\ell+1)$	4	3.5	2	0.875
ℓ	1.5616	1.4365	1	0.5607

(`code/ch11_small/verify_ch11a.py`). A solution with a *small* κ has a tiny w_0, i.e. a potential close to critical; this is why the plateau has to be $M = 4$.

Example 11.4 (the objects at $c' = (0,0,-8)$, $\vartheta = 1$, $f_- = 0$)**.** Here $g = \operatorname{sech}\xi$, $\gamma_c = 2$, $W_{\rm crit} = 1 - 2\operatorname{sech}^2\xi$, $M = 4$, $J_* = 8$, $A_0 = \frac14 \int \operatorname{sech}^4 = \frac13$ and $\mathcal{G}_{\rm int}(\xi) = \int_0^\xi \cosh^2 = \frac{\xi}{2} + \frac14 \sinh 2\xi$. We call this datum *set A*; other data used below are listed in Table 11.1. The potential $W_e = W_{\rm crit} - 2\operatorname{sech}^2\xi$ is supercritical, and the claim of Theorem 11.6 is that for small κ the swirl is ≈ 4 in the bulk, climbs from 0 to 4 near the south pole and from 4 to 8 near the north pole (Figure 11.2).

Lemma 11.5 (a priori bounds for $\kappa \le 1$)**.** *Assume* (H) *and let* m_*, w_*, J_1 *be the constants of Lemma 8.39 (they depend on* $\boldsymbol{c}'$, ϑ *and* f_- *only). Put* $\bar J := \max\bigl(J_1,\, 1/(\vartheta w_*)\bigr)$ *and* $S := f_- + \bar J$. *Every solution with* $0 < \kappa \le 1$ *satisfies:*

(a) f *is nondecreasing,* $f_- \le f \le f_- + J$, *and* $M - f_- \le J \le \bar J$; *in particular* $0 \le f \le S$ *and* $w_0 = \kappa/(\vartheta J) \le \kappa/(\vartheta(M - f_-))$.

set	$\boldsymbol{c}'$	ϑ	f_-	(k_1,k_2)	M	J_*
A	$(0,0,-8)$	1	0	$(1,1)$	4	8
B	$(-\frac12,-\frac12,-4)$	1	0	$(0.707,0.707)$	2.5186	5.0371
C	$(-0.6,-0.3,-5)$	2	0.3	$(0.632,0.837)$	3.1316	8.4948
D	$(1,1,-9)$	1	0	$(1.414,1.414)$	2.9473	5.8945
F	$(-\frac12,0,-5)$	1	0	$(0.707,1)$	2.7098	5.4196
G	$(-0.8,-0.8,-2)$	$\frac14$	0	$(0.447,0.447)$	1.6800	2.1000

TABLE 11.1. Data used in the examples and numerical observations of this chapter (all satisfy (H)); M and J_* computed by `verify_ch11a.py`.

(b) *$W = W_f \in \mathcal{K}(\underline{k}, B, C_S, m, x_\pm)$, the class of Lemma 3.28, with $\underline{k} = \min(k_1,k_2)$, $C_S = |c_1'-c_2'|+2|c_3'|+\frac12 S^2$, $B = (\sup W_e + \frac18 S^2)^{1/2}$, $m = \underline{k}/\sqrt{2}$ and $x_\pm \ge 0$ chosen so that $W_{\mathrm e} \ge m^2$ off $[-x_-, x_+]$; all these constants depend only on $(\boldsymbol{c}', \vartheta, f_-)$. In particular $z \ge (1+\vartheta)e^{-B|\xi|}$.*

(c) *$0 \le f(\xi) - f(0) \le \min\big(\kappa e^{2B\xi}/(2B(1+\vartheta)^2),\ S\big)$ for $\xi \ge 0$ and $0 \le f(0) - f(\xi) \le \min\big(\kappa e^{2B|\xi|}/(2B(1+\vartheta)^2),\ S\big)$ for $\xi \le 0$.*

PROOF. (a) By Lemma 8.10 (equation (11.1)) $f = f_- + J\Sigma$ with $0 < \Sigma < 1$ increasing. By Lemma 8.39 (second assertion, with $s = 0$) $J \le J_{\max} = \max(J_1, \kappa/(\vartheta w_*)) \le \bar J$ for $\kappa \le 1$. For the lower bound, $f_- \ge 0$ and f increasing give $\max(|f_-|,|f_+|) = f_+ = f_- + J$; by Proposition 8.18 (necessary condition) $c_3' + \frac14 f_+^2 \ge \bar c_3$, i.e. $f_+^2 \ge 4(\bar c_3 - c_3') = M^2$, so $f_- + J \ge M$. The bound for w_0 follows from $J = \kappa/(\vartheta w_0)$. (b) $0 \le \frac12 \mathcal{Q}[f^2] \le \frac18 S^2 \operatorname{sech}^2 \xi$ by Lemma 8.2(a), so $W_{\mathrm e} \le W \le W_{\mathrm e} + \frac18 S^2 \operatorname{sech}^2\xi \le B^2$ on $\mathbb{R}$, $W \ge W_{\mathrm e} \ge m^2$ off $[-x_-, x_+]$ (such $x_\pm$ exist because $W_{\mathrm e} \to k_i^2 \ge \underline{k}^2 > m^2$ at $\mp\infty$), and $W \in \mathcal{W}(k_1, k_2; C_S)$ by Lemma 8.5(a). The limits k_i lie in $[\underline{k}, B]$ because $B^2 \ge \sup W_{\mathrm e} \ge \max k_i^2$. Now Lemma 3.28(a) gives $u_\pm \ge e^{-B|\xi|}$, so $z = u_- + \vartheta u_+ \ge (1+\vartheta)e^{-B|\xi|}$. (c) For $\xi \ge 0$, $f(\xi) - f(0) = \kappa \int_0^\xi z^{-2} \le \kappa(1+\vartheta)^{-2} \int_0^\xi e^{2B\eta} d\eta \le \kappa e^{2B\xi}/(2B(1+\vartheta)^2)$, and $f(\xi) - f(0) \le f_+ - f_- \le S$; f is nondecreasing. The case $\xi \le 0$ is the same. □

THEOREM 11.6 (the critical bubble). *Assume* (H). *Let f_n be solutions with effective data $(\boldsymbol{c}', \vartheta, f_-, \kappa_n)$, $\kappa_n \downarrow 0$. Then*

(a) *$J_n \to J_* = (1+\vartheta)(M - f_-)$ and $f_n \to M$ locally uniformly on $\mathbb{R}$;*

(b) *$W_n \to W_{\mathrm{crit}}$ locally uniformly and $u_\pm^{(n)} \to g$ in $C^1_{\mathrm{loc}}(\mathbb{R})$; also $\gamma_{\mathrm c}\varphi_\pm^{(n)}(0) \to 1$ and $z_n/(1+\vartheta) \to g$ in C^1_{loc};*

(c) *$(U_\theta, U_\varphi) = (2(z_n'/z_n - \tanh\xi), f_n) \to (U^*, M)$ locally uniformly on $(-1,1)$, where $U^* = (1+k_1)(1-x) - (1+k_2)(1+x)$ is the critical no-swirl solution (Theorem 3.44(iv)).*

Equivalently: for every $\varepsilon > 0$ and $L > 0$ there is $\kappa_{\varepsilon,L} > 0$ such that every solution with $\kappa < \kappa_{\varepsilon,L}$ has $|J - J_*| < \varepsilon$ and $\sup_{|\xi|\le L} |f - M| < \varepsilon$.

PROOF. It suffices to show that *every subsequence has a further subsequence* along which (a)–(c) hold: if the assertions failed for the whole sequence there would be $\varepsilon > 0$, a compact set and a subsequence staying ε away from the limits, and

no sub-subsequence of it could converge to them. The same argument gives the "equivalently" formulation (negate it and extract a sequence). Let n be so large that $\kappa_n \le 1$; Lemma 11.5 applies to all f_n with constants independent of n.

Step 1: the swirl is flat on compact sets. By Lemma 11.5(a) $J_n \in [M - f_-, \bar J]$; pass to a subsequence with $J_n \to J_\infty \in [M - f_-, \bar J]$. By Lemma 11.5(c), $\sup_{|\xi|\le L} |f_n(\xi) - f_n(0)| \le \kappa_n e^{2BL}/(2B(1+\vartheta)^2) \to 0$ for every L, and $f_n(0) = f_- + J_n/(1+\vartheta) \to C := f_- + J_\infty/(1+\vartheta)$. Hence $f_n \to C$ locally uniformly, and $0 \le f_- \le C \le S$.

Step 2: the potentials converge. Let $W_C = W_{\mathrm e} + \frac12 \mathcal Q[C^2]$ be the potential of Lemma 11.2. For $|\xi| \le L$ and $\varrho > L$, by the definition of $\mathcal Q$ and $\int e^{-2|s|} ds = 1$,

$$\begin{aligned} |W_n(\xi) - W_C(\xi)| &= \tfrac12 \big|\mathcal Q[f_n^2 - C^2](\xi)\big| \\ &\le \frac18 \Big(\sup_{|\eta|\le\varrho} |f_n^2 - C^2| + S^2 \int_{|\eta|>\varrho} e^{-2|\xi-\eta|} d\eta \Big) \\ &\le \frac18 \Big(\sup_{|\eta|\le\varrho} |f_n^2 - C^2| + S^2 e^{-2(\varrho - L)} \Big), \end{aligned}$$

since $|f_n^2 - C^2| \le S^2$, $\int e^{-2|s|} ds = 1$ and, for $|\xi| \le L < \varrho$, $\int_{|\eta|>\varrho} e^{-2|\xi-\eta|} d\eta = \frac12 e^{-2(\varrho-\xi)} + \frac12 e^{-2(\varrho+\xi)} \le e^{-2(\varrho-L)}$. Let first $n \to \infty$ (Step 1) and then $\varrho \to \infty$: $W_n \to W_C$ locally uniformly. All W_n and W_C lie in $\mathcal W(k_1, k_2; C_S)$.

Step 3: the limit potential is critical, and $C = M$. Each W_n is admissible, so $\varphi_-^{W_n} > 0$: $W_n \in \mathcal P$ in the notation of Lemma 3.25(c). By that lemma $W_C \in \mathcal P$, $\varphi_\pm^{W_n} \to \varphi_\pm^{W_C}$ in C^1_{loc} (part (b)) and

$$w_0(W_C) = \lim w_0(W_n) = \lim \frac{\kappa_n}{\vartheta J_n} = 0, \qquad \text{since } J_n \ge M - f_- > 0.$$

By Lemma 3.25(c), W_C is critical, hence $C = M$ by Lemma 11.2 (here $C \ge 0$ is used). So $f_- + J_\infty/(1+\vartheta) = M$, i.e. $J_\infty = J_*$, and $f_n \to M$ locally uniformly: this is (a) along the subsequence, and $W_n \to W_M = W_{\mathrm{crit}}$ is the first half of (b).

Step 4: the Jost solutions. By Lemma 11.2, $\varphi_\pm^{W_n} \to g/\gamma_{\mathrm c}$ in C^1_{loc}. So $\varphi_\pm^{(n)}(0) \to g(0)/\gamma_{\mathrm c} = 1/\gamma_{\mathrm c} > 0$, and $u_\pm^{(n)} = \varphi_\pm^{(n)}/\varphi_\pm^{(n)}(0) \to g$ in C^1_{loc}; hence $z_n \to (1+\vartheta)g$ in C^1_{loc}, $z_n'/z_n \to g'/g$ locally uniformly ($g > 0$), and

$$U_n = 2\big(z_n'/z_n - \tanh\xi\big) \to 2\big(g'/g - \tanh\xi\big) = (k_1 - k_2) - (k_1 + k_2 + 2)x = U^*$$

(using $g'/g = \frac{k_1 - k_2}{2} - \bar k x$). Since $x = \tanh\xi$ is a homeomorphism of $\mathbb R$ onto $(-1,1)$ that maps compact sets onto compact sets, local uniform convergence in ξ is local uniform convergence on $(-1,1)$. □

Corollary 11.7. *For every sequence of solutions with $\kappa_n \to 0$: $w_0^{(n)} = \dfrac{\kappa_n}{\vartheta J_*}(1+o(1))$. Moreover $\gamma_{\mathrm c}\varphi_+(0) \to 1$, and $\sup_{|\xi|\le L} |f_n - M| \to 0$ for every L.*

Proof. $w_0 = \kappa/(\vartheta J)$ and $J_n \to J_*$ by Theorem 11.6(a); the other statements are in Theorem 11.6(b),(a). □

Remark 11.8 (what is, and is not, claimed)**.** (i) This is the theorem announced in Remark 8.55: Proposition 8.18 and Theorem 8.53(c) give only the lower bound $J \ge M - f_-$; here it is shown that every solution with small κ is close to the *bubble* $f \approx M$. Together with Theorem 8.42 (a solution exists for every $\kappa > 0$) it shows that J_* is the limit of *some* solution as $\kappa \to 0$, and of every solution. (ii) Uniqueness

is not claimed, nor is any rate: they are the subject of Section 11.3 (rate) and of Chapter 9, Chapter 10 and Theorem 11.58 (uniqueness). (iii) Hypothesis (H) is used at two places: $f_- \geq 0$ gives $\sup|f| = f_+$ and $C \geq 0$; $f_- < M$ makes $J_* > 0$ (if $f_- \geq M$ the datum is above the no-swirl set, where Theorem 8.53(a),(b) applies). For $f_- < 0$ the plateau $-M$ is also possible; this case is only discussed numerically in Numerical observation 11.34, and the limits of solutions are listed as an open question in Chapter 13.

11.2. The pole layers

In words. *By (11.1), $R = u_-/u_+$ satisfies $R - 1 = w_0 \int_0^\xi u_+^{-2}$ for $\xi \geq 0$. Near the north pole $u_+ \approx \varphi_+(0)^{-1} e^{-k_2\xi}$, so $R - 1 \approx w_0 \varphi_+(0)^2 e^{2k_2\xi}/(2k_2)$, which becomes of order 1 when $e^{2k_2\xi} \approx 1/w_0 \approx 1/\kappa$, that is at $\xi \approx \frac{1}{2k_2} \ln \frac{1}{\kappa}$. Inserting this into $f = f_- + J\,R/(R+\vartheta)$ gives a logistic step; the same happens at the south pole with $1/R$. The only inputs are the exact formulas (11.1) and the uniform Jost asymptotics, together with the convergence $u_\pm \to g$ of Theorem 11.6.*

Toy example 11.9 (a logistic layer from an exponentially small coefficient)**.** Let $z = ae^{k\xi} + be^{-k\xi}$ with $a, b > 0$ (this is the shape of $z = u_- + \vartheta u_+$ at a constant potential $W \equiv k^2$) and $f' = \kappa z^{-2}$. With $v = e^{2k\xi}$,

$$\int_\xi^\infty z^{-2}\,ds = \int_{e^{2k\xi}}^\infty \frac{dv}{2k(av+b)^2} = \frac{1}{2ka\,(ae^{2k\xi}+b)} \implies$$

$$f(+\infty) - f(\xi) = \frac{\kappa}{2kab}\ \lg\bigl(-2k(\xi-\xi_c)\bigr), \qquad e^{2k\xi_c} = \frac{b}{a}.$$

So f is a logistic step of height $\kappa/(2kab)$ and half-width $1/(2k)$, centred at $\xi_c = \frac{1}{2k}\ln\frac{b}{a}$. For $\xi \ll \xi_c$ the term $ae^{k\xi}$ is negligible, $z \approx be^{-k\xi}$ and f is constant up to $\kappa e^{2k\xi}/(2kb^2)$ (the *outer* solution); in the *inner* variable $\tau = 2k(\xi - \xi_c)$ the profile is exactly a logistic function. If $a = O(\kappa)$ then $\xi_c = \frac{1}{2k}\ln\frac{1}{\kappa} + O(1)$. For $k = 1$, $b = 1$, $a = 10^{-3}$: $\xi_c = \frac{1}{2}\ln 1000 = 3.454$. Theorem 11.10 says that this is the picture of a solution at each pole, with $a \leftrightarrow w_0/(2k_i\varphi_\pm(0)^{-1})$ playing the role of the small coefficient.

Theorem 11.10 (position and shape of the pole layers; solvability identity and rate). *Hypothesis (H): $c_1', c_2' > -1$, $c_3' < \bar{c}_3(c_1', c_2')$, $\vartheta > 0$, $0 \leq f_- < M = 2\sqrt{\bar{c}_3 - c_3'}$; $J_* = (1+\vartheta)(M - f_-)$; g the critical function, $\gamma_c^2 = 2^{k_1+k_2}$, $\delta = W - W_{\mathrm{crit}}$, ϕ_+ the recessive solution at $+\infty$, $A_0 = \frac{1}{4}\int g^2\,\mathrm{sech}^2$, $I_G = \int_{\mathbb{R}} \mathcal{G}_{\mathrm{int}}\, q_g\, d\eta$ with $\mathcal{G}_{\mathrm{int}}(\xi) = \int_0^\xi g^{-2}$ and $q_g(\eta) = \frac{1}{4}\int_{\mathbb{R}} g^2\,\mathrm{sech}^2\,\xi\, e^{-2|\xi-\eta|}d\xi$; $\kappa = \kappa_{\mathrm{D}}$. Under (H), for solutions with $\kappa_{\mathrm{D},n} \to 0$ put $\xi_+(\kappa_{\mathrm{D}}) = \frac{1}{2k_2}\ln\frac{2k_2\vartheta(1+\vartheta)J_*\gamma_c^2}{\kappa_{\mathrm{D}}}$, $\xi_-(\kappa_{\mathrm{D}}) = \frac{1}{2k_1}\ln\frac{2k_1(1+\vartheta)J_*\gamma_c^2}{\kappa_{\mathrm{D}}}$. For every T, uniformly in $|\tau| \leq T$: $f_n(\xi_+(\kappa_{\mathrm{D},n}) + \frac{\tau}{2k_2}) \to M + \vartheta(M - f_-)\lg(\tau)$, $f_n(-\xi_-(\kappa_{\mathrm{D},n}) + \frac{\tau}{2k_1}) \to f_- + (M - f_-)\lg(\tau)$, and $|f_n - M| \leq \varepsilon$ between the layers. Solvability identity and rate: (i) $\int \delta\, g\, u_- = \gamma_{\mathrm{c}}\phi_+(0)w_0$; (ii) $|J - J_*| \leq C\varepsilon_k(\kappa_{\mathrm{D}})$ with $\varepsilon_k = \kappa_{\mathrm{D}}$ $(k<1)$, $\kappa_{\mathrm{D}}\ln^2(1/\kappa_{\mathrm{D}})$ $(k=1)$, $\kappa_{\mathrm{D}}^{1/k}\ln(1/\kappa_{\mathrm{D}})$ $(k>1)$, $k = \max k_i$; (iii) for $k_1, k_2 < 1$: $(J - J_*)/\kappa_{\mathrm{D}} \to \Gamma_{\mathrm{rate}} = \frac{1+\vartheta}{MA_0}\bigl[\frac{1}{\vartheta J_*} - \frac{MI_G}{(1+\vartheta)^2}\bigr]$.*

Reading of the statement. The sequence f_n consists of arbitrary solutions with $\kappa_n = \kappa_{\mathrm{D},n} \to 0$, $k_i = \sqrt{1 + c_i'}$; "$|f_n - M| \leq \varepsilon$ between the layers" means: for every $\varepsilon > 0$ there are T and n_0 such that $|f_n - M| \leq \varepsilon$ on $[-\xi_-(\kappa_n) + T/(2k_1),\ \xi_+(\kappa_n) - T/(2k_2)]$ for $n \geq n_0$. $\phi_+ = \varphi_+$ is the Jost solution of the potential W of the solution,

$e^{k_2\xi}\phi_+ \to 1$ at $+\infty$. Part (i) holds for every solution and every $\kappa > 0$; in (ii) there are C, κ_0 (depending on the data) such that the bound holds for *every* solution with $\kappa < \kappa_0$; (iii) holds along every sequence of solutions with $\kappa_n \to 0$. The layer centres are $\pm\xi_\pm$ and the layers have half-width $1/(2k_i)$; at the north pole the swirl jumps by $\vartheta(M - f_-)$, at the south pole by $M - f_-$. The first part is proved in this section, (i)–(iii) in Section 11.3.

Proof of the layer assertions. Let f_n be solutions with $\kappa_n \to 0$; we may take $\kappa_n \le 1$. Put $w_n = w_0^{(n)} = \kappa_n/(\vartheta J_n)$, $R_n = u_-^{(n)}/u_+^{(n)}$ and $\beta_n = \varphi_+^{(n)}(0)$. By Theorem 11.6(b), $\gamma_{\mathrm c}\beta_n \to 1$ and $J_n \to J_*$. The constants $B, C_S, c_{\mathrm{lo}}, \dots$ below do not depend on n (Lemma 11.5(b) and Lemma 3.28).

Upper layer. Write $\varphi_+^{(n)} = e^{-k_2\xi}v_n$. By Lemma 3.21(a) (mirror image), $|v_n - 1| \le e^{\beta(\xi)} - 1$ with $\beta(\xi) = C_S e^{-2\xi}/(4k_2)$ for $\xi \ge 0$. Fix $\xi_1 \ge 0$ with $e^{\beta(\xi_1)} \le \frac{3}{2}$. For $\eta \ge \xi_1$ then $v_n \in [\frac12, \frac32]$ and

$$|v_n^{-2} - 1| = \frac{|1 - v_n||1 + v_n|}{v_n^2} \le 10\,|v_n - 1| \le 10\,\beta(\eta)e^{\beta(\eta)} \le c_1 e^{-2\eta},$$
$$c_1 := \frac{15 C_S}{4k_2}.$$

Since $u_+ = \varphi_+/\beta_n$, (11.1) gives, for $\xi \ge \xi_1$,

$$R_n(\xi) - 1 = w_n \int_0^\xi u_+^{-2} = w_n \int_0^{\xi_1} u_+^{-2} + w_n\beta_n^2 \int_{\xi_1}^\xi e^{2k_2\eta} v_n^{-2}\, d\eta.$$

By Lemma 3.28(e), $u_+ \ge c_{\mathrm{lo}} e^{-k_2\eta}$ on $[0,\infty)$, so the first integral is at most $A_1 := c_{\mathrm{lo}}^{-2}\int_0^{\xi_1} e^{2k_2\eta}d\eta$, independent of n. In the second, $e^{2k_2\eta}v_n^{-2} = e^{2k_2\eta} + O(c_1 e^{(2k_2-2)\eta})$, and $\int_{\xi_1}^\xi e^{(2k_2-2)\eta}d\eta$ is at most $\frac{1}{2-2k_2}$ if $k_2 < 1$, ξ if $k_2 = 1$ and $\frac{e^{(2k_2-2)\xi}}{2k_2-2}$ if $k_2 > 1$. Hence

$$\int_{\xi_1}^\xi e^{2k_2\eta} v_n^{-2}\, d\eta = \frac{e^{2k_2\xi}}{2k_2}\big(1 + r_n(\xi)\big),$$
$$|r_n(\xi)| \le e^{-2k_2(\xi-\xi_1)} + c_2(1+\xi)e^{-2\min(k_2,1)\xi},$$

with c_2 independent of n; so $\sup_n |r_n(\xi)| \to 0$ as $\xi \to \infty$. Now fix T and put $\xi = \xi_+(\kappa_n) + \tau/(2k_2)$, $|\tau| \le T$. As $n \to \infty$, $\xi \to \infty$ uniformly in τ, so $r_n \to 0$ and $w_n A_1 \to 0$ uniformly; by the definition of ξ_+, $e^{2k_2\xi} = e^\tau\, 2k_2\vartheta(1+\vartheta)J_*\gamma_{\mathrm c}^2/\kappa_n$, hence

$$w_n\beta_n^2\frac{e^{2k_2\xi}}{2k_2} = \frac{\kappa_n}{\vartheta J_n}\,\beta_n^2\,\frac{e^\tau}{2k_2}\cdot\frac{2k_2\vartheta(1+\vartheta)J_*\gamma_{\mathrm c}^2}{\kappa_n}$$
$$= \frac{J_*}{J_n}\,(\gamma_{\mathrm c}\beta_n)^2(1+\vartheta)\,e^\tau \;\longrightarrow\; (1+\vartheta)e^\tau.$$

Thus $R_n \to 1 + (1+\vartheta)e^\tau$ uniformly in $|\tau| \le T$, and by $f_n = f_- + J_n R_n/(R_n + \vartheta)$ and $J_* = (1+\vartheta)(M - f_-)$,

$$f_n \longrightarrow f_- + J_*\frac{1 + (1+\vartheta)e^\tau}{(1+\vartheta)(1+e^\tau)} = f_- + (M - f_-)\Big(1 + \vartheta\,\frac{e^\tau}{1+e^\tau}\Big)$$
$$= M + \vartheta(M - f_-)\lg(\tau)$$

uniformly in $|\tau| \le T$ (a computer-algebra check of this algebra is in `verify_ch11a.py`).

Lower layer. From $R' = w_0 u_+^{-2}$ one gets $(1/R)' = -R'/R^2 = -w_0 u_-^{-2}$, so $\frac{1}{R_n(\xi)} - 1 = w_n \int_\xi^0 u_-^{-2}$ for $\xi \le 0$. The mirror image of the argument above (with $\varphi_-^{(n)} = e^{k_1\xi}\tilde v_n$, $u_- \ge c_{\rm lo} e^{k_1\xi}$ on $(-\infty, 0]$ and $\varphi_-^{(n)}(0) \to 1/\gamma_{\rm c}$) gives, at $\xi = -\xi_-(\kappa_n) + \tau/(2k_1)$, where $e^{-2k_1\xi} = e^{-\tau}\, 2k_1(1+\vartheta) J_* \gamma_{\rm c}^2/\kappa_n$,

$$\frac{1}{R_n} - 1 \longrightarrow \frac{1+\vartheta}{\vartheta}\, e^{-\tau} \quad \text{uniformly in } |\tau| \le T.$$

Hence $\Sigma_n = \big(1 + \vartheta/R_n\big)^{-1} \to \big((1+\vartheta)(1+e^{-\tau})\big)^{-1} = \lg(\tau)/(1+\vartheta)$ and $f_n = f_- + J_n \Sigma_n \to f_- + (M - f_-)\lg(\tau)$.

Plateau. f_n is nondecreasing. Let $\varepsilon > 0$ and choose T with $\max(1,\vartheta)(M - f_-)e^{-T} < \varepsilon/2$. At $\xi = \xi_+(\kappa_n) - T/(2k_2)$ the limit value above is $M + \vartheta(M - f_-)\lg(-T) \le M + \vartheta(M-f_-)e^{-T}$; at $\xi = -\xi_-(\kappa_n) + T/(2k_1)$ it is $f_- + (M - f_-)\lg(T) \ge M - (M-f_-)e^{-T}$ (because $1 - \lg(T) = \lg(-T) \le e^{-T}$). By monotonicity f_n lies between the values at the two endpoints on the interval between them; for $n \ge n_0$ these are within $\varepsilon/2$ of the limits, and $|f_n - M| \le \varepsilon$ there. (The points $-\xi_- + T/(2k_1)$ and $\xi_+ - T/(2k_2)$ satisfy $-\xi_- + T/(2k_1) < \xi_+ - T/(2k_2)$ for n large because $\xi_\pm \to \infty$.) □

Remark 11.11 (the composite profile)**.** The proof gives more than the statement: up to $o(1)$ in the layers and $O(\kappa)$ in the bulk, the whole swirl profile is $f = f_- + J\,R/(R+\vartheta)$ with $R = 1 + w_0 \int_0^\xi u_+^{-2}$ for $\xi \ge 0$, $1/R = 1 + w_0 \int_\xi^0 u_-^{-2}$ for $\xi \le 0$ and $u_\pm \approx g$. Replacing $u_\pm$ by g gives the *composite profile* $R_{\rm c} = 1 + w_0 \mathcal{G}_{\rm int}^+$ $(\xi \ge 0)$, $R_{\rm c} = (1 + w_0 \mathcal{G}_{\rm int}^-)^{-1}$ $(\xi \le 0)$, with $\mathcal{G}_{\rm int}^\pm = \max(\pm\mathcal{G}_{\rm int}, 0)$, that is used in Section 11.4.

Example 11.12 (layer centres for set A)**.** For set A ($k_1 = k_2 = 1$, $\gamma_{\rm c} = 2$, $J_* = 8$, $\vartheta = 1$) $\xi_+(\kappa) = \frac12 \ln \frac{2\cdot1\cdot2\cdot8\cdot4}{\kappa} = \frac12 \ln \frac{128}{\kappa}$ and $\xi_-(\kappa) = \frac12 \ln \frac{2\cdot2\cdot8\cdot4}{\kappa} = \frac12 \ln \frac{128}{\kappa}$: for $\vartheta = 1$ the layers are symmetric. At $\kappa = 10^{-4}$, $\xi_\pm = \frac12 \ln(1.28 \cdot 10^6) = 7.0312$. The heights are $M - f_- = 4$ (south, $0 \to 4$) and $\vartheta(M - f_-) = 4$ (north, $4 \to 8$). For $\vartheta \ne 1$ and $k_1 = k_2 = 1$ the layers are no longer symmetric: $\xi_+ - \xi_- = \frac12 \ln \vartheta$.

Numerical observation 11.13 (layer positions and the limit $J \to J_*$)**.** *For the data of Table 11.1 we followed the solution branch down to* $\kappa = 10^{-9}$ *(code:* `code/ch11_small/`*; the recorded continuation is read by* `verify_ch11a.py`*). The solution profiles for set A are shown in Figure 11.2;* $J \to J_* = 8$*, and the plateau* $M = 4$ *with two logistic layers appears as predicted. The centre of the upper layer (the point where* $R = 2 + \vartheta$*) is compared with* $\xi_+(\kappa)$ *in Table 11.2: the difference is* $o(1)$*, as proved, and falls roughly like* $\kappa^{0.87}$ *for set A (fitted exponent on* $10^{-6} \le \kappa \le 10^{-3}$*;* 0.95 *for B and* 0.94 *for C). At the grid value* $\kappa = 1.34 \cdot 10^{-4}$ *the computed centres are* 6.8842 *(north) and* −6.8846 *(south) against* ±6.8845*.*

11.3. The solvability identity and the rate

In words. *Multiply* $z'' = Wz$ *by* g *and* $g'' = W_{\rm crit} g$ *by* z *and subtract:* $(gz' - g'z)' = \delta\, gz$*. Integrating over the line gives an* exact *formula: the weighted integral of the potential correction* $\delta = W - W_{\rm crit}$ *against* $g\,u_-$ *equals the "non-recessiveness"* w_0 *of* u_- *at* $+\infty$ *(times a constant), and* $w_0 = \kappa/(\vartheta J)$ *is small. Since* $\delta = \frac12 \mathcal{Q}[f^2 - M^2]$ *is, to first order,* $M(f - M)\mathcal{Q}[1]$*, the identity becomes a scalar equation for* $J - J_*$*. All remaining terms are errors coming from the layers, and they are controlled by how far the exponential tail of the kernel reaches into the region where* $f \ne f(0)$*.*

κ	$J - J_*$	ξ_+ computed	$\xi_+(\kappa)$	difference
$1.0 \cdot 10^{0}$	$6.51 \cdot 10^{-2}$	2.1524	2.4260	$-2.7 \cdot 10^{-1}$
$1.4 \cdot 10^{-2}$	$1.62 \cdot 10^{-3}$	4.5554	4.5661	$-1.1 \cdot 10^{-2}$
$1.9 \cdot 10^{-4}$	$2.39 \cdot 10^{-5}$	6.7058	6.7061	$-3.0 \cdot 10^{-4}$
$2.7 \cdot 10^{-6}$	$3.31 \cdot 10^{-7}$	8.8462	8.8462	$-7.4 \cdot 10^{-6}$
$3.7 \cdot 10^{-8}$	$4.59 \cdot 10^{-9}$	10.9862	10.9862	$-7.0 \cdot 10^{-7}$

TABLE 11.2. Set A: Richardson-extrapolated values (step sizes 0.01 and 0.005) of $J - J_*$ and of the centre of the upper layer, against $\xi_+(\kappa) = \frac{1}{2}\ln(128/\kappa)$ of Theorem 11.10.

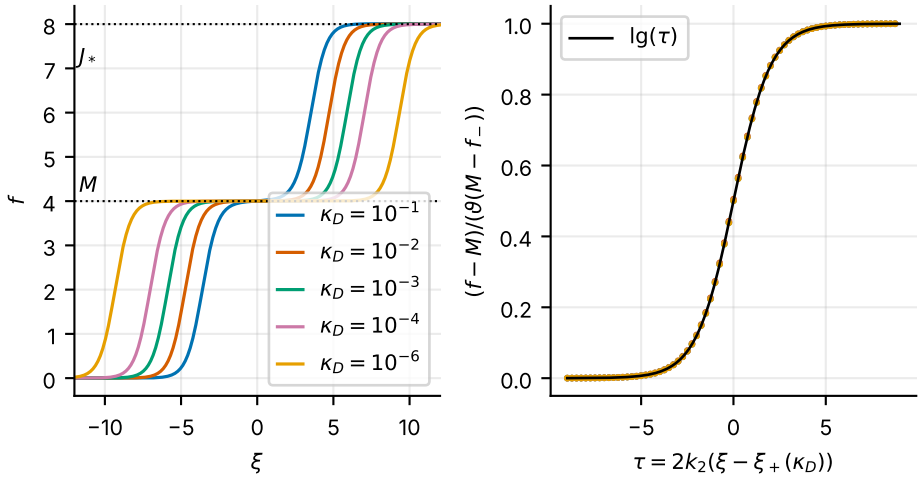


FIGURE 11.2. Set A ($\boldsymbol{c}' = (0, 0, -8)$, $\vartheta = 1$, $f_- = 0$, $M = 4$, $J_* = 8$). Left: swirl profiles for $\kappa = 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-6}$; the plateau $M = 4$ and the two layers move out like $\frac{1}{2}\ln(1/\kappa)$. Right: the north layer in the inner variable $\tau = 2k_2(\xi - \xi_+(\kappa))$ for $\kappa = 10^{-2}, \dots, 10^{-6}$, rescaled to $(f - M)/(\vartheta(M - f_-))$, against $\lg(\tau)$ (Theorem 11.10). Code: `code/ch11_small/make_a_layers.py`.

Toy example 11.14 (Lyapunov–Schmidt reduction in two lines)**.** The idea is to solve the equation in the directions where the linearisation is invertible and to keep one scalar *bifurcation equation* for the degenerate direction. Example: $F(x, y; \varepsilon) = \big(\varepsilon\beta_0 - x^2 + xy,\ y - \varepsilon b - x^2\big) = 0$. At $\varepsilon = 0$ the derivative at the origin is $\mathrm{diag}(0, 1)$: the x-direction is degenerate. The second equation gives $y = \varepsilon b + x^2$; inserting it into the first yields $\varepsilon\beta_0 - x^2 + \varepsilon b x + x^3 = 0$, so $x = \pm\sqrt{\varepsilon\beta_0}\,(1 + O(\sqrt{\varepsilon}))$: the small parameter enters through one scalar solvability condition, and its square root appears. In our problem the degenerate direction is the critical function g (the solution of the linearised problem $z'' = W_{\mathrm{crit}}\, z$), and the scalar condition is the

Wronskian identity Lemma 11.15: it is exact (not just first order) and it contains the unknown $J - J_*$ linearly in the bulk.

Lemma 11.15 (the exact identity)**.** *Assume* (H) *and let* $W = W_{\mathrm{e}} + \frac{1}{2}\mathcal{Q}[h]$ *be admissible, with* $h \in \mathrm{BC}$, $0 \le h \le S^2$ *(for a solution,* $h = f^2$*). Put* $\delta = W - W_{\mathrm{crit}} = \frac{1}{2}\mathcal{Q}[h - M^2]$*. Then* $\delta\, g\, u_- \in L^1(\mathbb{R})$ *and*

$$\int_{\mathbb{R}} \delta\, g\, u_-\, d\xi = \gamma_{\mathrm{c}}\, \varphi_+(0)\, w_0.$$

Proof. *Integrability.* By Lemma 8.2(a), $|\delta| \le \frac{1}{8}\max(S^2, M^2)\operatorname{sech}^2\xi \le \frac{1}{2}\max(S^2, M^2)e^{-2|\xi|}$. As in Lemma 11.5(b), W lies in a class $\mathcal{K}$ of Lemma 3.28 (with the present S), so $c_{\mathrm{lo}}e^{-k_2\xi} \le u_+ \le c_{\mathrm{hi}}e^{-k_2\xi}$ for $\xi \ge 0$ and $c_{\mathrm{lo}}e^{k_1\xi} \le u_- \le c_{\mathrm{hi}}e^{k_1\xi}$ for $\xi \le 0$ (Lemma 3.28(e)). For $\xi \ge 0$, (11.1)'s $R' = w_0 u_+^{-2}$ gives $R = 1 + w_0\int_0^\xi u_+^{-2} \le 1 + w_0 c_{\mathrm{lo}}^{-2}\frac{e^{2k_2\xi}}{2k_2}$ and $u_- = Ru_+ \le c_{\mathrm{hi}}e^{-k_2\xi} + Ce^{k_2\xi} \le C'e^{k_2\xi}$. With $g \le \gamma_{\mathrm{c}}e^{-k_2\xi}$ we get $gu_- \le C''$ on $[0,\infty)$; on $(-\infty, 0]$, $g \le \gamma_{\mathrm{c}}e^{k_1\xi}$ and $u_- \le c_{\mathrm{hi}}e^{k_1\xi}$ give $gu_- \le Ce^{2k_1\xi}$. So gu_- is bounded and $\delta g u_- \in L^1$.

Computation. $\mathrm{Wr}(g, u_-)' = gu_-'' - g''u_- = (W - W_{\mathrm{crit}})gu_- = \delta g u_-$, which is integrable, so $\int_{\mathbb{R}} \delta g u_- = \mathrm{Wr}(g,u_-)(+\infty) - \mathrm{Wr}(g,u_-)(-\infty)$ (both limits exist). Write $\mathrm{Wr}(g,u_-) = gu_-\,(\zeta_- - g'/g)$, where g'/g is bounded with limits k_1 and $-k_2$. At $-\infty$: $gu_- \le Ce^{2k_1\xi} \to 0$ and ζ_- is bounded (Lemma 3.28(a)), so the limit is 0. At $+\infty$: $u_- = w_0/(\rho\, u_+)$ by $\rho = w_0/(u_-u_+)$, hence

$$\mathrm{Wr}(g,u_-) = \frac{w_0}{\rho}\,\frac{g}{u_+}\left(\zeta_- - \frac{g'}{g}\right) \longrightarrow \frac{w_0}{2k_2}\cdot\gamma_{\mathrm{c}}\varphi_+(0)\cdot(k_2 + k_2) = \gamma_{\mathrm{c}}\varphi_+(0)\, w_0,$$

because $\rho \to 2k_2$, $\zeta_- \to k_2$ (Lemma 3.21(b)(iv)), $g'/g \to -k_2$, and $g/u_+ = \varphi_+(0)\, g/\varphi_+ \to \varphi_+(0)\gamma_{\mathrm{c}}$ ($g \sim \gamma_{\mathrm{c}}e^{-k_2\xi}$, $\varphi_+ \sim e^{-k_2\xi}$). □

Toy example 11.16 (the identity at a constant swirl above M)**.** Take set A and the constant swirl $h \equiv C^2 = M^2 + \varepsilon$ ($\varepsilon > 0$; $f \equiv C$ is not a solution, but Lemma 11.15 holds for every admissible $W = W_{\mathrm{e}} + \frac{1}{2}\mathcal{Q}[h]$). Then $W = 1 - (2 - \frac{\varepsilon}{8})\operatorname{sech}^2\xi$ is admissible (Lemma 11.2), $\delta = \frac{\varepsilon}{8}\operatorname{sech}^2\xi$, $g = \operatorname{sech}\xi$, $\gamma_{\mathrm{c}} = 2$, and to first order ($u_- \approx g$, $\gamma_{\mathrm{c}}\varphi_+(0) \approx 1$) the identity predicts $w_0 \approx \frac{\varepsilon}{8}\int \operatorname{sech}^4 = \frac{\varepsilon}{8}\cdot\frac{4}{3} = \frac{\varepsilon}{6}$. A direct Riccati computation (`verify_ch11a.py`) gives

ε	1	0.1	0.01
w_0	0.16294	0.016628	0.0016663
$\varepsilon/6$	0.16667	0.016667	0.0016667
relative difference of the sides	$2\cdot10^{-12}$	$4\cdot10^{-12}$	$6\cdot10^{-11}$

so the identity is exact (the sides agree to the accuracy of the quadrature) while its first-order consequence $w_0 \approx \varepsilon A_0/2$ is only asymptotic.

Lemma 11.17 (the kernel q and the layer estimates)**.** *Assume* (H)*, let* f *be a solution with* $\kappa \le 1$*, and put* $\Theta := g\, u_-$*,*

$$q(\eta) := \tfrac{1}{4}\int_{\mathbb{R}} \operatorname{sech}^2\xi\, e^{-2|\xi-\eta|}\,\Theta(\xi)\, d\xi,$$

$$\alpha := \int_{\mathbb{R}} q = \tfrac{1}{4}\int_{\mathbb{R}} \operatorname{sech}^2\xi\,\Theta(\xi)\, d\xi, \qquad k := \max(k_1, k_2).$$

Then, with constants C *depending only on* $(\boldsymbol{c}', \vartheta, f_-)$*:*

(a) $\displaystyle\int_{\mathbb{R}} \delta\, g\, u_-\, d\xi = \tfrac{1}{2}\int_{\mathbb{R}} (f^2 - M^2)\, q\, d\eta$ *and* $0 \le q(\eta) \le C(1+|\eta|)e^{-2|\eta|}$.

(b) $0 \le f(\xi) - f(0) \le \min\bigl(C\kappa e^{2k\xi},\, S\bigr)$ *for* $\xi \ge 0$ *and* $0 \le f(0) - f(\xi) \le \min\bigl(C\kappa e^{2k|\xi|},\, S\bigr)$ *for* $\xi \le 0$.

(c) *Along every sequence of solutions with* $\kappa_n \to 0$*:* $\alpha_n \to A_0 = \tfrac{1}{4}\int g^2 \operatorname{sech}^2$, $\sup_\eta |q_n - q_g| \to 0$ *and* $\int |f_n - M|\, q_n\, d\eta \to 0$.

Proof. (a) The first formula is Fubini:

$$\int \delta\Theta\, d\xi = \int \frac{1}{2}\frac{1}{4\cosh^2\xi}\int e^{-2|\xi-\eta|}(f^2-M^2)(\eta)\, d\eta\, \Theta(\xi)\, d\xi,$$

and all integrands are absolutely integrable because Θ is bounded (proof of Lemma 11.15; the bound depends only on the constants of Lemma 11.5 when $\kappa \le 1$) and $\operatorname{sech}^2\xi\, e^{-2|\xi-\eta|}$ is integrable in (ξ,η). Since $\Theta \le C$ and $\operatorname{sech}^2 \le 4e^{-2|\xi|}$, $q(\eta) \le C\int e^{-2|\xi|-2|\xi-\eta|}d\xi = C(|\eta| + \tfrac{1}{2})e^{-2|\eta|}$ (split the integral at $\xi = 0$ and $\xi = \eta$). (b) $f' = \kappa z^{-2}$ and $z \ge \vartheta u_+ \ge \vartheta c_{\mathrm{lo}} e^{-k_2\xi}$ for $\xi \ge 0$ (Lemma 3.28(e)), so $f(\xi) - f(0) \le \kappa\vartheta^{-2}c_{\mathrm{lo}}^{-2}\int_0^\xi e^{2k_2\eta}d\eta \le C\kappa e^{2k\xi}$; also $f(\xi) - f(0) \le f_+ - f_- \le S$, and f is nondecreasing. For $\xi \le 0$ use $z \ge u_- \ge c_{\mathrm{lo}}e^{k_1\xi}$. (c) By Theorem 11.6(b), $\Theta_n = g u_-^{(n)} \to g^2$ locally uniformly, and $0 \le \Theta_n \le C$. Since $e^{-2|\xi-\eta|} \le 1$, $\sup_\eta |q_n - q_g| \le \tfrac{1}{4}\int \operatorname{sech}^2\xi\, |\Theta_n - g^2|\, d\xi \to 0$ by dominated convergence, and likewise $\alpha_n \to A_0$. Finally $|f_n - M| \le S + M$, $f_n \to M$ locally uniformly (Theorem 11.6(a)) and $q_n \le C(1+|\eta|)e^{-2|\eta|} \in L^1$, so $\int |f_n - M| q_n \to 0$ by dominated convergence. □

Proof of Theorem 11.10 (i)–(iii). (i) is Lemma 11.15 applied to $W = W_f$ ($h = f^2$, $S = \sup f$).

For (ii) and (iii) let $\kappa \le 1$ and put $\eta_0 := f(0) - M = \frac{J - J_*}{1+\vartheta}$ (by $f(0) = f_- + \frac{J}{1+\vartheta}$ and $M = f_- + \frac{J_*}{1+\vartheta}$). Then

$$f^2 - M^2 = (f-M)(f+M) = 2M\eta_0 + \eta_0(f-M) + (f - f(0))(f+M),$$

and Lemma 11.15, Lemma 11.17(a) and $w_0 = \kappa/(\vartheta J)$ give the *bifurcation equation*

$$\begin{aligned} \eta_0\, b_\kappa + \frac{1}{2}\int_{\mathbb{R}} \bigl(f - f(0)\bigr)(f+M)\, q\, d\eta &= \gamma_{\mathrm{c}}\varphi_+(0)\,\frac{\kappa}{\vartheta J}, \\ b_\kappa := M\alpha + \frac{1}{2}\int_{\mathbb{R}} (f-M)\, q\, d\eta. & \end{aligned} \tag{11.2}$$

For every sequence with $\kappa_n \to 0$ we have $b_{\kappa_n} \to MA_0 > 0$ (Lemma 11.17(c)), $\gamma_{\mathrm{c}}\varphi_+(0) \to 1$ (Corollary 11.7) and $J_n \to J_*$. Hence there is $\kappa_0 \in (0,1]$ such that $b_\kappa \ge \tfrac{1}{2}MA_0$ for *every* solution with $\kappa < \kappa_0$ (otherwise a sequence of solutions with $\kappa_n \to 0$ and $b_{\kappa_n} < \tfrac{1}{2}MA_0$ would contradict the convergence) and $\gamma_{\mathrm{c}}\varphi_+(0)/(\vartheta J) \le C$.

(ii) For $\kappa < \kappa_0$, $\tfrac{1}{2}MA_0|\eta_0| \le C\kappa + \tfrac{1}{2}(S+M)E$ with $E := \int |f - f(0)|\, q\, d\eta$, and by Lemma 11.17(a),(b)

$$E \le C\int_{\mathbb{R}} \min\bigl(\kappa e^{2k|\eta|}, 1\bigr)(1+|\eta|)e^{-2|\eta|}\, d\eta.$$

Let $\xi_\kappa = \frac{1}{2k}\ln\frac{1}{\kappa}$. The part $|\eta| \ge \xi_\kappa$ contributes at most $C(1+\xi_\kappa)e^{-2\xi_\kappa} = C(1+\xi_\kappa)\kappa^{1/k}$. The part $|\eta| \le \xi_\kappa$ contributes $C\kappa\int_0^{\xi_\kappa}(1+\eta)e^{(2k-2)\eta}d\eta$, which is at most $C\kappa$ if $k < 1$, at most $C\kappa(1+\xi_\kappa)^2$ if $k = 1$, and at most $C\kappa(1+\xi_\kappa)e^{(2k-2)\xi_\kappa} = C(1+\xi_\kappa)\kappa^{1/k}$ if $k > 1$. For $k < 1$, $\kappa^{1/k}(1+\xi_\kappa) \le C\kappa$. Also $\kappa \le \varepsilon_k(\kappa)$ in all cases. Thus $|\eta_0| \le C\varepsilon_k(\kappa)$, and $|J - J_*| = (1+\vartheta)|\eta_0|$.

(iii) Let $k<1$ and let f_n be solutions with $\kappa_n\to 0$. By (ii), $\eta_0^{(n)}/\kappa_n$ is bounded. By Theorem 11.6(b), $z_n\to(1+\vartheta)g$ locally uniformly, so $(f_n-f_n(0))/\kappa_n=\int_0^\xi z_n^{-2}\to(1+\vartheta)^{-2}\mathcal{G}_{\mathrm{int}}(\xi)$ locally uniformly; and $f_n+M\to 2M$ locally uniformly. By Lemma 11.17(b),(a) the integrand $|(f_n-f_n(0))(f_n+M)q_n|/\kappa_n$ is bounded by $C(1+|\eta|)e^{-(2-2k)|\eta|}\in L^1$ (here $k<1$ is used), so dominated convergence and $q_n\to q_g$ give

$$\frac{1}{\kappa_n}\int \big(f_n-f_n(0)\big)(f_n+M)q_n\,d\eta \;\longrightarrow\; \frac{2M}{(1+\vartheta)^2}\int \mathcal{G}_{\mathrm{int}}\,q_g\,d\eta=\frac{2MI_G}{(1+\vartheta)^2}.$$

Dividing (11.2) by κ_n and using $b_{\kappa_n}\to MA_0$, $\gamma_{\mathrm{c}}\varphi_+(0)\to 1$, $J_n\to J_*$ we get

$$\frac{\eta_0^{(n)}}{\kappa_n}\,MA_0 \;\longrightarrow\; \frac{1}{\vartheta J_*}-\frac{MI_G}{(1+\vartheta)^2},$$

and $(J_n-J_*)/\kappa_n=(1+\vartheta)\eta_0^{(n)}/\kappa_n\to\Gamma_{\mathrm{rate}}$. □

Remark 11.18 (symmetric data; what (ii) does not say)**.** If $k_1=k_2$ then $g=(\cosh\xi)^{-\bar k}$ is even, so q_g is even and $\mathcal{G}_{\mathrm{int}}$ is odd: $I_G=0$ and $\Gamma_{\mathrm{rate}}=\frac{1+\vartheta}{\vartheta MA_0J_*}=\frac{1}{\vartheta(M-f_-)MA_0}$. For $\max k_i\ge 1$ the bound (ii) is only an upper bound; Section 11.4 shows formally that it is sharp up to the logarithm for $k>1$ and that for $k_1=k_2=1$ the true size is κ (the logarithms of the two layers cancel). The convergence of the whole profile in (ii) is uniform only for $J\le\bar J$, i.e. for $\kappa<\kappa_0$. Nothing is said about uniqueness: Theorem 11.10 holds for every solution, and the proofs never use that the solution is unique.

Example 11.19 (the constants for sets B, C, G)**.** For symmetric data, $\Gamma_{\mathrm{rate}}=\frac{1}{\vartheta(M-f_-)\,MA_0}$ (set B: $\vartheta=1$, $f_-=0$, $M=2.5186$, $A_0=0.364560$ give $\Gamma_{\mathrm{rate}}=0.432441$; set G: $M=1.6800$, $\vartheta=\frac14$, $A_0=0.400955$ give 3.534781). For the asymmetric set C one finds $A_0=0.364185$, $I_G=0.162325$ and $\Gamma_{\mathrm{rate}}=0.006254$ (numerical quadrature of the formulas of Theorem 11.10, `verify_ch11a.py`).

Numerical observation 11.20 (the exact identity on computed fixed points)**.** *For sets A, B, C, D (Table 11.1) and two values of J each (κ between 0.4 and 18), Theorem 11.10(i) was evaluated on solutions computed by a Newton method for the J-formulation (code: `verify_ch11a.py`, Part 4); $\varphi_+(0)$ was read off from $e^{-k_2\xi}/u_+$ at large ξ. The relative differences between the two sides are between $1.3\cdot10^{-10}$ and $3.2\cdot10^{-9}$, i.e. at the level of the discretisation.*

Numerical observation 11.21 (the limit Γ_{rate} for $\max k_i<1$)**.** *Set B: $(J-J_*)/\kappa=0.432407$ at $\kappa=1.3\cdot10^{-8}$ against $\Gamma_{\mathrm{rate}}=0.432441$ (relative difference $8\cdot10^{-5}$). Set G: 3.534780 at $\kappa=1.1\cdot10^{-7}$ against 3.534781. Set C (asymmetric, $I_G\neq0$): $\Gamma_{\mathrm{rate}}=0.006254$, but the computed ratio is still 0.0100 at $\kappa=1.3\cdot10^{-8}$; the difference decays like $\kappa^{0.194}$ between $3.1\cdot10^{-7}$ and $1.3\cdot10^{-8}$, matching the exponent $1/k_2-1=0.195$ of the layer correction ($k_2=0.837$ is close to 1). This is a case where numerics alone would give a wrong limit. Figure 11.3.*

11.4. Formal predictors: the composite profile and the bubble eigenvalue

In words. *Everything in this section is a* computation*, not a proof. The identity of Lemma 11.15 is exact; what is not controlled is the error made when the unknown profile f and the unknown Jost solution u_- in it are replaced by the composite profile*

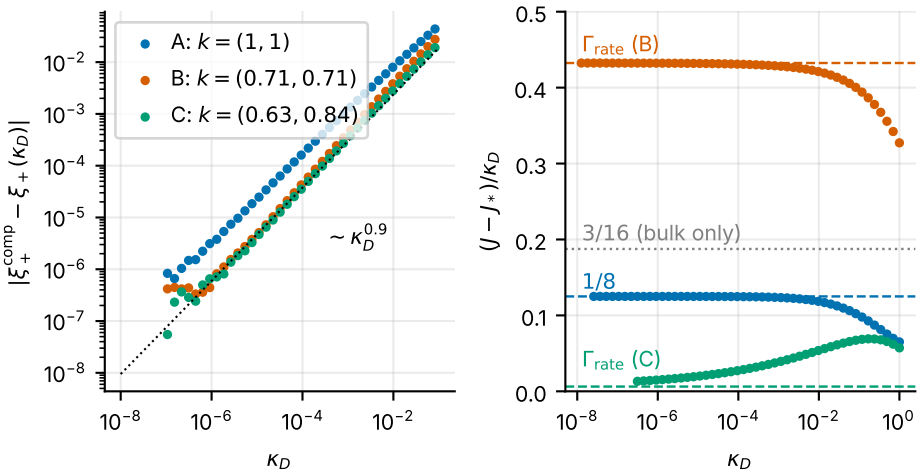


FIGURE 11.3. Sets A, B, C (Table 11.1). Left: the error $|\xi_+^{\mathrm{comp}} - \xi_+(\kappa)|$ of the layer centre (dotted: slope 0.9). Right: $(J - J_*)/\kappa$ against κ; dashed: the limits Γ_{rate} of Theorem 11.10(iii) for B and C, the value $\frac{1}{8}$ of Formal computation 11.25 for set A (formal) and the bulk-only value $\frac{3}{16}$ for A. Code: `code/ch11_small/make_a_rate.py`.

of Remark 11.11. For $\max k_i < 1$ *this replacement was shown to be harmless at the order* κ *(Theorem 11.10(iii)); for* $\max k_i \geq 1$ *the layers contribute at the order of the answer, and we only* predict *the result. We state each prediction, give its derivation, and compare it with the continuation of the fixed points (numerical observations). A second prediction concerns the one large multiplier of the fixed-point map at a bubble.*

Toy example 11.22 (a cut-off integral and the exponent $1/k$)**.** Take $k > 1$ and consider $E(\kappa) = \int_0^\infty \min(\kappa e^{2k\eta}, 1)\, e^{-2\eta}\, d\eta$, the model of the layer contribution in the proof of Theorem 11.10(ii). The two branches meet at $\eta_\kappa = \frac{1}{2k} \ln \frac{1}{\kappa}$; the first gives $\kappa \int_0^{\eta_\kappa} e^{(2k-2)\eta} d\eta = \frac{\kappa}{2k-2}\big(\kappa^{-(k-1)/k} - 1\big)$ and the second $\frac{1}{2} e^{-2\eta_\kappa} = \frac{1}{2}\kappa^{1/k}$; so $E(\kappa) = \big(\frac{1}{2k-2} + \frac{1}{2}\big)\kappa^{1/k} - \frac{\kappa}{2k-2}$. The exponent $1/k$ is that of the point where the layer sits; the term κ coming from the bulk is smaller by the factor $\kappa^{1-1/k}$. For $k = 1$ the two terms collapse into $\kappa \ln(1/\kappa)$-type terms, and only the *sign structure* of the layer integrand (below) decides whether they cancel.

Formal computation 11.23 (the composite-profile predictor)**.** *Let* f *be a solution with small* κ *and put* $s = f - f(0)$, $\eta_0 = f(0) - M$. *Then* $\frac{1}{2}(f^2 - M^2) = M(\eta_0 + s) + \frac{1}{2}(\eta_0 + s)^2$, *and Lemma 11.15 with Lemma 11.17(a) reads*

$$M\,\eta_0\,Q_1 + M\int s\,q\,d\eta + \frac{1}{2}\int (\eta_0 + s)^2 q\,d\eta = \gamma_{\mathrm{c}}\varphi_+(0)\,w_0,$$

$$Q_1 := \int q\,d\eta = \alpha.$$

Prediction. *Replace in this equation (a)* $\gamma_{\mathrm{c}}\varphi_+(0)$ *by* 1*, (b)* f *by* $f_- + J\,\Sigma_{\mathrm{c}}$ *with* $\Sigma_{\mathrm{c}} = R_{\mathrm{c}}/(R_{\mathrm{c}}+\vartheta)$ *and* R_{c} *the composite function of Remark 11.11 with* $w_0 = \kappa/(\vartheta J)$*,* $J = J_* + (1+\vartheta)\eta_0$*, and (c)* $\Theta = gu_-$ *by* $g^2 R_{\mathrm{c}}^+$*, where* $R_{\mathrm{c}}^+ = R_{\mathrm{c}}$ *for* $\xi \geq 0$ *and* 1 *for* $\xi \leq 0$*. The result is a scalar quadratic equation for* η_0 *(solved iteratively, with* J *updated), whose solution we call* $\eta_0^{\mathrm{pred}}(\kappa)$*,* $J^{\mathrm{pred}} - J_* = (1+\vartheta)\eta_0^{\mathrm{pred}}$*.* Why: *by Theorem 11.6 and Theorem 11.10 each replacement is correct up to a relative error* $o(1)$ *in the region* $|\xi| \leq \xi_\pm$ *and in the layers; the formula is not claimed to hold beyond the leading order of the answer. This is not a proof, because the errors, although small pointwise, are integrated against weights of size* κ*, i.e. of the order of the answer itself, when* $\max k_i \geq 1$*.*

Numerical observation 11.24 (the predictor against the continuation)**.** *For the sets A, B, C, D, F of Table 11.1 the ratio* $(J-J_*)/(J^{\mathrm{pred}}-J_*)$ *of the computed value to the predictor (code:* `verify_ch11a.py`*, Part 3.4) is, at* $\kappa \approx 1,\ 1.4\cdot10^{-2},\ 1.9\cdot10^{-4},\ 2.7\cdot10^{-6},\ 3.7\cdot10^{-8}$*:*

set	1	$1.4\cdot10^{-2}$	$1.9\cdot10^{-4}$	$2.7\cdot10^{-6}$	$3.7\cdot10^{-8}$
A	0.952	0.956	0.9973	0.99991	1.00000
B	1.171	1.0012	1.00001	1.00000	1.00000
D	1.212	0.831	1.0079	1.0004	1.00001 *(*$5.2\cdot10^{-8}$*)*
F	1.183	0.9954	1.0009	0.99998	0.99997
C	0.934	0.9924	0.99975	1.00012	1.00023

For set A the computed $(J-J_*)/\kappa$ *is* 0.124999 *at* $\kappa = 3.7\cdot10^{-8}$*, against* $\frac{1}{8}$ *of Formal computation 11.25. For* $\kappa \leq 2\cdot10^{-5}$ *the worst deviation of the ratio from* 1 *over all five sets is* $1.4\cdot10^{-3}$*, and for* $\kappa \leq 3\cdot10^{-7}$ *it is* $6.8\cdot10^{-5}$*. (For set C the ratio drifts away from* 1 *at the smallest* κ *shown,* $1.2\cdot10^{-3}$ *at* $1.3\cdot10^{-8}$*: the predictor is a leading-order statement, and the quantity compared is a small difference of two* $O(\kappa)$ *terms for this set.)*

Formal computation 11.25 (the case $k_1 = k_2 = 1$: the value $\frac{1}{8}$ for set A)**.** *Let* $c' = (0,0,c_3')$*,* $\vartheta = 1$*,* $f_- = 0$*, so* $g = \operatorname{sech}\xi$*,* $\gamma_{\mathrm{c}} = 2$*,* $A_0 = \frac{1}{3}$*,* $J_* = 2M$*. Then* R_{c} *and* $1/R_{\mathrm{c}}$ *are mirror images,* $s = f - f(0)$ *is odd (a computer-algebra check,* `verify_ch11a.py`*),* q *is even, and the term* $\int sq$ *of Formal computation 11.23 vanishes. Hence* $M\eta_0 A_0 = w_0 - \frac{1}{2}\int s^2 q$ *and* $w_0 = \kappa/J_* = \kappa/(2M)$ *to leading order. In the layers,* $q(\eta) \approx c\,e^{-2|\eta|}$ *with* $c = \frac{1}{4}\int \operatorname{sech}^4\xi\, e^{2\xi}d\xi = \frac{2}{3}$*;* $s = M\,\lg(\tau)$ *at the north layer (*$\tau = 2(\eta - \xi_+)$*) and* $e^{2\xi_+} = 32M/\kappa$ *(Theorem 11.10). With* $\int_{\mathbb{R}} \lg(\tau)^2 e^{-\tau}d\tau = 1$ *(substitute* $u = \lg(\tau)$*,* $du = u(1-u)d\tau$*) each layer gives* $\frac{1}{2}\int s^2 q \approx \frac{1}{2}M^2\cdot\frac{2}{3}\cdot\frac{\kappa}{32M}\cdot\frac{1}{2} = \frac{M\kappa}{192}$*, and the two layers* $\frac{M\kappa}{96}$*. So*

$$\frac{J-J_*}{\kappa} \;\to\; \Gamma(M) = \frac{2}{MA_0}\Big(\frac{1}{2M} - \frac{M}{96}\Big) = \frac{3}{M^2} - \frac{1}{16}. \tag{11.3}$$

For set A ($M = 4$*) this gives* $\Gamma = \frac{3}{16} - \frac{1}{16} = \frac{1}{8}$*: the bulk alone would give* $\frac{3}{16}$*, and the layers reduce it to* $\frac{1}{8}$*. The constant (the algebra is checked by computer algebra,* `verify_ch11a.py`*) does* not *contain a logarithm even though* $\max k_i = 1$*: the logarithmic term* $\kappa\ln(1/\kappa)$ *that the bound of Theorem 11.10(ii) allows is absent because the two integrals over* $|\eta| \leq \xi_\pm$ *that would produce it are of opposite sign (the odd term* $\int sq$ *vanishes). Formula (11.3) also predicts that* $J - J_*$ *changes sign at* $M^2 = 48$ *(*$c_3' = -12$ *for these data); this has not been tested.*

Heuristic 11.26 (why the layers lower J)**.** *The identity says that a weighted average of* $\frac{1}{2}(f^2-M^2)$ *equals the* positive *number* $w_0\gamma_{\rm c}\varphi_+(0)$*. The bulk part of the average is* $M\eta_0A_0$ *(positive iff* $f(0)>M$*). In the layers* f *deviates from* M *by* $\pm O(1)$ *and* $f^2-M^2=2M(f-M)+(f-M)^2$*; the quadratic term is positive on both sides and has to be paid for by a* smaller η_0*. The linear terms cancel if the data are symmetric, and for* $\max k_i>1$ *the layers sit where the tail of* q *is not negligible and the quadratic term dominates* w_0*; then* $\eta_0<0$ *and* $f(0)<M$*. Not a proof: the sign of the linear terms is not controlled for asymmetric data.*

Formal computation 11.27 (the case $\max k_i>1$: the leading term)**.** *Let* $k_2>1$*. The north layer lies in the region where* $q(\eta)\approx c_+e^{-2\eta}$ *with* $c_+=\frac{1}{4}\int g^2\operatorname{sech}^2\xi\,e^{2\xi}d\xi<\infty$*, and* $f-M=a\,\lg(\tau)$ *with* $a=\vartheta(M-f_-)$*,* $\tau=2k_2(\eta-\xi_+)$*. The layer integral is, with* $\sigma=1/k_2$ *and the Beta integrals* $\int\lg(\tau)e^{-\sigma\tau}d\tau=\pi/\sin\pi\sigma$*,* $\int\lg^2 e^{-\sigma\tau}d\tau=(1-\sigma)\pi/\sin\pi\sigma$ *(convergent at* $-\infty$ *exactly because* $k_2>1$*; checked numerically in* `verify_ch11a.py`*),*

$$\frac{1}{2}\int(f^2-M^2)q\,d\eta\approx\frac{c_+e^{-2\xi_+}}{4k_2}\cdot a\,\frac{\pi}{\sin(\pi/k_2)}\Big(2M+a\big(1-\tfrac{1}{k_2}\big)\Big),$$

$$e^{-2\xi_+}=\Big(\frac{\kappa}{2k_2\vartheta(1+\vartheta)J_*\gamma_{\rm c}^2}\Big)^{1/k_2},$$

which is of order $\kappa^{1/k_2}\gg\kappa$*. Balancing it against* $M\eta_0A_0$ *(the terms* w_0 *and* κ *are of lower order) gives* $J-J_*\approx-(1+\vartheta)\,[\text{that integral}]/(MA_0)<0$*, of exact order* κ^{1/k_2} *with no logarithm; the south layer adds a term of the same form with* $(k_1,1+\vartheta$ *in place of* ϑ*,* $-b)$*,* $b=M-f_-$*, if* $k_1>1$*. The* relative *error of this leading term is of order* $\kappa^{1-1/k}$ *(the bulk term of Toy example 11.22).*

Numerical observation 11.28 ($k>1$)**.** *Set* D *(*$c'=(1,1,-9)$*,* $k_1=k_2=\sqrt{2}$*):* $J-J_*>0$ *for* $\kappa\geq 9.69\cdot10^{-3}$ *and negative below, with minimum* $-3.23\cdot10^{-4}$ *at* $\kappa=2.33\cdot10^{-3}$*; the local slope of* $\ln|J-J_*|$ *against* $\ln\kappa$ *at the smallest computed value* $\kappa=5.2\cdot10^{-8}$ *is* 0.697*, against* $1/k=0.707$*. The ratio of the computed value to the closed form of Formal computation 11.27 is* $0.643,\ 0.764,\ 0.844,\ 0.897,\ 0.932,\ 0.955$ *for* $\kappa=1.9\cdot10^{-4},\ 4.6\cdot10^{-5},\ 1.1\cdot10^{-5},\ 2.7\cdot10^{-6},\ 6.4\cdot10^{-7},\ 1.5\cdot10^{-7}$*: it tends to* 1*, slowly, as the predicted relative correction* $\kappa^{1-1/k}$ *(exponent* 0.293*) requires. At* $\kappa=5.2\cdot10^{-8}$ *the leading term is* $-0.0862\,\kappa^{1/k}$*. For set* F *(*$k=(0.707,1)$*)* $(J-J_*)/\kappa=-0.722$ *at* $\kappa=1.3\cdot10^{-8}$*. The sets* E*,* H*,* I *(only* $\kappa\geq3.3\cdot10^{-3}$*,* $9.1\cdot10^{-7}$*,* $8.0\cdot10^{-4}$ *computed) are too far from the asymptotic range to test anything.*

Formal computation 11.29 (summary of the small-κ law of $J-J_*$)**.** *Under (H), for* $\kappa\to0$*:*

- *(a)* $\max k_i<1$*:* $J-J_*=\Gamma_{\rm rate}\,\kappa+o(\kappa)$ *with* $\Gamma_{\rm rate}$ *of Theorem 11.10(iii).* Proved. *The convergence is slow when* $\max k_i$ *is close to* 1 *(the correction is* $O(\kappa^{1/k-1})$ *relative, formally; set C).*
- *(b)* $k_1=k_2=1$*:* $J-J_*=\Gamma\kappa+o(\kappa)$*, with* $\Gamma=\frac{3}{M^2}-\frac{1}{16}$ *when* $c'=(0,0,c_3')$*,* $\vartheta=1$*,* $f_-=0$ *(Formal computation 11.25). Formal.*
- *(c)* $\max k_i>1$*:* $J-J_*\sim-C_k\kappa^{1/k}$ *with* $C_k>0$ *from Formal computation 11.27,* $k=\max k_i$*. Formal. In all three cases the predictor of Formal computation 11.23 reproduces the computed values (Numerical observation 11.24).*

Numerical observation 11.30 (the decay of the correction for set C)**.** *For set C the difference* $(J - J_*)/\kappa - \Gamma_{\mathrm{rate}}$ *is* 0.0070, 0.0056, 0.0046, 0.0037 *at* $\kappa = 3.12\cdot10^{-7}$, $1.07\cdot10^{-7}$, $3.67\cdot10^{-8}$, $1.26\cdot10^{-8}$*. For set B the exponent* $1/k-1 = 0.414$ *is observed in the same way (code:* `verify_ch11a.py`*, Part 3.2).*

Toy example 11.31 (a steep branch gives a huge multiplier)**.** Take the scalar map $T(y) = f_- + \varepsilon/(y - M)$, with $\varepsilon > 0$ small and $y > M$. The equation $y = T(y)$ has the solution $y_* = M + \varepsilon/(M - f_-) + O(\varepsilon^2)$ and there $T'(y_*) = -\varepsilon/(y_* - M)^2 \approx -(M - f_-)^2/\varepsilon$: the multiplier is large and negative although the fixed point is an ordinary one. Next we show that the fixed-point map near a bubble has this structure.

Formal computation 11.32 (the large eigenvalue of a bubble)**.** *Let* f *be the bubble solution with* $f_- \in \mathbb{R}$*,* $C = M$ *(positive) or* $C = -M$ *(negative), and write* $J_C = (1 + \vartheta)(C - f_-)$ *for the bubble jump* ($J_C > 0$ *is needed). Consider the map* $y \mapsto T$ *that sends the equatorial value* $y = f(0)$ *to the new value* $f_- + J_{\mathrm{new}}/(1 + \vartheta)$*, where* $J_{\mathrm{new}} = \kappa/(\vartheta w_0)$ *and* $w_0 = \pm MA_0(y \mp M)$ *is given by the bulk part of the identity of Lemma 11.15 (the layer terms are of higher order for* $k < 1$*). For* $C = M$*,*

$$T(y) = f_- + \frac{\kappa}{\vartheta(1+\vartheta)MA_0\,(y - M)},$$

$$\textit{so}\quad \nu_b = T'(y_*) = -\frac{\vartheta J_C^2\, MA_0}{(1+\vartheta)\,\kappa}\quad (\textit{positive bubble}),$$

at the fixed point $y_* - M = \kappa/(\vartheta J_C MA_0)$ *(algebra checked in* `verify_ch11a.py`*); this is Toy example 11.31 with* $\varepsilon = \kappa/(\vartheta(1 + \vartheta)MA_0)$*. For* $C = -M$ *the sign of the bulk term is reversed and* $\nu_b = +\vartheta J_C^2 MA_0/((1 + \vartheta)\kappa)$ *(negative bubble). So* $|\nu_b|\kappa \to \vartheta J_C^2 MA_0/(1 + \vartheta)$ finite*: the multiplier blows up like* $1/\kappa$*. Its sign decides the Leray–Schauder index:* $\nu_b < 0$ *(positive bubble) gives a real eigenvalue below* 1*,* $\nu_b > 0$ *large (negative bubble) an eigenvalue above* 1*, so index* -1 *(Theorem 8.42 counts the indices with this convention). Not a proof: the one-dimensional reduction is not derived from the linearisation of* T_D*, and the remaining spectrum is not controlled.*

Numerical observation 11.33 (the bubble eigenvalue)**.** *Positive bubbles of* $c' = (0, 0, -8)$*,* $\vartheta = 1$ *(*$M = 4$*,* $A_0 = \frac{1}{3}$*) with* $f_- = 0, -2, -5$ *(*$J_C = 8, 12, 18$*): the predicted* $\nu_b\kappa = -42.667, -96.000, -216.000$ *against the computed eigenvalue of largest modulus* $\nu\kappa = -42.669, -95.996, -215.980$ *at* $\kappa = 3.0 \cdot 10^{-3}$ *(computed spectra of the continuation; the largest* positive *eigenvalues there are* 0.0029, 0.0033, 0.0039*). At larger* κ *the agreement is* $-42.787, -93.471, -205.700$ *at* $\kappa = 1$*,* $-42.698, -95.878, -215.448$ *at* $\kappa = 0.0713$*,* $-42.677, -95.972, -215.864$ *at* $\kappa = 0.019$*. Negative bubble,* $f_- = -5$ *(*$J_C = 2$*): predicted* $+2.667$*, computed* 2.660 *at* $\kappa = 3.16 \cdot 10^{-3}$*. Figure 11.4.*

Numerical observation 11.34 (three solutions at $f_- = -5$ and the index count)**.** *For* $c' = (0, 0, -8)$*,* $\vartheta = 1$*,* $f_- = -5$ *(outside (H):* $f_- < 0$*) and* $\kappa \approx 3 \cdot 10^{-3}$ *the continuation shows three solutions: a* trivial *one with* $J/\kappa = 0.7975$ *(at* $\kappa = 2.0\cdot10^{-3}$*) and largest eigenvalue* $7.4 \cdot 10^{-4} < 1$ *(index* $+1$*); a negative bubble with largest eigenvalue* $842.0 > 1$ *(index* -1*); and a positive bubble with largest eigenvalue* $0.0039 < 1$ *(index* $+1$*). The sum of the indices is* 1*, in agreement with Theorem 8.42. The trivial and the negative branch merge in a fold at* $\kappa \approx 0.650$*. Neither the existence*

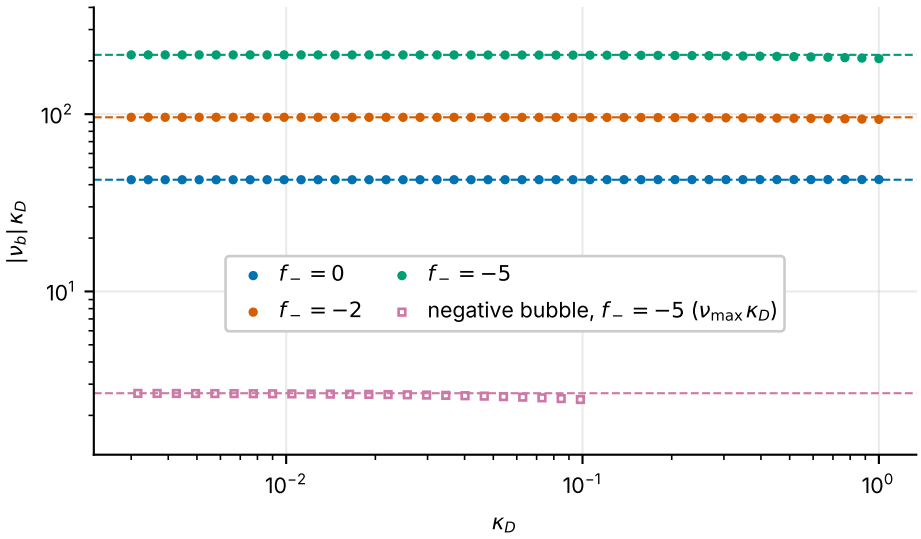


FIGURE 11.4. $|\nu_b|\kappa$ against κ for the bubbles of Numerical observation 11.33: curves computed from the spectra of the continuation; dashed horizontal lines: $\vartheta J_C^2 M A_0/(1+\vartheta)$ of Formal computation 11.32. Code: `code/ch11_small/make_a_eigen.py`.

of the negative bubble nor its limits are proved here (cf. Theorem 11.6, which needs $0 \le f_-$*).*

11.5. The cut lemma at the lower edge of the lens

In words. *The first half of the chapter looked at data* below *the no-swirl set, where a swirl is forced. The second half looks at the opposite end: data in the interior of the no-swirl set, where the swirl may be as small as we like, and where one asks how the lens of Definition 5.55 opens up when* κ *leaves* 0*. At* $\kappa = 0$ *the lens is the window* $[U_{\boldsymbol{c}}^-, U_{\boldsymbol{c}}^+]$ *of Theorem 3.44; for* $\kappa > 0$ *its lower edge* $\gamma^-(\kappa)$ *is the smallest initial slope for which the solution survives up to the north pole. The identity (E) of Proposition 3.51 turns the distance* $\gamma^- - U_{\boldsymbol{c}}^-$ *into an integral* $\int \psi u_+ z$ *of the swirl integral* ψ*, which is of order* $\varkappa^2$*. If* ψ *were exactly* $\varkappa^2$ *times its first-order shape and* z *exactly* u_+*, we would be done. Two things can go wrong near the pole: the first-order shape* $\hat{\psi}_+ u_+^2$ *may fail to be integrable (and then the integral lives far out where first order is false), and* z *may drift away from* u_+*. The* cut lemma *of this section says that both are controlled up to a distance* T *from the equator at which the swirl integral is still small,* $\Pi(T) \ll 1$*, and that beyond* T *the rest of the integral is exponentially small. Everything else in Sections 11.6–11.7 is an application.*

Setting of Sections 11.5–11.7. Here and in the next two sections we work in the *Cauchy normalisation* of Chapters 5 and 7: the datum is a point $(x_0, \boldsymbol{c}, \kappa, \gamma)$ with $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$ the Taylor constants of Q at x_0, x_0 the zero of f, $\kappa = \kappa_x = f_x(x_0)$ (normalisation 1 of Table 2.1) and $\gamma = U(x_0)$. A bare κ is therefore κ_x in these

three sections, and the slope of the swirl in the line variable at the base point is $\varkappa = (1-x_0^2)\kappa_x$. In Section 11.8 we return to the effective data $(\boldsymbol{c}', \vartheta, f_-, \kappa = \kappa_{\mathrm{D}})$ of the first half. The exponent at the north pole is $k := k_2 = \sqrt{1+c_2}$; the south pole is the mirror image (Proposition 5.56).

Definition 11.35 (the edge solution and its objects)**.** Let $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$, $x_0 \in (-1,1)$, $\xi_0 = \operatorname{artanh} x_0$, $\varepsilon = c_3 - \bar{c}_3 > 0$, $W_0 = W_{\mathrm{crit}} + \frac{\varepsilon}{2}\operatorname{sech}^2\xi$ (the potential of the no-swirl problem, W_{crit} as in Definition 11.1 with (c_1, c_2) in place of (c_1', c_2')), and let $u_\pm$ be its Jost solutions normalised by $u_\pm(\xi_0) = 1$, $\zeta_\pm = u_\pm'/u_\pm$. (Part a normalises Jost solutions at $\xi = 0$; in these sections the base point is ξ_0.) For $0 < \kappa \le \kappa_{\max}(x_0, \boldsymbol{c})$ let S be the class-Z solution with data $(x_0, \boldsymbol{c}, \kappa, \gamma^-(\kappa))$ (the point of the lower edge of $\mathrm{Lens}(x_0, \boldsymbol{c})$). In the gauge $z(\xi_0) = 1$ it has

$$f(\xi_0) = 0, \quad f' = \varkappa z^{-2}, \quad z'' = \big(W_0 - \tfrac{1}{2}\psi\big)z, \quad h = z'/z,$$
$$L[\psi] = f^2, \quad \psi(\xi_0) = \psi'(\xi_0) = 0,$$

with $L\psi = \cosh^2\xi\,\psi'' + 2\sinh 2\xi\,\psi' - 2\psi$ (the operator $(1-x^2)\partial_x^2 + 2x\partial_x - 2$ of Remark B.7 in the variable ξ). We write

$$t = \xi - \xi_0, \qquad \Delta(\kappa) := \gamma^-(\kappa) - U_{\boldsymbol{c}}^-(x_0) \ > 0,$$
$$\mathfrak{a} := z/u_+, \quad \mathfrak{l} := h - \zeta_+, \quad \mathcal{T}(t) := \int_{\xi_0+t}^{\infty} \psi\, u_+ z\, d\xi,$$

for the *amplification*, the *lean* and the *tail mass*. All functions of ξ are also regarded as functions of t, and $u := u_+(\xi_0 + t)$.

Facts used. (F1) By Corollary 3.52, $\zeta_+ \le h \le \zeta_-$ on $\mathbb{R}$. Since $(z/u_+)' = \mathrm{Wr}(u_+, z)/u_+^2 = (h - \zeta_+)\, z/u_+$, the amplification $\mathfrak{a} = z/u_+$ is nondecreasing, $\mathfrak{a}(0) = 1$, and $z \ge u_+$ on $[\xi_0, \infty)$. Moreover $\mathrm{Wr}(u_+, z)' = -\frac{1}{2}\psi u_+ z$ and $\mathrm{Wr}(u_+, z) \to 0$ at $+\infty$ ((3.9) and the proof of Proposition 3.51), so $\mathrm{Wr}(u_+, z)(\xi) = \frac{1}{2}\int_\xi^\infty \psi u_+ z$. In terms of the objects of Definition 11.35 this reads, for all $t \ge 0$,

$$\mathcal{T} = 2u^2\,\mathfrak{a}\,\mathfrak{l}, \qquad \mathfrak{a}' = \frac{\mathcal{T}}{2u^2},$$
$$\Delta = \mathcal{T}(0) = \int_{\xi_0}^{\infty} \psi\, u_+ z\, d\xi \quad (\text{in } U\text{-units, Proposition 3.51}). \tag{11.4}$$

(F2) By Proposition 5.59(c) the north pole of S is of type R, L_1 or L_2; by the definition of the types (Theorem 4.5) $U(1) = 2h(+\infty) - 2 \le -2$, i.e. $h(+\infty) \le 0$. (F3) By Lemma 5.41(a),(e): ψ is nondecreasing and $0 \le \psi \le 2 + 2c_2 = 2k^2$. (F4) By Lemma 3.21: $u > 0$, $u\,e^{kt} \to G_+ \in (0, \infty)$, $\zeta_+ \to -k$ and $W_0 \to k^2$ as $t \to \infty$. Hence there is $C_u \ge 1$, depending on $(\boldsymbol{c}, x_0)$ only, with $C_u^{-1}e^{-kt} \le u \le C_u e^{-kt}$ for $t \ge 0$.

First order, growth and cut.

Definition 11.36 (first order)**.** Put $F_+(t) := \int_0^t u(s)^{-2}ds$ (the swirl profile at first order: $f \approx \varkappa F_+$), let $\hat{\psi}_+$ solve $L[\hat{\psi}_+] = F_+^2$ with $\hat{\psi}_+(\xi_0) = \hat{\psi}_+'(\xi_0) = 0$, and

$$A_+(T) := \int_0^T \hat{\psi}_+\, u^2\, dt, \qquad A_+^{\lim} := \lim_{T\to\infty} A_+(T) \in (0, \infty].$$

The *growth function* ω_k and the *cut parameter* Π are

$$\omega_k(t)=\begin{cases} e^{(4k-2)t} & k>\tfrac12\\ 1+t & k=\tfrac12\\ 1 & k<\tfrac12\end{cases},\qquad \Pi(T)=\begin{cases}\varkappa^2 e^{(4k-2)T} & k>\tfrac12,\\ \varkappa^2(1+T)^2 & k\le\tfrac12.\end{cases}$$

$A_+^{\lim}$ is the coefficient of the small-swirl law (Theorem 11.44); it is finite iff $k<1$ (Lemma 11.38). $\Pi(T)$ is, up to a constant, the size of the swirl integral at distance T from the equator (Lemma 11.39).

Lemma 11.37 (the kernel of ψ)**.** *Let f be C^1 with $f(\xi_0)=0$ and let ψ solve $L[\psi]=f^2$, $\psi(\xi_0)=\psi'(\xi_0)=0$.*

(a) $\psi(\xi)=\displaystyle\int_{\xi_0}^{\xi}\frac{\sinh 2(\xi-s)}{2\cosh^2\xi}\,f(s)^2ds=\int_{\xi_0}^{\xi}K(\xi,\sigma)f(\sigma)f'(\sigma)\,d\sigma$, *with*

$$K(\xi,\sigma)=\frac{\sinh^2(\xi-\sigma)}{\cosh^2\xi}=e^{-2\sigma}\frac{(1-e^{-2(\xi-\sigma)})^2}{(1+e^{-2\xi})^2}.$$

(b) (*monotonicity*) *If $0\le f_1\le f_2$ on $[\xi_0,\xi]$, then $\psi[f_1]\le\psi[f_2]$ on $[\xi_0,\xi]$.*

(c) $0\le K(\xi,\sigma)\le e^{-2\sigma}$ *for $\sigma\le\xi$, and $K(\xi,\sigma)\ge\frac14(1-e^{-2})^2e^{-2\sigma}$ if $\xi-\sigma\ge1$ and $\xi\ge0$.*

(d) $\displaystyle\int_0^\infty e^{-2d}(1-e^{-2d})^2\,dd=\frac16.$

Proof. (a) For $\phi=\cosh^2\xi\,\psi$ one computes $\phi''-4\phi=\cosh^2\xi\,\psi''+2\sinh 2\xi\,\psi'-2\psi=L[\psi]=f^2$ with $\phi(\xi_0)=\phi'(\xi_0)=0$; variation of constants gives $\phi=\int_{\xi_0}^{\xi}\frac12\sinh 2(\xi-s)f(s)^2ds$, the first formula. Since $\frac{d}{ds}\sinh^2(\xi-s)=-\sinh 2(\xi-s)$ and $f(\xi_0)=0$, integration by parts has vanishing boundary terms and turns $\int\frac12\sinh 2(\xi-s)f^2ds$ into $\int\sinh^2(\xi-s)ff'ds$. The exponential form of K follows from $\sinh^2 d=\frac14e^{2d}(1-e^{-2d})^2$ and $\cosh^2\xi=\frac14e^{2\xi}(1+e^{-2\xi})^2$. (b) The kernel $\sinh 2(\xi-s)/(2\cosh^2\xi)$ is ≥0 for $s\le\xi$. (c) With $d=\xi-\sigma\ge0$: $\sinh d=\frac12(e^d-e^{-d})\le\frac12(e^d+e^{-\xi-\sigma})=e^{-\sigma}\cosh\xi$, which gives $K\le e^{-2\sigma}$. If $d\ge1$ and $\xi\ge0$: $\sinh^2 d\ge\frac14e^{2d}(1-e^{-2})^2$ and $\cosh^2\xi\le e^{2\xi}$, so $K\ge\frac14(1-e^{-2})^2e^{2d-2\xi}=\frac14(1-e^{-2})^2e^{-2\sigma}$. (d) $\int_0^\infty(e^{-2d}-2e^{-4d}+e^{-6d})dd=\frac12-\frac12+\frac16$. □

Lemma 11.38 (growth of the first-order objects)**.** *There are constants $C_F,C_\psi\ge1$ (depending on $\mathbf{c},x_0$ only) such that for $t\ge0$:* (a) $F_+(t)\le C_Fe^{2kt}$, $u(t)^2F_+(t)\le C_F$, $u(t)^{-2}\le C_u^2e^{2kt}$; (b) $\hat\psi_+(t)\le C_\psi\,\omega_k(t)$, *and* $\int_0^T\omega_k\le C_\psi\,\Pi(T)/\varkappa^2$; (c) $\hat\psi_+u^2\le C_\psi\,\omega_k(t)e^{-2kt}$; *hence $A_+^{\lim}<\infty$ if $k<1$, with $A_+(T)\le A_+^{\lim}$ and $A_+^{\lim}-A_+(T)\le C\,(1+T)^2e^{-2\min(k,1-k)T}$;* (d) *if $k>1$ there are $c_0>0$, $t_1\ge0$ with $\hat\psi_+(t)u(t)^2\ge c_0e^{(2k-2)t}$ for $t\ge t_1$; hence $A_+(T)\to\infty$ and $A_+(T)\ge c_0'e^{(2k-2)T}$ for large T;* (e) *if $k=1$ then $\hat\psi_+u^2\to e^{-2\xi_0}/(12G_+^2)$ as $t\to\infty$, and $A_+(T)=\frac{e^{-2\xi_0}}{12G_+^2}T+o(T)$.*

Proof. (a) By (F4), $F_+\le C_u^2\int_0^t e^{2ks}ds\le C_u^2e^{2kt}/(2k)$, $u^2F_+\le C_u^4/(2k)$, and $u^{-2}\le C_u^2e^{2kt}$. (b) By Lemma 11.37(a),(c) with $f=F_+$, $F_+'=u^{-2}$:

$$\hat\psi_+(t)\le e^{-2\xi_0}\int_0^t e^{-2s}F_+u^{-2}ds\le e^{-2\xi_0}C_FC_u^2\int_0^t e^{(4k-2)s}ds,$$

and $\int_0^t e^{(4k-2)s}ds\le\omega_k(t)\cdot\max(\frac{1}{4k-2},1)$ for $k>\frac12$ (by $e^{(4k-2)t}$), $\le t\le\omega_k$ for $k=\frac12$, and $\le\frac{1}{2-4k}$ for $k<\frac12$. Also $\int_0^T\omega_k\le\frac{e^{(4k-2)T}}{4k-2}$, $\frac12(1+T)^2$, $1+T$ in the three

cases, each $\le C\Pi(T)/\varkappa^2$. (c) The first claim is (a), (b); for $k<1$ the exponent $(4k-2)-2k=2k-2<0$ (case $k>\frac12$) and $-2k<0$ (cases $k\le\frac12$) make the integrand integrable, and $A_+^{\lim}-A_+(T)=\int_T^\infty \hat\psi_+u^2\le C(1+T)^2e^{-2\min(k,1-k)T}$ in all cases. (d) For $t\ge t_1$ so large that $t_1\ge1$ and $\xi_0+t\ge0$, and $k>1$: $F_+(s)\ge C_u^{-2}(e^{2ks}-1)/(2k)\ge c\,e^{2ks}$ for $s\ge1$ and $u^{-2}\ge C_u^{-2}e^{2ks}$, so by Lemma 11.37(c) (valid for $\xi-\sigma\ge1$, i.e. $s\le t-1$)

$$\hat\psi_+(t)\ge\tfrac14(1-e^{-2})^2e^{-2\xi_0}\int_1^{t-1}e^{-2s}\,c\,e^{2ks}C_u^{-2}e^{2ks}ds\ge c'e^{(4k-2)t},$$

and $u^2\ge C_u^{-2}e^{-2kt}$. Integrating $e^{(2k-2)t}$ gives the last claim. (e) Let $k=1$, $G=G_+$ and $\varphi(s):=e^{-4s}F_+(s)u(s)^{-2}$. By (F4), $u^{-2}e^{-2s}\to G^{-2}$, hence $e^{-2s}F_+(s)\to\frac{1}{2G^2}$ and $\varphi(s)\to\frac{1}{2G^4}$; φ is bounded by (a). The exponential form of K with $\sigma=\xi_0+s$, $\xi=\xi_0+t$ gives

$$\begin{aligned}\hat\psi_+(t)&=\frac{e^{-2\xi_0}}{(1+e^{-2\xi})^2}\int_0^t e^{-2s}\big(1-e^{-2(t-s)}\big)^2F_+u^{-2}\,ds\\&=\frac{e^{-2\xi_0}e^{2t}}{(1+e^{-2\xi})^2}\int_0^t e^{-2d}(1-e^{-2d})^2\varphi(t-d)\,dd,\end{aligned}$$

because $e^{-2s}F_+u^{-2}=e^{2s}\varphi(s)$ and $s=t-d$. The integrand is dominated by $\sup\varphi\cdot e^{-2d}$, so by dominated convergence and Lemma 11.37(d) the integral tends to $\frac{1}{2G^4}\cdot\frac16$, and $\hat\psi_+e^{-2t}\to\frac{e^{-2\xi_0}}{12G^4}$. With $u^2e^{2t}\to G^2$ this is the first claim; the second follows because the Cesàro means of a convergent function converge to the same limit. □

Lemma 11.39 (edge a priori bounds and the cut lemma)**.** *In the setting of Definition 11.35 let* $0<\varkappa\le1$.

(a) (Sandwich.) *For every* $T\ge0$, *with* $q:=\mathfrak a(T)$:

$$f\le\varkappa F_+\ \text{ on }[0,\infty),\qquad \varkappa^2q^{-4}\hat\psi_+\le\psi\le\varkappa^2\hat\psi_+\ \text{ on }[0,T],$$
$$\varkappa^2q^{-4}A_+(T)\ \le\ \Delta\ \le\ q\,\varkappa^2A_+(T)+\mathcal T(T).$$

(b) (Cut.) *There are* $\lambda_0\in(0,1]$, $C_1\ge1$ *and, for every* $\lambda\in(0,\lambda_0]$, *numbers* $\eta(\lambda)\in(0,\lambda]$ *and* $T_0(\lambda)$ *with the following dependence, and none on* κ. *Let* C_u *be the constant of* (F4) *and, for* $\lambda>0$, *let* $T_W(\lambda)\ge0$ *be a time such that* $W_0\ge k^2-\lambda k/16$ *and* $\zeta_+\ge-k-\lambda/4$ *on* $[T_W(\lambda),\infty)$ (*it exists by* (F4)). *Then* λ_0 *and* C_1 *depend only on* (k,x_0,C_u); $\eta(\lambda)$ *depends only on* (λ,k,x_0,C_u); *and* $T_0(\lambda)$ *may be taken to be* $T_W(\lambda)$. *If* $T\ge T_0(\lambda)$ *and* $\Pi(T)\le\eta(\lambda)$, *then*

$$\mathfrak l(T)<\lambda,\qquad 1\le\mathfrak a(T)\le1+C_1\lambda,\qquad \mathcal T(T)\le2C_u^2(1+C_1\lambda)\,\lambda\,e^{-2kT}.$$

Proof. (a) By (F1), $z\ge u_+$, so $f(t)=\varkappa\int_0^t z^{-2}\le\varkappa F_+(t)$. For $t\le T$ we have $z\le\mathfrak a(T)u_+=qu_+$ (monotonicity of $\mathfrak a$), so $f(t)\ge\varkappa q^{-2}F_+(t)$. By Lemma 11.37(a),(b), $L[\psi]=f^2$ and $L[\varkappa^2\hat\psi_+]=\varkappa^2F_+^2$ give $\psi\le\varkappa^2\hat\psi_+$, and $\psi\ge\varkappa^2q^{-4}\hat\psi_+$ on $[0,T]$. By (11.4), $\Delta=\int_0^T\psi uz\,dt+\mathcal T(T)$ with $u^2\le uz\le qu^2$ on $[0,T]$, which gives the third chain. (b) Fix $\lambda\in(0,\lambda_0]$; $\lambda_0\le\min(1,k)$ and the constants C_1,λ_0 are fixed in Step 2. Put $m:=k-\lambda/8$, $\psi_1:=\lambda k/4$, put $T_0(\lambda):=T_W(\lambda)$, so that for $t\ge T_0$: $W_0\ge k^2-\lambda k/16$ and $\zeta_+\ge-k-\lambda/4$. Then for $t\ge T_0$ and $\psi\le\psi_1$ one has $W_0-\frac12\psi\ge k^2-\frac{3\lambda k}{16}\ge m^2$ (because $m^2=k^2-\frac{\lambda k}{4}+\frac{\lambda^2}{64}$ and $\lambda\le k$). Fix $T\ge T_0$ with $\Pi(T)\le\eta$, where $\eta=\eta(\lambda)$ is determined in Step 1. In

what follows C_F, C_ψ and the constants $C_2, C_3, C_4, C_\Upsilon$ are those of Lemma 11.38 and of the steps below; they are fixed by (k, x_0, C_u) and, for C_3, C_4, also by λ, as recorded in the remark after the proof.

Step 1: $\mathfrak{l}(T) < \lambda$. Suppose $\mathfrak{l}(T) \geq \lambda$, i.e. $h(T) \geq \zeta_+(T) + \lambda \geq -k + \frac{3}{4}\lambda =: \theta_0$; note $\theta_0 > -m$. Let θ solve $\theta' = m^2 - \theta^2$, $\theta(T) = \theta_0$: it is increasing, $\theta(T + s) = m\tanh(ms + a)$ with $a = \operatorname{artanh}(\theta_0/m)$ depending on λ and k only, and $\theta \to m$. Let $t_2(\lambda)$ be such that $\theta(T + s) \geq m/2$ for $s \geq t_2$ (independent of T). Let $\Xi := \sup\{t \geq T : \psi \leq \psi_1 \text{ on } [T, t]\}$ (we show below that $\Xi = \infty$). On $[T, \Xi)$, $h' = W_0 - \frac{1}{2}\psi - h^2 \geq m^2 - h^2$, so by comparison $h \geq \theta$ there. Hence $z(T + s) \geq z(T)e^{-ks}$ for $s \leq t_2$ and $z(T + s) \geq z(T)e^{-kt_2}e^{\frac{m}{2}(s-t_2)}$ for $s \geq t_2$, and

$$f(t) - f(T) = \varkappa \int_T^t z^{-2} \leq \varkappa z(T)^{-2} e^{2kt_2}\big(t_2 + \tfrac{1}{m}\big) \leq C_4(\lambda)\, \varkappa\, e^{2kT} \quad (T \leq t < \Xi),$$

by $z(T) \geq u(T) \geq C_u^{-1}e^{-kT}$; with $f(T) \leq \varkappa F_+(T) \leq \varkappa C_F e^{2kT}$ we get $f \leq C_4 \varkappa e^{2kT}$ on $[T, \Xi)$ (enlarging C_4). By Lemma 11.37(a),(c), for $T \leq t < \Xi$,

$$\psi(t) \leq e^{-2\xi_0}\Big[\int_0^T e^{-2s} f f'\, ds + \int_T^t e^{-2s} f f'\, ds\Big].$$

Since $f \leq \varkappa F_+$ and $f' = \varkappa z^{-2} \leq \varkappa u^{-2}$, the first integral is $\leq \varkappa^2 \int_0^T e^{-2s} F_+ u^{-2} ds \leq \varkappa^2 C_F C_u^2 \int_0^T e^{(4k-2)s} ds \leq C\,\Pi(T)$ (as in Lemma 11.38(b)); the second is $\leq e^{-2T} \sup f \cdot (f(t) - f(T)) \leq C_4^2 \varkappa^2 e^{(4k-2)T} \leq C_4^2\, \Pi(T)$ (equal to Π if $k > \frac{1}{2}$; for $k \leq \frac{1}{2}$ it is $\leq C_4^2 \varkappa^2 \leq C_4^2 \Pi$). So $\psi \leq C_3(\lambda)\, \Pi(T)$ on $[T, \Xi)$, and also at $t = T$, where only the first term occurs. Define $\eta(\lambda) := \min\big(\lambda,\ \psi_1/(2C_3(\lambda))\big)$. If $\Pi(T) \leq \eta$, then $\psi(T) \leq \psi_1/2$, so $\Xi > T$, and $\psi \leq \psi_1/2$ on $[T, \Xi)$, so $\Xi = \infty$ by continuity; hence $h \geq \theta$ on $[T, \infty)$ and $h(+\infty) \geq m > 0$, contradicting (F2). Thus $\mathfrak{l}(T) < \lambda$.

Step 2: $\mathfrak{a}(T)$. By (11.4), $\mathfrak{a}(T) - 1 = \int_0^T \mathcal{T}(t)/(2u^2)\, dt$. For $t \leq T$, $\mathcal{T}(t) = \int_t^T \psi u z + \mathcal{T}(T) \leq q \int_t^T \psi u^2 + \mathcal{T}(T)$, so by Fubini

$$q - 1 \leq \frac{q}{2} \int_0^T \psi(s) u(s)^2 F_+(s)\, ds + \frac{1}{2} F_+(T)\, \mathcal{T}(T).$$

By (a) and Lemma 11.38, $\psi u^2 F_+ \leq C\varkappa^2 \omega_k$, so the first integral is $\leq C_\Upsilon \Pi(T)$; and $\frac{1}{2}F_+(T)\mathcal{T}(T) = F_+(T)u(T)^2 \mathfrak{a}(T)\mathfrak{l}(T) \leq C_F q\, \mathfrak{l}(T)$ by (11.4). Hence $q - 1 \leq C_2\, q\, (\Pi(T) + \mathfrak{l}(T)) \leq 2C_2\, q\, \lambda$ once $\Pi \leq \eta \leq \lambda$ and $\mathfrak{l} < \lambda$ (Step 1). Put $\lambda_0 := \min(1, k, 1/(4C_2))$ and $C_1 := 4C_2$: then $2C_2\lambda \leq \frac{1}{2}$ and $q \leq (1 - 2C_2\lambda)^{-1} \leq 1 + 4C_2\lambda = 1 + C_1\lambda$. (Step 1 does not use q, so there is no circularity.)

Step 3: $\mathcal{T}(T)$. By (11.4) and (F4): $\mathcal{T}(T) = 2u(T)^2 \mathfrak{a}\,\mathfrak{l} \leq 2C_u^2 e^{-2kT}(1 + C_1\lambda)\lambda$. □

Remark 11.40 (what the constants depend on)**.** Inspection of the proofs gives the dependence announced in Lemma 11.39(b). In Lemma 11.38, (a) holds with $C_F = \max(1, C_u^4/(2k))$, and in (b) the constant C_ψ is $e^{-2\xi_0} C_F C_u^2$ times a number depending on k only, so C_F, C_ψ depend on (k, x_0, C_u) only. In the proof of Lemma 11.39, Step 1 uses C_u, C_F, C_ψ, k, x_0 and λ (through a, t_2, C_4, C_3), and Step 2 uses C_F, C_ψ, k (through C_Υ and C_2); the threshold $T_W(\lambda)$ enters only as the lower limit $T \geq T_0$ and never in a bound. Hence $\lambda_0 = \min(1, k, 1/(4C_2))$ and $C_1 = 4C_2$ depend on (k, x_0, C_u), $\eta(\lambda)$ on (λ, k, x_0, C_u), and $T_0(\lambda) = T_W(\lambda)$ on the pair (W_0, ζ_+) only through the time after which the two inequalities of the lemma hold.

The datum c_3, and with it the distance $\varepsilon = c_3 - \bar{c}_3$ to the critical surface, and the slope κ enter the lemma *only* through C_u and $T_W(\lambda)$. The lemma is therefore uniform over every family of data with the same (k, x_0) for which C_u and $T_W(\lambda)$, $\lambda \in (0, \lambda_0]$, stay bounded; in particular its conclusions are uniform as $\varepsilon \downarrow 0$ as soon as these two quantities are. For the family $c_3 \downarrow \bar{c}_3$ with (c_1, c_2, x_0) fixed this is checked in Proposition 12.10 of Chapter 12, where the lemma is used.

Toy example 11.41 (the lean of a recessive solution under a small perturbation)**.** Take $W_0 \equiv k^2$ on $t \geq 0$ and replace ψ by the prescribed profile $\psi = b\,e^{-\beta t}$, $b > 0$ small. The recessive solution of $z'' = (k^2 - \frac{1}{2}\psi)z$ is the one whose logarithmic derivative h solves the Riccati equation $h' = k^2 - \frac{1}{2}\psi - h^2$ and stays bounded at $+\infty$ ($h \to -k$); its unperturbed counterpart is $u = e^{-kt}$, $\zeta_+ = -k$. The three behaviours of $h' = m^2 - h^2$ (for constant m) are the whole story: $h \equiv -m$ is a repelling separatrix; $h > -m$ flows to $+m$ (the solution z grows: type G); $h < -m$ runs to $-\infty$ in finite time (z has a zero). The edge of the lens is exactly the separatrix. First order in b: the lean $\mathfrak{l}(0) = h(0) - \zeta_+(0)$ is $\frac{1}{2}\int_0^\infty \psi u^2 = \frac{b}{2(\beta+2k)}$, i.e. $\Delta = 2\mathfrak{l}(0) = b/(\beta + 2k)$. For $k = 1$, $\beta = 1$, $b = 10^{-4}$, integrating the Riccati equation backwards from the separatrix at $t = 40$ gives $\Delta = 3.33335 \cdot 10^{-5}$, against $b/(\beta + 2k) = 3.33333 \cdot 10^{-5}$; for $k = 0.7$, $\beta = 2$ the ratio of the two is 1.000003 (`code/ch11_small/verify_ch11b.py`). The integral $\int \psi u^2$ is the first-order version of (11.4); what makes the real problem harder is that ψ is not given but is $\varkappa^2$ times a profile that *grows* like $e^{(4k-2)t}$ until the nonlinearity stops it.

Toy example 11.42 (where the integral lives: exponent bookkeeping for $k = 2$)**.** Let $k = 2$ and $u = e^{-2t}$ (the case $W_0 \equiv 4$, $x_0 = 0$). Then $F_+ \approx e^{4t}/4$, $\hat{\psi}_+ \asymp e^{6t}$ and $\hat{\psi}_+ u^2 \sim e^{2t}$, so the first-order mass $\varkappa^2 A_+(T) \sim \varkappa^2 e^{2T}$ is dominated by the far end T. First order stops being valid when the swirl integral reaches order 1, i.e. $\varkappa^2 e^{6T} \sim \eta$. At that distance $e^{2T} = (\eta/\varkappa^2)^{1/3}$ and the mass is $\sim \varkappa^2(\eta/\varkappa^2)^{1/3} = \eta^{1/3}\varkappa^{4/3}$: the exponent $\frac{4}{3} = \frac{2k}{2k-1}$ of Theorem 11.44(c) appears as the *balance of growth against cut*, and the beyond-the-cut mass obeys $\mathcal{T}(T) \lesssim e^{-2kT} = e^{-4T} = (\varkappa^2/\eta)^{2/3} = \eta^{-2/3}\varkappa^{4/3}$, which is of the same order $\varkappa^{4/3}$. For $k < 1$ the mass $\int e^{(2k-2)t}dt$ converges at $t \sim 1$ and $\Pi \to 0$ at the natural cut $e^{-2kT} = \varkappa^2$; for $k = 1$ it grows linearly, which is the logarithm.

Figure 11.5 shows the cut for a computed edge solution.

Numerical observation 11.43 (the cut lemma on computed edge solutions)**.** *For $\boldsymbol{c} = (k^2 - 1, k^2 - 1, 0)$, $x_0 = 0$ the edge solution was computed by shooting in γ (bisection to 10^{-15}) and the quantities of Definition 11.35 were read off at $T = \ln(1/\varkappa)/k$ for $k < 1$ and at T with $\Pi(T) = \eta := 0.05$ for $k \geq 1$. Table 11.3 gives $\mathfrak{a}(T) - 1$, $\mathfrak{l}(T)$, the fraction $\mathcal{T}(T)/\Delta$ of the integral that lies beyond T, and Δe^{2kT}. For $k < 1$ the amplification and the lean tend to 0 with $\varkappa$ and the beyond-the-cut fraction too, so that $\Delta \approx \varkappa^2 A_+^{\lim}$ (the number $\Delta e^{2kT} = \Delta/\varkappa^2$ is $A_+^{\lim}$: 0.2517 against $A_+^{\lim} = 0.25168$). For $k \geq 1$ the amplification and the lean stay at a fixed small level, as the lemma predicts (both are below $\eta = 0.05$), and Δe^{2kT} becomes independent of $\varkappa$ for $k > 1$: for $k = \sqrt{2}$ and $k = 2$ the fraction $\mathcal{T}(T)/\Delta$ of the integral that lies* beyond *the cut is* 0.96 *and* 0.996*, so that the upper bound of Lemma 11.39(b) is the whole story there; for $k = 1$ the fraction beyond the cut decreases like $1/\ln(1/\varkappa)$. Code:* `code/ch11_small/gen_b_data.py`*,* `verify_ch11b.py`*.*

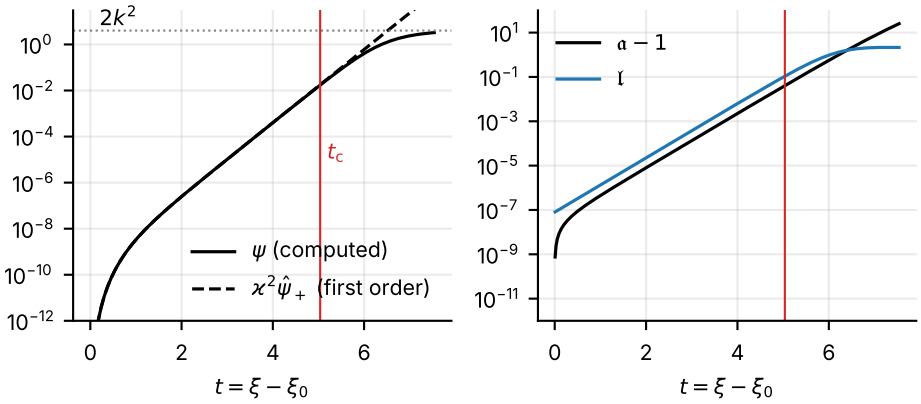


FIGURE 11.5. The cut for $\boldsymbol{c} = (1, 1, 0)$, $x_0 = 0$ ($k = \sqrt{2}$), $\varkappa = 10^{-4}$, computed edge solution. Left: the swirl integral ψ (solid) follows its first-order form $\varkappa^2\hat{\psi}_+$ (dashed) until it saturates below the bound $2k^2$ of (F3); the red line is the position $t_c = \ln(1/\varkappa)/(2k-1)$ at which $\Pi = 1$. Right: the amplification $\mathfrak{a}(t) - 1$ (black) and the lean $\mathfrak{l}(t)$ (blue): both are tiny up to the cut and then grow.

k	$\varkappa$	T	$\mathfrak{a}(T) - 1$	$\mathfrak{l}(T)$	$\mathcal{T}(T)/\Delta$	Δe^{2kT}
0.707	10^{-2}	6.51	$1.1 \cdot 10^{-2}$	$9.1 \cdot 10^{-3}$	0.073	0.2513
	10^{-3}	9.77	$1.7 \cdot 10^{-3}$	$1.4 \cdot 10^{-3}$	0.011	0.2517
	10^{-4}	13.03	$2.5 \cdot 10^{-4}$	$2.0 \cdot 10^{-4}$	0.0016	0.2517
1	10^{-2}	3.11	$6.05 \cdot 10^{-3}$	$1.1 \cdot 10^{-2}$	0.77	0.029
	10^{-3}	5.41	$6.14 \cdot 10^{-3}$	$1.1 \cdot 10^{-2}$	0.58	0.039
	10^{-4}	7.71	$6.15 \cdot 10^{-3}$	$1.1 \cdot 10^{-2}$	0.47	0.048
1.414	10^{-2}	1.70	$4.1 \cdot 10^{-3}$	$1.15 \cdot 10^{-2}$	0.98	0.0235
	10^{-3}	2.96	$4.1 \cdot 10^{-3}$	$1.15 \cdot 10^{-2}$	0.97	0.0240
	10^{-4}	4.22	$4.1 \cdot 10^{-3}$	$1.15 \cdot 10^{-2}$	0.96	0.0241
2	10^{-2}	1.04	$3.05 \cdot 10^{-3}$	$1.2 \cdot 10^{-2}$	0.9985	0.0248
	10^{-3}	1.80	$3.1 \cdot 10^{-3}$	$1.2 \cdot 10^{-2}$	0.9968	0.0249
	10^{-4}	2.57	$3.1 \cdot 10^{-3}$	$1.2 \cdot 10^{-2}$	0.9961	0.0249

TABLE 11.3. The cut lemma on computed edge solutions ($\boldsymbol{c} = (k^2 - 1, k^2 - 1, 0)$, $x_0 = 0$; $k = 0.707$ means $k = 1/\sqrt{2}$, $k = 1.414$ means $\sqrt{2}$).

11.6. Three laws at the edge of the no-swirl window

In words. *By the sandwich of Lemma 11.39(a), $\Delta = \gamma^- - U_{\boldsymbol{c}}^-$ is trapped between $\varkappa^2 A_+(T)/q^4$ and $q\varkappa^2 A_+(T) + \mathcal{T}(T)$, and the cut lemma says that at the cut T the factor q is $1 + O(\lambda)$ and the tail $\mathcal{T}(T)$ is exponentially small. So everything*

depends on how fast $A_+(T)=\int_0^T \hat\psi_+ u_+^2$ grows, and this is a one-line computation: $\hat\psi_+ u_+^2 \approx e^{(2k-2)t}$. For $k<1$ the integral converges and $\Delta \approx A_+^{\lim}\varkappa^2$; for $k=1$ it grows linearly, and the only thing to determine is the slope, $\frac{e^{-2\xi_0}}{12G_+^2}$ (the number $\frac{1}{12}$ comes from $\int_0^\infty e^{-2d}(1-e^{-2d})^2=\frac16$ halved by $F_+\approx e^{2t}/2$); for $k>1$ it is dominated by the cut, and balancing the cut position against the exponent gives the rate $\varkappa^{2k/(2k-1)}$, with no constant. We then record the closed forms that exist (digamma for $k<1$; a rest-state formula at $k=1$).

THEOREM 11.44 (three laws at the lower edge). *Let $\boldsymbol{c}\in\operatorname{int}\mathcal{J}$, $x_0\in(-1,1)$, $k=\sqrt{1+c_2}$, $\varkappa=(1-x_0^2)\kappa_x$ and $\Delta(\kappa_x)=\gamma^-(x_0,\boldsymbol{c},\kappa_x)-U_{\boldsymbol{c}}^-(x_0)$, in the setting of Definition 11.35. As $\kappa_x\downarrow 0$:*

(a) *($k<1$) $\Delta=a_*\varkappa^2+o(\varkappa^2)$ with $a_*:=A_+^{\lim}=\int_0^\infty \hat\psi_+u_+^2\,dt\in(0,\infty)$; in the variable κ_x, $\Delta=A_+^{\lim}(1-x_0^2)^2\kappa_x^2+o(\kappa_x^2)$. For $x_0=0$, $c_1=c_2=k^2-1$, $c_3=0$ the constant is the digamma expression of Proposition B.12.*

(b) *($k=1$) $\Delta=\dfrac{e^{-2\xi_0}}{12\,G_+^2}\;\varkappa^2\ln\dfrac{1}{\varkappa}\,(1+o(1))$, where $G_+=\lim_{t\to\infty}e^t u_+(\xi_0+t)>0$.*

(c) *($k>1$) There are $0<c_-\le c_+<\infty$ and $\varkappa_0>0$ such that $c_-\,\varkappa^{2k/(2k-1)}\le\Delta\le c_+\,\varkappa^{2k/(2k-1)}$ for $\varkappa\le\varkappa_0$.*

PROOF. All three use the sandwich (a) and the cut (b) of Lemma 11.39.

(a) Let $\lambda\in(0,\lambda_0]$. Put $T_*:=\ln(1/\varkappa)/k$. Then $e^{-2kT_*}=\varkappa^2$ and $\Pi(T_*)=\varkappa^{(2-2k)/k}$ if $k>\frac12$, $\Pi(T_*)=\varkappa^2(1+T_*)^2$ if $k\le\frac12$; in both cases $\Pi(T_*)\to0$ as $\varkappa\to0$ since $k<1$. So for $\varkappa$ small $T_*\ge T_0(\lambda)$ and $\Pi(T_*)\le\eta(\lambda)$, and Lemma 11.39(b) gives $q:=\mathfrak{a}(T_*)\le1+C_1\lambda$ and $\mathcal{T}(T_*)\le 2C_u^2(1+C_1\lambda)\lambda\,\varkappa^2$. By Lemma 11.39(a),

$$(1+C_1\lambda)^{-4}A_+(T_*)\;\le\;\frac{\Delta}{\varkappa^2}\;\le\;(1+C_1\lambda)A_+(T_*)+2C_u^2(1+C_1\lambda)\lambda.$$

As $\varkappa\to0$, $T_*\to\infty$ and $A_+(T_*)\to A_+^{\lim}<\infty$ (Lemma 11.38(c)); hence $(1+C_1\lambda)^{-4}A_+^{\lim}\le\liminf\Delta/\varkappa^2\le\limsup\Delta/\varkappa^2\le(1+C_1\lambda)A_+^{\lim}+C\lambda$ for every λ, and $\Delta/\varkappa^2\to A_+^{\lim}$. $A_+^{\lim}>0$ since $\hat\psi_+>0$ for $t>0$. The conversion to κ_x is $\varkappa=(1-x_0^2)\kappa_x$. The digamma value is Proposition B.12.

(b) Let $k=1$, $s_1:=e^{-2\xi_0}/(12G_+^2)$, $\lambda\in(0,\lambda_0]$, $\eta=\eta(\lambda)$ and choose $T:=\ln(1/\varkappa)-K_\lambda$ with $K_\lambda:=\frac12\ln(1/\eta)$, so that $\Pi(T)=\varkappa^2e^{2T}=e^{-2K_\lambda}=\eta$. For $\varkappa$ small $T\ge T_0(\lambda)$. By Lemma 11.39(b), $q\le1+C_1\lambda$ and $\mathcal{T}(T)\le2C_u^2(1+C_1\lambda)\lambda e^{-2T}=2C_u^2(1+C_1\lambda)\lambda\,\varkappa^2e^{2K_\lambda}$. By Lemma 11.38(e), $A_+(T)=s_1T+o(T)$, so Lemma 11.39(a) yields

$$\begin{aligned}(1+C_1\lambda)^{-4}\,s_1\big(\ln\tfrac{1}{\varkappa}-K_\lambda\big)(1+o(1))\;&\le\;\frac{\Delta}{\varkappa^2}\\ \le\;(1+C_1\lambda)\,s_1\big(\ln\tfrac{1}{\varkappa}-K_\lambda\big)(1+o(1))+C(\lambda).&\end{aligned}$$

Dividing by $\ln\frac1\varkappa\to\infty$ (the constants K_λ, $C(\lambda)$ do not depend on $\varkappa$): $(1+C_1\lambda)^{-4}s_1\le\liminf\Delta/(\varkappa^2\ln\frac1\varkappa)\le\limsup\le(1+C_1\lambda)s_1$. Letting $\lambda\downarrow0$ proves (b). Note that only $o(T)$ in Lemma 11.38(e) was needed: the constant term of $\Delta/\varkappa^2$ is *not* determined by this argument.

(c) Let $k>1$, fix $\lambda=\lambda_0$, $\eta_0:=\eta(\lambda_0)$ and let $t_a=t_a(\varkappa)$ be defined by $\varkappa^2e^{(4k-2)t_a}=\eta_0$, i.e. $t_a=\frac{1}{4k-2}\ln\frac{\eta_0}{\varkappa^2}\to\infty$. Then $\Pi(t_a)=\eta_0$, $e^{-2kt_a}=(\varkappa^2/\eta_0)^{k/(2k-1)}$ and $\varkappa^2e^{(2k-2)t_a}=\eta_0e^{-2kt_a}$. For $\varkappa$ small, $t_a\ge T_0(\lambda_0)$, so the cut lemma gives $q\le1+C_1\lambda_0\le2$ (as $C_1\lambda_0=4C_2\lambda_0\le1$) and $\mathcal{T}(t_a)\le Ce^{-2kt_a}$.

Upper bound. By Lemma 11.38(c), $\hat{\psi}_+ u^2 \le C\, e^{(2k-2)t}$, so $A_+(t_a) \le C e^{(2k-2)t_a}$ (here $2k-2>0$), and by the sandwich $\Delta \le q\varkappa^2 A_+(t_a) + \mathcal{T}(t_a) \le C\varkappa^2 e^{(2k-2)t_a} + Ce^{-2kt_a} \le C' e^{-2kt_a} = C'(\varkappa^2/\eta_0)^{k/(2k-1)}$. *Lower bound.* By the sandwich and Lemma 11.38(d), $\Delta \ge q^{-4}\varkappa^2 A_+(t_a) \ge \frac{1}{16}\varkappa^2 c_0' e^{(2k-2)t_a} = \frac{c_0'\eta_0}{16} e^{-2kt_a}$. Both bounds are constants times $e^{-2kt_a} \asymp \varkappa^{2k/(2k-1)}$. □

Remark 11.45 (what is and is not proved)**.** (i) Only the north exponent $k = k_2$ enters: the upper edge γ^+ of the lens is the mirror image (Proposition 5.56) and obeys the same laws with k_1. (ii) In (a) the next term is not claimed; the numbers of Numerical observation 11.50 show a relative deficit that decreases like $\varkappa^{(2-2k)/k}$, which is the size of the beyond-the-cut mass $\int_{T_*}^{\infty} \hat{\psi}_+ u^2$ (Lemma 11.38(c)); it is an observation. (iii) In (b) the constant term of $\Delta/\varkappa^2$ is not determined; in (c) the constant is not determined at all, the proof gives only the two-sided bound. Whether $\Delta\, \varkappa^{-2k/(2k-1)}$ has a limit, and which constant, is discussed in Chapter 13. (iv) For $k = \frac{1}{2}$ the formula for Π has a $(1+T)^2$; the proof of (a) is unchanged.

Corollary 11.46 (regularity of the lower edge at $\kappa = 0$)**.** *In the setting of Theorem 11.44, extend γ^- by $\gamma^-(0) := U_{\boldsymbol{c}}^-(x_0)$. Then:*

(a) $\gamma^-(\kappa) - \gamma^-(0) = o(\kappa)$ *as $\kappa \downarrow 0$, for every $k > 0$: the right derivative at 0 exists and is 0.*

(b) *If $k < 1$, then $\gamma^-(\kappa) = \gamma^-(0) + (1-x_0^2)^2 A_+^{\lim}\, \kappa^2 + o(\kappa^2)$ (this is Theorem 11.44(a), an expansion in the sense of Peano); so γ^- is twice differentiable at 0 from the right in the sense of a second-order Taylor expansion, with $\gamma^{-\prime\prime}(0) = 2(1-x_0^2)^2 A_+^{\lim}$.*

(c) *If $k \ge 1$, γ^- has no expansion $\gamma^-(0) + a\kappa + b\kappa^2 + o(\kappa^2)$ with finite a, b: indeed $(\gamma^- - \gamma^-(0))/\kappa^2 \to +\infty$.*

Proof. (a) By Theorem 11.44, $\Delta = O(\varkappa^p)$ with $p = 2$ for $k < 1$, $\Delta = O(\varkappa^2 \ln \frac{1}{\varkappa})$ for $k = 1$ and $p = \frac{2k}{2k-1}$ for $k > 1$; in all cases the exponent exceeds 1 ($\frac{2k}{2k-1} > 1$ because $2k > 2k-1$; and $\varkappa^2 \ln \frac{1}{\varkappa} = o(\varkappa)$), so $\Delta/\kappa \to 0$. (b) is Theorem 11.44(a): $\Delta/\kappa^2 \to (1-x_0^2)^2 A_+^{\lim}$, and $\gamma^-(0) + a\kappa + b\kappa^2 + o(\kappa^2)$ with $a = 0$ is the stated expansion. (c) Since $\Delta = o(\kappa)$ by (a) the only possible expansion has $a = 0$, and then Δ/κ^2 would be bounded; but $\Delta/\varkappa^2 \ge c \ln \frac{1}{\varkappa}$ for $k = 1$ and $\Delta/\varkappa^2 \ge c_- \varkappa^{\frac{2k}{2k-1}-2}$ for $k > 1$, and $\frac{2k}{2k-1} < 2$ for $k > 1$, so $\Delta/\varkappa^2 \to \infty$. □

Toy example 11.47 (the rest state: why a logarithm appears at $k = 1$)**.** Take $\boldsymbol{c} = 0$, $x_0 = 0$, the rest state of Example 2.32: $W_0 \equiv 1$, $k = 1$, $u_+ = e^{-\xi}$, $G_+ = 1$. Then $F_+ = \int_0^\xi e^{2\eta} d\eta = \frac{1}{2}(e^{2\xi} - 1)$, $\hat{\psi}_+ \approx \frac{1}{12} e^{2\xi}$ and $\hat{\psi}_+ u_+^2 \to \frac{1}{12}$: the integrand of A_+ tends to a *constant*, so $A_+(T) \approx T/12$ grows linearly. Cutting at $T = \ln \frac{1}{\varkappa} - K$ gives $\Delta \approx \frac{\varkappa^2}{12} \ln \frac{1}{\varkappa}$: the slope $\frac{1}{12}$ is exactly the first-order coefficient, and the finite cut only shifts the constant. For $\varkappa = 10^{-3}$ the cut of Numerical observation 11.43 is at $T = 5.41$, where the closed form of Proposition 11.48(c) gives $A_+(T) = 0.324$; the computed $\Delta/\varkappa^2$ is 0.774, and the difference 0.45 is the mass beyond the cut ($\mathcal{T}(T)/\Delta = 0.58$ in Table 11.3). So at $k = 1$ the part beyond the cut is of order 1 in $\Delta/\varkappa^2$ and shifts the constant term, while it cannot change the slope: this is why Theorem 11.44(b) has no constant.

Proposition 11.48 (closed forms of the first-order integrals)**.** (a) (Digamma, $k < 1$.) *For $x_0 = 0$, $\boldsymbol{c} = (k^2-1, k^2-1, 0)$ and $0 < k < 1$, $k \ne \frac{1}{2}$, the coefficient*

$A_+^{\lim}$ of Theorem 11.44(a) is the digamma expression of Proposition B.12; $A_+^{\lim} = 0.366725$ at $k = \sqrt{0.1}$, 0.251682 at $k = \sqrt{0.5}$ and 0.342534 at $k = 0.837$.

(b) (Landau solutions, interior of the window.) *For $\boldsymbol{c} = 0$, $x_0 = 0$ and the Landau solution z_λ with $|\lambda| > 1$ of Example B.6, the first-order swirl integral $\hat\psi_\lambda$ of Proposition B.8 has finite limits $\hat\psi_\lambda(\pm1)$, given there in closed form; $\hat\psi_\lambda(1) \to \frac{1}{24}$ as $\lambda \downarrow 1$ while $\hat\psi_\lambda(-1) \sim \frac{1}{4(\lambda-1)} \to \infty$.*

(c) (Rest state, $k = 1$.) *For $\boldsymbol{c} = 0$, $x_0 = 0$, with $x = \tanh\xi$ and $e = 1 - x$:*

$$\begin{aligned}\hat\psi_+(x) &= \frac{(1+x)^2}{16}\ln\frac{1-x}{1+x} + \frac{1}{6(1-x)} + \frac{2x^2-x-4}{24}\\ &= \frac{1}{6e} + \frac{1}{4}\ln e - \frac{1}{8} - \frac{\ln 2}{4} + o(1) \qquad (x \uparrow 1),\end{aligned}$$

and, with $X = \tanh T$,

$$\begin{aligned}A_+(T) &= \left[\tfrac{X-1}{16} - \tfrac{1}{24}\right]\ln(1-X) - \left[\tfrac{X+1}{16} + \tfrac{1}{6}\right]\ln(1+X) + \tfrac{X}{12} + \tfrac{1}{24} - \tfrac{1}{24(1+X)}\\ &= \frac{T}{12} + \frac{5}{48} - \frac{\ln 2}{3} + O(Te^{-2T}).\end{aligned}$$

Proof. (a) and (b) are Proposition B.12 and Proposition B.8 with its remark, which are proved in Appendix B; the three numerical values of (a) are the formula evaluated at those k (checked against a direct quadrature of $\int \hat\psi_+ u_+^2$ in `code/ch11_small/verify_ch11b.py`). (c) With $u_+ = e^{-\xi}$ we have $u_+^{-2} = e^{2\eta} = \frac{1+r}{1-r}$ and $F_+ = \frac{1}{2}(e^{2\eta}-1) = \frac{r}{1-r}$ in the variable $r = \tanh\eta$, $d\eta = dr/(1-r^2)$. By Lemma 11.37(a), $\hat\psi_+(\xi) = \int_0^\xi \frac{(x-r)^2}{1-r^2} F_+F_+'\,d\eta$ (because $\sinh^2(\xi-\eta)/\cosh^2\xi = (x-r)^2/(1-r^2)$, Remark B.7(ii)) and $F_+F_+' = \frac{r(1+r)}{(1-r)^2}$, so that

$$\hat\psi_+ = \int_0^x \frac{(x-r)^2\,r}{(1-r)^4(1+r)}\,dr.$$

Partial fractions in r give

$$\begin{aligned}\frac{(x-r)^2 r}{(1-r)^4(1+r)} &= \frac{(x+1)^2}{16}\Big(\frac{1}{r-1} - \frac{1}{r+1}\Big) - \frac{x^2+2x-7}{8(r-1)^2}\\ &\quad + \frac{(x-5)(x-1)}{4(r-1)^3} + \frac{(x-1)^2}{2(r-1)^4},\end{aligned}$$

and integrating from 0 to x and collecting terms gives the stated closed form (a computer-algebra check: `code/ch11_small/verify_ch11b.py`). The expansion at $x = 1-e$ is Taylor expansion: $(1+x)^2 = 4-4e+e^2$, $\ln\frac{1-x}{1+x} = \ln e - \ln 2 - \frac{e}{2} + O(e^2)$, so $\frac{(1+x)^2}{16}\ln\frac{1-x}{1+x} = \frac{1}{4}\ln e - \frac{\ln 2}{4} + O(e\ln e)$, $\frac{2x^2-x-4}{24} = -\frac{1}{8} + O(e)$. For A_+: $u_+^2 d\xi = e^{-2\xi}d\xi = \frac{1-x}{1+x}\frac{dx}{1-x^2} = \frac{dx}{(1+x)^2}$, so $A_+(T) = \int_0^X \hat\psi_+(x)(1+x)^{-2}dx$. The three terms of $\hat\psi_+/(1+x)^2$ are $\frac{1}{16}\ln\frac{1-x}{1+x}$, $\frac{1}{6(1-x)(1+x)^2} = \frac{1}{24(1+x)} + \frac{1}{12(1+x)^2} + \frac{1}{24(1-x)}$ and $\frac{2x^2-x-4}{24(1+x)^2} = \frac{1}{12} - \frac{5}{24(1+x)} - \frac{1}{24(1+x)^2}$; their primitives vanishing at 0 are elementary (the first is $\frac{x}{16}\ln\frac{1-x}{1+x} - \frac{1}{16}\ln(1-x^2)$), and adding them gives the displayed formula. Since $1 - X = \frac{2e^{-2T}}{1+e^{-2T}}$, $\ln(1-X) = \ln 2 - 2T + O(e^{-2T})$; at $X \to 1$ the coefficient of $\ln(1-X)$ tends to $-\frac{1}{24}$, that of $\ln(1+X)$ to $-\frac{7}{24}$ and the rational part to $\frac{5}{48}$, so $A_+ = \frac{1}{24}(2T - \ln 2) - \frac{7}{24}\ln 2 + \frac{5}{48} + O(Te^{-2T}) = \frac{T}{12} + \frac{5}{48} - \frac{\ln 2}{3} + O(Te^{-2T})$. □

Remark 11.49. The slope $\frac{1}{12}$ of $A_+(T)$ is the slope of the law Theorem 11.44(b); the intercept $\frac{5}{48} - \frac{\ln 2}{3} = -0.1269$ is *not* the intercept $b = 0.198$ of the law (Numerical observation 11.51) because the part of Δ beyond the cut, and the nonlinear layer, contribute at order $\varkappa^2$ too.

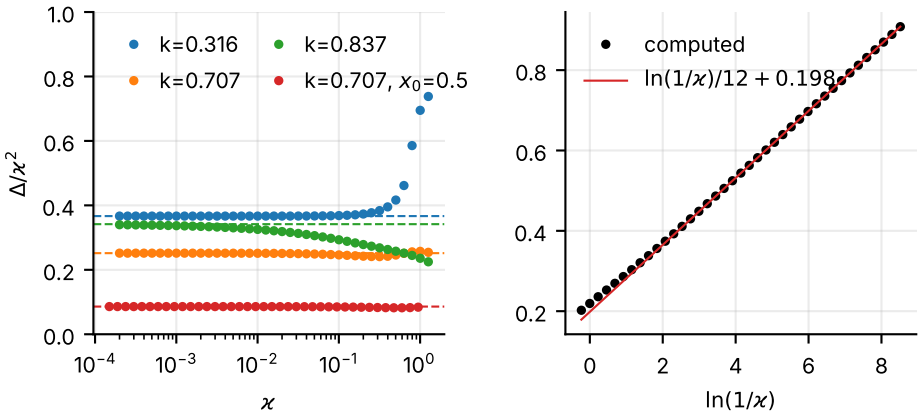


FIGURE 11.6. The laws at the edge, computed. Left: $\Delta/\varkappa^2$ against $\varkappa$ for $k < 1$ (points: $k = 0.316$, 0.707, 0.837 at $x_0 = 0$ and $k = 0.707$ at $x_0 = 0.5$); the dashed lines are the limits $A_+^{\lim}$ of Theorem 11.44(a) computed from Proposition B.12 and by quadrature. Right: the rest state $k = 1$: $\Delta/\varkappa^2$ against $\ln(1/\varkappa)$ follows a line of slope $\frac{1}{12}$ (red), with intercept 0.198.

Numerical observation 11.50 ($k < 1$: the limit $A_+^{\lim}$)**.** *The lower edge $\gamma^-(\kappa_x)$ was computed by shooting (bisection in γ to 10^{-15}, step $dt = 0.004$, shooting interval $\xi \le 40$) on a geometric grid $\kappa_x = 2 \cdot 10^{-4} \cdot 10^{j/10}$, and $\Delta = \gamma^-(\kappa_x) - \gamma^-(0)$ was formed with the same solver at $\kappa_x = 0$ so that discretisation errors cancel. Table 11.4 compares $\Delta/\varkappa^2$ with the limit $A_+^{\lim}$ (digamma values of Proposition 11.48(a) for $x_0 = 0$, direct quadrature of $\int \hat{\psi}_+ u_+^2$ for $x_0 = 0.5$). The quotient approaches*

k	x_0	$A_+^{\lim}$	$\Delta/\varkappa^2$	*deficit*	*deficit′*
$k = 0.316$	0	0.366725	0.366725	$< 10^{-7}$	$1.7 \cdot 10^{-6}$
$k = 0.707$	0	0.251682	0.251675	$2.5 \cdot 10^{-5}$	$1.6 \cdot 10^{-4}$
$k = 0.837$	0	0.342534	0.339942	$7.6 \cdot 10^{-3}$	$2.3 \cdot 10^{-2}$
$k = 0.707$	0.5	0.085891	0.085888	$3 \cdot 10^{-5}$	$6 \cdot 10^{-5}$

TABLE 11.4. $\Delta/\varkappa^2$ and the relative deficit $1 - \Delta/(\varkappa^2 A_+^{\lim})$ at $\kappa_x = 2 \cdot 10^{-4}$ (columns 4, 5) and the deficit′ at $\kappa_x = 2 \cdot 10^{-3}$, against the limit $A_+^{\lim}$ of Theorem 11.44(a); $\boldsymbol{c} = (k^2 - 1, k^2 - 1, 0)$.

$A_+^{\lim}$ from below with a deficit that shrinks with $\varkappa$; for $k = 0.707$ the deficit grows

by the factor 6.6 *between* $\kappa_x = 2\cdot 10^{-4}$ *and* $2\cdot 10^{-3}$*, and* $10^{(2-2k)/k} = 6.7$ *(for* $k = 0.837$*:* 3.0 *against* 2.5*), consistent with the size* $\varkappa^{(2-2k)/k} = \Pi(T_*)$ *of the cut (Remark 11.45(ii)); for* $x_0 = 0.5$ *the deficit is at the noise level of the computation (*$\lesssim 4\cdot 10^{-5}$*). In the variable* κ_x *the limit is* $A_+^{\lim}(1-x_0^2)^2$*:* 0.048314 *for* $x_0 = 0.5$*, and* $\Delta/\kappa_x^2 = 0.048312$ *at* $\kappa_x = 2\cdot 10^{-4}$*. Code:* `gen_b_data.py grid`*,* `verify_ch11b.py`*.*

Numerical observation 11.51 ($k = 1$: the slope $\frac{e^{-2\xi_0}}{12G_+^2}$ and the intercept)**.** *For six values* $\varkappa \in \{10^{-2}, 3\cdot 10^{-3}, 10^{-3}, 3\cdot 10^{-4}, 10^{-4}, 3\cdot 10^{-5}\}$ *the difference quotients* $\big(\Delta/\varkappa^2\big)'$ *with respect to* $\ln\frac{1}{\varkappa}$ *were formed and the mean of the first three is compared in Table 11.5 with the slope* $s_1 = \frac{e^{-2\xi_0}}{12G_+^2}$ *of Theorem 11.44(b) (*G_+ *computed from the Jost solution of* W_0 *at* $\xi_0 + 22$*). Agreement is within* $2\cdot 10^{-3}$ *relative in every row (best* $1.2\cdot 10^{-4}$*). The intercept* $b := \Delta/\varkappa^2 - \frac{1}{12}\ln\frac{1}{\varkappa}$ *at the rest state is* 0.1985*,* 0.1984*,* 0.1984*,* 0.1984 *for* $\varkappa = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$ *(Figure 11.6, right); it is a numerical value only, the proof of Theorem 11.44(b) does not determine it. The prediction* $2 - 1/(\ln\frac{1}{\varkappa} + 12b)$ *for the local exponent* $\frac{d\ln\Delta}{d\ln\varkappa}$ *is* 1.9073 *at* $\varkappa = 2.2\cdot 10^{-4}$*, equal to the computed value (Numerical observation 11.56).*

$\boldsymbol{c}$	x_0	G_+	s_1	*measured slope*
$(0,0,0)$	0	1	0.083333	0.083307
$(0,0,0)$	0.5	1	0.027778	0.027774
$(0,0,0)$	-0.5	1	0.250000	0.249765
$(0.5,0,0)$	0.5	0.98139	0.028841	0.028837
$(-0.5,0,1)$	-0.3	0.87171	0.203664	0.203320

TABLE 11.5. The slope of $\Delta/\varkappa^2$ in $\ln(1/\varkappa)$ at $k = 1$ (the exponent at the south pole, k_1, is $\sqrt{1+c_1} \neq 1$ in the last two rows; only k_2 matters).

Example 11.52 (an endpoint example with $k_2 > 1$, $x_0 \neq 0$)**.** Take $\boldsymbol{c} = (0, \frac{1}{2}, 0)$ (so $k_1 = 1$, $k_2 = \sqrt{3/2} = 1.2247$) and $x_0 = 0.3$. Here $U_{\boldsymbol{c}}^-(x_0) = -2.959796$, and the computed distances of the lower edge are

κ_x	10^{-4}	10^{-3}	10^{-2}	10^{-1}
Δ	$2.7272\cdot 10^{-8}$	$1.3075\cdot 10^{-6}$	$6.1237\cdot 10^{-5}$	$2.7237\cdot 10^{-3}$

The successive local exponents $\frac{d\ln\Delta}{d\ln\varkappa}$ are 1.681, 1.671, 1.648, to be compared with $\frac{2k_2}{2k_2-1} = 1.690$ of Theorem 11.44(c): the rate is that of the theorem although neither $x_0 = 0$ nor $c_1 = c_2$ holds.

11.7. The exponents and the convexity of the lens near the floor

In words. *The three laws say that near the floor of the lens the lower edge rises like* $\varkappa^2$*, like* $\varkappa^2\ln\frac{1}{\varkappa}$*, or like* $\varkappa^p$ *with* $1 < p = \frac{2k}{2k-1} < 2$*. All three are convex functions of* $\varkappa$ *near* 0*, so they are compatible with the numerical observation that* γ^- *is convex in* κ_x *on the whole range computed; but one has to be careful, because convexity is not invariant under the substitution* $\sigma = \varkappa^2$ *(in which the same laws are* concave *for* $k \geq 1$*) and because only the leading term is proved: the second derivative of*

γ^- is not controlled. This section collects the elementary facts, the measured local exponents, and the numerical positivity of $\gamma^{-\prime\prime}$.

Lemma 11.53 (the exponents)**.** (a) *For $p \geq 1$ the function $\varkappa \mapsto \varkappa^p$ is convex on $[0,\infty)$; for $1 < p < 2$ its second derivative $p(p-1)\varkappa^{p-2} \to +\infty$ at 0.*

(b) *$\varphi(\varkappa) = a\varkappa^2 \ln \frac{1}{\varkappa}$ ($a > 0$) has $\varphi'' = a\big(2\ln\frac{1}{\varkappa} - 3\big) > 0$ for $\varkappa < e^{-3/2}$, and $\varphi'' \to +\infty$ as $\varkappa \downarrow 0$.*

(c) *In the variable $\sigma = \varkappa^2$: $\sigma^{p/2}$ is concave for $p < 2$, linear for $p = 2$, and $\varkappa^2 \ln \frac{1}{\varkappa} = \frac{1}{2}\sigma \ln \frac{1}{\sigma}$ is concave ($(\sigma \ln \frac{1}{\sigma})'' = -\frac{1}{\sigma}$).*

(d) *$p_k := \frac{2k}{2k-1}$ satisfies $p_k > 1$ for all $k > \frac{1}{2}$, $p_k < 2$ iff $k > 1$, $p_k \downarrow 1$ as $k \to \infty$, $p_1 = 2$; in particular $p_{3/2} = \frac{3}{2}$, $p_2 = \frac{4}{3}$, $p_{5/2} = \frac{5}{4}$.*

Proof. (a), (b), (c) are the computations $(\varkappa^p)'' = p(p-1)\varkappa^{p-2}$, $(\varkappa^2 \ln \frac{1}{\varkappa})' = 2\varkappa \ln \frac{1}{\varkappa} - \varkappa$, $(\varkappa^2 \ln \frac{1}{\varkappa})'' = 2\ln \frac{1}{\varkappa} - 3$, $(\sigma^{p/2})'' = \frac{p}{2}(\frac{p}{2} - 1)\sigma^{\frac{p}{2}-2}$, $(\sigma \ln \frac{1}{\sigma})' = \ln \frac{1}{\sigma} - 1$, $(\sigma \ln \frac{1}{\sigma})'' = -\frac{1}{\sigma}$. (d) $p_k - 1 = \frac{1}{2k-1} > 0$, $2 - p_k = \frac{2k-2}{2k-1}$. □

Remark 11.54 (what this does and does not give)**.** By Theorem 11.44, $\Delta = \gamma^- - U^-_{\boldsymbol{c}}$ has the *order* $\varkappa^2$, $\varkappa^2 \ln \frac{1}{\varkappa}$, $\varkappa^{p_k}$, and Lemma 11.53 shows that the model functions are convex near 0. This is *not* a proof that γ^- is convex near the floor: the theorem controls Δ, not Δ'', and a function with the order $\varkappa^p$ may oscillate in its second derivative. The convexity of the lens (of γ^- in κ_x) is stated, with its proof status, in Chapter 13; what is proved here is the exponent algebra and the corollary Corollary 11.46.

Toy example 11.55 (the false alarm)**.** Let $k = 2$, $\Delta = \varkappa^{4/3}$. In $\varkappa$ it is convex ($\Delta'' = \frac{4}{9}\varkappa^{-2/3} > 0$). In $\sigma = \varkappa^2$ it is $\sigma^{2/3}$, whose second derivative $-\frac{2}{9}\sigma^{-4/3}$ is negative: a plot of Δ against κ_x^2 would suggest concavity. Since κ_x is the natural variable of the lens (and of Definition 5.55), convexity is a statement about $\varkappa$, not about $\varkappa^2$. The same computation shows why one should not infer a sign of $\gamma^{-\prime\prime}$ from a plot against $\ln \kappa_x$ either.

Numerical observation 11.56 (local exponents)**.** *From the geometric grid $\kappa_x = 2 \cdot 10^{-4} \cdot 10^{j/10}$ the local exponent $\frac{d\ln\Delta}{d\ln\varkappa}$ was formed from consecutive values (Figure 11.7, left). At $\varkappa \approx 2.2 \cdot 10^{-4}$ it is 2.0000 for $k = 0.316$ and $k = 0.707$, 1.9073 for $k = 1$ (the law $\varkappa^2 \ln \frac{1}{\varkappa}$ has $2 - 1/(\ln \frac{1}{\varkappa} + 12b) = 1.9073$ with the intercept of Numerical observation 11.51), and for $k > 1$:*

k	$\sqrt{2}$	1.5	2	2.5
$p_k = \frac{2k}{2k-1}$	1.5469	1.5000	1.3333	1.2500
computed	1.5444	1.4984	1.3329	1.2497

the computed exponents are within 0.003 of p_k (at $\varkappa \approx 10^{-2}$ they are 1.5334, 1.4902, 1.3303, 1.2483). The theorem proves only the two-sided bound for Δ; whether $\Delta\varkappa^{-p_k}$ has a limit is recorded in Chapter 13.

Numerical observation 11.57 (positivity of $\gamma^{-\prime\prime}$)**.** *The second derivative of γ^- in κ_x was estimated by second differences on the same grid ($2.5 \cdot 10^{-4} \leq \kappa_x \leq 1.26$) for nine data ($x_0 = 0$, $\boldsymbol{c} = (k^2-1, k^2-1, 0)$ with $k = 0.316$, 0.707, 0.837, 1, 1.414, 1.5, 2, 2.5, and $k = 0.707$ at $x_0 = 0.5$). It is positive everywhere; the smallest values are 0.733 ($k = 0.316$; equal to $2A_+^{\lim}(1-x_0^2)^2 = 0.7334$ at small κ_x, as Corollary 11.46(b) says), 0.414 ($k = 0.707$), 0.299 ($k = 0.837$), 0.214 ($k = 1$), 0.115 ($k = \sqrt{2}$), 0.104*

($k = 1.5$), 0.063 ($k = 2$), 0.043 ($k = 2.5$) and 0.090 ($k = 0.707$, $x_0 = 0.5$), all attained at $\kappa_x = 1$ except for $k = 0.316$ (at the lower end, $\kappa_x = 3.2 \cdot 10^{-4}$) and for $x_0 = 0.5$ (at $\kappa_x = 0.32$). For $k \geq 1$ the values at the lower end are large and grow like $\kappa_x^{p_k - 2}$ (Figure 11.7, right). Numerical, not part of any proof.

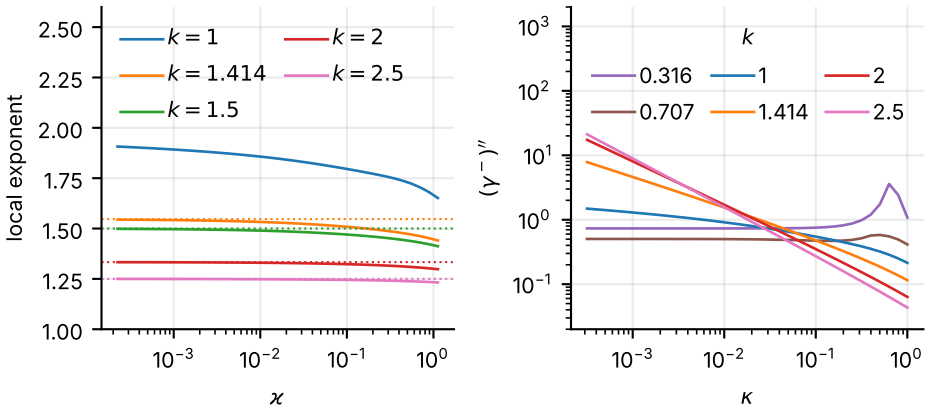


FIGURE 11.7. Left: the local exponent $d \ln \Delta / d \ln \varkappa$ for $k \geq 1$ ($x_0 = 0$, $\boldsymbol{c} = (k^2 - 1, k^2 - 1, 0)$); dotted lines: $p_k = \frac{2k}{2k-1}$ for $k > 1$. Right: second differences of γ^- in κ_x on the grid, all positive.

11.8. The combined regimes

In words. *The monograph proves uniqueness of the solution of the effective problem in two regimes: for* small *swirl κ_{D} (Theorem 8.34: a contraction) and for* large *swirl (Theorem 10.39); in between lies a window where uniqueness may fail, and Theorem 9.51 says that if it fails there are a first and a last fold in κ_{D}. The conditional form of that theorem used a hypothesis on large swirl that was proved only in Chapter 10; this section removes the hypothesis. The statement is made on a compact set K of the parameters (ϑ, f_-) with the exponents c' fixed, because the constants of the two regimes depend continuously on the data; the one new ingredient is the observation that the* small-κ_{D} *regime comes in two flavours. If c_3' is above the critical value (data in or near the no-swirl set), the solution is unique and nondegenerate for small κ_{D} (a contraction). If c_3' is below it (hypothesis (H) of this chapter), the contraction is impossible—there is no solution near f_-—but the critical bubble of Theorem 11.6 describes all solutions, uniformly on K. Uniqueness in the second flavour is* not *claimed.*

Conventions. In this section $\kappa = \kappa_{\mathrm{D}}$ again, and the data are the effective data $D = (\boldsymbol{c}', \vartheta, f_-, \kappa)$ of Chapter 8; $\mathrm{Fix}_\kappa = \mathrm{Fix}_\kappa(\boldsymbol{c}', \vartheta, f_-)$ is the set of fixed points of T_D (Proposition 8.13), $\mathcal{K}_{\mathrm{deg}}$ the set of κ for which some fixed point has $1 \in \operatorname{spec} \mathrm{D}T_D$, $\kappa_{\mathrm{first}} = \inf \mathcal{K}_{\mathrm{deg}}$, $\kappa_{\mathrm{last}} = \sup \mathcal{K}_{\mathrm{deg}}$ (Theorem 9.51); $\mathcal{K}_{\mathrm{deg}} = \emptyset$ is allowed. $\boldsymbol{c}'$ is fixed with $c_1', c_2' > -1$, and $K \subset (0, \infty) \times \mathbb{R}$ is compact; $\theta_K = \min_K \vartheta$,

$\theta^K = \max_K \vartheta$, $F_K = \max_K |f_-|$. (Adm_K) denotes the condition $c_3' + \frac{1}{4}m_K^2 > \bar{c}_3$ with $m_K := \min_K f_-^+$ (it holds for every K if $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$).

THEOREM 11.58 (combined regimes). *Let $\boldsymbol{c}'$ and K be as above.*

(a) (Small swirl, data not below the no-swirl set.) *Assume* (Adm_K). *Let* $W_K = W_{\mathrm{e}} + \frac{1}{8}m_K^2 \operatorname{sech}^2\xi$, $w_K = w_0(W_K)$, $\Gamma_K = \Gamma_{\ker}(W_K)$ *and let* $\kappa_{\mathrm{lo}}(K) > 0$ *be the unique root of*

$$L_K(\kappa) := \frac{3\,\kappa\,\big(F_K + \kappa/(\theta_K w_K)\big)\,\Gamma_K}{4\,\theta_K w_K} = 1.$$

Then for every $(\vartheta, f_-) \in K$ and $0 < \kappa < \kappa_{\mathrm{lo}}(K)$: $\#\mathrm{Fix}_\kappa = 1$; the fixed point is nondegenerate with $\|\mathrm{D}T_D(f)\| < \frac{1}{2}$ and depends analytically on κ; and for each fixed (ϑ, f_-), $f_\kappa = f_- + \kappa f_1 + o(\kappa)$ with $f_1 = \int_{-\infty}^{\xi}(u_- + \vartheta u_+)^{-2}$ the first-order swirl of Theorem 8.53(a). For $\boldsymbol{c}' = 0$ and $K \subset (0,\infty)\times\{0\}$ (rest state): $\kappa_{\mathrm{lo}}(K) = 4\theta_K/\sqrt{3}$.

(b) (Small swirl, data below the no-swirl set.) *Assume* (H) *on K: $c_3' < \bar{c}_3$ and $K \subset (0,\infty)\times[0,M)$, $M = 2\sqrt{\bar{c}_3 - c_3'}$. Then $\mathrm{Fix}_\kappa \neq \emptyset$ and $J \geq M - f_-$ for every $\kappa > 0$ and $(\vartheta, f_-) \in K$; and for every $\varepsilon, L > 0$ there is $\kappa_{\varepsilon,L}(K) > 0$ such that for all $(\vartheta, f_-) \in K$, $0 < \kappa < \kappa_{\varepsilon,L}(K)$ and every solution: $|J - J_*(\vartheta, f_-)| < \varepsilon$ and $\sup_{|\xi|\leq L}|f - M| < \varepsilon$, $J_* = (1+\vartheta)(M - f_-)$.*

(c) (Large swirl.) *Assume $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$. There is $\kappa_{**}(K) < \infty$ such that for all $(\vartheta, f_-) \in K$ and $\kappa \geq \kappa_{**}(K)$: $\#\mathrm{Fix}_\kappa = 1$, $\operatorname{spec}\mathrm{D}T_D(f) \cap [\frac{1}{2}, \infty) = \emptyset$, and $(\vartheta, f_-, \kappa) \mapsto f$ is real-analytic on $\operatorname{int}K \times (\kappa_{**}(K), \infty)$.*

(d) (Folds.) *Assume $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$, and fix $(\vartheta, f_-) \in K$. Then $\mathcal{K}_{\deg}$ is closed and $\mathcal{K}_{\deg} \subset [\kappa_{\mathrm{lo}}(K), \kappa_{**}(K))$; $\#\mathrm{Fix}_\kappa = 1$ for $\kappa < \kappa_{\mathrm{first}}$ and for $\kappa > \kappa_{\mathrm{last}}$ (for all κ if $\mathcal{K}_{\deg} = \emptyset$); $\#\mathrm{Fix}_\kappa$ is odd for $\kappa \notin \mathcal{K}_{\deg}$. If $\#\mathrm{Fix}_{\kappa_1} \geq 2$ for some κ_1, then $\mathcal{K}_{\deg} \neq \emptyset$, $\kappa_{\mathrm{first}}, \kappa_{\mathrm{last}} \in \mathcal{K}_{\deg}$,*

$$\kappa_{\mathrm{lo}}(K) \leq \kappa_{\mathrm{first}} \leq \kappa_1 \leq \kappa_{\mathrm{last}} < \kappa_{**}(K), \qquad \{\kappa : \#\mathrm{Fix}_\kappa \geq 2\} \subset [\kappa_{\mathrm{first}}, \kappa_{\mathrm{last}}].$$

Table 11.6 lists the hypotheses of the parts of Theorem 11.58.

part	κ	hypothesis on $\boldsymbol{c}'$	hypothesis on K	proved from
(a)	$\kappa < \kappa_{\mathrm{lo}}(K)$	$c_3' + \frac{1}{4}m_K^2 > \bar{c}_3$	compact	Theorem 8.34, Theorem 8.53
(b)	$\kappa < \kappa_{\varepsilon,L}(K)$	$c_3' < \bar{c}_3$	compact, $f_- \in [0, M)$	Theorem 11.6 (uniform on K), Corollary 8.45
(c)	$\kappa \geq \kappa_{**}(K)$	$\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$	compact	Theorem 10.39
(d)	all κ	$\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$	compact	(a), (c), Theorem 9.51

TABLE 11.6. Hypotheses of the parts of Theorem 11.58. Uniqueness is claimed for small κ only in (a) (above the critical value), never under (H).

PROOF. *(a).* Fix $(\vartheta, f_-) \in K$, and put $m_- = f_-^+ \geq m_K$. The reference potential $\widetilde{W} = W_{\mathrm{e}} + \frac{1}{8}m_-^2 \operatorname{sech}^2$ of Theorem 8.34 satisfies $\widetilde{W} \geq W_K$ pointwise, both are in $\mathcal{W}(k_1, k_2)$, and $W_K = W_{(c_1', c_2', c_3' + m_K^2/4)}$ is admissible by (Adm_K) and Lemma 3.35(b).

By Lemma 3.23, $\tilde{w}_0 := w_0(\widetilde{W}) \geq w_K$; by Lemma 8.23(e), $\widetilde{\Gamma} := \Gamma_{\ker}(\widetilde{W}) \leq \Gamma_K$. Therefore $B = \kappa/(\vartheta\tilde{w}_0) \leq \kappa/(\theta_K w_K)$, $S_1 = \max(|f_-|, f_- + B) \leq F_K + \kappa/(\theta_K w_K)$, and the number of Theorem 8.34 satisfies

$$\Lambda_{\mathrm{mod}} = \frac{3\kappa S_1 \widetilde{\Gamma}}{4\vartheta\tilde{w}_0} \leq L_K(\kappa).$$

L_K is continuous and strictly increasing with $L_K(0) = 0$, $L_K(\infty) = \infty$, so $\kappa_{\mathrm{lo}}(K)$ exists, and for $\kappa < \kappa_{\mathrm{lo}}$ we have $\Lambda_{\mathrm{mod}} < 1$; Theorem 8.34 gives the uniqueness, $\|\mathrm{D}T_D(f)\| \leq \frac{1}{2}\Lambda_{\mathrm{mod}} < \frac{1}{2}$ and nondegeneracy. Analytic dependence on κ is Theorem 9.51(ii) (its hypothesis is (Adm) for the single datum, implied by (Adm_K)). The expansion is Theorem 8.53(a),(b): for κ below κ_2 of that theorem the unique solution is f^κ, a C^1 curve with $f^0 = f_-$ and derivative f_1. At the rest state $\widetilde{w}_0 = 2$, $\widetilde{\Gamma} = 1$, $F_K = 0$, so $L_K = 3\kappa^2/(16\theta_K^2)$ and $\kappa_{\mathrm{lo}}(K) = 4\theta_K/\sqrt{3}$ (Remark 8.35); here $W_K = \widetilde{W}$ and $w_K = 2$, $\Gamma_K = 1$ because $m_K = 0$.

(b). Existence and $J \geq M - f_-$ are Corollary 8.45, pointwise on K. For the uniform bubble we argue by contradiction, following the proof of Theorem 11.6. Suppose there are $\varepsilon, L > 0$, points $(\vartheta_n, f_{-,n}) \in K$, $\kappa_n \downarrow 0$ and solutions f_n with $|J_n - J_*(\vartheta_n, f_{-,n})| \geq \varepsilon$ or $\sup_{|\xi|\leq L} |f_n - M| \geq \varepsilon$. By compactness of K pass to a subsequence with $(\vartheta_n, f_{-,n}) \to (\vartheta, f_-) \in K$, so $0 \leq f_- < M$. The constants of Lemma 11.5 are bounded uniformly on K: m_*, w_* depend on $\boldsymbol{c}'$ only (Lemma 8.39), $J_1 = (1+\vartheta)(m_* - f_-)^+ \leq (1+\theta^K)m_*$, $1/(\vartheta w_*) \leq 1/(\theta_K w_*)$, hence $\bar{J} \leq \bar{J}_K := \max((1+\theta^K)m_*, 1/(\theta_K w_*))$, $S \leq \bar{J}_K + \max_K f_-$, and in part (c) of that lemma $(1+\vartheta)^{-2} \leq 1$. *Step 1:* $J_n \in [M - f_{-,n}, \bar{J}_K + \max_K f_-]$, $M - f_{-,n} \geq \delta_K := M - \max_K f_- > 0$; pass to $J_n \to J_\infty \geq \delta_K$; by Lemma 11.5(c), $f_n \to C := f_- + J_\infty/(1+\vartheta)$ locally uniformly (using $f_n(0) = f_{-,n} + J_n/(1+\vartheta_n)$). *Step 2:* $W_n \to W_C$ locally uniformly exactly as in Theorem 11.6, the class $\mathcal{W}(k_1, k_2; C_S)$ being uniform. *Step 3:* $w_0(W_n) = \kappa_n/(\vartheta_n J_n) \leq \kappa_n/(\theta_K \delta_K) \to 0$, so W_C is critical by Lemma 3.25(c), $C = M$ by Lemma 11.2, $J_\infty = (1+\vartheta)(M - f_-) = J_*(\vartheta, f_-)$. Since J_* is continuous, $J_n - J_*(\vartheta_n, f_{-,n}) \to 0$ and $f_n \to M$ locally uniformly: a contradiction.

(c). This is Theorem 10.39(a),(b),(c) for $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$ and the compact set K.

(d). Since $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$, $c_3' > \bar{c}_3$ and (Adm_K) holds, so (a) applies and (c) gives (H_∞) of Theorem 9.51(iii) with $\kappa_{**} = \kappa_{**}(K)$ for the line $\kappa \mapsto (\boldsymbol{c}', \vartheta, f_-, \kappa)$. Hence Theorem 9.51(i)–(iv) hold without further hypothesis. By (a) no $\kappa < \kappa_{\mathrm{lo}}(K)$ lies in $\mathcal{K}_{\mathrm{deg}}$ (there $\|\mathrm{D}T_D\| < \frac{1}{2}$), by (c) none with $\kappa \geq \kappa_{**}$ (since $\operatorname{spec} \cap [\frac{1}{2}, \infty) = \emptyset$), and $\mathcal{K}_{\mathrm{deg}}$ is closed by Theorem 9.51(i); the rest is Theorem 9.51(ii)–(iv), where $\kappa_{\mathrm{last}} \leq \kappa_{**}$ is improved to $\kappa_{\mathrm{last}} < \kappa_{**}$ because $\mathcal{K}_{\mathrm{deg}}$ is a closed subset of $(0, \kappa_{**})$ in part (iii). □

Toy example 11.59 (a one-variable caricature of the three regimes)**.** Let Fix be the set of real roots x of $x^3 - 3x = \kappa - 3a$, where $a \in K = [1, 2]$ plays the role of the compact parameter and κ of the swirl. The function $x^3 - 3x$ has critical values ± 2, so there are three roots iff $|\kappa - 3a| < 2$, i.e. for $\kappa \in (3a - 2, 3a + 2)$, and one root otherwise; the folds ($\mathcal{K}_{\mathrm{deg}}$: the roots where the derivative $3x^2 - 3$ vanishes) sit at $\kappa = 3a \mp 2$. Uniformly on K: one root for every $\kappa < \kappa_{\mathrm{lo}} := 1$ (since $3a - 2 \geq 1$) and for every $\kappa > \kappa_{**} := 8$ (since $3a + 2 \leq 8$), and $\mathcal{K}_{\mathrm{deg}} \subset [1, 8]$, with $\kappa_{\mathrm{first}} = 3a - 2$, $\kappa_{\mathrm{last}} = 3a + 2$. The theorem is the statement that for the effective problem the two outer regimes exist uniformly on compact sets, so that all non-uniqueness is confined to a bounded window; it does not say that the window is non-empty (the example $a = 1$, $\kappa = 2.99$ has three roots, the example

$x^3 + 3x = \kappa$ has one for every κ), nor where its ends are (Remark 11.61). Checked in `code/ch11_small/verify_ch11b.py`.

Corollary 11.60 (the three-solution window at the certified datum, unconditionally)**.** *Let* $D^* = (\boldsymbol{c}' = (\frac{5}{4}, -\frac{15}{16}, -\frac{517}{160}), \vartheta = 40, f_- = 0, \kappa = 1400)$ *be the datum of Corollary 9.55; its exponents are* $(k_1', k_2') = (\frac{3}{2}, \frac{1}{4})$, $\bar{c}_3 = -\frac{105}{32} = -3.28125$ *and* $c_3' - \bar{c}_3 = \frac{1}{20}$, *so* $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$. *On the line* $\kappa \mapsto (\boldsymbol{c}', 40, 0, \kappa)$, *with* $K = \{(40, 0)\}$: $\kappa_{\mathrm{lo}}(K) \le \kappa_{\mathrm{first}} \le 1400 \le \kappa_{\mathrm{last}} < \kappa_{**}(K)$, *both* κ_{first} *and* κ_{last} *carry a degenerate fixed point, and* $\{\kappa : \#\mathrm{Fix}_\kappa \ge 2\} \subset [\kappa_{\mathrm{first}}, \kappa_{\mathrm{last}}]$; *this includes* $\kappa = 1400$. *The numerical turning points are* $\kappa = 1300.03$ *and* 1462.16 *(Numerical observation 9.75).*

Proof. $\#\mathrm{Fix}_{1400} \ge 3$ by Theorem 9.68; apply Theorem 11.58(d) with $K = \{(40, 0)\}$. □

Remark 11.61 (what is not claimed)**.** (i) For data below the no-swirl set, (b) describes *every* solution at small κ but does not say there is only one; uniqueness for $\kappa \to 0$ under (H) is not known (Chapter 13). A degenerate or non-unique small-κ regime would not contradict (b): it would be a family of solutions all close to the bubble. Under (H) one has $c_3' < \bar{c}_3$, so $\boldsymbol{c}' \notin \operatorname{int} \mathcal{J}$ and (c), (d) are not claimed there; for such $\boldsymbol{c}'$ the hypothesis (H_∞) is not known (Remark 10.41). (ii) For $f_- < 0$ below the no-swirl set, the limits of solutions as $\kappa \to 0$ are not described (Remark 11.8(iii)). (iii) The constants $\kappa_{\mathrm{lo}}(K)$ and $\kappa_{**}(K)$ are not sharp (Λ_{mod} is a contraction bound; κ_{**} comes from Proposition 10.35); they are not claimed to bracket the folds closely, and $\kappa_{**}(K)/\kappa_{\mathrm{lo}}(K)$ is large. At the rest state with $\vartheta = 1$ no fold is found numerically (Example 9.54). (iv) (a) does not claim uniform $o(\kappa)$ in the expansion of f_κ: the remainder is for each datum.

Atlas

Atlas 11.62 (Chapter 11: small swirl, the bubble and the edge of the window)**.** *All numerical results of this chapter are collected here; none is used in a proof.* Method (Sections 11.5–11.7). *The class-*Z *solution is found by shooting in* $\xi = \operatorname{artanh} x$ *from the base point (Runge–Kutta of order* 4, *step parameter* 0.004, *interval* $\xi \le 40$*) with bisection in* γ *to* 10^{-15}; $\gamma^-(\kappa_x)$ *is the smallest* γ *for which the solution reaches the north pole, and* $\Delta = \gamma^-(\kappa_x) - \gamma^-(0)$ *is formed with the* same *solver at* $\kappa_x = 0$ *so that the discretisation error cancels (an interval* $\xi \le 14$ *cuts the tail* $\int \hat{\psi}_+ u_+^2$ *for* $k < 1$ *and was found to produce a constant relative offset of* $5 \cdot 10^{-4}$; $\xi \le 40$ *removes it). The no-swirl Jost solutions* u_+ *(hence* G_+*) are obtained from the Riccati equation for* ζ_+ *with an adaptive Runge–Kutta method;* A_+^{lim} *by the digamma formula (Proposition B.12) and by double quadrature, which agree to* 10^{-8}. Method (Section 11.8). *No computation is needed: the theorem is proved; the toy and the corollary are checked by computer algebra and by root counting in* `verify_ch11b.py`. Code. `code/ch11_small/`: `gen_b_data.py` *(the heavy runs, recorded in* `data_b_grid.json`, `data_b_misc.json`*),* `verify_ch11b.py` *(prints every number of Sections 11.5–11.8 and "ALL CHECKS PASSED"),* `make_b_*.py` *(the three figures).*

Table 11.7 is the numerical atlas of the chapter.

Reading the table. *(1) For* $k < 1$ *the quotient* $\Delta/\varkappa^2$ *is within* 10^{-2} *of* A_+^{lim} *at* $\kappa_x = 2 \cdot 10^{-4}$ *in every row, and the gap behaves like the predicted size* $\varkappa^{(2-2k)/k}$ *of the beyond-the-cut mass: the closer* k *is to* 1, *the slower the approach (*$7.6 \cdot 10^{-3}$

statement	*datum*	*result*	*where*
$\Delta/\varkappa^2 \to A_+^{\lim}$ *(*$k<1$*)*	$\boldsymbol{c}=(k^2-1,k^2-1,0)$, $x_0=0$*:* $k=0.316, 0.707, 0.837$*;* $k=0.707$*,* $x_0=0.5$	$A_+^{\lim}=0.366725$*,* 0.251682*,* 0.342534*,* 0.085891*;* $\Delta/\varkappa^2$ *at* $\kappa_x=2\cdot10^{-4}$ *below it by* $<10^{-7}$*,* $2.5\cdot10^{-5}$*,* $7.6\cdot10^{-3}$*,* $3\cdot10^{-5}$	*Numerical observation 11.50, Table 11.4*
slope $\frac{e^{-2\xi_0}}{12G_+^2}$ *(*$k=1$*)*	*five data,* $x_0\in\{0,\pm0.5,-0.3\}$	*agreement within* $1.7\cdot10^{-3}$ *(best* $1.2\cdot10^{-4}$*); intercept* $b=0.1984$ *at the rest state*	*Numerical observation 11.51, Table 11.5*
rate $\varkappa^{2k/(2k-1)}$ *(*$k>1$*)*	$k=\sqrt{2}, 1.5, 2, 2.5$*;* $\boldsymbol{c}=(0,\frac{1}{2},0)$*,* $x_0=0.3$	*local exponents* 1.5444*,* 1.4984*,* 1.3329*,* 1.2497 *(predicted* 1.5469*,* 1.5*,* 1.3333*,* 1.25*);* 1.681 *against* 1.690	*Numerical observation 11.56, Example 11.52*
cut lemma	$k=0.707, 1, \sqrt{2}, 2$*;* $\varkappa=10^{-2}, 10^{-3}, 10^{-4}$	$\mathfrak{a}(T)-1\le1.1\cdot10^{-2}$*,* $\mathfrak{l}(T)\le1.3\cdot10^{-2}$ *for* $\Pi(T)\le0.05$	*Numerical observation 11.43, Table 11.3*
$\gamma^{-\prime\prime}>0$	*nine data*	*positive on* $2.5\cdot10^{-4}\le\kappa_x\le1.26$*, minimum* 0.043	*Numerical observation 11.57*

TABLE 11.7. Numerical atlas of Sections 11.5–11.7.

at $k=0.837$*). (2) At* $k=1$ *the slope* $\frac{e^{-2\xi_0}}{12G_+^2}$ *changes by a factor* 9 *between* $x_0=0.5$ *and* $x_0=-0.5$ *(*0.0278 *against* 0.25*) and the computed slopes follow it. (3) For* $k>1$ *the local exponents are within* 0.003 *of* $p_k=\frac{2k}{2k-1}$ *at* $\kappa_x\approx2\cdot10^{-4}$*, also for* $x_0\neq0$ *and* $c_1\neq c_2$ *(Example 11.52). (4) The cut lemma is not sharp:* $\mathfrak{a}-1$ *and* $\mathfrak{l}$ *at* $\Pi(T)=0.05$ *are of order* 10^{-2}*, below the threshold* λ *it provides; the constants of Lemma 11.39(b) are not computed.*

First half of the chapter. *The numerical observations of Sections 11.1–11.4 are not repeated; they are the following, each* below *the no-swirl set, all with the data of Table 11.1. Numerical observation 11.13: the solution branch followed down to* $\kappa=10^{-8}$ *shows the plateau* M *with two logistic layers and* $J\to J_*$*. Numerical observation 11.20: the exact identity of Lemma 11.15 holds on computed fixed points to rounding level. Numerical observations 11.21 and 11.30:* $(J-J_*)/\kappa\to\Gamma_{\mathrm{rate}}$ *for* $\max k_i<1$*, with the slow approach for the asymmetric set C. Numerical observation 11.24: the composite-profile predictor Formal computation 11.23 against the continuation. Numerical observation 11.28: for* $\max k_i>1$ *the sign change of* $J-J_*$ *and the exponent* $1/k$ *of Formal computation 11.27. Numerical observation 11.33: the large negative eigenvalue of a bubble against Formal computation 11.32. Numerical observation 11.34: three solutions at* $f_-=-5$ *(outside (H)) with indices summing to* $+1$*. The case* $k_1=k_2=1$ *is Formal computation 11.25.*

What is not in the atlas. *The uniqueness question for small κ under (H), and the behaviour of the lens for $k > 1$ beyond the two-sided bound, are not tested numerically here beyond the observations above; they are discussed in Chapter 13.*

CHAPTER 12

Thresholds: how large can the swirl be?

In words. *Chapters 5 and 7 showed that, for fixed Taylor constants $\boldsymbol{c}$ in the interior of the admissible set $\mathcal{J}$ and a fixed base point, the admissible data (κ, γ) form a lens whose tip $\kappa_{\max}$ is the* largest swirl *for which a global solution exists. This chapter asks how large that threshold is, and what the lens looks like. Three groups of answers are given.* Part A *(this part) treats the approach to the critical surface $c_3 \downarrow \bar{c}_3$: with $\varepsilon = c_3 - \bar{c}_3$ and $k = \max(k_1, k_2)$, the swirl threshold behaves like $\kappa_{\max} \asymp \varepsilon^{\max(1/2,\, 1-1/(2k))}$ (with a logarithm at $k = 1$), and for $k < 1$ even $\kappa_{\max}^2/\varepsilon$ has a limit, which is an explicit integral. The engine is one exact identity, the balance (T0): $\varepsilon \int \operatorname{sech}^2 \xi \, g \, z = \int \psi \, g \, z$, obtained by integrating a Wronskian of the critical function g against the solution z.* Part B *turns the same identity into upper and lower bounds for $\kappa_{\max}$ for* all *constants, into the large-constant and cone limits, and proves that the lens is a convex body in the coordinates $(\boldsymbol{c}, \ln \kappa, \gamma)$.* Part C *does for the class-*N *threshold (the extreme recessive shift) what Parts A and B do for $\kappa_{\max}$.*

Roadmap. Part A: Section 12.1 fixes the setting (critical function g, Jost solutions $u_\pm$, ε) and works the Landau example and two toys. Section 12.2 proves the window width (F1), the Jost bounds and the sandwich (F2). Section 12.3 proves (T0), the half-line identities and the effective-constant relation (F4). Section 12.4 proves the upper bounds on $\kappa_{\max}$ and the lower bound $\sqrt{\varepsilon}$ for $k_1, k_2 < \frac{1}{2}$. Section 12.5 proves, with the cut lemma of Chapter 11, that the tails of (T0) are small at the tip, and from this the two-sided law $\kappa_{\max} \asymp \varepsilon^\beta$. Section 12.6 proves the limit $\kappa_{\max}^2/\varepsilon \to A$ for $k < 1$ and the logarithmic law at $k = 1$. Section 12.7 explains, with a toy and a formal computation, why $k > 1$ is different, and Section 12.8 collects the numerical observations. Part B: bounds in terms of $(\boldsymbol{c}, x_0, f_-)$, large constants, cone limits, the shape of the lens, log-convexity. Part C: the class-N threshold law and the atlas of the chapter.

12.1. The setting, and the Landau example

In words. *Everything in Part A is the Schrödinger equation $z'' = Wz$ for the solution z of a class-*Z *datum, with the potential lowered by the swirl integral $\psi \geq 0$. We fix the notation once: the distance ε to the critical surface, the critical function g (the zero-energy bound state at $\varepsilon = 0$) and the Jost solutions $u_\pm$ of the no-swirl potential. The Landau case $\boldsymbol{c} = 0$ is the running example.*

12.1.1. Notation. Fix $x_0 \in (-1, 1)$, $\xi_0 = \operatorname{artanh} x_0$, $c_1, c_2 > -1$, $k_i = \sqrt{1 + c_i}$, $k = \max(k_1, k_2)$, and let $c_3 = \bar{c}_3 + \varepsilon$ with $0 < \varepsilon \leq \varepsilon_0$ (so $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$; ε_0 is chosen below, finitely often). We write $t = \xi - \xi_0$. A *class-*Z *datum* is $(\gamma, \varkappa)$ with $\varkappa > 0$ admissible in the sense of Section 3.9: $z > 0$ solves $z'' = Wz$, $z(\xi_0) = 1$, $h = z'/z$,

$h(\xi_0)=\frac12\gamma+x_0$, $W=W_0-\frac12\psi$, $f'=\varkappa z^{-2}$, $f(\xi_0)=0$ and $\psi=\psi_{\boldsymbol c}$ solves $\mathsf L[\psi]=f^2$, $\psi(\xi_0)=\psi'(\xi_0)=0$; $\varkappa=(1-x_0^2)\kappa$ with $\kappa=\kappa_x$ the swirl in the Cauchy normalisation (Lemma 2.15). Here $W_0=W_{\mathrm{crit}}+\frac{\varepsilon}{2}\operatorname{sech}^2\xi$ is the no-swirl potential. We use, from earlier chapters:

(Z1) (Section 3.9, (Z1)–(Z3)) $\psi\ge0$, ψ is nondecreasing in $|\xi-\xi_0|$, and $\psi(\xi)=\int_{\xi_0}^{\xi}K(\xi,s)ff'(s)ds$, $K=\sinh^2(\xi-s)/\cosh^2\xi\in[0,e^{-2s}]$ for $\xi\ge\xi_0$ (Lemma 11.37); $K(\xi,s)\ge c_0e^{-2s}$ if $\xi-s\ge1$ and $\xi\ge0$, $c_0=\frac14(1-e^{-2})^2$.

(Z2) (Lemma 5.41(e)) $\psi_+:=\lim_{+\infty}\psi\in(0,2k_2^2]$, and $\psi_-:=\lim_{-\infty}\psi\in(0,2k_1^2]$ by reflection.

(Z3) (Proposition 5.59) $\kappa_{\max}=\kappa_{\max}(x_0,\boldsymbol c)$ is attained; the tip is on the lower edge, $\gamma_*=\gamma^-(\kappa_{\max})$, and both poles of the tip solution are of type $\mathsf R,\mathsf L_1$ or $\mathsf L_2$, i.e. $h(+\infty)\le0\le h(-\infty)$ (as $U=2(h-\tanh\xi)$ and $U(1)\le-2$, $U(-1)\ge2$ for these types, Theorem 4.5). Also $\varkappa_{\max}=(1-x_0^2)\kappa_{\max}$ (Lemma 2.15).

(Z4) (Reflection) $\xi\mapsto-\xi$, $(x_0,c_1,c_2)\mapsto(-x_0,c_2,c_1)$ exchanges the poles and $\kappa_{\max}$ is invariant (Proposition 5.59(d)); statements are proved at the + pole and the mirror statement at the − pole is obtained by this reflection.

The critical function and the Jost solutions. Let g be the critical function of Lemma 3.30 *normalised* by $g(\xi_0)=1$: $g''=W_{\mathrm{crit}}g$, $h_g:=g'/g=\hat k-\bar k\tanh\xi=-k_2+\frac K2(1-\tanh\xi)=k_1-\frac K2(1+\tanh\xi)$ with $K=k_1+k_2$, so $h_g>-k_2$, $h_g<k_1$ and $h_g\le-k_2+Ke^{-2\xi}$, $h_g\ge k_1-Ke^{2\xi}$. Integrating from ξ_0,

$$e^{-k_2t}\le g\le G_0e^{-k_2t}\ (t\ge0),\qquad e^{-k_1|t|}\le g\le G_0e^{-k_1|t|}\ (t\le0),\qquad G_0:=\exp\bigl(\tfrac K2\max(e^{2\xi_0},e^{-2\xi_0})\bigr).\tag{12.1}$$

(For $t\ge0$: $\ln g+k_2t=\int_{\xi_0}^{\xi}(h_g+k_2)\in[0,K\int_{\xi_0}^{\infty}e^{-2s}ds]=[0,\frac K2e^{-2\xi_0}]$; for $t\le0$: $\ln g+k_1|t|=\int_{\xi}^{\xi_0}(k_1-h_g)\in[0,\frac K2e^{2\xi_0}]$.) Let $u_\pm$ be the Jost solutions of $u''=W_0u$ with $u_\pm(\xi_0)=1$ and $\zeta_\pm=u_\pm'/u_\pm$ (Lemma 3.21); $U_{\boldsymbol c}^{\mp}=2(\zeta_\pm-\tanh\xi)$ (Theorem 3.44). *Constants.* C,c denote positive constants that depend only on (x_0,c_1,c_2) and may change from line to line; they stay bounded when (c_1,c_2) ranges in a compact subset of $(-1,\infty)^2$ (every constant below is a continuous function of (x_0,c_1,c_2)). Numbered constants keep their meaning. Since $W_0\le k_2^2+Ae^{-2\xi}$ and $W_0\ge k_2^2-Ae^{-2\xi}$ on $\xi\ge0$ (Lemma 3.29, $\varepsilon\le\varepsilon_0\le1$), these constants do not depend on ε.

Toy example 12.1 (a zero-energy bound state and first-order perturbation)**.** At $\varepsilon=0$ the function g solves $-g''+W_{\mathrm{crit}}g=0$ and decays at both ends: it is a bound state at the bottom of the spectrum. Adding the small bump $\frac{\varepsilon}{2}\operatorname{sech}^2$ moves the bottom to $E_1\approx\langle g,\frac{\varepsilon}{2}\operatorname{sech}^2 g\rangle/\langle g,g\rangle$; in the language of this chapter, the log-derivatives ζ_-,ζ_+ of the two Jost solutions, which agree with h_g at $\varepsilon=0$, separate by an amount of order ε (Proposition 12.3), and the no-swirl window $[U_{\boldsymbol c}^-,U_{\boldsymbol c}^+]$ opens. By hand for $\boldsymbol c=(0,0)$, $x_0=0$: $g=\operatorname{sech}\xi$ (indeed $\operatorname{sech}''=(1-2\operatorname{sech}^2)\operatorname{sech}$) and $\int\operatorname{sech}^2\cdot g^2=\int\operatorname{sech}^4=[\tanh-\frac13\tanh^3]_{-\infty}^{\infty}=\frac43$, so the window has width $\approx\varepsilon\cdot\frac43$, half-width $\frac23\varepsilon$.

Example 12.2 (the Landau case $\boldsymbol c=(0,0,c_3)$, $x_0=0$)**.** Here $k_1=k_2=1$, $\bar c_3=-4$, $\varepsilon=c_3+4$, $g=\operatorname{sech}\xi$, $W_{\mathrm{crit}}=1-2\operatorname{sech}^2\xi$ (the Pöschl–Teller potential of Toy example 3.34), $W_0=1-(2-\frac{\varepsilon}{2})\operatorname{sech}^2\xi$. At $\varepsilon=4$ ($c_3=0$) $W_0\equiv1$, $u_\pm=e^{\mp\xi}$, the

window is $[-2,2]$ and the lens closes at $\kappa_{\max}=4.1608$ (Figure 5.5). As $\varepsilon\downarrow 0$ the computed thresholds are $\kappa_{\max}=0.72753,\ 0.21022,\ 0.06140,\ 0.01815,\ 0.00541,\ 0.00162$ for $\varepsilon=10^{-1},\dots,10^{-6}$, so that

$$\kappa_{\max}^2/\varepsilon = 5.2930,\ 4.4191,\ 3.7695,\ 3.2942,\ 2.9290,\ 2.6383$$

keeps decreasing, slowly: the signature of the logarithm of Theorem 12.15 (`code/ch12_thresh/verify_ch12a.py`).

12.2. (F1) and (F2): the window, the Jost bounds and the sandwich

In words. *Three facts, all consequences of the Wronskian of g with a solution. (F1): the no-swirl window has width ε times an order-one integral. Jost bounds: for $t\ge 0$ the Jost solutions are squeezed, $(1-C\varepsilon)g\le u_+\le g\le u_-$. (F2): every global swirling solution is squeezed between the two Jost solutions, grows at most like $e^{k_2 t}$, and its swirl integral is at most $\varkappa^2$ times the exponential $E_{k_2}(t)$.*

Proposition 12.3 (window, Jost bounds, sandwich)**.** *Let $0<\varepsilon\le\varepsilon_0$, $t=\xi-\xi_0$, g the critical function, $u_\pm$ the Jost solutions of $W_{\mathrm{crit}}+\frac{\varepsilon}{2}\operatorname{sech}^2$. The no-swirl window is $U_{c+}-U_{c-}=\varepsilon\big[\int_{-\infty}^{\xi_0}\operatorname{sech}^2 g\,u_- + \int_{\xi_0}^{\infty}\operatorname{sech}^2 g\,u_+\big]$ and $C_{\mathrm{w}}\varepsilon\le U_{c+}-U_{c-}\le\varepsilon\int\operatorname{sech}^2 g^2$ (width $\frac{4}{3}\varepsilon(1+O(\varepsilon))$ at $c=(0,0)$, $x_0=0$). Jost bounds: $(1-C_1\varepsilon)g\le u_+\le g\le u_-$ on $t\ge 0$. Lemma F2: let $(\gamma,\varkappa)$ be admissible with $\varkappa>0$ and z the corresponding solution of Section 12.1.1, $h=z'/z$, $\psi=\psi_{\boldsymbol{c}}$. Then $\zeta_+\le h\le\zeta_-$ on $\mathbb{R}$; $\limsup_{+\infty}h\le k_2''\coloneqq\sqrt{k_2^2-\psi_+/4}<k_2$; $gz\le C$ on $\mathbb{R}$; and on $t\ge 0$: $z\le Ce^{k_2 t}$ and $\psi\le C_\psi\varkappa^2 E_{k_2}(t)$ with $E_k(t)=\frac{e^{(4k-2)t}-1}{4k-2}$ ($k>\frac12$), t ($k=\frac12$), 1 ($k<\frac12$). Moreover, with $\varepsilon e^{2k_2 t}\le 1$ and $t\ge 0$: $u_-\le C_3 g$ and $z\le C_3 g$; and $u_+\ge\frac12 g$, $z\ge\frac12 g$ on $t\ge 0$. Mirror statements hold on $t\le 0$ with k_1, $u_-\le g\le u_+$ there.*

Proof. *Wronskians.* By Lemma 3.4 with $W_0-W_{\mathrm{crit}}=\frac{\varepsilon}{2}\operatorname{sech}^2$,

$$\operatorname{Wr}(g,u_-)'=\tfrac{\varepsilon}{2}\operatorname{sech}^2 g\,u_-,\qquad \operatorname{Wr}(g,u_+)'=\tfrac{\varepsilon}{2}\operatorname{sech}^2 g\,u_+. \tag{12.2}$$

At $-\infty$: $g\,u_-\to 0$ (both are $O(e^{k_1\xi})$) and $\zeta_- - h_g$ is bounded ($\zeta_-\to k_1$, $h_g\to k_1$), so $\operatorname{Wr}(g,u_-)=gu_-(\zeta_- - h_g)\to 0$; at $+\infty$ likewise $\operatorname{Wr}(g,u_+)\to 0$. Integrating (12.2), and using $g(\xi_0)=u_\pm(\xi_0)=1$,

$$\zeta_-(\xi_0)-h_g(\xi_0)=\tfrac{\varepsilon}{2}\int_{-\infty}^{\xi_0}\operatorname{sech}^2 g u_-,\qquad h_g(\xi_0)-\zeta_+(\xi_0)=\tfrac{\varepsilon}{2}\int_{\xi_0}^{\infty}\operatorname{sech}^2 g u_+. \tag{12.3}$$

Since $U_{c+}-U_{c-}=2(\zeta_- - \zeta_+)(\xi_0)$, adding gives the formula for the window. Moreover $\operatorname{Wr}(g,u_-)>0$ and $\operatorname{Wr}(g,u_+)<0$ on $\mathbb{R}$, so $(u_-/g)'=\operatorname{Wr}(g,u_-)/g^2>0$ and $(u_+/g)'<0$: with the normalisation at ξ_0, $u_-\ge g\ge u_+$ for $t\ge 0$ and $u_-\le g\le u_+$ for $t\le 0$. Hence $\int\operatorname{sech}^2 g u_\mp\le\int\operatorname{sech}^2 g^2$ over the corresponding half-lines: this is the upper bound of the window.

Lower bound of u_+. For $t\ge 0$, by (12.2), $(u_+/g)'=-\frac{\varepsilon}{2}g^{-2}(\xi)\int_\xi^\infty\operatorname{sech}^2 g u_+\ge -\frac{\varepsilon}{2}g^{-2}(\xi)\int_\xi^\infty\operatorname{sech}^2 g^2$. By (12.1), $g(s)^2/g(\xi)^2\le G_0^2 e^{-2k_2(s-\xi)}$ for $s\ge\xi\ge\xi_0$, and $\operatorname{sech}^2 s\le 4e^{-2s}$, so $g^{-2}(\xi)\int_\xi^\infty\operatorname{sech}^2 g^2\le 4G_0^2 e^{-2\xi}/(2+2k_2)\le Ce^{-2\xi}$. Integrating over $[\xi_0,\infty)$ (where $u_+/g=1$ at ξ_0): $u_+/g\ge 1-C_1\varepsilon$. The mirror gives $u_-\ge(1-C_1\varepsilon)g$ for $t\le 0$. Choose ε_0 with $C_1\varepsilon_0\le\frac12$: $u_+\ge\frac12 g$ on $t\ge 0$.

Lower bound of the window. On $[\xi_0,\xi_0+1]$: $\operatorname{sech}^2\xi\ge c$, $g\ge e^{-k_2}$, $u_+\ge\frac12 g$, so $\int_{\xi_0}^\infty\operatorname{sech}^2 g u_+\ge\frac{c}{2}e^{-2k_2}$; the same on the left. This is $U_{c+}-U_{c-}\ge C_{\mathrm{w}}\varepsilon$. For

$\boldsymbol{c} = (0,0)$, $x_0 = 0$: $g = \operatorname{sech}$, $(1 - C_1\varepsilon)g^2 \le gu_\pm \le g^2$ and $\int \operatorname{sech}^4 = \frac{4}{3}$ give $\frac{4}{3}\varepsilon(1 + O(\varepsilon))$.

$u_- \le C_3 g$. Let $r = u_-/g \ge 1$, $R(\xi) = \sup_{[\xi_0,\xi]} r$ and $t \ge 0$. By (12.2), $\operatorname{Wr}(g, u_-)(\xi) = \operatorname{Wr}(g, u_-)(\xi_0) + \frac{\varepsilon}{2}\int_{\xi_0}^{\xi} \operatorname{sech}^2 g^2 r$, and $\operatorname{Wr}(g, u_-)(\xi_0) \le \frac{\varepsilon}{2}\int \operatorname{sech}^2 g^2$ ((12.3), $u_- \le g$ on $t \le 0$). Hence $r' = \operatorname{Wr}/g^2 \le C\varepsilon e^{2k_2 t}(1 + R)$ (using $g^{-2} \le e^{2k_2 t}$). The right side is nondecreasing in ξ, so for $\xi' \le \xi$, $r(\xi') \le 1 + \int_{\xi_0}^{\xi} a(1 + R)$ with $a = C\varepsilon e^{2k_2 t}$; taking the supremum over ξ' and applying Gronwall's lemma to $1 + R \le 2 + \int a(1 + R)$ gives $1 + R \le 2\exp\int_{\xi_0}^{\xi} a \le 2\exp(C\varepsilon e^{2k_2 t}/2k_2) \le C_3'$ when $\varepsilon e^{2k_2 t} \le 1$. Thus $u_- \le C_3 g$ there, and integrating $r' \le C\varepsilon e^{2k_2 t}(1 + R) \le C\varepsilon e^{2k_2 t}$ once more gives $r - 1 \le C\varepsilon e^{2k_2 t}$ in the same range.

Lemma F2. (a) $\zeta_+ \le h \le \zeta_-$ is Corollary 3.52 (Chapter 3). In particular h is bounded on $\mathbb{R}$ ($\zeta_\pm$ are continuous with finite limits at $\pm\infty$), and $u_+ \le z \le u_-$ on $t \ge 0$ (as $(z/u_+)' = \operatorname{Wr}(u_+, z)/u_+^2 \ge 0$ and the mirror statement for u_-). So $z \ge u_+ \ge \frac{1}{2}g$ on $t \ge 0$, and $z \le C_3 g$ if $\varepsilon e^{2k_2 t} \le 1$. (b) $\limsup h \le k_2''$. Given $\delta > 0$ there is ξ_δ with $\psi \ge \psi_+ - \delta$ and $Ae^{-2\xi} \le \delta$ on $[\xi_\delta, \infty)$ (by (Z2)), so $W \le k_2^2 + \delta - \frac{1}{2}(\psi_+ - \delta) = k_2^2 - \frac{\psi_+}{2} + \frac{3}{2}\delta \le k_2''^2 + 2\delta$, where $k_2''^2 = k_2^2 - \psi_+/4 \ge \frac{1}{2}k_2^2$ (as $\psi_+ \le 2k_2^2$). Put $q = \sqrt{k_2''^2 + 2\delta}$. Where $h \ge q + \delta$: $h' = W - h^2 \le q^2 - (q+\delta)^2 \le -\delta^2$. So h (bounded) decreases at a definite rate until it is below $q + \delta$ and, since $h' < 0$ at $h = q + \delta$, stays below. Hence $\limsup h \le q + \delta \to k_2''$ as $\delta \downarrow 0$, and $k_2'' < k_2$ because $\psi_+ > 0$. (c) $z \le Ce^{k_2 t}$. Let $a = \min(1, k_2)$, $\hat{h} = k_2 + Be^{-at}$. By Lemma 3.29 $W \le W_0 \le k_2^2 + Ae^{-2\xi} \le k_2^2 + A'e^{-2at}$ ($A' = Ae^{-2\xi_0}$, $a \le 1$). Then $\hat{h}' - (W - \hat{h}^2) \ge -aBe^{-at} + 2k_2 Be^{-at} + B^2 e^{-2at} - A'e^{-2at} \ge 0$ if $B \ge \sqrt{A'}$ (as $2k_2 > a$). Take also $B \ge h(\xi_0) - k_2$ (possible: $h(\xi_0) \le \zeta_-(\xi_0) \le k_1 + C$ by (12.3), uniformly in ε). For $w = h - \hat{h}$: $w' = h' - \hat{h}' \le -(h^2 - \hat{h}^2) = -(h + \hat{h})w$, so $(w\, e^{\int (h+\hat{h})})' \le 0$ and $w(\xi_0) \le 0$ gives $w \le 0$ on $t \ge 0$. Integrating, $z \le \exp(k_2 t + B/a)$. With $g \le G_0 e^{-k_2 t}$ this gives $gz \le C$ on $t \ge 0$, and on $t \le 0$ by the mirror: $gz \le C$ on $\mathbb{R}$. (d) $\psi \le C_\psi \varkappa^2 E_{k_2}$. For $t \ge 0$: $f' = \varkappa z^{-2} \le 4\varkappa g^{-2} \le 4\varkappa e^{2k_2 t}$, $f \le 4\varkappa(e^{2k_2 t} - 1)/(2k_2) \le C\varkappa e^{2k_2 t}$, and by (Z1) $\psi(\xi) \le \int_{\xi_0}^{\xi} e^{-2s} f f' ds \le C\varkappa^2 \int_0^t e^{-2\tau} e^{4k_2 \tau} d\tau \le C_\psi \varkappa^2 E_{k_2}(t)$ (the integral is $\int_0^t e^{(4k_2 - 2)\tau} d\tau = E_{k_2}(t)$ for $k_2 > \frac{1}{2}$, equals t for $k_2 = \frac{1}{2}$, and is $\le (2 - 4k_2)^{-1}$ for $k_2 < \frac{1}{2}$.) □

12.3. (T0): the balance identity

In words. *Identity (E) of Chapter 3 measures the distance of the lens from one edge of the window with the Jost solution u_+ as reference. Take instead the critical function g as the reference solution and integrate over the whole line: the window mismatch, which is ε times an order-one integral, must equal the swirl mass $\int \psi g z$. The swirl must therefore be strong enough to carry mass ε, but $\psi \lesssim \varkappa^2 e^{(4k-2)t}$: this one sentence is the whole of Part A.*

Toy example 12.4 (the same computation with two exponentials)**.** If $a'' = \mu^2 a$ and $b'' = \nu^2 b$ on $[0, \infty)$ with $a = e^{-\mu s}$, $b = e^{-\nu s}$, then $\operatorname{Wr}(a, b) = (\mu - \nu)e^{-(\mu+\nu)s}$ and indeed $\int_0^\infty (\nu^2 - \mu^2)e^{-(\mu+\nu)s} ds = \nu - \mu = -\operatorname{Wr}(a, b)(0)$. (T0) is this computation with $a = g$, $b = z$ on the whole line, where both Wronskian boundary terms vanish.

THEOREM 12.5 (the balance identity (T0)). *Notation: class-Z base point ξ_0 ($f(\xi_0) = 0$, $\varkappa = \varkappa[\xi_0]$), g the critical function ($g(\xi_0) = 1$), $W = W_{\text{crit}} + \frac{\varepsilon}{2}\operatorname{sech}^2 - \frac{1}{2}\psi$,*

$\psi=\psi_{\boldsymbol{c}}$, $d_0=h(\xi_0)-h_g(\xi_0)$, $A_f=\int_{\xi_0}^{\infty}\operatorname{sech}^2 gz$, $B_f=\int_{\xi_0}^{\infty}\psi gz$ *(A_b, B_b the same over $(-\infty,\xi_0]$).* $\mathrm{Wr}(g,z)'=(\frac{\varepsilon}{2}\operatorname{sech}^2\xi-\frac12\psi)gz$. *For admissible $(\gamma,\varkappa)$ with $\varkappa>0$ of any pole types, gz is integrable,* $\mathrm{Wr}(g,z)\to0$ *at $\pm\infty$ and $\varepsilon\int\operatorname{sech}^2\xi\,g\,z=\int\psi\,g\,z$. Half-line identities $d_0=\frac12B_f-\frac{\varepsilon}{2}A_f=\frac{\varepsilon}{2}A_b-\frac12B_b$. Lem F4 ($A=0$): $c_i'=c_i-\psi_i/2$, $c_3'=c_3+(\psi_1+\psi_2)/4$ ($\psi_i=\psi(\mp\infty)$), $\varepsilon'=\varepsilon-\frac12[(k_1-k_1')(k_2+k_2'+2)+(k_2-k_2')(k_1+k_1'+2)]$.*

Proof. *Wronskian.* $z''=Wz$ and $g''=W_{\mathrm{crit}}g$, $W-W_{\mathrm{crit}}=\frac{\varepsilon}{2}\operatorname{sech}^2-\frac12\psi$, so Lemma 3.4 gives $\mathrm{Wr}(g,z)'=(\frac{\varepsilon}{2}\operatorname{sech}^2-\frac12\psi)gz$. *Integrability and decay.* By Proposition 12.3(F2), for every $\delta>0$, $z\le C_\delta e^{(k_2''+\delta)t}$ on $t\ge0$ (as $h\le k_2''+\delta$ for large t and h is continuous), and $g\le G_0e^{-k_2t}$; so $gz\le C_\delta e^{-(k_2-k_2''-\delta)t}$, integrable for $\delta<k_2-k_2''$ (this is where $\varkappa>0$, i.e. $\psi_+>0$, enters). The mirror estimate holds on $t\le0$. Since $h-h_g$ is bounded, $\mathrm{Wr}(g,z)=gz(h-h_g)\to0$ at $\pm\infty$. Also $0\le\psi\le2k^2$ and sech^2 is bounded, so both $\int\operatorname{sech}^2 gz$ and $\int\psi gz$ converge, and integrating the Wronskian identity over $\mathbb{R}$ gives (T0). *Half-lines.* Since $z(\xi_0)=g(\xi_0)=1$, $\mathrm{Wr}(g,z)(\xi_0)=h(\xi_0)-h_g(\xi_0)=d_0$. Integrating over $[\xi_0,\infty)$ and over $(-\infty,\xi_0]$ with $\mathrm{Wr}(\pm\infty)=0$: $d_0=-\int_{\xi_0}^{\infty}(\frac{\varepsilon}{2}\operatorname{sech}^2-\frac12\psi)gz=\frac12B_f-\frac{\varepsilon}{2}A_f$ and $d_0=\int_{-\infty}^{\xi_0}(\dots)=\frac{\varepsilon}{2}A_b-\frac12B_b$. *(F4).* This is Proposition 2.27(iii) with $A=0$: $\psi_-=\psi_{\boldsymbol{c}}(-1)$, $\psi_+=\psi_{\boldsymbol{c}}(1)$ are finite (Lemma 5.41(e)) and $Q(\mp1)=2c_{1,2}-\psi_\mp$, so $c_i'=c_i-\frac12\psi_i$ and $c_3'=-\frac14(\lambda+2c_1'+2c_2')=c_3+\frac14(\psi_1+\psi_2)$. For the formula for ε' put $S=k_1+k_2$, $S'=k_1'+k_2'$; $\psi_i=2(k_i^2-k_i'^2)$, and $\bar c_3=-\frac12S(S+2)$ give $\varepsilon'-\varepsilon=\frac12(k_1^2+k_2^2-k_1'^2-k_2'^2)-\frac12(S^2-S'^2)-(S-S')=-k_1k_2+k_1'k_2'-(S-S')$ (using $S^2-k_1^2-k_2^2=2k_1k_2$). Expanding $-\frac12[(k_1-k_1')(k_2+k_2'+2)+(k_2-k_2')(k_1+k_1'+2)]$ gives the same expression (computer-algebra check: `code/ch12_thresh/verify_ch12a.py`). □

Remark 12.6 (the size of the two sides)**.** Put $I_s:=\int\operatorname{sech}^2\xi\,gz\,d\xi$. Then $c_s\le I_s\le C$ for $\varepsilon\le\varepsilon_0$: the lower bound from $z\ge\frac12g$ on $[\xi_0,\xi_0+1]$ and $z\ge\frac12g$ on $[\xi_0-1,\xi_0]$ (Proposition 12.3), the upper bound from $gz\le C$. So (T0) reads $\int\psi gz=\varepsilon I_s\asymp\varepsilon$. The identity is exact, involves no smallness, and holds at the tip as well as in the interior of the lens. The three Wronskian computations (g,z), (u_+,z), (u_-,z) are those of identity (E) (Remark 3.53).

12.4. Upper bounds for the swirl threshold

In words. *From (T0): $\int\psi gz=\varepsilon I_s\le C\varepsilon$. A lower bound for ψ near the pole, which grows like $\varkappa^2e^{(4k-2)t}$, then bounds $\varkappa$ above: the larger k, the faster ψ grows and the smaller $\varkappa$ must be. If $k_1,k_2<\frac12$, a comparison with a no-swirl solution for slightly smaller constants gives the matching lower bound $\sqrt{\varepsilon}$.*

Put $\Lambda_k(t)=e^{(4k-2)t}$ $(k>\frac12)$, t $(k=\frac12)$, 1 $(k<\frac12)$, and $S_k(\varepsilon)=1$ $(k<1)$, $\ln\frac1\varepsilon$ $(k=1)$, $\varepsilon^{-(k-1)/k}$ $(k>1)$.

Lemma 12.7 (lower bound for ψ)**.** *Let $(\gamma,\varkappa)$ be admissible, $\varkappa>0$, and $t_\varepsilon=\ln(1/\varepsilon)/(2k_2)$. For $t\in[t_0,t_\varepsilon]$, $t_0:=\max(3,1-\xi_0)$: $\psi(\xi_0+t)\ge c\,\varkappa^2\Lambda_{k_2}(t)$.*

Proof. For $t\le t_\varepsilon$, $\varepsilon e^{2k_2t}\le1$, so $z\le C_3g\le C_3G_0e^{-k_2t}$ (Proposition 12.3) and $f'=\varkappa z^{-2}\ge c\varkappa e^{2k_2t}$; thus $f\ge c\varkappa(e^{2k_2t}-1)/2k_2\ge c'\varkappa e^{2k_2t}$ for $t\ge1$. For $\xi=\xi_0+t\ge0$ and $s=\xi_0+\tau\le\xi-1$, (Z1) gives $K\ge c_0e^{-2s}$, so $\psi(\xi)\ge c_0\int_1^{t-1}e^{-2\xi_0-2\tau}\,c\varkappa e^{2k_2\tau}\cdot c\varkappa e^{2k_2\tau}d\tau\ge c\varkappa^2\int_1^{t-1}e^{(4k_2-2)\tau}d\tau\ge c\varkappa^2\Lambda_{k_2}(t)$ for $t\ge3$ ($k_2>\frac12$: the integral is $\ge c\,e^{(4k_2-2)t}$; $k_2=\frac12$: $=t-2\ge t/3$; $k_2<\frac12$: $\ge\int_1^2$). □

THEOREM 12.8 (upper bounds). *There are ε_0, C depending only on (x_0, c_1, c_2) such that for $0 < \varepsilon \le \varepsilon_0$ and* every *admissible $(\gamma, \varkappa)$ with $\varkappa > 0$: $\varkappa^2 \le C\varepsilon/S_k(\varepsilon)$. In particular (with $\kappa_{\max} = \varkappa_{\max}/(1-x_0^2)$): $\kappa_{\max}^2 \le C\varepsilon$ for all k; $\kappa_{\max} \le C\varepsilon^{1-1/(2k)}$ for $k > 1$; and $\kappa_{\max}^2 \le C\varepsilon/\ln(1/\varepsilon)$ for $k = 1$.*

PROOF. Say $k = k_2$ (else reflect). By (T0) and Remark 12.6, $\int \psi g z \le C\varepsilon$. On $[t_0, t_\varepsilon]$ (nonempty for $\varepsilon \le \varepsilon_0$), $\psi \ge c\varkappa^2 \Lambda_k$ (Lemma 12.7) and $gz \ge \frac{1}{2}g^2 \ge \frac{1}{2}e^{-2kt}$, so

$$C\varepsilon \;\ge\; c\varkappa^2 \int_{t_0}^{t_\varepsilon} \Lambda_k(t) e^{-2kt} dt.$$

The integral is $\int e^{(2k-2)t} \ge c\, e^{(2k-2)t_\varepsilon} = c\,\varepsilon^{-(k-1)/k}$ for $k > 1$; $= t_\varepsilon - t_0 \ge c \ln \frac{1}{\varepsilon}$ for $k = 1$; and $\ge c$ for $k < 1$ (for $k \in [\frac{1}{2}, 1)$ the integrand is $\ge ce^{-(2-2k)t}$ resp. te^{-t}, and for $k < \frac{1}{2}$ it is e^{-2kt}; over $[t_0, t_0+1] \subset [t_0, t_\varepsilon]$ each is $\ge c$). Hence $\varkappa^2 \le C\varepsilon/S_k$. Apply to the tip. Finally $S_k \ge 1$ gives $\kappa_{\max}^2 \le C\varepsilon$, and for $k > 1$, $\varepsilon/S_k = \varepsilon^{(2k-1)/k}$. □

THEOREM 12.9 (a lower bound by comparison for $k_1, k_2 < \frac{1}{2}$). *If $k_1, k_2 < \frac{1}{2}$ then $\kappa_{\max} \ge c\sqrt{\varepsilon}$ for $\varepsilon \le \varepsilon_0$; with Theorem 12.8, $\kappa_{\max} \asymp \sqrt{\varepsilon}$.*

PROOF. *The input.* By Proposition 5.59(a) and Proposition 5.58(c), $\kappa_{\max} \ge \kappa_{\rm lo} = \sqrt{e/\max(B_+, B_-)}$ for every $e > 0$ with $\tilde{\boldsymbol{c}} = \boldsymbol{c} - \frac{e}{2}(1,1,0) \in \operatorname{int}\mathcal{J}$, every $\tilde\gamma$ in the open window of $\tilde{\boldsymbol{c}}$ at x_0, with B_+ of Lemma 5.48 (built from the no-swirl solution $\tilde U$ with $\tilde U(x_0) = \tilde\gamma$) and B_- its reflection. Let $\tilde z$ be the positive solution of $\tilde z'' = \tilde W_0 \tilde z$, $\tilde z(\xi_0) = 1$, belonging to $\tilde U$ ($\tilde U = 2(\tilde z'/\tilde z - \tanh\xi)$), and $\tilde Z = \int_{\xi_0}^{\xi} \tilde z^{-2}$. In the proof of Lemma 5.48, $\tilde g\, dx = d\xi/z^2$ for $z = y\cosh\xi$, $y(x_0) = 1$, i.e. $z = \cosh\xi_0\, \tilde z$; hence $\tilde f = \cosh^{-2}\xi_0 \tilde Z$, and with $\frac{1-s}{1+s} = e^{-2\xi}$, $B_+ = \cosh^{-4}\xi_0\, \hat B_+$, $\hat B_+ := \int_{\xi_0}^{\infty} e^{-2\xi} \tilde Z \tilde z^{-2} d\xi$, and likewise B_-. Since $\varkappa = \kappa/\cosh^2\xi_0$: $\varkappa_{\max}^2 \ge e/\max(\hat B_+, \hat B_-)$. *Choice of e.* Put $\tilde k_i = \sqrt{k_i^2 - e/2}$, $\tilde S = \tilde k_1 + \tilde k_2$, $S = k_1 + k_2$ and $\tilde\varepsilon = c_3 - \bar c_3(\tilde{\boldsymbol{c}}) = \varepsilon - \frac{1}{2}[(S^2 - \tilde S^2) + 2(S - \tilde S)]$. For $0 < e \le e_1 := \frac{1}{2}\min k_i^2$: $\tilde k_i \ge k_i/\sqrt{2}$ and $k_i - \tilde k_i = \frac{e/2}{k_i + \tilde k_i} \le \frac{e}{2k_i}$, so $\varepsilon - \tilde\varepsilon \le C_P e$ with $C_P = (S+1)\sum_i \frac{1}{2k_i}$; and $\varepsilon - \tilde\varepsilon(e_1) > 0$ is a fixed positive number, so for $\varepsilon \le \varepsilon_0$ small there is by continuity $e \in (0, e_1]$ with $\tilde\varepsilon = \varepsilon/2$, and then $\varepsilon/2 \le C_P e$: $e \ge \varepsilon/(2C_P)$. In particular $\tilde{\boldsymbol{c}} \in \operatorname{int}\mathcal{J}$. *Choice of $\tilde z$.* Let $\tilde u_\pm$ be the Jost solutions of $\tilde W_0 = W_{\rm crit}(\tilde k) + \frac{\tilde\varepsilon}{2}\operatorname{sech}^2$ and $\tilde z = \frac{1}{2}(\tilde u_- + \tilde u_+) > 0$. Its log-derivative $\tilde z'/\tilde z = \frac{1}{2}(\tilde\zeta_- + \tilde\zeta_+)$ at ξ_0 lies strictly inside the window, so $\tilde\gamma := 2(\tilde z'/\tilde z - x_0)$ is admissible in the sense above. By Proposition 12.3 for $\tilde{\boldsymbol{c}}$ (constants uniform since $\tilde k_i \in [k_i/\sqrt{2}, k_i]$), $\tilde z \ge \frac{1}{2}\max(\tilde u_+, \tilde u_-) \ge \frac{1}{2}(1 - C_1\tilde\varepsilon)\tilde g \ge \frac{1}{4}\tilde g$ on $\mathbb{R}$ (using $u_\mp \ge (1 - C_1\varepsilon)g$ on the respective half-lines and $u_- \ge g$, $u_+ \ge g$ on the other ones: on $t \ge 0$ $u_- \ge g$; on $t \le 0$ $u_+ \ge g$). *Bound of $\hat B_\pm$.* For $t \ge 0$: $\tilde z^{-2} \le 16\tilde g^{-2} \le 16 e^{2\tilde k_2 t}$, so $\tilde Z \le C e^{2\tilde k_2 t}$ and $\hat B_+ \le C\int_0^\infty e^{-2\xi_0 - 2t} e^{4\tilde k_2 t} dt \le B_* < \infty$, uniformly in ε, because $\tilde k_2 \le k_2 < \frac{1}{2}$ makes $4\tilde k_2 - 2 < 0$. The same holds for $\hat B_-$. Thus $\varkappa_{\max}^2 \ge e/B_* \ge c\varepsilon$. □

12.5. The tails are small at the tip: the two-sided law

In words. *(T0) integrates $\psi g z$ over the whole line. Split it at the Jost scale $\xi_a = \xi_0 + t_\varepsilon$, where $\varepsilon e^{2kt} = 1$: in the* bulk *$z \approx g$ (up to a constant), beyond it z may differ from g a lot. At the tip z is recessive-or-logarithmic at both poles, and the cut lemma of Chapter 11 says that the tail beyond ξ_a is a small fraction $\lambda\varepsilon$ of the total unless the swirl integral has already reached size $\Pi = O(1)$. Hence either the swirl*

is as large as the critical size $\varkappa^2 \sim \varepsilon^{(2k-1)/k}$, or the bulk carries the balance. Both alternatives give the lower bound.

For the two poles put $t_\varepsilon^i = \ln(1/\varepsilon)/(2k_i)$ and the *cut parameters* of Definition 11.36 $\Pi_i := \varkappa^2 e^{(4k_i-2)t_\varepsilon^i}$ ($k_i > \frac12$), $\Pi_i := \varkappa^2(1+t_\varepsilon^i)^2$ ($k_i \le \frac12$); $J_f := \int_{\xi_0+t_\varepsilon^2}^{\infty} \psi g z$, $J_b := \int_{-\infty}^{\xi_0-t_\varepsilon^1} \psi g z$.

Proposition 12.10 (tails)**.** *There are $C_*, \lambda_0 > 0$ and, for every $\lambda \in (0,\lambda_0]$, numbers $\eta(\lambda), \varepsilon_1(\lambda) > 0$ with the following property. If $\varepsilon \le \varepsilon_1(\lambda)$ and $(\gamma, \varkappa)$ is the tip solution (or any solution on the lower edge of the lens with $\varkappa \le 1$ and $h(+\infty) \le 0$) and $\Pi_2 \le \eta(\lambda)$, then $J_f \le C_*\lambda\varepsilon$. The same holds for J_b with Π_1 (at the tip).*

Proof. We use Lemma 11.39 (= Lemma 11.39) in the setting of Definition 11.35, with $k = k_2$ (its cut parameter is λ, as here). The tip solution is the edge solution at $\kappa = \kappa_{\max}$ (Proposition 5.59), and (F2) of Definition 11.35 is the property $h(+\infty) \le 0$ of (Z3). By Remark 11.40 the constants $\lambda_0^{\mathrm{cut}}, C_1, \eta(\lambda)$ and the threshold $T_0(\lambda) = T_W(\lambda)$ of the lemma depend on ε only through C_u and $T_W(\lambda)$, so it remains to bound these *uniformly in* $\varepsilon \in (0,\varepsilon_0]$: (i) $C_u^{-1}e^{-kt} \le u_+ \le C_u e^{-kt}$ on $t \ge 0$: by Proposition 12.3, $\frac12 g \le u_+ \le g$, so $C_u = 2G_0$ works; (ii) $W_0 \ge k^2 - \lambda k/16$ for $t \ge T_W$: since $W_0 \ge W_{\mathrm{crit}} \ge k^2 - Ae^{-2\xi}$; (iii) $\zeta_+ \ge -k - \lambda/4$ for $t \ge T_W$: by (12.3) applied at ξ in place of ξ_0 (the same computation), $\zeta_+(\xi) - h_g(\xi) = -\frac{\varepsilon}{2}\frac{\int_\xi^\infty \operatorname{sech}^2 g u_+}{g(\xi)u_+(\xi)} \ge -\frac{\varepsilon}{2}\frac{\int_\xi^\infty \operatorname{sech}^2 g^2}{\frac12 g(\xi)^2} \ge -C\varepsilon e^{-2\xi}$, and $h_g > -k_2$. So $T_W(\lambda)$ depends on λ and (x_0, c_1, c_2) only. Choose $\lambda_0 \le \lambda_0^{\mathrm{cut}}$ of the lemma. Take $\varepsilon_1(\lambda)$ so small that $t_\varepsilon^2 \ge T_0(\lambda)$ and $\varkappa \le 1$ (the latter by Theorem 12.8). With $T = t_\varepsilon^2$ the lemma gives, if $\Pi_2 = \Pi(T) \le \eta(\lambda)$, $\mathcal{T}(T) = \int_{\xi_a}^\infty \psi u_+ z \le 2C_u^2(1+C_1\lambda)\lambda e^{-2kT}$, and $e^{-2kT} = \varepsilon$. Finally $g \le u_+/(1-C_1\varepsilon) \le 2u_+$, so $J_f \le 2\mathcal{T}(T) \le C_*\lambda\varepsilon$. The statement for J_b is the reflection. □

Theorem 12.11 (the two-sided law). *For $x_0 \in (-1,1)$, $c_1, c_2 > -1$, $k = \max(k_1,k_2)$ there are $0 < c < C$ and $\varepsilon_0 > 0$ with*

$$c\,\varepsilon/S_k(\varepsilon) \;\le\; \varkappa_{\max}^2 \;\le\; C\,\varepsilon/S_k(\varepsilon), \qquad 0 < \varepsilon \le \varepsilon_0.$$

Equivalently $\kappa_{\max} \asymp \varepsilon^\beta$ with $\beta = \max(\frac12, 1-\frac{1}{2k})$ for $k \ne 1$, and $\kappa_{\max} \asymp (\varepsilon/\ln\frac{1}{\varepsilon})^{1/2}$ for $k = 1$.

The two-sided bound is illustrated by the numerical computation of $\kappa_{\max}$ against ε in Figure 12.2.

Proof. The upper bound is Theorem 12.8. For the lower bound work with the tip solution. Let $\lambda = \min(\lambda_0, c_s/(8C_*))$ (c_s of Remark 12.6) and $\eta = \eta(\lambda)$. *Case 1:* $\Pi_i > \eta$ *for some* i. Then $\varkappa^2 > \eta\,\varkappa^2/\Pi_i$ and $\Pi_i/\varkappa^2 = \varepsilon^{-(2k_i-1)/k_i} \le \varepsilon^{-(2k-1)/k}$ if $k_i > \frac12$ (the exponent $2-1/k_i$ increases with k_i), and $\Pi_i/\varkappa^2 = (1+t_\varepsilon^i)^2 \le C\ln^2\frac{1}{\varepsilon}$ if $k_i \le \frac12$. Hence $\varkappa^2 \ge c\,\varepsilon^{(2k-1)/k}$ or $\varkappa^2 \ge c/\ln^2(1/\varepsilon)$. Compare with ε/S_k: for $k > 1$, $\varepsilon/S_k = \varepsilon^{(2k-1)/k}$; for $k = 1$, $\varepsilon^{(2k-1)/k} = \varepsilon \ge \varepsilon/\ln\frac{1}{\varepsilon}$; for $k < 1$, $\varepsilon/S_k = \varepsilon \le \varepsilon^{(2k-1)/k}$ (the exponent is < 1 and $\varepsilon < 1$); and $c/\ln^2\frac{1}{\varepsilon} \ge \varepsilon \ge \varepsilon/S_k$ for small ε. In all cases $\varkappa^2 \ge c\varepsilon/S_k$. *Case 2:* $\Pi_1, \Pi_2 \le \eta$. Proposition 12.10 gives $J_f + J_b \le 2C_*\lambda\varepsilon \le \frac14 c_s\varepsilon$. By (T0) and Remark 12.6 the bulk integral $\mathcal{B} := \int_{\xi_0-t_\varepsilon^1}^{\xi_0+t_\varepsilon^2} \psi g z = \varepsilon I_s - J_f - J_b \ge \frac34 c_s\varepsilon$. On the bulk $z \le C_3 g$ (Proposition 12.3), $\psi \le C_\psi \varkappa^2 E_{k_i}(|t|)$ and $g^2 \le G_0^2 e^{-2k_i|t|}$ on the side of the ith pole, so $\mathcal{B} \le C\varkappa^2 \sum_i \int_0^{t_\varepsilon^i} E_{k_i}(t)e^{-2k_i t}dt \le C\varkappa^2 S_k(\varepsilon)$: for

$k_i > \frac{1}{2}$ the integrand is $\le Ce^{(2k_i-2)t}$, with integral $\le C\varepsilon^{-(k_i-1)/k_i}$ $(k_i > 1)$, $\le Ct^i_\varepsilon$ $(k_i = 1)$, $\le C$ $(k_i < 1)$; for $k_i \le \frac{1}{2}$ it is $\le C(1+t)e^{-2k_it}$ with bounded integral; and $S_{k_i} \le CS_k$ since $k_i \le k$. Hence $\varkappa^2 \ge c\varepsilon/S_k$. □

Remark 12.12 (the critical size)**.** For $k > 1$ the two cases meet: $\Pi_2 \asymp 1$ is exactly $\varkappa^2 \asymp \varepsilon^{(2k-1)/k}$, where the tail beyond ξ_a carries an order-one fraction of εI_s. So for $k > 1$ the *constant* in the law is decided beyond the Jost scale, in the pole layer of Section 12.7, not by the first-order profile; for $k < 1$ the bulk carries the balance and the constant is the first-order integral of Section 12.6.

12.6. The limit laws (d) and (e): $\kappa^2_{\max}/\varepsilon \to A$ for $k < 1$, the logarithm at $k = 1$

In words. *When both poles decay fast enough ($k < 1$), the swirl mass lives in the bulk, where $z \approx g$. Then (T0) becomes a statement about the* unperturbed *profile: $\varepsilon \int \operatorname{sech}^2 g^2 = \varkappa^2 \int \hat\psi[g]g^2$, where $\hat\psi[g]$ is the swirl integral produced by the unit-swirl flow $f' = g^{-2}$. For $k = 1$ that integral diverges logarithmically, which gives the logarithmic law at the end of the section.*

Toy example 12.13 ($A = 4$ at $k_1 = k_2 = \frac{1}{2}$, $x_0 = 0$)**.** Here $g^2 = \operatorname{sech}\xi$, $I_g := \int \operatorname{sech}^2 g^2 = \int \operatorname{sech}^3 = \frac{\pi}{2}$. The unit-swirl profile is $\hat F' = g^{-2} = \cosh\xi$, $\hat F = \sinh\xi$, and $\hat\psi(\xi) = \int_0^\xi \frac{\sinh^2(\xi-s)}{\cosh^2\xi} \sinh s \cosh s\, ds = \frac{1}{4}\tanh\xi(\xi - \tanh\xi)$ (a computer-algebra check, `code/ch12_thresh/verify_ch12a.py`, confirms $\mathsf{L}[\hat\psi] = \sinh^2\xi$). Then $P = \int \hat\psi g^2 = \int \frac{1}{4}\frac{\sinh\xi}{\cosh^2\xi}(\xi - \tanh\xi)d\xi = \frac{\pi}{8}$ (by parts: $\int_0^\infty \xi \frac{\sinh\xi}{\cosh^2\xi} = \int_0^\infty \operatorname{sech}\xi = \frac{\pi}{2}$ and $\int_0^\infty \frac{\sinh^2}{\cosh^3} = \frac{\pi}{2} - \frac{\pi}{4}$), so $A = I_g/P = 4$. Numerically $P = 0.3926990817 = \pi/8$ (`verify_ch12a.py`). This is the value that Theorem 12.15 predicts for $\kappa^2_{\max}/\varepsilon$; the shooting values at $\varepsilon = 10^{-3}, 10^{-4}$ are $3.9983, 3.9998$.

Let $\hat F(\xi) = \int_{\xi_0}^\xi g^{-2}$ and let $\hat\psi[g]$ solve $\mathsf{L}[\hat\psi] = \hat F^2$, $\hat\psi(\xi_0) = \hat\psi'(\xi_0) = 0$, i.e. $\hat\psi[g] = \int_{\xi_0}^\xi K\hat F\hat F'$ for $\xi \ge \xi_0$ and the mirror formula for $\xi \le \xi_0$. Then $\hat\psi[g] > 0$ for $\xi \ne \xi_0$.

Lemma 12.14 (growth of the first-order profile)**.** *For $t \ge 0$: $\hat\psi[g] \le CE_{k_2}(t)$, hence $\hat\psi[g]g^2 \le C(1+t)e^{-2\min(k_2,1-k_2)t}$ if $k_2 < 1$; the mirror on $t \le 0$.*

Proof. $\hat F \le \int_0^t e^{2k_2 s} \le Ce^{2k_2 t}$, $\hat F' = g^{-2} \le e^{2k_2 t}$ ((12.1)), so $\hat\psi \le \int_{\xi_0}^\xi e^{-2s}\hat F\hat F' \le C\int_0^t e^{(4k_2-2)\tau}d\tau \le CE_{k_2}(t)$ (as in Proposition 12.3(d)). With $g^2 \le G_0^2 e^{-2k_2 t}$: for $k_2 > \frac{1}{2}$, $\hat\psi g^2 \le Ce^{(2k_2-2)t}$; for $k_2 = \frac{1}{2}$, $\le Cte^{-t}$; for $k_2 < \frac{1}{2}$, $\le Ce^{-2k_2 t}$. In each case the bound is $C(1+t)e^{-2\min(k_2,1-k_2)t}$. □

Theorem 12.15 (laws (d) and (e)). *Notation: $\hat F(\xi) = \int_{\xi_0}^\xi g^{-2}$, $\hat\psi[g]$ the solution of $L[\hat\psi] = \hat F^2$ with zero Cauchy data at ξ_0, $k = \max(k_1, k_2)$, G_i the pole constants of g, $\kappa_{\max} = \kappa_{x,\max}$ in normalisation 1 and $\varkappa_{\max}$ in normalisation 2 ($\kappa_x = \varkappa\cosh^2\xi_0$). If $\max(k_1,k_2) < 1$ then $\hat\psi[g]g^2 \le C(1+|t|)e^{-2\min(k,1-k)|t|}$ is integrable, $P = \int\hat\psi[g]g^2 \in (0,\infty)$, $I_g = \int\operatorname{sech}^2 g^2$ and $\lim_{\varepsilon\downarrow 0}\varkappa^2_{\max}/\varepsilon = I_g/P$ (so $\kappa^2_{x,\max}/\varepsilon \to \cosh^4\xi_0 I_g/P$). For $k_2 = 1$: $\hat\psi[g]g^2 \to \frac{1}{12}G_2^{-2}$ at $+\infty$ (mirror at $-\infty$). For $k = 1$: $c_* = \frac{1}{12}\sum_{k_i=1} G_i^{-2}$ and $\varepsilon/\varkappa^2_{\max} = \frac{c_*}{2I_g}\ln\frac{1}{\varepsilon}(1+o(1))$; at $c = 0$, $x_0 = 0$: $\varepsilon/\kappa^2_{x,\max} = \frac{1}{64}\ln\frac{1}{\varepsilon}(1+o(1))$; numerically $\kappa_{x,\max}(0,0,0) = 4.1608$. Here G_i is defined by*

$g \sim G_2 e^{-\xi}$ at $+\infty$ ($k_2 = 1$) and $g \sim G_1 e^{\xi}$ at $-\infty$ ($k_1 = 1$); explicitly $G_2 = 2^{(k_1+1)/2}(1+x_0)^{-k_1/2}(1-x_0)^{-1/2}$ and $G_1 = 2^{(k_2+1)/2}(1+x_0)^{-1/2}(1-x_0)^{-k_2/2}$.

The behaviour of the integrand $\hat\psi[g]g^2$ in the three cases $k<1$, $k=1$, $k>1$ is plotted in Figure 12.1.

PROOF. *Part 1: the bound and positivity.* The bound is Lemma 12.14 on each side, with $\min(k_i, 1-k_i) > 0$ for $k_i < 1$; so $\hat\psi g^2$ is integrable, and $P > 0$ because $\hat\psi > 0$ off ξ_0 ($K > 0$, $\hat F\hat F' > 0$ for $\xi > \xi_0$).

Part 2: the limit for $k < 1$. Let z_ε, $\varkappa = \varkappa_\varepsilon$ belong to the tip. By Theorem 12.8, $\varkappa_\varepsilon^2 \le C\varepsilon \to 0$. (1) *Local convergence.* On $[\xi_0 - R, \xi_0 + R]$, $u_+ \le z \le u_-$ on the right and $u_- \le z \le u_+$ on the left, and by Proposition 12.3 $(1 - C_1\varepsilon)g \le u_+$, $u_- \le C_3 g$ with $u_-/g - 1 \le C\varepsilon e^{2kR}$ (proof of the proposition): so $z_\varepsilon \to g$ uniformly on compact sets. Hence $f/\varkappa = \int z^{-2} \to \hat F$, $f'/\varkappa \to g^{-2}$, and by the kernel formula $\psi/\varkappa^2 = \int K(f/\varkappa)(f/\varkappa)' \to \hat\psi[g]$, all uniformly on compact sets. (2) *Domination.* On the bulk $\{-t_\varepsilon^1 \le t \le t_\varepsilon^2\}$: $\psi/\varkappa^2 \le CE_{k_i}(|t|)$ and $gz \le C_3 g^2 \le Ce^{-2k_i|t|}$, so $\psi g z/\varkappa^2 \le D(t) := C(1+|t|)e^{-2\min(k_i, 1-k_i)|t|}$ (with i the pole on the side of t), which is integrable and independent of ε. Fix R. Then $\varkappa^{-2}\int_{|t|\le R}\psi g z \to \int_{|t|\le R}\hat\psi g^2$ by (1), and the rest of the bulk is $\le \varkappa^2 \int_{|t|>R} D$. Letting $\varepsilon \to 0$ and then $R \to \infty$: $\mathcal{B}/\varkappa^2 \to P$ for the bulk integral $\mathcal{B}$. (3) *Tails.* For $\frac12 < k_2 < 1$: $\Pi_2 \le C\varepsilon \cdot \varepsilon^{-(2k_2-1)/k_2} = C\varepsilon^{(1-k_2)/k_2} \to 0$; for $k_2 \le \frac12$, $\Pi_2 \le C\varepsilon(1+t_\varepsilon^2)^2 \to 0$; same for Π_1. By Proposition 12.10, for every λ small $J_f + J_b \le 2C_*\lambda\varepsilon$ once $\varepsilon \le \varepsilon(\lambda)$: $J_f + J_b = o(\varepsilon)$. (4) *Limit of I_s.* $\operatorname{sech}^2 gz \le C\operatorname{sech}^2$ (Proposition 12.3(F2)) and $z \to g$ pointwise, so $I_s \to I_g$ by dominated convergence. By (T0), $\varepsilon(I_g + o(1)) = \varkappa^2(P + o(1)) + o(\varepsilon)$, and $P > 0$: $\varkappa_{\max}^2/\varepsilon \to I_g/P$. The conversion $\kappa_x = \varkappa\cosh^2\xi_0$ gives the second form.

Part 3: the pole constant. Let $k_2 = 1$, $g \sim G_2 e^{-\xi}$ (from $g = g_{\mathrm c}/g_{\mathrm c}(\xi_0)$ and $g_{\mathrm c}e^{\xi} \to 2^{\bar k}$, Lemma 3.30; $g_{\mathrm c}(\xi_0) = (1+x_0)^{k_1/2}(1-x_0)^{1/2}$). Then $g^{-2} = G_2^{-2}e^{2s}(1 + O(e^{-2s}))$, $\hat F(s) = \frac12 G_2^{-2}e^{2s} + O(s)$, so $\varphi(s) := e^{-4s}\hat F(s)g^{-2}(s) \to \frac12 G_2^{-4}$ and φ is bounded. Using the exponential form $K = e^{-2\sigma}(1-e^{-2(\xi-\sigma)})^2/(1+e^{-2\xi})^2$ with $\sigma = s$, $d = \xi - s$:

$$\hat\psi(\xi) = \frac{e^{2\xi}}{(1+e^{-2\xi})^2}\int_0^{\xi-\xi_0} e^{-2d}(1-e^{-2d})^2\varphi(\xi-d)\,dd.$$

(because $e^{-2s}\hat F\hat F' = e^{-2s}\hat F g^{-2} = e^{2s}\varphi(s)$.) The integrand is dominated by $\sup\varphi \cdot e^{-2d}$, so by dominated convergence and $\int_0^\infty e^{-2d}(1-e^{-2d})^2 dd = \frac12 - \frac12 + \frac16 = \frac16$ the integral tends to $\frac{1}{12}G_2^{-4}$. With $g^2e^{2\xi} \to G_2^2$: $\hat\psi g^2 \to \frac{1}{12}G_2^{-2}$. (For $\boldsymbol{c} = 0$, $x_0 = 0$: $G_2 = 2$, limit $\frac{1}{48}$.) The mirror statement at $-\infty$ is the reflection.

Part 4: the logarithmic law, $k = 1$. Let $L = \ln(1/\varepsilon)$ and $I_1 \subset \{1, 2\}$ the set of poles with $k_i = 1$ (nonempty). By Theorem 12.8, $\varkappa^2 \le C\varepsilon/L$, so for a pole with $k_i = 1$, $\Pi_i = \varkappa^2/\varepsilon \le C/L \to 0$, and for $k_i < 1$, $\Pi_i \to 0$ as in Part 2(3). By Proposition 12.10, $J_f + J_b = o(\varepsilon)$; by (4), $I_s \to I_g$. So $\mathcal{B} = \varepsilon I_g(1 + o(1))$, and it remains to compute $\mathcal{B} = \sum_i \mathcal{B}_i$ (the two sides of the bulk). For a pole with $k_i < 1$, $\mathcal{B}_i \le C\varkappa^2 = o(\varkappa^2 L)$ (Part 2(2)). Consider the $+$pole with $k_2 = 1$, so $t_\varepsilon^2 = L/2$. Fix $\delta \in (0,1)$ and $T > 0$, and let $t_\delta = \frac12(L - \ln\frac{1}{\delta})$, so that $\varepsilon e^{2t_\delta} = \delta$. Split $[0, t_\varepsilon^2]$ at T and t_δ. On $[0, T]$ and on $[t_\delta, t_\varepsilon^2]$ the integrand $\psi g z$ is $\le C\varkappa^2$ (bulk bound of Part 2(2) with $E_1(t) \le e^{2t}/2$, $z \le C_3 g$, $g^2 \le Ce^{-2t}$), so these pieces contribute $\le C\varkappa^2(T + \ln\frac{1}{\delta})$. On $[T, t_\delta]$: $\varepsilon e^{2s} \le \delta$ for $s \le t_\delta$, so by the proof of Proposition 12.3, $(1 - C_1\varepsilon)g \le z \le (1 + C\delta)g$ and $z^{-2} = g^{-2}(1 + O(\delta))$. Hence $f' = \varkappa g^{-2}(1 + O(\delta))$

and, as the integral $\int_0^t e^{2s}\,ds$ is dominated by its upper end, $f = \varkappa\hat{F}(1+O(\delta))$ (integrate the pointwise relative bound $O(\delta)$ of z^{-2}/g^{-2}). Since $K \geq 0$ and both f and f' are within a factor $1+O(\delta)$ of $\varkappa\hat{F}$, $\varkappa g^{-2}$, $\psi = \varkappa^2\hat{\psi}(1+O(\delta))$ there, and $\psi g z/\varkappa^2 = \hat{\psi}g^2(1+O(\delta)) = \frac{1}{12}G_2^{-2}(1+O(\delta)+o_T(1))$ by Part 3. So

$$\begin{aligned}\varkappa^{-2}\mathcal{B}_+ &= \tfrac{1}{12}G_2^{-2}\big(\tfrac{L}{2} - \tfrac{1}{2}\ln\tfrac{1}{\delta} - T\big)(1+O(\delta)+o_T(1)) + O(T+\ln\tfrac{1}{\delta}) \\ &= \tfrac{L}{2}\cdot\tfrac{1}{12}G_2^{-2}\big(1+O(\delta)+o_T(1)+o_L(1)\big).\end{aligned}$$

Letting $\varepsilon \to 0$ ($L\to\infty$), then $T\to\infty$, $\delta\to 0$: $\mathcal{B}_+ = \varkappa^2\frac{L}{2}\frac{1}{12}G_2^{-2}(1+o(1))$; the same at $-\infty$. Hence $\mathcal{B} = \varkappa^2\frac{L}{2}c_*(1+o(1)) = \varepsilon I_g(1+o(1))$, which is $\varepsilon/\varkappa^2 = \frac{c_*}{2I_g}L(1+o(1))$. For $\boldsymbol{c}=0$, $x_0=0$: $G_i=2$, $c_* = \frac{1}{24}$, $I_g = \int\operatorname{sech}^4 = \frac{4}{3}$, so $\frac{c_*}{2I_g} = \frac{1}{64}$, and $\varkappa = \kappa_x$ at $x_0 = 0$. □

Corollary 12.16 (slope in $\ln(1/\kappa_{\max})$)**.** *For $k=1$, $\varepsilon/\varkappa_{\max}^2 = \frac{c_*}{I_g}\ln\frac{1}{\varkappa_{\max}}(1+o(1))$; at $\boldsymbol{c}=0$, $x_0=0$: $\varepsilon/\kappa_{\max}^2 = \frac{1}{32}\ln\frac{1}{\kappa_{\max}}(1+o(1))$.*

Proof. By Theorem 12.11, $\varkappa^2 \in [c\varepsilon/L, C\varepsilon/L]$, so $\ln\frac{1}{\varkappa} = \frac{1}{2}(L+O(\ln L)) = \frac{L}{2}(1+o(1))$; insert in Theorem 12.15. For $\kappa_{\max} = \varkappa_{\max}\cosh^2\xi_0$ the difference $\ln\cosh^2\xi_0$ is bounded. □

12.7. The pole layer for $k>1$: a toy and a formal computation

In words. *For $k>1$ the first-order integral $\int\hat{\psi}[g]g^2$ diverges like $e^{(2k-2)t}$ and the bulk carries only a vanishing part of (T0): the balance is decided in a thin layer near the pole, where the swirl integral has grown from ≈ 0 to order one and has lowered the potential from k^2 to $k'^2 < k^2$. Theorem 12.11 gives the order of $\kappa_{\max}$; this section only explains the mechanism that sets the constant. A step potential shows in two lines what such a layer does.*

Toy example 12.17 (a step potential)**.** Let $W = k^2$ for $\xi < \xi_L$ and $W = k'^2$ for $\xi > \xi_L$ ($0<k'<k$), and take the solution that is recessive at $+\infty$ ($\propto e^{-k'\xi}$ there). For $\xi<\xi_L$ write $z = e^{-k\xi} + ae^{k\xi}$. Matching z'/z at ξ_L:

$$\frac{-k+ake^{2k\xi_L}}{1+ae^{2k\xi_L}} = -k' \iff a = \theta e^{-2k\xi_L}, \qquad \theta = \frac{k-k'}{k+k'} \in (0,1)$$

(computer-algebra check in `verify_ch12a.py`). The layer leaves a *dominant* component of relative size $\theta e^{-2k\xi_L}$: exponentially small in the layer position and proportional to the drop $k-k'$. Compare the half-line identity $d_0 = \frac{1}{2}B_f - \frac{\varepsilon}{2}A_f$: the Wronskian at ξ_0 is fed by exactly such admixtures.

Formal computation 12.18 (layer scaling)**.** *Let $k=k_2>1$, $\xi_0=0$, and $\xi_L := \ln(1/\varkappa)/(2k-1)$, so $\varkappa^2e^{(4k-2)\xi_L}=1$ and $e^{-2k\xi_L} = \varkappa^{2k/(2k-1)}$. For $\xi < \xi_L - O(1)$ the tip solution is $z\approx g$ and $\psi \asymp \varkappa^2e^{(4k-2)\xi}$ (Lemma 12.7), which becomes $O(1)$ at ξ_L. In the variable $\tau = \xi-\xi_L$ put $z = e^{-k\xi_L}Z(\tau)$, $f = e^{\xi_L}F(\tau)$, $\psi = \tilde{\psi}(\tau)$ and neglect $\varepsilon\operatorname{sech}^2$ and the corrections $O(e^{-2\xi})$ (small in the layer). The problem becomes parameter free: $Z'' = (k^2 - \frac{1}{2}\tilde{\psi})Z$, $F' = Z^{-2}$, $\frac{1}{4}e^{2\tau}(\tilde{\psi}''+4\tilde{\psi}') \approx F^2$ (from $\mathsf{L}[\psi]=f^2$, $\cosh^2\xi \approx \frac{1}{4}e^{2\xi}$), with $Z\sim G^{-1}e^{-k\tau}$, $\tilde{\psi}\sim c_ke^{(4k-2)\tau}$ as $\tau\to-\infty$ and $\tilde{\psi}\to\psi_+$, Z recessive or linear as $\tau\to+\infty$ (tip). By Toy example 12.17 the layer feeds the outer region with a dominant admixture $a_\infty = \theta_\star e^{-2k\xi_L} = \theta_\star\varkappa^{2k/(2k-1)}$, $\theta_\star$ determined by the layer problem alone, and the half-line balance equates a_∞ to a multiple of*

ε. This predicts $\varkappa_{\max}^{2k/(2k-1)}/\varepsilon \to \mathrm{const}(k, G)$. This is a formal computation: a proof would need uniqueness of the layer problem and matching of the outer solution to it beyond the Jost scale; it is not proved here. Theorem 12.11 proves $\liminf > 0$ and $\limsup < \infty$ for this ratio; the limit is taken up in Chapter 13.

12.8. Numerical observations for Part A

In words. *This section puts numbers on the Part A theorems: the constant A for $k < 1$, the logarithmic law at $k = 1$, the exponents and ratios for $k > 1$, and the dependence on the base point. The numbers confirm the predicted exponents and slopes, and show how slowly the asymptotic regime is approached when the exponent is close to $\frac{1}{2}$ or 1. None of it is used in a proof.*

All numbers below come from the shooting and bisection code of the repository (relative tolerance 10^{-5} on $\kappa_{\max}$, `rtol` 10^{-11} in the integrations) driven by `code/ch12_thresh/verify_ch12a.py`; the runs for $k \neq 1$ (the pairs $k_1 = k_2 = k$ at $\varepsilon = 10^{-2}, \dots, 10^{-5}$) are those plotted by `make_a_kmax.py`.

Numerical observation 12.19 (the constant A for $k < 1$)**.** *At $x_0 = 0$ the first-order coefficient $A = I_g/P$ of Theorem 12.15 and the full computation of $\kappa_{\max}^2/\varepsilon$ agree:*

(c_1, c_2)	(k_1, k_2)	$A = I_g/P$	$\kappa_{\max}^2/\varepsilon$ *(shooting)*
$(-.75, -.75)$	$(.500, .500)$	4.00000	3.99983 *($\varepsilon = 10^{-4}$)*
$(-.6, -.6)$	$(.632, .632)$	4.72670	—
$(-.5, -.5)$	$(.707, .707)$	4.92704	4.93071 *($\varepsilon = 10^{-4}$)*
$(-.8, -.9)$	$(.447, .316)$	3.03761	—
$(-.9, -.9)$	$(.316, .316)$	2.61228	—
$(-.5, -.9)$	$(.707, .316)$	3.40694	—

For $c = (-\frac{1}{2}, -\frac{1}{2})$ the shooting ratio at $\varepsilon = 10^{-3}$ is 4.94290*: it approaches A from above, consistent with a correction of the size $\Pi_2 \sim \varepsilon^{(1-k)/k}$ of the tail parameter.*

Numerical observation 12.20 (the logarithmic law at $k = 1$)**.** *For $\boldsymbol{c} = 0$, $x_0 = 0$ and $\varepsilon = 10^{-2}, \dots, 10^{-6}$ a least-squares fit gives $\varepsilon/\kappa_{\max}^2 = 0.03140 \ln \frac{1}{\kappa_{\max}} + 0.1775$; the increments of $\varepsilon/\kappa_{\max}^2$ per unit of $\ln(1/\kappa_{\max})$ over successive decades are* 0.03168, 0.03141, 0.03129, 0.03126*, converging to $\frac{1}{32} = 0.03125$ (Corollary 12.16). The intercept* 0.1775 *is a numerical value only: the theorem determines the slope, and the intercept is not proved (see Chapter 13). The* relative *error of the leading term is still large at these ε because the intercept is not small against $\frac{1}{32} \ln \frac{1}{\kappa_{\max}}$, which is why the slope, not the value, is reported.*

Numerical observation 12.21 (exponents and the ratio for $k > 1$)**.** *The local exponents $d \ln \kappa_{\max}/d \ln \varepsilon$ over the last two decades ($\varepsilon = 10^{-4} \to 10^{-5}$ for the runs of `make_a_kmax.py`), against $\beta = \max(\frac{1}{2}, 1 - \frac{1}{2k})$: $k = 0.5$: 0.5000 ($\beta = 0.5$); $k = 0.9$: 0.5080 (0.5); $k = 1.2$: 0.5823 (0.5833); $k = 1.5$: 0.6627 (0.6667); $k = 3$: 0.8311 (0.8333). The compensated ratios $\kappa_{\max}^2 S_k/\varepsilon$ for $\varepsilon = 10^{-2}, \dots, 10^{-5}$ are* 7.739, 7.792, 7.841, 7.880 *($k = 1.2$),* 12.236, 13.130, 13.624, 13.877 *($k = 1.5$),* 23.567, 25.082, 25.704, 25.965 *($k = 3$): for $k = 1.5$ and $k = 3$ the increments shrink by about a factor two per decade (*0.894, 0.494, 0.253 *and* 1.515, 0.622, 0.261*), consistent with a finite limit (which Theorem 12.11 does not prove); for $k = 1.2$*

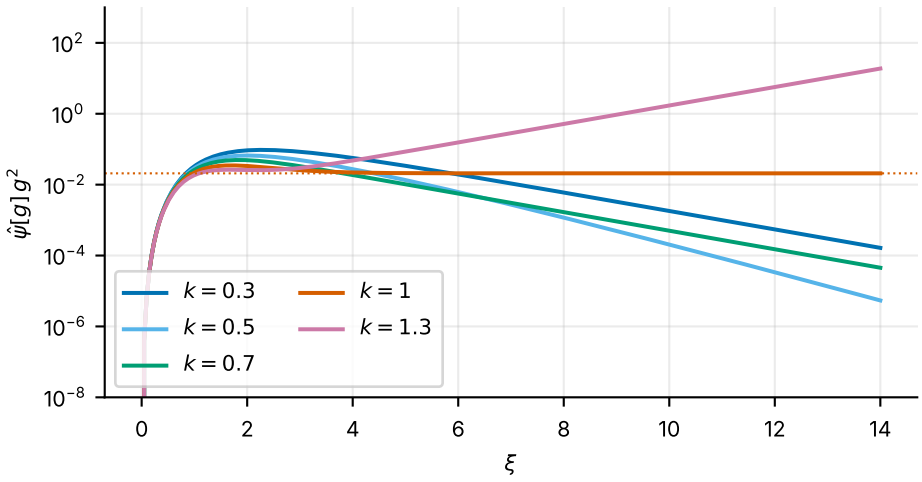


FIGURE 12.1. The first-order integrand $\hat{\psi}[g]g^2$ for $x_0 = 0$, $k_1 = k_2 = k$. For $k < 1$ it decays (like $e^{-2k\xi}$ for $k < \frac{1}{2}$, $\xi e^{-\xi}$ at $\frac{1}{2}$, $e^{-(2-2k)\xi}$ for $\frac{1}{2} < k < 1$) and is integrable; at $k = 1$ it tends to the constant $\frac{1}{48}$ (dotted, Theorem 12.15); for $k > 1$ it grows. Code: `make_a_psihat.py`.

the increments $(0.053, 0.049, 0.039)$ *shrink much more slowly, as expected when the correction exponent is small. For* $k = 0.9$ *the ratios* $4.805, 4.452, 4.226, 4.073$ *fall slowly towards the limit* A*, since the correction* $\varepsilon^{(1-k)/k} = \varepsilon^{0.11}$ *is small only very slowly.*

Numerical observation 12.22 (the base point $x_0 \to \pm 1$)**.** *For* $\boldsymbol{c} = (0, 0, 0)$ *(*$k = 1$*,* $\varepsilon = 4$*) one finds* $\kappa_{\max}(x_0) = 4.16081, 4.30666, 4.92483, 6.24344, 8.30044$ *at* $x_0 = 0, 0.3, 0.6, 0.8, 0.9$*: it grows, but* $(1-x_0^2)\kappa_{\max} = 4.1608, 3.9191, 3.1519, 2.2476, 1.5771$ *decreases. All results of Part A are for fixed* $x_0 \in (-1, 1)$ *with constants depending on* x_0*; the* ε*-law as* $x_0 \to \pm 1$ *is not treated here (see Chapter 13).*

12.9. Bounds for the threshold from the no-swirl problem

In words. *Part A found the size of* $\kappa_{\max}$ *near the critical surface. Here the constants* $\boldsymbol{c}$ *and the base point are arbitrary, and the aim is a pair of* explicit *bounds, one from above and one from below, that involve only the no-swirl problem: the Jost solutions* $u_\pm$ *and the width* w_0 *of the window. The upper bound is a one-line consequence of the sandwich* $z \le u_-$*: it makes the swirl* f at least *an explicit function, hence the swirl integral* ψ_+ *at least an explicit multiple of* $\varkappa^2$*, and* ψ_+ *may not exceed its ceiling* $2k_2^2$*. The lower bound is a continuity argument: compare the solution with a no-swirl solution for* lowered *constants* $\tilde{\boldsymbol{c}}$*, which lie on the other side of the swirl; as long as* ψ *stays below the lowering,* z *stays above the comparison solution and cannot reach zero.*

12.9.1. Setting and three elementary facts. Throughout this part $x_0 \in (-1, 1)$, $\xi_0 = \operatorname{artanh} x_0$, $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$, $k_i = \sqrt{1 + c_i}$, and $\kappa = \kappa_x$ is the swirl in the

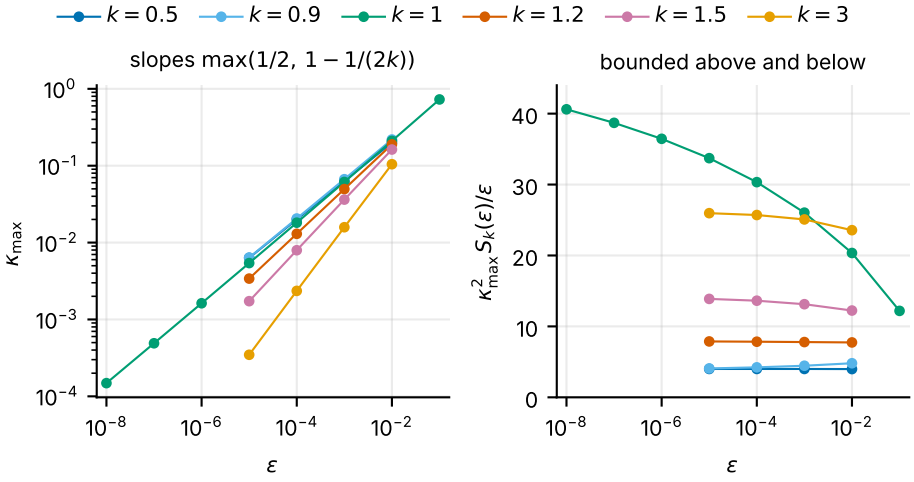


FIGURE 12.2. Left: $\kappa_{\max}$ against ε for $x_0 = 0$, $k_1 = k_2 = k$ (log–log). Right: the compensated ratio $\kappa_{\max}^2 S_k(\varepsilon)/\varepsilon$, which Theorem 12.11 says is bounded above and below; it tends to the constant A for $k < 1$ (for $k = \frac{1}{2}$, $A = 4$), keeps falling for $k = 0.9$ as $\varepsilon \downarrow 0$, rises slowly for $k = 1$ (towards 64 at $\boldsymbol{c} = 0$, by Corollary 12.16, since $\ln \frac{1}{\kappa_{\max}} \sim \frac{1}{2} \ln \frac{1}{\varepsilon}$; the vertical range is large because the approach is logarithmic), and is nearly flat for $k = 1.2$ and rising slowly for $k = 1.5, 3$. Code: `make_a_kmax.py`.

Cauchy normalisation, $\varkappa = (1 - x_0^2)\kappa = f'(\xi_0)$ (Lemma 2.15), so $\kappa = \cosh^2\xi_0\, \varkappa$ and $\sqrt{1 - x_0^2}\,\kappa = \varkappa \cosh \xi_0$; $\varkappa_{\max} = (1 - x_0^2)\kappa_{\max}$. The Jost solutions $u_\pm$ of $u'' = W_0 u$, $W_0 = 1 + \frac{1}{2} P_{\boldsymbol{c}}(\tanh \xi)$, are normalised by $u_\pm(\xi_0) = 1$, $\zeta_- = u_-'/u_-$, $\zeta_+ = u_+'/u_+$, and the *window width*

$$w_0 = w_0(\xi_0, \boldsymbol{c}) := \zeta_-(\xi_0) - \zeta_+(\xi_0) = \mathrm{Wr}(u_+, u_-) > 0$$

is the Wronskian of the pair (the sign convention of Chapter 3); in U-units the window is $[U_{\boldsymbol{c}}^-, U_{\boldsymbol{c}}^+] = [2(\zeta_+ - x_0), 2(\zeta_- - x_0)]$ of length $2w_0$. A datum $(\gamma, \varkappa)$, $\varkappa > 0$, is *admissible* if the solution (z, f, ψ) of the class-Z system (Z1)–(Z3) of Section 3.9 ($z(\xi_0) = 1$, $h = z'/z$, $f' = \varkappa z^{-2}$, $f(\xi_0) = 0$, $\mathsf{L}[\psi] = f^2$, $\psi(\xi_0) = \psi'(\xi_0) = 0$, $h(\xi_0) = \frac{1}{2}\gamma + x_0$) is global, i.e. $z > 0$ on $\mathbb{R}$ (Lemma 2.24); this is the same as $(\kappa, \gamma) \in \mathrm{Lens}(x_0, \boldsymbol{c})$ (Definition 5.55). Three facts are used all the time.

(Y1) Sandwich. For an admissible datum, $\zeta_+ \le h \le \zeta_-$ on $\mathbb{R}$ (Corollary 3.52); hence $(z/u_+)' = (h - \zeta_+)\, z/u_+ \ge 0$ and $(z/u_-)' = (h - \zeta_-)\, z/u_- \le 0$, and since all three functions equal 1 at ξ_0,

$$u_+ \le z \le u_- \quad \text{on } [\xi_0, \infty), \qquad u_- \le z \le u_+ \quad \text{on } (-\infty, \xi_0].$$

(Y2) Ceilings. For an admissible datum, $\psi_+ := \lim_{+\infty} \psi \in (0, 2k_2^2]$ and $\psi_- := \lim_{-\infty} \psi \in (0, 2k_1^2]$ (Lemma 5.41(e) and the reflection of Proposition 5.56).

(Y3) Monotonicity of the window. If $W \ge W_{\mathrm{ref}}$ have the same limits $k_1^2, k_2^2 > 0$ at $\mp\infty$ (at the rates of Definition 3.20) and W_{ref} is admissible, then W

is admissible and $w_0(W) \geq w_0(W_{\mathrm{ref}})$ (Lemma 3.23; the difference $\rho = \zeta_- - \zeta_+$ of the logarithmic derivatives does not depend on the normalisation of $u_\pm$, and after the normalisation $u_\pm(\xi_0) = 1$ one has $w_0 = \rho(\xi_0)$).

Lemma 12.23 (the swirl integral: kernel, ceilings, envelope)**.** *Let (z, f, ψ) solve (Z1)–(Z3) on an interval $I \ni \xi_0$ (with $z > 0$, $f' = \varkappa z^{-2}$, $\varkappa > 0$). Then for $\xi \in I$*

$$\cosh^2\xi\, \psi(\xi) = \int_{\xi_0}^{\xi} \sinh^2(\xi - s)\, f(s) f'(s)\, ds. \tag{12.4}$$

Consequently, with $x = \tanh\xi$:

(a) *for $\xi \geq s \geq \xi_0$: $0 \leq \dfrac{\sinh^2(\xi - s)}{\cosh^2\xi} \leq \dfrac{(1+x)^2}{4}\, e^{-2s} \leq e^{-2s}$, and the kernel increases to e^{-2s} as $\xi \to \infty$; for $\xi \leq \eta \leq \xi_0$: $0 \leq \dfrac{\sinh^2(\eta - \xi)}{\cosh^2\xi} \leq \dfrac{(1-x)^2}{4}\, e^{2\eta} \leq e^{2\eta}$;*

(b) *if $I = \mathbb{R}$ then $\psi_+ = \int_{\xi_0}^{\infty} e^{-2s} f f'\, ds$ and $\psi_- = \int_{-\infty}^{\xi_0} e^{2s} |f| f'\, ds$;*

(c) *if $I = \mathbb{R}$, then $0 \leq \psi \leq E := \frac{1}{4}\psi_-(1-x)^2 + \frac{1}{4}\psi_+(1+x)^2$ on $\mathbb{R}$. With $c_i' := c_i - \frac{1}{2}\psi_i$ ($\psi_1 = \psi_-$, $\psi_2 = \psi_+$), $c_3' := c_3 + \frac{1}{4}(\psi_1 + \psi_2)$ one has $P_{\boldsymbol{c}} - P_{\boldsymbol{c}'} = E$, so $W = W_0 - \frac{1}{2}\psi \;\geq\; W_{\mathrm{e}} := 1 + \frac{1}{2}P_{\boldsymbol{c}'}$.*

Proof. Write $\Phi = \cosh^2\xi\, \psi$. Since $(\cosh^2\xi)'' = 4\cosh^2\xi - 2$, one finds $\mathsf{L}[\psi] = \Phi'' - 4\Phi$ (a computer-algebra check: `verify_ch12b.py`, part `sym`), so $\Phi'' - 4\Phi = f^2$ with $\Phi(\xi_0) = \Phi'(\xi_0) = 0$. Let $\tilde\Phi(\xi)$ be the right-hand side of (12.4). Differentiating (the boundary term $\sinh^2 0 = 0$ vanishes) $\tilde\Phi' = \int_{\xi_0}^{\xi} \sinh(2(\xi - s)) f f'\, ds$ and $\tilde\Phi'' = 2\int_{\xi_0}^{\xi} \cosh(2(\xi - s)) f f'\, ds$, so $\tilde\Phi'' - 4\tilde\Phi = 2\int_{\xi_0}^{\xi} \big(\cosh 2u - 2\sinh^2 u\big) f f'\, ds = 2\int_{\xi_0}^{\xi} f f'\, ds = f(\xi)^2$ ($u = \xi - s$, $\cosh 2u - 2\sinh^2 u = 1$, $f(\xi_0) = 0$), and $\tilde\Phi(\xi_0) = \tilde\Phi'(\xi_0) = 0$. A linear equation with given data has one solution, so $\Phi = \tilde\Phi$.

(a) $\sinh^2(\xi - s) \leq \frac{1}{4}e^{2(\xi - s)}$ and $e^{2\xi}/\cosh^2\xi = (1+x)^2$ give the first bound; the identity $\frac{\sinh^2(\xi-s)}{\cosh^2\xi} = e^{-2s}\frac{(1-e^{-2(\xi-s)})^2}{(1+e^{-2\xi})^2}$ shows that the numerator increases and the denominator decreases in ξ, with limit e^{-2s}. The second statement is the mirror image ($\xi \mapsto -\xi$). (b) For $\xi \geq \xi_0$, $f f' \geq 0$ and the kernel increases to e^{-2s}: monotone convergence. For $\xi \leq \xi_0$, (12.4) reads $\cosh^2\xi\, \psi = \int_{\xi}^{\xi_0} \sinh^2(s - \xi)\, |f| f'\, ds$ (here $f \leq 0$), and the same argument applies. (c) Both $(1-x)^2\cosh^2\xi = e^{-2\xi}$ and $(1+x)^2\cosh^2\xi = e^{2\xi}$ satisfy $\Phi'' - 4\Phi = 0$, so $\mathsf{L}[E] = 0$. Put $v = \psi - E$: $\mathsf{L}[v] = f^2 \geq 0$, and $v \to 0$ at $\pm\infty$ because $\psi \to \psi_\pm$ and $E \to \psi_\pm$ ($(1 \pm x)^2 \to 4$ resp. 0). If $\sup v > 0$, it is attained at some ξ_1, where $v' = 0$, $v'' \leq 0$, so $\mathsf{L}[v](\xi_1) = \cosh^2\xi_1\, v'' - 2v \leq -2v(\xi_1) < 0 \leq f^2$, a contradiction. Finally $(1-x)^2 = 2(1-x) - (1-x^2)$ and $(1+x)^2 = 2(1+x) - (1-x^2)$ give $E = \frac{\psi_1}{2}(1-x) + \frac{\psi_2}{2}(1+x) - \frac{\psi_1+\psi_2}{4}(1-x^2) = P_{\boldsymbol{c}} - P_{\boldsymbol{c}'}$, hence $W_0 - \frac{1}{2}E = 1 + \frac{1}{2}P_{\boldsymbol{c}'}$ and $W = W_0 - \frac{1}{2}\psi \geq W_0 - \frac{1}{2}E = W_{\mathrm{e}}$. □

Remark 12.24. The last inequality, $W \geq W_{\mathrm{e}}$, is Proposition 2.27(i) (there from the explicit Green function of L); the maximum-principle proof above needs no kernel. It says: seen from the Schrödinger equation, a swirling solution is a no-swirl solution for the shifted constants $\boldsymbol{c}'$ plus a nonnegative remainder. The shift is the content of the term $+\frac{1}{4}(\psi_1 + \psi_2)$ in c_3'.

12.9.2. The theorem. Put $r_+ = u_+/u_-$ on $[\xi_0,\infty)$ and $r_- = u_-/u_+$ on $(-\infty,\xi_0]$ (both $= 1$ at ξ_0 and decreasing to 0, because $(u_+/u_-)' = -w_0/u_-^2$ and $(u_-/u_+)' = w_0/u_+^2$), and

$$I_+ = \int_{\xi_0}^{\infty} e^{-2s}\big(1-r_+(s)\big)^2 ds, \qquad I_- = \int_{-\infty}^{\xi_0} e^{2s}\big(1-r_-(s)\big)^2 ds.$$

For $e_S, e_N \ge 0$ with $e_S + e_N > 0$, $e_S \le 2k_1^2$, $e_N \le 2k_2^2$ let ("S" for the south pole $\xi = -\infty$, "N" for the north pole)

$$\tilde{\boldsymbol{c}} = \big(c_1 - \tfrac{e_S}{2},\ c_2 - \tfrac{e_N}{2},\ c_3 + \tfrac{e_S+e_N}{4}\big), \qquad E = \tfrac{e_S}{4}(1-x)^2 + \tfrac{e_N}{4}(1+x)^2,$$
$$\tilde W_0 = 1 + \tfrac12 P_{\tilde{\boldsymbol{c}}} = W_0 - \tfrac12 E$$

(the identity $P_{\boldsymbol{c}} - P_{\tilde{\boldsymbol{c}}} = E$ is that of Lemma 12.23(c)). Two cases of $\tilde{\boldsymbol{c}}$ are admitted: (i) $\tilde{\boldsymbol{c}} \in \operatorname{int}\mathcal{J}$ (no sign condition on $\tilde W_0$; it may be negative somewhere); or (ii) $\tilde c_1 = \tilde c_2 = -1$, $\tilde c_3 > 0$, where $\tilde W_0 = \frac{\tilde c_3}{2}\operatorname{sech}^2\xi \ge 0$. In both cases $\tilde u_\pm$ denote positive solutions of $\tilde u'' = \tilde W_0 \tilde u$ with $\tilde u_\pm(\xi_0) = 1$, $\tilde u_+$ recessive at $+\infty$ in case (i) and bounded there in case (ii) (mirror for $\tilde u_-$), and $\tilde w_0 = \operatorname{Wr}(\tilde u_+, \tilde u_-) > 0$; $\tilde r_+ = \tilde u_+/\tilde u_-$, $\tilde r_- = \tilde u_-/\tilde u_+$, $\tilde I_- = \int_{-\infty}^{\xi_0} e^{2s}(1-\tilde r_-)^2 ds$. (In case (ii) these objects exist and have $\tilde r_\pm \to 0$ at $\pm\infty$: see Step 0 below.)

THEOREM 12.25 (bounds for $\kappa_{\max}$ from the no-swirl problem). *Let $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ and $x_0 \in (-1,1)$.*

(a) (Upper bound.) *Every admissible swirl satisfies*

$$\varkappa \ \le\ w_0 \min\Big\{\frac{\sqrt2\,k_2}{\sqrt{I_+}},\ \frac{\sqrt2\,k_1}{\sqrt{I_-}}\Big\},$$

and in particular the same bound holds for $\varkappa_{\max}$.

(b) (Profile comparison: a lower bound.) *Let $e_S, e_N, \tilde{\boldsymbol{c}}$ be as above, with $\tilde{\boldsymbol{c}}$ of case* (i) *or* (ii)*. If there is $\alpha \in (0,1)$, $\beta = 1-\alpha$, with*

$$\text{(N)}\quad \frac{\varkappa^2 e^{-2\xi_0}}{2\beta^2\tilde w_0^2} < e_N, \qquad \text{(S)}\quad \frac{\varkappa^2\tilde I_-}{\alpha^3\tilde w_0^2} < e_S \quad \textit{or} \quad \frac{\varkappa^2 e^{2\xi_0}}{2\alpha^2\tilde w_0^2} < e_S,$$

then $\varkappa$ is admissible: $\varkappa_{\max} \ge \varkappa$.

(c) (Explicit form.) *If $e_0 > 0$ and $\tilde{\boldsymbol{c}}^0 := \boldsymbol{c} - \frac{e_0}{2}(1,1,0) \in \operatorname{int}\mathcal{J}$ with window width $\tilde w^0$, then*

$$\varkappa_{\max}\ \ge\ \sqrt{e_0/2}\,\tilde w^0/\cosh\xi_0, \qquad \textit{i.e.}\quad \sqrt{1-x_0^2}\,\kappa_{\max}\ \ge\ \sqrt{e_0/2}\,\tilde w^0.$$

Toy example 12.26 (a bootstrap)**.** $u' = -u + \delta u^2$, $u(0) = u_0 \in (0, \frac{1}{2\delta})$. Claim: $u \le u_0$ for all $t \ge 0$. Let $T = \sup\{t : u \le 2u_0 \text{ on } [0,t]\} > 0$. On $[0,T)$, $u' \le -u(1-2\delta u_0) < 0$, so $u \le u_0 < 2u_0$ on $[0,T]$; by continuity T cannot be finite. An *a priori* assumption ($u \le 2u_0$) is improved by the equation ($u \le u_0$), so it can never be violated first. Part (b) is this argument with the assumption $\psi \le E$ on the part of the half-line already examined, improved to $\psi < E$.

PROOF OF THEOREM 12.25. *(a)* Let (z,f,ψ) belong to an admissible datum. By (Y1), $z \le u_-$ on $[\xi_0,\infty)$, so $f' = \varkappa z^{-2} \ge \varkappa u_-^{-2}$. Since $(r_+)' = -w_0 u_-^{-2}$,

$$f(s) \ge \frac{\varkappa}{w_0}\big(1-r_+(s)\big) \ge 0, \qquad f'(s) \ge \frac{\varkappa}{w_0}\big(-r_+'(s)\big) \ge 0 \qquad (s \ge \xi_0).$$

By Lemma 12.23(b), $\psi_+ = \int_{\xi_0}^{\infty} e^{-2s} f f' ds \geq \frac{\varkappa^2}{w_0^2} \int_{\xi_0}^{\infty} e^{-2s}(1-r_+)(-r_+')ds$. As $(1-r_+)(-r_+') = \frac{1}{2}\big((1-r_+)^2\big)'$, integration by parts (boundary terms vanish: $1-r_+ = 0$ at ξ_0 and $e^{-2s} \to 0$) gives $\int_{\xi_0}^{\infty} e^{-2s}(1-r_+)(-r_+')\,ds = I_+$. So $\psi_+ \geq \varkappa^2 I_+/w_0^2$, and (Y2) gives $\varkappa^2 \leq 2k_2^2 w_0^2/I_+$. The second bound is the mirror image (use $z \leq u_+$ on $(-\infty, \xi_0]$, $(r_-)' = w_0 u_+^{-2}$, $\psi_- \leq 2k_1^2$). The statement on $\varkappa_{\max}$ holds because $\kappa_{\max}$ is attained (Proposition 5.59).

Step 0 (case (ii)). Let $\tilde{c}_1 = \tilde{c}_2 = -1$, $\tilde{c}_3 > 0$. The potential $\tilde{W}_0 = \frac{\tilde{c}_3}{2}\operatorname{sech}^2$ is integrable, so there is a solution $\tilde{u}_+$ with $\tilde{u}_+ \to 1$, $\tilde{u}_+' \to 0$ at $+\infty$: the equation $\tilde{u}_+(\xi) = 1 + \int_{\xi}^{\infty} (s-\xi)\tilde{W}_0(s)\tilde{u}_+(s)\,ds$ is a Volterra equation whose iteration converges on every $[\xi_1, \infty)$ because $\int^{\infty}(1+|s|)|\tilde{W}_0|\,ds < \infty$ (the n-th iterate is bounded by $\frac{1}{n!}\big(\int_{\xi}^{\infty}(s-\xi)|\tilde{W}_0|\,ds\big)^n$). Where $\tilde{u}_+ > 0$ it is convex ($\tilde{u}_+'' = \tilde{W}_0\tilde{u}_+ \geq 0$), so $\tilde{u}_+' \leq 0$ and $\tilde{u}_+ \geq 1$; hence $\tilde{u}_+ \geq 1 > 0$ on all of $\mathbb{R}$. The mirror solution $\tilde{u}_- \geq 1$ has $\tilde{u}_- \to 1$ at $-\infty$. They are independent: a multiple relation would give a bounded convex function on $\mathbb{R}$, constant, so $\tilde{W}_0\tilde{u} \equiv 0$, which is false. Hence $\tilde{w}_0 = \tilde{u}_+\tilde{u}_-(\tilde{\zeta}_- - \tilde{\zeta}_+) > 0$ ($\tilde{\zeta}_+ \leq 0 \leq \tilde{\zeta}_-$, $\tilde{w}_0 \neq 0$) and $\tilde{u}_- \to \infty$ at $+\infty$ (otherwise it would be bounded on $\mathbb{R}$), so $\tilde{r}_+ = \tilde{u}_+/\tilde{u}_- \to 0$ at $+\infty$; likewise $\tilde{r}_- \to 0$ at $-\infty$. After the normalisation $\tilde{u}_\pm(\xi_0) = 1$ these are the functions used below. In case (i) the same properties are those of Lemma 3.21(b).

(b) Setup. Put $\tilde{z} = \alpha\tilde{u}_+ + \beta\tilde{u}_-$. It is positive on $\mathbb{R}$, $\tilde{z}(\xi_0) = 1$, and $\tilde{h} := \tilde{z}'/\tilde{z}$ satisfies $\tilde{\zeta}_+(\xi_0) < \tilde{h}(\xi_0) < \tilde{\zeta}_-(\xi_0)$. Put $\gamma := 2(\tilde{h}(\xi_0) - x_0)$ and let (z, f, ψ) be the solution of (Z1)–(Z3) with these data and the given $\varkappa$, on its maximal interval (ξ_-^*, ξ_+^*) of positivity of z. Let $X' = \sup\{\xi \in [\xi_0, \xi_+^*) : \psi \leq E \text{ on } [\xi_0, \xi]\}$ and $X'' = \inf\{\xi \in (\xi_-^*, \xi_0] : \psi \leq E \text{ on } [\xi, \xi_0]\}$; since $\psi(\xi_0) = 0 < E(\xi_0)$, both are nontrivial.

Comparison. On (X'', X'), $W = W_0 - \frac{1}{2}\psi \geq W_0 - \frac{1}{2}E = \tilde{W}_0$, so by the Wronskian rule (Lemma 3.4) $\operatorname{Wr}(\tilde{z}, z)' = (W - \tilde{W}_0)\tilde{z}z \geq 0$, and $\operatorname{Wr}(\tilde{z}, z)(\xi_0) = 0$ (same data). As $\operatorname{Wr}(\tilde{z}, z) = \tilde{z}^2(z/\tilde{z})'$, the ratio $z/\tilde{z}$ is nondecreasing on $[\xi_0, X')$ and nonincreasing on $(X'', \xi_0]$; therefore $z \geq \tilde{z} > 0$ on (X'', X'). Hence $f' = \varkappa z^{-2} \leq \varkappa\tilde{z}^{-2}$ and, with $\tilde{F}(s) = \int_{\xi_0}^{s} \tilde{z}^{-2}$ ($s \geq \xi_0$), resp. $\int_s^{\xi_0} \tilde{z}^{-2}$ ($s \leq \xi_0$), $|f| \leq \varkappa\tilde{F}$ on (X'', X').

Right half-line. Write $\tilde{z} = \tilde{u}_-(\alpha\tilde{r}_+ + \beta)$ with $\tilde{r}_+' = -\tilde{w}_0\tilde{u}_-^{-2}$. With the elementary integral $\int_r^1 \frac{d\rho}{(\alpha\rho+\beta)^2} = \frac{1-r}{\beta+\alpha r}$ (using $\alpha + \beta = 1$; a computer-algebra check),

$$\tilde{F}(s) = \frac{1}{\tilde{w}_0}\int_{\tilde{r}_+(s)}^{1} \frac{d\rho}{(\alpha\rho+\beta)^2} = \frac{1-\tilde{r}_+(s)}{\tilde{w}_0\,(\beta + \alpha\tilde{r}_+(s))} \uparrow \frac{1}{\beta\tilde{w}_0} \quad (s \to \infty,\ \tilde{r}_+ \to 0).$$

By Lemma 12.23(a), for $\xi \in [\xi_0, X')$ and $x = \tanh\xi$,

$$\begin{aligned}\psi(\xi) &\leq \frac{(1+x)^2}{4}\int_{\xi_0}^{\xi} e^{-2s} f f' ds \leq \frac{(1+x)^2}{4}\,\varkappa^2 e^{-2\xi_0}\int_{\xi_0}^{\infty} \tilde{F}\tilde{F}' ds \\ &= \frac{(1+x)^2}{4}\cdot\frac{\varkappa^2 e^{-2\xi_0}}{2\beta^2\tilde{w}_0^2} < \frac{e_N}{4}(1+x)^2 \leq E,\end{aligned}$$

the strict inequality being (N).

Left half-line. Now $\tilde{z} = \tilde{u}_+(\alpha + \beta\tilde{r}_-)$ and $\tilde{r}_-' = \tilde{w}_0\tilde{u}_+^{-2}$; the same integral gives $\tilde{F}(s) = \frac{1-\tilde{r}_-(s)}{\tilde{w}_0(\alpha+\beta\tilde{r}_-(s))} \leq \frac{1-\tilde{r}_-(s)}{\alpha\tilde{w}_0}$ and $\tilde{z}^{-2} \leq \alpha^{-2}\tilde{u}_+^{-2} = \alpha^{-2}\tilde{r}_-'/\tilde{w}_0$. For $\xi \in (X'', \xi_0]$,

by Lemma 12.23(a), (12.4), $\xi \le \eta \le \xi_0$ and $f' \le \varkappa\tilde z^{-2}$,

$$\psi(\xi) \le \frac{(1-x)^2}{4}\int_\xi^{\xi_0} e^{2\eta}|f|f'd\eta \le \frac{(1-x)^2}{4}\cdot\frac{\varkappa^2}{\alpha^3\tilde w_0^{\,2}}\int_{-\infty}^{\xi_0} e^{2\eta}(1-\tilde r_-)\tilde r_-'\,d\eta$$
$$= \frac{(1-x)^2}{4}\cdot\frac{\varkappa^2\tilde I_-}{\alpha^3\tilde w_0^{\,2}},$$

where the last equality is the same integration by parts as in (a) ($(1-\tilde r_-)\tilde r_-' = -\frac12((1-\tilde r_-)^2)'$, the boundary terms vanish because $\tilde r_-(\xi_0)=1$ and $e^{2\eta}\to 0$). This is $< \frac{e_S}{4}(1-x)^2 \le E$ by the first form of (S). For the second form use instead $\int_\xi^{\xi_0} e^{2\eta}\tilde F|\tilde F'|\,d\eta \le e^{2\xi_0}\tilde F(-\infty)^2/2 = e^{2\xi_0}/(2\alpha^2\tilde w_0^{\,2})$.

Closing. On (X'',X') the inequality $\psi\le E$ has been improved to $\psi<E$ (strictly, with room at every point). If $X'<\xi_+^*$, then by continuity $\psi(X')=E(X')$ would hold at the supremum, a contradiction; so $X'=\xi_+^*$, and likewise $X''=\xi_-^*$. On $[\xi_0,\xi_+^*)$ we have $z\ge\tilde z\ge\delta>0$ if $\xi_+^*<\infty$ (the continuous positive function $\tilde z$ has a positive minimum on compact intervals), which contradicts Lemma 2.24 (a finite endpoint of the maximal interval requires $\liminf z=0$). So $\xi_\pm^*=\pm\infty$: $z>0$ on $\mathbb{R}$, the datum $(\gamma,\varkappa)$ is admissible.

(c) Apply (b) with $e_S=e_N=e_0$ (so $e_0<2\min k_i^2$ since $\tilde k_i^2=k_i^2-\frac{e_0}{2}>0$ in $\tilde{\boldsymbol c}^{\,0}\in\operatorname{int}\mathcal J$). The new constants are $\tilde{\boldsymbol c}=(c_1-\frac{e_0}{2},c_2-\frac{e_0}{2},c_3+\frac{e_0}{2})$, with $E=\frac{e_0}{2}(1+x^2)$, $\tilde W_0=W_0-\frac{e_0}{4}(1+x^2)$, whereas $\tilde W_0^{\,0}=W_0-\frac{e_0}{2}$; so $\tilde W_0-\tilde W_0^{\,0}=\frac{e_0}{4}(1-x^2)=\frac{e_0}{4}\operatorname{sech}^2\xi\ge 0$. Both potentials tend to $k_i^2-\frac{e_0}{2}$ at $\mp\infty$ at the rate $e^{-2|\xi|}$, and $\tilde W_0^{\,0}$ is admissible, so by (Y3) $\tilde W_0$ is admissible ($\tilde{\boldsymbol c}\in\operatorname{int}\mathcal J$) and $\tilde w_0\ge\tilde w^{\,0}$. Choose $\beta=e^{-\xi_0}/(2\cosh\xi_0)$, $\alpha=e^{\xi_0}/(2\cosh\xi_0)$ ($\alpha+\beta=1$). Then both (N) and the second form of (S) read $\frac{2\cosh^2\xi_0\,\varkappa^2}{\tilde w_0^{\,2}}<e_0$, which is implied by $\varkappa<\sqrt{e_0/2}\,\tilde w^{\,0}/\cosh\xi_0$. So every such $\varkappa$ is admissible. □

Example 12.27 (the rest state)**.** $\boldsymbol c=0$, $x_0=0$: $W_0\equiv 1$, $u_\pm=e^{\mp\xi}$, $w_0=2$, $r_+=e^{-2\xi}$ and $I_+=\int_0^\infty e^{-2s}(1-e^{-2s})^2ds=\frac12-\frac12+\frac16=\frac16$, likewise $I_-=\frac16$. Theorem 12.25(a): $\varkappa_{\max}\le 2\sqrt2/\sqrt{1/6}=4\sqrt3=6.9282$. For (c): $\tilde{\boldsymbol c}^{\,0}=-\frac{e_0}{2}(1,1,0)$ has $\tilde W_0^{\,0}\equiv 1-\frac{e_0}{2}$ and $\tilde w^{\,0}=2\sqrt{1-e_0/2}$, so $\sqrt{e_0/2}\,\tilde w^{\,0}=2\sqrt{\frac{e_0}{2}(1-\frac{e_0}{2})}\le 1$ with equality at $e_0=1$. Hence $1\le\kappa_{\max}(0,0)\le 4\sqrt3$, and the computed value is $\kappa_{\max}=4.1608$ (the constants of the computer-algebra check are in `verify_ch12b.py`).

Numerical observation 12.28 (how good are the bounds)**.** *Table 12.1 compares the two bounds with the computed threshold at $x_0=0$ (the lower bound of part (c) maximised over a grid of e_0, hence* 0.9999 *rather than the supremum* 1 *in the first row; the upper bound with $I_\pm$ by quadrature). The upper bound is within a factor* 1.26–1.77 *of the truth and the lower bound is far weaker,* 0.067–0.487 *of it; the lower bound is good only when $\boldsymbol c$ is far from the critical surface. In the cone limit the upper bound becomes sharp (Section 12.11): for $\boldsymbol c=0$,* $\cosh\xi_0\cdot$UB *against the computed* $\sqrt{1-x_0^2}\,\kappa_{\max}$ *is* 3.64642 *against* 3.61808 *at* $x_0=0.9$ *(relative difference* $7.8\cdot10^{-3}$*),* 3.48151 *against* 3.48108 *at* 0.99 *(*$1.3\cdot10^{-4}$*), and* 3.46583 *against* 3.46583 *at* 0.999 *(*$1.8\cdot10^{-6}$*). (`code/ch12_thresh/verify_ch12b.py`, part `rest`.)*

Proposition 12.29 (the tip is not above the critical surface for its effective constants)**.** *Let $(\gamma_*,\varkappa_{\max})$ be the tip, with pole values $\psi_-\le 2k_1^2$, $\psi_+\le 2k_2^2$ ((Y2))*

$\boldsymbol{c}$	$\kappa_{\max}$	lower bound (c)	upper bound (a)	lower/true	upper/true
$(0,0,0)$	4.1608	0.9999	6.9282	0.240	1.665
$(-\frac{1}{2},-\frac{1}{2},0)$	2.5737	0.4999	4.0602	0.194	1.578
$(2,-\frac{1}{2},0)$	4.7071	1.4160	5.9274	0.301	1.259
$(0,0,5)$	5.7586	2.8067	10.1744	0.487	1.767
$(0,0,-2.9)$	2.4281	0.1621	3.7686	0.067	1.552

TABLE 12.1. Theorem 12.25 against the computed $\kappa_{\max}$ at $x_0 = 0$ ($\kappa_{\max} = \varkappa_{\max}$). Code: `verify_ch12b.py`, part `rest`.

and effective constants $\boldsymbol{c}'$ of Lemma 12.23(c) (so that $k_i'^2 = k_i^2 - \frac{1}{2}\psi_i \geq 0$). Then $\boldsymbol{c}' \notin \operatorname{int}\mathcal{J}$: $c_3' \leq \bar{c}_3(c_1', c_2')$.

PROOF. If $\psi_+ = 2k_2^2$ (resp. $\psi_- = 2k_1^2$) then $k_2' = 0$, i.e. $c_2' = -1$ (resp. $c_1' = -1$), and $\boldsymbol{c}' \notin \operatorname{int}\mathcal{J}$ because points of $\operatorname{int}\mathcal{J}$ have $c_1', c_2' > -1$ (Theorem 3.44); so the claim holds. Otherwise $\psi_\pm < 2k_\pm^2$, that is $k_1', k_2' > 0$, and we argue by contradiction. If $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$, let $\tilde{u}_\pm$ be the Jost solutions of $W_{\mathrm{e}} = 1 + \frac{1}{2}P_{\boldsymbol{c}'}$ (normalised at ξ_0) and z the tip solution. By Lemma 12.23(c), $\operatorname{Wr}(\tilde{u}_+, z)' = (W - W_{\mathrm{e}})\tilde{u}_+ z \geq 0$. At the tip the lower-edge solution has a pole of type R, L_1 or L_2 at $+\infty$ (Proposition 5.59(c)), so $h(+\infty) \leq 0$ (as $U = 2(h - \tanh\xi) \to U(1) \leq -2$), and z grows more slowly than any exponential $e^{\mu\xi}$, $\mu > 0$, while $\tilde{u}_+ \leq Ce^{-k_2'\xi}$ with $k_2' > 0$; so $\operatorname{Wr}(\tilde{u}_+, z) = \tilde{u}_+ z\,(h - \tilde{\zeta}_+) \to 0$ at $+\infty$ and $\operatorname{Wr}(\tilde{u}_+, z)(\xi_0) \leq 0$, i.e. $h(\xi_0) \leq \tilde{\zeta}_+(\xi_0)$. The mirror argument (south pole, type of the tip is the same on both sides) gives $h(\xi_0) \geq \tilde{\zeta}_-(\xi_0)$. Hence $\tilde{w}_0 = \tilde{\zeta}_- - \tilde{\zeta}_+ \leq 0$, contradicting $\tilde{w}_0 > 0$ for $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$. The relation with $\bar{c}_3$ is the equivalence $\boldsymbol{c}' \in \operatorname{int}\mathcal{J} \iff c_3' > \bar{c}_3(c_1', c_2')$ for $c_i' > -1$ (Theorem 3.44). □

By Part A's identity, $\varepsilon' := c_3' - \bar{c}_3(c_1', c_2')$ equals $\varepsilon - \frac{1}{2}[(k_1 - k_1')(k_2 + k_2' + 2) + (k_2 - k_2')(k_1 + k_1' + 2)]$ (Theorem 12.5; the algebra is checked in `verify_ch12b.py`), so the proposition reads $\varepsilon' \leq 0$: the tip uses up the distance ε to the critical surface.

12.10. Large constants

In words. *If a constant c_i is large, W_0 is a deep and slowly varying well and the no-swirl window has width $w_0 \approx 2\sqrt{W_0(\xi_0)}$. Only a big swirl can close a big window, and how big is decided by the ceilings $\psi_\pm \leq 2k_\pm^2$ (the swirl may not push the potential at the poles below zero). We first prove two-sided bounds with the right* exponents*: $\kappa_{\max} \asymp \sqrt{c_3}$ if only c_3 grows, $\sqrt{c_i}$ if only c_1 or c_2 grows, and $\asymp p$ along a ray with $c_1 c_2 \neq 0$. Then we prove the sharp law*

$$\varkappa_{\max} \simeq \varkappa_{\mathrm{WKB}} := \frac{4\,a\,k_1 k_2}{k_1 e^{-\xi_0} + k_2 e^{\xi_0}}, \qquad a = \sqrt{W_0(\xi_0)},$$

a harmonic mean of the two pole terms, whenever $R_0/a^2 \to 0$, e.g. when $c_3 \to \infty$.

Toy example 12.30 (a deep well)**.** Let $W \equiv a^2$ near ξ_0. The solutions are $z = Ae^{aS} + Be^{-aS}$, $S = \xi - \xi_0$, $A + B = 1$, and $\int_0^\infty \frac{dS}{(Ae^{aS} + Be^{-aS})^2} = \frac{1}{2aA}$, $\int_{-\infty}^0 = \frac{1}{2aB}$ (computer-algebra check). So the swirl $f = \varkappa \int z^{-2}$ saturates, to the right and left, at $f_+ = \varkappa/(2aA)$, $|f_-| = \varkappa/(2aB)$ within a distance $O(1/a)$ of the cone. If it then enters the ceiling regime $\psi_+ \approx \frac{1}{2}e^{-2\xi_0} f_+^2 = 2k_2^2$ and $\psi_- \approx \frac{1}{2}e^{2\xi_0} f_-^2 = 2k_1^2$, then

$f_+ = 2k_2e^{\xi_0}$, $|f_-| = 2k_1e^{-\xi_0}$ and, adding $\frac{\varkappa}{2aA}\cdot\frac{2aA}{\varkappa}\ldots$, i.e. $\frac{2a(A+B)}{\varkappa} = \frac{1}{f_+}+\frac{1}{|f_-|}$, one gets exactly $\varkappa = \varkappa_{\mathrm{WKB}}$. The theorem below makes this rigorous (and the algebra is confirmed by a computer-algebra check, `code/ch12_thresh/verify_ch12b.py`).

For $\boldsymbol{c}\in[0,\infty)^3$ put $m = \min(k_1^2,k_2^2) = 1+\min(c_1,c_2)$, $a=\sqrt{W_0(\xi_0)}$, $W_{\mathrm{loc}} = \min_{|\xi-\xi_0|\le 1} W_0(\xi)$, and

$$R_0 = \tfrac14\big[k_1^2(1-x_0)^2 + k_2^2(1+x_0)^2\big],$$

the value at ξ_0 of the half-envelope $\frac12 E$ of Lemma 12.23(c) for the full ceilings $\psi_\pm = 2k_{1,2}^2$. By Lemma 3.10(f1), $W_0\ge m\ge 1$, $a\ge 1$, $W_{\mathrm{loc}}\ge e^{-2}a^2$, and $a^2-R_0 = \frac12\tilde c_3\operatorname{sech}^2\xi_0>0$, where $\tilde c_3 = c_3+\frac12(k_1^2+k_2^2)>0$ (because $W_0-\frac12 E = \frac12\tilde c_3\operatorname{sech}^2\xi$ for the full ceilings, Lemma 12.23(c)).

THEOREM 12.31 (large constants). *Let $\boldsymbol{c}\in[0,\infty)^3$ and $x_0\in(-1,1)$.*

(a) (Exponents.)

$$\frac{0.60}{\cosh\xi_0}\sqrt{m\,W_{\mathrm{loc}}} \;\le\; \varkappa_{\max} \;\le\; 27.9\,a\,\min\{k_2e^{\xi_0},k_1e^{-\xi_0}\} \;\le\; 28\,e^{|\xi_0|}\sqrt{m}\,a,$$

in particular $\dfrac{0.22}{\cosh\xi_0}\sqrt{m\,W_0(\xi_0)}\le\varkappa_{\max}\le 28\,e^{|\xi_0|}\sqrt{m\,W_0(\xi_0)}$, *and* $\kappa_{\max} = \varkappa_{\max}\cosh^2\xi_0$.

(b) (The sharp law.) *For every $M\ge 1$ (the lower bound for every $M>0$), with $\tilde a^2 := a^2-R_0>0$,*

$$\sqrt{1-\tfrac{R_0}{a^2}}\;e^{-M/\tilde a}\tanh\big(Me^{-M/\tilde a}\big) \;\le\; \frac{\varkappa_{\max}}{\varkappa_{\mathrm{WKB}}} \;\le\; e^{2M/a}\Big(1+\frac{2}{e^{2M}-1}\Big).$$

Consequently $\varkappa_{\max}/\varkappa_{\mathrm{WKB}}\le 1+O(\ln a/a)$ uniformly in x_0 and $\boldsymbol{c}\in[0,\infty)^3$ with $a\ge e^2$, and $\varkappa_{\max}/\varkappa_{\mathrm{WKB}}\to 1$ whenever $a\to\infty$ and $R_0/a^2\to 0$ (for instance $\boldsymbol{c}=(c_1,c_2,p)$ with $c_1,c_2\ge 0$ fixed, $p\to\infty$, any fixed x_0).

Corollary 12.32 (exponents)**.** *Fix $x_0\in(-1,1)$. (a) If $c_1,c_2\ge 0$ are fixed and $c_3\to\infty$, then $\varkappa_{\max}\asymp\sqrt{c_3}$; if $c_2,c_3\ge 0$ are fixed and $c_1\to\infty$, or $c_1,c_3\ge 0$ are fixed and $c_2\to\infty$, then $\varkappa_{\max}\asymp\sqrt{c_1}$, resp. $\sqrt{c_2}$. (b) Along a ray $\boldsymbol{c}=p\,\mathbf{a}$, $\mathbf{a}\in[0,\infty)^3\setminus\{0\}$: $\varkappa_{\max}\asymp p$ if $a_1a_2>0$ and $\varkappa_{\max}\asymp\sqrt{p}$ if $a_1a_2=0$. The constants depend only on x_0 and on the fixed parts, and $\kappa_{\max}=\varkappa_{\max}\cosh^2\xi_0$ has the same exponent.*

PROOF OF THEOREM 12.31(A). *Lower bound.* $\boldsymbol{c}\ge 0$ gives $\boldsymbol{c}\in\operatorname{int}\mathcal{J}$ ($c_i>-1$, $c_3\ge 0>\bar c_3$). Take $e_0=m$ in Theorem 12.25(c): $\tilde{\boldsymbol{c}}^{\,0} = \boldsymbol{c}-\frac{m}{2}(1,1,0)$ has $\tilde k_i^2 = k_i^2-\frac{m}{2}\ge\frac{m}{2}>0$ and $\tilde c_3 = c_3\ge 0>\bar c_3(\tilde c_1,\tilde c_2)$, so $\tilde{\boldsymbol{c}}^{\,0}\in\operatorname{int}\mathcal{J}$, and $\tilde W_0^{\,0} = W_0-\frac{m}{2}$. By Lemma 3.10(f1)(i), $W_0\ge m$ everywhere; therefore $\tilde W_0^{\,0}\ge\frac{m}{2}>0$ on $\mathbb{R}$ and $\tilde W_0^{\,0}\ge W_{\mathrm{loc}}-\frac{m}{2}\ge\frac12 W_{\mathrm{loc}}$ on $|\xi-\xi_0|\le 1$. The barrier bound Lemma 3.10(e) (with $\ell=1$, $b^2=\frac12 W_{\mathrm{loc}}\ge\frac12$) gives $\tilde w^{\,0}\ge 2b\tanh b\ge\sqrt{2}\tanh(2^{-1/2})\sqrt{W_{\mathrm{loc}}}\ge 0.861\sqrt{W_{\mathrm{loc}}}$ ($b\mapsto 2b\tanh b$ increases and $b\ge 2^{-1/2}$). Then Theorem 12.25(c): $\varkappa_{\max}\ge\sqrt{m/2}\cdot 0.861\sqrt{W_{\mathrm{loc}}}/\cosh\xi_0\ge 0.60\sqrt{mW_{\mathrm{loc}}}/\cosh\xi_0$ (the unrounded constant is 0.60886).

Upper bound. By Lemma 3.10(f1)(iii), $w_0<2a+2\le 4a$ ($a\ge 1$). For every σ the barrier bound with $\ell=1$ and $b=1$ ($W_0\ge m\ge 1$) gives $\rho(\sigma) := \zeta_-(\sigma)-\zeta_+(\sigma)\ge 2\tanh 1>1.52$. Since $(\ln r_+)' = \zeta_+-\zeta_- = -\rho$, $r_+(\xi_0+1)\le e^{-2\tanh 1}<0.2181$. As $I_+ = \int e^{-2s}(1-r_+)(-r_+')ds$ and $e^{-2s}\ge e^{-2\xi_0-2}$ on $[\xi_0,\xi_0+1]$,

$$I_+\ge e^{-2\xi_0-2}\cdot\tfrac12\big(1-r_+(\xi_0+1)\big)^2\ge\tfrac12 e^{-2}(0.7819)^2e^{-2\xi_0}>0.0413\,e^{-2\xi_0},$$

so $\sqrt{I_+} \ge 0.2032\, e^{-\xi_0}$. Theorem 12.25(a) gives

$$\varkappa_{\max} \le 4a \cdot \frac{\sqrt{2}\, k_2}{0.2032\, e^{-\xi_0}} \le 27.9\, a k_2 e^{\xi_0} \qquad (4\sqrt{2}/0.2032 = 27.84),$$

and the mirror bound is $27.9\, a k_1 e^{-\xi_0}$. Finally $\min\{k_2 e^{\xi_0}, k_1 e^{-\xi_0}\} \le \sqrt{m} e^{|\xi_0|}$ (if $m = k_2^2$ use the first entry, otherwise the second), and $W_{\mathrm{loc}} \ge e^{-2} a^2$ together with $0.60/e > 0.22$ gives the statement with $W_0(\xi_0)$. The unrounded values 0.861057, 0.608859, 0.21802, 0.04138, 0.20342 and 27.809 are printed by `verify_ch12b.py`, part `const`; the constants in the text are rounded in the safe direction. □

Proof of Corollary 12.32. Apply (a). (i) c_1, c_2 fixed, $c_3 \to \infty$: m is fixed and $a^2 = W_0(\xi_0) = \frac{c_3}{2}(1 - x_0^2) + O(1) \asymp c_3$. (ii) $c_1 \to \infty$: $m = k_2^2$ once $c_1 \ge c_2$ and $a^2 = \frac{1}{2} c_1 (1 - x_0) + O(1) \asymp c_1$; (iii) is the mirror. (iv) $\boldsymbol{c} = p\mathbf{a}$: $m = 1 + p \min(a_1, a_2) \asymp p$ if $a_1 a_2 > 0$, $m = 1$ otherwise, and $a^2 = 1 + \frac{p}{2} P_{\mathbf{a}}(x_0) \asymp p$ ($P_{\mathbf{a}}(x_0) > 0$ for $\mathbf{a} \ne 0$, $\mathbf{a} \ge 0$); so $\sqrt{ma} \asymp p$, resp. $\sqrt{p}$. □

Proof of Theorem 12.31(b). *Upper bound.* Let $(\gamma, \varkappa)$ be admissible (e.g. $\varkappa = \varkappa_{\max}$), (z, f, ψ) its solution, $h_0 = h(\xi_0)$, $S = \xi - \xi_0$. (i) By (Y1) and Lemma 3.10(f1)(iii), $|h_0| < a + 1$. (ii) For $\boldsymbol{c} \ge 0$, $|W_0'| \le 2W_0$ (Lemma 3.10(f1)(ii)), so $W \le W_0 \le \bar{a}^2 := a^2 e^{2M/a}$ on $I_M = \{|S| \le M/a\}$, and $\bar{a} \ge a + M > |h_0|$ for $M \ge 1$. (iii) Let $z_{\bar{a}} = \cosh \bar{a} S + \frac{h_0}{\bar{a}} \sinh \bar{a} S = A e^{\bar{a} S} + B e^{-\bar{a} S}$, $A = \frac{1}{2}(1 + h_0/\bar{a}) \in (0, 1)$, $B = 1 - A$, a positive solution of $z_{\bar{a}}'' = \bar{a}^2 z_{\bar{a}}$ with the data of z at $S = 0$. Then $\mathrm{Wr}(z_{\bar{a}}, z)' = (W - \bar{a}^2) z_{\bar{a}} z \le 0$ on I_M and $\mathrm{Wr}(0) = 0$; so $\mathrm{Wr} \le 0$ for $S \ge 0$ and ≥ 0 for $S \le 0$, i.e. $(z/z_{\bar{a}})' \le 0$ for $S \ge 0$ and ≥ 0 for $S \le 0$, whence $z \le z_{\bar{a}}$ on I_M. (iv) With $\varphi = e^{2\bar{a} M/a} - 1$ and $\int_0^S \frac{dS'}{(A e^{\bar{a} S'} + B e^{-\bar{a} S'})^2} = \frac{1}{2\bar{a}} \frac{\phi}{1 + A\phi}$, $\phi = e^{2\bar{a} S} - 1$ (computer-algebra check), and $f' = \varkappa z^{-2} \ge \varkappa z_{\bar{a}}^{-2}$,

$$f\big(\xi_0 + \tfrac{M}{a}\big) \ge \frac{\varkappa \varphi}{2\bar{a}(1 + A\varphi)}, \qquad \Big| f\big(\xi_0 - \tfrac{M}{a}\big) \Big| \ge \frac{\varkappa \varphi}{2\bar{a}(1 + B\varphi)}$$

(the second by $S \mapsto -S$, which exchanges A and B). (v) By Lemma 12.23(b) and (Y2), and because $f f' \ge 0$ on $[\xi_0, \infty)$ and $e^{-2s} \ge e^{-2\xi_0 - 2M/a}$ on $[\xi_0, \xi_0 + M/a]$, $2k_2^2 \ge \psi_+ \ge \int_{\xi_0}^{\xi_0 + M/a} e^{-2s} f f'\, ds \ge \frac{1}{2} e^{-2\xi_0 - 2M/a} f(\xi_0 + \frac{M}{a})^2$, i.e. $f(\xi_0 + \frac{M}{a}) \le 2k_2 e^{\xi_0 + M/a}$; likewise $|f(\xi_0 - \frac{M}{a})| \le 2k_1 e^{-\xi_0 + M/a}$. (vi) Combining, $\frac{\varkappa e^{-\xi_0 - M/a}}{4\bar{a} k_2} \le \frac{1}{\varphi} + A$ and $\frac{\varkappa e^{\xi_0 - M/a}}{4\bar{a} k_1} \le \frac{1}{\varphi} + B$; adding, $\varkappa \le 4\bar{a} e^{M/a}(1 + 2/\varphi) \big/ \big(\frac{e^{-\xi_0}}{k_2} + \frac{e^{\xi_0}}{k_1}\big) = e^{2M/a}(1 + 2/\varphi)\, \varkappa_{\mathrm{WKB}}$ because $\bar{a} = a e^{M/a}$, and $\varphi \ge e^{2M} - 1$.

Lower bound. Apply Theorem 12.25(b), case (ii), with the full ceilings $e_S = 2k_1^2$, $e_N = 2k_2^2$ (allowed: the conditions (N),(S) are strict, open conditions): $\tilde{\boldsymbol{c}} = (-1, -1, \tilde{c}_3)$, $\tilde{c}_3 = c_3 + \frac{1}{2}(k_1^2 + k_2^2) > 0$, $\tilde{W}_0 = \frac{\tilde{c}_3}{2\cosh^2 \xi} \ge 0$ and $\tilde{W}_0(\xi_0) = a^2 - R_0 = \tilde{a}^2$ ($W_0 - \tilde{W}_0 = \frac{1}{2} E$ and $\frac{1}{2} E(\xi_0) = R_0$). Since $|\tilde{W}_0'| = 2|\tanh \xi| \tilde{W}_0 \le 2\tilde{W}_0$, $\tilde{W}_0 \ge \tilde{a}^2 e^{-2M/\tilde{a}}$ on $|S| \le M/\tilde{a}$, and the barrier bound with $b = \tilde{a} e^{-M/\tilde{a}}$, $\ell = M/\tilde{a}$ gives $\tilde{w}_0 \ge 2\tilde{a} e^{-M/\tilde{a}} \tanh(M e^{-M/\tilde{a}})$ (all bounded solutions are positive and $\tilde{W}_0 \ge 0$, Step 0 of the proof of Theorem 12.25). With the second form of (S), the conditions read $\beta > \frac{\varkappa e^{-\xi_0}}{2k_2 \tilde{w}_0}$ and $\alpha > \frac{\varkappa e^{\xi_0}}{2k_1 \tilde{w}_0}$ (note $e_N = 2k_2^2$, $e_S = 2k_1^2$), and $\alpha + \beta = 1$ is possible iff $\varkappa < \frac{2\tilde{w}_0}{e^{-\xi_0}/k_2 + e^{\xi_0}/k_1} = \frac{\tilde{w}_0}{2a} \varkappa_{\mathrm{WKB}}$. Hence $\varkappa_{\max} \ge \frac{\tilde{w}_0}{2a} \varkappa_{\mathrm{WKB}} \ge \frac{\tilde{a}}{a} e^{-M/\tilde{a}} \tanh(M e^{-M/\tilde{a}}) \varkappa_{\mathrm{WKB}}$.

Consequences. For $M = \frac{1}{2}\ln a$ the upper bound is $e^{\ln a/a}(1+\frac{2}{a-1}) = 1+O(\ln a/a)$. If $R_0/a^2 \to 0$ then $\tilde a/a \to 1$, and with $M = \frac{1}{2}\ln\tilde a$ one has $e^{-M/\tilde a}\tanh(Me^{-M/\tilde a}) \ge 1 - O(\ln\tilde a/\tilde a)$. □

Remark 12.33. The upper bound is sharp in every direction; the lower bound loses the factor $\sqrt{1-R_0/a^2}$, which is 1 only when the envelope E is negligible at the cone (large c_3); for $\boldsymbol{c} = (p,p,0)$, $x_0 = 0$ the factor is 0.707. Whether a sharper envelope closes this gap is posed in Chapter 13 (Open problem 13.11).

Numerical observation 12.34 (large constants)**.** *Table 12.2 lists $\varkappa_{\max}/\varkappa_{\mathrm{WKB}}$ for four families at $x_0 = 0$, and, at $p = 10^4$, the proved bounds of Theorem 12.31(b) (optimised over M). All ratios decrease to 1, roughly like $1+O(p^{-1/2})$; the measured ratio always lies between the proved bounds (asserted in the script). The computed $\kappa_{\max}$ of the script grows per decade of p by factors tending to $\sqrt{10}$ for $(0,0,p)$ and $(0,p,0)$ and to 10 for $(p,p,0)$ and $p(1,2,3)$, as in Corollary 12.32. (Figure 12.3(a); `verify_ch12b.py`, part `large`.)*

family	$p=10$	10^2	10^3	10^4	lower at 10^4	upper at 10^4
$(0,0,p)$	1.4062	1.1383	1.0444	1.0141	0.952	1.088
$(0,p,0)$	1.3959	1.1216	1.0360	1.0109	0.662	1.088
$(p,p,0)$	1.3077	1.1001	1.0317	1.0100	0.674	1.065
$p(1,2,3)$	1.1786	1.0573	1.0182	1.0057	0.845	1.040

TABLE 12.2. $\varkappa_{\max}/\varkappa_{\mathrm{WKB}}$ at $x_0 = 0$; the last two columns are the proved bounds of Theorem 12.31(b) at $p = 10^4$.

12.11. Cones near a pole

In words. *Fix $\boldsymbol{c}$ and let the cone x_0 move to the north pole. The same swirl, measured at the cone, is spread out by the factor $\cosh^2\xi_0$ when one passes from ξ to x, so $\kappa_{\max} = \varkappa_{\max}\cosh^2\xi_0$ blows up; but the combination $\sqrt{1-x_0^2}\,\kappa_{\max} = \varkappa_{\max}\cosh\xi_0$ stays bounded and positive. Its limsup is bounded by the constant $\Lambda_1(\boldsymbol{c})$ of the upper bound Theorem 12.25(a), because the window width and the integral I_- have explicit limits as $\xi_0\to\infty$ (Lemma 12.36); its liminf is bounded below by the profile-comparison bound Theorem 12.25(b), in which the swirl is allowed to lower the south-pole constant c_1 as far as the critical surface permits. If $\boldsymbol{c}$ is far enough from the critical surface ($\varepsilon \ge \varepsilon_*$) the two bounds meet.*

Toy example 12.35 (the rest state, every ξ_0)**.** At $\boldsymbol{c} = 0$ one has $u_\pm = e^{\mp(\xi-\xi_0)}$, $w_0 = 2$, $r_- = e^{2(s-\xi_0)}$ and $I_- = e^{2\xi_0}\int_0^\infty e^{-2d}(1-e^{-2d})^2 dd = e^{2\xi_0}/6$ for *every* ξ_0. Theorem 12.25(a) gives $\varkappa_{\max} \le 2\sqrt{2}/\sqrt{I_-} = 4\sqrt{3}\,e^{-\xi_0}$, that is

$$\sqrt{1-x_0^2}\,\kappa_{\max} \;\le\; 4\sqrt{3}\,e^{-\xi_0}\cosh\xi_0 = 2\sqrt{3}\,(1+e^{-2\xi_0}), \qquad e^{-2\xi_0} = \frac{1-x_0}{1+x_0},$$

which tends to $\Lambda_1(0) = 2\sqrt{3} = 3.4641$ as $x_0 \to 1$. The bound is the number 3.64642 at $x_0 = 0.9$, 3.48151 at 0.99 and 3.46583 at 0.999 (Numerical observation 12.28).

For $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ put

$$\Lambda_1(\boldsymbol{c}) = \sqrt{2}\,k_1\sqrt{(1+k_2)(1+2k_2)},$$
$$\Lambda_2(\boldsymbol{c}) = \Lambda_1(\boldsymbol{c}_m) = \sqrt{2}\,k_2\sqrt{(1+k_1)(1+2k_1)},$$

$\varepsilon = c_3 - \bar{c}_3(c_1,c_2) > 0$ (the distance from the critical surface), $\varepsilon_* = k_1(1+k_2)$,

$$e_S^{\max}(\varepsilon) = \begin{cases} 2\Big(k_1^2 - \big(k_1 - \dfrac{\varepsilon}{1+k_2}\big)^2\Big), & \varepsilon \le \varepsilon_*, \\ 2k_1^2, & \varepsilon \ge \varepsilon_*, \end{cases}$$

and $\Lambda_*(\boldsymbol{c})^2 = (1+k_2)(1+2k_2)\,e_S^{\max}(\varepsilon)$. A direct expansion (a computer-algebra check, `verify_ch12b.py`, part `sym`) gives, for $\varepsilon \le \varepsilon_*$, $\Lambda_*^2 = 2(1+2k_2)\,\varepsilon\big(2k_1 - \frac{\varepsilon}{1+k_2}\big) = \Lambda_1^2\big[1-(1-\varepsilon/\varepsilon_*)^2\big]$, and $\Lambda_* = \Lambda_1$ for $\varepsilon \ge \varepsilon_*$.

Lemma 12.36 (two limits as the cone approaches the north pole)**.** *Let $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ be fixed and $\rho = \zeta_- - \zeta_+$ (so $w_0(\xi_0) = \rho(\xi_0)$). As $\xi_0 \to +\infty$, $w_0(\xi_0) \to 2k_2$ and*

$$e^{-2\xi_0} I_-(\xi_0) \;\to\; \frac{k_2^2}{(1+k_2)(1+2k_2)}, \qquad I_-(\xi_0) = \int_{-\infty}^{\xi_0} e^{2s}\big(1-r_-\big)^2 ds,$$

with $r_- = u_-/u_+$.

Proof. The Jost solutions $u_\pm$ are normalised at ξ_0, but $\zeta_\pm$ and ρ do not depend on the normalisation, and ρ is continuous and positive with limits $2k_1$ at $-\infty$ and $2k_2$ at $+\infty$ (Lemma 3.21(b)(iv)); in particular $w_0(\xi_0) = \rho(\xi_0) \to 2k_2$ and $\rho_{\sup} := \sup\rho < \infty$. The function $r_- = u_-/u_+$ with $r_-(\xi_0) = 1$ satisfies $r_-' = \rho\, r_-$ (as $(\ln r_-)' = \zeta_- - \zeta_+$), so $r_-(\xi_0 - d) = \exp\big(-\int_{\xi_0-d}^{\xi_0}\rho\big) \in (0,1]$, and for every fixed $d \ge 0$, $r_-(\xi_0 - d) \to e^{-2k_2 d}$ as $\xi_0 \to \infty$ (the interval $[\xi_0 - d, \xi_0]$ runs off to $+\infty$, where $\rho \to 2k_2$). Integrating by parts as in the proof of Theorem 12.25(a), $I_- = \int_{-\infty}^{\xi_0} e^{2s}(1-r_-)r_-'\,ds$ (the boundary terms vanish), hence with $s = \xi_0 - d$

$$e^{-2\xi_0} I_- = \int_0^\infty e^{-2d}\big(1 - r_-(\xi_0 - d)\big)\,\rho(\xi_0 - d)\,r_-(\xi_0 - d)\,dd.$$

The integrand is bounded by $\rho_{\sup}e^{-2d}$ and tends pointwise to $e^{-2d}(1 - e^{-2k_2 d})\,2k_2\,e^{-2k_2 d}$; by dominated convergence the limit is $2k_2\big[\frac{1}{2+2k_2} - \frac{1}{2+4k_2}\big] = \frac{k_2^2}{(1+k_2)(1+2k_2)}$. □

Theorem 12.37 (cones near a pole). *Let $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ be fixed and let x_0 vary.*

(a) $0 < \liminf_{x_0\to 1}\sqrt{1-x_0^2}\,\kappa_{\max} \le \limsup_{x_0\to 1}\sqrt{1-x_0^2}\,\kappa_{\max} \le \Lambda_1(\boldsymbol{c})$. *In particular $\kappa_{\max} \asymp (1-x_0^2)^{-1/2}$ and $\varkappa_{\max} = (1-x_0^2)\kappa_{\max} \asymp (1-x_0^2)^{1/2}$.*

(b) $\Lambda_*(\boldsymbol{c}) \le \liminf_{x_0\to 1}\sqrt{1-x_0^2}\,\kappa_{\max}$. *Consequently, if $\varepsilon \ge \varepsilon_* = k_1(1+k_2)$,*

$$\lim_{x_0\to 1}\sqrt{1-x_0^2}\,\kappa_{\max}(x_0,\boldsymbol{c}) = \Lambda_1(\boldsymbol{c}) \quad (= \sqrt{2}\,k_1\sqrt{(1+k_2)(1+2k_2)}).$$

(c) *The mirror statements hold as $x_0 \to -1$, with $\Lambda_2(\boldsymbol{c}) = \Lambda_1(\boldsymbol{c}_m)$ and $\Lambda_*(\boldsymbol{c}_m)$.*

Proof. Recall $\sqrt{1-x_0^2}\,\kappa_{\max} = \varkappa_{\max}\cosh\xi_0$.

(a) Lower bound. Choose $e_0 \in (0, 2\min k_i^2)$ so small that $\tilde{\boldsymbol{c}}^{\,0} = \boldsymbol{c} - \frac{e_0}{2}(1,1,0) \in \operatorname{int}\mathcal{J}$ ($\operatorname{int}\mathcal{J}$ is open). By Theorem 12.25(c), $\varkappa_{\max}\cosh\xi_0 \ge \sqrt{e_0/2}\;\tilde{w}^{\,0}(\xi_0)$, and $\tilde{w}^{\,0}(\xi_0) \to 2\tilde{k}_2 > 0$ by Lemma 12.36 applied to $\tilde{\boldsymbol{c}}^{\,0}$. So the lim inf is at least $\sqrt{2e_0}\,\tilde{k}_2 > 0$. *Upper bound.* By Theorem 12.25(a), $\varkappa_{\max} \le w_0\sqrt{2}k_1/\sqrt{I_-}$, so $\varkappa_{\max}e^{\xi_0} \le$

$w_0\sqrt{2}k_1/\sqrt{e^{-2\xi_0}I_-} \to 2k_2\sqrt{2}k_1\sqrt{(1+k_2)(1+2k_2)}/k_2 = 2\Lambda_1$ by Lemma 12.36; and $\cosh\xi_0 \le \frac{1}{2}e^{\xi_0}(1+e^{-2\xi_0})$. The statement on the orders follows from $\sqrt{1-x_0^2}\,\kappa_{\max} = \varkappa_{\max}\cosh\xi_0$.

(b) Let $e_S \in (0, 2k_1^2)$ and $e_N > 0$ be such that $\tilde{\boldsymbol{c}} = \big(c_1 - \frac{e_S}{2}, c_2 - \frac{e_N}{2}, c_3 + \frac{e_S+e_N}{4}\big) \in \operatorname{int}\mathcal{J}$; then $\tilde{k}_1^2 = k_1^2 - \frac{e_S}{2}$, $\tilde{k}_2^2 = k_2^2 - \frac{e_N}{2}$, and by the algebraic identity (F4) of Theorem 12.5, $\tilde{\varepsilon} = \varepsilon - \frac{1}{2}\big[(k_1-\tilde{k}_1)(k_2+\tilde{k}_2+2) + (k_2-\tilde{k}_2)(k_1+\tilde{k}_1+2)\big]$ (checked in `verify_ch12b.py`); $\tilde{\boldsymbol{c}} \in \operatorname{int}\mathcal{J}$ is equivalent to $\tilde{k}_i > 0$ and $\tilde{\varepsilon} > 0$. Let $\Lambda > 0$ with $\Lambda^2 < (1+\tilde{k}_2)(1+2\tilde{k}_2)\,e_S$ and put $\varkappa = \Lambda/\cosh\xi_0$, $\beta = e^{-\xi_0}$, $\alpha = 1-\beta$. Apply Theorem 12.25(b), case (i), the first form of (S): by Lemma 12.36 for $\tilde{\boldsymbol{c}}$, $\tilde{w}_0 \to 2\tilde{k}_2$ and $e^{-2\xi_0}\tilde{I}_- \to \tilde{k}_2^2/((1+\tilde{k}_2)(1+2\tilde{k}_2))$, while $\varkappa^2 e^{2\xi_0} \to 4\Lambda^2$, so

$$\frac{\varkappa^2\tilde{I}_-}{\alpha^3\tilde{w}_0^2} \to \frac{4\Lambda^2\tilde{k}_2^2/((1+\tilde{k}_2)(1+2\tilde{k}_2))}{4\tilde{k}_2^2} = \frac{\Lambda^2}{(1+\tilde{k}_2)(1+2\tilde{k}_2)} < e_S,$$

$$\frac{\varkappa^2 e^{-2\xi_0}}{2\beta^2\tilde{w}_0^2} = \frac{\varkappa^2}{2\tilde{w}_0^2} \to 0 < e_N.$$

Hence for ξ_0 large, $\varkappa$ is admissible and $\sqrt{1-x_0^2}\,\kappa_{\max} \ge \Lambda$: $\liminf \ge \Lambda$. Now let $e_N \downarrow 0$ (then $\tilde{k}_2 \to k_2$ and $\tilde{\varepsilon} \to \varepsilon - (k_1-\tilde{k}_1)(1+k_2)$) and $e_S \uparrow e_S^{\max}(\varepsilon)$: for every $e_S < e_S^{\max}(\varepsilon)$ one has $\tilde{\varepsilon}(e_S, 0) > 0$ (that is $\tilde{k}_1 > k_1 - \varepsilon/(1+k_2)$, together with $\tilde{k}_1 > 0$), hence by continuity $\tilde{\boldsymbol{c}} \in \operatorname{int}\mathcal{J}$ for all small $e_N > 0$. So $\liminf \ge \Lambda_*(\boldsymbol{c})$; for $\varepsilon \ge \varepsilon_*$, $\Lambda_* = \Lambda_1$ and the bounds of (a) and (b) coincide.

(c) By Proposition 5.59(d), $\kappa_{\max}(-x_0, \boldsymbol{c}_m) = \kappa_{\max}(x_0, \boldsymbol{c})$; apply (a), (b) to $(\boldsymbol{c}_m, -x_0)$. □

Numerical observation 12.38 (the cone limit)**.** *Table 12.3 gives $\sqrt{1-x_0^2}\,\kappa_{\max}$ for five constants, all with $\varepsilon \ge \varepsilon_*$, so that Theorem 12.37(b) identifies the limit Λ_1. The ratio to Λ_1 is* 1.04445, 1.00490, 1.00050, 1.00005 *at $x_0 = 0.9, 0.99, 0.999, 0.9999$ for $\boldsymbol{c} = 0$, i.e. it tends to* 1 *like $1 - x_0$; for $\boldsymbol{c} = (2,0,0)$ it is* 1.05999, 1.00774, 1.00081, 1.00008. *(Figure 12.3(b); `verify_ch12b.py`, part `cone`.) For $\varepsilon < \varepsilon_*$ the proved liminf $\Lambda_* < \Lambda_1$ is the best known lower bound; a law for the true limit is posed in Chapter 13 (Open problem 13.12).*

$\boldsymbol{c}$	ε	ε_*	$x_0 = 0.9$	0.99	0.999	0.9999	$\Lambda_1(\boldsymbol{c})$
$(0,0,0)$	4.0000	2.0000	3.61808	3.48108	3.46583	3.46427	3.46410
$(-\frac{1}{2},-\frac{1}{2},0)$	2.4142	1.2071	2.12992	2.04020	2.03112	2.03021	2.03010
$(2,0,0)$	6.4641	3.4641	6.35995	6.04646	6.00484	6.00049	6.00000
$(0,2,0)$	6.4641	2.7321	5.09034	4.95593	4.94066	4.93904	4.93886
$(1,\frac{1}{2},0)$	6.1210	3.1463	5.77660	5.56943	5.54352	5.54079	5.54048

TABLE 12.3. $\sqrt{1-x_0^2}\,\kappa_{\max}(x_0, \boldsymbol{c})$ and its limit $\Lambda_1(\boldsymbol{c})$ (Theorem 12.37).

12.12. The shape of the lens

In words. *Fix x_0 and $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$. The lens of Proposition 5.59 is the region between a rising lower wall γ^- and a falling upper wall γ^+ over $0 \le \kappa \le \kappa_{\max}$. Two questions: how can one read the curvature of a wall from the differential equation,*

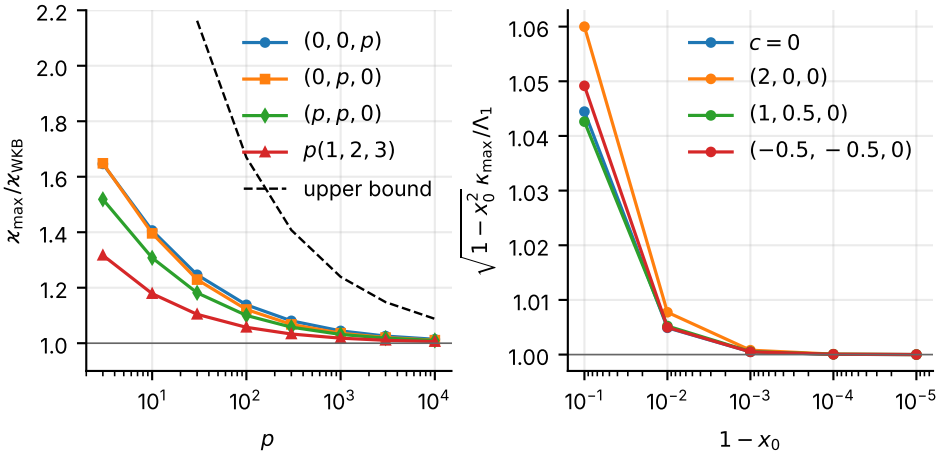


FIGURE 12.3. (a) $\varkappa_{\max}/\varkappa_{\mathrm{WKB}}$ against p for four families of large constants at $x_0 = 0$ (Theorem 12.31; dashed: the proved upper bound at the largest p of the figure). (b) $\sqrt{1-x_0^2}\,\kappa_{\max}/\Lambda_1(\boldsymbol{c})$ against $1 - x_0$ for four constants (Theorem 12.37). Code: `code/ch12_thresh/make_b_large_cone.py`.

and what does one have to prove to show that the lens is convex? The first is answered by the implicit function theorem for curves: the wall is the zero set of a smooth function H (the value of a defining function of the edge chart at a late cut), so its curvature is $-D^2H[\tau,\tau]/H_\gamma$ and the second derivative D^2H is computed by the second variation *of the flow, a linear equation with a quadratic source. The second is answered by an elementary reduction: the lens is convex iff the lower wall is convex and the upper wall concave. Which signs the quadratic source has decides what one can prove; this is the content of the last two statements of the section and of Section 12.13.*

12.12.1. The implicit function theorem for curves.

Lemma 12.39 (IFT for curves)**.** *Let F be C^2 near (κ_0,γ_0), $F(\kappa_0,\gamma_0) = 0$ and $F_\gamma(\kappa_0,\gamma_0) \neq 0$. Then $\{F = 0\}$ is near this point the graph of a C^2 function $\gamma(\kappa)$, and with $\tau = (1,\gamma')$*

$$\gamma' = -\frac{F_\kappa}{F_\gamma}, \qquad \gamma'' = -\frac{F_{\kappa\kappa} + 2F_{\kappa\gamma}\gamma' + F_{\gamma\gamma}\gamma'^2}{F_\gamma} = -\frac{D^2F[\tau,\tau]}{F_\gamma}.$$

If $F_\gamma > 0$ and the admissible region is $\{F \geq 0\}$, then the region lies above the graph, and the graph is convex iff $D^2F[\tau,\tau] \leq 0$.

PROOF. The implicit function theorem gives the C^2 graph. Differentiating $F(\kappa,\gamma(\kappa)) = 0$ gives $F_\kappa + F_\gamma\gamma' = 0$, and once more $F_{\kappa\kappa} + 2F_{\kappa\gamma}\gamma' + F_{\gamma\gamma}\gamma'^2 + F_\gamma\gamma'' = 0$. The last sentence is the sign of γ'' in the formula. □

Toy example 12.40 (the circle)**.** $F = \kappa^2 + \gamma^2 - 1$ and the branch $\gamma = \sqrt{1-\kappa^2}$: $\gamma' = -\kappa/\gamma$, $D^2F[\tau,\tau] = 2 + 2\gamma'^2 = 2/\gamma^2$, so $\gamma'' = -\gamma^{-3} < 0$. The region $\{F \leq 0\}$

(the disc) lies *below* the graph, and the graph is concave: exactly the convex-region test of the lemma, with the orientation reversed.

12.12.2. The second variation. Write the Q-system (5.2) as $Y' = F_{\boldsymbol{c}}(x, Y)$, $Y = (U, g, f, p_2, p_1, p_0)$, $d = 1 - x^2$; the data at x_0 are $Y(x_0) = (\gamma, \kappa, 0, 0, 0, 0)$, affine in (κ, γ).

Lemma 12.41 (second variation)**.** *Let $\kappa = \kappa_0 + \tau t$, $\gamma = \gamma_0 + \sigma t$ be a straight line in the data, $Y(x;t)$ the solution, $J = \partial_t Y$, $Z = \partial_t^2 Y$ at $t = 0$. Then $J' = D_Y F\, J$, $Z' = D_Y F\, Z + S$, with $J(x_0) = (\sigma, \tau, 0, 0, 0, 0)$, $Z(x_0) = 0$ and $S = D_Y^2 F[J, J]$ has the components*

$$S_U = -\frac{J_U^2}{d}, \quad S_g = -\frac{2J_U J_g}{d}, \quad S_{p_2} = \frac{4J_f J_g}{d}, \quad S_f = S_{p_1} = S_{p_0} = 0.$$

Proof. Differentiate $Y' = F(x, Y(x;t))$ once and twice in t (the data are affine in t, so $Z(x_0) = 0$; smooth dependence on the data is Lemma 2.24 together with the ODE theory of Appendix A). Only the three nonlinear terms $-\frac{U^2}{2d}$, $-\frac{Ug}{d}$, $\frac{2fg}{d}$ contribute to $D^2F[J, J]$, and the second derivative of a product ab in direction J is $2J_aJ_b$, of $-\frac{U^2}{2d}$ is $-J_U^2/d$. (Checked in `verify_ch12b.py`, part `sym`.) □

Toy example 12.42 (a Riccati toy)**.** For $y' = y^2$, $y(0) = a$: $y = \frac{a}{1-at}$, $J = \partial_a y = (1-at)^{-2} > 0$ and $Z = \partial_a^2 y = \frac{2t}{(1-at)^3}$; and $Z' = 2yZ + 2J^2$ with the source $2J^2 \geq 0$ a *square*. In the Q-system the sources are products of *different* components of J, so they need not have a sign.

12.12.3. The proposition.

Proposition 12.43 (the shape of the lens)**.** *Let $x_0 \in (-1, 1)$, $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$,* $\mathrm{Lens} = \mathrm{Lens}(x_0, \boldsymbol{c})$*, $U_{\boldsymbol{c}}^{\pm} = U_{\boldsymbol{c}}^{\pm}(x_0)$, γ_* the tip, and extend γ^-, γ^+ to $\kappa = 0$ by $U_{\boldsymbol{c}}^{\mp}$ (continuous by Proposition 5.59(e)).*

(a) (Unimodal hypograph.) *The function $K := (\gamma^-)^{-1}$ on $[U_{\boldsymbol{c}}^-, \gamma_*]$, $K := (\gamma^+)^{-1}$ on $[\gamma_*, U_{\boldsymbol{c}}^+]$ is continuous, $K(U_{\boldsymbol{c}}^{\pm}) = 0$, $K(\gamma_*) = \kappa_{\max}$, strictly increasing and then strictly decreasing, and the lens with its floor is*

$$\overline{\mathrm{Lens}} := \mathrm{Lens} \cup \big(\{0\} \times [U_{\boldsymbol{c}}^-, U_{\boldsymbol{c}}^+]\big) = \{(\kappa, \gamma) : U_{\boldsymbol{c}}^- \leq \gamma \leq U_{\boldsymbol{c}}^+,\ 0 \leq \kappa \leq K(\gamma)\}.$$

(b) (Two walls.) *$\overline{\mathrm{Lens}}$ is convex iff γ^- is convex and γ^+ concave on $[0, \kappa_{\max}]$; it is then the intersection of $\{\gamma \geq \gamma^-\}$ and $\{\gamma \leq \gamma^+\}$, contains the triangle with vertices $(0, U_{\boldsymbol{c}}^-)$, $(0, U_{\boldsymbol{c}}^+)$, $(\kappa_{\max}, \gamma_*)$ and satisfies $\gamma^-(\kappa) \leq U_{\boldsymbol{c}}^- + (\gamma_* - U_{\boldsymbol{c}}^-)\kappa/\kappa_{\max}$.*

(c) (The tip is a convex corner.) *If the two edge solutions at the tip are of type* R *or* L_1*, then $s^- = \gamma^{-\prime}(\kappa_{\max}) \in (0, \infty)$, $s^+ = \gamma^{+\prime}(\kappa_{\max}) \in (-\infty, 0)$, K has the one-sided derivatives $1/s^{\mp}$ at γ_*, and the interior angle of the lens at the tip is $\arctan s^- + \arctan |s^+| \in (0, \pi)$.*

(d) (Curvature as a second variation.) *Let $0 < \kappa_0 < \kappa_{\max}$ and let the lower edge solution at $(x_0, \boldsymbol{c}, \kappa_0)$ be of type* R *or* L_1*. Then there are a neighbourhood N of $(\kappa_0, \gamma^-(\kappa_0))$, a cut $x_* < 1$ and a C^∞ function s near the edge state at x_*, with $ds \neq 0$, such that $H(\kappa, \gamma) = s\big(\Theta(\kappa, \gamma)\big)$ (Θ the scale-free state at x_* of the solution with data (κ, γ)) satisfies $H_\gamma > 0$ and $\mathrm{FS} \cap N = \{H \geq 0\}$.*

Consequently γ^- is C^∞ near κ_0 and, with $\tau = (1, \gamma^{-\prime})$,

$$\gamma^{-\prime\prime}(\kappa_0) = -\frac{D^2H[\tau,\tau]}{H_\gamma}, \qquad D^2H[\tau,\tau] = D^2s[J_\tau, J_\tau] + ds\,(Z_{\tau\tau}),$$

where J_τ, $Z_{\tau\tau}$ are the first and second variation of Lemma 12.41 along the line through $(\kappa_0, \gamma^-(\kappa_0))$ with direction τ, evaluated at x_*. The sign of $D^2H[\tau,\tau]$ on $\{H = 0\}$ does not depend on the choice of s.

The mirror statements hold for γ^+.

Proof. (a) By Proposition 5.59(a),(e), $\gamma^- : [0, \kappa_{\max}] \to [U_{\boldsymbol{c}}^-, \gamma_*]$ is a continuous strictly increasing bijection (its value at $\kappa_{\max}$ is γ_* by Proposition 5.59(b)), hence a homeomorphism, and likewise γ^+ onto $[\gamma_*, U_{\boldsymbol{c}}^+]$ (decreasing). The inverses are continuous and monotone. Let $\gamma \in [U_{\boldsymbol{c}}^-, \gamma_*]$: for every $\kappa \in [0, \kappa_{\max}]$, $\gamma \le \gamma_* \le \gamma^+(\kappa)$, so (κ, γ) lies in the lens iff $\gamma^-(\kappa) \le \gamma$ iff $\kappa \le K(\gamma)$ (γ^- increasing). The case $\gamma \ge \gamma_*$ is symmetric. (b) By Proposition 5.59(a) the set is $\{0 \le \kappa \le \kappa_{\max},\ \gamma^- \le \gamma \le \gamma^+\}$. If γ^- is convex and γ^+ concave, it is the intersection of the strip with the epigraph of γ^- and the hypograph of γ^+, which is convex. Conversely, if the set is convex, the chord joining two points of the graph of γ^- lies in the set, so above the graph: γ^- is convex; similarly for γ^+. A convex function lies below its chord, which gives the last inequality (chord from $(0, U_{\boldsymbol{c}}^-)$ to $(\kappa_{\max}, \gamma_*)$), and the convex set contains the triangle spanned by the three points $(0, U_{\boldsymbol{c}}^\mp)$, $(\kappa_{\max}, \gamma_*)$. (c) By Corollary 7.14 (its hypotheses are the hypotheses of (c)) the walls are C^∞ near $\kappa_{\max}$, meet transversally and $\partial_\kappa\gamma^- > 0 > \partial_\kappa\gamma^+$. The inverse function theorem gives $K'(\gamma_*^\mp) = 1/s^\mp$. The edge tangents at the tip have the directions $(s^\mp, 1)$ in the (γ, κ)-plane, and the interior angle between the downward rays is $\arctan s^- - \arctan s^+$. (d) This is the construction in the proof of Theorem 7.12 (with $p = (x_0, \boldsymbol{c}, \kappa)$ and κ as the parameter): $H = s \circ \Theta$ with $\partial_\gamma H \neq 0$ by Proposition 7.7, $H(\kappa, \gamma^-(\kappa)) = 0$ and, after replacing s by $-s$, $H_\gamma > 0$; $H(\kappa, \cdot)$ is then strictly increasing near $\gamma^-(\kappa)$, so $\{H \ge 0\} = \{\gamma \ge \gamma^-\} = \mathrm{FS}$ locally. Apply Lemma 12.39 to $F = H$; by the chain rule $H_\tau = ds(J_\tau)$ and $D^2H[\tau,\tau] = D^2s[J_\tau, J_\tau] + ds(Z_{\tau\tau})$ because the second derivative of Θ along the straight line is the second variation at x_* (the data are affine in (κ, γ); Lemma 12.41). If $\tilde{s} = \theta s$ with a smooth $\theta > 0$, then on $\{H = 0\}$ one has $\tilde{H}_\gamma = \theta H_\gamma$ and $D^2\tilde{H}[\tau,\tau] = \theta D^2H[\tau,\tau]$ (the other terms contain H or $H_\tau = 0$), so the ratio is unchanged; any other defining function is of this form. □

Example 12.44 (the rest state, $x_0 = 0$)**.** At $\boldsymbol{c} = 0$, $x_0 = 0$ one has $\bar{c}_3 = -4$, $U^\mp = \mp 2$, $\gamma^+ = -\gamma^-$, $\gamma_* = 0$, $\kappa_{\max} = 4.16081$. The computed lower wall has $\gamma^-(1) = -1.78060$, $\gamma^-(1.66) = -1.49853$, $\gamma^-(3) = -0.75480$, with $\gamma^{-\prime\prime} = 0.2166,\ 0.1398,\ 0.0714$ there (central differences, step 0.02), and at the tip $s^- = 0.6812$, interior angle $2\arctan s^- = 68.5°$. The wall lies to the *left* of the straight line from $(\kappa, \gamma) = (0, -2)$ to $(\kappa_{\max}, 0)$: at $\kappa = 1.66$ the wall is at -1.49853, the line at -1.20208 (Figure 12.4). So the lens is a rounded diamond, with a vertical tangent at the floor ends (Theorem 12.55) and corners at the tips only. (`verify_ch12b.py`, part `shape`.)

12.12.4. Which signs does the source have? By Proposition 12.43(d), convexity of the wall is the sign of $ds(Z_{\tau\tau}) + D^2s[J_\tau, J_\tau]$, and $Z_{\tau\tau}$ solves $Z' = D_Y F\, Z + S$. The linearisation is cooperative in the order $\Sigma = \mathrm{diag}(+, -, -, -, -, -)$ along solutions (Lemma 5.21, Lemma 5.27), so a one-signed source would give a

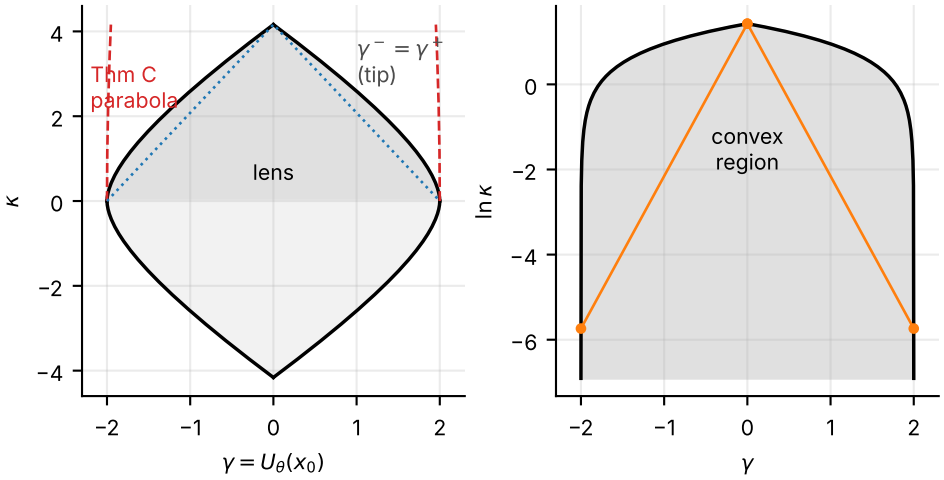


FIGURE 12.4. Left: the lens at $x_0 = 0$, $\boldsymbol{c} = 0$ and its mirror image $\kappa \to -\kappa$ (the diamond); dashed: the parabola $U^-_{\boldsymbol{c}} + A_{\text{low}}\varkappa^2$ of Theorem 12.53; dotted: the straight diamond. Right: the same lens in the coordinates $(\gamma, \ln\kappa)$, where it is convex (Theorem 12.48); the chords join two points of each wall. Code: `code/ch12_thresh/make_b_lens.py`.

one-signed ΣZ. The next statement says that this does not happen in the variables (U, g, f).

Proposition 12.45 (the source has no sign in (U, g, f, p); it has one in $(U, \ln g, \ln f, p)$)**.** *(a) Along a line in the data with $J(x_0) = (\sigma, \tau, 0, 0, 0, 0)$, $\sigma, \tau \neq 0$ of the same sign (as for the tangent $(1, \gamma^{-\prime})$ of the lower wall, $\gamma^{-\prime} > 0$), the vectors $\Sigma J(x_0) = (\sigma, -\tau, 0, 0, 0, 0)$ and $\Sigma S(x_0) = \left(-\frac{\sigma^2}{d_0}, \frac{2\sigma\tau}{d_0}, 0, 0, 0, 0\right)$ have components of both signs ($d_0 = 1 - x_0^2$). Hence the condition "$\Sigma S \leq 0$ on $[x_0, 1)$" fails at x_0 in these variables, for every lower-wall point. (b) In the variables (U, a, b, p_2, p_1, p_0), $a = \ln g$, $b = \ln f$ on $x > x_0$, the system reads $U' = \frac{P - p_0 - 2xU - U^2/2}{d}$, $a' = -\frac{U}{d}$, $b' = e^{a-b}$, $p_2' = \frac{2e^{a+b}}{d}$, $p_1' = p_2$, $p_0' = p_1$; its linearisation is Metzler in the order Σ with no sign condition on the solution, and the second-variation source has the components $S_U = -\frac{J_U^2}{d} \leq 0$, $S_a = 0$, $S_b = e^{a-b}(J_a - J_b)^2 \geq 0$, $S_{p_2} = \frac{2e^{a+b}}{d}(J_a + J_b)^2 \geq 0$; so $\Sigma S \leq 0$ on all of $(x_0, 1)$.*

PROOF. (a) Lemma 12.41 at x_0 with $J_U = \sigma$, $J_g = \tau$, $J_f = 0$ gives the components of S; apply Σ. (b) $\ln g$ and $\ln f$ satisfy the stated equations ($g' = -Ug/d$, $f' = g$). The off-diagonal entries of the Jacobian are $\partial_{p_0} U' = -\frac{1}{d}$, $\partial_U a' = -\frac{1}{d}$, $\partial_a b' = e^{a-b}$, $\partial_a p_2' = \partial_b p_2' = \frac{2e^{a+b}}{d}$, $\partial_{p_2} p_1' = \partial_{p_1} p_0' = 1$ (and $\partial_b b' = -e^{a-b}$, $\partial_U U' = -\frac{2x+U}{d}$ on the diagonal); multiplying entry (i, j) by $\Sigma_i \Sigma_j$ makes each off-diagonal entry ≥ 0, whatever the solution is. The nonlinear terms are $-\frac{U^2}{2d}$, e^{a-b} and $\frac{2e^{a+b}}{d}$, whose second derivatives in the direction J are $-\frac{J_U^2}{d}$, $e^{a-b}(J_a - J_b)^2$ and $\frac{2e^{a+b}}{d}(J_a + J_b)^2$ (a computer-algebra check: `verify_ch12b.py`, part `sym`).

Hence $\Sigma S = (S_U, -S_a, -S_b, -S_{p_2}, 0, 0) = (-\frac{J_U^2}{d}, 0, -e^{a-b}(J_a - J_b)^2, -\frac{2e^{a+b}}{d}(J_a + J_b)^2, 0, 0) \le 0$. □

Heuristic 12.46 (why $\ln\kappa$ and not κ)**.** *Part (b) says that in the logarithmic variables the infinitesimal convexity argument closes, for lines in the data $(\ln\kappa, \gamma)$: the line is straight in $(\ln\kappa, \gamma)$, not in (κ, γ), and the passage from the first to the second is exactly the loss of the sign of $S_g = -2J_U J_g/d$. (The system is singular at x_0 in these variables, $f(x_0) = 0$, so this is only the infinitesimal heuristic; the rigorous version, which does not differentiate at all, is Theorem 12.48.) Whether the lens is convex in (κ, γ) is posed in Chapter 13 (Conjecture 13.8).*

12.13. Log-convexity of the admissible set

In words. *Convexity of the lens in the plane (κ, γ) is not proved in this book (see Chapter 13). If one replaces κ by $\ln\kappa$, the answer is yes, and even jointly in the constants: the set of admissible triples $(\boldsymbol{c}, \ln\kappa, \gamma) \in \mathbb{R}^5$ is convex. The proof is the toy of Toy example 5.35 for the six-component system. From two admissible data one builds the "geometric–arithmetic mean" V (arithmetic mean for U and the swirl integrals, geometric mean for g and f); it is not a solution but a sub-solution, because the three defects are squares by the inequality between the arithmetic and the geometric mean. By the comparison principle the true solution with the mean data lies above V in the cooperative order, so it cannot blow up: the mean data are admissible. Part of the work was done in Chapter 5 (Lemma 5.36); here it is assembled with the backward half-line and with closedness, and the consequences for the lens and for $\kappa_{\max}$ are drawn.*

Toy example 12.47 (the ellipse)**.** For the fixed-potential model of Toy example 5.57 the lens is the ellipse $\frac{\gamma^2}{4} + \frac{b}{2}\kappa^2 \le 1$, $K(\gamma) = \sqrt{\frac{2}{b}(1 - \gamma^2/4)}$, and $\ln K = \frac{1}{2}\ln(1 - \gamma^2/4) + \text{const}$ is concave: the region $\{0 < \kappa \le K(\gamma)\}$ is convex in $(\gamma, \ln\kappa)$ (an unbounded convex region going down to $\ln\kappa \to -\infty$). Theorem 12.48 says the same for the true lens, and it also says more: the region is convex when the constants $\boldsymbol{c}$ are moved as well.

THEOREM 12.48 (log-convexity). *Fix $x_0 \in (-1, 1)$ and let*

$$\mathcal{G} = \mathcal{G}(x_0) = \big\{(\boldsymbol{c}, t, \gamma) \in \mathbb{R}^3 \times \mathbb{R} \times \mathbb{R} : \ (e^t, \gamma) \in \mathrm{Lens}(x_0, \boldsymbol{c})\big\}$$

($t = \ln\kappa$; $\mathrm{Lens}(x_0, \boldsymbol{c}) = \{\kappa > 0 : \gamma^- \le \gamma \le \gamma^+\}$, empty if $\boldsymbol{c} \notin \operatorname{int}\mathcal{J}$ or $\kappa > \kappa_{\max}$). Then $\mathcal{G}$ is a closed convex subset of $\mathbb{R}^5$.

PROOF. Write $\Phi(\boldsymbol{c}, t, \gamma) = (x_0, \boldsymbol{c}, 0, e^t, \gamma)$; then $\mathcal{G} = \Phi^{-1}(G^+ \cap G^-)$ (the definition of Lens, Proposition 5.42), where $G^\pm$ are the sets of data whose forward, resp. backward, solution exists up to $x = \pm 1$.

Step 1 (forward half: the mean data are forward admissible). Let $(\boldsymbol{c}^i, t_i, \gamma_i)$, $i = 1, 2$, be such that the forward solutions Y^i with data $(\gamma_i, \kappa_i, 0, 0, 0, 0)$, $\kappa_i = e^{t_i}$, exist on $[x_0, 1)$. Put $\bar{\boldsymbol{c}} = \frac{1}{2}(\boldsymbol{c}^1 + \boldsymbol{c}^2)$, $\bar\gamma = \frac{1}{2}(\gamma_1 + \gamma_2)$, $\bar\kappa = \sqrt{\kappa_1\kappa_2}$ (so $\ln\bar\kappa = \frac{1}{2}(t_1 + t_2)$) and let $\hat Y$ be the maximal solution with parameter $\bar{\boldsymbol{c}}$ and data $(\bar\gamma, \bar\kappa, 0, 0, 0, 0)$ on $[x_0, \hat b)$. Suppose $\hat b < 1$ and choose $b \in (\hat b, 1)$. By Lemma 5.36, applied on $[x_0, b]$ (where both Y^i exist), $\hat U \ge V_U = \frac{1}{2}(U^1 + U^2)$ on $[x_0, \hat b)$, and V_U is bounded below on $[x_0, b]$ (it is continuous). But a solution that ceases to exist before $x = 1$ has $U \to -\infty$ (Lemma 5.41(d)): contradiction. So $\hat b = 1$.

Step 2 (backward half). By Lemma 5.40, $\gamma \in \mathrm{BS}(x_0, \boldsymbol{c}, 0, \kappa)$ iff $-\gamma \in \mathrm{FS}(-x_0, \boldsymbol{c}_m, 0, \kappa)$, and $(\boldsymbol{c}, t, \gamma) \mapsto (\boldsymbol{c}_m, t, -\gamma)$ is linear. Apply Step 1 to the mirrored data (base point $-x_0$): if $(\boldsymbol{c}^i, t_i, \gamma_i)$ are backward admissible, then $(\bar{\boldsymbol{c}}_m, \bar{t}, -\bar{\gamma})$ is forward admissible at $-x_0$, i.e. $(\bar{\boldsymbol{c}}, \bar{t}, \bar{\gamma})$ is backward admissible. With Step 1, $\mathcal{G}$ is midpoint convex: the midpoint of two points of $\mathcal{G}$ lies in $\mathcal{G}$.

Step 3 (closedness). G^+ is closed in $(-1,1) \times \mathbb{R}^3 \times [0,\infty)^2 \times \mathbb{R}$ (Proposition 5.42(a)), G^- is its image under a linear homeomorphism (Lemma 5.40), hence closed, and Φ is continuous, so $\mathcal{G}$ is closed.

Step 4. A closed midpoint-convex set is convex: by repeated bisection it contains all points $(1-\lambda)P + \lambda Q$ with dyadic λ, which are dense on the segment, and closedness gives the rest. □

Corollary 12.49 (consequences)**.** *Let* $x_0 \in (-1,1)$.

(a) (The lens is log-convex.) *For fixed* $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$ *the set* $\{(\gamma, \ln\kappa) : 0 < \kappa \le K(\gamma)\}$ *is convex; equivalently* $\ln K$ *is concave on* $(U^-_{\boldsymbol{c}}, U^+_{\boldsymbol{c}})$, $t \mapsto \gamma^-(e^t)$ *is convex and* $t \mapsto \gamma^+(e^t)$ *is concave on* $(-\infty, \ln\kappa_{\max}]$.

(b) ($\kappa_{\max}$.) $\boldsymbol{c} \mapsto \ln\kappa_{\max}(x_0, \boldsymbol{c})$ *is concave on* $\operatorname{int}\mathcal{J}$.

(c) (The admissible constants.) $\operatorname{int}\mathcal{J}$ *and* J *are convex subsets of* $\mathbb{R}^3$*; hence* $\bar{c}_3(c_1,c_2) = -\frac{1}{2}(k_1+k_2)(k_1+k_2+2)$ *is a convex function on* $\{c_1, c_2 \ge -1\}$.

Proof. (a) The slice of $\mathcal{G}$ at fixed $\boldsymbol{c}$ is convex; by Proposition 12.43(a) it is $\{(t,\gamma) : e^t \le K(\gamma)\}$, and the hypograph of $\ln K$ is convex iff $\ln K$ is concave. The statement on the walls is the same slice described as $\{(t,\gamma) : \gamma^-(e^t) \le \gamma \le \gamma^+(e^t)\}$ (as in Proposition 12.43(b)). (b) For $\boldsymbol{c}^1, \boldsymbol{c}^2 \in \operatorname{int}\mathcal{J}$ pick γ_i with $(\boldsymbol{c}^i, \ln\kappa_{\max}(\boldsymbol{c}^i), \gamma_i) \in \mathcal{G}$ ($\kappa_{\max}$ is attained, Proposition 5.59(a)); the midpoint is in $\mathcal{G}$, so $\ln\kappa_{\max}(\bar{\boldsymbol{c}}) \ge \frac{1}{2}(\ln\kappa_{\max}(\boldsymbol{c}^1) + \ln\kappa_{\max}(\boldsymbol{c}^2))$. This midpoint concavity of the continuous function $\ln\kappa_{\max}$ (Proposition 5.59(d)) is concavity. (c) By Proposition 5.58(a) and Proposition 5.59(a), $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ iff $\mathrm{Lens}(x_0, \boldsymbol{c}) \ne \emptyset$, so $\operatorname{int}\mathcal{J}$ is the projection of the convex set $\mathcal{G}$ to $\boldsymbol{c}$-space, hence convex. Every $\boldsymbol{c} \in J$ is a limit of points $\boldsymbol{c} + \frac{1}{n}(1,1,1) \in \operatorname{int}\mathcal{J}$ ($\bar{c}_3$ is decreasing in c_1, c_2), so $J = \overline{\operatorname{int}\mathcal{J}}$ is convex; $\{c_3 \ge \bar{c}_3(c_1,c_2),\ c_1, c_2 \ge -1\} = J$ is the epigraph of $\bar{c}_3$. □

Remark 12.50 (what is, and is not, claimed)**.** In the coordinates (γ, t) the lens is convex, so $G(t) = \gamma^-(e^t)$ satisfies $G'' \ge 0$; convexity of γ^- in κ is the stronger statement $G'' \ge G'$, which is *not* proved here (Chapter 13, Conjecture 13.8). The loss comes from the Jensen gap between $\sqrt{\kappa_1\kappa_2}$ and $\frac{1}{2}(\kappa_1+\kappa_2)$ (Proposition 12.45). The convexity of J (part (c)) is the convexity of the set of constants for which swirl-free global solutions exist, here obtained as a by-product of a statement about swirling ones.

Numerical observation 12.51 (midpoint test)**.** *For* 120 *random pairs of data (base point* $x_0 \in (-0.6, 0.6)$, $c_{1,2} \in (-0.7, 3)$, $\varepsilon \in (0.3, 4)$, κ *between* 5% *and* 90% *of* $\kappa_{\max}$*) the computed forward threshold at the mean data* $(\bar{\boldsymbol{c}}, \sqrt{\kappa_1\kappa_2})$ *lies below the mean of the two thresholds (and the backward threshold above the mean): no violation, smallest relative slack* $1.2 \cdot 10^{-2}$ *(lower wall) and* $1.3 \cdot 10^{-2}$ *(upper wall). (*`verify_ch12b.py`*, part* `logconv`*; random seed fixed in the script.) Figure 12.4(right) shows the lens of the rest state in* $(\gamma, \ln\kappa)$.

12.14. The size of the lens: a parabola below the wall

In words. *Identity (E) writes the distance $\gamma^- - U_{\boldsymbol{c}}^-$ of the lower wall from the no-swirl endpoint as the integral of the swirl integral ψ against the positive function $u_+ z$. The sandwich $u_+ \le z \le u_-$ bounds z on both sides by* explicit *no-swirl solutions: the smallest possible swirl profile f is a fixed multiple of $\varkappa$, so ψ is at least a fixed multiple of $\varkappa^2$, and so is the distance. The conclusion is a global parabola $\gamma^- \ge U_{\boldsymbol{c}}^- + A_{\mathrm{low}}\varkappa^2$, valid for* all $\kappa \le \kappa_{\max}$. *Two consequences: an upper bound for $\kappa_{\max}$ (the two parabolas must not cross), and the sign of the quadratic coefficient at the floor, which gives convexity near the floor (Theorem 12.55) and, near the critical surface, on the whole lens (Theorem 12.57).*

Let $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$, $\xi_0 = \operatorname{artanh} x_0$, $\varkappa = (1-x_0^2)\kappa$, and with the kernel $k(\xi,s) = \sinh^2(\xi-s)/\cosh^2\xi$ of Lemma 12.23 put

$$G(\xi) = \int_{\xi_0}^{\xi} u_-^{-2}\,ds, \qquad \check\psi(\xi) = \int_{\xi_0}^{\xi} k(\xi,s)\,G(s)G'(s)\,ds,$$
$$A_{\mathrm{low}} = A_{\mathrm{low}}(x_0,\boldsymbol{c}) = \int_{\xi_0}^{\infty} \check\psi\,u_+^2\,d\xi.$$

Lemma 12.52 (A_{low} is finite and positive)**.** $0 < A_{\mathrm{low}} \le \dfrac{e^{-2\xi_0}}{2w_0^2}\displaystyle\int_{\xi_0}^{\infty} u_+^2\,d\xi < \infty.$

Proof. Since $(u_+/u_-)' = -w_0 u_-^{-2}$, $(u_+/u_-)(\xi_0) = 1$ and $u_+/u_- \to 0$ at $+\infty$ (u_+ is recessive, u_- is dominant, Lemma 3.21), $G(\infty) = 1/w_0$, so $0 \le G \le 1/w_0$. By Lemma 12.23(a), $0 \le k(\xi,s) \le e^{-2s} \le e^{-2\xi_0}$ for $\xi_0 \le s \le \xi$, hence $\check\psi(\xi) \le e^{-2\xi_0}\int_{\xi_0}^{\xi} GG' = \frac12 e^{-2\xi_0}G(\xi)^2 \le \frac{e^{-2\xi_0}}{2w_0^2}$. Moreover $u_+ \le Ce^{-k_2\xi}$ is square integrable. For positivity: $G > 0$, $G' > 0$ and $k(\xi,s) > 0$ for $s < \xi$, so $\check\psi > 0$ for $\xi > \xi_0$. □

Theorem 12.53 (a parabola below the wall). *Let $x_0 \in (-1,1)$ and $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$.*

(a) *For every $\kappa \in (0, \kappa_{\max}]$,*

$$\gamma^-(\kappa) - U_{\boldsymbol{c}}^-(x_0) \;\ge\; A_{\mathrm{low}}\,\varkappa^2 = (1-x_0^2)^2 A_{\mathrm{low}}\,\kappa^2,$$

and $U_{\boldsymbol{c}}^+ - \gamma^+(\kappa) \ge A_{\mathrm{low}}^+\varkappa^2$, where $A_{\mathrm{low}}^+ = A_{\mathrm{low}}(-x_0,\boldsymbol{c}_m)$.

(b) (Size.) $\varkappa_{\max}^2 \le \dfrac{2w_0}{A_{\mathrm{low}} + A_{\mathrm{low}}^+}$.

Proof. (a) Let (z,f,ψ) be the solution belonging to the admissible datum $(\gamma^-(\kappa),\kappa)$, $\varkappa > 0$. *(1)* By (Y1), $z \le u_-$ on $[\xi_0,\infty)$, so $f' = \varkappa z^{-2} \ge \varkappa u_-^{-2} = \varkappa G' \ge 0$ and, integrating from $f(\xi_0) = 0$, $f \ge \varkappa G \ge 0$. *(2)* By (12.4), $\psi(\xi) = \int_{\xi_0}^{\xi} k(\xi,s) f f'\,ds \ge \varkappa^2\check\psi(\xi)$, because $k \ge 0$ and $ff' \ge \varkappa^2 GG' \ge 0$. *(3)* By identity (E) (Proposition 3.51) and $z \ge u_+ > 0$ (Y1),

$$\gamma^- - U_{\boldsymbol{c}}^- = \int_{\xi_0}^{\infty} \psi\,u_+ z\,d\xi \;\ge\; \int_{\xi_0}^{\infty} \psi\,u_+^2\,d\xi \;\ge\; \varkappa^2 A_{\mathrm{low}}.$$

The statement for γ^+ is the mirror image (Proposition 5.56: $\gamma^+(x_0,\boldsymbol{c},\kappa) = -\gamma^-(-x_0,\boldsymbol{c}_m,\kappa)$ and $U_{\boldsymbol{c}}^+(x_0) = -U_{\boldsymbol{c}_m}^-(-x_0)$). (b) For $\kappa \le \kappa_{\max}$, $\gamma^- \le \gamma^+$. By (a), $U_{\boldsymbol{c}}^- + A_{\mathrm{low}}\varkappa^2 \le \gamma^- \le \gamma^+ \le U_{\boldsymbol{c}}^+ - A_{\mathrm{low}}^+\varkappa^2$, and $U_{\boldsymbol{c}}^+ - U_{\boldsymbol{c}}^- = 2w_0$ (the window is $[2(\zeta_+ - x_0), 2(\zeta_- - x_0)]$). Put $\kappa = \kappa_{\max}$. □

Numerical observation 12.54 (the parabola and the size bound)**.** *Table 12.4 lists A_{low} (a quadrature of no-swirl solutions), the computed ratio $(\gamma^- - U_{\boldsymbol{c}}^-)/\varkappa^2$ at four values of κ, the computed $\varkappa_{\max}$ and the bound (b). In every row and at every κ the ratio is at least A_{low}, also at the tip, as proved. The constant is far from sharp (the proof discards $z \geq u_+$ and $z \leq u_-$) and the size bound overshoots the computed $\varkappa_{\max}$ by a factor between 4.99 and 14.5. (Figure 12.4(left); `verify_ch12b.py`, part `thmC`.)*

$\boldsymbol{c}$	x_0	A_{low}	$(\gamma^- - U_{\boldsymbol{c}}^-)/\varkappa^2$ at $\kappa =$ 10^{-3}	$\frac{\kappa_{\max}}{10}$	$\frac{\kappa_{\max}}{2}$	$\kappa_{\max}$	$\varkappa_{\max}$	bound (b)
$(0,0,0)$	0	0.00272	0.774	0.2831	0.1652	0.1155	4.1608	27.13
$(-\frac{1}{2},-\frac{1}{2},0)$	0	0.00858	0.2514	0.2411	0.2538	0.2135	2.5737	12.84
$(2,-\frac{1}{2},0)$	$\frac{1}{2}$	0.00170	0.0996	0.0955	0.0944	0.0741	4.7189	34.77
$(0,-0.9,0)$	0	0.03010	0.4067	0.4145	0.8035	0.7851	1.7048	8.93
$(0,0,5)$	0	0.00051	2.3683	0.4595	0.1774	0.1078	5.7586	83.42

TABLE 12.4. Theorem 12.53 against computed values.

12.14.1. Convexity near the floor and near the critical surface.

THEOREM 12.55 (convexity near the floor). *Let $x_0 \in (-1,1)$ and $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ with $-1 < c_2 < -\frac{3}{4}$. By Theorem 7.21, $\gamma^-(\kappa) = \Gamma(\kappa^2)$ on $[0,\kappa_0)$ with $\Gamma \in C^\infty$. Then*

$$a := \Gamma'(0) \;\geq\; (1-x_0^2)^2 A_{\mathrm{low}} \;> 0,$$

$\gamma^{-\prime\prime}(\kappa) = 2\Gamma'(\kappa^2) + 4\kappa^2\Gamma''(\kappa^2) \to 2a$ as $\kappa \to 0$, and γ^- is strictly convex on $[0,\kappa_1]$ for some $\kappa_1 = \kappa_1(x_0,\boldsymbol{c}) > 0$, with $\gamma^{-\prime}(0) = 0$. If $-1 < c_1 < -\frac{3}{4}$ the same holds for γ^+ (strictly concave).

PROOF. By Theorem 12.53, $\Gamma(\sigma) - \Gamma(0) \geq (1-x_0^2)^2 A_{\mathrm{low}}\,\sigma$ for $\sigma = \kappa^2 \in (0,\kappa_0^2)$; dividing by σ and letting $\sigma \to 0$ gives $\Gamma'(0) \geq (1-x_0^2)^2 A_{\mathrm{low}} > 0$ (Lemma 12.52). The chain rule gives $\gamma^{-\prime} = 2\kappa\Gamma'(\kappa^2)$ and $\gamma^{-\prime\prime} = 2\Gamma'(\kappa^2) + 4\kappa^2\Gamma''(\kappa^2)$, which is continuous with value $2a$ at 0; hence $\gamma^{-\prime\prime} > a$ on some $[0,\kappa_1]$. The mirror statement is Proposition 5.56. □

Numerical observation 12.56 (the quadratic coefficient)**.** *For $\boldsymbol{c} = (0,-0.9,0)$, $x_0 = 0$: $A_{\mathrm{low}} = 0.03010$, and $(\gamma^- - U_{\boldsymbol{c}}^-)/\varkappa^2 = 0.4067$ at $\kappa = 10^{-2}$ and 0.40667 at 10^{-3}, so $a \approx 0.4067$ and $\gamma^{-\prime\prime}(0) \approx 0.8133$, more than ten times larger than the proved bound $2A_{\mathrm{low}} = 0.0602$; central differences give $\gamma^{-\prime\prime} = 0.8212,\ 0.9568,\ 3.3065$ at $\kappa = 0.05,\ 0.2,\ 0.5$. (`verify_ch12b.py`, part `thmB`.) The radius κ_1 of Theorem 12.55 is not uniform as $c_2 \uparrow -\frac{3}{4}$ (Remark 7.23); Theorem 12.57 obtains uniformity near $\partial\mathcal{J}$.*

THEOREM 12.57 (convexity near the critical surface, $k_1, k_2 < \frac{1}{2}$). *Let $x_0 \in (-1,1)$, $c_1, c_2 \in (-1,-\frac{3}{4})$ (that is $k_1, k_2 < \frac{1}{2}$) and $\bar{\boldsymbol{c}} = (c_1, c_2, \bar{c}_3(c_1,c_2))$. There are $\varepsilon_0 > 0$ and $a_0 > 0$ such that for $c_3 \in (\bar{c}_3, \bar{c}_3 + \varepsilon_0)$, with $\boldsymbol{c} = (c_1,c_2,c_3)$: $\gamma^{-\prime\prime} \geq a_0$ and $\gamma^{+\prime\prime} \leq -a_0$ on $(0,\kappa_{\max}]$. Hence the lens $\mathrm{Lens}(x_0,\boldsymbol{c})$ is strictly convex (and so is the diamond), and the tip is a convex corner.*

PROOF. *Step 1 (a uniform chart).* Since $\bar{\boldsymbol{c}} \in \partial\mathcal{J} \subset J$ with $c_1 > -1$ and $-1 < c_2 < -\frac{3}{4}$, Corollary 7.22 applies at $\bar{\boldsymbol{c}}$: there are a neighbourhood N of $\bar{\boldsymbol{c}}$ in $\mathbb{R}^3$, $\kappa_0 > 0$ and $g \in C^\infty(N \times (-\kappa_0, \kappa_0))$, even in κ, with $g(\boldsymbol{c}', 0) = U^-_{\boldsymbol{c}'}(x_0)$ on $N \cap J$, such that for $0 < \kappa < \kappa_0$ and $\boldsymbol{c}' \in N \cap \operatorname{int}\mathcal{J}$ the lower edge is of type R and $\gamma^- = g(\boldsymbol{c}', \kappa)$. By Whitney's theorem (as in Theorem 7.21) $g(\boldsymbol{c}', \kappa) = \Gamma(\boldsymbol{c}', \kappa^2)$ with $\Gamma \in C^\infty(N \times [0, \kappa_0^2))$, so $\gamma^-(x_0, \boldsymbol{c}', \kappa) = \Gamma(\boldsymbol{c}', \kappa^2)$ for $\boldsymbol{c}' \in N \cap \operatorname{int}\mathcal{J}$; shrink N to a compact ball (closure still inside the domain) and κ_0 so that Γ is smooth on a neighbourhood of $N \times [0, \kappa_0^2]$; let M bound $|\partial_\sigma^2 \Gamma|$ there.

Step 2 (a uniform lower bound for $\Gamma'(\boldsymbol{c}', 0)$). By Theorem 12.53, $\partial_\sigma \Gamma(\boldsymbol{c}', 0) \geq (1 - x_0^2)^2 A_{\mathrm{low}}(\boldsymbol{c}')$ for $\boldsymbol{c}' \in N \cap \operatorname{int}\mathcal{J}$. We claim $\liminf_{\boldsymbol{c}' \to \bar{\boldsymbol{c}}} A_{\mathrm{low}}(\boldsymbol{c}') > 0$. As $\boldsymbol{c}' \to \bar{\boldsymbol{c}}$ within $\operatorname{int}\mathcal{J}$, $w_0 \to 0$ and, by continuity of the Jost solutions in the potential (Lemma 3.25), $u^{\boldsymbol{c}'}_\pm \to u^{\bar{\boldsymbol{c}}}_\pm$ locally uniformly, where $u^{\bar{\boldsymbol{c}}}_- = u^{\bar{\boldsymbol{c}}}_+ =: u$ (at the critical surface the two recessive solutions coincide). So $G_{\boldsymbol{c}'} \to \int_{\xi_0}^{\xi} u^{-2}$ locally uniformly with $G'_{\boldsymbol{c}'} \to u^{-2}$, $\check{\psi}_{\boldsymbol{c}'} \to \check{\psi}_{\bar{\boldsymbol{c}}} > 0$ pointwise on (ξ_0, ∞), and Fatou's lemma gives $\liminf A_{\mathrm{low}}(\boldsymbol{c}') \geq \int_{\xi_0}^{\infty} \check{\psi}_{\bar{\boldsymbol{c}}} u^2 \, d\xi > 0$ (the right side may be $+\infty$, the positivity of the integrand is all that matters). Hence there are $a_1 > 0$ and a smaller ball N' with $\partial_\sigma \Gamma(\boldsymbol{c}', 0) \geq 2a_1$ on $N' \cap \operatorname{int}\mathcal{J}$, hence on its closure (which contains $\bar{\boldsymbol{c}}$). Only $N' \cap \operatorname{int}\mathcal{J}$ is used below.

Step 3 (the wall). For $\sigma = \kappa^2 \leq \kappa_0^2$ and $\boldsymbol{c}' \in N' \cap \operatorname{int}\mathcal{J}$, $\gamma^{-\prime\prime} = 2\partial_\sigma \Gamma + 4\sigma \partial_\sigma^2 \Gamma \geq 2(2a_1 - M\sigma) - 4M\sigma = 4a_1 - 6M\sigma \geq 2a_1$ if $\sigma \leq a_1/(3M)$. By Proposition 5.59(d), $\kappa_{\max}(x_0, \boldsymbol{c}) \to 0$ as $c_3 \downarrow \bar{c}_3$, so there is ε_0 with $(c_1, c_2, c_3) \in N'$ and $\kappa^2_{\max} \leq \min(\kappa_0^2, a_1/(3M))$ for $c_3 \in (\bar{c}_3, \bar{c}_3 + \varepsilon_0)$; then $\gamma^{-\prime\prime} \geq 2a_1$ on $(0, \kappa_{\max}]$. For γ^+ use the mirror ($c_1 < -\frac{3}{4}$, i.e. $k_1 < \frac{1}{2}$, Proposition 5.56) and shrink ε_0. Let a_1' be the constant obtained in the same way for the mirrored wall and put $a_0 = 2\min(a_1, a_1')$. By Proposition 12.43(b), the lens is convex. For $0 < \kappa < \kappa_0$ the lower edge is of type R (Corollary 7.22, at points of $N \cap \operatorname{int}\mathcal{J}$, which contains (c_1, c_2, c_3) for $c_3 \in (\bar{c}_3, \bar{c}_3 + \varepsilon_0)$), and the upper edge is of type R at the south pole $x = -1$: it is the image under $\sigma_1\sigma_2$ of the lower edge at $(-x_0, \boldsymbol{c}_m, \kappa)$ (Proposition 5.56), to which Corollary 7.22 applies because $c_1 \in (-1, -\frac{3}{4})$, i.e. $k_1 < \frac{1}{2}$; so both edges are of the types required in Proposition 12.43(c) at the tip, and the tip is a convex corner. □

Numerical observation 12.58 (the thin lens)**.** *For $\boldsymbol{c} = (-0.84, -0.84, \bar{c}_3 + \varepsilon)$, $x_0 = 0$ ($k_1 = k_2 = 0.4 < \frac{1}{2}$): $\kappa_{\max} = 0.55304,\ 0.18044,\ 0.05723$ for $\varepsilon = 0.1,\ 0.01,\ 0.001$; the smallest computed $\gamma^{-\prime\prime}$ on the grid $\kappa/\kappa_{\max} \in (0, 1]$ is $0.50744,\ 0.49978,\ 0.49903$, against the proved $2A_{\mathrm{low}} = 0.36084,\ 0.47596,\ 0.49616$ (Theorem 12.53), so the wall is nearly exactly a parabola with $\gamma^{-\prime\prime} \to 2a$. The range $k_i < \frac{1}{2}$ of Theorem 12.57 comes from Theorem 7.21; for $k_i \geq \frac{1}{2}$ nothing is proved here. (`verify_ch12b.py`, part `thmD`.)*

Remark 12.59 (what Part B leaves open)**.** The lens is log-convex (Theorem 12.48), near the floor and near the critical surface it is convex in (κ, γ) (Theorems 12.55 and 12.57) and the tip is a convex corner when the edge types allow (Proposition 12.43(c)); convexity of the whole lens in (κ, γ), the sharp constants of Theorem 12.31 when the large constants sit at the poles, and the law of the cone limit below ε_* are collected in Chapter 13 (Conjecture 13.8, Open problem 13.11, Open problem 13.12).

12.15. The class-N threshold law

In words. *A swirling flow of class* N_0 *is described from the south pole by a* germ, *and two commuting one-parameter groups act on germs (Chapter 6): the* t^2*-shift* T_c*, which makes a germ global once* c *is large, and the* recessive shift $\mathcal{R}_C$*, which adds a multiple* C *of the decaying mode of* $z'' = Wz$ *at the pole. For* $C \ll 0$ *the shifted germ is far from global and a very large* c *is needed to repair it; the threshold is* $\alpha_-(\mathcal{R}_C S)$*. This part proves that it grows like* $|C|^{2/\mu}$ *and computes the constant. The mechanism is simple. The repair* ct^2 *is felt only where* $ct^2 \gtrsim 1$*, i.e. at* $E = e^{2\xi} \approx c^{-1/2}$*; there, after rescaling, the equation is a modified Bessel equation with no parameter left, so the constant is a pure number: the demand that the shifted solution be the decaying function* $K_{\mu/2}$*. The delicate point is that the swirl is changed by the shifts and feeds back into the potential; we show that this feedback is negligible at a fraction* $1-\beta$ *of the critical shift, and squeeze.*

12.15.1. Setting and statement. In this part a bare κ is the germ constant κ_{ch} of Chapter 6 (normalisation 6 of Lemma 2.15); no conversion to κ_x or $\varkappa$ is needed. Throughout this part $E = E = e^{2\xi}$ for $\xi \le 0$, $t = 1 + x \in [E, 2E]$, $\nu_{\mathrm{B}} = \mu/2$, and (as in Equation (B.4))

$$\ell_*(\mu) = \frac{8^{\nu_{\mathrm{B}}}\,\Gamma(\nu_{\mathrm{B}})}{4\,\Gamma(1-\nu_{\mathrm{B}})}, \qquad \gamma_*(\mu) = \ell_*(\mu)^{2/\mu}. \tag{12.5}$$

For example $\gamma_*(1) = \frac{1}{2}$, and $\gamma_*(\mu)/(2-\mu) \to 1$ as $\mu \to 2$ (because $\Gamma(1-\nu_{\mathrm{B}}) \sim (1-\nu_{\mathrm{B}})^{-1}$; `verify_ch12c.py`, Part 1). Let $\mathcal{G}_{\frac{1}{2},2}$ be the set of germs of class N_0 that are of type G at the south pole with exponent $\mu \in (\frac{1}{2}, 2)$ (Corollary 6.55); this is exactly the set on which both T_c and $\mathcal{R}_C$ are defined (Theorems 6.19 and 6.22). The representative z of a germ is the one with $ze^{\mu\xi} \to 1$ at $\xi \to -\infty$, $f' = \kappa z^{-2}$, $\kappa > 0$ (the pole normalisation of Chapter 6); $l(\xi) = \int_{-\infty}^{\xi} z^{-2}$ and $\ell_\infty(S) = l(\infty) \in (0, \infty]$. A germ is *global* if its maximal interval is $\mathbb{R}$ in ξ, equivalently $z > 0$ on $\mathbb{R}$ (Corollary 2.25). By Theorem 6.32, $T_c S$ is global iff $c \ge \alpha_-(S)$.

Key result 12.60. The class-N threshold law. *For every germ* $S \in \mathcal{G}_{\frac{1}{2},2}$*,*

$$\alpha_-(\mathcal{R}_C S)/|C|^{2/\mu} \;\longrightarrow\; \gamma_*(\mu) \qquad (C \to -\infty).$$

THEOREM 12.61 (the class-N threshold). *Let* $S \in \mathcal{G}_{\frac{1}{2},2}$ *be fixed, with exponent* $\mu \in (\frac{1}{2}, 2)$*.*

(a) (Lower bound.) $c^{\mu/2}\ell_\infty(T_c S) \ge (1 - o(1))\,\ell_*(\mu)$ *as* $c \to \infty$*, and, as* $C \to -\infty$*,* $\liminf \alpha_-(\mathcal{R}_C S)/|C|^{2/\mu} \ge \gamma_*(\mu)$*.*

(b) (Upper bound.) *For every* $\beta \in (0,1)$ *there is* c_0 *such that for all* $c \ge c_0$ *the germ* $\mathcal{R}_C T_c S = T_c \mathcal{R}_C S$ *with* $C = -(1-\beta)c^{\mu/2}/\ell_*(\mu)$ *is global. Hence* $\limsup_{C\to-\infty} \alpha_-(\mathcal{R}_C S)/|C|^{2/\mu} \le \gamma_*(\mu)$*. Moreover* $c^{\mu/2}\ell_\infty(T_c S) \to \ell_*(\mu)$*.*

(c) (Limit law.) $\alpha_-(\mathcal{R}_C S)/|C|^{2/\mu} \to \gamma_*(\mu)$ *as* $C \to -\infty$*. Equivalently, for the depth* $\Lambda(S,c)$ *of Definition 6.56:* $\Lambda(S,c)\,\ell_*\,c^{-\mu/2} \to 1$ *and* $\Lambda(S,c)\,\ell_\infty(T_c S) \to 1$ *as* $c \to \infty$*.*

The hypotheses are exactly the ones proved: S is fixed (the constants depend on S through its envelope data), $\frac{1}{2} < \mu < 2$ (Theorem 6.19 needs $\mu < 2$, Theorem 6.22 needs $\mu > \frac{1}{2}$), $\kappa > 0$ and $f_- \ge 0$ (class N_0). The proof has four ingredients. (1) The shifts commute (Theorem 6.54), which gives the inverse relation between $\alpha_-(\mathcal{R}_{-d} S)$

and the depth $\Lambda(S,c)$ (Proposition 6.57). (2) The lower bound of Proposition 6.49, $|C| \le 1/\ell_\infty$, turns the threshold into a statement about $\ell_\infty(T_cS)$. (3) A lower bound for $\ell_\infty(T_cS)$ by positive-kernel comparison with Bessel models (Section 12.18). (4) The matching upper bound: at the fraction $1-\beta$ of the critical shift the swirl feedback is controlled *a priori*, by the sign structure of Proposition 6.23, up to a macroscopic point, and a forward Riccati barrier finishes (Section 12.19). Questions left open by the theorem are collected in Chapter 13.

Toy example 12.62 (the exponent $2/\mu$ by scaling)**.** Let $y'' = (\mu^2 + 2cE^2)y$ and put $\xi_c = -\frac{1}{4}\ln c$, $\theta = \xi - \xi_c$, so that $2cE^2 = 2e^{4\theta}$ and the equation $y_{\theta\theta} = (\mu^2 + 2e^{4\theta})y$ no longer contains c. The solution that is $e^{-\mu\xi}$ at the pole is $c^{\mu/4}g(\theta)$ with a c-free function $g \sim e^{-\mu\theta}$ (because $e^{-\mu\xi} = c^{\mu/4}e^{-\mu\theta}$). Hence $\int_{\mathbb{R}} y^{-2}\,d\xi = c^{-\mu/2}\int g^{-2}\,d\theta =: \ell_* c^{-\mu/2}$. The shifted solution $y(1+Cl)$ is positive iff $|C|\ell_* c^{-\mu/2} \le 1$, i.e. $c \ge (\ell_*|C|)^{2/\mu}$: the exponent $2/\mu$ is pure scaling, and $\ell_* = \int g^{-2}$ is the only quantity to compute (Proposition 12.66). At $\mu = 1$ everything is elementary (Example B.16): $\ell_* = 1/\sqrt{2}$, $\gamma_* = \frac{1}{2}$.

The pole zone, the Bessel window and the bulk, which organise the proofs of this chapter, are drawn in Figure 12.5.

12.16. Tools

In words. *Five elementary devices are used: Volterra equations with a parameter, comparison of Volterra solutions when the kernel is positive, scaling and Bessel functions (the toys above and Section 12.17), Riccati (Sturm-type) inequalities, and a convexity bound for tails. The one new idea is the second:* positivity replaces smallness*, so that two-sided bounds for a solution of $\psi = 1 + \mathcal{K}\psi$ follow from two Bessel models, with no contraction.*

Toy example 12.63 (a Volterra equation with a parameter)**.** Let $u(\xi) = \varepsilon\int_{-\infty}^{\xi}(\xi-\eta)e^{2\eta}(1+u(\eta))\,d\eta$, i.e. $u'' = \varepsilon e^{2\xi}(1+u)$ with $u, u' \to 0$. Seeking $1 + u = \sum b_n e^{2n\xi}$ gives $4n^2 b_n = \varepsilon b_{n-1}$, $b_0 = 1$, so $1 + u = \sum_n \frac{\varepsilon^n e^{2n\xi}}{4^n (n!)^2} = I_0(\sqrt{\varepsilon}\, e^{\xi})$ (checked in `verify_ch12c.py`, Part 2). The Picard iterates are the partial sums, each non-decreasing in ε: a larger coefficient gives a larger solution. This is the mechanism of Lemma 12.64(a).

Lemma 12.64 (tools)**.** (a) (Positive-kernel comparison.) *Let $0 \le \underline{k} \le k \le \bar{k}$ be kernels on $\{\eta < \xi \le \xi_1\}$ with Volterra operators $\underline{\mathcal{K}}\varphi = \int_{-\infty}^{\xi}\underline{k}(\xi,\eta)\varphi(\eta)d\eta$, etc., and suppose $\underline{\psi}, \bar{\psi} > 0$ are bounded solutions of $\underline{\psi} = 1 + \underline{\mathcal{K}}\underline{\psi}$ and $\bar{\psi} = 1 + \bar{\mathcal{K}}\bar{\psi}$. Then every bounded solution $\psi \ge 0$ of $\psi = 1 + \mathcal{K}\psi$ satisfies $\underline{\psi} \le \psi \le \bar{\psi}$.*

(b) (Tail bound.) *If $z > 0$, $z'' \ge 0$ on $[\xi_1, \xi_2)$ and $z'(\xi_1) > 0$, then $\int_{\xi_1}^{\xi_2} z^{-2} \le 1/(z(\xi_1)z'(\xi_1))$.*

(c) (Riccati lower bounds.) *Let $z > 0$, $z'' = Wz$ on $[\alpha,\beta]$, $r = z'/z$, $r(\alpha) \ge 0$. (i) If $W \ge w_0 > 0$ on $[\alpha,\beta]$, then $r \ge \min\{r(\alpha), \sqrt{w_0}\}$ there. (ii) If $W \ge k^2 > 0$ on $[\alpha, \alpha+1] \subset [\alpha,\beta]$ then $r(\alpha+1) \ge k\tanh k$; in particular $r(\alpha+1) \ge k/2$ if $k^2 \ge \frac{1}{2}$.*

(d) (Quasi-static bound.) *If $z > 0$ on $[\alpha,\beta]$, $z'/z \ge m_0 > 0$ there and $g \ge 0$ is non-decreasing, then $z(\eta)^{-2}\int_\alpha^\eta z^2 g\,d\zeta \le g(\eta)/(2m_0)$ for $\eta \in [\alpha,\beta]$.*

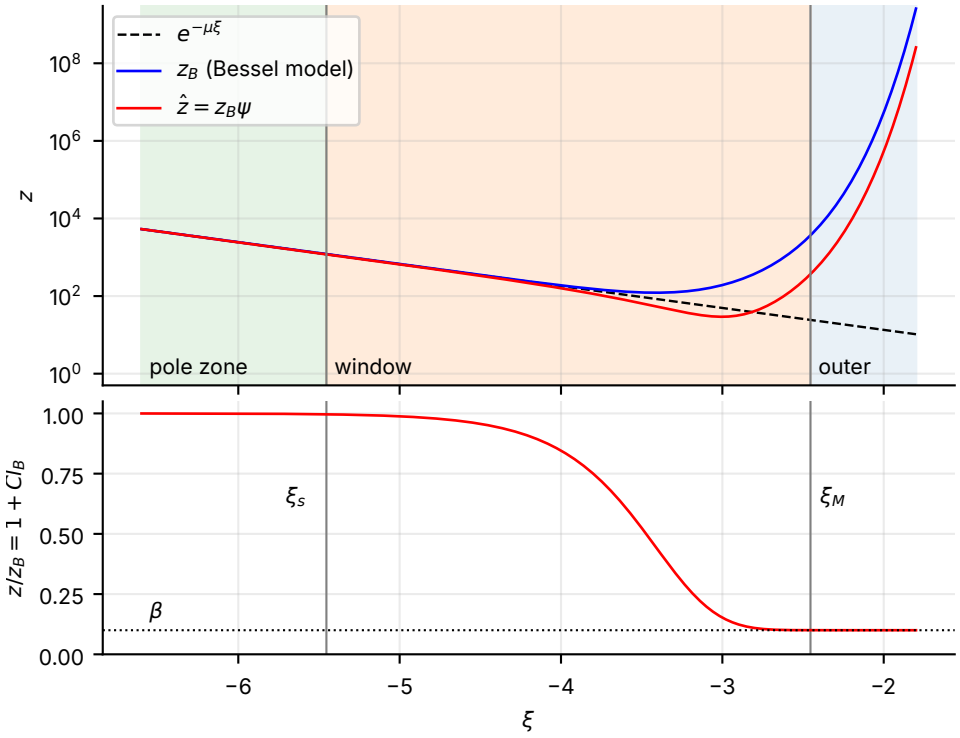


FIGURE 12.5. The three zones, for $\mu = 1.3$ and $c = 10^6$, in the Bessel model ($z_B = \zeta_c$ of Proposition 12.66; $\xi_s = \xi_c + \theta_1$, $\xi_M = \xi_c + M$ with $\theta_1 = -2$, $M = 1$ for the picture). *Pole zone* $\xi \le \xi_s$: $z \approx e^{-\mu\xi}$, the shifts are small perturbations and the contractions of Chapter 6 apply with constants independent of c. *Window* $\xi_s \le \xi \le \xi_M$: the potential is $\mu^2 + 2cE^2$, the solution follows the Bessel function, and the shifted $\hat{z} = z_B\psi$, $\psi = 1 + Cl_B$, dips (red) but stays $\ge \beta$ ($C = -(1-\beta)c^{\mu/2}/\ell_*$, $\beta = 0.1$). *Outer zone* $\xi \ge \xi_M$: z is convex and grows super-exponentially, so it contributes nothing to $\int z^{-2}$, and a forward barrier gives globality. Code: `make_c_zones.py`.

PROOF. (a) For $\bar{\psi}$: $\bar{\psi} = \sum_{n<N} \bar{\mathcal{K}}^n[1] + \bar{\mathcal{K}}^N[\bar{\psi}] \ge \sum_{n<N} \bar{\mathcal{K}}^n[1]$, because $\bar{\mathcal{K}}^N\bar{\psi} \ge 0$; so the non-negative series $\sum \bar{\mathcal{K}}^n[1]$ converges pointwise and its terms tend to 0, in particular $\bar{\mathcal{K}}^N[1] \to 0$. Since $0 \le \underline{k} \le k \le \bar{k}$ gives $\underline{\mathcal{K}}^N[1] \le \mathcal{K}^N[1] \le \bar{\mathcal{K}}^N[1]$ (each iterate is monotone in the kernel because all kernels are ≥ 0), the remainders $\mathcal{K}^N[\psi] \le (\sup\psi)\,\mathcal{K}^N[1]$ and $\underline{\mathcal{K}}^N[\underline{\psi}] \le (\sup\underline{\psi})\,\bar{\mathcal{K}}^N[1]$ tend to 0. Hence $\psi = \sum_n \mathcal{K}^n[1]$, $\underline{\psi} = \sum_n \underline{\mathcal{K}}^n[1]$, $\bar{\psi} = \sum_n \bar{\mathcal{K}}^n[1]$, and termwise $\underline{\mathcal{K}}^n[1] \le \mathcal{K}^n[1] \le \bar{\mathcal{K}}^n[1]$. (b) $z' \ge z_1'$, so $z \ge z_1 + z_1'(\xi - \xi_1)$ and $\int_{\xi_1}^{\infty}(z_1 + z_1' s)^{-2} ds = 1/(z_1 z_1')$. (c) (i) Put $m = \min\{r(\alpha), \sqrt{w_0}\} \in [0, \sqrt{w_0}]$ and $d = r - m$, $d(\alpha) \ge 0$. Then $d' = W - r^2 = (W - m^2) - (r + m)d \ge -(r + m)d$, since $m^2 \le w_0 \le W$;

so $\big(d\,e^{\int_\alpha^\xi (r+m)}\big)' \ge 0$ and $d \ge 0$. (ii) Let $\varrho(\tau) = k\tanh k\tau$, so $\varrho' = k^2 - \varrho^2$, $\varrho(0) = 0 \le r(\alpha)$, and $d(\tau) = r(\alpha+\tau) - \varrho(\tau)$: $d' \ge k^2 - r^2 - (k^2 - \varrho^2) = -(r+\varrho)d$, whence $d \ge 0$. Finally $\tanh k \ge \tanh 2^{-1/2} = 0.6089 > \frac12$ for $k \ge 2^{-1/2}$ (checked in `verify_ch12c.py`, Part 4, together with a test of (i), (ii) on 300 random potentials). (d) $z(\zeta)^2/z(\eta)^2 \le e^{-2m_0(\eta-\zeta)}$ for $\zeta \le \eta$ (integrate $z'/z \ge m_0$), and $g(\zeta) \le g(\eta)$; so the integral is at most $g(\eta)\int_0^\infty e^{-2m_0 s}ds$. □

Lemma 12.65 (Green kernel at the south pole)**.** *Let $-1 < s < x \le 0$, $t_x = 1+x$, $t_s = 1+s$, and $K_{\mathsf L}$ the Cauchy kernel of $\mathsf L$ (Lemma 2.18). Then (a) $0 \le K_{\mathsf L}(x,s) \le (t_x/t_s)^2$ and $ds \le 4E_s\,d\eta$ ($\eta = \operatorname{artanh} s$); (b) $0 \le \partial_x K_{\mathsf L}(x,s) = \dfrac{2t_x(1-t_s)+t_s^2}{t_s^2(2-t_s)^2} \le \dfrac{2t_x}{t_s^2} + 1$; (c) for continuous $F \ge 0$ with $\int^\xi FE^{-1} < \infty$: $\mathsf G_{\mathsf L}[F] \le 4E^2\int_{-\infty}^\xi FE^{-1}d\eta$ and $\partial_x \mathsf G_{\mathsf L}[F] \le 4t_x\int_{-\infty}^\xi FE^{-1}d\eta + 4\int_{-\infty}^\xi FE\,d\eta$.*

Proof. (a) and the first bound of (c) are in the proof of Lemma 2.23 (Steps 1, 2). (b) $\partial_x K_{\mathsf L} = (1-2xs+s^2)/(1-s^2)^2$ (Lemma 2.18(c)); with $x = t_x - 1$, $s = t_s - 1$ the numerator is $2t_x(1-t_s)+t_s^2$ and $(1-s^2)^2 = t_s^2(2-t_s)^2$ (a computer-algebra check, `verify_ch12c.py`, Part 4); use $0 \le 1-t_s \le 1$ and $(2-t_s)^2 \ge 1$. (c) $\partial_x \mathsf G_{\mathsf L}[F] = \int \partial_x K_{\mathsf L} F\,ds$ (Lemma 2.23); by (b), $(2t_x/t_s^2)\,ds \le (2t_x/t_s^2)\,2t_s\,d\eta \le 4t_x E_s^{-1}d\eta$ (as $1-s^2 \le 2t_s$, $t_s \ge E_s$), and $1\cdot ds \le 4E_s d\eta$. □

12.17. The Bessel model, and a model without swirl

In words. *Near the pole, with the swirl switched off, the equation of the shifted germ is $z'' = (\mu^2 + 2cE^2)z$, a modified Bessel equation after the substitution $y = \sqrt{c/2}\,E$ (Toy example B.14). Everything is explicit: the Bessel function $I_{-\nu_{\mathrm B}}$ is the dominant mode, $\ell_* c^{-\mu/2}$ is the total integral $\int z^{-2}$, and at the threshold the shifted solution is $K_{\nu_{\mathrm B}}$. These closed forms are in Appendix B (Proposition B.15); here they are used as* comparison functions*, with the Volterra form that the comparison needs. A second exact model, a genuine solution at $\mu = 1$ without swirl, shows what the law looks like when the Bessel approximation is replaced by the full potential.*

Proposition 12.66 (the Bessel model)**.** *Let $0 < \mu < 2$ and, for $c' > 0$, let $\zeta_{c'}$ solve $\zeta'' = (\mu^2 + 2c'E^2)\zeta$ with $\zeta_{c'} = e^{-\mu\xi}(1 + O(E^2))$, $l_{c'} = \int_{-\infty}^\xi \zeta_{c'}^{-2}$, and $\psi_{c'} := e^{\mu\xi}\zeta_{c'}$.*

(i) (Closed forms.) *$\zeta_{c'} = \Gamma(1-\nu_{\mathrm B})(a/2)^{\nu_{\mathrm B}} I_{-\nu_{\mathrm B}}(aE) > 0$, $a = \sqrt{c'/2}$; $\zeta_{c'}(\xi_{c'} + \theta) = c'^{\mu/4} g(\theta)$ with $\xi_{c'} = -\frac14 \ln c'$ and $g(\theta) = \Gamma(1-\nu_{\mathrm B})8^{-\mu/4} I_{-\nu_{\mathrm B}}(e^{2\theta}/\sqrt{2})$ independent of c'; $\int_\xi^\infty \zeta_{c'}^{-2} = \ell_* c'^{-\mu/2}\tau(aE)$ with $\tau = 1 - I_{\nu_{\mathrm B}}/I_{-\nu_{\mathrm B}}$ decreasing from 1 to 0; in particular $\ell_\infty = \ell_* c'^{-\mu/2}$, $\int g^{-2} = \ell_*$.*

(ii) (Threshold.) *For $C < 0$, $\tilde z = \zeta_{c'}(1 + C l_{c'}) > 0$ on $\mathbb R$ iff $|C| \le c'^{\mu/2}/\ell_*$, and at equality $\tilde z$ is a multiple of $K_{\nu_{\mathrm B}}(aE)$. So the smallest c' with $\tilde z > 0$ is $\gamma_* |C|^{2/\mu}$ exactly.*

(iii) (Volterra form.) *$\psi_{c'}$ is positive, bounded on $(-\infty, \xi_1]$, tends to 1 at $-\infty$, and $\psi = 1 + \int_{-\infty}^\xi k_0(\xi-\zeta)\,2c'E(\zeta)^2\psi(\zeta)\,d\zeta$, $k_0(s) = (e^{2\mu s} - 1)/(2\mu)$; moreover $\psi_{c'}' = e^{2\mu\xi}\int_{-\infty}^\xi e^{-2\mu\eta} 2c'E^2\psi_{c'}\,d\eta \ge 0$.*

(iv) (The profile g.) *$g > 0$, $g = e^{-\mu\theta}(1 + o(1))$ as $\theta \to -\infty$, and $g'/g \to +\infty$, $gg' \to +\infty$ as $\theta \to +\infty$.*

PROOF. (i), (ii): Proposition B.15(i)–(iii). (iii) Put $\zeta = e^{-\mu\xi}\psi$ in the equation: $\psi'' - 2\mu\psi' = 2c'E^2\psi$, i.e. $(e^{-2\mu\xi}\psi')' = e^{-2\mu\xi}2c'E^2\psi$. By (i), ψ is a power series in E^2 with constant term 1, so $\psi = 1+O(E^2)$, $\psi' = O(E^2)$ and $e^{-2\mu\xi}\psi' = O(e^{(4-2\mu)\xi}) \to 0$ ($\mu < 2$). Integrating, $\psi' = e^{2\mu\xi}\int_{-\infty}^{\xi} e^{-2\mu\eta}2c'E^2\psi$, and once more, with the order of integration exchanged and $\int_{\zeta}^{\xi} e^{2\mu(\eta-\zeta)}d\eta = k_0(\xi-\zeta)$, the Volterra equation. Positivity: the series of $I_{-\nu_B}$ has positive coefficients; boundedness: ψ is continuous with limit 1. (iv) The first two claims are (i). Since $g'' = (\mu^2 + 2e^{4\theta})g > 0$, g is convex; $g \to \infty$ (a series with positive coefficients, not a polynomial), so $g' > 0$ for large θ. Then $r = g'/g$ satisfies $r' = \mu^2 + 2e^{4\theta} - r^2$, and Lemma 12.64(c)(ii) gives $r(\theta+1) \geq \frac{1}{2}\sqrt{2}e^{2\theta} \to \infty$; hence $gg' = g^2 r \to \infty$. □

Figure 12.6 shows the profiles of the limit problem.

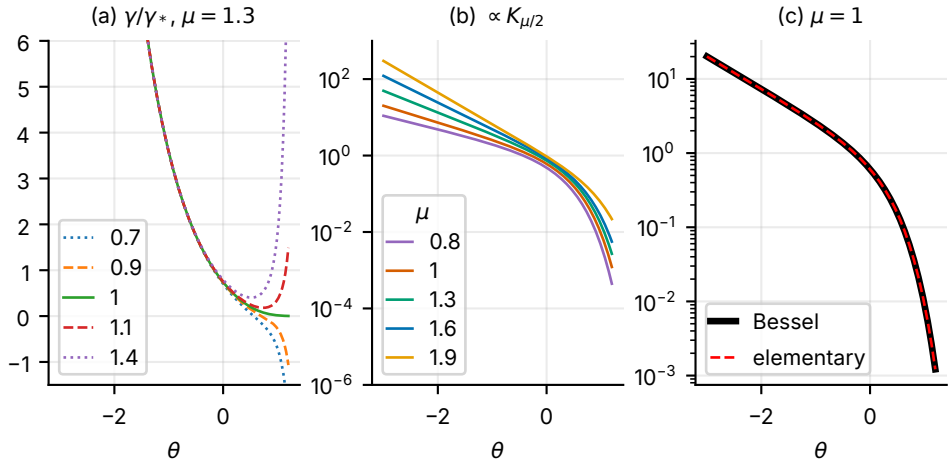


FIGURE 12.6. The limit problem $\zeta'' = (\mu^2 + 2\gamma e^{4\theta})\zeta$. (a) For $\gamma < \gamma_*$ the profile (dashed, dotted) crosses zero; for $\gamma \geq \gamma_*$ it stays positive. (b) The profile at $\gamma = \gamma_*$, a multiple of $K_{\mu/2}$, for several μ. (c) $\mu = 1$: the Bessel formula coincides with the elementary $\exp(-e^{2\theta}/2 - \theta)$ (Example B.16). Code: `make_c_profiles.py`.

Proposition 12.67 (a model without swirl: the rest state)**.** *Take $\mu = 1$, no swirl, $Q \equiv 0$, $z_0 = e^{-\xi}$, so that T_c adds ct^2 to Q ($\mathsf{L}[t^2] = 0$) and $z'' = (1 + \frac{c}{2}t^2)z$. Let $c > 0$, $r = \sqrt{1+2c}$, $x = \tanh\xi$ and*

$$z_c = 2^{(1-r)/2}(1+x)^{-1/2}(1-x)^{r/2}\Big(1 + \tfrac{r-1}{2r}\big(2^r(1-x)^{-r} - 1\big)\Big).$$

Then z_c solves $z'' = (1 + \frac{c}{2}t^2)z$, $z_c = e^{-\xi}(1 + O(E^2))$ (so $\mathrm{Wr}(z_0, z_c) \to 0$), and (i) $\ell_\infty(c) = \int_{\mathbb{R}} z_c^{-2}\,d\xi = 1/(r-1)$; (ii) $\tilde{z} = z_c(1 + Cl_c) > 0$ on $\mathbb{R}$ ($C < 0$) iff $|C| \leq r - 1$; (iii) hence the smallest c with a positive shifted state is $\alpha_- = |C|(|C|+2)/2$, and $\alpha_-/C^2 \to \frac{1}{2} = \gamma_*(1)$ with correction $|C|^{-1}$.

PROOF. Let $D = d/d\xi = (1-x^2)\partial_x$. For $z_e = (1-x)^p(1+x)^q$: $Dz_e/z_e = (q-p) - (p+q)x$ and $D^2 z_e/z_e = -(p+q)(1-x^2) + ((q-p) - (p+q)x)^2$. With

$p=\frac{r}{2}$, $q=-\frac{1}{2}$, $s=p+q=\frac{r-1}{2}$, $d=q-p=-\frac{r+1}{2}$ this is $1+\frac{c}{2}(1+x)^2$ coefficientwise: $s^2+s=\frac{r^2-1}{4}=\frac{c}{2}$, $-2ds=\frac{r^2-1}{2}=c$, $-s+d^2=\frac{r^2+3}{4}=1+\frac{c}{2}$. A second solution is $z_e l$ with $l=\int z_e^{-2}d\xi=\int(1-x)^{-r-1}dx=\frac{1}{r}((1-x)^{-r}-2^{-r})$ (reduction of order, Corollary 3.6, $l(-1)=0$). So every solution is $\lambda z_e(1+C_0 l)$, and z_c is the case $C_0=2^r(r-1)/2$, $\lambda=2^{(1-r)/2}$. Near the pole, with $t=2E/(1+E)$: $t^{-1/2}=(2E)^{-1/2}(1+\frac{E}{2}+\dots)$, $z_e=2^{r/2}t^{-1/2}(1-\frac{rt}{4}+\dots)$, $l=2^{-r}\frac{t}{2}+O(t^2)$, hence $z_e(1+C_0 l)=2^{r/2}(2E)^{-1/2}\big[1+E(\frac{1}{2}+C_0 2^{-r}-\frac{r}{2})+O(E^2)\big]$. The bracket's E-coefficient vanishes for our C_0, and $\lambda 2^{r/2}2^{-1/2}=1$; as $z_c=E^{-1/2}\varphi(E)$ with φ analytic, $\varphi(0)=1$, $\varphi'(0)=0$, we get $z_c=e^{-\xi}(1+O(E^2))$ and $z_c-z_0=O(e^{3\xi})$, so $\mathrm{Wr}(z_0,z_c)\to 0$. (i) $z_c^{-2}d\xi=\lambda^{-2}dl/(1+C_0 l)^2$ and l runs over $[0,\infty)$, so $\ell_\infty=\lambda^{-2}/C_0=2^{r-1}\cdot 2/(2^r(r-1))=1/(r-1)$. (ii) As in Proposition 12.66(ii) $1+Cl_c$ decreases from 1 to $1+C\ell_\infty$. (iii) $|C|\le\sqrt{1+2c}-1$ iff $c\ge((1+|C|)^2-1)/2$. All identities were also checked numerically (`verify_ch12c.py`, Part 3). □

Remark 12.68 (the Fuchsian model)**.** If the whole polynomial potential $\mu^2+p_1t+2p_2t^2$ (no swirl) is kept instead of its top term, ℓ_∞ of the solution $z_F+\alpha u_P$ is a ratio of Gamma functions (Proposition B.18) whose leading term for $p_2\to\infty$ is exactly the Bessel value $\ell_* p_2^{-\mu/2}$. So for that family the Bessel model of Proposition 12.66 is the correct leading-order model, and Proposition B.18 shows the first corrections. At $\mu=1$ the closed form degenerates and Proposition 12.67 is the elementary substitute.

12.18. The pole zone and the Bessel window: the lower bound

In words. *Why is $\alpha_-(\mathcal{R}_C S)\gtrsim\gamma_*|C|^{2/\mu}$? By Proposition 6.49 and commutation, $|C|\le 1/\ell_\infty(T_c S)$ at $c=\alpha_-(\mathcal{R}_C S)$. So it suffices to show that $\ell_\infty(T_c S)$ is* not *smaller than the Bessel value $\ell_* c^{-\mu/2}$. Since $\ell_\infty\ge\int_{-\infty}^{\xi_M} z_B^{-2}$, we need an* upper *bound for $z_B=z\psi_B$ on the pole zone and the window. The equation for ψ_B is a Volterra equation with a* non-negative *coefficient, so a larger coefficient gives a larger solution (Lemma 12.64(a)): we compare with Bessel models with slightly larger constants. The swirl only raises the coefficient, by an amount we show is negligible.*

12.18.1. Parameters. Fix $S\in\mathcal{G}_{\frac{1}{2},2}$. By Lemma 6.11(a), after the change of representative (the data ρ_0,k_0 below are invariant under $(z,\kappa)\mapsto(sz,s^2\kappa)$) there are $\xi_{\mathrm{env}}\le 0$, $0<m_-\le m_+$ and $f_{\max}$ with

$$m_- e^{-\mu\xi}\le z\le m_+ e^{-\mu\xi},\quad 0\le f\le f_{\max}\quad\text{on }(-\infty,\xi_{\mathrm{env}}].$$

Put $\rho_0=(m_+/m_-)^2$, $k_0=\kappa m_-^{-2}$, $\bar f=2f_{\max}+2$, $g_0=k_0 f_{\max}/\mu$, $\delta_T=\min\{\frac{1}{4},\frac{2-\mu}{2}\}$, $a_0=\rho_0\mathcal{M}(4-2\mu)$, $a_1=\rho_0\mathcal{M}(4-2\delta_T-2\mu)$ ($\mathcal{M}$ of Lemma 6.5). Since $Z=e^{-\mu\xi}\ge 1$ is an envelope of rate μ in the sense of Definition 6.10, Propositions 6.18 and 6.21 apply to S with these data. For $c>0$ put $\xi_c=-\frac{1}{4}\ln c$, so $e^{2\xi_c}=c^{-1/2}$. For $\theta_1\le 0$ and $M\ge 0$ let

$$\xi_s=\xi_c+\theta_1,\quad \xi_M=\xi_c+M,\quad E_s=e^{2\theta_1}c^{-1/2},\quad E_M=e^{2M}c^{-1/2};$$

$[\xi_s,\xi_M]$ is the Bessel window and $(-\infty,\xi_s]$ the pole zone. Let $\theta_1^*\le 0$ be such that

$$a_0 e^{4\theta_1^*}\le\tfrac{1}{3},\qquad \tfrac{15}{4}a_0 k_0 e^{4\theta_1^*}\le 1,\qquad \frac{2a_1 e^{4\theta_1^*}}{4-2\delta_T}\le\tfrac{1}{4},\tag{12.6}$$

and fix $\theta_1 \le \theta_1^*$. Finally, with $B_c := T_cS$, $z_B = z\psi_B$,

$$\delta_1 = \delta_1(c,M) := \sup_{\xi\le\xi_M}\big(|ze^{\mu\xi}-1| + |z'/z+\mu|\big), \qquad \bar\kappa := \Big(\frac{1+\delta_1}{1-\delta_1}\Big)^2.$$

For fixed M, $\delta_1 \to 0$ as $c\to\infty$: the *only* input about S here is the pole behaviour $ze^{\mu\xi}\to 1$, $z'/z\to-\mu$ (type G; no rate is needed). We always take c so large that $\xi_M \le \min\{\xi_{\mathrm{env}},0\}$ and $\delta_1 \le \frac12$.

Lemma 12.69 (the pole zone)**.** *There is $c_1 = c_1(S,\theta_1)$ such that for $c\ge c_1$:*

(a) *B_c exists on $(-\infty,\xi_s]$, $\psi_B = 1+u_T$ with $|u_T| \le \hat B E^2 \le \frac12$, $\hat B = \frac32 a_0 c$; $|f_B - f| \le \frac52 k_0 \hat B E^{2+\mu} \le 1$; and B_c has the envelope $\frac12 m_- e^{-\mu\xi} \le z_B \le \frac32 m_+ e^{-\mu\xi}$, $0 \le f_B \le f_{\max}+1$, with data $(m_-^B, m_+^B, k_0^B, \rho_0^B, \bar f^B) = (\frac12 m_-, \frac32 m_+, 4k_0, 9\rho_0, 2f_{\max}+4)$.*

(b) *Let $\gamma_0 > 0$ with $\Gamma_0 := \dfrac{4\gamma_0 e^{2\mu\theta_1}}{\mu m_-^2} \le \min\{\frac12, \frac{\mu}{10k_0}\}$ and $|C| \le \gamma_0 c^{\mu/2}$. Then $\mathcal{R}_C B_c$ exists on $(-\infty,\xi_s]$, $\mathcal{R}_C B_c = \hat S$ has $\hat z = z_B(1+\hat u)$ with $|\hat u| \le \frac{4|C|}{\mu m_-^2}E^\mu \le \frac12$ and $|\hat f - f_B| \le \frac{10k_0}{\mu}\cdot\frac{4|C|}{\mu m_-^2}E^{2\mu} = \frac{40k_0|C|}{\mu^2 m_-^2}E^{2\mu}$.*

Proof. (a) We verify (6.5) at $\xi_2 = \xi_s$ with $\bar c = c$, $\bar\mu = \mu$, $\delta = \delta_T$. Here $\hat B E_s^2 = \frac32 a_0 e^{4\theta_1} \le \frac12$ and $\frac52 k_0 \hat B E_s^2 = \frac{15}{4} a_0 k_0 e^{4\theta_1} \le 1$ by (12.6); $E_s \le 1$; the condition $a_0 k_0 \bar f E_s \le \frac14$ holds for c large; and $\Theta = \frac{24k_0\bar f E_s}{(1-\delta_T)(4-2\delta_T)} + 2cE_s^2 + \hat\Phi E_s^3$ equals $2e^{4\theta_1} + O(c^{-1/2})$, so $a_1\Theta/(4-2\delta_T) \le \frac12$ by the last inequality of (12.6) for c large. Proposition 6.18 and Theorem 6.19(a),(b) give B_c on $(-\infty,\xi_s]$ with the stated bounds ($Z^{-2}E^2 = E^{2+\mu}$). The envelope follows from $|u_T| \le \frac12$ and $|f_B - f| \le 1$. (b) Apply Proposition 6.21 to the germ B_c with the data of (a) ($\nu = \mu$, $\varsigma = \mu - \frac12 > 0$, $\varpi_1 = 1$, $\bar C = |C|$), so $\hat B_A = 4|C|/(\mu m_-^2)$. The four conditions of (6.7) at $E_* = E_s$ read $\hat B_A E_s^\mu \le \frac12$, $\frac{5}{2\mu} 4k_0 \hat B_A E_s^\mu \le 1$ (both are $\Gamma_0 \le \min\{\frac12, \frac{\mu}{10k_0}\}$, since $\hat B_A E_s^\mu \le \Gamma_0$), and $\Lambda_0(E_s) \le \frac12$, $\Lambda_1 + \Lambda_2 \le \frac12$ with

$$\Lambda_0 = \frac{135\rho_0 k_0 \bar f^B E_s}{8\mu\varsigma(\mu+1)(2\mu+1)}, \quad \Lambda_1 = \frac{288\rho_0 k_0 \bar f^B E_s}{\varsigma(2\mu+1)(2\mu+3)},$$
$$\Lambda_2 = \frac{45\rho_0 k_0 \bar f^B \hat B_A E_s^{\mu+1}}{\mu\varsigma(2\mu+3)(4\mu+3)},$$

which tend to 0 as $c\to\infty$ because $E_s \to 0$ and $\hat B_A E_s^\mu \le \Gamma_0$. The fixed point $\hat u$ of $\mathcal{F}_{0,C}$ in the ball solves the system on $(-\infty,\xi_s]$ (Lemma 6.17); it lies in $C_A(C)(B_c)$, so by Theorem 6.22(b) the germ $\mathcal{R}_C B_c$ is the maximal continuation of this solution. The swirl bound is $|\tilde f - f| \le \frac{5}{2\nu} k_0^B \hat B_A E^{2\mu}$ (proof of Proposition 6.21). □

12.18.2. Comparison with Bessel models.

Lemma 12.70 (comparison with Bessel models)**.** *Let $\varepsilon_B = \frac12(ct^2 + R_B)$, $R_B = Q_B - Q - ct^2$. For c large (depending on S,θ_1,M):*

(a) *$\psi_B = 1 + \int_{-\infty}^{\xi} k(\xi,\zeta)\varepsilon_B(\zeta)\psi_B(\zeta)\,d\zeta$, $k(\xi,\zeta) = z(\zeta)^2\int_\zeta^\xi z^{-2}$, on $(-\infty,\xi_a]$; $\psi_B \ge 1$, $\psi_B' = z^{-2}\int_{-\infty}^{\xi} z^2\varepsilon_B\psi_B \ge 0$.*

(b) *$0 \le R_B \le C_* c^{(1-\mu)/2}E^2$ on $(-\infty,\xi_M]$, with $C_* = 4\big[\frac{15k_0 a_0 \bar f}{8(1+\mu)}e^{2(1+\mu)\theta_1} + g_0\int_{\theta_1}^M e^{2(\mu-1)\theta}d\theta\big]$ independent of c.*

(c) *With* $\rho_c = \frac14 C_* c^{-(1+\mu)/2}$, $\bar c = \bar\kappa(1+\rho_c)c$, $\underline{c} = \bar\kappa^{-1}(1-2E_M)c$: $\psi_{\underline{c}} \le \psi_B \le \psi_{\bar c}$, $\psi_B' \ge \psi_{\underline{c}}'$ *on* $(-\infty,\xi_M]$, *and* $(1-\delta_1)\zeta_{\underline{c}} \le z_B \le (1+\delta_1)\zeta_{\bar c}$.

Here $\xi_a < \min\{0, \operatorname{artanh} x_S^*\}$ is a fixed cut (see Section 12.19), and B_c exists on $(-\infty,\xi_a]$ by Proposition 6.23.

Proof. (a) By Lemma 6.17 and Theorem 6.19(b), $z^2\psi_B' = \mathrm{Wr}(z,z_B) = \int_{-\infty}^{\xi} z^2\varepsilon_B\psi_B$ (absolutely convergent since $\mu<2$); integrate once more and exchange the order. By Proposition 6.23, $\varepsilon_B \ge \frac12 ct^2 \ge 0$, so $\psi_B' \ge 0$, $\psi_B \ge 1$. (b) $R_B = \mathsf{G}_{\mathsf{L}}[F_B]$, $F_B = f^2 - f_B^2 \ge 0$ (Lemma 6.17, Proposition 6.23), so by Lemma 12.65(c) $R_B \ge 0$ and $R_B(\xi) \le 4E^2(I_1+I_2)$, $I_1 = \int_{-\infty}^{\xi_s} F_B E^{-1}$, $I_2 = \int_{\xi_s}^{\xi} F_B E^{-1}$. *Pole zone:* $F_B \le \bar f|f-f_B| \le \frac{15}{4}k_0a_0\bar f\, c\, E^{2+\mu}$ (Lemma 12.69(a)), so $I_1 \le \frac{15k_0a_0\bar f}{8(1+\mu)} cE_s^{1+\mu} = \frac{15k_0a_0\bar f}{8(1+\mu)} e^{2(1+\mu)\theta_1} c^{(1-\mu)/2}$. *Window:* since $f_B \ge f_- \ge 0$, $F_B \le f^2 - f_-^2 \le 2f_{\max}(f-f_-) \le g_0E^\mu$ (indeed $f-f_- = \kappa\int_{-\infty}^{\xi} z^{-2} \le k_0E^\mu/(2\mu)$), so $I_2 \le g_0\int_{\xi_s}^{\xi_M} E^{\mu-1}d\eta = g_0c^{(1-\mu)/2}\int_{\theta_1}^{M} e^{2(\mu-1)\theta}d\theta$. (For $\mu \le 1$ the integral $\int_{-\infty} E^{\mu-1}$ diverges: this is why the cut ξ_s and the improved estimate of Lemma 12.69 are needed.) (c) For $\zeta<\eta\le\xi_M$, $z(\zeta)/z(\eta) = e^{\mu(\eta-\zeta)}\omega(\zeta)/\omega(\eta)$ with $\omega = ze^{\mu\xi} \in [1-\delta_1, 1+\delta_1]$, so $\bar\kappa^{-1}e^{2\mu(\eta-\zeta)} \le z(\zeta)^2/z(\eta)^2 \le \bar\kappa e^{2\mu(\eta-\zeta)}$; integrating in $\eta\in(\zeta,\xi)$, $\bar\kappa^{-1}k_0(\xi-\zeta) \le k(\xi,\zeta) \le \bar\kappa k_0(\xi-\zeta)$ (k_0 of Proposition 12.66(iii)). By (b), $t^2 \ge 4E^2(1-2E)$ (since $(1+E)^{-2} \ge 1-2E$) and $t \le 2E$: $2c(1-2E_M)E^2 \le \varepsilon_B \le 2c(1+\rho_c)E^2$. So the kernel $k\varepsilon_B$ lies between $k_0\cdot 2\underline{c}E^2$ and $k_0\cdot 2\bar cE^2$ (after multiplying by $\bar\kappa^{\mp1}$), the kernels of the Bessel models $\psi_{\underline{c}}, \psi_{\bar c}$ of Proposition 12.66(iii). ψ_B is bounded and ≥ 1 on $(-\infty,\xi_M]$, so Lemma 12.64(a) gives $\psi_{\underline{c}} \le \psi_B \le \psi_{\bar c}$. For the derivative, $\psi_B'(\xi) = \int_{-\infty}^{\xi} \frac{z(\eta)^2}{z(\xi)^2}\varepsilon_B\psi_B\,d\eta \ge \bar\kappa^{-1}e^{2\mu\xi}\int e^{-2\mu\eta}\,2c(1-2E_M)E^2\psi_{\underline{c}}\,d\eta = \psi_{\underline{c}}'$. Finally $z_B = z\psi_B$ and $(1\mp\delta_1)e^{-\mu\xi} \lessgtr z$. □

Proposition 12.71 (the lower bound for ℓ_∞)**.** *For* $S\in\mathcal{G}_{\frac12,2}$ *and every* $M\ge0$: $\liminf_{c\to\infty} c^{\mu/2}\ell_\infty(T_cS) \ge \ell_*\big(1-\tau(e^{2M}/\sqrt2)\big)$. *Hence* $c^{\mu/2}\ell_\infty(T_cS) \ge (1-o(1))\ell_*$.

Proof. Since $z_B>0$,

$$\ell_\infty(B_c) \ge \int_{-\infty}^{\xi_M} z_B^{-2} \ge (1+\delta_1)^{-2}\int_{-\infty}^{\xi_M}\zeta_{\bar c}^{-2}$$

by Lemma 12.70(c). By Proposition 12.66(i) the last integral is $\ell_*\bar c^{-\mu/2}(1-\tau(\bar aE_M))$, $\bar a = \sqrt{\bar c/2}$, and $\bar aE_M = \sqrt{\bar\kappa(1+\rho_c)/2}\,e^{2M} \ge e^{2M}/\sqrt2$; τ decreases. So $c^{\mu/2}\ell_\infty \ge (1+\delta_1)^{-2}(\bar\kappa(1+\rho_c))^{-\mu/2}\ell_*(1-\tau(e^{2M}/\sqrt2))$. Let $c\to\infty$ ($\delta_1\to0$, $\bar\kappa\to1$, $\rho_c\to0$), then $M\to\infty$ ($\tau\to0$). □

Proof of Theorem 12.61(a). First part: Proposition 12.71.

Let $C<0$ and $c = \alpha_-(\mathcal{R}_CS)$, finite by Theorem 6.32 ($\mathcal{R}_CS\in\mathcal{G}_{\frac12,2}$). *Step 1 ($c\to\infty$ as $C\to-\infty$).* By Proposition 6.57(ii) the function $d\mapsto\alpha_-(\mathcal{R}_{-d}S)$ is increasing with limit $+\infty$. *Step 2 (reduction).* $\hat S := T_c\mathcal{R}_CS = \mathcal{R}_CT_cS$ is global (Theorem 6.54, Theorem 6.32(b)), so $B_c = \mathcal{R}_{|C|}\hat S$ is global and of type G at $x=1$ (Lemma 6.26, $\mu>\frac12$), whence $\ell_\infty(B_c)<\infty$; Proposition 6.49(i) applied to B_c and $\mathcal{R}_CB_c = \hat S\in\mathcal{S}$ gives $|C| \le 1/\ell_\infty(B_c)$. *Step 3.* Given $\delta>0$ let $c_0(\delta)$ with $c^{\mu/2}\ell_\infty(T_cS) \ge (1-\delta)\ell_*$ for $c\ge c_0(\delta)$. For $|C|$ so large that $c\ge c_0(\delta)$ (Step

1): $|C| \le c^{\mu/2}/((1-\delta)\ell_*)$, i.e. $c \ge ((1-\delta)\ell_*|C|)^{2/\mu}$. Hence $\liminf \alpha_-/|C|^{2/\mu} \ge ((1-\delta)\ell_*)^{2/\mu} \to \ell_*^{2/\mu} = \gamma_*$ as $\delta \to 0$. □

12.19. The upper bound: the forward barrier and Stage A

In words. *We must show that for c large the doubly shifted germ $\hat S = \mathcal{R}_C T_c S$ with $C = -(1-\beta)c^{\mu/2}/\ell_*$ is* global. *Write $\hat z = z_B\psi$. In the Bessel model $\psi = 1 + Cl_B$ decreases from 1 to β: positive (Figure 12.5). In truth $\psi = 1 + Cl_B - D$, where $D \ge 0$ is the effect of the swirl feedback (the shift lowers the swirl potential: $\hat R \le 0$). The trick is to work at the fraction $1-\beta$ of the critical shift: as long as $\psi \ge \beta/2$, the feedback is bounded* a priori *by $\hat f - f_B \le \frac{4\kappa}{\beta^2} l_B = O(c^{-\mu/2})$, with no knowledge of the non-perturbative profile at threshold. This bound gives $D = O(c^{-\mu/2})$ because z_B grows super-exponentially after the window (the integral $\int z_B^2 g$ is quasi-static, Lemma 12.64(d)), which closes a bootstrap up to a macroscopic point x_a. (One cannot have $\psi \ge \beta/2$ for all ξ, since $\psi \to 0$ at $+\infty$: $\hat R(1) < 0$.) Beyond x_a positivity of $\hat z$ itself follows from a forward Riccati barrier, Lemma 12.72.*

12.19.1. A forward barrier.

Lemma 12.72 (forward barrier)**.** *Let $\hat S$ be a solution of the Q-system with $\kappa > 0$ on $(-1, x_a]$, $x_a \in (-1,1)$; $t_a = 1 + x_a$, $c_a = \frac{1-x_a}{2(1+x_a)}$; $\hat z_a = \hat z(\xi_a) > 0$, $\hat r_a = \hat z'/\hat z(\xi_a) \ge 0$, $\hat f_a = \hat f(\xi_a) \ge 0$, $F = \hat f_a + 1$. Assume $\hat Q(x_a) \ge q_a \ge 0$, $\hat Q_x(x_a) \ge 0$,*

$$s^2 := 1 + \tfrac12\big(q_a - (|\lambda| + F^2)c_a\big) > 0, \qquad 4\kappa \le s\hat z_a^2.$$

Then $\hat S$ is global, $\hat z(\xi) \ge \frac12 \hat z_a e^{s(\xi-\xi_a)}$ and $\hat f \le \hat f_a + \frac12$ on $[\xi_a, \infty)$; in particular $\int_{\xi_a}^\infty \hat z^{-2} \le 2/(s\hat z_a^2)$.

Proof. Let $\mathcal{N} = \{\xi \ge \xi_a :\ \hat S$ exists on $[\xi_a, \xi]$ and $\hat f \le F$ there$\} \ni \xi_a$. On $\mathcal{N}$ the Cauchy formula (2.34) for $\mathsf{L}[\hat Q] = \lambda - \hat f^2$ gives $\hat Q(x) \ge \hat Q(x_a)A_1 + \hat Q_x(x_a)A_2 - (|\lambda| + F^2)\int_{x_a}^x K_{\mathsf{L}} \ge q_a - (|\lambda| + F^2)c_a$ ($A_1 \ge 1$, $A_2 \ge 0$, $0 \le \int K_{\mathsf{L}} \le c_a$, Lemma 2.18(d)); so $\hat W = 1 + \frac12 \hat Q \ge s^2$. Let $\zeta = \hat z_a \cosh s\tau + (\hat z_a'/s)\sinh s\tau$, $\tau = \xi - \xi_a$, and $y = \hat z - \zeta$: $y'' - s^2 y = \hat z'' - s^2 \hat z \ge 0$, $y(\xi_a) = y'(\xi_a) = 0$, so $y(\tau) = \frac1s \int_0^\tau \sinh s(\tau - \sigma)\,(y'' - s^2 y)(\sigma)\,d\sigma \ge 0$. As $\hat z_a' \ge 0$, $\hat z \ge \hat z_a \cosh s\tau \ge \frac12 \hat z_a e^{s\tau}$. Then $\hat f(\xi) - \hat f_a = \kappa \int_{\xi_a}^\xi \hat z^{-2} \le 4\kappa \hat z_a^{-2} \int_0^\infty e^{-2s\tau} d\tau = \frac{2\kappa}{s\hat z_a^2} \le \frac12$. So $\hat f < F$ on $\mathcal{N}$: $\mathcal{N}$ is closed (continuity) and open (strict inequality), hence equals the maximal interval $[\xi_a, \xi^*)$. There $\hat z$ is bounded below by $\frac12 \hat z_a$ and the right end is ∞ by Lemma 2.24. □

This is the mechanism of Steps 2–4 in the proof of Theorem 6.32(a), with the data at ξ_a as hypotheses instead of a comparison with a fixed germ.

12.19.2. The germ $B_c = T_c S$ beyond the pole zone. *Fixed data.* Fix a cut $x_a \in (-1, \min\{0, x_S^*\})$, $\xi_a = \operatorname{artanh} x_a \le 0$, $t_a = 1 + x_a$, $E_a = e^{2\xi_a}$, $f_a = f(\xi_a)$. Q is continuous on $(-1, x_a]$ with finite limit q_0 at -1, so $q_{\min} := \max\{0, -\inf_{(-1,x_a]} Q\} < \infty$. By Proposition 6.23, for $c > 0$ the germ B_c exists on $(-\infty, \xi_a]$, $z_B \ge z$, $f_B \le f \le f_a$, $Q_B \ge Q + ct^2 \ge -q_{\min} + ct^2$ and $\partial_x(Q_B - Q) \ge 2ct$. Put $W_*(\xi) = 1 + \frac12(ct^2 - q_{\min})$ (non-decreasing) and $W_B = 1 + \frac12 Q_B \ge W_*$. Let M_0 be such that $e^{4M_0} \ge 2q_{\min} + 4$ and $g'(M_0) > 0$ (Proposition 12.66(iv)), and $M \ge M_0$. For $\xi \ge \xi_M$: $ct^2 \ge cE_M^2 = e^{4M}$, hence $W_* \ge 3 + \frac12 q_{\min} \ge 3$ and $W_* \ge \frac14 ct^2$ (because

$1+\frac14 ct^2 \ge \frac12 q_{\min}$). Write $r_B = z_B'/z_B$, $l_B = \int_{-\infty}^{\xi} z_B^{-2}$, $\Phi(\eta) = \int_{-\infty}^{\eta} z_B^2 E^2$ and $b_1 = e^{-2}/4$, $b_2 = e^{-4}/4$ (numerically 0.0338, 0.00458).

Proposition 12.73 (B_c beyond the pole zone)**.** *Let $M \ge M_0$. For $c \ge c_0(S,\theta_1,M)$:*

(i) $W_B \ge W_* \ge \max\{3, \frac14 ct^2\}$ *on* $[\xi_M,\xi_a]$ *and* z_B *is convex there.*

(ii) $r_B(\xi_M) \ge g'(M)/g(M) - \eta_c$ *with* $\eta_c \to 0$ *as* $c\to\infty$*; in particular* $r_B(\xi_M) > 0$.

(iii) $r_B \ge r_1 := \min\{r_B(\xi_M), \sqrt3\} > 0$ *on* $[\xi_M,\xi_a]$*;* $r_B(\xi) \ge b_1\sqrt{c}E_\xi$ *for* $\xi \in [\xi_M+1,\xi_a]$*; if* $\eta \ge \xi_M+2$ *then* $r_B \ge b_2\sqrt{c}E_\eta$ *on* $[\eta-1,\eta]$*; and* $z_B(\xi_a) \ge \exp(\frac14 b_1\sqrt{c}E_a)$.

(iv) $\limsup_{c\to\infty} c^{\mu/2} l_B(\xi_a) \le \ell_* + \dfrac{1}{g(M)g'(M)}$.

(v) $z_B(\eta)^{-2}\Phi(\eta) \le C_\Phi c^{-1/2}$ *for* $\eta \le \xi_a$*, and* $J(\xi_a) := \int_{-\infty}^{\xi_a} z_B^{-2}\Phi\,d\eta \le C_J c^{-1/2}$*, with* $C_\Phi = \frac{\bar\kappa E_M^2 c^{1/2}}{4-2\mu} + \frac{e^{4M+8}}{2r_1c^{1/2}} + \frac{1}{2b_2} + \frac{e^{-2}}{4b_2^2c^{1/2}}$ *and* $C_J = \frac{\bar\kappa e^{4M}}{4(4-2\mu)c^{1/2}} + \frac{\bar\kappa e^{4M}}{2r_1(4-2\mu)c^{1/2}} + \frac{e^{4M+8}}{r_1c^{1/2}} + \frac{1}{4b_2} + \frac{(\xi_a-\xi_M)e^{-2}}{4b_2^2c^{1/2}}$ (*bounded in c for $c\ge c_0$*).

Proof. (i) $W_B = 1+\frac12 Q_B \ge W_*$; use $t \ge E \ge E_M$ and the choice of M_0 above. $z_B'' = W_B z_B > 0$. (ii) $r_B = z'/z + \psi_B'/\psi_B \ge -\mu - \delta_1 + \psi_{\underline c}'(\xi_M)/\psi_{\bar c}(\xi_M)$ by Lemma 12.70(c). By Proposition 12.66(i), $\zeta_{c'}(\xi_c+\theta) = c'^{\mu/4} g(\theta + \frac14 \ln\frac{c'}{c})$ and $\psi_{c'}' = e^{\mu\xi}(\zeta_{c'}' + \mu\zeta_{c'})$, so the quotient is $(\underline c/\bar c)^{\mu/4}\big[g'(M+\underline\alpha) + \mu g(M+\underline\alpha)\big]/g(M+\bar\alpha)$ with $\underline\alpha = \frac14\ln\frac{\underline c}{c}$, $\bar\alpha = \frac14\ln\frac{\bar c}{c}$; it tends to $g'(M)/g(M)+\mu$ as $c\to\infty$ (then $\bar\kappa,\delta_1,\rho_c,E_M$ tend to $1,0,0,0$). Positivity: $M \ge M_0$. (iii) Lemma 12.64(c)(i) with $w_0 = W_*(\xi_M) \ge 3$ gives $r_B \ge r_1$ on $[\xi_M,\xi_a]$. For $\xi \in [\xi_M+1,\xi_a]$ put $\xi_0 = \xi - 1$: $r_B(\xi_0) \ge r_1 > 0$ and $W_B \ge W_*(\xi_0) =: k^2 \ge 3$ on $[\xi_0,\xi]$, so part (c)(ii) gives $r_B(\xi) \ge k/2 \ge \frac14\sqrt{c}\,t_{\xi_0} \ge \frac14\sqrt{c}E_{\xi_0} = b_1\sqrt{c}E_\xi$ ($E_{\xi-1} = e^{-2}E_\xi$). If $\eta \ge \xi_M+2$ and $s \in [\eta-1,\eta]$, then $s \ge \xi_M+1$ and $r_B(s) \ge \frac12\sqrt{W_*(s-1)} \ge \frac12\sqrt{W_*(\eta-2)} \ge \frac14\sqrt{c}\,E_{\eta-2} = b_2\sqrt{c}E_\eta$. Finally $z_B(\xi_a) \ge z_B(\xi_M)\exp\int_{\xi_M+1}^{\xi_a} r_B \ge z_B(\xi_M)\exp\big(\frac12 b_1\sqrt{c}(E_a - E_{M+1})\big)$, and $z_B(\xi_M) \ge z(\xi_M) \ge (1-\delta_1)e^{\mu M}c^{\mu/4} \ge 1$, $E_{M+1} \le \frac12 E_a$ for c large. (iv) $l_B(\xi_a) = \int_{-\infty}^{\xi_M} + \int_{\xi_M}^{\xi_a}$. By Lemma 12.70(c) and Proposition 12.66(i), the first is $\le (1-\delta_1)^{-2}\ell_*\underline c^{-\mu/2}$. By (i), (ii) and Lemma 12.64(b) the second is $\le 1/(z_B(\xi_M)^2 r_B(\xi_M)) \le \big[(1-\delta_1)^2\underline c^{\mu/2} g(M+\underline\alpha)^2(g'(M)/g(M) - \eta_c)\big]^{-1}$, since $z_B(\xi_M) \ge (1-\delta_1)\zeta_{\underline c}(\xi_M) = (1-\delta_1)\underline c^{\mu/4} g(M+\underline\alpha)$. Multiply by $c^{\mu/2}$ and let $c\to\infty$. (v) *Pole zone, $\eta \le \xi_M$.* ψ_B is non-decreasing, so for $\zeta \le \eta$, $z_B(\zeta)^2/z_B(\eta)^2 \le z(\zeta)^2/z(\eta)^2 \le \bar\kappa e^{2\mu(\eta-\zeta)}$, and $z_B(\eta)^{-2}\Phi(\eta) \le \bar\kappa\int_{-\infty}^{\eta} e^{2\mu(\eta-\zeta)}e^{4\zeta}d\zeta = \bar\kappa E_\eta^2/(4-2\mu)$ (here $\mu<2$ is used). *Beyond,* $\eta \in [\xi_M,\xi_a]$. z_B is increasing; split

$$\Phi(\eta) = \Phi(\xi_M) + \int_{\xi_M}^{\eta} z_B^2 E^2.$$

By (iii), $z_B(\xi_M)^2/z_B(\eta)^2 \le e^{-2r_1(\eta-\xi_M)}$, so the first part contributes at most $e^{-2r_1(\eta-\xi_M)}\bar\kappa E_M^2/(4-2\mu)$. For the second: if $\eta \le \xi_M+2$, Lemma 12.64(d) with $m_0 = r_1$, $g = E^2$ gives $\le E_\eta^2/(2r_1) \le e^{4M+8}/(2r_1c)$. If $\eta \ge \xi_M+2$, split at $\eta-1$: on $[\eta-1,\eta]$ Lemma 12.64(d) with $m_0 = b_2\sqrt{c}E_\eta$ gives $\le E_\eta^2/(2m_0) = E_\eta/(2b_2\sqrt{c})$; on $[\xi_M,\eta-1]$, $z_B(\zeta)^2/z_B(\eta)^2 \le e^{-2m_0}$, so the contribution is $\le e^{-2m_0}\int^{\eta-1} E^2 \le \frac14 E_\eta^2 e^{-2b_2\sqrt{c}E_\eta} \le e^{-2}/(4b_2^2c)$ (maximise x^2e^{-ax}: the maximum is $4e^{-2}/a^2$). With $E_\eta \le 1$ this gives $z_B^{-2}\Phi \le C_\Phi c^{-1/2}$. Integrating over η: the pole zone gives

$\bar\kappa E_M^2/(4(4-2\mu))$; the first parts beyond give $\bar\kappa E_M^2/(2r_1(4-2\mu))$; the second parts give $\le e^{4M+8}/(r_1c)$ on $[\xi_M,\xi_M+2]$ and $\le \frac{1}{2b_2\sqrt{c}}\int E_\eta d\eta + (\xi_a-\xi_M)\frac{e^{-2}}{4b_2^2c} \le \frac{1}{4b_2\sqrt{c}}+\dots$ on $[\xi_M+2,\xi_a]$ ($\int E \le \frac12$); note $\xi_a-\xi_M = O(\ln c)$. Since $E_M^2 = e^{4M}/c$, this is $J \le C_J c^{-1/2}$. □

Remark 12.74. (iv)–(v) are the quantitative form of "the outer zone contributes nothing to $\int z_B^{-2}$ and $\int^\eta z_B^2E^2$ is quasi-static". The decisive point in (v) is that the Riccati bound $r_B \gtrsim \sqrt{c}\,E$ holds *locally* at every η: a uniform bound $r_B \ge r_1 = O(1)$ would only give $J = O(1)$, which is not enough for $\mu \le 1$.

12.19.3. Stage A: the bootstrap up to x_a. Fix $\beta \in (0,1)$ and $c>0$, put $C = -(1-\beta)c^{\mu/2}/\ell_*$ and $\hat S = \mathcal{R}_C B_c$ (defined for every C, Theorem 6.22, since $B_c \in \mathcal{G}_{\frac12,2}$). Write $\hat z = z_B(1+\hat u) = z_B\psi$, $\hat f$, $\hat Q = Q_B + \hat R$, $\hat\varepsilon = \frac12\hat R$. By Corollary 6.24 (applied to $\hat S$ and the shift $-C>0$, which gives B_c), on the interval of existence of $\hat S$:

$$\hat u \le 0, \quad \hat f \ge f_B, \quad \hat R = \mathsf{G}_{\mathsf{L}}[f_B^2 - \hat f^2] \le 0, \quad z_B^2\hat u' \le C;$$

(the last because $\mathrm{Wr}(\hat z, z_B) = -z_B^2\hat u'$ is positive, increasing, with limit $-C$). By Lemma 6.53(a) and $\mathrm{Wr}(z_B,\hat z)\to C$,

$$\psi = 1 + Cl_B - D, \qquad D(\xi) = \int_{-\infty}^{\xi} z_B^{-2}(\eta)\int_{-\infty}^{\eta} z_B^2\,|\hat\varepsilon|\,\psi\,d\zeta\,d\eta \;\ge 0, \tag{12.7}$$

and $\hat f - f_B = \kappa\int z_B^{-2}(\psi^{-2}-1)$.

Proposition 12.75 (Stage A). *Let $\beta\in(0,1)$. There are $\theta_1 \le \theta_1^*$, $M \ge M_0$ and c_0 such that for every $c \ge c_0$ ($C = -(1-\beta)c^{\mu/2}/\ell_*$): $\hat S$ exists on $(-\infty,\xi_a]$ and*

(A1) *$\psi \ge \frac34\beta$ on $(-\infty,\xi_a]$; in particular $\hat z(\xi_a) \ge \frac34\beta z(\xi_a)$;*

(A2) *$0 \le \hat f - f_B \le 8\kappa\ell_*\beta^{-2}c^{-\mu/2}$ on $(-\infty,\xi_a]$, hence $\hat f(\xi_a) \le f_a + 1$;*

(A3) *$|\hat R| \le 4K_1c^{(1-\mu)/2}E^2$ on $(-\infty,\xi_a]$ and $|\hat R_x(x_a)| \le K_2c^{(1-\mu)/2}$, with K_1,K_2 independent of c (given in the proof);*

(A4) *$\hat r(\xi_a) = \hat z'/\hat z(\xi_a) \ge \frac12 b_1\sqrt{c}E_a > 0$.*

Proof. *Choice of parameters, in this order.* (1) $\theta_1 \le \theta_1^*$ such that $\Gamma_0 := \dfrac{4(1-\beta)e^{2\mu\theta_1}}{\ell_*\mu m_-^2} \le \min\{\frac12, \frac{\mu}{10k_0}\}$ (this is Lemma 12.69(b) with $\gamma_0 = (1-\beta)/\ell_*$; it does not involve c). (2) $M \ge M_0$ with $1/(g(M)g'(M)) \le \beta\ell_*/16$ (Proposition 12.66(iv); a numerical example is in Numerical observation 12.78). (3) c_0 so large that: $c_0 \ge c_1(S,\theta_1)$ of Lemma 12.69; Proposition 12.73 and the standing smallness ($\delta_1 \le \frac12$, $\xi_M \le \min\{\xi_{\mathrm{env}},0\}$) hold; $c^{\mu/2}l_B(\xi_a) \le (1+\frac{\beta}{8})\ell_*$ (by Proposition 12.73(iv) and (2): $\limsup \le \ell_*(1+\frac{\beta}{16})$); the numbers $\frac{8\kappa\ell_*}{\beta^2}c^{-\mu/2}$ and $C_2e^{4\mu\theta_1}c^{-\mu/2}$ (C_2 below) are ≤ 1; $2K_1C_Jc^{-\mu/2} \le \frac{\beta}{8}$ and $2K_1C_\Phi c^{-\mu/2} \le o(1)$ as needed in Step 3; and $c^{-\mu/2} \le K_1$-dependent bound in the definition of K_2. No constant depends on a later parameter.

Step 0: the pole zone. By Lemma 12.69(a),(b) with $\gamma_0 = (1-\beta)/\ell_*$: $\hat S$ exists on $(-\infty,\xi_s]$, $|\hat u| \le \Gamma_0 \le \frac12$, so $\psi \ge \frac12 > \frac{\beta}{2}$, and

$$0 \le \hat f - f_B \le C_2c^{\mu/2}E^{2\mu}, \qquad C_2 = \frac{40(1-\beta)k_0}{\ell_*\mu^2m_-^2}. \tag{12.8}$$

Step 1: the bootstrap hypothesis and its consequences. Let $\bar f_a = 2f_a + 1$ and, for $\xi \in [\xi_s, \xi_a]$,

$$(\mathrm{H})_\xi: \quad \hat S \text{ exists on } (-\infty,\xi] \text{ and } \psi > \beta/2 \text{ on } (-\infty,\xi].$$

Assume $(\mathrm{H})_\xi$. *(a) A priori swirl bound.* Since $\frac{\beta}{2} < \psi \le 1$, $0 \le \hat f - f_B = \kappa\int_{-\infty}^{\eta} z_B^{-2}(\psi^{-2}-1) \le \frac{4\kappa}{\beta^2} l_B(\eta) \le \frac{4\kappa}{\beta^2} l_B(\xi_a) \le \frac{8\kappa\ell_*}{\beta^2} c^{-\mu/2}$ for all $\eta \le \xi$ (using $l_B(\xi_a) \le (1+\frac{\beta}{8})\ell_* c^{-\mu/2} \le 2\ell_* c^{-\mu/2}$). This is (A2) on $(-\infty,\xi]$ and gives $\hat f - f_B \le 1$ and, as $f_B \le f \le f_a$, $0 \le \hat f^2 - f_B^2 \le \bar f_a(\hat f - f_B)$. *(b) The Green correction.* Put $\mathcal I(\zeta) = \int_{-\infty}^{\zeta}(\hat f^2 - f_B^2)E^{-1}d\eta$. By (12.8) on $\eta \le \xi_s$ and (a) on $[\xi_s,\zeta]$ ($\int_{-\infty}^{\xi_s} E^{2\mu-1} = E_s^{2\mu-1}/(2(2\mu-1))$, where $2\mu>1$ is used, and $\int_{\xi_s}^{\zeta} E^{-1} \le E_s^{-1}/2$),

$$\mathcal I(\zeta) \le \bar f_a\Big[C_2 c^{\mu/2}\frac{E_s^{2\mu-1}}{2(2\mu-1)} + \frac{8\kappa\ell_*}{\beta^2}c^{-\mu/2}\frac{E_s^{-1}}{2}\Big] = K_1 c^{(1-\mu)/2},$$

$$K_1 = \bar f_a\Big[\frac{C_2 e^{2(2\mu-1)\theta_1}}{2(2\mu-1)} + \frac{4\kappa\ell_*}{\beta^2}e^{-2\theta_1}\Big],$$

because $c^{\mu/2}E_s^{2\mu-1} = e^{2(2\mu-1)\theta_1}c^{(1-\mu)/2}$ and $c^{-\mu/2}E_s^{-1} = e^{-2\theta_1}c^{(1-\mu)/2}$. By Lemma 12.65(c), $|\hat R| = \mathsf{G}_{\mathsf L}[\hat f^2 - f_B^2] \le 4E^2\mathcal I \le 4K_1 c^{(1-\mu)/2}E^2$ and $|\hat R_x(x_a)| \le 4t_a\mathcal I + 4\int(\hat f^2 - f_B^2)E\,d\eta \le 4t_aK_1c^{(1-\mu)/2} + O(c^{-\mu/2}) \le K_2 c^{(1-\mu)/2}$, $K_2 := 4t_aK_1+1$ for $c \ge c_0$ (the E-weighted integral is $\le \bar f_a[C_2c^{\mu/2}E_s^{2\mu+1}/(2(2\mu+1)) + \frac{8\kappa\ell_*}{\beta^2}c^{-\mu/2}\frac12]$). *(c) Bound on D.* Since $\psi \le 1$ and $|\hat\varepsilon| = \frac12|\hat R| \le 2K_1c^{(1-\mu)/2}E^2$: $D(\xi') \le 2K_1c^{(1-\mu)/2}J(\xi') \le 2K_1C_Jc^{-\mu/2}$ for $\xi' \le \xi$ (Proposition 12.73(v)). *(d) Closing.* By (12.7), $l_B \le (1+\frac{\beta}{8})\ell_* c^{-\mu/2}$ and $(1-\beta)(1+\frac{\beta}{8}) \le 1 - \frac78\beta$ (checked numerically in `verify_ch12c.py`, Part 4): $\psi \ge 1 - (1-\beta)(1+\frac{\beta}{8}) - D \ge \frac78\beta - 2K_1C_Jc^{-\mu/2} \ge \frac34\beta$ on $(-\infty,\xi]$.

Step 2: continuity. Let $J_H = \{\xi \in [\xi_s,\xi_a] : (\mathrm{H})_\xi\}$, an interval containing ξ_s (Step 0). If $\xi_m = \sup J_H$, then by Step 1 $\psi \ge \frac34\beta$ on $(-\infty,\xi_m)$, so $\hat z \ge \frac34\beta z_B \ge \frac34\beta\min_{[\xi_s,\xi_a]} z_B > 0$ on $[\xi_s,\xi_m)$; by Lemma 2.24 $\hat S$ exists beyond ξ_m with $\psi(\xi_m) \ge \frac34\beta > \frac{\beta}{2}$, and by continuity $\psi > \frac{\beta}{2}$ slightly beyond ξ_m. So $\xi_m \in J_H$ and $\xi_m = \xi_a$. (A1)–(A3) follow from Steps 0, 1 ($\hat z(\xi_a) = \psi z_B \ge \frac34\beta z$ since $z_B \ge z$).

Step 3: (A4). $\hat r = r_B + \hat u'/\psi$, and $z_B^2\hat u' = C + \int_{-\infty}^{\xi} z_B^2\hat\varepsilon\psi \ge C - 2K_1c^{(1-\mu)/2}\Phi(\xi)$, so $|\hat u'| \le z_B^{-2}(|C| + 2K_1c^{(1-\mu)/2}\Phi)$. At ξ_a: $z_B^{-2}\Phi \le C_\Phi c^{-1/2}$ and $z_B(\xi_a)^{-2}|C| \le \ell_*^{-1}c^{\mu/2}\exp(-\frac12 b_1\sqrt{c}E_a)$ (Proposition 12.73(iii),(v)); the first gives $2K_1C_\Phi c^{-\mu/2}$, the second tends to 0. With $\psi \ge \frac34\beta$: $\hat r(\xi_a) \ge r_B(\xi_a) - o(1) \ge b_1\sqrt{c}E_a - o(1) \ge \frac12 b_1\sqrt{c}E_a$ for $c \ge c_0$. □

Proof of Theorem 12.61(b),(c). Let $\beta \in (0,1)$ and θ_1, M, c_0 as in Proposition 12.75; let $c \ge c_0$, $C = -(1-\beta)c^{\mu/2}/\ell_*$, $\hat S = \mathcal R_C T_c S$. We verify the hypotheses of Lemma 12.72 at x_a. By (A1), (A4): $\hat z_a \ge \frac34\beta z(\xi_a) =: \hat z_{\min} > 0$ and $\hat r_a > 0$; by (A2) $\hat f_a \le f_a + 1$. By Proposition 6.23, $Q_B(x_a) \ge Q(x_a) + ct_a^2$ and $\partial_x Q_B(x_a) \ge Q_x(x_a) + 2ct_a$; with (A3),

$$\hat Q(x_a) \ge ct_a^2 - q_{\min} - 4K_1c^{(1-\mu)/2} =: q_a,$$
$$\hat Q_x(x_a) \ge Q_x(x_a) + 2ct_a - K_2c^{(1-\mu)/2}.$$

Since $\mu > -1$, $c^{(1-\mu)/2} = o(c)$, so for c large $q_a \ge \frac12 ct_a^2 > 0$, $\hat Q_x(x_a) \ge ct_a > 0$, $s^2 \ge \frac18 ct_a^2$, and $s\hat z_a^2 \ge \frac13\sqrt{c}\,t_a\hat z_{\min}^2 \ge 4\kappa$. Lemma 12.72 applies: $\hat S$ is global. By

Theorem 6.54, $T_c\mathcal{R}_C S = \hat{S}$ is global, i.e. $c \geq \alpha_-(\mathcal{R}_C S)$ (Theorem 6.32(b)), with $|C| = (1-\beta)c^{\mu/2}/\ell_*$, i.e. $\alpha_-(\mathcal{R}_C S) \leq (\ell_*|C|/(1-\beta))^{2/\mu}$. Every $C \leq C_0 := -(1-\beta)c_0^{\mu/2}/\ell_*$ arises from some $c \geq c_0$, so $\limsup_{C\to-\infty} \alpha_-/|C|^{2/\mu} \leq \gamma_*(1-\beta)^{-2/\mu}$; let $\beta \to 0$. This is the first part of (b).

The statement on ℓ_∞. Apply Lemma 12.72 to B_c itself ($C = 0$): $\hat{z}_a = z_B(\xi_a) \geq z(\xi_a)$, $r_B(\xi_a) > 0$, $Q_B(x_a) \geq ct_a^2 - q_{\min}$, $\partial_x Q_B(x_a) \geq Q_x(x_a) + 2ct_a > 0$ for c large. So $\int_{\xi_a}^{\infty} z_B^{-2} \leq 2/(sz_B(\xi_a)^2) \leq \exp(-\frac{1}{2}b_1\sqrt{c}E_a)$ (Proposition 12.73(iii); $s \geq 2$), and $\ell_\infty(B_c) = l_B(\xi_a) + O(e^{-\frac{1}{2}b_1\sqrt{c}E_a})$. By Proposition 12.73(iv), $\limsup c^{\mu/2}\ell_\infty \leq \ell_* + 1/(gg')(M)$ for every $M \geq M_0$, i.e. $\leq \ell_*$ (Proposition 12.66(iv)). With Proposition 12.71, $c^{\mu/2}\ell_\infty(T_c S) \to \ell_*$. This is (b).

(c) By (a) and (b) the limit law holds. By Proposition 6.57(iii), $c = \alpha_-(\mathcal{R}_{-d}S)$ for $d = \Lambda(S,c)$, and $\Lambda(S,c) \to \infty$ as $c \to \infty$ (it is the inverse of an increasing unbounded function). So $c \sim \gamma_*\Lambda^{2/\mu}$, i.e. $\Lambda \sim (c/\gamma_*)^{\mu/2} = c^{\mu/2}/\ell_*$; then $\Lambda\,\ell_\infty(T_cS) \sim \ell_*^{-1}c^{\mu/2}\cdot \ell_* c^{-\mu/2} = 1$. The converse implications are the same computation read backwards. □

Remark 12.76 (where each hypothesis enters)**.** $\mu < 2$: the integrability of $e^{(4-2\mu)\eta}$ in Proposition 12.73(v), in Lemma 12.70(a) and in T_c itself. $\mu > \frac{1}{2}$: the existence of $\mathcal{R}_C$, Proposition 6.49, and $2\mu > 1$ in Step 1(b) of Proposition 12.75. Fixed S: the one non-quantified input is the modulus $\delta_1 \to 0$, from $ze^{\mu\xi} \to 1$; all other constants are envelope data. The exact value of ℓ_* enters only through $\int g^{-2} = \ell_*$ (Proposition 12.66); the proof would identify any other limit constant in the same way. The bound $D = O(c^{-\mu/2})$ is crude: the window contribution to D is $O(c^{-(1+\mu)/2})$.

12.20. Atlas of the chapter

In words. *This atlas lists, in one place, every numerical observation of the chapter and the code that produced it, and states how accurate the class-*N *computations are. The numbers support the theorems of the chapter but prove nothing; the tolerances tell the reader which digits to trust.*

Atlas 12.77 (numerical atlas of Chapter 12)**.** Contents. *The numerical observations of the chapter are: Part A (swirl threshold $\kappa_{\max}$ near the critical surface): Numerical observation 12.19 (the constant A for $k < 1$), Numerical observation 12.20 (the logarithmic law at $k = 1$), Numerical observation 12.21 (exponents and ratios for $k > 1$), Numerical observation 12.22 (the base point $x_0 \to \pm 1$), with Figures 12.1 and 12.2; Part B (constants, cone limits and the shape of the lens): Numerical observations 12.28, 12.34, 12.38, 12.51, 12.54, 12.56 and 12.58; Part C (this part): Numerical observations 12.78 to 12.82, with Figures 12.7 and 12.8. The numbers of Parts A and B are not repeated here.*

Method and tolerance for Part C. *The base germs have $\kappa = 1$, $\lambda = 0$, $f = 2$, $z'/z = -0.4$, Q_x-data fixed at $\xi_a = -1$, and the exponent μ prescribed by shooting on $Q(\xi_a)$. The shifted germ is computed from a deep cut $-X$ by integrating the exact difference system between $\mathcal{R}_C T_c S$ and S, with initial data from the leading terms of Theorems 6.19 and 6.22; the depth rule is tolerance 10^{-6}; a pole is declared when $\hat{z}/z < 10^{-6}$; α_- is found by geometric bisection (relative tolerance 10^{-9}), the classification (pole versus global) by forward integration; solver tolerances 10^{-13} (base) and 10^{-12} (difference system). This is one base family: nothing*

here tests uniformity in the germ. Limits of the method: floating-point cancellation in $1 + Cl$ for $c \gtrsim 10^{11}$, and slow convergence for $\mu \lesssim 0.7$. Code and data: `code/ch12_thresh/`, `verify_ch12c.py` *(all numbers below),* `make_c_atlas.py` *(figures),* `make_c_globality.py` *(slow run).*

Numerical observation 12.78 (accuracy of the limit law)**.** *Table 12.5 gives $\alpha_-(\mathcal{R}_C S)/|C|^{2/\mu}$ for $|C| = 10^2, \dots, 10^7$. At the best point per exponent ($\mu = 0.7, 0.8, 1, 1.3, 1.6, 1.9$) the relative error against $\gamma_*(\mu)$ is $3.7 \cdot 10^{-4}$ ($|C| = 10^3$), $4.0 \cdot 10^{-5}$ ($3.2 \cdot 10^3$), $8.4 \cdot 10^{-6}$ (10^5), $4.2 \cdot 10^{-7}$ (10^7), $4.8 \cdot 10^{-6}$ (10^8), $2.8 \cdot 10^{-5}$ (10^8). At $|C| = 10^7$ it is $8.3 \cdot 10^{-4}$ at $\mu = 1$ (already affected by cancellation) and $9.9 \cdot 10^{-5}$ at $\mu = 1.9$; the $\mu = 0.6$ column is not a reliable approximation (the search did not converge for $|C| \geq 10^4$) and no accuracy is claimed for $\mu < 0.7$. (The theorem is proved for every $\mu \in (\frac{1}{2}, 2)$; the numerics do not test $\mu < 0.8$.) The constant ℓ_* itself is checked to 10^{-30} (quadrature of $\int g^{-2}$) and to 10^{-9} against a bisection on the limit ODE ($\mu = 0.55, \dots, 1.9$). As an illustration of the choice of M in Stage A: for $\mu = 1.3$, $\ell_* = 0.5254$, the condition $1/(g(M)g'(M)) \leq \beta\ell_*/16$ holds from $M = 0.64$ ($\beta = 0.5$) and $M = 0.77$ ($\beta = 0.1$).*

μ	$\gamma_*(\mu)$	10^2	10^3	10^4	10^5	10^6	10^7
0.6	*1.2732*	*1.2934*	*1.4041*	–	–	–	–
0.7	*0.8682*	*0.8718*	*0.8685*	*0.8865*	–	–	–
0.8	*0.6769*	*0.6705*	*0.6767*	*0.6771*	*0.6771*	–	–
1	*0.5000*	*0.4717*	*0.4980*	*0.4999*	*0.5000*	*0.5000*	*0.4996*
1.3	*0.3715*	*0.2981*	*0.3645*	*0.3709*	*0.3714*	*0.3715*	*0.3715*
1.6	*0.2545*	*0.1376*	*0.2412*	*0.2530*	*0.2543*	*0.2545*	*0.2545*
1.9	*0.0844*	*0.0086*	*0.0750*	*0.0833*	*0.0842*	*0.0844*	*0.0844*

TABLE 12.5. $\alpha_-(\mathcal{R}_C S)/|C|^{2/\mu}$ for the base family at $|C| = 10^2, \dots, 10^7$ (columns), against $\gamma_*(\mu)$. '–': not computed or the search did not converge ($\mu = 0.6$ at 10^4; entries at $\mu = 0.6$ and $\mu = 0.7$ at 10^4 are not reliable). $\mu = 1$ at 10^7 (0.4996) shows the floating-point limit.

Numerical observation 12.79 (the intermediate limits and the feedback defect)**.** *For $\mu = 1, 1.3, 1.6, 1.9$ and c from 10^2 to 10^{12}: $c^{\mu/2}\ell_\infty(T_c S) \to \ell_*$, $\Lambda \to c^{\mu/2}/\ell_*$ and $\Lambda\,\ell_\infty(T_c S) \to 1$ from below (Figure 12.7(b)); for $\mu = 1.3$ the product is 0.9573 at $c = 10^2$ and 0.9947 at 10^3. On the window above the noise floor ($c \geq 10^3$, $1 - \Lambda\ell_\infty \geq 10^{-8}$) the defect $1 - \Lambda\ell_\infty$ behaves like c^s with $s = -0.79, -0.94, -1.10, -1.26$ for $\mu = 1, 1.3, 1.6, 1.9$ (least-squares fits on $8, 6, 5, 4$ points). The proof of Theorem 12.61(b) controls the feedback D by $c^{-\mu/2}$ ($-0.50, -0.65, -0.80, -0.95$) and the window part of D by $c^{-(1+\mu)/2}$ ($-1.00, -1.15, -1.30, -1.45$); the observed defect decays at a rate between these.*

Numerical observation 12.80 (globality just below and above the critical shift)**.** *For $\mu = 1, 1.3, 1.6$ and $c = 10^2, \dots, 10^6$, $\mathcal{R}_C T_c S$ with $C = -(1 - \beta)c^{\mu/2}/\ell_*$ is global at $\beta = 0.2$ and 0.05 (all 15 combinations of μ, c), and for $c \geq 10^4$ not global at the shift 1.05 times critical ($\beta = -0.05$; $c = 10^2, 10^3$ are still global: the*

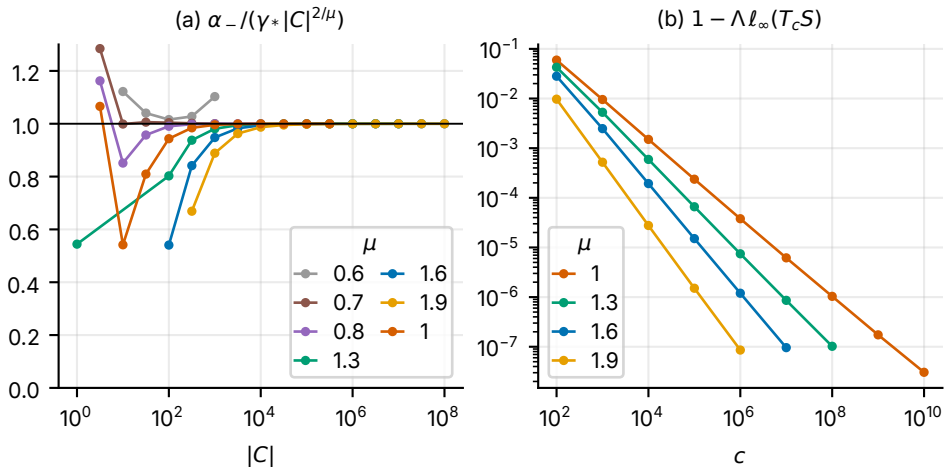


FIGURE 12.7. (a) $\alpha_-(\mathcal{R}_C S)/(\gamma_*|C|^{2/\mu})$ against $|C|$, converging to 1. (b) The defect $1-\Lambda(c)\ell_\infty(T_cS)$ against c (points above the solver noise floor 10^{-8}). Code: `make_c_atlas.py`.

correction is of the size of the defect of Numerical observation 12.79). This is the content of Theorem 12.61(a),(b) on the base family (`make_c_globality.py`, output `verify_ch12c_glob.out`).

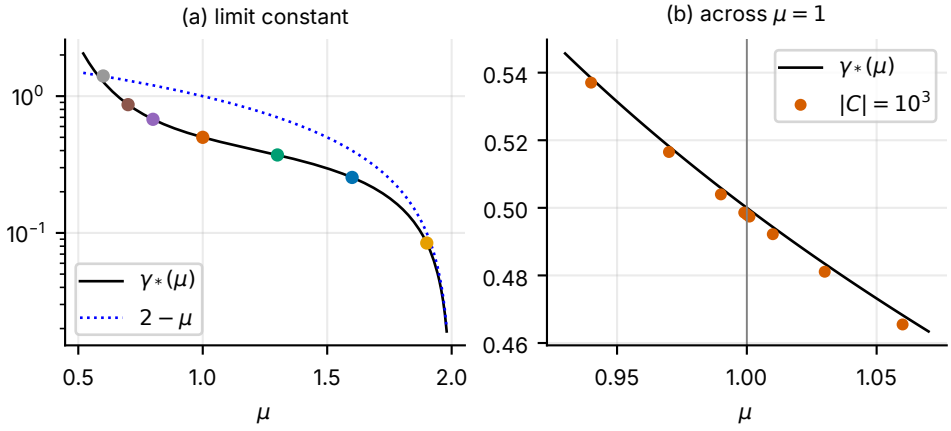


FIGURE 12.8. (a) The limit constant $\gamma_*(\mu)$, the line $2-\mu$, and the numerical limits (dots: last computed point per μ). (b) $\alpha_-/|C|^{2/\mu}$ at $|C|=10^3$ across $\mu=1$ against $\gamma_*(\mu)$: no jump at the resonant exponent. Code: `make_c_atlas.py`.

Numerical observation 12.81 (amplitude of the t^2-shift; the resonance $\mu=1$)**.** *For $c=1$ the amplitude $u_T/(cE^2)$ at $\xi=-6$ is* 0.17859, 0.25000, 0.35714, 0.62500, 2.49998 *for $\mu=0.6,1,1.3,1.6,1.9$, against $1/(4(2-\mu))=$* 0.17857, 0.25000, 0.35714,

0.62500, 2.50000 *(Toy example 6.15). For* $\mu \in [0.94, 1.06]$ *(nine exponents) and* $|C| = 10^3$ *the ratio* $\alpha_-/|C|^{2/\mu}$ *is smooth in* μ *and tracks* $\gamma_*(\mu)$ *with a relative offset between* -0.59% *and* -0.25% *(Figure 12.8(b)); at* $\mu = 0.999, 1, 1.001$ *the ratios are* 0.49859, 0.49800, 0.49741 *against* $\gamma_* = 0.50058$, 0.50000, 0.49942*, a relative offset* -0.40% *in each case. Every bisection found a single switch from pole to global as* c *increases, as in Theorem 6.32(b).*

Numerical observation 12.82 (the rest state)**.** *For the swirl-free model of Proposition 12.67, numerical integration gives* $\ell_\infty \sqrt{c} = 0.70761$ *at* $c = 10^6$ *(limit* $1/\sqrt{2} = 0.70711$*), and the thresholds* $\alpha_- = 7.5000004,\ 60.0000007,\ 5099.99997,\ 501000.022$ *at* $|C| = 3, 10, 10^2, 10^3$ *against* $|C|(|C| + 2)/2$ *(relative deviations* $\leq 5.2 \cdot 10^{-8}$*).*

Part 6

What is not known

CHAPTER 13

Conjectures and open problems at the edge of the picture

In words. *The first twelve chapters prove a classification: every axisymmetric* (-1)*-homogeneous solution with swirl is described by explicit parameters in a region whose boundary is understood, solutions exist for every strength of the swirl, and the asymptotic regimes of small and large swirl are controlled. This chapter collects what is* not *known. It is the only place in the book where a conjecture or an open problem is stated, and each is stated exactly once, with a label that the earlier chapters use when they point forward. For each item we say precisely what is asked, what evidence the book has (always by reference to a numerical observation of an earlier chapter), what is proved, why the problem is hard and what a proof might look like. The last of these is a suggestion and nothing more.*

Roadmap. Section 13.1 fixes the vocabulary and lists all items in one table. Section 13.2 is about the global picture: the interval property of the set Ω of global solutions (Conjecture H), the regularity of its edges, the types of poles on the edges and the topology of Ω. Section 13.3 is about the thresholds of Chapters 6 and 12: convexity of the lens, sharp constants, the cone limit and the class-N law. Section 13.4 is about uniqueness: the number of solutions at the certified datum, the fold set and its cusps. Section 13.5 is about the two asymptotic regimes of Chapters 10 and 11. Section 13.6 is a short outlook on non-axisymmetric solutions and stability.

13.1. Vocabulary and overview

In words. *A* conjecture *is a precise statement that the evidence supports and nobody has proved; the book gives a way in which it could fail. An* open problem *is a question, or a family of questions, where the right statement is part of the problem. Everything that* is *proved about an item stays in the chapter where it belongs and is only quoted here by reference.*

Two conventions run through the chapter. First, evidence is always one of the numerical observations of the earlier chapters; the code that produced it is described there and in Appendix E, and no new number appears in this chapter. Second, the paragraph headed *Suggestion* in each item is exactly that: a direction that looks plausible to us, with no claim that it works.

item	what is asked	proved part
The set Ω of global solutions		
Conjecture 13.1	interval property of the fibres of Ω; monotonicity	Theorem 7.65
Conjecture 13.2	the wall of L_2 edges is smooth, γ^- piecewise smooth	Theorem 7.32
Conjecture 13.3	Θ^- is C^∞ where the edge is of type R or L_1	Theorem 7.19
Conjecture 13.4	the window of R edges closes as $c_3 \to \infty$	Theorem 7.21
Open problem 13.5	large swirl, $c_2 > -\frac{3}{4}$, connectedness of the edge types	Proposition 7.44
Open problem 13.6	the lens at the endpoints of the no-swirl window	Theorem 7.74
Open problem 13.7	closure of Ω_{G}, base for $\frac{1}{2} < \mu \le 1$, manifold with corners	Theorem 7.90
Thresholds		
Conjecture 13.8	the lens is convex in (κ, γ)	Theorem 12.48
Conjecture 13.9	limit constants for $k > 1$	Theorem 11.44, Theorem 12.11
Open problem 13.10	next terms at $k = 1$ and $k < 1$; derivatives of the edge	Corollary 11.46
Open problem 13.11	sharp constants at the poles	Theorem 12.31
Open problem 13.12	cone limit below ε_*; the double limit $x_0 \to \pm 1$, $\varepsilon \to 0$	Theorem 12.37
Open problem 13.13	rates, uniformity, range in μ for the class-N law	Theorem 12.61
Open problem 13.14	differentiability of α_- across $\mu = 0$	Theorem 6.32
Uniqueness		
Open problem 13.16	exactly three solutions at D^*	Theorem 9.68
Open problem 13.17	the fold set; uniqueness regions	Theorem 9.42
Conjecture 13.18	the folds end in Whitney cusps	Lemma 9.19
Open problem 13.19	scaling of the ridge	Numerical observation 9.63
Asymptotic regimes and beyond		
Open problem 13.20	large swirl for $\boldsymbol{c}' \notin \operatorname{int} \mathcal{J}$	Theorem 10.39
Open problem 13.21	the regime $\lvert\ln \vartheta\rvert \sim J$ and the plateau	Theorem 10.11
Conjecture 13.22	sharp rates for $J - J_*$ as $\kappa \to 0$	Theorem 11.10
Open problem 13.23	small swirl for $f_- < 0$	Theorem 11.58
Open problem 13.24	non-axisymmetric solutions, stability	–

13.2. The set of global solutions

In words. *Chapters 5 and 7 describe the set $\Omega \subset \mathbb{R}^6$ of Cauchy data at the equator whose solution is global, as a stratified closed set. The deepest structural question left is whether each fibre of Ω over $(\boldsymbol{c}, A, \kappa)$ is an interval; the others concern the smoothness of the edge, the types of poles that occur on it, and the topology of Ω.*

13.2.1. Interval property. Recall the setting of Chapter 7. Cauchy data at the equator are $(\boldsymbol{c}, \gamma, A, \kappa) \in \mathbb{R}^6$: $\boldsymbol{c}$ are the Taylor constants of Q, $\gamma = U(0)$, $A = f(0)$ and κ is the slope of the swirl in the normalisation of Table 2.1 that uses the equator. For fixed $(\boldsymbol{c}, A, \kappa) \in \mathbb{R}^5$ let

$$\mathrm{FS} = \{\gamma : \text{ the forward solution exists on } [0,1)\},$$
$$\mathrm{BS} = \{\gamma : \text{ the backward solution exists on } (-1,0]\},$$

so that the fibre is $\Omega_{\boldsymbol{c},A,\kappa} = \mathrm{FS} \cap \mathrm{BS}$ (Proposition 5.42, Theorem 7.90).

Conjecture 13.1 (Conjecture H: the interval property)**.** *For every $(\boldsymbol{c}, A, \kappa) \in \mathbb{R}^5$:*

(H1) FS *is empty or a half-line $[\gamma_{\mathrm{F}}, \infty)$, and* BS *is empty or a half-line $(-\infty, \gamma_{\mathrm{B}}]$. Consequently the fibre $\Omega_{\boldsymbol{c},A,\kappa} = [\gamma_{\mathrm{F}}, \gamma_{\mathrm{B}}] \cap \mathbb{R}$ is an interval, and $\Omega = \{(\boldsymbol{c}, \gamma, A, \kappa) : \ \Gamma^- \le \gamma \le \Gamma^+\}$ with $\Gamma^- = \gamma_{\mathrm{F}}$ and $\Gamma^+ = \gamma_{\mathrm{B}}$ on the non-empty fibres.*

(H2) *Ω is an up-set in $\boldsymbol{c}$: if $(\boldsymbol{c}, \gamma, A, \kappa) \in \Omega$ and $\boldsymbol{c}' \ge \boldsymbol{c}$ componentwise, then $(\boldsymbol{c}', \gamma, A, \kappa) \in \Omega$.*

(H3) *For fixed $(\boldsymbol{c}, \kappa)$ with $\kappa > 0$ the functions $A \mapsto \gamma_{\mathrm{F}}(\boldsymbol{c}, A, \kappa)$ and $A \mapsto \gamma_{\mathrm{B}}(\boldsymbol{c}, A, \kappa)$ are nondecreasing on all of $\mathbb{R}$ (they are nonincreasing if $\kappa < 0$).*

What is proved. The symmetries σ_1, σ_2 reduce everything to $\kappa \ge 0$. (H1) holds when $A\kappa = 0$ (Proposition 7.50); for $A\kappa > 0$ the forward set FS is a half-line (Proposition 5.42) and the backward set is a half-line near γ_{F}, by Proposition 7.64. The unproved half is therefore BS for $A\kappa > 0$ or, by the mirror symmetry of Proposition 5.42(d), FS for $A\kappa < 0$. (H2) holds when $A\kappa = 0$ (Proposition 7.50). (H3) holds for γ_{F} when $A \ge 0$ (Proposition 5.42(c)) and for γ_{B} when $A\kappa \le 0$ (by the mirror). Theorem 7.65 collects the rest: a conditional reduction to a cut-orthant hypothesis (E) at the edge points of BS (Theorem 7.54), the local structure at edges of type R and L_1 (Theorem 7.55), and a theorem for small A/κ (Theorem 7.60). The monotonicity in κ is not part of the conjecture, and it is false (Numerical observation 7.69).

Evidence. Numerical observation 7.92 summarises it: in 400 random and 899 adversarial data the forward set is always an up-set and the backward set always a down-set; 1900 fibres, 944 of them with $A\kappa > 0$, show no gap; a fine rescan of 1148 fibres finds none with two components (Numerical observation 7.67, Numerical observation 7.68); monotonicity in $\boldsymbol{c}$ was tested 3600 times and in A on 120 curves without a violation (Numerical observation 7.69); and at 450 edge points of BS the orthant hypothesis (E) and the sign condition (E′) both hold (Numerical observation 7.70). The formal computation Formal computation 7.63 explains the sign at small A/κ. What the evidence cannot exclude is a counterexample in a regime never sampled (very large $|A|$ together with large κ and $\boldsymbol{c}$ near $\partial\mathcal{J}$); such a counterexample would be a fibre with two components, found by scanning γ on a fine grid.

Why it is hard. Going backwards from the equator with $A\kappa > 0$, the swirl first moves in the wrong direction until it vanishes at some point $x_z < 0$; before x_z the system is not cooperative, so the comparison principle that gives the half-line for FS is not available, and pointwise monotonicity in γ is actually false there (Numerical observation 5.34). After x_z the Jacobi field enters the good orthant, but whether it does so, and early enough, is a quantitative question about a signed integral. The four known gaps (Remark 7.61, Remark 7.62, Remark 7.57) are: (i) a uniform bound $\xi_1 \geq -\Xi$ on the position of the unique sign change of $\partial_\gamma \psi$ along edge solutions, which is what continuation in A from the trivial case $A = 0$ would need; (ii) an a priori lower bound for the location of the edge points γ_{B}, so that the constant $a_0(\Gamma)$ of Theorem 7.60 is uniform; (iii) edges of type L_2, which are not covered by Theorem 7.55; (iv) edge points of BS without a zero of the swirl on $(-1, 0)$, which are not covered by the re-basing at that zero.

Suggestion. A single *interface lemma* across the zero of f would close (H1) and, by the same mechanism, (H2) and (H3) at $A\kappa > 0$ together with the open questions about the closure of Ω_{G} (Open problem 13.7). One candidate is to compare in the variable $F = \int z^{-2}$ instead of x, in which the zero of f is a regular point of a monotone change of variable; another is to prove the bound (i) by an energy identity for $\partial_\gamma \psi$, using that ψ_γ solves a linear equation with a positive source after x_z (Lemma 7.49).

13.2.2. Regularity of the edge.

Conjecture 13.2 (regularity at the wall)**.** *Let E_{L_2} be the wall of Theorem 7.32 (the set of edge points of type L_2, where the edge passes between L_1 and R). The wall is a C^∞ hypersurface, and on a full neighbourhood U of each of its points $\gamma^- = \gamma_L$ on $\{\omega_L \geq 0\}$ and $\gamma^- = \gamma_L + \Psi$ on $\{\omega_L < 0\}$ with $\gamma_L, \omega_L, \Psi \in C^\infty(U)$ and $\Psi = \frac{2}{\ell_\gamma}\omega_L^2(1 + O(\omega_L))$. In particular γ^- is piecewise C^∞ with the Hessian jump of Theorem 7.32(c), and all higher jumps are determined by ℓ_γ and ω_L.*

What is proved. Theorem 7.32 gives $\gamma^- \in C^{1,1}(U)$, C^∞ off the wall, not C^2 at it, and the formula for the Hessian jump; the wall is a C^3 hypersurface and $\gamma_L, \Psi \in C^3$. The neighbourhood U shrinks with the order of differentiability (Remark 7.33). Away from walls γ^- is C^∞ (Theorem 7.12), and walls are nowhere dense (Corollary 7.35).

Evidence. The computation of the Hessian jump at $\boldsymbol{c} = (0, -0.9, 0)$ agrees with the prediction of the theorem (Numerical observation 7.37), and the variational formulas of Numerical observation 7.15 agree with finite differences.

Why it is hard. The unified corner chart at a wall (Theorem 4.45) loses an exponential weight with each derivative: the radius of the chart is $\mu_* = 1/(40(k+3))$ for k derivatives, so no neighbourhood works for all k.

Suggestion. Replace the finite-order contraction by analyticity in the chart variables (ω, σ), with Cauchy estimates on a nested family of domains (the loss per derivative becomes a loss of radius); or use a scale of weighted spaces indexed by a continuous parameter.

Conjecture 13.3 (smoothness of the recessive threshold)**.** *The function $\Theta^- : B \to \mathbb{R}$ of Theorem 6.50 is C^∞ on the open subset of B where the edge germ is of type R or L_1 at the north pole. Equivalently, the section $y \mapsto S^y$ of Lemma 6.47 can be chosen C^∞ in chart coordinates.*

What is proved. Θ^- is continuous and tends to $+\infty$ at ∂B (Theorem 6.50). Along any C^∞ family of germs of type G with $\mu > 1$ the quantity $\theta(S) = \Theta^-(\Pi(S)) - \mathfrak{a}(S)$ is C^∞ near every germ whose edge is of type R or L_1, and the edge is a C^∞ hypersurface in chart coordinates (Theorem 7.19(b), Remark 7.20(4)). So Θ^- is C^∞ at y_0 as soon as the section is a C^∞ family near y_0. The analogue for the t^2-shift is Theorem 7.19(a).

Evidence. None beyond the smoothness along families; no numerical test of the section is in the book.

Why it is hard. The section is obtained from a contraction in weighted spaces with weight exponent $2 + \mu_{\min} - \varepsilon_0$ (Lemma 6.47); smooth dependence on y needs C^∞ dependence of u_P^y, of the normalising constants and of the contraction map on the base point, and each y-derivative costs weight.

Suggestion. Choose the section by a contraction that is analytic in y on a scale of weights, or avoid sections altogether by writing $\{\mathfrak{a} = \Theta^-\}$ as the zero set of a defining function whose gradient does not vanish, which is what Theorem 7.19(b) already gives locally on the product $B \times \mathbb{R}$.

13.2.3. Types of poles on the edge, and the topology of Ω.

Conjecture 13.4 (the window of R edges closes)**.** *Fix $x_0 \in (-1, 1)$ and c_1, c_2 with $-1 < c_2 < -\frac{3}{4}$, and let $\kappa_0(\boldsymbol{c})$ be the supremum of the κ_1 for which the lower edge of the lens at $(x_0, \boldsymbol{c}, \kappa)$ is of type R for all $0 < \kappa < \kappa_1$; thus*

$$\kappa_0(\boldsymbol{c}) = \sup\{\kappa_1 : \text{ lower edge of type } \mathsf{R} \text{ for all } 0 < \kappa < \kappa_1\}.$$

Then $\kappa_0(c_1, c_2, c_3) \to 0$ as $c_3 \to \infty$.

What is proved. For every $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$ with $-1 < c_2 < -\frac{3}{4}$ the window is non-empty, $\kappa_0(\boldsymbol{c}) > 0$ (Theorem 7.21, Corollary 7.22), and it degenerates as $c_2 \uparrow -\frac{3}{4}$ (Remark 7.23). For $c_2 > -\frac{3}{4}$ the limit $\kappa \to 0$ is singular (Proposition 7.44).

Evidence. Numerical observation 7.45: for $\boldsymbol{c} = (0, -0.8, c_3)$ type R is seen at $c_3 = 3$ and, of the 18 edges scanned, nowhere else; at $c_3 = 30$ and 300 even the smallest swirl scanned is already of type L_1 to 10^{-8}.

Why it is hard. The proof of Theorem 7.21 uses the implicit function theorem in the pole chart and gives no quantitative lower or upper bound for κ_0; a bound as $c_3 \to \infty$ needs the chart constants tracked through the equatorial well of depth $\sim c_3$.

Suggestion. A criterion for R that is stable under rescaling by c_3, obtained from the Wronskian v of Lemma 7.1: type R requires $v \to 0$ at the pole, and a large equatorial potential should force v to be bounded below.

Open problem 13.5 (types of poles at the edge)**.** *Describe where the edge $\mathfrak{E}^-$ has type R, L_1, L_2. Specifically: (a) are there edges of type R at large κ, for any $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$? (b) For $c_2 > -\frac{3}{4}$, what is the type of the lower edge as $\kappa \to 0$ (the layer problem of Corollary 7.81)? (c) Is E_{R} connected, and is E_{L_1}? (d) Does the one-sided estimate $q(\gamma) + 2 \le C(\gamma - \gamma^-)$, with $C = \ell_\gamma$, hold at every edge of type L_1 and not only at walls?*

What is known. E_{R} and E_{L_1} are open, E_{L_2} is closed and nowhere dense, and a connected subset of the edge that meets both open types meets the wall (Proposition 7.44(a)); $Q(1)$ is continuous along the edge, so $E_{\mathsf{R}} = \{Q(1) > -2\}$

(Proposition 7.26); at a wall both types occur on its two sides (Theorem 7.32(b)); the linear lower bound for $q+2$ is Theorem 7.12(iv), and the matching upper bound is the content of Theorem 7.32(a) at a wall. For (a) the scan Numerical observation 7.45 stops at $\kappa = 14.7$ and the heuristic Heuristic 7.46 says that a type R edge at large κ would need a second, pole, layer in addition to the equatorial step.

Why it is hard. The type is decided by a balance between how fast z decays and how much swirl energy $\int f^2(1+s)^{-2}$ reaches the pole, and this is a global quantity along the solution (Lemma 7.25).

Suggestion. For (a) work in the effective coordinates of Chapter 10: the layer position of the large-swirl solution moves toward a pole as $\vartheta \to 0$ or ∞, and the edge of type R would be the limit; the scan needs $\kappa \sim 10^3$ with both a pole and an equatorial layer resolved.

Open problem 13.6 (attachment at the endpoints of the no-swirl window)**.** *Describe how the lens attaches to the face $\kappa = 0$ at the endpoints $\gamma = U_{\boldsymbol{c}}^{\mp}$ of the no-swirl window. (a) In the mixed quadrants $A_0 > 0 > \kappa$ and $A_0 = 0$, is* $\mathrm{FS} = [\gamma_{\mathrm{F}}, \infty)$*, and does the edge pass through the face smoothly? (b) For $k_2 \geq \frac{1}{2}$, where the passage is through a pole layer (Corollary 7.81), what are the width of the layer and the profile of ψ across it? (c) Does the answer change at edges of type L_2 or at edges without a zero of f?*

What is known. For $k_2 < \frac{1}{2}$ and $A, \kappa \geq 0$ the edge passes smoothly through the face (Theorem 7.74, Corollary 7.76); the layer statements of Corollary 7.81 are proved only for $k_2 > \frac{1}{2}$ ($c_2 > -\frac{3}{4}$), at $A = 0$, $x_0 = 0$ and on the lower edge, and the exponents of the distance $\Delta = \gamma^- - U_{\boldsymbol{c}}^-$ are proved in Chapter 11 in the same setting (Theorem 11.44); the width of the layer and the profile of ψ are not. The formal computations Formal computation 7.82, Formal computation 7.84 and Formal computation 7.86 are the candidates for (b), and Numerical observation 7.77 shows the double wedge. A proof of (a) would also settle Formal computation 7.80, the non-star-shapedness of Ω in κ.

Why it is hard. In the mixed quadrants the comparison argument of Theorem 7.74 loses its cooperative structure at the zero of f, the same obstruction as in Conjecture 13.1.

Open problem 13.7 (the topology of Ω)**.** *(a) Is Ω the closure of Ω_{G} where $A\kappa \neq 0$, and is $\Omega \setminus \Omega_{\mathsf{G}}$ nowhere dense there? (b) For $\frac{1}{2} < \mu \leq 1$, describe the base of the recessive-shift chart: for which Π does the orbit of the recessive shift contain global germs? (c) What happens to class-Z solutions whose south pole is of type R or L_2 as $x_0 \to -1$? (d) Is Ω, with its strata, a manifold with corners?*

What is known. Ω_{G} is open and contained in the interior of Ω, and Ω is the closure of Ω_{G} on the faces $A = 0$ and $\kappa = 0$ (Theorem 7.90, Proposition 5.69, Remark 7.91(3)). For $\mu > 1$ the base is $B = \{\Pi : P_d \in \operatorname{int} \mathcal{J}\}$ and the half-cylinder structure is proved, including $\mu \geq 2$ (Theorem 6.50, Theorem 6.61); on $\frac{1}{2} < \mu \leq 1$ the recessive shift exists and preserves the order (Theorem 6.22), but the explicit base uses $\mu > 1$ in the polynomial part of the Green representation (Proposition 6.37) and in the pole-region estimate, and the formal coefficient of the free mode is not defined (Numerical observation 6.68). For (c) the proofs of the gluing of Z and N_0 use the envelope of type G (Proposition 6.59, Numerical observation 6.60); if the south pole is of type R or L_2 then $f(-1) = -\infty$ and f_- is no longer available as a

gluing parameter. For (d), γ^- is $C^{1,1}$ at walls and C^∞ elsewhere, and charts are homeomorphisms, but no smooth structure on the whole of Ω is proved.

Why it is hard. In (b) three separate ingredients degrade at $\mu = 1$ simultaneously; in (a) and (d) one lacks the interval property (Conjecture 13.1) and uniform regularity of the edge (Conjecture 13.2, Conjecture 13.3).

Suggestion. For (b), replace the formal coefficient by a flow-defined coordinate: fix a section of the recessive flow, measure the t^2-content of a germ by the parameter of the t^2-shift relative to it, and look for a criterion analogous to $P_d \in \operatorname{int} \mathcal{J}$ that decides when the orbit contains global germs. For (c), use the pole charts mirrored to $x = -1$ to locate the zero of f inside the chart, as in the proof of the gluing for type G.

13.3. Thresholds, constants and rates

In words. *Chapters 6 and 12 locate the thresholds of the classification: the lens $\{\gamma^- \le \gamma \le \gamma^+,\ \kappa \le \kappa_{\max}\}$ for class Z, and the half-lines of the shifts for class N. The orders of magnitude are proved; this section asks for the shape of the lens, for the constants in the laws, and for the finer behaviour of the class-N law.*

13.3.1. The shape of the lens.

Conjecture 13.8 (convexity of the lens)**.** *For every $x_0 \in (-1, 1)$ and $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$ the lower wall γ^- is convex and the upper wall γ^+ is concave on $[0, \kappa_{\max}]$ as functions of $\kappa = \kappa_x$; equivalently the closed lens $\overline{\text{Lens}}$ of Proposition 12.43(a) is a convex subset of the (κ, γ)-plane. In terms of $G(t) = \gamma^-(e^t)$ this says $G'' \ge G'$, i.e. $n := \kappa {\gamma^-}''/{\gamma^-}' \ge 0$.*

What is proved. The lens is convex in the logarithmic variable: $\mathcal{G} = \{(\boldsymbol{c}, \ln \kappa, \gamma)\}$ is a convex subset of $\mathbb{R}^5$ (Theorem 12.48), so $G'' \ge 0$, equivalently $n \ge -1$ (Corollary 12.49). Convexity in κ itself is proved near the floor for $c_2 < -\frac{3}{4}$ (Theorem 12.55), near the critical surface for $k_1, k_2 < \frac{1}{2}$ (Theorem 12.57), and the tip is a convex corner when the edge types at the tip are R or L_1 (Proposition 12.43(c)); that the tip edges are of these types is verified only numerically. The orders of the distance to the floor give only the size of $\Delta = \gamma^- - U^-_{\boldsymbol{c}}$, not of Δ'' (Theorem 11.44).

Evidence. The midpoint test of Numerical observation 12.51 finds no violation of the log-convexity inequality with smallest relative slack 10^{-2}; the second difference of γ^- is positive on nine data over $2.5 \cdot 10^{-4} \le \kappa \le 1.26$, with minimum 0.043 (Numerical observation 11.57), and the quadratic coefficient at the floor exceeds the proved lower bound ten-fold (Numerical observation 12.56). The conjecture would fail if one found a pair of points of the lower wall whose midpoint lies strictly above the chord.

Why it is hard. Convexity is second order, and the nonlinear terms of the cooperative system have a sign only for ordered pairs of solutions (the identity $\frac{1}{2}(a_1 b_1 + a_2 b_2) - \frac{1}{4}(a_1 + a_2)(b_1 + b_2) = \frac{1}{4}(a_1 - a_2)(b_1 - b_2)$). Two points of the lower wall are not ordered: a larger κ moves the solution down, a larger γ moves it up. In the variables (U, g, f, p) the source of the second variation has no sign at x_0 (Proposition 12.45(a)); the geometric mean in $(U, \ln g, \ln f, p)$ removes the obstruction but delivers only log-convexity (Proposition 12.45(b), Heuristic 12.46).

Convexity is not invariant under reparametrising κ, so no change of variable can be used.

Suggestion. Differentiate the identity (T0) twice and look for a monotone Wronskian-type quantity for the pair of Jacobi fields (J_t, J_γ) along the edge, sharp in the large-constants limit; or complete the squares in (J_U, J_g) in the second variation; or find a pair of variables in which the two lower-wall solutions *are* ordered.

13.3.2. Constants in the laws near the floor and the critical surface.

Conjecture 13.9 (the limit constants for $k > 1$)**.** *Let $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$, $x_0 \in (-1, 1)$ and $k > 1$. (a) With $k = k_2$ and $p = 2k/(2k-1)$, the limit $\lim_{\varkappa \downarrow 0} \Delta\, \varkappa^{-p}$ exists in $(0, \infty)$, where $\Delta = \gamma^- - U^-_{\boldsymbol{c}}$. (b) With $k = \max(k_1, k_2)$ the limit $\lim_{\varepsilon \downarrow 0} \varkappa^p_{\max}/\varepsilon$ exists in $(0, \infty)$. Both limits are determined by the parameter-free layer problem of Formal computation 12.18.*

What is proved. Two-sided bounds with positive constants: $c_- \varkappa^p \le \Delta \le c_+ \varkappa^p$ (Theorem 11.44(c)) and $c\, \varepsilon^{(2k-1)/k} \le \varkappa^2_{\max} \le C\, \varepsilon^{(2k-1)/k}$, equivalently $\kappa_{\max} \asymp \varepsilon^{1-1/(2k)}$ (Theorem 12.11). For $k < 1$ the limit exists and is explicit (Theorem 11.44(a), Theorem 12.15).

Evidence. The local exponents of Δ at $\varkappa \approx 2 \cdot 10^{-4}$ are within 0.003 of p_k for $k = \sqrt{2}$, 1.5, 2, 2.5 (Numerical observation 11.56), and the compensated ratios $\kappa^2_{\max} S_k/\varepsilon$ of Numerical observation 12.21 have increments that halve per decade for $k = 1.5$ and $k = 3$ (consistent with a finite limit, not a proof; for asymmetric pairs the numerics do not discriminate). The conjecture would fail if such a ratio were found to drift without bound or to oscillate.

Why it is hard. The constant is not determined by the cut argument of Chapter 11, which controls the solution only up to the cut, where ψ is still small. The constant lives in a *nonlinear* layer at the cut (the swirl layer of width ξ_L meeting the Jost scale ξ_ε), and one needs existence and uniqueness for the layer problem and matching to the outer solution beyond the Jost scale.

Suggestion. Prove uniqueness for the layer problem $Z'' = (k^2 - \frac{1}{2}\tilde{\psi})Z$, $F' = Z^{-2}$, $\frac{1}{4}e^{2\tau}(\tilde{\psi}'' + 4\tilde{\psi}') = F^2$ with the prescribed behaviour at $\pm\infty$ (the step-potential toy Toy example 12.17 shows how the admixture $a_\infty = \theta_\star e^{-2k\xi_L}$ enters), and match it to the outer solution by the sandwich of Proposition 12.3.

Open problem 13.10 (the next terms)**.** *(a) For $k = 1$, determine the constant term b in $\Delta/\varkappa^2 = \frac{e^{-2\xi_0}}{12G_+^2} \ln\frac{1}{\varkappa} + b + o(1)$ and the constant term in $\varepsilon/\varkappa^2_{\max} = \frac{c_*}{2I_g} \ln\frac{1}{\varepsilon} + b' + o(1)$. (b) For $k < 1$, determine the second term of $\Delta/\varkappa^2 - A^{\lim}_+$. (c) Prove the expansions of the derivatives of γ^- at the floor: $\gamma^{-\prime\prime}(0) = 2(1 - x_0^2)^2 A^{\lim}_+$ for $k < 1$, $\gamma^{-\prime\prime} \sim \ln(1/\kappa)$ for $k = 1$, and $\gamma^{-\prime\prime} \sim \kappa^{p-2}$ for $k > 1$.*

What is known. The slope in $\ln(1/\varepsilon)$ at $k = 1$ is proved (Corollary 12.16, Theorem 11.44(b)), but the constant term is not determined by the argument (Remark 11.45(iii)): at $\boldsymbol{c} = 0$, $x_0 = 0$ it is observed to be 0.1775 for $\varepsilon/\kappa^2_{x,\max}$ (Numerical observation 12.20) and 0.1984 at the rest state for the edge (Numerical observation 11.51), and the first-order closed form of Proposition 11.48(c) is *not* it, since part of the mass lies beyond the cut. For $k < 1$ the relative deficit of the second term is observed to decay like $\varkappa^{(2-2k)/k}$ (Numerical observation 11.50), with

sign and coefficient not proved. For (c), Corollary 11.46(b) is conditional on twice differentiability, and the numerical second differences of Numerical observation 11.57 agree with the formulas.

Why it is hard. All three need the *layer* at the cut, not only the first-order integral; (c) additionally needs the expansions to be differentiable in κ.

Suggestion. Solve the layer problem of Formal computation 12.18 at $k = 1$ (a Pöschl–Teller case, where closed forms are plausible), and differentiate the matched expansion using the Hadamard-type formula for variations in the cut.

Open problem 13.11 (sharp constants at the poles)**.** *With the notation of Theorem 12.31, prove that* $\varkappa_{\max}/\varkappa_{\mathrm{WKB}} \to 1$ *whenever* $a \to \infty$*, without the hypothesis* $R_0/a^2 \to 0$*; in particular along* $\boldsymbol{c} = (p, p, 0)$*,* $x_0 = 0$ *and* $\boldsymbol{c} = (0, p, 0)$ *as* $p \to \infty$*.*

What is known. The upper bound $\varkappa_{\max}/\varkappa_{\mathrm{WKB}} \le 1 + O(\ln a/a)$ holds for all $\boldsymbol{c} \in [0, \infty)^3$; the lower bound carries the factor $\sqrt{1 - R_0/a^2}$ and is sharp only when $R_0/a^2 \to 0$ (Theorem 12.31(b)). For $(p, p, 0)$ at $x_0 = 0$ this factor is 0.707 and for $(0, p, 0)$ it is $\sqrt{(1 - x_0)/2}$. Numerically the ratio decreases to 1 in all four families (Numerical observation 12.34), roughly like $1 + O(p^{-1/2})$, and at $p = 10^4$ the proved lower bound reaches 0.66–0.95 of the value.

Why it is hard. The envelope for the swirl integral ψ near the cone, which the lower bound needs, is too crude when $\boldsymbol{c}$ is large at a pole.

Suggestion. Use the saturation route: $W \ge W_{\mathrm{eff}} \ge 0$ forces a logarithmic (L_1 or L_2) pole type and $\psi_\pm = 2k_\pm^2$ there, so that the tail ratio is $2/(Ae^{2M})$ and controlled when $A \lesssim M/a$; the sharp law is then the harmonic-mean ceiling seen through the values of f at $\pm M/a$.

Open problem 13.12 (the cone limit below ε_* and the double limit)**.** *(a) For* $\boldsymbol{c} \in \operatorname{int} \mathcal{J}$ *with* $\varepsilon < \varepsilon_* = k_1(1 + k_2)$ *determine* $\lim_{x_0 \to 1} \sqrt{1 - x_0^2}\, \kappa_{\max}(x_0, \boldsymbol{c})$*. (b) Determine the behaviour of* $\kappa_{\max}(x_0, \boldsymbol{c})$ *jointly as* $\varepsilon \downarrow 0$ *and* $x_0 \to \pm 1$*, i.e. the* ε*-law with constants uniform in* x_0*.*

What is known. Theorem 12.37 gives $\Lambda_* \le \liminf \le \limsup \le \Lambda_1$ with $\Lambda_*^2 = \Lambda_1^2[1 - (1 - \varepsilon/\varepsilon_*)^2]$ for $\varepsilon \le \varepsilon_*$, and the exact limit Λ_1 for $\varepsilon \ge \varepsilon_*$ (confirmed in Numerical observation 12.38). For $\varepsilon < \varepsilon_*$ the book does not compute the limit; the natural guess is Λ_*, that is, that the proved lower bound is the answer, and the missing half is $\limsup \le \Lambda_*$. For (b), all results of Part A of Chapter 12 have constants depending on x_0 (Numerical observation 12.22).

Why it is hard. One needs an envelope for ψ on $\xi \le -L$ that does not see the far pole layer, i.e. a plateau version of the swirl-kernel estimate. The mirror problem $x_0 \to -1$ is equivalent.

Suggestion. Prove that, below ε_*, the tip is exactly as large as the effective constants allow (the converse of Proposition 12.29) by locating the swirl plateau at the south pole through the maximum-principle envelope $\psi \le \frac{\psi_-}{4}(1-x)^2 + \frac{\psi_+}{4}(1+x)^2$.

13.3.3. The class-N law.

Open problem 13.13 (rates, uniformity and range for the class-N threshold)**.** *Let* $\gamma_*(\mu)$ *be the constant of Theorem 12.61. (a)* Rate*: find the rate in* $\alpha_-(\mathcal{R}_C S)/|C|^{2/\mu} \to \gamma_*(\mu)$ *as* $C \to -\infty$*, and the correct decay of the defect* $1 - \Lambda\, \ell_\infty$*. (b)* Uniformity*: is the convergence uniform for germs* S *in a compact subset of*

$\mathcal{G}_{\frac{1}{2},2}$*? (c)* Range*: what is the law for* $\mu \le \frac{1}{2}$ *and for* $\mu \ge 2$*? (d)* Labels*: state the law in the Frobenius parametrisation, i.e. as a function of the label of* S *rather than in the germ form.*

What is known. The theorem holds for fixed S with $\frac{1}{2} < \mu < 2$ and gives no rate. The proof controls the feedback defect by $c^{-\mu/2}$ and the window part by $c^{-(1+\mu)/2}$; the observed defect decays like c^s with $s = -0.79, -0.94, -1.10, -1.26$ for $\mu = 1, 1.3, 1.6, 1.9$, steeper than the proved $-0.50, \dots, -0.95$ (Numerical observation 12.79). Uniformity is tested on one base family only (Numerical observation 12.78). The range $\frac{1}{2} < \mu < 2$ is the range in which T_c (needs $\mu < 2$) and $\mathcal{R}_C$ (needs $\mu > \frac{1}{2}$) are both defined (Theorem 6.19, Theorem 6.22); numerics for $\mu < 0.7$ do not converge (Numerical observation 12.78), and $\gamma_*(\mu) \to 0$ as $\mu \uparrow 2$ because of the factor $1/\Gamma(1-\mu/2)$ in its closed form. The recessive chart itself covers $\mu \ge 2$ (Theorem 6.50).

Why it is hard. (a) needs the nonlinear feedback of the swirl beyond the a priori bound used in the upper bound; (c) leaves the range of the shifts and needs a different normal form.

Suggestion. The exact Fuchsian model without swirl (Proposition 12.67, Proposition 12.66) is a benchmark for (a); for (b) track the dependence of the constants on the envelope data and on the modulus of $ze^{\mu\xi} \to 1$.

Open problem 13.14 (differentiability across $\mu = 0$)**.** *Is* α_- *differentiable* (C^1) *as a function of the state at the wall* $\mu = 0$ *between the strata of type* G *and* L_1 *at the south pole? Is* f_-*, which is continuous at* L_1 *(Proposition 6.13), differentiable there?*

What is known. The threshold α_- is continuous across $\mu = 0$ (Theorem 6.32(d)). Numerically $\mu^2/\sigma \to 0.5647$ along family B, $(\alpha_- - \alpha_-^{\mathrm{wall}})/\mu^2 = -4.057$ at $\sigma = 10^{-2}$, and along family A the difference quotient -8.327 agrees with the gradient-times-tangent value -8.368 at $\mu = 0.01$, so the apparent linear rate along A is the signature of a C^1 function (Numerical observation 6.67).

Why it is hard. The pole charts at the south pole are not unified across $\mu = 0$ (Remark 7.20(1)); smoothness along chart families is proved only on each stratum.

Suggestion. Use the unified corner chart $(\beta, \omega, \sigma = \mu^2)$ of Theorem 4.45, in which the stratum change is a regular change of coordinates, together with the uniform envelope of Theorem 6.29.

Open problem 13.15 (explicit thresholds)**.** *Find explicit, or closed-form, descriptions of the two threshold functions: of* $\kappa_{\max}(x_0, \boldsymbol{c})$ *(equivalently* $\varkappa_{\max}$*), the supremum of the swirl constants for which the lens of class* Z *is nonempty, and of the class-*N *threshold* $\alpha_- = c^-$ *of the shifts, as functions of the base point and of the germ* S *(or at least of the invariants* $(x_0, \boldsymbol{c})$*, resp.* (μ, S)*).*

What is proved. The thresholds are *characterised* and *estimated*, not given by formulas. For class Z the lens, hence $\kappa_{\max}$, is characterised as the extent of the set of data admitting a global solution (Proposition 5.59); $\kappa_{\max}$ is two-sided estimated by the pole laws near the floor and the critical surface (Theorem 11.44, Theorem 12.11) and asymptotically in the large-constants regime (Theorem 12.31). For class N the threshold c^- is finite, a half-line threshold, continuous across $\mu = 0$ (Theorem 6.32) and has the asymptotic law $\alpha_-(\mathcal{R}_C S) \sim \gamma_*(\mu)|C|^{2/\mu}$ (Theorem 12.61). Closed

forms exist only in the models of Appendix B (no swirl, and the rest state to first order).

Why it is hard. The thresholds are defined through the global behaviour of the nonlinear problem: Q is a polynomial only without swirl, and as soon as $f \not\equiv 0$ the potential $W = 1 + \frac{1}{2}Q$ contains the non-polynomial centrifugal part Q_{sw} (the solution of $\mathsf{L}[Q_{\mathrm{sw}}] = -f^2$ with $f' = \kappa z^{-2}$), so $z'' = Wz$ is no longer a hypergeometric or Bessel equation and the threshold, a condition on where z first vanishes, depends on z itself. Every proved statement above is therefore an inequality or a limit law.

Suggestion. Look for a one-parameter closure in which Q_{sw} is explicit, for example by matching a Pöschl–Teller or Fuchsian model (Proposition B.15, Proposition B.18) to the nonlinear layer at the cut, and test any candidate formula against the benchmarks of Conjecture 13.8, Open problem 13.10 and the observed class-N constants; a formula that reproduces the proved laws and the numerical atlas would be a strong candidate to prove.

13.4. Uniqueness, folds and the number of solutions

In words. *For $\boldsymbol{c}' \in \operatorname{int}\mathcal{J}$ the number of solutions with prescribed effective data D is odd (Theorem 9.42) and changes only at degenerate fixed points; solutions are unique for small swirl, for large swirl, and outside a window of folds. The computer-assisted Theorem 9.68 shows that there is no uniqueness in general. What is missing is the exact description of the window.*

Open problem 13.16 (exactly three solutions at D^*)**.** *Show that T_{D^*} has* exactly *three fixed points, that each of them is nondegenerate ($1 \notin \operatorname{spec} \mathrm{D}T_{D^*}(f_k)$), and compute enclosures of the leading eigenvalues. Equivalently: the effective-data map has exactly three preimages of D^*, with indices $(+1, -1, +1)$.*

What is known. Theorem 9.68 proves the existence of three distinct solutions with certified enclosures of J_1, J_2, J_3, and Corollary 11.60 shows that the line through D^* contains degenerate fixed points at some $\kappa_{\mathrm{D}} \le 1400$ and some $\kappa_{\mathrm{D}} \ge 1400$. The number of solutions is odd at regular data (Theorem 9.42(Bb)), so at least three means three, five, ... Numerically there are exactly three, with indices $(+1, -1, +1)$ and leading eigenvalues $0.3504 \pm 0.1247\,\mathrm{i}$, 1.2818, 0.6296 (Numerical observation 9.75); these eigenvalues are not certified.

Why it is hard. Uniqueness of the zero of the four-dimensional shooting map Φ is certified only inside the three boxes; a proof of "exactly three" needs a global exclusion argument over a non-compact set of Cauchy data in $\mathbb{R}^4$ and a Lipschitz bound for T_D that is uniform on it.

Suggestion. Cover the complement of the three boxes inside a ball of a-priori-bounded Cauchy data by a finite number of boxes on which $\Phi \neq 0$ is certified by interval arithmetic (the a priori bounds of Proposition 9.38 give the ball), and use the Newton–Kantorovich method of Appendix C for the eigenvalues.

Open problem 13.17 (the fold set and the regions of uniqueness)**.** *Describe $\{D \in \mathbb{D} : \#E^{-1}(D) \ge 2\}$. (a) For which $(\boldsymbol{c}', \vartheta, f_-)$ is the fixed point unique for* every *κ_{D}, and what is the shape of the set of bad $(\vartheta, f_-, \kappa_{\mathrm{D}})$ for fixed $\boldsymbol{c}'$ (the "fold window" $[\kappa_{\mathrm{first}}, \kappa_{\mathrm{last}}]$, and the fold values $E(\Phi_{\mathrm{fold}})$)? (b) Is the set connected, and how does it end as $\vartheta \to \infty$ and $|f_-| \to \infty$? (c) Below the no-swirl set ($\boldsymbol{c}' \notin \operatorname{int}\mathcal{J}$,*

hypothesis (H) of Theorem 11.6), is the solution with small κ_{D} unique, i.e. is every small-κ_{D} solution the bubble solution, and is there only one?

What is known. E is proper and the count $N(D)$ is finite and locally constant off the fold values, odd at regular values, and equal to 1 as long as no degenerate fixed point is crossed (Theorem 9.42). Uniqueness holds for $\kappa_{\mathrm{D}} < \kappa_{\mathrm{lo}}$ (explicit, Theorem 11.58(a)) and for $\kappa_{\mathrm{D}} \geq \kappa_{**}$ (Theorem 10.39), and $\{\#\mathrm{Fix} \geq 2\} \subset [\kappa_{\mathrm{first}}, \kappa_{\mathrm{last}}]$ with $\kappa_{\mathrm{lo}} \leq \kappa_{\mathrm{first}}$ and $\kappa_{\mathrm{last}} < \kappa_{**}$ (Theorem 9.51, Theorem 11.58(d)); the constants are not sharp. Folds exist in class Z at moderate data (Numerical observation 9.59), in class N_0 at large ϑ (Numerical observation 9.63) and at D^* (Theorem 9.68). No fold was found at the rest state for $\vartheta = 1$ and for $\vartheta \leq 2300$ (Numerical observation 9.62, Numerical observation 9.63) or for a datum whose ridge comes within 1% of the threshold (Numerical observation 9.65). A fixed point in class N_0 at a fold has a sign-changing kernel vector (Remark 9.67). For (c), Theorem 11.58(b) describes every solution at small κ_{D} as close to the bubble but does not say that there is one; a non-unique regime would be a family of solutions all close to the bubble (Chapter 11).

Why it is hard. The operator $\mathrm{D}T_D$ is not positive (its leading eigenvalues can be complex), so Perron–Frobenius methods do not apply; the fold set has no a priori bound that is sharp; and the sharpness at $f_- = 0$, the boundary between classes N and Z, is visible in the numerics, where the leading eigenvalue approaches 1 to within 1% on a narrow ridge.

Suggestion. Study the fold hypersurface by pseudo-arclength continuation of the bordered system, which is cheap, in the variables $(\vartheta, \kappa_{\mathrm{D}})$ for several $(\boldsymbol{c}', f_-)$; once its shape is known, formulate a criterion in terms of $\ln \vartheta/J$, the ratio on which the ridge depends (Heuristic 9.66), and test it by the same continuation.

Conjecture 13.18 (Whitney cusps)**.** *The fold curves found numerically end in Whitney cusps: at the cusp point ω_c of Numerical observation 9.59 (class Z, in the plane $(c_3' - \bar{c}_3, \kappa_{\mathrm{D}})$) and at the birth of the fold pair of Numerical observation 9.63 (class N_0, in the plane $(\vartheta, \kappa_{\mathrm{D}})$ at $\boldsymbol{c}' = 0$, $f_- = 0$) one has $k(\omega_c) = 1$, transversality* (T) *holds, $\chi_{aa}(0) = 0$, $\chi_{aaa}(0) \neq 0$ and the cusp curve is transverse (that is $\nabla a(0) \neq 0$), in the notation of Lemma 9.19*(c).

What is proved. The normal forms: at a fold point with (T) the solution count changes by two across a smooth hypersurface (Proposition 9.34), and, if the cusp conditions hold, the width of the three-solution region is $\frac{4\sqrt{2}}{3}|a|^{3/2}/\sqrt{|\chi_{aaa}(0)|}$ (Corollary 9.32, Toy example 9.33). The fold set is the singular set of E where (T) holds (Theorem 9.42(A1)), and the fold set is non-empty (Corollary 9.46).

Evidence. Numerical observation 9.59 gives the cusp of the class-Z fold curve and the three-solution wedge; Numerical observation 9.63(2) gives the widths $162, 557, 1781, 4492$ for $\vartheta = 2400, 2500, 2700, 3000$, for which $w^{2/3}$ is linear in ϑ (Figure 9.4(b)). This is four points and one fitted pair of parameters: consistent with a Whitney cusp, not a demonstration of one. The conjecture would fail if the leading cubic coefficient χ_{aaa} vanished at the cusp, which would show as a width law with an exponent different from 3/2 on finer data.

Why it is hard. It needs second and third derivatives of T_D in f, along a kernel vector with a sign change, at a point only known numerically; no certified Newton–Kantorovich computation of such a point exists.

Suggestion. Extend the validated shooting of Appendix D by the bordered system for the fold and the cusp (the Lyapunov–Schmidt data χ_{aa}, χ_{aaa} are finite-dimensional once the shooting chart of Lemma 9.11 is used) and certify $\chi_{aaa} \neq 0$ at an enclosed cusp.

Open problem 13.19 (the scaling of the ridge)**.** *At the rest state $\boldsymbol{c}' = 0$, $f_- = 0$: (a) determine the asymptotics as $\vartheta \to \infty$ of $\Lambda(\vartheta) = \max_{\kappa_\mathrm{D}} \nu_{\max}$, of the lower and upper fold positions $\kappa_\mathrm{D}^{\pm}(\vartheta)$ and of the width of the window; is Λ unbounded like $\ln \vartheta$? (b) Determine the critical value ϑ_c at which the fold pair is born. (c) Give a mechanism for the ridge that survives a quantitative test.*

What is known. Numerically $\Lambda = 1 + 0.206 \ln(\vartheta/2320)$ fits $\vartheta \geq 2500$ within $4 \cdot 10^{-4}$, but the constant 2320 is forced by $\Lambda = 1$ at birth and the range is one decade; the folds move like $\vartheta^{1.22}$ and $\vartheta^{1.37}$, which is not established as a power law (a form $\vartheta \ln^2 \vartheta$ is compatible with one value); birth occurs between $\vartheta = 2300$ and 2400, with $\vartheta_c \approx 2.3 \cdot 10^3$ (Numerical observation 9.63). At D^* the exponents are 1.31 (lower fold) and 1.94 (upper fold) over $[100, 160]$ and Λ crosses 1 near $\vartheta \approx 25.6$ (Numerical observation 9.64). The plateau picture of Heuristic 9.66 explains the growth like $\ln \vartheta$ and nothing more: a quantitative test of the plateau value of ρ fails. If Conjecture 13.18 holds, the $\frac{3}{2}$ law of the width near ϑ_c follows (Corollary 9.32); the questions here are about the *large-ϑ* side.

Why it is hard. The constant $\kappa_{**}(\vartheta)$ of Theorem 10.39 cannot be uniform in ϑ (the window moves to infinity with ϑ), so one needs the large-swirl theory with $\ln \vartheta \sim J$, the same regime as Open problem 13.21.

Suggestion. Derive the amplification Λ from the rank-one structure of Proposition 10.35 with the second term of (8.6) kept (the response of ρ accumulated over a distance $\sim \ln \vartheta$), and compare with the profiles at several ϑ.

13.5. The regimes of large and small swirl

In words. *Chapter 10 proves the $\sqrt{\kappa_\mathrm{D}}$ law and the layer structure, Chapter 11 the critical bubble as $\kappa \to 0$. What is left are the edges of the regimes where the theorems apply.*

Open problem 13.20 (large swirl outside the admissible set)**.** *For $\boldsymbol{c}' \notin \operatorname{int} \mathcal{J}$, which does include the data below the no-swirl set, prove the large-swirl theory: uniqueness of the fixed point and non-degeneracy as $\kappa_\mathrm{D} \to \infty$, i.e. hypothesis ($H_\infty$) of Remark 9.52.*

What is known. For $\boldsymbol{c}' \in \operatorname{int} \mathcal{J}$, Theorem 10.39 proves ($\mathrm{H}_\infty$) and Theorem 11.58(c),(d) follow. For $\boldsymbol{c}' \notin \operatorname{int} \mathcal{J}$ the potential W_e is not admissible, T_D is defined only on an open subset of $\mathrm{BC}(\mathbb{R})$, and the argument is not claimed (Remark 10.41); existence of a solution for every D is proved in all cases (Theorem 8.20).

Why it is hard. The $\sqrt{\kappa}$ law uses a positive solution of $z'' = W_\mathrm{e} z$ with controlled Wronskian, which does not exist when W_e is not admissible.

Suggestion. Replace W_e by W_f with the centrifugal term and use that the large swirl itself makes the potential admissible: the plateau $W \approx J^2/16$ at the equator (Atlas 10.48).

Open problem 13.21 (the plateau and the regime $|\ln\vartheta| \sim J$)**.** *Extend the $\sqrt{\kappa_D}$ law (Theorem 10.11) and its uniqueness statement to (a) the regime $|\ln\vartheta| \sim J$, which contains the ridge of Open problem 13.19, and (b) the data with $f_- < 0$ for which the solution may sit on a pinned plateau $J \approx (1+\vartheta)|f_-|$ before the $\sqrt{\kappa_D}$ law takes over. Give also the explicit constants at the next order beyond the $O(\kappa_D^{-1/2})$ of Theorem 10.22.*

What is known. Theorem 10.11 and Theorem 10.22 need $J \gtrsim (1+\vartheta)|f_-|$ and $|\ln\vartheta| \lesssim J$; both conditions are real and not only artefacts of the proof (Remark 10.41, Atlas 10.48(6),(7)): a class-Z solution may sit on the plateau, and in the second regime the leading eigenvalue comes within $1.02 \cdot 10^{-2}$ of 1. The remainder of the next-order expansion is observed in Numerical observation 10.25.

Why it is hard. In both regimes the logistic layer of width $O(1/J)$ interacts with another scale: the pinned value of f in (b), the distance $\ln\vartheta$ of the layer from the equator in (a).

Suggestion. Treat the layer position as an additional unknown in the fixed-point problem (a collective coordinate) and apply the Lyapunov–Schmidt reduction in it.

Conjecture 13.22 (sharp rates for the bubble)**.** *Under (H) of Theorem 11.6, as $\kappa \downarrow 0$: (a) for $k_1 = k_2 = 1$ and $\boldsymbol{c}' = (0, 0, c_3')$, $\vartheta = 1$, $f_- = 0$, $(J - J_*)/\kappa \to \Gamma(M) = \frac{3}{M^2} - \frac{1}{16}$, which changes sign at $M^2 = 48$; (b) for $k = \max k_i > 1$, $J - J_* = -C_k \kappa^{1/k}(1 + o(1))$ with $C_k > 0$ the constant of Formal computation 11.27.*

What is proved. The bound $|J - J_*| \le C\varepsilon_k(\kappa_D)$, with $\varepsilon_k = \kappa_D$, $\kappa_D \ln^2(1/\kappa_D)$, $\kappa_D^{1/k} \ln(1/\kappa_D)$ for $k < 1$, $k = 1$, $k > 1$ respectively, and the limit Γ_{rate} for $k_1, k_2 < 1$ (Theorem 11.10(ii),(iii)); the convergence of solutions to the bubble for all k_i (Theorem 11.6).

Evidence. The composite-profile predictor of Formal computation 11.23 reproduces the computed $J - J_*$ in the three cases (Numerical observation 11.24), and the case $k > 1$ is observed in Numerical observation 11.28. The predictions are formal: Formal computation 11.25 and Formal computation 11.27, summarised in Formal computation 11.29. The sign change at $M^2 = 48$ has not been tested.

Why it is hard. A rigorous rate at $k \ge 1$ needs a uniform comparison of $u_\pm$ with the critical function g inside the layers; for $k = 1$ the logarithm that the proved bound allows must be shown to cancel.

Suggestion. Use the solvability identity $\int \delta\, g\, u_- = \gamma_c \phi_+(0) w_0$ of Theorem 11.10(i) as the Lyapunov–Schmidt condition at the critical potential, and the Wronskian identities of Chapter 12 for the layer integrals.

Open problem 13.23 (small swirl for $f_- < 0$)**.** *For data below the no-swirl set with $f_- < 0$ (outside (H)), describe the limits of solutions as $\kappa_D \to 0$: which types occur (the trivial branch, the positive bubble, the negative bubble with $C = -M$), how many solutions of each type there are, and the bubble eigenvalue law $\nu_b \sim \mp c/\kappa_D$ of Formal computation 11.32.*

What is known. Existence for all $\kappa > 0$ is proved below the no-swirl set (Theorem 11.58(b)) but the limits are described only for $f_- \ge 0$. At $\boldsymbol{c}' = (0, 0, -8)$, $\vartheta = 1$, $f_- = -5$ the continuation shows three solutions, a trivial branch and a positive and a negative bubble, with indices $(+1, -1, +1)$ adding up to 1 in agreement

with the parity statement (Numerical observation 11.34). The eigenvalue law is a formal one-dimensional reduction, not derived from the linearisation of T_D (Formal computation 11.32, Numerical observation 11.33).

Why it is hard. The negative bubble needs $f_- < -M$ for $J_C > 0$, outside the sign structure on which Theorem 11.6 rests, and the remaining spectrum is not controlled.

Suggestion. Generalise the exit argument used for existence to the negative bubble, treating $f_- < -M$ as the base case; the Leray–Schauder degree count then predicts the number of solutions of each type.

13.6. Beyond axisymmetry: an outlook

Open problem 13.24 (non-axisymmetric solutions and stability)**.** *(a) Are there non-axisymmetric (-1)-homogeneous solutions near the large-swirl solutions of this book, or bifurcating from the fold points of Section 13.4? (b) Which of the solutions here are stable, for the time-dependent Navier–Stokes equations or at least within the class of conical flows?*

Everything in this book is about axisymmetric stationary solutions. The (-1)-homogeneous solutions that are smooth on the whole sphere are Landau's solutions [21]: this is the theorem of Tian and Xin [47] for axisymmetric solutions and of Šverák [49] in general; solutions with singular rays are the subject of [24–26], and asymptotic stability of some of them has been studied by Li and Yan [27]. The classical literature on conical swirling flows reports folds, hysteresis and symmetry breaking at finite swirl [10, 36, 40, 41, 44]. The S-shaped branches and the ridge of Chapter 9 indicate where bifurcations should be looked for: at a fold of the axisymmetric solution set the linearised stationary problem has a non-trivial axisymmetric kernel with $\delta D = 0$, and a kernel of this kind is where branches of solutions, and changes of stability, typically occur. The concentrated shear layer of the large-swirl regime (Chapter 10) suggests shear instabilities for large J, and a WKB analysis of the linearised operator in the layer variable $J\xi$ is a natural first step.

What is missing is a linearisation of the stationary equations at the solutions of this book, in a class of (-1)-homogeneous perturbations with the right behaviour at the singular rays, its spectrum in each azimuthal Fourier mode $\mathrm{e}^{\mathrm{i}m\varphi}$, and the corresponding time-dependent analysis. This is hard because the base solutions are known only implicitly, they have four different asymptotic types at the poles so that the function spaces must be weighted differently at each pole, and for $m \neq 0$ the perturbation equations no longer reduce to the scalar Riccati–Schrödinger structure used throughout. A Lyapunov–Schmidt reduction on the sphere with the two poles removed, in weighted spaces adapted to the pole charts of Chapter 4, would decide whether non-axisymmetric branches bifurcate; this is a suggestion only.

Where this leaves the picture. The classification is complete in the following sense: every solution is in exactly one of the families of Theorem 7.90, and the solution set is described by parameters in a region whose thresholds are characterised, continuous and estimated. What is open falls into three groups. The *structural* questions are the interval property (Conjecture 13.1), the topology and smoothness of Ω (Open problem 13.7, Conjecture 13.2, Conjecture 13.3), and the types on the edge (Open problem 13.5). The *quantitative* questions are the shape of the lens (Conjecture 13.8), the constants in the laws (Conjecture 13.9, Open problem 13.10,

Open problem 13.11, Open problem 13.12, Open problem 13.13), and the sharp rates in the two regimes (Conjecture 13.22, Open problem 13.21). The *global* questions are about uniqueness: the number of solutions (Open problem 13.16), the fold set (Open problem 13.17, Conjecture 13.18, Open problem 13.19) and the stability of the solutions (Open problem 13.24).

APPENDIX A

Background results

In words. *Every proof in this book uses, besides its own earlier results, only a short list of textbook theorems. This appendix states them* precisely*, with a reference, and shows each one at work on a toy example. Where the book uses a version that is not literally the textbook's (a contraction principle with parameters, the cone lemma and Kamke comparison with forcing, a Taylor lemma in weighted spaces, degree theory on non-cylindrical sets, the Whitney fold and cusp, Krawczyk's lemma) the proof is written out, so that no step is left to the reader. Nothing here depends on the rest of the book.*

Roadmap. Appendix A.1 collects calculus and ODE basics (implicit function theorem, existence and dependence, Gronwall, Volterra equations, the L^1 case of Levinson's theorem). Appendix A.2 is the uniform contraction principle with parameters. Appendix A.3 treats smoothness with a loss of weight and Whitney's extension theorem. Appendix A.4 proves the cone lemma and Kamke's comparison theorem with forcing. Appendix A.5 states Arzelà–Ascoli (also on the line), Schauder's and Dugundji's theorems and the Riesz–Schauder theory; Appendix A.6 the Leray–Schauder degree, its extension to non-cylindrical sets, and the continuation principle. Appendix A.7 proves the Whitney fold and cusp normal forms and the Lyapunov–Schmidt reduction; Appendix A.8 interval arithmetic and Krawczyk's lemma. Appendix A.9 is the index of all results with their uses. The toy examples are reproduced by `make -C code/appA_background checks`.

A.1. Calculus and ODE basics

In words. *These are the facts every analyst knows and every argument uses: a map with invertible derivative can be inverted smoothly; an ODE has a unique maximal solution which depends smoothly on its data; a linear integral inequality has an explicit solution (Gronwall); a* Volterra *equation, whose kernel only looks forward (or backward) in the variable, always has a unique solution, with no smallness assumption. The last item, the L^1 case of Levinson's theorem, is what makes the exponents $e^{\pm k\xi}$ of a Schrödinger equation $z'' = (k^2+q)z$ with integrable q meaningful.*

Lemma A.1 (Neumann series)**.** *Let $\mathbb{B}$ be a Banach space and $B : \mathbb{B} \to \mathbb{B}$ bounded linear with $\|B\| < 1$. Then $I - B$ is invertible, $(I-B)^{-1} = \sum_{n\ge 0} B^n$ and $\|(I - B)^{-1}\| \le (1 - \|B\|)^{-1}$.*

Proof. $\sum \|B^n\| \le \sum \|B\|^n < \infty$, so $S = \sum B^n$ converges in operator norm, and $(I-B)S = S(I-B) = \lim(I - B^{N+1}) = I$. □

Theorem A.2 (Banach's fixed point theorem). *Let (M, d) be a nonempty complete metric space and $T : M \to M$ with $d(Tu, Tu') \le \theta d(u, u')$ for some $\theta < 1$.*

Then T has exactly one fixed point u^*, and $d(T^n u_0, u^*) \le \theta^n (1-\theta)^{-1} d(Tu_0, u_0)$ for every $u_0 \in M$.

PROOF. $d(T^{n+1}u_0, T^n u_0) \le \theta^n d(Tu_0, u_0)$, so $(T^n u_0)$ is Cauchy, with $d(T^n u_0, T^{n+m}u_0) \le \theta^n (1-\theta)^{-1} d(Tu_0, u_0)$; its limit u^* satisfies $Tu^* = u^*$ by continuity, and two fixed points u, u' have $d(u,u') \le \theta d(u,u')$, so $u = u'$. (Banach [1].) □

THEOREM A.3 (implicit function theorem in Banach spaces). *Let X, Y, Z be Banach spaces, $\mathcal{U} \subset X \times Y$ open, $k \in \{1, 2, \dots, \infty\}$, $F : \mathcal{U} \to Z$ of class C^k, $F(x_0, y_0) = 0$, and $\mathrm{D}_y F(x_0, y_0) : Y \to Z$ a bounded linear bijection. Then there are open neighbourhoods $X_0 \ni x_0$, $Y_0 \ni y_0$ and a unique map $g : X_0 \to Y_0$ with $\{(x,y) \in X_0 \times Y_0 : F(x,y) = 0\} = \{(x, g(x)) : x \in X_0\}$; g is of class C^k and $\mathrm{D}g = -(\mathrm{D}_y F)^{-1} \mathrm{D}_x F$ along the graph. If F is real-analytic, so is g.*

REFERENCE. C^k case: Dieudonné [8, Ch. X, (10.2.1)]; Deimling [7]. Analytic case: Whittlesey [54]; Zeidler [57, Thm. 4.B]. The inverse function theorem is the case $F(x,y) = x - \Phi(y)$. □

THEOREM A.4 (existence, continuation, dependence on data). *Let $\mathcal{U} \subset \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^m$ be open and $F : \mathcal{U} \to \mathbb{R}^n$ continuous with $\partial_y F$ continuous (so F is locally Lipschitz in y). For (t_0, y_0, p) with $(t_0, y_0, p) \in \mathcal{U}$ consider $y' = F(t, y, p)$, $y(t_0) = y_0$.*

- *(a) (Picard–Lindelöf) There is a unique maximal solution $y(\cdot; t_0, y_0, p)$, defined on an open interval $(\alpha, \beta) \ni t_0$.*
- *(b) (continuation) If $\beta < \infty$, then $(t, y(t), p)$ leaves every compact subset of $\mathcal{U}$ as $t \uparrow \beta$. In particular, if $\mathcal{U} = I \times \mathbb{R}^n \times P$ and $\beta < \sup I$, then $|y(t)| \to \infty$ as $t \uparrow \beta$. Likewise at α.*
- *(c) (dependence) The set $\{(t, t_0, y_0, p) : \alpha < t < \beta\}$ is open and $y(t; t_0, y_0, p)$ is continuous on it. If F is of class C^k $(k \ge 1)$ in (t, y, p), so is y in (t, t_0, y_0, p), and $J = \partial_{y_0} y$, $J_p = \partial_p y$ solve the variational equations $J' = \partial_y F(t, y, p) J$, $J(t_0) = I$; $J_p' = \partial_y F \, J_p + \partial_p F$, $J_p(t_0) = 0$.*

REFERENCE. Hartman [14, Ch. II, §1 and §3; Ch. V, §3 and §4]; Coddington–Levinson [6, Ch. 1–2]; Teschl [46, Ch. 2]. □

Lemma A.5 (Gronwall)**.** *Let u, α be continuous on $[t_0, t_1]$, $\beta \ge 0$ continuous, and $u(t) \le \alpha(t) + \int_{t_0}^t \beta u \, \mathrm{d}s$ for all t. Then $u(t) \le \alpha(t) + \int_{t_0}^t \alpha(s)\beta(s) \exp\big(\int_s^t \beta\big) \mathrm{d}s$. If α is nondecreasing, $u(t) \le \alpha(t) \exp \int_{t_0}^t \beta$. The same holds backward: if $u(t) \le \alpha(t) + \int_t^{t_1} \beta u \, \mathrm{d}s$ then $u(t) \le \alpha(t) + \int_t^{t_1} \alpha\beta \exp(\int_t^s \beta) \mathrm{d}s$, and $u \le \alpha(t) \exp \int_t^{t_1} \beta$ if α is nonincreasing.*

PROOF. Put $R(t) = \int_{t_0}^t \beta u \, \mathrm{d}s$ and $B(t) = \int_{t_0}^t \beta$. Then $R' = \beta u \le \beta\alpha + \beta R$ because $\beta \ge 0$, so $(Re^{-B})' \le \alpha\beta e^{-B}$; integrating from t_0 (where $R = 0$) gives $R(t) \le \int_{t_0}^t \alpha\beta e^{B(t)-B(s)} \mathrm{d}s$, and $u \le \alpha + R$. If α is nondecreasing, $\int_{t_0}^t \alpha(s)\beta(s) e^{B(t)-B(s)} \mathrm{d}s \le \alpha(t) \int_{t_0}^t \beta e^{B(t)-B(s)} \mathrm{d}s = \alpha(t)(e^{B(t)} - 1)$. The backward version is the forward version applied to $t \mapsto u(t_0 + t_1 - t)$. □

Lemma A.6 (Volterra equations)**.** *Let $-\infty < a < b \le \infty$, $g : [a, b) \to \mathbb{R}^n$ bounded and continuous, $k(\xi, \eta)$ a continuous $n \times n$-matrix function on $\{a \le \xi \le \eta < b\}$ with*

$|k(\xi,\eta)| \le m(\eta)$, where $m \ge 0$ is continuous and $M(\xi) := \int_\xi^b m < \infty$. Then

$$u(\xi) = g(\xi) + \int_\xi^b k(\xi,\eta)u(\eta)\,\mathrm{d}\eta \qquad (a \le \xi < b)$$

has a unique bounded continuous solution; it satisfies $|u(\xi)| \le \|g\|_\infty e^{M(\xi)}$ and $|u(\xi) - g(\xi)| \le \|g\|_\infty e^{M(\xi)} M(\xi)$. The same holds for the forward equation $u(\xi) = g(\xi) + \int_a^\xi k(\xi,\eta)u(\eta)\mathrm{d}\eta$ with $|k(\xi,\eta)| \le m(\eta)$ on $\{a \le \eta \le \xi\}$, $M(\xi) = \int_a^\xi m$ (reflect $\xi \mapsto -\xi$).

Proof. Let $(Ku)(\xi) = \int_\xi^b k(\xi,\eta)u(\eta)\mathrm{d}\eta$ and $u_0 = g$, $u_{j+1} = g + Ku_j$ (continuous by dominated convergence). Claim: $|u_{j+1} - u_j|(\xi) \le \|g\|_\infty M(\xi)^{j+1}/(j+1)!$. For $j = 0$: $|Kg| \le \|g\|_\infty M$. If it holds for $j-1$, then, since $-M' = m$, $|u_{j+1} - u_j|(\xi) \le \int_\xi^b m(\eta)\|g\|_\infty \frac{M(\eta)^j}{j!}\mathrm{d}\eta = \|g\|_\infty \frac{M(\xi)^{j+1}}{(j+1)!}$. Hence $u_j \to u$ uniformly, $u = g + Ku$, $|u| \le \|g\|_\infty \sum_j M^j/j! = \|g\|_\infty e^M$, and $|u - g| \le \|g\|_\infty (e^M - 1) \le \|g\|_\infty M e^M$. If d is a bounded solution of $d = Kd$, the same induction gives $|d|(\xi) \le \|d\|_\infty M(\xi)^j/j!$ for every j, so $d = 0$. □

Toy example A.7 (Volterra)**.** $u(\xi) = 1 + p\int_0^\xi u$ on $[0,1]$ is the forward equation with $k \equiv p$, $m \equiv |p|$, and the lemma gives $|u| \le e^{|p|}$; the solution is $e^{p\xi}$. For $b = \infty$ take $u(\xi) = 1 - \int_\xi^\infty e^{-\eta}u(\eta)\mathrm{d}\eta$ ($k = -e^{-\eta}$, $M(\xi) = e^{-\xi}$): $u = \exp(-e^{-\xi}) \to 1$ as $\xi \to \infty$. This is the shape of the Jost and pole-chart equations of Chapters 3 and 4: the kernel is bounded by an integrable weight, no smallness is needed.

Proposition A.8 (Levinson, L^1 case)**.** *Let $k > 0$, $R \in \mathbb{R}$, $q : [R,\infty) \to \mathbb{R}$ continuous with $\int_R^\infty |q| < \infty$. Then $z'' = (k^2 + q)z$ has a solution Ψ with*

$$\Psi(\xi) = e^{-k\xi}(1 + o(1)), \qquad \Psi'(\xi) = -ke^{-k\xi}(1 + o(1)) \qquad (\xi \to \infty),$$

it is the only solution with $\Psi e^{k\xi} \to 1$. Every solution z not proportional to Ψ satisfies, with $c = \mathrm{Wr}(\Psi, z)/(2k) \ne 0$, $z = ce^{k\xi}(1 + o(1))$ and $z' = kce^{k\xi}(1 + o(1))$. In particular Ψ is, up to a factor, the unique solution that does not grow.

Proof. *Existence.* Put $\Psi = e^{-k\xi}v$. Then $\Psi'' = (k^2+q)\Psi$ means $v'' - 2kv' = qv$. Consider the Volterra equation

$$v(\xi) = 1 + \int_\xi^\infty \frac{1 - e^{-2k(\eta-\xi)}}{2k}\, q(\eta)v(\eta)\,\mathrm{d}\eta.$$

Its kernel lies in $[0, \frac{1}{2k}]$, so Lemma A.6 (with $m = |q|/2k$) gives a unique bounded solution, $|v| \le e^{\|q\|_1/2k}$. Differentiating (the kernel vanishes at $\eta = \xi$): $v'(\xi) = -\int_\xi^\infty e^{-2k(\eta-\xi)}qv\,\mathrm{d}\eta$, i.e. $v' = -e^{2k\xi}\int_\xi^\infty e^{-2k\eta}qv$, hence $v'' = 2kv' + qv$. Moreover $|v - 1| \le \frac{1}{2k}\|v\|_\infty \int_\xi^\infty |q| \to 0$ and $|v'| \le \|v\|_\infty \int_\xi^\infty |q| \to 0$. So $\Psi = e^{-k\xi}v$ has $\Psi e^{k\xi} \to 1$ and $\Psi' = e^{-k\xi}(v' - kv) = -ke^{-k\xi}(1 + o(1))$. *Other solutions.* Let z be a solution, $W = \mathrm{Wr}(\Psi, z) = \Psi z' - \Psi' z$ (constant since both solve the same equation). If $W = 0$ then $z \propto \Psi$. Otherwise $(z/\Psi)' = W/\Psi^2$, and $\Psi^{-2} = e^{2k\eta}(1 + o(1))$ gives $\int_{R'}^\xi \Psi^{-2} = \frac{e^{2k\xi}}{2k}(1+o(1))$ for R' large. Hence $z = \Psi\,(c_0 + W\int_{R'}^\xi \Psi^{-2}) = \frac{W}{2k}e^{k\xi}(1+o(1))$ and $z' = \Psi'\frac{z}{\Psi} + \frac{W}{\Psi} = \big(-\frac{1}{2}+1\big)We^{k\xi}(1 + o(1)) = \frac{W}{2}e^{k\xi}(1 + o(1)) = kc\,e^{k\xi}(1 + o(1))$ with $c = W/2k$. *Uniqueness.* If $\tilde\Psi$ is another solution with $\tilde\Psi e^{k\xi} \to 1$, then $d = \tilde\Psi - \Psi$ satisfies $de^{k\xi} \to 0$. By the last paragraph either $d = s\Psi$, and then $se^{-k\xi}e^{k\xi} = s \to 0$, so $d = 0$; or d is not proportional to Ψ and $|d| \sim |c|e^{k\xi}$ with $c \ne 0$, impossible. □

Remark A.9. This is the scalar case of Levinson's theorem on asymptotic integration [23], [6, Ch. 3, Thm. 8.1], [14, Ch. X]; the book needs the weighted variants (e.g. $\int |q|\, e^{2\epsilon\xi} < \infty$ gives corrections $O(e^{-2\epsilon\xi})$), which follow from the same Volterra equation and Lemma A.6 with $m(\eta) = |q|(\eta)/2k$; see Chapter 3.

Toy example A.10 (Levinson)**.** Take $q = 1/(1+\xi^2) \in L^1$, $k = 1$. The Volterra equation for v is solved numerically; $v(0) \approx 1.75$, $v(10) \approx 1.05$, $v(50) \approx 1.01$, so $v \to 1$ but only like $\int_\xi^\infty q \sim 1/\xi$, and the residual of $v'' - 2v' - qv$ is 10^{-5} (`toy_levinson.py`). The convergence $v \to 1$ is slow because q is merely integrable, not exponentially decaying.

A.2. The uniform contraction principle with parameters

In words. *A contraction has a unique fixed point; if the contraction depends smoothly on a parameter, so does the fixed point,* provided the contraction constant and the ball do not degenerate as the parameter varies. *This is the engine behind every smoothness statement of the pole charts (Chapter 4) and of the thresholds (Chapter 7): smoothness of a solution in its parameters is a corollary of this one principle. The only danger is a parameter that changes the function space; Toy example A.14 shows the cure.*

THEOREM A.11 (uniform contraction principle with parameters). *Let X be a Banach space, P an open subset of $\mathbb{R}^m$ (or of a Banach space), $O \subset X$ open, $k \in \{1, 2, \dots, \infty\}$, and $T : P \times O \to X$ of class C^k. Let $B \subset O$ be closed in X, and assume*

(i) $T(p, B) \subset B$ for every $p \in P$;

(ii) $\|T(p,u) - T(p,u')\| \le \theta \|u - u'\|$ for all $u, u' \in B$, $p \in P$, with one $\theta < 1$ for all p;

(iii) for every p, the fixed point $u(p) \in B$ of $T(p,\cdot)$ (it exists and is unique by (i), (ii)) lies in the interior of B.

(Condition (iii) holds automatically if $T(p, B) \subset \operatorname{int} B$, since $u(p) = T(p, u(p))$.) Then $p \mapsto u(p)$ is of class C^k,

$$\mathrm{D}_p u = (I - \mathrm{D}_u T)^{-1} \mathrm{D}_p T \qquad \text{evaluated at } (p, u(p)),$$

and $\|\mathrm{D}_u T(p, u(p))\| \le \theta$. If T is real-analytic, so is u.

PROOF. *Existence and continuity.* B is a nonempty closed subset of a Banach space, hence complete; by (i), (ii) Banach's fixed point theorem (Theorem A.2) gives a unique fixed point $u(p) \in B$ for each p. For $p, p' \in P$,

$$\begin{aligned} \|u(p) - u(p')\| &= \|T(p, u(p)) - T(p', u(p'))\| \\ &\le \theta \|u(p) - u(p')\| + \|T(p, u(p')) - T(p', u(p'))\|, \end{aligned}$$

so $\|u(p) - u(p')\| \le (1-\theta)^{-1} \|T(p, u(p')) - T(p', u(p'))\| \to 0$ as $p \to p'$ (continuity of T at $(p', u(p'))$).

The derivative is small. Fix p_0 and $u_0 = u(p_0)$, an interior point of B. For $h \in X$ and small $t > 0$, $u_0 + th \in B$, so by (ii) $\|T(p_0, u_0 + th) - T(p_0, u_0)\| \le \theta t \|h\|$. Dividing by t and letting $t \to 0$ gives $\|\mathrm{D}_u T(p_0, u_0) h\| \le \theta \|h\|$.

Smoothness. Put $F(p, u) = u - T(p, u)$, a C^k map $P \times O \to X$ with $F(p_0, u_0) = 0$. Since $\|\mathrm{D}_u T(p_0, u_0)\| \le \theta < 1$, $\mathrm{D}_u F = I - \mathrm{D}_u T$ is invertible by Lemma A.1, with

inverse of norm $\le (1-\theta)^{-1}$. By Theorem A.3 there are neighbourhoods $P_0 \ni p_0$, $O_0 \ni u_0$ and a C^k map $\tilde{u} : P_0 \to O_0$ such that $\{F = 0\} \cap (P_0 \times O_0)$ is the graph of $\tilde{u}$. By continuity of u we may shrink P_0 so that $u(p) \in O_0$ for $p \in P_0$; as $F(p, u(p)) = 0$, $u = \tilde{u}$ on P_0. Thus u is C^k near each p_0. Differentiating $F(p, u(p)) = 0$ gives the formula. The analytic case is the analytic implicit function theorem (Theorem A.3, analytic case). □

Remark A.12 (why "uniform")**.** Both θ and the ball B must be independent of p (or at least locally uniform). If $T(p, \cdot)$ contracts only with a constant $\theta(p) \to 1$, or the fixed point approaches ∂B, the principle no longer *applies*; it does not follow that the fixed point is non-smooth (it is still C^k wherever $\mathrm{I} - \mathrm{D}_u T$ is invertible, by the implicit function theorem), but the bounds that the principle supplies, $\|\mathrm{D}_p u\| \le (1-\theta)^{-1} \|\mathrm{D}_p T\|$, blow up. See the end of Toy example A.13.

Toy example A.13 ($u = p + u^2/4$)**.** $T(p, u) = p + u^2/4$ on $P = (0, 0.9)$, $B = [0, \frac{3}{2}]$. Then $T(p, B) \subset [0, 0.9 + 0.5625] \subset B$ and $|\partial_u T| = u/2 \le \frac{3}{4} = \theta$ on B, and the fixed point $u(p) = 2 - 2\sqrt{1-p} \in (0, 1.37]$ is interior. The theorem gives $u'(p) = (1 - u/2)^{-1} \cdot 1 = (1-p)^{-1/2}$, which is indeed the derivative of the closed form. For $p \to 1$ the two roots of $u = p + u^2/4$ merge, and $\partial_u T \to 1$ at the fixed point: no uniform $\theta < 1$ exists, and indeed $u'(p) = (1-p)^{-1/2} \to \infty$: the derivative bound is lost. (Check: `toy_ucp.py`.) For a Volterra equation, $u(\xi) = 1 + p \int_0^\xi u$ on $C[0, 1]$ with $|p| < \frac{1}{2}$: $T(p, u) = 1 + p \int_0^\xi u$ is affine in u with $\|\mathrm{D}_u T\| \le |p| \le \frac{1}{2}$, so B may be any closed ball; the fixed point $e^{p\xi}$ has $\mathrm{D}_p u = (I - p\textstyle\int)^{-1} \int_0^\xi u = \xi e^{p\xi}$, as the theorem predicts. This is the structure of all chart equations: a Volterra-type operator whose kernel depends smoothly on p.

Toy example A.14 (parameters that enter the weights)**.** Let $u_p(\xi) = e^{-p\xi}$ and $\|u\|_a = \sup_{\xi \ge 0} e^{a\xi} |u(\xi)|$. For every $p > a$: $\|u_p - u_a\|_a = \sup_\xi (1 - e^{-(p-a)\xi}) = 1$, although $p \to a$. So $p \mapsto u_p$ is not even continuous into the fixed-weight space (numerically 0.9975, 0.45, 0.058 at $\xi \le 60$ for $p = 1.1, 1.01, 1.001$; the supremum is attained only at $\xi \to \infty$). The cure used in Chapter 4: rescale, $\hat{u}_p = e^{p\xi} u_p \equiv 1$, so that the unknown lives in a space *not* depending on p; or use one fixed weaker weight for all p in a compact range. Theorem A.11 is then applied to the rescaled equation.

A.3. Smoothness with loss of weight; Whitney's extension theorem

In words. *Maps that send a parameter to a* function *are often smooth only if each derivative is measured in a weaker weight: differentiating* $(1 + te^\xi)^{-2}$ *in* t *costs a factor* e^ξ *each time. The Taylor lemma says that this is the only loss. Whitney's theorem lets us treat a function that is smooth up to the boundary of a closed convex set (such as the closed Jost region of Chapter 4) as the restriction of a smooth function on the whole space, so that the inverse function theorem may be used at boundary points.*

For an interval $J \subset \mathbb{R}$ and a continuous weight $\mathfrak{w} : J \to (0, \infty)$ let $\mathcal{X}_\mathfrak{w} = \{\varphi \in C(J) : \|\varphi\|_\mathfrak{w} := \sup_J |\varphi|/\mathfrak{w} < \infty\}$; it is a Banach space.

Lemma A.15 (Taylor lemma)**.** *Let* $\mathcal{D}$ *be a convex open subset of a normed space* V*,* $k \ge 0$*, and* $G : \mathcal{D} \to C(J)$ *such that for every* $\xi \in J$ *the map* $p \mapsto G(p)(\xi)$ *is of*

class C^{k+1} and

$$|\mathrm{D}^j G(p)[h^j](\xi)| \le K_j\, \mathfrak{w}(\xi)\, \|h\|^j \qquad (p \in \mathcal{D},\ h \in V,\ 0 \le j \le k+1,\ \xi \in J).$$

Then $G : \mathcal{D} \to \mathcal{X}_{\mathfrak{w}}$ is of class C^k, its derivatives are the pointwise ones, and $\mathrm{D}^k G$ is Lipschitz, with a constant $c_k K_{k+1}$ where c_k depends only on k.

Proof. *Step 0: polarisation.* A symmetric j-linear form S satisfies $\sup_{\|h_i\|\le 1}|S[h_1,\dots,h_j]| \le \frac{j^j}{j!}\sup_{\|h\|\le 1}|S[h^j]|$. Hence every $\mathrm{D}^j G(p)$ $(j \le k+1)$ has the pointwise multilinear bound

$$|\mathrm{D}^j G(p)[h_1,\dots,h_j](\xi)| \le \frac{j^j}{j!} K_j \mathfrak{w}(\xi) \prod \|h_i\|.$$

In particular $\Gamma_j(p) : (h_1,\dots,h_j) \mapsto \mathrm{D}^j G(p)[h_1,\dots,h_j](\cdot)$ is a bounded j-linear map into $\mathcal{X}_{\mathfrak{w}}$, *provided* the function $\xi \mapsto \mathrm{D}^j G(p)[h^j](\xi)$ is continuous; this follows because it is the $\mathcal{X}_{\mathfrak{w}}$-limit of the continuous difference quotients, by the estimate of Step 1. *Step 1: Fréchet differentiability of order k.* Fix p, h with $p + h \in \mathcal{D}$ (the segment lies in $\mathcal{D}$). Taylor's formula with integral remainder for the C^{k+1} function $t \mapsto G(p+th)(\xi)$ gives

$$\Big| G(p+h)(\xi) - \sum_{j\le k} \tfrac{1}{j!}\mathrm{D}^j G(p)[h^j](\xi) \Big| \le \frac{K_{k+1}}{(k+1)!}\mathfrak{w}(\xi)\|h\|^{k+1}.$$

Dividing by $\mathfrak{w}(\xi)$ and taking the supremum over ξ, the remainder in $\mathcal{X}_{\mathfrak{w}}$ is $O(\|h\|^{k+1})$, so G is Fréchet differentiable at p as a map into $\mathcal{X}_{\mathfrak{w}}$ to the order k (Taylor polynomial with the pointwise derivatives). *Step 2: higher derivatives and regularity.* Apply Step 1 to $\mathrm{D}^j G$ in place of G $(j < k)$: the map $p \mapsto \mathrm{D}^j G(p)[h_1,\dots,h_j]$ satisfies the hypotheses with $k - j$ in place of k for each fixed h_i, with constants multiplied by $\prod\|h_i\|$; hence $p \mapsto \Gamma_j(p)$ is differentiable with derivative Γ_{j+1}. The remainder estimate is uniform for $\|h_i\| \le 1$ (it is a bound on a multilinear form, controlled by polarisation through K_{j+1}, K_{j+2}), so the differentiability holds in the operator norm of the space of j-linear maps $V^j \to \mathcal{X}_{\mathfrak{w}}$, which is what C^k means. The Lipschitz bound is the mean value inequality: for $\|e\| = 1$, $|(\mathrm{D}^k G(p+h) - \mathrm{D}^k G(p))[e^k](\xi)| \le \int_0^1 |\mathrm{D}^{k+1} G(p+th)[h, e^k](\xi)|\mathrm{d}t \le \frac{(k+1)^{k+1}}{(k+1)!} K_{k+1}\mathfrak{w}(\xi)\|h\|$, and polarisation in e (Step 0) gives the claim. Thus $\mathrm{D}^k G$ is Lipschitz, in particular continuous, and $G \in C^k$. □

Toy example A.16 (positivity and exponential loss)**.** Let $J = [0,\infty)$ and $G(t)(\xi) = (1 + te^{\xi})^{-2}$, $t \in \mathcal{D} = (0,\infty)$ (an open set, as the lemma requires; the derivatives extend continuously to $t = 0$) (the model of z^{-2} for $z = 1 + te^{\xi}$: a perturbation growing as fast as a dominant solution). In the sup norm G is *not continuous* at $t = 0$: $\sup_\xi |G(t) - G(0)| = 1$ for every $t > 0$. But $\partial_t^j G(t)(\xi) = (-1)^j (j+1)!\, e^{j\xi}(1+te^{\xi})^{-2-j}$, so $|\partial_t^j G| \le (j+1)!\, e^{j\xi} \le (j+1)!\, e^{(k+1)\xi}$ for $j \le k+1$, and Lemma A.15 gives $G \in C^k$ into $\mathcal{X}_{\mathfrak{w}}$, $\mathfrak{w} = e^{(k+1)\xi}$: each derivative costs one more factor e^{ξ}. Numerically the weighted Taylor remainder is $\le (k+2)h^{k+1}$ for $k = 1, 2, 3$ (`toy_taylor.py`), the constant $K_{k+1}/(k+1)!$ of the proof. In the pole charts of Chapter 4 the same happens with e^{ξ} replaced by a power of the dominant solution, which is harmless once the exponent is small compared with $1/k$.

Theorem A.17 (Whitney's extension theorem). *Let $K \subset \mathbb{R}^n$ be closed, $k \ge 0$, and $(f^\alpha)_{|\alpha|\le k}$ continuous functions on K ("a k-jet"). Put, for $x, y \in K$ and $|\alpha| \le k$, $R_\alpha(y;x) = f^\alpha(y) - \sum_{|\beta|\le k-|\alpha|} f^{\alpha+\beta}(x)\frac{(y-x)^\beta}{\beta!}$. Assume* Whitney's condition*: for*

every compact $K' \subset K$ and every α, $R_\alpha(y;x) = o(|y-x|^{k-|\alpha|})$ as $|y-x| \to 0$, uniformly in $x, y \in K'$. Then there is $F \in C^k(\mathbb{R}^n)$ with $\mathrm{D}^\alpha F = f^\alpha$ on K for all $|\alpha| \le k$.

Reference. Whitney [51]. (For C^∞ extension one needs the condition for every k.) □

Corollary A.18 (convex sets: C^k up to the boundary)**.** *Let $K \subset \mathbb{R}^n$ be closed and convex with nonempty interior, $f \in C^k(\operatorname{int} K)$ such that every derivative $\mathrm{D}^\alpha f$, $|\alpha| \le k$, extends continuously to K and is uniformly continuous on bounded subsets of $\operatorname{int} K$. Then f is the restriction to K of some $F \in C^k(\mathbb{R}^n)$, and $\mathrm{D}^\alpha F = \mathrm{D}^\alpha f$ on K (with $\mathrm{D}^\alpha f$ understood as the continuous extension).*

Proof. Let f^α be the continuous extension of $\mathrm{D}^\alpha f$ to K. We verify Whitney's condition. Let $x, y \in \operatorname{int} K$; the segment $[x,y]$ lies in $\operatorname{int} K$ (convexity), so Taylor's formula with integral remainder for $f^\alpha = \mathrm{D}^\alpha f$ along it gives, with $r = k - |\alpha| \ge 1$, $R_\alpha(y;x) = \sum_{|\beta|=r} \frac{r}{\beta!}(y-x)^\beta \int_0^1 (1-t)^{r-1}\big[f^{\alpha+\beta}(x+t(y-x)) - f^{\alpha+\beta}(x)\big]\mathrm{d}t$ (and $R_\alpha = f^\alpha(y) - f^\alpha(x)$ if $r = 0$). Hence $|R_\alpha(y;x)| \le C_n\,\omega(|y-x|)|y-x|^r$, where ω is the modulus of continuity of the top-order derivatives on the bounded set under consideration. Since $\operatorname{int} K$ is dense in K (closed convex, nonempty interior), and both sides are continuous in $(x,y) \in K \times K$, the estimate extends to all $x, y \in K \cap$ bounded set. Theorem A.17 applies. □

Toy example A.19 (boundary regularity)**.** On $K = [0,\infty)$: $u(t) = t^2 \cos t$ is C^k up to 0 for every k; extend it to $t < 0$ by its Taylor polynomial of order k. The function $u(t) = t^{3/2}$ is C^1 up to 0 but $u'' = \frac{3}{4}t^{-1/2} \to \infty$, so it violates the uniform-continuity hypothesis and has no C^2 extension at all. The closed Jost region of Chapter 4, $\{\sigma \ge (\omega_-)^2\}$, is convex (it is the epigraph of a $C^{1,1}$ function whose boundary is not C^2 at one point); Corollary A.18 applies to it, so a function that is C^k up to the boundary is the restriction of a C^k function on the whole plane. For a non-convex region (for instance an inward cusp $\{|y| \le x^{3/2}\}$) the corollary does not apply, and Theorem A.17 with its condition checked by hand is needed.

Theorem A.20 (Whitney's theorem on even functions). *Let $U \subset \mathbb{R}^m$ be open, $\delta > 0$, and $g \in C^\infty(U \times (-\delta,\delta))$ with $g(p,-t) = g(p,t)$. Then there is $h \in C^\infty(U \times [0,\delta^2))$ (smooth up to $s = 0$, i.e. the restriction of a C^∞ function on an open set) with $g(p,t) = h(p,t^2)$.*

Reference: [52], where the theorem is proved for one variable; the parameter p is carried along unchanged by the same proof, because every estimate in it is uniform on compact subsets of U. (Formally: the Taylor coefficients $\partial_t^{2j} g(p,0)/(2j)!$ are C^∞ in p, and Whitney's construction of h from them is linear and continuous in the C^∞ topology.)

A.4. The cone lemma and Kamke's comparison theorem

In words. *A comparison principle says: if two solutions start in the order $Y \le \hat{Y}$, they stay in that order. For one equation this is automatic (graphs cannot cross). For a system it holds if every component is pushed* up*, never down, by every other component: the system is* cooperative *(Kamke, Müller). The linear core is the cone lemma: if the matrix has nonnegative off-diagonal entries (Metzler) the nonnegative orthant is invariant, and if the couplings form a cycle through all components,*

positivity of one component is transmitted *to the next, so that strict positivity follows at once. A forcing term ≥ 0 is allowed; it is what lets us compare a solution with a sub-solution.*

A real square matrix A is *Metzler* if $A_{ij} \geq 0$ for all $i \neq j$.

Lemma A.21 (cone lemma)**.** *Let $A : [a,b] \to \mathbb{R}^{n\times n}$ and $\rho : [a,b] \to \mathbb{R}^n$ be continuous, $A(x)$ Metzler and $\rho(x) \geq 0$ (componentwise) for all x, and let $W \in C^1([a,b];\mathbb{R}^n)$ solve $W' = AW + \rho$. Put $\iota_i(x) = \exp(-\int_a^x A_{ii}) > 0$.*

(a) If $W(a) \geq 0$ then $W \geq 0$ on $[a,b]$, and each $\iota_i W_i$ is nondecreasing.

(b) (strict version) Suppose $W(a) \geq 0$ and in addition that the directed graph on $\{1,\dots,n\}$ with an arrow $j \to i$ whenever $A_{ij} > 0$ on $(a,b]$ is strongly connected (for instance, after relabelling, $A_{i,i-1} > 0$ for $i = 2,\dots,n$ and $A_{1,n} > 0$ on $(a,b]$: a cycle through all indices), and that either $W(a) \neq 0$ or ρ does not vanish identically on any interval $[a,x]$, $x > a$. Then $W_i > 0$ on $(a,b]$ for every i.

Proof. Put $V_i = \iota_i W_i$. Then $V_i' = \sum_{j\neq i} B_{ij} V_j + g_i$, with $B_{ij} = \iota_i A_{ij}/\iota_j \geq 0$ ($i \neq j$) and $g_i = \iota_i \rho_i \geq 0$: the diagonal has disappeared. (a) The Picard iterates $V^{(0)} \equiv V(a)$, $V^{(m+1)}(x) = V(a) + \int_a^x (BV^{(m)} + g)\mathrm{d}s$ are ≥ 0 by induction (a matrix with entries ≥ 0 applied to a vector ≥ 0, plus ≥ 0 terms). They converge uniformly to the solution of this linear system with continuous coefficients (Lemma A.6, forward version). Hence $V \geq 0$, then $V_i' \geq 0$. (b) Suppose $W_i(x_*) = 0$ for some i and some $x_* \in (a,b]$. Since $V_i \geq 0$ is nondecreasing, $V_i \equiv 0$ on $[a,x_*]$, so $0 = V_i' = \sum_{j\neq i} B_{ij}V_j + g_i$ there; all terms are ≥ 0, so $B_{ij}V_j = 0$ and $g_i = 0$ on $[a,x_*]$. For every $j \to i$ we have $B_{ij} > 0$ on $(a,x_*]$, so $V_j \equiv 0$ on $(a,x_*]$, hence on $[a,x_*]$ by continuity. Repeating the argument with j in place of i and using strong connectivity, $V_l \equiv 0$ on $[a,x_*]$ for all l. Then $W(a) = 0$ and $g_l = \iota_l\rho_l \equiv 0$ for all l, i.e. $\rho \equiv 0$ on $[a,x_*]$, contradicting the hypothesis. □

Remark A.22 (sharpness)**.** The Metzler condition is also necessary for invariance of the orthant: if $W' = AW$ and $W(a) = e_j$, then $W_i'(a) = A_{ij}$ must be ≥ 0 for $i \neq j$. Without it the lemma fails: for the rotation $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $W(0) = (1,0)$ one gets $W = (\cos t, \sin t)$, whose first component becomes negative.

Toy example A.23 (2×2 systems)**.** (1) $w_1' = -w_1 + w_2$, $w_2' = w_1 - 2w_2$, $w(0) = (1,0)$. A is Metzler and $1 \leftrightarrow 2$ is a cycle. Indeed $w_2(t) = \int_0^t e^{-2(t-s)} w_1(s)\mathrm{d}s > 0$ for $t > 0$ although $w_2(0) = 0$: positivity is *pushed* to the second component. (2) Mutually inhibiting quantities $p' = -q$, $q' = -p$ have Jacobian $\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right)$, not Metzler. For two solutions put $w = (\hat p - p,\ q - \hat q) = \Sigma(\hat Y - Y)$, $\Sigma = \mathrm{diag}(1,-1)$: then $w_1' = -\hat q + q = w_2$, $w_2' = -p + \hat p = w_1$, so $w' = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) w$ is Metzler, with $w_1 = w_1(0)\cosh t + w_2(0)\sinh t$, $w_2 = w_2(0)\cosh t + w_1(0)\sinh t$. Hence "larger p and smaller q" is an order preserved by the flow; and if $w(0) \geq 0$, $w(0) \neq 0$, both components are > 0 for $t > 0$ (verified numerically in `toy_kamke.py`).

For a sign matrix $\Sigma = \mathrm{diag}(s_1,\dots,s_n)$, $s_i = \pm 1$, write $Y \leq_\Sigma \hat Y$ iff $\Sigma(\hat Y - Y) \geq 0$. Note $(\Sigma M \Sigma)_{ij} = s_i s_j M_{ij}$: the diagonal is unchanged and an off-diagonal entry changes sign iff i, j carry opposite signs.

Theorem A.24 (Kamke–Müller comparison with forcing). *Let $D \subset \mathbb{R}^n$ be* convex *(open or closed or neither, e.g. a closed orthant such as $\{y_2, y_3 \geq 0\}$), $I \subset \mathbb{R}$*

an interval, and $\tilde D \supset D$ an open set. Let $F : I \times \tilde D \to \mathbb{R}^n$ be continuous with $\partial_y F$ continuous, and $\hat F : I \times D \to \mathbb{R}^n$ continuous, such that

$$\partial_{y_j} F_i(t,y) \geq 0 \quad \text{for } i \neq j,\ (t,y) \in I \times D \qquad (F \text{ is cooperative on } D),$$
$$F \leq \hat F \quad \text{on } I \times D.$$

Let $Y, \hat Y : [a,b] \to D$ be C^1 (with $[a,b] \subset I$) satisfying the differential inequalities $Y' \leq F(t,Y)$, $\hat Y' \geq \hat F(t,\hat Y)$, and $Y(a) \leq \hat Y(a)$. Then $Y \leq \hat Y$ on $[a,b]$. Strict version. *Put $A(t) = \int_0^1 \partial_y F(t, Y + s(\hat Y - Y))\mathrm{d}s$ and $\rho = (\hat Y - Y)' - A(\hat Y - Y)$ (≥ 0, see the proof). If A satisfies the strong-connectivity hypothesis of Lemma A.21(b) on $(a,b]$, and either $Y(a) \neq \hat Y(a)$ or ρ does not vanish identically on any interval $[a,x]$, $x > a$, then $Y_i < \hat Y_i$ on $(a,b]$ for every i.*

Proof. Put $W = \hat Y - Y$. By convexity of D the segment from $Y(t)$ to $\hat Y(t)$ lies in $D \subset \tilde D$, and by the fundamental theorem of calculus $F(t,\hat Y) - F(t,Y) = A(t)W$ with $A(t) = \int_0^1 \partial_y F(t, Y + sW)\mathrm{d}s$ continuous. $A(t)$ is Metzler, because each off-diagonal entry is an integral of a function that is ≥ 0 on D. Then

$$W' \ \geq\ \hat F(t,\hat Y) - F(t,Y) = \big[\hat F(t,\hat Y) - F(t,\hat Y)\big] + A(t)W,$$

i.e. $W' = AW + \rho$ with $\rho := W' - AW \geq \hat F(t,\hat Y) - F(t,\hat Y) \geq 0$ continuous. Lemma A.21(a) gives $W \geq 0$, and (b) the strict statement. □

For a sign matrix Σ write $F \leq_\Sigma \hat F$ iff $\Sigma(\hat F - F) \geq 0$.

Corollary A.25 (signed orders, with strictness)**.** *Let Σ be a sign matrix, D convex, $\tilde D \supset D$ open, $F, \hat F$ as in Theorem A.24 with "cooperative" replaced by: $\Sigma\,\partial_y F\,\Sigma$ is Metzler on $I \times D$, and $F \leq_\Sigma \hat F$ on $I \times D$. If $Y \leq_\Sigma \hat Y$ at $t = a$, $\Sigma Y' \leq \Sigma F(t,Y)$ and $\Sigma \hat Y' \geq \Sigma \hat F(t,\hat Y)$, then $Y \leq_\Sigma \hat Y$ on $[a,b]$.* Strict version. *Let $A_\Sigma(t) = \Sigma\Big(\int_0^1 \partial_y F(t, Y + s(\hat Y - Y))\mathrm{d}s\Big)\Sigma$ and $\rho_\Sigma = (\Sigma(\hat Y - Y))' - A_\Sigma \Sigma(\hat Y - Y)$ (≥ 0). If A_Σ satisfies the strong-connectivity hypothesis of Lemma A.21(b) on $(a,b]$, and either $Y(a) \neq \hat Y(a)$ or ρ_Σ does not vanish identically on any $[a,x]$, $x > a$, then $\Sigma(\hat Y - Y)_i > 0$ on $(a,b]$ for every i.*

Proof. Apply Theorem A.24 to $\tilde Y = \Sigma Y$, $\tilde F(t,\tilde y) = \Sigma F(t, \Sigma\tilde y)$, $\hat{\tilde F}(t,\tilde y) = \Sigma \hat F(t, \Sigma\tilde y)$ on ΣD (convex) and $\Sigma\tilde D$: $\partial_{\tilde y}\tilde F = \Sigma\,\partial_y F\,\Sigma$, whose average along the segment from $\tilde Y$ to $\hat{\tilde Y}$ is A_Σ. The two differential inequalities are the hypotheses; the strict statement is the strict statement of the theorem for $\tilde Y, \hat{\tilde Y}$. □

Remark A.26. Cooperativity is the classical "quasimonotone" (Kamke) or "type K" condition; for merely locally Lipschitz F the theorem is Kamke's [18], Müller's [32] (see [50], [43, Ch. 3], [16]). The proof above is the C^1 case, which is all the book uses: the nonlinearities there are polynomial. Convexity of D is needed because the proof integrates along the segment between the two solutions. If the Metzler property holds only on a convex closed set such as $D_+ = \{f \geq 0, g \geq 0\}$ (the situation of Chapters 5 and 12), take $D = D_+$: the theorem then applies to two solutions that both stay in D_+ (the open neighbourhood $\tilde D$ only provides differentiability of F at boundary points of D_+; for polynomial F one takes $\tilde D = \mathbb{R}^n$). In the six-component application $\Sigma = \mathrm{diag}(+,-,-,-,-,-)$ and ΣD_+ is again a closed convex cone.

Toy example A.27 (Riccati: the geometric mean is a sub-solution)**.** $y' = y^2$ is cooperative ($n = 1$, no off-diagonal entries). Its solutions are $y(t; a) = a/(1-at)$. For $y_i = y(\cdot; a_i)$ the geometric mean $V = \sqrt{y_1 y_2}$ has $V' = \frac{1}{2}V(y_1 + y_2) \geq V\sqrt{y_1 y_2} = V^2$ (AM–GM), so V is a supersolution with $V(0) = \sqrt{a_1 a_2}$. Theorem A.24 (with $Y = y(\cdot; \sqrt{a_1 a_2})$, $\hat{Y} = V$, $\hat{F} = F = y^2$, forcing $\rho = V' - V^2 \geq 0$) gives $y(t; \sqrt{a_1 a_2}) \leq \sqrt{y(t; a_1) y(t; a_2)}$, i.e. $s \mapsto \ln y(t; e^s)$ *is midpoint-convex*; directly, $\ln y = s - \ln(1-e^s t)$ has second derivative $e^s t/(1 - e^s t)^2 > 0$. The same comparison-with-a-sub-solution mechanism, in six components and with $\Sigma = \mathrm{diag}(+, -, -, -, -, -)$, is the log-convexity argument of Chapters 5 and 12. (Numerically checked in `toy_kamke.py`, together with the nonlinear supersolution $y \equiv \frac{1}{2} + \frac{t}{5} \geq$ the solution of $y' = y^2 - y$, $y(0) = \frac{1}{2}$.)

A.5. Compactness and fixed points

In words. *Existence for every swirl strength is a fixed point problem for a* compact *map on a bounded convex set of bounded continuous functions. Three facts are needed: a criterion for relative compactness in* BC($\mathbb{R}$) *(functions on the whole line, so the classical Arzelà–Ascoli needs a tail condition), Schauder's theorem (a continuous self-map of a closed convex set with relatively compact image has a fixed point), and the Riesz–Schauder spectral theory of compact linear maps (so that "eigenvalue 1" is a finite-dimensional condition).*

Theorem A.28 (Arzelà–Ascoli). *Let S be a compact metric space. A subset of $C(S)$ is relatively compact iff it is bounded and equicontinuous.*

Reference. Rudin [37, Thm. 7.25]. □

Lemma A.29 (Arzelà–Ascoli on the line)**.** *A set $\mathcal{F} \subset \mathrm{BC}(\mathbb{R})$ is relatively compact in the sup norm if (i) it is bounded, (ii) it is equicontinuous, and (iii) it has* uniform tails*: for every $\epsilon > 0$ there is L with $|g(\xi) - g(\eta)| \leq \epsilon$ for all $g \in \mathcal{F}$ and all $\xi, \eta \geq L$, and also for all $\xi, \eta \leq -L$.*

Proof. Let $(g_j) \subset \mathcal{F}$. By Theorem A.28 on each $[-N, N]$ and a diagonal argument there is a subsequence (still g_j) converging uniformly on every compact interval. It is Cauchy in $\|\cdot\|_\infty$: given ϵ, take L from (iii) and N_0 with $|g_j - g_l| \leq \epsilon$ on $[-L, L]$ for $j, l \geq N_0$. For $\xi \geq L$, $|g_j(\xi) - g_l(\xi)| \leq |g_j(\xi) - g_j(L)| + |g_j(L) - g_l(L)| + |g_l(L) - g_l(\xi)| \leq 3\epsilon$, and the same for $\xi \leq -L$. Since BC($\mathbb{R}$) is complete, the subsequence converges. □

Toy example A.30 (uniform tails)**.** $\mathcal{F}_1 = \{\tanh(\xi/s) : s \in [1, 2]\}$ is bounded and equicontinuous with uniform tails (the limits ± 1 are approached uniformly in s): relatively compact. $\mathcal{F}_2 = \{\tanh(\xi - s) : s \in \mathbb{N}\}$ is bounded and equicontinuous but violates (iii): $\|\tanh(\cdot - s) - \tanh(\cdot - s')\|_\infty \to 2$, no convergent subsequence. Monotone profiles with fixed limits (the swirl f of the book, $f_- \leq f \leq f_+$ with bounded slope) satisfy (iii) after an equicontinuity bound.

Theorem A.31 (Schauder's fixed point theorem). *Let K be a nonempty closed convex subset of a Banach space $\mathbb{B}$ and $T : K \to K$ continuous with $T(K)$ relatively compact in $\mathbb{B}$. Then T has a fixed point. (K itself need not be compact.)*

Reference. Schauder [38]; Deimling [7]; Granas–Dugundji [12]. Brouwer's theorem [4] (a continuous self-map of a compact convex subset of $\mathbb{R}^n$ has a fixed point) is the finite-dimensional case. □

THEOREM A.32 (Dugundji's extension theorem). *Let A be a closed subset of a metric space S, $\mathbb{B}$ a normed space and $\Phi : A \to \mathbb{B}$ continuous. Then Φ has a continuous extension $\tilde{\Phi} : S \to \mathbb{B}$ with $\tilde{\Phi}(S) \subset \operatorname{conv} \Phi(A)$. Consequently every nonempty closed convex $K \subset \mathbb{B}$ is a retract of $\mathbb{B}$ (apply it to $\Phi = \mathrm{id}_K$), and a compact map defined on a closed set extends to a compact map on S (by Mazur's theorem [29], the closed convex hull of a relatively compact set in a Banach space is compact).*

REFERENCE. Dugundji [9]; Deimling [7]; Granas–Dugundji [12]. □

THEOREM A.33 (Riesz–Schauder theory). *Let A be a compact linear operator on a Banach space $\mathbb{B}$. Then $L = I - A$ has finite-dimensional kernel and closed range,* $\operatorname{codim} \operatorname{ran} L = \dim \ker L$ *(Fredholm of index 0); in particular L is injective iff surjective. The spectrum of A is countable,* $\operatorname{spec} A \setminus \{0\}$ *consists of eigenvalues of finite algebraic multiplicity, and 0 is its only possible accumulation point.*

REFERENCE. Brezis [3, Ch. 6, Thm. 6.6 and Thm. 6.8]; Deimling [7]. □

Toy example A.34 (a compact self-map)**.** $T(f) = \frac{1}{2}\cos f$ maps $K = [-1, 1] \subset \mathbb{R}$ into $[-\frac{1}{2}, \frac{1}{2}]$; Brouwer/Schauder gives a fixed point, here the unique one $f^* \approx 0.4502$ (the iteration converges geometrically, $|T'| \le \frac{1}{2}$). A genuinely infinite-dimensional instance: $(Tu)(x) = \frac{2}{3}(1 + x)\big(b\, m(u) - m(u)^3 + \tau\big)$ on $C[0, 1]$ with $m(u) = \int_0^1 u$ has rank one, hence is compact; its derivative has the single nonzero eigenvalue $\nu_{\max} = b - 3m^2$ with eigenfunction $1 + x$, in the pattern of Theorem A.33 (see Toy example A.44).

A.6. The Leray–Schauder degree

In words. *The Leray–Schauder degree counts the fixed points of a compact map, with signs, in a way that is invariant under deformation. If it is non-zero there* is *a fixed point; if it equals $+1$ for one value of the data, it equals $+1$ for all values reachable by a deformation in which no fixed point escapes through the boundary. For a map with finitely many nondegenerate fixed points, the degree is the sum of the indices $(-1)^\beta$ (β is the number of real eigenvalues of the derivative above 1, with multiplicity), so the number of fixed points is odd as soon as the degree is ± 1. Two extensions are used in the book: a homotopy invariance on sets that move with the parameter (property (D6), which we prove), and the connectedness of the set of fixed points (the Leray–Schauder continuation principle, which we also prove, from (D6) and a lemma of Whyburn).*

Let $\mathbb{B}$ be a real Banach space and $\mathcal{O} \subset \mathbb{B}$ open and bounded. A map $F : \overline{\mathcal{O}} \to \mathbb{B}$ (or $F : [a, b] \times \overline{\mathcal{O}} \to \mathbb{B}$) is *compact* if it is continuous with relatively compact image.

THEOREM A.35 (Leray–Schauder degree). *For $\mathcal{O} \subset \mathbb{B}$ open bounded and $F : \overline{\mathcal{O}} \to \mathbb{B}$ compact with $x \neq F(x)$ on $\partial\mathcal{O}$ there is an integer $\deg_{\mathrm{LS}}(I - F, \mathcal{O})$ with the following properties.*

- *(D1) (normalisation)* $\deg_{\mathrm{LS}}(I - y_0, \mathcal{O}) = 1$ *if $y_0 \in \mathcal{O}$ (the constant map $F \equiv y_0$).*
- *(D2) (homotopy invariance) If $H : [0, 1] \times \overline{\mathcal{O}} \to \mathbb{B}$ is compact and $x \neq H(s, x)$ for all $s \in [0, 1]$, $x \in \partial\mathcal{O}$, then $s \mapsto \deg_{\mathrm{LS}}(I - H(s, \cdot), \mathcal{O})$ is constant.*
- *(D3) (existence) If $\deg_{\mathrm{LS}}(I - F, \mathcal{O}) \neq 0$ then F has a fixed point in $\mathcal{O}$.*

(D4) (additivity–excision) If $\mathcal{O}_1, \dots, \mathcal{O}_N \subset \mathcal{O}$ are pairwise disjoint open sets (possibly $N = 0$ or $N = 1$) and F has no fixed point in $\overline{\mathcal{O}} \setminus \bigcup_j \mathcal{O}_j$, then $\deg_{\mathrm{LS}}(I - F, \mathcal{O}) = \sum_j \deg_{\mathrm{LS}}(I - F, \mathcal{O}_j)$. In particular, if F has no fixed point in $\overline{\mathcal{O}}$ the degree is 0.

(D5) (index) For an isolated fixed point x_0 put $\operatorname{ind}(F, x_0) := \deg_{\mathrm{LS}}(I - F, B_r(x_0))$, r small (independent of r by (D4)). If F is Fréchet differentiable at x_0 with derivative $F'(x_0)$ and $1 \notin \operatorname{spec} F'(x_0)$, then x_0 is an isolated fixed point and $\operatorname{ind}(F, x_0) = (-1)^\beta$, where β is the sum of the algebraic multiplicities of the real eigenvalues of $F'(x_0)$ in $(1, \infty)$ (finite, since $F'(x_0)$ is compact).

Reference. Leray–Schauder [22]; Deimling [7] (axioms and index formula); Nirenberg [34]; Granas–Dugundji [12]; Krasnosel'skii–Zabreiko [19]. . □

Corollary A.36 (finite fixed-point sets; parity)**.** *If F has finitely many fixed points $x_1, \dots, x_N$ in $\mathcal{O}$, then $\deg_{\mathrm{LS}}(I - F, \mathcal{O}) = \sum_j \operatorname{ind}(F, x_j)$. If, in addition, every $F'(x_j)$ exists and $1 \notin \operatorname{spec} F'(x_j)$, then $\deg_{\mathrm{LS}}(I - F, \mathcal{O}) \equiv N \pmod 2$. In particular, if the degree is odd there are an odd number of fixed points, and if the degree is $+1$ and all fixed points are nondegenerate then the number of fixed points with β even exceeds the number with β odd by exactly one.*

Proof. Choose disjoint balls $B_r(x_j) \subset \mathcal{O}$ and apply (D4) with $\mathcal{O}_j = B_r(x_j)$: the sum is $\sum \operatorname{ind}(F, x_j)$. By (D5) each index is ± 1 so $\sum \operatorname{ind} \equiv N \pmod 2$; the last statement counts signs. □

Corollary A.37 (self-maps of convex sets have degree one)**.** *Let $\mathcal{O}$ be open, bounded and convex and $F : \overline{\mathcal{O}} \to \mathbb{B}$ compact with $F(\overline{\mathcal{O}}) \subset \mathcal{O}$. Then $x \neq F(x)$ on $\partial\mathcal{O}$ and $\deg_{\mathrm{LS}}(I - F, \mathcal{O}) = 1$.*

Proof. A fixed point on $\partial\mathcal{O}$ would lie in $F(\overline{\mathcal{O}}) \subset \mathcal{O}$, impossible. Fix $y_0 \in \mathcal{O}$ and let $H(s, x) = sF(x) + (1 - s)y_0$. It is compact; for $x \in \partial\mathcal{O}$, $H(s, x) \in \mathcal{O}$ by convexity ($F(x), y_0 \in \mathcal{O}$ and $\mathcal{O}$ is convex), so $x \neq H(s, x)$. By (D2) and (D1), $\deg_{\mathrm{LS}}(I - F, \mathcal{O}) = \deg_{\mathrm{LS}}(I - y_0, \mathcal{O}) = 1$. □

Toy example A.38 (degree $+1$, three fixed points)**.** (1) $T(f) = \frac{1}{2}\cos f$ on $\mathcal{O} = (-1, 1) \subset \mathbb{R}$ maps $\overline{\mathcal{O}}$ into $(-\frac{1}{2}, \frac{1}{2}) \subset \mathcal{O}$, so the degree is 1 (Corollary A.37); the unique fixed point has $T'(f^*) = -\frac{1}{2}\sin f^* \approx -0.22 < 1$: $\beta = 0$, index $+1$, consistent with (D5). (2) $F(x) = 2x - x^3$ on $\mathcal{O} = (-2, 2)$. Then $I - F = x^3 - x$; the fixed points are 0 ($F' = 2 > 1$: $\beta = 1$, index -1) and ± 1 ($F' = -1$, no eigenvalue above 1: index $+1$ each). Sum $= +1$, and indeed $\deg_{\mathrm{LS}}(I - F, \mathcal{O}) = \frac{1}{2}[\operatorname{sgn} g(2) - \operatorname{sgn} g(-2)] = 1$ for $g = x^3 - x$. The count $N = 3$ is odd, as Corollary A.36 predicts.

We now prove the version of homotopy invariance for sets that move with the parameter. Let $[a, b] \subset \mathbb{R}$ and let $\mathcal{U} \subset [a, b] \times \mathbb{B}$ be relatively open and bounded, with slices $\mathcal{U}_t = \{x : (t, x) \in \mathcal{U}\}$ (open in $\mathbb{B}$) and relative boundary $\partial\mathcal{U} = \overline{\mathcal{U}} \setminus \mathcal{U}$ (closure in $[a, b] \times \mathbb{B}$). Distances in $[a, b] \times \mathbb{B}$ are measured by $|t - t'| + \|x - x'\|$.

Lemma A.39 ((D6): homotopy invariance on non-cylindrical sets)**.** *Let $F : \overline{\mathcal{U}} \to \mathbb{B}$ be compact and $x \neq F(t, x)$ for $(t, x) \in \partial\mathcal{U}$. Then $\partial(\mathcal{U}_t) \subset (\partial\mathcal{U})_t$ for each t (so $\deg_{\mathrm{LS}}(I - F(t, \cdot), \mathcal{U}_t)$ is defined), and it does not depend on $t \in [a, b]$.*

PROOF. *Boundary of a slice.* If $x \in \partial(\mathcal{U}_t)$ then $(t,x) \in \overline{\mathcal{U}}$; if $(t,x) \in \mathcal{U}$, then $x \in \mathcal{U}_t$, which is open, contradicting $x \in \partial(\mathcal{U}_t)$. So $(t,x) \in \partial\mathcal{U}$ and $x \neq F(t,x)$.

Fixed-point set. Let $S = \{(t,x) \in \overline{\mathcal{U}} : x = F(t,x)\}$. It is closed (as F is continuous on the closed set $\overline{\mathcal{U}}$) and contained in $[a,b] \times \overline{F(\overline{\mathcal{U}})}$, a compact set; so S is compact. By hypothesis $S \cap \partial\mathcal{U} = \emptyset$, i.e. $S \subset \mathcal{U}$. If $S = \emptyset$, every slice degree vanishes by (D4) and the claim holds. Otherwise let $\delta > 0$ be the distance from S to the closed set $([a,b] \times \mathbb{B}) \setminus \mathcal{U}$ (> 0 since S is compact and disjoint from it); every point of $[a,b] \times \mathbb{B}$ at distance $< \delta$ from S lies in $\mathcal{U}$.

Local constancy. Fix a partition $a = t_0 < t_1 < \cdots < t_N = b$ of mesh $< \delta/3$. Let $S_i = S \cap ([t_{i-1}, t_i] \times \mathbb{B})$ (compact, perhaps empty), $\pi(S_i)$ its projection to $\mathbb{B}$, and $\mathcal{O}^i = \{x \in \mathbb{B} : \operatorname{dist}(x, \pi(S_i)) < \delta/2\}$ (if $S_i = \emptyset$ then $\mathcal{O}^i = \emptyset$ and the degree over it is 0). For $t \in [t_{i-1}, t_i]$ and $x \in \overline{\mathcal{O}^i}$ there is $(s,y) \in S_i$ with $\|x - y\| \leq \delta/2$ (compactness) and $|t - s| < \delta/3$, so (t,x) has distance $< \delta$ from S: $(t,x) \in \mathcal{U}$. Hence $[t_{i-1}, t_i] \times \overline{\mathcal{O}^i} \subset \mathcal{U}$ and F is defined and compact there. A fixed point (t,x) with $t \in [t_{i-1}, t_i]$, $x \in \overline{\mathcal{O}^i}$ belongs to S_i, so $x \in \pi(S_i) \subset \mathcal{O}^i$, i.e. none lies on $\partial\mathcal{O}^i$. By (D2) (with the reparametrised $H(s,x) = F(t_{i-1} + s(t_i - t_{i-1}), x)$), $\deg_{\mathrm{LS}}(I - F(t,\cdot), \mathcal{O}^i)$ is constant for $t \in [t_{i-1}, t_i]$. Now compare with $\mathcal{U}_t$: for $t \in [t_{i-1}, t_i]$ the fixed points of $F(t,\cdot)$ in $\mathcal{U}_t$ are the x with $(t,x) \in S_i$ (so $x \in \mathcal{O}^i$), and those in $\mathcal{O}^i$ also lie in $\mathcal{U}_t$ (shown above). Hence $F(t,\cdot)$ has no fixed point in $\overline{\mathcal{U}_t} \setminus (\mathcal{U}_t \cap \mathcal{O}^i)$, nor in $\overline{\mathcal{O}^i} \setminus (\mathcal{U}_t \cap \mathcal{O}^i)$, and (D4) with $N = 1$ gives $\deg_{\mathrm{LS}}(I - F(t,\cdot), \mathcal{U}_t) = \deg_{\mathrm{LS}}(I - F(t,\cdot), \mathcal{U}_t \cap \mathcal{O}^i) = \deg_{\mathrm{LS}}(I - F(t,\cdot), \mathcal{O}^i)$. So the slice degree is constant on each $[t_{i-1}, t_i]$; adjacent intervals share an endpoint, so it is constant on $[a,b]$. □

Toy example A.40 (a moving set)**.** $F(t,x) = 2x - x^3 + t$ and $\mathcal{U} = \{(t,x) : t \in [-1,1],\ |x| < 2 + \frac{1}{4}t\}$, a wedge-shaped set whose slices grow with t. The fixed points solve $x^3 - x = t$; on $\partial\mathcal{U}$ one has $|x| \geq \frac{7}{4}$, hence $|x^3 - x| \geq \frac{231}{64} > 1 \geq |t|$ (the function $x^3 - x$ is increasing for $x \geq 1$ and equals $\frac{231}{64}$ at $x = \frac{7}{4}$), so $\partial\mathcal{U}$ contains no fixed point, and the slice degree equals 1 for all t (at $t = -1$ there is the single fixed point $x \approx -1.32$ with index $+1$; at $t = 0$ three, with indices $+1, -1, +1$; the fixed-point set is the S-shaped graph $t = x^3 - x$; see Figure A.1, left).

THEOREM A.41 (Whyburn's lemma). *Let $\mathcal{K}$ be a compact metric space and $\mathcal{A}, \mathcal{B} \subset \mathcal{K}$ closed. If no connected component of $\mathcal{K}$ meets both $\mathcal{A}$ and $\mathcal{B}$, then $\mathcal{K} = \mathcal{K}_{\mathcal{A}} \sqcup \mathcal{K}_{\mathcal{B}}$ with $\mathcal{K}_{\mathcal{A}}, \mathcal{K}_{\mathcal{B}}$ disjoint and open-and-closed (hence compact), $\mathcal{A} \subset \mathcal{K}_{\mathcal{A}}$, $\mathcal{B} \subset \mathcal{K}_{\mathcal{B}}$.*

PROOF. (Classical, [55, Ch. I]; we give the compactness argument.) *Step 1: components are intersections of clopen sets.* For $x \in \mathcal{K}$ let Q_x be the intersection of all clopen (open and closed) subsets of $\mathcal{K}$ containing x, and C_x the component of x; clearly $C_x \subset Q_x$. We claim Q_x is connected, so $Q_x = C_x$. Suppose $Q_x = P_1 \sqcup P_2$ with P_1, P_2 closed, $x \in P_1$. Choose disjoint open $U_1 \supset P_1$, $U_2 \supset P_2$. The compact set $\mathcal{K} \setminus (U_1 \cup U_2)$ is disjoint from Q_x, hence covered by the complements of clopen sets containing x, hence by finitely many of them; the intersection N of those finitely many clopen sets is clopen, contains x, and $N \subset U_1 \cup U_2$. Then $N \cap U_1 = N \setminus U_2$ is open and closed, contains x, so $Q_x \subset N \cap U_1$, whence $P_2 \subset U_1 \cap U_2 = \emptyset$. *Step 2.* Let $a \in \mathcal{A}$. By hypothesis $Q_a = C_a$ misses $\mathcal{B}$; $\mathcal{B}$ is compact and covered by the complements of the clopen sets containing a, so finitely many suffice, and their intersection N_a is a clopen set with $a \in N_a$, $N_a \cap \mathcal{B} = \emptyset$. Cover the compact $\mathcal{A}$ by finitely many $N_{a_1}, \ldots, N_{a_m}$ and put $\mathcal{K}_{\mathcal{A}} = \bigcup N_{a_l}$ (empty if $\mathcal{A} = \emptyset$), $\mathcal{K}_{\mathcal{B}} = \mathcal{K} \setminus \mathcal{K}_{\mathcal{A}}$. Both are clopen, $\mathcal{A} \subset \mathcal{K}_{\mathcal{A}}$ and $\mathcal{B} \subset \mathcal{K}_{\mathcal{B}}$. □

THEOREM A.42 (Leray–Schauder continuation principle). *Let $\mathcal{O} \subset \mathbb{B}$ be open and bounded, $F : [a,b] \times \overline{\mathcal{O}} \to \mathbb{B}$ compact, $x \neq F(t,x)$ for all $t \in [a,b]$, $x \in \partial\mathcal{O}$, and $\deg_{\mathrm{LS}}(I - F(a,\cdot), \mathcal{O}) \neq 0$. Then the fixed-point set* $\mathrm{Fix} = \{(t,x) \in [a,b] \times \mathcal{O} : x = F(t,x)\}$ *is compact and contains a connected set that meets both $\{a\} \times \mathcal{O}$ and $\{b\} \times \mathcal{O}$. In particular, if λ is a continuous real function on* Fix *and $\mathcal{C}$ is this connected set, then $\lambda(\mathcal{C})$ is an interval, so every value between a value of λ on $\mathcal{C} \cap \{t = a\}$ and a value on $\mathcal{C} \cap \{t = b\}$ is attained by λ at some fixed point.*

PROOF. *Compactness.* Fix is closed in $[a,b] \times \overline{\mathcal{O}}$ and, by hypothesis, contained in $[a,b] \times \mathcal{O}$; it lies in $[a,b] \times \overline{F([a,b] \times \overline{\mathcal{O}})}$, a compact set. So Fix is compact. Let $\mathcal{A} = \mathrm{Fix} \cap (\{a\} \times \mathbb{B})$, $\mathcal{B} = \mathrm{Fix} \cap (\{b\} \times \mathbb{B})$; $\mathcal{A} \neq \emptyset$ by (D3). *Contradiction hypothesis.* Suppose no connected subset of Fix meets both $\mathcal{A}$ and $\mathcal{B}$ (if $\mathcal{B} = \emptyset$ this is vacuous). By Theorem A.41, $\mathrm{Fix} = \mathcal{K}_{\mathcal{A}} \sqcup \mathcal{K}_{\mathcal{B}}$ with $d := \mathrm{dist}(\mathcal{K}_{\mathcal{A}}, \mathcal{K}_{\mathcal{B}}) > 0$ (or $\mathcal{K}_{\mathcal{B}} = \emptyset$). Put

$$\mathcal{U} = \{(t,x) \in [a,b] \times \mathcal{O} : \ \mathrm{dist}((t,x), \mathcal{K}_{\mathcal{A}}) < d/2\}$$

(any $d/2 > 0$ if $\mathcal{K}_{\mathcal{B}} = \emptyset$): relatively open and bounded, $\mathcal{K}_{\mathcal{A}} \subset \mathcal{U}$. Then $\overline{\mathcal{U}} \subset [a,b] \times \overline{\mathcal{O}}$ and F is defined and compact on it. A fixed point in $\overline{\mathcal{U}}$ is in $\mathcal{K}_{\mathcal{A}} \cup \mathcal{K}_{\mathcal{B}}$ and at distance $\leq d/2$ from $\mathcal{K}_{\mathcal{A}}$, hence in $\mathcal{K}_{\mathcal{A}} \subset \mathcal{U}$: so $\partial\mathcal{U}$ has no fixed point, and Lemma A.39 applies. *The two end slices.* At $t = b$: $\mathcal{K}_{\mathcal{A}} \cap (\{b\} \times \mathbb{B}) \subset \mathcal{B} \subset \mathcal{K}_{\mathcal{B}}$ is empty, so $F(b,\cdot)$ has no fixed point in $\mathcal{U}_b$, whence $\deg_{\mathrm{LS}}(I - F(b,\cdot), \mathcal{U}_b) = 0$ by (D4). At $t = a$: the fixed points of $F(a,\cdot)$ in $\mathcal{O}$ are $\mathcal{A} \subset \mathcal{K}_{\mathcal{A}}$, all lying in $\mathcal{U}_a \subset \mathcal{O}$, so by (D4) $\deg_{\mathrm{LS}}(I - F(a,\cdot), \mathcal{U}_a) = \deg_{\mathrm{LS}}(I - F(a,\cdot), \mathcal{O}) \neq 0$. This contradicts Lemma A.39. Hence a connected subset of Fix meets $\mathcal{A}$ and $\mathcal{B}$; the last claim follows because the continuous image of a connected set is an interval. □

Toy example A.43 (continuation through a fold)**.** $F(t,x) = 2x - x^3 + t$ on $\mathcal{O} = (-2,2)$, $t \in [-1,1]$, $\deg_{\mathrm{LS}}(I - F(-1,\cdot), \mathcal{O}) = 1$ (one fixed point $x \approx -1.32$). The fixed-point set is the S-curve $\{t = x^3 - x\}$. It is connected and reaches $t = 1$, but *t is not monotone along it*: at $t = 0$ there are three points. This is exactly why the continuation principle is stated for a connected set and not for a graph over the parameter: the parameter κ of the book is, along the set of solutions, only guaranteed to take every value, not to be a single-valued function of the solution. (`toy_degree.py`, Figure A.1.)

A.7. Folds, cusps and the Lyapunov–Schmidt reduction

In words. *A* fold *is where a branch of solutions turns back; a* cusp *is where a fold curve ends and the turning-back disappears. Near such a point a problem in infinitely many unknowns behaves like a problem in* one *unknown, because only one direction (the kernel of the linearisation) is degenerate and all other directions can be eliminated by the implicit function theorem: that elimination is the Lyapunov–Schmidt reduction. After it the fold is "a function of one variable has a non-degenerate critical point" and the cusp is "a function of one variable has a degenerate critical point". Everything is visible in the cubic $m^3 + am = \tau$, which is the whole story in miniature. We prove the normal forms from scratch; the full classification theorem of Whitney [53] is not needed.*

Toy example A.44 (the cubic)**.** Consider $m^3 + am = \tau$ for unknown $m \in \mathbb{R}$ and parameters (a,τ). For $a \geq 0$ the left side is strictly increasing in m: one solution for each τ. For $a < 0$ there are three solutions for $|\tau| < \frac{2}{3\sqrt{3}}(-a)^{3/2}$ and one for

larger $|\tau|$. The number of solutions changes by 2 exactly where $\partial_m(m^3+am) = 3m^2+a=0$, i.e. on the *fold curve* $4a^3+27\tau^2=0$ ($a\le 0$), which has a cusp at the origin; the width of the three-solution interval is $\frac{4}{3\sqrt{3}}(-a)^{3/2}$ (Figure A.1, middle and right). Along $\tau\mapsto m$ at fixed $a<0$ one traces an S; the two turning points approach each other and merge as $a\uparrow 0$. In function space: on $C[0,1]$ let $(T_{b,\tau}u)(x) = \frac{2}{3}(1+x)\big(b\,m(u)-m(u)^3+\tau\big)$, $m(u)=\int_0^1 u$. T has rank one, so is compact; a fixed point is $u=c(1+x)$ with $m=\frac{3}{2}c$ and $m^3+(1-b)m=\tau$ (the cubic with $a=1-b$). The derivative is $\mathrm{D}T(u)\delta u = \frac{2}{3}(1+x)(b-3m^2)m(\delta u)$ with the single non-zero eigenvalue $\nu_{\max}=b-3m^2$. Hence $1\in\operatorname{spec}\mathrm{D}T$ *iff* $3m^2+a=0$, *i.e. exactly at the folds*, and the left eigenfunctional is $m(\cdot)$ with $m(\partial_\tau T)=\frac{2}{3}\cdot\frac{3}{2}=1\neq 0$ (transversality). This is the pattern proved below in general. (`toy_fold.py` checks the width formula and the eigenvalue.)

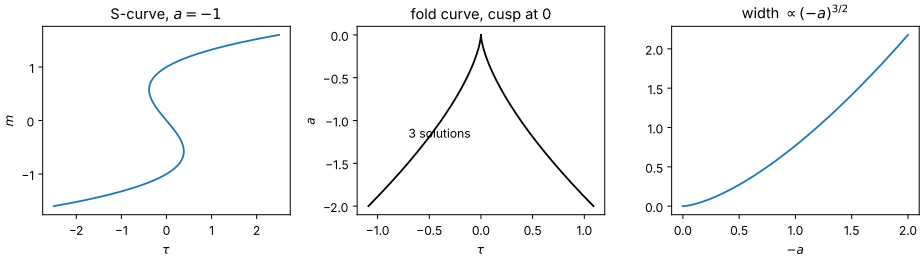


FIGURE A.1. Toy example A.44: the S-curve $\tau=m^3-m$ ($a=-1$; it is also the fixed-point set of Toy example A.43, with $t=\tau$), the fold curve $4a^3+27\tau^2=0$ with its cusp, and the width $\frac{4}{3\sqrt{3}}(-a)^{3/2}$ of the three-solution interval. Code: `code/appA_background/toy_fold.py`.

In this section $\Xi:\mathbb{R}^n\to\mathbb{R}^n$ is a smooth (C^∞) map with $\Xi(0)=0$ and $\operatorname{rank}\mathrm{D}\Xi(0)=n-1$ (*corank one*); $\Delta=\det\mathrm{D}\Xi$.

Lemma A.45 (scalar form)**.** *There are smooth local coordinates $(x,y)\in\mathbb{R}\times\mathbb{R}^{n-1}$ near 0 in the source and (u,y) near 0 in the target such that $\Xi(x,y)=(\varphi(x,y),y)$ with $\varphi(0)=0$, $\varphi_x(0)=0$. In these coordinates $\Delta=j\,\varphi_x$ with a smooth nowhere vanishing j, and at a point where $\varphi_x=0$ the kernel of $\mathrm{D}\Xi$ is spanned by ∂_x.*

PROOF. The range H of $\mathrm{D}\Xi(0)$ is a hyperplane. Choose a linear map $\pi:\mathbb{R}^n\to\mathbb{R}^{n-1}$ whose kernel is a line not contained in H; then $\pi|_H$ is injective, hence bijective, so $\pi\circ\mathrm{D}\Xi(0)$ is onto. So $y:=\pi\circ\Xi$ is a submersion at 0 and $\ker\mathrm{d}y(0)=\ker(\pi\circ\mathrm{D}\Xi(0))$ is a line; it contains the line $\ker\mathrm{D}\Xi(0)$, hence equals it. Choose a linear function x nonzero on this line; (x,y) are coordinates near 0. In the target use the linear coordinates (u,π) for any linear u independent of π. Then $\Xi=(u\circ\Xi,y)=(\varphi(x,y),y)$. The line $\{y=0\}$ is tangent to $\ker\mathrm{D}\Xi(0)$, so $\varphi_x(0)=0$. The Jacobian of $(x,y)\mapsto(\varphi,y)$ is $\left(\begin{smallmatrix}\varphi_x & \nabla_y\varphi\\ 0 & I\end{smallmatrix}\right)$ with determinant φ_x; changes of coordinates multiply Δ by nonvanishing smooth factors. Where $\varphi_x=0$ the matrix is $\left(\begin{smallmatrix}0 & \nabla_y\varphi\\ 0 & I\end{smallmatrix}\right)$, whose kernel is spanned by ∂_x. □

Lemma A.46 (Whitney fold)**.** *In the situation of Lemma A.45 assume $\varphi_{xx}(0) \neq 0$. Then there are smooth coordinates (η, y) near 0 in the source and (w, y) in the target with $\Xi(\eta, y) = (s_\varphi \eta^2, y)$, $s_\varphi = \operatorname{sgn} \varphi_{xx}(0)$. Consequently the singular set $\{\Delta = 0\}$ is a smooth hypersurface near 0, Ξ maps it diffeomorphically onto a smooth hypersurface Γ of the target, and a point near 0 in the target has no preimage near 0 on one side of Γ and exactly two on the other.*

Proof. By the implicit function theorem (Theorem A.3) there is a smooth $x^*(y)$ with $\varphi_x(x^*(y), y) = 0$ (here $\varphi_{xx}(0) \neq 0$). Put $m(y) = \varphi(x^*(y), y)$. Taylor's formula with integral remainder at x^* gives $\varphi(x, y) = m(y) + (x - x^*(y))^2 a(x, y)$ with $a = \int_0^1 (1-s)\varphi_{xx}(x^* + s(x - x^*), y)\mathrm{d}s$, $a(0) = \frac{1}{2}\varphi_{xx}(0) \neq 0$. Set $\eta = (x - x^*(y))\sqrt{|a(x, y)|}$. The map $(x, y) \mapsto (\eta, y)$ has Jacobian $\sqrt{|a|} \neq 0$ at 0, so it is a coordinate change, and $\varphi = m(y) + s_\varphi \eta^2$. In the target put $w = u - m(y)$. Now $\{\Delta = 0\} = \{\eta = 0\}$ (as $\Delta = j\varphi_x$ and $\varphi_x = 2 s_\varphi \eta\, \partial_x \eta$), $\Gamma = \{w = 0\}$, and the equation $s_\varphi \eta^2 = w$ has two solutions if $s_\varphi w > 0$, one if $w = 0$, none if $s_\varphi w < 0$. □

Definition A.47 (regular fold)**.** A point s_0 with $\operatorname{corank} \mathrm{D}\Xi(s_0) = 1$ is a *regular fold* if $\varphi_{xx}(s_0) \neq 0$ in the coordinates of Lemma A.45. Invariantly: if v spans $\ker \mathrm{D}\Xi(s_0)$ and ℓ spans the left null space ($\ell\, \mathrm{D}\Xi(s_0) = 0$), the condition is $\ell \cdot \mathrm{D}^2\Xi(s_0)[v, v] \neq 0$. (Equivalently $\mathrm{d}\Delta(s_0)v \neq 0$.)

Invariance of the condition. In the coordinates of Lemma A.45, $v = \partial_x$, $\mathrm{D}\Xi = \left(\begin{smallmatrix} 0 & \nabla_y \varphi \\ 0 & I \end{smallmatrix}\right)$, $\ell = (1, -\nabla_y \varphi)$ and $\ell \cdot \mathrm{D}^2\Xi[v, v] = \varphi_{xx}$. Under a change of source coordinates S, $\mathrm{D}^2(\Xi \circ S)[w, w] = \mathrm{D}^2\Xi[Sw, Sw] + \mathrm{D}\Xi\, \mathrm{D}^2 S[w, w]$ and $\ell\, \mathrm{D}\Xi = 0$ kills the second term; under a change Υ of target coordinates, $\mathrm{D}^2(\Upsilon \circ \Xi)[v, v] = \mathrm{D}\Upsilon\, \mathrm{D}^2\Xi[v, v] + \mathrm{D}^2\Upsilon[\mathrm{D}\Xi\, v, \mathrm{D}\Xi\, v]$ with $\mathrm{D}\Xi\, v = 0$, and the left null vector transforms by $\mathrm{D}\Upsilon^{-\top}$. Hence $\ell \cdot \mathrm{D}^2\Xi[v, v]$ changes by a nonzero factor only.

Lemma A.48 (Whitney cusp: the part of the normal form that is used)**.** *In the situation of Lemma A.45 let $\varphi_{xx}(0) = 0$, $h_0 := \varphi_{xxx}(0) \neq 0$ and $\nabla_y \varphi_x(0) \neq 0$. Let $\tau(y)$ be the root of $\varphi_{xx}(\tau, y) = 0$ near 0 and $a(y) := \varphi_x(\tau(y), y)$. Then a is a submersion at 0 ($\nabla a(0) = \nabla_y \varphi_x(0)$) and, near the origin,*

(i) $\varphi_x(\tau + X, y) = a(y) + X^2 H(X, y)$ with H smooth, $H(0) = h_0/2$;

(ii) if $a(y) h_0 \geq 0$, then $x \mapsto \varphi(x, y)$ is strictly monotone near 0: every target point has at most one preimage;

(iii) if $a(y) h_0 < 0$, then $x \mapsto \varphi(x, y)$ has exactly two critical points $x_\pm(y)$ near 0, with critical values $u_{\max}(y) > u_{\min}(y)$ (for $h_0 > 0$; the roles swap for $h_0 < 0$) and

$$u_{\max}(y) - u_{\min}(y) = \frac{4}{3} \frac{|a(y)|^{3/2}}{\sqrt{|H(0)|}} \big(1 + o(1)\big) \qquad (y \to 0) \qquad (|H(0)| = |h_0|/2).$$

A target point (u, y) has three preimages near 0 if $u_{\min}(y) < u < u_{\max}(y)$, two if $u \in \{u_{\min}, u_{\max}\}$, one otherwise.

Thus the three-preimage region is bounded by two fold hypersurfaces which meet along $\{a(y) = 0\}$ and approach each other like $|a|^{3/2}$ (a semicubical cusp).

Proof. $\varphi_{xxx}(0) \neq 0$ and the implicit function theorem give a smooth $\tau(y)$, $\tau(0) = 0$. (i) Taylor in X at 0: $\varphi_x(\tau + X, y) = \varphi_x(\tau, y) + X\varphi_{xx}(\tau, y) + X^2 \int_0^1 (1 - s)\varphi_{xxx}(\tau + sX, y)\mathrm{d}s$, and $\varphi_{xx}(\tau, y) = 0$. Differentiating $a(y) = \varphi_x(\tau(y), y)$: $\nabla a = \varphi_{xx} \nabla\tau + \nabla_y \varphi_x = \nabla_y \varphi_x$ at 0. (ii), (iii) Let $s_0 = \operatorname{sgn} h_0$, so $s_0 H > 0$ near 0. The map

$X \mapsto \rho = X\sqrt{s_0 H(X,y)}$ is, for each small y, a diffeomorphism near 0 (derivative $\sqrt{s_0 H(0)} \neq 0$ at 0), and $\varphi_x = a + s_0\rho^2$. If $s_0 a \geq 0$ then $s_0\varphi_x \geq 0$ with equality at most at $\rho = 0$: strictly monotone. If $s_0 a < 0$, φ_x has exactly the two zeros $\rho = \pm r$, $r = \sqrt{|a|}$, and for $h_0 > 0$ ($s_0 = 1$, $a < 0$) $\varphi_x > 0$ for $|\rho| > r$, < 0 for $|\rho| < r$: a local maximum at $\rho = -r$, a local minimum at $\rho = r$. Using ρ as the variable of integration, $X = X(\rho, y)$ and $X_\rho(\rho, y) = |H(0)|^{-1/2}(1 + O(|\rho| + |y|))$ (recall $\rho = X\sqrt{s_0 H}$, so $\rho_X = \sqrt{|H|}(1 + O(\cdot))$); for $h_0 > 0$

$$\begin{aligned} u_{\max} - u_{\min} &= -\int_{-r}^{r} (a + \rho^2) X_\rho \, \mathrm{d}\rho \\ &= |H(0)|^{-1/2}\big(1 + O(r + |y|)\big) \int_{-r}^{r} (r^2 - \rho^2)\mathrm{d}\rho = \tfrac{4}{3}\, r^3 |H(0)|^{-1/2}(1 + o(1)), \end{aligned}$$

and $r^3 = |a|^{3/2}$. The count of preimages is the count for a function with one local maximum and one local minimum. For $h_0 < 0$ replace φ by $-\varphi$ (which changes the sign of h_0, H and a, swaps $u_{\max}$ and $u_{\min}$, and leaves $|a|$, $|H(0)|$ and the width unchanged). □

Definition A.49 (cusp point)**.** Let s_0 have corank 1 and kernel v. It is a *cusp point* if (C1) $\mathrm{d}\Delta(s_0)v = 0$, (C2) $\mathrm{d}\Delta(s_0) \neq 0$ (so $\{\Delta = 0\}$ is a smooth hypersurface $\mathcal{H}$ near s_0, to which v is tangent), and (C3) the function $m = \mathrm{d}\Delta \cdot v$ on $\mathcal{H}$ has $\mathrm{d}m(s_0)v \neq 0$. In the coordinates of Lemma A.45, $\Delta = j\varphi_x$, so (C1) is $\varphi_{xx} = 0$, (C2) is $\nabla_y \varphi_x \neq 0$ (given (C1)), and (C3) is $\varphi_{xxx} \neq 0$: indeed on $\mathcal{H} = \{\varphi_x = 0\}$ one has $m = j\varphi_{xx}$ up to a nonvanishing factor, and along v (tangent to $\mathcal{H}$ at s_0), $\frac{\mathrm{d}}{\mathrm{d}x}\varphi_{xx}(x, Y(x,y'), y') = \varphi_{xxx} + \varphi_{xxy_1} Y_x = \varphi_{xxx}$ at the point, because the slope of $\mathcal{H}$, $Y_x = -\varphi_{xx}/\varphi_{xy_1}$, vanishes there. So Lemma A.48 applies at a cusp point.

Let now X be a Banach space, $\Pi \cong \mathbb{R}^d$ a finite-dimensional parameter space and $G : \Pi \times X \to X$ of class C^1, $G(\lambda, f) = f - T(\lambda, f)$ with T compact (so $\mathrm{D}_f T$ and $\mathrm{D}_\lambda T$ are compact operators: the derivative of a compact C^1 map is compact, see [7]). Let $G(\lambda_0, f_0) = 0$ and $A = \mathrm{D}_f T(\lambda_0, f_0)$, $L = I - A$, $B = \mathrm{D}_\lambda T(\lambda_0, f_0)$. By Theorem A.33, $\dim\ker L = \operatorname{codim}\operatorname{ran} L < \infty$.

Proposition A.50 (Lyapunov–Schmidt reduction, corank one)**.** *Assume* $\ker L = \operatorname{span}(v)$ *is one-dimensional. Let* $\ell^* \in X^*$ *with* $\ker \ell^* = \operatorname{ran} L$*, fix* $e \in X$ *with* $\ell^*(e) = 1$ *and a closed complement* Y *of* $\operatorname{span}(v)$*. Then there is a* C^1 *map* $y(\lambda, a) \in Y$ *with* $y(\lambda_0, 0) = 0$ *such that, near* (λ_0, f_0)*,*

$$G(\lambda, f) = 0 \iff f = f_0 + av + y(\lambda, a) \quad \text{and} \quad h(\lambda, a) := \ell^* G\big(\lambda, f_0 + av + y(\lambda, a)\big) = 0.$$

The reduced function h *is* C^1*, with* $h(\lambda_0, 0) = 0$*,* $h_a(\lambda_0, 0) = 0$*,* $\mathrm{D}_\lambda h(\lambda_0, 0) = -\ell^* B$*; if* T *is* C^2 *then* h *is* C^2 *and* $h_{aa}(\lambda_0, 0) = -\ell^* \mathrm{D}_f^2 T[v, v]$*. Consequently:*

- *(a) (transversality) if* $\ell^* B \neq 0$*, the solution set is near* (λ_0, f_0) *a* C^1 *manifold of dimension* d*, parametrised by* (a, λ') *where* $\lambda = (s, \lambda')$ *and* s *is a coordinate with* $\ell^* B \partial_s \neq 0$*, and its projection to* Π *is* $(a, \lambda') \mapsto (\chi(a, \lambda'), \lambda')$*, a map of the scalar form of Lemma A.45;*
- *(b) (regular fold) if* T *is of class* C^∞ *(so that Lemma A.46 applies to* χ*) and* $\ell^* B \neq 0$*, the projection has a regular fold at the point iff* $\ell^* \mathrm{D}_f^2 T[v, v] \neq 0$*; the two branches (the two preimages) are then the two solutions* $a_\pm$ *of*

$h = 0$ for the same λ on the side of Γ where solutions exist. (If T is only C^2, the formula $\chi_{aa} = -h_{aa}/h_s$ still holds, and $\chi_{aa} \neq 0$ makes $a \mapsto \chi(a, \lambda')$ strictly convex or concave near 0, which gives the same count of at most two solutions, but not the normal form.)

PROOF. Let $P = I - e\ell^*$, a projection onto $\ker \ell^* = \operatorname{ran} L$. The map $\Theta(\lambda, a, y) = P\,G(\lambda, f_0 + av + y)$ from $\Pi \times \mathbb{R} \times Y$ to $\operatorname{ran} L$ is C^1 and $\mathrm{D}_y\Theta(\lambda_0, 0, 0) = PL|_Y = L|_Y : Y \to \operatorname{ran} L$. $L|_Y$ is injective ($Y \cap \ker L = 0$) and onto ($X = \operatorname{span}(v) \oplus Y$ and $Lv = 0$, so $L(X) = L(Y)$), with bounded inverse by the open mapping theorem ($\operatorname{ran} L$ is closed). By Theorem A.3 there is $y(\lambda, a)$ with $\Theta(\lambda, a, y(\lambda, a)) = 0$, locally unique. Then $G = 0$ iff in addition $(I - P)G = e\,\ell^*G = 0$, i.e. $h = 0$. Differentiating $\Theta(\lambda, a, y(\lambda, a)) = 0$ at the base point: $Ly_a = -PLv = 0$ so $y_a = 0$ (injectivity on Y); $Ly_\lambda = -PG_\lambda = PB$. Then $h_a = \ell^*G_f(v + y_a) = \ell^*Lv = 0$, and, since $\ell^*L = 0$, $\mathrm{D}_\lambda h = \ell^*(G_\lambda + Ly_\lambda) = \ell^*G_\lambda = -\ell^*B$. For h_{aa} differentiate $h_a = \ell^*G_f[v + y_a]$ once more: $h_{aa} = \ell^*\big(G_{ff}[v{+}y_a, v{+}y_a] + G_f y_{aa}\big) = \ell^*\mathrm{D}_f^2G[v, v]$ at the base point (the term ℓ^*Ly_{aa} vanishes), and $\mathrm{D}_f^2G = -\mathrm{D}_f^2T$. (a) $\mathrm{D}h(\lambda_0, 0) = (h_a, \mathrm{D}_\lambda h) = (0, -\ell^*B) \neq 0$; in coordinates $\lambda = (s, \lambda')$ with $h_s \neq 0$ the implicit function theorem gives $s = \chi(a, \lambda')$ with $h(\chi(a, \lambda'), \lambda', a) = 0$; this parametrises the solution set, and its projection to Π is $(a, \lambda') \mapsto (\chi, \lambda')$, which has the scalar form of Lemma A.45 with $x = a$, $\varphi = \sigma$, $\chi_a(0) = -h_a/h_s = 0$. (b) Differentiating $h(\chi(a, \lambda'), \lambda', a) = 0$ twice in a at the base point (where $\chi_a = 0$, $h_a = 0$): $h_s\chi_{aa} + h_{aa} = 0$, i.e. $\varphi_{xx} = \chi_{aa} = -h_{aa}/h_s$, nonzero iff $h_{aa} \neq 0$; Lemma A.46 gives the two-preimage structure. □

Remark A.51 (smoothness in the application, and the next coefficient)**.** When $G(D, \cdot) = I - T_D$ is real-analytic in all arguments (as it is in Chapter 8), y and h are C^∞ and the formulas for h_{aa}, h_{aaa} and the normal forms of Lemmas A.46 and A.48 are legitimate; with only C^1 the condition $h_{aa} \neq 0$ would not even make sense. If $h_{aa} = 0$ then $h_{aaa} = \ell^*\big(\mathrm{D}_f^3G[v, v, v] + 3\mathrm{D}_f^2G[v, y_{aa}]\big)$ with $y_{aa} = -(L|_Y)^{-1}P\,\mathrm{D}_f^2G[v, v]$; up to the factor $-1/h_s$ this is the cusp coefficient φ_{xxx} of Lemma A.48, and the unfolding condition (C2) is $\nabla_{\lambda'}h_a \neq 0$ on $\ker \mathrm{D}_\lambda h$.

A.8. Interval arithmetic and Krawczyk's lemma

In words. *A computer cannot check "f has a zero near x_0" by evaluating f, but it can check it by evaluating f* on intervals*: replacing every number by an interval that surely contains it and every operation by the interval operation, the result is an interval that surely contains the true value. Krawczyk's lemma turns such a computation into a* proof *that a smooth map has exactly one zero in a box: it is Newton's method written so that the output box is a contraction-like image of the input box, and Brouwer's theorem does the rest. It is used in Appendix D for the computer-assisted certificate of Chapter 9; the lemma is proved here, with its toy, so that the appendix only has to explain the software.*

An *interval* is $[\underline{s}, \overline{s}] \subset \mathbb{R}$; a *box* $\mathbf{X} \subset \mathbb{R}^n$ is a product of intervals, $\operatorname{mid}\mathbf{X}$ its midpoint, $\operatorname{rad}\mathbf{X} \in \mathbb{R}^n_{\geq 0}$ its componentwise radius. An *interval matrix* $\mathbf{M} = [\underline{M}, \overline{M}]$ is the set of real matrices N with $\underline{M} \leq N \leq \overline{M}$ entrywise; put $|\mathbf{M}|_{ij} = \max(|\underline{M}_{ij}|, |\overline{M}_{ij}|)$. For intervals $\mathbf{s}, \mathbf{t}$ and $\circ \in \{+, -, \times\}$ define $\mathbf{s} \circ \mathbf{t} = \{s \circ t : s \in \mathbf{s}, t \in \mathbf{t}\}$ (computable from the endpoints; for $\times$ the min and max of the four endpoint products), and, if $0 \notin \mathbf{t}$, $\mathbf{s}/\mathbf{t} = \{s/t : s \in \mathbf{s}, t \in \mathbf{t}\}$ (the min and max of the four endpoint quotients). For a function ϕ that is continuous and monotone on an

interval $\mathbf{s}$ (for instance exp on $\mathbb{R}$, tanh on $\mathbb{R}$, $\sqrt{\cdot}$ on $[0,\infty)$, sech^2 on $[0,\infty)$ and on $(-\infty,0]$) put $\phi(\mathbf{s}) = \{\phi(s) : s \in \mathbf{s}\}$, the interval between the values at the endpoints; a non-monotone function is split into monotone pieces.

Lemma A.52 (fundamental theorem of interval arithmetic)**.** *Let $\phi : \mathbb{D}_\phi \subset \mathbb{R}^n \to \mathbb{R}$ be given by a finite expression in the variables, real constants, the operations $+, -, \times, /$ and a finite list of monotone elementary functions as above, and let $\boldsymbol{\phi}(\mathbf{X})$ be the same expression evaluated in interval arithmetic on a box $\mathbf{X}$,* provided every divisor interval met on the way does not contain 0 and every elementary function is evaluated inside its domain of monotonicity *(otherwise $\boldsymbol{\phi}(\mathbf{X})$ is undefined and the computation fails, it never returns a wrong enclosure). Then $\mathbf{X} \subset \mathbb{D}_\phi$ and*

$$\{\phi(x) : x \in \mathbf{X}\} \subset \boldsymbol{\phi}(\mathbf{X}).$$

The same holds in floating point, with these conventions: (i) each real constant c is replaced by an interval $[\underline{c}, \overline{c}] \ni c$ whose endpoints are machine numbers (for a decimal such as 0.1, which is not a binary number, one takes the two neighbouring machine numbers, or a ball $[m \pm r]$ with m the midpoint in the working precision); (ii) every endpoint computed by $+, -, \times, /$ is rounded outward (the lower end down, the upper end up); (iii) for an elementary function the endpoints of the output are rounded outward after evaluating the function with a certified error bound (as arbitrary-precision ball arithmetic libraries do [17]).

Proof. Induction on the expression. If $s \in \mathbf{s}$, $t \in \mathbf{t}$ then $s \circ t \in \mathbf{s} \circ \mathbf{t}$ for $\circ \in \{+, -, \times\}$, and $s/t \in \mathbf{s}/\mathbf{t}$ when $0 \notin \mathbf{t}$, by definition; and $\phi(s) \in \phi(\mathbf{s})$ for a monotone elementary function. All operations are inclusion-monotone, so an enclosure of the arguments gives an enclosure of the result. Replacing the interval constants and the endpoints by outward-rounded machine numbers only enlarges each interval, so inclusions persist. □

The enclosure is in general larger than the range (the dependency problem: $\mathbf{x} - \mathbf{x} = [-w, w]$ for $\mathbf{x} = [0, w]$), but its overestimate shrinks linearly with the width of $\mathbf{X}$ for C^1 expressions [30, 33]. The same applies to the Jacobian: $\mathrm{D}f(\mathbf{X})$ denotes the interval matrix obtained by evaluating the entries of the Jacobian matrix of f in interval arithmetic; it contains $\mathrm{D}f(x)$ for all $x \in \mathbf{X}$.

Lemma A.53 (mean value form)**.** *Let $f : \mathbf{U} \to \mathbb{R}^m$ be C^1 on an open set containing the box $\mathbf{X} \subset \mathbb{R}^n$, $\hat{x} \in \mathbf{X}$, and $\mathbf{J} \supset \{\mathrm{D}f(x) : x \in \mathbf{X}\}$ an interval matrix. Then*

$$f(\mathbf{X}) \subset f(\hat{x}) + \mathbf{J}\,(\mathbf{X} - \hat{x}).$$

Proof. For $x \in \mathbf{X}$ the segment $[\hat{x}, x]$ lies in the convex box and $f(x) - f(\hat{x}) = N_x(x - \hat{x})$ with $N_x = \int_0^1 \mathrm{D}f(\hat{x} + s(x - \hat{x}))\mathrm{d}s$; each entry of N_x lies between the endpoints of the corresponding entry of $\mathbf{J}$, so $N_x \in \mathbf{J}$. □

The mean value form has *quadratic* overestimate in the width of $\mathbf{X}$ (the dependency error sits in $\mathbf{J}$ and is multiplied by $\mathbf{X} - \hat{x}$), which is what keeps enclosures of contracting flows from blowing up: see the interval ODE enclosure below.

Lemma A.54 (a priori enclosure and a validated Taylor step)**.** *Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a polynomial map, $y' = F(y)$, and define the Taylor coefficient polynomials $c_0(y) = y$, $c_{k+1}(y) = \frac{1}{k+1}\mathrm{D}c_k(y)F(y)$, so that every solution satisfies $\frac{\mathrm{d}}{\mathrm{d}t}c_k(y(t)) =$*

$(k+1)c_{k+1}(y(t))$ *and* $y(t)=\sum_{k\le N}c_k(y(0))t^k+$ *(remainder). Let* $h>0$, $N\ge 0$, $\mathbf{Y}$ *a box of initial values,* $\mathbf{B}$ *a box, and put (in interval arithmetic)*

$$\mathbf{P}=\sum_{k=0}^{N}c_k(\mathbf{Y})\,[0,h]^k,\qquad \mathbf{R}=c_{N+1}(\mathbf{B})\,[0,h]^{N+1}.$$

If $\mathbf{P}+\mathbf{R}\subset\operatorname{int}\mathbf{B}$*, then every solution with* $y(0)\in\mathbf{Y}$ *exists on* $[0,h]$*, stays in* $\mathbf{B}$*, and*

$$y(h)\in\sum_{k=0}^{N}c_k(\mathbf{Y})\,h^k+c_{N+1}(\mathbf{B})\,h^{N+1}.$$

(Combined with Lemma A.53 applied to the flow map and its variational equation, this gives the one-step enclosure used for the certificate of Appendix D.)

Proof. Let $y(0)\in\mathbf{Y}$; note $\mathbf{Y}\subset\mathbf{P}\subset\operatorname{int}\mathbf{B}$ (the term $k=0$), so $y(0)\in\operatorname{int}\mathbf{B}$. Let $T^*=\sup\{T\in[0,h]: y \text{ exists on } [0,T] \text{ and } y([0,T])\subset\mathbf{B}\}$. Since y stays in the compact set $\mathbf{B}$ on $[0,T^*)$, it extends to $[0,T^*]$ (Theorem A.4(b)) and $y([0,T^*])\subset\mathbf{B}$. Taylor's formula with Lagrange remainder, applied componentwise to $t\mapsto y_i(t)$ on $[0,T^*]$ (which is C^{N+1} because F is polynomial), gives $y_i(T^*)=\sum_{k\le N}c_k(y(0))_iT^{*k}+c_{N+1}(y(\tau_i))_iT^{*N+1}$ for some $\tau_i\in[0,T^*]$. Since $y(0)\in\mathbf{Y}$, $T^*\in[0,h]$, $y(\tau_i)\in\mathbf{B}$, Lemma A.52 gives $y(T^*)\in\mathbf{P}+\mathbf{R}\subset\operatorname{int}\mathbf{B}$. If $T^*<h$ the solution would stay in $\mathbf{B}$ a little beyond T^* (continuity and $y(T^*)\in\operatorname{int}\mathbf{B}$), contradicting the definition of T^*; so $T^*=h$. The final enclosure is the same Taylor formula at $T^*=h$. □

Toy example A.55 ($y'=y^2$, $y(0)=\frac12$, exact $y=1/(2-t)$)**.** With $h=\frac14$ and $\mathbf{B}=[\frac25,1]$ the hypothesis holds for every $N\ge2$ (here $c_k(y)=y^{k+1}$), and the enclosures of $y(\frac14)=\frac47=0.571428\ldots$ for $N=2,4,8,12$ are $0.5783\pm7.6\cdot10^{-3}$, $0.57190\pm4.9\cdot10^{-4}$, $0.5714305\pm1.9\cdot10^{-6}$, $0.57142858\pm7.5\cdot10^{-9}$, all containing $\frac47$ (exact rational arithmetic). The mean value form: for $y'=-4y$ and a box of initial values of radius ϱ, the naive interval Taylor step multiplies the radius by $\sum_k(4h)^k/k!\approx e^{4h}$ (growth although the flow contracts), the mean value form by e^{-4h} (`toy_apriori.py`).

Theorem A.56 (Brouwer's fixed point theorem). *A continuous map of a nonempty compact convex subset of* $\mathbb{R}^n$ *into itself has a fixed point.*

Reference. Brouwer [4]; the finite-dimensional case of Theorem A.31. □

Lemma A.57 (Krawczyk)**.** *Let* $f:\mathbf{U}\to\mathbb{R}^n$ *be* C^1 *on an open set* $\mathbf{U}$ *containing a box* $\mathbf{X}$ *with* $r=\operatorname{rad}\mathbf{X}>0$ *in every component,* $x_0=\operatorname{mid}\mathbf{X}$*,* $C\in\mathbb{R}^{n\times n}$ *any matrix, and* $\mathbf{J}\supset\{\mathrm{D}f(x):x\in\mathbf{X}\}$ *an interval matrix (for example* $\mathrm{D}f(\mathbf{X})$*). Put* $\mathbf{M}=I-C\mathbf{J}$ *(the interval matrix* $\{I-CN:N\in\mathbf{J}\}$ *enclosed by interval arithmetic) and*

$$\mathbf{K}(\mathbf{X})=x_0-Cf(x_0)+\mathbf{M}\,(\mathbf{X}-x_0).$$

If $\mathbf{K}(\mathbf{X})\subset\operatorname{int}\mathbf{X}$*, then* C *and every* $N\in\mathbf{J}$ *are invertible, and* f *has* exactly one *zero in* $\mathbf{X}$*. Moreover, the condition is equivalent to the vector inequality* $|Cf(x_0)|+|\mathbf{M}|\,r<r$ *(componentwise).*

Proof. *Equivalence.* $\mathbf{X}-x_0=[-r,r]$, and for an interval matrix the product $\mathbf{M}[-r,r]$ is the symmetric box of radius $|\mathbf{M}|r$ (the i-th entry is $\sum_j\mathbf{M}_{ij}[-r_j,r_j]=\sum_j[-|\mathbf{M}|_{ij}r_j,|\mathbf{M}|_{ij}r_j]$). So $\mathbf{K}(\mathbf{X})_i=[x_{0,i}-(Cf(x_0))_i-(|\mathbf{M}|r)_i,\ x_{0,i}-(Cf(x_0))_i+$

$(|\mathbf{M}|r)_i]$, and this lies in $(x_{0,i}-r_i, x_{0,i}+r_i)$ iff $|(Cf(x_0))_i| + (|\mathbf{M}|r)_i < r_i$. *Invertibility.* Define $\|v\|_r = \max_i |v_i|/r_i$. The inequality gives $|\mathbf{M}|r < r$, hence $\lambda := \max_i (|\mathbf{M}|r)_i/r_i < 1$. For any matrix N' with $|N'| \le |\mathbf{M}|$ entrywise, $\|N'v\|_r \le \max_i \sum_j |\mathbf{M}|_{ij}|v_j|/r_i \le \|v\|_r \max_i (|\mathbf{M}|r)_i/r_i = \lambda\|v\|_r$. Every $I - CN$ ($N \in \mathbf{J}$) belongs to $\mathbf{M}$, so its $\|\cdot\|_r$-operator norm is $\le \lambda < 1$ and $CN = I - (I - CN)$ is invertible (Lemma A.1); since C and N are square, both are invertible. *Existence.* Let $g(x) = x - Cf(x)$, $x \in \mathbf{X}$. For $x \in \mathbf{X}$ let $N_x = \int_0^1 \mathrm{D}f(x_0 + s(x - x_0))\mathrm{d}s$; the segment lies in the convex box $\mathbf{X}$, so every entry of N_x lies between the endpoints of the corresponding entry of $\mathbf{J}$: $N_x \in \mathbf{J}$. The fundamental theorem of calculus gives $f(x) - f(x_0) = N_x(x - x_0)$ and

$$g(x) = x_0 - Cf(x_0) + (I - CN_x)(x - x_0) \in \mathbf{K}(\mathbf{X}) \subset \operatorname{int}\mathbf{X}.$$

So g maps the compact convex set $\mathbf{X}$ into itself; by Theorem A.56 it has a fixed point x^*, i.e. $Cf(x^*) = 0$, hence $f(x^*) = 0$ since C is invertible. *Uniqueness.* If $f(x) = f(y) = 0$ with $x, y \in \mathbf{X}$, then $0 = f(x) - f(y) = N(x - y)$ with $N = \int_0^1 \mathrm{D}f(y + s(x - y))\mathrm{d}s \in \mathbf{J}$ invertible, so $x = y$. □

Remark A.58. In practice C is an (approximate) inverse of $\mathrm{D}f(x_0)$ and the verification is done with outward-rounded interval arithmetic (Lemma A.52), so the computed $\mathbf{K}(\mathbf{X}) \subset \operatorname{int}\mathbf{X}$ is a proof. The statement is due to Krawczyk [20]; see [30, 31, 33, 48] and, for the software used in the book, Johansson [17] and Lohner [28]. This lemma is the statement used as the key step of the certificate in Appendix D. For the cross-check by the Newton–Kantorovich theorem see Appendix C.

Corollary A.59 (Krawczyk with a non-differentiable perturbation)**.** *Let* $\Phi = \Phi_0 + \mathsf{Z} : \mathbf{X} \to \mathbb{R}^n$ *be continuous on the box* $\mathbf{X}$ *(*$x_0 = \operatorname{mid}\mathbf{X}$*,* $r = \operatorname{rad}\mathbf{X} > 0$*), where* Φ_0 *is* C^1 *near* $\mathbf{X}$ *and* $\mathsf{Z}(x) \in \mathbf{T}$ *for all* $x \in \mathbf{X}$ *(a box). Let* $C \in \mathbb{R}^{n\times n}$*,* $\mathbf{J} \supset \{\mathrm{D}\Phi_0(x) : x \in \mathbf{X}\}$*,* $\mathbf{V} \ni \Phi_0(x_0)$*, and*

$$\mathbf{K} = x_0 - C(\mathbf{V} + \mathbf{T}) + (I - C\mathbf{J})(\mathbf{X} - x_0).$$

If $\mathbf{K} \subset \operatorname{int}\mathbf{X}$*, then* C *is invertible and* Φ *has a zero in* $\mathbf{K} \subset \mathbf{X}$*. (Uniqueness is claimed only when* $\mathsf{Z} \equiv 0$*, by Lemma A.57.)*

Proof. As in Lemma A.57: for $x \in \mathbf{X}$, $g(x) = x - C\Phi(x) = x_0 - C\Phi_0(x_0) - C\mathsf{Z}(x) + (I - CN_x)(x - x_0) \in \mathbf{K}$ with $N_x \in \mathbf{J}$, so g is a continuous self-map of $\mathbf{X}$ and Brouwer gives a fixed point, i.e. $C\Phi(x^*) = 0$. The inclusion $\mathbf{K} \subset \operatorname{int}\mathbf{X}$ forces, componentwise, $|I - C\mathbf{J}|\,r < r$ (the radius of $\mathbf{K}$ is at least that of $(I - C\mathbf{J})(\mathbf{X} - x_0)$, namely $|I - C\mathbf{J}|r$), so every CN, $N \in \mathbf{J}$, is invertible as in the lemma, hence C is, and $\Phi(x^*) = 0$. □

This is the form used in Appendix D, where the pole tails enter as the enclosed, non-differentiated perturbation Z.

Toy example A.60 ($x^2 - 2$ and a 2×2 system)**.** $f(x) = x^2 - 2$ on $\mathbf{X} = [1.4, 1.5]$: $x_0 = 1.45$, $C = 1/f'(x_0) = 1/2.9$, $\mathrm{D}f(\mathbf{X}) = [2.8, 3.0]$, $\mathbf{M} = 1 - [2.8, 3.0]/2.9 = [-0.0345, 0.0345]$, and $\mathbf{K}(\mathbf{X}) = 1.45 - \frac{0.1025}{2.9} + \mathbf{M}\cdot[-0.05, 0.05] = 1.41466 \pm 0.00173 = [1.4129, 1.4164] \subset (1.4, 1.5)$ (exact rational arithmetic, `toy_krawczyk.py`): a proof that there is exactly one zero in $\mathbf{X}$ (it is $\sqrt{2} = 1.41421\ldots$). On $[1.5, 1.6]$, where there is no zero, the test fails (as it must). For $f(x, y) = (x^2 + y^2 - 1, x - y)$ on $[0.6, 0.8]^2$ the test passes with $\mathbf{K} = [0.6929, 0.7214]^2$: the unique zero is $(1/\sqrt{2}, 1/\sqrt{2})$.

A.9. Index of the standard results

Remark A.61 (standard theorems used in the book, with references)**.** The table lists every result of this appendix, its status (ref: textbook theorem used as a black box, with reference; proved: proof given above) and the chapters that use it.

Result	Status	Reference	Used in
Implicit/inverse function (Theorem A.3)	ref	[8, 54, 57]	Chapters 4, 7 and 9
ODE: existence, continuation, smooth dependence (Theorem A.4)	ref	[6, 14, 46]	everywhere
Gronwall, Volterra (Lemmas A.5 and A.6)	proved	[13, 14]	Chapters 3 and 4
Levinson, L^1 case (Proposition A.8)	proved	[6, 23]	Chapters 3 and 4
Banach, uniform contraction (Theorems A.2 and A.11)	proved	[1, 8]	Chapters 4, 6 and 7
Taylor lemma, loss of weight (Lemma A.15)	proved	—	Chapters 3 and 4
Whitney extension (Theorem A.17, Corollary A.18)	ref / proved	[51]	Chapters 4 and 7
Even functions (Theorem A.20)	ref	[52]	Chapter 7
Cone lemma, Kamke–Müller (Lemma A.21 and Theorem A.24)	proved	[18, 32, 43, 50]	Chapters 5, 6 and 12
Arzelà–Ascoli, on the line (Theorem A.28, Lemma A.29)	ref / proved	[37]	Chapter 8
Schauder, Dugundji, Riesz–Schauder (Theorems A.31 to A.33)	ref	[3, 7, 9, 38]	Chapters 8 and 9
Leray–Schauder degree (D1)–(D5) (Theorem A.35)	ref	[7, 22]	Chapters 8 and 9
(D6), continuation principle, Whyburn (Lemma A.39 and Theorems A.41 and A.42)	proved	[55]	Chapter 8
Fold, cusp, Lyapunov–Schmidt (Lemmas A.46 and A.48 and Proposition A.50)	proved	[5, 11, 53]	Chapter 9
Interval arithmetic, mean value form, validated Taylor step, Krawczyk (Lemmas A.52 to A.54 and A.57 and Corollary A.59)	proved	[20, 30, 33]	Chapter 9, Appendix D

Not included here: Sturm comparison, the Riccati toolbox and the Wronskian identities (these are proved in Chapter 3); Hadamard's lemma on $C^{1,1}$ functions with a rank-one Hessian jump (proved where it is used, in Chapter 7); Newton–Kantorovich (Appendix C).

APPENDIX B

Special functions and closed forms

In words. *For the models of this appendix the Wronskians, integrals and thresholds used in the monograph are* explicit*: the no-swirl equation $z'' = W_{\boldsymbol{c}} z$ is Gauss's hypergeometric equation in disguise, so Jost solutions and the window width are Gamma-function quotients; at the rest state the first-order swirl integral is elementary and the small-swirl edge coefficient is a combination of digamma values; near a pole the equation is Bessel's, and keeping the whole polynomial potential gives an exact Fuchsian model. Each identity is proved and checked numerically.*

Roadmap. This appendix is read after Chapter 3 and uses its Jost solutions (existence and positivity, Lemma 3.21); nothing in Chapter 3 depends on the propositions below. Appendix B.1: textbook facts. Appendix B.2: Jost solutions in ${}_2F_1$ form and their Wronskian (Proposition B.3); Appendix B.3: the Landau family and Pöschl–Teller (Example B.6). Appendices B.4 and B.5: the Landau swirl integral (Proposition B.8) and the digamma formula (Proposition B.12). Appendix B.6: Bessel and Fuchsian models (Propositions B.15 and B.18). Appendix B.7: numerical confirmation.

B.1. Conventions and textbook facts

Throughout, $x = \tanh\xi$, $' = d/d\xi$, and

$$t := \tfrac{1}{2}(1+x) = \frac{1}{1+e^{-2\xi}} \in (0,1), \qquad 1-t = \frac{1}{1+e^{2\xi}}, \qquad \frac{d}{d\xi} = 2t(1-t)\frac{d}{dt} \tag{B.1}$$

(because $dt/d\xi = \tfrac{1}{2}(1-x^2) = 2t(1-t)$). For $\boldsymbol{c} = (c_1, c_2, c_3)$ with $c_1, c_2 > -1$ put $k_i = \sqrt{1+c_i} > 0$ and $s = \sqrt{1-2c_3}$, with $s \in i\,(0,\infty)$ if $c_3 > \tfrac{1}{2}$. Since $1-x = 2(1-t)$, $1+x = 2t$, $1-x^2 = 4t(1-t)$,

$$W_{\boldsymbol{c}} = 1 + \tfrac{1}{2}P_{\boldsymbol{c}}(\tanh\xi) = k_1^2 + (k_2^2 - k_1^2 + 2c_3)\,t - 2c_3\,t^2, \quad W_{\boldsymbol{c}}(\mp\infty) = k_{1,2}^2. \tag{B.2}$$

Note $1 - 2\bar{c}_3 = 1 + (k_1+k_2)(k_1+k_2+2) = (k_1+k_2+1)^2$, i.e. *the critical surface $c_3 = \bar{c}_3$ is $s = k_1 + k_2 + 1$*. We write $u_\mp$ for the Jost solutions of $W_{\boldsymbol{c}}$ normalised by $u_\mp(\xi_0) = 1$ and ζ_-, ζ_+ for their logarithmic derivatives (Lemma 3.21), and $w_0(\xi_0) := \zeta_-(\xi_0) - \zeta_+(\xi_0) = \mathrm{Wr}(u_+, u_-)(\xi_0)$ for the width of the no-swirl window in the h-variable (the window in U-units has width $2w_0(\xi_0)$). We use the following textbook facts, with the references of [35].

(F1) *Gamma and digamma* (DLMF §§5.5, 5.15, 5.11, 5.7): $\Gamma(z+1) = z\Gamma(z)$; $\Gamma(z)\Gamma(1-z) = \pi/\sin\pi z$; duplication $\Gamma(2z) = 2^{2z-1}\pi^{-1/2}\Gamma(z)\Gamma(z+\tfrac{1}{2})$. The digamma function $\psi^{(0)} = \Gamma'/\Gamma$ satisfies $\psi^{(0)}(z) = -\gamma_{\mathrm{E}} + \sum_{m\ge0}\big(\frac{1}{m+1} - \frac{1}{m+z}\big)$ (so $\psi^{(0)}(z) - \psi^{(0)}(w) = \sum_{m\ge0}\big(\frac{1}{m+w} - \frac{1}{m+z}\big)$) and $\psi^{(0)}(1-z) - \psi^{(0)}(z) =$

$\pi\cot\pi z$. For fixed real a,b, as $X\to+\infty$, $\Gamma(X+a)/\Gamma(X+b) = X^{a-b}\big[1+\frac{(a-b)(a+b-1)}{2X}+O(X^{-2})\big]$.

(F2) *Gauss functions* (DLMF §§15.2, 15.4, 15.5): for $\mathfrak{c}\notin\{0,-1,-2,\dots\}$ the series ${}_2F_1(a,b;\mathfrak{c};t)$ converges for $|t|<1$, is the solution of $t(1-t)w''+[\mathfrak{c}-(a+b+1)t]w'-ab\,w=0$ with $w(0)=1$, and $\frac{d}{dt}{}_2F_1(a,b;\mathfrak{c};t) = \frac{ab}{\mathfrak{c}}{}_2F_1(a+1,b+1;\mathfrak{c}+1;t)$. As $t\uparrow 1$, ${}_2F_1(a,b;\mathfrak{c};t)$ is $O(1)$, $O(|\ln(1-t)|)$, $O((1-t)^{\mathfrak{c}-a-b})$ according as $\mathfrak{c}-a-b>0,=0,<0$, and *Gauss's summation theorem*: if $\mathfrak{c}-a-b>0$ then ${}_2F_1(a,b;\mathfrak{c};1)=\Gamma(\mathfrak{c})\Gamma(\mathfrak{c}-a-b)/[\Gamma(\mathfrak{c}-a)\Gamma(\mathfrak{c}-b)]$.

(F3) *Modified Bessel functions* (DLMF §§10.25, 10.27, 10.40): for $\nu_{\mathrm{B}}\in(0,1)$, $I_{\pm\nu_{\mathrm{B}}}(y) = (y/2)^{\pm\nu_{\mathrm{B}}}\sum_{m\ge0}(y/2)^{2m}/[m!\,\Gamma(m+1\pm\nu_{\mathrm{B}})]$ solve $y^2w''+yw'-(y^2+\nu_{\mathrm{B}}^2)w=0$; $K_{\nu_{\mathrm{B}}} = \frac{\pi}{2\sin\nu_{\mathrm{B}}\pi}(I_{-\nu_{\mathrm{B}}}-I_{\nu_{\mathrm{B}}})$; and as $y\to\infty$, $I_{\pm\nu_{\mathrm{B}}}(y)\sim e^y/\sqrt{2\pi y}$, $K_{\nu_{\mathrm{B}}}(y)\sim\sqrt{\pi/2y}\,e^{-y}$.

B.2. Hypergeometric Jost solutions

Lemma B.1 (Gauss reduction)**.** *Let $\alpha\in\{\pm k_1/2\}$, $\beta\in\{\pm k_2/2\}$, $\delta=\alpha+\beta$, $\varphi=t^\alpha(1-t)^\beta$ and $w\in C^2(0,1)$. Then, with $\partial_\xi = d/d\xi = 2t(1-t)\,d/dt$,*

$$\partial_\xi^2(\varphi w) - W_{\boldsymbol{c}}\,\varphi w = 4t(1-t)\varphi\Big[t(1-t)w''+\big(1+2\alpha-(2\delta+2)t\big)w'-\big(\delta^2+\delta+\tfrac{c_3}{2}\big)w\Big]. \tag{B.3}$$

The bracket is Gauss's operator with $\mathfrak{c}=1+2\alpha$, $a+b=2\delta+1$, $ab=\delta^2+\delta+\frac{c_3}{2}$, i.e. $\{a,b\}=\{\delta+\frac{1+s}{2},\delta+\frac{1-s}{2}\}$.

Proof. Put $\chi=\alpha-\delta t$. From $\varphi'/\varphi=\alpha/t-\beta/(1-t)$ we get $\partial_\xi\varphi=2\chi\varphi$, and $\partial_\xi\chi=-2\delta t(1-t)$, hence $\partial_\xi^2\varphi=(4\chi^2-4\delta t(1-t))\varphi$. Therefore $\partial_\xi(\varphi w)=\varphi(\partial_\xi w+2\chi w)$ and $\partial_\xi^2(\varphi w)=\varphi\big[\partial_\xi^2 w+4\chi\,\partial_\xi w+(4\chi^2-4\delta t(1-t))w\big]$. Insert $\partial_\xi w=2t(1-t)w'$ and $\partial_\xi^2 w=4t^2(1-t)^2w''+4t(1-t)(1-2t)w'$:

$$\partial_\xi^2(\varphi w)-W_{\boldsymbol{c}}\varphi w = 4t(1-t)\varphi\Big[t(1-t)w''+(1-2t+2\chi)w'+\Big(\tfrac{4\chi^2-W_{\boldsymbol{c}}}{4t(1-t)}-\delta\Big)w\Big],$$

and $1-2t+2\chi=1+2\alpha-(2\delta+2)t$. It remains to show $4\chi^2-W_{\boldsymbol{c}}=4t(1-t)(\delta-ab)$ with $ab=\delta^2+\delta+c_3/2$. Both sides are polynomials of degree ≤2 in t. At $t=0$: $4\alpha^2-W_{\boldsymbol{c}}(0)=k_1^2-k_1^2=0$; at $t=1$: $4(\alpha-\delta)^2-W_{\boldsymbol{c}}(1)=4\beta^2-k_2^2=0$ (use (B.2)); the coefficients of t^2 are $4\delta^2+2c_3$ and $-4(\delta-ab)=4\delta^2+2c_3$. A polynomial of degree ≤2 whose values at $0,1$ and whose t^2-coefficient vanish is zero. Finally a,b are the roots of $\lambda^2-(2\delta+1)\lambda+\delta^2+\delta+\frac{c_3}{2}$, whose discriminant is $(2\delta+1)^2-4\delta^2-4\delta-2c_3=1-2c_3=s^2$. □

Toy example B.2 (the rest state in hypergeometric dress)**.** For $\boldsymbol{c}=0$: $k_1=k_2=s=1$, $\{a,b\}=\{1,0\}$, so ${}_2F_1(a,b;2;t)\equiv1$ and $u=t^{1/2}(1-t)^{-1/2}=e^{\xi}$ solves $u''=u$ (as $t/(1-t)=e^{2\xi}$); with $\alpha=-\frac12,\beta=\frac12$ one gets $e^{-\xi}$.

Proposition B.3 (Jost solutions of $W_{\boldsymbol{c}}$)**.** *Let $c_1,c_2>-1$, $c_3\in\mathbb{R}$, $a_\pm=\frac12(k_1-k_2+1\pm s)$, $b_\pm=\frac12(k_2-k_1+1\pm s)$, and for $\xi\in\mathbb{R}$ (t as in (B.1))*

$$u_-^{\mathrm{H}}(\xi) = t^{k_1/2}(1-t)^{-k_2/2}\,{}_2F_1(a_+,a_-;1+k_1;t)\,,$$
$$u_+^{\mathrm{H}}(\xi) = (1-t)^{k_2/2}\,t^{-k_1/2}\,{}_2F_1(b_+,b_-;1+k_2;1-t)\,.$$

(a) $u^{\mathrm{H}}_{\mp}$ *solve* $u'' = W_{\boldsymbol{c}}u$ *on* $\mathbb{R}$*, are real, and* $u^{\mathrm{H}}_{-}e^{-k_1\xi} \to 1$ *as* $\xi \to -\infty$*,* $u^{\mathrm{H}}_{+}e^{k_2\xi} \to 1$ *as* $\xi \to +\infty$*. Hence* u^{H}_{-} *(*u^{H}_{+}*) is the solution recessive at* $-\infty$ *(*$+\infty$*), and if* $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ *then* $u_{\mp} = u^{\mathrm{H}}_{\mp}/u^{\mathrm{H}}_{\mp}(\xi_0)$*.*

(b) Where $u^{\mathrm{H}}_{-} \neq 0$ *(everywhere if* $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$*),*

$$\zeta_- = k_1(1-t) + k_2 t + 2t(1-t)\frac{a_+a_-}{1+k_1}\frac{{}_2F_1(a_+ +1, a_- +1; k_1+2; t)}{{}_2F_1(a_+, a_-; k_1+1; t)},$$

and ζ_+ *is given by the mirror formula* $\zeta_+(\xi) = -\zeta_-\big|_{\xi\to-\xi,\ c_1\leftrightarrow c_2}$*, i.e. with* $b_\pm$*,* $k_1 \leftrightarrow k_2$ *and* $t \leftrightarrow 1-t$ *in the displayed expression and an overall minus sign.*

*(c) (*Gamma connection formula*)* $\operatorname{Wr}(u^{\mathrm{H}}_{+}, u^{\mathrm{H}}_{-}) = \dfrac{2\,\Gamma(1+k_1)\,\Gamma(1+k_2)}{\Gamma(\frac{k_1+k_2+1+s}{2})\,\Gamma(\frac{k_1+k_2+1-s}{2})}$.

Proof. (a) The mirror $\xi \mapsto -\xi$ maps $t \mapsto 1-t$ and $W_{(c_1,c_2,c_3)}$ to $W_{(c_2,c_1,c_3)}$, and $u^{\mathrm{H}}_{+}[\boldsymbol{c}](\xi) = u^{\mathrm{H}}_{-}[c_2,c_1,c_3](-\xi)$; it suffices to treat u^{H}_{-}. Apply Lemma B.1 with $\alpha = \frac{k_1}{2}$, $\beta = -\frac{k_2}{2}$: then $\delta = \frac{k_1-k_2}{2}$, $\mathfrak{c} = 1+k_1 \notin \{0,-1,\dots\}$, $\{a,b\} = \{a_+,a_-\}$, and $w = {}_2F_1(a_+,a_-;1+k_1;t)$ makes the bracket in (B.3) vanish by (F2). As $\xi \to -\infty$, $t = e^{2\xi}/(1+e^{2\xi})$, so $t^{k_1/2} = e^{k_1\xi}(1+O(e^{2\xi}))$, $(1-t)^{-k_2/2} \to 1$ and $w \to 1$. The other solution behaves like $e^{-k_1\xi}$ (the exponents of $z'' = W_{\boldsymbol{c}}z$ at $-\infty$ are $\pm k_1$), so u^{H}_{-} is the recessive one; if $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ the Jost solution is positive (Lemma 3.21) and unique up to a factor. Reality: $a_\pm$ are real or complex conjugates, so the series has real sum. (b) $\partial_\xi \ln u^{\mathrm{H}}_{-} = 2\chi + 2t(1-t)w'/w$ with $2\chi = 2(\alpha - \delta t) = k_1 - (k_1-k_2)t = k_1(1-t) + k_2 t$ and $w' = \frac{a_+a_-}{1+k_1}{}_2F_1(a_++1, a_-+1; k_1+2; t)$ by (F2). The mirror statement follows from $u^{\mathrm{H}}_{+}(\xi) = u^{\mathrm{H}}_{-}[c_2,c_1,c_3](-\xi)$ and the chain rule. (c) $\operatorname{Wr}(u^{\mathrm{H}}_{+}, u^{\mathrm{H}}_{-})$ is constant ($W_{\boldsymbol{c}}$ is the same potential for both), so we let $\xi \to +\infty$ ($t \uparrow 1$). Put $F = {}_2F_1(a_+,a_-;1+k_1;t)$. Here $\mathfrak{c} - a_+ - a_- = 1 + k_1 - (k_1 - k_2 + 1) = k_2 > 0$, so by Gauss's theorem $F(t) \to F(1) = \Gamma(1+k_1)\Gamma(k_2)/[\Gamma(1+k_1-a_+)\Gamma(1+k_1-a_-)]$, and $1+k_1-a_\pm = \frac{1}{2}(k_1+k_2+1\mp s)$. By (b), $u^{\mathrm{H}\prime}_{-} = u^{\mathrm{H}}_{-}\,[k_2 + (k_1-k_2)(1-t)] + (1-t)^{-k_2/2}t^{k_1/2}\cdot 2t(1-t)F'(t)$ and by (F2) $(1-t)F'(t) = O\big((1-t)^{\min(1,k_2)}(1+|\ln(1-t)|)\big) \to 0$ since $k_2 > 0$. Since $(1-t)^{-k_2/2} = e^{k_2\xi}(1+o(1))$, $u^{\mathrm{H}}_{-} = e^{k_2\xi}F(1)(1+o(1))$ and $u^{\mathrm{H}\prime}_{-} = e^{k_2\xi}[k_2F(1)+o(1)]$. By (a) for the mirror, $u^{\mathrm{H}}_{+} = e^{-k_2\xi}(1+o(1))$, and likewise $u^{\mathrm{H}\prime}_{+} = -k_2e^{-k_2\xi}(1+o(1))$ (the mirror of (b): the factor $2t(1-t)w'/w$ is $O(1-t)$ at $t=1$). Then $\operatorname{Wr}(u^{\mathrm{H}}_{+}, u^{\mathrm{H}}_{-}) = u^{\mathrm{H}}_{+}u^{\mathrm{H}\prime}_{-} - u^{\mathrm{H}\prime}_{+}u^{\mathrm{H}}_{-} \to k_2F(1) + k_2F(1) = 2k_2F(1)$, and $2k_2\Gamma(k_2) = 2\Gamma(1+k_2)$. □

Corollary B.4 (the no-swirl window in closed form)**.** *For* $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$ *and* $\xi_0 \in \mathbb{R}$*,*

$$w_0(\xi_0) = \zeta_-(\xi_0) - \zeta_+(\xi_0) = \frac{2\,\Gamma(1+k_1)\Gamma(1+k_2)}{\Gamma\big(\frac{k_1+k_2+1+s}{2}\big)\Gamma\big(\frac{k_1+k_2+1-s}{2}\big)\,u^{\mathrm{H}}_{-}(\xi_0)\,u^{\mathrm{H}}_{+}(\xi_0)} > 0,$$

and the endpoints of the window $U^{+}_{c}(x_0) = 2(\zeta_-(\xi_0) - x_0)$*,* $U^{-}_{c}(x_0) = 2(\zeta_+(\xi_0) - x_0)$ *(Theorem 3.44) are given by Proposition B.3(b). Moreover, for all* $c_1, c_2 > -1$*,* $c_3 \in \mathbb{R}$*, the Wronskian* $\operatorname{Wr}(u^{\mathrm{H}}_{+}, u^{\mathrm{H}}_{-})$ *of Proposition B.3(c) vanishes exactly when* $s = k_1 + k_2 + 1 + 2n$*,* $n \in \{0,1,2,\dots\}$*; the value* $n = 0$ *is the critical surface* $c_3 = \bar{c}_3$*, where the Jost solutions become proportional.*

Proof. $\operatorname{Wr}(u_+,u_-)(\xi_0) = \operatorname{Wr}(u^{\mathrm{H}}_{+}, u^{\mathrm{H}}_{-})/(u^{\mathrm{H}}_{+}u^{\mathrm{H}}_{-})(\xi_0) = u_+u_-(\zeta_- - \zeta_+)(\xi_0) = \zeta_- - \zeta_+$ because $u_\pm(\xi_0) = 1$. Positivity: $u^{\mathrm{H}}_{\mp} > 0$ for $\boldsymbol{c} \in \operatorname{int}\mathcal{J}$; and for $c_3 > \bar{c}_3$ either $s \in [0, k_1+k_2+1)$, both Gamma arguments are positive, or s is imaginary and the product is $|\Gamma(\cdot)|^2 > 0$. The last sentence: $1/\Gamma$ vanishes exactly at $0, -1, -2, \dots$,

so the Wronskian vanishes iff $\frac{1}{2}(k_1+k_2+1-s)=-n$, and $s=k_1+k_2+1$ is $c_3=\bar{c}_3$. □

Remark B.5. Parts (a), (b) restate Proposition 3.47 (which allows $c_i\geq -1$ and normalises by the principal solution, $u^{\mathrm{hyp}}=2^{-(k_1+k_2)/2}u_-$ of Lemma 3.42) with pole normalisation; (c) is new relative to it. For $c_i>-1$ (c) re-proves the positivity of the Jost Wronskian above the critical surface (cf. Lemma 3.35).

B.3. Worked examples: the Landau family and the Pöschl–Teller potential

Example B.6 (the Landau family; Pöschl–Teller)**.** *(a) The rest state* $\boldsymbol{c}=0$. Here $W_{\boldsymbol{c}}\equiv 1$, $u^{\mathrm{H}}_{\mp}=e^{\pm\xi}$ (Toy example B.2), $\mathrm{Wr}(u^{\mathrm{H}}_{+},u^{\mathrm{H}}_{-})=2$, in agreement with Proposition B.3(c): $2\Gamma(2)^2/[\Gamma(2)\Gamma(1)]=2$ ($k_1=k_2=s=1$). Every positive solution $Ae^{\xi}+Be^{-\xi}$ ($A,B\geq 0$) is, up to a positive factor, $\cosh\xi$ (the case $A=B$: $h=x$, $U\equiv 0$) or $z_\lambda=\lambda\cosh\xi+\sinh\xi$ with $|\lambda|\geq 1$ (replace z_λ by $-z_\lambda$ if $\lambda\leq -1$), and its Riccati log is

$$h=\frac{z_\lambda'}{z_\lambda}=\frac{\lambda x+1}{x+\lambda},\qquad U=2(h-x)=\frac{2(1-x^2)}{x+\lambda}\qquad (x=\tanh\xi),$$

the *Landau family*: $x\lambda+1-x^2-\lambda x=1-x^2$. For $\lambda\downarrow 1$, $h\to 1=\zeta_-$ and $U\to 2(1-x)=U_c^{+}$; for $\lambda\uparrow -1$ (use $-z_\lambda$), $h\to -1=\zeta_+$ and $U\to -2(1+x)=U_c^{-}$; the window at x_0 is $[-2(1+x_0),2(1-x_0)]$, of width $2w_0=4$. *(b) Pöschl–Teller.* Take $c_1=c_2=k^2-1$, $c_3=-2\ell(\ell+1)$, so $W_{\boldsymbol{c}}=k^2-\ell(\ell+1)\operatorname{sech}^2\xi$, $s=2\ell+1$, $a_\pm=\ell+1,-\ell$ and (since $k_1=k_2=k$) $u^{\mathrm{H}}_{-}=e^{k\xi}{}_2F_1(\ell+1,-\ell;1+k;t)$. For $\ell\in\mathbb{N}$ the series terminates; for $\ell=1$, ${}_2F_1(2,-1;1+k;t)=1-\frac{2t}{1+k}=\frac{k-\tanh\xi}{1+k}$, so

$$u^{\mathrm{H}}_{-}=\frac{e^{k\xi}(k-\tanh\xi)}{1+k},\qquad u^{\mathrm{H}}_{+}=\frac{e^{-k\xi}(k+\tanh\xi)}{1+k},$$
$$\mathrm{Wr}(u^{\mathrm{H}}_{+},u^{\mathrm{H}}_{-})=\frac{2\Gamma(1+k)^2}{\Gamma(k+\ell+1)\Gamma(k-\ell)}\Big|_{\ell=1}=\frac{2k(k-1)}{k+1}.$$

(Direct check: as $\xi\to+\infty$, $u^{\mathrm{H}}_{-}\sim\frac{k-1}{k+1}e^{k\xi}$.) The Wronskian vanishes exactly at $k=1=\ell$, the critical case of Toy example 3.34 ($\Gamma(k-\ell)$ has its first pole at $k=\ell$), and for $k>1$ it is positive.

B.4. The first-order swirl integral along a Landau solution

Fix $|\lambda|>1$ and $x_0\in(-1,1)$, so $x+\lambda\neq 0$ on $[-1,1]$. Along the Landau solution z_λ of Example B.6, $(1-x^2)z_\lambda^2=(x+\lambda)^2$, hence the swirl slope $f_x=f'/(1-x^2)=\kappa/[(1-x^2)z^2]$ is proportional to $(x+\lambda)^{-2}$. Normalise

$$g(\eta)=\Big(\frac{x_0+\lambda}{\eta+\lambda}\Big)^2\ (g(x_0)=1),\qquad \hat{f}(\eta)=\int_{x_0}^{\eta}g=\frac{(x_0+\lambda)(\eta-x_0)}{\eta+\lambda},$$
$$\hat{\psi}_\lambda(x)=\int_{x_0}^{x}\frac{(x-\eta)^2}{1-\eta^2}\,\hat{f}(\eta)g(\eta)\,d\eta.$$

Remark B.7 (meaning of $\hat{\psi}_\lambda$)**.** (i) $\hat{\psi}_\lambda$ is the solution of $\mathsf{L}[\hat{\psi}]=\hat{f}^2$, $\hat{\psi}(x_0)=\hat{\psi}'(x_0)=0$ with $\mathsf{L}=(1-x^2)\partial_x^2+2x\partial_x-2$: indeed $\partial_x\hat{\psi}=\int\frac{2(x-\eta)}{1-\eta^2}\hat{f}\hat{f}_\eta$, $\partial_x^2\hat{\psi}=\int\frac{2}{1-\eta^2}\hat{f}\hat{f}_\eta$ (the boundary terms vanish because $(x-\eta)^2$ vanishes to second order at $\eta=x$), and the kernel of L collects to $\frac{2(1-x^2)+4x(x-\eta)-2(x-\eta)^2}{1-\eta^2}=2$, so $\mathsf{L}[\hat{\psi}]=2\int_{x_0}^{x}\hat{f}\hat{f}_\eta\,d\eta=\hat{f}^2$.

(ii) Since $\sinh(\xi-\eta)=\cosh\xi\cosh\eta\,(x-r)$ $(r=\tanh\eta)$, $\sinh^2(\xi-\eta)/\cosh^2\xi=(x-r)^2/(1-r^2)$, the kernel of the swirl integral ψ in the variable ξ; so $\hat\psi_\lambda$ is what ψ/κ^2 becomes for the first-order swirl profile $\kappa\hat f$ (Chapter 11).

Proposition B.8 (Landau first-order swirl integral)**.** *Put $m=x_0+\lambda$, $X=x+\lambda$, $\Delta=\lambda^2-1$. For $x\in(-1,1)$, $\hat\psi_\lambda(x)=G(x)-G(x_0)$ with*

$$G(\eta)=r_+\ln|\eta-1|+r_-\ln|\eta+1|+\rho_1\ln|\eta+\lambda|-\frac{\rho_2}{\eta+\lambda}-\frac{\rho_3}{2(\eta+\lambda)^2},$$

$$r_+=\frac{m^3(x-1)^2(x_0-1)}{2(\lambda+1)^3},\qquad r_-=-\frac{m^3(x+1)^2(x_0+1)}{2(\lambda-1)^3},$$

$$\rho_3=\frac{X^2m^4}{\Delta},\qquad \rho_2=\frac{X\,m^3\,p_2}{\Delta^2},\qquad \rho_1=\frac{m^3\,p_1}{\Delta^3},$$

$$p_2=-\lambda^3+\lambda^2x+2\lambda xx_0+3\lambda+x+2x_0,$$

$$p_1=(\lambda^3{+}3\lambda)(x^2{+}2xx_0{+}1)+(3\lambda^2{+}1)(x^2x_0{+}2x{+}x_0).$$

(The coefficients depend on x,x_0,λ but not on η.) The limits $\hat\psi_\lambda(\pm1)=\lim_{x\to\pm1}\hat\psi_\lambda(x)$ exist, and for $x_0=0$, $|\lambda|>1$,

$$\hat\psi_\lambda(1)=\frac{\lambda^2\big[4\lambda(\lambda+1)\ln\big|\frac{\lambda+1}{2\lambda}\big|+(3\lambda+1)(\lambda-1)\big]}{2(\lambda-1)^3(\lambda+1)},$$

$$\hat\psi_\lambda(-1)=\frac{\lambda^2\big[4\lambda(\lambda-1)\ln\big|\frac{\lambda-1}{2\lambda}\big|+(3\lambda-1)(\lambda+1)\big]}{2(\lambda+1)^3(\lambda-1)}.$$

Proof. *Partial fractions.* Since $\hat fg=m^3(\eta-x_0)/(\eta+\lambda)^3$, the integrand is $R(\eta)=m^3N(\eta)/[(\eta+\lambda)^3(1-\eta)(1+\eta)]$ with $N(\eta)=(x-\eta)^2(\eta-x_0)$, a rational function whose numerator has degree $3<5$. Its poles $1,-1,-\lambda$ are distinct ($|\lambda|>1$), so $R=\frac{r_+}{\eta-1}+\frac{r_-}{\eta+1}+\frac{\rho_1}{\eta+\lambda}+\frac{\rho_2}{(\eta+\lambda)^2}+\frac{\rho_3}{(\eta+\lambda)^3}$ with unique constants. At the simple poles, $(\eta-1)/(1-\eta^2)\to-\frac12$ and $(\eta+1)/(1-\eta^2)\to\frac12$ give $r_+=-\frac{m^3N(1)}{2(1+\lambda)^3}$, $r_-=\frac{m^3N(-1)}{2(\lambda-1)^3}$, with $N(1)=(x-1)^2(1-x_0)$, $N(-1)=-(x+1)^2(1+x_0)$: the stated $r_\pm$. At the triple pole write $\eta=-\lambda+y$ and $R=m^3\phi(\eta)/y^3$ with $\phi=N/(1-\eta^2)$ holomorphic at $y=0$, $\phi=\phi_0+\phi_1y+\phi_2y^2+O(y^3)$; then $\rho_3=m^3\phi_0$, $\rho_2=m^3\phi_1$, $\rho_1=m^3\phi_2$. Now $x-\eta=X-y$, $\eta-x_0=y-m$, $1-\eta^2=-(\Delta-2\lambda y+y^2)$, so $\phi=-n(y)/d(y)$ with $n=(X-y)^2(y-m)=n_0+n_1y+n_2y^2+\cdots$, $n_0=-mX^2$, $n_1=X^2+2Xm$, $n_2=-2X-m$, and $1/d=\Delta^{-1}[1+2\lambda y/\Delta+(4\lambda^2/\Delta^2-1/\Delta)y^2+O(y^3)]$. Multiplying out,

$$\phi_0=\frac{mX^2}{\Delta},\quad \phi_1=-\frac{n_1+2\lambda n_0/\Delta}{\Delta},$$

$$\phi_2=-\frac{\Delta^2n_2+2\lambda\Delta n_1+(3\lambda^2+1)n_0}{\Delta^3}.$$

This gives $\rho_3=m^4X^2/\Delta$; $\rho_2=m^3Xp_2/\Delta^2$ with $p_2=-[(X+2m)\Delta-2\lambda mX]$, which expands to the stated polynomial: $(X+2m)\Delta=(x+2x_0+3\lambda)(\lambda^2-1)=3\lambda^3+(x+2x_0)\lambda^2-3\lambda-x-2x_0$ and $2\lambda mX=2\lambda^3+2(x+x_0)\lambda^2+2xx_0\lambda$, whose difference is $\lambda^3-x\lambda^2-3\lambda-x-2x_0-2xx_0\lambda=-p_2$; and $\rho_1=m^3p_1/\Delta^3$ with $p_1=\Delta^2(2X+m)-2\lambda\Delta X(X+2m)+(3\lambda^2+1)mX^2$, where $4\lambda^2-\Delta=3\lambda^2+1$ was used in ϕ_2. To expand p_1 put $\sigma=2x+x_0$, $\tau=x+2x_0$, so $X=\lambda+x$, $m=\lambda+x_0$, $2X+m=3\lambda+\sigma$, $X+2m=3\lambda+\tau$, $\Delta^2=\lambda^4-2\lambda^2+1$. The coefficients by powers of λ of the three pieces are

piece	λ^5	λ^4	λ^3	λ^2	λ^1	λ^0
$\Delta^2(2X+m)$	3	σ	-6	-2σ	3	σ
$-2\lambda\Delta X(X+2m)$	-6	-4σ	$-2(x^2{+}2xx_0{-}3)$	4σ	$2x(x{+}2x_0)$	0
$(3\lambda^2{+}1)mX^2$	3	3σ	$3x^2{+}6xx_0{+}1$	$3x^2x_0{+}2x{+}x_0$	$x(x{+}2x_0)$	x^2x_0

(the rows come from $\Delta^2(3\lambda+\sigma)$; from $-2\lambda(\lambda^2-1)\big(3\lambda^2+2\sigma\lambda+x\tau\big)$, since $X(X+2m)=(\lambda+x)(3\lambda+\tau)=3\lambda^2+(\tau+3x)\lambda+x\tau$ and $\tau+3x=2\sigma$; and from $(3\lambda^2+1)\big(\lambda^3+\sigma\lambda^2+x\tau\,\lambda+x^2x_0\big)$, since $mX^2=(\lambda+x_0)(\lambda^2+2x\lambda+x^2)$; here $x\tau=x^2+2xx_0$). The column sums are: λ^5: $3-6+3=0$; λ^4: $\sigma-4\sigma+3\sigma=0$; λ^3: $-6-2(x^2+2xx_0-3)+3x^2+6xx_0+1=x^2+2xx_0+1$; λ^2: $-2\sigma+4\sigma+3x^2x_0+2x+x_0=3(x^2x_0+2x+x_0)$; λ^1: $3+2x(x+2x_0)+x(x+2x_0)=3(x^2+2xx_0+1)$; λ^0: $\sigma+x^2x_0=x^2x_0+2x+x_0$. These are the coefficients of the stated $p_1=(\lambda^3+3\lambda)(x^2+2xx_0+1)+(3\lambda^2+1)(x^2x_0+2x+x_0)$. (Table and sums are also checked by computer algebra, Numerical observation B.20.) *Integration.* On $[x_0,x]$ none of $\pm1,-\lambda$ occurs, so $\int\frac{d\eta}{\eta-a}=\ln|\eta-a|$, $\int\frac{d\eta}{(\eta+\lambda)^2}=-\frac{1}{\eta+\lambda}$, $\int\frac{d\eta}{(\eta+\lambda)^3}=-\frac{1}{2(\eta+\lambda)^2}$ yield $\hat\psi_\lambda(x)=G(x)-G(x_0)$. *The poles,* $x_0=0$. As $x\to1$, $r_+\propto(x-1)^2$ so $r_+\ln|x-1|\to0$ and all other terms are continuous; the limit exists and equals the value of $G(1)-G(0)$ with the coefficients at $x=1$ (and $r_+=0$): $m=\lambda$, $X=\lambda+1$, $r_-=-2\lambda^3/(\lambda-1)^3$, $p_1=2(\lambda+1)^3$, $\rho_1=2\lambda^3/(\lambda-1)^3=-r_-$, $\rho_2=(\lambda+1)\lambda^3p_2/\Delta^2$ with $p_2=-\lambda^3+\lambda^2+3\lambda+1$, $\rho_3=\lambda^4(\lambda+1)/(\lambda-1)$. The logarithms combine to $r_-(\ln2-\ln1)+\rho_1(\ln|1+\lambda|-\ln|\lambda|)=\frac{2\lambda^3}{(\lambda-1)^3}\ln\big|\frac{\lambda+1}{2\lambda}\big|$. The rational part is $-\rho_2\big(\frac{1}{1+\lambda}-\frac{1}{\lambda}\big)-\frac{\rho_3}{2}\big(\frac{1}{(1+\lambda)^2}-\frac{1}{\lambda^2}\big)=\frac{\rho_2}{\lambda(\lambda+1)}+\frac{\rho_3(2\lambda+1)}{2\lambda^2(\lambda+1)^2}=\frac{\lambda^2[2p_2+(2\lambda+1)\Delta]}{2\Delta^2}$ and $2p_2+(2\lambda+1)\Delta=3\lambda^2+4\lambda+1=(3\lambda+1)(\lambda+1)$, so it equals $\frac{\lambda^2(3\lambda+1)}{2(\lambda-1)^2(\lambda+1)}$. Adding gives the formula for $\hat\psi_\lambda(1)$. For -1 use the symmetry $(x,x_0,\lambda)\mapsto(-x,-x_0,-\lambda)$: the substitution $\eta\to-\eta$ leaves g invariant, changes $\hat f\to-\hat f$ and $d\eta\to-d\eta$, hence $\hat\psi$ is invariant. Thus $\hat\psi_\lambda(-1)=\hat\psi_{-\lambda}(1)$, and replacing λ by $-\lambda$ in the first formula (it holds for all $|\lambda|>1$, with absolute values) gives the second. □

Remark B.9. At $\lambda=\frac32$, $x_0=0$: $\hat\psi_\lambda(1)=0.0546359331265$, $\hat\psi_\lambda(-1)=0.4859599092935$. As $\lambda\downarrow1$ (upper window edge) $\hat\psi_\lambda(1)\to\frac{1}{24}$ but $\hat\psi_\lambda(-1)\sim\frac{1}{4(\lambda-1)}\to\infty$ (the bracket tends to 4, the denominator is $2(\lambda+1)^3(\lambda-1)$): the first-order trace of the change of pole type at the edge.

B.5. The edge coefficient of the small-swirl law: digamma closed form

Let $c_1=c_2=c>-1$, $c_3=0$, $k=\sqrt{1+c}$, $x_0=0$. Then $W_{\boldsymbol c}\equiv k^2$, $u_+=e^{-k\xi}$, $F_+(\xi)=\int_0^\xi u_+^{-2}=(e^{2k\xi}-1)/(2k)$, and the first-order coefficient of Chapter 11 is

$$A_+:=\int_0^\infty\hat\psi_+u_+^2\,d\xi,\qquad F_+F_+'=\frac{e^{4k\eta}-e^{2k\eta}}{2k},$$

$$\hat\psi_+(\xi)=\frac{1}{\cosh^2\xi}\int_0^\xi\sinh^2(\xi-\eta)\,F_+F_+'(\eta)\,d\eta.$$

By the symmetry $\xi\mapsto-\xi$ also $A_-=A_+$. Using $\sinh^2(\xi-\eta)/\cosh^2\xi\le e^{-2\eta}$, $\hat\psi_+\le\frac{1}{2k}\int_0^\xi e^{(4k-2)\eta}d\eta\le\frac{1+\xi}{2k}e^{\max(0,4k-2)\xi}$, so the integrand is $\le C(1+\xi)e^{-2\min(k,1-k)\xi}$ and $A_+<\infty$ if $k<1$ (only this is used below; for $k\ge1$ the integral diverges, cf. the Remark below).

Lemma B.10 (the weight integral)**.** *For $b > -2$ let $\Phi(b) = \int_0^\infty e^{-b\xi}\operatorname{sech}^2\xi\,d\xi$. Then $\Phi(b) = 1 - \frac{b}{2}\big[\psi^{(0)}(\frac{b+4}{4}) - \psi^{(0)}(\frac{b+2}{4})\big]$.*

Proof. Let $b > 0$. Integration by parts ($\tanh' = \operatorname{sech}^2$, $\tanh 0 = 0$) gives $\Phi(b) = b\int_0^\infty e^{-b\xi}\tanh\xi\,d\xi$. Now $\tanh\xi = 1 - \frac{2e^{-2\xi}}{1+e^{-2\xi}} = 1 + 2\sum_{n\ge1}(-1)^n e^{-2n\xi}$ for $\xi > 0$, with partial sums bounded by 3, so termwise integration is justified by dominated convergence: $\Phi(b) = 1 + 2b\sum_{n\ge1}\frac{(-1)^n}{b+2n}$. Pairing consecutive terms $n = 2j-1, 2j$ ($j \ge 1$) and writing $\frac{1}{b+2n} = \frac{1}{2}\frac{1}{n+b/2}$: $\sum_{n\ge1}\frac{(-1)^n}{b+2n} = \frac{1}{2}\sum_{j\ge1}\big[\frac{1}{2j+b/2} - \frac{1}{2j-1+b/2}\big] = \frac{1}{4}\sum_{j\ge1}\big[\frac{1}{j+b/4} - \frac{1}{j-\frac{1}{2}+b/4}\big] = \frac{1}{4}\sum_{m\ge0}\big[\frac{1}{m+\frac{b}{4}+1} - \frac{1}{m+\frac{b}{4}+\frac{1}{2}}\big] = -\frac{1}{4}\big[\psi^{(0)}(\frac{b}{4}+1) - \psi^{(0)}(\frac{b}{4}+\frac{1}{2})\big]$ by (F1). Hence $\Phi(b) = 1 - \frac{b}{2}\big[\psi^{(0)}(\frac{b+4}{4}) - \psi^{(0)}(\frac{b+2}{4})\big]$. Both sides are holomorphic in $\Re b > -2$ (the integral converges there since $\operatorname{sech}^2\xi \le 4e^{-2\xi}$; $\psi^{(0)}(\frac{b+2}{4})$ has its first pole at $b = -2$), so the identity extends from $b > 0$ to $b > -2$. □

Toy example B.11 (the lemma at $b = 2$)**.** With $u = e^{-2\xi}$, $\Phi(b) = 2\int_0^1 u^{b/2}(1+u)^{-2}du$, so $\Phi(2) = 2[\ln(1+u) + \frac{1}{1+u}]_0^1 = 2\ln 2 - 1$. The lemma gives $1 - [\psi^{(0)}(\frac{3}{2}) - \psi^{(0)}(1)] = 1 - (2 - 2\ln 2) = 2\ln 2 - 1$ since $\psi^{(0)}(\frac{3}{2}) = 2 - \gamma_{\mathrm{E}} - 2\ln 2$ and $\psi^{(0)}(1) = -\gamma_{\mathrm{E}}$.

Proposition B.12 (digamma closed form of A_+)**.** *Let $0 < k < 1$, $k \ne \frac{1}{2}$. With Φ as in Lemma B.10,*

$$A_+ = A_- = \frac{1}{2k}\sum_{(a,\sigma)\in\{(4k,+1),\,(2k,-1)\}} \sigma\Big[\frac{2\,\Phi(2k-a)}{a(a^2-4)} - \frac{\Phi(2k-2)}{4(a-2)} - \frac{\Phi(2k+2)}{4(a+2)} + \frac{\Phi(2k)}{2a}\Big].$$

The right-hand side extends continuously to $k = \frac{1}{2}$ (the singularity of the term $a = 4k = 2$ is removable), where it equals A_+. Values: $A(k = \sqrt{0.1}) = 0.366725$, $A(\sqrt{0.5}) = 0.251682$.

Proof. For $a \notin \{0, \pm2\}$ put $\Psi_a(\xi) = \int_0^\xi \sinh^2(\xi-\eta)e^{a\eta}d\eta$. With $\sinh^2 r = \frac{1}{4}(e^{2r} + e^{-2r} - 2)$,

$$\Psi_a = \frac{1}{4}\Big[e^{2\xi}\frac{e^{(a-2)\xi}-1}{a-2} + e^{-2\xi}\frac{e^{(a+2)\xi}-1}{a+2} - 2\,\frac{e^{a\xi}-1}{a}\Big] = \frac{2\,e^{a\xi}}{a(a^2-4)} - \frac{e^{2\xi}}{4(a-2)} - \frac{e^{-2\xi}}{4(a+2)} + \frac{1}{2a},$$

because $\frac{1}{4}[\frac{1}{a-2} + \frac{1}{a+2} - \frac{2}{a}] = \frac{1}{4}\cdot\frac{8}{a(a^2-4)}$. Since $F_+F_+' = (e^{4k\eta} - e^{2k\eta})/(2k)$, $\hat\psi_+ = \frac{1}{2k}\operatorname{sech}^2\xi\,[\Psi_{4k} - \Psi_{2k}]$ (both $a = 4k, 2k$ avoid $\{0, \pm2\}$ for $k \ne \frac{1}{2}, 1$), and $A_+ = \frac{1}{2k}\int_0^\infty \operatorname{sech}^2\xi\,e^{-2k\xi}[\Psi_{4k} - \Psi_{2k}]\,d\xi$. Each of the four terms of Ψ_a times $e^{-2k\xi}\operatorname{sech}^2\xi$ integrates to $\Phi(2k-a)$, $\Phi(2k-2)$, $\Phi(2k+2)$, $\Phi(2k)$ with the displayed coefficients; the arguments $-2k$, 0, $2k-2$, $2k$, $2k+2$ all exceed -2 because $k < 1$, so every integral converges and the splitting is legitimate. Evaluating Φ by Lemma B.10 gives the formula. A_+ is continuous in $k \in (0,1)$ by dominated convergence (by the bound in the first paragraph of this section, $\hat\psi_+ u_+^2 \le C\,(1+\xi)\,e^{-2\min(k,1-k)\xi}$ with C locally bounded in $k \in (0,1)$), so the right-hand side, which equals A_+ for $k \ne \frac{1}{2}$, has the continuous extension $A_+(\frac{1}{2})$. □

Remark B.13. As $k \uparrow 1$ one has $\Phi(b) = \frac{4}{b+2} + O(1)$ as $b \downarrow -2$ (from $\Phi(b) = 2\int_0^1 u^{b/2}(1+u)^{-2}du$), so $\Phi(-2k) = \frac{2}{1-k} + O(1)$, and at $a = 4k$ the coefficient is $2/(a(a^2-4)) \to \frac{1}{24}$. The terms at $a = 2k$ are not singular in the limit: the two apparently singular pieces combine as $\frac{1}{a-2}\big[\frac{2\Phi(0)}{a(a+2)} - \frac{\Phi(2k-2)}{4}\big]$, and the bracket vanishes at $a = 2$ because $\Phi(2k-2) \to \Phi(0) = 1$ and $\frac{2}{a(a+2)} \to \frac{1}{4}$ (it is a difference quotient of a smooth function). Hence $A_+ \sim \frac{1}{2k}\cdot\frac{1}{24}\cdot\frac{2}{1-k} = \frac{1}{24(1-k)} \to \infty$ (numerically $24(1-k)A_+ = 1.0016$ at $k = 0.999$), the divergence behind the logarithmic law at $k = 1$ of Chapter 11.

B.6. Near the pole: the Bessel model and the exact Fuchsian model

Let $E = e^{2\xi}$, $0 < \mu < 2$ and $\nu_{\mathrm{B}} = \mu/2 \in (0,1)$ (the Bessel order). The constants of the class-N threshold law (Chapter 12) are

$$\ell_*(\mu) = \frac{8^{\nu_{\mathrm{B}}}\,\Gamma(\nu_{\mathrm{B}})}{4\,\Gamma(1-\nu_{\mathrm{B}})}, \qquad \gamma_*(\mu) = \ell_*(\mu)^{2/\mu}. \tag{B.4}$$

Toy example B.14 (the equation is Bessel's)**.** For $c > 0$, $q = \sqrt{c/2}$, $y = qE$: since $d/d\xi = 2E\,d/dE = 2y\,d/dy$ and $2cE^2 = 4y^2$, the equation $\zeta'' = (\mu^2 + 2cE^2)\zeta$ becomes $(y\partial_y)^2\zeta = (\nu_{\mathrm{B}}^2 + y^2)\zeta$, i.e. $y^2\zeta_{yy} + y\zeta_y - (y^2+\nu_{\mathrm{B}}^2)\zeta = 0$. The solution $\sim y^{-\nu_{\mathrm{B}}} \propto e^{-\mu\xi}$ is the *dominant* mode at the pole, $\sim y^{\nu_{\mathrm{B}}} \propto e^{\mu\xi}$ the recessive one. For $\nu_{\mathrm{B}} = \frac{1}{2}$: $I_{-1/2} = \sqrt{2/\pi y}\cosh y$, $I_{1/2} = \sqrt{2/\pi y}\sinh y$, $K_{1/2} = \sqrt{\pi/2y}\,e^{-y}$, so $\zeta_c = e^{-\xi}\cosh(qE)$.

Proposition B.15 (Bessel model)**.** *Let $0 < \mu < 2$, $c > 0$, and let ζ_c be the solution of $\zeta'' = (\mu^2 + 2cE^2)\zeta$ with $\zeta_c = e^{-\mu\xi}(1 + O(E^2))$ at $-\infty$, $l_c(\xi) = \int_{-\infty}^{\xi}\zeta_c^{-2}$ and $\ell_\infty = l_c(\infty)$. Then*

- *(i) $\zeta_c(\xi) = \Gamma(1-\nu_{\mathrm{B}})(q/2)^{\nu_{\mathrm{B}}} I_{-\nu_{\mathrm{B}}}(qE) > 0$; with $\theta = \xi + \frac{1}{4}\ln c$ one has $\zeta_c = c^{\mu/4} g(\theta)$, $g(\theta) = \Gamma(1-\nu_{\mathrm{B}})8^{-\mu/4} I_{-\nu_{\mathrm{B}}}(e^{2\theta}/\sqrt{2})$ independent of c.*
- *(ii) $\int_{\xi}^{\infty}\zeta_c^{-2} = \ell_*(\mu)\,c^{-\mu/2}\,\tau(qE)$ with $\tau(y) = 1 - I_{\nu_{\mathrm{B}}}(y)/I_{-\nu_{\mathrm{B}}}(y)$, which decreases from 1 to 0, $\tau(y) \sim 2\sin(\nu_{\mathrm{B}}\pi)e^{-2y}$. In particular $\ell_\infty = \ell_*(\mu)\,c^{-\mu/2}$, i.e. $\int_{\mathbb{R}} g^{-2} = \ell_*(\mu)$.*
- *(iii) For $C < 0$ the function $\tilde{z} = \zeta_c(1 + C l_c)$ is positive on $\mathbb{R}$ iff $|C| \le c^{\mu/2}/\ell_*(\mu)$; for equality $\tilde{z} = \frac{2\sin\nu_{\mathrm{B}}\pi}{\pi}\Gamma(1-\nu_{\mathrm{B}})(q/2)^{\nu_{\mathrm{B}}} K_{\nu_{\mathrm{B}}}(qE)$ (the decaying state).*
- *(iv) The smallest c for which $\tilde{z} > 0$ on $\mathbb{R}$ is $c^- = \gamma_*(\mu)|C|^{2/\mu}$.*

Proof. (i) By Toy example B.14 and (F3), $I_{\pm\nu_{\mathrm{B}}}(qE)$ solve the equation; $I_{-\nu_{\mathrm{B}}}(y) = (y/2)^{-\nu_{\mathrm{B}}}[\Gamma(1-\nu_{\mathrm{B}})^{-1} + O(y^2)]$, so $\Gamma(1-\nu_{\mathrm{B}})(q/2)^{\nu_{\mathrm{B}}} I_{-\nu_{\mathrm{B}}}(qE) = E^{-\nu_{\mathrm{B}}}(1 + O(E^2)) = e^{-\mu\xi}(1 + O(E^2))$ (while $I_{\nu_{\mathrm{B}}} \sim y^{\nu_{\mathrm{B}}}$ is the $e^{\mu\xi}$ mode, so this normalisation singles out the solution). Positivity: all coefficients $1/[m!\Gamma(m+1-\nu_{\mathrm{B}})]$ are positive. With $e^{2\xi_c} = c^{-1/2}$, $qE = e^{2\theta}/\sqrt{2}$ and $(q/2)^{\nu_{\mathrm{B}}} = (c/8)^{\mu/4}$. (ii) Let $\mathrm{Wr}_y = I_{-\nu_{\mathrm{B}}} I'_{\nu_{\mathrm{B}}} - I'_{-\nu_{\mathrm{B}}} I_{\nu_{\mathrm{B}}}$. By Abel's formula for $y^2w'' + yw' - \cdots = 0$, $y\,\mathrm{Wr}_y$ is constant; letting $y \to 0$ with the leading terms, $y\,\mathrm{Wr}_y \to \frac{\nu_{\mathrm{B}}+\nu_{\mathrm{B}}}{\Gamma(1-\nu_{\mathrm{B}})\Gamma(1+\nu_{\mathrm{B}})} = \frac{2\sin\nu_{\mathrm{B}}\pi}{\pi}$ (F1). So $(I_{\nu_{\mathrm{B}}}/I_{-\nu_{\mathrm{B}}})' = \mathrm{Wr}_y/I_{-\nu_{\mathrm{B}}}^2 = \frac{2\sin\nu_{\mathrm{B}}\pi}{\pi y} I_{-\nu_{\mathrm{B}}}^{-2}$. By (F3), $I_{-\nu_{\mathrm{B}}} - I_{\nu_{\mathrm{B}}} = \frac{2}{\pi}\sin(\nu_{\mathrm{B}}\pi)K_{\nu_{\mathrm{B}}}$ is $O(e^{-y}/\sqrt{y})$ while $I_{-\nu_{\mathrm{B}}} \sim e^y/\sqrt{2\pi y}$, so $I_{\nu_{\mathrm{B}}}/I_{-\nu_{\mathrm{B}}} \to 1$ at ∞, and $\to 0$ at 0. Hence

$\int_y^\infty \frac{dy'}{y' I_{-\nu_B}^2} = \frac{\pi}{2\sin\nu_B\pi}\tau(y)$. Now $d\xi = dy/(2y)$, so

$$\int_\xi^\infty \zeta_c^{-2} = \Gamma(1-\nu_B)^{-2}(q/2)^{-2\nu_B}\tfrac12\int_{qE}^\infty \frac{dy}{yI_{-\nu_B}^2} = \frac{(q/2)^{-2\nu_B}\,\pi}{4\sin(\nu_B\pi)\Gamma(1-\nu_B)^2}\,\tau(qE),$$

and $(q/2)^{-2\nu_B} = 8^{\nu_B}c^{-\nu_B}$, $\pi/\sin\nu_B\pi = \Gamma(\nu_B)\Gamma(1-\nu_B)$ give the constant $8^{\nu_B}\Gamma(\nu_B)/(4\Gamma(1-\nu_B))\cdot c^{-\mu/2}$. Monotonicity of τ is clear from $\tau' = -(I_{\nu_B}/I_{-\nu_B})' < 0$; $\tau(y) = \frac{2\sin\nu_B\pi}{\pi}K_{\nu_B}/I_{-\nu_B} \sim 2\sin(\nu_B\pi)e^{-2y}$ by (F3). (iii) l_c increases from 0 to ℓ_∞, so $1+Cl_c$ decreases strictly from 1 to $1+C\ell_\infty$, and $\tilde z > 0$ on $\mathbb{R}$ iff $1+C\ell_\infty \geq 0$, i.e. $|C|\ell_* c^{-\mu/2} \leq 1$. At equality $1+Cl_c = |C|\int_\xi^\infty \zeta_c^{-2} = \tau(qE)$, so $\tilde z = \zeta_c\tau = \Gamma(1-\nu_B)(q/2)^{\nu_B}(I_{-\nu_B} - I_{\nu_B})(qE)$, which is the stated multiple of $K_{\nu_B}(qE)$. (iv) $|C| \leq c^{\mu/2}/\ell_*$ iff $c \geq (\ell_*|C|)^{2/\mu} = \gamma_*|C|^{2/\mu}$. □

Example B.16. $\mu = 1$: $\ell_*(1) = \frac{\sqrt8\,\Gamma(\frac12)}{4\Gamma(\frac12)} = \frac{1}{\sqrt2}$, $\gamma_*(1) = \frac12$. Direct check: $\zeta_c = e^{-\xi}\cosh(qE)$, $\ell_\infty = \int_0^\infty \frac{dE}{2E}\,E\,\operatorname{sech}^2(qE) = \frac{1}{2q} = \frac{1}{\sqrt{2c}}$, so $c^- = C^2/2$; the threshold state is $K_{1/2}(y) \propto y^{-1/2}e^{-y}$.

Toy example B.17 (the Fuchsian model is the hypergeometric class)**.** With $t = \frac12(1+\tanh\xi)$ as in (B.1), put $W = \mu^2 + p_1 t + 2p_2t^2$. Comparing with (B.2) this is $W_{\mathfrak{c}}$ with $k_1 = \mu$, $k_2 = \mu_+ := \sqrt{\mu^2+p_1+2p_2}$ and $c_3 = -p_2$ (so the t of Appendix B.1 is $r_2 := \sqrt{1+2p_2}$). Thus Lemma B.1 applies, now also with the *other* sign $\alpha = -\mu/2$, which gives the dominant solution at $-\infty$. In the Bessel limit $p_2 \to \infty$ the whole polynomial is replaced by its top term.

Proposition B.18 (exact Fuchsian model)**.** *Let $\mu \in (0,1)\cup(1,2)$, $p_1 \in \mathbb{R}$, $p_2 > 0$, $\mu_+ = \sqrt{\mu^2+p_1+2p_2} > 0$, $r_2 = \sqrt{1+2p_2}$, $t = \frac12(1+\tanh\xi)$, $W = \mu^2 + p_1 t + 2p_2t^2$. Put $a_\mp = \frac12(1\mp r_2 - \mu - \mu_+)$, $b_\mp = \frac12(1\mp r_2 + \mu - \mu_+)$ and*

$$z_F = t^{-\mu/2}(1-t)^{-\mu_+/2}\,{}_2F_1(a_-, a_+; 1-\mu; t)\,,$$
$$u_P = \frac{1}{2\mu}\,t^{\mu/2}(1-t)^{-\mu_+/2}\,{}_2F_1(b_-, b_+; 1+\mu; t)\,.$$

Then $z_F'' = Wz_F$, $u_P'' = Wu_P$, $z_F \sim e^{-\mu\xi}$, $u_P \sim e^{\mu\xi}/(2\mu)$ as $\xi \to -\infty$, and $\operatorname{Wr}(z_F, u_P) = 1$. For $z = z_F + \alpha u_P > 0$ on $\mathbb{R}$ the integral $\ell_\infty(z) = \int_{\mathbb{R}} z^{-2} \in (0,\infty]$ equals

$$\ell_\infty(z) = \frac{A_u}{2\mu A_F + \alpha A_u} \quad (= \infty \text{ if the denominator is } 0),$$
$$A_F = \frac{\Gamma(1-\mu)\Gamma(\mu_+)}{\Gamma\big(\frac{1-\mu+r_2+\mu_+}{2}\big)\Gamma\big(\frac{1-\mu-r_2+\mu_+}{2}\big)},$$
$$A_u = \frac{\Gamma(1+\mu)\Gamma(\mu_+)}{\Gamma\big(\frac{1+\mu+r_2+\mu_+}{2}\big)\Gamma\big(\frac{1+\mu-r_2+\mu_+}{2}\big)},$$

and, as $p_2 \to \infty$ with p_1, μ, α fixed and whenever $z > 0$ on $\mathbb{R}$,

$$\ell_\infty(z) = \ell_*(\mu)\,p_2^{-\mu/2}\Big(1 - \frac{\pi}{4\sqrt2}\tan\frac{\pi\mu}{2}\,(\mu^2 - 1 + p_1)\,p_2^{-1/2} - \alpha\,\ell_*(\mu)\,p_2^{-\mu/2} + O(p_2^{-1} + p_2^{-\mu})\Big).$$

Proof. *Solutions.* By Toy example B.17 Lemma B.1 applies with $c_3 = -p_2$, $\delta = \alpha_0 + \beta_0$ where $\beta_0 = -\mu_+/2$. For z_F take $\alpha_0 = -\mu/2$: $\delta = -\frac{\mu+\mu_+}{2}$, $\mathfrak{c} = 1-\mu$ (not

in $\{0,-1,\dots\}$ because $\mu\notin\mathbb{N}$), and $\{\delta+\frac{1\pm r_2}{2}\}=\{a_-,a_+\}$; for u_P take $\alpha_0=\mu/2$: $\mathfrak{c}=1+\mu$, $\{\delta+\frac{1\pm r_2}{2}\}=\{b_-,b_+\}$. Hence both solve $z''=Wz$ by (F2). Near $t=0$, $t=e^{2\xi}(1+O(e^{2\xi}))$ and the hypergeometric factors are $1+O(t)$, with derivatives $O(1)$, so $z_F=e^{-\mu\xi}(1+O(E))$, $u_P=\frac{e^{\mu\xi}}{2\mu}(1+O(E))$ with the corresponding derivatives, and $\mathrm{Wr}(z_F,u_P)=\frac{1}{2\mu}(\mu+\mu)=1$, constant. *Connection.* For $z=z_F+\alpha u_P$ positive, $(u_P/z)'=\mathrm{Wr}(z,u_P)/z^2=z^{-2}$ and $u_P/z\to 0$ at $-\infty$, hence $\int_{-\infty}^{\xi}z^{-2}=u_P/z$ and $\ell_\infty=\lim_{\xi\to\infty}u_P/z$. As $t\uparrow 1$ both hypergeometric factors have $\mathfrak{c}-a-b=\mu_+>0$ (for z_F: $1-\mu-(1-\mu-\mu_+)$; for u_P: $1+\mu-(1+\mu-\mu_+)$), so by Gauss's theorem they tend to $A_F={}_2F_1(a_-,a_+;1-\mu;1)$ and $A_u={}_2F_1(b_-,b_+;1+\mu;1)$ with the Gamma values displayed ($\mathfrak{c}-a_\pm=\frac12(1-\mu\pm r_2+\mu_+)$, and similarly for u_P). The common factor $(1-t)^{-\mu_+/2}$ cancels in u_P/z, so $\ell_\infty=\frac{A_u/(2\mu)}{A_F+\alpha A_u/(2\mu)}$. *Asymptotics.* For $\alpha=0$ put $\ell_0=A_u/(2\mu A_F)$, $R=\sqrt{2p_2}$, $X=\frac12(r_2+\mu_+)$, $Y=\frac12(\mu_+-r_2)$. Cancelling $\Gamma(\mu_+)$,

$$\ell_0=\frac{\Gamma(1+\mu)}{2\mu\,\Gamma(1-\mu)}\cdot\frac{\Gamma(X+\frac{1-\mu}{2})}{\Gamma(X+\frac{1+\mu}{2})}\cdot\frac{\Gamma(Y+\frac{1-\mu}{2})}{\Gamma(Y+\frac{1+\mu}{2})}.$$

We have $r_2=R+O(R^{-1})$, $\mu_+=R+O(R^{-1})$, so $X=R+O(R^{-1})$, and $Y=\frac{\mu_+^2-r_2^2}{2(\mu_++r_2)}=\frac{\mu^2+p_1-1}{4R}(1+O(R^{-2}))$. By (F1) with $a=\frac{1-\mu}{2}$, $b=\frac{1+\mu}{2}$ ($a+b=1$, so the $1/X$ term vanishes) the first ratio is $X^{-\mu}(1+O(X^{-2}))=R^{-\mu}(1+O(R^{-2}))$. The second ratio is analytic near $Y=0$ since $a,b\notin\{0,-1,\dots\}$ (here $\mu\neq1$ is used), and its logarithmic derivative at 0 is $\psi^{(0)}(a)-\psi^{(0)}(b)=\psi^{(0)}(a)-\psi^{(0)}(1-a)=-\pi\cot\pi a=-\pi\tan\frac{\pi\mu}{2}$ by (F1); so it equals $\frac{\Gamma(a)}{\Gamma(b)}\big(1-\pi\tan\frac{\pi\mu}{2}\,Y+O(Y^2)\big)$. The prefactor: by duplication (F1) applied to $\Gamma(1+\mu)=\Gamma(2(\nu_{\mathrm B}+\frac12))$ and $\Gamma(1-\mu)=\Gamma(2(\frac12-\nu_{\mathrm B}))$,

$$\frac{\Gamma(1+\mu)\Gamma(a)}{\Gamma(1-\mu)\Gamma(b)}=2^{4\nu_{\mathrm B}}\frac{\Gamma(\nu_{\mathrm B}+\frac12)\Gamma(\nu_{\mathrm B}+1)}{\Gamma(\frac12-\nu_{\mathrm B})\Gamma(1-\nu_{\mathrm B})}\cdot\frac{\Gamma(\frac12-\nu_{\mathrm B})}{\Gamma(\frac12+\nu_{\mathrm B})}=2^{2\mu}\frac{\Gamma(1+\nu_{\mathrm B})}{\Gamma(1-\nu_{\mathrm B})},$$

so, with $\Gamma(1+\nu_{\mathrm B})=\nu_{\mathrm B}\Gamma(\nu_{\mathrm B})$ and $R^{-\mu}=2^{-\mu/2}p_2^{-\mu/2}$, the leading term is $\frac{2^{2\mu-\mu/2}\Gamma(\nu_{\mathrm B})}{4\Gamma(1-\nu_{\mathrm B})}p_2^{-\mu/2}=\ell_*(\mu)p_2^{-\mu/2}$ ($2^{3\mu/2}=8^{\nu_{\mathrm B}}$). The Y-term contributes $-\pi\tan\frac{\pi\mu}{2}\cdot\frac{\mu^2+p_1-1}{4\sqrt2\sqrt{p_2}}$, which gives the displayed $p_2^{-1/2}$ correction, with remainder $O(R^{-2}+Y^2)=O(p_2^{-1})$. Finally $\ell_\infty=\ell_0/(1+\alpha\ell_0)=\ell_0-\alpha\ell_0^2+O(\ell_0^3)$ and $\ell_0=O(p_2^{-\mu/2})$, $\ell_0^2=\ell_*^2p_2^{-\mu}(1+O(p_2^{-1/2}))$; the cross term is $O(p_2^{-(1+\mu)/2})\le O(p_2^{-\min(1,\mu)})$. □

Remark B.19 ($\mu=1$, and what the correction means)**.** At $\mu=1$ the exponents $\pm\frac{\mu}{2}$ differ by an integer, $\mathfrak{c}=1-\mu=0$, the Frobenius series of z_F acquires a logarithm, and $\tan\frac{\pi\mu}{2}$ is infinite; the expansion degenerates. For $\mu=1$ the Bessel limit itself is elementary (Example B.16). In the swirl-free model the two corrections to the limit law are thus the shape term $p_2^{-1/2}$, with the factor $\tan\frac{\pi\mu}{2}(\mu^2-1+p_1)$, and the label term $\alpha\ell_*p_2^{-\mu/2}$; the transfer of these rates to genuine germs with swirl is an expectation, not a theorem, and is discussed in Chapter 12.

B.7. Numerical confirmation

Numerical observation B.20 (checks of Appendix B)**.** *`check_appB.py` in `code/appB_special/` (run by `make checks` there; 30-digit mpmath, exact computer*

algebra; 142 residuals, all passing) verifies every identity above: the Gauss reduction (exactly, all four sign choices); for seven potentials (including $c_3 > \frac{1}{2}$, $c_1 < 0$, $c_3 = -2.9$*) the ODE residuals of* $u_{\mp}^{\mathrm{H}}$ *(*$< 10^{-18}$*) and the Gamma formula (c) against the numerical Wronskian (*10^{-20}*); the Pöschl–Teller forms (*$\ell = 1, 2, 3$*); the partial fractions, the* p_1 *table and the pole values (*10^{-20}*,* $\lambda = 1.5, 2.7, -1.8$*;* $\hat{\psi}_\lambda(0.3) = 8.2198709957 \cdot 10^{-6}$ *at* $(x_0, \lambda) = (\frac{1}{5}, \frac{7}{5})$*); Lemma B.10 (*$10^{-20}$*) and* $A_+ = 0.366725,\ 0.251682,\ 0.385440,\ 0.240330$ *at* $k = \sqrt{0.1}, \sqrt{0.5}, 0.3, 0.6$ *against double quadrature (*10^{-9}*), and* $24(1-k)A_+ \to 1$*; for* $\mu \in \{0.6, 1, 1.4, 1.9\}$ *the Bessel model (*10^{-18}*) and the Fuchsian closed form of* ℓ_∞*; the rescaled defect* $(\ell_\infty p_2^{\mu/2}/\ell_* - 1)\sqrt{p_2}$ *at* $p_2 = 10^8$ *is* $0.6153, 0.7521, 0.5348$ *for* $\mu = 0.8, 1.3, 1.7$*, as predicted.* `make_appB_compare.py` *compares* (U_c^-, U_c^+, w_0) *with an independent Riccati integration: deviation* $\le 3.7 \cdot 10^{-13}$*.*

APPENDIX C

Numerical methods and reproducibility

In words. *Every number in a* numerical observation*, every figure and every atlas of this book is produced by a small set of numerical methods: shooting for the ODE system in the variable* $\xi = \operatorname{artanh} x$*, stable Riccati integration for the Jost solutions, a fixed-point solver for the effective map* T_D *with damping, Newton's method and continuation to follow folds, a few quadratures, closed forms by hypergeometric and digamma functions, and interval arithmetic for the one computer-assisted proof. This appendix lists them in one table, and for each method it says where it is used, with which tolerances and cut-offs, how it was validated, and which traps the computations of the chapters found. None of these computations is part of a proof, except those of Appendix D, whose rigour does not depend on the floating-point methods described here.*

Roadmap. Appendix C.1 is the method table. Appendix C.2 to Appendix C.8 treat the methods one by one. Appendix C.9 collects the pitfalls in a second table and records one measurement made for this appendix (Numerical observation C.1). Appendix C.10 states what "reproduce" means in this book. Dependencies, running times and the layout of the repository are in Appendix E.

C.1. The methods at a glance

Table C.1 lists the methods, the chapters and atlases in which each is used, the code and the kind of validation reported. The tolerances and cut-offs are in Table C.2. All floating-point computations are in IEEE double precision (binary64), except those that say otherwise (30-digit `mpmath` for the closed forms and for the residuals of identities; 160-bit Arb balls in Appendix D).

TABLE C.1. The numerical methods of the book. "Used in" lists chapters and atlases; the last column names the validation that the book reports. The scripts are named in the sections below.

method	used in	validation
S. Shooting in ξ (fixed-step RK4 or adaptive DOP853) with a terminal test for the type of the pole; bisection on γ for the edges γ^-, γ^+ and on κ for $\kappa_{\max}$	Chapter 4 (Atlas 4.50), Chapter 5 (Atlas 5.73), Chapter 6 (Atlas 6.65), Chapter 7 (Atlases 7.39 and 7.93), Chapter 11 (Atlas 11.62), Chapter 12 (Atlas 12.77)	$\kappa_{\max} = 4.1608149$ at $x_0 = 0$, $\boldsymbol{c} = 0$; $\gamma^-(0,0,0) = -2$; mirror symmetry; Jacobi fields against central differences
R. Stable Riccati integration of $\zeta' = W - \zeta^2$ for the Jost solutions and w_0; transfer matrix per panel in the fixed-point solver	Chapter 3 (Numerical observation 3.49), Chapter 8, Chapter 9 (Atlas 9.76), Chapter 10 (Atlas 10.48), Chapter 11	closed form of Proposition 3.47: differences $\le 2.3 \cdot 10^{-13}$ and $3 \cdot 10^{-11}$
F. Fixed-point solver for T_D: damped Picard iteration, or Newton's method with a dense Jacobian assembled from (8.6)	Chapter 8 (Numerical observations 8.36 and 8.58), Chapter 9 (Atlas 9.76), Chapter 10 (Atlas 10.48), Chapter 11 (Atlas 11.62)	residual $\le 2 \cdot 10^{-13}$; agreement with the certificate to $2 \cdot 10^{-13}$ at D^*; refinement in order, panel, L
N. Newton's method with pseudo-arclength continuation in $\ln \kappa_{\mathrm{D}}$ or in a second parameter; fold conditions $E(\omega) = D$, $\det \mathrm{D}E = 0$	Chapter 9 (Atlas 9.76), Chapter 10, Chapter 11	folds from two independent solvers, equal to six digits; three grids
B. Bisection for thresholds (α_-, κ_t, $\kappa_{\max}$, $\gamma^{\mp}$)	Chapter 6, Chapter 7, Chapter 12	tolerances of Table C.2; same solver at the reference point
Q. Quadratures: trapezoid rule on a window; Gauss–Legendre on panels; exact recursion for the kernel $e^{-2\lvert\xi-\eta\rvert}$; adaptive quadrature (`scipy`, `mpmath`)	Chapter 4, Chapter 9, Chapter 11, Chapter 12, Appendix B	digamma form against double quadrature (10^{-8}, 10^{-9}); ℓ_* to 10^{-30}
H. Closed forms: ${}_2F_1$, Γ, ψ at 30 digits (`mpmath`); exact algebra (`sympy`)	Chapter 3, Chapter 11, Appendix B; identities in every chapter	142 residuals in Numerical observation B.20, all passing

method	used in	validation
E. Eigenvalues: dense solver, Arnoldi on the analytic derivative, GMRES for the secular function	Chapter 9, Chapter 10, Chapter 11	refinement to 10^{-4}; secular function vanishes to $5 \cdot 10^{-14}$
I. Interval arithmetic (Arb balls, Taylor models, Krawczyk's test)	Chapter 9 (Theorem 9.68), Appendix D, Appendix A	independent double-precision run, a second solver, three precisions (Numerical observations 9.74 and D.12)

where	tolerance	cut-off
Chapter 4	trapezoid rule, $N = 900$ points, double precision	window of length 16 after the cut
Chapter 5	RK4 step $0.02/(1 + \lvert h\rvert + \sqrt{\lvert W\rvert})$; γ^- by bisection to 10^{-11}; $\kappa_{\max}$ to relative 10^{-9}	$\xi \le \xi_0 + 14$; $\xi_0 + 30$ for the non-uniform limit
Chapter 6	DOP853, relative tolerance 10^{-13}; α_- by bisection to 10^{-10}	backward to $\xi = -9$; error of the pure-exponential tail $O(e^{-18}) \approx 10^{-8}$
Chapter 7	edges by bisection to 10^{-14}; $Q(1)$ read at $\gamma^- + 10^{-12}$ (wall) or 10^{-10} (map of types)	$s_{\max} = 40$ (60 for $c_2 = -0.8$ in Atlas 7.39), $s = -\ln(1 - x)$ (30 for the Jacobi fields)
Chapter 9	residual $\le 2 \cdot 10^{-13}$; J to 10^{-9}; eigenvalues to 10^{-4}; folds on three grids	$L = 14$ to 24; Gauss–Legendre order 6 to 10; panel width 0.25 to 1.5
Chapter 10	damped Picard until $\sup \lvert T_D f - f\rvert < 10^{-11}$	box $[-12, 12]$; finest spacing $\ge 0.1/J$ at the layer
Chapter 11	RK4, step parameter 0.004; bisection in γ to 10^{-15}	$\xi \le 40$ (the interval $\xi \le 14$ is not enough)
Chapter 12	$\kappa_{\max}$ to relative 10^{-5}, `rtol` 10^{-11} (Part A); α_- to relative 10^{-9}, solver tolerance 10^{-13} and 10^{-12} (Part C)	depth rule 10^{-6}; pole declared at $\hat{z}/z < 10^{-6}$
Appendix B	30 digits, residuals below 10^{-18} to 10^{-20}	–
Appendix D	160 bits, Taylor degree 24, box radius 10^{-19}	cuts $X_R = 26$, $X_L = 12$

TABLE C.2. Tolerances and cut-offs, by chapter. These are the values stated in the atlases of the chapters, collected here; the observations of each chapter remain the authority.

C.2. Shooting in ξ and bisection on the edges

The method. The class-Z problem is integrated from the base point x_0 in the line variable $\xi = \operatorname{artanh} x$ (or in $s = -\ln(1-x)$ near the pole) for the state $(h, \ln z, f, \psi, \psi')$ with $h = z'/z$, $W = 1 + \frac{1}{2}(P_{\mathbf{c}} - \psi)$ and $h' = W - h^2$ (Chapter 2); γ parametrises the data at x_0. The solution is *global* if it reaches the pole of the correct type, and the lower edge $\gamma^-(\kappa)$ is the smallest such γ, found by bisection; the upper edge γ^+ follows by the mirror symmetry at $x_0 = 0$ (otherwise by shooting in the other direction), and $\kappa_{\max}$ by bisection on the sign of $\gamma^+ - \gamma^-$. Two integrators are used: a fixed-step RK4 with a step parameter (in Chapter 5 the step is $0.02/(1+|h|+\sqrt{|W|})$, so that the stiff phase of a solution close to a pole is resolved; in Chapter 11 it is 0.004), and DOP853 with the defaults `rtol` 10^{-12}, `atol` 10^{-14} (`rtol` $= 10^{-13}$ in Chapter 6). Sensitivities with respect to the data are obtained from the variational equations integrated together with the solution.

The terminal test. A finite window cannot show whether a trajectory that survives to ξ_{end} would later die. The shooting runs of Chapter 5 and Chapter 7 therefore end with the classification of the pole by (q, h) from Theorem 4.5 and Lemma 5.46: with $q = 2c_2 - \psi(\xi_{\text{end}})$ and $D = 1 + \frac{q}{2}$, the trajectory is unsafe if $D < 0$ or $h \le -\sqrt{D}$, and safe if $D \ge 0$ and $h > -\sqrt{D}$. Without this test the tips of the lens would be computed as about 4.27, 2.70, 12.24 instead of 4.1608, 2.5737, 12.148 (Atlas 5.73).

Validation. The reference points are $\kappa_{\max}(0,0) = 4.1608149$ with $\gamma_* = 0$ to within 10^{-9} and $\gamma^-(0,0,0) = -2$ (Atlas 5.73, Example 5.71); the monotonicity in c_i and the mirror symmetry $\gamma^- = -\gamma^+$ at $x_0 = 0$ are tested in every run; derivatives from Jacobi fields are compared with central differences of the bisected edge (step 10^{-4}) in `verify_ch07a.py`; and the lens values of Example 5.71 are reproduced by `verify_ch05b.py`. The pole limits $Q(1) + 2$ of the type map in Atlas 7.39 are cross-checked by the logarithmic law $U + 2 \sim 4/s$ at the type L_1 (Numerical observation 7.24).

Pitfalls found. (i) A finite window cuts the tail of $\int \hat{\psi}_+ u_+^2$: with $\xi \le 14$ the computed $\Delta = \gamma^-(\kappa) - \gamma^-(0)$ for $k < 1$ carries a constant relative offset of $5 \cdot 10^{-4}$, and $\xi \le 40$ removes it (Atlas 11.62). (ii) The quotient Δ/κ^2 is a difference of nearly equal numbers; it is only meaningful if $\gamma^-(0)$ is computed *by the same solver with the same settings* (Numerical observation C.1). (iii) Readings at a finite window of a slowly relaxing layer are not edge values: for $-\frac{3}{4} < c_2 < -0.4$ and small κ the entries of the type map are read at $s_{\max} = 40$ before the layer is crossed (Atlas 7.39, Atlas 4.50); such entries are never used. (iv) The classification without the terminal test returns wrong tips (Atlas 5.73).

C.3. Jost solutions: the Riccati equation, transfer matrices, closed forms

The method. For a potential W with limits k_1^2, k_2^2 the log-derivatives $\zeta_\mp = u'_\mp/u_\mp$ solve $\zeta' = W - \zeta^2$. The recessive solution at $-\infty$ is obtained by integrating forward from a point $-L$, where $\zeta_- = k_1$ up to an error of the size of the tail of $W - k_1^2$, and similarly ζ_+ backward from $+L$; in these directions the Riccati equation is attractive, so that errors decay, while the integration in the opposite direction amplifies them like $e^{2k|\xi|}$. In the fixed-point solver the same is done panel by panel: on each panel of Gauss–Legendre nodes a spectral collocation matrix gives a 2×2 transfer matrix for (u, u'), with logarithmic scaling so that growing and decaying

solutions are tracked without overflow, started at $\pm L$ from the exact exponentials. The quantity $w_0 = \zeta_-(0) - \zeta_+(0)$ and its derivative with respect to f follow by differentiating the whole chain ((8.6)).

Cut-offs and tolerances. Backward integration to $\xi = -9$ (Chapter 6) leaves a tail error $O(e^{-18}) \approx 10^{-8}$; the Riccati sweeps of Chapter 10 use a grid with spacing $\le 2/J$ away from the layer so that the explicit RK4 steps of the stable sweeps are stable; the boxes $[-L, L]$ with $L = 14$ to 24 in Chapter 9 are chosen larger when k_2 is small, because the tail of the swirl f saturates like $e^{-2k_2|\xi|}$ and dominates the cut-off error.

Validation. The closed forms of Proposition 3.47 and its mirror, evaluated at 30 digits, agree with the Riccati integration (tolerance 10^{-12}) to at most $2.3 \cdot 10^{-13}$ on six parameter sets and two base points (Numerical observation 3.49); the Riccati computation of $\tilde{w}_0(s)$ is within $3 \cdot 10^{-11}$ of the closed form at $s = 9, 16, 40, 100$ (Chapter 8).

Pitfalls found. Integrating ζ_- backward or ζ_+ forward is unstable (errors grow like $e^{2k|\xi|}$), so the direction is never chosen otherwise. At $\mu \lesssim 0.7$ the convergence of the exponent in the class-N computations is slow, and for $|C| \gtrsim 10^{11}$ the quantity $1 + C\ell$ suffers floating-point cancellation (Atlas 12.77).

C.4. The fixed-point solver for T_D

Two solvers. *Damped Picard iteration* is used for the large-swirl atlas (Chapter 10) and the profiles of Chapter 8: the map $f \mapsto f + \omega(T_D f - f)$ on a non-uniform grid of $[-12, 12]$, $\omega = \frac{1}{2}$ (and $\omega = 0.7$ in Numerical observation 8.58), until $\sup|T_D f - f| < 10^{-11}$ (more iterations are needed near the ridge: 36–37 iterations at $\kappa_{\mathrm{D}} = 10^2, \dots, 10^5$ at the rest state). The operator $\mathcal{Q}$ is applied exactly for piecewise linear data by a two-sided recursion for the kernel $e^{-2|\xi-\eta|}$. *Newton's method* with a dense Jacobian is used for the atlas of Chapter 9: panels with Gauss–Legendre nodes of order 6 to 10, panel width 0.25 to 0.6 near the equator and up to 1.5 far away, $L = 14$ to 24; $\mathrm{D}T_D$ is assembled by differentiating the whole chain and diagonalised by a dense eigensolver. Pseudo-arclength continuation in $\ln \kappa_{\mathrm{D}}$ follows branches through folds, which are detected by sign changes of $\mathrm{d}\kappa_{\mathrm{D}}$ along the arclength and refined by a cubic fit (scripts `code/ch09_uniq/_w_classZ.py`, `_w_ridge.py`).

Validation. (i) Residual $\|f - T_D f\|_\infty \le 2 \cdot 10^{-13}$; J converged to 10^{-9} or better under refinement of the order, the panel size and L; eigenvalues to 10^{-4}. (ii) At the certified datum D^* the three fixed points agree with the centres of the certificate to $5.3 \cdot 10^{-14}$, $2.1 \cdot 10^{-13}$, $1.1 \cdot 10^{-13}$ (Numerical observation 9.74). (iii) At the rest state the J values 0.49806, 4.01393, 14.84365, 45.64194, 448.20504 at $\kappa_{\mathrm{D}} = 1, 10, 10^2, 10^3, 10^5$ are stable under halving the finest spacing and enlarging the box from 12 to 16 (relative change $5 \cdot 10^{-8}$ at $\kappa_{\mathrm{D}} = 10^4$). (iv) The certificate (shooting in the original variables) and the fixed-point solver share no code (Numerical observation 9.74). (v) The derivative formula (8.6) is tested against finite differences in `verify_ch08a.py`.

Pitfalls found. (i) *Damped iteration cannot converge to an unstable fixed point*: at $\kappa_{\mathrm{D}} = 1400$ only the middle solution of Theorem 9.68 ($J = 33.5892$) has a real multiplier above 1 (1.28184; the others have $0.35 \pm 0.12\mathrm{i}$ and 0.62962, Numerical observation 9.75), so it cannot be an attracting fixed point of the damped iteration; the three solutions are found by Newton's method with arclength continuation, and multistart searches (at least 40 initial profiles) are not exhaustive. (ii) The

undamped iteration converges only for moderate κ_{D}: its leading multiplier tends to -1 (-0.94 at $\kappa = 256$), which is why $\omega = 0.7$ is used. (iii) In the scan of Chapter 9 the continuation did not complete at two entries (one fold of the pair is missing; the cause was not investigated), and these entries are not used.

C.5. Newton's method and continuation for folds

A fold of the effective-data map is a solution of $E(\omega) = D$, $\det \mathrm{D}E(\omega) = 0$. There are two independent computations: the fixed-point solver of Appendix C.4 with arclength continuation, and Newton's method on the fold conditions in the shooting chart (Cauchy data of Lemma 9.11) with Runge–Kutta step $h = 0.005, 0.0025, 0.00125$ and box $L = 14, 18$. At the datum of Chapter 9 with $\vartheta = 4.49$ the folds are $\kappa_{\mathrm{D}} = 6.30477$ and 11.35302 from the first and 6.304766, 11.353022 from the second: equal to six digits for all steps and boxes. The fold curve is continued in the second parameter $c_3' - \bar{c}_3$ with ϑ and f_- fixed, and the cusp is located by the meeting of the two branches ($\approx (0.16, 13.74)$). For the thresholds of Chapter 6 and Chapter 12 the analogue is bisection (Table C.2) together with a *globality test on the forward continuation*.

Pitfalls found. At the rest state with $f_- = 0$ a window of folds opens at $\vartheta_c \approx 2330$ (its width to the power $\frac{2}{3}$ is linear in ϑ): there is no fold at $\vartheta = 2300$ and there are folds at $\vartheta = 2400$, so that a continuation at one value of ϑ near ϑ_c is not evidence about its neighbours (Atlas 9.76).

C.6. Quadratures

Four quadratures occur. (a) The *trapezoid rule* on a fixed window: $N = 900$ points on a window of length 16 after the cut for the loop and contraction constants of Chapter 4; and double quadrature for $A_+^{\lim}$ of Chapter 11. (b) *Gauss–Legendre* nodes on panels, order 6 to 10 (fixed-point solver). (c) The *exact recursion* for the convolution with $e^{-2|\xi-\eta|}$ on piecewise linear data, which has no quadrature error at all. (d) *Adaptive quadrature* (`scipy.integrate.quad`, `mpmath.quad`) for the constants of Chapter 12 and Appendix B.

Validation. The digamma form of $A_+^{\lim}$ (Proposition B.12) and double quadrature agree to 10^{-8} in Atlas 11.62 and to 10^{-9} in Numerical observation B.20; ℓ_* of Chapter 12 is computed to 10^{-30} by quadrature of $\int g^{-2}$ and compared to 10^{-9} with a bisection on the limit ODE; the integrals of Chapter 4 such as $\int_0^\infty (1+s)e^{-2s}\mathrm{d}s = \frac{3}{4}$ are checked symbolically.

Pitfalls found. Truncating a tail integral at a window that is too short is a systematic error, not a random one: the tail $\int \hat{\psi}_+ u_+^2$ for $k < 1$ is not negligible at $\xi = 14$, which is the origin of the offset of Appendix C.2.

C.7. Closed forms, computer algebra and residual checks

The closed forms of Appendix B (Gauss reduction, ${}_2F_1$ Jost solutions, the Landau integral, the digamma form of $A_+^{\lim}$, the Bessel and the exact Fuchsian model) are evaluated with `mpmath` at 30 digits and every identity is checked by a residual: 142 residuals in `check_appB.py`, all below 10^{-18} to 10^{-20} except the comparison with double quadrature (10^{-9}) (Numerical observation B.20). Algebraic identities used in proofs of the chapters (`verify_*.py`) are verified exactly with `sympy`; a check is a `simplify(...)==0` or a rational-arithmetic equality, and each

script prints one line per identity and exits with a non-zero code on failure. The kappa-conversions of Chapter 2 have in addition a test suite (`code/common/tests`) with exact rational arithmetic.

C.8. Interval arithmetic

The only rigorous computation of the book is the certificate of Theorem 9.68, described in Appendix D: Arb ball arithmetic (python-flint) at 160 bits, Taylor steps of degree 24 in mean value form with a validated remainder, enclosures of the two infinite tails at the cuts $X_R = 26$, $X_L = 12$, and Krawczyk's test in a box of radius 10^{-19} (Lemma A.57). The floating-point methods of the other sections play no role in it, but they supply its starting points and its cross-checks: an independent double-precision Dormand–Prince integration (differences from the enclosure midpoints $4.6{\cdot}10^{-14}$, $6.4{\cdot}10^{-14}$, $1.2{\cdot}10^{-12}$ in J, Numerical observation 9.74) and the independent fixed-point solver (Appendix C.4). The enclosures were also computed at 128 bits (degree 16) and 256 bits (degree 32, cuts 30 and 14), and the three Krawczyk tests passed in each (Numerical observation D.12). The hash of the result files is reproduced exactly by `make -C code/appD_cap reproduce` (Appendix D.6.1).

Pitfalls found. Naive interval evaluation of a Taylor step wraps (the dependency problem): evaluating $\sum(-4h)^k y_0/k!$ with the same ball for each occurrence of y_0 inflates the result, and the mean value form is needed (Toy example D.7). The enclosure width of the tail decays like $e^{-2.5X}$ and is always 7.11 times the true truncation error; beyond $X = 14$ the enclosures are narrower than the double-precision rounding of f, so that the comparison with a double-precision run stops being informative.

C.9. Pitfalls and a measurement

Table C.3 collects the pitfalls of the sections above in one place, with the size of the effect and the remedy used in the book.

Numerical observation C.1 (an adaptive integrator at $\kappa = 0$)**.** *At $x_0 = 0$, $\boldsymbol{c} = 0$, $\kappa = 0$ the lower edge is $\gamma^- = -2$ exactly. The adaptive Dormand–Prince solver of order 8 (DOP853, absolute tolerance 10^{-14}, bisection to 10^{-13}) returns $\gamma^- + 2 = 4.6 \cdot 10^{-12}$ at `rtol` 10^{-12} and $5.4 \cdot 10^{-13}$ at `rtol` 10^{-13}, for $s_{\max} = 25$ and 40 alike; the fixed-step RK4 solver of Appendix C.2 returns -2 to $4.4 \cdot 10^{-16}$ for $\xi_{\text{end}} = 14, 25, 40$ and steps $0.02, 0.01, 0.004$. The error of the adaptive solver is of the size of the tolerance. At $\kappa = 1$ the difference of γ^- between $s_{\max} = 25$ and $s_{\max} = 40$ is $1.1 \cdot 10^{-10}$. For a difference such as $\Delta = \gamma^-(\kappa) - \gamma^-(0) \approx A_+^{\lim}\kappa^2$ at $\kappa = 10^{-3}$, that is $\Delta \approx 3 \cdot 10^{-7}$ (with $A_+^{\lim} \approx 0.25$ to 0.37, Atlas 11.62), an error of $5 \cdot 10^{-12}$ in the reference value is a relative error of about $2 \cdot 10^{-5}$ in Δ, and for $\kappa = 10^{-4}$ of about $2 \cdot 10^{-3}$. Code: `code/appC_numerics/verify_appC.py` (part 1; the two underlying solvers).*

C.10. What "reproduce" means

Three levels of reproduction are distinguished. (1) *Exact*: the hash of the certificate output of Appendix D is reproduced by a rerun with the software of Appendix E; ball arithmetic is deterministic. (2) *To the digits quoted*: every number of a numerical observation is printed by a script, and the book quotes digits that

pitfall	size	remedy
window $\xi \le 14$ cuts the tail of $\int \hat{\psi}_+ u_+^2$ ($k < 1$)	relative offset $5 \cdot 10^{-4}$ in Δ/κ^2	$\xi \le 40$; Δ formed with the same solver at $\kappa = 0$
adaptive solver at $\kappa = 0$	$\gamma^- + 2 \approx 5 \cdot 10^{-12}$ at `rtol` 10^{-12} (Numerical observation C.1)	same solver and settings at the reference point, or a fixed-step solver for which the exact value is reproduced
finite window for a slowly relaxing layer	reading is not the pole limit	such entries are marked and never used (Atlases 4.50 and 7.39)
trajectory that survives the window but dies later	tips 4.27, 2.70, 12.24 instead of 4.1608, 2.5737, 12.148	terminal test by (q, h)
damped iteration	the unstable (middle, $J = 33.5892$) solution is not an attracting fixed point	Newton's method with dense Jacobian; multistart is not exhaustive
undamped Picard at large κ_{D}	multiplier $\to -1$	damping $\omega = 0.7$ or $\frac{1}{2}$
cancellation in $1 + C\ell$	$\lvert C\rvert \gtrsim 10^{11}$	range of C limited; error at $\lvert C\rvert = 10^7$ is $8.3 \cdot 10^{-4}$ at $\mu = 1$
slow convergence for $\mu \lesssim 0.7$	search does not converge for $\lvert C\rvert \ge 10^4$ at $\mu = 0.6$	no accuracy claimed for $\mu < 0.7$

TABLE C.3. Pitfalls found by the computations of the chapters.

were stable under the refinements listed in the atlases; a different release of `scipy` or of the C library may in principle change the last digit of a floating-point result. (3) *Qualitative*: the shape of a figure. The random samples and the multistart searches use fixed seeds. Wall-clock times depend on the machine; those in Appendix E are approximate.

APPENDIX D

The computer-assisted proof

In words. *Theorem 9.68 says that the effective data D^* are the data of at least three different solutions. Its proof has a finite-dimensional heart: a solution is determined by four numbers $s = (h_0, f_0, Q_0, Q_1)$ read at the equator, the conditions "effective data $= D^*$" are four equations $\Phi(s) = 0$, and we must show that Φ has three zeros. A computer cannot evaluate Φ, which involves the limits $\xi \to \pm\infty$ of a solution of an ODE, but it can do something a person cannot do with pencil and paper: carry out several million arithmetic operations* with a guarantee, *namely with every number replaced by a ball that is certain to contain the true value. This appendix explains how such guarantees are produced and combined into a proof. It has one rule:* everything the computer does is first stated as a theorem on paper whose hypotheses are inequalities between balls, *and the computer's only role is to check those inequalities. The statements proved on paper are the formulas of the system and its variational equation (Lemma D.6), the guarantee of ball arithmetic and of automatic differentiation (Lemma D.3), Krawczyk's test (Lemma D.5), the enclosure of one Taylor step of the ODE (Theorem D.8), and the enclosure of the two infinite tails at the poles by a comparison argument (Proposition D.10). The computer then verifies, for three small boxes in $\mathbb{R}^4$, that all the hypotheses hold (Proposition D.11). We also say exactly what is* not *proved.*

Roadmap. Appendix D.1 lists what the computer proves and what the proof trusts. Appendix D.2 explains ball arithmetic and Krawczyk's test on toys done by hand and proves Lemmas D.3 and D.5. Appendix D.3 states the system and the four equations (Lemma D.6). Appendix D.4 proves the one-step enclosure of the flow (Theorem D.8). Appendix D.5 handles the poles (Proposition D.10). Appendix D.6 gives the certificate, the reproduction record (software versions, run time, hash) and the sanity checks. Appendix D.7 is the table of what is and is not certified. All numbers below are produced by the scripts of `code/appD_cap/` (`make -C code/appD_cap checks`, `figures`, `reproduce`).

D.1. What is certified, and what the proof rests on

Fix the datum $D^* = \big(\boldsymbol{c}' = (\frac{5}{4}, -\frac{15}{16}, -\frac{517}{160}),\ \vartheta = 40,\ f_- = 0,\ \kappa_{\mathrm{D}} = 1400\big)$ of Theorem 9.68 and the two rational constants

$$\lambda^* = \tfrac{123}{10}, \qquad \tilde{\kappa}^* = \tfrac{1400}{41^2} = \tfrac{1400}{1681}.$$

Here $\tilde{\kappa} = \kappa_0 = f'(0) = f_x(0)$ for the representative z with $z(0) = 1$ (normalisation 3 of Table 2.1); it is related to the swirl constant κ_{D} of the datum by $\kappa_{\mathrm{D}} = (1+\vartheta)^2\tilde{\kappa}$, which for $\vartheta = 40$ gives $\tilde{\kappa}^* = 1400/41^2$. The proof of Theorem 9.68 in Chapter 9 is a chain of four statements.

(C1) *The equations* (Lemma D.6): the system (9.8), its variational equation, the truncated map Φ_0, and the equivalence $\Phi(s) = 0 \iff E(\omega(s)) = D^*$.

(C2) *The tails* (Proposition D.10): if the state at a cut satisfies three explicit inequalities (H1)–(H3), the solution continues to the pole, is of type G there, and the limits are enclosed.

(C3) *Krawczyk* (Lemma D.5): a ball computation that implies the existence of a zero.

(C4) *The certificate* (Theorem D.8, Proposition D.11): the validated integration over three boxes, and the computer's output.

Only (C4) involves a computer, and only in the sense that a program evaluates finitely many expressions in ball arithmetic and reports that certain inequalities hold. Whoever distrusts the program can re-run it (Appendix D.6) or re-implement it: every inequality it checks is displayed in the statements of this appendix.

Remark D.1 (the trust base)**.** The conclusion rests on: (1) the correctness of the ball arithmetic of the Arb library [17], called through python-flint (Lemma A.52 states what is guaranteed); (2) the correctness of the three modules that implement Theorem D.8, Proposition D.10 and Lemma D.5 (about 400 lines of Python; guarded by regression tests, Numerical observation D.12); (3) the hand proofs of this appendix and of Chapter 9; (4) the standard theorems quoted from Appendix A (Taylor's formula, dependence of ODE solutions on data, Brouwer's theorem). The centres $\bar{s}_k$ of the three boxes are *inputs*: they were obtained by Newton's method, but the proof does not depend on how, since the Krawczyk test is applied to the box and is either passed or not.

D.2. Ball arithmetic and Krawczyk's test

In words. *A computer represents a number as a binary fraction with finitely many digits, so* 0.1 *is not represented exactly and the result of* $0.1+0.2-0.3$ *is not* 0. Ball arithmetic *represents instead a pair* $[m \pm \rho]$*: a binary number* m *and a radius* $\rho \geq 0$*, with the promise that the true value lies in* $[m-\rho, m+\rho]$*. Every operation returns a ball that contains all possible results, rounding* outwards *whenever the exact result cannot be stored. The price of the promise is that balls may be larger than necessary.*

Toy example D.2 (balls by hand)**.** Represent a quantity known to lie between 0.9 and 1.1 by the interval $X = [0.9, 1.1]$, and use the endpoint rules $[a,b] \pm [c,d]$, $[a,b] \cdot [c,d] = [\min, \max]$ of the four products. Then

$$X - X = [-0.2,\, 0.2], \qquad (X-1)\cdot(X-1) = [-0.01,\, 0.01],$$
$$X \cdot X - 2X + 1 = [-0.39,\, 0.41],$$

although $x - x = 0$ and $x^2 - 2x + 1 = (x-1)^2 \in [0, 0.01]$ for every $x \in X$. The enclosures are correct and the same function written in three ways gives three different enclosures: this is the *dependency problem*: the interval rules do not know that the two copies of x are the same number. In floating-point arithmetic $0.1 + 0.2 - 0.3 = 5.55 \cdot 10^{-17}$; with 160-bit balls the result is a ball containing 0 and no wrong claim is ever made. (`toy_hand.py`, part 1.)

The dependency problem is the reason for the mean value form below: it moves the overestimate into a derivative that multiplies a *tiny* quantity.

Lemma D.3 (what is guaranteed)**.** (a) (*expressions*) *If φ is an expression in variables, constants, $+,-,\times,/$ and $\sqrt{\ }$,* exp, tanh, cosh (*the enclosure of an elementary function on a ball is taken piecewise on its monotone pieces:* exp *and* tanh *on* $\mathbb{R}$, cosh *on* $[0,\infty)$ *and on* $(-\infty,0]$, sech^2 *likewise; a non-monotone function on a ball containing its turning point is split at that point, and in the programs* cosh *and* sech^2 *are only applied to the exact points X and ξ_j, i.e. to degenerate balls, where the library's own certified enclosure of the value is used), and $\boldsymbol{\varphi}(\mathbf{X})$ is the same expression evaluated in ball arithmetic on a box $\mathbf{X}$ (the computation fails if a divisor ball contains* 0 *or a square root is taken of a ball that is not positive), then $\varphi(x)\in\boldsymbol{\varphi}(\mathbf{X})$ for every $x\in\mathbf{X}$.* (b) (*automatic differentiation*) *Let B be a set, $u_i : B\to\mathbb{R}$ $(i=1,\dots,m)$ be C^1 in a parameter $s\in\mathbb{R}^4$, and let balls $\mathbf{U}_i$ and ball vectors $\mathbf{G}_i\in\mathbb{R}^4$ satisfy $u_i(s)\in\mathbf{U}_i$ and $\nabla_s u_i(s)\in\mathbf{G}_i$ for all $s\in B$. Carry pairs $(\mathbf{U},\mathbf{G})$ through $+,-,\times,/,\sqrt{\ }$ by the rules*

$$(\mathbf{U},\mathbf{G})\pm(\mathbf{V},\mathbf{H})=(\mathbf{U}\pm\mathbf{V},\mathbf{G}\pm\mathbf{H}),\quad (\mathbf{U},\mathbf{G})(\mathbf{V},\mathbf{H})=(\mathbf{UV},\ \mathbf{VG}+\mathbf{UH}),$$

$$\frac{(\mathbf{U},\mathbf{G})}{(\mathbf{V},\mathbf{H})}=\Big(\mathbf{q},\ \frac{\mathbf{G}-\mathbf{qH}}{\mathbf{V}}\Big),\ \mathbf{q}=\frac{\mathbf{U}}{\mathbf{V}},\qquad \sqrt{(\mathbf{U},\mathbf{G})}=\Big(\mathbf{r},\ \frac{\mathbf{G}}{2\mathbf{r}}\Big),\ \mathbf{r}=\sqrt{\mathbf{U}}.$$

If ϕ is an expression in $u_1,\dots,u_m$ and rational constants, the resulting pair $(\boldsymbol{\phi},\mathbf{G}_\phi)$ satisfies $\phi(u(s))\in\boldsymbol{\phi}$ and $\nabla_s\big(\phi\circ u\big)(s)\in\mathbf{G}_\phi$ for all $s\in B$. (c) (*products of interval matrices*) *If $M_1\in\mathbf{J}$ and $M_2\in\mathbf{A}$ entrywise for interval matrices, then $M_1M_2\in\mathbf{JA}$, where the product of interval matrices is evaluated entry by entry with (a).*

Proof. (a) is Lemma A.52. (b) Induction on the expression. For a sum or a difference this is the linearity of the gradient. For a product, $\nabla(uv)=v\nabla u+u\nabla v$ and $v(s)\in\mathbf{V}$, $\nabla u(s)\in\mathbf{G}$ etc., so (a) gives the claim. For a quotient, put $q(s)=u(s)/v(s)\in\mathbf{q}$ by (a); then $\nabla q=(\nabla u-q\nabla v)/v$, and the right-hand side lies in $(\mathbf{G}-\mathbf{qH})/\mathbf{V}$ by (a) applied to the numbers $\nabla u(s), q(s), \nabla v(s), v(s)$. For the square root $r=\sqrt{u}$, $\nabla r=\nabla u/(2r)$ and $r(s)\in\mathbf{r}$. (c) Each entry of M_1M_2 is a sum of products of an entry of M_1 by an entry of M_2; apply (a). □

Toy example D.4 (Krawczyk's test on a one-dimensional shooting problem, by hand)**.** Let $y'=y^2$, $y(0)=s$, and ask for s such that $y(1)=2$. The flow is $y(1;s)=s/(1-s)$, so $\Phi(s)=s/(1-s)-2$ and the answer is $s=\frac{2}{3}$; for the toy we pretend that we do not know it, and that we only have the approximation $x_0=0.665$. Take the box $\mathbf{X}=[0.66,0.67]=x_0+[-r,r]$, $r=0.005$. The three ingredients of the test (Lemma A.57) are computed with the interval rules of Toy example D.2:

(1) $\Phi(x_0)=\frac{0.665}{0.335}-2=-\frac{1}{67}=-0.014925\ldots$;
(2) an enclosure of the derivative over the box: $1-\mathbf{X}=[0.33,0.34]$, its square is $[0.1089,0.1156]$, so $\Phi'(\mathbf{X})=1/(1-\mathbf{X})^2\subset[8.6505,\ 9.1828]$;
(3) the factor $C=1/\Phi'(x_0)=(0.335)^2=0.112225$, and $\mathbf{M}=1-C\,\Phi'(\mathbf{X})\subset[-0.0306,\ 0.0292]$.

Then

$$\begin{aligned}\mathbf{K}&=x_0-C\Phi(x_0)+\mathbf{M}(\mathbf{X}-x_0)\\&=0.665+0.001675+\mathbf{M}\cdot[-0.005,0.005]\subset[0.66652,\ 0.66683],\end{aligned}$$

and $\mathbf{K}\subset(0.66,0.67)=\operatorname{int}\mathbf{X}$. Lemma A.57 now says that Φ has exactly one zero in $\mathbf{X}$, and the zero is in $\mathbf{K}$: indeed $\frac{2}{3}=0.66667\in\mathbf{K}$. Equivalently $|C\Phi(x_0)|+|\mathbf{M}|\,r=0.001675+0.000153<0.005=r$. The test is a sufficient condition and can fail for a good reason: for $r=0.1$ it returns $\mathbf{K}=[0.5635,0.7699]\not\subset\operatorname{int}\mathbf{X}$ (too big a

box: inconclusive, not a disproof), and for the box of radius 0.01 around 0.8, where there is no zero, it returns $[0.7189, 0.7211] \not\subset [0.79, 0.81]$. The numbers are from `toy_hand.py`, part 2; Figure D.2(a) shows the picture. The certificate has the same structure with Φ replaced by the four-dimensional map Φ of Lemma D.6, whose value, derivative and non-differentiable perturbation are enclosed by the computer.

Lemma D.5 (Krawczyk–Brouwer for a perturbed map)**.** *Let* $\mathbf{X} = \bar{s} + r[-1,1]^4 \subset \mathbb{R}^4$ $(r > 0)$ *and let* $\Phi : \mathbf{X} \to \mathbb{R}^4$ *be continuous with* $\Phi = \Phi_0 + \mathsf{Z}$*, where* Φ_0 *is* C^1 *in a neighbourhood of* $\mathbf{X}$ *and* $\mathsf{Z}(s) \in \mathbf{T}$ *for all* $s \in \mathbf{X}$*,* $\mathbf{T}$ *a box. Let* $C \in \mathbb{R}^{4\times 4}$ *be any matrix,* $\mathbf{J}$ *an interval matrix containing* $\mathrm{D}\Phi_0(s)$ *for all* $s \in \mathbf{X}$*,* $\mathbf{V}$ *a box containing* $\Phi_0(\bar{s})$*, and*

$$\mathbf{K} := \bar{s} - C(\mathbf{V} + \mathbf{T}) + (I - C\mathbf{J})(\mathbf{X} - \bar{s}).$$

(a) *If* $\mathbf{K} \subset \operatorname{int}\mathbf{X}$*, then* C *is invertible and* Φ *has a zero in* $\mathbf{K}$*.* (b) *The inclusion* $\mathbf{K} \subset \operatorname{int}\mathbf{X}$ *is equivalent to the componentwise inequality* $\big|C(\mathbf{V}+\mathbf{T})\big| + |I - C\mathbf{J}|\, r < r$*, where* $|\cdot|$ *of a box or interval matrix is the entrywise maximum of the absolute values of its elements; in particular it implies* $Y/r + Z_1 < 1$ *for* $Y := \max_i |C(\mathbf{V}+\mathbf{T})|_i$ *and* $Z_1 := \max_i \sum_j |I - C\mathbf{J}|_{ij}$ *(the quantities printed in Table D.1).*

Proof. (a) is Corollary A.59 (proved there by Brouwer's theorem applied to $s \mapsto s - C\Phi(s)$, a continuous self-map of $\mathbf{X}$). (b) For an interval matrix $\mathbf{M} = I - C\mathbf{J}$ the product $\mathbf{M}(\mathbf{X} - \bar{s}) = \mathbf{M}[-r, r]^4$ is the symmetric box with radii $|\mathbf{M}|r$, so $\mathbf{K}$ is the box centred at $\bar{s} - \operatorname{mid}(C(\mathbf{V}+\mathbf{T}))$ whose i-th radius is $\operatorname{rad}(C(\mathbf{V}+\mathbf{T}))_i + (|\mathbf{M}|r)_i$. It lies in the open interval $(\bar{s}_i - r, \bar{s}_i + r)$ iff $|C(\mathbf{V}+\mathbf{T})|_i + (|\mathbf{M}|r)_i < r$, where $|C(\mathbf{V}+\mathbf{T})|_i$ is the largest modulus of an element of the i-th entry of $C(\mathbf{V}+\mathbf{T})$. Taking the maximum over i gives $Y + Z_1 r < r$. □

D.3. The system, its variational equation and the four equations

We write $' = \mathrm{d}/\mathrm{d}\xi$, $\xi = \operatorname{artanh} x$, $T = \tanh\xi$, $S = \operatorname{sech}^2\xi$ ($T' = S$, $S' = -2ST$), and use the notation of Section 9.8: $s = (h_0, f_0, Q_0, Q_1) \in \mathbb{R}^4$, the system (9.8) with $\lambda = \lambda^*$, $\tilde{\kappa} = \tilde{\kappa}^*$, the set $\mathcal{S}$ and the map Φ of (9.9), the cuts $X_R = 26$, $X_L = 12$, and, for a cut X,

$$\varepsilon = e^{-2X}, \qquad c = \cosh^2 X, \qquad x_a = \tanh X, \qquad \varepsilon = \frac{1 - x_a}{1 + x_a},$$

(the last identity because $\frac{1-\tanh X}{1+\tanh X} = e^{-2X}$). Write $u = (h, E, f, Q, P)$ for the state and $u(\xi; s)$ for the solution with $u(0; s) = (h_0, 1, f_0, Q_0, Q_1)$.

Lemma D.6 (the equations)**.** (a) (*the system*) *The right-hand side of* (9.8) *is a polynomial in* (h, E, f, Q, P, T, S)*. For every* $s \in \mathbb{R}^4$ *there is a unique maximal solution* $u(\cdot; s)$*, defined on an open interval* $I_s \ni 0$*, the set* $\{(\xi, s) : \xi \in I_s\}$ *is open and* u *is* C^∞ *on it. On* I_s *one has* $E > 0$*; with* $z := E^{-1/2}$*,* $z(0) = 1$*, one has* $z'/z = h$*,* $z'' = (1 + \frac{1}{2}Q)z$*,* $f' = \tilde{\kappa} z^{-2}$*, and the last equation of* (9.8) *is equivalent to* $\mathsf{L}[Q] = \lambda - f^2$ *in the line form. If* $I_s = \mathbb{R}$*, then* $u(\cdot; s)$ *is a global solution of the reduced system.* (b) (*the variational system*) *The* 5×4 *matrix* $M(\xi) = \partial_s u(\xi; s)$ *solves* $M' = \mathcal{A}(\xi)M$*, where* $M(0)$ *is the matrix whose four columns are the unit*

vectors $e_h, e_f, e_Q, e_P \in \mathbb{R}^5$, *and*

$$\mathcal{A} = \begin{pmatrix} -2h & 0 & 0 & \frac{1}{2} & 0 \\ -2E & -2h & 0 & 0 & 0 \\ 0 & \tilde{\kappa} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -2Sf & 2S & -4T \end{pmatrix}$$

(*rows and columns in the order* h, E, f, Q, P); *in components* $\delta h' = \frac{1}{2}\delta Q - 2h\delta h$, $\delta E' = -2(E\delta h + h\delta E)$, $\delta f' = \tilde{\kappa}\delta E$, $\delta Q' = \delta P$, $\delta P' = S(2\delta Q - 2f\delta f) - 4T\delta P$. *The same matrix* $\mathcal{A}$ *governs the* 5×5 *derivative of the flow with respect to the initial state at any node* ξ_j. (c) (*the truncated map*) *For a cut pair* (X_R, X_L) *let* Ω_0 *be the set of* s *for which* $u(\cdot; s)$ *exists on* $[-X_L, X_R]$ *and the quantities*

$$Q^{\mathrm{tr}}_{\pm}, \quad a_{\pm} := \sqrt{1 + \tfrac{1}{2}Q^{\mathrm{tr}}_{\pm}}, \quad h_R + a_+, \quad a_- - h_L$$

defined before the list (C1)–(C4) *in Section 9.8 satisfy* $1 + \frac{1}{2}Q^{\mathrm{tr}}_{\pm} > 0$, $h_R + a_+ > 0$, $a_- - h_L > 0$. *Then* Ω_0 *is open,* $\Phi_0 : \Omega_0 \to \mathbb{R}^4$ *is* C^∞, *and* $\mathrm{D}\Phi_0(s) = \partial_1\Theta\,\partial_s u_R + \partial_2\Theta\,\partial_s u_L + \partial_3\Theta\, e_{f_0}^{\mathsf{T}}$, *where* Θ *is the algebraic map from* (u_R, u_L, f_0) *to* Φ_0. (d) (*the equivalence*) *For* $s \in \mathcal{S}$: $\Phi(s) = 0$ *if and only if* $E(\omega(s)) = D^*$, *where* $\omega(s) \in \mathrm{Sol}$ *is the point of Lemma 9.71*(a). (e) (*the pole values*) *At a zero of* Φ *the right pole has* $\mu_R = \frac{1}{4}$ *and the left pole* $\mu_L = \frac{3}{2}$, *i.e.* $h(\xi) \to \frac{1}{4}$ *as* $\xi \to +\infty$ *and* $h(\xi) \to -\frac{3}{2}$ *as* $\xi \to -\infty$.

Proof. (a) The right-hand side is polynomial, hence smooth, so Theorem A.4 gives the unique maximal solution and its smooth dependence. $E' = -2hE$ gives $E(\xi) = \exp(-2\int_0^\xi h) > 0$. With $z = E^{-1/2} = \exp\int_0^\xi h$ one has $z' = hz$ and $z'' = (h' + h^2)z = (1 + \frac{1}{2}Q)z$ by the first equation; $f' = \tilde{\kappa}E = \tilde{\kappa}z^{-2}$. Multiplying $\mathsf{L}[Q] = \cosh^2\xi\, Q'' + 4\sinh\xi\cosh\xi\, Q' - 2Q = \lambda - f^2$ (Lemma 2.18(a)) by $\operatorname{sech}^2\xi$ gives $Q'' = S(\lambda - f^2 + 2Q) - 4TQ'$, which is the last equation with $P = Q'$. So the system is the system on the line (2.29) in the variables $h = z'/z$, $E = z^{-2}$, and a solution on $\mathbb{R}$ has $z > 0$ everywhere, hence is global (Lemma 2.24). (b) By Theorem A.4(c) the derivative with respect to the data solves the variational equation $M' = (\partial_u G)M$, with G the right-hand side; differentiating the five components of G gives $\mathcal{A}$ (a computer-algebra check is in `verify_appD.py`, part 1). The initial state is $(h_0, 1, f_0, Q_0, Q_1)$, so $M(0)$ is as stated ($E(0) = 1$ does not depend on s). The derivative with respect to the initial state at a node is obtained in the same way, with identity initial value. (c) The set of s for which the solution exists on the compact interval $[-X_L, X_R]$ is open by Theorem A.4(c), and the strict inequalities are open conditions on continuous functions of s. On Ω_0 the map Φ_0 is a composition of the smooth map $s \mapsto (u_R, u_L)$ with algebraic expressions whose denominators ($h_R + a_+$, $a_- - h_L$, $a_\pm$) are positive; the chain rule gives $\mathrm{D}\Phi_0$ (f_0 enters Φ_0 directly through $-41f_0$). (d) This is Lemma 9.71(b). (e) By Lemma 9.71(b), $\Phi(s) = 0$ means $Q(-\infty) = \frac{5}{2}$ and $Q(+\infty) = -\frac{15}{8}$; the pole is of type G, so $h \to +\mu_R$ with $\mu_R^2 = 1 + \frac{1}{2}Q(1) = 1 + \frac{1}{2}(-\frac{15}{8}) = \frac{1}{16}$ at the right, and $h \to -\mu_L$ with $\mu_L^2 = 1 + \frac{1}{2}\cdot\frac{5}{2} = \frac{9}{4}$ at the left (mirror image, Definition 4.16). Hence $\mu_R = \frac{1}{4}$ and $\mu_L = \frac{3}{2}$. □

The numbers $\mu_R = \frac{1}{4}$, $\mu_L = \frac{3}{2}$ are not needed in the proof; they are a check on the certificate: the tail enclosures of Table D.2 must, and do, contain these values.

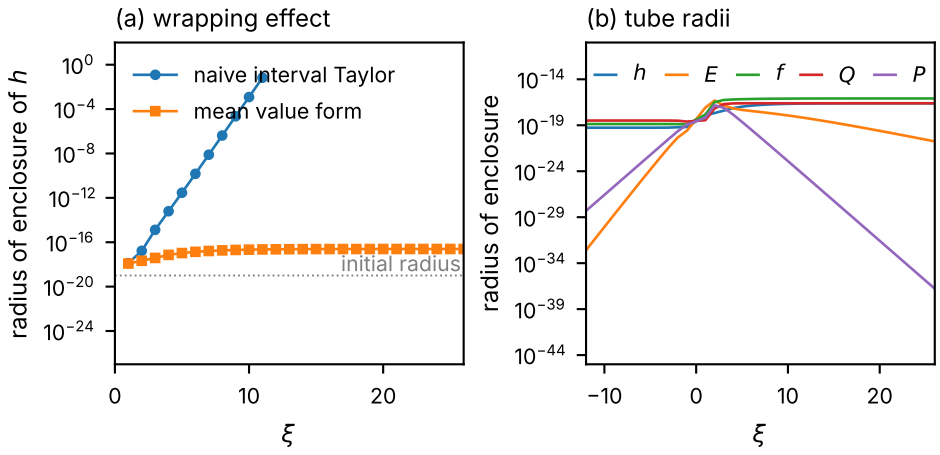


FIGURE D.1. (a) Radius of the enclosure of h along the flow over the box $\bar{s}_1 + 10^{-19}[-1,1]^4$: naive interval Taylor steps against the mean value form (the naive integrator stops at $\xi \approx 12$, its radius being of order 1). (b) Radii of the enclosures of all five state components (h, E, f, Q, P) in the mean value form, from $\xi = -12$ to $\xi = 26$. Code: `code/appD_cap/make_appD_figs.py`.

D.4. Enclosing the flow: one Taylor step in mean value form

In words. *To enclose the solution of an ODE one advances in small steps. In one step the solution is a Taylor polynomial plus a remainder; the remainder is controlled by knowing in advance a box* $\mathbf{B}$ *that the solution cannot leave during the step. If one then takes a* box *of initial values and evaluates the Taylor polynomial with interval arithmetic, the boxes grow exponentially even for a contracting equation (the wrapping effect). Lohner's remedy is to follow the flow of the midpoint of the box exactly (with a tiny ball) and to carry along, in addition, an enclosure of the derivative of the flow; the box is then* $\varphi(\hat{y}) + \mathbf{J} \cdot (\mathbf{Y} - \hat{y})$.

Toy example D.7 (wrapping, by numbers)**.** For $y' = -4y$ and $y(0) \in 1 \pm 10^{-20}$ the exact flow shrinks the radius of the set of solutions by e^{-4t}. The naive interval Taylor step of degree 20 and length $h = \frac{1}{16}$ evaluates $\sum_{k \le 20} (-4h)^k y_0 / k!$ with the *same* ball y_0 in every term; the interval rules do not know this, so the terms do not cancel but add their radii and the radius is multiplied by $\sum_k (4h)^k/k! \approx e^{+4h}$ instead of e^{-4h}. After 64 steps ($t = 4$) the naive radius is $8.9 \cdot 10^{-14}$, that is $e^{16} = 8.9 \cdot 10^6$ times the initial radius, while the mean value form, which multiplies the radius by the single number $\sum_k (-4h)^k/k! \approx e^{-4h}$, ends at $1.1 \cdot 10^{-27}$ (`toy_hand.py`, part 3). In our problem the unstable direction is $P = Q'$ with $P' = -4 \tanh \xi \, P + \ldots$; Figure D.1(a) shows the radius of the enclosure of h over the box of radius 10^{-19} around the first solution: the naive method fails at $\xi \approx 12$ with a radius of order 1, the mean value form stays at 10^{-16} up to $\xi = 26$.

The step. Put $w = (u, T, S) \in \mathbb{R}^7$ and $\frac{\mathrm{d}}{\mathrm{d}\xi} w = \mathbb{G}(w)$ for the autonomous polynomial system consisting of (9.8) and $T' = S$, $S' = -2ST$; the constants $\lambda^*, \tilde{\kappa}^*$ are carried as two further components with derivative 0, or equivalently as ball coefficients. The solution with $(T, S)(\xi_j) = (\tanh \xi_j, \operatorname{sech}^2 \xi_j)$ has $(T, S)(\xi) = (\tanh \xi, \operatorname{sech}^2 \xi)$ for all ξ (uniqueness), so the u-component is the solution of (9.8). Let $\Psi_t^j(y)$ denote the value at $\xi_j + t$ of the solution with $u(\xi_j) = y$. Following Lemma A.54 let $c_0(w) = w$, $c_{k+1}(w) = \frac{1}{k+1} \mathrm{D}c_k(w)\, \mathbb{G}(w)$ be the Taylor coefficient polynomials (computed by Picard sweeps in truncated power series arithmetic: if a truncated series agrees with the solution up to order m, one sweep $y \mapsto y_0 + \int \mathbb{G}(y)$ makes it agree up to order $m + 1$, because $\mathbb{G}$ is polynomial).

THEOREM D.8 (one validated step). *Let ξ_j be a node, $\eta \neq 0$ a step and $N \geq 1$ (in the program ξ_j and $\xi_j + \eta$ are multiples of $\frac{1}{64}$, and $N = 24$). Let $\mathbf{Y} \subset \mathbb{R}^5$ be a box with midpoint $\hat{y}$ (a binary number), let $\mathbf{T}_j \ni \tanh \xi_j$ and $\mathbf{S}_j \ni \operatorname{sech}^2 \xi_j$ be balls, and let $\mathbf{A}$ be an interval 5×4 matrix. Consider the augmented system consisting of*

- *$w = (u, T, S)$ started from $(y, \mathbf{T}_j, \mathbf{S}_j)$ with $y \in \mathbf{Y}$ (the box trajectories),*
- *w started from $(\hat{y}, \mathbf{T}_j, \mathbf{S}_j)$ (the midpoint trajectory),*
- *the 5×5 variational equation $M' = \mathcal{A}(w)M$, $M(0) = I$, along the box trajectories,*

and let $\mathbf{W}_0$ be its set of initial values. Compute in ball arithmetic, with $\mathbf{t} = \operatorname{hull}\{0, \eta\}$,

$$\mathbf{P} = \sum_{k \leq N} c_k(\mathbf{W}_0)\, \mathbf{t}^k, \qquad \mathbf{B}^0 \supset \mathbf{P} \text{ a box}, \qquad \mathbf{B}^1 = \mathbf{P} + c_{N+1}(\mathbf{B}^0)\, \mathbf{t}^{N+1},$$

and suppose $\mathbf{B}^1 \subset \operatorname{int} \mathbf{B}^0$ componentwise. Let $\hat{y}_{\mathrm{new}}$ and $\mathbf{J}$ be the u-component of the midpoint trajectory and the M-component of the enclosure $\sum_{k \leq N} c_k(\mathbf{W}_0)\eta^k + c_{N+1}(\mathbf{B}^0)\eta^{N+1}$ of the end value, and put

$$\mathbf{Y}_{\mathrm{new}} = \hat{y}_{\mathrm{new}} + \mathbf{J}(\mathbf{Y} - \hat{y}), \qquad \mathbf{A}_{\mathrm{new}} = \mathbf{J}\mathbf{A}.$$

Then:

1. *for every $y \in \mathbf{Y}$ the solution of (9.8) with $u(\xi_j) = y$ exists on the interval between ξ_j and $\xi_j + \eta$;*
2. *Ψ_η^j is C^∞ near $\mathbf{Y}$, $\Psi_\eta^j(\mathbf{Y}) \subset \mathbf{Y}_{\mathrm{new}}$, and $\mathrm{D}\Psi_\eta^j(y) \in \mathbf{J}$ for all $y \in \mathbf{Y}$;*
3. *if $\mathbf{X} \subset \mathbb{R}^4$ is a parameter box such that for every $s \in \mathbf{X}$ the solution $u(\cdot; s)$ exists from 0 to ξ_j, with $u(\xi_j; s) \in \mathbf{Y}$ and $\partial_s u(\xi_j; s) \in \mathbf{A}$, then it exists up to $\xi_j + \eta$, and $u(\xi_j + \eta; s) \in \mathbf{Y}_{\mathrm{new}}$, $\partial_s u(\xi_j + \eta; s) \in \mathbf{A}_{\mathrm{new}}$.*

PROOF. *Step 1: the augmented enclosure.* Take $y \in \mathbf{Y}$ and let w^{box}, w^{pt}, M be the solutions of the augmented system with the initial value $(y, \hat{y}, \tanh \xi_j, \operatorname{sech}^2 \xi_j, I) \in \mathbf{W}_0$.

The augmented system is polynomial, so Lemma A.54 (for $\eta < 0$ applied to the time-reversed system $\mathbb{G} \to -\mathbb{G}$, which is also polynomial) applies to it with the initial set $\mathbf{W}_0$ and the boxes $\mathbf{P}, \mathbf{B}^0$: by the hypothesis $\mathbf{B}^1 \subset \operatorname{int} \mathbf{B}^0$ the solution exists on $[0, \eta]$, stays in $\mathbf{B}^0$, and its end value lies in the enclosure $\sum_k c_k(\mathbf{W}_0)\eta^k + c_{N+1}(\mathbf{B}^0)\eta^{N+1}$. The three components of the augmented system decouple as follows: w^{box} is the solution through y, so $u^{\mathrm{box}} = \Psi_t^j(y)$; w^{pt} is the solution through $\hat{y}$; and $M(t) = \mathrm{D}\Psi_t^j(y)$, because M solves the variational equation along the trajectory through y with identity initial value (Lemma D.6(b), Theorem A.4(c)). This proves (1), $\mathrm{D}\Psi_\eta^j(y) \in \mathbf{J}$, and that the true value $\Psi_\eta^j(\hat{y})$ lies

in the computed enclosure $\hat{y}_{\mathrm{new}}$. The smoothness is Theorem A.4(c) (the existence on $[0,\eta]$ is an open condition in y).

Step 2: mean value form. $\mathbf{Y}$ is convex and Ψ_η^j is C^1 on a neighbourhood of it, so Lemma A.53 gives $\Psi_\eta^j(\mathbf{Y}) \subset \Psi_\eta^j(\hat{y}) + \mathbf{J}(\mathbf{Y}-\hat{y}) \subset \hat{y}_{\mathrm{new}} + \mathbf{J}(\mathbf{Y}-\hat{y})$, using $\Psi_\eta^j(\hat{y}) \in \hat{y}_{\mathrm{new}}$ from Step 1. This is (2).

Step 3: the parameter box. By (1) applied to $y = u(\xi_j;s) \in \mathbf{Y}$, the solution exists up to $\xi_j+\eta$, and $u(\xi_j+\eta;s) = \Psi_\eta^j(u(\xi_j;s)) \in \mathbf{Y}_{\mathrm{new}}$. By the chain rule $\partial_s u(\xi_j+\eta;s) = \mathrm{D}\Psi_\eta^j(u(\xi_j;s))\,\partial_s u(\xi_j;s)$, and the first factor lies in $\mathbf{J}$ by (2) and the second in $\mathbf{A}$; Lemma D.3(c) gives $\partial_s u(\xi_j+\eta;s) \in \mathbf{JA}$. □

How the theorem is used. Starting from $\mathbf{Y}_0 = u(0;\mathbf{X})$, i.e. $\mathbf{Y}_0 = \mathbf{X}$ placed in the components (h,f,Q,P) and $E=1$, and $\mathbf{A}_0 = \partial_s u(0;s)$ (the four unit columns of Lemma D.6(b)), the program takes dyadic steps $\eta = \pm\frac{1}{m}$, $m \in \{2,4,\dots,64\}$, halving the step whenever $\mathbf{B}^1 \subset \operatorname{int}\mathbf{B}^0$ fails and doubling it after three successes; it also refuses a step whose remainder term exceeds $2^{-80}\max(1,|u|)$, which only serves accuracy and plays no role in the proof. By induction on j, Theorem D.8(3) shows: *if the program reaches $\xi = 26$ (respectively -12), then for every $s \in \mathbf{X}$ the solution exists on $[0,26]$ (resp. $[-12,0]$), and the final boxes $\mathbf{Y},\mathbf{A}$ contain the state and its s-derivative.* The numbers of steps for the three boxes of Appendix D.6 are $162, 158, 151$ to the right and $93, 96, 96$ to the left. The values $\tanh\xi_j$, $\operatorname{sech}^2\xi_j$ at the dyadic nodes are computed by Arb with a certified error bound (Lemma D.3(a)).

D.5. The poles: enclosing the infinite tails

In words. *Beyond the cut X the solution is not computed, only enclosed. The idea is simple: Q solves a linear equation whose solution operator, the Cauchy kernel, is positive and explicit, so that from the values of Q, Q' at the cut and a crude bound on f^2 one knows Q on the whole tail up to an error of size e^{-2X}. Hence $W = 1+\frac{1}{2}Q$ is squeezed between two constants $a^2 \le W \le b^2$, and a Riccati equation with a potential squeezed between constants is squeezed between the Riccati equations with these constants, which are solved in closed form. Everything else follows. The condition for all this to work is a short list of inequalities on the state at the cut, (H1)–(H3).*

Toy example D.9 (the tail at the cut $X=8$, solution 1)**.** Take the first solution of Theorem 9.68, integrated in double precision to the cut $X=8$ (`verify_appD.py`, part 3): $h = 0.281524$, $E = 1.6385\cdot10^{-2}$, $f = 19.015066$, $Q = -1.874960$, $P = -7.94\cdot10^{-5}$. Then $\varepsilon = e^{-16} = 1.1254\cdot10^{-7}$, $c\varepsilon = \cosh^2(8)\,e^{-16} = 0.250000$, so that

$$Q^{\mathrm{tr}} = Q(1+\varepsilon) + c\varepsilon P + \tfrac{\varepsilon}{2}(\lambda^* - f^2) = -1.874999, \qquad a_0 = \sqrt{1+\tfrac{1}{2}Q^{\mathrm{tr}}} = 0.250000.$$

With $F_{\max} = f+1$ one has $G = |\lambda^*| + F_{\max}^2 = 412.9$ and $D = Q[0,\varepsilon] + cP[0,\varepsilon] + \varepsilon G[-1,1] = [-6.65\cdot10^{-5},\, 4.65\cdot10^{-5}]$, so that $a^2 = a_0^2 - \frac{1}{2}\sup D$, $b^2 = a_0^2 - \frac{1}{2}\inf D$ give the squeezing constants $a = 0.249954$, $b = 0.250067$. The Riccati variable satisfies $h \ge \hat{h}_a$, with $h_X + a = 0.531478 > 0$, hence

$$f(\infty) - f(8) \in \tilde{\kappa}^* E\Big[\frac{1}{b+h}, \frac{1}{a+h}\Big] \quad\text{and}\quad f(\infty) - f^{\mathrm{tr}} \in [-3.2\cdot10^{-6},\, +2.2\cdot10^{-6}]$$

with $f^{\mathrm{tr}} = f + \tilde{\kappa}^* E/(h+a_0) = 19.040739043$. The true value, from the integration to $\xi = 70$, is $f(\infty) = 19.040738660$, i.e. $f(\infty) - f^{\mathrm{tr}} = -3.8\cdot10^{-7}$, inside the enclosure.

At the cut $X = 26$ used in the certificate the width is 10^{-18} smaller (Numerical observation D.12).

Proposition D.10 (tails)**.** *Let $\lambda \in \mathbb{R}$, $\tilde\kappa > 0$ (in the certificate $\lambda^*, \tilde\kappa^*$), $X > 0$ a cut, $\varepsilon = e^{-2X}$, $c = \cosh^2 X$, $x_a = \tanh X$, and let $u = (h, E, f, Q, P)$ solve (9.8) on $[0, X]$. Write $(h_R, E_R, f_R, Q_R, P_R) = u(X)$.*

(A) Right pole. *Assume $f_R \geq 0$ and fix a constant $F_{\max} > f_R$; put $G = |\lambda| + F_{\max}^2$,*

$$Q^{\mathrm{tr}} = Q_R(1+\varepsilon) + c\varepsilon P_R + \tfrac{\varepsilon}{2}(\lambda - f_R^2), \qquad D = Q_R[0,\varepsilon] + cP_R[0,\varepsilon] + \varepsilon G[-1,1],$$

$a_0 = \sqrt{1 + \frac{1}{2}Q^{\mathrm{tr}}}$ (when $1 + \frac{1}{2}Q^{\mathrm{tr}} > 0$), $a^2 = 1 + \frac{1}{2}(Q^{\mathrm{tr}} - \sup D)$, $b^2 = 1 + \frac{1}{2}(Q^{\mathrm{tr}} - \inf D)$, and assume

$$\text{(H1) } a^2 > 0, \qquad \text{(H2) } h_R + a > 0, \qquad \text{(H3) } f_R + \frac{\tilde\kappa E_R}{a + h_R} < F_{\max}.$$

Then the solution extends to $[X, \infty)$ with $E > 0$, and, with $\tau = \tilde\kappa E_R/(a + h_R)$:

(A1) *$Q(x) \in Q^{\mathrm{tr}} - D$ for $x \in [x_a, 1)$ ($x = \tanh\xi$), hence $a^2 \leq W := 1 + \frac{1}{2}Q \leq b^2$ on $[X, \infty)$; the limit $Q(1) := \lim_{x\to 1} Q$ exists and lies in $Q^{\mathrm{tr}} - D$;*

(A2) *f is increasing and bounded, $f(\infty) - f_R \in \tilde\kappa E_R\big[\frac{1}{b+h_R}, \frac{1}{a+h_R}\big]$;*

(A3) *$h \to \mu := \sqrt{1 + \frac{1}{2}Q(1)} \geq a > 0$ as $\xi \to \infty$, so $Q(1) > -2$ and the pole is of type G; and $Q(1) - Q^{\mathrm{tr}} \in -\frac{\varepsilon}{2}\,[0,\ \tau(2f_R + \tau)]$.*

(B) Left pole. *Let u solve (9.8) on $[-X, 0]$, write $(h_L, E_L, f_L, Q_L, P_L) = u(-X)$ and fix constants $F_{\mathrm{lo}} < f_L < F_{\mathrm{hi}}$; put $G = |\lambda| + \max(F_{\mathrm{lo}}^2, F_{\mathrm{hi}}^2)$,*

$$Q_-^{\mathrm{tr}} = Q_L(1+\varepsilon) - c\varepsilon P_L + \tfrac{\varepsilon}{2}(\lambda - f_L^2), \qquad D_L = Q_L[0,\varepsilon] - cP_L[0,\varepsilon] + \varepsilon G[-1,1],$$

and a_0, a, b defined from Q_-^{tr} and D_L as in (A)*. Assume* (H1) *$a^2 > 0$,* (H2) *$a - h_L > 0$,* (H3) *$f_L - \tau_- > F_{\mathrm{lo}}$, $\tau_- := \tilde\kappa E_L/(a - h_L)$. Then the solution extends to $(-\infty, -X]$ with $E > 0$ and:* (B1) *$Q(x) \in Q_-^{\mathrm{tr}} - D_L$ for $x \in (-1, -x_a]$, so $a^2 \leq W \leq b^2$, and $Q(-1) := \lim_{x\to -1} Q$ exists and lies in $Q_-^{\mathrm{tr}} - D_L$;* (B2) *f is increasing in ξ and bounded below, $f_L - f(-\infty) \in \tilde\kappa E_L\big[\frac{1}{b-h_L}, \frac{1}{a-h_L}\big]$;* (B3) *$h \to -\mu_L$ with $\mu_L := \sqrt{1 + \frac{1}{2}Q(-1)} \geq a > 0$, so the pole at $x = -1$ is of type G, and $Q(-1) - Q_-^{\mathrm{tr}} \in -\frac{\varepsilon}{2}\Delta$, $\Delta := \{\varphi^2 - f_L^2 : \varphi \in [f_L - \tau_-, f_L]\}$ (if $f_L - \tau_- \geq 0$ this is $+\frac{\varepsilon}{2}[0,\ f_L^2 - (f_L - \tau_-)^2]$).*

(C) Enclosures. (i) (*any admissible constants*) *Let $[d_*, d^*] \supset D$ be any interval and put $\alpha^2 := a_0^2 - \frac{1}{2}d^*$, $\beta^2 := a_0^2 - \frac{1}{2}d_*$ and, for $\gamma > 0$ and $d \in \mathbb{R}$,*

$$\Lambda_R(\gamma, d) := \frac{d/2}{(a_0 + \gamma)(\gamma + h_R)(a_0 + h_R)}.$$

Then, if $\alpha^2 > 0$, $0 < \alpha \leq a \leq b \leq \beta$, and if moreover $h_R + \alpha > 0$,

$$f(\infty) - f_R^{\mathrm{tr}} \in \tilde\kappa E_R\,\big[\Lambda_R(\beta, d_*),\ \Lambda_R(\alpha, d^*)\big], \qquad f_R^{\mathrm{tr}} := f_R + \frac{\tilde\kappa E_R}{h_R + a_0}.$$

At the left pole, with $D_L \subset [d_*, d^*]$, α, β defined in the same way and $\alpha - h_L > 0$, put $\Lambda_L(\gamma, d) := \frac{d/2}{(a_0+\gamma)(\gamma - h_L)(a_0 - h_L)}$; then

$$f(-\infty) - f_L^{\mathrm{tr}} \in -\tilde\kappa E_L\,\big[\Lambda_L(\beta, d_*),\ \Lambda_L(\alpha, d^*)\big], \qquad f_L^{\mathrm{tr}} := f_L - \frac{\tilde\kappa E_L}{a_0 - h_L}.$$

(ii) (*families*) *Let $B \subset \mathbb{R}^4$ be compact, $u(\cdot; s)$ exist on $[-X_L, X_R]$ for all $s \in B$, and let* (A), (B) *hold for all $s \in B$ with constants $F_{\max}, F_{\mathrm{lo}}, F_{\mathrm{hi}}$ independent of s.*

Then $B \subset \mathcal{S}$, the maps $s \mapsto f(\pm\infty; s)$, $Q(\pm 1; s)$ are continuous on B, hence Φ is continuous on B, and $\Phi = \Phi_0 + \mathsf{Z}$ with $\mathsf{Z}(s) \in \mathbf{T}$ for the box $\mathbf{T}$ whose components are the enclosures (A3), (B3), (C)(i) *combined as in the definition of Φ: $\mathsf{Z}_1 \in -\frac{\varepsilon_L}{2}\Delta$, $\mathsf{Z}_2 \in -\frac{\varepsilon_R}{2}[0, \tau(2f_R + \tau)]$, $\mathsf{Z}_4 = f(-\infty) - f_L^{\mathrm{tr}}$ and $\mathsf{Z}_3 = (f(+\infty) - f_R^{\mathrm{tr}}) + 40\,\mathsf{Z}_4$.*

Proof. *Step 0: reduction of the left pole to the right pole.* If $u = (h, E, f, Q, P)$ solves (9.8) on $[-X', -X]$, then $\tilde{u}(\eta) = (-h, E, f, Q, -P)(-\eta)$ solves, on $[X, X']$, the system (9.8) with the equation $f' = \tilde{\kappa}E$ replaced by $f' = -\tilde{\kappa}E$: indeed $\tilde{h}'(\eta) = h'(-\eta) = W(-\eta) - \tilde{h}^2$, $\tilde{E}'(\eta) = -E'(-\eta) = 2h(-\eta)E = -2\tilde{h}\tilde{E}$, $\tilde{f}'(\eta) = -f'(-\eta) = -\tilde{\kappa}\tilde{E}$, $\tilde{Q}' = -Q'(-\eta) = \tilde{P}$ and, because T is odd and S is even, $\tilde{P}'(\eta) = P'(-\eta) = S(\lambda - \tilde{f}^2 + 2\tilde{Q}) - 4T(-\eta)P(-\eta) = S(\lambda - \tilde{f}^2 + 2\tilde{Q}) - 4T(\eta)\tilde{P}(\eta)$ (checked by computer algebra in `verify_appD.py`, part 1). The tilde state at $\eta = X$ is $(-h_L, E_L, f_L, Q_L, -P_L)$; the quantities Q^{tr}, D, a, b computed from it by the formulas of (A) are exactly $Q_-^{\mathrm{tr}}, D_L, a, b$, and (H2) of (A) is $a - h_L > 0$. So (B) is (A) for the system with $\varsigma = -1$ in $f' = \varsigma\tilde{\kappa}E$, with the roles: $F_{\mathrm{lo}}, F_{\mathrm{hi}}$ for the range of f, $h \to \tilde{h}$, $\mu = \mu_L$. We therefore prove (A) for $\varsigma \in \{+1, -1\}$ simultaneously; for $\varsigma = +1$, f stays in $[f_R, F_{\max})$ and for $\varsigma = -1$ in $(F_{\mathrm{lo}}, f_R]$, and in both cases $|\lambda - f^2| \le G$ (for $\varsigma = +1$ because $0 \le f_R \le f < F_{\max}$).

Step 1: the Cauchy formula at the cut. In the variable $x = \tanh\xi$ the equation for Q is $\mathsf{L}[Q] = g$, $g := \lambda - f^2$ (Lemma D.6(a)), and $Q_x = \cosh^2\xi\, Q'$, so $Q_x(x_a) = cP_R$. Lemma 2.18(d) gives, for $x \ge x_a$ as long as the solution exists,

$$\begin{aligned} Q(x) &= Q_R A_1(x) + cP_R A_2(x) + \int_{x_a}^{x} K_{\mathsf{L}}(x, s) g(s)\,\mathrm{d}s, \\ A_1 &= 1 + \tfrac{(x - x_a)^2}{1 - x_a^2}, \qquad A_2 = \tfrac{(x - x_a)(1 - x_a x)}{1 - x_a^2}. \end{aligned} \tag{D.1}$$

For $x_a \le x \le 1$: A_1 is increasing with $A_1(1) = 1 + \frac{1 - x_a}{1 + x_a} = 1 + \varepsilon$; $A_2' = \frac{1 + x_a^2 - 2x_a x}{1 - x_a^2} \ge \frac{(1 - x_a)^2}{1 - x_a^2} > 0$ since $1 + x_a^2 - 2x_a x - (1 - x_a)^2 = 2x_a(1 - x) \ge 0$, so A_2 is increasing from 0 to $A_2(1) = \frac{(1 - x_a)^2}{1 - x_a^2} = \varepsilon$; and $K_{\mathsf{L}} \ge 0$ with $\int_{x_a}^{x} K_{\mathsf{L}}(x, s)\mathrm{d}s = \frac{(x - x_a)^2}{2(1 - x_a^2)} \le \frac{\varepsilon}{2}$. Moreover, for $x_a \le \sigma \le x$ a direct computation gives

$$\begin{aligned} K_{\mathsf{L}}(1, \sigma) - K_{\mathsf{L}}(x, \sigma) &= \frac{(1 - x)\,[(1 - \sigma)^2 + \sigma(1 - x)]}{(1 - \sigma^2)^2} \ \ge 0, \\ K_{\mathsf{L}}(1, \sigma) &= (1 + \sigma)^{-2}, \qquad \int_{x_a}^{1} (1 + \sigma)^{-2}\mathrm{d}\sigma = \frac{\varepsilon}{2}. \end{aligned} \tag{D.2}$$

Step 2: the bootstrap. Let $\xi_1 \in (X, \infty]$ be the supremum of the ξ' such that the solution exists on $[X, \xi')$ and f stays in the open range fixed in Step 0 ($f < F_{\max}$, resp. $f > F_{\mathrm{lo}}$); $\xi_1 > X$ by (H3) and continuity. On $[X, \xi_1)$ we have $|g| \le G$ and $|g_R| \le G$, $g_R := \lambda - f_R^2$. Subtract (D.1) from $Q^{\mathrm{tr}} = Q_R(1 + \varepsilon) + c\varepsilon P_R + \frac{\varepsilon}{2}g_R$:

$$Q^{\mathrm{tr}} - Q(x) = Q_R\,(1 + \varepsilon - A_1(x)) + cP_R\,(\varepsilon - A_2(x)) + \Big[\tfrac{\varepsilon}{2}g_R - \int_{x_a}^{x} K_{\mathsf{L}} g\,\mathrm{d}s\Big].$$

The factors $1 + \varepsilon - A_1$ and $\varepsilon - A_2$ lie in $[0, \varepsilon]$ and the bracket in $\varepsilon G[-1, 1]$ (as $\frac{\varepsilon}{2}|g_R| + \int K_{\mathsf{L}}|g| \le \frac{\varepsilon}{2}G + \frac{\varepsilon}{2}G$). So $Q^{\mathrm{tr}} - Q(x) \in D$, hence $Q(x) \in Q^{\mathrm{tr}} - D$ and $a^2 \le W \le b^2$ on $[X, \xi_1)$ by the definitions of a, b; by (H1), $a > 0$. *Step 3: Riccati comparison.* Let $0 < \alpha \le \beta$ and suppose $\alpha^2 \le W \le \beta^2$ on $[X, \xi_1)$ and $\tilde{h}_R > -\alpha$ (the value of the Riccati variable at the cut: h_R for the right pole; $-h_L$ for the left pole).

Put $t=\xi-X$ and $\hat z_\gamma(t)=\cosh\gamma t+\frac{\tilde h_R}{\gamma}\sinh\gamma t=\frac12(1+\beta_\gamma)e^{\gamma t}+\frac12(1-\beta_\gamma)e^{-\gamma t}$, $\beta_\gamma=\tilde h_R/\gamma$, $\hat h_\gamma=\hat z_\gamma'/\hat z_\gamma$, for $\gamma\in\{\alpha,\beta\}$. Then $1+\beta_\gamma>0$ and

$$\begin{aligned}\hat z_\gamma(t)&=\tfrac12 e^{-\gamma t}\big[(1+\beta_\gamma)e^{2\gamma t}+(1-\beta_\gamma)\big]\\&\ge\ \tfrac12 e^{-\gamma t}\big[(1+\beta_\gamma)+(1-\beta_\gamma)\big]=e^{-\gamma t}>0\qquad(t\ge0),\end{aligned}$$

so $\hat h_\gamma$ is defined on $[0,\infty)$, solves $\hat h_\gamma'=\gamma^2-\hat h_\gamma^2$, $\hat h_\gamma(0)=\tilde h_R$. For $d=h-\hat h_\alpha$ ($h=\tilde h$ the Riccati variable of the tilde system): $d'=(W-\alpha^2)-(h+\hat h_\alpha)d\ge-(h+\hat h_\alpha)d$, so $\big(d\,e^{\int_X^\xi(h+\hat h_\alpha)}\big)'\ge0$, $d(X)=0$, and $h\ge\hat h_\alpha$. In the same way $\hat h_\beta-h$ has derivative $(\beta^2-W)-(h+\hat h_\beta)(\hat h_\beta-h)$, whence $h\le\hat h_\beta$. Since $z=E^{-1/2}=z_R\exp\int_X^\xi h$ and $\hat z_\gamma=\exp\int\hat h_\gamma$, we obtain $z_R\hat z_\alpha\le z\le z_R\hat z_\beta$ and

$$E_R\,\hat z_\beta(t)^{-2}\ \le\ E(\xi)\ \le\ E_R\,\hat z_\alpha(t)^{-2},\qquad t=\xi-X,\tag{D.3}$$

so in particular z stays bounded away from 0 and h finite on bounded intervals. Finally, for $\gamma>0$ and $\beta_\gamma>-1$,

$$\begin{aligned}&\frac{\mathrm d}{\mathrm dt}\frac{e^{-\gamma t}}{\hat z_\gamma(t)}=-\gamma\,\frac{e^{-\gamma t}(\hat z_\gamma+\hat z_\gamma'/\gamma)}{\hat z_\gamma^2}=-\frac{\gamma(1+\beta_\gamma)}{\hat z_\gamma^2},\\&\text{so}\qquad\int_0^\infty\hat z_\gamma^{-2}\,\mathrm dt=\frac{1}{\gamma(1+\beta_\gamma)}=\frac{1}{\gamma+\tilde h_R},\end{aligned}\tag{D.4}$$

because $\hat z_\gamma+\hat z_\gamma'/\gamma=(1+\beta_\gamma)e^{\gamma t}$ and $e^{-\gamma t}/\hat z_\gamma\to0$ ($\hat z_\gamma\ge\frac12(1+\beta_\gamma)e^{\gamma t}-\frac12|1-\beta_\gamma|$) and equals 1 at $t=0$. (A computer-algebra check of (D.2), (D.4): `verify_appD.py`, part 1.) *Step 4: closing the bootstrap; (A1)–(A2).* Apply Step 3 with $\alpha=a$, $\beta=b$ ($\tilde h_R+a>0$ is (H2)). Then by (D.3), (D.4), for $\xi\in[X,\xi_1)$, $|f(\xi)-f_R|=\tilde\kappa\int_X^\xi E\le\tilde\kappa E_R\int_0^\infty\hat z_a^{-2}=\tau$, so f stays between f_R and $f_R+\varsigma\tau$, which by (H3) ($f_R+\tau<F_{\max}$, resp. $f_R-\tau>F_{\mathrm{lo}}$) is strictly inside the range of Step 2. If $\xi_1<\infty$ the state (h,E,f,Q,P) would stay bounded on $[X,\xi_1)$: h and E by (D.3) and the comparison, f by the above, Q by Step 2, and P because $P'=S(\lambda-f^2+2Q)-4TP$ is a linear equation with bounded coefficients on a bounded interval; by Theorem A.4(b) the solution would extend beyond ξ_1 with f strictly inside its range, contradicting the maximality of ξ_1. So $\xi_1=\infty$, which proves the extension, $E>0$, and the first half of (A1). The monotone function f is bounded and has a limit, with $|f(\infty)-f_R|=\tilde\kappa\int_X^\infty E\in\tilde\kappa E_R\big[\int\hat z_b^{-2},\int\hat z_a^{-2}\big]$, which is (A2) by (D.4). *Step 5: the limit of Q; (A1), (A3).* Because $f\to f(\infty)$, g is a bounded continuous function on $[x_a,1)$ with a limit at 1. For $\sigma\le x<1$, (D.2) gives $0\le K_{\mathsf L}(x,\sigma)\le K_{\mathsf L}(1,\sigma)=(1+\sigma)^{-2}$, an integrable bound, and $K_{\mathsf L}(x,\sigma)\mathbb{1}_{\sigma\le x}\to K_{\mathsf L}(1,\sigma)$ as $x\to1$. By dominated convergence (D.1) has the limit

$$\begin{aligned}Q(1)&=Q_R(1+\varepsilon)+c\varepsilon P_R+\int_{x_a}^1(1+\sigma)^{-2}g(\sigma)\,\mathrm d\sigma,\\\text{so}\qquad Q(1)-Q^{\mathrm{tr}}&=\int_{x_a}^1(1+\sigma)^{-2}\,(g(\sigma)-g_R)\,\mathrm d\sigma,\end{aligned}$$

using $\int_{x_a}^1(1+\sigma)^{-2}\mathrm d\sigma=\frac{\varepsilon}{2}$. Since $Q(x)\in Q^{\mathrm{tr}}-D$ for $x<1$ and D is a closed interval, $Q(1)\in Q^{\mathrm{tr}}-D$. Now $g-g_R=-(f^2-f_R^2)$ with f between f_R and $f_R+\varsigma\tau$; for $\varsigma=+1$ and $f_R\ge0$ this lies in $-[0,\tau(2f_R+\tau)]$, for $\varsigma=-1$ in $-\Delta$ with Δ the hull of $\{\varphi^2-f_R^2\}$, $\varphi\in[f_R-\tau,f_R]$. The weight $(1+\sigma)^{-2}$ is positive with total mass $\frac{\varepsilon}{2}$, so $Q(1)-Q^{\mathrm{tr}}\in-\frac{\varepsilon}{2}[0,\tau(2f_R+\tau)]$, resp. $-\frac{\varepsilon}{2}\Delta$. *Step 6: the pole type; (A3), (B3).* By Step 3 on

$[X,\infty)$, $h \ge \hat h_a \to a$ (explicitly $\hat h_a(t) = a\frac{\sinh at+\beta_a\cosh at}{\cosh at+\beta_a\sinh at} \to a$), so $\liminf h \ge a > 0$. By Step 5, $W \to \mu^2 := 1 + \frac12 Q(1) \ge a^2 > 0$, so $Q(1) > -2$. Given $\delta \in (0, \mu^2/2)$ choose $X' \ge X$ such that $|W - \mu^2| \le \delta$ on $[X',\infty)$ and $h(X') > -\sqrt{\mu^2-\delta}$ (possible since $\liminf h > 0$). Step 3 on $[X',\infty)$ with $\alpha = \sqrt{\mu^2-\delta}$, $\beta = \sqrt{\mu^2+\delta}$ gives $\hat h_\alpha \le h \le \hat h_\beta$ there, so $\sqrt{\mu^2-\delta} \le \liminf h \le \limsup h \le \sqrt{\mu^2+\delta}$; let $\delta \to 0$. Thus $h \to \mu$, i.e. type G (Definition 4.16). For $\varsigma = -1$ the tilde variable is $\tilde h = -h$, so $h \to -\mu_L$ at the left pole; (B1)–(B3) are (A1)–(A3) for the tilde state, and Q_-^{tr}, D_L are as computed in Step 0. *Step 7: (C)(i).* Since $D \subset [d_*, d^*]$: $\sup D \le d^*$ and $\inf D \ge d_*$, so $a^2 = a_0^2 - \frac12 \sup D \ge \alpha^2$ and $b^2 \le \beta^2$. Also $0 \in D$, so $d_* \le 0 \le d^*$ and $\alpha \le a_0 \le \beta$. Then $\alpha^2 \le a^2 \le W \le b^2 \le \beta^2$, and Step 3 (needs $\tilde h_R + \alpha > 0$) gives $f(\infty) - f_R \in \tilde\kappa E_R[\frac{1}{\beta+\tilde h_R}, \frac{1}{\alpha+\tilde h_R}]$ for $\varsigma = +1$. Subtract $\tilde\kappa E_R/(a_0+h_R)$ and use $\frac{1}{\gamma+h} - \frac{1}{a_0+h} = \frac{a_0-\gamma}{(\gamma+h)(a_0+h)} = \frac{a_0^2-\gamma^2}{(a_0+\gamma)(\gamma+h)(a_0+h)}$ with $a_0^2 - \alpha^2 = \frac12 d^*$, $a_0^2 - \beta^2 = \frac12 d_*$. The left pole is the same with $\varsigma = -1$ and $\tilde h_R = -h_L$: $f_L - f(-\infty) \in \tilde\kappa E_L[\frac{1}{\beta-h_L}, \frac{1}{\alpha-h_L}]$, whence the stated enclosure of $f(-\infty) - f_L^{\mathrm{tr}}$. *Step 8: (C)(ii). Membership in $\mathcal S$.* For $s \in B$ the solution exists on $[-X_L, X_R]$, extends to $\mathbb R$ with $E > 0$ by (A), (B), and is of type G at both poles by (A3), (B3): $s \in \mathcal S$. *Uniform constants.* The maps $s \mapsto u(\pm X; s)$ are continuous (Theorem A.4(c)) on the compact B, hence so are $a(s)$ (the sup and inf of D are continuous functions of the state), $h_R + a$ and E_R; by (H1), (H2) and compactness there are $a_{\min} > 0$, $c_{\min} > 0$, $E_{\max}$, M with $a \ge a_{\min}$, $h_R + a \ge c_{\min}$, $E_R \le E_{\max}$, $a \le a_{\max}$, $|Q_R|, c|P_R| \le M$ on B, and likewise at the left. *Uniform convergence of f.* For $\xi' \ge X_R$ and $s \in B$, by (D.3), $0 \le f(\infty; s) - f(\xi'; s) \le \tilde\kappa E_{\max}\int_{\xi'-X_R}^\infty \hat z_a^{-2}\mathrm{d}t$, and $\hat z_a e^{-at} = \frac12(1+\beta_a) + \frac12(1-\beta_a)e^{-2at} \ge \min\{1, \frac12(1+\beta_a)\} \ge m := \min\{1, \frac{c_{\min}}{2a_{\max}}\}$, so the right side is $\le \tilde\kappa E_{\max}\frac{e^{-2a_{\min}(\xi'-X_R)}}{2a_{\min}m^2}$, which tends to 0 with ξ' uniformly in s. *Uniform convergence of Q.* From (D.1) and Step 5, for $x \in [x_a, 1)$,

$$|Q(1;s) - Q(x;s)| \le M\big(A_1(1) - A_1(x)\big) + M\big(A_2(1) - A_2(x)\big) + G\,\rho(x),$$
$$\rho(x) = \int_{x_a}^1 \big[K_{\mathsf L}(1,\sigma) - K_{\mathsf L}(x,\sigma)\mathbb{1}_{\sigma\le x}\big]\mathrm{d}\sigma = \frac{(1-x)(1+x-2x_a)}{2(1-x_a^2)},$$

by (D.2) (the two integrals are $\frac{\varepsilon}{2}$ and $\frac{(x-x_a)^2}{2(1-x_a^2)}$); the right side does not depend on s and tends to 0 as $x \to 1$. The left pole is the mirror image. A uniform limit of continuous functions is continuous, so $f(\pm\infty;\cdot)$ and $Q(\pm1;\cdot)$ are continuous on B (the functions $f(\xi';\cdot)$ and $Q(x;\cdot)$ are continuous for fixed ξ', x by Theorem A.4(c)), hence Φ is. Finally $\Phi - \Phi_0 = \mathsf Z$ has the components stated: $\Phi_1 - \Phi_{0,1} = Q(-1) - Q_-^{\mathrm{tr}}$, $\Phi_2 - \Phi_{0,2} = Q(1) - Q_+^{\mathrm{tr}}$, $\Phi_4 - \Phi_{0,4} = f(-\infty) - f_-^{\mathrm{tr}}$ and $\Phi_3 - \Phi_{0,3} = [f(\infty) - f_+^{\mathrm{tr}}] + 40[f(-\infty) - f_-^{\mathrm{tr}}]$, each enclosed by (A3), (B3), (C)(i). □

D.6. The certificate

In words. *Three approximate solutions of $\Phi(s) = 0$ were found by Newton's method. For each, the program builds the box $\mathbf X_k = \bar s_k + r[-1,1]^4$, $r = 10^{-19}$, encloses the flow on $[-12, 26]$ (Theorem D.8), the tails (Proposition D.10), the value and the Jacobian of Φ_0 (Lemma D.3), and tests $\mathbf K \subset \operatorname{int}\mathbf X$ (Lemma D.5). All three tests pass, with a margin of 4 to 5 orders of magnitude.*

Parameters. Precision 160 bits (about 48 decimal digits); Taylor degree $N = 24$; cuts $X_R = 26$, $X_L = 12$; box radius $r = 10^{-19}$ (the binary64 number nearest to 10^{-19}, which differs from it by a relative $3 \cdot 10^{-17}$; the box $\bar{s}_k + 10^{-19}[-1,1]^4$ of Theorem 9.68(b) contains the set $\mathbf{K}_k$ below, because the zero lies within $1.1 \cdot 10^{-23}$ of $\bar{s}_k$, see (b)); the constants $\lambda^*, \tilde{\kappa}^*$ are the balls $\frac{123}{10}$, $\frac{1400}{1681}$ computed at 160 bits; $F_{\max} = \sup f_R + 1$, $F_{\mathrm{hi}} = \sup f_L + 1$, $F_{\mathrm{lo}} = \inf f_L - 1$, with sup, inf over the box enclosure of the cut value. The centres $\bar{s}_k = (h_0, f_0, Q_0, Q_1)$ are the 160-bit numbers read from the decimal strings supplied with the book (60 digits, `data/sbar_k.json`); to 12 decimals:

k	h_0	f_0	Q_0	Q_1
1	−0.951474132355	0.464408260003	−1.213852381422	1.183825736761
2	−0.306353213704	0.819248654817	−2.340802175433	−1.187265002967
3	−0.034141977537	1.129720201281	−2.439732631127	−1.562887539432

Proposition D.11 (the certificate)**.** *For $k = 1, 2, 3$ and the box $\mathbf{X}_k = \bar{s}_k + r[-1,1]^4$ the program of Appendices D.4 and D.5 terminates successfully, and:*

(a) (*flow and hypotheses*) *for every $s \in \mathbf{X}_k$ the solution $u(\cdot; s)$ exists on $[-12, 26]$, $f_R > 0$, and* (H1)–(H3) *hold at both poles with the constants above; consequently $\mathbf{X}_k \subset \mathcal{S}$, Φ is continuous on $\mathbf{X}_k$, and $\Phi = \Phi_0 + \mathsf{Z}$ with $\mathsf{Z}(s) \in \mathbf{T}_k$, where $\mathbf{T}_k$ has components of modulus at most $(7.7 \cdot 10^{-42}, 1.5 \cdot 10^{-27}, 1.2 \cdot 10^{-24}, 2.7 \cdot 10^{-26})$ for $k = 1$, $(8.8 \cdot 10^{-41}, 2.0 \cdot 10^{-25}, 2.5 \cdot 10^{-23}, 9.1 \cdot 10^{-26})$ for $k = 2$ and $(3.2 \cdot 10^{-40}, 2.6 \cdot 10^{-24}, 3.7 \cdot 10^{-22}, 1.7 \cdot 10^{-25})$ for $k = 3$;*

(b) (*Krawczyk*) *with C the rounded midpoint inverse of $\mathrm{D}\Phi_0(\bar{s}_k)$ and $\mathbf{J}_k \supset \mathrm{D}\Phi_0(\mathbf{X}_k)$, $\mathbf{V}_k \ni \Phi_0(\bar{s}_k)$: $\mathbf{K}_k \subset \operatorname{int} \mathbf{X}_k$, with $Z_1 = 2.5 \cdot 10^{-14},\ 1.8 \cdot 10^{-14},\ 3.7 \cdot 10^{-15}$ and $Y/r = 7.6 \cdot 10^{-6},\ 2.1 \cdot 10^{-5},\ 1.1 \cdot 10^{-4}$; the radii of the components of $\mathbf{K}_k$ are at most $6.2 \cdot 10^{-6} r,\ 1.5 \cdot 10^{-5} r,\ 6.6 \cdot 10^{-5} r$; so Φ has a zero $s_k^* \in \mathbf{K}_k$, and $|s_k^* - \bar{s}_k| \le 1.1 \cdot 10^{-23}$ componentwise;*

(c) (*enclosures*) *for every $s \in \mathbf{X}_k$ (in particular at s_k^*) the jump $J(s) = f(+\infty; s) - f(-\infty; s)$ and $f(-\infty; s)$ lie in*

$$k = 1 : J \in 19.040738660117701 \pm 5.70 \cdot 10^{-16}, \qquad f(-\infty) \in \pm 1.36 \cdot 10^{-19},$$
$$k = 2 : J \in 33.58919484747842 \pm 1.94 \cdot 10^{-15}, \qquad f(-\infty) \in \pm 2.16 \cdot 10^{-19},$$
$$k = 3 : J \in 46.31852825252179 \pm 4.83 \cdot 10^{-15}, \qquad f(-\infty) \in \pm 3.22 \cdot 10^{-19};$$

(d) *the three boxes are pairwise disjoint: the first coordinates h_0 of the centres differ pairwise by more than 0.27, while $2r < 10^{-18}$.*

Proof. Everything in (a)–(d) is a numerical output of the program except the implications "output ⇒ statement", which we prove now; the outputs are those of Tables D.1 and D.2 and the record of Appendix D.6.1. *Step 1* (*flows*). The program integrates (9.8) with the data $s \in \mathbf{X}_k$ (the balls $\bar{s}_k + r[-1,1]$ in the four parameter components, $E = 1$) and the identity 4-frame $\mathbf{A}_0$, from $\xi = 0$ to 26 and to -12 by the steps of Theorem D.8, accepting a step only when $\mathbf{B}^1 \subset \operatorname{int} \mathbf{B}^0$ holds componentwise (the function `step` of `lohner.py` raises an exception otherwise). By Theorem D.8(3) and induction, for every $s \in \mathbf{X}_k$ the solution exists on $[-12, 26]$ and the final boxes $\mathbf{Y}_R, \mathbf{A}_R$ (at 26) and $\mathbf{Y}_L, \mathbf{A}_L$ (at -12) contain $u(\pm X; s)$ and $\partial_s u(\pm X; s)$. The same program run with the degenerate box $\{\bar{s}_k\}$ gives $\mathbf{V}$, $\mathbf{J}(\bar{s}_k)$, whose midpoint is inverted. *Step 2* (*hypotheses*). The module `tails.py` evaluates, in ball arithmetic and with $\mathbf{Y}_R$, the quantities of Proposition D.10(A): Q^{tr}, a_0^2,

k	$J=f(+\infty)-f(-\infty)$	$f(-\infty)$	$\max\operatorname{rad}\mathbf{K}_i/r$	Z_1	$\max Y/r$
1	$19.040738660117701\pm5.70\cdot10^{-16}$	$\pm1.36\cdot10^{-19}$	$6.2\cdot10^{-6}$	$2.5\cdot10^{-14}$	$7.6\cdot10^{-6}$
2	$33.58919484747842\pm1.94\cdot10^{-15}$	$\pm2.16\cdot10^{-19}$	$1.5\cdot10^{-5}$	$1.8\cdot10^{-14}$	$2.1\cdot10^{-5}$
3	$46.31852825252179\pm4.83\cdot10^{-15}$	$\pm3.22\cdot10^{-19}$	$6.6\cdot10^{-5}$	$3.7\cdot10^{-15}$	$1.1\cdot10^{-4}$

TABLE D.1. The certified enclosures over the boxes $\mathbf{X}_k$ (output of the certificate, `tab_cert.tex`) and the Krawczyk margins. The determinants of $\mathrm{D}\Phi_0(\bar{s}_k)$ are $-732.5372966\pm4.0\cdot10^{-8}$, $+1344.443346\pm1.2\cdot10^{-7}$, $-4863.013169\pm4.4\cdot10^{-7}$. Figure D.2(b).

the set D (with $G=|\lambda^*|+F_{\max}^2$) and its endpoints $d^*=\sup\mathbf{D}$, $d_*=\inf\mathbf{D}$ over the box, $\alpha^2=a_0^2-\frac12 d^*$ and $\beta^2=a_0^2-\frac12 d_*$ (so $\alpha(s)\le a(s)\le b(s)\le\beta(s)$ for all s as in (C)(i)), and it *aborts* unless the following balls are certainly positive, resp. certainly below $F_{\max}$: α^2 (the ball $a_0^2-\frac12 d^*$ contains $\alpha(s)^2$ for every s, so $a(s)^2\ge\alpha(s)^2>0$, which is (H1)), $h_R+\alpha$ (so $h_R+a\ge h_R+\alpha>0$, (H2), and also the hypothesis of (C)(i)), f_R, and $f_R+\tilde\kappa E_R/(\alpha+h_R)$ (an upper bound for $f_R+\tilde\kappa E_R/(a+h_R)$, so (H3)). The left pole is treated the same way with $a-h_L$ and $f_L-\tilde\kappa E_L/(\alpha-h_L)>F_{\mathrm{lo}}$. Since $\alpha\le a_0$ (because $d^*\ge0$, as $0\in D$), $h_R+a_0\ge h_R+\alpha>0$, and likewise $a_0-h_L\ge\alpha-h_L>0$ at the left pole, while $1+\frac12 Q_\pm^{\mathrm{tr}}=a_0^2\ge\alpha^2>0$; hence $\mathbf{X}_k$ lies in the set Ω_0 of Lemma D.6(c) and Φ_0 is C^1 near $\mathbf{X}_k$. Hence, by Lemma D.3(a) and Proposition D.10(C)(ii) ($B=\mathbf{X}_k$ is compact; $F_{\max},F_{\mathrm{lo}},F_{\mathrm{hi}}$ are constants), $\mathbf{X}_k\subset\mathcal{S}$, Φ is continuous and $\Phi=\Phi_0+\mathsf{Z}$ with $\mathsf{Z}(s)\in\mathbf{T}_k$, where $\mathbf{T}_k$ is the box assembled from the formulas of Proposition D.10(C)(i), (A3), (B3): $\mathsf{Z}_2\in-\frac{\varepsilon_R}{2}[0,\tau_{\mathrm{hi}}(f_R+f_{\mathrm{hi}})]$, $f_{\mathrm{hi}}=\sup f_R+\tau_{\mathrm{hi}}$, $\mathsf{Z}_1\in-\frac{\varepsilon_L}{2}\{\varphi^2-f_L^2:\varphi\in[\inf f_L-\tau_{\mathrm{hi}}^-,\sup f_L]\}$, and so on. This proves (a). *Step 3* (*Jacobian and Krawczyk*). The pairs $(u_R,\partial_s u_R)$, $(u_L,\partial_s u_L)$ are carried through the expressions of Φ_0 in the arithmetic of Lemma D.3(b) (the class `Dn` of `tails.py`; f_0 with gradient e_2), which by Lemma D.3(b) and Lemma D.6(c) encloses $\Phi_0(s)$ and $\mathrm{D}\Phi_0(s)$ for all $s\in\mathbf{X}_k$: this gives $\mathbf{J}_k$ and $\mathbf{V}$ (from the degenerate box). The matrix C is any real matrix (Lemma D.5 does not ask more). The program computes $\mathbf{K}_k=\bar{s}_k-C(\mathbf{V}+\mathbf{T}_k)+(I-C\mathbf{J}_k)[-r,r]^4$ in ball arithmetic and checks $\bar{s}_{k,i}-r<\inf\mathbf{K}_{k,i}$ and $\sup\mathbf{K}_{k,i}<\bar{s}_{k,i}+r$ for $i=1,\dots,4$. Lemma D.5(a) gives the zero $s_k^*\in\mathbf{K}_k$; its distance to $\bar{s}_k$ is at most $Y+Z_1 r\le(1.1\cdot10^{-4}+2.5\cdot10^{-14})\,r<1.1\cdot10^{-23}$. *Step 4* (*enclosures*). $J(s)=f(\infty;s)-f(-\infty;s)=[f_+^{\mathrm{tr}}(s)+\mathsf{Z}_{f,+}(s)]-[f_-^{\mathrm{tr}}(s)+\mathsf{Z}_{f,-}(s)]$ with $\mathsf{Z}_{f,\pm}$ in the enclosures of (C)(i); the program evaluates this with Lemma D.3(a). Likewise $f(-\infty;s)\in\mathbf{f}_-^{\mathrm{tr}}+\mathbf{T}_{f,-}$. This gives (c); the same computation shows that the enclosure of $f(-\infty)$ contains 0, as it must at a zero of Φ. *Step 5* (*disjointness*). The program compares the centres with ball arithmetic (`summarize.py`): $|h_0^{(1)}-h_0^{(2)}|=0.645\dots$, $|h_0^{(1)}-h_0^{(3)}|=0.917\dots$, $|h_0^{(2)}-h_0^{(3)}|=0.272\dots$, each certainly larger than $2r$; two points of boxes of radius r around centres at distance $>2r$ in one coordinate cannot coincide. □

What the output says about the reading of Theorem 9.68. Proposition D.11, Lemma D.6(d) and Lemma 9.71 give exactly the four certificate statements (C1)–(C4) of Section 9.8; the proof of Theorem 9.68 there uses them as follows: (a) of the

pole	k	certified $[a_{\mathrm{lo}}, b_{\mathrm{up}}] \ni \mu$	margin $\geq$	$\tau \leq$
right	1	$0.25 \pm 2.5 \cdot 10^{-17}$	0.500003	$2.98 \cdot 10^{-6}$
right	2	$0.25 \pm 5.9 \cdot 10^{-18}$	0.499999	$2.30 \cdot 10^{-4}$
right	3	$0.25 \pm 1.6 \cdot 10^{-18}$	0.499978	$2.13 \cdot 10^{-3}$
left	1	[1.49999999984, 1.50000000009]	2.9999999995	$5.2 \cdot 10^{-16}$
left	2	[1.49999999984, 1.50000000009]	2.9999999995	$1.8 \cdot 10^{-15}$
left	3	[1.49999999984, 1.50000000009]	2.9999999995	$3.4 \cdot 10^{-15}$

TABLE D.2. The squeezing constants of Proposition D.10 over the boxes $\mathbf{X}_k$: the certified lower bound a_{lo} for a and upper bound b_{up} for b at the two poles, so that $\mu \in [a_{\mathrm{lo}}, b_{\mathrm{up}}]$, with $\varepsilon_R = 2.6102791 \cdot 10^{-23}$, $\varepsilon_L = 3.7751345 \cdot 10^{-11}$. The values at a zero of Φ, $\mu_R = \frac{1}{4}$ and $\mu_L = \frac{3}{2}$ (Lemma D.6(e)), lie inside. The left enclosure is wider because D is of the size $\varepsilon_L G$ and $\varepsilon_L \gg \varepsilon_R$. The margin is $h_R + \alpha$ at the right pole and $\alpha - h_L$ at the left, which (H2) needs positive; the last column is the bound $\tau = \tilde{\kappa}^* E/(\alpha + h)$ (resp. $/(\alpha - h)$) for the change of f on the tail (`pole_constants.py`).

proposition gives $\mathbf{X}_k \subset \mathcal{S}$ and the splitting $\Phi = \Phi_0 + \mathsf{Z}$ of Step 1; (b) gives the zero of Step 2; (c) the enclosures of J_k; (d) distinctness.

D.6.1. Reproduction record. The certificate is reproduced by one command, `make -C code/appD_cap reproduce`, which stages a copy of the solver, runs `run_all.sh` (the computer-algebra identity checks, the regression tests, the toy computations, the three Krawczyk stages with stages 1 and 2 in parallel, the summary, and the sanity script of Numerical observation D.12), and compares the hash. It was run on a machine with 2 CPUs under Linux (kernel 6.18, x86_64) with

Python 3.13.18, python-flint 0.9.0 (Arb), mpmath 1.3.0, numpy 2.5.3, scipy 1.18.1, sympy 1.14.0, `PYTHONHASHSEED=0`.

The total wall-clock time was 284 s (4 min 44 s); stages 1, 2, 3 (the programs `prove.py`) took 56.9 s, 58.8 s and 40.5 s of wall-clock time, the first two running in parallel. A second complete run through the command supplied with the book (`make -C code/appD_cap reproduce`), with other jobs on the machine, took 293 s and gave the same hashes. All identity checks, the tests and the toy computations passed; the three Krawczyk tests passed; the boxes are pairwise disjoint. The result files `cert_1.json`, `cert_2.json`, `cert_3.json` contain two wall-clock fields (`time`, `total_time`) that change from run to run; with these removed, serialising with sorted keys and no white space, the SHA-256 hashes are

```
cert_1.json  4ee0cf90156ac62bafca581ea9191421
             ed9affac472e3eeec85a731dc1422bae
cert_2.json  77cbbc5f861ae1d8406ba29419ee28c0
             8924bad95ffcf44b72f67f0604ab80c7
cert_3.json  0ef3d517db8350c567ea40c56627acea
             10c47e24d76decd8f9bdce95f8501304
combined     6839ccbd6dacea72489d00d237c267f4
             c06367e094b0039a60c030c94bcf3d69
```

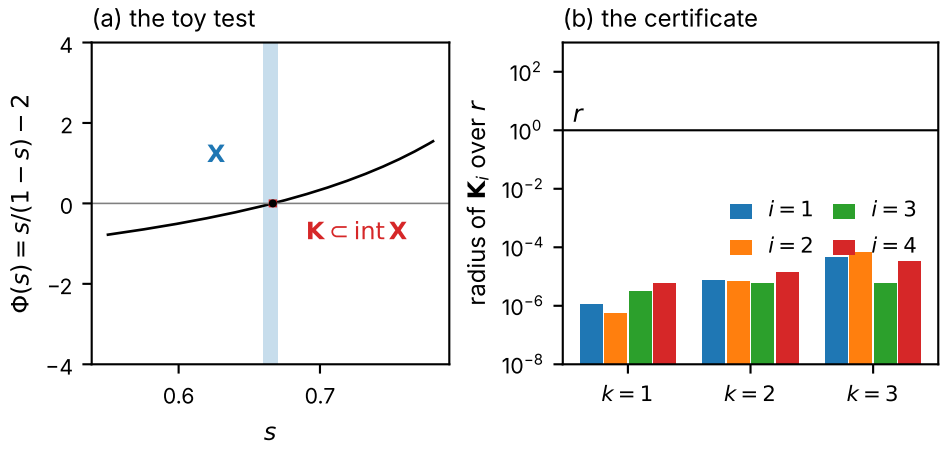


FIGURE D.2. (a) The one-dimensional Krawczyk test of Toy example D.4: the input box $\mathbf{X}$ (blue) and the image $\mathbf{K} \subset \operatorname{int} \mathbf{X}$ (red, 10^{-4} wide) around the zero $s = \frac{2}{3}$. (b) The same test for the three solutions of Theorem 9.68: the radii of the four components of the Krawczyk image $\mathbf{K}_k$ in units of the box radius r, on a logarithmic scale; the test needs them below 1. Code: `code/appD_cap/make_appD_figs.py`.

where "combined" is the SHA-256 of the three canonical byte strings concatenated. The hash of the fresh run is identical to the recorded one, and all other files of the run (the tables, the summary, the sanity output) agree with the recorded ones key by key. The files as written (timings included) differ between runs, so their hashes are not comparable; the centre files `sbar_k.json` are inputs and are not hashed as results.

Numerical observation D.12 (sanity checks of the program)**.** *These tests exercise the code; they are not part of the proof.* (1) Exact solution. *For* $\tilde{\kappa} = 0$, $\lambda = f_0^2$, $Q \equiv 0$ *the system has the exact solution* $h = \tanh(\xi + c)$, $c = \operatorname{artanh} h_0$, $E = \cosh^2 c / \cosh^2(\xi + c)$ *with exact derivatives in* h_0*; the enclosures at* $\xi = \pm 3$ *contain them, with radii between* $2 \cdot 10^{-40}$ *and* $2 \cdot 10^{-36}$ (`tests.py`). (2) Independent integration. *For solution* 1*, the midpoints agree with a double-precision DOP853 integration to a relative difference* $1.1 \cdot 10^{-11}$ *on* $[0, 12]$ *and* $3.8 \cdot 10^{-13}$ *on* $[-8, 0]$*, and the enclosed Jacobian with central finite differences to* $1.2 \cdot 10^{-6}$ *and* $9.4 \cdot 10^{-8}$ (`tests.py`). (3) Cut independence. *For solution* 1*, the tail enclosures of* $f(+\infty)$, $Q(+1)$, $f(-\infty)$, $Q(-1)$ *obtained with right cuts* $X_R = 7, 8, 10, 14$ (*left cut* $\min(X_R, 10)$) *all contain the midpoints obtained with the cuts* 26 *and* 14 (`numobs44.py`)*. The certified half-width of the enclosure of* $f(+\infty)$ *is* $1.84 \cdot 10^{-8}$, $1.24 \cdot 10^{-10}$, $8.4 \cdot 10^{-13}$, $5.6 \cdot 10^{-15}$ *at* $X = 10, 12, 14, 16$*, always* 7.11 *times the true truncation error, and decays like* $e^{-2.5X}$ ($2.5 = 2\mu_R + 2$*: the width is of the size of* $E_R \approx e^{-2\mu_R X}$ *times the width of* D*, which is of the size of* $\varepsilon = e^{-2X}$). (4) Tails against a long integration. *In double precision, for the first solution, the true corrections* $f(+\infty) - f^{\mathrm{tr}}$ *and* $Q(1) - Q^{\mathrm{tr}}$ *computed from an integration to* $\xi = 70$ *lie inside the intervals of Proposition D.10(A),*

Statement	Status	Basis
Φ has a zero in each box $\mathbf{X}_k$, $k = 1, 2, 3$	proved	Proposition D.11(b), Lemma D.5
the three zeros give three distinct solutions with data D^* in class N_0	proved	Theorem 9.68, Lemma D.6(d)
enclosures of J_k and $f_k(-\infty) = 0$	proved	Proposition D.11(c)
each solution is of type G at both poles, with $\mu_R \in 0.25 \pm 2.5 \cdot 10^{-17}$, $\mu_L \in 1.5 \pm 1.6 \cdot 10^{-10}$	proved	Proposition D.10, Table D.2
the zero of Φ in $\mathbf{X}_k$ is unique	**not claimed**: Φ is only enclosed, not differentiated, on the tails	Lemma D.5 gives existence
there are *exactly* three solutions with data D^*	**not proved**	would need a global exclusion over all Cauchy data in $\mathbb{R}^4$ (Open problem 13.16)
the fixed points of T_{D^*} are nondegenerate; the eigenvalues quoted in Chapter 9	**not certified**	the spectrum of $\mathrm{D}T_{D^*}$ is not enclosed
three solutions at nearby data, and the fold values 1300.03, 1462.16 of κ_{D}	**not certified**	numerical observation (Chapter 9); persistence of nondegenerate solutions follows from Corollary 9.31 if the fixed points are nondegenerate

TABLE D.3. Scope of the certificate.

(C)(i) at the cuts $X = 8, 10, 12, 14$ (*e.g.* $-3.8 \cdot 10^{-7} \in [-3.2 \cdot 10^{-6}, 2.2 \cdot 10^{-6}]$ *at* $X = 8$*, Toy example D.9), and the corrections* $f(-\infty) - f^{\mathrm{tr}}$ *lie inside the left-pole intervals at* $X_L = 4, 6, 8, 10$ (`verify_appD.py`*, part 3; beyond* $X = 14$ *the enclosures are narrower than the double-precision rounding error of* f*).* (5) Other settings. *The certificate was also run with* 128 *bits and degree* 16*, and with* 256 *bits and degree* 32 *and cuts* 30*,* 14*; all three Krawczyk tests passed in each case.* (*These runs are not part of the supplied command.*)

D.7. What is, and what is not, proved

Table D.3 summarises the scope of the certificate.

The one thing a reader who does not trust the computer must do is to re-run the program on the three boxes (Appendix D.6.1); the inequalities it checks are those displayed in Proposition D.10, Theorem D.8 and Lemma D.5, and the centres are the numbers of Appendix D.6. The sanity checks of Numerical observation D.12 show that the code agrees with exact solutions, with an independent solver and with long integrations, which is evidence about the code and does not enter the proof.

APPENDIX E

Code, data and the repository

In words. *The book ships with a directory* `code/` *that contains everything needed to regenerate its figures and to rerun every computation whose result is quoted: one directory for each chapter and each appendix, a small shared module of conversions, one directory for the computer-assisted proof of Theorem 9.68, and a frozen copy of the underlying solvers that generated the stored data. Three commands do the work:* `make -C code figures` *redraws all figures,* `make -C code checks` *reruns all verification scripts, and* `make -C code all` *does both and rebuilds the tables. This appendix describes the layout, the software on which the scripts were run, the running times, and which script produces which figure and which number.*

Roadmap. Appendix E.1 describes the layout of `code/`. Appendix E.2 lists the targets of the `Makefile`. Appendix E.3 gives the dependencies, measured on the machine on which the book was produced, and Appendix E.4 the running times. Appendix E.5 is the table of figures and scripts, Appendix E.6 explains how to find the script behind a number, and Appendix E.7 records the certificate reproduction. The numerical methods themselves are described in Appendix C.

E.1. The layout of the repository

The repository is the directory `code/` next to the sources of the book. It contains the following (Table E.1).

Conventions. A figure script writes one or more PDF files to `figures/`, named `chNN_<name>.pdf` or `appX_<name>.pdf` after the directory in which the script lives; the caption of each figure in the book names its script. Most figure scripts draw at width 5.0 in with fonts of 8 to 9 pt (a minority of multi-panel figures are drawn wider, up to about 13 in, and are scaled to the text width, so that their fonts print smaller than nominal). A verification script prints one line for each number or identity it checks; the numerical observations of the book quote these lines, and name the script and, where the script has parts, the part. The scripts are run as `python3 -I`, that is, in isolated mode (no user site directory, no environment variables, no script directory on the path), so that the only code that is imported is the standard library, the packages of Table E.3, and modules that the script itself puts on the path.Expensive computations are stored: `data_*.json` next to the scripts of Chapters 7, 11 and 5, a `_cache/` directory in Chapter 9 and a `cache/` directory in `appD_cap/`. The stored files are the output of the scripts that are named in the README of the directory and can be recomputed (Appendix E.2); removing the cache by `make clean` forces recomputation.

The directory `legacy/`. It holds the underlying solvers, in 15 subdirectories, 805 files plus four documentation files (`STATUS.md`, `PATCHES.md`, `3D_CERTIFICATE.txt` and the table `MANIFEST.tsv` itself), 809 in all; `legacy/MANIFEST.tsv` lists the

path	content
`Makefile`	the top-level driver (Table E.2)
`ch01_intro`, ..., `ch13_open`	one directory per chapter: the figure scripts `make_<fig>.py` (a letter `a`, `b`, `c` after `make_` marks the part of the chapter), the verification scripts `verify_*.py`, helper modules whose names begin with an underscore, the stored data `*.json`, a `Makefile` and a `README.md`
`appA_background`, `appB_special`, `appC_numerics`	the toy computations of Appendix A, the checks of the closed forms of Appendix B, and the measurement of Numerical observation C.1
`appD_cap`	the computer-assisted proof of Theorem 9.68: scripts for the figures and for the checks of Appendix D, the hash program `cert_hash.py`, a cache of the long runs
`common`	`swirl_conv.py`, the conversions between the normalisations of κ and between the data $\boldsymbol{c}$ and $\boldsymbol{c}'$ (Chapter 2); no solver, no integration, no input or output; and a test suite `tests/` with exact rational arithmetic
`legacy`	the underlying solvers: a frozen copy of the programs that generated the stored data; it is unedited, and the scripts of the book never write into it (Appendix E.1)
`build`	created on demand: private working copies of parts of `legacy/`, in which the solvers that compile C kernels or write caches are run; removed by `make clean`

TABLE E.1. The layout of `code/`.

SHA-256 hash of each of the 805 files. The subdirectory names (`0A`, `2C`, ...) are the labels of the stored computations. It is a frozen copy of the solvers used to generate the stored data and the data themselves, and it is *unedited*: nothing in the book changes it, and the hashes of its files in `MANIFEST.tsv` are those of the files, except for three files that carry a one-line path adjustment recorded in `PATCHES.md` (a clean tree contains no generated byte-code or cache files; `make clean` removes them if a script has written any). The figure and verification scripts of the chapters are thin wrappers that import a solver or call a script from `legacy/` in a private copy (a temporary directory, or `build/`), so that compiled kernels and caches are written there and, with byte-code writing disabled (`PYTHONDONTWRITEBYTECODE=1` in the Makefile), never into `legacy/`. The solvers of the book that are not wrappers are the ones listed in Appendix C: shooting in ξ (subdirectories named `0A`, `2C`, `4A`, `4B`), Jost solutions and the fixed-point solver for T_D (`3A`, `3B`, `P1`), the germ code of Chapters 6 and 12 (`0B`, `4C`) and the interval code of the certificate (`3D`: `lohner.py`, `tails.py`, `krawczyk.py`, `prove.py`). The subdirectory `3D` also holds the recorded output of the certificate, which Appendix E.7 compares with a fresh run.

E.2. The Makefile targets

The targets `legacy-*`, `reproduce`, `ch12c-glob` and `ch11b-data` are *not* part of `all`. Everything else in the book is, including all figures and all verification scripts

command (from the top directory)	effect
`make -C code figures`	regenerates every figure in `figures/` (calls the target `figures` in each directory)
`make -C code checks`	`pytest` on `common/tests`, then the verification scripts of every directory, then two computer-algebra checks of the underlying code (`0B`, `3B`); stops at the first failure
`make -C code tables`	recomputes the stored tables of the directories that have any (Chapter 9: the cached branches)
`make -C code all`	`checks`, `figures`, `tables`, in this order
`make -C code clean`	removes caches, `build/` and any byte-code or `numba` cache files in `legacy/`
`make -C code/chNN_key figures` (or `checks`)	the same for one chapter; the chapter makefiles of Chapters 5, 7, 10, 11, 12 offer finer targets such as `ch07b-checks`
`make -C code/appD_cap reproduce`	reruns the whole certificate of Theorem 9.68 and compares the hash (Appendix E.7); about 5 min
`make -C code/ch12_thresh ch12c-glob`	the globality run of Chapter 12 (Atlas 12.77); about 10 min; not part of `checks`
`make -C code/ch11_small ch11b-data`	recomputes the stored heavy runs `data_b_grid.json`, `data_b_misc.json` of Chapter 11; about 10 min; the stored files are used otherwise
`make -C code legacy-<dir>`	the own entry point of one subdirectory of `legacy/` in a staged copy in `build/<dir>/` (hours for some); `legacy-<dir>-quick` is a fast subset; `legacy-quick` runs all the quick subsets

TABLE E.2. The targets of the repository.

quoted in the numerical observations, except the certificate of Theorem 9.68, which is rerun by `reproduce` only. The target `checks` of `appD_cap` uses the recorded certificate (`cert_k.json`) and does not recompute it.

E.3. Dependencies

The versions in Table E.3 are those of the machine on which every number of this appendix was produced (measured with `python3 -V` and `pip show`), and the same as the software recorded in Appendix D.6.1. No LaTeX installation is needed to run the code, and no network access.

The machine has 2 CPUs (Intel Xeon, 2.1 GHz) and runs Linux (kernel 6.18, x86_64). Several scripts start two worker processes. The directory `code/` has about 28 MB, of which `legacy/` is 25 MB, plus about 3 MB of `build/`. The scripts were written for these versions; other recent versions are expected to work, but the digits of floating-point results that are quoted may then differ in the last place (Appendix C.10).

package	version	required for	remark
Python	3.13.18	everything	`python3 -I`
`numpy`	2.5.3	everything	
`scipy`	1.18.1	ODE integration (`solve_ivp`, DOP853), quadrature, `optimize`, sparse eigen- and linear solvers	
`mpmath`	1.3.0	closed forms (30 digits), Appendix B, Chapters 3, 6, 11	
`sympy`	1.14.0	the identities of every chapter	
`python-flint`	0.9.0	the certificate of Appendix D (Arb)	only in `appD_cap` and `legacy/3D`
`matplotlib`	3.11.2	the figures (backend Agg)	
`numba`	0.68.1	the shooting solver of `legacy/4A`, `4B` (Chapters 5, 11, 12)	the cache is written to a temporary or staged directory, not to `legacy/`
`pytest`	9.1.1	`common/tests`	
`gcc`	13.3.0	C kernels of `legacy/2C`, `3A`, `3C` (Chapters 5, 7, 8, 9)	compiled into private directories

TABLE E.3. Software on which the scripts were run.

E.4. Running times

The times of Table E.4 are wall-clock times of one complete run of `make -C code all` on the machine above, with the stored data and caches present, together with the separate runs of the slow targets. They are approximate (the machine was not otherwise idle for the shorter ones) and are given to show the orders of magnitude.

For orientation: a reader who wants only the figures should run `make -C code figures`, which takes about 10 minutes. The only long verification scripts are those of Chapters 7 and 12; the others together take about 10 minutes.

E.5. Which script produces which figure

Table E.5 lists, for every figure of the book, the script that draws it. Each script is in the directory of the chapter named in the first column. The figure files are named after the directory; for instance `make_a_wall.py` in `ch07_edges` writes `ch07_wall.pdf`.

directory	`checks`	`figures`	remark
`common` (`pytest`)	1 s	–	
`ch02_system`	18 s	4 s	
`ch03_line`	18 s	5 s	
`ch04_pole`	5 s	30 s	
`ch05_global`	1.5 min	7 s	`pointwise.py` about 1 min
`ch06_classN`	2.2 min	2.4 min	
`ch07_edges`	33 min	28 s	`verify_ch07a.py` 8.5 min, `verify_ch07b.py` 24 min; they write `data_*.json`
`ch08_effective`	5 s	12 s	
`ch09_uniq`	27 s	2.5 min	with the cached branches (`_cache/`) present
`ch10_large`	1.2 min	45 s	
`ch11_small`	24 s	9 s	with the stored `data_b_*.json`
`ch12_thresh`	7.7 min	2.0 min	`verify_ch12a.py` 3.1 min, `verify_ch12b.py` 3.4 min, `verify_ch12c.py` 1.2 min
`appA`, `appB`, `appC`	3 s, 9 s, 8 s	1 s	
`appD_cap`	2 s	1 s	with the cached long run (`cache/`); longer the first time
two checks of `legacy/`	6 s	–	
`make -C code all`	49 min	10 min	total 58 min 43 s
`appD_cap reproduce`	4 min 0 s		separate target
`ch12c-glob`	about 10 min		separate target; not run for this table
`ch11b-data`	about 10 min		separate target; not run for this table

TABLE E.4. Running times on 2 CPUs. The two columns are the targets of the same names in each directory.

TABLE E.5. The figure scripts.

directory	script	figures (without extension)
`appD_cap`	`make_appD_figs.py`	`appD_enclosure`, `appD_krawczyk`
`ch02_system`	`make_landau.py`	`ch02_landau`
	`make_qsw.py`	`ch02_qsw`
`ch03_line`	`make_noswirl_window.py`	`ch03_noswirl_window`
	`make_pt_toy.py`	`ch03_pt_toy`
	`make_riccati.py`	`ch03_riccati_toy`

directory	script	figures (without extension)
ch04_pole	make_a_green_check.py	ch04_green_check
	make_a_landau_family.py	ch04_landau_family
	make_a_pole_types.py	ch04_pole_types
	make_a_toy_euler.py	ch04_toy_euler
	make_qedge_small.py	ch04_qedge_small
	make_toy_rchart.py	ch04_toy_rchart
ch05_global	make_a_slice.py	ch05_slice_Akappa
	make_a_toy_kamke.py	ch05_toy_kamke
	make_b_bounds.py	ch05_bounds
	make_b_extra.py	ch05_edge_scan, ch05_lens_toy
	make_b_lens.py	ch05_lens
ch06_classN	make_a_envelope.py	ch06_envelope, ch06_normalization, ch06_weights
	make_b_Bset.py	ch06_Bset
	make_b_cylinder.py	ch06_cylinder
	make_b_gluing.py	ch06_gluing
	make_b_threshold.py	ch06_threshold
ch07_edges	make_a_mech.py	ch07_mech
	make_a_tmap.py	ch07_tmap
	make_a_wall.py	ch07_wall
	make_b_attach.py	ch07_attach
	make_b_interval.py	ch07_interval
	make_b_layer.py	ch07_layer
	make_b_omega.py	ch07_omega
ch08_effective	make_DTsign.py	ch08_DTsign
	make_iteration.py	ch08_iteration
	make_profiles.py	ch08_profiles
	make_wzero_s.py	ch08_wzero_s
ch09_uniq	make_a_foldnormal.py	ch09_foldnormal
	make_a_secular.py	ch09_secular
	make_b_cap.py	ch09_cap
	make_b_classZ.py	ch09_classZ
	make_b_ridge.py	ch09_ridge
	make_b_scurve.py	ch09_scurve
ch10_large	make_a_sqrtlaw.py	ch10_sqrtlaw
	make_a_tracking.py	ch10_tracking
	make_b_atlas.py	ch10_atlas
	make_b_layer.py	ch10_layer
	make_b_rank.py	ch10_rank
ch11_small	make_a_eigen.py	ch11_eigen
	make_a_layers.py	ch11_layers
	make_a_rate.py	ch11_rate

directory	script	figures (without extension)
	`make_b_cut.py`	`ch11_cut`
	`make_b_exponents.py`	`ch11_exponents`
	`make_b_laws.py`	`ch11_laws`
`ch12_thresh`	`make_a_kmax.py`	`ch12_kmax`
	`make_a_psihat.py`	`ch12_psihat`
	`make_b_large_cone.py`	`ch12_largecone`
	`make_b_lens.py`	`ch12_lens`
	`make_c_atlas.py`	`ch12_c_gamma`, `ch12_c_ratio`
	`make_c_profiles.py`	`ch12_c_profiles`
	`make_c_zones.py`	`ch12_c_zones`
`appA_background`	`toy_fold.py`	`appA_fold`

E.6. Which script produces which number

Every numerical observation and every atlas of the book ends with, or contains, the name of the script that prints its numbers, and the scripts print them in the order and with the labels in which the text quotes them. Table E.6 gives the verification scripts, the sections they serve and their content.

TABLE E.6. The verification scripts.

script	serves	content
`ch02_system/sympy_derivation.py`	Ch. 2	derivation of the reduced system and of the identities of the chapter, Landau family
`ch03_line/verify_ch03a.py`, `verify_ch03b.py`	Ch. 3	identities and toy computations of Sections 3.1–3.9 (Riccati, Pöschl–Teller, reduction of order)
`ch03_line/verify_hyp_window.py`, `verify_table_cases.py`	Ch. 3	Numerical observation 3.49; the table of the no-swirl cases
`ch04_pole/verify_a_ch04.py`, `verify_ch04b.py`	Ch. 4	identities of Sections 4.2–4.5; Atlas 4.50 (loops, constants)
`ch05_global/verify_ch05a.py`, `verify_ch05b.py`, `pointwise.py`	Ch. 5	identities of Sections 5.1–5.4; Atlas 5.73; Numerical observation 5.34
`ch06_classN/verify_ch06a.py`, `verify_ch06b.py`	Ch. 6	kernel and weight identities; Atlas 6.65 (writes `ch06b_numbers.json`)
`ch07_edges/verify_ch07a.py`, `verify_ch07b.py`	Ch. 7	Atlas 7.39, Atlas 7.93 and the numerical observations of Sections 7.1–7.9
`ch08_effective/verify_ch08a.py`, `verify_ch08b.py`	Ch. 8	the derivative formula (8.6) against finite differences; Numerical observations 8.36 and 8.58

script	serves	content
`ch09_uniq/verify_ch09a.py`, `verify_ch09b.py`, `secular_check.py`	Ch. 9	sign structure; Atlas 9.76, Numerical observation 9.74; the secular function
`ch10_large/verify_ch10a.py`, `verify_ch10b.py`	Ch. 10	lemmas of Sections 10.1–10.4 on computed solutions; Atlas 10.48
`ch11_small/verify_ch11a.py`, `verify_ch11b.py`	Ch. 11	the bubble (Numerical observation 11.13 ff.); Atlas 11.62; prints "ALL CHECKS PASSED"
`ch12_thresh/verify_ch12a.py`, `verify_ch12b.py`, `verify_ch12c.py`	Ch. 12	Part A (Numerical observations 12.19 and 12.20); Part B; Part C (Atlas 12.77)
`appA_background/toy_*.py`	App. A	the eight toys (each prints PASS)
`appB_special/check_appB.py`	App. B	Numerical observation B.20
`appC_numerics/verify_appC.py`	App. C	Numerical observation C.1
`appD_cap/toy_hand.py`, `verify_appD.py`	App. D	the hand toys; Numerical observation D.12 and the numbers of Appendix D
`common/tests`	Ch. 2	the conversions between the normalisations of κ and between $\boldsymbol{c}$ and $\boldsymbol{c}'$

Two kinds of numbers are *not* printed by a script in the same run. (1) Numbers that come from a long run are read from stored files by the verification script (for instance `data_b_grid.json` in Chapter 11, the cached branches in Chapter 9, `ch06b_numbers.json`); the file is itself produced by a script of the same directory. (2) The certificate of Theorem 9.68 is computed in `legacy/3D` and read by `appD_cap/verify_appD.py`. In both cases the script that quotes the number says where it reads it from.

E.7. The certificate reproduction

The command `make -C code/appD_cap reproduce` stages a copy of the certificate code from `legacy/3D` in `build/3D`, runs its driver `run_all.sh` (identity checks, tests, toy computations, three Krawczyk stages with the first two in parallel, summary, sanity script), strips the wall-clock fields from the result files, and compares the SHA-256 hash of the canonical serialisation with the recorded one. On the machine of Appendix E.3 the run took 240 s (4 min), and the output was:

```
cert_1.json  4ee0cf90156ac62bafca581ea9191421  match
             ed9affac472e3eeec85a731dc1422bae
cert_2.json  77cbbc5f861ae1d8406ba29419ee28c0  match
             8924bad95ffcf44b72f67f0604ab80c7
cert_3.json  0ef3d517db8350c567ea40c56627acea  match
             10c47e24d76decd8f9bdce95f8501304
combined     6839ccbd6dacea72489d00d237c267f4  match
             c06367e094b0039a60c030c94bcf3d69
```

This is the hash recorded in Appendix D.6.1; the three Krawczyk tests pass, and the three boxes are pairwise disjoint. The directory `legacy/3D` is not changed by the run.

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